

MIAMI DADE COLLEGE

2026-2028



CATALOG



Volume XLIX

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PURPOSE OF CATALOG

The Miami Dade College Catalog is a guide to the College's academic programs, courses, degree and certificate requirements, student services, and policies. It helps students understand their educational options, plan their studies, and stay informed about College requirements. Students are encouraged to use the Catalog along with academic advising and other support services to make informed decisions and stay on track toward completing their academic and career goals. By becoming familiar with College expectations and requirements, students can take full advantage of the opportunities available at Miami Dade College and prepare for successful graduation.

Accreditations

Miami Dade College is accredited by the [Southern Association of Colleges and Schools Commission on Colleges](http://www.sacscoc.org) to award associate and baccalaureate degrees. Miami Dade College also may offer credentials such as certificates and diplomas at approved degree levels. Questions about the status of Miami Dade College may be directed in writing to the Southern Association of Colleges and Schools Commission on Colleges at 1866 Southern Lane, Decatur, GA 30033-4097, by calling (404) 679-4500, or by using information available on SACSCOC's website (www.sacscoc.org).

In addition to institutional accreditation by SACSCOC, a number of Miami Dade College programs and facilities are accredited by nationally recognized specialized accrediting agencies charged with upholding the standards of the professions they represent. Specialized accreditors of MDC programs include:

Accreditation Commission for Education in Nursing (ACEN)

Accreditation Review Commission on Education for the Physician Assistant (ARC-PA)

Accreditation Review Council on Education in Surgical Technology and Surgical Assisting (ARC/STSA)

American Bar Association (ABA), Standing Committee on Paralegals

American Board of Funeral Service Education, Committee on Accreditation (ABFSE)

American Dental Association, Commission on Dental Accreditation (CODA)

American Veterinary Medical Association, Committee on Veterinary Technician Education and Activities (CVTEA)

Commission on Accreditation for Health Informatics and Information Management Education (CAHIIM)

Commission on Accreditation for Respiratory Care (CoARC)

Commission on Accreditation in Physical Therapy Education (CAPTE)

Commission on Accreditation of Allied Health Education Programs (CAAHEP)

Committee on Accreditation of Educational Programs for the Emergency Medical Services Professions (CoAEMSP)

Commission on Collegiate Nursing Education (CCNE)

Commission on Opticianry Accreditation (COA)

Joint Review Committee on Education in Diagnostic Medical Sonography (JRC-DMS)

Joint Review Committee on Education in Radiologic Technology (JRCERT)

Joint Review Committee on Educational Programs in Nuclear Medicine Technology (JRCNMT)

National Accrediting Agency for Clinical Laboratory Sciences (NAACLS)

National Association for the Education of Young Children (NAEYC)

Society for Simulation in Healthcare (SSH)

Certain programs require approval or certification from governmental oversight agencies, including:

Federal Aviation Administration (FAA)

Florida Board of Nursing

Florida Department of Health, Bureau of Emergency Medical Services

Florida Department of Law Enforcement, Criminal Justice Standards and Training Commission (CJSTC)

Florida Real Estate Commission, Department of Business and Professional Regulation - Division of Real Estate

Professional Organizations and Association Memberships

American Association of Collegiate Registrars and Admissions Officers (AACRAO)

American Association of Community Colleges (AACC)

American Association of Hispanics in Higher Education, Inc. (ACCHHE)

American Association of State Colleges and Universities (AASCU)

American Council on Education (ACE)
 Association of American Colleges and Universities (AAC&U)
 Association of Community College Trustees (ACCT)
 Association of Florida Colleges (AFC)
 Association of Governing Boards of Universities and Colleges (AGB)
 Business-Higher Education Forum (BHEF)
 College Consortium for International Studies (CCIS)
 Council for Higher Education Accreditation (CHEA)
 Council of Foreign Relations (CFR)
 EDUCAUSE -Association of Managing and Using Information Technology in Higher Education
 Florida Association of College Test Administrators (FACTA)
 Florida Association of Colleges and Universities (FACU)
 Florida Campus Compact
 Florida College System Activities Association (FCSAA)
 Florida Consortium for International Education (FCIE)
 Fulbright Association
 Global Community College Leadership Network
 Greater Miami Chamber of Commerce (GMCC)
 Hispanic Association of Colleges & Universities (HACU)
 National Association for Community College Entrepreneurship (NACCE)
 National Association of College and University Attorneys (NACUA)
 National Association of College and University Business Officers (NACUBO)
 National Association of Cuban American Educators (NACAE)
 National Association of Student Financial Aid Administrators (NASFAA)
 National College Testing Association (NCTA)
 National Collegiate Honors Council (NCHC)
 National Community College Hispanic Council (NCCHC)
 National Institute for Staff and Organizational Development (NISOD)
 South Florida Manufacturers Association (SFMA)
 Southern Association of Collegiate Registrars and Admissions Officers (SACRAO)
 The College Board
 The Commission on Colleges and Universities

Requests for review of letters of accreditation may be forwarded to the Office of the Vice Provost for Academic Affairs or Associate Provost for Academic Affairs.

Note: *In addition to the above, Miami Dade College administrators, faculty and staff members participate in numerous other international, national, state and regional organizations. Additional information regarding professional associations may be obtained from the College.*

EQUAL ACCESS/EQUAL OPPORTUNITY

Miami Dade College is committed to providing equal access to education and employment opportunities for students, employees, applicants for admission and employment, and in its activities for the general community. We are an equal access/equal opportunity institution which does not discriminate on the basis of sex, race, color, marital status, age, religion, national origin, disability, veteran's status, ethnicity, pregnancy, sexual orientation, or genetic information. The College's commitment to equal access and equal opportunity is contained in the District Board of Trustees policies and procedures based on the nondiscrimination provisions of federal and state laws and regulations, including the Civil Rights Acts of 1964, as amended; Title IX of the Education Amendments of 1972; Section 504 of the Rehabilitation Act of 1973; the Americans with Disabilities Act Amendments Act of 2008; the Florida Educational Equity Act (§ 1000.05, F.S.); The Jeanne Clery Disclosure of Campus Security Policy and Campus Crime Statistics Act (The Clery Act) as amended in 2008; and The Violence Against Women Act.

In accordance with these protections, Miami Dade College provides equal access/equal opportunity in admissions, recruitment, financial assistance, access to course offerings, participation in extracurricular programs and activities, access to and use of facilities, counseling, housing referral, guidance, advising, health services, athletics, employment and retention of personnel and students.

Responsibility for the implementation of the College's commitment to equal access and equal opportunity rests with the College president.

Consult the offices below for assistance or to obtain detailed information on equal access/equal opportunity:

District Administration

Cindy Lau Evans, Director,
Equal Opportunity Programs/ ADA Coordinator/ Title IX
Coordinator
Miami Dade College
11011 SW 104th St., Room 1102-01
Miami, FL 33176
Phone: (305) 237-2577 (Voice) or 711 (Relay Service);
EquityOff@mdc.edu

North Campus

Office of the Campus President
11380 N.W. 27th Ave.
Miami, FL 33167-3495

Kendall Campus

Office of the Campus President
11011 S.W. 104th St.
Miami, FL 33176-3393

Wolfson Campus

Office of the Campus President
300 N.E. Second Ave.
Miami, FL 33132-2296

Medical Campus

Office of the Campus President
950 N.W. 20th St.
Miami, FL 33127-4693

Homestead Campus

Office of the Campus President
500 College Terrace
Homestead, FL 33030-6009

Padrón Campus

Office of the Campus President
627 S.W. 27th Ave.
Miami, FL 33135

Hialeah Campus

Office of the Campus President
1780 W. 49th St.
Hialeah, FL 33012-2918

West Campus

Office of the Campus President
3800 N.W. 115th Ave.
Doral, FL 33178-4856

To obtain additional information about the College, including an Application for Admission/Readmission, contact any campus Admissions and Registration Office or visit the College's website at www.mdc.edu.

Non-Discrimination Statement

Miami Dade College is an equal access/equal opportunity institution which does not discriminate on the basis of sex, race, color, marital status, age, religion, national origin, disability, veteran's status, ethnicity, pregnancy, sexual orientation or genetic information. To obtain more information about the College's equal access and equal opportunity policies, procedures and practices, please contact the College's Equity Officer:

Cindy Lau Evans
Director, Equal Opportunity Programs/ ADA Coordinator/
Title IX Coordinator (305) 237-2577 (Voice) or 711 (Relay
Service)
11011 SW 104th St Room, 1102-01
Miami, FL 33176
EquityOff@mdc.edu

ACADEMIC CALENDARS

Academic Calendar 2026 - 2028

The Office of the College Registrar maintains several calendars, each of which provides details for a separate category of information. The Academic Calendar is published once each semester. It contains important dates for registration, fees, graduation, and other academic deadlines. For more information, please visit Miami Dade College's Academic Calendar

at <http://www.mdc.edu/academics/calendar>

Academic Calendar

<https://www.mdc.edu/academics/calendar/>

Registration Dates Calendar

<https://www.mdc.edu/academics/calendar/registration-calendar.aspx>

Holidays

<https://www.mdc.edu/academics/calendar/holidays.aspx>

ABOUT MIAMI DADE COLLEGE

Miami Dade College (MDC) offers a wide range of programs designed to meet the needs of Greater Miami. The College offers associate and baccalaureate degree options and a wide range of career and technical certificate and specialized programs. MDC currently offers five Bachelor of Applied Science (BAS) degrees and 12 Bachelor of Science (BS) degrees which are academically rigorous and innovative in Business, Education, Entertainment, Health Science, Nursing, Public Safety, and STEM related fields. The Associate in Arts (AA) degree, designed to prepare students for further study at four-year institutions, includes more than 70 pathways leading to the baccalaureate degree. The College also maintains more than 60 transfer agreements with colleges and universities across the state and country, guaranteeing entry for MDC students who meet the entry criteria. With more than 60 areas of study, the Associate in Science (AS) degree, prepares students for direct entry into the workforce. Our graduates take advantage of the College's numerous partnerships with innovative businesses throughout South Florida. In addition to these degrees, the College offers numerous short-term career and technical certificate programs which are stackable career ladders towards and associate in science degree as well as courses of study to enhance career knowledge through continuing education.

The Open-Door Policy

Miami Dade College's open-door admissions policy provides educational opportunities to community residents and to national and international applicants. Anyone seeking to benefit from the degree or short-term certificate programs, or from the College's student and community services, is encouraged to enroll. The College welcomes all students regardless of gender, race, color, marital status, age, religion, national origin, disability, veteran's status, sexual orientation or genetic information.

Mission Statement

As democracy's college, Miami Dade College changes lives through accessible, high-quality teaching and learning experiences. The College embraces its responsibility to serve as an economic, cultural and civic leader for the advancement of our diverse global community.

Vision Statement

To be the recognized leader in student learning, achievement and success while enriching our community.

Core Values

- An exceptional learning environment that challenges students and empowers them to attain their academic goals
- An international perspective that makes our students civically engaged and globally competitive
- A commitment to evidence-informed decision making and accountability
- Innovation and efficiency that ensure affordability while optimizing educational quality
- An exceptional work environment that engages an exemplary and diverse workforce
- Quality community partnerships that serve as the foundation for the development of relevant workforce, cultural and civic programs
- Cultural initiatives that capture the richness of Miami-Dade County's multicultural fabric
- Environmental awareness that results in intentional sustainability practices

MDC History

Miami Dade College PRESIDENTS

 Kenneth R. Williams Tenure: 1959 – 1962	 Peter Masiko Jr. Tenure: 1962 – 1980
 Robert H. McCabe Tenure: 1980 – 1995	 Eduardo J. Padrón Tenure: 1995 – 2019
 Rolando Montoya Interim President Tenure: 2019 – 2020	 Madeline Pumariega Tenure: 2020 – present

Opening Doors

Miami Dade College opened its doors in 1960 with just under 1,400 students. Dade County Junior College (DCJC), as it was known at the time, was founded amid the era of Civil Rights, desegregation and an influx of asylum-seekers from Cuba. The student body included seven black students who made DCJC the first integrated junior college in Florida, along with Cuban refugees seeking a better life in America. The College was open to any Miami-Dade County resident who had graduated from high school. Tuition was free to all county residents. The institution's first graduating class had fewer than 200 students and by 1963 was being referred to as Dade Junior College. By the mid-1960s, nearly 15,000 students were attending the original North Campus, its buildings bursting at the seams. While new construction got underway at the College's first campus, plans were made for a second campus in Kendall and a third in Downtown Miami. South Campus (Kendall) would open in 1964. With more than 23,000 students by 1967, the College was the fastest growing junior college in the U.S. and the largest institution of higher learning in the state of Florida. During this era of tremendous growth, the College enrolled more freshmen than the University of Florida, Florida State University and the University of South Florida combined. In 1968, the Florida Legislature separated community colleges from the public school system and the College was then governed by its own District Board of Trustees.

Access with Excellence

A philosophy of "access with excellence" defined the 1970s for the College. A bold institutional review reaffirmed its open-door policy, set new academic goals and raised the standard for all community colleges across the country. Dr. K. Patricia Cross, 20th century thought leader and then visiting professor at Harvard University's Graduate School of Education, called the College "the most exciting institution of higher education in the country." The campus in Downtown Miami officially opened in 1970 and would later be renamed for one of the College's founders, Mitchell Wolfson. In 1973, the College changed its name to Miami-Dade Community College (MDCC) to reflect its growing prominence in the community college movement, and in keeping with the county's name change, from Dade to Miami-Dade, which took advantage of the international name recognition of Miami. Medical Center Campus opened its doors in 1976 in the heart of Miami's flourishing medical district as the College's fourth campus. Today, this marvel of medical and health technologies is referred to as Medical Campus. Bilingual studies became a full-

fledged division in 1979, with more than 2,000 students enrolled in outreach centers in Little Havana, making it the largest bilingual program in all of higher education.

Unprecedented Achievement

In the 1980s, burgeoning MDCC reflected the greatly expanding diversity of South Florida. During this period, 30% of the College's students were immigrants; English was not the native language for 46%; two-thirds were minorities; and 56% were women. Part-time students were the norm, as those seeking credentials for greater earning potential continued to work to support themselves and their families. A modest College-sponsored book fest, "Books by the Bay," was held in 1983 at Wolfson Campus, and in 1984, the event officially became Miami Book Fair International. The annual eight-day festival would grow into the largest and most comprehensive community-rooted literary gathering in the United States. Today, Miami Book Fair stands as a cultural institution with a lasting and ever-expanding influence. In 1985, the MDCC became the first community college in the U.S. to graduate 100,000 students. That same year, the College celebrated its 25th anniversary and became nationally recognized when the prestigious University of Texas Community College Leadership Program identified the College as the No. 1 community college in America. New World School of the Arts (NWSA) at MDC opened in 1988 and would be recognized as one of the premier arts high school/college programs in the country. NWSA, in further partnership with Miami-Dade County Public Schools and the University of Florida, provides a comprehensive artistic, creative and academic program, conferring high school diplomas, Associate in Arts degrees, and bachelor's degrees in Fine Arts and Music in programs accredited by the National Association of Schools of Dance, Music, Theater and Art & Design.

Advancement and Opportunity

To kick off the 1990s, the College initiated comprehensive reforms to academic programs and administrative operations. Academic core and electives were modernized for a curriculum to meet the needs of a dynamic community. Advancements were not limited to academics; improvements were also made to human resources, maintenance operations and budgeting. In addition, the College's new Technology Master Plan emphasized both academic and administrative computing. Throughout the decade, the College sought to keep pace with the changing economy and workforce, developing strong partnerships in business and industry. More than 50 new degree and short-term certificate training programs were developed, all aimed at emerging industries and South Florida's job market. In 1990, a fifth campus opened in Homestead with just 350 students. Classes were held in the basement of the First Baptist Church. The next year, a modern campus facility would be built to serve the growing student population in southern Miami-Dade County. The College established its Cultural Affairs department in 1990, tasked with producing and presenting the newest, most challenging contemporary and cultural-specific works with an emphasis on creations reflective of Greater Miami's multi-ethnic community. Also in 1990, the College launched the Cultura del Lobo performing arts series along with visual arts programming and campus galleries. What would later become Live Arts Miami, these performing and visual arts programs are celebrated as the most eclectic arts offerings in Miami, presenting artists from across the globe. In 1992, the College became the first community college in the country to establish an Endowed Teaching Chair program. More than 300 faculty have been recognized with the honor since. In 1995, MDC's aviation school was officially renamed the Eig-Watson School of Aviation after Lois Eig-Watson, once a flight student herself and a former faculty member at Kendall Campus. The school had been a pioneer in aviation education since its beginnings in 1961. Jazz at Wolfson Presents started entertaining audiences in 1998 by showcasing renowned musicians to educate jazz students and to offer a cultural experience for the entire college community and the people of Downtown Miami. The College's sixth campus became a reality in 1998 when the InterAmerican Center in Little Havana was granted campus status by the District Board of Trustees. Virtual College began in 1997, allowing students to take classes remotely. Recognition soon followed when Yahoo! Internet Life, proclaimed the College "WIRED" and voted it "second best of all colleges and universities." The College's information technology efforts also gained residence in the Smithsonian's permanent collection with the nomination for the ComputerWorld Smithsonian Award for Technological Innovation in 1998. Remote learning at the College would greatly expand by the 2020s under a new name, MDC Online.

The New Learning Agenda

The College kicked off the new millennium by enrolling its 1 millionth student and obtaining state approval to offer bachelor's degrees. The College also launched its new "Learning Agenda" to advance contemporary learning models and student support programs, as well as campus, faculty and leadership development efforts. In 2002, the City of Miami turned the historic Tower Theater over to the College to operate as a movie theater and cultural center. Originally opened in 1926, the popular neighborhood theater was the first in the county to offer Spanish subtitles for English-language movies, beginning in 1960, in recognition of the rapidly rising Cuban refugee

population. In 2003, the Florida Legislature formally approved the school's name change to Miami Dade College (MDC). That was also the year that the College began offering four-year degrees. The Hialeah Center became the Hialeah Campus in 2004, MDC's seventh campus. Also in 2004, the College acquired Miami Film Festival from Florida International University. This international film event serves as a major film showcase for world cinema and has become renowned for championing films made by the South Florida community, first-time feature filmmakers and International Feature Film submissions to the Academy Awards. In 2005, MDC awarded its first bachelor's degrees. The Honors College opened to academically gifted students in 2005. Students receive the prestigious Honors College Fellow Award, which provides free tuition and fees as well as a generous stipend for books and other expenses. The Honors College prepares students for transfer to many of the most prestigious colleges and universities in the nation, and was once referred to by TIME magazine as an "Ivy Stepladder." Also in 2005, MDC became the proud steward of the historic Freedom Tower and would become home to the College's Museum of Art and Design (MOAD) and MDC Special Collections. Completed in 1925, the building was originally the home of the Miami Daily News, the city's oldest newspaper. The "Ellis Island of the South" served as the Cuban Assistance Center from 1962 through 1974, offering nationally sanctioned relief to the Cuban refugees seeking asylum. One of those refugees was then-teenager Eduardo J. Padrón, who would later become president of the College in 1995. The Freedom Tower was designated a National Historic Landmark in 2008. West Campus, the youngest of MDC's eight campuses, opened in 2006 in Doral with just 150 students. In 2008, MDC's Medical Campus began offering a Bachelor of Science in Nursing in response to workforce needs in that industry. The School of Nursing had been established in the early years of the College and has graduated more than 23,000 nurses since. In 2012, the school would be renamed the Benjamín León School of Nursing. MDC TV debuted in 2008 as the College's own television network. MDC TV is broadcast to millions of cable TV subscribers throughout Miami-Dade County and has won a total of seven Emmys and 14 nominations from the Suncoast Regional Chapter of the National Academy of Television Arts & Sciences. In 2009, The Carnegie Foundation for the Advancement of Teaching named MDC among the select group of institutions granted its prestigious Community Engagement Classification. Also in 2009, MDC took over the Lynn and Louis Wolfson II Florida Moving Image Archive, the nation's largest state archive of video and film. Opened in 2010, the Miami Culinary Institute (MCI) at MDC is at the epicenter of culinary education excellence. Mixing classic skills and innovative techniques used by the world's best chefs, the program is at the forefront of the industry. MCI has won several awards and distinctions, and has also been home to world-renowned chefs. In 2010, Miami Dade College celebrated its 50th anniversary.

Innovation • Expansion • Purpose

Over the next decade, MDC would take higher education innovation to entirely new levels with groundbreaking initiatives and curricula that would evolve in sync with local industry as well as the global workforce. By 2011, the College was offering more than 20 baccalaureate degrees including a new Bachelor of Science in Biological Sciences. MDC was selected for a Bill and Melinda Gates Foundation Completion by Design grant in 2011 to increase student success and boost graduation rates among low-income students. Also in 2011, the College began offering its unprecedented American Dream Scholarship, which pays for in-state tuition and fees for the first two years of college for all newly graduated high school students who could not otherwise afford it. Graduating seniors must have a minimum 3.0 GPA to qualify. The Koubek Center was acquired by MDC in 2011. Built in 1929, the South Florida landmark has a long and rich history as a cultural hub dedicated to building community through the arts with a 200-seat theater, an expansive garden, classrooms and multi-use spaces. In 2012, the College became the first institution of higher learning in Florida to offer a Bachelor of Applied Science in Public Safety Management. In 2013, the College enrolled its 2 millionth student. The President's Higher Education Community Service Honor Roll is the highest federal recognition a college or university can receive for its commitment to volunteering, service learning and civic engagement, and in 2013, MDC received its seventh consecutive accolade. In 2014, the Student Success Award of Excellence was conferred on MDC by the American Association of Community Colleges (AACC) for advancing diversity. In 2015, MDC opened the new 9,000-square-foot, state-of-the-art Miami Animation & Gaming International Complex (MAGIC). The facility is a working animation studio where students prepare for careers as creative directors, graphic artists, illustrators and 2D and 3D animators. MAGIC offers an intensive, hands-on program where students have the opportunity to work with industry powerhouses such as Nickelodeon and Telemundo. They also gain hands-on experience in augmented and virtual reality (AR/VR). Ashoka U designated MDC as a 2015 Changemaker Campus for its role in promoting social innovation, change-making and thought leadership in the Miami community and abroad. MDC launched its cutting-edge Miami Fashion Institute (MFI) in 2016. Located at the Wolfson Campus in the heart of Downtown Miami, it is the only facility of its kind at a public institution of higher education in Florida, and one of only a handful nationally. Real-world experience meets the classroom with faculty that includes renowned artists, designers and business professionals. MFI offers Associate in Science degrees in both Fashion Design and Fashion Merchandising, as well as a College Credit Certificate in Pattern

Making and Construction. The Idea Center at MDC opened in 2016 as an “industry-agnostic ecosystem of creativity, innovation and entrepreneurship for MDC students.” The Idea Center offers classroom education, professional development opportunities and experiential learning programs to MDC students and the community at large. Also in 2016, MDC acquired the historic David W. Dyer building, which is being restored for academic and civic purposes. Built in 1933, the building is on the National Registry of Historic Places and on the City of Miami’s list of historic places. In 2017 MDC was selected to join the exclusive Frontier Set, a group of higher education institutions aiming to eliminate the racial, ethnic and socio-economic barriers that bar students from completing college. MDC’s business school was renamed in 2018 the Miguel B. Fernandez Family School of Global Business, Trade and Transportation in honor of billionaire entrepreneur and philanthropist Miguel “Mike” B. Fernandez, a longtime friend and advocate of the College. MDC’s Cybersecurity Center of the Americas opened in 2018 to address the demand for cybersecurity professionals. MDC is the only institution of higher learning in Florida with a Cyberbit range, a training facility that provides students and working professionals access to hands-on cybersecurity education and simulated training that mirrors real-life cyberattack scenarios. In 2019, Tesla chose MDC as its first partner in the state of Florida for its Tesla START automotive training program. The 15-week Tesla START program trains students to become Tesla automotive technicians at a new state-of-the-art facility at West Campus, fully equipped with Tesla vehicles and equipment and built to Tesla specifications. Building on its outstanding record, MDC once again made The Chronicle of Higher Education’s Great Colleges to Work For in 2019, marking its 11th consecutive year on the list. MDC opened its new Cloud Computing Center in 2019. The state-of-the-art Center is the only one of its kind in Florida, and offers the state’s only College Credit Certificate and Associate in Science in Enterprise Cloud Computing. It also hosts the College’s first Cloud Computing Academy, which provides training for both MDC students and members of the community. In collaboration with Amazon Web Services (AWS), Microsoft, Azure and Google Cloud, among others, the new Center allows technology leaders and industry experts to engage with students who are learning about leading cloud platforms and receiving industry certifications. In 2019, MDC opened the new Center for Learning, Innovation and Simulation (CLIS) at Medical Campus. The five-floor Center features an entire floor dedicated to advanced medical simulation that facilitates the practice of working in real-world clinical environments such as operating and emergency rooms. CLIS is a one-of-a-kind health training venue in the region and rivals those at top medical schools around the nation. MDC was awarded the 2019 Aspen Prize for College Excellence, the nation’s highest recognition of achievement and performance among state colleges across America. The award is considered the highest and most sought after among American colleges. MDC was also awarded the 2019 Lumina Foundation Education Innovation Judges Choice Prize for its adult learner “earn and learn” Tesla START program. Also in 2019, after the College’s fourth president, Eduardo J. Padrón, announced he was stepping down, the InterAmerican Campus was renamed the Padrón Campus in honor of his role in developing the location as a campus. MDC opened its new cutting-edge BIT Center in 2020. The 6,000-square-foot facility is a cross between a Silicon Valley tech company and a futuristic classroom. The BIT Center examines and promotes the role of technology in digital transformation, process automation and analytics in the ever-changing business landscape. In 2020, Madeline Pumariaga was named the first female president to lead Miami Dade College. Adding to the historic nature of this appointment was that Pumariaga is an alumna of MDC. The School of Engineering and Technology (EnTec) at MDC announced two new degrees in 2021: Bachelor of Science in Information Technology and Bachelor of Science in Cybersecurity. The College has developed more than 50 new or revamped degrees and embedded over 30 industry certifications into tech-related courses for these sought-after skills. In 2021, MDC joined the AI4ALL College Pathways program, a nationwide initiative to foster AI-related careers and to embed AI concepts into all associate degree curricula in coming years. MDC launched Rising Black Scholars in 2021. The trailblazing initiative provides a full-ride scholarship and a host of curated services for two years to students of African descent who are graduates of a Miami-Dade County high school. In 2021, MDC’s award-winning newspaper, *The Reporter*, was named the top college student newspaper in Florida for the 10th year in a row by the Florida College System Publications Association (FCSPA). The print publication was launched in 2010 and is a product of three nowdefunct student newspapers at MDC: The Falcon Times launched in 1961, followed by the Catalyst in 1966 and The Downtowner in 1970, which was renamed the Metropolis — a magazine that still exists today. Through the years, the College’s student newspapers have won 21 National Pacemaker Awards and received hundreds of honors from the FCSPA. The Goldman Sachs 10KSB celebrated its 21st cohort at MDC in 2021. The practical education program provides entrepreneurs with valuable skills for growing their businesses at no cost. Since the program’s inception at MDC in 2013, 10KSB has graduated over 500 small-business owners. In 2021, Excelencia in Education, the nation’s premier authority on efforts accelerating Latino student success in higher education, recognized MDC with its Seal of Excelencia. That same year, the College was also recognized by the U.S. Department of Education as one of the leading institutions for Hispanics in the United States. MDC once again ranked No. 1 in the Florida College System (FCS) for reporting the most students earning industry certifications during the 2021 reporting year. In its continuing efforts to provide greater access, MDC launched the Presidential Scholars Program in 2021.

This initiative offers high-performing graduating high school students in Miami-Dade County the opportunity to complete an associate degree at no cost. MDC established a transfer articulation agreement in 2021 with the University of Miami (UM) that allows MDC graduates with a Bachelor of Science in Data Analytics to seamlessly transfer to a Master of Science in Data Science at UM. According to the 2021 rankings released by U.S. News & World Report, MDC's School for Advanced Studies (SAS) was ranked the fifth best high school in the United States and the best high school in Florida. SAS takes pride in its diverse student population, nurturing learning environment, and commitment to providing all students with a rich and rigorous liberal arts education, with many of its students specializing in STEM. In 2021, MDC's women's volleyball team won their 12th state championship in 16 years. The team has won a total of 48 state titles and 11 national titles overall with back-to-back NJCAA National Championships in 2017 and 2016. And, for the first time since 2014, MDC's baseball team was crowned state champions in 2021. Prior to that, MDC baseball won two national titles: in 1964 at Dade North and in 1981 at Dade South. In all, MDC baseball teams have made 13 trips to nationals since first qualifying in 1964. MDC's long history of athletic excellence boasts 35 NJCAA national titles, including 15 in women's sports, and dozens of All-Americans. The College joined the NASA Florida Space Grant Consortium (FSGC) in 2021, a statewide network of colleges and universities supporting the expansion and diversification of Florida's space industry through grants, scholarships and fellowships to students and educators in Florida. By 2021, MDC had conferred more associate degrees than any other community college in the country. And among all large postsecondary Institutions, MDC ranked No. 1 in associate degrees awarded to Hispanics; No. 4 in associate degrees awarded to Black students; No. 1 in associate degrees in Liberal Arts and Sciences, General Studies and Humanities; and No. 5 in associate degrees in Nursing and Health Professions & Related Sciences. As Miami's relevance in the technology and startup landscapes dramatically expands, the College is anticipating and meeting industry needs with a host of high-tech, in-demand career options, including Artificial Intelligence, Cybersecurity, Cloud Computing, Data Analytics, AR/VR, Tesla electric vehicle technology, Animation & Gaming, Business Entrepreneurship and more.

Nearly 14,000 graduates from MDC's eight campuses earned degrees for the 2020-21 academic school year. The College's eight campuses and outreach centers offer more than 300 academic pathways, associate and bachelor's degrees, and career certificates. MDC is among the most diverse institution in the nation with 164 nations and 53 languages represented in its student body.

In 2022, MDC was designated a National Center of Academic Excellence in Cyber Defense (CAE-CD) by the National Security Agency (NSA) for its robust cybersecurity curriculum, the very highest designation of its kind. The College also received validation from the NSA for the Associate of Science in Cybersecurity through 2027. Those developments took place as the College launched and continued expanding MDC Tech, its unified brand and comprehensive strategy that combines all its cutting-edge tech educational offerings with the growing tech community's needs. Also in 2022, MDC inaugurated its Artificial Intelligence (AI) Center at North Campus, a state-of-the-art facility designed to prepare tomorrow's AI workforce, upskill existing professional and elevate AI awareness in Miami through classes, workshops, lectures and other activations. Its conception and vision are the product of a strategic partnership with industry giants like Microsoft, Intel and IBM.

From its earliest beginnings, Miami Dade College has been at the forefront of innovation and cultivating opportunities in an ever-changing career outlook. The College's efforts have created new generations of professionals trained and encouraged to pursue their passions and discover their purpose for a life of career success.

CAMPUSES AND EDUCATIONAL CENTERS

MDC enrolls more than 170,000 students at its eight campuses and multiple outreach centers throughout Miami-Dade County. While each campus has developed its own distinct identity, the entire College is united around a fundamental mission: providing access to high-quality educational opportunities for all residents of the community.

Campuses

North

Located on 245 acres in northern Miami-Dade County, this beautifully landscaped campus was the College's first. It was built in 1960 on land that once hosted a World War II Naval air station. The main academic buildings of the Campus surround a serene lake and lush walking paths. North Campus is a major gateway for students wishing to upgrade skills, complete one-year certificate programs, prepare for licensing exams or complete a bachelor's degree. North Campus is also recognized for its unique programs. The School of Justice, Public Safety, and Law Studies offers the Bachelor of Applied Science with a major in public safety management and provides basic training for all police and correctional officers in Miami-Dade County as well as more than half the private security personnel. The Fire Science Program trains all Miami-Dade County firefighters and provides continuing education for municipalities throughout the county and the East Coast. A live fire training facility is the only one of its kind in South Florida. Additionally, the School offers programs in chemical and watershed management. North Campus also houses the Funeral Services program that trains morticians and funeral service directors, the only program of its kind in southeastern Florida. Those students wishing to pursue careers in film and digital imaging, television, and sound engineering, or in the radio and music business take advantage of the School of Entertainment Technologies. This School operates the cable station MDC-TV, and in 2008 it inaugurated the Televisa Centre for Film and Television Production, a hub for Latin American and Caribbean entertainment industries. North Campus also administers the Carrie P. Meek Entrepreneurial Education Center, which promotes excellence in education, entrepreneurship, and workforce preparation. In January 2010, MDC celebrated the official opening of the College's state-of-the-art science complex at North Campus. The first of its kind in north Miami-Dade County, the facility will create new educational and career-training opportunities for area residents and beyond. The 90,000 square foot complex features 21 laboratories with media stations, a Palmetum, a rooftop observatory, classrooms, lecture hall, faculty and administrative offices, botanical garden, greenhouse, an outdoor plaza and café. The new science complex is an invaluable tool for all MDC students who must meet science requirements to graduate. In particular, it will serve as a hands-on training venue for students interested in biotechnology, chemical technology, green and environmental sciences, and advanced manufacturing careers, among many other science, technology, engineering and mathematics (STEM) professions. Training in these areas can lead to an array of jobs, from research technologist to environmental scientist. Another goal is for the complex to become a top regional research facility. The School of Engineering, Technology, and Design at North Campus launched a Bachelor of Applied Science (B.A.S.) with a major in Film, Television and Digital Production in 2010. Designed to provide students with a seamless, workforce driven baccalaureate degree that will enable them to obtain immediate employment in the field of digital production, thereby meeting South Florida's workforce needs, the B.A.S. also serves those currently in the production workforce who need to update their skills to increase their career opportunities and wages.

Kendall

Kendall Campus, situated on a 185-acre tract of trees and lakes, opened in 1967, and has become home to a wide variety of academic programs and specialized institutes. The campus features 14 buildings equipped with the latest technologies, a wellness center with an Olympic-sized pool and several athletic fields. Kendall Campus offers a comprehensive range of learning opportunities. Kendall provides students with transfer programs designed to facilitate the move to four-year institutions, programs that enhance and modernize professional and technical skills, and preparatory programs for licensing or certification. Kendall Campus provides students with academic support services focused on enhancing student learning in science, mathematics and engineering. The Environmental Center is a 10-acre facility on campus that hosts Eco Tours for more than 10,000 school children each year. The Gourmet Academy is the culinary showpiece of Kendall Campus and offers a variety of non-credit programs and courses to the community. Located west of the main campus, the Landscape Technology Program maintains a large nursery and several greenhouses. One of the additions to the campus is the Earth Sciences Museum and Demonstration Center, which boasts one of the largest collections of geological specimens in the southeastern United States. The Kendall Campus art gallery provides the campus and surrounding community with several exhibitions each year and houses a permanent collection of more than 700 works. Also, award-winning publications such as Miambiance, Kendall's literary magazine, highlight students' literary excellence. Kendall Campus offers a Bachelor of Applied Science (B.A.S.) in Leadership and Management Innovation to prepare graduates for

leadership roles and positions in a variety of industries. The upper-level coursework includes applied management practices, prepares students for supervisory and management opportunities within their technical field, and affords students the opportunity to demonstrate the application of acquired knowledge, skills and competencies through internships and capstone experiences. After graduation, students will be prepared to advance to supervisory and managerial positions in a wide variety of industries, including financial services, retail, hospitality, marketing, aviation management, food service, insurance, real estate, office and administration, and sales.

Wolfson

Wolfson Campus opened in 1970 by holding classes in the storefronts of downtown Miami. With the completion of the campus' first permanent facility in 1973, Wolfson Campus catalyzed a downtown renaissance by hosting all manner of civic and cultural discourse. It is the only comprehensive urban campus in the city. Located within the city's financial, governmental, technological and cultural hubs, Wolfson Campus capitalizes on its unique geographic resource by offering programs in banking/financial services, business, computer technology, paralegal studies, architecture, economics, hospitality management, engineering, the arts, humanities and social sciences. The campus also houses the New World School of the Arts, a comprehensive high school and college program, recognized as one of the best performing and visual arts schools in the country. Each year Wolfson Campus hosts Miami Book Fair. This is the nation's largest and finest literary festival, bringing hundreds of renowned authors, publishing houses and hundreds of thousands of fairgoers to the Campus. The New York Times calls this Wolfson Campus event the model for all other book fairs. The School of Computer and Engineering Technologies at Wolfson Campus offers a Bachelor of Science in Electrical and Computer Engineering Technology. This degree addresses the local workforce need for baccalaureate-level engineers, providing job opportunities for MDC's Associate in Science graduates to earn, at a local institution, a degree that will support higher-paying careers. In the spring of 2011, MDC inaugurated the Miami Culinary Institute at its new, state-of-the-art, green facilities, training students in the latest sustainable food and energy technologies with unparalleled nutrition and culinary knowledge.

Medical

In 1976, Miami Dade College opened its Medical Campus on 4.3 acres within the city's medical/civic center complex. Along with the other members of this complex (the University of Miami Miller School of Medicine, UM/ Jackson Memorial Hospital, Veterans Administration Hospital and Miami-Dade County Public Health Service), Medical Campus forms the backbone of Miami's health care community. The campus offers specialty disciplines in nursing and allied health, and state-of-the-art technologies that help to ensure that students are prepared in these and other challenging medical careers. The Medical Campus continues to be one of the top producers of nurses in the South Florida area. The Bachelor of Science in Nursing was launched in 2008. More than 20 Allied Health programs are offered, including Physician's Assistant, Opticianry, Emergency Medical Technician, Veterinary Technology, Physical Therapist Assistant, Dental Hygiene and more. Quality medical faculty guide students with support from tutors, labs and the Student Success Center. Medical Campus launched a Bachelor of Applied Science (B.A.S.) in Health Sciences in 2010. Designed to provide students with entry-level skills specific to the allied health professions and to supplement the workforce-specific skills that are inherent in Miami Dade College's associate degree program, the B.A.S. in Health Sciences also offers students a Physician Assistant Studies option. The B.A.S. curriculum provides an interdisciplinary approach to health care by equipping practitioners with specialized health delivery system and patient management strategies, including medical knowledge and skills, the teaching of multicultural health care clients, leadership and management strategies, and research approaches that address health care issues, and alternative medicine.

Homestead

In 1990, Homestead became the fifth campus of Miami Dade College. It was opened in the historic downtown district of the city of Homestead with the mission to deliver a full range of higher education programs for the Homestead/Florida City communities. In fulfilling its mission, the campus enhances the community's capacity to meet cultural and social needs, in turn fostering a stronger sense of community. This togetherness was very important following the devastation of Hurricane Andrew and the closing of the Homestead Air Force Base. After Hurricane Andrew, the campus, like the city of Homestead, began to rebuild, adding four new facilities by 1996. In January of 2002, the College opened its Aviation Building, housing a simulator of an airport control tower and runways, as well as classrooms and avionics equipment to support the aviation program. The aviation program also extends to facilities at Miami International Airport and Miami Executive Airport. Today, Homestead Campus is a modern, six-building complex offering an array of academic programs, including aviation, entertainment technologies, arts and sciences, and nursing. The campus' award-winning structures include a computer courtyard, student learning lab, career center and specialized assessment facility. As the community continues to grow, Homestead Campus will also grow, expanding its horizons to meet the needs of the South Dade community.

Padrón

Padrón Campus is located in the heart of Little Havana, a colorful and lively neighborhood in Miami's historic Latin Quarter. The seed for Padrón Campus was planted in 1972 when the College offered two night courses at the Belen Jesuit Prep School. Sixty students enrolled. By 1979, the program had blossomed into the Wolfson Campus' Division of Bilingual Studies, enrolling 2,000 students. In the early 1980s, an influx of students from Latin America and the Caribbean led to the addition of day classes and full-time faculty. By 1986, the division had grown to "center" designation, and it moved into a building in Little Havana purchased by the College Foundation. What was then called InterAmerican Center became the largest bilingual learning environment in all of higher education. With enrollment at 5,500, the College District Board of Trustees petitioned the state of Florida for "campus" status. The request was approved and on March 27, 2001, InterAmerican Campus – its original name – was born. The District Board of Trustees pronounced InterAmerican Campus a full-fledged, full-service campus, the sixth of Miami Dade College. In May 2019, the campus was renamed the Padrón Campus in honor of the longtime College president, who announced he was stepping down effective August 2019. Today, the campus provides service to students in more than 200 programs. It is also home to the College's School of Education, which offers bachelor's degree programs in secondary mathematics education, exceptional student education, and secondary science education in the areas of biology, chemistry, physics and earth/space science.

Hialeah

Hialeah Campus became MDC's seventh campus, accorded official campus status by the Florida State Board of Education in 2005. The campus serves the Greater Hialeah-Miami Lakes area, offering day and evening classes six days a week. Courses leading to the Associate in Arts and Associate in Science degrees are offered. Educational opportunities are also available through career technical education programs, as well as through courses providing career entry in computer technology, office technology, electronics and early childhood development. Hialeah Campus houses a large and comprehensive English-language training program for speakers of other languages in various instructional formats.

West

The West Campus was approved by the Florida State Board of Education in 2005 and became a designated branch campus of Miami Dade College in July 2017, with the approval of the Southern Association of Colleges and Schools Commission on Colleges. Serving one of the fastest-growing locales in Miami-Dade County, including Doral and surrounding areas, the West Campus offers courses toward the Associate in Arts and Associate in Science degrees. Corporate training programs are also offered at the campus, which opened for classes on March 1, 2006, and promises to be the next exciting learning environment for the greater Miami community.

Education Centers

Gibson Education Center

MDC's Gibson Education Center is an outreach center of the Wolfson Campus offering a wide range of traditional credit courses, career readiness programs, workforce training, college preparedness programs and personal enrichment seminars at a location that is assessable to US-1, but especially convenient to the community of Coconut Grove and surrounding neighborhoods. The Gibson Educational Center is nestled in the heart of Coconut Grove just six blocks south from the business district of Coco Walk and equipped with three state of the art classrooms with a fully modern digital classroom. Whether you are a student that is looking to upskill, prepare for industry certifications and exams such as the ACT, SAT, or PERT, or simply get a head start through dual enrollment, the Gibson Education Center is here to help.

The Carrie P. Meek Entrepreneurial Education Center

The Carrie P. Meek Entrepreneurial Education Center is an outreach program of the North Campus. It opened its doors on Oct. 4, 1989, in the heart of Liberty City, a predominantly African-American community within the City of Miami. The mission of the Entrepreneurial Education Center is to implement the broader mission of the College while promoting entrepreneurship, business growth and economic revitalization for the local residents of Liberty City and the surrounding communities. The Entrepreneurial Education Center offers a vast array of college credit and noncredit courses. Students pursue certificate and vocational programs in

a number of fields; they participate in seminars and conferences that promote workforce training and business skills and facilitate entrepreneurship and entry into the labor market.

ADMISSIONS

Admissions Information

Admission is a simple process, requiring a completed application, the admission application fee, and official transcripts of high school or college studies. International applicants have additional entrance requirements based on U.S. immigration rules, and applicants for baccalaureate and limited-access programs may have additional entrance requirements. In accordance with Manual of Procedure 4000 – Admission Criteria and Guidelines <https://www.mdc.edu/procedures/Chapter4/4000.pdf> and section 1007.263 Florida Statutes, the following requirements have been established for college credit and vocational credit admission to Miami Dade College (MDC).

1. Apply and get admitted by completing the MDC admission application (<http://www.mdc.edu/admissions/>).
2. International students must submit the application by the published deadlines. See deadlines at: <https://www.mdc.edu/internationalstudents/admission/default.aspx>.
3. There is a \$30 non-refundable application fee for all new students applying to MDC, except for Dual Enrollment, Early Admission, and School for Advanced Studies. For International students, there is a \$50 non-refundable application fee, and Baccalaureate applicants must pay a \$25 non-refundable application fee. Returning students do not pay the application fee.
4. High school graduates who attended as dual enrollment students need to complete a new application.
5. Applicants who transfer to MDC from other postsecondary institutions must request final, official transcripts from all of those institutions to be sent directly to MDC Transcript Processing Services. Domestic (U.S.) high school and/or college/university transcripts must be sent to Miami Dade College either electronically (by a third party) or by mail directly from the institution(s).

A. College Readiness and Developmental Education

MDC reviews and publishes the Placement Criteria Manual as necessary, which adheres to 1008.30, Florida Statutes, and State Board of Education Rule 6A10-.0315 on common placement testing, alternative methods to common placement tests, and developmental instruction for public postsecondary education. The Placement Criteria Manual details all methods for assessing communication and computation skills and the minimum standards a student must achieve to demonstrate readiness to perform college-level work.

B. Admission Application Fees and Waivers

Students who fall into one of the categories listed below may request an Admission Application Fee Waiver.

1. Student is a U.S. Veteran or active-duty military.
2. The student or family is receiving public assistance.
3. Student is living in federally subsidized public housing or experiencing homelessness.
4. Student is a ward of the state, an orphan, or is in foster care.
5. Student is an MDC Employee or Retiree.

C. Admissions to College Credit Programs for the Associate in Arts, Associate in Science, and College Credit Certificate programs with special admission requirements shall include (some programs have additional admission requirements):

1. Students with a standard high school diploma from any public high school in the United States or territories, per 1007.263, Florida Statutes.
2. Students who are at least 16 years of age and are demonstrating qualification to obtain a higher education per Federal Student Aid eligibility requirements by:
 - Passing an approved ability-to-benefit test, or
 - Completing six credit hours toward a degree or certificate.
3. Students with a Florida public high school withdrawal (completion) code are eligible for college credit admission, as defined by the Florida Department of Education.
4. Students from accredited non-public high schools in the United States and its territories.

5. Students awarded a Department of Education high school equivalent (GED) diploma in the United States and its territories, the High School Equivalency Test (HiSET), Test Assessing Secondary Completion (TASC), or California High School Proficiency Exam.

6. Home-schooled students with a notarized signed affidavit from their parent or legal guardian attesting that they completed a Florida home education program under 1002.41, Florida Statutes. A Florida notary must notarize the signed affidavit. Students must also provide:

- a. A government-issued picture ID for the student and the parent or legal guardian.
- b. Proof of registration with the local secondary school board.

7. Students with the foreign equivalency of a United States high school diploma that meets the admission requirements to a recognized institution of higher education in their home country, with original documentation that shows all the requisite seals and apostilles. The College reserves the right to require a certified official English translation if the language used in the documents is not English. Students who do not have the requisite seals and apostilles on the original foreign high school documentation may opt to obtain an official evaluation of their credentials from a National Association of Credential Evaluation Services (NACES) or Association of International Credential Evaluators, Inc. (AICE) member organization. Students must present this official evaluation to their campus Admissions and Registration Office. MDC reserves the right to require NACES evaluations for any transcripts in which we are unable to verify the authenticity or legitimacy of the documents provided.

D. Cases that Require Additional Validation

Additional Validation for non-public high schools in compliance with 34 CFR 668.16, reasons that the high school diploma may not be accepted or may require additional validation include:

1. The issuance of the high school diploma was based only on a test and/or payment of fees.
2. The curriculum consisted of online/distance education instruction, and the high school is not listed as an accredited institution by the Distance Education Accrediting Commission (DEAC). The DEAC maintains a list of accredited institutions that offer high school via distance education.
3. There is conflicting high school information.
4. The high school was previously questioned as being a diploma mill by MDC or other accredited higher education institutions.
5. The high school is not listed in the U.S. Department of Education's National Center for Education Statistics (NCES). The list of public high schools is accessible at <http://nces.ed.gov/ccd/schoolsearch/>, and the list of private high schools is at <http://nces.ed.gov/surveys/pss/privateschoolsearch/>.
6. The private high school is not listed in the Florida Department of Education's Office of Independent Education and Parental Choice.
7. The high school or its course(s) is identified by the National Collegiate Athletic Association (NCAA) as not accepted for athletic eligibility.
8. Other evidence provides reason to believe that the diploma is not valid or was not earned from an entity that provides secondary school education. A High School Validation Checklist should be utilized to evaluate institutions uniformly. Once MDC renders a decision on the validity of a high school diploma, the decision is final and not subject to appeal.

E. International students who meet the admission requirements specified in section C must provide all of the following supplementary admission documents to the International Students Office to receive an I-20 Certificate of Eligibility, which is needed to apply for the F1 student visa before the beginning of the term for which the students seek admission:

1. Copy of a valid passport.
2. Statement of financial resources available to support their educational expenses.
3. Certificate of health and accident insurance, before enrollment.

F. Transfer Students

Transfer Students Applicants who transfer to MDC from other postsecondary institutions must request final, official transcripts from all of those institutions to be sent directly to MDC Transcript Processing Services. Additional requirements are as follows:

1. Students who transfer with fewer than six (6) college-level credits earned with "C" or higher grades must submit proof of valid high school graduation.
2. Courses taken at an institution accredited by a former regional or national accrediting agency recognized by the United States Department of Education (USDOE) that adheres to the Florida Statewide Course Numbering System will be accepted.
3. Courses taken at an institution accredited by a national accrediting agency recognized by the USDOE will be reviewed through an evaluation process to determine the transferability of courses and credit hours.
4. Students who transfer from foreign postsecondary institutions approved by the country's Ministry of Education who provide original documentation showing all the requisite seals and apostilles must provide a certified official English translation if the language used in the documents is not English. Students with fewer than six (6) college-level credits earned with "C" or higher grades must submit proof of valid high school graduation as listed in Section II.C. MDC reserves the right to require evaluations for any transcripts in which we are unable to verify the authenticity or legitimacy of the documents provided.

G. Dual Enrollment and Early Admission

Student's access to dual enrollment and early admission is authorized pursuant to 1007.271, Florida Statutes. Students in grades 6-12 who are enrolled in Miami-Dade County public schools, participating private and charter schools, or home schools are eligible to participate in the MDC dual enrollment program. Students must have a minimum 3.0 high school unweighted grade point average and demonstrated readiness for college coursework.

H. Admissions to Baccalaureate Degree Programs

Baccalaureate degree applicants must meet all general and program-specific admission requirements and pay a non-refundable \$25 application fee.

I. Admission to Career and Technical Education Certificate Programs

Students who meet the admission requirements may be admitted to Career and Technical Education programs as well as the following (some programs have additional admission requirements):

1. Applicants who are at least 16 years of age and have left high school prior to completion.
2. Applicants who have been awarded a special diploma, as defined in section 1003.438 Florida Statute, or a certificate of completion, as defined in section 1003.428 Florida Statute.
3. International students with a vocational credit student visa (M1) and the supplementary admission documents indicated in II.C.7 above.

J. Recruitment of Students

1. Student recruitment promotes enrollment by presenting the learning options available at MDC. The recruitment team develops and implements activities consistent with the mission of the College. Activities include providing prospective students, parents, and other members of the community with current and accurate information regarding admission and registration, testing requirements, and program offerings, as well as other resources and related opportunities available at the College.
2. MDC refrains from unfair, deceptive, and abusive marketing tactics in the recruitment of all students. In addition, MDC will refrain from using third-party lead generators or marketing firms aimed at service members (veterans/military students). MDC will:
 - Refrain from providing any commission, bonus, or other incentive payment based directly or indirectly on securing enrollments or federal financial aid, including tuition assistance (TA) for members, to any persons or entities engaged in any student recruiting or admission activities, or making decisions regarding the award of student financial assistance.
 - Ban inducements such as gratuity, favor, discount, entertainment, hospitality, transportation, lodging, meals, or other items of monetary value to any individual or entity, or its agents, including third-party lead generators or marketing firms. Only salaries paid to employees or fees paid to contractors in conformity with the applicable laws are allowable for the purpose of securing enrollments of service members (veterans/military students) or obtaining access to TA funds.

K. Other Considerations

1. Certain academic programs have additional program admission requirements. Students need to check with the respective academic department for more information.
2. Applicants who have been convicted of a felony or are the subject of an arrest pertaining to a controlled substance and who wish to apply for a program that leads to licensure should confer with the regulatory/licensing agency to determine eligibility for future credentialing and practice. Applicants who are determined to be not eligible for licensing for any reason may apply for admission to that program, but must recognize that program completion may not result in licensure or employment. Miami Dade College has determined that the presence of students officially designated as Sexual Offenders/Sexual Predators on campus may be disruptive to the College's programs and/or would interfere with the rights and privileges of other students. In accordance with 1001.64, Florida Statutes, the College may consider the past actions of any person applying for admission or enrollment and may deny admission or enrollment because of misconduct if determined to be in the best interest of the College. Therefore, MDC reserves the right to deny admission/enrollment to students who are officially designated as Sexual Offenders/Sexual Predators.
3. MDC reserves the right to deny admission to applicants who have been incarcerated, convicted of a felony, experienced disciplinary problems at another educational institution, or who may pose a threat to the life and/or safety of its students, faculty, staff, community, or guests, as determined by the College administration. In accordance with Section 1001.64, Florida Statutes, the College may consider the past actions of any person applying for admission or enrollment and may deny admission or enrollment because of misconduct if determined to be in the best interest of the College.
4. Applications received by the College will be reviewed. The College may designate any application for further review at any time and for any lawful reason. The College may require that applicants provide supplemental documents or information, participate in enhanced verification protocols, or undergo additional review before an application is approved. The provisions of this section will also apply to previously approved applications. In the event an applicant or student does not respond to the College's requests, whether in full or in part, the College reserves the right to take any appropriate action, including, but not limited to, rejecting an application or reversing an earlier approval.

Transient Student Information

Transient students are enrolled at their "home institution" and are enrolled at MDC for a term.

For Students from Florida Public Institutions

Transient students are advised to use Florida's official online student advising system at www.floridashines.org.

1. To request and submit a Transient Student Form, visit <https://www.floridashines.org>.
2. Under 'Succeed in College,' click on 'Take a Course at Another School.'
3. At the bottom of the page, click on 'Start or check the status of your Transient Student Admission Application now.'
4. 'Select Your School' and enter your 'Student ID' and 'PIN/Password.'
5. Select the college of your major in the first drop-down; then select your major from the second drop-down.
6. Complete the transient application-- be sure the course information is accurate-- and click 'Send.'
7. Check the status of your application and keep in contact with the transient student advisor on the [Florida Shines site](http://www.floridashines.org) until registration is completed.
8. Please contact the [Admissions Office](#) with any questions regarding your application. Contact the [Advisement Office](#) for inquiries related to course approval.

For students from Florida Private Institutions and Out-of-State Institutions

1. [Apply for Admissions](#) and select "Non-Degree Transient" as your program objective. Program code: 39001.
2. If applicable, complete the [Florida Residency for Tuition Purposes Form](#) (in person via [electronic form](#)).
3. Submit the transient form to the [Advisement Office](#) via email or in person. The form may be submitted by either the student or the institution representative.

Non-Degree Applicants

Non-degree applicants are students who wish to take selected college courses without the intent of completing a college credit certificate, associate or baccalaureate degree program. These students must fill out an application for admission and are not required to provide evidence of high school graduation. Many students attend the College to upgrade their job skills, for transfer credit purposes, or for their own personal interest and enjoyment. Non-degree students who wish to enroll in a Math (excluding MTB 1103) or English course, or any course that requires college-level Reading and/or Writing placement, are required to demonstrate college readiness through traditional placement tests or alternate methods. If, at a later time, these students become associate or baccalaureate degree candidates, regular admissions procedures regarding all transcript(s) requirements will apply.

Special Admissions Categories

In each of the following categories, the regular admissions procedures apply:

1. **Dual Enrollment** – The Dual Enrollment program allows eligible high school, middle school, and home education students in grades 6-12 to simultaneously earn college credit and credit toward a high school diploma. The college credit may be applied toward a postsecondary certificate or degree at a Florida public institution. The Dual Enrollment program provides an opportunity to take challenging courses and accelerate education opportunities. Students who complete dual enrollment courses will save time in obtaining their college degree. They will also save money, as these students are exempt from the payment of registration, tuition, and laboratory fees. To enroll in courses through the dual enrollment program, students must demonstrate readiness for college-level coursework. Eligibility criteria include both a GPA requirement and assessment of communication and computational skills. The student's school must grant permission for the student to enroll in these courses, thereby agreeing to accept these college courses to meet high school graduation requirements. Students participating in dual enrollment may begin their studies in any term, provided that they complete the dual enrollment admission, advisement, and registration procedures. Early admission is a form of dual enrollment through which eligible high school seniors enroll at the college on a full-time basis.
2. The courses these students take are creditable toward a postsecondary certificate or associate degree and meet the requirements for the student's senior year and high school graduation. Early admission students are required to enroll in a minimum of 24 college credits (12 credits per semester) during their senior year.
3. **Early Admission** –The Early Admission Program is a form of dual enrollment in which eligible high school students, including home school students, enroll at MDC on a full-time basis during their senior year in courses that are creditable toward the high school diploma and the associate degree. To be considered a full-time dual enrollment early admission student, the student must enroll in a minimum of twelve (12) college credit hours per term beginning in the fall term of their senior year. It is recommended that a student enroll in no more than fifteen (15) college credit hours per term or the equivalent.

Student Eligibility

- Must have a minimum 3.5 high school unweighted GPA.
- Must have completed the 11th grade year successfully.
- Must have provide required common placement test scores or proof of eligibility using alternate methods.
- Must have authorization from their high school principal, counselor, and parent/guardian.
- Must express intent to pursue a postsecondary degree.
- Must enroll in a minimum of college credit courses which meet the requirements for the student's senior year and high school graduation (24 credits).
- Must maintain an unweighted 3.0 high school GPA and a 3.0 college GPA in order to continue participating in the Early Admission Program.

Readmission to the College

Students who have not enrolled within the past 12 months and are not enrolled in a future term are required to submit an application for readmission (<http://www.mdc.edu/admissions/>), along with a new residency form if applicable. Students who completed coursework at another institution must submit official transcripts from each institution attended so that prior coursework can be reviewed and evaluated.

Developmental Education Courses

With the exception of students who meet the criteria for an exemption from common placement testing <https://www.mdc.edu/aet/placement-criteria/default.aspx> and developmental education instruction, the State Board of Education requires demonstration of readiness for degree-seeking students and students who have not met college-level competency either through the completion of developmental education requirements in the Florida College System or have not been awarded credit for college-level coursework in the area of deficiency. Non-exempt students will be placed into developmental education courses in the subjects where scores indicate a need for this instruction, along with a Student Life Skills course (SLS). Enrollment in certain other courses may be restricted until all developmental education courses have been completed.

If scores on one or more of the subtests require developmental education placement, students must enroll in at least one developmental education course during their first term. Students who test into developmental education instruction and subsequently enroll in developmental education instruction must complete the required developmental education studies by the time they have accumulated twelve (12) credits of college credit coursework, or they must maintain continuous enrollment in developmental education coursework each semester until the requirements are completed while performing satisfactorily in the degree.

Florida Board of Education rules limit the number of times a student can take a developmental education or credit courses. Enrollment beyond the 100 percent refund deadline is considered an "attempt," and students can attempt a course only three times. ***Contact the Advisement Office for additional information**

Admissions to Select College Programs and Programs Leading to Licensure

Admissions to specialized programs, such as those offered by Medical Campus, have specific eligibility requirements due to enrollment limitations imposed by physical facilities, state licensure regulations, or other criteria. Students interested in admission to such programs have specific eligibility requirements. Students should visit the program website and the campus for additional information on those requirements.

The academic department will determine final approval for placement into these specific programs. Students are notified of admission into these programs. Students who are not selected for a specific program are encouraged to continue their studies in other courses and programs at the College. Advisement offices will assist all such students to determine alternative educational objectives.

A limited number of programs have supplementary admissions requirements. Applicants who have been convicted of a felony and/or subjected to an arrest pertaining to a controlled substance and are applying to a program that leads to licensure may be ineligible for that license. Applicants in this situation should check with the appropriate regulatory/licensing agency to determine whether this would be the case. These students still can be admitted to the program; however, they should understand that program completion may not result in licensure or employment. Additionally, there are usually other requirements for licensure, such as physical and psychological criteria, completion of unpaid internships, criminal history verification, and other background checks. It is the student's responsibility to understand and meet these requirements.

General Educational Development (GED) Tests and Diploma

An equivalent to a Florida high school diploma may be obtained by completing the General Educational Development (GED) test. A GED holder is eligible for admission to associate degree programs at the College. To qualify to take the test, individuals must be at least 16 years old and reside in the state. A 16 or 17-year-old must meet College criteria to be eligible to prepare for and take the GED test. Preparation for the GED test is available at all MDC campuses with the exception of Medical and West. Individuals should contact campus Continuing Education and Professional Development departments for assistance and further information. The GED test covers writing skills, reading skills, social studies, science and mathematics. A fee is charged to take the test battery, and there is an additional charge, although nominal, to retake subtests.

Teacher Certification Information

Before taking courses to meet Teacher Certification requirements, teachers should confirm from their public school district's certification office or the Florida Department of Education's Office of Teacher Education, Certification and Staff Development, that the courses in which they wish to enroll meet specific certification requirements. College credit courses offered by Miami Dade College may be used for extension, reissuing, other vocational certificates, reinstatement of certificates, and for recency of credit. Additionally, information about courses required for general

and professional preparation certification is available at the School of Education or campus Academic Advisement offices.

Florida Residency for Tuition Purposes

Miami Dade College policy concerning Florida residency requirements complies with the laws of Florida s. 1009.21, F.S., and Rule 6A10.044, F.A.C., which are reprinted as follows: s. 1009.21, F.S., determination of resident status for tuition purposes (<https://www.mdc.edu/admissions-info/tuition/florida-residency.aspx>).

Determination of resident status for tuition purposes: Students shall be classified as residents or nonresidents for the purpose of assessing tuition in post-secondary educational programs offered by charter technical career centers or career centers operated by school districts, in Florida College System institutions, and in state universities.

A. As used in this section, the term:

1. "Dependent child" means any person under the age of 24, whether or not living with his or her parent, who is eligible to be claimed by his or her parent as a dependent under the federal income tax code.
2. "Initial enrollment" means the first day of class at an institution of higher education.
3. "Institution of higher education" means any charter technical career center as defined in s. 1002.34, career center operated by a school district as defined in 1001.44, Florida College System institutions as defined in s. 1000.21(3), or state university as defined in s. 1000.21(4).
4. "Legal resident" or "resident" means a person who has maintained his or her residence in this state for the preceding year, has purchased a home which is occupied by him or her as his or her residence, or has established a domicile in this state pursuant to s. 222.1(5).
5. "Nonresident for tuition purposes" means a person who does not qualify for the instate tuition rate.
6. "Parent" means either or both parents of a student, any guardian of a student, or any person in a parental relationship to a student.
7. "Resident for tuition purposes" means a person who qualifies as provided in this section for the instate tuition rate.

B. To qualify as a resident for tuition purposes:

1. A person or, if that person is a dependent child, his or her parent or parents must have established legal residence in this state and must have maintained legal residence in this state for at least 12 consecutive months immediately prior to his or her initial enrollment in an institution of higher education.
2. Every applicant for admission to an institution of higher education shall be required to make a statement as to his or her length of residence in the state and, further, shall establish that his or her presence or, if the applicant is a dependent child, the presence of his or her parent or parents in the state currently is, and during the requisite 12-month qualifying period was, for the purpose of maintaining a bona fide domicile, rather than for the purpose of maintaining a mere temporary residence or abode incident to enrollment in an institution of higher education. An individual shall not be classified as a resident for tuition purposes and, thus, shall not be eligible to receive the instate tuition rate until he or she has provided such evidence related to legal residence and its duration or, if that individual is a dependent child, evidence of his or her parent's legal residence and its duration, as may be required by law and by officials of the institution of higher education from which he or she seeks the in state tuition rate.
 - However, with respect to a dependent child living with an adult relative other than the child's parent, such child may qualify as a resident for tuition purposes if the adult relative is a legal resident who has maintained legal residence in this state for at least 12 consecutive months immediately before the child's initial enrollment in an institution of higher education, provided the child has resided continuously with such relative for the 3 years immediately before the child's initial enrollment in an institution of higher education, during which time the adult relative has exercised day-to-day care, supervision, and control of the child.
 - The legal residence of a dependent child whose parents are divorced, separated, or otherwise living apart will be deemed to be this state if either parent is a legal resident of this state, regardless of which parent is entitled to claim, and does in fact claim, the minor as a dependent pursuant to federal individual income tax provisions.
 - A dependent child who is a United States citizen may not be denied classification as a resident for tuition purposes based solely upon the immigration status of his or her parent.

3. An individual shall not be classified as a resident for tuition purposes and, thus, shall not be eligible to receive the in-state tuition rate until he or she has provided such evidence related to legal residence and its duration or, if that individual is a dependent child, evidence of his or her parent's legal residence and its duration, as may be required by law and by officials of the institution of higher education from which he or she seeks the in-state tuition rate.
 - Except as otherwise provided in this section, evidence of legal residence and its duration shall include clear and convincing documentation that residency in this state was for a minimum of 12 consecutive months prior to a student's initial enrollment in an institution of higher education.
 - Each institution of higher education shall affirmatively determine that an applicant who has been granted admission to that institution as a Florida resident meets the residency requirements of this section at the time of initial enrollment. The residency determination must be documented by the submission of written or electronic verification that includes two or more of the documents identified in this paragraph, unless the document provided is the homestead exemption in Florida, which is deemed a single, conclusive piece of evidence proving residency.
 - The documents must include at least one of the following:
 - A Florida voter's registration card;
 - A Florida driver's license;
 - A State of Florida identification card;
 - A Florida vehicle registration;
 - Proof of a permanent home in Florida which is occupied as a primary residence by the individual or by the individual's parent if the individual is a dependent child;
 - Proof of a homestead exemption in Florida;
 - Transcripts from a Florida high school for multiple years if the Florida high school diploma or high school equivalency diploma was earned within the last 12 months;
 - Proof of permanent fulltime employment in Florida for at least 30 hours per week for a 12-month period.
 - The documents may include one or more of the following:
 - A declaration of domicile in Florida;
 - A Florida professional or occupational license;
 - Florida incorporation;
 - A document evidencing family ties in Florida;
 - Proof of membership in a Florida-based charitable or professional organization;
 - Any other documentation that supports the student's request for resident status, including, but not limited to, utility bills and proof of 12 consecutive months of payments; a lease agreement and proof of 12 consecutive months of payments; or an official state, federal, or court document evidencing legal ties to Florida.
4. With respect to a dependent child, the legal residence of the dependent child's parent or parents is prima facie evidence of the dependent child's legal residence, which evidence may be reinforced or rebutted, relative to the age and general circumstances of the dependent child, by the other evidence of legal residence required of or presented by the dependent child. However, the legal residence of a dependent child's parent or parents who are domiciled outside this state is not prima facie evidence of the dependent child's legal residence if that dependent child has lived in this state for 5 consecutive years prior to enrolling or reregistering at the institution of higher education at which resident status for tuition purposes is sought.
5. A person who physically resides in this state may be classified as a resident for tuition purposes if he or she marries a person who meets the 12-month residency requirement under subsection (2) and who is a legal resident of this state.
6. Except as otherwise provided in this section, a person who is classified as a nonresident for tuition purposes may become eligible for reclassification as a resident for tuition purposes if that person or, if that person is a dependent child, his or her parent presents clear and convincing documentation that supports permanent legal residency in this state for at least 12 consecutive months rather than temporary residency for the purpose of pursuing an education, such as documentation of fulltime permanent employment for the prior 12 months or the purchase of a home in this state and residence therein for the prior 12 months while not enrolled in an institution of higher education.

- If a person who is a dependent child and his or her parent move to this state while such child is a high school student and the child graduates from a high school in this state, the child may become eligible for reclassification as a resident for tuition purposes when the parent submits evidence that the parent qualifies for permanent residency.
 - If a person who is a dependent child and his or her parent moves to this state after such child graduates from high school, the child may become eligible for reclassification as a resident for tuition purposes after the parent submits evidence that he or she has established legal residence in the state and has maintained legal residence in the state for at least 12 consecutive months.
 - A person who is classified as a nonresident for tuition purposes and who marries a legal resident of the state or marries a person who becomes a legal resident of the state may, upon becoming a legal resident of the state, become eligible for reclassification as a resident for tuition purposes upon submitting evidence of his or her own legal residency in the state, evidence of his or her marriage to a person who is a legal resident of the state, and evidence of the spouse's legal residence in the state for at least 12 consecutive months immediately preceding the application for reclassification.
7. A person shall not lose his or her resident status for tuition purposes solely by reason of serving, or, if such person is a dependent child, by reason of his or her parent's or parents' serving, in the Armed Forces outside this state.
 8. A person who has been properly classified as a resident for tuition purposes but who, while enrolled in an institution of higher education in this state, loses his or her resident tuition status because the person or, if he or she is a dependent child, the person's parent or parents establish domicile or legal residence elsewhere shall continue to enjoy the instate tuition rate for a statutory grace period, which period shall be measured from the date on which the circumstances arose that culminated in the loss of resident tuition status and shall continue for 12 months. However, if the 12 month grace period ends during a semester or academic term for which such former resident is enrolled, such grace period shall be extended to the end of that semester or academic term.
 9. Any person who ceases to be enrolled at or who graduates from an institution of higher education while classified as a resident for tuition purposes and who subsequently abandons his or her domicile in this state shall be permitted to reenroll at an institution of higher education in this state as a resident for tuition purposes without the necessity of meeting the 12 month durational requirement of this section if that person has reestablished his or her domicile in this state within 12 months of such abandonment and continuously maintains the reestablished domicile during the period of enrollment. The benefit of this subsection shall not be accorded more than once to any one person.
 10. The following persons shall be classified as residents for tuition purposes:
 - Active duty members of the Armed Services of the United States residing or stationed in this state, their spouses, and dependent children, and active drilling members of the Florida National Guard;
 - Active duty members of the Armed Services of the United States and their spouses and dependents attending a Florida College System institution or state university within 50 miles of the military establishment where they are stationed, if such military establishment is within a county contiguous to Florida;
 - United States citizens living on the Isthmus of Panama, who have completed 12 consecutive months of college work at the Florida State University Panama Canal Branch, and their spouses and dependent children;
 - Fulltime instructional and administrative personnel employed by state public schools and institutions of higher education and their spouses and dependent children;
 - Students from Latin America and the Caribbean who receive scholarships from the federal or state government. Any student classified pursuant to this paragraph shall attend, on a fulltime basis, a Florida institution of higher education;
 - Southern Regional Education Board's Academic Common Market graduate students attending Florida's state universities;
 - Fulltime employees of state agencies or political subdivisions of the state when the student fees are paid by the state agency or political subdivision for the purpose of job-related law enforcement or corrections training;
 - McKnight Doctoral Fellows and Finalists who are United States citizens;
 - United States citizens living outside the United States who are teaching at a Department of Defense Dependent School or in an American International School and who enroll in a graduate level education program which leads to a Florida teaching certificate;

- Active duty members of the Canadian military residing or stationed in this state under the North American Air Defense (NORAD) agreement, and their spouses and dependent children, attending a Florida College System institution or state university within 50 miles of the military establishment where they are stationed;
 - Active duty members of a foreign nation's military who are serving as liaison officers and are residing or stationed in this state, and their spouses and dependent children, attending a Florida College System institution or state university within 50 miles of the military establishment where the foreign liaison officer is stationed.
11. Once a student has been classified as a resident for tuition purposes, an institution of higher education to which the student transfers is not required to reevaluate the classification unless inconsistent information suggests that an erroneous classification was made or the student's situation has changed. However, the student must have attended the institution making the initial classification within the prior 12 months, and the residency classification must be noted on the student's transcript. The Higher Education Coordinating Council shall consider issues related to residency determinations and make recommendations relating to efficiency and effectiveness of current law.
 12. Each institution of higher education shall establish a residency appeal committee comprised of at least three members to consider student appeals of residency determinations, in accordance with the institution's official appeal process. The residency appeal committee must render to the student the final residency determination in writing. The institution must advise the student of the reasons for the determination.
 13. The State Board of Education and the Board of Governors shall adopt rules to implement this section.

6A - 10.044 Residency for Tuition Purposes

1. The purpose of this rule is to establish consistent policies to determine the residency status for tuition purposes of students who are citizens of the United States or lawfully present in the United States. The determination of residency is to be made in accordance with criteria set forth in Federal law and Section (s.) 1009.21, Florida Statutes (F.S.), after the student has been admitted to a Florida College System institution.
2. Definitions:
 - "Dependent student" has the same meaning as "dependent child" as defined in s. 1009.21, F.S.
 - "Independent student" for the purposes of residency determination shall be defined as a student meeting one of the following criteria:
 1. The student is 24 years old or older by the first day of classes of the term for which residency status is sought at the institution.
 2. The student is married.
 3. The student has dependents that live with him or her, and the student provides more than half of the income to support those dependents.
 4. The student is a graduate student or professional student.
 5. The student is actively serving in the United States Armed Forces, the National Guard, or is a veteran.
 6. The student is not eligible to be claimed as a dependent by his or her parent or legal guardian for federal income tax purposes according to the rules and regulations established by the United States Internal Revenue Service.
 7. The student can demonstrate that he or she pays more than half of his or her tuition and required fees pursuant to s. 1009.23, F.S.
 8. Both parents of the student are deceased.
 - "Resident for tuition purposes" means an admitted or currently enrolled student who meets the residency requirements in s. 1009.21, F.S., and the provisions of this rule, and is therefore eligible to pay the resident tuition and fee rate as described in s. 1009.23, F.S.
 - "Non-resident for tuition purposes" means an admitted or currently enrolled student who does not meet the residency requirements in s. 1009.21, F.S., and the provisions of this rule, and must therefore be charged the non-resident tuition and fee rate as described in s. 1009.23, F.S.
3. Initial Determination of Residency. Each Florida College System institution shall develop policies and procedures for determining each admitted student's Florida residency status for tuition purposes. Policies and procedures shall be consistent with the criteria set forth in s. 1009.21, F.S. Each institution's residency appeal process established pursuant to s. 1009.21(12), F.S., shall be in writing in the institution's catalog and prominently displayed on the institution's website. Such policies and procedures shall, at a minimum, adhere to the following requirements:

- Each admitted student seeking to declare residency for tuition purposes shall submit a residency declaration, as prescribed by the institution, and shall submit the documentation required by the institution to establish Florida residency for tuition purposes.
 - Each Florida College System institution shall establish submission deadlines for all documentation used to determine residency for tuition purposes.
 - Each Florida College System institution shall provide written notice to admitted students that the burden of providing clear and convincing documentation to justify the institution's classification of a student's residency status for tuition purposes rests with the student or, if the student is a dependent, with the student's parent or legal guardian. For documentation to be "clear and convincing," it must be credible, precise, and compelling enough to persuade the institution that the student or, if that student is a dependent, the student's legal guardian has established legal residency in Florida.
 - Each Florida College System institution shall provide written notice requesting additional documentation to affirmatively determine residency for tuition purposes, including documentation required to review an admitted student's citizenship status for the purposes of determining residency in accordance with s. 1009.21 (2)(d), F.S.
 - Each Florida College System institution shall ensure that the student, parent or legal guardian has resided in Florida for at least twelve (12) consecutive months immediately preceding the term in which the student is seeking residency classification, and that their purpose for residence in the State shall not be solely for the purpose of pursuing an education, except as otherwise provided in s. 1009.21, F.S.
 - Each Florida College System institution shall provide written notice disclosing the penalties for intentional fraud or misrepresentation of the student's residency status, or if that student is a dependent, the student's parent or legal guardian's residency status.
4. Residency Determination Documents. If a declaration of domicile, pursuant to s. 222.17, F.S., is being used as one of the documents to establish residency for tuition purposes, the date that an applicant shall be deemed as establishing residency for tuition purposes shall be twelve (12) months after the date that the Clerk of Circuit Court notes the declaration was sworn and subscribed to them. Nothing in this subsection shall prevent the use of additional documentation as evidence that legal residency was established by other means pursuant to s. 1009.21(3)(c), F.S., as of a date earlier than that established by the declaration of domicile.
5. Residency Reclassification Determination. A currently enrolled student who is classified as a non-resident for tuition purposes must be allowed to apply for reclassification as a resident for tuition purposes if the student, or their parent or legal guardian in the case of a dependent student, is able to meet the residency requirements set forth in s. 1009.21, F.S., to be considered a resident of Florida for tuition purposes. Each institution's residency reclassification process established pursuant to s. 1009.21(6), F.S., shall be in writing in the institution's catalog and prominently displayed on the institution's website.
- Reclassified students are not entitled to reimbursement of any non-resident tuition or fees properly assessed prior to reclassification.
 - Reclassified students shall be assessed the resident tuition and fee rate at the start of the next academic term after the reclassification occurs based on the deadlines established by the institution.
 - Nothing in this rule precludes an institution from requesting additional documentation as defined in s. 1009.21(3), F.S., to support a student's request for reclassification of residency status.

Guidelines for Appeal of Florida Residency Classification for Tuition Purposes

- Appeals must be submitted no later than the end of the current term.
- The student shall exhaust all appeals at the campus level prior to submitting a request to the college-wide appeal committee. The student must submit the appeal to the campus Admissions and Registration office and meet with the director or designee prior to requesting an appeal from the college-wide committee.
- The student must complete the College-wide Appeal for Florida Residency Classification for Tuition Purposes Form and state the reason(s) for appealing the campus' decision. The explanation should address any information that was not previously submitted or considered by the campus.
- A copy of all new and previously submitted documents supporting the Florida residency claim must be attached to the form.
- The form must be signed by the student and submitted with required documents to the Campus Admissions and Registration office.
- The Campus director or designee will review the request to explore all the possibilities to assist the student.

- The Campus Admissions and Registration director or designee must initiate the process, sign the appeal form and forward it to the college-wide committee with all attached documents and any necessary comments.
- At its regularly scheduled meetings, the college-wide committee will review the appeals submitted. The committee may request additional information from the student or the campus in order to make its decision.
- Upon review, the college-wide committee will inform the student of its decision in writing.
- The college-wide committee's decision is final and may not be appealed.

International Student Admissions

Admission – Miami Dade College is authorized under United States Federal Law, Immigration and Nationality Act, §(101)(a)(15) (F or M) to enroll nonimmigrant alien students. In addition to following the regular admission procedures, international students are required to provide English language placement test scores, such as TOEFL if a non-native speaker, proof of mandatory health insurance coverage, and official bank letter of financial resources to support education costs. Registration and placement into available courses and programs is dependent on English language proficiency, advisement and counseling, assessment/placement testing and course or program requirements. Academic transcript(s) of secondary school, college, university, technical and other post-secondary schools attended must be certified as official. Transcript(s) in languages other than English must include official certified English translations, authentic verifying statements and signatures.

Deadlines – International applicants should apply at least three months prior to enrollment at the College. International mail, transcript verifications, international money transfers, consular appointments, travel and housing arrangements and advisement/testing requirements all take a great deal of time and may cause delays. Applications for admission, including all admissions credentials English proficiency examinations (if available), must be received at least 45 days prior to the start of the term in which the applicant plans to enroll.

Deadlines for International Student Admissions

Fall Term.....July 1

Spring Term.....November 1

Summer Term.....March 25

Deadlines and additional information for international students can be found on their website (<https://www.mdc.edu/internationalstudents/>).

Readmission – Readmission to the College for the international student requires submitting a new application for admission, new official transcripts of postsecondary education attempted since last attendance at Miami Dade College, and an official bank letter of financial resources to support education costs. Transcript(s) in languages other than English shall include official certified English translations, authentic verifying statements and signatures provided by members of the National Association of Credential Evaluation Services (NACES) <http://www.naces.org>.

English Language Requirements – Miami Dade College courses are taught in the English language. The College will provide English-language training for students who have insufficient English language skills. English-language test scores determine placement into college courses. Students with TOEFL, IELTS, or PTE scores meeting MDC's English proficiency placement test exemption requirements regardless of the test date are eligible to take the Basic Skills Assessment Test to determine placement in courses leading to an associate degree. Other English proficiency examinations may also be used if the Common European Framework of Reference for Languages (CEFR) level is B2 or higher. Students requiring English language training may need to attend additional semesters at the College in order to complete all associate degree requirements.

Financial Requirements – All international students must have sufficient funds to pay full college matriculation and nonresident fees, textbooks, living expenses, transportation expenses, health insurance coverage and other incidental expenses while attending college in the United States. Financial requirements are included with the application for admissions. Documentary evidence of means of financial support must be provided to the College to be issued a Certificate of Eligibility (SEVIS I-20). This evidence is also required by the American Embassy or Consulate when applying for a student visa to enter the United States. Students must have these funds available when they register for classes each term. College financial aid is not available to students on visa. See the "Fees" section in this catalog for details concerning matriculation, non-resident and other fee requirements.

Employment – Visa students in the United States are not allowed to be employed outside the College, unless permission has been granted by the United States Citizenship and Immigration Services (USCIS). On-campus employment may be authorized by the International Student Services advisors.

Health and Accident Insurance Certificate – Prior to registration, international students must purchase the mandatory health insurance policy available in the International Student Services Office. This insurance coverage must continue for the entire period of enrollment at the College.

Duration of Status – International students on a visa are admitted to the United States for the entire time

estimated for them to complete their approved program of study as indicated on the SEVIS I-20. Students must fulfill the following conditions to maintain Duration of Status: pursue a full course of study at the educational institution they are authorized to attend, make normal progress, keep a current passport that is valid for at least six months, maintain a valid SEVIS I-20 and not accept off-campus employment without USCIS approval.

Arrival in Miami – International students should arrive in Miami approximately 30 days before the beginning of the first term of enrollment based upon the program start date on the I-20. Students need the time to obtain housing, provide a local address to the College, participate in new student orientation, take English-language and placement assessment tests, obtain advisement and counseling and register for courses.

Housing in the Community – Miami Dade College does not provide or supervise student housing. International students must bring sufficient funds to pay three months' rent in advance (first and last month's rent, plus a security deposit equal to one month's rent). The estimated expense information provided with the application for admission provides important details.

Transportation – International students must provide their own transportation or use public transportation (buses or rail) to travel between home and the campus(es).

School Transfer – Completion of a degree program at the designated educational institution is recommended. International students who wish to transfer to another school must officially do so by requesting a release of their SEVIS record to the school they wish to transfer to and by providing an admission letter. The institution will notify Immigration of the student's school transfer. A student who transfers to a different school without completing this process is considered to be out of status.

Passport Validity – International students on a visa must have and maintain a current passport valid for a period of not less than six (6) months into the future. It is the student's responsibility to meet this requirement.

Full-Time Enrollment – International students are required by USCIS regulations to be enrolled full-time. Students should make satisfactory progress in their approved program each term, otherwise the continuation of study on an International Student Visa may be jeopardized and the Certificate of Eligibility (SEVIS I-20) rescinded. See Standards of Academic Progress in "Academic Regulations" section.

United States Department of Homeland Security Laws and Regulations – It is the student's responsibility to comply with all non-immigrant alien requirements as stated under the United States statutes I.N.A. 101(a) (15) (F); I.N.A. 214(m); IIRIRA 641. The College is required to report to the Department of Homeland Security international students who:

1. Do not register at the College by the first day of the semester.
2. Do not carry a full course of studies.
3. Do not enroll in the minimum required credits of in-person classes.
4. Do not attend classes to the extent normally required.
5. Become employed without authorization.
6. Terminate their attendance at the College.

Visa Student Advisement – Advisors are available at each campus to advise international students concerning academic programs and course objectives. Students on an International Visa should contact the International Student Services advisor each term for a review of the student's progress and for updates and compliance with immigration regulations.

Admission to Continuing Education (Non-College Credit) Programs and Courses

Miami Dade College, through its Continuing Education Program, offers students opportunities for enrollment in Continuing Workforce Education Training and recreation and leisure courses. Admission requirements are established by the nature of the particular program or course. A student who plans to register only for continuing education non-college credit courses need not apply for regular College admission.

1. Continuing Workforce Education Courses – These courses are for those students who have had prior employment in jobs related to the enrolled course or are presently employed in a career related to the Continuing Workforce Education course. Students enroll in the courses to upgrade their current skills, for re-employment purposes, or to enhance their current employability. For purposes of state certification or registration and updating to meet various professional organization requirements, the College student registration system allows for the award of Continuing Education Units (CEUs) on the student's transcript. These units may be awarded when a Continuing Workforce Education course is completed, and the course has been designated for the award of CEUs. Ten contact hours of classroom instruction equal one CEU.

2. Recreation and Leisure Courses – These non-credit courses are self-supporting with the total program costs being paid by the students who are enrolled. There are no state or College funds provided to support these activities. The College offers these courses on demand from students and the community, as space is available. The range of activities and courses is unlimited and is determined by the students enrolled. For further information, please consult the website at www.mdc.edu/ce.

Fees and Refunds

Fees are contingent upon approval of the District Board of Trustees and are subject to change. Special fees may also apply. Important note: Tuition and fee rates are determined annually by state and Board of Trustee processes. The best way to determine current tuition and fee rates is to check the Miami Dade College website, www.mdc.edu/student-financial-services/tuition-fees/, or to check the fee invoice on MDConnect Student Center. The fees listed below are an example – for planning purposes only – of rates for the 2024-25 year only.

1. Registration Fees 2025-26 – College Credit Courses

- Florida Residents* Matriculation
- Total: \$118.22 per credit
- Non-Florida Residents* Matriculation
- Total: \$402.51 per credit

2. Registration Fees 2025-26 – Baccalaureate Courses

- Florida Residents*
- Total: \$129.89 per credit
- Non-Florida Residents*
- Total: \$535.97 per credit

3. Registration Fees 2025-26 – Career and Technical Education Courses

- Florida Residents* Matriculation
- Total: \$91.08 per vocational credit (Special fees may also apply)
- Non-Florida Residents*
- Total: \$355.31 per vocational credit

◦ **See Florida Residency for Tuition Purposes section for definitions. Note: Fees are subject to change.*

4. Special Fees and Charges Special Registration Fees: Some courses carry special fees in addition to the regular registration fees. Special fees in music courses that offer private lessons range from \$60 to \$300.

- *\$30 Admission Application Processing Fee:* All new college credit students are assessed a \$30 non-refundable admission application processing fee. This fee must be paid when you submit the application.
- *\$25 Bachelor's Degree In-Program Admission Application Processing Fee:* All students admitted to an in-program Bachelor's degree are assessed a \$25 non-refundable admission application processing fee.
- *\$50 International Student Admission Application Processing Fee:* All new international students are assessed a \$50 non-refundable admission application processing fee.
- *\$15 Per Credit - MDC Online Fee:* MDC Online classes have a distance learning fee of \$15 per credit. (i.e., \$45 for a 3-credit course). To comply with federal requirements, MDC Online uses secure login and password to verify the identity of online students. There are no additional student charges associated with verification of student identification.
- *Resident Students/Non-Resident Students:* Review information about Florida Residency for Tuition Purposes online (<https://www.mdc.edu/student-financial-services/tuition-fees/>).
- *Full Cost of Instruction:* Out-of-State fee charged for students repeating courses more than allowed by state law (This is on a third or subsequent attempt).
- *Examination Fee:* A \$30 per credit nonrefundable fee is charged for institutional credit by exam.

5. Registration Fees – Continuing Education & Professional Development and Non-Credit Courses

- Continuing Workforce Education (CWE) – Fees are variable and calculated to cover the cost of the course.
- Recreation and Leisure Courses – Fees are charged to cover all expenses for providing the course.
- Adult Education Courses, which are considered Adult Basic Ed, Adult High School, GED, and VPI course fees:
 - \$30.00 per term for In-State Residency

Note: All fees are subject to change without notice.

Refunds of matriculation and tuition fees are made only if official drop or withdrawal cards are turned in at the campus Admissions and Registration Office by the published deadlines (see Academic Calendar), or if you drop via the web (and the drop is confirmed) by the deadline. If the student withdraws from the College as a result of administrative action or for the convenience of the College, except for disciplinary reasons, the student is entitled to a full refund of matriculation and tuition fees. If the student is dropped from a class due to cancellation of that class, the student is entitled to a full refund of matriculation and tuition fees. If the student is withdrawn from a course or courses for disciplinary reasons, the student is not entitled to a refund of matriculation, tuition or special fees. The admissions application fees (for credit, bachelor's, and international student admissions) are not refundable. Refunds for payments made with cash/checks will be refunded via the MDC One Card. Payments made with Visa/MasterCard/American Express will be refunded to the credit card account.

Fee Policy for Repeated Courses

Sections 1009.28 and 1009.285, Florida Statute require the assessment of fees for community college students who repeat a course due to withdrawal or failure. The fee for a third attempt of the same course is equal to 100 percent of the cost of instruction. Since state law prescribes student fees to equal 25 percent of the cost of instruction, the fee for a repeated course is approximately four times that of an initial attempt. Sections 1009.28 and 1009.285, Florida Statutes, and College policy allow one-time exceptions to the increased fees for courses. Students assessed such a fee should consult an advisor for more information (www.mdc.edu/smart).

Excess Hours Advisory

Section 1009.286, Florida Statutes, establishes an "excess hour" surcharge for a student seeking a baccalaureate degree at a state university. It is critical that students, including those entering Florida colleges, are aware of the potential for additional course fees. "Excess hours" are defined as hours that go beyond 120% of the hours required for a baccalaureate degree program. For example, if the length of the program is 120 credit hours, the student may be subject to an excess hour surcharge for any credits attempted beyond 144 credit hours (120% x 120). All students whose educational plan may include earning a baccalaureate degree should make every effort to enroll in and successfully complete those courses that are required for their intended major on their first attempt. Florida college students intending to transfer to a state university should identify a major or "transfer program" early and be advised of admission requirements for that program, including the approved common prerequisites. Course withdrawals and/or repeats, as well as enrollment in courses nonessential to the intended major, may contribute to a potential excess hours surcharge.

Per Section 1009.286(5), Florida Statute, it is recommended that students who intend to earn credit hours at the institution in excess of the credit hours required for baccalaureate degrees in which students are enrolled meet with their academic advisor.

Refund Policy

- Refunds of tuition and fees are made only if official drop or withdrawal cards are turned in at the campus Admissions and Registration Office by the published deadlines (see Academic Calendar), or if you drop via the web (and the drop is confirmed) by the deadline.
- If the student withdraws from the College as a result of administrative action or for the convenience of the College, except for disciplinary reasons, the student is entitled to a full refund of tuition and fees.
- If the student is dropped from a class due to cancellation of that class, the student is entitled to a full refund of tuition and fees.
- If the student is withdrawn from a course or courses for disciplinary reasons, the student is not entitled to a refund of tuition and fees.
- If the student does not utilize the College parking facility and a parking decal is not obtained, the college will refund the term parking fees after the specified term.
- The admissions application fees (for credit, bachelor's, and international student admissions) and late registration fees are not refundable.
- Refunds for payments made with cash/checks/wire transfers will be refunded via Bank Mobile.
- Debit card payment made at the Bursar's office are refunded via check (USPS mail).
- Payments made with Visa/MasterCard/American Express/Discover will be refunded to the credit card account. The Credit Card Service fee 2.0% is nonrefundable.
- Tuition payments made via Third Party agency will be refunded based on the conditions of the Third Party agency*.

Number of Weeks in Term	Number of Class Days Student has to drop class(es) and have all student fees removed
1-3	1
4-5	2
6-10	3
11-14	4
15-16	5
17-20	6
21-23	7
24-26	8
27-29	9
30-32	10

A procedure exists for handling specified exceptions to the refund policy. See the "Petitions Procedure" in the Students' Rights and Responsibilities Handbook (<https://www.mdc.edu/procedures/Chapter4/4018.pdf>).

F. **Refund Deadlines** – Continuing Education & Professional Development Courses

For one-day courses and workshops, the student must have paid in full and must make an official withdrawal at least one day prior to the day of class. For courses meeting for two or more days, the student must have paid in full and must make an official withdrawal at least one day prior to the second class meeting. A procedure exists for handling specified exceptions to the refund policy. Students should see the Continuing Education chairperson on their campus.

Payment Policy

1. All fees are due and payable in full by the due date posted on the fee invoice. Fees and charges are subject to change without notice. Cash is not to be sent by mail.
2. Payment of Fees by Check - Checks may be remitted to Miami Dade College for payment of fees owed. Check payments are also accepted via the MDC Web page. All checks accepted in payment for fees must be drawn on a United States bank and must be payable to the College. If a student submits a check exceeding the amount owed to the College, he or she will not get cash back. The College will issue the refund through Bank Mobile. (<https://www.refundselection.com/refundselection/#/welcome/continue>).
3. Payment by Credit Card - Miami Dade College will accept American Express, Discover, MasterCard and Visa. There is a 2.0% service fee on each credit card transaction made to the college. Credit card payments can be made over the telephone, and via the MDC Web page, <https://www.mdc.edu/student-financial-services/payment-options/online-payments/default.aspx>.
4. Payment by an Employer, Company or Other Agency - Miami Dade College accepts payment for tuition and fees from employers, companies and other agencies.
5. Payment Via Wire Transfer by International Students – Prospective and current international students may utilize the Nelnet Payment Plan as an option for paying tuition and fees. There is a \$40 per semester sign-up fee for using the Nelnet Payment Plan. For more information, please visit MDC's Nelnet Payment page (<https://www.mdc.edu/student-financial-services/payment-options/nelnet-payment-plan/default.aspx>).

6. Payment Via Florida Pre-Paid Tuition Program -The Florida Pre-Paid Tuition Program covers only defined registration, tuition, scholarship, and capital improvement fees. Students are required to pay any special fees and other local service fees, which include student service fees, parking fees, and technology fees. Students may submit a copy of their Florida Prepaid recipient card or letter to the Admissions office to help establish Florida Residency for Tuition Purposes.

For further information, contact Student Financial Services at asksfs@mdc.edu or (305) 237-2141.

Florida Pre-Paid Tuition Program

The Florida Pre-Paid Tuition Program covers only defined matriculation, scholarship and capital improvement fees. Students are required to pay any special fees and other local service fees, which include student service fees and technology fees. Students may submit a copy of their Florida Prepaid recipient card or letter to the Admissions office to help establish Florida Residency for Tuition Purposes.

FINANCIAL AID

Student Financial Aid

Financial aid is any grant, scholarship, loan or employment offered to assist a student to meet college expenses. Funding is usually provided by federal and state agencies, foundations, corporations, private donors and/or the College itself. Most financial aid is based on financial "need" as determined by the federal government's system of needs analysis. The amounts and types of financial aid that a student can receive are determined by federal, state and institutional guidelines. Financial aid is usually offered in "packages," which may consist of a combination of grants, loans, employment and scholarships. Grants and scholarships are regarded as a "gift" and need not be repaid. Loans are usually offered at low interest rates and can be repaid over an extended time period. When aid is offered in the form of employment, the student is paid an hourly rate for work performed (usually minimum wage). Students who wish to be considered for financial assistance offered by or through the College, including scholarships, must complete and submit the FAFSA (Free Application for Federal Student Aid, see "How to Apply"). The availability of certain types of financial aid is dependent upon the student's immigration status. Financial aid is available for approved and/or certified credit and vocational certificate programs of study.

Philosophy of Financial Aid

The objective of the student financial aid program at Miami Dade College is to provide financial assistance to students who need assistance in funding their educational goals. Financial aid officers are trained and available to counsel and assist students and parents seeking additional or alternative sources of aid. Prospective students and parents are strongly encouraged to contact the Financial Aid Office at any one of our campuses to obtain additional information regarding financial aid opportunities.

What is Financial Need?

Financial need is defined as the difference between the cost of education and the amount the student (and parents) can be expected to contribute to offset educational expenses. Financial need is based on federal regulations and information provided by the student and/or student's family on the Free Application for Federal Student Aid (FAFSA, see below).

How to Apply

To be considered for most types of financial assistance, a student must complete the Free Application for Federal Student Aid (FAFSA). The FAFSA is available online at <https://studentaid.gov>. The application process begins Oct. 1 for the academic year that begins in August. The results of the federal analysis are transmitted electronically to the College and are also sent to the student in the form of a Student Aid Report (SAR) via email or regular mail. Students should carefully read all notifications and communications from the U.S. Department of Education, Federal Student Aid offices and in a timely manner, provide information to the College or on the FAFSA, if the information originally submitted has to be corrected. Miami Dade College reserves the right to request supplemental information from parent(s), guardian(s), spouse and/or student as required by the financial aid staff to assess the need of the student. Students who are eligible to receive outside educational assistance such as Veterans Administration benefits and vocational rehabilitation assistance are expected to apply for this assistance through the appropriate agencies.

Verification

The Department of Education selects applicants for verification randomly, to determine the accuracy of the information provided on the FAFSA. If selected for verification, a student will be asked to provide additional information, such as but not limited to federal, tax return transcripts. Student files will not be processed until all required documentation is received, verification is complete and all corrections have been made.

Reapplying

Financial aid is not automatically renewed each year. To be considered for financial assistance from one year to the next, all students must reapply. Since the amount and type of aid are based upon the family's financial situation each year, it is quite possible that financial aid awards may change from one year to the next.

Basis on Which Financial Aid is Granted

The amount of financial assistance a student receives is generally determined by the need of the applicant, the availability of funds from federal, state, institutional and private sources, as well as the order in which the applications were completed (first-come, first-served basis). Students receiving federal financial aid are required to achieve and maintain an acceptable level of academic progress to receive financial aid. Specific eligible categories are posted on the Financial Aid Web page, and information is available in the Financial Aid Office.

Who Qualifies for Financial Aid

To be considered for most need-based assistance, you must meet the following basic eligibility requirements:

- Demonstrate financial need
- Be a U.S. citizen or eligible non-citizen
- Be registered with selective service, if required
- Not be in default on a previous student loan or owe a repayment on previous federal financial aid received at any institution
- Be enrolled at least half-time in an eligible program of study (some aid is available only to full-time students)
- Maintain satisfactory academic progress.
- Additional requirements may apply depending on the financial aid awarded to you.

Refunds and Repayments (Return to Title IV)

Federal regulations mandate that financial aid recipients who drop all courses or officially withdraw from the College before completing 60 percent of their enrollment period for the semester may be liable to repay a portion of the federal aid disbursed. The amount of the return is calculated using a federal formula that depends on the date the student ceased attendance. A student who owes a repayment will not be eligible for additional financial aid until the repayment is made in full.

Miami Dade College Student Assistance Programs

Scholarships and Grants

Scholarships and grants are available annually for students who require additional financial assistance beyond that received from federal and state sources. College funds for scholarships and grants are provided by businesses, clubs and organizations, agencies and from individual friends of the College through the Miami Dade College Foundation Inc. The primary criterion on which grant and scholarship recipients are selected is financial need. However, academic achievement is strongly considered during scholarship recipient selection. A limited number of grants are made available annually for service to the College and to students who may not be eligible for other types of financial assistance. Students who complete the FAFSA will be considered for a College grant. Students must complete an MDC Scholarship Application online at <http://www.mdc.edu/financialaid/scholarships/default.aspx> to be considered for a scholarship. Scholarship candidates may be required to submit additional documentation and/or information.

Tuition Payment Plan

A Tuition Payment Plan may be offered to students who are unable to pay the full amount of their schedule. Students should review their fee invoice for the term to determine the payment due date and to apply for the Tuition Payment Plan.

Tax Help for Educational Expenses

The Taxpayers Relief Act of 1997 offers several tax credits and deductions for educational expenses. For more information regarding these programs, go to the IRS webpage at: www.irs.gov.

Other Sources of Financial Assistance

Benefits for the Disabled – The state of Florida provides funding for the purchase of special equipment and services for all persons with disabilities enrolled in public postsecondary institutions.

Accessing the Financial Aid Office

- Counseling – Financial Aid counselors are available at all MDC campuses, on a walk-in basis, to assist students.
- Online – You can access the Financial Aid Office webpage at www.mdc.edu/financialaid to obtain more detailed information about financial aid programs, procedures and to check the status of your application and financial aid award.
- Email Communications – Regardless of the campus you attend, you can communicate with the Financial Aid Office via email at: finaid@mdc.edu

Student Complaint Procedures

Prospective or current students may voice their concerns about college rules, regulations, procedures or experience. Students must first voice concerns to the department staff and supervisor. Students who are unable to resolve

any concerns on their own, may contact the Federal Student Aid Ombudsman via <https://studentaid.gov/feedback-ombudsman/disputes/prepare>.

VETERAN AND MILITARY SERVICES

Military and Veteran Educational Benefits

Miami Dade College supports eligible veterans, service members, and dependents who use military and veteran educational benefits to pursue their educational goals. The College is approved for training under programs administered by the **U.S. Department of Veterans Affairs (VA)**, the **Department of Defense (DoD)**, and applicable state military education programs.

Through the **Veteran and Military Services (VMS) Office**, Miami Dade College assists students utilizing a variety of education benefits, including **VA education benefits**, **Military Tuition Assistance (TA)** for active service members, the **My Career Advancement Account (MyCAA)** program for eligible military spouses, and the **Educational Dollars for Duty (EDD)** program for members of the Florida National Guard. The College provides assistance with benefit eligibility, required documentation, and enrollment certification. Additional services may include academic advising, tuition deferments when applicable, tutorial assistance, and participation in the **VA Work-Study Program**.

In accordance with **Florida Statute §1004.07**, students who are called to active military service during an academic term may be eligible to withdraw from courses without academic penalty. Students may receive a full refund of tuition and fees or may request an incomplete grade or other appropriate academic accommodation, consistent with institutional policies.

In accordance with **Title 38 U.S. Code § 3679(e)**, this school adopts the following additional provisions for any students using U.S. Department of Veterans Affairs (VA) assistance, to include, **Post-9/11 GI Bill® (CH 33)**, **Veteran Readiness & Employment (CH 31)**, **Survivors' and Dependents' Educational Assistance (DEA - CH 35)**, or **Montgomery GI Bill® Selected Reserve (MGIB-SR - CH 1606)**, of **Title 10**, while payment to the institution is pending from VA. As such, following certification of veteran beneficiary enrollment to VA, and for up to 90 days, this school will **NOT**:

☐	Prevent the student's enrollment;
☐	Assess a late penalty fee to the student;
☐	Require the student to secure alternative or additional funding;
☐	Deny the student access to any resources (access to classes, libraries, or other institutional facilities) available to other students who have satisfied their tuition and fee bills to the institution.

However, to qualify for this provision, such students will be required to:

Provide the enrolling institution with a copy of his/her VA Certification of Eligibility (COE) - A "certificate of eligibility" can also be satisfied by a "Statement of Benefits" obtained from the U.S. Department of Veterans Affairs' (VA) website; eBenefits; or an Electronic Authorization from Veteran Readiness & Employment.

Reserve Officers Training Corps

Miami Dade College, in cooperation with the University of Miami and Florida International University, permits students to enroll in Air Force ROTC (through the University of Miami) and Army ROTC (through Florida International University). An application for admission to the ROTC program, including eligibility information for new and currently enrolled students, may be obtained from the ROTC offices at the University of Miami or Florida

International University. MDC credit is awarded for successful completion of ROTC courses. For further information, see "Military Science" in the Course Description section.

Military Training and Credit for Prior Learning

Miami Dade College may award academic credit toward a degree for eligible military education and training. Credit is evaluated based on recommendations published by the **American Council on Education (ACE)** in the **ACE Military Guide**, which provides standardized evaluations of military training and occupational experiences. Students who have served in the United States Armed Forces should submit an official **Joint Services Transcript (JST)** or **Community College of the Air Force (CCAF)** transcript for evaluation. These transcripts document military coursework and occupational training completed during service. Credit may also be awarded for approved **College Level Examination Program (CLEP)** or **DSST (formerly DANES Subject Standardized Tests)** scores in accordance with College policies for credit by examination. In accordance with **Florida Statute §1004.096**, Miami Dade College evaluates and may award academic credit for college-level training and education acquired during military service. Military training and examination credit that is accepted by the College will be recorded on the student's academic record as transfer or earned credit. Official transcripts must be submitted to the College for evaluation after the student has been admitted. All credit is evaluated in accordance with Miami Dade College policies and applicable state and accreditation guidelines. Students seeking additional information regarding credit for military training should contact the **Veteran and Military Services Office** or the **Admissions Office**.

STUDENT SERVICES

Advisement and Career Services

Advisement and Career Services support all student populations by facilitating an effective decision-making process regarding educational, transfer and career goals, advisors promote appropriate course selection and assist students with referrals to internal and external resources and support services. Advisement and Career Services supports first-time in-college and non-traditional student populations by offering office hours until 7:00pm Monday to Thursday and until 4:30 on Fridays at all of our campuses which facilitates access to the full scope of services provided by advisors even for those students attending classes during non-traditional times. Access to advisors is also available via phone, virtually, and by email.

All degree-seeking students will be assigned an advisor after enrolling in their first semester at the College. At that time, students and their assigned advisors will begin to chart an appropriate choice of courses based on the student's academic pathway, placement scores, transcripts, non-cognitive factors, and transfer institution of choice. In addition, advisors provide students assistance with career exploration, skills assessments, and guidance about how to best match student's skills and abilities with the right career path. Once a career path is chosen, advisors will provide important information regarding required courses, program information, graduation status and much more. All students are encouraged to see an advisor after admission to the College, after assessment testing has been completed (for those who require testing) and before first term registration. During these advisement sessions, the student and the advisor can chart an appropriate choice of courses based on the student's chosen pathway. Conferring of graduation eligibility at this time may be crucial to a student's success in meeting their career goals. During enrollment at Miami Dade College, students are encouraged, and sometimes required, to see an advisor when they encounter academic problems or contemplate a change in educational goals. In addition to helping students chart their educational and professional careers, advisors work with students to resolve problems affecting academic performance. Students may be referred for testing or to community agencies when appropriate, as a means to aid decision-making.

The Advisement and Career Services department also assists students with career planning, resume building and interviewing skills, and other employment needs. Career-related events, including seminars, career exploration workshops, and job and college fairs, are scheduled throughout the academic year. Students are also provided with guidance and information about transfer options and transfer assistance in preparation for the completion of their degree at the College. Transfer resources, including college catalogs, scholarship information and information on the College's Articulation Agreements with local, in-state and out-of-state institutions are available through the Department and on the MDC website. The department offers a variety of online tools that assist students with job placement, through a feedback system on resumes, cover letters, and interviewing skills assessments.

Academic Requirement Report

An Academic Requirements Report is the official record keeper of graduation requirements. This is different than a MAP which is intended to help you organize and schedule your pathway/program courses. An advisor will help you understand both your [Academic Requirements](#) and MAP. Reviewing these documents once per semester is recommended. The catalog and/ or the MDC website should be consulted for program/degree requirements.

My Academic Plan (MAP)

Once you decide on an academic pathway or program of study, you will need a MAP (My Academic Plan). This semester-by-semester plan outlines the courses you need to complete your degree or certificate. Your advisor will help you develop a MAP to stay on track.

Bookstore

The college bookstore is your on-campus resource for everything you need to thrive academically and feel connected to campus life. From textbooks and study materials to technology accessories, testing preparations, nursing supplies and apparel, snacks, graduation supplies, athletic apparel and MDC spirit gear, the bookstore offers a wide range of products and services at all campus locations and online. Students can take advantage of digital textbook options, rental programs, and online ordering for added convenience. For the most up-to-date store hours and location details, visit <https://www.mdc.edu/bookstore/>, as hours vary by term and may be extended during peak periods.

Bookstore Name	Store Phone	Bookstore Email	Store Manager
MDC Hialeah Campus Bookstore 1780 W 49th St, Room #1103 Hialeah, FL 33012-2918	305.237.8806	1184mgr@follett.com	Beatrice Flores
MDC Homestead Campus Bookstore 500 College Terrace, Room #F102 Homestead, FL 33030-6009	305.237.5042	1182mgr@follett.com	Nathaniel Austin
MDC Kendall Campus Bookstore 11011 SW 104th St, Room #8105 Miami, FL 33176-3393	305.237.2361	0835mgr@follett.com	Jessenia Delgado
MDC Medical Campus Bookstore 950 NW 20th St, Room #1180 Miami, FL 33127-4622	305.237.4178	0838mgr@follett.com	Chiquita Roberson
MDC North Campus Bookstore 11850 NW 27th Ave, Room #4101 Miami, FL 33167-3418	305.237.1247	0836mgr@follett.com	Cecilia Armstrong
MDC Online Fulfillment Center	305.237.1455	1214mgr@follett.com	Celeste Alvarez
MDC Padrón Campus Bookstore 701 SW 27th Ave, Room #6017 Miami, FL 33135	305.237.6019	1183mgr@follett.com	Salmayen Delos Angeles Davila Velasquez
MDC West Campus Bookstore 3800 NW 115th Ave Doral, FL 33178	305.237.8953	1234mgr@follett.com	Jordan Rodriguez
MDC Wolfson Bookstore 500 NE 2nd Ave, Room #7174	305.237.3236	0837mgr@follett.com	Carlos Guerra

Textbook Affordability and Transparency

Miami Dade College is committed to helping students manage the cost of their education by providing information about required textbooks and instructional materials before registration. In accordance with **Section 1004.085, Florida Statutes**, the College promotes textbook and instructional material affordability and transparency and provides access to lists of required and recommended course materials for the vast majority of courses offered each term. MDC Textbook and Instructional Materials published list for the preceding five (5) academic years: <https://app.smartsheet.com/b/publish?EQBCT=73402d80749748a497083944b53cc41b>

First Year Experience

Associate of Arts degree-seeking students must enroll in a Student Life Skills course, such as SLS1106 (or approved equivalent courses). In this course, students will learn skills which will assist in a successful transition into college. Students will learn to develop the skills required for success in college and beyond.

New Student Center

The New Student Center is the first point of contact for prospective and new students who are attending college for the first time or who are transferring from another institution. Prospective students are encouraged to meet with a college admissions counselor to obtain information about degree and vocational program options, admissions requirements, assistance with the admissions process and the steps a new student will take from admission through course registration. The New Student Center conducts orientation sessions prior to each semester. All new degree-seeking students are required to participate in an orientation program. The objective of the new student orientation sessions is to provide practical information to assist new students in transitioning to college life. The New Student Center at the Medical Campus (MC) also assists students with selection into limited access programs as they transition from other campuses. Staff help guide students with course selection, the development of educational plans, and the application process for the limited access programs at MC.

Services for Students with Disabilities

ACCESS – A Comprehensive Center for Exceptional Students' Services

Federal and state laws and regulations guarantee students with disabilities equal access and equal opportunity in post-secondary education. The College provides auxiliary aids and services to assist students with disabilities in achieving equal opportunity. These services include, but are not limited to, assistance with registration, advisement, financial aid, and sign language interpreting services, note takers, adaptive or assistive technology, testing accommodations and more. The ACCESS department works to promote awareness of disability issues, federal and state regulations, and College procedures that ensure accessible and supportive learning environments. Under certain circumstances, ACCESS can arrange for program modifications, course substitutions, and waivers, in accordance with the College's Manual of Procedures. Students may find out about additional services and eligibility by contacting the ACCESS office or representative in the Division of Student Services at their campus and visiting on line at (www.mdc.edu/ACCESS/).

Student Health Services

Miami Dade College is not legally or financially responsible for medical care and does not provide the services of a physician on any campus. At the time of application, each student should provide, on the appropriate line of the application form, the name of a person to contact in an emergency. If that contact person changes while the student is attending the College, the student should update that information through the Registrar's Office. Students should carry emergency information at all times, as well as any medical insurance card(s).

Single Stop

Single Stop aims to connect students with valuable resources, so they can concentrate on what matters most: their coursework, families, and futures. It is a one-stop source for students and immediate family members to be connected to public benefits and local resources. Single Stop offers students a wide array of services including benefits screening, individual counseling, free tax preparation, financial coaching, food pantry, health insurance assistance, and various other services (<http://www.mdc.edu/singlestop/>).

Student Wellness

Miami Dade College offers a range of resources to help students emotionally and physically navigate the demanding pace of modern life. From food pantries to financial planning to mental health counseling, Miami Dade College can help students get the help they need (<https://www.mdc.edu/student-wellness/>). The College offers the MDC community with the necessary tools and knowledge to make health and fitness part of a successful lifestyle with fitness, aquatics, and wellness centers at the Kendall, North, and Wolfson campuses. Additionally, the College is deeply committed to supporting student mental well-being by offering comprehensive resources that go beyond traditional counseling services. Licensed mental health professionals provide confidential, one-on-one counseling sessions across all campuses, ensuring students have timely access to qualified care. By combining direct counseling, innovative digital services, and preventive community-based supports, the College fosters a campus culture where mental health is prioritized as an essential part of student success.

Information and Policies

Automobiles on Campus

All vehicles parked on College property must be registered with MDC and have a valid parking decal prominently displayed on the rear bumper or left rear window. MDC Staff, On-Campus and Hybrid Students will receive an MDC parking decal in the mail. MDC Online and MDC Live students may obtain a temporary (daily) parking pass by visiting their campus Student Life or Public Safety office. Please visit the [Parking Decals page](#) for further details and to register your vehicle.

Vehicles may only be parked in clearly designated legal parking spaces and should park head-in only; do not back in the space. Areas which are posted with signs or have yellow pavement painting indicating a restriction or prohibition of parking must be observed and complied with. Parking in the roadway, taking more than one parking space, blocking another vehicle or parking on the lawn are all prohibited, except when specifically authorized. Driving or parking on any sidewalks or pedestrian ways is strictly prohibited. Vehicles operated by students and visitors are restricted or prohibited from parking in faculty/staff parking lots and other designated areas. These regulations are in effect during ALL HOURS of parking operation, and they apply equally to all students, faculty, staff, and visitors.

Parking Tickets

Parking tickets may be issued throughout the day to those vehicles parked illegally. They are issued by off-duty Police officers working on campus, or by any other law enforcement agency authorized to issue tickets. The Public Safety Department is not responsible for and cannot resolve any disputes between the owner of a ticketed vehicle and law enforcement officials. Parking violation tickets are issued on a regular basis. Additionally, violators may be towed at their own expense.

Parking Safety

Our parking safety is everyone's concern. Please follow the guidelines, rules and regulations to prevent crime and accidents. Below are some tips to observe for your safety.

- Never leave personal belongings unattended
- Keep your valuables out of sight and reach.
- In order for people not to know where your valuables are, secure them before you park on campus.
- Report any suspicious activity on campus, even if you are not the victim.
- Lock your car and keep your keys in a safe place.
- Report lost or stolen properties immediately. The faster you report, the better the chances to recover them.
- Stay alert and tuned into your surroundings. Be aware who is around and prepared to react accordingly.
- If you are a victim of a crime, please report it immediately to the appropriate Campus Public Safety Department.

Student and faculty parking areas are designated on each campus. The MDCard may be required for access to a lot or a garage. All vehicles must have a valid MDC parking decal prominently displayed on the rear bumper or left

rear window. Parking decals are good for one year. Failing to register a vehicle may result in the denial of parking privileges or other enforcement measures.

Although campus security officers patrol parking areas, the College assumes no responsibility for the care or protection of a vehicle or its contents at any time. If a vehicle must be left on campus overnight, students should notify the Campus Security Office. Visitor parking policies vary by campus, so visitors should phone ahead for information. Visitors parked in unauthorized spaces may be subject to traffic citations and towing at the owner's expense.

Family Educational Rights and Privacy Act (FERPA) - Information Statement

Release of Student Information

Miami Dade College has a longstanding commitment to protect students' rights and privacy of information. This commitment will continue as a matter of College practice. The College complies with the provisions of the federal Family Educational Rights and Privacy Act (FERPA), State of Florida law, and Florida State Department of Education, Florida College System rules. These federal and state requirements concern accessibility and confidentiality of student records. Miami Dade College Procedure 4085, Release of Student Information, provides pertinent details concerning classifications of student records and access and release provisions. The College procedure is available to students, faculty, administration and staff in the Dean of Student Services Office, as well as other offices and departments at each campus. In addition, the complete procedures are published in the Student's Rights and Responsibilities Handbook. In accordance with U.S. Public Law 93380 (FERPA), students at Miami Dade College have the right to inspect their educational records and to correct such records if warranted. All student records are open for inspection and review by the student unless he or she waives this right. These records are protected from release of information without written consent. The parent(s) of a dependent student, as defined in Title 26 U.S.C. §152 of the Internal Revenue Code, also has the right to inspect records which are maintained by the College on behalf of the student. Directory Information, which may be made public, includes:

1. Student name,
2. Major field of study,
3. Participation in officially recognized activities and sports,
4. Weight and height of members of athletic teams,
5. Degrees, honors and awards received,
6. Enrollment status (fulltime, halftime, not enrolled).

The office of the Dean of Student Services or designee will release this information only after the requestor has demonstrated a legitimate need to have such information. Students not wishing the dissemination of Directory Information must complete a statement in the Registrar's Office, otherwise Directory Information may be disclosed for legitimate purposes by the College. Additional details concerning the release of student information, including exceptions, challenges to the content of records and related matters, may be obtained by consulting the Dean of Student Services, the Registrar's Office or designee at any campus. FERPA information can be found on our website at: www.mdc.edu/main/ferpa/.

Grievance Policy

In compliance with federal and state requirements, the College has an institutional grievance policy as outlined in Procedure 4032. Prospective or enrolled students may voice a complaint about a College rule/regulation, procedure, or experience; including but not limited to discriminatory practices or sexual harassment.

Students are encouraged to first voice their complaint to the front-line staff and/or their supervisors. If the complaint cannot be resolved at that level, then supervisors will communicate it to the next leadership level and notify the student(s). Students who are unable to resolve their concerns/issues with the front-line staff and/or their supervisors may file a complaint in writing to the appropriate division head as listed in the table of this procedure. If the complaint is against a first or second-level individual in the chain of command, the student should present the concern to the next level. Students may also submit a complaint at any time by visiting the Student's Rights and Responsibilities website at <http://www.mdc.edu/rightsandresponsibilities/>.

Housing

Miami Dade College does not provide or supervise housing facilities. Out-of-area students should arrive with sufficient time in advance of registration to locate suitable housing.

Identification

The MDID is the official identification card for students and employees. Students are required to wear visibly and present their MDID when requested by authorized College officials. Any misrepresentation, alteration or misuse of identification is prohibited. This card will provide access to Learning Resources, laboratories and parking lots. Students with questions should contact the Student Life Office at any campus for details.

Students' Rights and Responsibilities

The Students' Rights and Responsibilities publication, available to all students, sets forth the rights of students with corresponding responsibilities. This document details the relationship between student and College. The document covers protection in academic pursuits and privacy of records, sets forth the conditions for responsible behavior on the campus and lists the various appeal mechanisms and grievance procedures available to students. The section on student discipline complies with Rule 6A14.56, F.A.C., and §240.132, §240.133 and §877.13, F.S. This section concerns control and discipline of college students. The document complies with relevant federal regulations such as the award of financial aid, protection of privacy of records and equal access/equal opportunity. For more information, visit <https://www.mdc.edu/procedures/Chapter4/4009.pdf> and <http://www.mdc.edu/rightsandresponsibilities/>.

Safety and Security

As required by the Federal Student Right to Know Legislation, the College publishes the annual crime statistics for each campus. These statistics may be obtained at the campus Public Safety Office.

Student Life

Campus Activities, Clubs and Organizations

Student Life at Miami Dade College is where leadership, connection, and campus culture come to life. With each campus offering a unique and vibrant community, there's something for everyone, whether you're looking to join a student organization, attend engaging events, build your leadership skills, or give back through service. Student Life is also your one-stop hub for getting connected on campus!

There are many opportunities for students to get involved in campus activities. Each year, outstanding artists, musicians, singers, dancers, lecturers and other performers share their talents and expertise with students. Student Life committees, composed of representatives from student groups, assist with the establishment of these programs and the policies governing these activities. In addition, there are on-campus art exhibits, dance programs, music concerts and theatrical productions presented by different campus departments. Students have the opportunity to join 85+ clubs chartered on the various campuses. The best time to find out about clubs and organizations on each campus is at the beginning of the semester, when most campuses hold special events to publicize the various clubs. Students may also visit the campus Student Life Department to find out how to get involved. All students are encouraged to participate actively in clubs and organizations.

Intercollegiate Athletics

Miami Dade College teams, all known as The Sharks, compete at the highest level of the National Junior College Athletic Association. Home to 39 national championships, the Sharks travel around the state to compete against other college teams, and they consistently finish in the higher rounds of conference and state events. Sharks also have the opportunity to compete for the National Junior College Championship and have the chance to be selected for NJCAA All American teams and other special awards. MDC offers first-rate athletic facilities, training and conditioning services and a talented coaching staff. For information on trying out for an athletic team, contact the college director of athletics, based at Kendall Campus.

Students with outstanding athletic abilities may try out for one of the following intercollegiate sports teams: men's basketball, baseball, and soccer; women's, basketball, volleyball, soccer, and softball. <https://www.mdcathletics.com/>

Student Government Association

The Student Government Association is the voice of the student body and represents students' interests and concerns to the campus leadership. Through Student Government, students build leadership, organizational and communication skills. They also have the opportunity to represent the student body in campus, college, and statewide meetings.

Student Publications

The award winning student publication, The Reporter, is the student newspaper at Miami Dade College. It was launched on Oct. 4, 2010 and is augmented by a website with video and audio content. The Reporter is distributed collegewide and has newsrooms at the North, Wolfson, Kendall, Hialeah, Homestead, and Padron campuses. These publications serve as the media for student expression on matters involving the curricular and extracurricular activities of the College. These publications also provide training for those interested in journalism. <https://www.mdc.edu/studentlife/publications/>.

The Students' Rights and Responsibilities Handbook provides students on each campus with basic information about collegewide policies and procedures.

MDC Shark Pack Program Overview

Shark Pack – Digital Materials

Miami Dade College complies with Florida Statute (F.S.). Section 1004.085, F.S. and Rule 6A-14.092, Florida Administrative Code (F.A.C.), and has established guidance in the adoption, posting, and reporting requirements related to textbook and course material affordability and transparency. The College strives for transparency in the adoption of textbook and course materials, minimizing cost of instructional materials for students, addressing the need to make required textbooks and course materials available to students who otherwise are unable to afford them, while maintaining the highest quality of education and academic freedom. The Shark Pack program is designed to guarantee that every student has the textbook and course materials they need on day one. In partnership with our bookstore, Follett Higher Education, we have created a textbook savings initiative designed to offer our students high-quality, affordable learning materials with reliable first-day availability. Students are automatically enrolled in the program when registering for participating credit sections and may review the associated digital material fees on MDConnect Student Center. Students will receive direct emails from our bookstore partner before each term with course material access instructions. Course materials are delivered directly through our Learning Management System, Canvas, and are available in digital formats such as ebooks or courseware. In select cases, print materials may be made available when they best support the program's pedagogical needs. If students choose to participate in Shark Pack, they will receive a list of all the courses in which they are enrolled, detailing what materials are required, how to access those materials, as well as access to any digital materials they have been provided as part of the program. Additionally, if students wish to obtain materials another way, the program offers flexibility to opt out by individual class section before the 100% refund deadline. An opt-out option is provided to all students through the posted add/drop date, per [MDC's Academic Calendar](#), for each class section for which they do not want the digital materials. All materials they had access to on the first day of class will be deactivated and the digital material fee removed from their account. If physical prints were received, these will need to be returned to the bookstore once Follett reaches out requesting the return of materials. For more information on the Shark Pack program, visit the website at <https://www.mdc.edu/sharkpack>.

1. Digital Materials Fee: All participating credit courses with textbook adoptions will have a pricing of \$32.50 per credit.
 - Participating Medical Campus course materials have a flat rate per class with an average savings of 25% off the current retail price.
 - English for Academic Programs (EAP) courses have a flat rate per course at \$121.00.

Note: Fees are subject to change without notice.

Shark Pack Resources			
Carmen G. Velasquez	Associate Dean, Academic & Faculty Affairs	cvelasqu@mdc.edu 305.237.3361	College-wide Shark Pack Liaison

Basic Skills Assessment Program

In an effort to provide more effective educational services for students, the College has established a Basic Skills Assessment Program. Through this program, students can demonstrate readiness and the College can identify academic strengths and weaknesses in reading, writing and mathematics. Methods for assessing

communication and computation skills may be through any combination of approved common placement tests or alternate methods. These assessment results are used to advise students on how best to take advantage of their strengths. Regarding weaknesses, assessment results are used to guide a student into courses designed toward improvement in the respective discipline.

MDC administers the ACCUPLACER Next-Generation and the Florida Postsecondary Education Readiness Test (PERT) as placement testing options for students. Both are not timed, and each consists of three sections: reading, writing and mathematics. With the exception of students who meet the criteria for an exemption from common placement testing and develop mental education instruction, the State Board of Education (SBOE) requires demonstration of readiness for degree seeking students and students who have not met college level competency either through the completion of developmental education requirements in the Florida College System or have not been awarded credit for college-level coursework in the area of deficiency. Methods for assessing communication and computation skills may be through any combination of approved common placement tests or alternate methods. Non-exempt students whose assessment results indicate a need for developmental education shall enroll in developmental education in the area of the deficiency. Students whose native language is not English may be required to take another test to measure their English proficiency and subsequently matriculate through EAP to demonstrate readiness. Students with documented disabilities are provided appropriate testing modifications or accommodations.

The SBOE also requires institutions offering Postsecondary Career and Technical Education and Adult Education programs to test students pursuing these programs. Miami Dade College (MDC) may accept official test scores from approved academic institutions and approved regional workforce boards. If a student presents valid test scores or high school course grade criteria that meet or exceed the state minimum score and/or grade requirements, he or she does not have to take the placement test in the related sub test area. All test scores presented must have been obtained within the past two years and course grades are limited to currently enrolled secondary school students and students who graduated from a high school within a two (2) year period. To find out what minimum scores or course grade a student needs to be exempt from taking the placement test, or for other reasons why a student may not be required to take the placement test, students are asked to contact the campus Testing and Assessment Department. This information may also be acquired by visiting the Testing and Assessment website (<https://www.mdc.edu/testing/testing-criteria/>).

Students who have to take the placement test should utilize the 'Test Preparation' resources available on the specific test webpage before they take the placement test. If a non-exempt student's scores on one or more of the subtests of the placement test fall below the minimum passing scores established by the SBOE, he or she must enroll for at least one developmental education course during their first term. Further evaluation may be conducted in classes, and developmental education course placement changed, based on the results of the additional assessment.

The SBOE also requires institutions offering Postsecondary Career and Technical Education and Adult Education programs to test students pursuing these programs. The TABE and CASAS Steps are used for this purpose at MDC. Adult Education students without English proficiency are administered the CASAS Steps for placement into appropriate Adult English for Speakers of Other Languages (ESOL) program courses.

If a student has any questions regarding these assessments, including exemptions, he or she should contact the campus Testing and Assessment Department. This information may also be acquired by visiting the Testing Criteria webpage referenced above in the third paragraph.

Learning Resources

Learning Resources houses the Library, Computer Courtyard, and various multi-disciplinary tutoring centers. Students have access to comfortable study spaces for individual or collaborative learning with librarians and other academic experts within reach. At each campus, students will find an extensive collection of books (both academic and bestsellers), periodicals, videos, and mobile devices such as laptops and headsets. Additionally, students have access to eBooks, streaming videos, and an large selection of journal and magazine articles available online. All Learning Resources computers are equipped with the latest educational software applications required by faculty at MDC. All items within the collection are available to check out at no cost to students and staff. Librarians are available to assist students during the research process both on a one-on-one basis and in the classroom. In addition, Learning Resources works closely with the Academic Disciplines to provide tutoring in a wide variety of subjects. Supplemental instruction sessions from embedded tutors, learning assistants, and educational technologists can also extend the tutoring services into the classroom upon faculty request. Students can take

advantage of hundreds of educational workshops designed to enhance valuable skills while in college, and later in the job place. All library and instructional support services are available on campus and online. For more information, visit <https://www.mdc.edu/learning-resources/>

Registration and Records

Students may register online by going to the College's Homepage (www.mdc.edu/registration). Students may also register for courses in person at each campus Admissions and Registration. The College Registrar's Office is the designated custodian of all official academic records. The College Registrar's Office maintains official student transcripts, processes final grades at the end of each term and updates student records with address, name and approved grade changes. The campus Admissions and Registration offices maintains official student transcripts, processes final grades at the end of each term and updates student records with address, name and approved grade changes. The Collegewide Transcript Processing Services area provides official copies of student transcripts to students, or to institutions or agencies upon request from students. The College also participates in the electronic transmission of student transcripts (to other participating institutions). Students may request academic transcript online at www.mdc.edu/transcripts.

ACADEMIC REGULATIONS

Attendance in Class

Students are expected to attend every class meeting and to arrive on time. Students who expect to miss a class, or those anticipating tardiness, should let the instructor know. In most courses, attendance requirements are listed on the syllabus. It is the responsibility of the student to make up work missed.

Audit

Students desiring to enroll in a course at Miami Dade College, but who do not wish to receive a grade or credit for the class, may elect to audit. Students will not be allowed to change from an audit status to a credit status (or from credit to audit) after the 100 percent refund date for each term. Audit courses will be included in the student's academic record with a non-punitive grade of "Z"; however any audit courses taken prior to fall 2016 will show a grade of "X". Courses and credits enrolled for audit purposes do not count in the computation of a student's full-time or part-time enrollment status. College Preparatory students, who are required to be certified as completing competency-based College Preparatory instruction, cannot be enrolled under audit status. Auditing a class costs the same as enrolling for a credit course.

Course Load

All credit courses carry a specified number of credits. A 3-credit lecture course normally meets three hours per week during the 16-week terms, and eight hours per week during the six-week terms. Lab classes generally meet for two hours per credit. The fall and spring terms are called "major terms" and are approximately 16 weeks long. During a major term, a full course load is considered to be 12 or more credits. The summer term consists of two six-week summer sessions (first six-weeks/second six-weeks). Some courses are scheduled for the combined summer sessions of 12 weeks. During the six-week summer session a full load is considered to be 6 or more credits. It is suggested that students who are employed should reduce their college load as follows:

Work Hours Per Week	# Credits Fall/Spring	# Credits Summer A/B
20	12-15	6-7
25	8-11	5-6
40	6-7	3

Grading System

Students in college credit and vocational credit courses are graded according to the following grade point average (GPA) system:

A. Used in GPA computation:		
Grade	Interpretation	Point Value
A	Excellent	4
B	Good	3
C	Average	2
D	Poor	1
F	Failure	0

B. Not used in GPA computation:	
Code	Interpretation
I	Incomplete
W	Withdrew
Z	Audit
S	Satisfactory
P	Progress - course requirements not completed, student must repeat

U	Unsatisfactory
NC	No Credit
NG	No grade assigned
NR	Grade not recorded by instructor

Final grades are available on the student portal following the end of the term.

Grade Point Average (GPA)

Each letter grade has a point value (see above). To compute the grade points for a course, multiply the grade point value by the number of credits. For example, a "B" in a 3-credit course, is worth 9 points. A "B" in a 4-credit course is worth 12 points. To calculate a GPA, add the total grade point values for all courses and divide that figure by the total number of credits attempted. In order to graduate from any credit program at MDC and/or to qualify for entry into a bachelor's degree program, a student must have a minimum 2.0 GPA.

Course Code	Credits	Grade	Total Points
ENC 1101	3	A	12
HUM 120	3	C	6
ISS 1120	3	F	0
ISS 1161	3	B	9
ART 1300C	3	C	6
DAA 1160	1	B	3
Total	16		36
<i>Divide 36 points by 16 credits = 2.25 GPA</i>			

Repeating Courses

Students may repeat courses taken at MDC if they received a "W," "U," "D," or "F" grade. State rule (F.A.C. 6A-14.0301) limits the number of repeat attempts to three per course. The third and final repeat attempt (i.e., the fourth time a student attempts the course) may be granted only if the student petitions (<https://www.mdc.edu/smart/>) through an appeals process, and if the student has documentation to convey extenuating circumstances. However, a student is not permitted to withdraw during the third or fourth attempt (i.e., a grade must be assigned). Repeated surcharges apply to any third or fourth attempt. All courses originally taken and then repeated will appear on the student's transcript with assigned grades, but the GPA will be recomputed to average the third and subsequent attempts of computable grades. Specific courses, as identified in the course description section, may be repeated multiple times for additional credit. All attempts of these courses will be included within the GPA. Students should note that some state universities and colleges may not accept courses repeated for additional credit. Students should also be aware that some private colleges or universities might not accept the grade of a repeated course, and that some institutions compute the grade originally assigned.

Incomplete "I" Grade

When a student is unable to complete the requirements of a course by the end of the semester, the student may be assigned an "Incomplete" or "I" grade. The "I" grade is recorded by the instructor if the student has valid reasons for not being able to finish the work. The student and instructor complete an "Agreement for Grade of Incomplete" form, which stipulates the work to be completed for a grade. If the student has not completed the required coursework after 180 calendar days, the incomplete grade will be changed to a failing grade.

Grade Appeals

The responsibility for the academic evaluation and assignment of grades is that of the faculty member teaching the course. A student who believes that he or she has been unfairly graded should first appeal the grade to the faculty

member. If satisfaction is not achieved, the student may appeal through administrative channels (Department Chair, Academic Dean or the grade appeals committee <http://www.mdc.edu/procedures/Chapter8/8301.pdf>).

Academic Amnesty

Students with credits more than 10 years old may petition to have these grades excluded from cumulative GPA calculation. This is a one-time privilege. Students may not request specific courses to be removed; it must be the entire prior record. Students may obtain a Request for Academic Amnesty form at any Advisement & Career Services Department. Since academic amnesty does not remove courses from the academic record (it only excludes them from cumulative grade point average calculations at MDC), state/federal regulations concerning course attempts will still apply. Therefore, if a student has three or more attempts in the same course, state regulations mandate that the student be assessed the full cost of instruction and not be permitted to withdraw after the 100 percent refund deadline for the course. Academic amnesty does not apply to federal and state financial aid regulations. The academic record impacted by academic amnesty is not excluded from federal and state financial aid policies. Students should consult an MDC financial aid advisor prior to requesting academic amnesty. Academic amnesty requests that include courses that were used as part of the requirements for a previously awarded program of study will not be processed.

Petitions Committee

The Petitions Committee (www.mdc.edu/rightsandresponsibilities/) considers exceptions to financial and withdrawal policies as stated in this catalog. Students should submit a written petition to the committee. The committee will make a recommendation to the Dean of Student Services for approval and implementation. The decision of the Dean is final. Petitions should identify the student (complete name and student number), and clearly and concisely state the request (by writing a personal letter and supplying supporting documentation for the reason stated in the letter). Students should address the petition to: Petitions Committee, Dean of Student's Office, and submit the petition at the campus at which they are registered for courses. Petitions must be made by the end of the next major term (fall and spring).

Student Ombudsman

MDC has a student ombudsman who serves as the initial point of contact for students who have concerns, complaints or issues related to College processes, policies and procedures. The Ombudsman listens to student concerns; directs students to the appropriate MDC office, policies and procedures; and, if requested, assists students in completing the forms required to obtain a resolution. The Student Ombudsman has the authority to investigate issues and arrange meetings among the involved parties in order to reach a resolution. MDC has designated the Student Ombudsman as the Assistant Dean of Student Services or designee at each campus (<https://www.mdc.edu/financialaid/how-aid-works/access-consumer-resources.aspx>).

Standards of Academic Progress

The main purpose for the Standards of Academic Progress (SOAP) Procedure is to establish a formal process through which the faculty, staff, and administration at Miami Dade College may identify and provide support to students who experience academic difficulty and fall below a Combined Cumulative Grade Point Average (GPA) of 2.0 (Calculated from the combined graded units for GPA). The combined Cumulative GPA includes computation of grades for both MDC and posted transfer courses. Good Academic Standing is defined as 2.0 or higher for the Combined Cumulative GPA. SOAP is not intended to discourage or penalize students. Rather, SOAP reflects the commitment of the College's faculty, staff, and administration to provide students with assistance and support to ensure success in achieving their educational goals. Students have available to them a variety of means to remedy their cognitive and non-cognitive challenges and to be academically successful. When academic progress has not been satisfactory, SOAP requires students to meet with an academic and career advisor to develop an academic improvement plan and discuss support services that may assist them in achieving good academic standing. Students who are not in Good Academic Standing are ineligible to run for executive board positions in student organizations unless special permission is granted by the Dean of Students at their home campus. The overall objective of SOAP is to improve the performance of students experiencing academic difficulty by connecting them to academic and student support services (<https://www.mdc.edu/procedures/Chapter4/4010.pdf>).

Categories for Standards of Academic Progress

- *Warning:* from 8 to 16.99 combined graded units/credits for GPA with less than 2.0 for the Combined Cumulative GPA.

- *Probation*: from 17 to 45.99 combined graded units/credits for GPA with less than 2.0 for the Combined Cumulative GPA or previously in Probation and Term GPA is less than 2.0.
- *Course Load Reduction Warning*: from 30 to 45.99 combined graded units/credits for GPA less than 1.5 Combined Cumulative GPA and previously in Good Academic Standing or 46 or more combined graded units/credits for GPA less than 2.0 for the Combined Cumulative GPA and previously in Good Academic Standing.
- *Course Load Reduction 1*: from 30 to 45.99 combined graded units/credits for a GPA of less than 1.5 for the Combined Cumulative GPA and previously not in Good Academic Standing.
- *Course Load Reduction 2*: 46 or more combined graded units/credits for GPA with less than 2.0 for the Combined Cumulative GPA and previously not in Good Academic Standing.
- *Extended Course Load Reduction*: 30 or more combined graded units/credits for GPA and previously on Course Load Reduction Warning, 1, 2, or 3 with a Term GPA of 2.0 or higher.
- *Course Load Reduction 3*: 30 or more combined graded units/credits for GPA and failure to meet the minimum requirements of maintaining a Term GPA of 2.0 or higher during Extended Course Load Reduction or Course Load Reduction 1, 2, or 3.

Interventions for each Category of Standard of Academic Progress (SOAP)

- *Warning*: Students in this category are encouraged to meet with College staff to discuss enrollment and academic planning. The College provides support with registration, updating academic plans, and development of educational and career goals as well as connections with academic resources and holistic student support services. In an effort to support student success, course load reductions for the subsequent term may be placed to ensure the student makes adequate progress.
- *Course Load Reduction Warning*: Students in this category must meet with a Student Success Advisor to discuss enrollment and academic planning. This plan may include referrals to academic support services and/or holistic student support services. In an effort to support student success, course load reductions for the subsequent term may be placed to ensure the student makes adequate progress.
- *Probation*: Students in this category may register for up to 12 units/credits per semester and must meet with College staff to discuss enrolling beyond the unit limit. The College provides support with registration, updating academic plans, and development of educational and career goals as well as connections with academic resources and holistic student support services. In an effort to support student success, course load reductions for the subsequent term may be placed to ensure the student makes adequate progress.
- *Extended Course Load Reduction*: Students in this category may register for up to 9 units/credits and must meet with College staff to discuss enrolling beyond the unit/credit limit. The College provides support with registration, updating academic plans, and development of educational and career goals as well as connections with academic resources and holistic student support services. In an effort to support student success, course load reductions for the subsequent term may be placed to ensure the student makes adequate progress.
- *Course Load Reduction 1 or 2*: Students in this category may register for up to 9 units/credits and must meet with the Director of Advisement or their designee to discuss enrolling beyond the unit/credit limit. Students are also encouraged to meet with a Student Success Advisor to develop an academic improvement plan. This plan may include referrals to academic support services and/or holistic student support services. In an effort to support student success, course load reductions for the subsequent term may be placed to ensure the student makes adequate progress.
- *Course Load Reduction 3*: Students in this category may register for up to 7 units/credits and must meet with the Director of Advisement or their designee to discuss enrolling beyond the unit/credit limit. Students are also encouraged to meet with a Student Success Advisor to develop an academic improvement plan. This plan may include referrals to academic support services and/or holistic student support services. In an effort to support student success, course load reductions for the subsequent term may be placed to ensure the student makes adequate progress.

Advisement at their home campus for consideration of continued enrollment:

1. Academic Suspension 1 or 2 Students are required to pause enrollment in credit courses at the College for the next term (fall, spring, or summer).
2. Extended Academic Probation Students who have successfully appealed Academic Suspension or who have met the time conditions for suspension may continue enrollment in credit courses at the College under the category of Extended Academic Probation, provided that they maintain a Combined Cumulative GPA of 2.0 or higher. Students must meet with an academic and career advisor prior to enrolling in courses to develop an

academic improvement plan which may include referrals to academic support /tutoring and/or student support services, career advising, learning style assessment, and/or enrollment in a Student Life Skills (SLS) course. Students may also be required to reduce their course load.

3. Academic Dismissal Students are required to discontinue enrollment in credit courses at the College for at least twelve months. After the twelve-month dismissal period, students are required to meet with the Dean of Students or designee at their home campus to petition for re-admission.

Guidelines for Appeal of Standards of Academic Progress

Students may request an exception to course load reductions according to MDC Procedure 4015 Guidelines for Appeal of the Standards of Academic Progress (<https://www.mdc.edu/procedures/Chapter4/4015.pdf>) as follow:

1. Probation and Extended Course Load Reduction statuses Students in these statuses may meet with college staff to request to exceed course load reductions. Students not approved to exceed course load reductions by college staff may appeal to the Director of Advisement or designee. Students not approved to exceed course load reductions by the Director of Advisement or designee may appeal the decision to their home campus' Dean of Students or designee and their decision is final.
2. Course Load Reduction 1, 2, or 3 statuses Students in these statuses may submit a request to exceed course load reductions to the Director of Advisement or designee at their home campus. Students not approved to exceed course load reduction by the Director of Advisement or designee may appeal the decision to their home campus Dean of Students or designee and their decision is final.
 - Documentation Students will be encouraged to complete the "Exception to Course Load Reduction Questionnaire" form and present evidence/supporting documentation that reflects a change in circumstances in order to enroll beyond course load reduction.
 - Academic Improvement Plan Students must meet with a Student Success Advisor prior to enrolling in courses to develop an academic improvement plan which may include a referral(s) to academic support/tutoring and/or student support services, career advising, learning style assessment, and/or enrollment in a Student Life Skills (SLS) or other prescribed intervention course. Students may also be required to reduce their course load.
 - Decision and Record of the Appeal The decision of the Dean of Students or designee is final on behalf of the College. Students will be informed of the decision in five (5) business days.

Standards of Progress for Students Receiving Financial Aid

A student receiving financial aid must meet "Standards of Academic Progress." Federal regulations state that students are eligible to receive financial aid benefits for up to 150 percent of the number of credits registered to complete the degree or certificate. After the 150-percent mark, benefits will terminate. This applies to all registered credits, including courses that were attempted or withdrawn from, but not including "I" grades or audits. Thirty credits of College Preparatory and AP credits are exempted from this 150-percent rule. Students who meet or exceed the 150 percent are no longer eligible to receive federal/state financial aid. For extenuating circumstances, students may appeal through the Petition for Financial Aid Waiver.

Code of Conduct

By the act of registering at Miami Dade College, a student agrees to abide by the Code of Conduct of the College. A student who violates the Code of Conduct while on College property or while participating at a College sponsored event may be suspended (<http://www.mdc.edu/procedures/Chapter4/4025.pdf>). Refer to Petitions Committee (<https://www.mdc.edu/rightsandresponsibilities/>) for more information.

Transcript of Records

A transcript (<http://www.mdc.edu/transcripts/>) is a printed list of all the courses taken, the number of credits and grade earned. Transcripts summarize the GPA and also indicate the receipt of any certificates or degrees. Students must request their official transcripts via the student portal (<https://www.mdc.edu/transcripts/official.aspx>) or in-person at the Office of the College Registrar.

Drops and Withdrawals

Dropping Courses

Students may drop courses within the drop period indicated on their class schedule. They may do so online using

their MyMDC account, or in person at any campus Admissions and Registration Office. Note that a reduction in course load may impact athletic eligibility, financial aid, scholarships, and veteran benefits.

Administrative Withdrawal from Courses

Miami Dade College reserves the right to cancel courses and/or programs for which there is insufficient enrollment, to close a course when the enrollment limit in that course is reached and to make any schedule changes as necessary, including a change in time, days, credit, location or instructor. In the event of course cancellation, the College will notify each registrant by email and/or by telephone and will issue a full refund for the course. Miami Dade College is not responsible for any other related expenses. Students may see an academic advisor regarding selection of another course. Faculty have the right to withdraw a student from their course for lack of attendance ("no show") or excessive absences as determined by established departmental guidelines. If a student is withdrawn from a course or courses for disciplinary reasons, the student is not entitled to a refund of matriculation, tuition or special fees.

College Withdrawal Policy for Credit and Developmental Education Courses

Students who enroll in the same course for the third time (or subsequent time) will not be permitted to withdraw from (drop) the course. Attempts taken by students prior to the Fall Term 1997 will not be counted as an attempt for the purposes of this policy. An attempt is counted any time students officially enroll, have a paid fee invoice for a course, and do not withdraw from (drop) the course with a refund. For example, a student enrolled in a course in the Fall Term 2015 and received an "F" grade in the course. The student enrolled again in the same course in the Spring Term 2016 and found it necessary to withdraw from (drop) the course with a grade of "W" (Withdrawal). The student enrolled again for the third time in the course for the Summer Term 2016. The student needed to withdraw again from the course. This was not permitted, and the student received a valid grade of A, B, C, D, F, S, P or U at the end of the term.

Cost to Re-Enroll in a Course

Florida Statutes 1009.28 (applies to developmental education courses) and 1009.285 (applies to college credit courses) state that students who enrolled in the same course twice, received a grade of W, D, F, P, U or X and wish to re-enroll for the third time must pay the full cost of instruction for this attempt and any later attempts. This fee is equivalent to the cost of the course for a student paying fees as a nonresident of the state of Florida for tuition purposes. Students who are assessed the higher fee on the third attempt only may appeal to have the fee lowered. Contact the Dean of Student Services Office (<https://www.mdc.edu/remote-learningstudents/student-support-services/dean-of-students>) at the campus where the course is offered for more information. A student cannot re-enroll in a course for credit if the student previously earned a grade of I, S, C or better. This policy is a result of 6A-14.0301 Florida Administrative Code.

Simple Syllabus

Simple Syllabus is Miami Dade College's official syllabus platform. It provides students with easy access to current course information, including learning outcomes, course expectations, required materials, grading policies, and important College resources. By using Simple Syllabus, students can stay informed, understand course requirements, and access key information from any device throughout the term. MDC Syllabus Library: <https://mdc.simplesyllabus.com/>

GRADUATION REQUIREMENTS AND TRANSFER INFORMATION

Graduation Requirements

Miami Dade College offers programs leading to the Associate in Arts (AA), Associate in Science (AS), and baccalaureate degrees, as well as Certificate of Professional Preparation (CPP), Advanced Technical Certificates (ATC), College Credit Certificates (CCC), and Career Technical Certificates (CTC). Students must satisfy general education and/or program requirements to earn a credential. Information about academic offerings is available through the Miami Dade College Academic Programs page: <https://www.mdc.edu/academics/programs/default.aspx>.

Continuous Enrollment for Graduation Requirements

The College graduation requirements are based upon the term of admission or readmission to Miami Dade College. Those requirements apply as long as the student has continuous annual enrollment. If a student does not register for a period exceeding one year, he or she is subject to the graduation requirements in effect for the year and term of re-admission to the College.

Baccalaureate Degree

Required Hours and GPA

Successful completion of a minimum of 120 semester hours or as approved by the State Board of Education, and a minimum GPA of 2.0 is required to earn a baccalaureate degree.

Note: A higher GPA may be required for specific majors.

General Education

Satisfactory completion of General Education courses (36 semester hours) as follows Per 6A-10.030, Florida Administrative Code:

- Area I. English Composition (6 semester hours)
- Area II. Oral Communication (3 semester hours)
- Area III. Humanities/Fine Arts (6 semester hours)
- Area IV. Behavioral/Social Science (6 semester hours)
- Area V. Natural Science (6 semester hours)
- Area VI. Mathematics (6 semester hours)
- Area VII. General Education Elective (3 semester hours)

Civics Literacy Competency

Prior to the award of an associate in arts or baccalaureate degree, new and returning students entering a Florida College System institution in the 2021-22 catalog year, and thereafter must demonstrate competency in civic literacy. This requirement may be satisfied by passing AMH 2010 or AMH 2020 or POS 2041 AND an approved exam (AP, CLEP, or FCLE). For more information regarding the Florida Civic Literacy Requirement, go to [Civic Literacy Competency](#).

Computer Skills Competency

By the 16th earned college-level credit, students must attempt the computer competency requirement OR by the 31st earned college-level credit, students must satisfy the requirement (CGS1060C, an equivalent college credit course or the College's approved computer competency test). For more information, see [Computer Competency](#).

Requirements for Admission to Upper Division

Students should review the Baccalaureate Degree section of this catalog and contact the Academic Department responsible for the specific Baccalaureate admissions criteria.

Foreign Language Requirement

In accordance with Rule 6A-10.02412, Florida Administrative Code and pursuant to Section 1007.262, Florida Statute, all AA and baccalaureate degree seeking students must demonstrate Foreign Language Competence (FLC) by: (a) successfully completing the elementary 2 level (i.e., 2 years of secondary/high school level) or postsecondary/

college level equivalent in one (1) foreign language or American Sign Language OR (b) successfully completing a standardized examination that documents the required FLC. Students who demonstrate proficiency in a native language other than English are exempt. For more information, please visit the MDC Credit-by-Exam website or contact the MDC World Languages department. For more information, refer to [Foreign Language Competency](#).

Associate in Science Degree

Associate in Science degree: Awarded to students who successfully complete a program of career and technical instruction consisting of lower division college credit courses to prepare for entry into employment. The associate in science degree is a transfer degree and a basis for admission to a related bachelor's degree. The associate in science degree shall be awarded upon satisfactory completion of a planned program of instruction comprised of the standard credit hour length established, after demonstration of the attainment of predetermined and specified performance requirements. The standard credit hour length of all associate in science degree programs as defined in Rule 6A-6.0571, F.A.C.

Requirements for Associate in Science Degree

1. Complete an approved program of study with 60 credits or as approved by the State Board of Education.
2. Earn a minimum 2.0 cumulative GPA in the 60 credits or as approved by the State Board of Education required for graduation.
3. Complete a minimum of 15 credits of general education courses.
4. First-time-in-college students must demonstrate competency in civic literacy.
5. Meet MDC Residency for Graduation requirements.

General Education and Miami Dade College Student Learning Outcomes

General Education: Student Learning

General education courses should provide broad foundational knowledge to help students develop intellectual skills and habits that enable them to become more effective and lifelong learners and contain high-level academic and critical thinking skills.

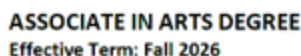
Student Learning Outcomes: A Promise between Students and Faculty

All students who graduate from MDC - regardless of major or degree type - have ten things in common: the college-wide student learning outcomes (CSLO). Developed collaboratively by students, faculty, alumni and industry partners, and adopted in 2007, the outcomes are part of the Miami Dade College learning experience. The Learning Outcomes Committee (LOC) provides oversight of MDC's assessment and student learning improvement activities. The CSLOs summarized below will help students succeed in their chosen field, strengthen critical life skills, and encourage lifelong learning:

1. Communicate effectively using listening, speaking, reading, and writing skills
2. Use quantitative analytical skills to evaluate and process numerical data
3. Solve problems using critical and creative thinking and scientific reasoning
4. Formulate strategies to locate, evaluate, and apply information
5. Demonstrate knowledge of diverse cultures, including global and historical perspectives
6. Create strategies that can be used to fulfill personal, civic, and social responsibilities
7. Demonstrate knowledge of ethical thinking and its application to issues in society
8. Use computer and emerging technologies effectively
9. Demonstrate an appreciation for aesthetics and creative activities
10. Describe how natural systems function and recognize the impact of humans on the environment

General Education Requirements for the Associate in Arts

To receive an Associate in Arts, students must complete 36 Units of General Education courses and 24 Units of Pathway Elective courses. Total credits required for the degree: 60. Students must complete the following pathway/program information located at <https://www.mdc.edu/academics/programs/ps/associate-in-arts-pathway-guide-effective-2026.pdf>:



*Effective term depends on your most recent term of admission to the College. Students admitted to the College prior to Fall 2024 can view their effective term on the Academic Requirements page in MDConnect at mdconnect.mdc.edu.

NATURAL SCIENCES 6 Credits* Select at least 1 course from the 13 State Core options. Lab Credits are not allowed in this area.		GENERAL EDUCATION ELECTIVE 3 Credits Select at least 1 course from the following options.				
State Core 3 Credits 1. AST 1002 - Descriptive Astronomy 2. BSC 1005 - General Education Biology 3. BSC 2010 - Principles of Biology 4. BSC 2085 - Human Anatomy and Physiology I 5. CHM 1020 - General Education Chemistry 6. CHM 1045 - General Chemistry & Qualitative Analysis 7. ESC 1000 - General Education Earth Science 8. EVR 1001 - Introduction to Environmental Science 9. GLY 1010 - Physical Geology 10. OCE 1001 - Introduction to Oceanography 11. PHY 1020 - General Education Physics 12. PHY 2048 - Physics with Calculus 1 13. PHY 2053 - Physics (without Calculus) 1		AMH2010 AMH2020 AMH2035 AML2010 AML2020 ANT2000 ARC2701 ARC2702 ARH1000 ARH2050 ARH2051 ARH2740 ASL1140C ASL1150C ASL2210 ASL2220 ASL2400 ASL2430 ASL2510 AST1002 BOT1010 BOT1010L BOT2150C BSC1005 BSC1005L BSC1084 BSC2010 BSC2010L BSC2011 BSC2011L BSC2020	BSC2085 BSC2085L BSC2086 BSC2086L CHI1120 CHI1121 CHM1020 CHM1025 CHM1025L CHM1033 CHM1033L CHM1045 CHM1045L CHM1046 CHM1046L CHM2200 CHM2200L CHM2210 CHM2210L CHM2211 CHM2211L CLP2000 CRW2001 CRW2002 DAN2100 DAN2130 DAN2131 DEP2000 DEP2100 ECO2013 ECO2023	ECO2301 EDF1005 ENC1101 ENC1102 ENC2300 ENG2012 ENL2012 ENL2022 ESC1000 EUH2032 EVR1001 FRE1120 FRE1121 GER1120 GER1121 GLY1010 GLY1010L GLY1100 HLP1081 HUM1020 HUN1201 INR2002 ITA1120 ITA1121 JPN1120 JPN1121 LAH2021 LIT2000 LIT2090 LIT2120 MAC1105	MAC1105L MAC1106 MAC1114 MAC1140 MAC1147 MAC2233 MAC2311 MAC2312 MAC2313 MAD1100 MAD2104 MAP2302 MAS2103 MCB2010 MCB2010L MET1010 MGF1130 MGF1131 MUH2111 MUH2112 MUL1010 MUL2380 OCB1010 OCE1001 PHI1100 PHI2010 PHI2600 PHY1004 PHY1004L PHY1020 PHY1025	PHY2048 PHY2048L PHY2049 PHY2049L PHY2053 PHY2053L PHY2054 PHY2054L POR1120 POR1121 POS2041 POS2112 PSC1121 PSC1515 PSY2012 QMB2100 REL1210 REL1240 RUS1120 RUS1121 SOP2002 SPC1017 SPC2601 SPC2608 SPN1120 SPN1121 STA2023 THE2000 WOH2012 WOH2022
MDC Core 3 Credits AST 1002 - Descriptive Astronomy BOT 1010 - Botany BSC 1005 - General Education Biology BSC 1084 - Functional Human Anatomy BSC 2010 - Principles of Biology BSC 2020 - Human Biology: Fundamentals of Anatomy/Physiology BSC 2085 - Anatomy and Physiology I CHM1020 - General Education Chemistry CHM1025 - Introductory Chemistry CHM1033 - Chemistry for Health Sciences CHM1045 - General Chemistry and Qualitative Analysis CHM1046 - General Chemistry and Qualitative Analysis CHM2200 - Survey of Organic Chemistry CHM2210 - Organic Chemistry 1 CHM2211 - Organic Chemistry 2 ESC 1000 - General Education Earth Science EVR 1001 - Introduction to Environmental Science GLY 1010 - Physical Geology GLY1100 - Historical Geology HUN 1201 - Essentials of Human Nutrition MET1010 - Introduction to Weather OCB 1010 - Introduction to Marine Biology OCE 1001 - Introduction to Oceanography PHY1004 - Physics with Applications 1 PHY1020 - General Education Physics PHY1025 - Basic Physics PHY2048 - Physics with Calculus 1 PHY2049 - Physics with Calculus 2 PHY2053 - Physics (without Calculus) 1 PHY2054 - Physics (without Calculus) 2 PSC 1121 - General Education Physical Science PSC 1515 - Energy in the Natural Environment		FIRST YEAR EXPERIENCE SLS 1106 - First Year Experience Seminar OR One of the courses below based on advisor's recommendation: IDH 1001 - Honors Leadership Seminar 1 IDH 1002 - Honors Leadership Seminar 2 IDH 2003 - Honors Leadership Seminar 3 IDH 2004 - Honors Leadership Seminar 4 SLS 1125 - Student Support Seminar SLS 1401 - Psychology of Career Adjustment SLS 1502 - College Study Skills SLS 1505 - College Survival Skills SLS 1510 - Preparing for Student Success				
PATHWAY ELECTIVES 24 Credits Elective courses should be selected by pathway and/or specialization. Consult with an advisor. Also refer to information available at your Transfer Institution of choice. General education courses that are not used to meet general education requirements may be used for pathway electives in this block.						
FOREIGN LANGUAGE COMPETENCY May be satisfied by Foreign Language Competency (FLC) standardized examinations. For more information, refer to Foreign Language Competency . OR						
ASL 1150C CHI 1121 FRE 1121		GER 1121 ITA 1121 JPN 1121		POR 1121 RUS 1121 SPN 1121		

COMPUTER COMPETENCY		
By the 16th earned college-level credit, students must attempt the computer competency requirement OR by the 31st earned college-level credit, students must satisfy the requirement (CGS1060C, an equivalent college credit course or the College's approved computer competency test). For more information, see Computer Competency .		
CHM 1025		
The Chemistry Advanced Readiness Test (CART) is an opportunity for eligible students to bypass CHM1025. Review the MDC CART webpage for eligibility.		
CIVIC LITERACY COMPETENCY		
Associate in arts or Baccalaureate degree students who entered a Florida College System (FCS) or State University System (SUS) institution in the 2021-2022 academic school year and thereafter, and associate in science degree students entering a Florida College System (FCS) or State University System (SUS) institution in the 2022-2023 academic school year and thereafter, must demonstrate competency through successful completion of a civic literacy course (AMH 2010 or AMH 2020 or POS 2041) <u>and</u> by achieving a passing score on the Florida Civic Literacy Examination (FCLE). For more information regarding the Florida Civic Literacy Requirement, visit the MDC Civic Literacy Competency resources.		
60 CREDITS REQUIRED FOR GRADUATION		
General Education: 36 Credits		Pathway Electives
State Core: 15 Credits	MDC Core: 21 Credits	24 Credits
For more information regarding General Education Course Options, refer to Rule 6A-14.0303, General Education Course Options .		
GENERAL EDUCATION CORE COURSE STANDARDS		
1. Communication courses must afford students the ability to communicate effectively, including the ability to write clearly and engage in public speaking. 2. Humanities courses must afford students the ability to think critically through the mastering of subjects concerned with human culture, especially literature, history, art, music, and philosophy, and must include selections from the Western canon. 3. Mathematics courses must afford students a mastery of foundational mathematical and computation models and methods by applying such models and methods in problem solving. 4. Natural Science courses must afford students the ability to critically examine and evaluate the principles of the scientific method, model construction, and use the scientific method to explain natural experiences and phenomena. 5. Social Science courses must afford students an understanding of the basic social and behavioral science concepts and principles used in the analysis of behavior and past and present social, political, and economic issues.		
MDC Advisement & Career Services Offices		
Hialeah Campus Room 2101 305-237-8794	Homestead Campus Room C210 305-237-5046	
Padrón Campus Room 1101 305-237-6133	Kendall Campus Room R243 305-237-2125	
Medical Campus Room 1223 305-237-4141	North Campus Room 1104 305-237-1425	
Wolfson Campus Room 2301 305-237-3077	West Campus Room 2114 305-237-8947	
Meek Center Room 1102-02 305-237-1900		
Call Center 305-237-8888 mdcinfo@mdc.edu		

Important Information

- The official graduation requirements are on the Academic Requirements page in MDConnect at mdconnect.mdc.edu. You are encouraged to visit Advisement for assistance with your degree requirements.
- Other Assessment Procedures for College-Level Communication and Computation Skills (6A-10.030) (often referenced as Gordon Rule) requires:
 - o **W** = Writing Intensive Course: Six (6) semester hours of English coursework and six (6) semester hours of additional coursework in which the student is required to demonstrate college-level writing skills through multiple assignments.
 - o **C** = Computational Course: Six (6) semester hours of mathematics coursework at the level of college algebra or higher.
- *General education courses require a grade of C or higher to satisfy the requirement.
- W = Writing Intensive Course
- C = Computational Course

Other Assessment Procedures for College-Level Communication and Computation Skills (6A-10.030) (often referenced as Gordon Rule)

1. In addition to assessments that may be adopted by the State Board of Education or Board of Governors to measure student achievement in college-level communication and computation skills, other assessment requirements shall be met by successful completion of coursework in English and mathematics. For the purposes of this rule, a grade of C or higher shall be considered successful completion.
2. Prior to receipt of an Associate in Arts degree from a public community college or university or prior to entry into the upper division of a public university or college, a student shall complete successfully the following:
 - Six (6) semester hours of English coursework and six (6) semester hours of additional coursework in which the student is required to demonstrate college-level writing skills through multiple assignments. Each institution shall designate the courses that fulfill the writing requirements of this section. These course designations shall be submitted to the Statewide Course Numbering System. An institution to which a student transfers shall accept courses so designated by the sending institution as meeting the writing requirements outlined in this section.
 - Six (6) semester hours of mathematics coursework at the level of college algebra or higher. For the purposes of this rule, applied logic, statistics and other such computation coursework which may not be placed within a mathematics department may be used to fulfill three (3) hours of the six (6) hours required by this section.
 - (c) Students awarded college credit in English based on their demonstration of writing skills through dual enrollment, advanced placement, or international baccalaureate instruction pursuant to Rule 6A-10.024, F.A.C., and students awarded college credit based on their demonstration of mathematics skills at the level of college algebra or higher through one (1) or more of the acceleration mechanisms in Rule 6A-10.024, F.A.C., shall be considered to have satisfied the requirements in subsection 6A-10.030(2), F.A.C., to the extent of the college credit awarded.
3. Exemptions and Waivers. Any public community college or university desiring to exempt its students from the requirements of subsection 6A-10.030(2), F.A.C., shall submit an alternative plan to the Department of Education. Upon approval of the plan by the Department, the plan shall be submitted to the State Board of Education or the Board of Governors as appropriate. Upon approval by the State Board of Education or the Board of Governors, said plan shall be deemed effective in lieu of the requirements of subsection 6A-10.030(2), F.A.C.

Note: A list of MDC courses that fulfill this requirement, can be found at http://www.mdc.edu/asa/faculty/gordon_rule_courses.asp.

Advanced Technical Certificate Programs

Awarded to students who successfully complete a program of career and technical instruction consisting of nine (9) hours or more but less than forty-five (45) college credits of lower division and/or upper division courses. Florida College System institutions offering advanced technical certificates with upper division courses must be approved to offer baccalaureate programs containing those courses. An advanced technical certificate may be awarded to students who have already received an associate in science degree or an associate in applied science degree and are seeking an advanced specialized program of instruction to supplement their associate degree (information located at www.mdc.edu/academics/programs/default.aspx).

College Credit Certificate Programs

Awarded to students who successfully complete a program of career and technical instruction consisting of less than sixty (60) college credits which are part of an associate in science degree or an associate in applied science degree program and prepare students for entry into employment. The standard credit hour length of all technical certificate programs as defined in Rule 6A-6.0571, F.A.C. (information located at www.mdc.edu/academics/programs/default.aspx).

Career Technical Certificate Programs

Awarded to students who successfully complete a program of career and technical instruction consisting of clock hour courses to prepare for entry into employment. Upon satisfactory completion of a planned program of instruction, after the demonstration of the attainment of predetermined and specified performance requirements, the career and technical certificate shall be awarded. The standard clock hour length of all career and technical

certificate programs as defined in Rule 6A-6.0571, F.A.C. (updated information located at www.mdc.edu/academics/programs/default.aspx).

Commencement (Graduation Ceremony)

Students who anticipate completing their program during the academic year should meet with an academic advisor to ensure that all graduation requirements will be met. Also, students must apply for graduation by the deadline published in the academic calendar. Students planning to graduate in spring or summer terms should note that the deadline is very early in the spring term. The commencement ceremony is held once a year, at the end of spring term (late April or early May). Caps and gowns are available at campus bookstores for those who have applied for graduation. There is no cost for these items.

Special Recognition for Outstanding Academic Performance (College Credit Students Only)

The College gives special recognition to students who demonstrate outstanding academic performance while working toward a degree. Students are eligible for the following recognition:

- *Dean's List* – recognizes students who have a term combined cumulative GPA of 3.5 or above for 12 or more credits earned in the fall, spring, or summer term.
- *Letter of Congratulations* – the campus dean of faculty or equivalent sends a special letter of congratulations to students who earn a term grade point average of 4.0 for 12 or more credits earned in the fall or spring terms (excluding courses that do not satisfy degree requirements).

In addition, special designations are entered on transcripts of students awarded a degree program as follows:

- *Honors* - A combined cumulative GPA of 3.5–3.69 is required to graduate with Honors.
- *Honors and Distinction* - A combined cumulative GPA of 3.5–3.69 and at least 15 credits earned in honors courses is required to graduate with Honors and Distinction.
- *Highest Honors* - To graduate with Highest Honors, a student must achieve a combined cumulative GPA of 3.7 or higher.
- *Highest Honors and Distinction* - A combined cumulative GPA of 3.7 or higher and at least 15 credits earned in honors courses is required to graduate with Highest Honors and Distinction.
- *Phi Theta Kappa* - To be eligible for induction into Phi Theta Kappa, the International Honor Society of the Two Year College, a student must have completed a minimum of 12 college-level credits leading to an Associate degree, with a minimum 3.5 combined cumulative GPA and the student must be currently enrolled. Upon graduation, initiated students will have the designation noted on their transcript.

Transfer Information

Transfer information and resources, including transfer agreements and transfer tips, can be found at: <https://www.mdc.edu/transfer-information/>. Graduating with an Associate in Arts degree guarantees MDC graduates numerous transfer benefits. Certain Associate in Science degrees are also transferable to related baccalaureate degree programs. Additional information about transfer guarantees and agreements can be found in the Transfer Agreement section. Miami Dade College offers baccalaureate degrees in several areas of study: <http://www.mdc.edu/academics/programs/bachelors.aspx>. Students who meet the admissions criteria for these programs may apply at: <https://mdcwap.mdc.edu/admission/>. Campus Advisement & Career Services Offices offer students a variety of academic advising and career-related services. Students who are undecided about their academic major or career goals, or who are interested in a systematic investigation of the universities best suited for their needs, should visit the Advisement & Career Services Office at their campus.

Articulation

Articulation is a system designed to provide for seamless transfer pathways for students from high school, through the Florida College System and into a baccalaureate degree program at a State University, Florida College System institution, or other postsecondary institution. There are a number of articulation agreements which create special transfer opportunities for students (updated information located at <https://www.mdc.edu/transfer-information/>).

Articulation Agreements with Miami-Dade County Public Schools

Miami Dade College and Miami-Dade County Public Schools have entered into several articulation agreements. These range from the formalized Early College Program agreements with New World School of the Arts and the School for Advanced Studies, to agreements for transfer of specific career pathways and career and technical education credits to Associate in Science degrees and certificate programs.

State of Florida Articulation Agreement

Section 1007.23, Florida Statutes, states the following:

1. The State Board of Education and the Board of Governors shall enter into a statewide articulation agreement which the State Board of Education shall adopt by rule. The agreement must preserve Florida's "2+2" system of articulation, facilitate the seamless articulation of student credit across and among Florida's educational entities, and reinforce the provisions of this chapter by governing:
 - Articulation between secondary and postsecondary education;
 - Admission of Associate in Arts degree graduates from Florida College System institutions and state universities;
 - Admission of applied technology diploma program graduates from Florida College System institutions or career centers;
 - Admission of Associate in Science degree and Associate in Applied Science degree graduates from Florida College System institutions;
 - The use of acceleration mechanisms, including nationally standardized examinations through which students may earn credit;
 - General education requirements and statewide course numbers as provided for in ss. 1007.24 and 1007.25; and
 - Articulation among programs in nursing.
2. The articulation agreement must specifically provide that every associate in arts graduate of a Florida College System institution shall have met all general education requirements and must be granted admission to the upper division of a:
 - State university, except for a limited access or teacher certification program or a major program requiring an audition.
 - Florida College System institution if it offers baccalaureate degree programs, except for a limited access or teacher certification program or a major program requiring an audition. (b) Florida College System institution Associate in Arts graduates shall receive priority for admission to the upper division of a Florida College System institution or to a state university over out-of-state students. Orientation programs, catalogs, and student handbooks provided to freshman enrollees and transfer students at Florida College System institutions and state universities must include an explanation of this provision of the articulation agreement.
3. To improve articulation and reduce excess credit hours, beginning with students initially entering a Florida College System institution in 2013-2014 and thereafter, the articulation agreement must require each student who is seeking an Associate in Arts degree to indicate a baccalaureate degree program offered by an institution of interest by the time the student earns 30 semester hours. The institution in which the student is enrolled shall inform the student of the prerequisites for the baccalaureate degree program.
4. The articulation agreement must guarantee the statewide articulation of appropriate workforce development programs and courses between school districts and Florida College System institutions and specifically provide that every applied technology diploma graduate must be granted the same amount of credit upon admission to an Associate in Science degree or Associate in Applied Science degree program unless it is a limited access program. Preference for admission must be given to graduates who are residents of Florida.
5. The articulation agreement must guarantee the statewide articulation of appropriate courses within Associate in Science degree programs to baccalaureate degree programs. Courses within an Associate in Applied Science degree program may articulate into a baccalaureate degree program on an individual or block basis as authorized in local interinstitutional articulation agreements.
6. The articulation agreement must guarantee the articulation of 9 credit hours toward a postsecondary degree in early childhood education for programs approved by the State Board of Education and the Board of Governors which: (a) Award a child development associate credential issued by the National Credentialing Program of the Council for Professional Recognition or award a credential approved under s.1002.55(3) (c)1.b. or s. 402.305(3) (c) as being equivalent to the child development associate credential; and (b) Include training in emergent literacy which meets or exceeds the minimum standards for training courses for prekindergarten instructors of the Voluntary Prekindergarten Education Program in s. 1002.59.

Independent Colleges and Universities of Florida (ICUF)

There is also an articulation agreement between the Florida College System and the Independent Colleges and Universities of Florida (ICUF). Under the agreement, Florida College System graduates holding an Associate in Arts

degree are guaranteed junior standing in any member institution, recognition of their completed general education core and the application of a minimum of 60 earned credit hours toward a baccalaureate degree.

Additional Agreements

In addition, Miami Dade College has developed several unique arrangements with local and out-of-state colleges and universities that make it possible for a student to apply for admission toward a baccalaureate degree. As a general rule, participating institutions will accept associate degree credits and work out a schedule for the additional bachelor's degree requirements. Miami Dade College also has agreements with universities that facilitate the transfer of MDC baccalaureate graduates to graduate programs. For additional information relating to articulation agreements, contact the Office of Articulation and Academic Pathways or visit <https://www.mdc.edu/transfer-information/transfer-agreements/>.

MIAMI DADE COLLEGE (MDC) GENERAL EDUCATION CORE

DIGITAL BADGES

What is a Digital Badge

A digital badge is an award that is given to students upon completion of general education core courses that demonstrate career readiness. It is intended to help the student translate what they learned in general education courses to knowledge, skills and abilities that employers require.

Who's Eligible

Pursuant to s. 1007.25(4), F.S., all students initially entering a public postsecondary institution in the 2022-23 academic year (fall 2022) and thereafter and enroll in general education core courses are eligible for the awarding of a digital badge. This is inclusive of dual enrollment students who complete core courses while in high school in the 2022-23 academic year and thereafter. Minimally, institutions must award badges to students meeting these criteria.

Digital Badges Offered at MDC

	Fundamentals of Written Communication	Students will demonstrate effective reading, writing, speaking, listening, and nonverbal communication skills.
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What is Effective Written Communication?

Effective written communication is the ability to communicate ideas, information, and perspectives clearly, adapting a message to different audiences and situations, and using the appropriate style to convey meaning in various written contexts.

What are you learning by earning this badge?

By completing the college course associated with this badge, you will demonstrate information literacy, comprehension of written material, and conveying information in writing for a variety of rhetorical purposes and audiences. Some of the learning outcomes associated with the college course that results in this badge include:

- Examining and analyzing written material
- Synthesizing information and ideas
- Developing content relevant to the purpose
- Demonstrating the ability to write to a specific audience
- Presenting a perspective informed by research and critical thinking
- Revising written communication based on feedback

Why is Effective Written Communication important in the workplace?

- Enhances an employee's ability to interpret and evaluate a wide variety of written material
- Promotes research, critical thinking, and problem solving
- Advances the development of clearly written material relevant to the intent
- Enhances the use of appropriate language for an intended audience
- Increases the competent, effective, and responsible use of information

PROGRAMS

Bachelor of Applied Sciences (BAS)

Program Description

Film, Television & Digital Production

Bachelor of Applied Science | Code: P9220 | 120 credits

The Bachelor of Applied Science with a major in Film, Television & Digital Production degree is designed to provide the South Florida workforce with a highly skilled applicant pool to fulfill job demands and to provide an affordable opportunity for students completing either an Associate in Science or an Associate in Arts degree to seamlessly continue and complete a baccalaureate degree. Graduates of the Bachelor of Applied Science with a major in Film, Television, and Digital Production degree will be prepared with the abilities and skills needed to succeed in the film and television industry. The Bachelor of Applied Science with a major in Film, Television & Digital Production degree will prepare graduates with the hands-on training and business acumen necessary to meet workforce demands and to operate within the entertainment business environment.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
DIG 3347C	Advanced Cinematography	3.0
FIL 3602	Production Management	3.0
FIL 4164	Advanced Writing for Film and Television	3.0
RTV 3810C	Broadcast Design & On Air Promotions	3.0
Semester 2		Units: 13.0
DIG 4345C	Digital Effects and Compositing	3.0
ENC 1102	English Composition 2	3.0
FIL 4585C	Production Workshop 1	4.0
Humanities		3.0
ARH 1000	Art Appreciation	3.0
OR		
ARH 2740	Cinema Appreciation	3.0
OR		
MUL 1010	Music Appreciation	3.0
Semester 3		Units: 6.0
DIG 3255C	Advanced Sound Design	3.0
Social Science		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
Foreign Language Competence*		

*Note: Foreign language is a graduation requirement for the baccalaureate met through 8 credit hours at the elementary 2 level in one foreign language or equivalent. Certain foreign language courses count towards program electives. Students may satisfy equivalence through standardized examinations or successful completion of two credits (two years) in one foreign language at the secondary (high school) level. For additional information, including exemptions for students whose native language is not English, see the Testing and Assessment Department.

Semester 4		Units: 13.0
MGF 1131	Mathematics in Context	3.0
FIL 4586C	Production Workshop 2	4.0
RTV 3277C	Television Studio Production Workshop	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Semester 5		Units: 12.0
FIL 3651	Business Proposals for Film & Television	3.0
<i>Upper Division Electives</i>		3.0
DIG 3940	Upper Division Internship	3.0
OR		
RTV 3203	The Telemundo Academy	3.0
OR		
RTV 3256C	Advanced Post Production	3.0
OR		
RTV 3408	Ethics & Research for Non-Fiction Scripts	3.0
<i>Natural Science</i>		3.0
AST 1002	Descriptive Astronomy	3.0
OR		
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
GLY 1010	Physical Geology	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0
OR		
PHY 1020	General Education Physics	3.0
<i>General Education Elective*</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
ARH 2740	Cinema Appreciation	3.0
OR		
MUL 1010	Music Appreciation	3.0

*Note: Please see your academic advisor and/or degree audit for appropriate course options.

Total: 56.0

DIG3347C Advanced Cinematography 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn the technical and aesthetic principles of advanced cinematography techniques.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: RTV2246C OR FIL2515C

Course User Fee: \$100.00

FIL3602 Production Management 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn the theory and practice of managing film and television production with an emphasis on pre-production processes and software.

Lecture Hours: 48.000

Prerequisite: FIL2611 OR MMC2000

FIL4164 Advanced Writing for Film and Television 3.000 credits

In this upper division course in Film, Television & Digital Production students will learn the process of completing a long form motion picture or television script.

Lecture Hours: 48.000

Prerequisite: FIL2131 OR RTV2300

RTV3810C Broadcast Design & On Air Promotions 3.000 credits

In this upper division course in BAS in Film, Television & Digital Production students will learn about on-air promotions with specific emphasis on layout, color, and composition.

Lecture Hours: 32.000, Lab Hours: 32.000

Recommended Preparation: BAS Students ONLY

Course User Fee: \$100.00

DIG4345C Digital Effects and Compositing 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn the theory and practice of video compositing and motion graphics.

Lecture Hours: 32.000, Lab Hours: 32.000

Recommended Preparation: BAS Students ONLY

Course User Fee: \$100.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

FIL4585C Production Workshop 1 4.000 credits

In this upper division workshop for BAS students in Film, Television & Digital Production students will learn to apply industry-standard pre-production and production techniques to Production Workshop projects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: DIG3347C AND FIL3602

Course User Fee: \$200.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARH2740 Cinema Appreciation 3.000 credits

An analysis of the cinema as an important social force and an artistic medium. Significant American, British, and foreign language films will be shown and discussed. Writing intensive course.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: HUM1020

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

DIG3255C Advanced Sound Design 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn advanced audio production, emphasizing audio recording, mixing, editing, overdubbing, and aesthetics.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: RTV1240C

Course User Fee: \$100.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

FIL4586C Production Workshop 2 4.000 credits

In this upper division workshop for BAS Students in Film, Television & Digital Production students will learn to apply industry-standard post-production techniques to complete Production Workshop 1 projects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL4585C

Course User Fee: \$200.00

RTV3277C Television Studio Production Workshop 3.000 credits

In this upper division workshop class for BAS students in Film, Television & Digital Production students will learn advanced television studio production techniques for pre-scripted programs.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2515C OR RTV2246C

Course User Fee: \$75.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

FIL3651 Business Proposals for Film & Television 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn the theory and practice of business plans/grant proposals in media production funding.

Lecture Hours: 48.000

Prerequisite: FIL2611 OR MMC2000

DIG3940 Upper Division Internship 3.000 credits

In this upper division internship for BAS students in Film, Television & Digital Production students will learn to apply their knowledge and skills at an established film or television entertainment company (144 hr. Internship).

Other Hours: 144.000

Recommended Preparation: BAS Students ONLY.

Must have at least a 3.0 GPA in order to register.

Must obtain professor permission in order to register.

RTV3203 The Telemundo Academy 3.000 credits

The Telemundo Academy is a project-based learning course that provides students with a fundamental understanding of the concepts and processes of short-form content production for social media and mobile video platforms. The course will be divided into four phases: development, pre-production, production, and post-production. The first portion of the course will focus on pre-production given that the quality of the final projects will be correlated to students' understanding of all elements and practices in this phase. Students will learn how research, script development, and storyboarding are critical to transforming their original concepts into successful productions. The course will cover additional fundamental skills related to: communications, leadership, marketing, and digital and social media best practices.

Lecture Hours: 48.000

Course User Fee: \$80.00

RTV3256C Advanced Post Production 3.000 credits

An advanced course in which students will learn complex digital media post production techniques.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2553C

RTV3408 Ethics & Research for Non-Fiction Scripts 3.000 credits

In this upper division course students will learn research methods applicable to documentary production.

Lecture Hours: 48.000

Recommended Preparation: Student must be in BAS Program AND Prerequisite: FIL2131 OR RTV2300

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARH2740 Cinema Appreciation 3.000 credits

An analysis of the cinema as an important social force and an artistic medium. Significant American, British, and foreign language films will be shown and discussed. Writing intensive course.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: HUM1020

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

Program Description

Health Science - Histotechnology Concentration

Bachelor of Applied Science | Code: P9212 | 130 credit

The Bachelor of Applied Science (B.A.S.) in Health Sciences with a concentration in Histotechnology is designed to accommodate the unique demands for entry and advancement within the health science industry.

Histotechnologist process patient samples for the detection of tissue abnormalities in order to determine the best treatment for the diseases causing the abnormalities. In addition to the dyes and chemicals used to prepare and stain tissues for microscopy, immunological and DNA techniques are utilized to determine specific cell types in tumors. Histotechnologist work in conjunction with the pathology department at a variety of hospital, surgical and clinical locations. Upon completion of the program and the attainment of certification and license, Histotechnologist are employed in a variety of health-care settings with the opportunity of upward mobility to supervisory and management positions. Histotechnologist program graduates are eligible for certification by the American Society for Clinical Pathology (ASCP).

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 17.0

ENC 1102	English Composition 2	3.0
HSC 3655	Theoretical Foundations of Health Care Ethics	3.0
MLS 4621	Clinical Biochemistry	4.0
MLS 4705	Laboratory Operations and Management	3.0
<i>Lower Division Electives</i>		4.0
BSC 1005	General Education Biology	3.0
BSC 1005L	General Education Biology Laboratory	1.0
BSC 1084	Functional Human Anatomy	3.0

BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
BSC 2020	Human Biology: Fundamentals of Anatomy/Physiology	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy & Physiology Lab 1	1.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0
BSC 2426	Biotechnology Methods and Applications 1	3.0
BSC 2426L	Biotechnology Methods and Applications 1 Laboratory	2.0
BSC 2427	Biotechnology Methods and Applications 2	3.0
BSC 2427L	Biotechnology Methods and Applications 2 Laboratory	2.0
BSC 2943L	Bioscience Internship	3.0

Semester 2

Units: 15.0

<i>Oral Communications</i>		3.0
SPC 1017	Introduction to Communication	3.0
SPC 2608	Introduction to Public Speaking	3.0

<i>Social Science (GE)</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
DEP 2000	Human Growth and Development	3.0
ECO 2013	Principles of Economics (Macro)	3.0
PSY 2012	Introduction to Psychology	3.0

MLS 4181C	Immunohistochemistry	3.0
MLS 4195C	Enzyme Histochemistry	3.0
MLS 4196C	In Situ Hybridization (FISH)	3.0

Semester 3

Units: 11.0

<i>Humanities (GE)</i>		3.0
HUM 1020	Humanities	3.0
PHI 2010	Introduction to Philosophy	3.0
PHI 2600	Introduction to Ethics	3.0

HSC 3701	Leadership and Management in Healthcare	3.0
MLS 4198	Immunohistochemistry Clinical Concentration Capstone	5.0

Semester 4

Units: 16.0

<i>Mathematics (GE)*</i>		3.0
MGF 1131	Mathematics in Context	3.0
STA 2023	Statistical Methods	3.0

HSC 3057	Introduction to Research Methods in Health Care	3.0
MLS 3150	Special Topics in Medical Laboratory Sciences	3.0
MLS 4193	Clinical Molecular Diagnostics	3.0

<i>Lower Division Elections</i>		4.0
CHM 1020	General Education Chemistry	3.0
CHM 1033	Chemistry for Health Sciences	3.0

CHM 1033L	Chemistry for Health Sciences Lab	1.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
CHM 1046	General Chemistry And Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
CHM 2200	Survey of Organic Chemistry	3.0
CHM 2200L	Survey of Organic Chemistry Laboratory	1.0
CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0
CHM 2211	Organic Chemistry 2	3.0
CHM 2211L	Organic Chemistry 2 Laboratory	2.0

*Recommended Course: STA 2023 - Statistical Methods

Semester 5		Units: 15.0
MLS 4910	Advances in Histotechnology Capstone	7.0
<i>Electives</i>		8.0
AST 1002	Descriptive Astronomy	3.0
ESC 1000	General Education Earth Science	3.0
EVR 1001	Introduction to Environmental Science	3.0
GLY 1010	Physical Geology	3.0
MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0
OCE 1001	Introduction to Oceanography	3.0
PHY 1020	General Education Physics	3.0

Semester 6		Units: 56.0
MLS 3997	Validated Medical Laboratory Technology Course*	27.0
<i>Electives</i>		29.0
Please consult your academic advisor to determine appropriate course selections.		

*(This course awards articulated credit for students who have successfully completed an accredited Associate in Science (AS) degree in Histotechnology. Emphasis is placed on validating skills aligned with program outcomes and professional standards in histotechnology.)

Total: 130.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HSC3655 Theoretical Foundations of Health Care Ethics 3.000 credits

This course will cover the Theoretical Foundations of Health Care Ethics. The student will learn how to apply the core principles of bioethics to the medical and health care decision-making process. (3 cr. Lecture)

Lecture Hours: 48.000

MLS4621 Clinical Biochemistry 4.000 credits

This course provides the student with the knowledge and understanding of clinical disorders and how biochemical factors and laboratory methods are used for the investigation, diagnosis and management of patients. (4 hour lecture)

Lecture Hours: 64.000

Course User Fee: \$41.00

MLS4705 Laboratory Operations and Management 3.000 credits

This course provides students with Quality Management skills necessary in the medical laboratory. Students are exposed to organizational structure along with principles for leadership and managerial decision making and process improvement along with different principles used in Laboratory Instrumentation. (3 Hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC1005L General Education Biology Laboratory 1.000 credits

This is an optional one-credit lab to provide students with experience in the scientific process. This course applies the scientific method to critically examine and explain the natural world, including but not limited to cells, organisms, genetics, evolution, ecology, and behavior.

Lab Hours: 32.000

Pre/Corequisite: BSC1005

Course User Fee: \$25.00

BSC1084 Functional Human Anatomy 3.000 credits

This course covers basic human anatomy for students in allied health and mortuary science programs. It includes the dynamics of gross and functional anatomy, terminology, body orientation, and systematic relationships.

Lecture Hours: 48.000

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

**BSC2011L Principles of Biology 2 Laboratory
2.000 credits**

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

BSC2020 Human Biology: Fundamentals of Anatomy/Physiology 3.000 credits

This course provides a basic understanding of the human body, its systems, and their functions. It includes the dynamics of physiology, terminology, and physiological relationships of the body systems.

Lecture Hours: 48.000

Course User Fee: \$10.00

**BSC2085 Human Anatomy and Physiology 1
3.000 credits**

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

**BSC2085L Human Anatomy & Physiology Lab 1
1.000 credits**

In this laboratory course, student will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

**BSC2086L Human Anatomy & Physiology 2 Lab
1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

BSC2426 Biotechnology Methods and Applications 1 3.000 credits

This course addresses the basic principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical, biotechnology and/or research laboratory setting(s). Practical applications of biotechnology are explored.

Lecture Hours: 48.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426L

BSC2426L Biotechnology Methods and Applications 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2426 Biotechnology Methods and Applications 1. This is a hands-on course that emphasizes the basic laboratory principles, techniques, and instrumentation, necessary for effective work in pharmaceutical, biotechnology, and/or research laboratory settings(s).

Lab Hours: 64.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426

Course User Fee: \$50.00

BSC2427 Biotechnology Methods and Applications 2 3.000 credits

This course addresses advanced principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical-biotechnology and/or research-laboratory setting(s). The following areas of contemporary biotechnology are covered: forensics, bioremediation, and animal-, plant-, and marine biotechnology.

Lecture Hours: 48.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427L

BSC2427L Biotechnology Methods and Applications 2 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2427 Biotechnology Methods and Applications 2. This is a hands-on course that emphasizes advanced laboratory principles, techniques, and instrumentation necessary for effective work in a pharmaceutical, biotechnology, and/or research-laboratory setting(s).

Lab Hours: 64.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427

Course User Fee: \$50.00

BSC2943L Bioscience Internship 3.000 - 6.000 credits

This internship course is a capstone for students majoring in bioscience and related programs. Students will learn to apply acquired knowledge and skills to gain experience in the bioscience workplace.

Other Hours: 144.000 - 288.000

Course User Fee: \$21.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan. Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence. Lecture Hours: 48.000

MLS4181C Immunohistochemistry 3.000 credits

This course introduces the various techniques that are used in the preparation and evaluation of immunohistochemistry slides. Procedures and terminology related to immunohistochemistry are also discussed and strategies for troubleshooting problems are presented. (2 hr. lecture, 3 hr. clinical) Lecture Hours: 32.000, Clinical Hours: 48.000
Course User Fee: \$499.15

MLS4195C Enzyme Histochemistry 3.000 credits

Introduction to advanced techniques and special procedures. Students will learn procedures for, muscle enzyme Histochemistry and molecular histology. The course will include tissue preparation, staining technology, quality control and troubleshooting. (2 hr. lecture, 3 hr. clinical) Lecture Hours: 32.000, Clinical Hours: 48.000
Course User Fee: \$499.15

MLS4196C In Situ Hybridization (FISH) 3.000 credits

This course will explore the theoretical concepts used in fluorescence in-situ hybridization (FISH) testing. Commonly used FISH methodologies, necessary equipment, and the enumeration of FISH signals will also be discussed. (2 hr. lecture, 6 hr. clinical) Lecture Hours: 32.000, Clinical Hours: 48.000
Course User Fee: \$499.15

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

HSC3701 Leadership and Management in Healthcare 3.000 credits

This is a leadership and management course that will examine leadership as a process with a tri-fold focus: the leader, the followers, and the situation. The student will learn leadership theories and build leadership skills. (3 hr. lecture)

Lecture Hours: 48.000

MLS4198 Immunohistochemistry Clinical Concentration Capstone 5.000 credits

This clinical capstone course will introduce the students to the basic immunohistochemistry techniques as applied to the routine anatomical pathology laboratory.

Clinical Hours: 240.000

Course User Fee: \$499.15

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

HSC3057 Introduction to Research Methods in Health Care 3.000 credits

This course will provide an overview of research methods used in healthcare. Students will learn the use of effective inquiry through research strategies that address healthcare issues with logical and observational rigor. Students will learn the rudiments relative to the evaluation of research literature, research design and the application of research methods to the clinical setting. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: STA2023

MLS3150 Special Topics in Medical Laboratory Sciences 3.000 credits

This course stresses the importance of evidence based practice in the medical laboratory sciences field. Students will be presenting case studies to the faculty and peers in the program. Instruction will emphasize professional, legal and ethics issues affecting the medical laboratory science field. Students will review the material covered in the program to prepare for the comprehensive assessment. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$191.14

MLS4193 Clinical Molecular Diagnostics 3.000 credits

Clinical molecular diagnostics course provides an introduction to molecular analysis of biological markers in clinical samples to aid in the diagnosis, monitoring and treatment of diseases (3 Hour Lecture).

Lecture Hours: 48.000

Course User Fee: \$41.20

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions. Corequisites: CHM 1033L, MAT 1033 (3 hr. lecture)

Lecture Hours: 48.000

Corequisite: CHM1033L, Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite:

MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

CHM 1045 is the first semester of a two-semester general chemistry course for science, premedical science and engineering students. Students will learn stoichiometry, atomic structure, completing and balancing chemical reactions, nomenclature, bonding, acid-base theories, solutions, gas laws and beginning thermodynamics. Special fee.

Lecture Hours: 48.000

Prerequisite: CHM1025or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

CHM1046 General Chemistry And Qualitative Analysis 3.000 credits

CHM 1046 is the second course in the CHM 1045-1046 sequence. Students will learn major topics in modern chemistry including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria and electrochemistry. Special fee. Prerequisite: CHM1045; Corequisite: CHM1046L (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

CHM2200 Survey of Organic Chemistry 3.000 credits

XXXX This one-semester course briefly examines the structure, synthesis, nomenclature and reactivity of selected mono-and-poly-functional organic compounds. Theories that relate the structure of organic molecules to their chemically reactivity will be presented as a unifying principle. Prerequisite: CHM 1046 and CHM 1046L; Corequisite: CHM 2200L. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2200L

CHM2200L Survey of Organic Chemistry Laboratory 1.000 credits

Experiments and exercises will be conducted to introduce students to the basic laboratory techniques that are used in organic chemistry and that re-enforce and illustrate several important topics in organic chemistry. Prerequisite: CHM 1046 and CHM 1046L; Corequisite: CHM 2200. Special fee. (2 hr. lab)

Lab Hours: 32.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2200

Course User Fee: \$46.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds. Prerequisite: CHM1046 and CHM 1046L; Corequisite: CHM2210L

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

CHM2211 Organic Chemistry 2 3.000 credits

In organic chemistry 2, students will learn about nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds. Prerequisite: CHM 2210 and CHM 2210L; Corequisite: CHM 2211L. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211L

Course User Fee: \$15.00

CHM2211L Organic Chemistry 2 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2211. Topics include nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds will be performed in a laboratory setting.

Lab Hours: 64.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211

Course User Fee: \$45.00

MLS4910 Advances in Histotechnology Capstone 7.000 credits

This course will support the educational development of the histotechnology students by providing an opportunity for in-depth learning in one of the following domains Cytopreparatory techniques; Digital pathology; Electron microscopy; Micro-wave (MOHS) pathology; Ocular pathology histotechnology techniques; which will result in a scholarly project underlying its relevance in today's advanced pathology laboratories.

Lecture Hours: 112.000

Course User Fee: \$499.15

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms. Lecture Hours: 48.000

Course User Fee: \$10.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues. Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues. Lecture Hours: 48.000

Course User Fee: \$5.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity.

Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

MLS3997 Validated Medical Laboratory Technology Course 27.000 credits

This course awards articulated credit for students who have successfully completed an Associate in Science (AS) degree in Histotechnology. It recognizes prior learning and competencies in histological techniques, including tissue processing, sectioning, staining, and laboratory safety, as applied in clinical practice. Emphasis is placed on validating skills aligned with program outcomes and professional standards in histotechnology.

Program Description

Bachelor of Applied Science | Code: P9210 | 130 credits

The Bachelor of Applied Science in Health Science - Physician Assistant Studies Concentration is designed to address the critical shortage of healthcare professionals by providing graduates a continuance of technical, supervisory, and management skills necessary to meet workforce needs and labor market projections. Students will be provided with opportunities to develop discipline-specific medical skills and to expand their knowledge by participating in structured clinical experiences under the supervision of Physician Assistants and physicians.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
MAC 1105	College Algebra	3.0
Computer Competency*		

*Computer Skills Placement (CSP) Test ---or--- CGS1060C. Departmental Recommendation: Computer Skills Placement (CSP)

Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
<i>Oral Communications</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
PHI 2600	Introduction to Ethics	3.0

Semester 3		Units: 11.0
STA 2023	Statistical Methods	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Social Science</i>		3.0
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4		Units: 13.0
MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0
PSY 2012	Introduction to Psychology	3.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0

Semester 5		Units: 7.0
HSA 2532	Medical Documentation in Health Care	1.0
PAS 1800C	Physical Diagnosis I	2.0
PAS 1803	Clinical Anatomy and Physiology	2.0
PAS 1831	Clinical Diagnostic Imaging	1.0
PAS 2936	Contemporary Issues for the PA	1.0
Semester 6		Units: 16.0
HSA 3533	Medical Documentation in Health Care II	1.0
PAS 1801C	Physical Diagnosis 2	2.0
PAS 1811C	Introduction to Medicine I for PAS	5.0
PAS 1812	Behavioral and Community Medicine	1.0
PAS 1813	Pathophysiological Basis of Disease I	2.0
PAS 1822C	Electrocardiography	1.0
PAS 1823	Pharmacology 1	4.0
Semester 7		Units: 15.0
DEH 2810L	Interprofessional Practice and Education Lab	1.0
HSC 3202	Introduction to Public Health	1.0
PAS 1820C	Clinical Medicine 2 for Physician Assistants	5.0
PAS 1824	Pathophysiological Basis of Disease II	2.0
PAS 3038C	Physical Diagnosis III	2.0
PAS 3075	Pharmacotherapeutics	4.0
Semester 8		Units: 17.0
PAS 3019	Pathophysiological Basis of Disease III	2.0
PAS 3042C	Clinical Medicine III for Physician Assistants	5.0
PAS 3070	Clinical Pharmacotherapeutics	3.0
PAS 3140	Genetics	2.0
PAS 3203C	Surgical Problems and Procedures	5.0
Fall – Summer Semester: Clinical Rotations		Units: 25.0
HSC 4942	Community Service-Learning Practicum	3.0
PAS 4191L	Internal Medicine Clerkship	4.0
PAS 4290	Surgery Clerkship	2.0
PAS 4391	Pediatrics Clerkship	4.0
PAS 4493	Family Medicine Clerkship	4.0
PAS 4590	OBSTETRICS/GYNECOLOGY	2.0
PAS 4690	Emergency Medicine Clerkship	2.0
PAS 4940	Psychiatry Clerkship	2.0
PAS 4943	Selective Clerkship	2.0

Total: 130.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better. Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)
Lecture Hours: 48.000
Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.
Lecture Hours: 48.000
Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."
Course User Fee: \$15.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Lecture Hours: 48.000
Prerequisite: BSC2085, Corequisite: BSC2086L
Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086. Lab Hours: 32.000
Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086
Course User Fee: \$25.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course. Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

HSA2532 Medical Documentation in Health Care 1.000 credits

Medical Documentation in Health Care will introduce the student to documentation in the written patient chart or electronic medical record. Through case discussions and in-class writing assignments, the student will acquire the necessary skills to document in the patient's medical record utilizing medico-legal principles and evaluation and management criteria. Patient confidentiality, billing, and coding will also be discussed.

Lecture Hours: 16.000

PAS1800C Physical Diagnosis I 2.000 credits

This course provides students with the critical basis for and clinical exposure to techniques used in the proper performance and recording of the physical examination of patients. Prerequisites: BSC2085, BSC2085L, BSC2086, BSC2086L, CHM1045, CHM1045L, CHM1046, CHM1046L, MCB2010, and MCB2010L

Lecture Hours: 16.000, Clinical Hours: 96.000

Prerequisite: BSC2085 AND BSC2085L AND BSC2086 AND BSC2086L AND CHM1045 AND CHM1045L AND CHM1046 AND CHM1046L AND MCB2010 AND MCB2010L

Course User Fee: \$560.00

PAS1803 Clinical Anatomy and Physiology 2.000 credits

This course is designed for students accepted into the Physician Assistant Program. The course will review basic Anatomy and Physiology principles, while integrating important clinical concepts.

Students will learn to transition from Anatomy and Physiology to Pathophysiology.

Lecture Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L AND BSC2086 AND BSC2086L AND CHM1045 AND CHM1045L AND CHM1046 AND CHM1046L AND MCB2010 AND MCB2010L

PAS1831 Clinical Diagnostic Imaging 1.000 credits

The course introduces the PA student to principles of diagnostic imaging and a familiarity with the many diagnostic techniques available, their values and limitations, and how they may be best used in the management of the patient in clinical practice.

Lecture Hours: 16.000

Prerequisite: BSC2085 AND BSC2085L AND BSC2086 AND BSC2086L AND CHM1045 AND CHM1045L AND CHM1046 AND CHM1046L AND MCB2010, MCB2010L

Course User Fee: \$280.00

PAS2936 Contemporary Issues for the PA 1.000 credits

In this course the student will examine current issues, challenges, and practices influencing the physician assistant profession. Historical perspectives will be examined and the contributions of physician assistants in the healthcare system will be researched. The student will learn to use evidenced based medicine to research topics including the changing nature of health care delivery system in the United States; demographic, economic, ethical, and political factors influencing the practice of the physician assistant.

Lecture Hours: 16.000

HSA3533 Medical Documentation in Health Care II 1.000 credits

Medical Documentation in Health Care II is a continuation of HSA 2532. This course will teach the student to documentation in the written patient chart or electronic medical record for special populations. Through case discussions and in-class writing assignments, the student will acquire the necessary skills to document in the patient's medical record utilizing medico-legal principles and evaluation and management criteria. Patient confidentiality, billing, and coding will also be discussed. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822C, 1823, 1831. (1 hour lecture)

Lecture Hours: 16.000

Prerequisite: HSA2532, PAS1801C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

PAS1801C Physical Diagnosis 2 2.000 credits

This course will build upon the skills learned in Physical Diagnosis I and will prepare the physician assistant student with the essential skills for entry into clinical practice. Students will learn to perform medical interviews and physical examinations on live and professional patients, and high-fidelity simulators. Prerequisite: PAS1800C , PAS1803, PAS1831 , PAS2936. (1 hr. lecture, 3 hr. clinical)

Lecture Hours: 16.000, Clinical Hours: 48.000

Prerequisite: PAS1800C, PAS1803, PAS1831, PAS2936

Course User Fee: \$560.00

PAS1811C Introduction to Medicine I for PAS 5.000 credits

Students will learn the signs, symptoms, and pathophysiology of common diseases affecting pediatric, adult, and geriatric patients. In addition, diagnosis, therapeutic interventions, and follow-up will be learned. Patient education and preventative medicine will also be included. 3 hr. lecture, 2 hr. clinical)

Lecture Hours: 48.000, Clinical Hours: 96.000

Prerequisite: PAS 1800C, PAS 1803, PAS 1831, PAS 2936

Course User Fee: \$1400.00

PAS1812 Behavioral and Community Medicine 1.000 credits

This course utilizes a biopsychosocial system approach to identify social determinants within the health care delivery system. Students will participate in outreach projects to implement concepts learned in the classroom within a community environment. Prerequisites:

Lecture Hours: 16.000

Prerequisite: PAS1800C AND PAS1831 AND PAS2936, PAS 2YYY

Course User Fee: \$280.00

PAS1813 Pathophysiological Basis of Disease I 2.000 credits

The clinical pathophysiology course serves as a transition from the basic medical sciences to clinical medicine. Students will learn the study of disease, both congenital and acquired with an emphasis on providing understanding of pathologic physiology in conjunction with information regarding medical history and laboratory data to solve case based clinical problems. Students are taught how pathophysiology translates into patient signs, symptoms. Prerequisites: PAS1800C, 2936. (2 hr. clinical)

Clinical Hours: 96.000

Prerequisite: PAS1800C AND PAS2936

PAS1822C Electrocardiography 1.000 credits

Students will learn the basic knowledge to understand, interpret and utilize an electrocardiogram (EKG) as a diagnostic tool. This course provides the foundation for learning to interpret 12-lead EKG tracings and applying those principles to evaluate the EKG tracings of common cardiac disease.

Lab Hours: 32.000

Prerequisite: PAS 1800C, PAS 1803, PAS 1831, PAS 2936

Course User Fee: \$560.00

PAS1823 Pharmacology 1 4.000 credits

This course will focus on the preparation, uses and action of drugs. Students will learn about Pharmacokinetics and Pharmacodynamics. Students will be introduced to Clinical Pharmacology following an organ system approach, covering the major classes of therapeutically relevant drugs, and how they work at the molecular and cellular levels.

Prerequisites: PAS1800C, PAS1803, PAS1831, and PAS2936 (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: HSA2532, PAS1801C, PAS1803, PAS1831, PAS2936

Course User Fee: \$560.00

DEH2810L Interprofessional Practice and Education Lab 1.000 credits

Students will learn to examine the Dental Hygiene scope of practice through professional identity by way of interprofessional collaboration. Students will also develop knowledge and skills to serve on health professional teams to improve health outcomes.

Lab Hours: 32.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L , Corequisite: DEH 2702L, DEH 2808, DEH 2808L, DES 2100, DES 2100L

HSC3202 Introduction to Public Health 1.000 credits

This course is designed to give students a foundation in the core functions of population-based health consisting of assessment, policy development and assurance. Students will gain an understanding of public health as a broad field which applies the benefits of current biomedical, environmental, and behavioral knowledge in ways that maximize the health status of all populations. This course is blended. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822C, 1823, 1831. (1 hour lecture)

Lecture Hours: 16.000

Prerequisite: HSA2532, PAS1800C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

PAS1820C Clinical Medicine 2 for Physician Assistants 5.000 credits

The second course in the sequence. Students will learn the signs, symptoms, and pathophysiology of common diseases affecting pediatric, adult, and geriatric patients. In addition, diagnosis, therapeutic interventions, and follow-up will be learned. Patient education and preventative medicine will also included. Prerequisite: HSA2532, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823. (4 hr. lecture, 3 hr. clinical)

Lecture Hours: 64.000, Clinical Hours: 48.000

Prerequisite: HSA2532, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823

PAS1824 Pathophysiological Basis of Disease II 2.000 credits

This course is a continuation of PAS 1813 Pathophysiological Basis of Disease I. This course establishes scientific core knowledge and bridges the basic medical sciences with clinical medicine. The course covers the pathophysiology of human diseases that appear as a result of structural and functional alterations of the human body systems. The course begins with the study of human anatomy and physiology pertinent to the pathological conditions presented and progresses to the pathophysiological topics needed by physician assistant students. The course will enhance decision-making ability when working as a PA in clinical practice. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822L, 1823, 1831. (2 hr. lecture)

Lecture Hours: 32.000

Prerequisite: HSA2532, PAS1800C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

Course User Fee: \$560.00

PAS3038C Physical Diagnosis III 2.000 credits

This course will build upon skills learned in Physical Diagnosis I and II and will prepare the physician assistant student with the essential skills for entry into clinical practice. Students will learn to perform medical interviews and physical examinations on professional patients and high-fidelity simulators. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822C, 1823, 1831. (1 hour lecture; 3 hour clinical)

Lecture Hours: 16.000, Clinical Hours: 48.000

Prerequisite: HSA2532, PAS1800C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

PAS3075 Pharmacotherapeutics 4.000 credits

The second course in the sequence pas 1823, 3070. The study of the use of drugs to treat disease, including contraindication and incompatibilities; drug interactions; side effects and their treatment, and dosages and calculations. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822C, 1823, 1831. (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: HSA2532, PAS1800C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

Course User Fee: \$1120.00

PAS3019 Pathophysiological Basis of Disease III 2.000 credits

This course establishes scientific core knowledge and bridges the basic medical sciences with clinical medicine. The course covers the pathophysiology of human diseases that appear as a result of structural and functional alterations of the human body systems. The course begins with the study of human anatomy and physiology pertinent to the pathological conditions presented and progresses to the pathophysiological topics needed by physician assistant students. The course will enhance decision-making ability when working as a Physician Assistant (PA) in clinical practice. Prerequisite: PAS 1821, 1824, 3038C, 3075. (2 hour lecture)

Lecture Hours: 32.000

Prerequisite: DEH2810L, HSA3533, HSC3202, PAS1820C, PAS1824, PAS3038C, PAS3075

PAS3042C Clinical Medicine III for Physician Assistants 5.000 credits

Students will learn the signs and symptoms, diagnosis, therapeutic and non-therapeutic interventions of common diseases affecting pediatric, adult, and geriatric patients. Patient education and preventive medicine will also be included.

Prerequisites: DEH 2801L, HSA 3533, HSC 3202, PAS 1824, 1820C, 3038C, 3075. (4 hour lecture; 3 hour clinical)

Lecture Hours: 64.000, Clinical Hours: 48.000

Prerequisite: DEH2810L, HSA3533, HSC3202, PAS1820C, PAS1824, PAS3038C, PAS3075

PAS3070 Clinical Pharmacotherapeutics 3.000 credits

This course will provide a team-based, active, and applied learning environment to solve patient cases related to the management of pharmacotherapy.

Students will work within specified learning groups to solve patient cases. Students will learn from their future colleagues by working together to develop treatment plans and answer questions which are designed to promote active learning and critical thinking skills. Following each case, facilitated discussion will occur to promote student learning and retention of material. Real world examples will be incorporated into discussions and explanations of patient cases. Prerequisites: DEH 2801L, HSA 3533, HSC 3202, PAS 1824, 1820C, 3038C, 3075. (3 hour lecture)

Lecture Hours: 48.000

Prerequisite: PAS1824, DEH2810L, HSA3533, PAS3038C, PAS3075, HSC3202, PAS1820C

PAS3140 Genetics 2.000 credits

This course will prepare the PA student for human genetics and medical genomics, including Mendelian and complex diseases, chromosomal abnormalities, heritability, genetic variations and methods for investigating associations with diseases, gene-environment interactions, population genetics, cancer genetics, and epigenetics. Prerequisites: DEH 2801L, HSA 3533, HSC 3202, PAS 1824, 1820C, 3038C, 3075. (2 hour lecture)

Lecture Hours: 32.000

Prerequisite: DEH2810L, HSA3533, HSC3202, PAS1820C, PAS1824, PAS3038C, PAS3075

PAS3203C Surgical Problems and Procedures 5.000 credits

During this course the student will be exposed to the various aspects of general, orthopedic, cardiovascular, thoracic, ENT, neurologic, urologic, and pediatric surgical problems, their diagnosis and treatment. Laboratory components of this course will include learning fundamental techniques necessary in preoperative and postoperative care, including nasogastric intubation, central venous line placement, arterial and venous punctures and sterile techniques. Prerequisites: DEH 2801L, HSA 3533, HSC 3202, PAS 1824, 1820C, 3038C, 3075. (4 hr. lecture; 3 hr. clinical)

Lecture Hours: 64.000, Clinical Hours: 48.000

Prerequisite: DEH2810L, HSA3533, HSC3202, PAS1820C, PAS1824, PAS3038C, PAS3075

Course User Fee: \$1400.00

HSC4942 Community Service-Learning Practicum 3.000 credits

This course is designed to engage students in meaningful healthcare related community service learning activities. Students will learn to explore new collaborations in service-learning, cultural diversity and community healthcare development. These experiences will help to and prepare them for lives as civically-engaged local, national, and global citizens. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: HSC3057, HSC 3231

PAS4191L Internal Medicine Clerkship 4.000 credits

The clinical course focuses on basic medical practice. The student is exposed to common medical problems encountered on in-patient and out-patient medical services. Emphasis is placed on the history and physical examination and the process required in the proper work-up and management of the patient. Patient care experience in the various subdivisions of internal medicine including oncology, hematology, neurology, nephrology, gastroenterology, rheumatology, pulmonology, cardiology, and infectious diseases may be required. Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C. (192 hr. Clinical)

Clinical Hours: 192.000

Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

PAS4290 Surgery Clerkship 2.000 credits

During the clinical course the student will be exposed to a variety of clinical problems routinely seen on the surgical service. Emphases will be placed on preoperative, intraoperative and postoperative management of the patient. In the operating room the student will practice aseptic technique, operating room principles, and assist in surgery. Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C. (96 hr. Clinical)

Clinical Hours: 96.000

Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

Course User Fee: \$560.00

PAS4391 Pediatrics Clerkship 4.000 credits

This clinical course in pediatric care settings will introduce students to childhood illnesses and normal variations of growth and development. Students will perform histories and physical examinations and manage patients in the newborn nursery, pediatric out-patient clinic and emergency room.

Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C. (192 hr. Clinical)

Clinical Hours: 192.000

Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

Course User Fee: \$1120.00

PAS4493 Family Medicine Clerkship 4.000 credits

This clinical course introduces the student to the family practice setting where emphasis is placed on the common diseases treated by primary care practitioners in conjunction with other members of the health care team. The student is exposed to rural epidemiology, cultural diversity, and problems that affect delivery of health care in rural and under-served areas. Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C. (192 hr. clinical)

Clinical Hours: 192.000

Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

Course User Fee: \$1120.00

PAS4590 OBSTETRICS/GYNECOLOGY 2.000 credits

During this clinical course the student will participate on the obstetrical service managing pregnancy, labor and delivery and be introduced to pre-and postnatal complications. The student will also participate in the management of common gynecologic problems.

Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C. (96 hr. clinical)

Clinical Hours: 96.000

Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

Course User Fee: \$560.00

PAS4690 Emergency Medicine Clerkship 2.000 credits

This clinical course in an emergency care setting will provide opportunities for the student to manage the acutely ill and traumatized patient. The student will learn to perform history and physical examination on the acutely ill patient with emphasis being placed on the management and support measures necessary in situations which are life threatening. Prerequisites: PAS 3019, 3042C, 3070, 3140, 3203C.

Clinical Hours: 96.000

Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

Course User Fee: \$560.00

PAS4940 Psychiatry Clerkship 2.000 credits

This clinical course in a psychiatric care setting will allow students to participate in daily rounds and become knowledgeable of the use of psychotropic medications for psychiatric disorders. Group therapy sessions will be a major part of the learning experience. Prerequisites: PAS 3019, 3042C, 3070, 3140, 3203C.

Clinical Hours: 96.000

Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

Course User Fee: \$560.00

PAS4943 Selective Clerkship 2.000 credits

This clinical course provides the opportunity for students to become familiar with a medical specialty of their choice. The student will have the opportunity to familiarize with the role of the physician assistant in that specialty and reinforce the most common examination techniques, diagnoses and treatments within the selected specialty. Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C

Clinical Hours: 96.000

Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

Course User Fee: \$560.00

Program Description

Leadership and Management Innovation

Bachelor of Applied Science | Code: P9200 | 120 credit

The Bachelor of Applied Science (BAS) in Leadership and Management Innovation prepares associate degree graduates from a variety of disciplines for work as a supervisor or manager. The programs prepare students with the hands-on training necessary to meet the workforce demands of today and tomorrow while providing the skills needed to compete for advancement.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
MAN 3025	Organization Management	3.0
MAN 3065	Business Ethics	3.0
MAN 3301	Human Resource Management	3.0
Upper Division Elective*		3.0

Select 3 credits from the following course options: ACG3, ACG4*, BUL4*, FIN3*, FIN4*, GEB3*, GEB4*, HFT3*, HFT4*, ISM4*, MAN3*, MAN4*, MAR3*, MAR4*, TAX4*, TRA3*, TRA4*

Semester 2		Units: 12.0
MAN 3240	Organizational Behavior	3.0
MAR 3803	Marketing for Managers	3.0
Upper Division Elective*		3.0
Elective**		3.0

Select 3 credits from the following course options: ACG3, ACG4*, BUL4*, FIN3*, FIN4*, GEB3*, GEB4*, HFT3*, HFT4*, ISM4*, MAN3*, MAN4*, MAR3*, MAR4*, TAX4*, TRA3*, TRA4*

**Select 3 credits from the following course options: ACG*, BUL*, CGS 1060C, ECO*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET*, PUR*, QMB*, SBM*, SCM*, SPM*, TAX*, TRA*

Semester 3		Units: 12.0
MAN 4113	Managing in a Multifaceted Environment	3.0
MAN 4120	Leadership Challenges and Supervision	3.0
Elective*		6.0

Select 3 credits from the following course options: ACG, BUL*, CGS 1060C, ECO*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET*, PUR*, QMB*, SBM*, SCM*, SPM*, TAX*, TRA*

Semester 4		Units: 12.0
FIN 3400	Finance for Non-Financial Managers	3.0
MAN 4720	Strategic Management Decision Making	3.0
Elective*		6.0

Select 3 credits from the following course options: ACG, BUL*, CGS 1060C, ECO*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET*, PUR*, QMB*, SBM*, SCM*, SPM*, TAX*, TRA*

Semester 5		Units: 12.0
MAN 4900	Capstone Project in Supervision and Management	3.0
MAN 4941	Management Internship	3.0

Select 3 credits from the following course options: ACG, BUL*, CGS 1060C, ECO*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET*, PUR*, QMB*, SBM*, SCM*, SPM*, TAX*, TRA*

Total: 60.0

MAN3025 Organization Management 3.000 credits

The student will learn the major functions of supervision and management including the functions of planning, organizing, leading and controlling. Emphasis is placed on learning how to communicate more effectively with employees and how to increase leadership effectiveness. Major topics include goal setting and attainment, organizational structure, decision-making, strategic planning, managing stress, and ethical behavior and ethical role modeling. Cases that present the student with opportunities to make supervisory and management decisions, along with timely feedback on their effectiveness, will be used. Must pass this course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3065 Business Ethics 3.000 credits

In this course the student will learn how personal values and ethics influence the decision-making capacity of the manager within an organization. The student will learn about ethical theories and the roles they play in the social and corporate behavior of an organization. Issues such as diversity in the workplace, intuition, technology, and the global environment of business, will be included in case examples. The student will acquire an understanding of how an organization can institutionalize its ethical system. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3301 Human Resource Management 3.000 credits

The student will learn the functions of Human Resource Management including human resource planning, strategic development of human resources, recruitment techniques, selection and hiring processes, compensation systems, development of policy and procedures for effective and ethical human resource management, performance review and evaluation systems, working effectively with organized labor, retention of employees, and current issues in human resource management. The course will also include an exploration of human resources within the global business environment of a boundless organization. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3240 Organizational Behavior 3.000 credits

The students will learn about social behavioral sciences that can be applied to supervision and management through major topics include motivation, conflict, corrective actions and rewards, job related stress, organizational dynamics, the evolving global environment, and the responsibility to stakeholders and the planetary environment. The student will examine organizational behavior and how to integrate behavioral concepts in an effective managerial decision-making process. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR3803 Marketing for Managers 3.000 credits

Students will learn how managers implement the marketing elements within a strategic planning framework. Emphasis is on decision-making, and the development and execution of marketing strategies related to product and brand development, channels of distribution, pricing, and promotional efforts under varying marketplace conditions. This course incorporates the use of case studies. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$45.00

MAN4113 Managing in a Multifaceted Environment 3.000 credits

Managing in a Multifaceted Environment is a thorough course that offers a management perspective on making the most of the various qualities of today's workforce. This course explores the value that comes from human capital and other distinguishing factors in the modern workplace. Furthermore, it delves into new leadership styles as solutions for effectively managing and leading people. Students will gain insights into utilizing the strengths of the workforce and creating a dynamic work environment.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4120 Leadership Challenges and Supervision 3.000 credits

The student will learn to analyze leadership theories and will acquire an awareness of the dynamics of supervisory and managerial decision-making. Emphasis will be placed on team building, crisis management, social and environmental responsibility, developing and communicating a vision, developing a full set of managerial and leadership skills. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair, MAN4894

Course User Fee: \$70.00

FIN3400 Finance for Non-Financial Managers 3.000 credits

The students will learn to apply their financial skills and decision-making abilities to address financial issues in a business environment. They will learn how financial markets influence their decisions and the role of financial intermediaries in these markets. Emphasis will be placed on financial and capital budgeting, working capital management, short and long term financing, valuation of the firm, and overall capital structure. The course will require the student to apply the time value of money through the use of present and future value scenarios. This course must be met with a grade of "C" or higher. (3 hour lecture)

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN4720 Strategic Management Decision Making 3.000 credits

The student will learn the designing, planning, and implementation of strategic decision making in a business organization by identifying problems and designing possible solutions, formulating plans, goals, and feedback mechanisms. Through case studies, the student will conduct internal and external assessments to analyze effective strategic choices for companies.

Lecture Hours: 48.000

Course User Fee: \$70.00

MAN4900 Capstone Project in Supervision and Management 3.000 credits

Students will apply and integrate the knowledge and skills learned throughout the program by completing a capstone project. Student will identify a business opportunity; conduct an industry feasibility assessment and operational mapping; evaluate market condition; construct a marketing plan, and formulate a financial plan for implementing and managing an existing or new business venture; and evaluate the role of ethic, social, and environmental responsibility within a business. Must pass course with a grade of "C" or higher. Prerequisite: Senior Status or Departmental Permission.

Lecture Hours: 48.000

Prerequisite: Senior Status or Departmental Permission.(3 hr. lecture)

Course User Fee: \$90.00

MAN4941 Management Internship 3.000 credits

The student will learn management techniques by becoming an employee or intern (on a paid or unpaid basis) at either a not-for-profit or for-profit organization. The student will be required to work at least 144 hours required by the state to earn the credit for the internship. The student will work with their MDC Faculty and Supervising Employer to establish a set of assignments / learning goals that will be achieved during the semester. This course must be met with a grade of "C" or higher. Prerequisite: Senior Status or Departmental Permission.

Other Hours: 144.000

Prerequisite: Senior status and permission of department chair., Pre/Corequisite: MAN4900

Course User Fee: \$45.00

Program Description

Leadership and Management Innovation - Accounting Concentration

Bachelor of Applied Science | Code: P9203 | 120 credit

The Bachelor in Applied Science (BAS) in Leadership and Management Innovation prepares associate degree graduates from a variety of disciplines for work as a supervisor or manager. The programs prepares students with the hands-on training necessary to meet the workforce demands of today and tomorrow while providing the skills needed to compete for advancement. In addition, students may complete the general degree or one of three concentrations in Accounting, Digital Marketing, Hospitality Management, or Human Resource Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ACG 3103	Intermediate Accounting I	3.0
MAN 3025	Organization Management	3.0
MAN 3065	Business Ethics	3.0
MAN 3301	Human Resource Management	3.0

Semester		Units: 12.0
ACG 3113	Intermediate Financial Accounting II	3.0
MAN 3240	Organizational Behavior	3.0
MAR 3803	Marketing for Managers	3.0
<i>Elective</i>		3.0
TAX 2000	Income Tax	3.0
OR		
Elective Course Options*		

Select 3 credits from the following course options: ACG, BUL*, CGS 1060C, ECO*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET*, PUR*, QMB*, SBM*, SCM*, SPM*, TAX*, TRA*

Semester 3		Units: 12.0
ACG 3343	Cost Accounting and Controls	3.0
BUL 4320	Business Law	3.0
MAN 4120	Leadership Challenges and Supervision	3.0
MAN 4720	Strategic Management Decision Making	3.0

Semester 4		Units: 12.0
ACG 4632	Auditing	3.0
FIN 3400	Finance for Non-Financial Managers	3.0
MAN 4113	Managing in a Multifaceted Environment	3.0
TAX 4001	Federal Income Tax I	3.0

Semester 5		Units: 12.0
ACG 4401	Accounting Information Systems	3.0
MAN 4900	Capstone Project in Supervision and Management	3.0
MAN 4941	Management Internship	3.0
TAX 4011	Federal Income Tax II	3.0

Total: 60.0

ACG3103 Intermediate Accounting I 3.000 credits

Theory and methodology underlying financial reporting, including the FASB's conceptual framework, the accounting process, financial statements, accounting changes, present value applications, and current assets. Prerequisites: ACG2071, MAC2233, QMB2100

Lecture Hours: 48.000

Course User Fee: \$98.00

MAN3025 Organization Management 3.000 credits

The student will learn the major functions of supervision and management including the functions of planning, organizing, leading and controlling. Emphasis is placed on learning how to communicate more effectively with employees and how to increase leadership effectiveness. Major topics include goal setting and attainment, organizational structure, decision-making, strategic planning, managing stress, and ethical behavior and ethical role modeling. Cases that present the student with opportunities to make supervisory and management decisions, along with timely feedback on their effectiveness, will be used. Must pass this course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3065 Business Ethics 3.000 credits

In this course the student will learn how personal values and ethics influence the decision-making capacity of the manager within an organization. The student will learn about ethical theories and the roles they play in the social and corporate behavior of an organization. Issues such as diversity in the workplace, intuition, technology, and the global environment of business, will be included in case examples. The student will acquire an understanding of how an organization can institutionalize its ethical system. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3301 Human Resource Management 3.000 credits

The student will learn the functions of Human Resource Management including human resource planning, strategic development of human resources, recruitment techniques, selection and hiring processes, compensation systems, development of policy and procedures for effective and ethical human resource management, performance review and evaluation systems, working effectively with organized labor, retention of employees, and current issues in human resource management. The course will also include an exploration of human resources within the global business environment of a boundless organization. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

ACG3113 Intermediate Financial Accounting II 3.000 credits

Continuation of ACG3103. Particular emphasis on analysis of balance sheet accounts through problem solving. Provides students with a more in-depth knowledge of Generally Accepted Accounting Principles (GAAP), including the advanced study of long-term assets, current and long-term liabilities and equities. Prerequisite: ACG3103.

Lecture Hours: 48.000

Prerequisite: ACG3103

MAN3240 Organizational Behavior 3.000 credits

The students will learn about social behavioral sciences that can be applied to supervision and management through major topics include motivation, conflict, corrective actions and rewards, job related stress, organizational dynamics, the evolving global environment, and the responsibility to stakeholders and the planetary environment. The student will examine organizational behavior and how to integrate behavioral concepts in an effective managerial decision-making process. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR3803 Marketing for Managers 3.000 credits

Students will learn how managers implement the marketing elements within a strategic planning framework. Emphasis is on decision-making, and the development and execution of marketing strategies related to product and brand development, channels of distribution, pricing, and promotional efforts under varying marketplace conditions. This course incorporates the use of case studies. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$45.00

TAX2000 Income Tax 3.000 credits

Federal income tax fundamentals with emphasis on individual returns. Topics considered include gross income, capital gains and losses, deductions and exemptions, and tax credits.

Lecture Hours: 48.000

Course User Fee: \$15.00

ACG3343 Cost Accounting and Controls 3.000 credits

This course covers the preparation of accounting information for use in management decision making process. It contains information on budgeting, standard costing, direct costing, performance evaluation, and use of accounting information.

Prerequisite: ACG2701

Lecture Hours: 48.000

Prerequisite: ACG2071

Course User Fee: \$98.00

BUL4320 Business Law 3.000 credits

This course covers the basics of business law for accountants which include the topics of business organizations, securities law and corporate governance, agency and employment, sales and lease contracts, creditors' rights and bankruptcy, professional liability, government regulation, and property. The course also focuses on the nature of legal relationships in business with a particular emphasis on the government regulation of business, ethical, global, and corporate issues while addressing core curriculum requirements. Prerequisites: ACG3113 and ACG3343. Students must pass this course with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: ACG3113 AND ACG3343

MAN4120 Leadership Challenges and Supervision 3.000 credits

The student will learn to analyze leadership theories and will acquire an awareness of the dynamics of supervisory and managerial decision-making.

Emphasis will be placed on team building, crisis management, social and environmental responsibility, developing and communicating a vision, developing a full set of managerial and leadership skills. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair, MAN4894

Course User Fee: \$70.00

MAN4720 Strategic Management Decision Making 3.000 credits

The student will learn the designing, planning, and implementation of strategic decision making in a business organization by identifying problems and designing possible solutions, formulating plans, goals, and feedback mechanisms. Through case studies, the student will conduct internal and external assessments to analyze effective strategic choices for companies.

Lecture Hours: 48.000

Course User Fee: \$70.00

ACG4632 Auditing 3.000 credits

This course provides a sound conceptual foundation of basic auditing processes from the perspective of the public accounting profession. Professional standards, ethics, legal responsibilities, and utilization of technology are addressed. Principles and procedures of internal and public auditing are discussed, including professional standards, ethics, legal responsibilities, and the utilization of technology.

Lecture Hours: 48.000

Prerequisite: ACG3113

Course User Fee: \$98.00

FIN3400 Finance for Non-Financial Managers 3.000 credits

The students will learn to apply their financial skills and decision-making abilities to address financial issues in a business environment. They will learn how financial markets influence their decisions and the role of financial intermediaries in these markets.

Emphasis will be placed on financial and capital budgeting, working capital management, short and long term financing, valuation of the firm, and overall capital structure. The course will require the student to apply the time value of money through the use of present and future value scenarios. This course must be met with a grade of "C" or higher. (3 hour lecture)

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN4113 Managing in a Multifaceted Environment 3.000 credits

Managing in a Multifaceted Environment is a thorough course that offers a management perspective on making the most of the various qualities of today's workforce. This course explores the value that comes from human capital and other distinguishing factors in the modern workplace. Furthermore, it delves into new leadership styles as solutions for effectively managing and leading people. Students will gain insights into utilizing the strengths of the workforce and creating a dynamic work environment.

Lecture Hours: 48.000

Course User Fee: \$35.00

TAX4001 Federal Income Tax I 3.000 credits

A survey of federal income tax with emphasis of taxation of individuals and the ethics of income tax accounting. Students will gain the basic theory, concepts, practice, and methods of determining taxable income and tax liabilities. Prerequisite: ACG3103.

Lecture Hours: 48.000

Course User Fee: \$98.00

ACG4401 Accounting Information Systems 3.000 credits

This is an information systems course for accounting students, and not a traditional accounting course, nor a traditional MIS course. The course aims to provide students with a broad conceptual knowledge of accounting information systems, the need for and control of accounting information systems, current topics in accounting information systems, and discussions of specific transaction-cycle based accounting information systems. The course seeks to examine the linkages between information systems and accounting, and it also provides an overview of how to store, retrieve, analyze, and control data using information systems. Students must pass the course with a grade of "C" or higher. Prerequisites: ACG3113 and ACG3343.

Lecture Hours: 48.000

Prerequisite: ACG3113 AND ACG3343

MAN4900 Capstone Project in Supervision and Management 3.000 credits

Student will apply and integrate the knowledge and skills learned throughout the program by completing a capstone project. Student will identify a business opportunity; conduct an industry feasibility assessment and operational mapping; evaluate market condition; construct a marketing plan, and formulate a financial plan for implementing and managing an existing or new business venture; and evaluate the role of ethic, social, and environmental responsibility within a business.

Lecture Hours: 48.000

Course User Fee: \$90.00

MAN4941 Management Internship 3.000 credits

The student will learn management techniques by becoming an employee or intern (on a paid or unpaid basis) at either a not-for-profit or for-profit organization. The student will be required to work at least 144 hours required by the state to earn the credit for the internship. The student will work with their MDC Faculty and Supervising Employer to establish a set of assignments / learning goals that will be achieved during the semester. This course must be met with a grade of "C" or higher.

Prerequisite: Senior Status or Departmental Permission.

Other Hours: 144.000

Prerequisite: Senior status and permission of department chair., Pre/Corequisite: MAN4900

Course User Fee: \$45.00

TAX4011 Federal Income Tax II 3.000 credits

A survey of federal income tax with emphasis of taxation of business entities and the ethics of income tax accounting, student of the basic theory, concepts, practice and methods of determining the taxable income and tax liabilities. Prerequisite: TAX4001.

Lecture Hours: 48.000

Prerequisite: TAX4001

Course User Fee: \$98.00

Program Description

Leadership and Management Innovation – Digital Marketing Concentration

Bachelor of Applied Science | Code: P9204

The Bachelor in Applied Science (BAS) in Leadership and Management Innovation prepares associate degree graduates from a variety of disciplines for work as a supervisor or manager. The programs prepares students with the hands-on training necessary to meet the workforce demands of today and tomorrow while providing the skills needed to compete for advancement. In addition, students may complete the general degree or one of three concentrations in Accounting, Digital Marketing, Hospitality Management, or Human Resource Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1 **Units: 12.0**

MAN 3025	Organization Management	3.0
MAN 3065	Business Ethics	3.0
MAN 3301	Human Resource Management	3.0
MAR 3803	Marketing for Managers	3.0

Semester 2 **Units: 12.0**

MAN 3240	Organizational Behavior	3.0
MAR 3325	Digital Advertising	3.0
MAR 4233	Social Media Marketing Application	3.0
Business Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, ECO*, ENT*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET* PUR*, QMB*, SBM*, SPM* TAX*, TRA*

Semester 3 **Units: 12.0**

MAN 4113	Managing in a Multifaceted Environment	3.0
MAN 4120	Leadership Challenges and Supervision	3.0
MAR 4327	Search Engine Optimization (SEO) and Search Engine Marketing (SEM)	3.0
Business Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, ECO*, ENT*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET* PUR*, QMB*, SBM*, SPM* TAX*, TRA*

Semester 4 **Units: 12.0**

FIN 3400	Finance for Non-Financial Managers	3.0
MAN 4720	Strategic Management Decision Making	3.0
MAR 4674	Marketing Analytics	3.0
Business Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, ECO*, ENT*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET* PUR*, QMB*, SBM*, SPM* TAX*, TRA*

Semester 5 **Units: 12.0**

MAR 4721	Digital Marketing Strategy	3.0
MAR 4860	Customer Relationship Management	3.0
MAN 4900	Capstone Project in Supervision and Management	3.0
MAN 4941	Management Internship	3.0

Total: 60.0

MAN3025 Organization Management 3.000 credits

The student will learn the major functions of supervision and management including the functions of planning, organizing, leading and controlling. Emphasis is placed on learning how to communicate more effectively with employees and how to increase leadership effectiveness. Major topics include goal setting and attainment, organizational structure, decision-making, strategic planning, managing stress, and ethical behavior and ethical role modeling. Cases that present the student with opportunities to make supervisory and management decisions, along with timely feedback on their effectiveness, will be used. Must pass this course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3065 Business Ethics 3.000 credits

In this course the student will learn how personal values and ethics influence the decision-making capacity of the manager within an organization. The student will learn about ethical theories and the roles they play in the social and corporate behavior of an organization. Issues such as diversity in the workplace, intuition, technology, and the global environment of business, will be included in case examples. The student will acquire an understanding of how an organization can institutionalize its ethical system. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3301 Human Resource Management 3.000 credits

The student will learn the functions of Human Resource Management including human resource planning, strategic development of human resources, recruitment techniques, selection and hiring processes, compensation systems, development of policy and procedures for effective and ethical human resource management, performance review and evaluation systems, working effectively with organized labor, retention of employees, and current issues in human resource management. The course will also include an exploration of human resources within the global business environment of a boundless organization. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR3803 Marketing for Managers 3.000 credits

Students will learn how managers implement the marketing elements within a strategic planning framework. Emphasis is on decision-making, and the development and execution of marketing strategies related to product and brand development, channels of distribution, pricing, and promotional efforts under varying marketplace conditions. This course incorporates the use of case studies. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$45.00

MAN3240 Organizational Behavior 3.000 credits

The students will learn about social behavioral sciences that can be applied to supervision and management through major topics include motivation, conflict, corrective actions and rewards, job related stress, organizational dynamics, the evolving global environment, and the responsibility to stakeholders and the planetary environment. The student will examine organizational behavior and how to integrate behavioral concepts in an effective managerial decision-making process. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR3325 Digital Advertising 3.000 credits

In this course students learn about the principles and practices of digital advertising, as well as hands-on experience with the tools and platforms used in the industry. Students will learn about search engine advertising, social media advertising, e-commerce advertising, video advertising and effective implementation of advertising campaigns.

Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAR4233 Social Media Marketing Application 3.000 credits

This course introduces students to social media and marketing functions and strategies that are essential to consumer involvement, community engagement, and customer relationship management. Students will learn about the dynamics of social marketing, trends in new technology, and integrated marketing through online platforms. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAN4113 Managing in a Multifaceted Environment 3.000 credits

Managing in a Multifaceted Environment is a thorough course that offers a management perspective on making the most of the various qualities of today's workforce. This course explores the value that comes from human capital and other distinguishing factors in the modern workplace. Furthermore, it delves into new leadership styles as solutions for effectively managing and leading people. Students will gain insights into utilizing the strengths of the workforce and creating a dynamic work environment.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4120 Leadership Challenges and Supervision 3.000 credits

The student will learn to analyze leadership theories and will acquire an awareness of the dynamics of supervisory and managerial decision-making. Emphasis will be placed on team building, crisis management, social and environmental responsibility, developing and communicating a vision, developing a full set of managerial and leadership skills. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair, MAN4894

Course User Fee: \$70.00

MAR4327 Search Engine Optimization (SEO) and Search Engine Marketing (SEM) 3.000 credits

In this course students will learn how to optimize websites for search engines and search engine ranking. Students will learn how to use social media to complement and enhance traditional SEO strategies. Students will apply their knowledge of search and search optimization tools and technics using leading industry certification. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

FIN3400 Finance for Non-Financial Managers 3.000 credits

The students will learn to apply their financial skills and decision-making abilities to address financial issues in a business environment. They will learn how financial markets influence their decisions and the role of financial intermediaries in these markets.

Emphasis will be placed on financial and capital budgeting, working capital management, short and long term financing, valuation of the firm, and overall capital structure. The course will require the student to apply the time value of money through the use of present and future value scenarios. This course must be met with a grade of "C" or higher. (3 hour lecture)

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN4720 Strategic Management Decision Making 3.000 credits

The student will learn the designing, planning, and implementation of strategic decision making in a business organization by identifying problems and designing possible solutions, formulating plans, goals, and feedback mechanisms. Through case studies, the student will conduct internal and external assessments to analyze effective strategic choices for companies.

Lecture Hours: 48.000

Course User Fee: \$70.00

MAR4674 Marketing Analytics 3.000 credits

A study of the metrics and systems needed to receive a return on every sale and marketing investment made. This course focuses on common calculations used in business to determine market share, penetration, brand/category development, awareness, margin, price per unit, break-even. The focus of this course is to enable students to take the vast amount of "data" available within companies and turn that data into "information" making them that much more valuable to potential employers.

Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAR4721 Digital Marketing Strategy 3.000 credits

Students will learn the process of creating a plan and developing a strategy for promoting a business or organization using digital channels. Students will learn how to apply their knowledge and skills in a marketing campaign project. Students will plan, implement and evaluate a digital marketing campaign using case studies and collaborative projects. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAR4860 Customer Relationship Management 3.000 credits

In this course students will learn how to effectively manage relationships with customers to increase customer satisfaction, loyalty, and retention. They will learn about customer behavior, communication strategies, data analysis techniques and CRM tools and technics to better understand and meet the needs of their customers. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAN4900 Capstone Project in Supervision and Management 3.000 credits

Student will apply and integrate the knowledge and skills learned throughout the program by completing a capstone project. Student will identify a business opportunity; conduct an industry feasibility assessment and operational mapping; evaluate market condition; construct a marketing plan, and formulate a financial plan for implementing and managing an existing or new business venture; and evaluate the role of ethic, social, and environmental responsibility within a business.

Lecture Hours: 48.000

Course User Fee: \$90.00

MAN4941 Management Internship 3.000 credits

The student will learn management techniques by becoming an employee or intern (on a paid or unpaid basis) at either a not-for-profit or for-profit organization. The student will be required to work at least 144 hours required by the state to earn the credit for the internship. The student will work with their MDC Faculty and Supervising Employer to establish a set of assignments / learning goals that will be achieved during the semester. This course must be met with a grade of "C" or higher.

Prerequisite: Senior Status or Departmental Permission.

Other Hours: 144.000

Prerequisite: Senior status and permission of department chair., Pre/Corequisite: MAN4900

Course User Fee: \$45.00

Program Description

Leadership and Management Innovation - Hospitality Management Concentration

Bachelor of Applied Science | Code: P9201 | 120 credits

The Bachelor in Applied Science (BAS) in Leadership and Management Innovation prepares associate degree graduates from a variety of disciplines for work as a supervisor or manager. The programs prepares students with the hands-on training necessary to meet the workforce demands of today and tomorrow while providing the skills needed to compete for advancement. In addition, students may complete the general degree or one of three concentrations in Accounting, Digital Marketing, Hospitality Management, or Human Resource Management. In addition, students may complete the general degree or one of three concentrations in Accounting, Digital Marketing, Hospitality Management, or Human Resource Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

GEB 3213	Advanced Communications in Business	3.0
MAN 3025	Organization Management	3.0
MAN 3065	Business Ethics	3.0
MAN 3301	Human Resource Management	3.0

Semester 2		Units: 12.0
HFT 3263	Restaurant Management	3.0
HFT 3603	Hospitality Law	3.0
MAN 3240	Organizational Behavior	3.0
MAR 3803	Marketing for Managers	3.0
Semester 3		Units: 12.0
HFT 4253	Hotel Management	3.0
HFT 4468	Revenue Management in the Hospitality Industry	3.0
ISM 4011	Management Information Systems	3.0
MAN 4120	Leadership Challenges and Supervision	3.0
Semester 4		Units: 12.0
FIN 3400	Finance for Non-Financial Managers	3.0
HFT 4064	Bar and Beverage Management	3.0
MAN 4113	Managing in a Multifaceted Environment	3.0
MAN 4720	Strategic Management Decision Making	3.0
Semester 5		Units: 12.0
HFT 4809	Food Service Management	3.0
<i>Business Elective</i>		3.0
MAN 4162	Customer Relations for Managers	3.0
OR		
MAN 4741	Change & Innovation Management	3.0
MAN 4900	Capstone Project in Supervision and Management	3.0
MAN 4941	Management Internship	3.0

Total: 60.0

**GEB3213 Advanced Communications in Business
3.000 credits**

Student will develop effective and efficient oral and written communications skills that can be applied in professional business settings. Topics include formal and informal writing, preparation of reports, creation of business proposal, written correspondence, and presentations. 3 hours Lecture, 3 credits

Lecture Hours: 48.000

Course User Fee: \$35.00

**MAN3025 Organization Management 3.000
credits**

The student will learn the major functions of supervision and management including the functions of planning, organizing, leading and controlling. Emphasis is placed on learning how to communicate more effectively with employees and how to increase leadership effectiveness. Major topics include goal setting and attainment, organizational structure, decision-making, strategic planning, managing stress, and ethical behavior and ethical role modeling. Cases that present the student with opportunities to make supervisory and management decisions, along with timely feedback on their effectiveness, will be used. Must pass this course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3065 Business Ethics 3.000 credits

In this course the student will learn how personal values and ethics influence the decision-making capacity of the manager within an organization. The student will learn about ethical theories and the roles they play in the social and corporate behavior of an organization. Issues such as diversity in the workplace, intuition, technology, and the global environment of business, will be included in case examples. The student will acquire an understanding of how an organization can institutionalize its ethical system. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3301 Human Resource Management 3.000 credits

The student will learn the functions of Human Resource Management including human resource planning, strategic development of human resources, recruitment techniques, selection and hiring processes, compensation systems, development of policy and procedures for effective and ethical human resource management, performance review and evaluation systems, working effectively with organized labor, retention of employees, and current issues in human resource management. The course will also include an exploration of human resources within the global business environment of a boundless organization. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

HFT3263 Restaurant Management 3.000 credits

Students will examine restaurant management and operations. Topics include operational issues, customer satisfaction, restaurant trends and challenges, new technologies, financial accountability, cost controls, service issues, regulatory and safety compliance, leadership, training and staff development. Upon completion, students will be able to articulate knowledge of all aspects of a restaurant operation.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT3603 Hospitality Law 3.000 credits

Students will examine the legal rights and responsibilities related to the hospitality industry. Topics include federal and state regulations, historical and current practices, safety and security, risk management, loss prevention, liability, relevant torts and contracts. Upon completion, students will be able to articulate knowledge of the legal issues in the hospitality industry, and methods of preventing or minimizing organizational liability.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN3240 Organizational Behavior 3.000 credits

The students will learn about social behavioral sciences that can be applied to supervision and management through major topics include motivation, conflict, corrective actions and rewards, job related stress, organizational dynamics, the evolving global environment, and the responsibility to stakeholders and the planetary environment. The student will examine organizational behavior and how to integrate behavioral concepts in an effective managerial decision-making process. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR3803 Marketing for Managers 3.000 credits

Students will learn how managers implement the marketing elements within a strategic planning framework. Emphasis is on decision-making, and the development and execution of marketing strategies related to product and brand development, channels of distribution, pricing, and promotional efforts under varying marketplace conditions. This course incorporates the use of case studies. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$45.00

HFT4253 Hotel Management 3.000 credits

Students will examine the working components of a hotel and their interrelationships. Topics include hotel organization, management and leadership, human resources, food and beverage service, meeting industry, marketing and selling, communications, management companies, and ethics in hospitality management. Upon completion, students will be able to articulate knowledge of the responsibilities of each hotel department and how they interact to provide the overall guest experience.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT4468 Revenue Management in the Hospitality Industry 3.000 credits

Students will examine the basics of revenue management in the hotel industry. Topics include pricing and length-of-stay tools, revenue management performance measures, capacity management, demand and revenue forecasting, discounting, reservation systems, and pricing execution. Upon completion, students will be able to articulate knowledge of how to accurately forecast hotel guest arrivals, examine pricing models in accordance with revenue management principles, and how to manage overbooking. Prerequisite: ACG2021 , ACG2021L, or HFT2421.

Lecture Hours: 48.000

Pre/Corequisite: HFT2421, Prerequisite: ACG2021, ACG2021L

Course User Fee: \$35.00

ISM4011 Management Information Systems 3.000 credits

The student will use information technology software to assist in making decisions of a business nature. The course will examine the use of computer systems and information technology and their applications to make more effective business decisions. The course will include the latest terminology, techniques and applications of information systems in a business organization. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$90.00

MAN4120 Leadership Challenges and Supervision 3.000 credits

The student will learn to analyze leadership theories and will acquire an awareness of the dynamics of supervisory and managerial decision-making. Emphasis will be placed on team building, crisis management, social and environmental responsibility, developing and communicating a vision, developing a full set of managerial and leadership skills. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair, MAN4894

Course User Fee: \$70.00

FIN3400 Finance for Non-Financial Managers 3.000 credits

The students will learn to apply their financial skills and decision-making abilities to address financial issues in a business environment. They will learn how financial markets influence their decisions and the role of financial intermediaries in these markets. Emphasis will be placed on financial and capital budgeting, working capital management, short and long term financing, valuation of the firm, and overall capital structure. The course will require the student to apply the time value of money through the use of present and future value scenarios. This course must be met with a grade of "C" or higher. (3 hour lecture)

Lecture Hours: 48.000

Course User Fee: \$60.00

HFT4064 Bar and Beverage Management 3.000 credits

Students will examine the management of beverages served in hospitality operations. Topics include history and trends; service, procurement and storage; knowledge and control of wines and fermented/ distilled beverages; and non-alcoholic beverages, coffees, and teas. Upon completion, students will be able to articulate knowledge of responsible alcohol service and knowledge of beverages consumed in a hospitality operation.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4113 Managing in a Multifaceted Environment 3.000 credits

Managing in a Multifaceted Environment is a thorough course that offers a management perspective on making the most of the various qualities of today's workforce. This course explores the value that comes from human capital and other distinguishing factors in the modern workplace. Furthermore, it delves into new leadership styles as solutions for effectively managing and leading people. Students will gain insights into utilizing the strengths of the workforce and creating a dynamic work environment.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4720 Strategic Management Decision Making 3.000 credits

The student will learn the designing, planning, and implementation of strategic decision making in a business organization by identifying problems and designing possible solutions, formulating plans, goals, and feedback mechanisms. Through case studies, the student will conduct internal and external assessments to analyze effective strategic choices for companies.

Lecture Hours: 48.000

Course User Fee: \$70.00

HFT4809 Food Service Management 3.000 credits

Students will examine food service management and operations. Topics include the management process, marketing, sales techniques, cost and pricing strategies, food purchase controls, production and service, sanitation and safety, management of bar and beverage design and financial management. Upon completion, students will be able to articulate knowledge of the financial relationship of the food and beverage aspect of the hotel industry.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4162 Customer Relations for Managers 3.000 credits

Students will learn Customer Relations for Managers skills by exploring the dynamics of building solid and lasting relationships with customers. Topics will include doing business in a global environment, cultural diversity, the diversity of customs and global etiquette, negotiation tactics, global promotional tactics, and acceptable professional and corporate behaviors in a global business environment. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair., MAN4894

Course User Fee: \$70.00

MAN4741 Change & Innovation Management 3.000 credits

In this course, the student will study the concepts and strategies for change management and diffusion of innovation. Topics covered include, but are not limited to, implementing change, becoming a learning organization, innovation process management, and technology forecasting.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4900 Capstone Project in Supervision and Management 3.000 credits

Student will apply and integrate the knowledge and skills learned throughout the program by completing a capstone project. Student will identify a business opportunity; conduct an industry feasibility assessment and operational mapping; evaluate market condition; construct a marketing plan, and formulate a financial plan for implementing and managing an existing or new business venture; and evaluate the role of ethic, social, and environmental responsibility within a business.

Lecture Hours: 48.000

Course User Fee: \$90.00

MAN4941 Management Internship 3.000 credits

The student will learn management techniques by becoming an employee or intern (on a paid or unpaid basis) at either a not-for-profit or for-profit organization. The student will be required to work at least 144 hours required by the state to earn the credit for the internship. The student will work with their MDC Faculty and Supervising Employer to establish a set of assignments / learning goals that will be achieved during the semester. This course must be met with a grade of "C" or higher.

Prerequisite: Senior Status or Departmental Permission.

Other Hours: 144.000

Prerequisite: Senior status and permission of department chair., Pre/Corequisite: MAN4900

Course User Fee: \$45.00

Program Description

Leadership and Management Innovation - Human Resource Management Concentration

Bachelor of Applied Science | Code: P9202 | 120 credit

The Bachelor in Applied Science (BAS) in Leadership and Management Innovation prepares associate degree graduates from a variety of disciplines for work as a supervisor or manager. The program prepares students with the hands-on training necessary to meet the workforce demands of today and tomorrow while providing the skills needed to compete for advancement. In addition, students may complete the general degree or one of three concentrations in Accounting, Digital Marketing, Hospitality Management, or Human Resource Management. In addition, students may complete the general degree or one of three concentrations in Accounting, Digital Marketing, Hospitality Management, or Human Resource Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
MAN 3025	Organization Management	3.0
MAN 3065	Business Ethics	3.0
MAN 3301	Human Resource Management	3.0
Business Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, ECO*, ENT*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET*, PUR*, QMB*, SBM*, SPM* TAX*, TRA*

Semester 2		Units: 12.0
MAN 3240	Organizational Behavior	3.0
MAN 4361	Organizational Staffing	3.0
MAN 4402	Employment Law and Regulation	3.0
MAR 3803	Marketing for Managers	3.0

Semester 3		Units: 12.0
MAN 4120	Leadership Challenges and Supervision	3.0
MAN 4330	Compensation Management	3.0
MAN 4335	Employee Benefit Planning	3.0
MAN 4720	Strategic Management Decision Making	3.0

Semester 4		Units: 12.0
FIN 3400	Finance for Non-Financial Managers	3.0
MAN 3322	Human Resources Information Systems	3.0
MAN 4113	Managing in a Multifaceted Environment	3.0
Business Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, ECO*, ENT*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET*, PUR*, QMB*, SBM*, SPM* TAX*, TRA*

Semester 5		Units: 12.0
MAN 4352	Effective Employee Training	3.0
MAN 4900	Capstone Project in Supervision and Management	3.0
MAN 4941	Management Internship	3.0
Business Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, ECO*, ENT*, FIN*, GEB*, HFT*, ISM*, MAN*, MAR*, MKA*, MNA*, OST*, PET*, PUR*, QMB*, SBM*, SPM* TAX*, TRA*

Total: 60.0

MAN3025 Organization Management 3.000 credits

The student will learn the major functions of supervision and management including the functions of planning, organizing, leading and controlling. Emphasis is placed on learning how to communicate more effectively with employees and how to increase leadership effectiveness. Major topics include goal setting and attainment, organizational structure, decision-making, strategic planning, managing stress, and ethical behavior and ethical role modeling. Cases that present the student with opportunities to make supervisory and management decisions, along with timely feedback on their effectiveness, will be used. Must pass this course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3065 Business Ethics 3.000 credits

In this course the student will learn how personal values and ethics influence the decision-making capacity of the manager within an organization. The student will learn about ethical theories and the roles they play in the social and corporate behavior of an organization. Issues such as diversity in the workplace, intuition, technology, and the global environment of business, will be included in case examples. The student will acquire an understanding of how an organization can institutionalize its ethical system. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3301 Human Resource Management 3.000 credits

The student will learn the functions of Human Resource Management including human resource planning, strategic development of human resources, recruitment techniques, selection and hiring processes, compensation systems, development of policy and procedures for effective and ethical human resource management, performance review and evaluation systems, working effectively with organized labor, retention of employees, and current issues in human resource management. The course will also include an exploration of human resources within the global business environment of a boundless organization. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3240 Organizational Behavior 3.000 credits

The students will learn about social behavioral sciences that can be applied to supervision and management through major topics include motivation, conflict, corrective actions and rewards, job related stress, organizational dynamics, the evolving global environment, and the responsibility to stakeholders and the planetary environment. The student will examine organizational behavior and how to integrate behavioral concepts in an effective managerial decision-making process. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN4361 Organizational Staffing 3.000 credits

This course provides the student with an overview of the staffing function in organizations, including the topics of job analysis, forecasting, recruitment, selection, retention and turnover. It serves as an introductory course for the prospective human resources manager and as a survey of responsibility and activities of any manager with supervisory responsibilities. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4402 Employment Law and Regulation 3.000 credits

This course analyzes the federal and state regulation of the employment relationship, including wage and hour laws, EEO, and Affirmative Action programs. The student will address human resource issues such as employee benefits, insurance, workers' compensation, safety, health, employees' personal rights and collective bargaining legislation.

Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAR3803 Marketing for Managers 3.000 credits

Students will learn how managers implement the marketing elements within a strategic planning framework. Emphasis is on decision-making, and the development and execution of marketing strategies related to product and brand development, channels of distribution, pricing, and promotional efforts under varying marketplace conditions. This course incorporates the use of case studies. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$45.00

**MAN4120 Leadership Challenges and Supervision
3.000 credits**

The student will learn to analyze leadership theories and will acquire an awareness of the dynamics of supervisory and managerial decision-making. Emphasis will be placed on team building, crisis management, social and environmental responsibility, developing and communicating a vision, developing a full set of managerial and leadership skills. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair, MAN4894

Course User Fee: \$70.00

MAN4330 Compensation Management 3.000 credits

This course is a study of the strategic use of compensation system for the purposes of attracting, retaining, and motivating a competitive workforce. The student will address topics such as designing compensation systems, bases for pay, employee benefit programs, laws affecting compensation practices, and compensation challenges for various employee groups. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4335 Employee Benefit Planning 3.000 credits

This course focuses on the both wage and nonwage related benefits made available to employees by the firm and various related social and governmental programs. The student will examine various state and federal laws regarding employee benefits planning.

Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4720 Strategic Management Decision Making 3.000 credits

The student will learn the designing, planning, and implementation of strategic decision making in a business organization by identifying problems and designing possible solutions, formulating plans, goals, and feedback mechanisms. Through case studies, the student will conduct internal and external assessments to analyze effective strategic choices for companies.

Lecture Hours: 48.000

Course User Fee: \$70.00

**FIN3400 Finance for Non-Financial Managers
3.000 credits**

The students will learn to apply their financial skills and decision-making abilities to address financial issues in a business environment. They will learn how financial markets influence their decisions and the role of financial intermediaries in these markets.

Emphasis will be placed on financial and capital budgeting, working capital management, short and long term financing, valuation of the firm, and overall capital structure. The course will require the student to apply the time value of money through the use of present and future value scenarios. This course must be met with a grade of "C" or higher. (3 hour lecture)

Lecture Hours: 48.000

Course User Fee: \$60.00

**MAN3322 Human Resources Information Systems
3.000 credits**

This course examines the role of human resources information system (HRIS) in today's organizations and human resources departments. The student will address topics such as human resource information systems design, acquisition, and implementation. The role of these systems in talent acquisition and management is also examined. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4113 Managing in a Multifaceted Environment 3.000 credits

Managing in a Multifaceted Environment is a thorough course that offers a management perspective on making the most of the various qualities of today's workforce. This course explores the value that comes from human capital and other distinguishing factors in the modern workplace. Furthermore, it delves into new leadership styles as solutions for effectively managing and leading people. Students will gain insights into utilizing the strengths of the workforce and creating a dynamic work environment.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4352 Effective Employee Training 3.000 credits

This course focuses on professional development activities as performed by human resources specialists or organizational specialists. The student will analyze the benefits of employee training, establish employee development programs, and address the theories, issues, practices and problems regarding employee training and development.

Prerequisites: MAN3025, MAN3301

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4900 Capstone Project in Supervision and Management 3.000 credits

Student will apply and integrate the knowledge and skills learned throughout the program by completing a capstone project. Student will identify a business opportunity; conduct an industry feasibility assessment and operational mapping; evaluate market condition; construct a marketing plan, and formulate a financial plan for implementing and managing an existing or new business venture; and evaluate the role of ethic, social, and environmental responsibility within a business.

Lecture Hours: 48.000

Course User Fee: \$90.00

MAN4941 Management Internship 3.000 credits

The student will learn management techniques by becoming an employee or intern (on a paid or unpaid basis) at either a not-for-profit or for-profit organization. The student will be required to work at least 144 hours required by the state to earn the credit for the internship. The student will work with their MDC Faculty and Supervising Employer to establish a set of assignments / learning goals that will be achieved during the semester. This course must be met with a grade of "C" or higher.

Prerequisite: Senior Status or Departmental Permission.

Other Hours: 144.000

Prerequisite: Senior status and permission of department chair., Pre/Corequisite: MAN4900

Course User Fee: \$45.00

Program Description

Public Safety Management - Crime Scene Investigation Concentration

Bachelor of Applied Science | Code: P9106 | 120 credits

The four-year Bachelor of Applied Science degree is a workforce-driven baccalaureate degree in Public Safety Management designed to provide education and training, resulting in immediate employment possibilities for students in numerous careers in Public Safety. This comprehensive curriculum emphasizes critical thinking, analytical, written and oral communications, as well as research skills designed to prepare today's learners for entry-level, mid-management, and upper level supervisory positions within Public Safety agencies such as the FBI, DEA, ICE, U.S. Marshals Service, U.S. Secret Service and others.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1102	English Composition 2	3.0
PHI 2600	Introduction to Ethics	3.0
CCJ 3032	Crime and the Media	3.0
CJL 4133	Criminal Evidence	3.0

Semester 2

Units: 12.0

CCJ 3663	Female Crime and Delinquency	3.0
CCJ 3666	Victimology	3.0
CJE 4648	Crime Scene Safety	3.0

STA 2023	Statistical Methods	3.0
Semester 3		Units: 12.0
CCJ 3700	Methods of Research in Criminal Justice	3.0
CJE 4647	Advanced Crime Scene Investigations	3.0
CJE 4650	Advanced Crime Scene Investigations	3.0
<i>Natural Science (GE)</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
Semester 4		Units: 12.0
CCJ 4054	Ethics in The Criminal Justice System	3.0
CCJ 4450	Criminal Justice Administration	3.0
CCJ 4641	Organized Crime	3.0
CJE 4675	Modern Fingerprint Technology	3.0
Semester 5		Units: 12.0
CCJ 4651	Drugs and Crime	3.0
CJE 3110	Law Enforcement Systems	3.0
CJE 4615	Advanced Criminal Investigations	3.0
CJE 4668	Computer Crime	3.0

Total: 60.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

CCJ3032 Crime and the Media 3.000 credits

An examination of the inter-relationship among the mass media, crime, and criminal justice. Includes media and the social construction of crime and criminal justice; media effects on attitudes toward crime and justice; and media as a cause of crime. Lecture Hours: 48.000

CJL4133 Criminal Evidence 3.000 credits

A study of evidentiary principles and rules of evidence, and their application in a courtroom setting.-

Lecture Hours: 48.000

CCJ3663 Female Crime and Delinquency 3.000 credits

A study of females in society and the criminal justice system. Includes the female delinquent, females as criminals, females as victims, and the impact of females as professionals in the Criminal Justice System. Prerequisites: CCJ1191, CJJ2002.

Lecture Hours: 48.000

Prerequisite: CCJ1191, CJJ2002

CCJ3666 Victimology 3.000 credits

A comprehensive study of victimization; analysis of contemporary victim-assistance and victim compensation programs and related research; review of the historical importance of victim restitution as a basis for punitive criminal law.

Lecture Hours: 48.000

CJE4648 Crime Scene Safety 3.000 credits

A study of how to properly handle crime scenes and hazardous crime scenes relative to various hazardous materials, to include chemical and biological.

Lecture Hours: 48.000

Course User Fee: \$71.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CCJ3700 Methods of Research in Criminal Justice 3.000 credits

Evaluates the application of research methodologies as applied to the study of Public Safety Management.

Prerequisite: STA2023

Lecture Hours: 48.000

Prerequisite: STA2023

CJE4647 Advanced Crime Scene Investigations 3.000 credits

A study of advanced search techniques, crime scenes reconstruction, computer sketching, laser mapping. DNA evidence, trajectory, and blood spatter evidence.

Lecture Hours: 48.000

Prerequisite: CJE2600

Course User Fee: \$59.00

CJE4650 Advanced Crime Scene Investigations 3.000 credits

A study of advanced search techniques, crime scenes reconstruction, computer sketching, laser mapping. DNA evidence, trajectory, and blood spatter evidence.

Lecture Hours: 48.000

Course User Fee: \$105.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

CCJ4054 Ethics in The Criminal Justice System 3.000 credits

An in depth study of moral, ethical, legal, and professional issues and dilemmas facing individuals and organizations within the Criminal Justice systems. Prerequisites: CCJ2053 or PHI2600.

Lecture Hours: 48.000

Prerequisite: CCJ2053, PHI2600

CCJ4450 Criminal Justice Administration 3.000 credits

An analysis of leadership styles, management principles, supervisory techniques, policies and procedures within Law Enforcement agencies.

Prerequisite: CCJ1020.

Lecture Hours: 48.000

Prerequisite: CCJ1020

CCJ4641 Organized Crime 3.000 credits

An analysis of organized crime in today's society, as well as, past, present, and future perspectives of the topic.

Lecture Hours: 48.000

CJE4675 Modern Fingerprint Technology 3.000 credits

A study of the detection, preservation, and removal of fingerprint evidence pertaining to latent, patent, and plastic prints.

Lecture Hours: 48.000

Prerequisite: CJE2600

Course User Fee: \$32.00

CCJ4651 Drugs and Crime 3.000 credits

An analysis of the interrelationship among drug usage, crime and the criminal justice system.

Lecture Hours: 48.000

CJE3110 Law Enforcement Systems 3.000 credits

An analysis of the different law enforcement systems in Criminal Justice. Focuses on the different law agencies and their mission at the local, state, and federal levels.

Lecture Hours: 48.000

CJE4615 Advanced Criminal Investigations 3.000 credits

The understanding, interpretation, and application of criminal investigative procedures in the U.S., based upon constitutional issues and legal precedent. (3 hr. Lecture)

Lecture Hours: 48.000

Prerequisite: CJT 2100

CJE4668 Computer Crime 3.000 credits

Synthesizes knowledge of crime elements, legal issues, investigative techniques, and computer skills used in the prevention and investigation of computer-generated crime.

Lecture Hours: 48.000

Program Description

Public Safety Management - Criminal Justice Concentration

Bachelor of Applied Science | Code: P9110 | 120 credits

The four-year Bachelor of Applied Science degree is a workforce-driven baccalaureate degree in Public Safety Management designed to provide education and training, resulting in immediate employment possibilities for students in numerous careers in Public Safety. This comprehensive curriculum emphasizes critical thinking, analytical, written and oral communications, as well as research skills designed to prepare today's learners for entry-level, mid-management, and upper level supervisory positions within Public Safety agencies such as the FBI, DEA, ICE, U.S. Marshals Service, U.S. Secret Service and others.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1102	English Composition 2	3.0
PHI 2600	Introduction to Ethics	3.0
CCJ 3032	Crime and the Media	3.0
CCJ 4641	Organized Crime	3.0

Semester 2

Units: 12.0

CCJ 3663	Female Crime and Delinquency	3.0
CCJ 3666	Victimology	3.0
CJC 4163	Advanced Probation & Parole	3.0
STA 2023	Statistical Methods	3.0

Semester 3

Units: 12.0

CCJ 3700	Methods of Research in Criminal Justice	3.0
CJE 3115	Police and Society	3.0
DSC 4012	Terrorism	3.0
<i>Natural Science (GE)</i>		3.0
BOT 1010	Botany	3.0
OR		
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0

Semester 4

Units: 12.0

CCJ 4054	Ethics in The Criminal Justice System	3.0
CCJ 4450	Criminal Justice Administration	3.0
CCJ 4641	Organized Crime	3.0
CJE 4615	Advanced Criminal Investigations	3.0

Semester 5

Units: 12.0

CCJ 4651	Drugs and Crime	3.0
CCJ 4660	Crime, Violence, and Schools	3.0
CJE 3110	Law Enforcement Systems	3.0
CJE 4668	Computer Crime	3.0

Total: 60.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.
Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

CCJ3032 Crime and the Media 3.000 credits

An examination of the inter-relationship among the mass media, crime, and criminal justice. Includes media and the social construction of crime and criminal justice; media effects on attitudes toward crime and justice; and media as a cause of crime.
Lecture Hours: 48.000

CCJ4641 Organized Crime 3.000 credits

An analysis of organized crime in today's society, as well as, past, present, and future perspectives of the topic.

Lecture Hours: 48.000

CCJ3663 Female Crime and Delinquency 3.000 credits

A study of females in society and the criminal justice system. Includes the female delinquent, females as criminals, females as victims, and the impact of females as professionals in the Criminal Justice System. Prerequisites: CCJ1191, CJJ2002.

Lecture Hours: 48.000

Prerequisite: CCJ1191, CJJ2002

CCJ3666 Victimology 3.000 credits

A comprehensive study of victimization; analysis of contemporary victim-assistance and victim compensation programs and related research; review of the historical importance of victim restitution as a basis for punitive criminal law.

Lecture Hours: 48.000

CJC4163 Advanced Probation & Parole 3.000 credits

A study of the process in which a convicted person can be released into society by means of probation or parole.

Lecture Hours: 48.000

Prerequisite: CJC1162

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CCJ3700 Methods of Research in Criminal Justice 3.000 credits

Evaluates the application of research methodologies as applied to the study of Public Safety Management.

Prerequisite: STA2023

Lecture Hours: 48.000

Prerequisite: STA2023

CJE3115 Police and Society 3.000 credits

Identifies police roles and philosophies, the nature of police work, community policing, and the debates pertaining to police discretion, community relations, and police misconduct.

Lecture Hours: 48.000

DSC4012 Terrorism 3.000 credits

A study of domestic and international terrorism, using current events and past incidents for analysis, to include the events, the responses, and the outcomes.

Lecture Hours: 48.000

BOT1010 Botany 3.000 credits

This course is a survey of the plant kingdom based on a detailed study of the morphology, anatomy and physiology of selected representative specimens.

Corequisite: BOT1010L

Lecture Hours: 48.000

Corequisite: BOT1010L

Course User Fee: \$8.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

CCJ4054 Ethics in The Criminal Justice System 3.000 credits

An in depth study of moral, ethical, legal, and professional issues and dilemmas facing individuals and organizations within the Criminal Justice systems. Prerequisites: CCJ2053 or PHI2600.

Lecture Hours: 48.000

Prerequisite: CCJ2053, PHI2600

CCJ4450 Criminal Justice Administration 3.000 credits

An analysis of leadership styles, management principles, supervisory techniques, policies and procedures within Law Enforcement agencies.

Prerequisite: CCJ1020.

Lecture Hours: 48.000

Prerequisite: CCJ1020

CCJ4641 Organized Crime 3.000 credits

An analysis of organized crime in today's society, as well as, past, present, and future perspectives of the topic.

Lecture Hours: 48.000

CJE4615 Advanced Criminal Investigations 3.000 credits

The understanding, interpretation, and application of criminal investigative procedures in the U.S., based upon constitutional issues and legal precedent. (3 hr. Lecture)

Lecture Hours: 48.000

Prerequisite: CJT 2100

CCJ4651 Drugs and Crime 3.000 credits

An analysis of the interrelationship among drug usage, crime and the criminal justice system.

Lecture Hours: 48.000

CCJ4660 Crime, Violence, and Schools 3.000 credits

An examination of comprehensive and proven theoretical models of explaining, predicting, and preventing school-based violence.

Lecture Hours: 48.000

CJE3110 Law Enforcement Systems 3.000 credits

An analysis of the different law enforcement systems in Criminal Justice. Focuses on the different law agencies and their mission at the local, state, and federal levels.

Lecture Hours: 48.000

CJE4668 Computer Crime 3.000 credits

Synthesizes knowledge of crime elements, legal issues, investigative techniques, and computer skills used in the prevention and investigation of computer-generated crime.

Lecture Hours: 48.000

Program Description

Supply Chain Management

Bachelor of Applied Science | Code: P9300 | 120 credits

Supply chain management is an interdisciplinary field that emphasizes cross-functional integration of diverse business functions, facilities, and activities and seeks to manage those activities to enhance a company's competitive advantage. The Bachelor of Applied Science with a major in Supply Chain Management (BAS-SCM) is designed to provide knowledge of the supply chain while exploring the risks, operations, logistics, economics, regulatory issues, change management, forecasting, resource allocation, production planning, inventory management, customer delivery, after-sales support and service and other functions basic to business.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1102	English Composition 2	3.0
GEB 3358	International Negotiations and Transactions	3.0
MAN 3506	Operations Management	3.0
MAN 3562	Purchasing, Inventory and Warehouse Management	3.0
Semester 2		Units: 12.0
FIN 3403	Financial Management	3.0
MAN 3583	Project Management	3.0
SPC 2608	Introduction to Public Speaking	3.0
<i>Humanities (GE)</i>		3.0
HUM 1020	Humanities	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
Semester 3		Units: 12.0
BUL 4461	Law of International Trade	3.0
MAN 4597	Global Supply Chain Management	3.0
<i>Natural Science (GE)</i>		3.0
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
GLY 1010	Physical Geology	3.0
OR		
MET 1010	Introduction to Weather	3.0
<i>Upper Division Elective</i>		3.0
GEB 4363	Import Export Management	3.0
OR		
MAN 3504	Production Operations and Logistics Management	3.0
OR		
MAN 3786	Sustainable Enterprise Planning	3.0
OR		
MAN 4593	Supply Chain Management Theory & Methodology	2.0
OR		
TRA 3034	Transportation and Traffic Management	3.0
OR		
TRA 3132	Purchasing and Inventory Management	3.0
OR		
TRA 4234	Warehouse Management	3.0
Semester 4		Units: 12.0
MAN 4520	Quality Management	3.0
MAN 4523	Production Information Systems	3.0
MAR 4203	Supply Chain Marketing	3.0

<i>Upper Division Elective</i>		3.0
GEB 4363	Import Export Management	3.0
OR		
MAN 3504	Production Operations and Logistics Management	3.0
OR		
MAN 3786	Sustainable Enterprise Planning	3.0
OR		
MAN 4593	Supply Chain Management Theory & Methodology	2.0
OR		
TRA 3034	Transportation and Traffic Management	3.0
OR		
TRA 3132	Purchasing and Inventory Management	3.0
OR		
TRA 4234	Warehouse Management	3.0

Semester 5

Units: 12.0

ECO 2013	Principles of Economics (Macro)	3.0
MAN 3554	Safety and Risk Management	1.0
MAN 4940	Field Study and Research	2.0
General Education Elective*		3.0
<i>Upper Division Elective</i>		3.0
GEB 4363	Import Export Management	3.0
OR		
MAN 3504	Production Operations and Logistics Management	3.0
OR		
MAN 3786	Sustainable Enterprise Planning	3.0
OR		
MAN 4552	Supply Chain Analytics & Decision Making	3.0
OR		
TRA 3034	Transportation and Traffic Management	3.0
OR		
TRA 3132	Purchasing and Inventory Management	3.0
OR		
TRA 4234	Warehouse Management	3.0

Select 3 credits from the following course options: ECO, MAC*, MET*, STA 2023 or Consult with an advisor for appropriate selection

Total: 60.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

GEB3358 International Negotiations and Transactions 3.000 credits

This course presents business negotiations in selected regions of the world. Students will learn the skills of negotiation by analyzing international business cases and developing solutions to situations. Topics include overseas market research using both domestic and international sources and the application of theoretical and practical business knowledge to foreign situations and trade negotiations. The importance of culture, language, and values in international negotiations is emphasized.

Lecture Hours: 48.000

Prerequisite: TRA1154

Course User Fee: \$69.00

MAN3506 Operations Management 3.000 credits

The course emphasizes the application of operational decision-making techniques to improve process, productivity, and the effective utilization of resources within organizations. Students will learn to recognize the trade-offs associated with operations management decisions and their effect on resource allocation. Topics include production processes, operations strategies, quantitative techniques, quality, performance, capacity planning, efficiency, forecasting, resource management, statistical process control, project management, and supply chains.

Prerequisite: QMB2100 or STA2023.

Lecture Hours: 48.000

Prerequisite: QMB2100 OR STA2023

MAN3562 Purchasing, Inventory and Warehouse Management 3.000 credits

This course explores the integration of production operations and logistics management. It will also examine purchasing, Inventory and Warehouse Management concepts and theory. Special emphasis is placed on the relationships that exist between operations and the supply chain to include operations and supply chain strategies, business processes, project management, product design and development, inventory management and effective warehouse and stores management.

Lecture Hours: 48.000

Course User Fee: \$62.76

FIN3403 Financial Management 3.000 credits

Students will learn the sources and uses of funds as they relate to financial decision making in the corporate form of enterprise. Emphasis is placed on working capital management, capital budgeting techniques, short and long term financing, and capital structure and the value of the firm. This course emphasizes the use of formulas and spreadsheets through Excel.

Lecture Hours: 48.000

Prerequisite: ACG2071, ACG2071L, QMB2100

MAN3583 Project Management 3.000 credits

Students will learn project management concepts and theory, including attributes for a project life cycle, global project management, benefits of project management, project management environments, planning and managing risk, project planning controls, and terminating a project.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

BUL4461 Law of International Trade 3.000 credits

Students will learn to interpret laws of international trade, licensing, and customer's regulation. This course also addresses U.S. customs regulations, classification of merchandise, application of tariff rules, duty free treatment, importing and exporting, liquidation inspection, search and seizure, and fines and penalties. In addition, the course will review the general rules and how to interpret the harmonized tariff schedule. Prerequisites: MAN2021 and GEB3358.

Lecture Hours: 48.000

Prerequisite: GEB3358

MAN4597 Global Supply Chain Management 3.000 credits

This course presents an overview of the management of sourcing, operations, and distribution processes along a supply chain in domestic and international markets. Students will learn how firms gain a competitive advantage through supply chain activities. Topics include: supply chain network design, purchasing, forecasting, inventory management, globalization and outsourcing, logistics, and information technology.

Lecture Hours: 48.000

Prerequisite: MAN2021, MAN3506

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

MET1010 Introduction to Weather 3.000 credits

An introduction to fundamentals of weather and their impact on human activities. Topics include temperature, humidity, clouds, precipitation, air masses, fronts, and storms. Emphasis is on understanding how these processes take place and their results.

Lecture Hours: 48.000

Pre/Corequisite: PSC1515

Course User Fee: \$10.00

GEB4363 Import Export Management 3.000 credits

Students will learn the functions and range of traffic management services performed by freight forwarders, including changes in governmental restrictions, rules, and regulations applicable to different countries, ports, and trade routes. Students will also learn the documentation needed to facilitate and coordinate the movement of goods in international trade and supply chain management systems. Prerequisites: MAN 2021 and TRA 2010.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA2010

Course User Fee: \$69.00

MAN3504 Production Operations and Logistics Management 3.000 credits

This course examines the integration of production operations and logistics management and how they enable an organization to compete successfully in business. Students will learn the relationships that exist between operations and the supply chain, including operations and supply chain strategies, business processes, project management, product design and development, and inventory management. This course uses case study methodology. Prerequisite: TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA1154 with a minimum grade of C

MAN3786 Sustainable Enterprise Planning 3.000 credits

Students will learn the assessment tools, design and construction considerations, and operating planning requirements for sustainable enterprises. Students will also learn the ecological and economic benefits of sustainability practices, including construction, operations, supply chain decisions, recycling, reusing, and reconditioning to preserve the environment by increasing revenues. Best practices, case studies, evolving trends and experimental efforts are also covered.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4593 Supply Chain Management Theory & Methodology 2.000 credits

This course presents a range of advanced topics in integrated logistics and supply chain management. Students will learn new theoretical and methodological developments in the field of supply chain. Specific topics vary depending upon current industry developments and will incorporate the use of speakers in the classroom. Prerequisites:

MAN2021, MAN3583, MAN4523.

Lecture Hours: 32.000

Prerequisite: MAN2021, MAN3583, MAN4523

TRA3034 Transportation and Traffic Management 3.000 credits

This course covers developments leading to national and federal regulations, division of territories, official descriptions, etc. Students will learn the scope of authority of territorial associations, factors controlling traffic flows, basic governing classification rules, principals of freight rates and tariffs, and elements of rate making. Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA3132 Purchasing and Inventory Management 3.000 credits

This course provides a comprehensive introduction to the purchasing/procurement and supply chain management field. Students will learn purchasing and supply chain issues in a variety of settings, from process industries to high-tech manufacturing and services as well as public institutions. Emphasis is on the purchasing process as it relates to such topics as inventory control procedures, price/cost analysis, laws and ethics, negotiations, vendor selection and the development of vendor relationships.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA4234 Warehouse Management 3.000 credits

Students will learn warehousing functions, facility operations, and operational productivity improvements and measurements with the inclusion of concepts from marketing, finance, statistics, operations management, and human resources. The course presents an integrated business approach to the detailed operational aspects of logistics facilities such as warehouse and distribution centers. The course will use real-world warehouse data to design the layout and operations requirements for a local warehousing facility and to identify the operational equipment used in warehousing. Prerequisites: MAN2021, MAN3065, QMB2100.

Lecture Hours: 48.000

Prerequisite: MAN2021 AND MAN3065 AND QMB2100

MAN4520 Quality Management 3.000 credits

This course provides an understanding of various theories of quality management (QM). Students will learn how organizations can develop excellence through the adoption of continuous improvement and process management. The course analyzes and uses various process management techniques, continuous improvement tools, and strategies to improve quality. The conceptual and analytical skills acquired in this course enable students to provide leadership in shaping a culture for quality within an organization and determining the effectiveness of quality initiatives such as Total QM, Six Sigma, Process Capability, Process Control, and Customer Relationships.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4523 Production Information Systems 3.000 credits

This course presents the fundamental aspects of computer technology required by the systems that provide data to, and derive information from, production in manufacturing. Students will learn the techniques to organize, store, manipulate data, report, derive and analyze production information, basics networking used in production, as well as various forms of information systems.

Lecture Hours: 48.000

MAR4203 Supply Chain Marketing 3.000 credits

Students will learn the management of traditional and emerging marketing channels, with emphasis on legal, economic, and ethical considerations in wholesale and retail inventory control, raw goods, finished product transportation and relationship management. Prerequisite: MAN3506. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MAN3506

GEB4363 Import Export Management 3.000 credits

Students will learn the functions and range of traffic management services performed by freight forwarders, including changes in governmental restrictions, rules, and regulations applicable to different countries, ports, and trade routes. Students will also learn the documentation needed to facilitate and coordinate the movement of goods in international trade and supply chain management systems. Prerequisites: MAN 2021 and TRA 2010.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA2010

Course User Fee: \$69.00

MAN3504 Production Operations and Logistics Management 3.000 credits

This course examines the integration of production operations and logistics management and how they enable an organization to compete successfully in business. Students will learn the relationships that exist between operations and the supply chain, including operations and supply chain strategies, business processes, project management, product design and development, and inventory management. This course uses case study methodology. Prerequisite: TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA1154 with a minimum grade of C

MAN3786 Sustainable Enterprise Planning 3.000 credits

Students will learn the assessment tools, design and construction considerations, and operating planning requirements for sustainable enterprises. Students will also learn the ecological and economic benefits of sustainability practices, including construction, operations, supply chain decisions, recycling, reusing, and reconditioning to preserve the environment by increasing revenues. Best practices, case studies, evolving trends and experimental efforts are also covered.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4593 Supply Chain Management Theory & Methodology 2.000 credits

This course presents a range of advanced topics in integrated logistics and supply chain management. Students will learn new theoretical and methodological developments in the field of supply chain. Specific topics vary depending upon current industry developments and will incorporate the use of speakers in the classroom. Prerequisites:

MAN2021, MAN3583, MAN4523.

Lecture Hours: 32.000

Prerequisite: MAN2021, MAN3583, MAN4523

TRA3034 Transportation and Traffic Management 3.000 credits

This course covers developments leading to national and federal regulations, division of territories, official descriptions, etc. Students will learn the scope of authority of territorial associations, factors controlling traffic flows, basic governing classification rules, principals of freight rates and tariffs, and elements of rate making. Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA3132 Purchasing and Inventory Management 3.000 credits

This course provides a comprehensive introduction to the purchasing/procurement and supply chain management field. Students will learn purchasing and supply chain issues in a variety of settings, from process industries to high-tech manufacturing and services as well as public institutions. Emphasis is on the purchasing process as it relates to such topics as inventory control procedures, price/cost analysis, laws and ethics, negotiations, vendor selection and the development of vendor relationships.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA4234 Warehouse Management 3.000 credits

Students will learn warehousing functions, facility operations, and operational productivity improvements and measurements with the inclusion of concepts from marketing, finance, statistics, operations management, and human resources. The course presents an integrated business approach to the detailed operational aspects of logistics facilities such as warehouse and distribution centers. The course will use real-world warehouse data to design the layout and operations requirements for a local warehousing facility and to identify the operational equipment used in warehousing. Prerequisites: MAN2021, MAN3065, QMB2100.

Lecture Hours: 48.000

Prerequisite: MAN2021 AND MAN3065 AND QMB2100

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

MAN3554 Safety and Risk Management 1.000 credits

This course will focus on safety and risk management with emphasis on how it applies to supply chain management. Students will learn the processes used to characterize and manage risk, as well as maintain a safe operating environment with the protection of personnel, assets, and services. Hazard characteristics of products, appropriate modes of transport, carrier selection and qualifications, packaging and container specifications, driver training requirements, financial issues, security precautions, warehousing risk, OSHA, hazardous materials handling, and customer assessment will be addressed in this course.

Lecture Hours: 16.000

MAN4940 Field Study and Research 2.000 credits

Students will learn to apply information and skills studied in core and concentration program courses through various special projects, field research, or internships designed for cogitative learning in a student-centered manner which requires the command, analysis, and synthesis of knowledge and skills.

Lecture Hours: 32.000

Prerequisite: MAN3583, TRA1154

GEB4363 Import Export Management 3.000 credits

Students will learn the functions and range of traffic management services performed by freight forwarders, including changes in governmental restrictions, rules, and regulations applicable to different countries, ports, and trade routes. Students will also learn the documentation needed to facilitate and coordinate the movement of goods in international trade and supply chain management systems. Prerequisites: MAN 2021 and TRA 2010.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA2010

Course User Fee: \$69.00

MAN3504 Production Operations and Logistics Management 3.000 credits

This course examines the integration of production operations and logistics management and how they enable an organization to compete successfully in business. Students will learn the relationships that exist between operations and the supply chain, including operations and supply chain strategies, business processes, project management, product design and development, and inventory management. This course uses case study methodology. Prerequisite: TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA1154 with a minimum grade of C

MAN3786 Sustainable Enterprise Planning 3.000 credits

Students will learn the assessment tools, design and construction considerations, and operating planning requirements for sustainable enterprises. Students will also learn the ecological and economic benefits of sustainability practices, including construction, operations, supply chain decisions, recycling, reusing, and reconditioning to preserve the environment by increasing revenues. Best practices, case studies, evolving trends and experimental efforts are also covered.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4552 Supply Chain Analytics & Decision Making 3.000 credits

Through this course students will focus on the science of learning from data to support decision making. The objective of this course is to explore and apply basic statistical concepts and procedures that are used to collect, analyze, summarize, and report data in a supply chain management role. Course topics covered include methods for: collecting, analyzing, summarizing data; making statistical inferences about populations and exploring the relationship between variables. The course introduces analytical tools and techniques that provide a cause and effect understanding linking operational plans with corporate objectives.

Lecture Hours: 48.000

Course User Fee: \$62.76

TRA3034 Transportation and Traffic Management 3.000 credits

This course covers developments leading to national and federal regulations, division of territories, official descriptions, etc. Students will learn the scope of authority of territorial associations, factors controlling traffic flows, basic governing classification rules, principals of freight rates and tariffs, and elements of rate making. Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA3132 Purchasing and Inventory Management 3.000 credits

This course provides a comprehensive introduction to the purchasing/procurement and supply chain management field. Students will learn purchasing and supply chain issues in a variety of settings, from process industries to high-tech manufacturing and services as well as public institutions. Emphasis is on the purchasing process as it relates to such topics as inventory control procedures, price/cost analysis, laws and ethics, negotiations, vendor selection and the development of vendor relationships.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA4234 Warehouse Management 3.000 credits

Students will learn warehousing functions, facility operations, and operational productivity improvements and measurements with the inclusion of concepts from marketing, finance, statistics, operations management, and human resources. The course presents an integrated business approach to the detailed operational aspects of logistics facilities such as warehouse and distribution centers. The course will use real-world warehouse data to design the layout and operations requirements for a local warehousing facility and to identify the operational equipment used in warehousing. Prerequisites: MAN2021, MAN3065, QMB2100.

Lecture Hours: 48.000

Prerequisite: MAN2021 AND MAN3065 AND QMB2100

Program Description

Supply Chain Management – Procurement Management Concentration

Bachelor of Applied Science | Code: P9301 | 120 credits

Supply chain management is an interdisciplinary field that emphasizes cross-functional integration of diverse business functions, facilities, and activities and seeks to manage those activities to enhance a company's competitive advantage. The Bachelor of Applied Science in Supply Chain Management (BAS-SCM) is designed to provide knowledge of the supply chain while exploring the risks, operations, logistics, economics, regulatory issues, change management, forecasting, resource allocation, production planning, inventory management, customer delivery, after-sales support and service and other functions basic to business.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

MAN 3506	Operations Management	3.0
MAN 3562	Purchasing, Inventory and Warehouse Management	3.0
MAN 4520	Quality Management	3.0
MAN 4523	Production Information Systems	3.0

Semester 2		Units: 12.0
FIN 3403	Financial Management	3.0
MAN 3583	Project Management	3.0
MAN 4570	Purchasing for Industry	3.0
Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, CHI*, FIN*, FRE*, FRW*, GEB*, GER*, HBR*, ISM*, ITA*, JPN*, MAC*, MAN*, MAR*, PET*, POR*, PUR*, QMB*, SPM*, SPN*, STA*, TAX*, TRA*

Semester 3		Units: 12.0
BUL 4461	Law of International Trade	3.0
MAN 3504	Production Operations and Logistics Management	3.0
MAN 4597	Global Supply Chain Management	3.0
<i>Upper Division Elective</i>		3.0
GEB 4363	Import Export Management	3.0
OR		
MAN 3786	Sustainable Enterprise Planning	3.0
OR		
MAN 4593	Supply Chain Management Theory & Methodology	2.0
OR		
TRA 3034	Transportation and Traffic Management	3.0
OR		
TRA 3132	Purchasing and Inventory Management	3.0
OR		
TRA 4234	Warehouse Management	3.0

Semester 4		Units: 12.0
GEB 3358	International Negotiations and Transactions	3.0
MAN 3577	Procurement for Major Projects	3.0
MAN 3578	Global Procurement Management	3.0
Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, CHI*, FIN*, FRE*, FRW*, GEB*, GER*, HBR*, ISM*, ITA*, JPN*, MAC*, MAN*, MAR*, PET*, POR*, PUR*, QMB*, SPM*, SPN*, STA*, TAX*, TRA*

Semester 5		Units: 12.0
MAN 3554	Safety and Risk Management	1.0
MAN 4940	Field Study and Research	2.0
MAR 4203	Supply Chain Marketing	3.0
<i>Upper Division Elective</i>		6.0
GEB 4363	Import Export Management	3.0
OR		
MAN 3786	Sustainable Enterprise Planning	3.0
OR		
MAN 4593	Supply Chain Management Theory & Methodology	2.0
OR		
TRA 3034	Transportation and Traffic Management	3.0
OR		

TRA 3132	Purchasing and Inventory Management	3.0
OR		
TRA 4234	Warehouse Management	3.0

Total: 60.0

MAN3506 Operations Management 3.000 credits

The course emphasizes the application of operational decision-making techniques to improve process, productivity, and the effective utilization of resources within organizations. Students will learn to recognize the trade-offs associated with operations management decisions and their effect on resource allocation. Topics include production processes, operations strategies, quantitative techniques, quality, performance, capacity planning, efficiency, forecasting, resource management, statistical process control, project management, and supply chains. Prerequisite: QMB2100 or STA2023.

Lecture Hours: 48.000

Prerequisite: QMB2100 OR STA2023

MAN3562 Purchasing, Inventory and Warehouse Management 3.000 credits

This course explores the integration of production operations and logistics management. It will also examine purchasing, Inventory and Warehouse Management concepts and theory. Special emphasis is placed on the relationships that exist between operations and the supply chain to include operations and supply chain strategies, business processes, project management, product design and development, inventory management and effective warehouse and stores management.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4520 Quality Management 3.000 credits

This course provides an understanding of various theories of quality management (QM). Students will learn how organizations can develop excellence through the adoption of continuous improvement and process management. The course analyzes and uses various process management techniques, continuous improvement tools, and strategies to improve quality. The conceptual and analytical skills acquired in this course enable students to provide leadership in shaping a culture for quality within an organization and determining the effectiveness of quality initiatives such as Total QM, Six Sigma, Process Capability, Process Control, and Customer Relationships.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4523 Production Information Systems 3.000 credits

This course presents the fundamental aspects of computer technology required by the systems that provide data to, and derive information from, production in manufacturing. Students will learn the techniques to organize, store, manipulate data, report, derive and analyze production information, basics networking used in production, as well as various forms of information systems.

Lecture Hours: 48.000

FIN3403 Financial Management 3.000 credits

Students will learn the sources and uses of funds as they relate to financial decision making in the corporate form of enterprise. Emphasis is placed on working capital management, capital budgeting techniques, short and long term financing, and capital structure and the value of the firm. This course emphasizes the use of formulas and spreadsheets through Excel.

Lecture Hours: 48.000

Prerequisite: ACG2071, ACG2071L, QMB2100

MAN3583 Project Management 3.000 credits

Students will learn project management concepts and theory, including attributes for a project life cycle, global project management, benefits of project management, project management environments, planning and managing risk, project planning controls, and terminating a project.

Lecture Hours: 48.000

MAN4570 Purchasing for Industry 3.000 credits

Students will learn purchasing for industry concepts and theory, including negotiation skills: mastering the art of deal making strategies, contracts: bidding, evaluation, negotiation and award, purchasing management, tendering and supplier selection, vendor qualification: managing performance and contract compliance.

Lecture Hours: 48.000

Course User Fee: \$62.76

BUL4461 Law of International Trade 3.000 credits

Students will learn to interpret laws of international trade, licensing, and customer's regulation. This course also addresses U.S. customs regulations, classification of merchandise, application of tariff rules, duty free treatment, importing and exporting, liquidation inspection, search and seizure, and fines and penalties. In addition, the course will review the general rules and how to interpret the harmonized tariff schedule. Prerequisites: MAN2021 and GEB3358.

Lecture Hours: 48.000

Prerequisite: GEB3358

MAN3504 Production Operations and Logistics Management 3.000 credits

This course examines the integration of production operations and logistics management and how they enable an organization to compete successfully in business. Students will learn the relationships that exist between operations and the supply chain, including operations and supply chain strategies, business processes, project management, product design and development, and inventory management. This course uses case study methodology. Prerequisite: TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA1154 with a minimum grade of C

MAN4597 Global Supply Chain Management 3.000 credits

This course presents an overview of the management of sourcing, operations, and distribution processes along a supply chain in domestic and international markets. Students will learn how firms gain a competitive advantage through supply chain activities. Topics include: supply chain network design, purchasing, forecasting, inventory management, globalization and outsourcing, logistics, and information technology.

Lecture Hours: 48.000

Prerequisite: MAN2021, MAN3506

GEB4363 Import Export Management 3.000 credits

Students will learn the functions and range of traffic management services performed by freight forwarders, including changes in governmental restrictions, rules, and regulations applicable to different countries, ports, and trade routes. Students will also learn the documentation needed to facilitate and coordinate the movement of goods in international trade and supply chain management systems. Prerequisites: MAN 2021 and TRA 2010.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA2010

Course User Fee: \$69.00

MAN3786 Sustainable Enterprise Planning 3.000 credits

Students will learn the assessment tools, design and construction considerations, and operating planning requirements for sustainable enterprises. Students will also learn the ecological and economic benefits of sustainability practices, including construction, operations, supply chain decisions, recycling, reusing, and reconditioning to preserve the environment by increasing revenues. Best practices, case studies, evolving trends and experimental efforts are also covered.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4593 Supply Chain Management Theory & Methodology 2.000 credits

This course presents a range of advanced topics in integrated logistics and supply chain management. Students will learn new theoretical and methodological developments in the field of supply chain. Specific topics vary depending upon current industry developments and will incorporate the use of speakers in the classroom. Prerequisites: MAN2021, MAN3583, MAN4523.

Lecture Hours: 32.000

Prerequisite: MAN2021, MAN3583, MAN4523

TRA3034 Transportation and Traffic Management 3.000 credits

This course covers developments leading to national and federal regulations, division of territories, official descriptions, etc. Students will learn the scope of authority of territorial associations, factors controlling traffic flows, basic governing classification rules, principals of freight rates and tariffs, and elements of rate making. Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA3132 Purchasing and Inventory Management 3.000 credits

This course provides a comprehensive introduction to the purchasing/procurement and supply chain management field. Students will learn purchasing and supply chain issues in a variety of settings, from process industries to high-tech manufacturing and services as well as public institutions. Emphasis is on the purchasing process as it relates to such topics as inventory control procedures, price/cost analysis, laws and ethics, negotiations, vendor selection and the development of vendor relationships.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA4234 Warehouse Management 3.000 credits

Students will learn warehousing functions, facility operations, and operational productivity improvements and measurements with the inclusion of concepts from marketing, finance, statistics, operations management, and human resources. The course presents an integrated business approach to the detailed operational aspects of logistics facilities such as warehouse and distribution centers. The course will use real-world warehouse data to design the layout and operations requirements for a local warehousing facility and to identify the operational equipment used in warehousing. Prerequisites: MAN2021, MAN3065, QMB2100.

Lecture Hours: 48.000

Prerequisite: MAN2021 AND MAN3065 AND QMB2100

GEB3358 International Negotiations and Transactions 3.000 credits

This course presents business negotiations in selected regions of the world. Students will learn the skills of negotiation by analyzing international business cases and developing solutions to situations. Topics include overseas market research using both domestic and international sources and the application of theoretical and practical business knowledge to foreign situations and trade negotiations. The importance of culture, language, and values in international negotiations is emphasized.

Lecture Hours: 48.000

Prerequisite: TRA1154

Course User Fee: \$69.00

MAN3577 Procurement for Major Projects 3.000 credits

Students will learn the principles of procurement for major projects concepts and theory, including advanced purchasing and cost savings techniques, contracts administration from award to completion, strategic procurement management.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3578 Global Procurement Management 3.000 credits

Students will learn Global Procurement Management concepts and theory, including a good understanding in the differences in cultures, law, currency, communications and buying from foreign suppliers, international procurement is a challenge, buying internationally and to help support international procurement efforts.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3554 Safety and Risk Management 1.000 credits

This course will focus on safety and risk management with emphasis on how it applies to supply chain management. Students will learn the processes used to characterize and manage risk, as well as maintain a safe operating environment with the protection of personnel, assets, and services. Hazard characteristics of products, appropriate modes of transport, carrier selection and qualifications, packaging and container specifications, driver training requirements, financial issues, security precautions, warehousing risk, OSHA, hazardous materials handling, and customer assessment will be addressed in this course.

Lecture Hours: 16.000

MAN4940 Field Study and Research 2.000 credits

Students will learn to apply information and skills studied in core and concentration program courses through various special projects, field research, or internships designed for cogitative learning in a student-centered manner which requires the command, analysis, and synthesis of knowledge and skills.

Lecture Hours: 32.000

Prerequisite: MAN3583, TRA1154

MAR4203 Supply Chain Marketing 3.000 credits

Students will learn the management of traditional and emerging marketing channels, with emphasis on legal, economic, and ethical considerations in wholesale and retail inventory control, raw goods, finished product transportation and relationship management. Prerequisite: MAN3506. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MAN3506

GEB4363 Import Export Management 3.000 credits

Students will learn the functions and range of traffic management services performed by freight forwarders, including changes in governmental restrictions, rules, and regulations applicable to different countries, ports, and trade routes. Students will also learn the documentation needed to facilitate and coordinate the movement of goods in international trade and supply chain management systems. Prerequisites: MAN 2021 and TRA 2010.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA2010

Course User Fee: \$69.00

MAN3786 Sustainable Enterprise Planning 3.000 credits

Students will learn the assessment tools, design and construction considerations, and operating planning requirements for sustainable enterprises. Students will also learn the ecological and economic benefits of sustainability practices, including construction, operations, supply chain decisions, recycling, reusing, and reconditioning to preserve the environment by increasing revenues. Best practices, case studies, evolving trends and experimental efforts are also covered.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4593 Supply Chain Management Theory & Methodology 2.000 credits

This course presents a range of advanced topics in integrated logistics and supply chain management. Students will learn new theoretical and methodological developments in the field of supply chain. Specific topics vary depending upon current industry developments and will incorporate the use of speakers in the classroom. Prerequisites: MAN2021, MAN3583, MAN4523.

Lecture Hours: 32.000

Prerequisite: MAN2021, MAN3583, MAN4523

TRA3034 Transportation and Traffic Management 3.000 credits

This course covers developments leading to national and federal regulations, division of territories, official descriptions, etc. Students will learn the scope of authority of territorial associations, factors controlling traffic flows, basic governing classification rules, principals of freight rates and tariffs, and elements of rate making. Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA3132 Purchasing and Inventory Management 3.000 credits

This course provides a comprehensive introduction to the purchasing/procurement and supply chain management field. Students will learn purchasing and supply chain issues in a variety of settings, from process industries to high-tech manufacturing and services as well as public institutions. Emphasis is on the purchasing process as it relates to such topics as inventory control procedures, price/cost analysis, laws and ethics, negotiations, vendor selection and the development of vendor relationships.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA4234 Warehouse Management 3.000 credits

Students will learn warehousing functions, facility operations, and operational productivity improvements and measurements with the inclusion of concepts from marketing, finance, statistics, operations management, and human resources. The course presents an integrated business approach to the detailed operational aspects of logistics facilities such as warehouse and distribution centers. The course will use real-world warehouse data to design the layout and operations requirements for a local warehousing facility and to identify the operational equipment used in warehousing. Prerequisites: MAN2021, MAN3065, QMB2100.

Lecture Hours: 48.000

Prerequisite: MAN2021 AND MAN3065 AND QMB2100

Program Description

Supply Chain Management – Project Management Concentration

Bachelor of Applied Science | Code: P9302 | 120 credits

Supply chain management is an interdisciplinary field that emphasizes cross-functional integration of diverse business functions, facilities, and activities and seeks to manage those activities to enhance a company's competitive advantage. The Bachelor of Applied Science in Supply Chain Management (BAS-SCM) with a concentration in Project Management, is designed to provide knowledge of the supply chain while exploring the risks, operations, logistics, economics, regulatory issues, change management, forecasting, resource allocation, production planning, inventory management, customer delivery, after-sales support and service and other functions basic to business with particular emphasis in Project Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
MAN 3506	Operations Management	3.0
MAN 4520	Quality Management	3.0
MAN 4523	Production Information Systems	3.0
Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, CHI*, FIN*, FRE*, FRW*, GEB*, GER*, HBR*, ISM*, ITA*, JPN*, MAC*, MAN*, MAR*, PET*, POR*, PUR*, QMB*, SPM*, SPN*, STA*, TAX*, TRA*

Semester 2		Units: 12.0
FIN 3403	Financial Management	3.0
MAN 3583	Project Management	3.0
MAN 4887	Project Planning and Control Systems for Supply Chain Management	3.0
Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, CHI*, FIN*, FRE*, FRW*, GEB*, GER*, HBR*, ISM*, ITA*, JPN*, MAC*, MAN*, MAR*, PET*, POR*, PUR*, QMB*, SPM*, SPN*, STA*, TAX*, TRA*

Semester 3		Units: 12.0
BUL 4461	Law of International Trade	3.0
MAN 3504	Production Operations and Logistics Management	3.0
MAN 4597	Global Supply Chain Management	3.0
<i>Upper Division Elective</i>		3.0
GEB 4363	Import Export Management	3.0
OR		
MAN 3786	Sustainable Enterprise Planning	3.0
OR		
MAN 4593	Supply Chain Management Theory & Methodology	2.0
OR		
TRA 3034	Transportation and Traffic Management	3.0
OR		
TRA 3132	Purchasing and Inventory Management	3.0
OR		
TRA 4234	Warehouse Management	3.0

Semester 4		Units: 12.0
GEB 3358	International Negotiations and Transactions	3.0
MAN 3562	Purchasing, Inventory and Warehouse Management	3.0
MAN 3888	Project Leadership	3.0
Upper Division Elective*		3.0

*Select 3 credits from the following course options: GEB 4363, MAN 3786, MAN 4593, TRA 3034, TRA 3132, TRA 4234

Semester 5		Units: 12.0
MAN 3554	Safety and Risk Management	1.0
MAN 3731	Assessing and Managing Project Risk	3.0
MAN 4940	Field Study and Research	2.0

MAR 4203	Supply Chain Marketing	3.0
Upper Division Elective*		3.0

*Select 3 credits from the following course options: GEB 4363, MAN 3786, MAN 4593, TRA 3034, TRA 3132, TRA 4234

Total: 60.0

MAN3506 Operations Management 3.000 credits

The course emphasizes the application of operational decision-making techniques to improve process, productivity, and the effective utilization of resources within organizations. Students will learn to recognize the trade-offs associated with operations management decisions and their effect on resource allocation. Topics include production processes, operations strategies, quantitative techniques, quality, performance, capacity planning, efficiency, forecasting, resource management, statistical process control, project management, and supply chains. Prerequisite: QMB2100 or STA2023.

Lecture Hours: 48.000

Prerequisite: QMB2100 OR STA2023

MAN4520 Quality Management 3.000 credits

This course provides an understanding of various theories of quality management (QM). Students will learn how organizations can develop excellence through the adoption of continuous improvement and process management. The course analyzes and uses various process management techniques, continuous improvement tools, and strategies to improve quality. The conceptual and analytical skills acquired in this course enable students to provide leadership in shaping a culture for quality within an organization and determining the effectiveness of quality initiatives such as Total QM, Six Sigma, Process Capability, Process Control, and Customer Relationships.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4523 Production Information Systems 3.000 credits

This course presents the fundamental aspects of computer technology required by the systems that provide data to, and derive information from, production in manufacturing. Students will learn the techniques to organize, store, manipulate data, report, derive and analyze production information, basics networking used in production, as well as various forms of information systems.

Lecture Hours: 48.000

FIN3403 Financial Management 3.000 credits

Students will learn the sources and uses of funds as they relate to financial decision making in the corporate form of enterprise. Emphasis is placed on working capital management, capital budgeting techniques, short and long term financing, and capital structure and the value of the firm. This course emphasizes the use of formulas and spreadsheets through Excel.

Lecture Hours: 48.000

Prerequisite: ACG2071, ACG2071L, QMB2100

MAN3583 Project Management 3.000 credits

Students will learn project management concepts and theory, including attributes for a project life cycle, global project management, benefits of project management, project management environments, planning and managing risk, project planning controls, and terminating a project.

Lecture Hours: 48.000

MAN4887 Project Planning and Control Systems for Supply Chain Management 3.000 credits

Students will learn about the complex supply-chain transformation that is required to manage resources from many different departments. Additionally, an exploration of how supply-chain project managers ensure internal and external stakeholder alignment, mitigate large amounts of risk, and implement communication, risk mitigation, and change management plans to ensure a successful project. Finally, the student will gain an integrated view of supply-chain transformation that incorporates elements of change management, test plan development, project management techniques, and establishing effective project management teams.

Lecture Hours: 48.000

Course User Fee: \$62.76

BUL4461 Law of International Trade 3.000 credits

Students will learn to interpret laws of international trade, licensing, and customer's regulation. This course also addresses U.S. customs regulations, classification of merchandise, application of tariff rules, duty free treatment, importing and exporting, liquidation inspection, search and seizure, and fines and penalties. In addition, the course will review the general rules and how to interpret the harmonized tariff schedule. Prerequisites: MAN2021 and GEB3358.

Lecture Hours: 48.000

Prerequisite: GEB3358

MAN3504 Production Operations and Logistics Management 3.000 credits

This course examines the integration of production operations and logistics management and how they enable an organization to compete successfully in business. Students will learn the relationships that exist between operations and the supply chain, including operations and supply chain strategies, business processes, project management, product design and development, and inventory management. This course uses case study methodology. Prerequisite: TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA1154 with a minimum grade of C

MAN4597 Global Supply Chain Management 3.000 credits

This course presents an overview of the management of sourcing, operations, and distribution processes along a supply chain in domestic and international markets. Students will learn how firms gain a competitive advantage through supply chain activities. Topics include: supply chain network design, purchasing, forecasting, inventory management, globalization and outsourcing, logistics, and information technology.

Lecture Hours: 48.000

Prerequisite: MAN2021, MAN3506

GEB4363 Import Export Management 3.000 credits

Students will learn the functions and range of traffic management services performed by freight forwarders, including changes in governmental restrictions, rules, and regulations applicable to different countries, ports, and trade routes. Students will also learn the documentation needed to facilitate and coordinate the movement of goods in international trade and supply chain management systems. Prerequisites: MAN 2021 and TRA 2010.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA2010

Course User Fee: \$69.00

MAN3786 Sustainable Enterprise Planning 3.000 credits

Students will learn the assessment tools, design and construction considerations, and operating planning requirements for sustainable enterprises. Students will also learn the ecological and economic benefits of sustainability practices, including construction, operations, supply chain decisions, recycling, reusing, and reconditioning to preserve the environment by increasing revenues. Best practices, case studies, evolving trends and experimental efforts are also covered.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4593 Supply Chain Management Theory & Methodology 2.000 credits

This course presents a range of advanced topics in integrated logistics and supply chain management. Students will learn new theoretical and methodological developments in the field of supply chain. Specific topics vary depending upon current industry developments and will incorporate the use of speakers in the classroom. Prerequisites: MAN2021, MAN3583, MAN4523.

Lecture Hours: 32.000

Prerequisite: MAN2021, MAN3583, MAN4523

TRA3034 Transportation and Traffic Management 3.000 credits

This course covers developments leading to national and federal regulations, division of territories, official descriptions, etc. Students will learn the scope of authority of territorial associations, factors controlling traffic flows, basic governing classification rules, principals of freight rates and tariffs, and elements of rate making. Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA3132 Purchasing and Inventory Management 3.000 credits

This course provides a comprehensive introduction to the purchasing/procurement and supply chain management field. Students will learn purchasing and supply chain issues in a variety of settings, from process industries to high-tech manufacturing and services as well as public institutions. Emphasis is on the purchasing process as it relates to such topics as inventory control procedures, price/cost analysis, laws and ethics, negotiations, vendor selection and the development of vendor relationships.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA4234 Warehouse Management 3.000 credits

Students will learn warehousing functions, facility operations, and operational productivity improvements and measurements with the inclusion of concepts from marketing, finance, statistics, operations management, and human resources. The course presents an integrated business approach to the detailed operational aspects of logistics facilities such as warehouse and distribution centers. The course will use real-world warehouse data to design the layout and operations requirements for a local warehousing facility and to identify the operational equipment used in warehousing. Prerequisites: MAN2021, MAN3065, QMB2100.

Lecture Hours: 48.000

Prerequisite: MAN2021 AND MAN3065 AND QMB2100

GEB3358 International Negotiations and Transactions 3.000 credits

This course presents business negotiations in selected regions of the world. Students will learn the skills of negotiation by analyzing international business cases and developing solutions to situations. Topics include overseas market research using both domestic and international sources and the application of theoretical and practical business knowledge to foreign situations and trade negotiations. The importance of culture, language, and values in international negotiations is emphasized.

Lecture Hours: 48.000

Prerequisite: TRA1154

Course User Fee: \$69.00

MAN3562 Purchasing, Inventory and Warehouse Management 3.000 credits

This course explores the integration of production operations and logistics management. It will also examine purchasing, Inventory and Warehouse Management concepts and theory. Special emphasis is placed on the relationships that exist between operations and the supply chain to include operations and supply chain strategies, business processes, project management, product design and development, inventory management and effective warehouse and stores management.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3888 Project Leadership 3.000 credits

Students will learn the fundamentals of management vs. leadership, monitoring/controlling, delegation, personal traits, communication, and people management when in a leadership role. The specifics include techniques and strategies using the company's vision and project charter needed to effectively manage, lead and direct a team to ensure reaching project objectives. As part of this process students will determine their personal leadership traits and approach.

Lecture Hours: 48.000, Lab Hours: 3.000

Course User Fee: \$62.76

MAN3554 Safety and Risk Management 1.000 credits

This course will focus on safety and risk management with emphasis on how it applies to supply chain management. Students will learn the processes used to characterize and manage risk, as well as maintain a safe operating environment with the protection of personnel, assets, and services. Hazard characteristics of products, appropriate modes of transport, carrier selection and qualifications, packaging and container specifications, driver training requirements, financial issues, security precautions, warehousing risk, OSHA, hazardous materials handling, and customer assessment will be addressed in this course.

Lecture Hours: 16.000

MAN3731 Assessing and Managing Project Risk 3.000 credits

This course explores project uncertainty, and how to manage risk to keep the project on track and meet project goals. Project risk management is a vital part of all projects, which requires a purposeful strategy to avoid any setbacks. In this course there will be an examination of what the risk management process involves through the identification, assessment, and response to project risks.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4940 Field Study and Research 2.000 credits

Students will learn to apply information and skills studied in core and concentration program courses through various special projects, field research, or internships designed for cogitative learning in a student-centered manner which requires the command, analysis, and synthesis of knowledge and skills.

Lecture Hours: 32.000

Prerequisite: MAN3583, TRA1154

MAR4203 Supply Chain Marketing 3.000 credits

Students will learn the management of traditional and emerging marketing channels, with emphasis on legal, economic, and ethical considerations in wholesale and retail inventory control, raw goods, finished product transportation and relationship management. Prerequisite: MAN3506. (3 hr. lecture)
 Lecture Hours: 48.000
 Prerequisite: MAN3506

Program Description

Supply Chain Management – Supply Chain Analytics Concentration

Bachelor of Applied Science | Code: P9303 | 120 credits

Supply chain management is an interdisciplinary field that emphasizes cross-functional integration of diverse business functions, facilities, and activities and seeks to manage those activities to enhance a company's competitive advantage. The Bachelor of Applied Science in Supply Chain Management (BAS-SCM) is designed to provide knowledge of the supply chain while exploring the risks, operations, logistics, economics, regulatory issues, change management, forecasting, resource allocation, production planning, inventory management, customer delivery, after-sales support and service and other functions basic to business.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
MAN 3506	Operations Management	3.0
MAN 4520	Quality Management	3.0
MAN 4523	Production Information Systems	3.0
Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, CHI*, FIN*, FRE*, FRW*, GEB*, GER*, HBR*, ISM*, ITA*, JPN*, MAC*, MAN*, MAR*, PET*, POR*, PUR*, QMB*, SPM*, SPN*, STA*, TAX*, TRA*

Semester 2		Units: 12.0
FIN 3403	Financial Management	3.0
MAN 3583	Project Management	3.0
MAN 4552	Supply Chain Analytics & Decision Making	3.0
Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, CHI*, FIN*, FRE*, FRW*, GEB*, GER*, HBR*, ISM*, ITA*, JPN*, MAC*, MAN*, MAR*, PET*, POR*, PUR*, QMB*, SPM*, SPN*, STA*, TAX*, TRA*

Semester 3		Units: 12.0
BUL 4461	Law of International Trade	3.0
MAN 3504	Production Operations and Logistics Management	3.0
MAN 4597	Global Supply Chain Management	3.0
<i>Upper Division</i>		3.0
GEB 4363	Import Export Management	3.0

OR		
MAN 3786	Sustainable Enterprise Planning	3.0
OR		
MAN 4593	Supply Chain Management Theory & Methodology	2.0
OR		
TRA 3034	Transportation and Traffic Management	3.0
OR		
TRA 3132	Purchasing and Inventory Management	3.0
OR		
TRA 4234	Warehouse Management	3.0

Semester 4

Units: 12.0

GEB 3358	International Negotiations and Transactions	3.0
MAN 3562	Purchasing, Inventory and Warehouse Management	3.0
MAN 4732	Business Intelligence for Supply Chain	3.0
Upper Division Elective*		3.0

*Select 3 credits from the following course options: GEB 4363, MAN 3786, MAN 4593, TRA 3034, TRA 3132, TRA 4234

Semester 5

Units: 12.0

MAN 3554	Safety and Risk Management	1.0
MAN 4719	Challenges in the Digital Supply Chain	3.0
MAN 4940	Field Study and Research	2.0
MAR 4203	Supply Chain Marketing	3.0
Upper Division Elective*		3.0

*Select 3 credits from the following course options: GEB 4363, MAN 3786, MAN 4593, TRA 3034, TRA 3132, TRA 4234

Total: 60.0

MAN3506 Operations Management 3.000 credits

The course emphasizes the application of operational decision-making techniques to improve process, productivity, and the effective utilization of resources within organizations. Students will learn to recognize the trade-offs associated with operations management decisions and their effect on resource allocation. Topics include production processes, operations strategies, quantitative techniques, quality, performance, capacity planning, efficiency, forecasting, resource management, statistical process control, project management, and supply chains.

Prerequisite: QMB2100 or STA2023.

Lecture Hours: 48.000

Prerequisite: QMB2100 OR STA2023

MAN4520 Quality Management 3.000 credits

This course provides an understanding of various theories of quality management (QM). Students will learn how organizations can develop excellence through the adoption of continuous improvement and process management. The course analyzes and uses various process management techniques, continuous improvement tools, and strategies to improve quality. The conceptual and analytical skills acquired in this course enable students to provide leadership in shaping a culture for quality within an organization and determining the effectiveness of quality initiatives such as Total QM, Six Sigma, Process Capability, Process Control, and Customer Relationships.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4523 Production Information Systems 3.000 credits

This course presents the fundamental aspects of computer technology required by the systems that provide data to, and derive information from, production in manufacturing. Students will learn the techniques to organize, store, manipulate data, report, derive and analyze production information, basics networking used in production, as well as various forms of information systems.

Lecture Hours: 48.000

FIN3403 Financial Management 3.000 credits

Students will learn the sources and uses of funds as they relate to financial decision making in the corporate form of enterprise. Emphasis is placed on working capital management, capital budgeting techniques, short and long term financing, and capital structure and the value of the firm. This course emphasizes the use of formulas and spreadsheets through Excel.

Lecture Hours: 48.000

Prerequisite: ACG2071, ACG2071L, QMB2100

MAN3583 Project Management 3.000 credits

Students will learn project management concepts and theory, including attributes for a project life cycle, global project management, benefits of project management, project management environments, planning and managing risk, project planning controls, and terminating a project.

Lecture Hours: 48.000

MAN4552 Supply Chain Analytics & Decision Making 3.000 credits

Through this course students will focus on the science of learning from data to support decision making. The objective of this course is to explore and apply basic statistical concepts and procedures that are used to collect, analyze, summarize, and report data in a supply chain management role. Course topics covered include methods for: collecting, analyzing, summarizing data; making statistical inferences about populations and exploring the relationship between variables. The course introduces analytical tools and techniques that provide a cause and effect understanding linking operational plans with corporate objectives.

Lecture Hours: 48.000

Course User Fee: \$62.76

BUL4461 Law of International Trade 3.000 credits

Students will learn to interpret laws of international trade, licensing, and customer's regulation. This course also addresses U.S. customs regulations, classification of merchandise, application of tariff rules, duty free treatment, importing and exporting, liquidation inspection, search and seizure, and fines and penalties. In addition, the course will review the general rules and how to interpret the harmonized tariff schedule. Prerequisites: MAN2021 and GEB3358.

Lecture Hours: 48.000

Prerequisite: GEB3358

MAN3504 Production Operations and Logistics Management 3.000 credits

This course examines the integration of production operations and logistics management and how they enable an organization to compete successfully in business. Students will learn the relationships that exist between operations and the supply chain, including operations and supply chain strategies, business processes, project management, product design and development, and inventory management. This course uses case study methodology. Prerequisite: TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA1154 with a minimum grade of C

MAN4597 Global Supply Chain Management 3.000 credits

This course presents an overview of the management of sourcing, operations, and distribution processes along a supply chain in domestic and international markets. Students will learn how firms gain a competitive advantage through supply chain activities. Topics include: supply chain network design, purchasing, forecasting, inventory management, globalization and outsourcing, logistics, and information technology.

Lecture Hours: 48.000

Prerequisite: MAN2021, MAN3506

GEB4363 Import Export Management 3.000 credits

Students will learn the functions and range of traffic management services performed by freight forwarders, including changes in governmental restrictions, rules, and regulations applicable to different countries, ports, and trade routes. Students will also learn the documentation needed to facilitate and coordinate the movement of goods in international trade and supply chain management systems. Prerequisites: MAN 2021 and TRA 2010.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA2010

Course User Fee: \$69.00

MAN3786 Sustainable Enterprise Planning 3.000 credits

Students will learn the assessment tools, design and construction considerations, and operating planning requirements for sustainable enterprises. Students will also learn the ecological and economic benefits of sustainability practices, including construction, operations, supply chain decisions, recycling, reusing, and reconditioning to preserve the environment by increasing revenues. Best practices, case studies, evolving trends and experimental efforts are also covered.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4593 Supply Chain Management Theory & Methodology 2.000 credits

This course presents a range of advanced topics in integrated logistics and supply chain management. Students will learn new theoretical and methodological developments in the field of supply chain. Specific topics vary depending upon current industry developments and will incorporate the use of speakers in the classroom. Prerequisites: MAN2021, MAN3583, MAN4523.

Lecture Hours: 32.000

Prerequisite: MAN2021, MAN3583, MAN4523

TRA3034 Transportation and Traffic Management 3.000 credits

This course covers developments leading to national and federal regulations, division of territories, official descriptions, etc. Students will learn the scope of authority of territorial associations, factors controlling traffic flows, basic governing classification rules, principals of freight rates and tariffs, and elements of rate making. Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA3132 Purchasing and Inventory Management 3.000 credits

This course provides a comprehensive introduction to the purchasing/procurement and supply chain management field. Students will learn purchasing and supply chain issues in a variety of settings, from process industries to high-tech manufacturing and services as well as public institutions. Emphasis is on the purchasing process as it relates to such topics as inventory control procedures, price/cost analysis, laws and ethics, negotiations, vendor selection and the development of vendor relationships.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA4234 Warehouse Management 3.000 credits

Students will learn warehousing functions, facility operations, and operational productivity improvements and measurements with the inclusion of concepts from marketing, finance, statistics, operations management, and human resources. The course presents an integrated business approach to the detailed operational aspects of logistics facilities such as warehouse and distribution centers. The course will use real-world warehouse data to design the layout and operations requirements for a local warehousing facility and to identify the operational equipment used in warehousing. Prerequisites: MAN2021, MAN3065, QMB2100.

Lecture Hours: 48.000

Prerequisite: MAN2021 AND MAN3065 AND QMB2100

GEB3358 International Negotiations and Transactions 3.000 credits

This course presents business negotiations in selected regions of the world. Students will learn the skills of negotiation by analyzing international business cases and developing solutions to situations. Topics include overseas market research using both domestic and international sources and the application of theoretical and practical business knowledge to foreign situations and trade negotiations. The importance of culture, language, and values in international negotiations is emphasized.

Lecture Hours: 48.000

Prerequisite: TRA1154

Course User Fee: \$69.00

MAN3562 Purchasing, Inventory and Warehouse Management 3.000 credits

This course explores the integration of production operations and logistics management. It will also examine purchasing, Inventory and Warehouse Management concepts and theory. Special emphasis is placed on the relationships that exist between operations and the supply chain to include operations and supply chain strategies, business processes, project management, product design and development, inventory management and effective warehouse and stores management.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4732 Business Intelligence for Supply Chain 3.000 credits

This course will enable students to synthesize their learning from program coursework. The global context within which firms operate will provide the foundation for understanding global supply chain strategy, from market entry through to the integration of demand management and supply management processes to meet corporate objectives. The course will utilize exercises and a simulation to allow students to analyze and synthesize program coursework. The course will help students understand how to best use program knowledge to maximize value creation for employers and in their own career.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3554 Safety and Risk Management 1.000 credits

This course will focus on safety and risk management with emphasis on how it applies to supply chain management. Students will learn the processes used to characterize and manage risk, as well as maintain a safe operating environment with the protection of personnel, assets, and services. Hazard characteristics of products, appropriate modes of transport, carrier selection and qualifications, packaging and container specifications, driver training requirements, financial issues, security precautions, warehousing risk, OSHA, hazardous materials handling, and customer assessment will be addressed in this course.

Lecture Hours: 16.000

MAN4719 Challenges in the Digital Supply Chain 3.000 credits

Through this course, students will analyze the social, ethical, and legal issues in supply chain management. Students will examine the importance of fairness, accountability, and transparency of data policies in various supply chain operations. Various issues governing the collection, storage, access, use, and sharing of data will also be explored. Prerequisites: MAN3506, QMB2100 or STA2023.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4940 Field Study and Research 2.000 credits

Students will learn to apply information and skills studied in core and concentration program courses through various special projects, field research, or internships designed for cogitative learning in a student-centered manner which requires the command, analysis, and synthesis of knowledge and skills.

Lecture Hours: 32.000

Prerequisite: MAN3583, TRA1154

MAR4203 Supply Chain Marketing 3.000 credits

Students will learn the management of traditional and emerging marketing channels, with emphasis on legal, economic, and ethical considerations in wholesale and retail inventory control, raw goods, finished product transportation and relationship management. Prerequisite: MAN3506. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MAN3506

Bachelor of Sciences (BS)

Program Description

Applied Artificial Intelligence

Bachelor of Science | Code: S9520 | 120 credits

The Bachelor of Science (BS) degree in Applied Artificial Intelligence (AI) offers a practical approach to using complex fields such as computer vision, natural language processing, and machine learning to transform large datasets into actionable outputs that can be used to increase productivity and operational efficiencies. The program is well-rounded and tailored to meet employers' needs, offering in-depth knowledge of artificial intelligence (AI) tools and their applications, as well as AI process automation and optimization. In addition, students learn to use ethical standards and socially responsible practices in the design and implementation of AI systems.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

CAI 3303C	Natural Language Processing	3.0
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CAI 3821C	Computational Methods and Applications for Artificial Intelligence I	3.0
COP 2800	Java Programming	4.0
ENC 1102	English Composition 2	3.0

Semester 2

Units: 13.0

CAI 3822C	Computational Methods and Applications for Artificial Intelligence II	3.0
COP 3530	Data Structures	4.0
<i>Oral Communications (GE)</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science (GE)</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
POS 2041	American Federal Government	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 3

Units: 10.0

CAI 4505C	Artificial Intelligence	3.0
<i>Statistics Option</i>		4.0
CAP 3330	Programming R for Statistics	4.0
OR		
STA 3164	Statistical Methods II	4.0
General Education Elective *		3.0
Foreign Language Competence **		

*Students are encouraged to consult with an academic advisor to select approved courses that align with their interests and fulfill remaining graduation requirements.

**Foreign language is a graduation requirement for the baccalaureate met through 8 credit hours at the elementary 2 level in one foreign language or equivalent. Certain foreign language courses count towards program electives. Students may satisfy equivalence through standardized examinations or successful completion of two credits (two years) in one foreign language at the secondary (high school) level. For additional information, including exemptions for students whose native language is not English, see the Testing and Assessment Department.

Semester 4

Units: 12.0

CAI 4420C	Applied Decision and Optimization Theory	3.0
CAI 4510C	Machine Intelligence	3.0
CAI 4830C	Simulation for Applied Artificial Intelligence	3.0
<i>Humanities (GE)</i>		3.0
HUM 1020	Humanities	3.0
OR		

MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0

Semester 5

Units: 12.0

CAI 4525C	Artificial Intelligence Systems Automation	3.0
CAI 4950C	Artificial Intelligence Capstone	3.0
<i>Natural Science (GE)</i>		3.0
PHY 1025	Basic Physics	3.0
OR		
PHY 2048	Physics with Calculus 1	4.0
OR		
PHY 2053	Physics (without Calculus) 1	3.0
<i>Program Elective</i>		3.0
<i>Business</i>		3.0
GE 1432	Applied Artificial Intelligence (AI) in Business	3.0
OR		
<i>Computingg</i>		4.0
CAP 1788	Introduction to Data Analytics	4.0
OR		
CAP 2761C	Intermediate Analytics	4.0
OR		
CAP 3321C	Data Wrangling	4.0
OR		
CAP 4767	Data Mining	4.0
OR		
CAP 4936	Special Topics in Data Analytics	4.0
OR		
CEN 2211	C/C++ Programming for Embedded Devices	4.0
OR		
CET 3383C	Software Engineering I	4.0
OR		
CGS 1540C	Database Concepts Design	4.0
OR		
CIS 2010C	Introduction to Internet of Things Networking and Security	4.0
OR		
CIS 2012C	Introduction to Data Analytics and AI for the Internet of Things	4.0
OR		
CIS 3360	Principles of Information Security	4.0
OR		
COP 2041C	Python Programming 2	4.0
OR		
COT 3100C	Discrete Structures	4.0
OR		
COT 4400	Design and Analysis of Algorithms	4.0
OR		
CTS 1145	Cloud Essentials	4.0
OR		

CTS 2192C	Microsoft Azure Administration	4.0
OR		
<i>Mathematics</i>		3.0-5.0
MAC 1114	Trigonometry	3.0
OR		
MAC 1140	Pre-Calculus Algebra	3.0
OR		
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
OR		
MAC 2311	Calculus and Analytical Geometry 1	5.0
OR		
MAC 2312	Calculus and Analytical Geometry 2	4.0
OR		
MAS 3105	Linear Algebra	3.0
OR		
<i>Robotics</i>		4.0
ETS 1603C	Introduction to Robotics	4.0
OR		
ETI 4480C	Applied Robotics	4.0

Total: 60.0

CAI3303C Natural Language Processing 3.000 credits

Students will further explore Natural Language Processing domains. Focus on deep learning application and use of the latest pre-trained models on real world applications. Prerequisite: CAI2300C. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: CAI2300C
Course User Fee: \$63.00

CAI3821C Computational Methods and Applications for Artificial Intelligence I 3.000 credits

Computational data analysis is an essential part of artificial intelligence (AI). This course is designed to help students develop programming skills for AI applications. Students will learn core concepts of computational methods to solve data analysis problems, AI algorithmic methodologies, and how to test AI models. Prerequisites: CAI2100C, COP1047C, MAC1105, and STA2023. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: CAI2100C AND COP1047C AND STA2023 AND MAC1105
Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles. Lecture Hours: 48.000, Lab Hours: 32.000
Corequisite: COP1047C OR COP1334 OR COP2270
Course User Fee: \$63.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better. Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

CAI3822C Computational Methods and Applications for Artificial Intelligence II 3.000 credits

This course is designed for students to acquire a deeper understanding of computational methods used in the applications of artificial intelligence (AI) with programming. The topics of this course will be a continuation of those covered in Computational Methods and Applications for Artificial Intelligence I, with added emphasis on case studies using machine learning. Prerequisite: CAI3821C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI3821C

Course User Fee: \$63.00

COP3530 Data Structures 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology. Students will learn the fundamentals of data structures using the Java programming language. The students will learn to design, implement and use data structures to organize and store data in a computer so that it can be accessed and modified efficiently.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

CAI4505C Artificial Intelligence 3.000 credits

The course provides an in-depth survey of important concepts, problems, and techniques in artificial intelligence, including heuristic and adversarial search, constraint satisfaction problems, logical reasoning, reasoning with uncertainty, and machine learning. A particular focus and a unifying theme of the course will be the notion of intelligent agents and their implementation. Prerequisites: COP3530 and CAI3822C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: COP3530 AND CAI3822C

Course User Fee: \$63.00

CAP3330 Programming R for Statistics 4.000 credits

This upper division course is for students majoring in data analytics. Students will learn the R programming language and use it to perform intermediate-level statistical analysis. Techniques used in data analysis, such as analysis of variance and regression, will be emphasized. Prerequisite: STA2023.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: STA2023

Course User Fee: \$63.00

STA3164 Statistical Methods II 4.000 credits

This course is for students majoring in Data Analytics, Systems Engineering, and related disciplines which require advanced skills in statistical analysis. Students will learn how to perform tests of variance, analysis of variance, analysis of covariance, regression, correlation, and non-parametric statistics.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: STA2023

Course User Fee: \$50.00

CAI4420C Applied Decision and Optimization Theory 3.000 credits

Students will learn how to formulate and optimize queuing, linear and nonlinear programming models. Students will understand how to apply decision-making logic and optimization techniques to artificial intelligence models and conduct decision analysis of real-world problems. Prerequisite: CAI4505C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI4505C

Course User Fee: \$63.00

CAI4510C Machine Intelligence 3.000 credits

The course will cover advanced modeling techniques, including ensemble learning, extended linear models and kernel methods (PCA, support vector machines), probabilistic graphical models, Bayesian networks, mixture and latent variable models, biologically inspired computing (neural networks), feature selection and feature engineering techniques, Markov models, and temporal modeling to find patterns over time. Prerequisites: COP3530 and CAI3822C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: COP3530 AND CAI3822C

Course User Fee: \$63.00

CAI4830C Simulation for Applied Artificial Intelligence 3.000 credits

Students will learn how to choose the right constructs of the modeling language to create a representation of a real-world system that is suitable for risk-free dynamic experiments. In addition, the students will learn how to build and deploy simulation models using the three major paradigms in simulation modeling for AI: agent-based, system dynamics, and discrete-event. Prerequisite: CAI4505C.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: CAI4505C

Course User Fee: \$63.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

CAI4525C Artificial Intelligence Systems Automation 3.000 credits

Students will learn how to use automated tools to develop enterprise automation projects. Students will understand how to create variables, arguments, workflows, build and publish automated processes and effective user interface automations.

Prerequisites: CAI4505C and CAI4510C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI4505C AND CAI4510C

Course User Fee: \$63.00

CAI4950C Artificial Intelligence Capstone 3.000 credits

Students will demonstrate competence to scope, acquire/explore data, model, evaluate, and deploy an AI/Machine Learning solution in a team environment. Students will create and present an AI solution.

Course must be taken during the last semester

before graduation. Prerequisites: CAI4510C,

CAI4830C, and CAI4420C; Pre/Corequisite: CAI4525C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI4510C AND CAI4830C AND

CAI4420C AND Pre/Corequisite: CAI4525C

Course User Fee: \$63.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves.

Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course. Lecture Hours: 48.000
Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L
Course User Fee: \$10.00

GEB1432 Applied Artificial Intelligence (AI) in Business 3.000 credits

This course illustrates how Artificial Intelligence (AI) is currently applied in industries and across business functions to support business strategy formulation, implementation and evaluation. This course introduces the data-driven outcomes and ethical issues to be considered when implementing responsible AI in business and industry. (3 hr. lecture)
Lecture Hours: 48.000
Course User Fee: \$35.20

CAP1788 Introduction to Data Analytics 4.000 credits

This course offers a broad introduction to data analytics and the role it plays in modern organizations. Students will use Excel and Tableau to import, clean, transform, and visualize data. Students will also learn how to effectively communicate findings to decision makers.
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

CAP2761C Intermediate Analytics 4.000 credits

This course offers an introduction to the SQL language and how it can be used by data analysts to query databases and gain valuable insights. Students will also gain extensive hands-on experience with Microsoft SQL Server, Azure, and GitHub.
Prerequisites: CAP1788, CGS1540C.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CGS1540C
Course User Fee: \$63.00

CAP3321C Data Wrangling 4.000 credits

This course offers a broad introduction to data wrangling, data retrieval, and the use of Python in data analytics. Students will use Python and other business intelligence tools to retrieve data from various sources, clean the dataset, and prepare it for data analysis tasks. Prerequisites: CAP1788 and CAP2704.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CAP1788 AND IDC2132
Course User Fee: \$63.00

CAP4767 Data Mining 4.000 credits

Students will learn how to extract useful data from an organization's data warehouse and analyze it to uncover hidden patterns and relationships. An emphasis will be placed on unsupervised data mining techniques, as well as the use of the RapidMiner software. Prerequisites: CAP1788 and CAP2761C.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CAP1788 AND CAP2761C
Course User Fee: \$63.00

CAP4936 Special Topics in Data Analytics 4.000 credits

This upper division course is for students majoring in data analytics. The course centers around topics of current interest or of special interest to students or instructors. Topics or focus may vary from semester to semester. Prerequisites: CAP1788, CAP2761C.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CAP1788, CAP2761C
Course User Fee: \$63.00

CEN2211 C/C++ Programming for Embedded Devices 4.000 credits

This course teaches the principles of programming in the C/C++ language for embedded devices. The student will learn how to create programs to control open-source hardware for building digital devices that can sense and control the physical world around them and communicate with the Internet.
Prerequisite: COP 1334; Corequisite: EET 1033C. (2 hr. lecture 4 hr. lab).
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: COP1334 AND Corequisite: EET1033C

CET3383C Software Engineering I 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology or B.S. in Electrical and Computer Engineering Technology. The student will learn the basic principles and concepts of software engineering; system requirements; modeling and testing; object oriented analysis and design; testing and validation; configuration management; and the analysis, design and programming of extensible software systems. Prerequisite: CET2369C or COP2800.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2369C, COP2800

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement. Lecture Hours: 48.000, Lab Hours: 32.000 Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CIS2010C Introduction to Internet of Things Networking and Security 4.000 credits

This course explores Internet of Things (IoT) networks' design, implementation, and security, providing a well-rounded understanding of protocols and standards governing the Internet and IoT. Students gain experience performing security assessments and vulnerability testing of IoT systems using industry-standard tools, focusing on identifying and mitigating risks to build secure and resilient IoT solutions.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS2012C Introduction to Data Analytics and AI for the Internet of Things 4.000 credits

This course equips students with the knowledge and skills necessary to navigate the complex landscape of data analytics and artificial intelligence within the context of the Internet of Things (IoT), empowering them to contribute to the development of smart, data-driven IoT solutions. Prerequisites: CIS 2010C, CIS 2011C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS2010C AND CIS2011C

Course User Fee: \$63.00

CIS3360 Principles of Information Security 4.000 credits

This upper division course provides an overview of information systems security principles, practices, methods, and tools. Students will learn about the threats against an organization's digital assets, as well as the tools and methods to mitigate those threats. Topics include cryptography, authentication and access control systems, infrastructure security, and ethical issues. Prerequisite: CTS1134 or CTS1650

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

COP2041C Python Programming 2 4.000 credits

This is an advanced level programming course using Python. Students will learn how to code, compile, and execute programs. Topics include problem solving, object-oriented programming, database programming, data visualization, and secure software development. Prerequisite: COP1047C; (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C

Course User Fee: \$63.00

COT3100C Discrete Structures 4.000 credits

This course delves into discrete structures with a strong focus on applications in computer science. It covers a range of topics including sets, functions and relations, propositional and predicate logic, basic proof techniques, logic, recursion and induction, graphs, Big-O notation, and various specific applications relevant to computer science.

Prerequisite: COP 2800; Corequisite: COP 3530 (3 lecture, 2 lab hours)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800 AND Corequisite: COP3530

Course User Fee: \$63.00

COT4400 Design and Analysis of Algorithms 4.000 credits

This upper division course is for students majoring in the B.S. in Information Systems Technology or the B.S. in Electrical and Computer Engineering Technology programs. This course covers general techniques in algorithm design (such as divide-and-conquer, greedy method, dynamic programming, search and traversal techniques, branch-and-bound) in the context of problem domains like graph, sorting and optimization problems. Prerequisite(s): COP1334 or COP2270. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800 AND Pre/Corequisite: COP3530

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2192C Microsoft Azure Administration 4.000 credits

This course is intended for students preparing for IT careers as cloud administrators, as well as candidates for industry certification. Students will learn the administration skills and knowledge required to implement, manage, and monitor identity, governance, storage, and compute virtual networks in the Microsoft Azure cloud environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS1145 or equivalent knowledge in Microsoft Azure Fundamentals

Course User Fee: \$63.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics.

Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

MAS3105 Linear Algebra 3.000 credits

This course is designed for students who are majoring in secondary mathematics education. Major topics include systems of linear equations, matrices, determinants, vector spaces, linear transformations, eigenvectors and eigenvalues, inner-product spaces and orthogonality.

Lecture Hours: 48.000

Prerequisite: MAC2312

ETS1603C Introduction to Robotics 4.000 credits

This is an introductory level course designed as an introduction to robotics and robotic applications. Students learn topics related to robotic design including robotic terminology, robotic programming, sensing and sensors, actuators, modeling and sensing, robotic platforms, and the application of artificial intelligence in robotics. Laboratory activities provide hands-on application of concepts and theories.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

ETI4480C Applied Robotics 4.000 credits

This is an upper division level course designed as an introduction to robotics programming and includes robotic applications for multifunction part manipulation and motion with stepper and servo-motors. Students learn topics related to robotic design including robotic vision, motion planning, sensing and sensors, actuators, navigation systems, mobility, forward and inverse kinematics, and path planning. Laboratory activities provide hands-on application of concepts and theories. Prerequisite: CET2123C or CET3383C.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2123C OR CET 3383C

Course User Fee: \$63.00

Program Description

Biological Sciences - Biopharmaceutical Sciences Concentration

Bachelor of Science | Code: S9121 | 120 credit

The Bachelor in Science (B.S.) in Biological Sciences is designed to provide students with a broad foundation in biology. Miami Dade College students will explore structure, function, genetics, and evolution of cells and organisms, while being provided the hands-on skills necessary to obtain jobs in the Biological/Life Science fields. Students take the core courses in biology, chemistry, physics and mathematics, and must meet the same General Education Requirements of all students seeking an AA degree. Students may choose from a variety of electives in science and general education to complete their degree requirements.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 13.0**

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 2053	Physics (without Calculus) 1	3.0
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0
BOT 3015	Survey of Plant Diversity	3.0
BOT 3015L	Survey of Plant Diversity Laboratory	1.0

Semester 2**Units: 13.0**

PHY 2054	Physics (Without Calculus) 2	3.0
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PHY 2054L	Physics (Without Calculus) 2 Lab	1.0
STA 2023	Statistical Methods	3.0
BSC 3930	Biological Sciences Seminar	1.0
PCB 3060	Principles of Genetics	3.0
PCB 3060L	Principles of Genetics Laboratory	2.0

Semester 3

Units: 9.0

PCB 3043	Fundamentals of Ecology	3.0
PCB 4023	Molecular and Cell Biology	3.0
ZOO 3021	Survey of Animal Diversity	3.0

Semester 4

Units: 13.0

BCH 3023	Introductory Biochemistry	3.0
BCH 3023L	Introductory Biochemistry Laboratory	2.0
MCB 3023	Principles of Microbiology	3.0
MCB 3023L	Principles of Microbiology Lab	2.0
PCB 4674	Evolution	3.0

Semester 5

Units: 13.0

BSC 4940	Senior Specialty Internship	3.0
MCB 4503	Virology	3.0
PCB 4097	Human Physiology	3.0
PCB 4233C	Fundamentals of Immunology	4.0

Total: 61.0

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

BOT3015 Survey of Plant Diversity 3.000 credits

This course explores the plant kingdom and emphasizes structure, function and genetics of plants. Students will learn the evolutionary relationships, natural history, ecological adaptations, physiology, morphology and reproductive biology of gymnosperms and angiosperms.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BOT3015L

Course User Fee: \$10.00

BOT3015L Survey of Plant Diversity Laboratory 1.000 credits

This course is designed to provide the necessary laboratory experiments and dissection exercises to supplement/accompany the BOT3015 Survey of Plant Diversity lecture course. Students will learn about the plant kingdom and with emphasis on structure, function and genetics of plants. Dissections and laboratory exercises are designed to explore the fundamental cell and tissue structures of both vascular and non-vascular plants.

Lab Hours: 32.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BOT3015

Course User Fee: \$40.00

PHY2054 Physics (Without Calculus) 2 3.000 credits

An introduction to the basic principles of physics. PHY 2053 covers mechanics, sound and thermodynamics. PHY 2054 includes electricity, magnetism and optics.

Lecture Hours: 48.000

Prerequisite: PHY2053, PHY2053L, Corequisite: PHY2054L

Course User Fee: \$10.00

PHY2054L Physics (Without Calculus) 2 Lab 1.000 credits

The physics lab courses are one-credit courses designed to be taken in conjunction with a physics lecture. A different experiment is performed each week with topics chosen to correspond with the material being studied in the lecture. Each experiment is designed to be completed in about 2 contact hours.

Lab Hours: 32.000

Prerequisite: PHY2053, PHY2053L, Corequisite: PHY2054

Course User Fee: \$20.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

BSC3930 Biological Sciences Seminar 1.000 credits

This course is designed for biological science majors. Students will gain an understanding of the broad range of career options within the biological sciences. Additionally, students will learn how to read, interpret, discuss, and cite selected examples of scientific literature in different areas of biology.

Lecture Hours: 16.000

Prerequisite: BSC2011, BSC2011L

Course User Fee: \$10.00

PCB3060 Principles of Genetics 3.000 credits

This course is an introduction to the mechanisms of transmission of hereditary information. Students will learn the classical Mendelian principles of heredity, deviation of Mendelian principles, genetic analysis, linkage and mapping, genetics of populations, gene regulation, mutation, the genetic bases of cancer and other genetic disorders will also be studied.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: PCB3060L

PCB3060L Principles of Genetics Laboratory 2.000 credits

This laboratory course is designed to complement PCB 3060 Principles of Genetics. Students will learn hands-on skills with emphasis on laboratory principles, techniques, and instrumentation within the field of genetics.

Lab Hours: 64.000

Prerequisite: BSC2011, BSC2011L, Corequisite: PCB3060

Course User Fee: \$110.00

PCB3043 Fundamentals of Ecology 3.000 credits

This is a foundations course in ecology. In this course, students will learn the basic principles of ecology at organismal, population, community, and ecosystem levels, including consideration of Florida's ecosystems and human impact on those systems.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L

PCB4023 Molecular and Cell Biology 3.000 credits

Students will learn the structure and function of cells and biological membranes, signal transduction pathways, cell cycle and cell division, the flow of genetic information and the regulation of gene expression. Exploration of laboratory techniques and discussion of the cellular basis of human diseases will also occur.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L

Course User Fee: \$25.00

ZOO3021 Survey of Animal Diversity 3.000 credits

This course presents zoology as a scientific discipline. Students will learn the basic principles of zoological nomenclature, taxonomy, systematic, and the basic understanding of the relationships of animals to one another, to humans, their environment and to society.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L

Course User Fee: \$10.00

BCH3023 Introductory Biochemistry 3.000 credits

This course surveys the fundamental components of biochemistry. In this course, students will learn concepts such as the structure and function of amino acids, proteins, carbohydrates, lipids, and nucleic acids, together with discussions of oxidative metabolism and regulation. Prerequisites: BSC2010, BSC2010L, BSC2011, BSC2011L, CHM2200 or CHM2211, CHM2211L; Corequisite: BCH3023L.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BCH3023L

Course User Fee: \$25.00

BCH3023L Introductory Biochemistry Laboratory 2.000 credits

This laboratory course complements the lecture corequisite BCH 3023, which involves the study of the fundamental components of biochemistry. In this laboratory course, students will learn and will be provided with hands-on experiences with the concepts addressed in the lecture course.

Prerequisites: BSC2010, BSC2010L, BSC2011, BSC2011L, CHM2200 or CHM2211, CHM2211L;

Corequisite: BCH3023.

Lab Hours: 64.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BCH3023L

Course User Fee: \$110.00

MCB3023 Principles of Microbiology 3.000 credits

This course offers an introduction to the principles of microbiology. Students will learn the taxonomy, biochemistry, genetics, and ecology of microorganisms and will have an understanding of the impact of microorganisms on the advancement of the biological sciences.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, CHM2211, CHM2211L, Corequisite: MCB3023L

Course User Fee: \$25.00

MCB3023L Principles of Microbiology Lab 2.000 credits

This Laboratory course accompanies MCB3023.

Students will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, enumeration, biochemistry, identification, genetics, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2011, BSC2011L, CHM2211, CHM2211L, Corequisite: MCB3023

Course User Fee: \$110.00

PCB4674 Evolution 3.000 credits

Students will learn the theory of evolution as it pertains to different fields of modern biology including the theory of natural selection, the evidence for evolution, microevolution, speciation, macroevolution, the origin of life on Earth, major evolutionary trends, and the evolution of humans.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, PCB3060, PCB3060L

Course User Fee: \$25.00

BSC4940 Senior Specialty Internship 3.000 credits

This course will provide students with hands-on experience in the biological science workplace by conducting an internship. The experience readies the individual for their first position in-field.

Other Hours: 144.000

Prerequisite: Departmental Approval

Course User Fee: \$181.00

MCB4503 Virology 3.000 credits

This course will cover general virology, including virus structure, replication cycles, infection and mode of transmission of human diseases. Student will learn the major families of the bacterial (bacteriophages), plant and animal viruses and how they influence infection.

Lecture Hours: 48.000

Prerequisite: MCB3023, MCB3023L

Course User Fee: \$25.00

PCB4097 Human Physiology 3.000 credits

The student studies the physiology of organisms major organ systems with emphasis on humans. Students will learn the principles of physics, cell biology, and anatomy in order to explain how the different organs systems work individually and in the context of the whole organism.

Lecture Hours: 48.000

Prerequisite: PHY2054, PHY2054L, BCH3023, BCH3023L

Course User Fee: \$25.00

PCB4233C Fundamentals of Immunology 4.000 credits

Students will learn the immunological processes and concepts as they pertain to human health, disease prevention, development, and treatment. Its primary emphasis is on the cellular and non-cellular components of the immune system, and the ways in which these components interact to provide immunity. This is a combination lecture and lab course.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: MCB3023, MCB3023L

Course User Fee: \$55.00

Program Description

Biological Sciences - Biotechnology Concentration

Bachelor of Science | Code: S9120 | 120 credit

The Bachelor in Science (B.S.) in Biological Sciences is designed to provide students with a broad foundation in biology. Miami Dade College students will explore structure, function, genetics, and evolution of cells and organisms, while being provided the hands-on skills necessary to obtain jobs in the Biological/Life Science fields. Students take the core courses in biology, chemistry, physics and mathematics, and must meet the same General Education Requirements of all students seeking an AA degree. Students may choose from a variety of electives in science and general education to complete their degree requirements.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1102	English Composition 2	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
PCB 3060	Principles of Genetics	3.0
PCB 3060L	Principles of Genetics Laboratory	2.0
Semester 2		Units: 14.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 2053	Physics (without Calculus) 1	3.0
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0
MCB 3023	Principles of Microbiology	3.0
MCB 3023L	Principles of Microbiology Lab	2.0
Semester 3		Units: 10.0
Oral Communications (GE)		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
PHI 2600	Introduction to Ethics	3.0
PHY 2054	Physics (Without Calculus) 2	3.0
PHY 2054L	Physics (Without Calculus) 2 Lab	1.0
Semester 4		Units: 12.0

DEP 2000	Human Growth and Development	3.0
BCH 3023	Introductory Biochemistry	3.0
BCH 3023L	Introductory Biochemistry Laboratory	2.0
BSC 3930	Biological Sciences Seminar	1.0
PCB 4674	Evolution	3.0
Foreign Language Competence*		

*Foreign language is a graduation requirement for the baccalaureate met through 8 credit hours at the elementary 2 level in one foreign language or equivalent. Certain foreign language courses count towards program electives. Students may satisfy equivalence through standardized examinations or successful completion of two credits (two years) in one foreign language at the secondary (high school) level. For additional information, including exemptions for students whose native language is not English, see the Testing and Assessment Department.

Semester 5

Units: 12.0

BSC 4422	Biotechnology Methods and Applications III	3.0
BSC 4422L	Biotechnology Methods and Applications III Lab	2.0
BSC 4940	Senior Specialty Internship	3.0
PCB 4233C	Fundamentals of Immunology	4.0

Total: 61.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

PCB3060 Principles of Genetics 3.000 credits

This course is an introduction to the mechanisms of transmission of hereditary information. Students will learn the classical Mendelian principles of heredity, deviation of Mendelian principles, genetic analysis, linkage and mapping, genetics of populations, gene regulation, mutation, the genetic bases of cancer and other genetic disorders will also be studied.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: PCB3060L

PCB3060L Principles of Genetics Laboratory 2.000 credits

This laboratory course is designed to complement PCB 3060 Principles of Genetics. Students will learn hands-on skills with emphasis on laboratory principles, techniques, and instrumentation within the field of genetics.

Lab Hours: 64.000

Prerequisite: BSC2011, BSC2011L, Corequisite: PCB3060

Course User Fee: \$110.00

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications.

Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

MCB3023 Principles of Microbiology 3.000 credits

This course offers an introduction to the principles of microbiology. Students will learn the taxonomy, biochemistry, genetics, and ecology of microorganisms and will have an understanding of the impact of microorganisms on the advancement of the biological sciences.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, CHM2211, CHM2211L, Corequisite: MCB3023L

Course User Fee: \$25.00

MCB3023L Principles of Microbiology Lab 2.000 credits

This Laboratory course accompanies MCB3023. Students will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, enumeration, biochemistry, identification, genetics, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2011, BSC2011L, CHM2211, CHM2211L, Corequisite: MCB3023

Course User Fee: \$110.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2054 Physics (Without Calculus) 2 3.000 credits

An introduction to the basic principles of physics. PHY 2053 covers mechanics, sound and thermodynamics.

PHY 2054 includes electricity, magnetism and optics.

Lecture Hours: 48.000

Prerequisite: PHY2053, PHY2053L, Corequisite: PHY2054L

Course User Fee: \$10.00

PHY2054L Physics (Without Calculus) 2 Lab 1.000 credits

The physics lab courses are one-credit courses designed to be taken in conjunction with a physics lecture. A different experiment is performed each week with topics chosen to correspond with the material being studied in the lecture. Each experiment is designed to be completed in about 2 contact hours.

Lab Hours: 32.000

Prerequisite: PHY2053, PHY2053L, Corequisite: PHY2054

Course User Fee: \$20.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

BCH3023 Introductory Biochemistry 3.000 credits

This course surveys the fundamental components of biochemistry. In this course, students will learn concepts such as the structure and function of amino acids, proteins, carbohydrates, lipids, and nucleic acids, together with discussions of oxidative metabolism and regulation. Prerequisites: BSC2010, BSC2010L, BSC2011, BSC2011L, CHM2200 or CHM2211, CHM2211L; Corequisite: BCH3023L.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BCH3023L

Course User Fee: \$25.00

BCH3023L Introductory Biochemistry Laboratory 2.000 credits

This laboratory course complements the lecture corequisite BCH 3023, which involves the study of the fundamental components of biochemistry. In this laboratory course, students will learn and will be provided with hands-on experiences with the concepts addressed in the lecture course. Prerequisites: BSC2010, BSC2010L, BSC2011, BSC 2011L, CHM2200 or CHM2211, CHM2211L; Corequisite: BCH3023.

Lab Hours: 64.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BCH3023L

Course User Fee: \$110.00

BSC3930 Biological Sciences Seminar 1.000 credits

This course is designed for biological science majors. Students will gain an understanding of the broad range of career options within the biological sciences. Additionally, students will learn how to read, interpret, discuss, and cite selected examples of scientific literature in different areas of biology.

Lecture Hours: 16.000

Prerequisite: BSC2011, BSC2011L

Course User Fee: \$10.00

PCB4674 Evolution 3.000 credits

Students will learn the theory of evolution as it pertains to different fields of modern biology including the theory of natural selection, the evidence for evolution, microevolution, speciation, macroevolution, the origin of life on Earth, major evolutionary trends, and the evolution of humans.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, PCB3060, PCB3060L

Course User Fee: \$25.00

BSC4422 Biotechnology Methods and Applications III 3.000 credits

This course will explore biotechnology as a science and its implications in modern society. Students will learn how to make well-designed and controlled experiments. Students will also demonstrate knowledge of data acquisition and interpretation.

Lecture Hours: 48.000

Prerequisite: BSC2427, BSC2427L, BCH3023, BCH3023L, PCB3060, PCB3060L, Corequisite: BSC4422L

Course User Fee: \$25.00

BSC4422L Biotechnology Methods and Applications III Lab 2.000 credits

This course provides students with hands-on laboratory experiences to supplement the BSC4422 lecture course. Students will learn how to perform advanced molecular biotechniques that build on previous knowledge. They will perform diagnostic assays, western blots, purifications, etc. and determine how to correlate findings with the basic research or clinical data.

Lab Hours: 64.000

Prerequisite: BSC2427, BSC2427L, BCH3023, BCH3023L, PCB3060, PCB3060L, Corequisite: BSC4422

Course User Fee: \$110.00

BSC4940 Senior Specialty Internship 3.000 credits

This course will provide students with hands-on experience in the biological science workplace by conducting an internship. The experience readies the individual for their first position in-field.

Other Hours: 144.000

Prerequisite: Departmental Approval

Course User Fee: \$181.00

PCB4233C Fundamentals of Immunology 4.000 credits

Students will learn the immunological processes and concepts as they pertain to human health, disease prevention, development, and treatment. Its primary emphasis is on the cellular and non-cellular components of the immune system, and the ways in which these components interact to provide immunity. This is a combination lecture and lab course.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: MCB3023, MCB3023L

Course User Fee: \$55.00

Program Description**Clinical Laboratory Science**

Bachelor of Science | Code: P9211 | 130 credits

The Bachelor of Science (BS) in Clinical Laboratory Science is designed to accommodate the unique demands for entry and advancement within the health science industry. Clinical Laboratory Scientists work with pathologists and other laboratory professionals in performing and analyzing laboratory testing used for diagnosis, treatment, and prevention of disease. Medical Laboratory Sciences program graduates are eligible for certification by the American Association of Bioanalysts (AAB), American Society for Clinical Pathology (ASCP), and American Medical Technologist (AMT). Upon completion of the program and the attainment of certification and license, medical laboratory scientists are employed in a variety of health-care settings as laboratory technologists with the opportunity of upward mobility to supervisory and management positions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 17.0**

HSC 3655	Theoretical Foundations of Health Care Ethics	3.0
MLS 4621	Clinical Biochemistry	4.0
MLS 4705	Laboratory Operations and Management	3.0
<i>Social Science (GE)</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
<i>Lower Division Eletives</i>		4.0
CHM 1020	General Education Chemistry	3.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
CHM 1046	General Chemistry And Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
CHM 2200	Survey of Organic Chemistry	3.0
CHM 2200L	Survey of Organic Chemistry Laboratory	1.0
CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0
CHM 2211	Organic Chemistry 2	3.0

CHM 2211L	Organic Chemistry 2 Laboratory	2.0
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Semester 2	Units: 17.0
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ENC 1102	English Composition 2	3.0
MLS 4306	Clinical Hematology	3.0
MLS 4335	Clinical Hemostasis	3.0
MLS 4506	Clinical Immunology	3.0
<i>Lower Division Eletives</i>		5.0
BSC 1005	General Education Biology	3.0
BSC 1005L	General Education Biology Laboratory	1.0
BSC 1084	Functional Human Anatomy	3.0
BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
BSC 2020	Human Biology: Fundamentals of Anatomy/Physiology	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0
BSC 2426	Biotechnology Methods and Applications 1	3.0
BSC 2426L	Biotechnology Methods and Applications 1 Laboratory	2.0
BSC 2427	Biotechnology Methods and Applications 2	3.0
BSC 2427L	Biotechnology Methods and Applications 2 Laboratory	2.0
BSC 2943L	Bioscience Internship	3.0

Semester 3	Units: 14.0
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HSC 3701	Leadership and Management in Healthcare	3.0
MLS 4221	Clinical Urinalysis	3.0
<i>Humanities (GE)</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
Lower Division Electives*		5.0

*Select 5 credits from the following course options: BSC 1005, BSC 1005L, BSC 1084, BSC 2010, BSC 2010L, BSC 2011, BSC 2011L, BSC 2020, BSC 2085, BSC 2085L, BSC 2086, BSC 2086L, BSC 2426, BSC 2426L, BSC 2427, BSC 2943L, BSC 2949

Semester 4	Units: 15.0
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MLS 4193	Clinical Molecular Diagnostics	3.0
MLS 4461C	Clinical Diagnostic Microbiology	3.0
MLS 4630	Clinical Chemistry	3.0
<i>Mathematics (GE)</i>		3.0
MGF 1131	Mathematics in Context	3.0

OR		
STA 2023	Statistical Methods*	3.0
<i>Oral Communications (GE)</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

*Recommended Course: STA 2023 - Statistical Methods

Semester 5		Units: 20.0
HSC 3057	Introduction to Research Methods in Health Care	3.0
MLS 3150	Special Topics in Medical Laboratory Sciences	3.0
MLS 4552	Clinical Immunohematology	3.0
Lower Division Electives*		5.0
Electives**		6.0

*Select 5 credits from the following course options: CHM 1020, CHM 1033, CHM 1033L, CHM 1045, CHM 1045L, CHM 1046, CHM 1046L, CHM 2200, CHM 2200L, CHM 2210, CHM 2210L, CHM 2211, CHM 2211L

**Select 6 credits from the following course options: AMH*, AML*, ANT*, ARC*, ARH*, ASL*, AST*, BOT*, BSC*, CHI*, CHM*, CLP*, CRW*, DAN*, DEP*, ECO*, EDF*, ENC*, ENG*, ENL*, ESC*, EUH*, EVR*, FRE*, GER*, GLY*, HLP*, HUM*, HUN*, INR*, ITA*, JPN*, LAH*, LIT*, MAC*, MAD*, MAP*, MAS*, MCB*, MET*, MGF*, MUH*, MUL*, OCB*, OCE*, PHI*, PHY*, POR*, POS*, PSC*, PSY*, QMB*, REL*, RUS*, SOP*, SPC*, SPN*, STA*, THE*, WOH*

Total: 83.0

HSC3655 Theoretical Foundations of Health Care Ethics 3.000 credits

This course will cover the Theoretical Foundations of Health Care Ethics. The student will learn how to apply the core principles of bioethics to the medical and health care decision-making process. (3 cr. Lecture)

Lecture Hours: 48.000

MLS4621 Clinical Biochemistry 4.000 credits

This course provides the student with the knowledge and understanding of clinical disorders and how biochemical factors and laboratory methods are used for the investigation, diagnosis and management of patients. (4 hour lecture)

Lecture Hours: 64.000

Course User Fee: \$41.00

MLS4705 Laboratory Operations and Management 3.000 credits

This course provides students with Quality Management skills necessary in the medical laboratory. Students are exposed to organizational structure along with principles for leadership and managerial decision making and process improvement along with different principles used in Laboratory Instrumentation. (3 Hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions.

Corequisites: CHM 1033L, MAT 1033 (3 hr. lecture)

Lecture Hours: 48.000

Corequisite: CHM1033L, Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite: MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

CHM 1045 is the first semester of a two-semester general chemistry course for science, premedical science and engineering students. Students will learn stoichiometry, atomic structure, completing and balancing chemical reactions, nomenclature, bonding, acid-base theories, solutions, gas laws and beginning thermodynamics. Special fee.

Lecture Hours: 48.000

Prerequisite: CHM1025or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

CHM1046 General Chemistry And Qualitative Analysis 3.000 credits

CHM 1046 is the second course in the CHM 1045-1046 sequence. Students will learn major topics in modern chemistry including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria and electrochemistry. Special fee. Prerequisite: CHM1045; Corequisite: CHM1046L (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

CHM2200 Survey of Organic Chemistry 3.000 credits

XXXX This one-semester course briefly examines the structure, synthesis, nomenclature and reactivity of selected mono-and-poly-functional organic compounds. Theories that relate the structure of organic molecules to their chemically reactivity will be presented as a unifying principle. Prerequisite: CHM 1046 and CHM 1046L; Corequisite: CHM 2200L. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2200L

CHM2200L Survey of Organic Chemistry Laboratory 1.000 credits

Experiments and exercises will be conducted to introduce students to the basic laboratory techniques that are used in organic chemistry and that re-enforce and illustrate several important topics in organic chemistry. Prerequisite: CHM 1046 and CHM 1046L; Corequisite: CHM 2200. Special fee. (2 hr. lab)

Lab Hours: 32.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2200

Course User Fee: \$46.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds. Prerequisite: CHM1046 and CHM 1046L; Corequisite: CHM2210L

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

CHM2211 Organic Chemistry 2 3.000 credits

In organic chemistry 2, students will learn about nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds. Prerequisite: CHM 2210 and CHM 2210L; Corequisite: CHM 2211L. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211L

Course User Fee: \$15.00

CHM2211L Organic Chemistry 2 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2211. Topics include nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds will be performed in a laboratory setting.

Lab Hours: 64.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211

Course User Fee: \$45.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MLS4306 Clinical Hematology 3.000 credits

This course is the study of the composition and function of blood; diseases related to blood disorders. Students will receive the necessary skills in the application of hematology diagnostic procedures, interpretation, problem solving and correlation of laboratory findings with disease states (3 Hour Lecture).

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4335 Clinical Hemostasis 3.000 credits

This course Provides an overview application of hemostasis (coagulation), as it relates to the medical laboratory. Presents coagulation laboratory principles with hemostasis diagnostic procedures, interpretation, problem solving and correlation of laboratory findings and results in accordance to the disease states. (3 Hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4506 Clinical Immunology 3.000 credits

Clinical Immunology will provide an overview of immunology concepts and the theory of some immunologic procedures. The immunologic manifestation of infectious disease and immune disorders will also be covered. (3 Hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC1005L General Education Biology Laboratory 1.000 credits

This is an optional one-credit lab to provide students with experience in the scientific process. This course applies the scientific method to critically examine and explain the natural world, including but not limited to cells, organisms, genetics, evolution, ecology, and behavior.

Lab Hours: 32.000

Pre/Corequisite: BSC1005

Course User Fee: \$25.00

BSC1084 Functional Human Anatomy 3.000 credits

This course covers basic human anatomy for students in allied health and mortuary science programs. It includes the dynamics of gross and functional anatomy, terminology, body orientation, and systematic relationships.

Lecture Hours: 48.000

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

BSC2020 Human Biology: Fundamentals of Anatomy/Physiology 3.000 credits

This course provides a basic understanding of the human body, its systems, and their functions. It includes the dynamics of physiology, terminology, and physiological relationships of the body systems.

Lecture Hours: 48.000

Course User Fee: \$10.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Lab 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

BSC2426 Biotechnology Methods and Applications 1 3.000 credits

This course addresses the basic principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical, biotechnology and/or research laboratory setting(s). Practical applications of biotechnology are explored.

Lecture Hours: 48.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426L

BSC2426L Biotechnology Methods and Applications 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2426 Biotechnology Methods and Applications 1. This is a hands-on course that emphasizes the basic laboratory principles, techniques, and instrumentation, necessary for effective work in pharmaceutical, biotechnology, and/or research laboratory settings(s).

Lab Hours: 64.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426

Course User Fee: \$50.00

BSC2427 Biotechnology Methods and Applications 2 3.000 credits

This course addresses advanced principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical-biotechnology and/or research-laboratory setting(s). The following areas of contemporary biotechnology are covered: forensics, bioremediation, and animal-, plant-, and marine biotechnology.

Lecture Hours: 48.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427L

BSC2427L Biotechnology Methods and Applications 2 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2427 Biotechnology Methods and Applications 2. This is a hands-on course that emphasizes advanced laboratory principles, techniques, and instrumentation necessary for effective work in a pharmaceutical, biotechnology, and/or research-laboratory setting(s).

Lab Hours: 64.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427

Course User Fee: \$50.00

BSC2943L Bioscience Internship 3.000 - 6.000 credits

This internship course is a capstone for students majoring in bioscience and related programs. Students will learn to apply acquired knowledge and skills to gain experience in the bioscience workplace.

Other Hours: 144.000 - 288.000

Course User Fee: \$21.00

HSC3701 Leadership and Management in Healthcare 3.000 credits

This is a leadership and management course that will examine leadership as a process with a tri-fold focus: the leader, the followers, and the situation. The student will learn leadership theories and build leadership skills. (3 hr. lecture)

Lecture Hours: 48.000

MLS4221 Clinical Urinalysis 3.000 credits

The study of body fluids for physical health and identification of abnormalities in relation to disease states. (3 hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

MLS4193 Clinical Molecular Diagnostics 3.000 credits

Clinical molecular diagnostics course provides an introduction to molecular analysis of biological markers in clinical samples to aid in the diagnosis, monitoring and treatment of diseases (3 Hour Lecture).

Lecture Hours: 48.000

Course User Fee: \$41.20

MLS4461C Clinical Diagnostic Microbiology 3.000 credits

Clinical Diagnostic Microbiology provides concepts in bacteriology identification methods, rapid identification methods for parasites and fungi and an overview of virology methodology.

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4630 Clinical Chemistry 3.000 credits

The study of the concepts and principles of Clinical Chemistry. Analytes and lab values are correlated to normal homeostasis and disease states. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

HSC3057 Introduction to Research Methods in Health Care 3.000 credits

This course will provide an overview of research methods used in healthcare. Students will learn the use of effective inquiry through research strategies that address healthcare issues with logical and observational rigor. Students will learn the rudiments relative to the evaluation of research literature, research design and the application of research methods to the clinical setting. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: STA2023

MLS3150 Special Topics in Medical Laboratory Sciences 3.000 credits

This course stresses the importance of evidence-based practice in the medical laboratory sciences field. Students will be presenting case studies to the faculty and peers in the program. Instruction will emphasize professional, legal and ethics issues affecting the medical laboratory science field. Students will review the material covered in the program to prepare for the comprehensive assessment. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$191.14

MLS4552 Clinical Immunohematology 3.000 credits

The study of concepts related to the blood group systems, blood antigens and antibodies. The student will analyze the principles, procedures and clinical significance of testing in genetics, pre-transfusion testing, adverse effects of transfusions, donor selection and components, and hemolytic disease of the newborn. (3 hour lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

Program Description

Cybersecurity

Bachelor of Science | Code: S9515 | 120 credits

The Bachelor of Science (BS) in Cybersecurity degree is designed to help meet the local need for cyber security professionals. Students in this program gain detailed understanding and hands-on skills regarding the tools and protocols needed to use and manage cybersecurity infrastructure, risks, and vulnerabilities in real-world situations. The program curriculum includes courses in network defense, penetration testing, computer and network forensics, risk management and ethics, among others. These courses prepare students for multiple industry certifications such as Certified Ethical Hacker, Computer Hacking Forensics Investigator and Certified Information Security Manager.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 14.0

ENC 1102	English Composition 2	3.0
<i>Mathematics (GE)</i>		3.0
MAC 1105	College Algebra *	3.0
OR		
STA 2023	Statistical Methods	3.0
CIS 3215	Ethics in CyberSecurity	4.0
CIS 3360	Principles of Information Security	4.0

Note: MAC 1105 may be replaced by higher-level mathematics with prefix MAC, MAD*, MAS*, or MAP*.

Semester 2

Units: 12.0

CIS 3361	Information Security Management	4.0
CIS 4204	Ethical Hacking I	4.0
CIS 4366	Computer Forensics	4.0

Semester 3 **Units: 8.0**

CIS 4378	Ethical Hacking II	4.0
CIS 4388	Advanced Computer Forensics	4.0
Foreign Language Competence*		

*Foreign language is a graduation requirement for the baccalaureate met through 8 credit hours at the elementary 2 level in one foreign language or equivalent. Certain foreign language courses count towards program electives. Students may satisfy equivalence through standardized examinations or successful completion of two credits (two years) in one foreign language at the secondary (high school) level. For additional information, including exemptions for students whose native language is not English, see the Testing and Assessment Department.

Semester 4 **Units: 14.0**

CIS 4364	Intrusion Detection and Incident Response	4.0
CNT 3409C	Network Security	4.0
<i>Oral Communications (GE)</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science (GE)</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
POS 2041	American Federal Government	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5 **Units: 13.0**

CIS 4891	Cyber Security Capstone	4.0
<i>Humanities (GE)</i>		3.0
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
<i>Natural Science (GE)</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		

*Students are encouraged to consult with an academic advisor to select approved courses that align with their interests and fulfill remaining graduation requirements.

Total: 61.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CIS3215 Ethics in CyberSecurity 4.000 credits

This course provides the study of the risk factors for digital and ethical misconduct, and it explores ethics, relevant laws, regulations, policies, standards, moral, and social issues and responsibilities faced by CyberSecurity professionals. Coverage includes examination of CyberSecurity policies; Federal Laws and Authorities and International Standards; ethical and legal compliance and enforcement; business issues; contractual management of assets and liabilities; and issues involving privacy, disclosure, free speech and individual rights.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS3360 Principles of Information Security 4.000 credits

This upper division course provides an overview of information systems security principles, practices, methods, and tools. Students will learn about the threats against an organization's digital assets, as well as the tools and methods to mitigate those threats. Topics include cryptography, authentication and access control systems, infrastructure security, and ethical issues. Prerequisite: CTS1134 or CTS1650
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS1134 OR CTS1650
Course User Fee: \$63.00

CIS3361 Information Security Management 4.000 credits

This course covers how to manage, design, oversee and assess an organization's information security. Students will learn how to develop an information security strategy, how to write information security policies, and how to manage information risk. Other topics include security program development and management, business continuity planning and disaster recovery planning. Prerequisite(s): CIS 3360 or CTS 1120. Laboratory fee. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS3360 OR CTS1120
Course User Fee: \$63.00

CIS4204 Ethical Hacking I 4.000 credits

This upper division course introduces students to penetration testing techniques. Students will learn how to footprint, scan, sniff, enumerate networks, and hack systems. Other topics include denial of service attacks, social engineering, malware, vulnerability analysis and relevant laws. Prerequisites: CIS3360
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS3360
Course User Fee: \$63.00

CIS4366 Computer Forensics 4.000 credits

This upper division course provides the student with knowledge and skills to conduct formal incident investigations. Students will learn how to collect and analyze evidence from Windows and Linux computer systems. Other topics include legal issues, evidence analysis, and report writing. Prerequisite(s): CIS 3360 or CET 2880C. Laboratory fee. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS3360 OR CET2880C
Course User Fee: \$63.00

CIS4378 Ethical Hacking II 4.000 credits

This upper division course is a continuation of Ethical Hacking I. Students will focus on how web applications, wireless networks, and mobile platforms can be hacked, and how intrusion detection systems (IDS), firewalls, and honeypots can be evaded. Other topics include cloud computing security, Internet of Things (IoT) security, and cryptography. Prerequisite: CIS4204.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS4204
Course User Fee: \$63.00

CIS4388 Advanced Computer Forensics 4.000 credits

This upper division course is a continuation of Computer Forensics. The course examines forensics techniques necessary to investigate and analyze network traffic. The course covers packet capture and analysis, log file analysis, and flow analysis. Other topics include mobile forensics, cloud forensics, malware forensics, database forensics, and investigating email crimes and web attacks. Prerequisite: CIS 4366 (3 hr. lecture, 2 hr. lab).
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS4366
Course User Fee: \$63.00

CIS4364 Intrusion Detection and Incident Response 4.000 credits

This upper division course addresses the underlying principles and techniques for detecting and responding to current and emerging cybersecurity threats. Students will learn how to handle various types of malware, email, web, network, cloud and internal network incidents, as well as risk assessment methodologies, and policies related to incident handling. Prerequisite: CIS 3360.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS3360
Course User Fee: \$63.00

CNT3409C Network Security 4.000 credits

This upper division course introduces students to current and emerging threats to the security of computer networks, as well as tools and techniques for the prevention, detection and recovery from such attacks. Topics include firewalls, intrusion detection and intrusion prevention systems, virtual private networks, remote authentication and authorization systems, and security protocols. Prerequisite(s): CIS 3360. Laboratory fee. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 27.000 - 32.000
Prerequisite: CIS3360
Course User Fee: \$63.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

CIS4891 Cyber Security Capstone 4.000 credits

This upper-division course requires students to apply the knowledge and skills acquired for a cyber security project. Students will assess risk and define the cyber security requirements for a real-world scenario. Then they will design, implement and test the necessary cyber defenses to reduce the risk to an acceptable level in an emulated IT environment. Must be taken during the last semester before graduation and with a departmental permission. Prerequisite(s): Senior status. Department approval required. Laboratory fee. (3 hr. lecture; 2 hr. lab). Lecture Hours: 48.000, Lab Hours: 32.000 Prerequisite: Departmental approval required. *Course User Fee: \$63.00*

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

Program Description

Data Analytics

Bachelor of Science | Code: S9510 | 120 credit

The Bachelor of Science (BS) in Data Analytics program is designed to train and supply a workforce of skilled graduates in data manipulation and analysis across a spectrum of industries. Through the cross-disciplinary curriculum, students will learn to clean, organize, analyze, and interpret unstructured data, to derive knowledge and communicate discoveries using sophisticated visualization techniques. Students will demonstrate competence with fundamental algorithmic approaches to analyzing large data sets.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 14.0

ENC 1102	English Composition 2	3.0
Oral Communications (GE)		3.0

SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
CAP 3321C	Data Wrangling	4.0
CAP 4767	Data Mining	4.0

Semester 2

Units: 12.0

CAP 4744	Data Visualization	4.0
CAP 4784	Big Data	4.0
<i>Statistics Selection</i>		4.0
CAP 3330	Programming R for Statistics	4.0
OR		
STA 3164	Statistical Methods II	4.0

Semester 3

Units: 7.0

<i>Humanities (GE)</i>		3.0
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
<i>Data Analytics Options</i>		4.0
CAP 4936	Special Topics in Data Analytics	4.0
OR		
CIS 3368	Data Security & Governance	4.0

Foreign Language Competence *

*Foreign language is a graduation requirement for the baccalaureate met through 8 credit hours at the elementary 2 level in one foreign language or equivalent. Certain foreign language courses count towards program electives. Students may satisfy equivalence through standardized examinations or successful completion of two credits (two years) in one foreign language at the secondary (high school) level. For additional information, including exemptions for students whose native language is not English, see the Testing and Assessment Department.

Semester 4

Units: 14.0

CAP 4631C	Machine Learning for Data Analytics I	4.0
CAP 4633C	Machine Learning for Data Analytics II	4.0
<i>Social Science (GE)</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
POS 2041	American Federal Government	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
General Education Elective (GE)*		3.0

*Students are encouraged to consult with an academic advisor to select approved courses that align with their interests and fulfill remaining graduation requirements.

Semester 5

Units: 13.0

<i>Natural Science (GE)</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1025	Basic Physics	3.0
CAP 4910	Data Analytics Capstone	4.0
Data Analytics Options*		4.0
Program Electives**		2.0

*Select 4 credits from the following course options: CAP 4936 or CIS 3368. Students may take CAP 4936 up to two times to fill the Topics in Data Analytics Upper Division Requirements (eight credits), just as long as the Topics are different. This course may additionally be taken under the Program Electives Block.

**Select 1-4 credits from the following course options: ACG 1403, ACG 2450, CAI 1001C, CAI 1320C, CAI 2100C, CAI 2300C, CAI 2840C, CAP 2743C, CAP 2791C, CAP 3794C, CAP 4936, COP 2041C, CIS 1949, CIS 2900, COP 3530, CTS 1145, CTS 2192C, FIN 3400, GEB 1011, GEB 1432, MAC 1114, MAC 1140, MAC 1147, MAC 2311, MAC 2312, MAN 3025, MAN 4720, MAS 3105, QMB 2100

Total: 60.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

CAP3321C Data Wrangling 4.000 credits

This course offers a broad introduction to data wrangling, data retrieval, and the use of Python in data analytics. Students will use Python and other business intelligence tools to retrieve data from various sources, clean the dataset, and prepare it for data analysis tasks. Prerequisites: CAP1788 and CAP2704.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND IDC2132

Course User Fee: \$63.00

CAP4767 Data Mining 4.000 credits

Students will learn how to extract useful data from an organization's data warehouse and analyze it to uncover hidden patterns and relationships. An emphasis will be placed on unsupervised data mining techniques, as well as the use of the RapidMiner software. Prerequisites: CAP1788 and CAP2761C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND CAP2761C

Course User Fee: \$63.00

CAP4744 Data Visualization 4.000 credits

This course focuses on creating complex and informative visualizations using Tableau. Students will learn how to formulate business questions, organize complex and large datasets, analyze and interpret data, deliver actionable insights, and publish the results. Coverage includes preparation for intermediate-level Tableau certifications.

Prerequisites: CAP1788 and CAP2761C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND IDC2132

Course User Fee: \$63.00

CAP4784 Big Data 4.000 credits

Course Description: This course focuses on the processing of massive datasets, both structured and unstructured. Students will learn how to use Databricks and Spark to manage and analyze large datasets from a variety of sources. In addition, student will gain an understanding of how Databricks supports the end-to-end data science workflows that allow users to extract and share business insights.

Prerequisites: CAP1788, CAP2761C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND IDC2132

Course User Fee: \$63.00

CAP3330 Programming R for Statistics 4.000 credits

This upper division course is for students majoring in data analytics. Students will learn the R programming language and use it to perform intermediate-level statistical analysis. Techniques used in data analysis, such as analysis of variance and regression, will be emphasized. Prerequisite: STA2023.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: STA2023

Course User Fee: \$63.00

STA3164 Statistical Methods II 4.000 credits

This course is for students majoring in Data Analytics, Systems Engineering, and related disciplines which require advanced skills in statistical analysis. Students will learn how to perform tests of variance, analysis of variance, analysis of covariance, regression, correlation, and non-parametric statistics.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: STA2023

Course User Fee: \$50.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy.

Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

CAP4936 Special Topics in Data Analytics 4.000 credits

This upper division course is for students majoring in data analytics. The course centers around topics of current interest or of special interest to students or instructors. Topics or focus may vary from semester to semester. Prerequisites: CAP1788, CAP2761C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788, CAP2761C

Course User Fee: \$63.00

CIS3368 Data Security & Governance 4.000 credits

This upper division course is for students majoring in Data Analytics. Students will gain an understanding of how analytics can be applied to a variety of security-related problems across organizations. In addition, students will explore various ethical, legal, and data governance issues that affect data analysts. (3 hr. lecture; 1 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

CAP4631C Machine Learning for Data Analytics I 4.000 credits

This upper division course is for students majoring in data analytics. Students will learn why machine learning is crucial for data analytics and why regression analysis is a foundation of supervised machine learning. Using Python programming, students will use a variety of packages to create regression models that make predictions.

Prerequisites: COP1047C, STA3164 or CAP3330.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND STA3164 OR CAP3330

Course User Fee: \$63.00

CAP4633C Machine Learning for Data Analytics II 4.000 credits

This upper division course is for students majoring in data analytics. In this second-level course, students will use the Python programming language to create additional machine learning models for classification. In addition, students will explore various applications of multi-layer neural networks. Prerequisites: CAP4631C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP4631C

Course User Fee: \$63.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

CAP4910 Data Analytics Capstone 4.000 credits

This upper division course is for students majoring in data analytics. Students will initiate a business-driven data analytics solution to a real-world problem utilizing acquired skills in statistical analysis, machine learning, data mining, and data visualization. Must be taken during the last semester before graduation.

Prerequisite: CAP4631C.

Lecture Hours: 48.000, Lab Hours: 32.000 - 28.000

Prerequisite: CAP4631C

Course User Fee: \$63.00

Program Description

Dental Hygiene

Bachelor of Science | Code: S9300 | 120 credits

The Bachelor of Science (B.S.) in Dental Hygiene is designed to accommodate the unique demands for entry and advancement within the health science industry. Upon completion of the program and the attainment of licensure, dental hygienists are employed in a variety of health-care settings with the opportunity of upward mobility to supervisory and management positions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 17.0
ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
DEH 3703	Oral Health Care: Determinants and Directions	3.0
DEH 3814	Dental Hygiene and Population Health Considerations	3.0
DEH 4740	Introduction to Health Concepts in Dental Hygiene	3.0
Electives*		2.0

*Note: Check with advisor for appropriate course (Credit Type 01 or Credit Type 02) selection.

Semester 2		Units: 18.0
<i>Oral Communications</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
DEH 4730	Dental Hygiene Educational Concepts	3.0
DEH 4851	Dental Hygiene Practice Management	3.0
DEH 4892	Principles of Professionalism in Dental Hygiene	3.0
HSC 3057	Introduction to Research Methods in Health Care	3.0
Electives*		3.0

*Note: Check with advisor for appropriate course (Credit Type 01 or Credit Type 02) selection.

Semester 3		Units: 15.0
STA 2023	Statistical Methods	3.0
DEH 4813	Dental Hygiene: Emerging Topics and Techniques	3.0
DEH 4930	Dental Hygiene Special Topics	3.0
HSC 3701	Leadership and Management in Healthcare	3.0
Electives*		3.0

*Note: Check with advisor for appropriate course (Credit Type 01 or Credit Type 02) selection.

Semester 4		Units: 17.0
<i>Social Science</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
DEH 4854	Leadership in Dental Hygiene	3.0
DEH 4947	Dental Hygiene Capstone	5.0
Electives*		6.0

*Note: Check with advisor for appropriate course (Credit Type 01 or Credit Type 02) selection.

Total: 67.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

DEH3703 Oral Health Care: Determinants and Directions 3.000 credits

This course introduces theoretical frameworks for understanding how social, economic, environmental, and political factors influence oral health outcomes and dental hygiene practice. Students will examine the relationship between these determinants and oral health, as well as their impact on access to care and community health status. The course emphasizes the development of policy at the community, state, and national levels, and includes analysis of current research and case studies relevant to dental hygiene and public health. Students will gain skills in assessing social issues affecting oral health and in understanding how to advocate for effective dental public health policies.

Lecture Hours: 48.000

DEH3814 Dental Hygiene and Population Health Considerations 3.000 credits

This course provides an in-depth exploration of the health and healthcare needs specific to groups at elevated risk for oral health challenges. Students will examine the unique circumstances affecting these groups at the psychosocial and physiological levels and their impact on oral health. Through analysis of current trends, students will identify factors contributing to compromised oral health and explore strategies aimed at reducing risk and supporting health promotion, maintenance, and restoration within the context of dental hygiene practice.

Lecture Hours: 48.000

DEH4740 Introduction to Health Concepts in Dental Hygiene 3.000 credits

This foundation course introduces the study of dental global health and its theories, perspectives, methods, and evidence. The course offers a basic overview of the methods for measuring population health, analytic tools for decision-making, and the evidence base for the effectiveness, risks, and efficiency of interventions, programs, and policies. The course focuses on challenges, diseases, and conditions of major public health importance, their patterns of global mortality and morbidity, their social, economic, and political determinants and consequences, their inequalities and inequities, and the population-based interventions as well as social and institutional responses to these challenges structured through communities, national health systems as well as international agencies, consortia, and alliances.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

DEH4730 Dental Hygiene Educational Concepts 3.000 credits

This course is designed to introduce the principles of learning and teaching, especially as they apply to allied health professions such as staff training and patient education. Topics include understanding various learning theories, learning styles, motivational strategies, collaborative learning, the influence of diversity, and using technology to support learning. The course introduces students to the career option of education as a profession and provides a foundation for understanding the relationship between planning, implementing, and assessing student learning. Dental hygiene and adult education theory and concepts are the major components of this specialization.

Lecture Hours: 48.000

DEH4851 Dental Hygiene Practice Management 3.000 credits

This course is designed to assist the dental hygienist in understanding and applying the principles of management related to a dental hygiene practice. The concepts of patient, business office, and dental team management will be explored. Strategies will be presented on the marketing and promoting of dental services and products to the general public.

Lecture Hours: 48.000

DEH4892 Principles of Professionalism in Dental Hygiene 3.000 credits

The student will be able to demonstrate knowledge of the dental hygiene code of ethics with emphasis on cultural competence as related to all professional encounters.

Lecture Hours: 48.000

HSC3057 Introduction to Research Methods in Health Care 3.000 credits

This course will provide an overview of research methods used in healthcare. Students will learn the use of effective inquiry through research strategies that address healthcare issues with logical and observational rigor. Students will learn the rudiments relative to the evaluation of research literature, research design and the application of research methods to the clinical setting. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: STA2023

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

DEH4813 Dental Hygiene: Emerging Topics and Techniques 3.000 credits

This course examines current societal and professional issues and their impact on dental hygiene practice. Students will examine the roles of the dental hygienist and discuss the dental hygienists' role in increasing access to dental care. Students will research and compare traditional and alternative practice models and propose changes to improve dental care delivery. It also discusses changing technology in dentistry and dental hygiene; political advocacy; demographic shifts; ethics and professionalism; interprofessional education; and global perspectives of dental hygiene.

Lecture Hours: 48.000

DEH4930 Dental Hygiene Special Topics 3.000 credits

This online course is designed to help the dental hygiene student identify and explore current issues and topics related to dentistry and specifically to dental hygiene. The course will take the format of a study club and will involve student participation in the identification and discussion of the topic selected.

Lecture Hours: 48.000

HSC3701 Leadership and Management in Healthcare 3.000 credits

This is a leadership and management course that will examine leadership as a process with a tri-fold focus: the leader, the followers, and the situation. The student will learn leadership theories and build leadership skills. (3 hr. lecture)

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

DEH4854 Leadership in Dental Hygiene 3.000 credits

This course focuses on the theories, concepts and principles of leadership. Emphasis will be on the development of leadership skills related to personal behavior, communication, organization and self-evaluation. This course explores opportunity to develop leadership roles appropriate to the dental hygiene profession.

Lecture Hours: 48.000

DEH4947 Dental Hygiene Capstone 5.000 credits

This course is designed to provide the student with an opportunity to apply their knowledge of a chosen professional role through teaching, internship, observation or work experience. This course will use selected sites on and off campus to develop the expanding role of the dental hygienist in the health care system.

Lecture Hours: 48.000

Program Description

Early Childhood Education

Bachelor of Science | Code: S9270 | 120 credit

Upon completion of the Bachelor of Science Degree with a major in Early Childhood Education program, the student will be eligible to obtain a Florida Educator Certification in Pre-school (Birth to Age 4) and Pre-Kindergarten/ Primary (Age 3 to Grade 3) with endorsements in English for Speakers of Other Languages (ESOL), Reading, and Pre-Kindergarten Disabilities.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Lower Division		Units: 6.0
EDF 2144	Maximizing Student Potential in the School Context	3.0
		3.0
EME 2040	Introduction to Technology for Educators	3.0
OR		
EME 2071	Educating Young Children for Digital Literacy in the 21st Century	3.0
Semester 1		Units: 12.0
EDF 3115	Child Development for ECE Teachers	3.0
EEC 3301	Teaching Methods in Early Childhood Education	3.0
EEC 3613	Observation and Assessment of Young Children	3.0
EEX 3201	Young Children with Special Needs	3.0
Semester 2		Units: 12.0
EEC 3211	STEM for Early Childhood Education I	3.0
EEC 3213	Language, Literature, and Emergent Literacy	3.0
EEC 3412	Family and Community Partnerships in Early Childhood Education	3.0
EEX 3603	Positive Behavior Supports in All Settings	3.0
Semester 3		Units: 12.0
EEC 4219C	Science, Technology, and Mathematics (STEM) Methods for ECE II	3.0
EEX 3226	Assessment of All Young Children	3.0
RED 3009	Emergent and Early Literacy	3.0
TSL 3080	ESOL in Early Childhood Education I	3.0
Semester 4		Units: 12.0
EEC 3212	Integrated Social Sciences, Humanities, and Arts	3.0
EEC 4268	Practicum in Early Childhood Education	3.0
LAE 4211	Methods and Resource for Literacy Development in Young Children	3.0
TSL 4310	ESOL in Early Childhood Education II	3.0
Semester 5		Units: 12.0
EEC 4936	Student Teaching Seminar: ECE	2.0
EEC 4940	Student Teaching in Early Childhood Education	10.0
		Total: 66.0

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

EME2040 Introduction to Technology for Educators 3.000 credits

The student will learn instructional design principles for the use of technology in teaching and learning. The student will participate in hands-on experiences with educational media, emerging technologies, hardware, software, and peripherals for the personal computer will be provided. The student will use data driven decision-making to identify appropriate software for classroom applications.

Lecture Hours: 48.000

Prerequisite: CGS1060C

Course User Fee: \$18.00

EME2071 Educating Young Children for Digital Literacy in the 21st Century 3.000 credits

This course provides an overview of appropriate use of technology in early childhood settings (ages 3-8). The student will learn how to make informed decisions regarding the intentional use of technology in ways that support young children's learning and development.

Lecture Hours: 48.000

Course User Fee: \$25.00

EDF3115 Child Development for ECE Teachers 3.000 credits

This course provides an overview of human life from fertilization through eight years of age. The student will examine growth and developmental characteristics during the prenatal, infancy, and early childhood periods. The student will learn to analyze typical and atypical development, developmental theories, learning theories, brain research, attachment, and relationships.

Lecture Hours: 48.000

Course User Fee: \$18.00

EEC3301 Teaching Methods in Early Childhood Education 3.000 credits

This course will introduce instructional models to design lesson plans and instruction based on state standards. The student will incorporate educational theories to develop strategies for early childhood serving all learners. (Ten hours of clinical experience required in an approved kindergarten-third grade setting.)

Lecture Hours: 48.000

Pre/Corequisite: EDF3115 and 10 hrs. of Field Experience Required.

Course User Fee: \$18.00

EEC3613 Observation and Assessment of Young Children 3.000 credits

This course will provide an overview of the process and importance of observing, documenting, and interpreting the behavior of young children. The student will learn and apply various methods to document the ongoing development of children and the value of using this information to support and plan for learning experiences. (Ten hours of clinical experience in an approved birth-age 1 least restrictive setting).

Lecture Hours: 48.000

Prerequisite: EEC1000

EEX3201 Young Children with Special Needs 3.000 credits

This course provides an overview of issues related to young children (birth through age eight) with special needs. The student will learn about possible causes and characteristics of exceptionalities, federal laws pertaining to students with disabilities, the referral process, educational interventions, available resources, and advocacy (Ten hours of clinical experience required in an approved birth - age four least restrictive setting).

Lecture Hours: 48.000

EEC3211 STEM for Early Childhood Education I 3.000 credits

The student will learn to use scientific and mathematical educational research-based methods and strategies to teach inquiry and problem-solving skills and plan activities for young children that foster exploration in the sciences. (Ten hours of clinical experience required in an approved prekindergarten: 1 observation required).

Lecture Hours: 48.000

Prerequisite: EEC3301 and 10 hrs. of Field Experience Required

Course User Fee: \$18.00

EEC3213 Language, Literature, and Emergent Literacy 3.000 credits

This course focuses on the development of language and literacy in children ages birth through age five. The student will learn the process of emergent literacy including vocabulary and language development, phonological awareness, alphabetic and print knowledge, and comprehension. The student will learn to design developmentally appropriate and effective emergent literacy activities in relation to the early learning and developmental standards. (Ten hours of clinical experience required in an approved pre-kindergarten inclusion setting.) This course focuses on the development of language and literacy in children ages birth through age five. The student will learn the process of emergent literacy including vocabulary and language development, phonological awareness, alphabetic and print knowledge, and comprehension. The student will learn to design developmentally appropriate and effective emergent literacy activities in relation to early learning and developmental standards. (Ten hours of clinical experience required in an approved pre-kindergarten least restrictive setting.)

Lecture Hours: 48.000

Pre/Corequisite: EEC1000

EEC3412 Family and Community Partnerships in Early Childhood Education 3.000 credits

This course focuses on family engagement and community involvement in early childhood education. The student will identify strategies for establishing and maintaining reciprocal relationships with all parents, families, and communities. The student will explore ways to connect students, parents, families, the school, and the community as a partnership to support student learning and well-being.

Lecture Hours: 48.000

Prerequisite: EEC1000

EEX3603 Positive Behavior Supports in All Settings 3.000 credits

This course provides a holistic approach to guiding young children's behavior and facilitating character education. The student will learn to utilize strategies that emphasize the importance of relationships to learning, self-awareness, and character based on best practices in character-based classroom management while focusing on individual needs of each child. (10 clinical experience hours in an approved pre-kindergarten least restrictive classroom).

Lecture Hours: 48.000

Pre/Corequisite: EEC3301, EEX3201 and 10 hrs. of Field Experience Required

Course User Fee: \$18.00

EEC4219C Science, Technology, and Mathematics (STEM) Methods for ECE II 3.000 credits

The student will learn to use scientific and mathematical research-based methods and strategies to teach inquiry and problem-solving skills and plan activities for young children that foster exploration in the nature of science, mathematics, and technology. (Ten hours of clinical experience required in a kindergarten-third grade setting with ESOL students during math and science instruction: 1 observation).

Lecture Hours: 48.000

Prerequisite: EEC3211

EEX3226 Assessment of All Young Children 3.000 credits

The student will utilize guidelines and techniques for observing, assessing, evaluating, and planning curriculum for young children. The student will learn to use formal and informal assessments to evaluate social-emotional, cognitive, language, and motor development and will use data to plan for instruction. (Ten hours of clinical experience required: 3 hours to include an observation of an evaluation of a young child at Early Steps, 3 hours to include a tour of the Mailman Center, Early Steps, and the FAAST Lab, and the remaining 4 hours are to be completed in an approved kindergarten-third grade inclusion setting with ESOL students).

Lecture Hours: 48.000

Prerequisite: EEC2271, EEC2601, Pre/Corequisite: EEC3301, EDF3115

Course User Fee: \$18.00

RED3009 Emergent and Early Literacy 3.000 credits

The teacher candidate will acquire the knowledge to apply the principles of explicit, systematic, and sequential evidence-based reading instruction and scaffold student learning while integrating the six components of reading (oral language, phonological awareness, phonics, fluency, vocabulary, and comprehension) in a K-2 setting. The teacher candidate will understand that teaching reading is an ongoing strategic process resulting in students comprehending various text. Reading competencies 1 and 2.

Lecture Hours: 48.000

Prerequisite: EEC3213, EEC3412

Course User Fee: \$18.00

**TSL3080 ESOL in Early Childhood Education I
3.000 credits**

The student will explore theories, research, and practices of English learners, including legal issues that have influenced the field. The student will analyze theories of first and second language acquisition, and concepts of global citizenship. (Ten hours of clinical experience requires in an approved kindergarten-third grade least restrictive environment with ESOL students.)

Lecture Hours: 48.000

Pre/Corequisite: EEC3301

Course User Fee: \$18.00

EEC3212 Integrated Social Sciences, Humanities, and Arts 3.000 credits

This course provides an overview of the role of social studies, the humanities, and the fine arts in early childhood classrooms. Students will learn to plan, implement, and assess developmentally appropriate activities that reflect the family and community influences on young children. Special fee. (10 hours of clinical experience and 1 observation required in grade 1–3 classroom with ESOL students).

Lecture Hours: 48.000

Prerequisite: EEC3301

Course User Fee: \$18.00

**EEC4268 Practicum in Early Childhood Education
3.000 credits**

The teacher candidate will use research methods to plan and implement an action research project to meet the needs of struggling readers. The student will attend professional development experiences designed to develop knowledge and pedagogy, prepare for teaching interviews, and entry into the profession.

Lecture Hours: 48.000

Recommended Preparation: Departmental Permission

Course User Fee: \$18.00

LAE4211 Methods and Resource for Literacy Development in Young Children 3.000 credits

The student will utilize a variety of assessment tools to measure and evaluate literacy in a K-3 setting. The student will learn to create and administer informal assessments, evaluate results, and differentiate instruction encompassing the reading components. Twenty hours of clinical experience and 1 observation required.

Lecture Hours: 48.000

Prerequisite: EEC4219C, EEX3226, RED3009, TSL3080,

Pre/Corequisite: EEC3212, TSL4310, Corequisite: EEC4268

Course User Fee: \$18.00

**TSL4310 ESOL in Early Childhood Education II
3.000 credits**

In this second TESOL course, the student will apply theories, principles, and current research on curriculum, instructional methods, and assessment for English learners in early childhood settings. The student will select modifications appropriate for language and content area teaching and learning. (Ten (10) hours of clinical experience in an approved first through third grade setting with ESOL students.)

Lecture Hours: 48.000

Prerequisite: TSL3080 and 10 hrs. of Field Experience Required

Course User Fee: \$18.00

EEC4936 Student Teaching Seminar: ECE 2.000 credits

The student will discuss and reflect on their development and mastery of the Florida Educator Accomplished Practices (FEAPS) during the completion of their student teaching experience in a classroom setting.

Lecture Hours: 32.000

Prerequisite: EEC4268, LAE4211, Corequisite: EEC4940

EEC4940 Student Teaching in Early Childhood Education 10.000 credits

The student will complete a full-time differentiated student teaching experience consisting of a traditional internship or teaching residency in a classroom setting. The student will learn and experience all the educational and professional responsibilities common to teachers within their area of expertise.

Internship Hours: 480.000

Corequisite: EEC4936

Program Description

Bachelor of Science | Code: S9100 | 134 credit

The Bachelor of Science in Electrical and Computer Engineering Technology degree is designed to prepare students for entry level engineering positions such as Electronics Engineers, Test Engineers, Project Engineers, Electronics Manufacturing Engineers, Electronics Systems Engineers, Electronics Hardware Engineers, Technical Support Engineers, Quality Control Engineers, Reliability Engineers, Field Engineers, Processing Engineers, and Sales Engineers.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 15.0
ENC 1102	English Composition 2	3.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
CET 3126C	Computer Architecture	4.0
<i>Program Elective</i>		3.0
ETP 3240	Power Systems	3.0
OR		
ETP 3320	Introduction to Renewable Energy Technology	3.0
Lower Division*		

*Note: This Course Sequence Guide is for students transitioning from the Associate in Science (AS) in Electronics Engineering Technology (Plan Code: 26039). It is based on students having completed 15 general education credits and 45 program core/major requirements. Additionally, students must have completed EET 2323C Analog Communications and EET 2351C Digital and Data Communications for a total of 8 credits as major course electives. If one of these courses were not chosen as electives, they should be completed this term.

Semester 2		Units: 13.0
MAC 2312	Calculus and Analytical Geometry 2	4.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0
ETI 4480C	Applied Robotics	4.0

Semester 3		Units: 12.0
PHY 2049	Physics with Calculus 2	4.0
PHY 2049L	Physics with Calculus 2 Lab	1.0
<i>Social Sciences</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
ANT 2000	Introduction to Anthropology	3.0
OR		
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
POS 2041	American Federal Government	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
EET 3716C	Advanced Systems Analysis	4.0

Foreign Language Competence*

*Note: Foreign language is a graduation requirement for the baccalaureate met through 8 credit hours at the elementary 2 level in one foreign language or equivalent. Certain foreign language courses count towards program electives. Students may satisfy equivalence through standardized examinations or successful completion of two credits (two years) in one foreign language at the secondary (high school) level. For additional information, including exemptions for students whose native language is not English, see the Testing and Assessment Department.

Semester 4		Units: 14.0
MAP 2302	Introduction to Differential Equations	3.0
EET 4158C	Linear Integrated Circuits and Devices	4.0
EET 4165C	Senior Design 1	3.0
EET 4732C	Signals & Systems	4.0
Semester 5		Units: 15.0
<i>Oral Communications</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Humanities</i>		3.0
ARC 2701	History of Architecture 1	3.0
OR		
ARC 2702	History of Architecture 2	3.0
OR		
HUM 1020	Humanities	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
EET 4166C	Senior Design 2	2.0
EET 4730C	Feedback Control Systems	4.0
<i>Program Elective</i>		3.0
CET 4190C	Applied Digital Signal Processing	4.0
OR		
CET 4663C	Electronic Security	3.0

Total: 69.0

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

**MAC2311 Calculus and Analytical Geometry 1
5.000 credits**

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

CET3126C Computer Architecture 4.000 credits

This course is intended for upper division students majoring in Electrical and Computer Engineering Technology as well as Information Systems Technology. This course introduces the study of advanced microprocessor design. Students will learn the basic organization of computer systems including instruction-set architecture, execution pipeline, memory hierarchy, virtual memory, and I/O subsystems. Students also learn advanced processor microarchitecture issues such as dynamic instruction scheduling, branch prediction, lock-up free caches, instruction-level parallelism, multiple instruction fetch/issuing, speculative execution, etc. to improve computer processor performance. Students will experimentally verify microarchitecture designs using industry standard microarchitecture simulators.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

ETP3240 Power Systems 3.000 credits

In this course, students will learn renewable energy theory and applications. This course focuses on solar photo-voltaics, solar power and tracking systems, charge controllers and inverters, wind power systems, biomass and geothermal power generation. In addition, this course covers the integration with electrical grid, production and end user systems.

Prerequisite: EET1025C.

Lecture Hours: 48.000

Prerequisite: EET1025C

**ETP3320 Introduction to Renewable Energy
Technology 3.000 credits**

In this course, students will learn renewable energy theory and applications. This course focuses on solar photo-voltaics, solar power and tracking systems, charge controllers and inverters, wind power systems, biomass and geothermal power generation. In addition, this course covers the integration with electrical grid, production and end user systems.

Prerequisite: EET2101C.

Lecture Hours: 48.000

Prerequisite: EET2101C

**MAC2312 Calculus and Analytical Geometry 2
4.000 credits**

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

ETI4480C Applied Robotics 4.000 credits

This is an upper division level course designed as an introduction to robotics programming and includes robotic applications for multifunction part manipulation and motion with stepper and servo-motors. Students learn topics related to robotic design including robotic vision, motion planning, sensing and sensors, actuators, navigation systems, mobility, forward and inverse kinematics, and path planning. Laboratory activities provide hands-on application of concepts and theories. Prerequisite: CET2123C or CET3383C.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2123C OR CET 3383C

Course User Fee: \$63.00

PHY2049 Physics with Calculus 2 4.000 credits

Foundation course for physical science and engineering majors. PHY 2048 covers classical mechanics and thermodynamics. PHY 2049 includes electricity, magnetism, waves and optics.

Lecture Hours: 64.000

Prerequisite: PHY2048, PHY2048L, Corequisite: MAC2312, PHY2049L

Course User Fee: \$10.00

PHY2049L Physics with Calculus 2 Lab 1.000 credits

Laboratory for PHY 2049.

Lab Hours: 32.000

Prerequisite: PHY2048, PHY2048L, Corequisite: MAC2312, PHY2049

Course User Fee: \$20.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan. Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

EET3716C Advanced Systems Analysis 4.000 credits

This is an upper division level course for students majoring in electronics engineering technology designed to prepare students to perform electrical circuit systems analysis using Laplace transforms and partial fraction expansion. Students will learn theorems, Fourier series, frequency response and bode plots, and their application towards practical systems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1025C AND MAC2312

MAP2302 Introduction to Differential Equations 3.000 credits

The student will analyze ordinary differential equations, methods of solution of first order linear and nonlinear equations and applications, homogeneous and non-homogeneous linear equations with constant coefficients, differential operator methods, higher order linear equations; the Laplace transform and its properties, elementary existence theorems, series solutions, numerical solutions of first order equations, initial and boundary value problems, vibrations and waves, and an introduction to autonomous systems.

Computational Course. Prerequisite: MAC 2312 with a "C" or better or equivalent.

Lecture Hours: 48.000

Prerequisite: MAC2312

Course User Fee: \$18.00

EET4158C Linear Integrated Circuits and Devices 4.000 credits

This is an upper division level course for students majoring in electronics engineering technology designed to provide students with practical skills and knowledge needed for application of operational amplifiers, comparators, phase-locked loops, timers, regulators, other integrated circuits in electronic systems. Students learn to apply these skills towards the design of amplifiers, active filters, oscillators, differentiators, integrators and other miscellaneous integrated circuit based systems . Prerequisite: EET 3716C. Laboratory fee. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET3716C

Course User Fee: \$63.00

EET4165C Senior Design 1 3.000 credits

This project-based course is designed to synthesize students' knowledge of the analysis, design, manufacturing, and testing of electronic systems. Students will design experiments, explore professional ethics, practice professional oral and written communications, conduct project feasibility studies, and perform project scheduling. Students learn about human factors, intellectual property, and liability issues.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: Department approval required

Course User Fee: \$63.00

EET4732C Signals & Systems 4.000 credits

This course is intended for upper division students majoring in Electronics Engineering Technology. Students will learn the theory and the mathematical techniques used in analyzing continuous-time linear systems. Students will learn continuous-time signal and systems analysis, the input-output relationships of linear time-invariant (LTI) systems, transient and steady state analysis, frequency domain analysis and Fourier analysis. Students will analyze and characterize LTI systems using Laplace transforms.

Prerequisite: EET 3716C.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET3716C

Course User Fee: \$63.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ARC2701 History of Architecture 1 3.000 credits

This course offers a survey of architecture from ancient times through the 18th century and includes works from the Western canon. Students will examine the integration of art forms, structural elements, and ornamental designs across world civilizations, analyzing their connections to broader aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societies throughout historical periods.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2702 History of Architecture 2 3.000 credits

This course provides a survey of architecture from the 19th century to the present and includes works from the Western canon and examining architectural developments. Students will analyze the integration of art forms, structural elements, and ornamental designs across civilizations, exploring their connections to human development, including art and history. Through this study, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societal changes and cultural expressions throughout modern and contemporary periods. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course. Lecture Hours: 48.000

EET4166C Senior Design 2 2.000 credits

Senior Design 2 is a project-based experience course in which students apply all of the skills they have acquired to analyze, design, simulate, synthesize, and test a complete system. Lecture Hours: 16.000, Lab Hours: 32.000
 Prerequisite: EET4165C
Course User Fee: \$63.00

EET4730C Feedback Control Systems 4.000 credits

This upper division course for students majoring in electronics and computer engineering technology, is designed to introduce students to the analysis of circuit networks and control systems. Students learn about stability and compensation considerations, using root locus, the Nichols chart, and Bode plots; simulation techniques; and how to apply these principles to build and test control systems. Lecture Hours: 32.000, Lab Hours: 64.000
 Prerequisite: EET3716C
Course User Fee: \$63.00

CET4190C Applied Digital Signal Processing 4.000 credits

This is an upper division level course for students majoring in electronics engineering technology. Students will learn how to model digital signal processing (DSP) systems, apply the Z transform, and develop algorithms for convolution, correlation, the Discrete Fourier Transform (DFT), and the Fast Fourier Transform (FFT). Students will apply these concepts in the design and implementation of digital filters and DSP algorithms in an embedded system. Prerequisites: COP2270, EET4732C, EET2323C or EET2351C. Special Fee. Lecture Hours: 32.000, Lab Hours: 64.000
 Prerequisite: COP2270 AND EET4732C AND EET2323C OR EET2351C
Course User Fee: \$63.00

CET4663C Electronic Security 3.000 credits

This is an upper division course for students who are majoring in electronics engineering technologies. The student will learn information and communication security principles for computer systems and networks, including authentication, protection, security models, cryptography, applications, and public policy, along with case studies. Prerequisites: CET2123C, COP2270. Lecture Hours: 32.000, Lab Hours: 32.000
 Prerequisite: CET2123C AND COP2270
Course User Fee: \$63.00

Program Description

Exceptional Student Education (K-12)

Bachelor of Science | Code: S9260 | 120 credit

The Bachelor of Science (BS) in Exceptional Student Education (K-12) prepares students to work with children with disabilities who need specially designed instruction and related services. Graduates of the program are eligible for Florida Professional Teacher Certification in ESE, Reading, and ESOL.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Prerequisite Courses

Units: 12.0

EDF 1005	Introduction to the Education Profession	3.0
EDF 2144	Maximizing Student Potential in the School Context	3.0
EEX 2000	Introduction to Exceptional Student Education	3.0
EME 2040	Creativity, Innovation, and Technology	3.0
Semester 1		Units: 15.0
EDF 4430	Measurement and Assessment in Education	3.0
EDG 3321	General Teaching Skills	3.0
EDG 3443	Classroom and Behavior Management	3.0
RED 3013	Foundations of Reading Instruction	3.0
TSL 3243	ESOL I: Second Language Acquisition and Communication	3.0
Semester 2		Units: 12.0
EEX 3071	Teaching Exceptional and Diverse Populations in Inclusive Settings	3.0
RED 3393	Differentiated Instruction in Content Reading	3.0
SCE 4362	Methods of Teaching Science	3.0
TSL 4311	ESOL II: Methods of Teaching and Assessing ESOL Students	3.0
Semester 3		Units: 9.0
EDG 4376	Integrating Language Arts and Social Sciences	3.0
EEX 4221	Assessment in Special Education	3.0
EEX 4932	Advanced Topics in Exceptional Student Education	3.0
Semester 4		Units: 12.0
EEX 3120	Language Development and Communication Disorders	3.0
EEX 4833	Practicum in Exceptional Student Education	3.0
MAE 4360	Methods of Teaching Mathematics	3.0
RED 4519	Diagnosis and Instructional Intervention in Reading	3.0
Semester 5		Units: 12.0
EEX 4930	Seminar in Exceptional Student Education	3.0
EEX 4940	Internship in Exceptional Student Education	9.0
		Total: 72.0

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000

Course User Fee: \$18.00

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

EEX2000 Introduction to Exceptional Student Education 3.000 credits

The student will learn about the history, legal issues, and legislation related to exceptional student education in K-12. The student will learn about the differing types of exceptionalities, supports, and accommodations provided for students with differing learning profiles from birth to adulthood.

Lecture Hours: 48.000

Prerequisite: EDF1005

Course User Fee: \$18.00

EME2040 Creativity, Innovation, and Technology 3.000 credits

The student will learn to manage a productive & safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Prerequisite: CGS1060 OR Passing Score on the Computer Competency Exam

Course User Fee: \$190.03

EDF4430 Measurement and Assessment in Education 3.000 credits

The student will learn research-based principles of assessment. The student will select specific standards and competencies and develop formative and summative traditional and alternative assessments. The student will interpret assessment data that will improve academic achievement and ensure equity in the application of quantitative and qualitative assessments.

Lecture Hours: 48.000

Prerequisite: EDG3321

Course User Fee: \$18.00

EDG3321 General Teaching Skills 3.000 credits

The student will learn how human development and learning theories, developments in educational neuroscience, and current research-based pedagogy apply to the teaching and learning process. The student will incorporate principles and skills of effective instructional planning, including differentiated instruction, aligning instructional teaching skills to practice, selecting appropriate formative assessments, and developing learning experiences that require students to demonstrate a variety of skills.

Lecture Hours: 48.000

Course User Fee: \$18.00

EDG3443 Classroom and Behavior Management 3.000 credits

The student will examine strategies for fostering a productive learning environment, informed by research-based frameworks and learning theories. The student will learn strategies for observing, assessing, modifying behavior, and communicating with stakeholders. The student will learn to create a learning environment that encourages positive interactions, self-regulation, and resiliency. Fifteen hours of clinical experience is required.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

RED3013 Foundations of Reading Instruction 3.000 credits

The teacher candidate will develop an understanding of reading components as a systematic process including oral language, phonological awareness, phonics, fluency, vocabulary, and comprehension. The teacher candidate will apply the principles of explicit, systematic and sequential evidence-based reading instruction and scaffold student learning while integrating the six components of reading in a K-12 setting. (Fifteen hour of clinical experience in a K-3 classroom is required.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

TSL3243 ESOL I: Second Language Acquisition and Communication 3.000 credits

The teacher candidate will critically examine and apply theoretical frameworks related to first and second language acquisition, literacy development, and the structural features of language; evaluate the impact of varied communication styles on student learning; and analyze legal and policy considerations pertinent to the education of English language learners (ELLs). Fifteen hours of clinical experience in a classroom with ELLs are required. Fifteen hours of clinical experience are required.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321, EEX3120

EEX3071 Teaching Exceptional and Diverse Populations in Inclusive Settings 3.000 credits

The student will learn current research-based instructional strategies, educational neuroscience, and legal and ethical issues necessary in addressing the needs of diverse learners in inclusive classrooms. The student will make informed decisions in adapting, accommodating, and modifying the curriculum for students with special needs. Fifteen hours of clinical experience are required. Special fee. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: EEX2000 AND EDG3321

Course User Fee: \$18.00

RED3393 Differentiated Instruction in Content Reading 3.000 credits

The teacher candidate will differentiate instruction for individual students based on their strengths and needs by applying the principles of research-based strategies and integrating the six components of reading: phonological awareness, phonics, oral language, fluency, vocabulary, and comprehension in the content area classroom. The teacher candidate will have a broad knowledge of students with differing profiles to understand and apply evidence-based instructional practices by differentiating process, product, and context. Reading competencies 2 and 4 (fifteen hours of clinical experience are required in a math or science 3-12 classroom.)

Lecture Hours: 48.000

Prerequisite: EDG3321, : Optional Prerequisite: RED3013.

Course User Fee: \$18.00

SCE4362 Methods of Teaching Science 3.000 credits

The teacher candidate will learn to design and implement science instruction utilizing the national framework for K-12 science education and educational neuroscience to provide all students with high-quality science education. The student will learn about the theoretical knowledge and skills essential for facilitating science instruction in a variety of classroom settings. Fifteen hours of clinical experience are required.

Lecture Hours: 48.000

Corequisite: EDF4430

Course User Fee: \$35.00

TSL4311 ESOL II: Methods of Teaching and Assessing ESOL Students 3.000 credits

The teacher candidate will apply TESOL instructional methods and strategies, as well as analyze, plan, design, and evaluate curriculum and materials. The teacher candidate will select, develop, and adapt assessment instruments and examine standardized ESOL measurement tools. Fifteen hours of clinical experience with English language learners are required.

Lecture Hours: 48.000

Prerequisite: TSL3243

EDG4376 Integrating Language Arts and Social Sciences 3.000 credits

The student will use knowledge, skills, and dispositions from the social sciences to organize and provide integrated instruction in the major themes, concepts, and modes of inquiry in grades K-12. The student will plan and integrate language arts and social science strategies and content to create accessibility of the curriculum to all learners.

Lecture Hours: 48.000

Prerequisite: EDG3321, Pre/Corequisite: EDF4430, RED3393

Course User Fee: \$18.00

EEX4221 Assessment in Special Education 3.000 credits

The student will study, analyze, and administer informal and formal assessments to K-12 learners with special needs. The student will learn to prepare and present assessment data for use in instructional planning and developing individualized educational plans for K-12 learners with disabilities.

Lecture Hours: 48.000

Prerequisite: EDG3321 AND EDF4430

Course User Fee: \$18.00

EEX4932 Advanced Topics in Exceptional Student Education 3.000 credits

The student will explore current issues and trends in Exceptional Student Education (ESE). The student will learn about the foundations of ESE, including instructional practices, differentiated instruction, positive behavioral supports, assessment, the transition process, and assistive technology. This course will help prepare students for the Florida Teacher Certification ESE exam.

Lecture Hours: 48.000

Prerequisite: EEX3071

EEX3120 Language Development and Communication Disorders 3.000 credits

The student will learn about typical language and speech development, characteristics and manifestations of communication disorders, and educational neuroscience research concerning first and second language acquisition. The student will learn about effective strategies and accommodations that can be used in planning instruction for P – 12 students with speech and language disorders and differences.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

EEX4833 Practicum in Exceptional Student Education 3.000 credits

The teacher candidate will plan, develop, and implement literacy pedagogic methods that meet the needs of students with differing profiles in a K-12 setting. The teacher candidate will utilize action research methodology, assessment principles, educational neuroscience research, and best practices to determine the effectiveness of a literacy strategy. Reading course competency 5. (Sixty hours of clinical experience are required).

Lecture Hours: 48.000

Prerequisite: EDF4430, EDG3321, EDG3443, EDG4376, EEX3071, EEX3120, RED3393, TSL3243, Pre/

Corequisite: EEX4221 OR MAE4360 OR RED4519 OR SCE4362 OR TSL4311

MAE4360 Methods of Teaching Mathematics 3.000 credits

The teacher candidate will utilize theory and educational neuroscience research in developing knowledge and pedagogy essential for K-12 mathematics instruction which accommodates the needs of all learners. The problem-solving approach will be used to design, implement, and assess mathematics instruction and curriculum. Fifteen hours of clinical experience in a mathematics classroom are required.

Lecture Hours: 48.000

Prerequisite: EDG3321, Pre/Corequisite: EDF4430

Course User Fee: \$25.00

RED4519 Diagnosis and Instructional Intervention in Reading 3.000 credits

The teacher candidate will apply the principles of explicit, systematic and sequential evidence-based reading instruction and scaffold student learning while integrating the six components of reading. Teacher candidates will engage in a systematic problem-solving process to identify and remediate reading difficulties, using the results of informal reading assessments, to plan interventions. Reading competencies 3 and 4 (fifteen hours of clinical experience required in an approved first-third grade setting with ESOL students.)

Lecture Hours: 48.000

Prerequisite: RED3393

Course User Fee: \$18.00

EEX4930 Seminar in Exceptional Student Education 3.000 credits

The teacher candidate will engage in professional dialogue and explore professional learning opportunities related to teaching students with exceptionalities. This seminar course is taken in conjunction with a full time, supervised teaching experience and provides an opportunity to examine and reflect on the daily experiences of becoming a highly effective teacher.

Lecture Hours: 48.000

Prerequisite: All Coursework, State Examinations, and E-Portfolio completed and passed., Corequisite: EEX4940

Course User Fee: \$18.00

EEX4940 Internship in Exceptional Student Education 9.000 credits

The teacher candidate will engage in the educational and professional responsibilities common to teachers in exceptional student education. This internship experience reinforces and augments teaching strategies that students have developed through their coursework and clinical experiences. Students participate in a full time, supervised teaching experience.

Internship Hours: 432.000

Prerequisite: All coursework and state exams completed and passed, Corequisite: EEX4930

Program Description

Information Systems Technology - Networking Concentration

Bachelor of Science | Code: S9500 | 120 credit

The Bachelor of Science (BS) in Information Systems Technology (IST) degree program prepares students with essential skills and knowledge to effectively support the design, planning and management of information infrastructures and information resources within diverse organizational settings. The Networking Concentration focuses on how to plan, design, implement and maintain network infrastructures to keep organizations running smoothly. Students learn how to select technologies that best suit the client's needs. Students also acquire the technical skills needed to install, maintain, and extend multi-user computer systems and how to develop administrative policies and procedures.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

CGS 3763	Operating System Principles	4.0
CIS 3360	Principles of Information Security	4.0
<i>Elective</i>		4.0
<i>Group A</i>		3.0-4.0
CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
OR		
CAI 2100C	Machine Learning Foundations	3.0
OR		
CAI 2300C	Introduction to Natural Language Processing	3.0
OR		
CAI 2820C	Artificial Intelligence Applications Solutions	3.0
OR		
CAI 2840C	Introduction to Computer Vision	3.0
OR		
CET 3126C	Computer Architecture	4.0
OR		
CIS 2010C	Introduction to Internet of Things Networking and Security	4.0
OR		

CIS 3215	Ethics in CyberSecurity	4.0
OR		
CIS 3361	Information Security Management	4.0
OR		
CIS 3368	Data Security & Governance	4.0
OR		
CIS 3641C	Cloud Developer	4.0
OR		
CIS 3644C	Cloud Security	4.0
OR		
CIS 3651C	Cloud System Administration and Operations	4.0
OR		
CTS 1145	Cloud Essentials	4.0
OR		
CTS 2375C	Cloud Infrastructure and Services	4.0
OR		
MAC 2233	Business Calculus	3.0
OR		
Group B*		

*Choose 4 credits from the following: CAI 2100C, CAI 2300C, CAI 2820C, CAI 2840C, CET 3126C, CIS 2010C, CIS 3215, CIS 3361, CIS 3368, CIS 3641C, CIS 3644C, CIS 3651C, CTS 2375C, MAC 2233

Semester 2		Units: 14.0
ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
CNT 4603	System Administration and Maintenance	4.0
CNT 4702	Network Design and Planning	4.0

Semester 3		Units: 8.0
CIS 4347	Information Storage Management	4.0
CNT 3409C	Network Security	4.0
Foreign Language Competence*		

*Note: Foreign language is a graduation requirement for the baccalaureate met through 8 credit hours at the elementary 2 level in one foreign language or equivalent. Certain foreign language courses count towards program electives. Students may satisfy equivalence through standardized examinations or successful completion of two credits (two years) in one foreign language at the secondary (high school) level. For additional information, including exemptions for students whose native language is not English, see the Testing and Assessment Department.

Semester 4		Units: 14.0
<i>Natural Science</i>		3.0

BSC 1005	General Education Biology	3.0
OR		
CHM 1025	Introductory Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1025	Basic Physics	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
CIS 3510	Information Technology Project Management	4.0
CNT 3526C	Wireless and Mobile Networking	4.0
General Education*		3.0

*Note: Students are encouraged to consult with an academic advisor to select approved courses that align with their interests and fulfill remaining graduation requirements.

Semester 5

Units: 12.0

CTS 4955	Networking Capstone	4.0
<i>Elective</i>		8.0
Group A*		
OR		
Group B**		

*Choose 8 credits from the following recommended options: CAI 1001C, CAI 2100C, CAI 2300C, CAI 2820C, CAI 2840C, CET 3126C, CIS 2010C, CIS 3215, CIS 3361, CIS 3368, CIS 3641C, CIS 3644C, CIS 3651C, CTS 1145, CTS 2375C, MAC 2233

**Group B - Choose 8 credits from the following 2000, 3000, or 4000 level recommended options: CAI 2100C, CAI 2300C, CAI 2820C, CAI 2840C, CET 3126C, CIS 2010C, CIS 3215, CIS 3361, CIS 3368, CIS 3641C, CIS 3644C, CIS 3651C, CTS 2375C, MAC 2233

Total: 60.0

CGS3763 Operating System Principles 4.000 credits

This upper division course, for students majoring in Information Systems Technology, introduces fundamental operating system topics and includes both computer system and operating system structure. Students will learn how processes, threads, concurrent programming, interrupt handling, CPU scheduling and process synchronization, and I/O system memory management affect the system structure. Additionally, students will learn how virtual memory, deadlocks, file system, and command interpreter relate to client/server systems.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: COP1334
Course User Fee: \$63.00

CIS3360 Principles of Information Security 4.000 credits

This upper division course provides an overview of information systems security principles, practices, methods, and tools. Students will learn about the threats against an organization's digital assets, as well as the tools and methods to mitigate those threats. Topics include cryptography, authentication and access control systems, infrastructure security, and ethical issues. Prerequisite: CTS1134 or CTS1650
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS1134 OR CTS1650
Course User Fee: \$63.00

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAI2100C Machine Learning Foundations 3.000 credits

Students are introduced to machine learning concepts and Python applications, including data acquisition, supervised, unsupervised, and reinforced learning. In addition, students will develop and deploy artificial intelligence (AI) models utilizing classification algorithms. Prerequisites: CAI1001C, COP1047C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI1001CCOP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

CAI2300C Introduction to Natural Language Processing 3.000 credits

Students will learn the fundamental concepts of Natural Language Processing (NLP) and text processing. In addition, focus will be on knowledge and skills necessary to create a language recognition application. Prerequisite: CAI2100C and recommended preparation: COP1047C or equivalent knowledge of Python programming language.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2100CCOP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

CAI2820C Artificial Intelligence Applications Solutions 3.000 credits

A lower division course for students majoring in Applied Artificial Intelligence (AI). Students will demonstrate competence to scope, acquire/explore data, model, evaluate, and deploy an AI/machine learning solution in a team environment. Students will create and present a code or no-code AI solution. Must be taken during the last semester before graduation. Prerequisites: CAI2300C and CAI2840C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2840C AND CAI2300C

Course User Fee: \$63.00

CAI2840C Introduction to Computer Vision 3.000 credits

Students will learn fundamental concepts in Computer Vision (CV) and image processing, including introduction to necessary proprietary and open-source Python libraries. Recommended preparation: COP1047C or equivalent knowledge of Python programming language. (2 hr. lecture, 2 hr. lab)

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2100C Recommended preparation: COP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

CET3126C Computer Architecture 4.000 credits

This course is intended for upper division students majoring in Electrical and Computer Engineering Technology as well as Information Systems Technology. This course introduces the study of advanced microprocessor design. Students will learn the basic organization of computer systems including instruction-set architecture, execution pipeline, memory hierarchy, virtual memory, and I/O subsystems. Students also learn advanced processor microarchitecture issues such as dynamic instruction scheduling, branch prediction, lock-up free caches, instruction-level parallelism, multiple instruction fetch/issuing, speculative execution, etc. to improve computer processor performance. Students will experimentally verify microarchitecture designs using industry standard microarchitecture simulators.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

CIS2010C Introduction to Internet of Things Networking and Security 4.000 credits

This course explores Internet of Things (IoT) networks' design, implementation, and security, providing a well-rounded understanding of protocols and standards governing the Internet and IoT. Students gain experience performing security assessments and vulnerability testing of IoT systems using industry-standard tools, focusing on identifying and mitigating risks to build secure and resilient IoT solutions.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS3215 Ethics in CyberSecurity 4.000 credits

This course provides the study of the risk factors for digital and ethical misconduct, and it explores ethics, relevant laws, regulations, policies, standards, moral, and social issues and responsibilities faced by CyberSecurity professionals. Coverage includes examination of CyberSecurity policies; Federal Laws and Authorities and International Standards; ethical and legal compliance and enforcement; business issues; contractual management of assets and liabilities; and issues involving privacy, disclosure, free speech and individual rights.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS3361 Information Security Management 4.000 credits

This course covers how to manage, design, oversee and assess an organization's information security. Students will learn how to develop an information security strategy, how to write information security policies, and how to manage information risk. Other topics include security program development and management, business continuity planning and disaster recovery planning. Prerequisite(s): CIS 3360 or CTS 1120. Laboratory fee. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS3360 OR CTS1120

Course User Fee: \$63.00

CIS3368 Data Security & Governance 4.000 credits

This upper division course is for students majoring in Data Analytics. Students will gain an understanding of how analytics can be applied to a variety of security-related problems across organizations. In addition, students will explore various ethical, legal, and data governance issues that affect data analysts. (3 hr. lecture; 1 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

CIS3641C Cloud Developer 4.000 credits

This course focuses on software development with cloud technologies. Students learn about core cloud services, basic architecture, and best practices used in the development and maintenance of applications in a cloud platform. Proficiency is gained in application lifecycle management, serverless computing, and containers using primary cloud platform tools, including IDE's, distributed version control systems, CI/CD pipelines, and security. API usage, command line interfaces, and software development kits (SDKs) will be emphasized. Prerequisites: COP 1047C and CTS 1145 (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND CTS1145

Course User Fee: \$63.00

CIS3644C Cloud Security 4.000 credits

This course equips students with essential principles, tools, and techniques to effectively secure, configure, and operate cloud-based information systems.

Students will learn cloud services essentials, such as computing, storage, databases, and networking.

Additionally, they'll delve into security tools, key management services, digital signatures, certificates, authentication protocols, SSO services, firewalls, and access control lists. Prerequisites: CTS 1120 and CTS 2375C (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120 AND CTS2375C

Course User Fee: \$63.00

CIS3651C Cloud System Administration and Operations 4.000 credits

This course offers a comprehensive exploration into the world of cloud system administration using cloud computing as the cornerstone of study. This intensive course seeks to equip students with hands-on knowledge and skill development in cloud infrastructure, tooling, computing, networking, storage, security, and deployment. From the fundamentals of Systems Operations on the cloud to intricate deployment strategies using CloudFormation, students will navigate various cloud services, ensuring a solid foundation in cloud administration. With a balance of theory, lab work, and activities, students will prepare to address real-world challenges helpful in the transition to cloud-based roles. Prerequisites: COP 1047C and CTS 1145. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND CTS1145

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2375C Cloud Infrastructure and Services 4.000 credits

This course helps students develop technical expertise in Cloud computing and prepares them for industry certification. Students will learn the essentials of Cloud computing, business security and compliance considerations, migrating to the Cloud, architecting a Cloud server, and how to troubleshoot Cloud services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2145

Course User Fee: \$63.00

MAC2233 Business Calculus 3.000 credits

This course introduces the basic concepts of differential and integral calculus for students majoring in business administration and related fields. Topics include limits, continuity, differentiation and integration of polynomials, logarithmic and exponential functions with applications to business, economics, and the life sciences. Prerequisite: MAC 1105 or MAC1106 with a grade of C or better or departmental permission. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

CNT4603 System Administration and Maintenance 4.000 credits

This upper division course, for students majoring in Information Systems Technology, explores UNIX and Microsoft Windows systems and their administration and maintenance within the network setting. Students will learn how to install, maintain, and extend multi-user computer systems and how to develop administrative policies and procedures. Students will also learn how to apply troubleshooting and problem solving skills to resolve user and system issues.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CNT4702 Network Design and Planning 4.000 credits

This upper division course, for students majoring in Information Systems Technology, presents network design using layering. Students will learn how to apply cabling, topology, and architecture to design systems. Students will also learn how design impacts network performance and control issues such as congestion control, error control, and contention resolution. Prerequisite: CIS 3360.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS4347 Information Storage Management 4.000 credits

This upper division course, for students majoring in Information Systems Technology, introduces challenges and solutions for data storage and data management. Students will learn how to manage advanced storage systems, protocols, and architectures, including storage area networks (SAN), network attached storage (NAS), fiber channel networks, internet protocol SANS (IPSAN), iSCSI, and content-addressable storage (CAS). Prerequisite: CTS1134 or CTS1650. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CNT3409C Network Security 4.000 credits

This upper division course introduces students to current and emerging threats to the security of computer networks, as well as tools and techniques for the prevention, detection and recovery from such attacks. Topics include firewalls, intrusion detection and intrusion prevention systems, virtual private networks, remote authentication and authorization systems, and security protocols. Prerequisite(s): CIS 3360. Laboratory fee. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 27.000 - 32.000

Prerequisite: CIS3360

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CIS3510 Information Technology Project Management 4.000 credits

This upper division course, for students majoring in Information Systems Technology, covers the general aspects of project management and emphasizes the pertinent considerations that apply to information technology projects. Students will learn the principles, processes and practices of information technology project management, including techniques for planning, organizing, scheduling, and controlling IT projects with a substantial focus on cost estimation and risk management. Students must enroll in this course one semester before taking CTS 4955.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CNT3526C Wireless and Mobile Networking 4.000 credits

This is upper division course for students majoring in Information Systems Technology introduces students to wireless and mobile network architecture, protocols and technologies. In particular, students will learn about Wireless Local Area Networks (WLANs), Wireless Personal Area Networks (WPANs) and Wireless Metropolitan Area Networks (WMANs) technologies. Other topics include antenna concepts, cellular networks, the 802.11 network architecture, and wireless security. Prerequisites: CTS 1134 or CTS 1650.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CTS4955 Networking Capstone 4.000 credits

This upper division course, for students majoring in the BS-IST Networking track, requires students to demonstrate their competence to analyze, design, develop, and test an information system in a team environment. Each team will create and present an information technology (IT) solution proposal that includes: design documentation, implementation plan, and project test plan to create an operational information system. Students will also implement a proof-of-concept in a real or simulated/virtualized environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: Student must be classified as a Senior and have completed at least 3 of the 5 concentration courses to obtain departmental approval.

Course User Fee: \$63.00

Program Description

Information Systems Technology - Software Engineering Concentration

Bachelor of Science | Code: S9501 | 120 credit

The Bachelor of Science (BS) in Information Systems Technology (IST) degree program prepares students with essential skills and knowledge to effectively support the design, planning and management of information infrastructures and information resources within diverse organizational settings. The Software Engineering concentration focuses on designing and creating software. Students learn how to specify software requirements from clients and how to design, implement and validate software solutions for real-world problems.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

CIS 3360	Principles of Information Security	4.0
COP 2800	Java Programming	4.0
COT 3100C	Discrete Structures	4.0

Semester 2

Units: 14.0

ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
CGS 3763	Operating System Principles	4.0
COP 3530	Data Structures	4.0

Semester 3

Units: 10.0

<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1025	Introductory Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1025	Basic Physics	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
CET 3126C	Computer Architecture	4.0
General Education Elective*		3.0
Foreign Language Competence**		

*Note: Students are encouraged to consult with an academic advisor to select approved courses that align with their interests and fulfill remaining graduation requirements.

**Note: Foreign language is a graduation requirement for the baccalaureate met through 8 credit hours at the elementary 2 level in one foreign language or equivalent. Certain foreign language courses count towards program electives. Students may satisfy equivalence through standardized examinations or successful completion of two credits (two years) in one foreign language at the secondary (high school) level. For additional information, including exemptions for students whose native language is not English, see the Testing and Assessment Department.

Semester 4

Units: 12.0

CEN 4065C	Software Architecture and Design	4.0
CET 3383C	Software Engineering I	4.0
<i>Elective</i>		4.0
<i>Group A</i>		3.0-4.0
CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
OR		
CAI 2100C	Machine Learning Foundations	3.0
OR		
CAI 2300C	Introduction to Natural Language Processing	3.0
OR		
CAI 2820C	Artificial Intelligence Applications Solutions	3.0
OR		

CAI 2840C	Introduction to Computer Vision	3.0
OR		
CAI 3303C	Natural Language Processing	3.0
OR		
CIS 3368	Data Security & Governance	4.0
OR		
CIS 3641C	Cloud Developer	4.0
OR		
CIS 3651C	Cloud System Administration and Operations	4.0
OR		
CIS 4347	Information Storage Management	4.0
OR		
CNT 4603	System Administration and Maintenance	4.0
OR		
CNT 4702	Network Design and Planning	4.0
OR		
COP 2658	iPhone Application Development 2	4.0
OR		
COP 2660	Android Application Development 1	4.0
OR		
COP 2662	Android Application Development 2	4.0
OR		
COP 2664	iPhone Application Development 1	4.0
OR		
CTS 1145	Cloud Essentials	4.0
OR		
MAC 2233	Business Calculus	3.0
OR		
Group*		

*Choose 4 credits from the following 2000, 3000, or 4000 level recommended options: CAI 2100C, CAI 2300C, CAI 2820C, CAI 2840C, CAI 3303C, CIS 3368, CIS 3641C, CIS 3651C, CIS 4347, CNT 4603, CNT 4702, COP 2658, COP 2660, COP 2662, COP 2664, MAC 2233

Total: 48.0

CIS3360 Principles of Information Security 4.000 credits

This upper division course provides an overview of information systems security principles, practices, methods, and tools. Students will learn about the threats against an organization's digital assets, as well as the tools and methods to mitigate those threats. Topics include cryptography, authentication and access control systems, infrastructure security, and ethical issues. Prerequisite: CTS1134 or CTS1650
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS1134 OR CTS1650
Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles. Lecture Hours: 48.000, Lab Hours: 32.000
Corequisite: COP1047C OR COP1334 OR COP2270
Course User Fee: \$63.00

COT3100C Discrete Structures 4.000 credits

This course delves into discrete structures with a strong focus on applications in computer science. It covers a range of topics including sets, functions and relations, propositional and predicate logic, basic proof techniques, logic, recursion and induction, graphs, Big-O notation, and various specific applications relevant to computer science.

Prerequisite: COP 2800; Corequisite: COP 3530 (3 lecture, 2 lab hours)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800 AND Corequisite: COP3530

Course User Fee: \$63.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

CGS3763 Operating System Principles 4.000 credits

This upper division course, for students majoring in Information Systems Technology, introduces fundamental operating system topics and includes both computer system and operating system structure. Students will learn how processes, threads, concurrent programming, interrupt handling, CPU scheduling and process synchronization, and I/O system memory management affect the system structure. Additionally, students will learn how virtual memory, deadlocks, file system, and command interpreter relate to client/server systems.

Prerequisite: COP1334. Special Fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1334

Course User Fee: \$63.00

COP3530 Data Structures 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology. Students will learn the fundamentals of data structures using the Java programming language. The students will learn to design, implement and use data structures to organize and store data in a computer so that it can be accessed and modified efficiently.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CET3126C Computer Architecture 4.000 credits

This course is intended for upper division students majoring in Electrical and Computer Engineering Technology as well as Information Systems Technology. This course introduces the study of advanced microprocessor design. Students will learn the basic organization of computer systems including instruction-set architecture, execution pipeline, memory hierarchy, virtual memory, and I/O subsystems. Students also learn advanced processor microarchitecture issues such as dynamic instruction scheduling, branch prediction, lock-up free caches, instruction-level parallelism, multiple instruction fetch/issuing, speculative execution, etc. to improve computer processor performance. Students will experimentally verify microarchitecture designs using industry standard microarchitecture simulators.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

CEN4065C Software Architecture and Design 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology or B.S. in Electrical and Computer Engineering Technology. The course covers the software engineering process of building the software architecture and designing the software product according to design criteria. This course covers in-depth topics in software process models, software architecture, requirements modeling with use-cases and class-based methods. Students will also learn design concepts including abstraction, component-level and OO architectural design, user interface analysis and design, and design patterns, and software project management.

Prerequisite: COP3530.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP3530

Course User Fee: \$63.00

CET3383C Software Engineering I 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology or B.S. in Electrical and Computer Engineering Technology. The student will learn the basic principles and concepts of software engineering; system requirements; modeling and testing; object oriented analysis and design; testing and validation; configuration management; and the analysis, design and programming of extensible software systems.

Prerequisite: CET2369C or COP2800.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2369C, COP2800

Course User Fee: \$63.00

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAI2100C Machine Learning Foundations 3.000 credits

Students are introduced to machine learning concepts and Python applications, including data acquisition, supervised, unsupervised, and reinforced learning. In addition, students will develop and deploy artificial intelligence (AI) models utilizing classification algorithms. Prerequisites: CAI1001C, COP1047C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI1001CCOP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

CAI2300C Introduction to Natural Language Processing 3.000 credits

Students will learn the fundamental concepts of Natural Language Processing (NLP) and text processing. In addition, focus will be on knowledge and skills necessary to create a language recognition application. Prerequisite: CAI2100C and recommended preparation: COP1047C or equivalent knowledge of Python programming language.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2100CCOP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

CAI2820C Artificial Intelligence Applications Solutions 3.000 credits

A lower division course for students majoring in Applied Artificial Intelligence (AI). Students will demonstrate competence to scope, acquire/explore data, model, evaluate, and deploy an AI/machine learning solution in a team environment. Students will create and present a code or no-code AI solution. Must be taken during the last semester before graduation. Prerequisites: CAI2300C and CAI2840C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2840C AND CAI2300C

Course User Fee: \$63.00

CAI2840C Introduction to Computer Vision 3.000 credits

Students will learn fundamental concepts in Computer Vision (CV) and image processing, including introduction to necessary proprietary and open-source Python libraries. Recommended preparation: COP1047C or equivalent knowledge of Python programming language. (2 hr. lecture, 2 hr. lab)

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2100C Recommended preparation: COP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

CAI3303C Natural Language Processing 3.000 credits

Students will further explore Natural Language Processing domains. Focus on deep learning application and use of the latest pre-trained models on real world applications. Prerequisite: CAI2300C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2300C

Course User Fee: \$63.00

CIS3368 Data Security & Governance 4.000 credits

This upper division course is for students majoring in Data Analytics. Students will gain an understanding of how analytics can be applied to a variety of security-related problems across organizations. In addition, students will explore various ethical, legal, and data governance issues that affect data analysts. (3 hr. lecture; 1 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

CIS3641C Cloud Developer 4.000 credits

This course focuses on software development with cloud technologies. Students learn about core cloud services, basic architecture, and best practices used in the development and maintenance of applications in a cloud platform. Proficiency is gained in application lifecycle management, serverless computing, and containers using primary cloud platform tools, including IDE's, distributed version control systems, CI/CD pipelines, and security. API usage, command line interfaces, and software development kits (SDKs) will be emphasized.

Prerequisites: COP 1047C and CTS 1145 (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND CTS1145

Course User Fee: \$63.00

CIS3651C Cloud System Administration and Operations 4.000 credits

This course offers a comprehensive exploration into the world of cloud system administration using cloud computing as the cornerstone of study. This intensive course seeks to equip students with hands-on knowledge and skill development in cloud infrastructure, tooling, computing, networking, storage, security, and deployment. From the fundamentals of Systems Operations on the cloud to intricate deployment strategies using CloudFormation, students will navigate various cloud services, ensuring a solid foundation in cloud administration. With a balance of theory, lab work, and activities, students will prepare to address real-world challenges helpful in the transition to cloud-based roles. Prerequisites: COP 1047C and CTS 1145. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND CTS1145

Course User Fee: \$63.00

CIS4347 Information Storage Management 4.000 credits

This upper division course, for students majoring in Information Systems Technology, introduces challenges and solutions for data storage and data management. Students will learn how to manage advanced storage systems, protocols, and architectures, including storage area networks (SAN), network attached storage (NAS), fiber channel networks, internet protocol SANS (IPSAN), iSCSI, and content-addressable storage (CAS). Prerequisite: CTS1134 or CTS1650. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CNT4603 System Administration and Maintenance 4.000 credits

This upper division course, for students majoring in Information Systems Technology, explores UNIX and Microsoft Windows systems and their administration and maintenance within the network setting. Students will learn how to install, maintain, and extend multi-user computer systems and how to develop administrative policies and procedures. Students will also learn how to apply troubleshooting and problem solving skills to resolve user and system issues.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CNT4702 Network Design and Planning 4.000 credits

This upper division course, for students majoring in Information Systems Technology, presents network design using layering. Students will learn how to apply cabling, topology, and architecture to design systems. Students will also learn how design impacts network performance and control issues such as congestion control, error control, and contention resolution. Prerequisite: CIS 3360.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2658 iPhone Application Development 2 4.000 credits

This intermediate iOS course teaches the principles of iPhone application development for majors in Computer Science, Computer Information Systems, and related disciplines. Students will learn how to create mobile applications that can be deployed to iPhone smartphones, tablets or simulators utilizing Cocoa and XCode for development. Emphasis will be placed on learning the underlying iPhone framework and components in order to create quality mobile applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2664

Course User Fee: \$63.00

COP2660 Android Application Development 1 4.000 credits

This course teaches the principles of Android application development for majors in Computer Science, Computer Information Systems, and related disciplines. Students will learn how to create mobile applications for deployment to Android smartphones, tablets or simulators utilizing open source software (Java, Eclipse IDE, Android Plug-In and Android SDK) for development. Emphasis will be placed on the underlying Android framework to create quality applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

COP2662 Android Application Development 2 4.000 credits

This course for majors in Computer Science, Computer Information Systems, and related disciplines teaches how to develop advanced Android applications. Students will learn how to create applications utilizing the advanced capabilities of Android smartphones, including interfacing the application to the device's content providers' databases, GPS and location based services, notifications, background threads, audio, video, SMS, motion sensors and network connectivity. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2660, COP2800

Course User Fee: \$58.00

COP2664 iPhone Application Development 1 4.000 credits

This is an introduction to iOS programming course using the Objective C computer language, recommended for Computer Science and Computer Information Systems majors. Students will learn to code, compile and execute mobile iOS applications while learning advanced programming concepts and object oriented programming design concepts and principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1332 OR COP1334

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

MAC2233 Business Calculus 3.000 credits

This course introduces the basic concepts of differential and integral calculus for students majoring in business administration and related fields. Topics include limits, continuity, differentiation and integration of polynomials, logarithmic and exponential functions with applications to business, economics, and the life sciences. Prerequisite: MAC 1105 or MAC1106 with a grade of C or better or departmental permission. Computational Course.
 Lecture Hours: 48.000
 Prerequisite: MAC1105 OR MAC1106
 Course User Fee: \$18.00

Program Description

Bachelor of Science in Nursing (RN to BSN)

Bachelor of Science in Nursing | Code: N9100 | 125 credit

The Bachelor of Science in Nursing (BSN) is designed for licensed RNs with an A.S. degree in Nursing from regionally accredited programs who wish to attain the next level of education in order to provide professional nursing care in all clinical practice settings around the world, or to be eligible for advanced nursing leadership, management, staff education and practice positions, in a multicultural society.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1 (In-Milestone)		Units: 12.0
NUR 3069L	Health Assessment Across the Lifespan Lab	1.0
NUR 3069	Health Assessment Across the Lifespan Theory	2.0
NUR 3125	Pathophysiology in Nursing Practice	3.0
NUR 3805	Foundations & Role Transitions of the Professional Nurse	3.0
NUR 4146	Pharmacology for Nursing	3.0
Semester 2 (In-Milestone)		Units: 12.0
NUR 3165	Evidence-Based Nursing Practice	3.0
NUR 4636	Community Health Nursing	3.0
NUR 4636L	Community Health Nursing Practicum	3.0
NUR 4667	Worldwide Trends in Nursing Care	3.0
Semester 3 (In-Milestone)		Units: 18.0
NUR 4945L	Transforming Practice Through Innovation and Leadership Capstone	3.0
Program Elective		3.0
NSP 3685	End-of-Life Nursing Care	3.0
OR		
NUR 3178	Complementary and Alternative Health Care	3.0
OR		
NUR 3289	Foundations of Gerontology	3.0
OR		
NUR 3674	Faith Based Community Nursing	3.0
Electives*		12.0

Select 12 credits from the following options: BSC, BCH*, CGS 1060C, CHM*, CHI1*, FRE1*, FRE2*, GER1*, GER2*, HBR1*, ITA1*, ITA2*, JPN1*, JPN2*, NUR 1002, NUR 1002L, PHY*, SPN1*, SPN2*

Total: 42.0

NUR3069 Health Assessment Across the Lifespan Theory 2.000 credits

This theory-based course introduces students to the principles and practices of comprehensive health assessment across the lifespan. Emphasis is placed on health history-taking, clinical reasoning, cultural sensitivity, and ethical standards. Students will learn to interpret subjective data, prioritize patient needs, and plan individualized health promotion strategies. The course culminates in the submission of a comprehensive health history and a narrated PowerPoint presentation featuring an age-appropriate health promotion intervention. (2.0 hours)

Lecture Hours: 32.000

NUR3069L Health Assessment Across the Lifespan Lab 1.000 credits

This hands-on lab course builds on the theory-based course by guiding students through the performance of comprehensive head-to-toe assessments across the lifespan. Students apply therapeutic communication, cultural humility, and ethical standards while practicing physical examination techniques and documentation. The course culminates in a video submission of a complete subjective and objective assessment, demonstrating clinical judgment and evidence-based practice.

Clinical Hours: 48.000

Corequisite: NUR3069 Graduate from an accredited RN Program

NUR3125 Pathophysiology in Nursing Practice 3.000 credits

This course builds upon prior knowledge in assessment, management of patients with commonly occurring diseases, and evaluation patient outcomes. The course will integrate the principles and rationales of pathophysiology at the cellular and systemic level using the nursing process. It will provide an in-depth theoretical and clinical principles of diseases encountered in healthcare settings. Health disparities resulting from social determinants of health, genetics, and stress related causes will also be explored.

Lecture Hours: 48.000

NUR3805 Foundations & Role Transitions of the Professional Nurse 3.000 credits

This course will introduce students to the theory and principles for delivering professional nursing practice. The course is designed to enhance the transition of registered nurses to the baccalaureate role and build on existing knowledge and professional experience. The course explores concepts of professional nursing as a way to guide and promote professional practice. The core components of this course focus on standards of nursing practice as it relates to the historical and theoretical framework for practice, moral, legal issues, evidence-based practice, quality and safety, leadership, economics, technology, information management, and trends in the future of healthcare. A minimum grade of "C" or better is required. Corequisite: NUR 3041. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

NUR4146 Pharmacology for Nursing 3.000 credits

This course is designed for the RN – BSN student and builds on the prior knowledge of entry level registered nurses. The focus is on the pharmacological treatment of various illnesses in diverse individuals across the lifespan. The emphasis is on the principles of pharmacokinetics, pharmacodynamics and pharmacogenetics in the safe treatment of selected illnesses.

Lecture Hours: 48.000

NUR3165 Evidence-Based Nursing Practice 3.000 credits

This introductory course emphasizes developing skills to translate empirical evidence into professional nursing practice, strongly focusing on evidence-based practice (EBP) knowledge. Students will explore best practice guidelines and empirical dissemination to improve patient and organizational outcomes in healthcare delivery. Throughout the course, students will use technology to access, retrieve, organize, and evaluate literature according to levels of evidence to enhance nursing practice. A key component of this course is the responsible use of Artificial Intelligence (AI) in gathering reliable information. Students will learn about AI applications in nursing, ethical considerations, and the importance of transparency, accountability, and fairness in AI systems (3-hour lecture)

Lecture Hours: 48.000 - 52.000

NUR4636 Community Health Nursing 3.000 credits

This course focuses on the holistic aspects of community nursing care applied to diverse global populations across the lifespan. The course introduces students to community nursing practice and formulates a paradigm shift from individual patients to the global community, addressing the history, evolution, theoretical framework, and purpose of community health nursing practice with an introduction to epidemiological principles, concepts of community assessment, health promotion, maintenance and education. The course involves the analysis of current knowledge and practice to illness prevention, health promotion, health restoration, community education and empowerment. Minimum grade of "C" or better required.

Lecture Hours: 48.000

Prerequisite: NUR3069, NUR3805, Corequisite: NUR4636L

Course User Fee: \$13.00

NUR4636L Community Health Nursing Practicum 3.000 credits

This course focuses on the clinical application of Community Health Nursing Theory. Students will utilize the nursing process in the delivery of healthcare within the community environment. Students will assess the individual, family, and/or community, develop a plan of care, and deliver care to an individual, family and/or community within a multicultural environment. Minimum grade of "C" or better required.

Clinical Hours: 144.000

Corequisite: NUR4636

NUR4667 Worldwide Trends in Nursing Care 3.000 credits

This course integrates the nursing process and theory to provide culturally sensitive, competent, and safe nursing care to diverse cultures. The goal of the course is to improve health outcomes by promoting quality of care for diverse populations through the development of self-evaluation and awareness of ethical, cultural practices, values, beliefs, including biases and prejudices. The course also provides the students with a worldview of the influence of culture on health care practices, barriers to transcultural communication, political, economic, social, and demographic issues in a diverse global society. A minimum grade of "C" or better.

Prerequisite: NUR 3069, NUR 3069L, NUR 3805 (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: NUR3069 AND NUR3069L AND NUR3805

NUR4945L Transforming Practice Through Innovation and Leadership Capstone 3.000 credits

This capstone practice experience empowers RN to BSN students to synthesize all previously acquired competencies gained throughout the program to a real-life clinical, patient, or population problem in a healthcare setting. Focus is on the application of nursing theory and principles, quality and safety standards, evidence-based practice guidelines, leadership skills, interprofessional collaboration, and innovative technologies to a quality improvement project. Students will present their findings through the development of an evidence-based poster.

Clinical Hours: 144.000

Prerequisite: NUR4636 AND NUR4636L

NSP3685 End-of-Life Nursing Care 3.000 credits

The purpose of this course is to provide healthcare professionals an overview of End-of-Life palliative and hospice care. Students will learn pain and symptom management, ethical and cultural considerations, assess psychosocial and emotional concerns of the patient and family, and describe the care of the patient during the last hours. Course is restricted to BSN students, requires departmental permission.

Lecture Hours: 48.000

NUR3178 Complementary and Alternative Health Care 3.000 credits

This is an upper-division course in Complementary and Alternative Health Care. This course evaluates complementary and alternative health care in diverse populations across the lifespan and around the globe. The course introduces students to complementary and alternative practice and formulates a paradigm shift from traditional patient care to holistic nursing care. The course addresses different complementary and alternative treatment practices through evidence-based research. The course involves the analysis of current knowledge and practice to illness prevention, health promotion, health restoration, patient education, and empowerment.

Lecture Hours: 48.000

NUR3289 Foundations of Gerontology 3.000 credits

This course focuses on the special healthcare needs of the geriatric population. Students will learn the physical, physiological, psychosocial and geropharmacologic implications related to aging. The trends in the changing demographics and the social consequences of aging will be identified. Established geriatric assessment and evaluative tools will also be discussed.

Lecture Hours: 48.000

Prerequisite: Admission to the BSN - RN program N9100

Course User Fee: \$125.00

NUR3674 Faith Based Community Nursing 3.000 credits

This course will provide education in faith-based community nursing to registered nurses. Students will learn the skills needed to integrate the care of mind, body, and spirit in faith-based communities. The intentional care of the spirit is part of the process in preventing and minimizing illness in a faith-based community.

Lecture Hours: 48.000

Prerequisite: RN with at least 2 years of experience and baccalaureate degree standing. Departmental permission required. (3 hr. lecture)

Program Description

Secondary Mathematics Education

Bachelor of Science | Code: S9250 | 120 credit

The four-year baccalaureate degrees in Education are designed to prepare students to become teachers and pass state professional certification exams. An internship in a school setting is required to provide practical experience. Additionally, individuals with Bachelor's degrees in other fields are able to earn teacher certification. Please refer to the College Catalog for specific program prerequisites (www.mdc.edu).

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Prerequisite Courses

EDF 1005	Introduction to the Education Profession	3.0
EDF 2144	Maximizing Student Potential in the School Context	3.0
EEX 2000	Introduction to Exceptional Student Education	3.0
EME 2040	Creativity, Innovation, and Technology	3.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
MAC 2312	Calculus and Analytical Geometry 2	4.0
MAC 2313	Calculus and Analytic Geometry 3	4.0
STA 2023	Statistical Methods	3.0

Semester 1

Units: 15.0

EDF 4430	Measurement and Assessment in Education	3.0
EDG 3321	General Teaching Skills	3.0
EDG 3443	Classroom and Behavior Management	3.0

MAD 3107	Discrete Structures	3.0
MAP 2302	Introduction to Differential Equations	3.0
Semester 2		Units: 12.0
EEX 3071	Teaching Exceptional and Diverse Populations in Inclusive Settings	3.0
MAE 4360	Methods of Teaching Mathematics	3.0
MAS 3301	Algebraic Structures	3.0
RED 3393	Differentiated Instruction in Content Reading	3.0
Semester 3		Units: 9.0
MAS 3105	Linear Algebra	3.0
MAS 4203	Number Theory	3.0
TSL 4324C	ESOL Strategies for Content Area Teachers	3.0
Semester 4		Units: 10.0
MAE 4940	Advanced Topics in Mathematics Education Practicum	3.0
MHF 4404	History of Mathematics	3.0
MTG 3214	Euclidean Geometry	4.0
Semester 5		Units: 2.0
MAE 3951	Problem-Based Learning in Mathematics Education	2.0
Semester 6		Units: 12.0
MAE 4942	Seminar in Mathematics Education	3.0
MAE 4945	Internship in Secondary Mathematics Education	9.0

Total: 60.0

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000

Course User Fee: \$18.00

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

EEX2000 Introduction to Exceptional Student Education 3.000 credits

The student will learn about the history, legal issues, and legislation related to exceptional student education in K-12. The student will learn about the differing types of exceptionalities, supports, and accommodations provided for students with differing learning profiles from birth to adulthood.

Lecture Hours: 48.000

Prerequisite: EDF1005

Course User Fee: \$18.00

**EME2040 Creativity, Innovation, and Technology
3.000 credits**

The student will learn to manage a productive & safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Prerequisite: CGS1060 OR Passing Score on the Computer Competency Exam

Course User Fee: \$190.03

**MAC2311 Calculus and Analytical Geometry 1
5.000 credits**

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

**MAC2312 Calculus and Analytical Geometry 2
4.000 credits**

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

EDF4430 Measurement and Assessment in Education 3.000 credits

The student will learn research-based principles of assessment. The student will select specific standards and competencies and develop formative and summative traditional and alternative assessments. The student will interpret assessment data that will improve academic achievement and ensure equity in the application of quantitative and qualitative assessments.

Lecture Hours: 48.000

Prerequisite: EDG3321

Course User Fee: \$18.00

EDG3321 General Teaching Skills 3.000 credits

The student will learn how human development and learning theories, developments in educational neuroscience, and current research-based pedagogy apply to the teaching and learning process. The student will incorporate principles and skills of effective instructional planning, including differentiated instruction, aligning instructional teaching skills to practice, selecting appropriate formative assessments, and developing learning experiences that require students to demonstrate a variety of skills.

Lecture Hours: 48.000

Course User Fee: \$18.00

EDG3443 Classroom and Behavior Management 3.000 credits

The student will examine strategies for fostering a productive learning environment, informed by research-based frameworks and learning theories. The student will learn strategies for observing, assessing, modifying behavior, and communicating with stakeholders. The student will learn to create a learning environment that encourages positive interactions, self-regulation, and resiliency. Fifteen hours of clinical experience is required.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

MAD3107 Discrete Structures 3.000 credits

Topics include sets, logic, switching circuits, Boolean Algebra, combinatorics, probability, mathematical proofs, mathematical induction, functions, relations, and graph theory. Credit is not also given for MAD 2104.

Lecture Hours: 48.000

Prerequisite: MAC2312

MAP2302 Introduction to Differential Equations 3.000 credits

The student will analyze ordinary differential equations, methods of solution of first order linear and nonlinear equations and applications, homogeneous and non-homogeneous linear equations with constant coefficients, differential operator methods, higher order linear equations; the Laplace transform and its properties, elementary existence theorems, series solutions, numerical solutions of first order equations, initial and boundary value problems, vibrations and waves, and an introduction to autonomous systems.

Computational Course. Prerequisite: MAC 2312 with a "C" or better or equivalent.

Lecture Hours: 48.000

Prerequisite: MAC2312

Course User Fee: \$18.00

EEX3071 Teaching Exceptional and Diverse Populations in Inclusive Settings 3.000 credits

The student will learn current research-based instructional strategies, educational neuroscience, and legal and ethical issues necessary in addressing the needs of diverse learners in inclusive classrooms. The student will make informed decisions in adapting, accommodating, and modifying the curriculum for students with special needs. Fifteen hours of clinical experience are required. Special fee. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: EEX2000 AND EDG3321

Course User Fee: \$18.00

MAE4360 Methods of Teaching Mathematics 3.000 credits

The teacher candidate will utilize theory and educational neuroscience research in developing knowledge and pedagogy essential for K-12 mathematics instruction which accommodates the needs of all learners. The problem-solving approach will be used to design, implement, and assess mathematics instruction and curriculum. Fifteen hours of clinical experience in a mathematics classroom are required.

Lecture Hours: 48.000

Prerequisite: EDG3321, Pre/Corequisite: EDF4430

Course User Fee: \$25.00

MAS3301 Algebraic Structures 3.000 credits

This course is designed for students who are majoring in secondary mathematics education, mathematics, science or engineering. Topics include set theory, basic properties of the integers, groups, rings, fields and the homomorphisms of these algebraic structures.

Lecture Hours: 48.000

Prerequisite: MAC2312

RED3393 Differentiated Instruction in Content Reading 3.000 credits

The teacher candidate will differentiate instruction for individual students based on their strengths and needs by applying the principles of research-based strategies and integrating the six components of reading: phonological awareness, phonics, oral language, fluency, vocabulary, and comprehension in the content area classroom. The teacher candidate will have a broad knowledge of students with differing profiles to understand and apply evidence-based instructional practices by differentiating process, product, and context. Reading competencies 2 and 4 (fifteen hours of clinical experience are required in a math or science 3-12 classroom.)
Lecture Hours: 48.000
Prerequisite: EDG3321, : Optional Prerequisite: RED3013.

Course User Fee: \$18.00

MAS3105 Linear Algebra 3.000 credits

This course is designed for students who are majoring in secondary mathematics education. Major topics include systems of linear equations, matrices, determinants, vector spaces, linear transformations, eigenvectors and eigenvalues, inner-product spaces and orthogonality.
Lecture Hours: 48.000
Prerequisite: MAC2312

MAS4203 Number Theory 3.000 credits

Topics include mathematical induction, divisibility, the Euclidean algorithm, primes, the Fundamental Theorem of Arithmetic, number-theoretic functions, congruence, linear Diophantine equations, linear congruences, the Chinese Remainder Theorem, and the theorems of Euler, Fermat, and Wilson.
Lecture Hours: 48.000
Prerequisite: MAC2312

TSL4324C ESOL Strategies for Content Area Teachers 3.000 credits

The student will learn topics related to teaching content area subjects to English Language Learners. The student will plan and implement curriculum, instruction, and assessment activities to meet the needs of culturally and linguistically diverse learners. Fifteen hours of clinical experience are required. Pre/Corequisite: EDG3321.
Lecture Hours: 48.000
Pre/Corequisite: EDG3321

MAE4940 Advanced Topics in Mathematics Education Practicum 3.000 credits

The teacher candidate will plan and implement mathematics instruction that meets the needs of all learners. The student will utilize action research methodology, assessment principles, educational neuroscience research, and best practices to identify and address issues related to mathematics learning in grades 6-12. Sixty (60) hours of clinical experience in a 6-12 mathematics classroom are required.
Lecture Hours: 48.000
Prerequisite: MAE4360, Select 6 credits (2 courses) from the following course option: MAD3103, MAS3105, MAS3301, MAS4203, MHF4404, MTG3214.

MHF4404 History of Mathematics 3.000 credits

A study of the development of mathematics from ancient civilizations to the present time.
Lecture Hours: 48.000
Prerequisite: MAC2312

MTG3214 Euclidean Geometry 4.000 credits

This course encompasses a range of geometry topics and pedagogical ideas for the teaching of geometry including properties of shapes, defined and undefined terms, postulates and theorems, logical thinking and proofs, constructions, patterns, and sequences, the coordinate plane, axiomatic nature of Euclidean geometry, and basic topics of non-Euclidean geometries. Prerequisite: MAC 1147. (4 hr. lecture)
Lecture Hours: 64.000
Prerequisite: MAC1147
Course User Fee: \$25.00

MAE3951 Problem-Based Learning in Mathematics Education 2.000 credits

The teacher candidate will learn and apply the principles of project-based learning by designing and implementing projects to explore real-world problems, questions, and challenges in the field of education. The student will develop their technology, critical, creative, and communication skills by producing products to share their findings and proposed solutions. (2-hour lecture; this course requires approved clinical hours).
Lecture Hours: 32.000
Prerequisite: Departmental Approval Required.

**MAE4942 Seminar in Mathematics Education
3.000 credits**

The teacher candidate will engage in professional dialogue and explore professional learning opportunities related to teaching in the secondary mathematics setting. This seminar course is taken in conjunction with a full time, supervised teaching experience and provides an opportunity to examine and reflect on the daily experiences of becoming a highly effective teacher.

Lecture Hours: 48.000

Prerequisite: All Coursework, State Examinations, and E-Portfolio completed and passed., Corequisite: MAE4945

Course User Fee: \$18.00

MAE4945 Internship in Secondary Mathematics Education 9.000 credits

The teacher candidate will complete a full time, supervised teaching experience. The student will learn and experience all of the educational and professional responsibilities common to mathematics teachers.

Internship Hours: 432.000

Prerequisite: All coursework and state exams completed and passed, Corequisite: MAE4942

Program Description

Secondary Science Education - Biology

Bachelor of Science | Code: S9210 | 120 credit

The four-year baccalaureate degrees in Education are designed to prepare students to become teachers and pass state professional certification exams. An internship in a school setting is required to provide practical experience. Additionally, individuals with Bachelor's degrees in other fields are able to earn teacher certification. Please refer to the College Catalog for specific program prerequisites (www.mdc.edu).

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Prerequisite Courses

BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
EDF 1005	Introduction to the Education Profession	3.0
EDF 2144	Maximizing Student Potential in the School Context	3.0
EEX 2000	Introduction to Exceptional Student Education	3.0
EME 2040	Creativity, Innovation, and Technology	3.0

Semester 1

Units: 15.0

<i>Biological Sciences - Anatomy</i>		3.0
BSC 1084	Functional Human Anatomy	3.0
OR		
BSC 2020	Human Biology: Fundamentals of Anatomy/Physiology	3.0
EDF 4430	Measurement and Assessment in Education	3.0
EDG 3321	General Teaching Skills	3.0
Science Electives*		6.0

*Select 6 credits from the following course options: BCH 3023, BCH 3023L, BOT 3015, BSC 3930, CHM 2210, CHM 2210L, CHM 2211, CHM 2211L, CHM 2200, CHM 2200L, PCB 3060L, PCB 4674, SCE 3893, ZOO 3021

Semester 2		Units: 12.0
EEX 3071	Teaching Exceptional and Diverse Populations in Inclusive Settings	3.0
RED 3393	Differentiated Instruction in Content Reading	3.0
SCE 4362	Methods of Teaching Science	3.0
Science Electives*		3.0

*Select 3 credits from the following course options: BCH 3023, BCH 3023L, BOT 3015, BSC 3930, CHM 2210, CHM 2210L, CHM 2211, CHM 2211L, CHM 2200, CHM 2200L, PCB 3060L, PCB 4674, SCE 3893, ZOO 3021

Semester 3		Units: 9.0
PCB 3043	Fundamentals of Ecology	3.0
<i>Microbiology</i>		5.0
		5.0
MCB 2010	Microbiology	3.0
AND		
MCB 2010L	Microbiology Laboratory	2.0
OR		
		5.0
MCB 3023	Principles of Microbiology	3.0
AND		
MCB 3023L	Principles of Microbiology Lab	2.0
Science Elective*		1.0

*Select 1 credit from the following course options: BCH 3023, BCH 3023L, BOT 3015, BSC 3930, CHM 2210, CHM 2210L, CHM 2211, CHM 2211L, CHM 2200, CHM 2200L, PCB 3060L, PCB 4674, SCE 3893, ZOO 3021

Semester 4		Units: 12.0
EDG 3443	Classroom and Behavior Management	3.0
PCB 3060	Principles of Genetics	3.0
SCE 4363	Addressing Scientific Misconceptions in the Secondary Science Classroom (Practicum)	3.0
TSL 4324C	ESOL Strategies for Content Area Teachers	3.0
Semester 5		Units: 12.0
SCE 4943	Seminar in Science Education	3.0
SCE 4945	Internship in Science Education	9.0

Total: 60.0

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000

Course User Fee: \$18.00

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

EEX2000 Introduction to Exceptional Student Education 3.000 credits

The student will learn about the history, legal issues, and legislation related to exceptional student education in K-12. The student will learn about the differing types of exceptionalities, supports, and accommodations provided for students with differing learning profiles from birth to adulthood.

Lecture Hours: 48.000

Prerequisite: EDF1005

Course User Fee: \$18.00

EME2040 Creativity, Innovation, and Technology 3.000 credits

The student will learn to manage a productive & safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Prerequisite: CGS1060 OR Passing Score on the Computer Competency Exam

Course User Fee: \$190.03

BSC1084 Functional Human Anatomy 3.000 credits

This course covers basic human anatomy for students in allied health and mortuary science programs. It includes the dynamics of gross and functional anatomy, terminology, body orientation, and systematic relationships.

Lecture Hours: 48.000

BSC2020 Human Biology: Fundamentals of Anatomy/Physiology 3.000 credits

This course provides a basic understanding of the human body, its systems, and their functions. It includes the dynamics of physiology, terminology, and physiological relationships of the body systems.

Lecture Hours: 48.000

Course User Fee: \$10.00

EDF4430 Measurement and Assessment in Education 3.000 credits

The student will learn research-based principles of assessment. The student will select specific standards and competencies and develop formative and summative traditional and alternative assessments. The student will interpret assessment data that will improve academic achievement and ensure equity in the application of quantitative and qualitative assessments.

Lecture Hours: 48.000

Prerequisite: EDG3321

Course User Fee: \$18.00

EDG3321 General Teaching Skills 3.000 credits

The student will learn how human development and learning theories, developments in educational neuroscience, and current research-based pedagogy apply to the teaching and learning process. The student will incorporate principles and skills of effective instructional planning, including differentiated instruction, aligning instructional teaching skills to practice, selecting appropriate formative assessments, and developing learning experiences that require students to demonstrate a variety of skills.

Lecture Hours: 48.000

Course User Fee: \$18.00

EEX3071 Teaching Exceptional and Diverse Populations in Inclusive Settings 3.000 credits

The student will learn current research-based instructional strategies, educational neuroscience, and legal and ethical issues necessary in addressing the needs of diverse learners in inclusive classrooms. The student will make informed decisions in adapting, accommodating, and modifying the curriculum for students with special needs. Fifteen hours of clinical experience are required. Special fee. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: EEX2000 AND EDG3321

Course User Fee: \$18.00

RED3393 Differentiated Instruction in Content Reading 3.000 credits

The teacher candidate will differentiate instruction for individual students based on their strengths and needs by applying the principles of research-based strategies and integrating the six components of reading: phonological awareness, phonics, oral language, fluency, vocabulary, and comprehension in the content area classroom. The teacher candidate will have a broad knowledge of students with differing profiles to understand and apply evidence-based instructional practices by differentiating process, product, and context. Reading competencies 2 and 4 (fifteen hours of clinical experience are required in a math or science 3-12 classroom.)

Lecture Hours: 48.000

Prerequisite: EDG3321, : Optional Prerequisite: RED3013.

Course User Fee: \$18.00

SCE4362 Methods of Teaching Science 3.000 credits

The teacher candidate will learn to design and implement science instruction utilizing the national framework for K-12 science education and educational neuroscience to provide all students with high-quality science education. The student will learn about the theoretical knowledge and skills essential for facilitating science instruction in a variety of classroom settings. Fifteen hours of clinical experience are required.

Lecture Hours: 48.000

Corequisite: EDF4430

Course User Fee: \$35.00

PCB3043 Fundamentals of Ecology 3.000 credits

This is a foundations course in ecology. In this course, students will learn the basic principles of ecology at organismal, population, community, and ecosystem levels, including consideration of Florida's ecosystems and human impact on those systems.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

MCB3023 Principles of Microbiology 3.000 credits

This course offers an introduction to the principles of microbiology. Students will learn the taxonomy, biochemistry, genetics, and ecology of microorganisms and will have an understanding of the impact of microorganisms on the advancement of the biological sciences.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, CHM2211, CHM2211L, Corequisite: MCB3023L

Course User Fee: \$25.00

MCB3023L Principles of Microbiology Lab 2.000 credits

This Laboratory course accompanies MCB3023. Students will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, enumeration, biochemistry, identification, genetics, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2011, BSC2011L, CHM2211, CHM2211L, Corequisite: MCB3023

Course User Fee: \$110.00

EDG3443 Classroom and Behavior Management 3.000 credits

The student will examine strategies for fostering a productive learning environment, informed by research-based frameworks and learning theories. The student will learn strategies for observing, assessing, modifying behavior, and communicating with stakeholders. The student will learn to create a learning environment that encourages positive interactions, self-regulation, and resiliency. Fifteen hours of clinical experience is required.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

PCB3060 Principles of Genetics 3.000 credits

This course is an introduction to the mechanisms of transmission of hereditary information. Students will learn the classical Mendelian principles of heredity, deviation of Mendelian principles, genetic analysis, linkage and mapping, genetics of populations, gene regulation, mutation, the genetic bases of cancer and other genetic disorders will also be studied.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: PCB3060L

SCE4363 Addressing Scientific Misconceptions in the Secondary Science Classroom (Practicum) 3.000 credits

The teacher candidate will design, implement, and examine the alignment of their personal instructional practices to the national framework for K-12 science education utilizing the action research process. The student will focus on identifying, analyzing, and addressing misconceptions in science in grades 6-12. Sixty hours of clinical.

Lecture Hours: 48.000

Prerequisite: EEX3071, SCE4362, Corequisite: RED3393

Course User Fee: \$18.00

TSL4324C ESOL Strategies for Content Area Teachers 3.000 credits

The student will learn topics related to teaching content area subjects to English Language Learners. The student will plan and implement curriculum, instruction, and assessment activities to meet the needs of culturally and linguistically diverse learners. Fifteen hours of clinical experience are required. Pre/Corequisite: EDG3321.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

SCE4943 Seminar in Science Education 3.000 credits

The teacher candidate will engage in professional dialogue and explore professional learning opportunities related to teaching in the secondary science setting. This seminar course is taken in conjunction with a full time, supervised teaching experience and provides an opportunity to examine and reflect on the daily experiences of becoming a highly effective teacher.

Lecture Hours: 48.000

Prerequisite: All coursework (EDG 3321, EDF 4430, BSC 1084 or BSC 2020, RED 3393, SCE 4362, EEX 3071, MCB 2010/MCB 2010L or MCB 3023/MCB 3023L, PCB 3043, PCB 3060, TSL 4324C, EDG 3443, SCE 4363, state exams, and E-Portfolio completed and passed., Corequisite: SCE4945

Course User Fee: \$18.00

SCE4945 Internship in Science Education 9.000 credits

The teacher candidate will complete a full time, supervised teaching experience. The student will learn and experience all of the educational and professional responsibilities common to science teachers.

Internship Hours: 432.000

Prerequisite: All coursework (EDG 3321, EDF 4430, BSC 1084 or BSC 2020, RED 3393, SCE 4362, EEX 3071, MCB 2010/MCB 2010L or MCB 3023/MCB 3023L, PCB 3043, PCB 3060, TSL 4324C, EDG 3443, SCE 4363) and state exams completed and passed,

Corequisite: SCE4943

Certificate of Professional Preparation (CPP)

Cyber Defense

Certificate of Professional Preparation | Code: PBC04 | 24 credits

Semester 1		Units: 4.0
CTS 1134	Networking Technologies	4.0
OR		
CTS 1650	CCNA 1: Cisco Fundamentals	4.0
Semester 2		Units: 4.0
CIS 3360	Principles of Information Security	4.0
Semester 3		Units: 8.0
CIS 3361	Information Security Management	4.0
CIS 4366	Computer Forensics	4.0
Semester 4		Units: 8.0
CIS 4364	Intrusion Detection and Incident Response	4.0
CIS 4388	Advanced Computer Forensics	4.0
		Total: 24.0

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1650 CCNA 1: Cisco Fundamentals 4.000 credits

This is the first course of the four-course Cisco curriculum that will prepare students for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn networking concepts and practices, network terminology and protocols, the OSI reference model, cabling, cabling tools, routers, router and switch configurations, LAN/WAN topologies, IP addressing, and network standards.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Knowledge of computer operating systems and hardware is required.

Course User Fee: \$63.00

CIS3360 Principles of Information Security 4.000 credits

This upper division course provides an overview of information systems security principles, practices, methods, and tools. Students will learn about the threats against an organization's digital assets, as well as the tools and methods to mitigate those threats. Topics include cryptography, authentication and access control systems, infrastructure security, and ethical issues. Prerequisite: CTS1134 or CTS1650

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CIS3361 Information Security Management 4.000 credits

This course covers how to manage, design, oversee and assess an organization's information security. Students will learn how to develop an information security strategy, how to write information security policies, and how to manage information risk. Other topics include security program development and management, business continuity planning and disaster recovery planning. Prerequisite(s): CIS 3360 or CTS 1120. Laboratory fee. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS3360 OR CTS1120

Course User Fee: \$63.00

CIS4366 Computer Forensics 4.000 credits

This upper division course provides the student with knowledge and skills to conduct formal incident investigations. Students will learn how to collect and analyze evidence from Windows and Linux computer systems. Other topics include legal issues, evidence analysis, and report writing. Prerequisite(s): CIS 3360 or CET 2880C. Laboratory fee. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS3360 OR CET2880C

Course User Fee: \$63.00

CIS4364 Intrusion Detection and Incident Response 4.000 credits

This upper division course addresses the underlying principles and techniques for detecting and responding to current and emerging cybersecurity threats. Students will learn how to handle various types of malware, email, web, network, cloud and internal network incidents, as well as risk assessment methodologies, and policies related to incident handling. Prerequisite: CIS 3360.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS3360

Course User Fee: \$63.00

CIS4388 Advanced Computer Forensics 4.000 credits

This upper division course is a continuation of Computer Forensics. The course examines forensics techniques necessary to investigate and analyze network traffic. The course covers packet capture and analysis, log file analysis, and flow analysis. Other topics include mobile forensics, cloud forensics, malware forensics, database forensics, and investigating email crimes and web attacks. Prerequisite: CIS 4366 (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS4366

Course User Fee: \$63.00

[Educator Preparation Institute](#)

Certificate of Professional Preparation | Code: 69000 | 21 credits

The Educator Preparation Institute (EPI) is a state approved competency-based alternative certification program for individuals who have a bachelor's degree or higher in a discipline other than education and are currently teaching on a Temporary Teaching Certificate or who wish to enter the teaching profession. It focuses on the skills and competencies identified by the state as necessary for a high quality teacher to possess. Emphasis is placed on the Sunshine State Standards, teaching methods and strategies, the integration of technology into instructional practice, literacy development, assessment techniques and integration of technology into instructional practice, literacy development, assessment techniques and analysis of data, classroom management and school safety. At the completion of these modules the student will have successfully demonstrated the Florida Educator Accomplished Practices and have provided documentation of mastery in a comprehensive professional portfolio.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 9.0
EDG 3443	Classroom and Behavior Management	3.0
EPI 0001	Classroom Management	3.0
EPI 0002	Instructional Strategies	3.0
EPI 0003	Technology	3.0
Semester 2		Units: 6.0
EPI 0004	The Teaching and Learning Process	3.0
EPI 0010	Research-Based Practices in Reading	3.0
Semester		Units: 6.0
EPI 0020	Professional Foundations	1.0
EPI 0030	Empowering All Learners	2.0
EPI 0940	Field Experience II	2.0
EPI 0945	Field Experience I	1.0
		Total: 21.0

EDG3443 Classroom and Behavior Management 3.000 credits

The student will examine strategies for fostering a productive learning environment, informed by research-based frameworks and learning theories. The student will learn strategies for observing, assessing, modifying behavior, and communicating with stakeholders. The student will learn to create a learning environment that encourages positive interactions, self-regulation, and resiliency. Fifteen hours of clinical experience is required.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

EPI0001 Classroom Management 3.000 credits

The student will learn how to plan and develop a positive and productive learning environment in P-12 settings grounded in research-based character development models and learning theories. The student will learn to create a learning environment that encourages positive interactions and self-regulation. Ten hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Recommended Preparation: (1) Bachelor degree earned from an accredited institution of higher education in any area except Education; (2) Cumulative bachelor degree GPA of 2.5 or higher on a 4.0 scale; and (3) Admitted to the Educator Preparation Institute program.

EPI0002 Instructional Strategies 3.000 credits

The student will connect human developmental theories and current educational neuroscience research to the planning of instructional activities for students in P-12 settings. The student will learn to develop lesson plans that incorporate varied teaching strategies and questioning skills at all levels, differentiated instruction, and appropriate formative and summative assessments. The student will incorporate learning experiences that require P-12 students to demonstrate a variety of skills. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Prerequisite: EPI0001 or an equivalent course.

EPI0003 Technology 3.000 credits

The student will learn about the use of instructional technology to teach P-12 students. The student will learn how technology can support classroom management, administration, and teaching. The student will learn the legal and ethical use of technology. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Prerequisite: EPI0002

EPI0004 The Teaching and Learning Process 3.000 credits

The teacher candidate will learn the formal philosophies of education: perennialism, essentialism, progressivism, and reconstructionism, and their application to the P-12 teaching and learning process. The teacher candidate will develop a substantive understanding of the six components of reading as a process: oral language, phonological awareness, phonics, fluency, vocabulary, and comprehension, select and administer appropriate assessments, and analyze data to inform reading instruction. Reading competencies 1 and 3. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Prerequisite: EPI0002

EPI0010 Research-Based Practices in Reading 3.000 credits

The teacher candidate will differentiate instruction for all learners by applying the principles of research-based strategies and integrating the six components of reading: phonological awareness, phonics, oral language, fluency, vocabulary, and comprehension. Teacher candidates will have a broad knowledge of students from differing profiles to understand and apply evidence-based instructional practices by differentiating process, product, and context. Reading competencies 2 and 4. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Prerequisite: EPI0004

EPI0020 Professional Foundations 1.000 credits

The student will learn to evaluate his/her role as a productive member of the teaching profession. The student will learn to reflect on his/her development and mastery of the pre-professional Florida educator accomplished practices during the completion of the clinical education experience in a P-12 classroom setting. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 16.000

Prerequisite: EPI0004

EPI0030 Empowering All Learners 2.000 credits

This course focuses on effective strategies for identifying and meeting the varied needs of students in the classroom. Students will examine teaching practices, legal and ethical responsibilities, and assessment methods to ensure all learners are supported. The course covers approaches to differentiated instruction, fostering positive learning environments, and utilizing assessment data to guide instructional decisions. Participants will also learn how to collaborate with families and communities to support student learning. Through reflective practice, students will develop actionable plans to enhance teaching performance and improve learner outcomes.

Lecture Hours: 32.000

Prerequisite: EPI0002

EPI0940 Field Experience II 2.000 credits

This is the second of two field experience courses that mark the culmination of the educator preparation institute program. In this course, the teacher will observe the classroom and develop and teach a unit lesson in a P-12 setting. Building on previous formal classroom observations and actionable feedback on their lesson plans and mini-teache, the teacher candidate will refine their instructional practices. They will also demonstrate knowledge of the components of reading and the ability to apply effective instructional reading practices. The teacher candidate will be formally observed teaching in a P-12 setting. Reading course competency 5. Sixty hours of clinical experience are required for successful completion of this course.

Lecture Hours: 32.000

Prerequisite: EPI0004

EPI0945 Field Experience I 1.000 credits

The student will observe effective teaching/learning techniques in P-12 settings with to collect and analyze observational data, and also plan/implement teaching strategies that meet the needs of all learners. A formal observation/assessment of the students teaching performance will be conducted. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 16.000

Prerequisite: EPI0004

Program Description

Future-Ready Teaching: AI Applications in the Classroom

Certificate of Professional Preparation | Code: PBC08 | 15 credits

The Certificate of Professional Preparation (CPP) in Future-Ready Teaching: AI Applications in the Classroom equips educators with essential skills to integrate artificial intelligence into teaching and learning environments. Through hands-on coursework, participants learn to apply AI tools for data-driven instruction, foster innovation, and demonstrate leadership in educational technology. Graduates will be prepared to lead the way in AI-enhanced education for their schools and communities.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 6.0
CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
PHI 2680	Artificial Intelligence and Ethics	3.0
Semester 2		Units: 9.0
ECW 4012	Leading the Future: AI Advocacy and Innovation in Education	3.0
ECW 4013	Teaching with AI: Designing Human-Centered Classrooms	3.0
ECW 4014	Reimagining Assessments to Maximize Student Potential	3.0
		Total: 15.0

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

ECW4012 Leading the Future: AI Advocacy and Innovation in Education 3.000 credits

This course empowers educators to lead ethical, inclusive AI integration in schools through systems thinking, human-centered advocacy, and collective leadership. Students will explore inclusive policy, responsible implementation, and community empowerment, while designing initiatives that promote lifelong leadership, voice, and agency for meaningful and sustainable local impact.

Lecture Hours: 48.000

ECW4013 Teaching with AI: Designing Human-Centered Classrooms 3.000 credits

This course invites learners to explore the integration of artificial intelligence through a pedagogical lens, where technology serves as a tool for reimagining teaching and learning. Learners will critically engage with AI tools by grounding their use in learning theory and evidence-based design practices. Through inquiry, collaboration, and hands-on experimentation, learners will examine how AI can amplify creativity, personalize learning, and promote human-centeredness.

Lecture Hours: 48.000

ECW4014 Reimagining Assessments to Maximize Student Potential 3.000 credits

The course will explore the interconnected relationship between lesson planning, assessment, feedback, and classroom management while maintaining pedagogical integrity. The student will develop reflective, inclusive, and ethical practices that support differentiated instruction, student engagement, and responsible AI use across diverse learning environments.

Lecture Hours: 48.000

Instructional Design and Technology

Certificate of Professional Preparation | Code: CPP03 | 15 credits

This program is designed to provide baccalaureate-prepared students with the knowledge and skills necessary to work as instructional designers and technologists. Instructional designers and technologists plan and conduct instructional design workshops, seminars, and trainings for learning and organizational development. Instructional designers and technologists develop and deliver comprehensive, highly-engaging and interactive training experiences and evaluation strategies for all learning solutions that demonstrate impact and effectiveness. Instructional designers promote collaboration, partnerships, and relationships among the participants and stakeholders, acquire and apply new technology skills to instructional design practice, and design learning that reflects an understanding and appreciation of diverse learners and cultural differences.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester		Units: 15.0
EME 4610	Introduction to Instructional Design	3.0
EME 4611	Instructional Design Development 1	3.0
EME 4612	Instructional Design Development 2	3.0
EME 4671	Instructional Design Analysis	3.0
EME 4683	Instructional Design Application	3.0

Total: 15.0

EME4610 Introduction to Instructional Design 3.000 credits

This is the first in a series of five courses that leads to a certificate in Instructional Design. The student will develop an overview of the field of instructional design as it relates to training, development, and education. The student will compare and contrast instructional design models, learning theories, and current technologies.

Lecture Hours: 48.000

Prerequisite: Bachelor degree

EME4611 Instructional Design Development 1 3.000 credits

In this course, the student will develop and launch a prototype of a training, development, or educational opportunity using different types of media tools.

Lecture Hours: 48.000

Recommended Preparation: Bachelor Degree., Pre/Corequisite: EME4683

EME4612 Instructional Design Development 2 3.000 credits

In this course in Instructional Design, the student will develop assessments and an evaluation plan for a training, development, or educational opportunity. The student will conduct User Acceptance Testing using established criteria, and utilize feedback to modify his or her prototype.

Lecture Hours: 48.000

Recommended Preparation: Bachelor degree, Pre/Corequisite: EME4611

EME4671 Instructional Design Analysis 3.000 credits

In this course, the student will evaluate and analyze training, development, and education opportunities and project development. The student will develop a needs analysis for an instructional design project based on research-based best practices in the field.

Lecture Hours: 48.000

Recommended Preparation: Bachelor degree, Pre/Corequisite: EME4610

EME4683 Instructional Design Application 3.000 credits

In this course, the student will utilize best practices of learning design, learning theories, and instructional strategies for adult learners to write objectives, develop an outline, and create storyboards that encompass the scope of a training, development, or educational opportunity.

Lecture Hours: 48.000

Recommended Preparation: Bachelor degree, Pre/Corequisite: EME4671

Networking

Certificate of Professional Preparation | Code: PBC05 | 28 credit

Modern computer networks are increasingly pervasive and complex. They help improve the speed and efficiency of daily business operations, allowing for easy information discovery, storage, sharing, and protection. As technology evolves, it is indispensable for organizations to maintain the efficiency, speed and security of computer networks, thus facilitating seamless collaboration among people and applications. The Certificate of Professional Preparation (CPP) in Networking is designed for current and pre-professionals interested in working in the field of networking. They will learn how to plan, design, implement and maintain network infrastructures.

Note: Program Admissions Requirements: Baccalaureate degree in any field of study from a regionally accredited institution.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 4.0**

CTS 1134	Networking Technologies	4.0
OR		4.0
CTS 1650	CCNA 1: Cisco Fundamentals	4.0

Semester 2**Units: 12.0**

CNT 3526C	Wireless and Mobile Networking	4.0
CNT 4603	System Administration and Maintenance	4.0
CIS 3360	Principles of Information Security	4.0

Semester 3

Units: 12.0

CNT 3409C	Network Security	4.0
CNT 4702	Network Design and Planning	4.0
CIS 4347	Information Storage Management	4.0

Total: 28.0

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1650 CCNA 1: Cisco Fundamentals 4.000 credits

This is the first course of the four-course Cisco curriculum that will prepare students for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn networking concepts and practices, network terminology and protocols, the OSI reference model, cabling, cabling tools, routers, router and switch configurations, LAN/WAN topologies, IP addressing, and network standards.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Knowledge of computer operating systems and hardware is required.

Course User Fee: \$63.00

CNT3526C Wireless and Mobile Networking 4.000 credits

This is upper division course for students majoring in Information Systems Technology introduces students to wireless and mobile network architecture, protocols and technologies. In particular, students will learn about Wireless Local Area Networks (WLANs), Wireless Personal Area Networks (WPANs) and Wireless Metropolitan Area Networks (WMANs) technologies. Other topics include antenna concepts, cellular networks, the 802.11 network architecture, and wireless security. Prerequisites: CTS 1134 or CTS 1650.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CNT4603 System Administration and Maintenance 4.000 credits

This upper division course, for students majoring in Information Systems Technology, explores UNIX and Microsoft Windows systems and their administration and maintenance within the network setting. Students will learn how to install, maintain, and extend multi-user computer systems and how to develop administrative policies and procedures. Students will also learn how to apply troubleshooting and problem solving skills to resolve user and system issues.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CIS3360 Principles of Information Security 4.000 credits

This upper division course provides an overview of information systems security principles, practices, methods, and tools. Students will learn about the threats against an organization's digital assets, as well as the tools and methods to mitigate those threats. Topics include cryptography, authentication and access control systems, infrastructure security, and ethical issues. Prerequisite: CTS1134 or CTS1650

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CNT3409C Network Security 4.000 credits

This upper division course introduces students to current and emerging threats to the security of computer networks, as well as tools and techniques for the prevention, detection and recovery from such attacks. Topics include firewalls, intrusion detection and intrusion prevention systems, virtual private networks, remote authentication and authorization systems, and security protocols. Prerequisite(s): CIS 3360. Laboratory fee. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 27.000 - 32.000

Prerequisite: CIS3360

Course User Fee: \$63.00

CNT4702 Network Design and Planning 4.000 credits

This upper division course, for students majoring in Information Systems Technology, presents network design using layering. Students will learn how to apply cabling, topology, and architecture to design systems. Students will also learn how design impacts network performance and control issues such as congestion control, error control, and contention resolution. Prerequisite: CIS 3360.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS4347 Information Storage Management 4.000 credits

This upper division course, for students majoring in Information Systems Technology, introduces challenges and solutions for data storage and data management. Students will learn how to manage advanced storage systems, protocols, and architectures, including storage area networks (SAN), network attached storage (NAS), fiber channel networks, internet protocol SANS (IPSAN), ISCSI, and content-addressable storage (CAS).

Prerequisite: CTS1134 or CTS1650. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

Reading Endorsement

Certificate of Professional Preparation | Code: CPP02 | 15 credits

The Reading Endorsement is a Certificate of Professional Preparation that adds licensure to professionally certified teachers so that they can teach reading. This program meets the specifications in State Board Rule 6A-4.0292 for the Reading Endorsement.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 15.0**

RED 4033	Teaching Foundations of Reading Instruction	3.0
RED 4342	Application of Research Based Practices in Reading Education	3.0
RED 4541	Foundations of Assessment	3.0
RED 4654	Foundations and Applications of Differentiated Instruction	3.0
RED 4854	Reading Practicum	3.0

Total: 15.0

RED4033 Teaching Foundations of Reading Instruction 3.000 credits

The student will develop a foundational understanding of the six components of reading as a systematic process: oral language, phonological awareness, phonics, fluency, vocabulary, and comprehension. The student will recognize the principles, techniques, and procedures required to develop the foundational reading skills that are essential to increasing reading proficiency in students from grades K-12. This is the first of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 1.

Lecture Hours: 48.000

RED4342 Application of Research Based Practices in Reading Education 3.000 credits

Building on the foundation of RED4033, the student will apply the principles of evidence-based research in comprehensive reading instruction. The student will use the reading skills, techniques, and strategies that facilitate reading comprehension. The student will apply the principles of research-based reading instruction, and integrate the six components of reading to facilitate the comprehension of different texts. This is the second of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 2.

Lecture Hours: 48.000

Prerequisite: RED4033Bachelor's Degree

RED4541 Foundations of Assessment 3.000 credits

The student will select and administer appropriate assessments and analyze data to inform reading instruction to meet the needs of all students. The student will engage in a systematic problem-solving process to remediate reading difficulties in emergent, beginning, and fluent readers and will plan effective instructional interventions. This is the third of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 3.

Lecture Hours: 48.000

Prerequisite: RED4342Bachelor's Degree

RED4654 Foundations and Applications of Differentiated Instruction 3.000 credits

The student will apply research-based best practices in educational neuroscience research related to the instruction of reading. Using knowledge of diverse learners, the student will apply intensive, explicit, multisensory sequential approaches and instructional practices by differentiating the process, product, or context of their reading instruction, and providing scaffolds to enhance comprehension in all content areas. This is the fourth of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 4.

Lecture Hours: 48.000

Prerequisite: RED4541Bachelor's Degree

RED4854 Reading Practicum 3.000 credits

The student will synthesize and apply knowledge of the six components of reading, formal and informal reading assessments, skills related to data analysis, and differentiation of instruction in order to plan and implement a comprehensive, intensive, effective, systematic, multisensory, research-based reading plan of instruction for all students. This is the fifth of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 5.

Lecture Hours: 48.000

Prerequisite: RED4654Bachelor's Degree

Software Engineering

Certificate of Professional Preparation | Code: PBC06 | 28 credits

Software engineers contribute to the development and maintenance of computer systems software and applications software. Computer systems software consists of programs that include computing utilities and operating systems. Applications software includes a variety of user-focused programs from web-browsers to multimedia applications. The Certificate of Professional Preparation in Software Engineering will provide students with the knowledge of software development, computer architecture, algorithm design, and data structures to apply software engineering principles to software creation. By applying these engineering principles to every stage of the development process, students learn how to specify software requirements from clients and how to design, implement and validate software solutions. Note: Program Admissions Requirements: Baccalaureate degree in any field of study from a regionally accredited institution.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 4.0

COP 1334	Introduction to C++ Programming	4.0
Semester		Units: 8.0
CET 3126C	Computer Architecture	4.0
COP 2800	Java Programming	4.0
Semester 3		Units: 8.0
CET 3383C	Software Engineering I	4.0
COP 3530	Data Structures	4.0
Semester 4		Units: 8.0
CEN 4025C	Software Engineering II	4.0
COT 4400	Design and Analysis of Algorithms	4.0
		Total: 28.0

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CET3126C Computer Architecture 4.000 credits

This course is intended for upper division students majoring in Electrical and Computer Engineering Technology as well as Information Systems Technology. This course introduces the study of advanced microprocessor design. Students will learn the basic organization of computer systems including instruction-set architecture, execution pipeline, memory hierarchy, virtual memory, and I/O subsystems. Students also learn advanced processor microarchitecture issues such as dynamic instruction scheduling, branch prediction, lock-up free caches, instruction-level parallelism, multiple instruction fetch/issuing, speculative execution, etc. to improve computer processor performance. Students will experimentally verify microarchitecture designs using industry standard microarchitecture simulators.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

CET3383C Software Engineering I 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology or B.S. in Electrical and Computer Engineering Technology. The student will learn the basic principles and concepts of software engineering; system requirements; modeling and testing; object oriented analysis and design; testing and validation; configuration management; and the analysis, design and programming of extensible software systems. Prerequisite: CET2369C or COP2800.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2369C, COP2800

Course User Fee: \$63.00

COP3530 Data Structures 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology. Students will learn the fundamentals of data structures using the Java programming language. The students will learn to design, implement and use data structures to organize and store data in a computer so that it can be accessed and modified efficiently.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

CEN4025C Software Engineering II 4.000 credits

This upper division course is for students majoring in the B.S. in Information Systems Technology or the B.S. in Electrical and Computer Engineering Technology programs. This course covers in-depth topics in software process structures, process models, requirements modeling with use-cases and class based methods. Students will also learn design concepts including abstraction, OOD concepts, component-level and architectural design, user interface analysis and design, and design patterns. Prerequisite: CET3383C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: Pre-requisite requirement CET3383C

Course User Fee: \$63.00

COT4400 Design and Analysis of Algorithms 4.000 credits

This upper division course is for students majoring in the B.S. in Information Systems Technology or the B.S. in Electrical and Computer Engineering Technology programs. This course covers general techniques in algorithm design (such as divide-and-conquer, greedy method, dynamic programming, search and traversal techniques, branch-and-bound) in the context of problem domains like graph, sorting and optimization problems.

Prerequisite(s): COP1334 or COP2270. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800 AND Pre/Corequisite: COP3530

Course User Fee: \$63.00

Vulnerability Assessment

Certificate of Professional Preparation | Code: PBC07 | 24 credits

Vulnerability Assessment is a cybersecurity specialty for professionals who protect and defend an organization's information system from various threats. They perform assessments of systems and networks against known vulnerabilities and recommend appropriate mitigation countermeasures against potential cyber-attacks and data breaches. The Certificate of Professional Preparation (CPP) in Vulnerability Assessment is designed for current and pre-professionals interested in working in the field of cybersecurity operations. Note: Program Admissions Requirements: Baccalaureate degree in any field of study from a regionally accredited institution.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 8.0
CTS 1134	Networking Technologies	4.0
OR		4.0
CTS 1650	CCNA 1: Cisco Fundamentals	4.0
CIS 3215	Ethics in CyberSecurity	4.0
Semester 2		Units: 4.0
CIS 3360	Principles of Information Security	4.0
Semester 3		Units: 8.0
CIS 4204	Ethical Hacking I	4.0

CIS 4364	Intrusion Detection and Incident Response	4.0
Semester 4		Units: 4.0
CIS 4378	Ethical Hacking II	4.0

Total: 24.0

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1650 CCNA 1: Cisco Fundamentals 4.000 credits

This is the first course of the four-course Cisco curriculum that will prepare students for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn networking concepts and practices, network terminology and protocols, the OSI reference model, cabling, cabling tools, routers, router and switch configurations, LAN/WAN topologies, IP addressing, and network standards.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Knowledge of computer operating systems and hardware is required.

Course User Fee: \$63.00

CIS3215 Ethics in CyberSecurity 4.000 credits

This course provides the study of the risk factors for digital and ethical misconduct, and it explores ethics, relevant laws, regulations, policies, standards, moral, and social issues and responsibilities faced by CyberSecurity professionals. Coverage includes examination of CyberSecurity policies; Federal Laws and Authorities and International Standards; ethical and legal compliance and enforcement; business issues; contractual management of assets and liabilities; and issues involving privacy, disclosure, free speech and individual rights.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS3360 Principles of Information Security 4.000 credits

This upper division course provides an overview of information systems security principles, practices, methods, and tools. Students will learn about the threats against an organization's digital assets, as well as the tools and methods to mitigate those threats. Topics include cryptography, authentication and access control systems, infrastructure security, and ethical issues. Prerequisite: CTS1134 or CTS1650

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CIS4204 Ethical Hacking I 4.000 credits

This upper division course introduces students to penetration testing techniques. Students will learn how to footprint, scan, sniff, enumerate networks, and hack systems. Other topics include denial of service attacks, social engineering, malware, vulnerability analysis and relevant laws. Prerequisites: CIS3360

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS3360

Course User Fee: \$63.00

CIS4364 Intrusion Detection and Incident Response 4.000 credits

This upper division course addresses the underlying principles and techniques for detecting and responding to current and emerging cybersecurity threats. Students will learn how to handle various types of malware, email, web, network, cloud and internal network incidents, as well as risk assessment methodologies, and policies related to incident handling. Prerequisite: CIS 3360.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS3360

Course User Fee: \$63.00

CIS4378 Ethical Hacking II 4.000 credits

This upper division course is a continuation of Ethical Hacking I. Students will focus on how web applications, wireless networks, and mobile platforms can be hacked, and how intrusion detection systems (IDS), firewalls, and honeypots can be evaded. Other topics include cloud computing security, Internet of Things (IoT) security, and cryptography. Prerequisite: CIS4204.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS4204

Course User Fee: \$63.00

Associate in Arts (AA) Degree Pathways

Program Description

Accounting

Associate in Arts | Code: 10504 | 60 credit

This program offers fundamental instruction in accounting and related subjects (such as economics or business). Students who wish to become an accountant may transfer to senior colleges or universities that offer baccalaureate degrees in accounting. Accountants work in a variety of settings such as corporations, small businesses, financial institutions, and government agencies.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
GEB 1011	Principles of Business	3.0
Computer Competency*		

*Computer Skills Placement (CSP) ---or--- CGS 1060C. Departmental Recommendation: Computer Skills Placement (CSP)

Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0

<i>Oral Communications</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
ECO 2013	Principles of Economics (Macro)	3.0
QMB 2100	Basic Business Statistics	3.0

Semester 3

Units: 9.0

BSC 1005	General Education Biology	3.0
ECO 2023	Principles of Economics (Micro)	3.0
PHI 2600	Introduction to Ethics	3.0

Semester 4

Units: 13.0

<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
PSC 1515	Energy in the Natural Environment	3.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
Electives*		3.0

Select 3 credits from the following course options: ACG 1403, ACG 2360, ACG 2450, ACG 2630, CHI 1120, COP 1332, COP 1334, COP 2270, FRE 1120, GER 1120, ITA 1120, JPN 1120, MAN, MAR*, OST 2335, POR 1120, RUS 1120, SPN 1120, STA 2023, TAX*

Semester 5

Units: 13.0

ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
MAC 2233	Business Calculus	3.0
Electives*		6.0

Select 3 credits from the following course options: ACG 1403, ACG 2360, ACG 2450, ACG 2630, CHI 1120, COP 1332, COP 1334, COP 2270, FRE 1120, GER 1120, ITA 1120, JPN 1120, MAN, MAR*, OST 2335, POR 1120, RUS 1120, SPN 1120, STA 2023, TAX*

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits
Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033Any Mathematics level higher than MAT 1033 will satisfy this requirement. MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites:

ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

MAC2233 Business Calculus 3.000 credits

This course introduces the basic concepts of differential and integral calculus for students majoring in business administration and related fields. Topics include limits, continuity, differentiation and integration of polynomials, logarithmic and exponential functions with applications to business, economics, and the life sciences. Prerequisite: MAC 1105 or MAC1106 with a grade of C or better or departmental permission.

Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

Program Description

Agriculture

Associate in Arts | Code: 10100 | 60 credit

Agriculture is the art, science and industry of managing the growth of plants and animals for human use. Study at MDC emphasizes a strong foundation in the sciences of biology (including botany), chemistry and/or physics. The range of careers in agriculture extends from rural farming to urban landscaping, with numerous specializations in areas such as hydroponics, agricultural engineering, animal husbandry, food packing and processing and soil chemistry.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
HOS 1010	Horticulture 1	3.0
Computer Competency*		

*Computer Skills Placement (CSP) ---or--- CGS 1060C. Departmental Recommendation: Computer Skills Placement (CSP)

Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
HUM 1020	Humanities	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
SLS 1106	First Year Experience Seminar	1.0

Semester 3		Units: 11.0
BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
PSY 2012	Introduction to Psychology	3.0
IPM 2112	Principles of Entomology	3.0

Semester 4		Units: 13.0
<i>Biological Science</i>		4.0
BSC 2011	Principles of Biology 2	3.0
AND		
BSC 2011L	Principles of Biology 2 Laboratory	2.0
SPC 2608	Introduction to Public Speaking	3.0
ORH 1251	Nursery Practices 1	3.0
<i>Program Elective</i>		3.0
IPM 2635	Introduction to Plant Pathology	3.0

OR		
HOS 2005	Hydroponic Systems	3.0

Semester 5

Units: 12.0

ECO 2013	Principles of Economics (Macro)	3.0
PHI 2600	Introduction to Ethics	3.0
STA 2023	Statistical Methods	3.0
ORH 2949	Landscape Technology Internship	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

HOS1010 Horticulture 1 3.000 credits

This is an introductory course on the principles of horticulture. Students will learn plant structure and function, plant propagation, plant nutrients and fertilizers, potting media, soils, pruning, and plant pests. A survey of various fields in ornamental horticulture will also be covered.

Lecture Hours: 48.000

ENC1102 English Composition 2 3.000 credits

This is the second required general education core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres, and/or nonfiction, and produce a documented paper based on research. Prerequisite: ENC1101. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

SL1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

IPM2112 Principles of Entomology 3.000 credits

This is an introductory course on the principles of entomology. Students will learn to identify characteristics of arthropods, the insect orders, and the growth cycle of insects. Students will also address insect pests specific to South Florida and methods to responsibly manage plant pests.

Lecture Hours: 48.000

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ORH1251 Nursery Practices 1 3.000 credits

The student will learn the techniques and practices in commercial production of ornamental plants.

Emphasis on types of nurseries.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: HOS1010

IPM2635 Introduction to Plant Pathology 3.000 credits

The students will learn to identify diseases that affect plants and management practices for different types of plant diseases. Environmental factors contributing to a plant's susceptibility to a particular disease will also be discussed. Methods of prevention, eradication, and control will be given for each specific disease.

Lecture Hours: 48.000

HOS2005 Hydroponic Systems 3.000 credits

This course will provide an overview of the different types of hydroponic systems. Students will learn about set-up requirements, maintenance, nutrient formulations, and growing plants in a soilless culture. The course will emphasize knowledge and skills required to run small hydroponic systems. (3 Hr. Lecture)

Lecture Hours: 48.000

Course User Fee: \$35.60

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

ORH2949 Landscape Technology Internship 3.000 credits

The internship will provide students with hands-on work experience in horticulture. Landscape, or related technology industries. Students will learn employability skills, and the specific skills and safety requirements necessary for effective work in this environment.

Other Hours: 144.000

Program Description

Associate in Arts | Code: 12200 | 60 credit

Anthropology studies all aspects of human life by evaluating society, evolution and culture. Course offerings prepare the student in the four fields of the discipline: cultural anthropology, physical anthropology, anthropological linguistics and archaeology. Most anthropologists are researchers who work in museums or educational institutions. Students majoring in anthropology should plan to obtain the Ph.D. degree to fully succeed in the field.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
ANT 2000	Introduction to Anthropology	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
ANT 2410	Introduction to Cultural Anthropology	3.0
<i>Pathway Elective</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		
SYG 2430	Marriage and The Family	3.0

Computer Competency*

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT).

Semester 3		Units: 12.0
ISS 1120	The Social Environment	3.0
PHI 2010	Introduction to Philosophy	3.0
SPC 1017	Introduction to Communication	3.0
SYG 2000	Introduction to Sociology	3.0

Semester 4		Units: 12.0
BSC 1005	General Education Biology	3.0
PHI 2600	Introduction to Ethics	3.0
SYG 2230	Multi-Ethnic America	3.0
<i>Pathway Elective</i>		3.0
EUH 2032	History of The Holocaust	3.0
OR		
PSY 2800	Psychology of Genocide	3.0

Semester 5**Units: 11.0**

ANT 2511	Introduction to Physical Anthropology	3.0
PSC 1515	Energy in the Natural Environment	3.0
<i>Pathway Elective</i>		5.0
CLP 2000	Dynamics of Behavior	3.0
OR		
PSY 2050	Introduction to Forensic Psychology	3.0
OR		
WOH 2012	History of World Civilization to 1789	3.0
OR		
WOH 2022	History of World Civilization From 1789	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ANT2410 Introduction to Cultural Anthropology 3.000 credits

This course focuses on the study of human organization and culture by exploring similarities and differences among a variety of cultures. Students will apply theories and methods developed by anthropologists to understand complex human behaviors revealed in the cultural expression of race, ethnicity, language, economic systems, political systems, families and kinship, marriage, gender, religion, ecology, art, transnational migration, and the globalization of the world system.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

SYG2430 Marriage and The Family 3.000 credits

The family as a social institution — its origin and development, its forms and functions, its interrelation with other social institutions, and its role in contemporary civilization. Areas of study include factors contributing to or acting against successful, stable marriage.

Lecture Hours: 48.000

ISS1120 The Social Environment 3.000 credits

The broader aspects of the social environment with a focus on the political and economic dimensions of culture; emphasis is given to a critical analysis of the roles that economics and government play in the life of the individual and the nation state, and their global implications. (3 hr lecture)

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

SYG2230 Multi-Ethnic America 3.000 credits

In this course, students will gain an understanding of the theory and problems of underrepresented populations in American society, with an emphasis on inequity, bias, and the evolving dynamics of bias and discrimination. Additionally, the course will critically examine the portrayal of underrepresented populations in the Western Canon, providing insight into the historical and cultural influences that have shaped societal perceptions and attitudes.

EUH2032 History of The Holocaust 3.000 credits

This is a foundation course in Holocaust Studies. Students will learn the historical origins, execution and consequences of the Holocaust. This course also examines the Holocaust's place in context of genocides past and present.

Lecture Hours: 48.000

PSY2800 Psychology of Genocide 3.000 credits

Students will learn the psychological, social, and cultural roots of genocide, human cruelty, and mass violence. Students will examine the various factors influencing such acts, and the emotional and psychological impact upon victims, perpetrators, rescuers, and society.

Lecture Hours: 48.000

Prerequisite: ENC1101

ANT2511 Introduction to Physical Anthropology 3.000 credits

This course introduces students to the origins of humanity from early primates, extinct hominins, and to modern people. Students in this course will learn the basics of evolutionary theory and genetics, investigate human evolutionary history through the fossil record, observe contemporary non-human primates, and apply this knowledge to a biocultural understanding of human variation, past and present.

Lecture Hours: 48.000

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CLP2000 Dynamics of Behavior 3.000 credits

Analysis of mechanisms of adjustment, motivation, frustration and conflict, learning personality and psychotherapy. Emphasis is on the psychological processes of the normal individual functioning in society rather than on behavior disorders.

Lecture Hours: 48.000

PSY2050 Introduction to Forensic Psychology 3.000 credits

This course explores the interaction between psychology and the legal system. Students will learn the foundations, history, and terminology of forensic psychology and the influence of media and cultural issues. Students will also learn about the forensic psychologists role in criminal and civil proceedings, public policy, law enforcement, and victimization.

Lecture Hours: 48.000

Prerequisite: PSY2012

WOH2012 History of World Civilization to 1789 3.000 credits

This course covers the history of World Civilizations from the prehistoric period to the 18th century. Students will learn of the major political, social, economic, and cultural structures of civilizations and their development through 1789.

Lecture Hours: 48.000

WOH2022 History of World Civilization From 1789 3.000 credits

The student will examine the historical development of world civilizations since 1789. Students will learn of historical processes and developments in social, cultural, political, and economic contexts since the 18th century.

Lecture Hours: 48.000

Program Description

Architecture

Associate in Arts | Code: 10200 | 60 credit

This program provides a foundation in areas such as architectural drawing, design and structure, as well as necessary courses in mathematics. Students may transfer to one of three universities in Florida that have accredited programs in architecture. An architect designs and oversees the construction or remodeling of buildings, working with engineers and contractors toward a prescribed goal.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

		Units: 12.0
ENC 1101	English Composition 1	3.0
ARH 1000	Art Appreciation	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
ARC 1115	Architectural Communications 1	2.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2

Units: 14.0

ENC 1102	English Composition 2	3.0
MAC 1105	College Algebra	3.0
ARC 1126C	Architectural Drawing 1	4.0
ARC 1301	Architectural Design 1	4.0
Computer Competency *		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3 **Units: 6.0**

<i>Natural Science</i>		3.0
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
ARC 2701	History of Architecture 1	3.0

Semester 4 **Units: 14.0**

MAC 1114	Trigonometry	3.0
SPC 2608	Introduction to Public Speaking	3.0
ARC 1302	Architectural Design 2	4.0
ARC 2171C	Computer Aided Drafting 1	4.0

Semester 5 **Units: 14.0**

<i>Natural Science</i>		4.0
PHY 2053	Physics (without Calculus) 1	3.0
AND		
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0
<i>Social Science</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
OR		
WOH 2012	History of World Civilization to 1789	3.0
OR		
WOH 2022	History of World Civilization From 1789	3.0
ARC 2461	Architectural Materials and Construction 1	4.0
ARC 2702	History of Architecture 2	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ARC1115 Architectural Communications 1 2.000 credits

Exercises in freehand drawing, sketching and linear perspective are designed to increase the student's awareness of the architectural environment. This is accomplished through a series of form studies of nature, architectural forms, and abstract elements of composition.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$25.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions. Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC1301 Architectural Design 1 4.000 credits

Introductory course to architectural design, its scope, methods and vocabulary interfacing graphics and design as a means towards an awareness and understanding of basic organizational principles. Design concepts analyzed through graphical representation and modeling. Pre/Corequisite: ARC1115.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

ARC2701 History of Architecture 1 3.000 credits

This course offers a survey of architecture from ancient times through the 18th century and includes works from the Western canon. Students will examine the integration of art forms, structural elements, and ornamental designs across world civilizations, analyzing their connections to broader aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societies throughout historical periods.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ARC1302 Architectural Design 2 4.000 credits

A continuation of ARC1301, emphasizing the application of ordering concepts, and aspects and determinants of form and space. An individual design process is developed by the student. Prerequisite: ARC1301; Pre/Corequisites: ARC1126, ARC2701.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC2171C Computer Aided Drafting 1 4.000 credits

Computer-aided drafting as it applies in the fields of architecture and interior design using office simulation. Emphasis is on the production of computer-aided drafting of working drawings involving different types of structure.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR ARC2461

Course User Fee: \$50.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

WOH2012 History of World Civilization to 1789 3.000 credits

This course covers the history of World Civilizations from the prehistoric period to the 18th century. Students will learn of the major political, social, economic, and cultural structures of civilizations and their development through 1789.

Lecture Hours: 48.000

WOH2022 History of World Civilization From 1789 3.000 credits

The student will examine the historical development of world civilizations since 1789. Students will learn of historical processes and developments in social, cultural, political, and economic contexts since the 18th century.

Lecture Hours: 48.000

ARC2461 Architectural Materials and Construction 1 4.000 credits

An introduction to basic materials and methods of building construction. Emphasis is on wood, concrete, unit masonry, and light steel construction. Laboratory projects may include working drawings' interpretation, sketching construction details, or field trips to construction sites and fabricant plants. Designed primarily as the initial materials and methods course for architectural transfer students.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR BCN1251

Course User Fee: \$25.00

ARC2702 History of Architecture 2 3.000 credits

This course provides a survey of architecture from the 19th century to the present and includes works from the Western canon and examining architectural developments. Students will analyze the integration of art forms, structural elements, and ornamental designs across civilizations, exploring their connections to human development, including art and history. Through this study, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societal changes and cultural expressions throughout modern and contemporary periods. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$10.00

Program Description

Art or Art Education

Associate in Arts | Code: 11000 | 60 credit

This program offers hands-on instruction in medias such as ceramics, jewelry making and metalsmithing, painting, photography, print making and sculpture. Additionally, the curriculum includes design, art history and education classes, so that students may work as artists or art teachers.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
ARH 1000	Art Appreciation	3.0
ART 1300C	Drawing	4.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 14.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
ART 1202C	Two-Dimensional Design	4.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
Semester 3		Units: 7.0
ARH 2050	Art History 1	3.0
ART 1202C	Two-Dimensional Design	4.0
Semester 4		Units: 13.0
ARH 2051	Art History 2	3.0
BSC 1005	General Education Biology	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
ART 1330C	Figure Drawing	4.0
Semester 5		Units: 12.0
ESC 1000	General Education Earth Science	3.0
SPC 1017	Introduction to Communication	3.0
<i>Social Science</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		
ANT 2000	Introduction to Anthropology	3.0
Studio Art*		3.0

*Select 3 credits from any Studio Art course with an ART prefix.

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

ART1300C Drawing 4.000 credits

Basic problems in freehand drawing, including perspective, still-life and landscape. Emphasis is on developing a sense of structure through line, form and texture.

Lecture Hours: 32.000, Lab Hours: 64.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ART1202C Two-Dimensional Design 4.000 credits

This course is designed to give students an understanding of advanced concepts of two-dimensional design and to give hands on opportunity to transform visual and experiential information into two-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals.

Lecture Hours: 32.000, Lab Hours: 64.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ARH2050 Art History 1 3.000 credits

This course offers a survey of visual arts from prehistory to 800 C.E. and includes works from the western canon. Students will examine artistic expressions across civilizations, analyzing their connections to historical contexts and aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how visual arts reflect and influence societal developments, ideas, and values throughout early history.

Lecture Hours: 48.000

Course User Fee: \$10.00

ART1202C Two-Dimensional Design 4.000 credits

This course is designed to give students an understanding of advanced concepts of two-dimensional design and to give hands on opportunity to transform visual and experiential information into two-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals.

Lecture Hours: 32.000, Lab Hours: 64.000

ARH2051 Art History 2 3.000 credits

In this course, students will embark on a world survey of the visual arts from 800 to 1850 A.D., exploring various artistic movements, styles, and cultural contexts. The course will delve into the rich and diverse artistic traditions from different regions, including works from the Western canon. Writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$5.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ART1330C Figure Drawing 4.000 credits

Drawing and painting from the live model with emphasis on structure, movement and expression.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$80.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

Program Description

Atmospheric Science & Meteorology

Associate in Arts | Code: 11903 | 60 credit

To transfer to a four-year program in atmospheric science and meteorology, students must take science and math courses as well as introductory courses in meteorology. Job opportunities may include weather forecasting in aviation, marine or shipping companies, government agencies, and broadcasting or transportation industries. Additionally, meteorologists may work with other scientists researching phenomena such as volcanoes, hurricanes and fluctuating climate systems.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
EVR 1001	Introduction to Environmental Science	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2		Units: 15.0
ENC 1102	English Composition 2	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
MET 1010	Introduction to Weather	3.0
GIS 1040	Introduction to GIS Technology	4.0

Semester 3		Units: 10.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
<i>Natural Science</i>		5.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
AND		
CHM 1045L	General Chemistry Lab	2.0

Semester 4		Units: 11.0
HUM 1020	Humanities	3.0
SPC 1017	Introduction to Communication	3.0
<i>Natural Science</i>		5.0
PHY 2048	Physics with Calculus 1	4.0
AND		
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 5		Units: 11.0
MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
<i>Natural Science</i>		5.0
PHY 2049	Physics with Calculus 2	4.0
AND		
PHY 2049L	Physics with Calculus 2 Lab	1.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

MET1010 Introduction to Weather 3.000 credits

An introduction to fundamentals of weather and their impact on human activities. Topics include temperature, humidity, clouds, precipitation, air masses, fronts, and storms. Emphasis is on understanding how these processes take place and their results.

Lecture Hours: 48.000

Pre/Corequisite: PSC1515

Course User Fee: \$10.00

GIS1040 Introduction to GIS Technology 4.000 credits

This course teaches the principles of Geographic Information Systems (GIS). The student will learn fundamental GIS concepts, GIS models and formats, as well as map creation and analysis. At the end of the course, the student will be able to perform basic operations in GIS software. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2049 Physics with Calculus 2 4.000 credits

Foundation course for physical science and engineering majors. PHY 2048 covers classical mechanics and thermodynamics. PHY 2049 includes electricity, magnetism, waves and optics.

Lecture Hours: 64.000

Prerequisite: PHY2048, PHY2048L, Corequisite: MAC2312, PHY2049L

Course User Fee: \$10.00

PHY2049L Physics with Calculus 2 Lab 1.000 credits

Laboratory for PHY 2049.

Lab Hours: 32.000

Prerequisite: PHY2048, PHY2048L, Corequisite: MAC2312, PHY2049

Course User Fee: \$20.00

Program Description

Biology

Associate in Arts | Code: 10400 | 60 credit

Biology, or life science, is the study of all aspects of living organisms, emphasizing the relationship of animals and plants to their environment. This program provides the first two years of a four-year curriculum for students planning to major in biology, botany, zoology, marine biology, ecological studies or microbiology. Biology majors may also enter professional schools in medicine, dentistry, veterinary medicine, optometry or podiatry.

Program Modality: Both [In-Person (IP) and On-line]**Contracted:** No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
SPC 1017	Introduction to Communication	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT).

Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
<i>Natural Science</i>		10.0
		5.0
BSC 2010	Principles of Biology	3.0
AND		
BSC 2010L	Principles of Biology 1 Laboratory	2.0
AND		
		5.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
AND		
CHM 1045L	General Chemistry Lab	2.0

Semester 3	Units: 13.0
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STA 2023	Statistical Methods	3.0
<i>Natrual Science</i>		10.0
BSC 2011	Principles of Biology 2	3.0
AND		
BSC 2011L	Principles of Biology 2 Laboratory	2.0
AND		
CHM 1046	General Chemistry and Qualitative Analysis	3.0
AND		
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0

Semester 4

Units: 13.0

ANT 2000	Introduction to Anthropology	3.0
PHI 2600	Introduction to Ethics	3.0
Math Elective Option*		3.0
CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0

*Select 3 credits from any course under the MAC prefix

Semester 5

Units: 8.0

HUM 1020	Humanities	3.0
CHM 2211	Organic Chemistry 2	3.0
CHM 2211L	Organic Chemistry 2 Laboratory	2.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

CHM2211 Organic Chemistry 2 3.000 credits

In organic chemistry 2, students will learn about nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds.

Lecture Hours: 48.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211L

Course User Fee: \$15.00

CHM2211L Organic Chemistry 2 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2211. Topics include nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds will be performed in a laboratory setting.

Lab Hours: 64.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211

Course User Fee: \$45.00

Program Description

Building Construction

Associate in Arts | Code: 10907 | 60 credit

This program is for students primarily interested in the construction of buildings rather than their architectural design. Coursework includes math and science subjects as well as courses in business and construction. A four-year degree in this program will prepare students to enter the building construction industry at the management level.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		

THE 2000	Theatre Appreciation	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2

Units: 12.0

ENC 1102	English Composition 2	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
GLY 1010	Physical Geology	3.0
GLY 1010L	Physical Geology Laboratory	1.0
Computer Competency*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Skills Placement (CSP)

Semester 3

Units: 10.0

ARC 1126C	Architectural Drawing 1	4.0
ARC 2701	History of Architecture 1	3.0
ECO 2013	Principles of Economics (Macro)	3.0

Semester 4

Units: 14.0

ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
BUL 2241	Business Law 1	3.0
SPC 2608	Introduction to Public Speaking	3.0
SUR 1101C	Surveying 1	4.0

Semester 5

Units: 11.0

PHY 2053	Physics (without Calculus) 1	3.0
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0
STA 2023	Statistical Methods	3.0
Pathway Electives*		4.0

*Select 4 credits from the following course options: BCN1272, BCN1275, BCT1743, BCT1750, BCT1770, BCT1771, or BCT2760

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon.

Written analysis of literary works may be required.

Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics.

Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

GLY1010L Physical Geology Laboratory 1.000 credits

Laboratory for GLY 1010. Studies of common minerals and rocks and topographic and geologic maps along with aerial photography.

Lab Hours: 32.000

Corequisite: GLY1010

Course User Fee: \$10.00

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions.

Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC2701 History of Architecture 1 3.000 credits

This course offers a survey of architecture from ancient times through the 18th century and includes works from the Western canon. Students will examine the integration of art forms, structural elements, and ornamental designs across world civilizations, analyzing their connections to broader aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societies throughout historical periods.

Lecture Hours: 48.000

Course User Fee: \$10.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

SUR1101C Surveying 1 4.000 credits

The theories and practices in surveying and the use of the principal types of surveying instruments in horizontal and vertical planes. Problems include the measurement of distance; the use of compass, sextant, transit traverse, stadia, and basic mapping. Field and laboratory practice are required. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147

Course User Fee: \$55.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

Program Description

Business Administration

Associate in Arts | Code: 10503 | 60 credits

Business Administration includes courses in accounting, business law and finance, as well as more generalized courses in mathematics. Students may transfer to senior colleges or universities that offer baccalaureate degrees in business administration. Ultimately, graduates may work in the fields of banking, finance, marketing, information systems or real estate.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
GEB 1011	Principles of Business	3.0
MAC 1105	College Algebra	3.0
SLS 1106	First Year Experience Seminar	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT).

Semester 2		Units: 12.0
ECO 2013	Principles of Economics (Macro)	3.0
ENC 1102	English Composition 2	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
QMB 2100	Basic Business Statistics	3.0

Semester 3		Units: 9.0
BSC 1005	General Education Biology	3.0
ECO 2023	Principles of Economics (Micro)	3.0
PHI 2600	Introduction to Ethics	3.0

Semester 4		Units: 13.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
<i>Humanities *</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0

OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
PSC 1515	Energy in the Natural Environment	3.0
Pathway Electives*		3.0

Select 3 credits from the following course option: ACG 1403, ACG 2360, ACG 2450, ACG 2630, CHI 1120, COP 1332, COP 1334, COP 2270, FRE 1120, GER 1120, ITA 1120, JPN 1120, MAN, MAR*, OST 2335, POR 1120, RUS 1120, SPN 1120, STA 2023, TAX*.

Semester		Units: 13.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
MAC 2233	Business Calculus	3.0
Pathway Elective*		6.0

Select 6 credits from the following course option: ACG 1403, ACG 2360, ACG 2450, ACG 2630, CHI 1120, COP 1332, COP 1334, COP 2270, FRE 1120, GER 1120, ITA 1120, JPN 1120, MAN, MAR*, OST 2335, POR 1120, RUS 1120, SPN 1120, STA 2023, TAX*.

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits
Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033 Any Mathematics level higher than MAT 1033 will satisfy this requirement. MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision- making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L
Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.
Lab Hours: 32.000
Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071
Course User Fee: \$32.00

MAC2233 Business Calculus 3.000 credits

This course introduces the basic concepts of differential and integral calculus for students majoring in business administration and related fields. Topics include limits, continuity, differentiation and integration of polynomials, logarithmic and exponential functions with applications to business, economics, and the life sciences. Prerequisite: MAC 1105 or MAC1106 with a grade of C or better or departmental permission.
Computational Course.
Lecture Hours: 48.000
Prerequisite: MAC1105 OR MAC1106
Course User Fee: \$18.00

Program Description

Chemistry

Associate in Arts | Code: 11901 | 60 credits

Chemistry is the science that investigates the composition, properties and change of properties of elementary forms of matter. In addition to coursework in chemistry, the A.A. is a science and math intensive program that includes courses in botany, biology, physics, geometry and calculus. Chemists may work as researchers, analysts, or quality control specialists in companies that manufacture anything from pharmaceuticals to food products. Additionally, students may pursue careers in medicine, environmental science, chemical engineering or many other fields.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester		Units: 13.0
ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 1105	College Algebra	3.0
SLS 1106	First Year Experience Seminar	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		

POS 2041	American Federal Government	3.0
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Semester 2	Units: 13.0
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CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
ENC 1102	English Composition 2	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0

Semester 3	Units: 11.0
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CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
<i>Oral Communic</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
PSY 2012	Introduction to Psychology	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT).

Semester 4	Units: 13.0
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CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
MAC 2311	Calculus and Analytical Geometry 1	5.0

Semester 5	Units: 10.0
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BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view. Lecture Hours: 48.000

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L. Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105
Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM 2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting.

Lab Hours: 64.000

Prerequisite: CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

Program Description

Computer Arts Animation

Associate in Arts | Code: 11005 | 60 credits

This program enables students develop creative and artistic skills conjunction with advanced computer skills. Studies include basic drawing and figure drawing, use of computer animation software and general education, well as evaluation of trends and standards in the animation industry for television and film.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: Yes

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
ARH 1000	Art Appreciation	3.0
ART 1300C	Drawing	4.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 14.0
ENC 1102	English Composition 2	3.0
MAC 1105	College Algebra	3.0
ART 1202C	Two-Dimensional Design	4.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
Semester 3		Units: 7.0
ARH 2050	Art History 1	3.0
ART 1203C	Three-Dimensional Design	4.0
Semester 4		Units: 13.0
BSC 1005	General Education Biology	3.0
ARH 2051	Art History 2	3.0
ART 1330C	Figure Drawing	4.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		

POS 2041	American Federal Government	3.0
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Semester 5

Units: 12.0

ESC 1000	General Education Earth Science	3.0
PSY 2012	Introduction to Psychology	3.0
SPC 2608	Introduction to Public Speaking	3.0
<i>Pathway Elective</i>		3.0
Electives*		3.0
OR		
<i>Foreign Languages</i>		4.0
ASL 1150C	American Sign Language 2	4.0
OR		
CHI 1121	Elementary Mandarin Chinese 2	4.0
OR		
FRE 1121	Elementary French 2	4.0
OR		
GER 1121	Elementary German 2	4.0
OR		
ITA 1121	Elementary Italian 2	4.0
OR		
JPN 1121	Elementary Japanese 2	4.0
OR		
POR 1121	Elementary Portuguese 2	4.0
OR		
RUS 1121	Elementary Russian 2	4.0
OR		
SPN 1121	Elementary Spanish 2	4.0

Select 4 credits from the following course option: ART, CAP*, DIG*, FIL*

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

ART1300C Drawing 4.000 credits

Basic problems in freehand drawing, including perspective, still-life and landscape. Emphasis is on developing a sense of structure through line, form and texture.

Lecture Hours: 32.000, Lab Hours: 64.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

ART1202C Two-Dimensional Design 4.000 credits

This course is designed to give students an understanding of advanced concepts of two-dimensional design and to give hands on opportunity to transform visual and experiential information into two-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals. Lecture Hours: 32.000, Lab Hours: 64.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement. Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

ARH2050 Art History 1 3.000 credits

This course offers a survey of visual arts from prehistory to 800 C.E. and includes works from the western canon. Students will examine artistic expressions across civilizations, analyzing their connections to historical contexts and aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how visual arts reflect and influence societal developments, ideas, and values throughout early history. Lecture Hours: 48.000
Course User Fee: \$10.00

ART1203C Three-Dimensional Design 4.000 credits

This course is designed to give students an understanding of the concepts of three-dimensional design and to provide hands-on opportunity to transform visual and experiential information into three-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals. Self-evaluation and safety skills will also figure prominently. Prerequisite: ART1202C. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ART1202C
Course User Fee: \$40.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method. Lecture Hours: 48.000
Course User Fee: \$8.00

ARH2051 Art History 2 3.000 credits

In this course, students will embark on a world survey of the visual arts from 800 to 1850 A.D., exploring various artistic movements, styles, and cultural contexts. The course will delve into the rich and diverse artistic traditions from different regions, including works from the Western canon. Writing intensive course. Lecture Hours: 48.000
Course User Fee: \$5.00

ART1330C Figure Drawing 4.000 credits

Drawing and painting from the live model with emphasis on structure, movement and expression. Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$80.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ASL1150C American Sign Language 2 4.000 credits

This course provides continued instruction in the linguistic principles of American Sign Language and an additional 500 sign concepts. The course includes lecture, discussion and lab practice which is conducted in ASL. Prerequisite: ASL1140C.

Lecture Hours: 64.000

CHI1121 Elementary Mandarin Chinese 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Mandarin Chinese. Note: students must pass this course with a "C" or better to continue to CHI2220.

Lecture Hours: 64.000

Prerequisite: CHI1120

Course User Fee: \$15.00

FRE1121 Elementary French 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in French. Note: Students must pass this course with a C or better to continue to FRE2220.

Lecture Hours: 64.000

Prerequisite: FRE1120

Course User Fee: \$15.00

GER1121 Elementary German 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in German. Note: Students must pass this course with a C or better to continue to GER2220.

Lecture Hours: 64.000

Prerequisite: GER1120

Course User Fee: \$15.00

ITA1121 Elementary Italian 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Italian. Note: Students must pass this course with a C or better to continue to ITA2220.

Lecture Hours: 64.000

Prerequisite: ITA1120

Course User Fee: \$15.00

JPN1121 Elementary Japanese 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Japanese. Note: Students must pass this course with a C or better to continue to JPN2220.

Lecture Hours: 64.000

Prerequisite: JPN1120

Course User Fee: \$15.00

POR1121 Elementary Portuguese 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Portuguese. Note: students must pass this course with a "C" or better to continue to POR2220.

Lecture Hours: 64.000

Prerequisite: POR1120

Course User Fee: \$15.00

RUS1121 Elementary Russian 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Russian.

Lecture Hours: 64.000

Prerequisite: RUS1120

Course User Fee: \$15.00

SPN1121 Elementary Spanish 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Spanish. Note: Students must pass this course with a C or better to continue to SPN2220.

Lecture Hours: 64.000

Prerequisite: SPN1120

Course User Fee: \$15.00

Program Description

Computer Information Systems

Associate in Arts | Code: 10702 | 60 credits

CIS focuses on the structure, management and control of information resources on computers. Coursework includes business and math classes, as well as courses in information systems and programming languages. Students transfer to four-year institutions and major in computer information systems, computer and information sciences, information sciences, or management information systems. Degrees lead to careers in systems analysis, computer application programming, database management, network services and IT support.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
MAC 1105	College Algebra	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

COP 1334	Introduction to C++ Programming	4.0
Semester 3		Units: 6.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
PHI 2600	Introduction to Ethics	3.0
Semester 4		Units: 13.0
MAD 2104	Discrete Mathematics	3.0
<i>Natural Science</i>		3.0
AST 1002	Descriptive Astronomy	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 2048	Physics with Calculus 1	4.0
OR		
PHY 2053	Physics (without Calculus) 1	3.0
PSY 2012	Introduction to Psychology	3.0
COP 2800	Java Programming	4.0
Semester 5		Units: 14.0
ECO 2013	Principles of Economics (Macro)	3.0
STA 2023	Statistical Methods	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
PHY 1020	General Education Physics	3.0
OR		
PHY 1025	Basic Physics	3.0
<i>Pathway Electives</i>		5.0
CAP 1788	Introduction to Data Analytics	4.0
OR		
CGS 1540C	Database Concepts Design	4.0
OR		
COP 1047C	Introduction to Python Programming	4.0
OR		
COP 2805C	Java Programming 2	4.0
OR		
CTS 1120	Cybersecurity Fundamentals	4.0
OR		
CTS 1134	Networking Technologies	4.0
OR		

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)
Lecture Hours: 48.000
Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.
Lecture Hours: 48.000
Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."
Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

MAD2104 Discrete Mathematics 3.000 credits

This course introduces the student to the principles of discrete mathematics that apply to computer science. The student will examine set theory, logic, Boolean algebra, number theory, vectors and matrices, combinatorics, probability, relations, functions, and basic graph theory. Computational Course.
Lecture Hours: 48.000
Prerequisite: MAC1106 OR MAC1140
Course User Fee: \$18.00

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.
Lecture Hours: 48.000
Course User Fee: \$10.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.
Lecture Hours: 48.000, Lab Hours: 32.000
Corequisite: COP1047C OR COP1334 OR COP2270
Course User Fee: \$63.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.
Lecture Hours: 48.000
Course User Fee: \$15.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.
Lecture Hours: 48.000
Recommended Preparation: Completion of MGF1131 prior to registration.
Course User Fee: \$18.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.
Lecture Hours: 48.000
Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

CAP1788 Introduction to Data Analytics 4.000 credits

This course offers a broad introduction to data analytics and the role it plays in modern organizations. Students will use Excel and Tableau to import, clean, transform, and visualize data. Students will also learn how to effectively communicate findings to decision makers.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2805C Java Programming 2 4.000 credits

This is an advanced level programming course using Java. Students will learn how to code, compile and execute programs. Topics include object serialization, Java Collection, sorting/searching algorithms, multithreading and networking capabilities, and Java databases.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional. Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

Program Description

Computer Science

Associate in Arts | Code: 10703 | 60 credits

As the name suggests, Computer Science is a more science-intensive program than CIS. In addition to courses in programming and applications, the program provides a thorough grounding in mathematics, biology, chemistry and physics. Computer scientists design technical programs, do research, create new technologies, develop operating systems, code device drivers, write specialized programming languages and implement complex applications in a variety of settings. Computer Science requires skills in mathematics and physics. Students must complete Calculus II and Physics with Calculus II before entering their junior year.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		

POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2

Units: 14.0

CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
COP 1334	Introduction to C++ Programming	4.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3

Units: 7.0

COP 2800	Java Programming	4.0
ENC 1102	English Composition 2	3.0

Semester 4

Units: 14.0

ECO 2013	Principles of Economics (Macro)	3.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
PHY 1025	Basic Physics	3.0

Semester 5

Units: 12.0

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

**COP1334 Introduction to C++ Programming
4.000 credits**

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

**MAC1147 Pre-Calculus Algebra and Trigonometry
5.000 credits**

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics.
Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.
Lecture Hours: 80.000
Prerequisite: MAC1105
Course User Fee: \$18.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.
Lecture Hours: 48.000, Lab Hours: 32.000
Corequisite: COP1047C OR COP1334 OR COP2270
Course User Fee: \$63.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.
Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.
Lecture Hours: 48.000
Course User Fee: \$15.00

**MAC2311 Calculus and Analytical Geometry 1
5.000 credits**

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

**MAC2312 Calculus and Analytical Geometry 2
4.000 credits**

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

Program Description

Criminal Justice Administration

Associate in Arts | Code: 12204 | 60 credits

In addition to coursework focusing on criminal justice and law, this program includes classes in history, sociology and political science. Thus, pre-law students will find this program suitable, as well as those seeking Bachelor's degrees in criminal justice. This program may lead to careers in law enforcement, corrections (including probation and parole), and security in private businesses or government.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
HUM 1020	Humanities	3.0
PHI 2600	Introduction to Ethics	3.0
SPC 1017	Introduction to Communication	3.0

Semester 2

Units: 12.0

BSC 1005	General Education Biology	3.0
ENC 1102	English Composition 2	3.0
POS 2041	American Federal Government	3.0
<i>Social Science</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		

ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 3 **Units: 12.0**

CCJ 1010	Introduction to Criminology	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
MAC 1105	College Algebra	3.0
PSC 1515	Energy in the Natural Environment	3.0

Semester 4 **Units: 12.0**

CCJ 1020	Introduction to Criminal Justice	3.0
CJC 1000	Introduction to Corrections	3.0
CJL 2062	Constitutional Law and Legal Procedure or Evidence	3.0
STA 2023	Statistical Methods	3.0

Semester 5 **Units: 12.0**

CCJ 1191	Human Behavior in Criminal Justice	3.0
CCJ 2840	Introduction to Criminal Justice Research with AI	3.0
CJE 2600	Criminal Investigation	3.0
CJJ 2002	Juvenile Delinquency	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

CCJ1010 Introduction to Criminology 3.000 credits

Theories and causes of criminal and delinquent behavior, including its variations, ramifications, explanations and measures of prevention, control and treatment.

Lecture Hours: 48.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CCJ1020 Introduction to Criminal Justice 3.000 credits

This course will explore and define the various components of the criminal justice system. It is designed for the entry level criminal justice practitioner and beginning student interested in entering the field.

Lecture Hours: 48.000

CJC1000 Introduction to Corrections 3.000 credits

A comprehensive view of the historical and philosophical treatment programs and developments in the field of adult institutional corrections. Emphasis is on understanding the offender in the correctional system; an examination of the correctional client and staff, agencies and recidivism; (3 hr. lecture).

Lecture Hours: 48.000

CJL2062 Constitutional Law and Legal Procedure or Evidence 3.000 credits

An examination of the United States and Florida Constitutions, with emphasis on leading cases dealing with arrest, search and seizure, confessions and the rules of evidence.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CCJ1191 Human Behavior in Criminal Justice 3.000 credits

Human behavior and how it relates to the duties and responsibilities of the criminal justice practitioner.

Lecture Hours: 48.000

CCJ2840 Introduction to Criminal Justice Research with AI 3.000 credits

This course provides an introduction to core research methods utilized in the field of Criminology and Criminal Justice. Students will learn foundational research concepts, including research design, data collection methods, and data analysis techniques. A key focus will be on the ethical and effective use of Artificial Intelligence (AI) tools to assist in various stages of the research process, such as data cleaning, analysis, and interpretation. Students will gain practical experience in applying these methods through hands-on projects and develop critical thinking skills necessary for evaluating research findings and understanding the social implications of criminal justice research. Recommended: Students are encouraged to take ENC1101 or CCJ1020 prior to registration.

Lecture Hours: 48.000

Recommended Preparation: ENC1101 OR CCJ1020

CJE2600 Criminal Investigation 3.000 credits

Fundamentals of criminal investigation, theory and practice, including crime scene search; preservation, collection and transportation of physical evidence; interviewing, interrogating; statement taking; and case preparation, with investigation of specific offenses; relationship with the police science laboratory.

Lecture Hours: 48.000

CJJ2002 Juvenile Delinquency 3.000 credits

A history of the juvenile court system in the United States. An analysis of delinquency and the family structure. Delinquency control including the police function with juveniles and their families, preventative measures by adequate court and police enforcement, rehabilitative processes, and the philosophy and programs of schools for boys and girls and detention homes.

Lecture Hours: 48.000

Program Description

Dance

Associate in Arts | Code: 11003 | 60 credits

Studio classes feature modern dance and ballet, and the program also includes theoretical courses. This curriculum meets the pre-professional and general education course requirements for transfer, but students should meet with an advisor to discuss the specific requirements of the four-year institution they plan to attend. Often,

departments in four-year institutions will require an audition. This program is designed to prepare students pursuing careers in choreography or the performance of ballet and jazz or contemporary forms of dance. The program is also suited for students wishing to become teachers of dance.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 11.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
DAA 1104	Modern 1	2.0
DAA 1204	Ballet 1	2.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
DAA 1105	Intermediate Modern	2.0
DAA 1205	Intermediate Ballet	2.0
DAN 2100	Dance Appreciation	3.0
Semester 3		Units: 13.0
DAN 2130	Dance History 1	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
MUL 1010	Music Appreciation	3.0
PSY 2012	Introduction to Psychology	3.0
Semester 4		Units: 12.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
DAA 2106	Modern2	2.0
DAA 2207	Advanced Ballet	2.0
<i>Pathway Elective</i>		2.0
DAA 1504	Jazz Dance 1	2.0
OR		
DAA 2610	Dance Composition and Improvisation 1	2.0
OR		
DAA 2680	Repertory 1	2.0
Semester 5		Units: 11.0
BSC 1005	General Education Biology	3.0

<i>Natural Science</i>		3.0
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
HUN 1201	Essentials of Human Nutrition	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
<i>Social Science</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
DEP 2000	Human Growth and Development	3.0
<i>Pathway Elective</i>		2.0
DAA 2107	Advanced Modern 2	2.0
OR		
DAA 2681	Repertory 2	2.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

DAA1104 Modern 1 2.000 - 3.000 credits

Beginning exploration of techniques, creative aspects, and theoretical concepts of modern dance which includes but is not limited to proper alignment and mechanics of breathing and phrasing, verbal and movement vocabulary, including structural improvisation, and exercises utilizing Laban's movement analysis. No previous experience required. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA1204 Ballet 1 2.000 - 3.000 credits

Beginning exploration of techniques and theoretical concepts of ballet increasing awareness of proper alignment, balance, coordination and application of various musical meters. No previous experience required. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

DAA1105 Intermediate Modern 2.000 - 3.000 credits

Further development of modern dance techniques, creative aspects, theoretical concepts emphasizing components based on Graham, Cunningham and Limon techniques. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA1104

DAA1205 Intermediate Ballet 2.000 - 3.000 credits

Continuing exploration of techniques and theoretical concepts of ballet placing further emphasis on precision of lines and exactness of movement. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA1204

DAN2100 Dance Appreciation 3.000 credits

This course provides an overview of dance as an art form, entertainment, and social activity and includes works from the Western canon to examine its significance from ancient times to the 21st century. Students will analyze dance genres such as ballet, modern dance, jazz dance, and world dance forms, exploring their connections to literature, music, and philosophy while considering the roles of dancers, choreographers, and audiences. Through this study, students will develop critical thinking skills and gain understanding of how dance reflects and influences culture across time periods and societies.

Lecture Hours: 48.000

DAN2130 Dance History 1 3.000 credits

In this course, students will engage in a comprehensive study of the origins and development of dance as an art form, tracing its evolution from its inception in primitive cultures to the present day. The course will explore the cultural, historical, and social contexts that shaped the various styles and techniques of dance throughout history. Students will analyze and critique significant dance works, including those from the western canon, to gain a deeper understanding of the artistic, aesthetic, and expressive elements of dance. Writing intensive course.

Lecture Hours: 48.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

DAA2106 Modern2 2.000 - 3.000 credits

Further development of modern dance techniques, creative aspects and theoretical concepts emphasizing components based on Graham, Cunningham and Limon techniques. The use of improvisation as an introduction to basic principles of form and their application to dance composition will be emphasized. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA1104or Departmental Permission.

Course User Fee: \$5.00

DAA2207 Advanced Ballet 2.000 - 3.000 credits

Continuing exploration of techniques and theoretical concepts of ballet placing further emphasis on precision of line and exactness of movement. May be repeated for credit. Dance majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA1205or Departmental Permission.

DAA1504 Jazz Dance 1 2.000 - 3.000 credits

This course is designed to introduce the student to the vocabulary and technique of jazz dance, incorporating a fusion of styles from popular, Afro-Caribbean, and contemporary modern jazz choreographers. For majors only. Audition required. May be repeated for credit.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA2610 Dance Composition and Improvisation 1 2.000 - 3.000 credits

Individual experience in developing movement phrases and combinations based on solving problems within a form and a movement framework, as well as the movement imagery designed to develop the dancer's creative imagination. Individuals will experience composition using the basic elements of movement theory in an improvisational framework.

Lecture Hours: 16.000 - 32.000, Lab Hours: 32.000 - 64.000

DAA2680 Repertory 1 2.000 - 3.000 credits

Dance works in both ballet and many different styles of modern and ethnic dance vocabularies are studied. Works include both standard repertory and commissioned dances. Students work with choreographers, directors and reconstructors of classic works, giving the dancer the experience of being choreographed on and being directed in repertory works. The works learned are performed by the students in workshop and public performances throughout the year.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan. Lecture Hours: 48.000

DAA2107 Advanced Modern 2 2.000 - 3.000 credits

Further development of modern dance techniques, creative aspects and theoretical concepts based on Graham, Cunningham, and Limon techniques. May be repeated for credit. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA2106or Departmental Permission

DAA2681 Repertory 2 2.000 - 3.000 credits

A continuation of DAA 2680.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA2680or Departmental Permission.

Program Description

Dietetics

Associate in Arts | Code: 11305 | 60 credits

This program provides the science education needed to transfer to a four-year program in dietetics. Chemistry, biology, anatomy and physiology are emphasized in this program. Dieticians and nutrition specialists may work as meal planners in institutions such as schools and hospitals, in the food products or health and fitness industry, or in a range of health and medical professions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
MAT 1033	Intermediate Algebra	3.0
MAT 1033L	Intermediate Algebra Recitation Hall	0.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
MAC 1105	College Algebra	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
CHM 1025	Introductory Chemistry	3.0
CHM 1025L	Introductory Chemistry Lab	1.0
Semester 3		Units: 13.0

BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Semester 4

Units: 13.0

CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0
PSY 2012	Introduction to Psychology	3.0

Semester 5

Units: 11.0

HUN 1201	Essentials of Human Nutrition	3.0
PHI 2600	Introduction to Ethics	3.0
STA 2023	Statistical Methods	3.0
ECO 2013	Principles of Economics (Macro)	2.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAT1033L Intermediate Algebra Recitation Hall 0.000 credits

The student will receive individualized, small group or whole group instruction to deepen their conceptual understanding of mathematics. The student will reinforce and apply content knowledge with effective problem-solving techniques and non-cognitive activities to make mathematics meaningful and relevant to their fields of study while strengthening the concepts needed to achieve the objectives of MAT1033. This lab session is associated and funded by the First in the World Grant, CCAT Math Project.

Lab Hours: 1.000

Corequisite: MAT1033

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.
Lecture Hours: 48.000

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Lab Hours: 32.000
Prerequisite: MAT1033, Corequisite: CHM1025
Course User Fee: \$30.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

Program Description

Drama or Drama Education

Associate in Arts | Code: 11002 | 60 credits

This is a comprehensive program in all aspects of theatrical production, including lighting, costuming, make-up and other aspects of stagecraft. Students participate in stage productions which are presented to the public throughout the academic year. While this program does provide the necessary coursework to transfer to a four-year institution, some departments in four-year colleges and universities will require an audition or portfolio, depending on the student's intended area of study. Careers in drama include education, theatrical production, casting, acting and a wide variety of stagecraft.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 15.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
TPP 1100	Acting 1	3.0
TPP 1120	Improvisation Ensemble	3.0
SLS 1510	Preparing for Student Success	3.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
TPP 1700	Voice for the Stage	3.0
TPP 2111	Acting 2	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
Semester 3		Units: 9.0
MGF 1131	Mathematics in Context	3.0
THE 2000	Theatre Appreciation	3.0
<i>Humanities</i>		3.0
DAN 2100	Dance Appreciation	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
MUL 2380	Jazz and Popular Music in America	3.0
Semester 4		Units: 12.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
TPP 1150	Scene Study 1	3.0
TPA 1248	Makeup for the Stage	3.0

Semester 5**Units: 12.0**

BSC 1005	General Education Biology	3.0
<i>Natural Science</i>		3.0
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
HUN 1201	Essentials of Human Nutrition	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
<i>Social Science</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
DEP 2000	Human Growth and Development	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
TPP 2260	Acting for the Camera 1	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

TPP1100 Acting 1 3.000 credits

The fundamentals of stage performance, stressing voice, movement, and the more formal and technical aspects of the actor's art. May be repeated for credit. Lecture Hours: 48.000

TPP1120 Improvisation Ensemble 3.000 credits

The student will develop the skills of improvisation for use in role development and for performance.

Lecture Hours: 48.000

SLS1510 Preparing for Student Success 3.000 credits

This course provides an orientation to college life and helps develop academic, career, and personal goals. Students will learn college success strategies, goal-setting, learning style assessments, as well as general and discipline-specific study skills in the context of various theoretical, practical, and experiential perspectives.

Lecture Hours: 48.000

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

TPP1700 Voice for the Stage 3.000 credits

The study and application of voice production, breathing, articulation, accents and movement in the actor's delivery. Emphasis is on clarity, precision, properly phrased and meaningful communication from the performer to the audience.

Lecture Hours: 48.000

TPP2111 Acting 2 3.000 credits

In this course, actors who have learned to express themselves freely now learn to adjust this expression to the demand of the role. Students begin to apply their skills for observation, imagination, and concentration to the study of roles close to themselves. Vocal and physical flexibility and expressiveness are now put to work in the realization of expectations of the playwright, here the student develops a systematic approach to creating a three-dimensional character. (3 hr. lecture)

Lecture Hours: 48.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society.

Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

DAN2100 Dance Appreciation 3.000 credits

This course provides an overview of dance as an art form, entertainment, and social activity and includes works from the Western canon to examine its significance from ancient times to the 21st century. Students will analyze dance genres such as ballet, modern dance, jazz dance, and world dance forms, exploring their connections to literature, music, and philosophy while considering the roles of dancers, choreographers, and audiences. Through this study, students will develop critical thinking skills and gain understanding of how dance reflects and influences culture across time periods and societies.

Lecture Hours: 48.000

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

MUL2380 Jazz and Popular Music in America 3.000 credits

A survey of the development of popular and jazz music with an emphasis on musical styles and outstanding artists. (3 hr. lecture)

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

TPP1150 Scene Study 1 1.000 - 3.000 credits

This course teaches the aspiring young theatre professional how to analyze a play in terms of the author's personal statement, the historical and social context within which it was written, the particular style used by the author, and the many options open to director and actor for bringing the work to stage life. A substantial portion of class time will be devoted to oral reading and interpretation of text.

Lecture Hours: 16.000 - 48.000

TPA1248 Makeup for the Stage 3.000 credits

An introduction to the art and techniques of makeup as used by the actor, theatrical designer, and technician. Special emphasis is given to straight makeup, age makeup, hair, character extension, and stylization.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

TPP2260 Acting for the Camera 1 3.000 credits

Acting students will attend lecture/lab to acquire the technical knowledge and training necessary for acting in the film and television industry. They will acquire a knowledge of the working procedure and terminology used in these media.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: TPP1100or Departmental Permission.

Program Description

Economics

Associate in Arts | Code: 12201 | 60 credits

Economics is the study of how people produce, trade and consume goods and services. The A.A. program emphasizes fundamental coursework in business and mathematics. While many students choose to obtain graduate degrees, economists with Bachelor's degrees can work in fields such as business economics and forecasting, urban real estate and regional planning, analysis of markets and industrial regulation, management consulting and in banking and financial services.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
GEB 1011	Principles of Business	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

* Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
ECO 2013	Principles of Economics (Macro)	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
QMB 2100	Basic Business Statistics	3.0

Semester 3		Units: 9.0
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BSC 1005	General Education Biology	3.0
ECO 2023	Principles of Economics (Micro)	3.0
PHI 2600	Introduction to Ethics	3.0

Semester 4

Units: 13.0

<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
PSC 1515	Energy in the Natural Environment	3.0
Pathway Elective*		3.0

Select 3 credits from the following course options: ACG 1403, ACG 2360, ACG 2450, ACG 2630, CHI 1120, COP 1332, COP 1334, COP 2270, FRE 1120, GER 1120, ITA 1120, JPN 1120, MAN, MAR*, OST 2335, POR 1120, RUS 1120, SPN 1120, STA 2023, TAX*

Semester 5

Units: 13.0

ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
MAC 2233	Business Calculus	3.0
Pathway Elective*		6.0

Select 6 credits from the following course options: ACG 1403, ACG 2360, ACG 2450, ACG 2630, CHI 1120, COP 1332, COP 1334, COP 2270, FRE 1120, GER 1120, ITA 1120, JPN 1120, MAN, MAR*, OST 2335, POR 1120, RUS 1120, SPN 1120, STA 2023, TAX*

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view. Lecture Hours: 48.000

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033Any Mathematics level higher than MAT 1033 will satisfy this requirement.

MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites:

ACG2001, ACG2021L, ACG2021, ACG2011;

Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

MAC2233 Business Calculus 3.000 credits

This course introduces the basic concepts of differential and integral calculus for students majoring in business administration and related fields. Topics include limits, continuity, differentiation and integration of polynomials, logarithmic and exponential functions with applications to business, economics, and the life sciences. Prerequisite: MAC 1105 or MAC1106 with a grade of C or better or departmental permission. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

Program Description

Engineering - Architectural

Associate in Arts | Code: 10905 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 13.0**

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester**Units: 11.0**

ENC 1102	English Composition 2	3.0
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MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3	Units: 6.0
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MAC 1114	Trigonometry	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4	Units: 14.0
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MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5	Units: 12.0
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MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6	Units: 4.0
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MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.
Lecture Hours: 64.000
Prerequisite: MAC2311
Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.
Lecture Hours: 64.000
Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L
Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.
Lab Hours: 32.000
Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048
Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better.
Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Biomedical

Associate in Arts | Code: 10904 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester		Units: 11.0
ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3		Units: 6.0
MAC 1114	Trigonometry	3.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4

Units: 14.0

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5

Units: 12.0

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6

Units: 4.0

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills. Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program. Lecture Hours: 16.000
Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better. Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Chemical

Associate in Arts | Code: 10906 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0

HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester **Units: 11.0**

ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3 **Units: 6.0**

MAC 1114	Trigonometry	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4 **Units: 14.0**

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5 **Units: 12.0**

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6 **Units: 4.0**

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.
Lecture Hours: 64.000
Prerequisite: MAC2311
Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.
Lecture Hours: 64.000
Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L
Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.
Lab Hours: 32.000
Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048
Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better.

Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Civil

Associate in Arts | Code: 10908 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester		Units: 11.0
ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3		Units: 6.0
MAC 1114	Trigonometry	3.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4

Units: 14.0

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5

Units: 12.0

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6

Units: 4.0

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills. Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program. Lecture Hours: 16.000
Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better. Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Computer

Associate in Arts | Code: 10705 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0

HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester **Units: 11.0**

ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3 **Units: 6.0**

MAC 1114	Trigonometry	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4 **Units: 14.0**

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5 **Units: 12.0**

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6 **Units: 4.0**

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.
Lecture Hours: 64.000
Prerequisite: MAC2311
Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.
Lecture Hours: 64.000
Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L
Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.
Lab Hours: 32.000
Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048
Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better.

Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Electrical

Associate in Arts | Code: 10910 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester		Units: 11.0
ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3		Units: 6.0
MAC 1114	Trigonometry	3.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4

Units: 14.0

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5

Units: 12.0

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6

Units: 4.0

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills. Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program. Lecture Hours: 16.000
Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better. Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Geomatics (Surveying and Mapping)

Associate in Arts | Code: 10909 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0

HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester **Units: 11.0**

ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3 **Units: 6.0**

MAC 1114	Trigonometry	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4 **Units: 14.0**

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5 **Units: 12.0**

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6 **Units: 4.0**

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.
Lecture Hours: 64.000
Prerequisite: MAC2311
Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.
Lecture Hours: 64.000
Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L
Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.
Lab Hours: 32.000
Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048
Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better.

Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Industrial

Associate in Arts | Code: 10912 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester		Units: 11.0
ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3		Units: 6.0
MAC 1114	Trigonometry	3.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4

Units: 14.0

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5

Units: 12.0

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6

Units: 4.0

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills. Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program. Lecture Hours: 16.000
Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better. Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Mechanical

Associate in Arts | Code: 10911 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0

HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester **Units: 11.0**

ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3 **Units: 6.0**

MAC 1114	Trigonometry	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4 **Units: 14.0**

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5 **Units: 12.0**

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6 **Units: 4.0**

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.
Lecture Hours: 64.000
Prerequisite: MAC2311
Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.
Lecture Hours: 64.000
Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L
Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.
Lab Hours: 32.000
Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048
Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better.

Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

Engineering - Ocean

Associate in Arts | Code: 10913 | 60 credits

Miami Dade College offers ten Engineering A.A. programs: architectural, chemical, civil, computer, electrical, industrial, mechanical, ocean, science, and surveying and mapping. Each has its own curriculum to best prepare students for transfer to a four-year institution. Interdisciplinary fields can include the study of biomechanics, kinesiology, nutrition and related areas.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Humanities		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EGN 1008C	Introduction to Engineering	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester		Units: 11.0
ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3		Units: 6.0
MAC 1114	Trigonometry	3.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4

Units: 14.0

MAC 2311	Calculus and Analytical Geometry 1	5.0
PHY 1025	Basic Physics	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0

Semester 5

Units: 12.0

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Semester 6

Units: 4.0

MAC 2313	Calculus and Analytic Geometry 3	4.0
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Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

EGN1008C Introduction to Engineering 3.000 credits

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills. Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program. Lecture Hours: 16.000
Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better. Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

Program Description

English/Literature & English Education

Associate in Arts | Code: 11500 | 60 credits

English/Literature is the study of great written works and how they were shaped by historical and cultural events. This program also includes education courses to prepare students for careers as teachers. However, students who graduate with a Bachelor's degree in English are equipped to work in publishing, and may be qualified to work as a writer in virtually any field.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
BSC 1005	General Education Biology	3.0

SLS 1510	Preparing for Student Success	3.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
LIT 2000	Introduction to Literature	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
Semester 3		Units: 12.0
ENC 2300	Advanced Composition and Communication	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
<i>Creative Writing</i>		3.0
CRW 2001	Creative Writing 1	3.0
OR		
CRW 2002	Creative Writing 2	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
Semester 4		Units: 12.0
ENG 2012	Literary Theory	3.0
LIT 2120	A Survey of World Literature	3.0
<i>Social Science</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
OR		
SYG 2000	Introduction to Sociology	3.0
<i>American Literature</i>		3.0
AML 2010	American Literature 1	3.0
OR		
AML 2020	American Literature 2	3.0
OR		
AML 2600	African American Literature	3.0
Semester 5		Units: 12.0
ESC 1000	General Education Earth Science	3.0
LIT 2090	Contemporary Literature	3.0
LIT 2480	Issues in Literature & Culture	3.0
<i>English Literature</i>		3.0

ENL 2012	English Literature 1	3.0
OR		
ENL 2022	English Literature 2	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

SLS1510 Preparing for Student Success 3.000 credits

This course provides an orientation to college life and helps develop academic, career, and personal goals. Students will learn college success strategies, goal-setting, learning style assessments, as well as general and discipline-specific study skills in the context of various theoretical, practical, and experiential perspectives.

Lecture Hours: 48.000

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon.

Written analysis of literary works may be required.

Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ENC2300 Advanced Composition and Communication 3.000 credits

This writing-based course addresses techniques of critical thinking, persuasion, and argumentation. Students will refine their composition skills and develop their oral communication skills by examining and discussing a range of issues. Prerequisites: ENC1101, 1102 or equivalent with a grade of "C" or better. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101 AND ENC1102

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CRW2001 Creative Writing 1 3.000 credits

Imaginative writing in selected genres.

Lecture Hours: 48.000

CRW2002 Creative Writing 2 3.000 credits

Imaginative writing in selected genres.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ENG2012 Literary Theory 3.000 credits

This course introduces students to the study of literary criticism. Students will discuss, analyze, and write about literature through the lens of contemporary critical theory.

Lecture Hours: 48.000

Prerequisite: ENC1101

LIT2120 A Survey of World Literature 3.000 credits

This course examines world literature from the mid-renaissance to the present and includes works from the Western canon. Students will analyze literary works that exemplify human experience, exploring their connections to historical contexts, ideas, and literary developments. Through this study, students will develop critical thinking skills and gain understanding of how literature reflects and influences societies across time periods and societies. This is a writing intensive course.

Lecture Hours: 48.000

Prerequisite: ENC1102 or equivalent.

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.
Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.
Lecture Hours: 48.000

AML2010 American Literature 1 3.000 credits

American Literature from Colonial times to the Civil War. Prerequisites: ENC1101, ENC1102.
Lecture Hours: 48.000
Prerequisite: ENC1101

AML2020 American Literature 2 3.000 credits

American literature from the Civil War to the present. Prerequisites: ENC1101, ENC1102.
Lecture Hours: 48.000
Prerequisite: ENC1101

AML2600 African American Literature 3.000 credits

A study of African American historical and contemporary literature from various genres such as poetry, fiction, narratives, speeches, films, and drama. Prerequisite of ENC1101 with a Grade of "C" or higher, or equivalent.
Lecture Hours: 48.000
Prerequisite: ENC1101

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.
Lecture Hours: 48.000
Course User Fee: \$10.00

LIT2090 Contemporary Literature 3.000 credits

This course is a survey of contemporary prose and poetry.
Lecture Hours: 48.000

LIT2480 Issues in Literature & Culture 3.000 credits

Students will learn about literature as a socio-cultural response by writers to the world in which they live; they will connect literary texts to cultural issues through completion of oral and written assignments, critical analysis and practical investigation. Fulfills Gordon Rule writing requirement. Prerequisite: ENC 1102. (3 hr. lecture)
 Lecture Hours: 48.000
 Prerequisite: ENC1102ENC1102

ENL2012 English Literature 1 3.000 credits

A survey of major British writers from Chaucer through the 18th century. Required of English majors.
 Lecture Hours: 48.000
 Prerequisite: ENC1102

ENL2022 English Literature 2 3.000 credits

A survey of major British writers from the 18th century through the contemporary period. Required of English majors.
 Lecture Hours: 48.000
 Prerequisite: ENC1102

Program Description

Environmental Sciences

Associate in Arts | Code: 10203 | 60 credits

Environmental Studies examines environmental issues from both ecological and sociological standpoints. Thus, it is an interdisciplinary major which combines life sciences, social sciences and the humanities. Students at MDC take mostly science and mathematics courses to prepare for transfer into a baccalaureate program. This field is projected to grow in the 21st century, as the need for environmental researchers, analysts, engineers and journalists will grow.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2

Units: 14.0

ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
CHM 1025	Introductory Chemistry	3.0
CHM 1025L	Introductory Chemistry Lab	1.0

EVR 1001	Introduction to Environmental Science	3.0
EVR 1001L	Introduction to Environmental Science Lab	1.0

Semester 3

Units: 13.0

BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Social Science</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
OR		
SYG 2000	Introduction to Sociology	3.0

Semester 4

Units: 13.0

BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Semester

Units: 10.0

GLY 1010	Physical Geology	3.0
GLY 1010L	Physical Geology Laboratory	1.0
PHI 2600	Introduction to Ethics	3.0
STA 2023	Statistical Methods	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1025

Course User Fee: \$30.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

EVR1001L Introduction to Environmental Science Lab 1.000 credits

This course is the laboratory component for EVR1001 - Introduction to Environmental Science. Students will learn how the human and physical/biological worlds affect global climate change, including human/non-human interactions with minerals and mining, landscape ecology, petroleum depletion, and alternative fuels with the understanding of the earth's environment.

Lab Hours: 32.000

Pre/Corequisite: EVR1001

Course User Fee: \$20.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.
Lecture Hours: 48.000

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.
Lecture Hours: 48.000
Prerequisite: BSC2010, BSC2010L, BSC2011L
Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.
Lab Hours: 64.000
Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011
Course User Fee: \$40.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.
Lecture Hours: 48.000
Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L
Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.
Lab Hours: 64.000
Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046
Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)
Lecture Hours: 48.000
Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

GLY1010L Physical Geology Laboratory 1.000 credits

Laboratory for GLY 1010. Studies of common minerals and rocks and topographic and geologic maps along with aerial photography.

Lab Hours: 32.000

Corequisite: GLY1010

Course User Fee: \$10.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

Program Description

Exercise Science

Associate in Arts | Code: 12103 | 60 credits

Exercise Science studies the relationship of physical exercise to human health and disease prevention. This program at MDC prepares students for transfer with coursework in human anatomy and physiology, nutrition, health and exercise. Exercise science is a growing field with professionals working in diverse settings, such as

hospitals and health clubs, research facilities and sports teams. Specialists also work in corporate, industrial and educational environments.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
MAT 1033	Intermediate Algebra	3.0
MAT 1033L	Intermediate Algebra Recitation Hall	0.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
MAC 1105	College Algebra	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy & Physiology Lab 1	1.0
CHM 1025	Introductory Chemistry	3.0

Semester 3		Units: 12.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
DEP 2000	Human Growth and Development	3.0
HUN 1201	Essentials of Human Nutrition	3.0
PHY 1025	Basic Physics	3.0

Semester 4		Units: 13.0
HLP 1081	Fitness & Wellness for Life	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
PHI 2010	Introduction to Philosophy	3.0
PHY 1004	Physics With Applications 1	3.0
PHY 1004L	Physics with Applications 1 Lab	1.0

Semester 5		Units: 12.0
GEB 1011	Principles of Business	3.0
<i>Humanities</i>		3.0

DAN 2100	Dance Appreciation	3.0
OR		
DAN 2130	Dance History 1	3.0
OR		
HUM 1020	Humanities	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
PET 1170	Fundamentals of Athletic Coaching and Management	3.0
STA 2023	Statistical Methods	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAT1033L Intermediate Algebra Recitation Hall 0.000 credits

The student will receive individualized, small group or whole group instruction to deepen their conceptual understanding of mathematics. The student will reinforce and apply content knowledge with effective problem-solving techniques and non-cognitive activities to make mathematics meaningful and relevant to their fields of study while strengthening the concepts needed to achieve the objectives of MAT1033. This lab session is associated and funded by the First in the World Grant, CCAT Math Project.

Lab Hours: 1.000

Corequisite: MAT1033

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

**BSC2085 Human Anatomy and Physiology 1
3.000 credits**

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

**BSC2085L Human Anatomy & Physiology Lab 1
1.000 credits**

In this laboratory course, student will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

**BSC2086 Human Anatomy & Physiology 2 3.000
credits**

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite:

BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

**BSC2086L Human Anatomy & Physiology 2
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites:

BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

**DEP2000 Human Growth and Development 3.000
credits**

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

**HUN1201 Essentials of Human Nutrition 3.000
credits**

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

HLP1081 Fitness & Wellness for Life 3.000 credits

In this course students will learn the roles of exercise, physical activity, diet, and stress management in achieving optimal wellness. Students will explore current developments in health and complete lab assignments, which will assist in the determination of their current health status. Individualized exercise and dietary protocols based on these assessments will be developed. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHY1004 Physics With Applications 1 3.000 credits

Emphasizes the basic concepts and principles and their practical applications. Designed specifically for students in technical studies and for others wishing to strengthen their physics background before taking advanced courses.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: PHY1004L

Course User Fee: \$10.00

PHY1004L Physics with Applications 1 Lab 1.000 credits

Laboratory for PHY 1004.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: PHY1004

Course User Fee: \$20.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

DAN2100 Dance Appreciation 3.000 credits

This course provides an overview of dance as an art form, entertainment, and social activity and includes works from the Western canon to examine its significance from ancient times to the 21st century. Students will analyze dance genres such as ballet, modern dance, jazz dance, and world dance forms, exploring their connections to literature, music, and philosophy while considering the roles of dancers, choreographers, and audiences. Through this study, students will develop critical thinking skills and gain understanding of how dance reflects and influences culture across time periods and societies.

Lecture Hours: 48.000

DAN2130 Dance History 1 3.000 credits

In this course, students will engage in a comprehensive study of the origins and development of dance as an art form, tracing its evolution from its inception in primitive cultures to the present day. The course will explore the cultural, historical, and social contexts that shaped the various styles and techniques of dance throughout history. Students will analyze and critique significant dance works, including those from the western canon, to gain a deeper understanding of the artistic, aesthetic, and expressive elements of dance. Writing intensive course.

Lecture Hours: 48.000

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PET1170 Fundamentals of Athletic Coaching and Management 3.000 credits

This course will introduce students to the basic principles of athletic coaching and team management. Students will discuss the fundamentals of developing an individual athlete and team, effective athlete training programming, and factors which impact the operations of athletic organizations. Students will develop an understanding of how policy and coaching decisions influence an organization at all levels (Credit hours 3.00).

Lecture Hours: 48.000

Course User Fee: \$55.83

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

Program Description

Foreign Language

Associate in Arts | Code: 10301 | 60 credits

Foreign language programs train students to achieve reading, writing and verbal fluency in one or more foreign languages. The demand for interpreters, translators and language instructors is projected to grow in the 21st

century, and graduates with Bachelor's degrees can work almost anywhere in the world for corporations, businesses, governments, non-profit agencies or schools.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
HUM 1020	Humanities	3.0
<i>Foreign Language Elementary 1</i>		4.0
CHI 1120	Elementary Mandarin Chinese 1	4.0
OR		
FRE 1120	Elementary French 1	4.0
OR		
GER 1120	Elementary German 1	4.0
OR		
ITA 1120	Elementary Italian 1	4.0
OR		
JPN 1120	Elementary Japanese 1	4.0
OR		
POR 1120	Elementary Portuguese 1	4.0
OR		
SPN 1120	Elementary Spanish 1	4.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
<i>Humanities</i>		3.0
PHI 2010	Introduction to Philosophy	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
<i>Foreign Language Elementary 2</i>		4.0
CHI 1121	Elementary Mandarin Chinese 2	4.0
OR		
FRE 1121	Elementary French 2	4.0
OR		
GER 1121	Elementary German 2	4.0
OR		
ITA 1121	Elementary Italian 2	4.0
OR		
JPN 1121	Elementary Japanese 2	4.0
OR		
POR 1121	Elementary Portuguese 2	4.0
OR		
SPN 1121	Elementary Spanish 2	4.0
Semester 3		Units: 7.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
<i>Foreign Language Literature</i>		3.0

FRW 2010	Introductory Readings in French Literature I	3.0
OR		
FRW 2011	Introductory Readings in French Literature II	3.0
OR		
HCW 2020	Selected Readings in Haitian-Creole Literature	3.0
OR		
SPW 2010	Introduction to Masterpieces of Spain	3.0
OR		
SPW 2020	Introduction to Masterpieces of Latin America	3.0

Semester 4

Units: 13.0

BSC 1005	General Education Biology	3.0
POS 2041	American Federal Government	3.0
<i>Social Science</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
WOH 2012	History of World Civilization to 1789	3.0
OR		
WOH 2022	History of World Civilization From 1789	3.0
<i>Foreign Language Intermediate</i>		4.0
FRE 2220	Intermediate French 1	4.0
OR		
GER 2220	Intermediate German 1	4.0
OR		
ITA 2220	Intermediate Italian 1	4.0
OR		
JPN 2220	Intermediate Japanese 1	4.0
OR		
SPN 2220	Intermediate Spanish 1	4.0

Semester 5

Units: 13.0

ESC 1000	General Education Earth Science	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Foreign Language Intermediate 2</i>		4.0
FRE 2221	Intermediate French 2	4.0
OR		
GER 2221	Intermediate German 2	4.0
OR		
ITA 2221	Intermediate Italian 2	4.0
OR		
JPN 2221	Intermediate Japanese 2	4.0
OR		
SPN 2221	Intermediate Spanish 2	4.0
Foreign Language Literature*		3.0

*Select 3 credits from the following course options: FRW 2010, FRW 2011, HCW 2020, SPW 2010, SPW 2020

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

CHI1120 Elementary Mandarin Chinese 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Mandarin Chinese. Note: students must pass this course with a "C" or better to continue to CHI1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

FRE1120 Elementary French 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in French. Note: Students must pass this course with a C or better to continue to FRE1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

GER1120 Elementary German 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in German. Note: Students must pass this course with a C or better to continue to GER1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

ITA1120 Elementary Italian 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Italian. Note: Students must pass this course with a C or better to continue to ITA1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

JPN1120 Elementary Japanese 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Japanese. Note: Students must pass this course with a C or better to continue to JPN1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

POR1120 Elementary Portuguese 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Portuguese. Note: Students must pass this course with a C or better to continue to POR1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

SPN1120 Elementary Spanish 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Spanish. Note: Students must pass this course with a C or better to continue to SPN1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

CHI1121 Elementary Mandarin Chinese 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Mandarin Chinese. Note: students must pass this course with a "C" or better to continue to CHI2220.

Lecture Hours: 64.000

Prerequisite: CHI1120

Course User Fee: \$15.00

FRE1121 Elementary French 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in French. Note: Students must pass this course with a C or better to continue to FRE2220.

Lecture Hours: 64.000

Prerequisite: FRE1120

Course User Fee: \$15.00

GER1121 Elementary German 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in German. Note: Students must pass this course with a C or better to continue to GER2220.

Lecture Hours: 64.000

Prerequisite: GER1120

Course User Fee: \$15.00

ITA1121 Elementary Italian 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Italian. Note: Students must pass this course with a C or better to continue to ITA2220.

Lecture Hours: 64.000

Prerequisite: ITA1120

Course User Fee: \$15.00

JPN1121 Elementary Japanese 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Japanese. Note: Students must pass this course with a C or better to continue to JPN2220.

Lecture Hours: 64.000

Prerequisite: JPN1120

Course User Fee: \$15.00

POR1121 Elementary Portuguese 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Portuguese. Note: students must pass this course with a "C" or better to continue to POR2220.

Lecture Hours: 64.000

Prerequisite: POR1120

Course User Fee: \$15.00

SPN1121 Elementary Spanish 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Spanish. Note: Students must pass this course with a C or better to continue to SPN2220.

Lecture Hours: 64.000

Prerequisite: SPN1120

Course User Fee: \$15.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

FRW2010 Introductory Readings in French Literature I 3.000 credits

This course introduces students to foundational texts of French literature from medieval times through the 17th century and includes works from the Western canon. Students will analyze major literary movements, themes, and stylistic developments through careful reading of representative authors and texts. The course develops critical thinking skills through discussion, analysis, and written interpretation of literary works that have shaped French intellectual tradition.

Lecture Hours: 48.000

FRW2011 Introductory Readings in French Literature II 3.000 credits

This course examines French literature from the 18th century to the present day, exploring major authors and literary movements, and includes works from the Western canon. Students will engage with texts that reflect evolving philosophical, social, and artistic perspectives in French literary tradition. Through close reading and critical analysis, students develop interpretive skills while examining the continued influence of these works on contemporary thought.

Lecture Hours: 48.000

HCW2020 Selected Readings in Haitian-Creole Literature 3.000 credits

This course surveys significant works in Haitian-Creole literature alongside foundational texts that influenced its development and includes works from the Western canon. Students will examine the literary traditions, themes, and artistic expressions that characterize this body of literature. The course emphasizes critical analysis and interpretation while exploring the historical and intellectual contexts that shaped these literary works.

Lecture Hours: 48.000

SPW2010 Introduction to Masterpieces of Spain 3.000 credits

This course presents major works of Spanish literature from medieval times through the present and includes works from the Western canon. Students will study representative texts that demonstrate the evolution of Spanish literary forms, themes, and artistic achievement. Through critical reading and analysis, students develop skills in literary interpretation while examining works that have significantly influenced Spanish intellectual tradition.

Lecture Hours: 48.000

Recommended Preparation: Oral and written fluency in Spanish and/or permission from faculty or Department Chairperson.

SPW2020 Introduction to Masterpieces of Latin America 3.000 credits

This course introduces students to significant works of Latin American literature alongside influential texts that shaped its development and includes works from the Western canon. Students will analyze diverse literary voices and movements that characterize Latin American artistic expression. The course emphasizes critical thinking through examination of texts that reflect the philosophical, historical, and intellectual foundations of Latin American literary tradition.

Lecture Hours: 48.000

Recommended Preparation: Oral and written native fluency in Spanish and/or permission from faculty or Department Chairperson.

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.
Lecture Hours: 48.000

WOH2012 History of World Civilization to 1789 3.000 credits

This course covers the history of World Civilizations from the prehistoric period to the 18th century. Students will learn of the major political, social, economic, and cultural structures of civilizations and their development through 1789.
Lecture Hours: 48.000

WOH2022 History of World Civilization From 1789 3.000 credits

The student will examine the historical development of world civilizations since 1789. Students will learn of historical processes and developments in social, cultural, political, and economic contexts since the 18th century.
Lecture Hours: 48.000

FRE2220 Intermediate French 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of french through a systematic review (using an integrated, multimedia approach) of reading grammar, and writing skills with emphasis on oral and written communication.
Lecture Hours: 64.000
Prerequisite: FRE1121
Course User Fee: \$15.00

GER2220 Intermediate German 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of German through a systematic review (using an integrated, multimedia approach) of reading, grammar, and writing skills with emphasis on oral and written communication.
Lecture Hours: 64.000
Prerequisite: GER1121
Course User Fee: \$15.00

ITA2220 Intermediate Italian 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of italian through a systematic review (using an integrated, multimedia approach) of reading, grammar, and writing skills with emphasis on oral and written communication.
Lecture Hours: 64.000
Prerequisite: ITA1121
Course User Fee: \$15.00

JPN2220 Intermediate Japanese 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of Japanese through a systematic review (using an integrated, multimedia approach) of reading, grammar, and writing skills with emphasis on oral and written communication.
Lecture Hours: 64.000
Prerequisite: JPN1121
Course User Fee: \$15.00

SPN2220 Intermediate Spanish 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of Spanish through a systematic review (using an integrated, multimedia approach) of reading, grammar, and writing skills with emphasis on oral and written communication.
Lecture Hours: 64.000
Prerequisite: SPN1121
Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

FRE2221 Intermediate French 2 4.000 credits

This is a continuation of Intermediate French 1. Students will learn to understand, speak, read, and write French. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: FRE1121

Course User Fee: \$15.00

GER2221 Intermediate German 2 4.000 credits

This is a continuation of Intermediate German 1. Students will learn to understand, speak, read, and write German. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: GER2220

Course User Fee: \$15.00

ITA2221 Intermediate Italian 2 4.000 credits

This is a continuation of Intermediate Italian 1. Students will learn to understand, speak, read, and write Italian. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: ITA2220

Course User Fee: \$15.00

JPN2221 Intermediate Japanese 2 4.000 credits

This is a continuation of Intermediate Japanese 1. Students will learn to understand, speak, read, and write Japanese. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: JPN2220

Course User Fee: \$15.00

SPN2221 Intermediate Spanish 2 4.000 credits

This is a continuation of Intermediate Spanish 1. Students will learn to understand, speak, read, and write Spanish. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: SPN2220

Course User Fee: \$15.00

Program Description

Forestry

Associate in Arts | Code: 10101 | 60 credits

Forestry studies the ecology and economy of forest management. Students should be aware that the University of Florida is the only in-state university offering this program, with majors in forest resources, and conservation and urban forestry. Foresters manage, develop, and protect woodlands and their resources (timber, water, wildlife, forage and recreational areas).

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
CHM 1025	Introductory Chemistry	3.0
CHM 1025L	Introductory Chemistry Lab	1.0
EVR 1001	Introduction to Environmental Science	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0

Semester 3		Units: 13.0
BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Social</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
OR		
SYG 2000	Introduction to Sociology	3.0

Semester 4	Units: 11.0
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BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
HOS 1010	Horticulture 1	3.0
PHI 2600	Introduction to Ethics	3.0

Semester 5

Units: 13.0

GIS 1040	Introduction to GIS Technology	4.0
HOS 1011	Horticulture 2	3.0
STA 2023	Statistical Methods	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1025

Course User Fee: \$30.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy.

Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

HOS1010 Horticulture 1 3.000 credits

This is an introductory course on the principles of horticulture. Students will learn plant structure and function, plant propagation, plant nutrients and fertilizers, potting media, soils, pruning, and plant pests. A survey of various fields in ornamental horticulture will also be covered.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

GIS1040 Introduction to GIS Technology 4.000 credits

This course teaches the principles of Geographic Information Systems (GIS). The student will learn fundamental GIS concepts, GIS models and formats, as well as map creation and analysis. At the end of the course, the student will be able to perform basic operations in GIS software. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

HOS1011 Horticulture 2 3.000 credits

The student will learn the maintenance and management aspects of horticulture business (nursery facility or landscape maintenance and design) including irrigation systems, plant growing facilities, plant propagation equipment, and landscape maintenance equipment. Hands-on practice in programming of plant production crops and nursery design in our nursery.

Lecture Hours: 48.000

Prerequisite: HOS1010

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

Program Description

Geology

Associate in Arts | Code: 11904 | 60 credits

Geologists study the structure, composition and history of the Earth. This program provides basic coursework in geology, calculus, biology and chemistry. Some examples of employers of geologists include agencies targeting pollution or urban waste, corporations searching for new sources of petroleum or natural gas and research organizations studying volcanoes or earthquakes.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101

English Composition 1

3.0

MAT 1033	Intermediate Algebra	3.0
MAT 1033L	Intermediate Algebra Recitation Hall	0.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2 **Units: 12.0**

ENC 1102	English Composition 2	3.0
MAC 1105	College Algebra	3.0
CHM 1025	Introductory Chemistry	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0

Semester 3 **Units: 13.0**

CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry and Qualitative Analysis Lab	2.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
<i>Social Science</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
OR		
SYG 2000	Introduction to Sociology	3.0

Semester 4 **Units: 12.0**

CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
PHY 2053	Physics (without Calculus) 1	3.0
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0

Semester 5 **Units: 13.0**

<i>Humanities</i>		3.0
ARH 2050	Art History 1	3.0
OR		
ARC 2701	History of Architecture 1	3.0
OR		
ARC 2702	History of Architecture 2	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
EVR 1001	Introduction to Environmental Science	3.0
GLY 1010	Physical Geology	3.0
PHY 2054	Physics (Without Calculus) 2	3.0
PHY 2054L	Physics (Without Calculus) 2 Lab	1.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAT1033L Intermediate Algebra Recitation Hall 0.000 credits

The student will receive individualized, small group or whole group instruction to deepen their conceptual understanding of mathematics. The student will reinforce and apply content knowledge with effective problem-solving techniques and non-cognitive activities to make mathematics meaningful and relevant to their fields of study while strengthening the concepts needed to achieve the objectives of MAT1033. This lab session is associated and funded by the First in the World Grant, CCAT Math Project.

Lab Hours: 1.000

Corequisite: MAT1033

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy.

Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: MAC1105, Corequisite: CHM1045, Prerequisite: CHM1025 Or an approved ACS Proficiency Exam score.

Course User Fee: \$30.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics.

Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

ARH2050 Art History 1 3.000 credits

This course offers a survey of visual arts from prehistory to 800 C.E. and includes works from the western canon. Students will examine artistic expressions across civilizations, analyzing their connections to historical contexts and aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how visual arts reflect and influence societal developments, ideas, and values throughout early history.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2701 History of Architecture 1 3.000 credits

This course offers a survey of architecture from ancient times through the 18th century and includes works from the Western canon. Students will examine the integration of art forms, structural elements, and ornamental designs across world civilizations, analyzing their connections to broader aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societies throughout historical periods.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2702 History of Architecture 2 3.000 credits

This course provides a survey of architecture from the 19th century to the present and includes works from the Western canon and examining architectural developments. Students will analyze the integration of art forms, structural elements, and ornamental designs across civilizations, exploring their connections to human development, including art and history. Through this study, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societal changes and cultural expressions throughout modern and contemporary periods. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$10.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon.

Written analysis of literary works may be required.

Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHY2054 Physics (Without Calculus) 2 3.000 credits

An introduction to the basic principles of physics. PHY 2053 covers mechanics, sound and thermodynamics. PHY 2054 includes electricity, magnetism and optics.

Lecture Hours: 48.000

Prerequisite: PHY2053, PHY2053L, Corequisite: PHY2054L

Course User Fee: \$10.00

PHY2054L Physics (Without Calculus) 2 Lab 1.000 credits

The physics lab courses are one-credit courses designed to be taken in conjunction with a physics lecture. A different experiment is performed each week with topics chosen to correspond with the material being studied in the lecture. Each experiment is designed to be completed in about 2 contact hours.

Lab Hours: 32.000

Prerequisite: PHY2053, PHY2053L, Corequisite: PHY2054

Course User Fee: \$20.00

Program Description

Graphic or Commercial Arts

Associate in Arts | Code: 11004 | 60 credits

Graphic Arts emphasizes studio courses in design, drawing and digital techniques. Graduates may work in advertising agencies, design studios, exhibit and display businesses, department stores and industrial organizations.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 14.0

ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
ARH 1000	Art Appreciation	3.0
ART 1300C	Drawing	4.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
MAC 1105	College Algebra	3.0
ART 1202C	Two-Dimensional Design	4.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
Semester 3		Units: 7.0
ARH 2051	Art History 2	3.0
ART 1203C	Three-Dimensional Design	4.0
Semester 4		Units: 13.0
BSC 1005	General Education Biology	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
ARH 2051	Art History 2	3.0
ART 1330C	Figure Drawing	4.0
Semester 5		Units: 13.0
ESC 1000	General Education Earth Science	3.0
PSY 2012	Introduction to Psychology	3.0
SPC 2608	Introduction to Public Speaking	3.0
Pathway Electives*		4.0

*Select 4 credits from the following course options: GRA 1111C, GRA 1113C, GRA 1206C, GRA 1280C, GRA 1750, GRA 1949, GRA 2117C, GRA 2121C, GRA 2151C, GRA 2203C, GRA 2207C, GRA 2305C, GRA 2545C, GRA 2546C, GRA 2949

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

ART1300C Drawing 4.000 credits

Basic problems in freehand drawing, including perspective, still-life and landscape. Emphasis is on developing a sense of structure through line, form and texture.

Lecture Hours: 32.000, Lab Hours: 64.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

ART1202C Two-Dimensional Design 4.000 credits

This course is designed to give students an understanding of advanced concepts of two-dimensional design and to give hands on opportunity to transform visual and experiential information into two-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals. Lecture Hours: 32.000, Lab Hours: 64.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement. Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

ARH2051 Art History 2 3.000 credits

In this course, students will embark on a world survey of the visual arts from 800 to 1850 A.D., exploring various artistic movements, styles, and cultural contexts. The course will delve into the rich and diverse artistic traditions from different regions, including works from the Western canon. Writing intensive course. Lecture Hours: 48.000
Course User Fee: \$5.00

ART1203C Three-Dimensional Design 4.000 credits

This course is designed to give students an understanding of the concepts of three-dimensional design and to provide hands-on opportunity to transform visual and experiential information into three-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals. Self-evaluation and safety skills will also figure prominently. Prerequisite: ART1202C. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ART1202C
Course User Fee: \$40.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method. Lecture Hours: 48.000
Course User Fee: \$8.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ARH2051 Art History 2 3.000 credits

In this course, students will embark on a world survey of the visual arts from 800 to 1850 A.D., exploring various artistic movements, styles, and cultural contexts. The course will delve into the rich and diverse artistic traditions from different regions, including works from the Western canon. Writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$5.00

ART1330C Figure Drawing 4.000 credits

Drawing and painting from the live model with emphasis on structure, movement and expression.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$80.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence. Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

Program Description

Health Services Administration

Associate in Arts | Code: 11200 | 60 credits

This program provides the fundamental science coursework for transfer to a baccalaureate program in health services administration. The Baccalaureate degree prepares students for entry-level management positions in health services delivery organizations. Persons licensed in clinical health often pursue this degree, as do medical care professionals who do not have an undergraduate degree. The Baccalaureate also prepares individuals for graduate study in this field.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
MAT 1033	Intermediate Algebra	3.0
MAT 1033L	Intermediate Algebra Recitation Hall	0.0
GEB 1011	Principles of Business	3.0
Oral Communication		3.0

SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
Semester		Units: 12.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
MAN 2021	Principles of Management	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 3		Units: 9.0
MAN 2300	Human Resources Management	3.0
STA 2023	Statistical Methods	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
Semester 4		Units: 14.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
PHI 2600	Introduction to Ethics	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
Semester 5		Units: 13.0
ACG 2071	Managerial Accounting	3.0

ACG 2071L	Managerial Accounting Lab	1.0
ECO 2013	Principles of Economics (Macro)	3.0
Natural Science		3.0
HUN 1201	Essentials of Human Nutrition	3.0
OR		
OCB 1010	Introduction to Marine Biology	3.0
OR		
PSC 1121	General Education Physical Science	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
Pathway Elective*		3.0

*Note: Check with advisor for any transferable Credit Type 1 or Credit Type 2 course.

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAT1033L Intermediate Algebra Recitation Hall 0.000 credits

The student will receive individualized, small group or whole group instruction to deepen their conceptual understanding of mathematics. The student will reinforce and apply content knowledge with effective problem-solving techniques and non-cognitive activities to make mathematics meaningful and relevant to their fields of study while strengthening the concepts needed to achieve the objectives of MAT1033. This lab session is associated and funded by the First in the World Grant, CCAT Math Project.

Lab Hours: 1.000

Corequisite: MAT1033

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written

Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAN2300 Human Resources Management 3.000 credits

Reviews how the personnel/human resources department contributes to overall planning and profitability of an organization. Major topics include typical personnel functions: recruitment and selection, training, performance appraisal, job analysis, and compensation and benefits administration. Class discussions will focus on changing value systems in the work force and the resulting challenges for managers.

Lecture Hours: 48.000

Course User Fee: \$20.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

OCB1010 Introduction to Marine Biology 3.000 credits

An introduction to the biology of the seas. Emphasis is placed on the variety of marine organisms and their structural, physiological, and behavioral adaptations within specific marine environments. Special attention is directed to marine communities, e.g., coral reefs and shallow grass flats, and the factors limiting the distribution of organisms within those communities. Discussions will also be directed towards geological, chemical and physical characteristics of the world's oceans. (3 hr. lecture)

Lecture Hours: 48.000

PSC1121 General Education Physical Science 3.000 credits

A study of the major concepts and principles from each of the following areas: physics, chemistry, and astronomy.

Lecture Hours: 48.000

Pre/Corequisite: MGF 1130 or MGF 1131 or MAC 1105 or MAC 1114 or MAC 1140 or MAC 2223 or MAT 1033

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

Program Description

History

Associate in Arts | Code: 12202 | 60 credits

History is the study of the events, patterns and cycles that have shaped our present world. Depending on the area of specialization, history may examine political events, social evolution, cultural developments or a combination of these. The two-year program at MDC prepares students for transfer with courses in American, African-American and Latin American history, and surveys of American, English and world literature. Professional historians (e.g. museum curators and educators) tend to pursue the Doctoral degree, but the Bachelor's degree in history can prepare students for graduate work in law or political science, and apply to careers requiring good writing or analytical skills.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
MGF 1131	Mathematics in Context	3.0
AMH 2010	History of the US to 1877	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
STA 2023	Statistical Methods	3.0
AMH 2020	History of the US since 1877	3.0
SPC 2608	Introduction to Public Speaking	3.0
Semester 3		Units: 12.0
AST 1002	Descriptive Astronomy	3.0
LAH 2021	Colonial Latin America	3.0
PHI 2010	Introduction to Philosophy	3.0
<i>World History</i>		3.0
WOH 2012	History of World Civilization to 1789	3.0
OR		
WOH 2022	History of World Civilization From 1789	3.0
Semester 4		Units: 12.0
ESC 1000	General Education Earth Science	3.0
EUH 2032	History of The Holocaust	3.0
PHI 2600	Introduction to Ethics	3.0
<i>Pathway Elective</i>		3.0
AMH 2035	Recent American History Since 1945	3.0
OR		
AMH 2079	History of South Florida	3.0
Semester 5		Units: 12.0
AMH 2090	History of Women in the United States	3.0
AMH 2091	Afro-American History	3.0
LAH 2025	History of Cuba	3.0
WOH 2003	History of Genocide	3.0
		Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

LAH2021 Colonial Latin America 3.000 credits

This course covers the history of Latin America, from the pre-Columbian civilizations of the region, to the Wars of Independence. Students will learn about the development of the political, social, economic and cultural structures of colonial Latin America to 1825.

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

WOH2012 History of World Civilization to 1789 3.000 credits

This course covers the history of World Civilizations from the prehistoric period to the 18th century. Students will learn of the major political, social, economic, and cultural structures of civilizations and their development through 1789.

Lecture Hours: 48.000

WOH2022 History of World Civilization From 1789 3.000 credits

The student will examine the historical development of world civilizations since 1789. Students will learn of historical processes and developments in social, cultural, political, and economic contexts since the 18th century.

Lecture Hours: 48.000

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EUH2032 History of The Holocaust 3.000 credits

This is a foundation course in Holocaust Studies. Students will learn the historical origins, execution and consequences of the Holocaust. This course also examines the Holocaust's place in context of genocides past and present.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

AMH2035 Recent American History Since 1945 3.000 credits

This course covers the internal development of the United States and the role of the U.S. in world affairs since World War II.

Lecture Hours: 48.000

AMH2079 History of South Florida 3.000 credits

A study of the history of South Florida (Lake Okeechobee south to Key West) including geological foundations exploration, settlement and contemporary cultural trends.

Lecture Hours: 48.000

AMH2090 History of Women in the United States 3.000 credits

This course will explore the history of women from precolonial times to the present, by examining how gender roles, class, and ethnicity defined women's experiences in the social, political, and economic contexts. It will also analyze how women shaped and contributed to historical events and movements within the United States.
Lecture Hours: 48.000

AMH2091 Afro-American History 3.000 credits

A survey, including the African background, of the Afro-Americans in the United States history, with emphasis on their economic, political, and cultural development.
Lecture Hours: 48.000

LAH2025 History of Cuba 3.000 credits

Historical analysis of the development of Cuban society, its culture and institutions.
Lecture Hours: 48.000

WOH2003 History of Genocide 3.000 credits

This course includes works from the Western canon and is designed to explore the history, causes, and consequences of genocide by identifying patterns of intentional mass killings. Students will examine case studies to understand the characteristics of both historical and contemporary incidents classified as genocide. Through this analysis, learners will gain insight into the social and behavioral science concepts that inform understanding of tragic events and their implications for society today.
Lecture Hours: 48.000

Program Description

[Hospitality Administration/Travel & Tourism](#)

Associate in Arts | Code: 10506 | 60 credits

This field combines traditional business and management education with training specific to the tourism, travel and hospitality industries. Careers in the hospitality/travel and tourism industry include hotel and restaurant management, and managerial positions with cruise ships, airlines, land-based tourism companies, as well as travel agencies.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
HFT 1000	Introduction to Hospitality	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
ECO 2013	Principles of Economics (Macro)	3.0
HFT 1210	Human Resources	3.0
Pathway Elective*		3.0

*Select 3 credits from the following course options: HFT 1212, HFT 1220, HFT 1300, HFT 1454, HFT 1611, HFT 1631, HFT 1841, HFT 1841L, HFT 1852, HFT 1949, HFT 2223, HFT 2241, HFT 2261, HFT 2410, HFT 2500, HFT 2501, HFT 2750, HFT 2772, HFT 2800, HFT 2949

Semester 3

Units: 10.0

PHI 2600	Introduction to Ethics	3.0
Pathway Electives ¹⁾		7.0
Computer Competency Test ²⁾		

¹⁾Select 7 credits from the following course options: HFT 1212, HFT 1220, HFT 1300, HFT 1454, HFT 1611, HFT 1631, HFT 1841, HFT 1841L, HFT 1852, HFT 1949, HFT 2223, HFT 2241, HFT 2261, HFT 2410, HFT 2500, HFT 2501, HFT 2750, HFT 2772, HFT 2800, HFT 2949

²⁾Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4

Units: 12.0

ESC 1000	General Education Earth Science	3.0
<i>Mathematics</i>		3.0
QMB 2100	Basic Business Statistics	3.0
OR		
STA 2023	Statistical Methods	3.0
HFT 2421	Managerial Accounting for Hospitality	3.0
HFT 2449	E-Business For The Hospitality Industry	3.0

Semester 5

Units: 13.0

HUN 1201	Essentials of Human Nutrition	3.0
PHI 2010	Introduction to Philosophy	3.0
SPC 1017	Introduction to Communication	3.0
Pathway Elective*		4.0

*Select 4 credits from the following course options: HFT 1212, HFT 1220, HFT 1300, HFT 1454, HFT 1611, HFT 1631, HFT 1841, HFT 1841L, HFT 1852, HFT 1949, HFT 2223, HFT 2241, HFT 2261, HFT 2410, HFT 2500, HFT 2501, HFT 2750, HFT 2772, HFT 2800, HFT 2949

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.
Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.
Lecture Hours: 16.000
Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.
Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.
Lecture Hours: 48.000
Course User Fee: \$15.00

HFT1210 Human Resources 3.000 credits

Provides information relating to the recruitment and selection of new staff, the handling of difficult employees, motivating employees and conducting performance evaluations. (3 hr. lecture)
Lecture Hours: 48.000
Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

HFT2421 Managerial Accounting for Hospitality 3.000 credits

Presents managerial accounting concepts and explains how they apply to specific operations within the hospitality industry.

Lecture Hours: 48.000

HFT2449 E-Business For The Hospitality Industry 3.000 credits

Prepares student to manage information systems within their organizations. Emphasis is on selecting the right computer systems technology and maximizing available technology in order to promote and sell services. Introduces the use of automation in the broad hospitality sector and exams technological applications ranging from distribution systems (GDS, CRS and Web based), Property Management Systems, and EPOS systems to developments in telecommunications, and assesses their effect on the tourism sector. While a broad interpretation of both technology and tourism will be used, particular emphasis will be placed on the hospitality sector (i.e. hotels and catering) and on distribution technology. A combination of lecture, case studies, seminars, visiting lecture and lab sessions are used.

Lecture Hours: 48.000

Prerequisite: HFT1000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

Program Description

Interior Design

Associate in Arts | Code: 10201 | 60 credits

Interior Design studies combine architecture, art and design courses, training the student to understand the relation of interior spaces to the total design of structures (including architecture, landscaping and lighting). An interior designer encounters a variety of challenging work, available in professional, institutional and private settings.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
ARH 1000	Art Appreciation	3.0
ARC 1115	Architectural Communications 1	2.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 14.0
ARC 1126C	Architectural Drawing 1	4.0
<i>Humanities</i>		3.0
ARC 2701	History of Architecture 1	3.0
OR		
ARC 2702	History of Architecture 2	3.0
OR		
HUM 1020	Humanities	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
IND 1020	Interior Design 1	4.0
MAC 1105	College Algebra	3.0
Computer Competency Test*		
*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)		
Semester 3		Units: 6.0
ENC 1102	English Composition 2	3.0
MAC 1140	Pre-Calculus Algebra	3.0
Semester 4		Units: 14.0
ARC 2171C	Computer Aided Drafting 1	4.0
IND 1200	Interior Design 2	4.0
MAC 1114	Trigonometry	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
Semester 5		Units: 14.0
ARC 2461	Architectural Materials and Construction 1	4.0
ECO 2013	Principles of Economics (Macro)	3.0
EVR 1001	Introduction to Environmental Science	3.0
PHY 2053	Physics (without Calculus) 1	3.0
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0
		Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC1115 Architectural Communications 1 2.000 credits

Exercises in freehand drawing, sketching and linear perspective are designed to increase the student's awareness of the architectural environment. This is accomplished through a series of form studies of nature, architectural forms, and abstract elements of composition.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions. Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC2701 History of Architecture 1 3.000 credits

This course offers a survey of architecture from ancient times through the 18th century and includes works from the Western canon. Students will examine the integration of art forms, structural elements, and ornamental designs across world civilizations, analyzing their connections to broader aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societies throughout historical periods.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2702 History of Architecture 2 3.000 credits

This course provides a survey of architecture from the 19th century to the present and includes works from the Western canon and examining architectural developments. Students will analyze the integration of art forms, structural elements, and ornamental designs across civilizations, exploring their connections to human development, including art and history. Through this study, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societal changes and cultural expressions throughout modern and contemporary periods. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

IND1020 Interior Design 1 4.000 credits

Student's projects develop the ability to plan simple interior floor plans and elevations.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1115

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

ARC2171C Computer Aided Drafting 1 4.000 credits

Computer-aided drafting as it applies in the fields of architecture and interior design using office simulation. Emphasis is on the production of computer-aided drafting of working drawings involving different types of structure.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR ARC2461

Course User Fee: \$50.00

IND1200 Interior Design 2 4.000 credits

Problems in room planning, correlation of color schemes and furnishings.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: IND1020

Course User Fee: \$25.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAC1105 OR MAC1106
Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)
Lecture Hours: 48.000
Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.
Lecture Hours: 48.000
Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."
Course User Fee: \$15.00

ARC2461 Architectural Materials and Construction 1 4.000 credits

An introduction to basic materials and methods of building construction. Emphasis is on wood, concrete, unit masonry, and light steel construction. Laboratory projects may include working drawings' interpretation, sketching construction details, or field trips to construction sites and fabricant plants. Designed primarily as the initial materials and methods course for architectural transfer students. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ARC1126C OR BCN1251
Course User Fee: \$25.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.
Lecture Hours: 48.000
Course User Fee: \$15.00

**EVR1001 Introduction to Environmental Science
3.000 credits**

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

**PHY2053 Physics (without Calculus) 1 3.000
credits**

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

**PHY2053L Physics (Without Calculus) 1 Lab 1.000
credits**

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

Program Description

International Relations

Associate in Arts | Code: 12205 | 60 credits

Students can obtain the coursework necessary to transfer to four-year programs in international relations, a major which usually includes political science and economics courses. Employment opportunities are available at the baccalaureate level in business, government, journalism and political organizations. Many students, however, go on to pursue graduate work or law school.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
MGF 1131	Mathematics in Context	3.0
POS 2041	American Federal Government	3.0
SPC 1017	Introduction to Communication	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
INR 2002	International Relations	3.0
PHI 2600	Introduction to Ethics	3.0
STA 2023	Statistical Methods	3.0
Semester 3		Units: 12.0
CPO 2100	Comparative European Government	3.0
ISS 2270	Multicultural Communications and Relations	3.0
PHI 2010	Introduction to Philosophy	3.0
SYG 2000	Introduction to Sociology	3.0
Semester 4		Units: 12.0
CPO 2408	Comparative Middle Eastern and North African Governments	3.0
EVR 1001	Introduction to Environmental Science	3.0
POS 2112	State and Local Government In America	3.0
POT 2014	European Political Theory 1	3.0
Semester 5		Units: 12.0
INR 2440	International Law and Organization	3.0
PSC 1515	Energy in the Natural Environment	3.0
<i>Pathway Elective</i>		6.0
CPO 2441	Islam in the Modern World	3.0
OR		
INS 2200	Introduction to Conflict Resolution and Peace Studies	3.0
OR		
INS 2300	Global Environmental Politics	3.0
OR		
INS 2310	Religion and Global Politics	3.0
OR		
POS 2293	Islam and America	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

INR2002 International Relations 3.000 credits

The nature of international relations, the causes of leading international problems, foreign policies of world powers, international political organizations, and the origins of war and peace in the international arena.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CPO2100 Comparative European Government 3.000 credits

This course discusses the structures and functioning of the systems of government in Western Europe. Students will analyze major policy issues and approaches to resolving them, and cultural differences and interdependence among the Western European countries in policy determination. Recommended prerequisite: a prior course in History or Social Science. Offered first semester. Given in English. Offered through overseas study program. Lecture Hours: 48.000

ISS2270 Multicultural Communications and Relations 3.000 credits

This course uses an interdisciplinary approach to examine the complex interactions among ethnicity, race, gender, age, and class, as well as other ways in which we differ as they pertain to shaping personal awareness, understanding, and skills that will allow them to interact more effectively with diverse populations, age, groups, and lifestyles and to think through and value human diversity. This course has an overriding principle based on the concept of human rights. (3 hr. lecture)

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

CPO2408 Comparative Middle Eastern and North African Governments 3.000 credits

The Middle East & North Africa (MENA) has uninterruptedly been at the forefront of international geopolitics for almost a century. This enduring geopolitical visibility has important cultural, societal, economic, and security implications at national, regional, and international levels. This course is designed to tackle some of these issues by providing students with a historico-cultural, politico-economic, and geo-strategic survey of the Middle East.

Lecture Hours: 48.000

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

POS2112 State and Local Government In America 3.000 credits

The typical state and local government organization, together with political practices in America, with special emphasis on governmental organizations and the major contemporary political problems of the State of Florida and of Florida communities.

Lecture Hours: 48.000

POT2014 European Political Theory 1 3.000 credits

This course covers the more important trends in European political thought from Plato to the present. It examines those ideas which have contributed to the shaping of the political cultures of Western and Eastern Europe. It discusses the historical evolution of key concepts of politics such as freedom, order, political obligations, justice, consent, rights and duties, power and authority. A prior course in Government, History or Philosophy is desirable. Given in English. Level 1. Offered through Overseas Study Program.

Lecture Hours: 48.000

INR2440 International Law and Organization 3.000 credits

International law and problems in world politics; a review of man's attempt to control international politics through international law and organizations, including the League of Nations, the United Nations, NATO, and the European Union. A prior course in History or Social Science is desirable. Offered second semester. Given in English. Offered through Overseas Study Program.

Lecture Hours: 48.000

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CPO2441 Islam in the Modern World 3.000 credits

The survey course is designed to grant students a broad-based exposure across a variety of conceptual debates, historical events, and policy issues at the intersection of Islam and the modern world and the paths forward. The issues especially religiously-inspired and informed violence/conflict and debates surrounding them have assumed prominence in the 21st century with the resurgence of religion in politics and the subsequent struggle for Islam. These issues will be contextualized in normative, empirical and historical frameworks to equip students with a sophisticated and multi-dimensional conceptual toolbox to analyze them.

Lecture Hours: 48.000

INS2200 Introduction to Conflict Resolution and Peace Studies 3.000 credits

Introduction to Conflict Resolution and Peace Studies is an interdisciplinary course exploring foundational concepts, methodologies, and practical approaches to interstate and intra-state conflicts and how to tackle them. The course has no specific prerequisites and will appeal to a broad audience as it analyzes a broad spectrum of perspectives in the field of conflict and peace studies. By contextualizing the challenge and transmitting the knowledge of peace in normative, empirical and historical frameworks, the course will equip students with a multi-dimensional conceptual toolbox to analyze it while inspiring them to create a more equitable and harmonious world.

INS2300 Global Environmental Politics 3.000 credits

Global Environmental Politics is an interdisciplinary course exploring the evolving relationship between humanity and the environment from diverse conceptual, normative, and empirical perspectives. The course has no specific prerequisites and will appeal to a broad audience as it analyzes a broad spectrum of perspectives in the field. The course will explore the most pressing contemporary environmental challenges, including climate change, loss of biodiversity, desertification, trade in dangerous waste, and over-exploitation of natural resources as well as various political and policy responses to deal with them.

INS2310 Religion and Global Politics 3.000 credits

The interdisciplinary course introduces students to the increasingly visible role of religious identity in international and global politics. It offers a macro-level overview of the intersection of religion and politics in a historical context and sheds light on the intertwined histories and trajectories connecting them. It explores how this nexus has been shaped by belief and value systems as well as the historical socio-economic and political circumstances and conditions. Among the themes addressed in the course are religious diversity and human rights, interfaith dialogue and conflict resolution, and religion in the age of globalization.

Lecture Hours: 48.000

POS2293 Islam and America 3.000 credits

Islam and the United States offers a survey treatment of the historically charged relationship between America and the Muslim World and the path forward. The course analyzes the intersection of religion, identity politics, geopolitics and economics of Islam-US relations in a historical context. These issue areas will be contextualized in normative, empirical and historical frameworks to equip the student with a sophisticated and multi-dimensional conceptual toolbox to analyze them.

Lecture Hours: 48.000

Program Description

Landscape Architecture

Associate in Arts | Code: 10202 | 60 credits

The A.A. in Landscape Architecture prepares students for transfer by offering courses in architecture, horticulture and botany. Landscape architects plan the arrangement of outdoor areas for public use and enjoyment, making recommendations for the types and location of plantings, circulation, drainage and other harmonizing improvements with existing land features and architectural structures. The University of Florida offers the only in-state program in landscape architecture.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
ARC 1115	Architectural Communications 1	2.0
ARH 1000	Art Appreciation	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2		Units: 14.0
ARC 1126C	Architectural Drawing 1	4.0
ARC 1301	Architectural Design 1	4.0
ARC 2701	History of Architecture 1	3.0
MAC 1105	College Algebra	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3		Units: 6.0
ENC 1102	English Composition 2	3.0
<i>Natural Science</i>		3.0
BOT 1010	Botany	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0

Semester 4		Units: 14.0
ARC 1302	Architectural Design 2	4.0
ARC 2171C	Computer Aided Drafting 1	4.0
MAC 1114	Trigonometry	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Semester 5		Units: 14.0
ARC 2461	Architectural Materials and Construction 1	4.0
ARC 2702	History of Architecture 2	3.0
PHY 2053	Physics (without Calculus) 1	3.0
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0
<i>Social Science</i>		3.0

ECO 2013	Principles of Economics (Macro)	3.0
OR		
SYG 2010	Social Problems	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARC1115 Architectural Communications 1 2.000 credits

Exercises in freehand drawing, sketching and linear perspective are designed to increase the student's awareness of the architectural environment. This is accomplished through a series of form studies of nature, architectural forms, and abstract elements of composition.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$25.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions.

Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC1301 Architectural Design 1 4.000 credits

Introductory course to architectural design, its scope, methods and vocabulary interfacing graphics and design as a means towards an awareness and understanding of basic organizational principles. Design concepts analyzed through graphical representation and modeling. Pre/Corequisite: ARC1115.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC2701 History of Architecture 1 3.000 credits

This course offers a survey of architecture from ancient times through the 18th century and includes works from the Western canon. Students will examine the integration of art forms, structural elements, and ornamental designs across world civilizations, analyzing their connections to broader aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societies throughout historical periods.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better. Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

BOT1010 Botany 3.000 credits

This course is a survey of the plant kingdom based on a detailed study of the morphology, anatomy and physiology of selected representative specimens. Corequisite: BOT1010L
Lecture Hours: 48.000
Corequisite: BOT1010L
Course User Fee: \$8.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

ARC1302 Architectural Design 2 4.000 credits

A continuation of ARC1301, emphasizing the application of ordering concepts, and aspects and determinants of form and space. An individual design process is developed by the student. Prerequisite: ARC1301; Pre/Corequisites: ARC1126, ARC2701. Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$25.00

ARC2171C Computer Aided Drafting 1 4.000 credits

Computer-aided drafting as it applies in the fields of architecture and interior design using office simulation. Emphasis is on the production of computer-aided drafting of working drawings involving different types of structure. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ARC1126C OR ARC2461
Course User Fee: \$50.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAC1105 OR MAC1106
Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ARC2461 Architectural Materials and Construction 1 4.000 credits

An introduction to basic materials and methods of building construction. Emphasis is on wood, concrete, unit masonry, and light steel construction. Laboratory projects may include working drawings' interpretation, sketching construction details, or field trips to construction sites and fabricant plants.

Designed primarily as the initial materials and methods course for architectural transfer students.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR BCN1251

Course User Fee: \$25.00

ARC2702 History of Architecture 2 3.000 credits

This course provides a survey of architecture from the 19th century to the present and includes works from the Western canon and examining architectural developments. Students will analyze the integration of art forms, structural elements, and ornamental designs across civilizations, exploring their connections to human development, including art and history. Through this study, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societal changes and cultural expressions throughout modern and contemporary periods. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$10.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

SYG2010 Social Problems 3.000 credits

An analysis of the major contemporary and recurring social problems, emphasizing scientific search for variables involved and exploring alternative solutions.

Lecture Hours: 48.000

Program Description

Mass Communications/Journalism

Associate in Arts | Code: 10600 | 60 credits

Mass Communication examines the role of media in society. Coursework includes media criticism and analysis, U.S. history and government, sociology, and a study of the broadcast, cable and Internet industries. Depending upon the student's area of interest, study may also include journalism, and television and radio production. A Bachelor's in mass communication equips one to work in journalism, corporate communication, or in certain business or managerial positions in television or radio.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 12.0**

ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
JOU 1100	Basic Reporting	3.0
SLS 1510	Preparing for Student Success	3.0

Semester 2**Units: 13.0**

ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
HUM 1020	Humanities	3.0

Semester 3

Units: 12.0

BSC 1005	General Education Biology	3.0
ENT 1501	Fundamentals of Changemaking and Social Innovation	3.0
MMC 2000	Introduction to Mass Communications	3.0
SPC 1017	Introduction to Communication	3.0

Semester 4

Units: 12.0

ESC 1000	General Education Earth Science	3.0
JOU 2200	Editing and Makeup	3.0
POS 2041	American Federal Government	3.0
PSY 2012	Introduction to Psychology	3.0

Semester 5

Units: 10.0

CRW 2001	Creative Writing 1	3.0
JOU 1946	Journalism Internship	1.0
PHI 2010	Introduction to Philosophy	3.0
SYG 2230	Multi-Ethnic America	3.0

Total: 59.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

JOU1100 Basic Reporting 3.000 credits

Journalistic writing emphasizing the elements of reporting with an emphasis on the modern news story, analysis of the elements of news, style structure of news stories, news sources, and the mechanics of newspaper production.

Lecture Hours: 48.000

SLS1510 Preparing for Student Success 3.000 credits

This course provides an orientation to college life and helps develop academic, career, and personal goals. Students will learn college success strategies, goal-setting, learning style assessments, as well as general and discipline-specific study skills in the context of various theoretical, practical, and experiential perspectives.

Lecture Hours: 48.000

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ENT1501 Fundamentals of Changemaking and Social Innovation 3.000 credits

This course introduces students to the work of changemaking and the field of social innovation. Students will explore principles of social innovation and social change, while developing the skills to analyze social issues, generate solutions to those issues, and become an effective social change agent.

Lecture Hours: 48.000

Course User Fee: \$43.00

MMC2000 Introduction to Mass Communications 3.000 credits

Development of a critical perception of the mass communications process and its results in both printed and electronic media. Applications of the ethics and codes of journalism to the changing roles and forms of journalistic media. MMC 2000 will transfer for mass communications majors to various universities within the Florida State System.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

JOU2200 Editing and Makeup 3.000 credits

The application of copy desk techniques, including evaluating and editing copy, correcting faulty news stories, handling wire copy, writing headlines, and designing page layouts.

Lecture Hours: 48.000

Prerequisite: JOU1100

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

CRW2001 Creative Writing 1 3.000 credits

Imaginative writing in selected genres.

Lecture Hours: 48.000

JOU1946 Journalism Internship 1.000 - 3.000 credits

Qualified students will receive practical experience working with local or college communications media under the supervision of professional media specialists and the journalism faculty.

Other Hours: 48.000 - 144.000

Prerequisite: JOU1100and Departmental Permission.

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

SYG2230 Multi-Ethnic America 3.000 credits

In this course, students will gain an understanding of the theory and problems of underrepresented populations in American society, with an emphasis on inequity, bias, and the evolving dynamics of bias and discrimination. Additionally, the course will critically examine the portrayal of underrepresented populations in the Western Canon, providing insight into the historical and cultural influences that have shaped societal perceptions and attitudes.

Program Description

Mathematics

Associate in Arts | Code: 11700 | 60 credits

The Mathematics A.A. emphasizes math and science training, and includes coursework in computer programming. Mathematics is both a science and a tool essential for many kinds of work in industry and business. As a result, employment opportunities for graduates trained in mathematics have expanded rapidly in industries such as aviation and communications, sciences such as oceanography and meteorology, and government agencies such as the U.S. Census Bureau.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 16.0

ENC 1101	English Composition 1	3.0
MAC 1106	Integrated College and Precalculus Algebra	5.0
POS 2041	American Federal Government	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2

Units: 12.0

ENC 1102	English Composition 2	3.0
MAC 1114	Trigonometry	3.0
PHI 2010	Introduction to Philosophy	3.0
PSY 2012	Introduction to Psychology	3.0

Semester 3

Units: 8.0

MAC 2311	Calculus and Analytical Geometry 1	5.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Semester 4

Units: 14.0

BSC 1005	General Education Biology	3.0
HUM 1020	Humanities	3.0
MAC 2312	Calculus and Analytical Geometry 2	4.0
<i>Pathway Elective</i>		4.0
COP 1047C	Introduction to Python Programming	4.0
OR		
COP 1334	Introduction to C++ Programming	4.0
OR		
COP 2800	Java Programming	4.0
OR		
MAS 2103	Elementary Linear Algebra	3.0
OR		
STA 2023	Statistical Methods	3.0

Semester 5

Units: 10.0

MAC 2313	Calculus and Analytic Geometry 3	4.0
MAP 2302	Introduction to Differential Equations	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1106 Integrated College and Precalculus Algebra 5.000 credits

The student will learn to analyze linear, quadratic, polynomial, rational, radical, absolute value, composite, inverse, piecewise, exponential, and logarithmic functions, conic sections, systems of equations/inequalities, matrices and determinants, sequences & series, the Binomial Theorem, and applications of mathematical modeling including exponential growth and decay. Prerequisite: MAT1033 with a grade of "B" or better. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAT1033 or appropriate placement scores.

Course User Fee: \$10.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

MAS2103 Elementary Linear Algebra 3.000 credits

This course introduces the student to linear algebra and its applications. The student will examine linear systems of equations, matrices, determinants, vectors spaces, inner product spaces, linear transformations, linear independence and basis, eigenvalues and eigenvectors, decomposition theorems, and elements of proof writing. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC2311

Course User Fee: \$18.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems. Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

MAP2302 Introduction to Differential Equations 3.000 credits

The student will analyze ordinary differential equations, methods of solution of first order linear and nonlinear equations and applications, homogeneous and non-homogeneous linear equations with constant coefficients, differential operator methods, higher order linear equations; the Laplace transform and its properties, elementary existence theorems, series solutions, numerical solutions of first order equations, initial and boundary value problems, vibrations and waves, and an introduction to autonomous systems. Computational Course. Prerequisite: MAC 2312 with a "C" or better or equivalent.

Lecture Hours: 48.000

Prerequisite: MAC2312

Course User Fee: \$18.00

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

Program Description

Music or Music Education

Associate in Arts | Code: 11001 | 60 credits

Music or Music Education students must be proficient in music theory and music history as well as be a skilled performer. Careers in music include individual and group performance, conducting, composing and teaching. Music graduates may also have jobs working in ancillary professions such as retail, publishing and recording.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 15.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
MVK 1111	Class Piano 1	1.0
MUT 1111	Theory 1	3.0
MUT 1241	Sightsinging & Ear Training 1 Year	1.0
SLS 1106	First Year Experience Seminar	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
MUT 1112	Theory 2	3.0
MUT 1242	Sightsinging & Ear Training 2 Year	1.0
Principal Instrument Applied Lessons*		2.0

Select 2 credits from the following course options: MVB, MVJ*, MVK*, MVO*, MVP*, MVS*, MVV*, MVW*

Semester 3		Units: 10.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
PSC 1515	Energy in the Natural Environment	3.0

PSY 2012	Introduction to Psychology	3.0
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Semester 4

Units: 9.0

MUH 2111	Survey of Music History 1	3.0
MUT 2116	Theory 3	3.0
MUT 2246	Sightsinging and Ear Training 2 Yr.	1.0
Ensemble ¹⁾		1.0
Principal Instrument Applied Lessons ²⁾		1.0

¹⁾Select 1 credit from the following course prefix: MUN*

²⁾Select 1 credits from the following course options: MVB*, MVJ*, MVK*, MVO*, MVP*, MVS*, MVV*, MVW*

Semester 5

Units: 14.0

BSC 1005	General Education Biology	3.0
MUL 1010	Music Appreciation	3.0
MUT 2117	Theory 4	3.0
MUT 2247	Sight Singing and Ear Training 2	1.0
SPC 1017	Introduction to Communication	3.0
Principal Instrument Applied Lessons*		1.0

Select 1 credits from the following course options: MVB, MVJ*, MVK*, MVO*, MVP*, MVS*, MVV*, MVW*

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MVK1111 Class Piano 1 1.000 credits

The secondary area of piano with emphasis on sight-reading, melody harmonization, and ensemble playing. Required of all music students except piano majors. May be repeated for credit.

Lab Hours: 32.000

MUT1111 Theory 1 3.000 credits

The techniques of writing four-part chord progressions using root position and inversions of the primary and secondary triads and the dominant and supertonic seventh; also, non-harmonic tones, melodic writing, and an introduction modulation.

Lecture Hours: 48.000

Prerequisite: MUT1001 or passing score on departmental placement exam., Corequisite: MUT1241

Course User Fee: \$15.00

MUT1241 Sightsinging & Ear Training 1 Year 1.000 credits

The development of aural skill by means of rhythmic and melodic dictation and sightsinging.

Lab Hours: 32.000

Corequisite: MUT1111, MUT1112

Course User Fee: \$10.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MUT1112 Theory 2 3.000 credits

The techniques of writing four-part chord progressions using root position and inversions of the primary and secondary triads and the dominant and supertonic seventh; also, non-harmonic tones, melodic writing, and an introduction modulation.

Lecture Hours: 48.000

Prerequisite: MUT1111, Corequisite: MUT1242

Course User Fee: \$15.00

MUT1242 Sightsinging & Ear Training 2 Year 1.000 - 2.000 credits

The development of aural skills by means of rhythmic and melodic dictation and sightsinging.

Lab Hours: 16.000 - 32.000

Prerequisite: MUT1241, Corequisite: MUT1112

Course User Fee: \$10.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MUH2111 Survey of Music History 1 3.000 credits

This course introduces the history of musical styles from antiquity through the Baroque period and includes works from the Western canon. Students will examine the development of music in relation to cultural, historical, and philosophical contexts. Through analysis of musical works and their significance, students will develop critical thinking skills and gain understanding of how music reflects and influences people across time periods.

Lecture Hours: 48.000

Course User Fee: \$15.00

MUT2116 Theory 3 3.000 credits

The continuation of modulation and the presentation of diatonic sevenths, secondary dominants, altered chords, augmented and Neapolitan sixths; melodic and harmonic analysis of selected works; ninth, eleventh, and thirteenth chords, and instrumental part writing.

Lecture Hours: 48.000

Prerequisite: MUT1112, Corequisite: MUT2246

Course User Fee: \$15.00

MUT2246 Sightsinging and Ear Training 2 Yr. 1.000 - 2.000 credits

Develops aural and visual skills by means of rhythmic, melodic and harmonic dictation and sightsinging. Emphasis is on chromatic materials.

Lab Hours: 32.000 - 64.000

Prerequisite: MUT1242, Corequisite: MUT2117

Course User Fee: \$10.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

MUT2117 Theory 4 3.000 credits

The continuation of modulation and the presentation of diatonic sevenths, secondary dominants, altered chords, augmented and Neapolitan sixths; melodic and harmonic analysis of selected works; ninth, eleventh, and thirteenth chords, and instrumental part writing.

Lecture Hours: 48.000

Prerequisite: MUT2116, Corequisite: MUT2247

Course User Fee: \$15.00

MUT2247 Sight Singing and Ear Training 2 1.000 - 2.000 credits

Develops aural and visual skills by means of rhythmic, melodic and harmonic dictation and sight singing. Emphasis is on chromatic materials.

Lab Hours: 32.000 - 64.000

Prerequisite: MUT2246, Corequisite: MUT2117

Course User Fee: \$10.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

Program Description

Philosophy

Associate in Arts | Code: 11502 | 60 credits

Philosophy investigates the fundamental principles of being, knowledge or conduct. There are numerous systems of philosophical discourse and the two-year program introduces students to many of these. Unless a student wishes to earn a Doctorate and teach at the college level, a Bachelor's degree in philosophy is generally useful

only in indirect ways. It can prepare students for graduate work in other fields such as law or theology, and the study of philosophy usually sharpens a student's analytic skills.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
LIS 2004	Digital Research Strategies	1.0
MGF 1130	Mathematical Thinking	3.0
PSY 2012	Introduction to Psychology	3.0
REL 2300	Survey of World Religions	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
MGF 1131	Mathematics in Context	3.0
PHI 2010	Introduction to Philosophy	3.0
Semester 3		Units: 9.0
BSC 1005	General Education Biology	3.0
ESC 1000	General Education Earth Science	3.0
SPC 1017	Introduction to Communication	3.0
Semester 4		Units: 12.0
HUM 1020	Humanities	3.0
PHI 2600	Introduction to Ethics	3.0
PHI 2680	Artificial Intelligence and Ethics	3.0
SYG 2000	Introduction to Sociology	3.0
Semester 5		Units: 12.0
PHI 1100	Introduction to Logic	3.0
POT 2014	European Political Theory 1	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
WOH 2012	History of World Civilization to 1789	3.0
		Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

LIS2004 Digital Research Strategies 1.000 credits

LIS 2004, Digital Research Strategies, equips students with information literacy skills to locate, evaluate, and ethically use information in an evolving digital landscape. Students will apply critical thinking and research techniques to explore traditional and emerging research tools. (1 hr. lecture)

Lecture Hours: 16.000

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

REL2300 Survey of World Religions 3.000 credits

A survey of the origins, beliefs and contemporary practices of the world's religions: Hinduism, Islam, Taoism, Zen Buddhism, Judaism, Christianity and Confucianism. Attention is given to the interactions between specific religions and the cultures in which they are practiced.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

PHI1100 Introduction to Logic 3.000 credits

This course serves as a solid foundation for the study of philosophy, focusing on both the fundamental principles of valid reasoning and the exploration of the Western Canon. Students will delve into the rich tradition of philosophical thought, honing their analytical skills through the application of various techniques.

Lecture Hours: 48.000

POT2014 European Political Theory 1 3.000 credits

This course covers the more important trends in European political thought from Plato to the present. It examines those ideas which have contributed to the shaping of the political cultures of Western and Eastern Europe. It discusses the historical evolution of key concepts of politics such as freedom, order, political obligations, justice, consent, rights and duties, power and authority. A prior course in Government, History or Philosophy is desirable. Given in English. Level 1. Offered through Overseas Study Program.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

WOH2012 History of World Civilization to 1789 3.000 credits

This course covers the history of World Civilizations from the prehistoric period to the 18th century. Students will learn of the major political, social, economic, and cultural structures of civilizations and their development through 1789.

Lecture Hours: 48.000

Program Description

Physics

Associate in Arts | Code: 11900 | 60 credits

Physics is the study of the motion and force of energy and matter. This science is applied to different kinds of energy and matter, as in thermodynamics, astrophysics, nuclear physics and wave motion analysis. The A.A. coursework provides a fundamental education in mathematics and science topics so that students may transfer to pursue their area of interest. Careers in research are available both in government agencies and private industries, as well as in educational institutions, though in most cases graduate degrees are required.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2

Units: 12.0

ENC 1102	English Composition 2	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0

Semester 3

Units: 12.0

CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
PSY 2012	Introduction to Psychology	3.0

Semester 4

Units: 12.0

MAC 2312	Calculus and Analytical Geometry 2	4.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Semester 5

Units: 12.0

MAC 2313	Calculus and Analytic Geometry 3	4.0
PHI 2600	Introduction to Ethics	3.0
PHY 2049	Physics with Calculus 2	4.0
PHY 2049L	Physics with Calculus 2 Lab	1.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC1114 both with a minimum grade of C

Course User Fee: \$18.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems. Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2049 Physics with Calculus 2 4.000 credits

Foundation course for physical science and engineering majors. PHY 2048 covers classical mechanics and thermodynamics. PHY 2049 includes electricity, magnetism, waves and optics.

Lecture Hours: 64.000

Prerequisite: PHY2048, PHY2048L, Corequisite: MAC2312, PHY2049L

Course User Fee: \$10.00

PHY2049L Physics with Calculus 2 Lab 1.000 credits

Laboratory for PHY 2049.

Lab Hours: 32.000

Prerequisite: PHY2048, PHY2048L, Corequisite: MAC2312, PHY2049

Course User Fee: \$20.00

Program Description

Political Science

Associate in Arts | Code: 12206 | 60 credits

Political science examines the role and effects of government actions on society. The A.A. program prepares students for transfer with coursework in history, literature, economics and government. Political scientists may work in various government jobs, or may work as lobbyists, researchers, political analysts or journalists. In addition to graduate work in the field, a Bachelor's degree in political science also prepares students for law school.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MGF 1131	Mathematics in Context	3.0
POS 2041	American Federal Government	3.0
SPC 1017	Introduction to Communication	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
INR 2002	International Relations	3.0
PHI 2600	Introduction to Ethics	3.0
STA 2023	Statistical Methods	3.0
Semester 3		Units: 12.0
CPO 2100	Comparative European Government	3.0
ISS 2270	Multicultural Communications and Relations	3.0
PHI 2010	Introduction to Philosophy	3.0
SYG 2000	Introduction to Sociology	3.0
Semester 4		Units: 12.0
CPO 2408	Comparative Middle Eastern and North African Governments	3.0
EVR 1001	Introduction to Environmental Science	3.0
POS 2112	State and Local Government In America	3.0
POT 2014	European Political Theory 1	3.0
Semester 5		Units: 12.0

INR 2440	International Law and Organization	3.0
PSC 1515	Energy in the Natural Environment	3.0
<i>Pathway Elective</i>		6.0
CPO 2441	Islam in the Modern World	3.0
INS 2200	Introduction to Conflict Resolution and Peace Studies	3.0
INS 2300	Global Environmental Politics	3.0
INS 2310	Religion and Global Politics	3.0
POS 2293	Islam and America	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Total: 61.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

INR2002 International Relations 3.000 credits

The nature of international relations, the causes of leading international problems, foreign policies of world powers, international political organizations, and the origins of war and peace in the international arena.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CPO2100 Comparative European Government 3.000 credits

This course discusses the structures and functioning of the systems of government in Western Europe. Students will analyze major policy issues and approaches to resolving them, and cultural differences and interdependence among the Western European countries in policy determination.

Recommended prerequisite: a prior course in History or Social Science. Offered first semester. Given in English. Offered through overseas study program.

Lecture Hours: 48.000

ISS2270 Multicultural Communications and Relations 3.000 credits

This course uses an interdisciplinary approach to examine the complex interactions among ethnicity, race, gender, age, and class, as well as other ways in which we differ as they pertain to shaping personal awareness, understanding, and skills that will allow them to interact more effectively with diverse populations, age, groups, and lifestyles and to think through and value human diversity. This course has an overriding principle based on the concept of human rights. (3 hr. lecture)

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

CPO2408 Comparative Middle Eastern and North African Governments 3.000 credits

The Middle East & North Africa (MENA) has uninterruptedly been at the forefront of international geopolitics for almost a century. This enduring geopolitical visibility has important cultural, societal, economic, and security implications at national, regional, and international levels. This course is designed to tackle some of these issues by providing students with a historico-cultural, politico-economic, and geo-strategic survey of the Middle East.

Lecture Hours: 48.000

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

POS2112 State and Local Government In America 3.000 credits

The typical state and local government organization, together with political practices in America, with special emphasis on governmental organizations and the major contemporary political problems of the State of Florida and of Florida communities.

Lecture Hours: 48.000

POT2014 European Political Theory 1 3.000 credits

This course covers the more important trends in European political thought from Plato to the present. It examines those ideas which have contributed to the shaping of the political cultures of Western and Eastern Europe. It discusses the historical evolution of key concepts of politics such as freedom, order, political obligations, justice, consent, rights and duties, power and authority. A prior course in Government, History or Philosophy is desirable. Given in English. Level 1. Offered through Overseas Study Program.

Lecture Hours: 48.000

INR2440 International Law and Organization 3.000 credits

International law and problems in world politics; a review of man's attempt to control international politics through international law and organizations, including the League of Nations, the United Nations, NATO, and the European Union. A prior course in History or Social Science is desirable. Offered second semester. Given in English. Offered through Overseas Study Program.

Lecture Hours: 48.000

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CPO2441 Islam in the Modern World 3.000 credits

The survey course is designed to grant students a broad-based exposure across a variety of conceptual debates, historical events, and policy issues at the intersection of Islam and the modern world and the paths forward. The issues especially religiously-inspired and informed violence/conflict and debates surrounding them have assumed prominence in the 21st century with the resurgence of religion in politics and the subsequent struggle for Islam. These issues will be contextualized in normative, empirical and historical frameworks to equip students with a sophisticated and multi-dimensional conceptual toolbox to analyze them.

Lecture Hours: 48.000

INS2200 Introduction to Conflict Resolution and Peace Studies 3.000 credits

Introduction to Conflict Resolution and Peace Studies is an interdisciplinary course exploring foundational concepts, methodologies, and practical approaches to interstate and intra-state conflicts and how to tackle them. The course has no specific prerequisites and will appeal to a broad audience as it analyzes a broad spectrum of perspectives in the field of conflict and peace studies. By contextualizing the challenge and transmitting the knowledge of peace in normative, empirical and historical frameworks, the course will equip students with a multi-dimensional conceptual toolbox to analyze it while inspiring them to create a more equitable and harmonious world.

INS2300 Global Environmental Politics 3.000 credits

Global Environmental Politics is an interdisciplinary course exploring the evolving relationship between humanity and the environment from diverse conceptual, normative, and empirical perspectives. The course has no specific prerequisites and will appeal to a broad audience as it analyzes a broad spectrum of perspectives in the field. The course will explore the most pressing contemporary environmental challenges, including climate change, loss of biodiversity, desertification, trade in dangerous waste, and over-exploitation of natural resources as well as various political and policy responses to deal with them.

INS2310 Religion and Global Politics 3.000 credits

The interdisciplinary course introduces students to the increasingly visible role of religious identity in international and global politics. It offers a macro-level overview of the intersection of religion and politics in a historical context and sheds light on the intertwined histories and trajectories connecting them. It explores how this nexus has been shaped by belief and value systems as well as the historical socio-economic and political circumstances and conditions. Among the themes addressed in the course are religious diversity and human rights, interfaith dialogue and conflict resolution, and religion in the age of globalization.

Lecture Hours: 48.000

POS2293 Islam and America 3.000 credits

Islam and the United States offers a survey treatment of the historically charged relationship between America and the Muslim World and the path forward. The course analyzes the intersection of religion, identity politics, geopolitics and economics of Islam-US relations in a historical context. These issue areas will be contextualized in normative, empirical and historical frameworks to equip the student with a sophisticated and multi-dimensional conceptual toolbox to analyze them.

Lecture Hours: 48.000

Program Description

Pre-Bachelor of Arts

Associate in Arts | Code: 14902 | 60 credits

The Pre-Bachelor of Arts program at MDC is designed for students seeking a broad-based degree program with greater freedom to explore intellectual fields that align with their interests. This program challenges students to take primary responsibility for directing their own education. It also provides a wider range of educational opportunities than more specialized

programs. At the upper-division level, students are typically required to complete a major theme or area of concentration.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
GEB 1000	Business Career Strategies	3.0
<i>Oral Communication</i>		3.0
ENC 2300	Advanced Composition and Communication	3.0
OR		
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 3		Units: 9.0
PHI 2600	Introduction to Ethics	3.0
<i>Natural Science</i>		3.0
AST 1002	Descriptive Astronomy	3.0
OR		
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		

ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
<i>Social Science</i>		3.0
ANT 2000	Introduction to Anthropology	3.0
OR		
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
OR		
PSY 1012	Introduction to Psychology	3.0
OR		
SYG 2000	Introduction to Sociology	3.0
OR		
WOH 1012	History of World Civilization to 1789	3.0
OR		
WOH 2022	History of World Civilization From 1789	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4		Units: 12.0
Natural Science		3.0
HUN 1201	Essentials of Human Nutrition	3.0
OR		
OCB 1010	Introduction to Marine Biology	3.0
OR		
PSC 1121	General Education Physical Science	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
Pathway Elective ¹⁾		9.0

¹⁾Select 9 credits from the following course options: ACG*, AFH*, AMH*, ARC*, ARH*, BUL*, CPO*, DAN*, ECO*, EUH*, GEB*, HUM*, IND*, INR*, LAH*, LIT*, MAN*, MAR*, MUH*, OST*, PHI*, POS*, POT*, PUR*, REL*, SBM*, TAX*, THE*, WOH*

Semester 5	Units: 14.0
Pathway Elective	14.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

GEB1000 Business Career Strategies 3.000 credits

This course is designed to enable students to thrive in a competitive business environment. Students will learn business etiquette, alternative career pathways, personal financial management, and budgets. Additionally, students will learn interviewing, networking, and career development skills.

Lecture Hours: 48.000

Course User Fee: \$24.00

ENC2300 Advanced Composition and Communication 3.000 credits

This writing-based course addresses techniques of critical thinking, persuasion, and argumentation. Students will refine their composition skills and develop their oral communication skills by examining and discussing a range of issues. Prerequisites: ENC1101, 1102 or equivalent with a grade of "C" or better. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101 AND ENC1102

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

WOH2012 History of World Civilization to 1789 3.000 credits

This course covers the history of World Civilizations from the prehistoric period to the 18th century. Students will learn of the major political, social, economic, and cultural structures of civilizations and their development through 1789.

Lecture Hours: 48.000

WOH2022 History of World Civilization From 1789 3.000 credits

The student will examine the historical development of world civilizations since 1789. Students will learn of historical processes and developments in social, cultural, political, and economic contexts since the 18th century.

Lecture Hours: 48.000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

OCB1010 Introduction to Marine Biology 3.000 credits

An introduction to the biology of the seas. Emphasis is placed on the variety of marine organisms and their structural, physiological, and behavioral adaptations within specific marine environments. Special attention is directed to marine communities, e.g., coral reefs and shallow grass flats, and the factors limiting the distribution of organisms within those communities. Discussions will also be directed towards geological, chemical and physical characteristics of the world's oceans. (3 hr. lecture)

Lecture Hours: 48.000

PSC1121 General Education Physical Science 3.000 credits

A study of the major concepts and principles from each of the following areas: physics, chemistry, and astronomy.

Lecture Hours: 48.000

Pre/Corequisite: MGF 1130 or MGF 1131 or MAC 1105 or MAC 1114 or MAC 1140 or MAC 2223 or MAT 1033

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

Program Description

Pre-Law

Associate in Arts | Code: 11400 | 60 credits

Although no specific area of study is mandatory for the Pre-Law major, the MDC program offers courses in criminal justice, government, history and business to best prepare a student for future coursework. Students should work with an advisor to determine the best four-year degree to pursue.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
BUL 2241	Business Law 1	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
<i>First Year Experience</i>		3.0
SLS 1401	Psychology of Career Adjustment	1.0
OR		
SLS 1510	Preparing for Student Success	3.0

Semester 2

Units: 12.0

ENC 1102	English Composition 2	3.0
<i>Mathe</i>		3.0
MAC 1105	College Algebra	3.0
OR		
MGF 1130	Mathematical Thinking	3.0
OR		
STA 2023	Statistical Methods	3.0
BUL 2241	Business Law 1	3.0
PLA 2890	Human Rights Law & Procedure	3.0

Semester 3

Units: 12.0

CJL 1100	Criminal Law	3.0
<i>Natural Science</i>		3.0

BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
WOH 2012	History of World Civilization to 1789	3.0
OR		
WOH 2022	History of World Civilization From 1789	3.0

Semester 4

Units: 12.0

INR 2002	International Relations	3.0
PLA 2704	Professionalism & Civility in the Law	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
<i>Mathematics</i>		3.0
QMB 2100	Basic Business Statistics	3.0
OR		
STA 2023	Statistical Methods	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 5

Units: 12.0

<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Natural Science</i>		3.0
HUN 1201	Essentials of Human Nutrition	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
CJL 2062	Constitutional Law and Legal Procedure or Evidence	3.0
PLA 2872	Artificial Intelligence (AI) and the Law	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1401 Psychology of Career Adjustment 1.000 - 6.000 credits

For students who have not decided, are having difficulty deciding, or need clarification in making a career choice. A format for a systematic investigation for career and life planning is included. It is concerned with "who you are," "where are you going," "how to get there," and "what's out there that fits you."

Lecture Hours: 16.000 - 96.000

Course User Fee: \$15.00

SLS1510 Preparing for Student Success 3.000 credits

This course provides an orientation to college life and helps develop academic, career, and personal goals. Students will learn college success strategies, goal-setting, learning style assessments, as well as general and discipline-specific study skills in the context of various theoretical, practical, and experiential perspectives.

Lecture Hours: 48.000

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

PLA2890 Human Rights Law & Procedure 3.000 credits

This course will address theoretical and practical questions the evolution and development of the laws addressing international human rights issues.

Emphasis will be placed on the core issues, debates, struggles, failures, and successes of legal claims. (3 hr. lecture)

Lecture Hours: 48.000

CJL1100 Criminal Law 3.000 credits

Historical background and foundations of American criminal law, including United States Constitution al requirements, Federal and State court organization and jurisdiction, criminal law basics, Florida statutes, rules of evidence and procedure.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

WOH2012 History of World Civilization to 1789 3.000 credits

This course covers the history of World Civilizations from the prehistoric period to the 18th century.

Students will learn of the major political, social, economic, and cultural structures of civilizations and their development through 1789.

Lecture Hours: 48.000

WOH2022 History of World Civilization From 1789 3.000 credits

The student will examine the historical development of world civilizations since 1789. Students will learn of historical processes and developments in social, cultural, political, and economic contexts since the 18th century.

Lecture Hours: 48.000

INR2002 International Relations 3.000 credits

The nature of international relations, the causes of leading international problems, foreign policies of world powers, international political organizations, and the origins of war and peace in the international arena.

Lecture Hours: 48.000

PLA2704 Professionalism & Civility in the Law 3.000 credits

This class will provide students guidance as they interact in a real-world setting as legal professionals while strengthening their communication skills. This course will cultivate the students' leadership capacities at the intra-personal, interpersonal, and industry levels. The coursework will proceed through experiential learning that is grounded in theory and includes reflection. The study material will develop students' knowledge in the context of the aspiration, relational, and practical aspects of professionalism and civility in the legal field (3 credits)

Lecture Hours: 48.000

Course User Fee: \$45.84

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CJL2062 Constitutional Law and Legal Procedure or Evidence 3.000 credits

An examination of the United States and Florida Constitutions, with emphasis on leading cases dealing with arrest, search and seizure, confessions and the rules of evidence.

Lecture Hours: 48.000

PLA2872 Artificial Intelligence (AI) and the Law 3.000 credits

Introduction to the intersection of artificial intelligence and the law, focusing on its challenges, opportunities, ethics, and impact on the law, the legal industry, and society. (3 credits)

Lecture Hours: 48.000

Recommended Preparation: Students are encouraged to enroll in ENC1101 prior to registering for PLA2872.

Program Description

Pre-Medical Science/Dentistry

Associate in Arts | Code: 11211 | 60 credits

This program is designed to meet the first two years of required courses for students planning careers in medicine and dentistry. Pre-medical education should include a foundation in chemistry, biology, mathematics, and physics, as well as a broad education in the humanities and social sciences. This program enables the student to transfer to colleges or universities that offer a Baccalaureate degree in physician assistant (PA), or other pre-medical degrees such as biology. Admission to a professional school is dependent upon academic coursework and scores on a national test. Applicants should have a minimum "B" average.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101

English Composition 1

3.0

MAT 1033	Intermediate Algebra	3.0
MAT 1033L	Intermediate Algebra Recitation Hall	0.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2 **Units: 13.0**

CHM 1025	Introductory Chemistry	3.0
CHM 1025L	Introductory Chemistry Lab	1.0
ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 1105	College Algebra	3.0

Semester 3 **Units: 13.0**

BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>So</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0

Semester 4 **Units: 13.0**

BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Semester 5 **Units: 11.0**

CHM 2210	Organic Chemistry 1	3.0
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CHM 2210L	Organic Chemistry 1 Laboratory	2.0
PHI 2600	Introduction to Ethics	3.0
STA 2023	Statistical Methods	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAT1033L Intermediate Algebra Recitation Hall 0.000 credits

The student will receive individualized, small group or whole group instruction to deepen their conceptual understanding of mathematics. The student will reinforce and apply content knowledge with effective problem-solving techniques and non-cognitive activities to make mathematics meaningful and relevant to their fields of study while strengthening the concepts needed to achieve the objectives of MAT1033. This lab session is associated and funded by the First in the World Grant, CCAT Math Project.

Lab Hours: 1.000

Corequisite: MAT1033

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1025

Course User Fee: \$30.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan. Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

Program Description

Pre-Medical Technology

Associate in Arts | Code: 11209 | 60 credits

This program provides the science coursework necessary to transfer to a four-year baccalaureate program. Students must transfer to an upper-division institution for the third year. Generally, the fourth year is spent in a clinical setting, usually in a hospital where students learn laboratory techniques. Members of this profession work in clinical laboratories performing the wide variety of tests which aid physicians in the diagnosis and treatment of patients. Most medical technologists work in hospitals, physician's public health laboratories, universities, or in industry.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
BSC 1084	Functional Human Anatomy	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test (CCT)*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2	Units: 15.0
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CHM 1025	Introductory Chemistry	3.0
CHM 1025L	Introductory Chemistry Lab	1.0
ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0

Semester 3	Units: 13.0
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CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
<i>Social Science</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0

Semester 4	Units: 11.0
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CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
PHY 1025	Basic Physics	3.0

Semester 5	Units: 8.0
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PHI 2600	Introduction to Ethics	3.0
PHY 2048	Physics with Calculus 1	4.0
PHY 2048L	Physics with Calculus 1 Lab	1.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BSC1084 Functional Human Anatomy 3.000 credits

This course covers basic human anatomy for students in allied health and mortuary science programs. It includes the dynamics of gross and functional anatomy, terminology, body orientation, and systematic relationships.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1025

Course User Fee: \$30.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics.

Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

**MAC2311 Calculus and Analytical Geometry 1
5.000 credits**

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan. Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves.

Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

Program Description

Associate in Arts | Code: 11203 | 60 credits

The A.A. degree does not prepare students to be eligible to take certification/licensure exams or to practice in the Nursing profession. This program includes the pre-professional courses necessary for admission to a Bachelor of Science degree program in nursing (BSN). The first two years at the community college level consist of general education and science courses. The professional nursing courses are taken in the last two years at the upper division. Upper-division programs are limited access, require an above average academic record, and have widely differing pre-professional course requirements. Therefore, students are advised to check with the Nursing Department of the senior institution they wish to attend. Most upper-division programs also offer a track for registered nurses (RNs) completing an Associate in Science degree to earn a BSN degree.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2		Units: 16.0
ENC 1102	English Composition 2	3.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0
PHI 2010	Introduction to Philosophy	3.0
SPC 2608	Introduction to Public Speaking	3.0
Pathway Elective*		3.0

*Note: Must take 3 credits. Check with advisor for any transferable Credit Type 1 or Credit Type 2 course.

Semester 3		Units: 13.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
DEP 2000	Human Growth and Development	3.0
PHI 2600	Introduction to Ethics	3.0
Pathway Elective*		3.0
Computer Competency Test ¹		

*Note: Must take 3 credits. Check with advisor for any transferable Credit Type 1 or Credit Type 2 course.

¹Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4		Units: 17.0
HUN 1201	Essentials of Human Nutrition	3.0

MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0
PSY 2012	Introduction to Psychology	3.0
STA 2023	Statistical Methods	3.0
Pathway Elective *		3.0

*Note: Must take 3 credits. Check with advisor for any transferable Credit Type 1 or Credit Type 2 course.

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

**BSC2085 Human Anatomy and Physiology 1
3.000 credits**

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

**BSC2085L Human Anatomy and Physiology 1
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

**AMH2020 History of the US since 1877 3.000
credits**

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Lab 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions.

Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite:

MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan. Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.
Lecture Hours: 48.000
Recommended Preparation: Completion of MGF1131 prior to registration.
Course User Fee: \$18.00

Program Description

Pre-Occupational Therapy

Associate in Arts | Code: 11204 | 60 credits

This program provides the fundamental science coursework necessary to transfer to a four-year institution, where students can obtain a degree in an appropriate field, such as biology. To be an optometrist, one must earn the Doctor of Optometry professional degree. A Bachelor's degree with a strong science background is required for admission. Graduates must pass a state licensure exam in order to practice. Optometrists prescribe glasses, contact lenses and visual therapy, and offer non-surgical treatment of eye diseases and the rehabilitation of patients with visual disabilities.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101	English Composition 1	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
MAT 1033	Intermediate Algebra	3.0
MAT 1033L	Intermediate Algebra Recitation Hall	0.0

SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
CHM 1025	Introductory Chemistry	3.0
CHM 1025L	Introductory Chemistry Lab	1.0
ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 1105	College Algebra	3.0
Semester 3		Units: 13.0
BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Social Science</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
Semester 4		Units: 13.0
BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
Semester 5		Units: 11.0
CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0
PHI 2010	Introduction to Philosophy	3.0
STA 2023	Statistical Methods	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAT1033L Intermediate Algebra Recitation Hall 0.000 credits

The student will receive individualized, small group or whole group instruction to deepen their conceptual understanding of mathematics. The student will reinforce and apply content knowledge with effective problem-solving techniques and non-cognitive activities to make mathematics meaningful and relevant to their fields of study while strengthening the concepts needed to achieve the objectives of MAT1033. This lab session is associated and funded by the First in the World Grant, CCAT Math Project.

Lab Hours: 1.000

Corequisite: MAT1033

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1025

Course User Fee: \$30.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

Program Description

Pre-Optometry

Associate in Arts | Code: 11205 | 60 credits

This program provides the fundamental science coursework necessary to transfer to a four-year institution, where students can obtain a degree in an appropriate field, such as biology. To be an optometrist, one must earn the Doctor of Optometry professional degree. A Bachelor's degree with a strong science background is required for admission. Graduates must pass a state licensure exam in order to practice. Optometrists prescribe glasses, contact lenses and visual therapy, and offer non-surgical treatment of eye diseases and the rehabilitation of patients with visual disabilities.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101	English Composition 1	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
MAT 1033	Intermediate Algebra	3.0
MAT 1033L	Intermediate Algebra Recitation Hall	0.0

SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
CHM 1025	Introductory Chemistry	3.0
CHM 1025L	Introductory Chemistry Lab	1.0
ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 1105	College Algebra	3.0
Semester 3		Units: 13.0
BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Social Science</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		
ECO 2013	Principles of Economics (Macro)	3.0
Semester 4		Units: 13.0
BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
Semester 5		Units: 11.0
CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0
PHI 2010	Introduction to Philosophy	3.0
STA 2023	Statistical Methods	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAT1033L Intermediate Algebra Recitation Hall 0.000 credits

The student will receive individualized, small group or whole group instruction to deepen their conceptual understanding of mathematics. The student will reinforce and apply content knowledge with effective problem-solving techniques and non-cognitive activities to make mathematics meaningful and relevant to their fields of study while strengthening the concepts needed to achieve the objectives of MAT1033. This lab session is associated and funded by the First in the World Grant, CCAT Math Project.

Lab Hours: 1.000

Corequisite: MAT1033

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1025

Course User Fee: \$30.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

Program Description

Pre-Pharmacy

Associate in Arts | Code: 11206 | 60 credits

The Pre-Pharmacy program provides the math and science education needed to transfer to a baccalaureate program. Career opportunities in pharmacy include positions in a hospital or institutional pharmacy, in industry or manufacturing, in a retail or clinical pharmacy, in government service, in pharmacy administration, in laboratories and in pharmaceutical journals.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 14.0
CHM 1025	Introductory Chemistry	3.0
ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
Semester 3		Units: 12.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 4		Units: 13.0
BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Oral Communication</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
Semester 5		Units: 11.0
CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0
PHI 2600	Introduction to Ethics	3.0
STA 2023	Statistical Methods	3.0
		Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy.

Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics.

Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

Program Description

Pre-Physical Therapy

Associate in Arts | Code: 11207 | 60 credits

The A.A. degree does not prepare students to be eligible to take certification/licensure exams or to practice in the Physical Therapy profession. This program prepares students for transfer by providing intensive coursework in mathematics and science. Most upper-division programs have selective admissions and transfer requirements vary, so students should work with an advisor in planning a program of study. Physical therapists help rehabilitate individuals who have been disabled by injury or disease. They usually work in healthcare settings such as hospitals or nursing homes.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2	Units: 13.0
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BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
CHM 1025	Introductory Chemistry	3.0
ENC 1102	English Composition 2	3.0
STA 2023	Statistical Methods	3.0

Semester 3	Units: 12.0
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BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
DEP 2000	Human Growth and Development	3.0

Semester 4	Units: 14.0
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CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 1114	Trigonometry	3.0
PSY 2012	Introduction to Psychology	3.0

Semester 5	Units: 11.0
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BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
<i>Social Science</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0
PHI 2600	Introduction to Ethics	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.
Lecture Hours: 48.000

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.
Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence. Lecture Hours: 48.000

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee. Lecture Hours: 48.000
Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee. Lab Hours: 64.000
Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010
Course User Fee: \$40.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)
Lecture Hours: 48.000
Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.
Lecture Hours: 48.000
Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."
Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.
Lecture Hours: 48.000

Program Description

Pre-Veterinary Medicine

Associate in Arts | Code: 11208 | 60 credits

Veterinary medicine is the study of the diagnosis, treatment and control of disease and injuries among animals. Veterinarians may specialize in the health and breeding of certain animals, performing surgery, prescribing and administering drugs and vaccines, and research. Veterinarians may also concentrate on the inspection of meat, poultry and other foods as part of federal and state public health programs. The University of Florida is the only state school that offers a veterinary program.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
CHM 1025	Introductory Chemistry	3.0
CHM 1025L	Introductory Chemistry Lab	1.0
ENC 1102	English Composition 2	3.0
MAC 1114	Trigonometry	3.0
PHI 2010	Introduction to Philosophy	3.0

Semester 3		Units: 13.0
BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
STA 2023	Statistical Methods	3.0

Semester 4		Units: 13.0
BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry & Qualitative Analysis Lab	2.0
<i>Oral Communications</i>		3.0
SPC 1017	Introduction to Communication	3.0
OR		
SPC 2608	Introduction to Public Speaking	3.0

Semester 5		Units: 11.0
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CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0
ECO 2013	Principles of Economics (Macro)	3.0
PHI 2600	Introduction to Ethics	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1025

Course User Fee: \$30.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry & Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students.

Lab Hours: 64.000

Prerequisite: CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting.

Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

Program Description

Psychology

Associate in Arts | Code: 12001 | 60 credits

Psychology is the science of human behavior and mental processes that affect mental and physical health. A.A. coursework covers the science and mathematics subjects needed to transfer to a four-year institution. While the Bachelor's degree in psychology could be useful in a number of careers, professional psychologists must continue to graduate study. Employment opportunities with a Doctorate or a Masters' degree include teaching or counseling in a wide variety of settings.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MGF 1131	Mathematics in Context	3.0
PSY 2012	Introduction to Psychology	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2

Units: 12.0

ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
ISS 1301	Intro. to Social Research	3.0
STA 2023	Statistical Methods	3.0

Semester 3

Units: 12.0

DEP 2000	Human Growth and Development	3.0
PHI 2600	Introduction to Ethics	3.0
SPC 2608	Introduction to Public Speaking	3.0
<i>Pathway Elective</i>		3.0
SOP 2002	Social Psychology	3.0
OR		
SOP 2772	Human Sexuality	3.0

Computer Competency Test*

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4

Units: 12.0

BSC 1005	General Education Biology	3.0
CLP 2140	Psychopathology	3.0
INP 2390	Psychology of Work	3.0
<i>Pathway Elective</i>		3.0
PSY 2050	Introduction to Forensic Psychology	3.0
OR		
PSY 2800	Psychology of Genocide	3.0

Semester 5

Units: 6.0

PSC 1515	Energy in the Natural Environment	3.0
<i>Pathway</i>		3.0
CLP 2000	Dynamics of Behavior	3.0

OR		
CLP 2470	Introduction to Childhood Psychopathology	3.0
OR		
DEP 2100	Child Growth And Development	3.0
OR		
DEP 2402	Psychology of Adulthood and Aging	3.0
OR		
ISS 2270	Multicultural Communications and Relations	3.0
OR		
PSY 2050	Introduction to Forensic Psychology	3.0
OR		
PSY 2800	Psychology of Genocide	3.0
OR		
PSB 2041	Behavioral Neuroscience	3.0
OR		
PSB 2442	Psychology of Addiction	3.0

Total: 55.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

ISS1301 Intro. to Social Research 3.000 credits

This course is a general introduction to research methodology in the Social Sciences, paying particular attention to research design, data collection and data analysis. This is a course in Applied Social Sciences that will provide students with a survey understanding of social scientific research. This course will include examination of empirical research, including literature reviews, theory, methodology, data collection, data analysis and presentation of results. (3 hr. lecture)

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan. Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

SOP2002 Social Psychology 3.000 credits

Combines a knowledge of psychology and sociology, in an interdisciplinary approach to the study of human interaction. Main themes deal with the nature of attitudes, how attitudes may be changed, the processes of interaction and the nature of group structures.

Lecture Hours: 48.000

SOP2772 Human Sexuality 3.000 credits

This course examines the biological, theoretical, social, psychological, and cultural aspects of human sexuality. Students will learn about sexual anatomy, sexual response cycle, sex and gender development, sexual attraction and relationships, reproductive health issues, and social issues in sexuality.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CLP2140 Psychopathology 3.000 credits

This course examines the major categories of mental disorders. Diagnostic criteria, treatment methods, cultural factors, public attitudes, community resources, ethical issues and legislation applicable to individuals with mental disorders are studied. The impact of mental disorders on individuals, families and society are discussed.

Lecture Hours: 48.000

Prerequisite: PSY2012

INP2390 Psychology of Work 3.000 credits

Applies the understanding of effective human relations to work situations. Personal dynamics for success are also considered. Students will be taught how to influence behavior on the job as they apply their knowledge and interpersonal skills to specific experiences in the workplace.

Lecture Hours: 48.000

PSY2050 Introduction to Forensic Psychology 3.000 credits

This course explores the interaction between psychology and the legal system. Students will learn the foundations, history, and terminology of forensic psychology and the influence of media and cultural issues. Students will also learn about the forensic psychologists role in criminal and civil proceedings, public policy, law enforcement, and victimization.

Lecture Hours: 48.000

Prerequisite: PSY2012

PSY2800 Psychology of Genocide 3.000 credits

Students will learn the psychological, social, and cultural roots of genocide, human cruelty, and mass violence. Students will examine the various factors influencing such acts, and the emotional and psychological impact upon victims, perpetrators, rescuers, and society.

Lecture Hours: 48.000

Prerequisite: ENC1101

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CLP2000 Dynamics of Behavior 3.000 credits

Analysis of mechanisms of adjustment, motivation, frustration and conflict, learning personality and psychotherapy. Emphasis is on the psychological processes of the normal individual functioning in society rather than on behavior disorders.

Lecture Hours: 48.000

CLP2470 Introduction to Childhood Psychopathology 3.000 credits

This course will satisfy a requirement for the Specialized Registered Behavior Technician Associate in Science program. This course is an introduction to the various forms of abnormal behavior in infancy, childhood, and adolescence within the context of traditional and contemporary psychological theory. Diagnostic criteria, treatment methods, cultural factors, public attitudes, community resources, ethical issues and legislation applicable to childhood mental disorders are studied. The impacts of mental disorders on individuals, families and society are discussed. Prerequisite: PSY 2012. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: PSY2012

DEP2100 Child Growth And Development 3.000 credits

This course in Child Growth and Development is designed especially for the student interested in the human life span from birth through the first eight years. The course is intended to acquaint the student with basic theoretical models of development and such specific topics as heredity teratogenic agents, learning, intelligence, socialization, personality, sex role identification, language acquisition and moral development.
Lecture Hours: 48.000

DEP2402 Psychology of Adulthood and Aging 3.000 credits

This course will examine adulthood and aging from a multidisciplinary and diversity perspective. This course will explore the biological, physical, psychological, cognitive, and cultural changes during adulthood and aging including contemporary aspects of work-family life, and retirement. Aspects of death and dying will also be discussed from a multidisciplinary perspective. (3 hr. lecture)
Lecture Hours: 48.000

ISS2270 Multicultural Communications and Relations 3.000 credits

This course uses an interdisciplinary approach to examine the complex interactions among ethnicity, race, gender, age, and class, as well as other ways in which we differ as they pertain to shaping personal awareness, understanding, and skills that will allow them to interact more effectively with diverse populations, age, groups, and lifestyles and to think through and value human diversity. This course has an overriding principle based on the concept of human rights. (3 hr. lecture)
Lecture Hours: 48.000

PSY2050 Introduction to Forensic Psychology 3.000 credits

This course explores the interaction between psychology and the legal system. Students will learn the foundations, history, and terminology of forensic psychology and the influence of media and cultural issues. Students will also learn about the forensic psychologists role in criminal and civil proceedings, public policy, law enforcement, and victimization.
Lecture Hours: 48.000
Prerequisite: PSY2012

PSY2800 Psychology of Genocide 3.000 credits

Students will learn the psychological, social, and cultural roots of genocide, human cruelty, and mass violence. Students will examine the various factors influencing such acts, and the emotional and psychological impact upon victims, perpetrators, rescuers, and society.
Lecture Hours: 48.000
Prerequisite: ENC1101

PSB2041 Behavioral Neuroscience 3.000 credits

This course presents an approach to the study of the biology of human behavior and aims to introduce the field of Biopsychology. It is designed to provide each student with comprehensive exposure to the nervous system and how it governs various behaviors. This is an introduction course that explores how our brains develop, how they work, and how they are changed by life experiences. Topics include neural communication, localization of brain function, neural systems, and control of behavior. Upon completion of the course, the student will have a solid foundation regarding the biological basis of behavior upon which to build on in more advanced courses of study.
Lecture Hours: 48.000

PSB2442 Psychology of Addiction 3.000 credits

This course will examine psychological, medical, pharmacological, legal, economic and sociological aspects of addiction to and use of various chemicals. The course will take an in-depth look at narcotic sedatives, and stimulants including alcohol, cocaine, heroin, cannabis, caffeine and tobacco.
Lecture Hours: 48.000

Program Description

Public Administration

Associate in Arts | Code: 12100 | 60 credits

This is an interdisciplinary program gearing the combined study of business, government and economics toward a career in the public sector. Although some students pursue graduate degrees, those with Bachelor's degrees may obtain work managing budgets, or developing programs and policies in government, education and non-profit settings.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
POS 2041	American Federal Government	3.0
SPC 2608	Introduction to Public Speaking	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
Humanities		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MGF 1130	Mathematical Thinking	3.0
POS 2112	State and Local Government In America	3.0
Semester 3		Units: 12.0
BSC 1005	General Education Biology	3.0
ECO 2013	Principles of Economics (Macro)	3.0
INR 2002	International Relations	3.0
PHI 2600	Introduction to Ethics	3.0
Semester 4		Units: 13.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
PSC 1515	Energy in the Natural Environment	3.0
PSY 2012	Introduction to Psychology	3.0
SYG 2000	Introduction to Sociology	3.0
Semester 5		Units: 12.0
MAN 2021	Principles of Management	3.0
MAR 1011	Principles of Marketing	3.0
PUR 2003	Public Relations	3.0
STA 2023	Statistical Methods	3.0
		Total: 63.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

POS2112 State and Local Government In America 3.000 credits

The typical state and local government organization, together with political practices in America, with special emphasis on governmental organizations and the major contemporary political problems of the State of Florida and of Florida communities.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

INR2002 International Relations 3.000 credits

The nature of international relations, the causes of leading international problems, foreign policies of world powers, international political organizations, and the origins of war and peace in the international arena.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

PUR2003 Public Relations 3.000 credits

This course provides students with a broad spectrum of topics related to the Public Relations profession. Current practices or organized programs used in business to earn public acceptance and good will for products, services, personnel, and policies are explored, studied and experienced. The course employs a hands-on approach to applying public relations techniques in hypothetical business situations. Students prepare press releases, brochures, and other collateral materials.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

To prepare for upper-division work in recreation, students take courses in accounting, economics, human anatomy and physiology, and health sciences. This curriculum meets the pre-professional and general education course requirements for transfer, but due to variations in upper-division requirements, students should confer with an advisor. Recreation professionals often work in youth agencies, but may also develop careers in industries such as healthcare, fitness, and travel and tourism.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Program Description

Recreation

Associate in Arts | Code: 12101 | 60 credits

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 12.0
BSC 1084	Functional Human Anatomy	3.0
CHM 1020	General Education Chemistry	3.0
ENC 1102	English Composition 2	3.0
STA 2023	Statistical Methods	3.0

Semester 3		Units: 11.0
DEP 2000	Human Growth and Development	3.0
HLP 1080	Wellness	2.0
HUN 1201	Essentials of Human Nutrition	3.0
Pathway Electives*		3.0

Select 3 credits from the following course options: ACG, FIN*, HFT*, MAN*, MAR*, SPM*

Semester 4		Units: 12.0
ECO 2013	Principles of Economics (Macro)	3.0
HLP 1081	Fitness & Wellness for Life	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
SYG 2000	Introduction to Sociology	3.0

Semester 5		Units: 12.0
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GEB 1011	Principles of Business	3.0
HUS 1001	Introduction to Human Services	3.0
PHI 2600	Introduction to Ethics	3.0
PSY 2012	Introduction to Psychology	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

BSC1084 Functional Human Anatomy 3.000 credits

This course covers basic human anatomy for students in allied health and mortuary science programs. It includes the dynamics of gross and functional anatomy, terminology, body orientation, and systematic relationships.

Lecture Hours: 48.000

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

HLP1080 Wellness 2.000 credits

This course enables students to assess their present aerobic fitness level, lung capacity, percentage of body fat, flexibility and strength. From data collected, the student will be able to set personal wellness goals. Lectures, demonstrations, and multi-media materials will be used to provide the scientific basis for meeting ones personal wellness goals. (2 hr. lecture/lab)

Lecture Hours: 32.000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

HLP1081 Fitness & Wellness for Life 3.000 credits

In this course students will learn the roles of exercise, physical activity, diet, and stress management in achieving optimal wellness. Students will explore current developments in health and complete lab assignments, which will assist in the determination of their current health status. Individualized exercise and dietary protocols based on these assessments will be developed. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUS1001 Introduction to Human Services 3.000 credits

An introduction to an overview of the field of Human Services, including the role of the human services worker as it relates to various agencies, counseling, interviewing and managing.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

Program Description

Social Work

Associate in Arts | Code: 12102 | 60 credits

This program prepares students for upper-division education in social work by offering courses in science and sociology. Social workers provide the link between organized social services and individuals and families unable to provide for themselves or needing assistance in problem solving. Potential employers include hospitals, mental health centers, rehabilitation centers, government agencies, schools and correctional institutions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
POS 2041	American Federal Government	3.0
ENC 1101	English Composition 1	3.0
SPC 2608	Introduction to Public Speaking	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		
MGF 1131	Mathematics in Context	3.0

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
PSY 2012	Introduction to Psychology	3.0
STA 2023	Statistical Methods	3.0

Semester 3	Units: 12.0
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BSC 1005	General Education Biology	3.0
ISS 1301	Intro. to Social Research	3.0
PHI 2600	Introduction to Ethics	3.0
SYG 2000	Introduction to Sociology	3.0

Semester 4

Units: 12.0

DEP 2000	Human Growth and Development	3.0
ECO 2023	Principles of Economics (Micro)	3.0
HUN 1201	Essentials of Human Nutrition	3.0
HUS 1302	Basic Counseling Skills	3.0

Semester 5

Units: 12.0

CLP 2140	Psychopathology	3.0
HUS 2303	Counseling Techniques	3.0
HUS 2500	Issues & Ethics in Human Services	3.0
SOP 2002	Social Psychology	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ISS1301 Intro. to Social Research 3.000 credits

This course is a general introduction to research methodology in the Social Sciences, paying particular attention to research design, data collection and data analysis. This is a course in Applied Social Sciences that will provide students with a survey understanding of social scientific research. This course will include examination of empirical research, including literature reviews, theory, methodology, data collection, data analysis and presentation of results. (3 hr. lecture)

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033 Any Mathematics level higher than MAT 1033 will satisfy this requirement.

MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

HUS1302 Basic Counseling Skills 3.000 credits

Development of the skills of observation, recording, reporting, interviewing and counseling. These skills are presented in the context of general counseling theory.

Lecture Hours: 48.000

CLP2140 Psychopathology 3.000 credits

This course examines the major categories of mental disorders. Diagnostic criteria, treatment methods, cultural factors, public attitudes, community resources, ethical issues and legislation applicable to individuals with mental disorders are studied. The impact of mental disorders on individuals, families and society are discussed.

Lecture Hours: 48.000

Prerequisite: PSY2012

HUS2303 Counseling Techniques 3.000 credits

Specific counseling techniques are introduced within the various counseling theories. Work involves both group and individual techniques.

Lecture Hours: 48.000

HUS2500 Issues & Ethics in Human Services 3.000 credits

This course is designed to familiarize students with the ethical problems that emerge from counseling the chemically dependent client. Emphasis will be placed on the following: the history and theory of ethics in health care; professionals' and patients' rights and responsibilities; the relationship between ethics and law; confidentiality and truth-telling in clinical relationships; technology; diagnostic testing and treatment; treatment of terminal illness; distribution of scarce medical resources and access to health care and systems payment. Prerequisite: PSB2442.

Lecture Hours: 48.000

SOP2002 Social Psychology 3.000 credits

Combines a knowledge of psychology and sociology, in an interdisciplinary approach to the study of human interaction. Main themes deal with the nature of attitudes, how attitudes may be changed, the processes of interaction and the nature of group structures.

Lecture Hours: 48.000

Program Description

Sociology

Associate in Arts | Code: 12203 | 60 credits

Sociology is the systematic study of human interaction, that is, society, social relationships, social structures and social change. Coursework emphasizes liberal arts topics such as literature, cultural anthropology, theatre appreciation and history, as well as introductory courses in sociology. Graduates with a Bachelor's degree can work within community organizations, government agencies and the criminal justice field. Many students go on to pursue graduate degrees and work in social policy, public administration, law, government or social services.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ANT 2000	Introduction to Anthropology	3.0
ENC 1101	English Composition 1	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
SYG 2000	Introduction to Sociology	3.0
SLS 1106	First Year Experience Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2

Units: 12.0

ENC 1102	English Composition 2	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0

OR			
HUM 1020	Humanities		3.0
OR			
LIT 2000	Introduction to Literature		3.0
OR			
MUL 1010	Music Appreciation		3.0
OR			
PHI 2010	Introduction to Philosophy		3.0
OR			
THE 2000	Theatre Appreciation		3.0
MGF 1131	Mathematics in Context		3.0
SPC 2608	Introduction to Public Speaking		3.0
Semester 3			Units: 12.0
BSC 1005	General Education Biology		3.0
PHI 2600	Introduction to Ethics		3.0
STA 2023	Statistical Methods		3.0
SYG 2010	Social Problems		3.0
Semester 4			Units: 12.0
EVR 1001	Introduction to Environmental Science		3.0
ISS 1301	Intro. to Social Research		3.0
SYG 2230	Multi-Ethnic America		3.0
SYG 2430	Marriage and The Family		3.0
Semester 5			Units: 11.0
PSC 1515	Energy in the Natural Environment		3.0
PaElective*			8.0

*Note: Check with advisor for any transferable Credit Type 1 or Credit Type 2 course.

Total: 60.0

ANT2000 Introduction to Anthropology 3.000 credits

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

SYG2010 Social Problems 3.000 credits

An analysis of the major contemporary and recurring social problems, emphasizing scientific search for variables involved and exploring alternative solutions.

Lecture Hours: 48.000

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

ISS1301 Intro. to Social Research 3.000 credits

This course is a general introduction to research methodology in the Social Sciences, paying particular attention to research design, data collection and data analysis. This is a course in Applied Social Sciences that will provide students with a survey understanding of social scientific research. This course will include examination of empirical research, including literature reviews, theory, methodology, data collection, data analysis and presentation of results. (3 hr. lecture)

Lecture Hours: 48.000

SYG2230 Multi-Ethnic America 3.000 credits

In this course, students will gain an understanding of the theory and problems of underrepresented populations in American society, with an emphasis on inequity, bias, and the evolving dynamics of bias and discrimination. Additionally, the course will critically examine the portrayal of underrepresented populations in the Western Canon, providing insight into the historical and cultural influences that have shaped societal perceptions and attitudes.

SYG2430 Marriage and The Family 3.000 credits

The family as a social institution — its origin and development, its forms and functions, its interrelation with other social institutions, and its role in contemporary civilization. Areas of study include factors contributing to or acting against successful, stable marriage.

Lecture Hours: 48.000

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

Program Description

Speech Pathology & Audiology

Associate in Arts | Code: 11501 | 60 credits

This program provides fundamental coursework in biology and communications so that students may transfer to a four-year institution. The curriculum leading to the Baccalaureate degree is usually designed as pre-professional education for a graduate program. Speech language pathologists and audiologists provide clinical services to individuals with speech, language and hearing impairments. Eligibility for the Certificate of Clinical Competence from the American Speech-Language-Hearing Association and state licensure are not possible until requirements for the graduate degree are met.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
BSC 1005	General Education Biology	3.0
CLP 2000	Dynamics of Behavior	3.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
Humanities		3.0
HUM 1020	Humanities	3.0
OR		
ARH 1000	Art Appreciation	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
PSY 2012	Introduction to Psychology	3.0
STA 2023	Statistical Methods	3.0
Semester 3		Units: 9.0
PHI 2600	Introduction to Ethics	3.0
SPC 1017	Introduction to Communication	3.0
Social Science		3.0

AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4

Units: 12.0

CLP 2140	Psychopathology	3.0
DEP 2000	Human Growth and Development	3.0
HUN 1201	Essentials of Human Nutrition	3.0
PHY 1020	General Education Physics	3.0

Semester 5

Units: 15.0

BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy & Physiology Lab 1	1.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
DEP 2100	Child Growth And Development	3.0
Pathway Elective*		4.0

*Note: Check with advisor for any transferable Credit Type 1 or Credit Type 2 course.

Total: 60.0

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CLP2000 Dynamics of Behavior 3.000 credits

Analysis of mechanisms of adjustment, motivation, frustration and conflict, learning personality and psychotherapy. Emphasis is on the psychological processes of the normal individual functioning in society rather than on behavior disorders.

Lecture Hours: 48.000

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context. Lecture Hours: 48.000

Lecture Hours: 48.000

Course User Fee: \$10.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course. Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CLP2140 Psychopathology 3.000 credits

This course examines the major categories of mental disorders. Diagnostic criteria, treatment methods, cultural factors, public attitudes, community resources, ethical issues and legislation applicable to individuals with mental disorders are studied. The impact of mental disorders on individuals, families and society are discussed.

Lecture Hours: 48.000

Prerequisite: PSY2012

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy & Physiology Lab 1 1.000 credits

In this laboratory course, student will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

DEP2100 Child Growth And Development 3.000 credits

This course in Child Growth and Development is designed especially for the student interested in the human life span from birth through the first eight years. The course is intended to acquaint the student with basic theoretical models of development and such specific topics as heredity teratogenic agents, learning, intelligence, socialization, personality, sex role identification, language acquisition and moral development.

Lecture Hours: 48.000

Program Description

Teaching (Pre-Elementary/Early Childhood)

Associate in Arts | Code: 10809 | 60 credits

This program prepares students to major in elementary education by presenting the science, liberal arts and education curriculum necessary for transfer to Florida colleges and universities. Areas of specialization include elementary, pre-elementary/early childhood, exceptional student and secondary education. Available areas of specialization in secondary education are biology, chemistry, earth/space science, English and foreign language, mathematics, physics and social science. Additionally, a specialization in vocational secondary education is available. This includes agriculture, business, home economics and technical coursework. A curriculum appropriate to each area of specialization is featured in the specific A.A. program. Students should work with an advisor to determine the appropriate coursework for transfer into their intended area of study.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
EEC 1000	Introduction to Early Childhood Education	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
DEP 2000	Human Growth and Development	3.0
EDF 1005	Introduction to the Education Profession	3.0
Semester 3		Units: 9.0
CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
SPC 2608	Introduction to Public Speaking	3.0

Program Elective		3.0
EEC 1200	Early Childhood Curriculum 1 (Birth - Age 5)	3.0
OR		
EEC 1311	Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5)	3.0
OR		
EEC 2407	Foundations of Learning and Executive Function in Early Childhood (Birth - 5)	3.0
OR		
EEC 2002	Operation of an Early Childhood Facility	3.0
OR		
EEC 2520	Early Childhood Organization Leadership	3.0
OR		
EEC 2523	Programming & Management for Early Childhood Admin	3.0
OR		
EEC 2527	Legal and Financial Issues in Childcare	3.0
OR		
ECW 2010	Teaching with AI: Designing Human-Centered Classrooms	3.0
OR		
PHI 2680	Artificial Intelligence and Ethics	3.0

Semester 4

Units: 12.0

BSC 1005	General Education Biology	3.0
Humanities		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EDF 2144	Maximizing Student Potential in the School Context	3.0
EME 2040	Creativity, Innovation, and Technology	3.0

Semester 5

Units: 14.0

PHI 2600	Introduction to Ethics	3.0
Natural Science		3.0
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
HUN 1201	Essentials of Human Nutrition	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
EDG 2943	Educational Service Field Work	2.0
Program Electives*		6.0

*Select 6 credits from the following course options: EEC 1200, EEC 1311, EEC 2407, EEC2002, EEC2520, EEC2523, EEC2527, ECW 2010, PHI

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

EEC1000 Introduction to Early Childhood Education 3.000 credits

Students will be introduced to the field of early childhood education with a focus on infants, toddlers, and preschool-aged children from birth through age five, highlighting play as a developmentally appropriate and research-supported approach to early learning. The course examines the purpose and scope of early childhood education and the roles of children, families, and educators. Students gain an overview of foundational theories and philosophies, family and societal influences, learning environments, classroom routines, and guidance practices that support development across domains. Common early childhood program types and curriculum models serving birth-five settings are also explored. Ten hours of service learning in an early childhood setting are required. Fingerprinting through MDC's school of education must be completed before starting this course. See the MDC SOE Work-Force handbook.

Lecture Hours: 48.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000

Course User Fee: \$18.00

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

EEC1200 Early Childhood Curriculum 1 (Birth - Age 5) 3.000 credits

Students will design and implement developmentally appropriate curriculum for children from birth to age five using the NAEYC Professional Standards as a guiding framework. Emphasis is placed on foundational curriculum planning grounded in children's interests, with focus on development, social studies, language and literacy, and creative expression through the arts. Course content includes observation-based planning, learning centers, classroom and outdoor environments, daily routines, and play-based experiences. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. This course is paired with EEC1311, which extends curriculum planning through integrated and inquiry-based applications. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000

EEC1311 Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5) 3.000 credits

Students will design and implement integrated, developmentally appropriate curriculum for children from birth to age five, building on foundational curriculum planning introduced in EEC 1200. Emphasis is placed on arts-integrated STEM learning through inquiry-based curriculum planning, with focus on scientific inquiry, mathematical thinking, building and engineering, technology, and maker spaces for creative and artistic expression. Course content includes inquiry-based planning, learning centers, enriching classroom and outdoor environments, and play-based investigations that extend children's thinking across domains. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000 AND EEC1200

Course User Fee: \$10.00

EEC2407 Foundations of Learning and Executive Function in Early Childhood (Birth - 5) 3.000 credits

Students will build foundational knowledge of executive-function life skills, including focus and self-control, perspective taking, communication, making connections, critical thinking, taking on challenges, and self-directed learning, as the basis for learning in early childhood (Birth–5). The course examines how these skills interact with interpersonal development and cognitive growth, drawing on neuroscience and research on child development. Reciprocal family partnerships, observation-based reflection, and ethical considerations guide the analysis of young children’s interactions and abilities. Emphasis is placed on research-based inhibitory control strategies that support neurocognitive development. Students will connect these executive-function skills and regulation strategies to developmentally appropriate teaching methods across home and early education settings. Ten hours of service learning in an approved early childhood setting are required. See the School of Education workforce handbook.

Lecture Hours: 48.000

EEC2002 Operation of an Early Childhood Facility 3.000 credits

This introductory course provides an overview for early childhood administrators to develop and enhance their leadership role in designing and implementing quality early care and education programs. The student will study the following topics: organizational leadership and management, programming, and financial and legal issues. This course meets the requirements for the Education Program Administrator Foundational Level Credential.

Lecture Hours: 48.000

EEC2520 Early Childhood Organization Leadership 3.000 credits

The student will develop skills needed by early childhood program administrators to manage a high-quality center. The student will learn about organizational structure of a center; ethics and professionalism; leadership strategies, skills, and competencies; self-reflection and work environment; quality improvement; staff recruitment, evaluation, and retention. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.)

Lecture Hours: 48.000

Prerequisite: EEC2002

Course User Fee: \$10.00

EEC2523 Programming & Management for Early Childhood Admin 3.000 credits

The student will learn about developmentally and culturally appropriate environments, curriculum, and professional standards for early childhood care program administrators. The student will develop an understanding of child observation, assessment, documentation, and referrals and their importance. The student will learn about program evaluation, health, safety, healthy nutrition practices, and the importance of partnerships with families. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.).

Lecture Hours: 48.000

Course User Fee: \$10.00

EEC2527 Legal and Financial Issues in Childcare 3.000 credits

The student will learn about the legal and financial issues related to operating a successful early childhood center. The student learns about financial planning, personnel cost and allocation, budgeting and accounting. The student will develop knowledge about compensation and benefit, risk management, marketing and public relations, regulatory requirements, and legal issues related to childcare management. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential.

Lecture Hours: 48.000

Prerequisite: EEC2002

ECW2010 Teaching with AI: Designing Human-Centered Classrooms 3.000 credits

This course invites learners to explore the integration of artificial intelligence through a pedagogical lens, where technology serves as a tool for reimagining teaching and learning. Learners will critically engage with AI tools by grounding their use in learning theory and evidence-based design practices. Through inquiry, collaboration, and hands-on experimentation, learners will examine how AI can amplify creativity, personalize learning, and promote human-centeredness.

Lecture Hours: 48.000

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized. Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

EME2040 Creativity, Innovation, and Technology 3.000 credits

The student will learn to manage a productive & safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Prerequisite: CGS1060 OR Passing Score on the Computer Competency Exam

Course User Fee: \$190.03

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

EDG2943 Educational Service Field Work 1.000 - 3.000 credits

The student will learn to compile the necessary documents and complete the process of obtaining a state and/or national early childhood credential. The student must complete the four courses in either the Infant/Toddler Specialization or the Preschool Specialization in order to take this course.

Lecture Hours: 16.000 - 48.000

Prerequisite: EEC1000, EEC1200, EEC1311, EEC2202, or EEC2002, EEC2520, EEC2523, EEC2527

Program Description

Teaching (Exceptional Student Education)

Associate in Arts | Code: 10804 | 60 credits

This program prepares students to major in elementary education by presenting the science, liberal arts and education curriculum necessary for transfer to Florida colleges and universities. Areas of specialization include elementary, pre-elementary/early childhood, exceptional student and secondary education. Available areas of specialization in secondary education are biology, chemistry, earth/space science, English and foreign language, mathematics, physics and social science. Additionally, a specialization in vocational secondary education is available. This includes agriculture, business, home economics and technical coursework. A curriculum appropriate to each area of specialization is featured in the specific A.A. program. Students should work with an advisor to determine the appropriate coursework for transfer into their intended area of study.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
EDF 1005	Introduction to the Education Profession	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0

PSY 2012	Introduction to Psychology	3.0
EDF 2144	Maximizing Student Potential in the School Context	3.0

Semester 3

Units: 9.0

SPC 2608	Introduction to Public Speaking	3.0
EME 2040	Creativity, Innovation, and Technology	3.0
<i>Computer Competence</i>		3.0
CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
OR		
CGS 1060C	Introduction to Computer Technology and Applications	4.0

Semester 4

Units: 12.0

BSC 1005	General Education Biology	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EEX 2000	Introduction to Exceptional Student Education	3.0
<i>Program Elective</i>		3.0
EEC 1000	Introduction to Early Childhood Education	3.0
OR		
EEC 1200	Early Childhood Curriculum 1 (Birth - Age 5)	3.0
OR		
EEC 1311	Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5)	3.0
OR		
EEC 2407	Foundations of Learning and Executive Function in Early Childhood (Birth - 5)	3.0
OR		
EEC 2002	Operation of an Early Childhood Facility	3.0
OR		
EEC 2520	Early Childhood Organization Leadership	3.0
OR		
EEC 2523	Programming & Management for Early Childhood Admin	3.0
OR		
EEC 2527	Legal and Financial Issues in Childcare	3.0
OR		
ECW 2010	Teaching with AI: Designing Human-Centered Classrooms	3.0
OR		
PHI 2680	Artificial Intelligence and Ethics	3.0

Semester 5

Units: 14.0

DEP 2000	Human Growth and Development	3.0
PHI 2600	Introduction to Ethics	3.0
<i>Natural Science</i>		3.0
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		

HUN 1201	Essentials of Human Nutrition	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
Program Elective*		5.0

*Select 2 courses from the following course options: EEC 1000, EEC 1200, EEC 1311, EEC 2407, EEC 2002, EEC 2520, EEC 2523, EEC 2527, ECW 2010, PHI 2680

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000

Course User Fee: \$18.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

EME2040 Creativity, Innovation, and Technology 3.000 credits

The student will learn to manage a productive & safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Prerequisite: CGS1060 OR Passing Score on the Computer Competency Exam

Course User Fee: \$190.03

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

EEX2000 Introduction to Exceptional Student Education 3.000 credits

The student will learn about the history, legal issues, and legislation related to exceptional student education in K–12. The student will learn about the differing types of exceptionalities, supports, and accommodations provided for students with differing learning profiles from birth to adulthood.

Lecture Hours: 48.000

Prerequisite: EDF1005

Course User Fee: \$18.00

EEC1000 Introduction to Early Childhood Education 3.000 credits

Students will be introduced to the field of early childhood education with a focus on infants, toddlers, and preschool-aged children from birth through age five, highlighting play as a developmentally appropriate and research-supported approach to early learning. The course examines the purpose and scope of early childhood education and the roles of children, families, and educators. Students gain an overview of foundational theories and philosophies, family and societal influences, learning environments, classroom routines, and guidance practices that support development across domains. Common early childhood program types and curriculum models serving birth–five settings are also explored. Ten hours of service learning in an early childhood setting are required. Fingerprinting through MDC's school of education must be completed before starting this course. See the MDC SOE Work-Force handbook.

Lecture Hours: 48.000

EEC1200 Early Childhood Curriculum 1 (Birth - Age 5) 3.000 credits

Students will design and implement developmentally appropriate curriculum for children from birth to age five using the NAEYC Professional Standards as a guiding framework. Emphasis is placed on foundational curriculum planning grounded in children's interests, with focus on development, social studies, language and literacy, and creative expression through the arts. Course content includes observation-based planning, learning centers, classroom and outdoor environments, daily routines, and play-based experiences. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. This course is paired with EEC1311, which extends curriculum planning through integrated and inquiry-based applications. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000

EEC1311 Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5) 3.000 credits

Students will design and implement integrated, developmentally appropriate curriculum for children from birth to age five, building on foundational curriculum planning introduced in EEC 1200.

Emphasis is placed on arts-integrated STEM learning through inquiry-based curriculum planning, with focus on scientific inquiry, mathematical thinking, building and engineering, technology, and maker spaces for creative and artistic expression. Course content includes inquiry-based planning, learning centers, enriching classroom and outdoor environments, and play-based investigations that extend children's thinking across domains. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000 AND EEC1200

Course User Fee: \$10.00

EEC2407 Foundations of Learning and Executive Function in Early Childhood (Birth - 5) 3.000 credits

Students will build foundational knowledge of executive-function life skills, including focus and self-control, perspective taking, communication, making connections, critical thinking, taking on challenges, and self-directed learning, as the basis for learning in early childhood (Birth–5). The course examines how these skills interact with interpersonal development and cognitive growth, drawing on neuroscience and research on child development. Reciprocal family partnerships, observation-based reflection, and ethical considerations guide the analysis of young children's interactions and abilities. Emphasis is placed on research-based inhibitory control strategies that support neurocognitive development. Students will connect these executive-function skills and regulation strategies to developmentally appropriate teaching methods across home and early education settings. Ten hours of service learning in an approved early childhood setting are required. See the School of Education workforce handbook.

Lecture Hours: 48.000

EEC2002 Operation of an Early Childhood Facility 3.000 credits

This introductory course provides an overview for early childhood administrators to develop and enhance their leadership role in designing and implementing quality early care and education programs. The student will study the following topics: organizational leadership and management, programming, and financial and legal issues. This course meets the requirements for the Education Program Administrator Foundational Level Credential.

Lecture Hours: 48.000

EEC2520 Early Childhood Organization Leadership 3.000 credits

The student will develop skills needed by early childhood program administrators to manage a high-quality center. The student will learn about organizational structure of a center; ethics and professionalism; leadership strategies, skills, and competencies; self-reflection and work environment; quality improvement; staff recruitment, evaluation, and retention. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.)

Lecture Hours: 48.000

Prerequisite: EEC2002

Course User Fee: \$10.00

EEC2523 Programming & Management for Early Childhood Admin 3.000 credits

The student will learn about developmentally and culturally appropriate environments, curriculum, and professional standards for early childhood care program administrators. The student will develop an understanding of child observation, assessment, documentation, and referrals and their importance. The student will learn about program evaluation, health, safety, healthy nutrition practices, and the importance of partnerships with families. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.).

Lecture Hours: 48.000

Course User Fee: \$10.00

EEC2527 Legal and Financial Issues in Childcare 3.000 credits

The student will learn about the legal and financial issues related to operating a successful early childhood center. The student learns about financial planning, personnel cost and allocation, budgeting and accounting. The student will develop knowledge about compensation and benefit, risk management, marketing and public relations, regulatory requirements, and legal issues related to childcare management. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential.

Lecture Hours: 48.000

Prerequisite: EEC2002

ECW2010 Teaching with AI: Designing Human-Centered Classrooms 3.000 credits

This course invites learners to explore the integration of artificial intelligence through a pedagogical lens, where technology serves as a tool for reimagining teaching and learning. Learners will critically engage with AI tools by grounding their use in learning theory and evidence-based design practices.

Through inquiry, collaboration, and hands-on experimentation, learners will examine how AI can amplify creativity, personalize learning, and promote human-centeredness.

Lecture Hours: 48.000

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

Program Description

Teaching Secondary (Biology)

Associate in Arts | Code: 10815 | 60 credits

This program prepares students to major in elementary education by presenting the science, liberal arts and education curriculum necessary for transfer to Florida colleges and universities. Areas of specialization include elementary, pre-elementary/early childhood, exceptional student and secondary education. Available areas of specialization in secondary education are biology, chemistry, earth/space science, English and foreign language, mathematics, physics and social science. Additionally, a specialization in vocational secondary education is available. This includes agriculture, business, home economics and technical coursework. A curriculum appropriate

to each area of specialization is featured in the specific A.A. program. Students should work with an advisor to determine the appropriate coursework for transfer into their intended area of study.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
EDF 1005	Introduction to the Education Profession	3.0
SLS 1106	First Year Experience Seminar	1.0
Semester 2		Units: 13.0
ENC 1102	English Composition 2	3.0
DEP 2000	Human Growth and Development	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
EDF 2144	Maximizing Student Potential in the School Context	3.0
Semester 3		Units: 9.0
SPC 2608	Introduction to Public Speaking	3.0
EME 2040	Creativity, Innovation, and Technology	3.0
<i>Computer Competence</i>		3.0
CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
OR		
CGS 1060C	Introduction to Computer Technology and Applications	4.0
Semester 4		Units: 13.0
BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
EEX 2000	Introduction to Exceptional Student Education	3.0
Semester 5		Units: 12.0
BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry & Qualitative Analysis Lab	2.0
PHI 2600	Introduction to Ethics	3.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000
Course User Fee: \$18.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000
Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000
Prerequisite: ENC1101
Course User Fee: \$15.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics.

Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000
Prerequisite: MAC1105
Course User Fee: \$18.00

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

EME2040 Creativity, Innovation, and Technology 3.000 credits

The student will learn to manage a productive & safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Prerequisite: CGS1060 OR Passing Score on the Computer Competency Exam

Course User Fee: \$190.03

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

EEX2000 Introduction to Exceptional Student Education 3.000 credits

The student will learn about the history, legal issues, and legislation related to exceptional student education in K-12. The student will learn about the differing types of exceptionalities, supports, and accommodations provided for students with differing learning profiles from birth to adulthood.

Lecture Hours: 48.000

Prerequisite: EDF1005

Course User Fee: \$18.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

**BSC2011L Principles of Biology 2 Laboratory
2.000 credits**

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

**CHM1046 General Chemistry and Qualitative
Analysis 3.000 credits**

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

**CHM1046L General Chemistry & Qualitative
Analysis Lab 2.000 credits**

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students.

Lab Hours: 64.000

Prerequisite: CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

Program Description

Teaching Secondary (Mathematics Education)

Associate in Arts | Code: 10812 | 60 credits

This program prepares students to major in elementary education by presenting the science, liberal arts and education curriculum necessary for transfer to Florida colleges and universities. Areas of specialization include elementary, pre-elementary/early childhood, exceptional student and secondary education. Available areas of specialization in secondary education are biology, chemistry, earth/space science, English and foreign language, mathematics, physics and social science. Additionally, a specialization in vocational secondary education is available. This includes agriculture, business, home economics and technical coursework. A curriculum appropriate to each area of specialization is featured in the specific A.A. program. Students should work with an advisor to determine the appropriate coursework for transfer into their intended area of study.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		

AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
EDF 1005	Introduction to the Education Profession	3.0
SLS 1106	First Year Experience Seminar	1.0

Semester 2 **Units: 12.0**

ENC 1102	English Composition 2	3.0
MAC 1147	Pre-Calculus Algebra and Trigonometry	5.0
STA 2023	Statistical Methods	3.0
DEP 2000	Human Growth and Development	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3 **Units: 11.0**

BSC 1005	General Education Biology	3.0
MAC 2311	Calculus and Analytical Geometry 1	5.0
SPC 2608	Introduction to Public Speaking	3.0

Semester 4 **Units: 12.0**

<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 2312	Calculus and Analytical Geometry 2	4.0
EDF 2144	Maximizing Student Potential in the School Context	3.0
EME 2040	Creativity, Innovation, and Technology	3.0

Semester 5 **Units: 12.0**

PHI 2600	Introduction to Ethics	3.0
MAC 2313	Calculus and Analytic Geometry 3	4.0
<i>Natural Science</i>		3.0
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
HUN 1012	Nutritional Counseling	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
EEX 2000	Introduction to Exceptional Student Education	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000

Course User Fee: \$18.00

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "C" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics.

Prerequisite: MAC1105 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

MAC2311 Calculus and Analytical Geometry 1 5.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC1114 both with a minimum grade of C

Course User Fee: \$18.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC2312 Calculus and Analytical Geometry 2 4.000 credits

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2311

Course User Fee: \$18.00

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

EME2040 Creativity, Innovation, and Technology 3.000 credits

The student will learn to manage a productive & safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Prerequisite: CGS1060 OR Passing Score on the Computer Competency Exam

Course User Fee: \$190.03

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

MAC2313 Calculus and Analytic Geometry 3 4.000 credits

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems.

Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course.

Lecture Hours: 64.000

Prerequisite: MAC2312

Course User Fee: \$18.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUN1012 Nutritional Counseling 3.000 credits

This course focuses on the application of scientific knowledge to actual case studies. Techniques for effective nutrition interviewing and counseling are covered. The goal of this course is to acquire the basic skills required for conducting interviews and counseling sessions related to the field of nutrition and dietetics. (3 credits)

Lecture Hours: 48.000

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

EEX2000 Introduction to Exceptional Student Education 3.000 credits

The student will learn about the history, legal issues, and legislation related to exceptional student education in K-12. The student will learn about the differing types of exceptionalities, supports, and accommodations provided for students with differing learning profiles from birth to adulthood.

Lecture Hours: 48.000

Prerequisite: EDF1005

Course User Fee: \$18.00

Associate in Science (AS)

Program Description

Accounting Technology

Associate in Science | Code: 25022 | 60 credits

The Accounting Technology Associate in Science (A.S.) program offers opportunities to earn multiple credentials that help students find jobs quickly while advancing both their academic and professional careers. The A.S. in Accounting Technology allows students to gain valuable experience in this in-demand field. Graduates of the Accounting program develop a strong, competitive accounting foundation.

Accounting and bookkeeping are essential functions of any business. Experienced accounting professionals have a high degree of job stability and are often top-tier salary earners. This A.S. allows for a seamless transition into the [Bachelor of Applied Science in Leadership Management and Innovation with a Concentration in Accounting](#).

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
PHI 2010	Introduction to Philosophy	3.0
MGF 1130	Mathematical Thinking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 2		Units: 11.0
		4.0
		4.0
ACG 2011	Principles of Accounting II	3.0
ACG 2011L	Principles of Accounting 2 Lab	1.0
OR		
		4.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
		3.0
BUL 2131	Legal Environment	3.0
OR		
BUL 2241	Business Law 1	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0

Semester 3		Units: 13.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
GEB 1011	Principles of Business	3.0
MAN 2021	Principles of Management	3.0
TAX 2000	Income Tax	3.0

Semester 4		Units: 12.0
<i>Natural Science</i>		3.0
AST 1002	Descriptive Astronomy	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
PHY 1020	General Education Physics	3.0
ACG 2450	Microcomputers in Accounting	3.0

FIN 2000	Principles of Finance	3.0
OST 2335	Business Writing	3.0

Semester 5

Units: 12.0

Major Course Elective*

12.0

*Select 12 credits from the following course option: ACG1403, ACG2031, ACG2100, ACG2170, ACG2360, ACG2630, MT1103, TAX2002, TAX2021, TAX2401.

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This is a required general education core course in college-level writing. The students will learn to compose essays and other works using various methods of development. This course fulfills the Gordon Rule writing requirement and must be completed with a grade "C" or better. Pre-requisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ACG2011 Principles of Accounting II 3.000 credits

This is the second part of the introductory financial accounting series, which includes ACG2001 and ACG2011, with emphasis on accounting owner's equity and corporate financial statements. Other topics include receivables, fixed assets, intangible assets, current and long-term liabilities. Co-requisite: ACG2011L

Lecture Hours: 48.000

ACG2011L Principles of Accounting 2 Lab 1.000 credits

Students will use personal computer software and financial accounting applications to analyze accounting records, prepare financial statements, and compile EXCEL spreadsheets. This course will reinforce, with tutorial help and problem-solving, the concepts needed to achieve the objectives of ACG2011. Corequisite: ACG2011.

Lab Hours: 32.000

Corequisite: ACG2001

Course User Fee: \$22.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

BUL2131 Legal Environment 3.000 credits

Law in relation to the proper conduct of business includes a consideration of the nature and sources of law, its legal environment and history. The topics of business torts, crimes, contracts, and forms of organizations are also covered.

Lecture Hours: 48.000

Course User Fee: \$10.00

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites:

ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

TAX2000 Income Tax 3.000 credits

Federal income tax fundamentals with emphasis on individual returns. Topics considered include gross income, capital gains and losses, deductions and exemptions, and tax credits.

Lecture Hours: 48.000

Course User Fee: \$15.00

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

ACG2450 Microcomputers in Accounting 3.000 credits

Accounting application of electronic data processing including the preparation, interpretation and use of computer information in financial decision-making. Pre/Corequisite: ACG2021. Special fee.

Lecture Hours: 48.000

Pre/Corequisite: ACG2021

Course User Fee: \$22.00

FIN2000 Principles of Finance 3.000 credits

The creation, allocation, and utilization of money, and the effect of monetary policy upon individuals, business, national and international economics. This course provides a basis for further study of monetary theory, banking, finance and securities.

Lecture Hours: 48.000

OST2335 Business Writing 3.000 credits

Covers the procedures for writing effective business letters and memorandums, a review of grammar, and the proper format of today's business correspondence. Students learn how to prepare inquiry letters, direct and indirect response letters, application letters and resumes, and short reports.

Lecture Hours: 48.000

Course User Fee: \$15.00

Program Description

Aerospace Technician

Associate in Science | Code: 25081 | 64 credits

The Associate of Science in Aerospace Technician program provides hands-on training across a range of critical skills needed by technicians in the aerospace industry. Students train in assembling, servicing, operating, and repairing aircraft, spacecraft, and non-flight hardware and systems that support a launch. This comprehensive program prepares graduates for careers in the commercial, government, and military sectors, equipping them with the technical knowledge and practical experience required to excel in the field of aerospace technology.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester		Units: 13.0
MAC 1105	College Algebra	3.0
EET 1033C	Electrical Fundamentals	4.0
ETI 1043C	Introduction to Aerospace	3.0
ETI 1853C	Aerospace Safety and Quality	3.0
Semester 2		Units: 14.0
CET 1110C	Digital Circuits	4.0
ETI 1830C	Materials and Processes 1	3.0
ETI 1851C	Applied Mechanics	4.0
ETS 2632C	Introduction to Computer Aided Manufacturing	3.0
Semester 3		Units: 9.0
ENC 1101	English Composition 1	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
Computer Competency*		

*CGS 1060C ---OR--- Computer Competency Test. Departmental Recommendation: Computer Skills Placement Test

Semester 4**Units: 14.0**

ETI 1855C	Aerospace Structural Fabrication 1	3.0
ETI 2321C	Aerospace Systems and Propulsion	4.0
ETI 2832C	Materials and Processes 2	3.0
ETS 1603C	Introduction to Robotics	4.0

Semester 5**Units: 14.0**

ETI 1852C	Aerospace Test and Measurements	4.0
ETI 2856C	Aerospace Structural Fabrication 2	4.0
ETI 2951C	Aerospace Technician Capstone	3.0
EVR 1001	Introduction to Environmental Science	3.0

Total: 64.0

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

EET1033C Electrical Fundamentals 4.000 credits

This course is designed for students obtaining a CCC or AS degree in Engineering Technology, and related disciplines. The student will learn the basic concepts of electronics principles including Direct Current, Alternating Current, Series and Parallel circuits topologies, Basic electronics components, electronics measure tools and software simulation tools. Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$103.00

ETI1043C Introduction to Aerospace 3.000 credits

This course includes an overview of aerospace industry terminology and acronyms, along with practical activities involving tools, procedures, and standard practices. Additionally, it covers inspection procedures, workplace rules and regulations, safety measures, maintenance protocols, and career options in the aeronautical technician workforce. (2 hr. lecture, 2 hr. lab)
Lecture Hours: 16.000, Lab Hours: 64.000
Course User Fee: \$63.00

ETI1853C Aerospace Safety and Quality 3.000 credits

This course delves into the identification of hazards, the use of personal protective equipment, and the implementation of safe practices and measures for the protection of personnel, property, and equipment in the aerospace sector. It also addresses safety procedures, including compliance with OSHA regulations and the handling of hazardous materials. The course additionally explores fundamental principles of quality assurance engineering as they pertain to work processes. It includes the use of computer-based tools for analysis and reporting. (2 hr. lecture, 2 hr. lab)
Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$63.00

CET1110C Digital Circuits 4.000 credits

This is an introductory course in digital circuits for engineering and other technology majors. Students learn how to apply electronic principles to digital computer circuits and systems. Students will also learn how to simplify logic circuits, build digital circuits, and perform other laboratory activities. Pre/Co-requisite: MAC1105. Laboratory fee. Lecture Hours: 32.000, Lab Hours: 64.000
Pre/Corequisite: MAC1105
Course User Fee: \$63.00

ETI1830C Materials and Processes 1 3.000 credits

In this course, students explore the physical properties and characteristics of materials and commodities commonly utilized in the aerospace industry. It delves into materials compatibility, basics of metallurgy, and treatment processes, offering a foundation of how these elements interact and are applied within aerospace contexts. (1 hr. lecture, 4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$63.00

ETI1851C Applied Mechanics 4.000 credits

This course adopts a practical approach to teaching the identification, usage, and maintenance of tools and equipment utilized in aerospace systems. It covers blueprint reading, geometric dimensioning, tolerancing, and using both English and metric measuring systems. (2 hr. lecture, 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

ETS2632C Introduction to Computer Aided Manufacturing 3.000 credits

An introduction into the fundamentals of Computer Integrated Manufacturing as it relates to theory, operation, setup, safety, and practices. Students will learn the application of Computer Aided Drawing (CAD) and Computer Aided Manufacturing (CAM) software to develop prototypes. Prerequisite: MAC1105.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.
Lecture Hours: 48.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.
Lecture Hours: 48.000
Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.
Lecture Hours: 48.000
Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.
Lecture Hours: 48.000
Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.
Lecture Hours: 48.000
Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

ETI1855C Aerospace Structural Fabrication 1 3.000 credits

This course covers essential skills in metalworking, machining, and aerospace fastener applications. Students will learn blueprint interpretation, sheet metal repair, and the use of tools like brakes, shears, and rivets, while gaining hands-on experience in project construction and layout design, following industry standards. Prerequisites: ETI 1830C and ETI 1851C. (1 hr. lecture, 4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: ETI1830C AND ETI1851C

Course User Fee: \$63.00

ETI2321C Aerospace Systems and Propulsion 4.000 credits

This course introduces aeronautical and spacecraft systems, including hydraulic, pneumatic, electrical, propulsion, mechanical, HVAC, and ECLSS (Environmental Control and Life Support Systems). It covers system interactions with computer and data acquisition systems, emphasizing safety, power sources, assembly, troubleshooting, and integration for aerospace operations. Prerequisites: AFR 1100C (2 hr. lecture, 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETI1043C

Course User Fee: \$63.00

ETI2832C Materials and Processes 2 3.000 credits

This course provides information in aerospace applications of non-metallic materials. The use of adhesives, coatings, sealing, and issues with delamination, and faulty bonds are covered.

Prerequisite: ETI 1830C (1 hr. lecture, 4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: ETI1830C

Course User Fee: \$98.00

ETS1603C Introduction to Robotics 4.000 credits

This is an introductory level course designed as an introduction to robotics and robotic applications. Students learn topics related to robotic design including robotic terminology, robotic programming, sensing and sensors, actuators, modeling and sensing, robotic platforms, and the application of artificial intelligence in robotics. Laboratory activities provide hands-on application of concepts and theories.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

ETI1852C Aerospace Test and Measurements 4.000 credits

This course covers the principles and applications of primarily nondestructive testing techniques used to examine materials and components of spacecraft systems. Students learn to interpret procedures, operate testing equipment, and apply safety measures while conducting visual, dye penetrant, magnetic particle, ultrasonic, and Eddy-current inspections. Emphasis is placed on accurate measurement, tool selection, and result evaluation.

Prerequisite: ETI 1830C. (2 hr. lecture, 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETI1830C

Course User Fee: \$63.00

**ETI2856C Aerospace Structural Fabrication 2
4.000 credits**

This course introduces the student to the theory, materials, and basic manufacturing processes of composite materials. The course focuses on basic composite theory, including fiber reinforcements, matrix systems, fabrication techniques, and safety. It also introduces common core materials that are used in composites manufacturing, and outlines inspection and repair of composites structures including damage detection and repair instructions. Prerequisites: ETI 1855C and ETI 2832C (2 hr. lecture, 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETI1855C AND ETI2832C

Course User Fee: \$98.00

**ETI2951C Aerospace Technician Capstone 3.000
credits**

This course guides students through a capstone project, emphasizing project management, testing, troubleshooting, and documentation. Students evaluate requirements, conduct testing, resolve issues, and complete failure analysis. The course culminates with a comprehensive project presentation and a "lessons learned" review.

Prerequisites: ETI 1855C and ETI 2832C; Pre/

Corequisites: ETI 1852C and ETI 2856C (1 hr. lecture, 4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: ETI1855C AND ETI2832C AND Pre/

Corequisite: ETI1852C AND ETI2856C

Course User Fee: \$98.00

**EVR1001 Introduction to Environmental Science
3.000 credits**

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

Program Description

Animation & Game Art

Associate in Science | Code: 25074 | 60 credits

The Animation & Game Art is an intensive hands-on program in which students learn narrative structure along with technical skills required in the animation industry. Students learn character design and animation, 3D modeling, storyboarding and environment design. Graduates are prepared for entry level jobs as 3D artist and animators.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 15.0**

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
DIG 1111	Digital Character Design	3.0
DIG 1132	Digital Art and Design	3.0
DIG 1430	Storyboarding	3.0

Computer Competency Test*

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT).

Semester 2		Units: 14.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
DIG 1302	3D Modeling	4.0
DIG 1437	Narrative Storytelling	3.0
DIG 2113	Post Production & Editing	4.0
Semester 3		Units: 10.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1020	General Education Physics	3.0
DIG 2396C	Motion Capture	4.0
Semester 4		Units: 11.0
		3.0
DIG 2318	Animation Studio 1	3.0
OR		
CAP 2048	Game Development Project I	3.0
DIG 2370	Character Modeling & Rigging	4.0

DIG 2790	Texturing & Environment Design	4.0
Semester 5		Units: 10.0
DIG 2304	Character Animation	3.0
		3.0
DIG 2319	Animation Studio 2	3.0
OR		
CAP 2920C	Game Development Project II	3.0
DIG 2391C	Animation Studio 3	4.0
		Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

DIG1111 Digital Character Design 3.000 credits

This course, for students majoring in Animation and Game Art, covers the observation and translation of three-dimensional form into two-dimensional drawings. Students will learn the interpretation of the human body, based on major masses organized by gestural lines. Students will create original characters and create design elements to support them. Students will transition to draw on digital tablets, knowledge or proficiency in Adobe Photoshop recommended. (3 hr. lecture)

Lecture Hours: 48.000, Other Hours: 3.000

Course User Fee: \$129.55

DIG1132 Digital Art and Design 3.000 credits

This course is for students majoring in Animation and Game Art and introduces environmental design. Students will learn the concepts, hardware, and software related to digital image acquisition, image editing, manipulation, color management basics, masking, layering, retouching, scanning and output, and color theory as it relates to digital media. Knowledge or proficiency in Adobe Photoshop and Illustrator recommended. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$129.55

DIG1430 Storyboarding 3.000 credits

This course is for students majoring in Animation and Game Art. It introduces the necessary tasks in the storytelling phase of an animation project. Students will learn how to develop and design visual storyboards and how to sell their storyboard ideas. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$97.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

DIG1302 3D Modeling 4.000 credits

This course, for students majoring in Animation & Game Art introduces students to the basic tools, techniques and applications for feature 3D animation and game development. Students will learn how to manipulate objects, build models, employ lighting, design movement, work with materials and textures and render a final image. (4 hr. lecture)

Lecture Hours: 64.000

Course User Fee: \$129.55

DIG1437 Narrative Storytelling 3.000 credits

This course is for students majoring in Animation and Game Art. It introduces the conceptual structure and design of visual storytelling. Students will learn principles of animation, mechanics, cinematics, character development, personality of place, exaggeration and effects, adapting movement for the animation medium, and how to create the illusion of life. Pre/Corequisite of DIG 1430 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000, Lab Hours: -2.000, Clinical Hours: -1.000, Practicum Hours: -2.000

Pre/Corequisite: DIG1430

Course User Fee: \$129.55

DIG2113 Post Production & Editing 4.000 credits

This course, for students majoring in Animation & Game Art, equips students with skills required in post-production editing. Students will learn how to combine computer-generated imagery with matte painting and backgrounds and the core principles of proper compositing, color correction, and editing. Pre/Corequisite of DIG 1430 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG1430

Course User Fee: \$129.55

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

DIG2396C Motion Capture 4.000 credits

This course is for students majoring in Animation and Game Art. Students will learn to digitize motion and clean-up and editing techniques. They will also learn how to set up motion capture and shooting, data tracking, skeleton retargeting, as well as animation correction and enhancements. Pre/Corequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG1302

Course User Fee: \$129.55

DIG2318 Animation Studio 1 3.000 credits

This course is for students majoring in Animation and Game Art. Students will learn to design and implement a project involving computer animation, game production, VFX or scientific/architecture visualization. Students will work in collaboration with faculty and industry mentors. Prerequisite of DIG 1430; Prerequisite of DIG 1302; Prerequisite of DIG 1437 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG1430 AND DIG1302 AND DIG1437

Course User Fee: \$129.55

CAP2048 Game Development Project I 3.000 credits

This capstone course is for students majoring in Game Development and Game Animation. Students will work in cross disciplinary teams to develop a working animated game or film. Students will learn how to apply the skills and knowledge they have acquired in a real-world working development environment. Prerequisites: CAP2047 or DIG1302, COP1334 or DIG1111, COP2335 or DIG1437, DIG1430, DIG1710 or DIG1132; Pre/Corequisite: DIG1712 or DIG2113.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAP2047, Corequisite: DIG2626, Pre/Corequisite: DIG1712

Course User Fee: \$129.55

DIG2370 Character Modeling & Rigging 4.000 credits

This course focuses on the modeling and rigging of characters for performance. Students will solve complex issues of character articulation with an emphasis on skeleton, skin, and binding techniques. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: DIG1302

Course User Fee: \$129.55

DIG2790 Texturing & Environment Design 4.000 credits

This course is for students majoring in Animation & Game Art. Students will learn advanced 3D animation job skills used in creating 3D feature animation and game development, including advanced texturing, lighting and rendering a final image. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: DIG1302

Course User Fee: \$129.55

DIG2304 Character Animation 3.000 credits

This course, for students majoring in Animation & Game Art, equips students with the skills needed to create animated characters. Students will learn in-depth character design, development, rigging, and animation techniques, how to create segmented and solid model mesh of bipeds and quadrupeds, and techniques used to create facial expressions and lip syncing. Prerequisite of DIG 2790 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG2790

Course User Fee: \$129.55

DIG2319 Animation Studio 2 3.000 credits

This is a capstone course for students majoring in Animation and Game Art. Building on skills learned in Animation Studio 1, students will learn enhanced skills in the areas of 3D modeling, texturing, lighting, and animation. Working in groups, students develop a project plan and produce a short, 3D animated movie. Prerequisite of DIG 2318 with a Grade of C or better, or equivalent or Prerequisite of CAP 2048 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG2318 OR CAP2048

Course User Fee: \$97.00

CAP2920C Game Development Project II 3.000 credits

This capstone course is for students majoring in Game Development and Game Animation. Students will work in cross disciplinary teams to develop a working 5-10 min interactive game experience. Students will learn how to apply the skills and knowledge they have acquired in a real world working development environment. Prerequisites: CAP2048, DIG1712 or DIG 2113.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAP2048 AND DIG1430 AND Corequisite: DIG1705

Course User Fee: \$129.55

DIG2391C Animation Studio 3 4.000 credits

This is a capstone course for students majoring in Animation and Game Art. Students develop a project plan and produce a short, 3D animated movie. Students also create a website for the project, social media and market campaigns, and submit the short animated movie to festivals. Pre/Corequisite of DIG 2319 with a Grade of C or better, or equivalent or Pre/Corequisite of CAP 2920C with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG2319 OR CAP2920C

Course User Fee: \$129.55

Program Description

Applied Artificial Intelligence

Associate in Science | Code: 25080 | 60 credits

The Associate in Science in Applied Artificial Intelligence introduces learners to the importance of managing the lifecycle of an artificial intelligence (AI) project and ethical considerations relevant to its design and implementation. It offers hands-on learning on the applications of AI and use cases. Students also learn concepts and popular tools such as machine learning, data collection, classification, natural language processing, the functions of AI virtual assistants, and techniques used in AI for computer vision as domains to build AI solutions. The program is geared for those who want to enter or upskill into the field as AI Developers or AI Specialists.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
COP 1047C	Introduction to Python Programming	4.0
MAC 1105	College Algebra	3.0
Semester		Units: 12.0
CAI 2100C	Machine Learning Foundations	3.0
ENC 1101	English Composition 1	3.0
PHI 2680	Artificial Intelligence and Ethics	3.0
STA 2023	Statistical Methods	3.0
Semester 3		Units: 7.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
CGS 1540C	Database Concepts Design	4.0
Semester 4		Units: 14.0
CAI 2300C	Introduction to Natural Language Processing	3.0

CAI 2840C	Introduction to Computer Vision	3.0
CTS 1145	Cloud Essentials	4.0
Elective*		4.0

Select 4 credits from the following course option: CAI, CAP*, CEN*, CET*, CGS*, CIS*, CNT*, COP*, CTS*, MAC*, MAD*, MAP*, CAP1788, COP2800, CTS1120, ETS1603C, GEB1432, HSC2060.

Semester 5

Units: 13.0

CAI 2820C	Artificial Intelligence Applications Solutions	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1020	General Education Physics	3.0
Elective*		4.0

Select 4 credits from the following course option: CAI, CAP*, CEN*, CET*, CGS*, CIS*, CNT*, COP*, CTS*, MAC*, MAD*, MAP*, CAP1788, COP2800, CTS1120, ETS1603C, GEB1432, HSC2060.

Total: 60.0

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CAI2100C Machine Learning Foundations 3.000 credits

Students are introduced to machine learning concepts and Python applications, including data acquisition, supervised, unsupervised, and reinforced learning. In addition, students will develop and deploy artificial intelligence (AI) models utilizing classification algorithms. Prerequisites: CAI1001C, COP1047C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI1001CCOP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CAI2300C Introduction to Natural Language Processing 3.000 credits

Students will learn the fundamental concepts of Natural Language Processing (NLP) and text processing. In addition, focus will be on knowledge and skills necessary to create a language recognition application. Pre/Corequisite: CAI2100C.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: CAI2100C

Course User Fee: \$63.00

CAI2840C Introduction to Computer Vision 3.000 credits

Students will learn fundamental concepts in Computer Vision (CV) and image processing, including introduction to necessary proprietary and open-source Python libraries. Pre/Corequisite: CAI2100C.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: CAI2100C

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAI2820C Artificial Intelligence Applications Solutions 3.000 credits

A lower division course for students majoring in Applied Artificial Intelligence (AI). Students will demonstrate competence to scope, acquire/explore data, model, evaluate, and deploy an AI/machine learning solution in a team environment. Students will create and present a code or no-code AI solution. Must be taken during the last semester before graduation. Prerequisites: CAI2300C and CAI2840C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2840C AND CAI2300C

Course User Fee: \$63.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

Program Description

[Aviation Administration](#)

Associate in Science | Code: 26028 | 60 credits

The Aviation Administration program is designed to prepare students to succeed in the dynamic aviation industry. The program focuses on the necessary entry-level skills for most aviation employment fields. The Air Traffic Control option provides students with the opportunity to be hired with the Federal Aviation Administration (FAA). Accordingly, graduates

find opportunities in airline sales and reservations, air cargo, airport operations and many data-entry positions required by the airline management. Additional Information: Contact the Aviation Department at (305) 237-5950 for information and advisement.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 14.0
ATF 1601L	Flight Orientation Simulator Lab	1.0
AVM 1010	Aviation Industry Operation	3.0
AVM 1022	Flight Operations	3.0
AVM 1062	Aviation Career Planning	1.0
ENC 1101	English Composition 1	3.0
STA 2023	Statistical Methods	3.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 12.0
ASC 1210	Aviation Meteorology	3.0
AVM 1121	Hazardous Materials/Dangerous Goods	3.0
HUM 1020	Humanities	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 3		Units: 9.0
ASC 2320	Aviation Laws and Regulations	3.0
AST 1002	Descriptive Astronomy	3.0
AVM 1440	Aviation/Airport Security	3.0

Semester 4		Units: 12.0
AVM 2120	Air Cargo	3.0
AVM 2410	Principles of Airport Management	3.0
AVM 2441	Aviation Safety and Human Factors	3.0
AVM 2450	Airport Facilities/Financial Planning	3.0

Semester 5		Units: 12.0
ATT 2820	Air Traffic Control	3.0
AVM 2431	Customer Service Agent	3.0
AVM 2510	Airline Management	3.0
AVM 2515	Airline Marketing	3.0

Total: 59.0

ATF1601L Flight Orientation Simulator Lab 1.000 credits

This course will provide the student with an introduction to the environment of operating an aircraft from a pilot's point of view. It is designed to provide this knowledge to those students such as Air Traffic Controllers and Aviation Administration Students who have no piloting experience. Special fee.

Lab Hours: 32.000

Course User Fee: \$200.00

AVM1010 Aviation Industry Operation 3.000 credits

The course provides insight into the development and present status of aircraft and air transportation, governmental organizations, controls and regulations, and career opportunities in the field.

Lecture Hours: 48.000

AVM1022 Flight Operations 3.000 credits

An investigation of the occupational duties, responsibilities, and physical facilities required by the positions of pilot, co-pilot, flight engineer, dispatcher and flight attendant.

Lecture Hours: 48.000

AVM1062 Aviation Career Planning 1.000 credits

This course provides direction and guidance in career planning for all aviation students. Topics of discussion will include the job search education and training requirements, resume writing, business etiquette, interview skills and follow-up techniques

Lecture Hours: 16.000

ENC1101 English Composition 1 3.000 credits

This is a required general education core course in college-level writing. The students will learn to compose essays and other works using various methods of development. This course fulfills the Gordon Rule writing requirement and must be completed with a grade "C" or better. Pre-requisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Lecture Hours: 48.000

Course User Fee: \$15.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

ASC1210 Aviation Meteorology 3.000 credits

This is a core aviation course. The student will be prepared to understand weather and environmental issues in commercial aviation. Topics covered will be atmospheric phenomena relating to aircraft operations, the analysis and use of weather data as presented by the U.S. National Weather Service. Prerequisite: ATT1100 or equivalent; Corequisite: ATT2110 or equivalent.

Lecture Hours: 48.000

AVM1121 Hazardous Materials/Dangerous Goods 3.000 credits

This course is designed to provide the student with knowledge of dangerous goods/ hazardous materials and their effect in air transportation and logistics. The students will be conversant in hazardous material regulations for cargo and passenger transportation. The course will encompass the identification, labeling, packaging and handling of 9 types of dangerous goods in air transportation and general logistics. Prerequisite: AVM2120. Special fee.

Lecture Hours: 48.000

Prerequisite: AVM2120

Course User Fee: \$25.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ASC2320 Aviation Laws and Regulations 3.000 credits

Insight pertinent to federal governing bodies, and current local, federal and international laws forming the present structure of aviation law.

Lecture Hours: 48.000

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

AVM1440 Aviation/Airport Security 3.000 credits

This course will provide the student with knowledge of the issues and strategies that are used to protect the national airspace system, airports and airlines from security threats. The various types of threats and responses to those threats will be covered. In addition, the legal requirements, planning issues, physical equipment and facility requirements and personnel issues will also be discussed.

Lecture Hours: 48.000

Course User Fee: \$50.00

AVM2120 Air Cargo 3.000 credits

The course develops a comprehensive grasp of the characteristics and evolution of air cargo, its impact on United States industry, inherent problems and future development.

Lecture Hours: 48.000

AVM2410 Principles of Airport Management 3.000 credits

This course provides the student with a broad background in the Principles of Airport Management. This includes the airport system and its history, planning, land use, community relation issues, financial issues, capacity and growth, operations, organization and administration. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$25.00

AVM2441 Aviation Safety and Human Factors 3.000 credits

This course will provide the student with an understanding of human factors and safety concepts as they apply to aviation. There will be an evaluation of aircraft accidents and their causal factors. Accident prevention measures are stressed as integral parts of an aviation safety program.

Lecture Hours: 48.000

Course User Fee: \$25.00

AVM2450 Airport Facilities/Financial Planning 3.000 credits

This course provides the student with an in-depth knowledge of the techniques and strategies of the airport master plan in planning airport facilities and financial resources. Forecasting, demand analysis, sources of funding, planning requirements, environmental issues and requirements and compliance issues will be discussed. Also, implementation and control issues, financial management, budgets, costs and revenues as well as airport economics will be discussed. Prerequisite: AVM2410. Special fee.

Lecture Hours: 48.000

Prerequisite: AVM2410

Course User Fee: \$25.00

ATT2820 Air Traffic Control 3.000 credits

The basic elements of air traffic control operations, providing the necessary foundation for successful completion of the Air Traffic Control Basic Certification Examination. Prerequisite: sophomore standing in major program.

Lecture Hours: 48.000

Prerequisite: Sophomore standing in major program.

AVM2431 Customer Service Agent 3.000 credits

This course covers the generic skills needed for any airline position involving regular contact with the traveling public. It includes human relations, personal appearance enhancement, etiquette, conflict management, speech skills, and the acquisition of attributes that would promote a proper professional image.

Lecture Hours: 48.000

AVM2510 Airline Management 3.000 credits

This course offers an insight relative to the business policies and the functions of management in airline operations. The course involves various internal managerial facets and the impact of external regulatory and economic implications.

Lecture Hours: 48.000

AVM2515 Airline Marketing 3.000 credits

A differentiation of the functions of marketing in airline operations; market research, demand analysis, advertising and promotion, sales, traffic, and the theory of price determination.

Lecture Hours: 48.000

Program Description

Aviation Maintenance Management

Associate in Science | Code: 26027 | 83 credits

The Aviation Maintenance Management is a special program in which 45 semester hours are awarded to students who possess the Federal Aviation Administration Aircraft and Powerplant (A & P) certificate. The 38 additional required credits consist of general education and aviation requirements needed by the licensee for the Associate in Science degree. Additional Information: Contact the Aviation Department at (305) 237-5950 for information and advisement.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: Yes

Semester 1

Units: 12.0

ASC 2320	Aviation Laws and Regulations	3.0
AVM 1160	Aviation Maintenance Programs and Inspections	3.0
ENC 1101	English Composition 1	3.0
PHY 1020	General Education Physics	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT).

Semester 2

Units: 12.0

AVM 1161	Aircraft Performance Measures and Maintenance Requirements	3.0
AVM 1162	Maintenance Repair and Overhaul (MRO) Interactions with Commercial Airline Operations	3.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 3

Units: 9.0

ASC 1610	Aircraft Engines and Structure Theory	3.0
AVM 2510	Airline Management	3.0
PHI 2010	Introduction to Philosophy	3.0

Semester 4

Units: 51.0

AVM 1163	Policies and Procedures for Commercial Airlines Maintenance Programs	3.0
AVM 1164	Logistics and Maintenance Programs for Commercial Airlines	3.0
AVM 9996	FAA Certification	45.0

Total: 84.0

ASC2320 Aviation Laws and Regulations 3.000 credits

Insight pertinent to federal governing bodies, and current local, federal and international laws forming the present structure of aviation law.

Lecture Hours: 48.000

AVM1160 Aviation Maintenance Programs and Inspections 3.000 credits

This course provides an in-depth study of aircraft inspection programs and maintenance scheduling procedures. Students will learn national and international regulations governing aircraft inspection, maintenance evaluation, and the required procedures to update airline and governmental maintenance technical manuals.

Lecture Hours: 48.000

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

AVM1161 Aircraft Performance Measures and Maintenance Requirements 3.000 credits

Students will learn aircraft performance measures and maintenance requirements for airplanes powered by reciprocating, turboprop, and/or jet turbine and turbofan engines. Topics include stability and control, weight and balance, performance charts and graphs, takeoff and cruise control, airplane performance characteristics, from which they will extract data that maximizes performance.

Lecture Hours: 48.000

AVM1162 Maintenance Repair and Overhaul (MRO) Interactions with Commercial Airline Operations 3.000 credits

Students will learn the Maintenance Repair and Overhaul (MRO) Procedures in Commercial Airline Operations. Topics include airline maintenance operations, engineering, maintenance, repair of structures, systems, and aircraft components. In addition, students will explore MRO financing, domestic and offshore operations, regulatory requirements, logistics, supply chain support, human resources and industry oversight.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view. Lecture Hours: 48.000

ASC1610 Aircraft Engines and Structure Theory 3.000 credits

This is a foundation course in aircraft engines and structure. Students will learn the elements of aircraft engines, engine theory, construction, systems, operating procedures, performance diagnosis, and aircraft structures. Lecture Hours: 48.000

AVM2510 Airline Management 3.000 credits

This course offers an insight relative to the business policies and the functions of management in airline operations. The course involves various internal managerial facets and the impact of external regulatory and economic implications. Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

AVM1163 Policies and Procedures for Commercial Airlines Maintenance Programs 3.000 credits

Students will learn the maintenance policies and programs for commercial airlines. Concepts of Maintenance Steering Group (MSG) and Reliability Centered Maintenance (RCM) programs, Maintenance Control by Reliability Methods (MCEM) program, and Operational Availability (OA) for Commercial Aircraft will be discussed.

Lecture Hours: 48.000

AVM1164 Logistics and Maintenance Programs for Commercial Airlines 3.000 credits

This course focuses on the concepts and application of logistics and supply chain management utilized within the aviation maintenance industry to increase efficiency in production and maintenance. Students will learn the logistics support from Maintenance Repair Operators and Original Equipment Manufacturers in aviation maintenance operations.

Lecture Hours: 48.000

AVM9996 FAA Certification 45.000 credits

The Federal Aviation Administration (FAA) Certification placeholder awards equivalent college credit for non-collegiate training based on Technical Manual of Procedure Number 1100:815701(A) and 1100:815701B.

Students must provide original FAA certificate in order to receive credit.

Lecture Hours: 720.000

Program Description

Biomedical Engineering Technology

Associate in Science | Code: 26053 | 62 credits

The Biomedical Engineering Technology program prepares students for employment as Biomedical Engineering Technicians/Technologists and in related occupations in health-related fields. The program also provides supplemental training for persons currently or previously employed in these occupations. The program focuses on the understanding and applying of concepts in electronics, in addition to troubleshooting techniques, to digital, microprocessor, or computer-based systems as they relate to medical devices. Assembly, installation, operations maintenance, calibration, trouble-shooting, repairing and elementary design on medical systems are taught using an integrated, applied and theoretical approach.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
CET 1487C	Network+	3.0
COP 2270	C for Engineers	4.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Semester 2		Units: 13.0

EET 1015C	Direct Current Circuits	4.0
HIM 2472	Medical Terminology	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
MAC 1114	Trigonometry	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT).

Semester **Units: 8.0**

CET 1110C	Digital Circuits	4.0
EET 1025C	Alternating Current Circuits	4.0

Semester 4 **Units: 14.0**

		3.0
BSC 1005	General Education Biology	3.0
OR		
BSC 2085	Human Anatomy and Physiology 1	3.0
EET 1141C	Electronics 1	4.0
ETS 2673C	Programmable Logic Controls	4.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 5 **Units: 14.0**

EET 2101C	Electronics 2	4.0
Electives*		10.0

*Select 8 credits from the following course option: CET2123C, EET2323C, EET2351C, ETM1315C, ETS2163C.

Total: 62.0

CET1487C Network+ 3.000 credits

This is an intermediate level course designed for students preparing for the hardware component of the Network+ certification. Students will learn how to install, configure, manage, troubleshoot and upgrade network devices including network interface cards, switches, hubs, wireless access points, routers, and patch panels. They will also learn about the construction, installation, testing and repair of the physical layer of the network, including wired cables, fiber optic media, wireless transmitters and antennas. Demonstrated knowledge of microcomputer fundamentals and system components is required. Recommended Preparation: CET1178C or A+ certification. Laboratory fee. A.S. degree credit only.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2270 C for Engineers 4.000 credits

This course is intended for students majoring in Computer Engineering Technology, Electronics Engineering Technology, or any engineering discipline. Students will learn the "C" programming language, MATLAB, and the Engineering Problem Solving Method to analyze, design, code, compile and execute programs that solve engineering related problems. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105, Recommended Preparation: CGS1060 OR knowledge of computer skills

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

EET1015C Direct Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn electrical components, inductance, capacitance, and elementary network analysis. Students will additionally learn to verify and apply basic theories and principles through hands-on, laboratory experiments utilizing modern testing equipment. Pre/corequisite: MAT 1033. Laboratory fee. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 48.000

Pre/Corequisite: MAT1033

Course User Fee: \$63.00

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

CET1110C Digital Circuits 4.000 credits

This is an introductory course in digital circuits for engineering and other technology majors. Students learn how to apply electronic principles to digital computer circuits and systems. Students will also learn how to simplify logic circuits, build digital circuits, and perform other laboratory activities. Pre/Co-requisite: MAC1105. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

EET1025C Alternating Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn inductance, capacitance, vector notation, AC circuits, impedance, phase shift, networks, transformers, and resonance. Students will apply and verify theories and principles through hands-on, laboratory experiments utilizing modern testing equipment.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1015C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

EET1141C Electronics 1 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology or related fields. Students will learn how to apply electronic principles to analog circuits and systems, including semiconductor diodes, applying the fundamental theory of transistors and other solid-state devices; analysis of amplifiers, oscillators, and other applications using a sinusoidal wave. Students also learn basic safety procedures to follow when working in an electronics laboratory and with electronic circuits and systems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1025C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

ETS2673C Programmable Logic Controls 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology and Industrial Engineering Technology. Students will learn the principals of Programmable Logic Controls (PLC) including hardware, programming, and troubleshooting. Students will also develop advanced working programs, and troubleshoot hardware and software communication problems. Prerequisite: CET110C

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET1110C

Course User Fee: \$63.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

EET2101C Electronics 2 4.000 credits

This course is intended for students majoring in electronics or computer engineering technologies. Students will learn how to apply electronic principles to analog circuits including transistor amplifiers, feedback and frequency response of linear circuits, operational amplifiers, MOSFET and oscillators.

Lecture Hours: 32.000, Lab Hours: 48.000

Prerequisite: EET1141C

Course User Fee: \$63.00

Program Description

Biotechnology

Associate in Science | Code: 22027 | 61 credits

The Biotechnology Program exposes students to a breadth of topics and emphasizes hands-on learning in a variety of techniques and procedures necessary for employment in the bioscience industry. The program includes modules designed to enhance critical thinking and technical communication skills. It focuses on developing broad transferable skills and stresses understanding and demonstration of laboratory/industry protocols and regulations, bio-safety and safe operating procedures, ethical and environmental issues, product generation/formulation, quality control, validation, instrumentation, and computing.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 14.0

BSC 2010	Principles of Biology	3.0
BSC 2010L	Principles of Biology 1 Laboratory	2.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Social Science*		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Computer Competency Test*

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
 Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
BSC 2011	Principles of Biology 2	3.0
BSC 2011L	Principles of Biology 2 Laboratory	2.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0

Semester 3		Units: 10.0
BSC 2426	Biotechnology Methods and Applications 1	3.0
BSC 2426L	Biotechnology Methods and Applications 1 Laboratory	2.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0

Semester 4		Units: 13.0
BSC 2427	Biotechnology Methods and Applications 2	3.0
BSC 2427L	Biotechnology Methods and Applications 2 Laboratory	2.0
CHM 2210	Organic Chemistry 1	3.0
CHM 2210L	Organic Chemistry 1 Laboratory	2.0
STA 2023	Statistical Methods	3.0

Semester 5		Units: 11.0
BSC 2943L	Bioscience Internship	3.0
CHM 2211	Organic Chemistry 2	3.0
CHM 2211L	Organic Chemistry 2 Laboratory	2.0
COP 1047C	Introduction to Python Programming	4.0

Total: 61.0

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L, Corequisite: BSC2010

Course User Fee: \$40.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

CHM 1045 is the first semester of a two-semester general chemistry course for science, premedical science and engineering students. Students will learn stoichiometry, atomic structure, completing and balancing chemical reactions, nomenclature, bonding, acid-base theories, solutions, gas laws and beginning thermodynamics.

Special fee.

Lecture Hours: 48.000

Prerequisite: CHM1025or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC2426 Biotechnology Methods and Applications 1 3.000 credits

This course addresses the basic principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical, biotechnology and/or research laboratory setting(s). Practical applications of biotechnology are explored.

Lecture Hours: 48.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426L

BSC2426L Biotechnology Methods and Applications 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2426 Biotechnology Methods and Applications 1. This is a hands-on course that emphasizes the basic laboratory principles, techniques, and instrumentation, necessary for effective work in pharmaceutical, biotechnology, and/or research laboratory settings(s).

Lab Hours: 64.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426

Course User Fee: \$50.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

BSC2427 Biotechnology Methods and Applications 2 3.000 credits

This course addresses advanced principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical-biotechnology and/or research-laboratory setting(s). The following areas of contemporary biotechnology are covered: forensics, bioremediation, and animal-, plant-, and marine biotechnology.

Lecture Hours: 48.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427L

BSC2427L Biotechnology Methods and Applications 2 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2427 Biotechnology Methods and Applications 2. This is a hands-on course that emphasizes advanced laboratory principles, techniques, and instrumentation necessary for effective work in a pharmaceutical, biotechnology, and/or research-laboratory setting(s).

Lab Hours: 64.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427

Course User Fee: \$50.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives. Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM 2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting.

Lab Hours: 64.000

Prerequisite: CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

BSC2943L Bioscience Internship 3.000 - 6.000 credits

This internship course is a capstone for students majoring in bioscience and related programs. Students will learn to apply acquired knowledge and skills to gain experience in the bioscience workplace.

Other Hours: 144.000 - 288.000

Course User Fee: \$21.00

CHM2211 Organic Chemistry 2 3.000 credits

In organic chemistry 2, students will learn about nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds.

Lecture Hours: 48.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211L

Course User Fee: \$15.00

CHM2211L Organic Chemistry 2 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2211. Topics include nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds will be performed in a laboratory setting.

Lab Hours: 64.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211

Course User Fee: \$45.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

Program Description

Building Construction Technology

Associate in Science | Code: 26033 | 60 credits

The Building Construction Technology program is designed to furnish technically trained personnel for the building construction industry. The graduate may work with a contractor as part of the administrative team in such entry-level job positions as those leading to estimators, job coordinators, or project managers. Technical jobs may also be available in the following areas: land and project developers; technical sales for building materials, systems, and equipment; with local, state, and federal government agencies; as well as various financial institutions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ARC 1126C	Architectural Drawing 1	4.0
BCN 1272	Building Construction Plans Interpretation 1	3.0
BCN 1275	Building Construction Plans Interpretation 2	3.0
ENC 1101	English Composition 1	3.0
Computer Competency Test*		

* Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 12.0
BCT 1770	Building Construction Estimating Fundamentals	3.0
BCT 1771	Building Construction Advanced Estimating	3.0
MAC 1105	College Algebra	3.0
Electives *		3.0

Select 3 credits from the following course option: ARC, BCN*, BCT*, CGS1060C, ETM1700.

Semester 3	Units: 10.0
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ARC 2171C	Computer Aided Drafting 1	4.0
MAC 1114	Trigonometry	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 4		Units: 13.0
ARC 2461	Architectural Materials and Construction 1	4.0
BCT 1743	Building Construction Law	3.0
BCT 1750	Building Construction Financing	3.0
BCT 2760	Building Code Regulations	3.0
Semester 5		Units: 12.0
ARC 2681	Environmental Technology	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Natural Science</i>		3.0
AST 1002	Descriptive Astronomy	3.0
OR		
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
GLY 1010	Physical Geology	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0
OR		
PHY 1020	General Education Physics	3.0
SUR 1001C	Construction Survey	3.0
		Total: 60.0

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions. Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

BCN1272 Building Construction Plans Interpretation 1 3.000 credits

Develops the students' ability to interpret working drawings. Students will learn the conventions of graphic and symbolic language used by construction professionals to communicate information on drawings. Emphasis is on architectural and structural details with limited coverage on mechanical and electrical aspects. Develops the students' ability to interpret working drawings. Students will learn the conventions of graphic and symbolic language used by construction professionals to communicate information on drawings. Emphasis is on architectural and structural details with limited coverage on mechanical and electrical aspects.

Lecture Hours: 48.000

Course User Fee: \$10.00

BCN1275 Building Construction Plans Interpretation 2 3.000 credits

In this course, students will plan interpretation of complex working drawings for multi-story residential and commercial buildings. Students will learn to read and understand construction working drawings, identify structural systems and their details. Familiarity with all aspects of complex working drawings will be addressed.

Prerequisite: BCN1272.

Lecture Hours: 48.000

Prerequisite: BCN1272

Course User Fee: \$10.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BCT1770 Building Construction Estimating Fundamentals 3.000 credits

This course is an analysis and calculation of building construction costs. Students will learn the classification of materials, labor, and subcontracted work into the smallest manageable units. Students will also develop a simple estimate for a residential structure. Prerequisite: BCN1272.

Lecture Hours: 48.000

Prerequisite: BCN1272

Course User Fee: \$10.00

BCT1771 Building Construction Advanced Estimating 3.000 credits

Estimating more advanced elements of building construction. Students will learn to calculate direct, indirect, and overhead costs, as well as prepare bid proposals and related documents for commercial buildings.

Prerequisites: BCN1275, BCT1770.

Lecture Hours: 48.000

Prerequisite: BCN1275, BCT1770

Course User Fee: \$10.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

ARC2171C Computer Aided Drafting 1 4.000 credits

Computer-aided drafting as it applies in the fields of architecture and interior design using office simulation. Emphasis is on the production of computer-aided drafting of working drawings involving different types of structure.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR ARC2461

Course User Fee: \$50.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ARC2461 Architectural Materials and Construction 1 4.000 credits

An introduction to basic materials and methods of building construction. Emphasis is on wood, concrete, unit masonry, and light steel construction. Laboratory projects may include working drawings' interpretation, sketching construction details, or field trips to construction sites and fabricant plants. Designed primarily as the initial materials and methods course for architectural transfer students.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR BCN1251

Course User Fee: \$25.00

BCT1743 Building Construction Law 3.000 credits

In this course, students will learn the legal aspects of construction contracts and the responsibilities arising particularly from the field operations. This course also includes the relationship of the general contractor to owner, architect, and subcontractor; material men and mechanics lien law; bonds; labor law; and other statutes and ordinances regulating contractors.

Lecture Hours: 48.000

Course User Fee: \$10.00

BCT1750 Building Construction Financing 3.000 credits

This course is a study of building construction financing and related contract requirements. Topics include construction loans, permanent building mortgages, construction bids and contracts, penalty and incentive provisions, progress payments and retention, escalation, escalation provisions, costs extras, performance and bid bonds, company profits, cash flow, and business loans.

Lecture Hours: 48.000

Course User Fee: \$10.00

BCT2760 Building Code Regulations 3.000 credits

In this course, students will learn the restrictions and limitations of the various agencies concerned with the building industry. Provisions of the South Florida Building Code are stressed.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2681 Environmental Technology 3.000 credits

An introduction to technological aspects of building design, which relates to human comfort, safety, and building performance. Includes a survey of the fundamentals of water supply, waste lines, plumbing equipment, heat and air conditioning, solar applications, and electrical components and equipment in the design and construction of buildings. Prerequisite: ARC1126C.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

SUR1001C Construction Survey 3.000 credits

Practice of surveying as related to the building and construction industry. Includes a combination of classroom instruction and practical field problems with the tape, level and transit.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147

Course User Fee: \$55.00

Program Description

Associate in Science | Code: 25051 | 60 credits

This program transfers to four-year institutions. See department for information. The Business Administration program trains individuals to assume management or supervisory positions in business, industry, and government. It provides basic skills in a broad range of business functions including accounting, computer usage, management and marketing. Successful completion of this program earns the student entry into any university in the State University System as part of the A.S. to B.S. program.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 15.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
PHI 2010	Introduction to Philosophy	3.0
ECO 2023	Principles of Economics (Micro)	3.0
GEB 1011	Principles of Business	3.0

Semester 2		Units: 14.0
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<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

<i>Financial Accounting</i>		4.0
<i>Option 1</i>		4.0
ACG 2011	Principles of Accounting II	3.0
ACG 2011L	Principles of Accounting 2 Lab	1.0
OR		
<i>Option 2</i>		4.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0

CGS 1060C	Introduction to Computer Technology and Applications	4.0
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Semester 3		Units: 7.0-10.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
Program Electives*		3.0-6.0

Select 6-9 credits from the following course options: ACG, BAN*, BUL*, ECO*, ENC*, FIN*, GEB*, HFT*, LIS*, MAC*, MAN*, MAR*, MKA*, MNA*, MTB*, OST*, PUR*, QMB*, SBM*, SPC*, SPM*, TAX*, TRA

Semester 4**Units:** 12.0-18.0**Program Core Requirements**

12.0-18.0

Track 1 – Business Management (15.00 credits)¹⁾

OR

Track 2 – Marketing (18.00 credits)²⁾

OR

Track 3 – Human Resources (15.00 credits)³⁾

OR

Track 4 – International Business (12.00 credits)⁴⁾

OR

Track 5 – Small Business (12.00 credits)⁵⁾¹⁾MAN 2021, MAN 2604, MAR 1011, MNA 1345, SBM 1000²⁾MAR 1011, MAR 1720, MAR 2101, MAR 2150, MAR 2520, MKA 1511³⁾MAN 2021, MAN 2300, MAR 2101, MNA 1345, MNA 2120⁴⁾FIN 2051, GEB 2350, MAN 2604, MAR 2150⁵⁾GEB 2112, MAN 2021, MAR 1011, SBM 1000**Semester 5****Units:** 12.0-3.0

Program Elective*

12.0-3.0

Select 3-12 credits from the following course options: ACG, BAN*, BUL*, ECO*, ENC*, FIN*, GEB*, HFT*, LIS*, MAC*, MAN*, MAR*, MKA*, MNA*, MTB*, OST*, PUR*, QMB*, SBM*, SPC*, SPM*, TAX*, TRA

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033 Any Mathematics level higher than MAT 1033 will satisfy this requirement.

MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law. Lecture Hours: 48.000

Course User Fee: \$20.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ACG2011 Principles of Accounting II 3.000 credits

This is the second part of the introductory financial accounting series, which includes ACG2001 and ACG2011, with emphasis on accounting owner's equity and corporate financial statements. Other topics include receivables, fixed assets, intangible assets, current and long-term liabilities. Co-requisite: ACG2011L

Lecture Hours: 48.000

ACG2011L Principles of Accounting 2 Lab 1.000 credits

Students will use personal computer software and financial accounting applications to analyze accounting records, prepare financial statements, and compile EXCEL spreadsheets. This course will reinforce, with tutorial help and problem-solving, the concepts needed to achieve the objectives of ACG2011. Corequisite: ACG2011.

Lab Hours: 32.000

Corequisite: ACG2001

Course User Fee: \$22.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

Program Description

Business Intelligence Specialist

Associate in Science | Code: 25073 | 60 credits

The Associate in Science in Business Intelligence Specialist prepares students for employment as business intelligence analysts and related occupations and/or for upper division studies in data analytics. Business intelligence is a broad category of application programs and technologies for gathering, storing, analyzing, and providing access to data to help enterprise users make improved business decisions. Students will learn how to build business intelligence applications and how to manipulate massive amounts of data, turning it into useful information and reports.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

CGS 1060C	Introduction to Computer Technology and Applications	4.0
ENC 1101	English Composition 1	3.0
GEB 1011	Principles of Business	3.0
MAC 1105	College Algebra	3.0

Semester 2

Units: 14.0

CAP 1788	Introduction to Data Analytics	4.0
CGS 1540C	Database Concepts Design	4.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 3

Units: 7.0

STA 2023	Statistical Methods	3.0
Electives*		4.0

* Select 4 credits from the following course option: ACG1403, ACG2450, CAI1001C, CAI1320C, CAI2100C, CAI2300C, CAI2840C, COP2041C, CTS1145, CTS2192C, GEB1432, MAC1114, MAC1140, MAC1147, MAC2311, MAC2312, QMB2100.

Semester 4

Units: 12.0

CAP 2761C	Intermediate Analytics	4.0
COP 1047C	Introduction to Python Programming	4.0
Electives*		4.0

* Select 4 credits from the following course option: ACG1403, ACG2450, CAI1001C, CAI1320C, CAI2100C, CAI2300C, CAI2840C, COP2041C, CTS1145, CTS2192C, GEB1432, MAC1114, MAC1140, MAC1147, MAC2311, MAC2312, QMB2100.

Semester 5

Units: 14.0

CAP 2743C	Power BI - Data Visualization and Analysis	4.0
CAP 2791C	Power BI - Data Preparation and Modeling	4.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0

OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
Electives *		3.0

* Select 3 credits from the following course option: ACG1403, ACG2450, CAI1001C, CAI1320C, CAI2100C, CAI2300C, CAI2840C, COP2041C, CTS1145, CTS2192C, GEB1432, MAC1114, MAC1140, MAC1147, MAC2311, MAC2312, QMB2100.

Total: 60.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CAP1788 Introduction to Data Analytics 4.000 credits

This course offers a broad introduction to data analytics and the role it plays in modern organizations.

Students will use Excel and Tableau to import, clean, transform, and visualize data. Students will also learn how to effectively communicate findings to decision makers.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CAP2761C Intermediate Analytics 4.000 credits

This course offers an introduction to the SQL language and how it can be used by data analysts to query databases and gain valuable insights. Students will also gain extensive hands-on experience with Microsoft SQL Server, Azure, and GitHub. Prerequisites: CAP1788, CGS1540C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1540C

Course User Fee: \$63.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAP2743C Power BI - Data Visualization and Analysis 4.000 credits

This is the second of two introductory courses aimed at preparing students for the Microsoft Power BI certification exam. Students will learn how to create informative data visualizations and leverage Power BI's analytic capabilities to provide meaningful business insights. Prerequisite: CAP2791C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP2791C

Course User Fee: \$63.00

CAP2791C Power BI - Data Preparation and Modeling 4.000 credits

This is one of two introductory courses aimed at preparing students for the Microsoft Power BI certification exam. Students will learn how to get data from different sources, clean and transform the data, and design a data model that will be used in analysis and reports. Prerequisites: CAP1788 and CGS1540C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND CGS1540C

Course User Fee: \$63.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

Program Description

Civil Engineering Technology

Associate in Science | Code: 26035 | 63 credits

The Civil Engineering Technology program is designed for those students who wish immediate job placement prior to or after graduation. This program also satisfies many of the civil engineering freshman and sophomore requirements for the Bachelor of Engineering Technology degree offered by certain universities. Consult your Civil Engineering advisor prior to registration.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
ETD 1110	Technical Drawing	4.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
MAC 1105	College Algebra	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
 Departmental Recommendation: Computer Competency Test (CCT).

Semester 2		Units: 14.0
ETD 1542	Structural Drafting	4.0
ETG 2502	Statics	3.0
MAC 1114	Trigonometry	3.0
Major Course Elective*		4.0

*Select 4 credits from the following course option: EGN2200, EGS1111C, EGS1220C, ETC1250, ETC1930, ETC2201, ETC2207, ETC2210C, ETI2121, ETM1313C.

Semester 3		Units: 9.0
<i>Major Course Elective</i>		3.0
SUR 1001C	Construction Survey	3.0
<i>Natural Science</i>		3.0
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
GLY 1010	Physical Geology	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0
OR		
PHY 1020	General Education Physics	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4		Units: 13.0
ETC 2450	Concrete Construction	3.0
ETG 2530C	Strength of Materials	3.0
SUR 1101C	Surveying 1	4.0
SUR 2462C	Subdivisions	3.0

Semester 5		Units: 14.0
SUR 1202	Surveying 2	4.0
SUR 2301C	Topography and Mapping	4.0
SUR 2400C	Land Surveying 1	3.0
SUR 2404C	Land Surveying 2	3.0

Total: 63.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ETD1110 Technical Drawing 4.000 credits

Introduces students to the principles of instrument drawing, orthographic projection, visualization, specialized computer processes and introductory computer aided drawing (CAD). Students develop drawing and sketching techniques common to industry.

Lecture Hours: 32.000, Lab Hours: 64.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

ETD1542 Structural Drafting 4.000 credits

Development of structural, fabrication and erecting drawings. Course involves study of structural shapes, their properties, and methods of developing connections, as well as study of common reinforced concrete practices. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETD1110

Course User Fee: \$63.00

ETG2502 Statics 3.000 credits

The application of dead and live loads to rigid bodies at rest, including the force and moment of laws of equilibrium, determination of the direction and intensity of reactions, moments and stress in the design of engineering and architectural structures.

Lecture Hours: 48.000

Prerequisite: MAC1105

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

SUR1001C Construction Survey 3.000 credits

Practice of surveying as related to the building and construction industry. Includes a combination of classroom instruction and practical field problems with the tape, level and transit.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147

Course User Fee: \$55.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ETC2450 Concrete Construction 3.000 credits

The use of concrete in construction to include foundations, columns, beams, slabs, hydraulic conduits.

Lecture Hours: 48.000

Prerequisite: ETG2502

ETG2530C Strength of Materials 3.000 credits

The elastic behavior of materials under tension, compression, direct shear, torsion, bending, and column action. Problems in analysis and design of members under single and combined loading. Laboratory tests and demonstrations to illustrate the theory.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: ETG2502

SUR1101C Surveying 1 4.000 credits

The theories and practices in surveying and the use of the principal types of surveying instruments in horizontal and vertical planes. Problems include the measurement of distance; the use of compass, sextant, transit traverse, stadia, and basic mapping. Field and laboratory practice are required. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147

Course User Fee: \$55.00

SUR2462C Subdivisions 3.000 credits

Actual plant construction from tract description to final record plat. Research current plat laws and regulations, meaning of legal parts, necessary computations and graphical representation.

Lecture Hours: 64.000

SUR1202 Surveying 2 4.000 credits

Advanced study in route, land, and mapping surveying to include triangulation, astronomic observations, topographic and photogrammetric mapping. Field demonstrations and surveys performed with many modern types of survey instruments. Laboratory fee.

Lecture Hours: 32.000 - 30.000, Lab Hours: 64.000 - 6.000

Prerequisite: SUR1001C

Course User Fee: \$55.00

SUR2301C Topography and Mapping 4.000 credits

Field and engineering graphic methods of map construction. Mapping principles and their application in producing topographic, land survey, hydrographic, and photographic maps. Field work includes control and detail surveys using transit-tape and transit-stadia, plane table and alidade applications. Measuring areas of straight or irregular boundaries with the planimeter.

Lecture Hours: 96.000

Prerequisite: Permission of Department Chairperson.

SUR2400C Land Surveying 1 3.000 credits

The theory and practice of land surveying, subdivision, filing and recording deeds; United States Government survey of public land; Florida laws governing land surveys, descriptions, coordinate systems and professional licensing. Field surveys are performed.

Lecture Hours: 64.000

Prerequisite: SUR1101C

SUR2404C Land Surveying 2 3.000 credits

Emphasizes the practice of surveying and the writing of legal descriptions. Areas studied include legal principles of retracement under Federal rules, combination of sequence and simultaneous conveyances, locating reversion rights, riparian and littoral owners, state statutes and regulations, standards of practice, field and office guidelines, performing the survey, legal authority and liability of the surveyor.

Lecture Hours: 64.000

Prerequisite: SUR2400C

Program Description

Clinical Laboratory Science

Associate in Science | Code: 23024 | 76 credits

The Clinical Laboratory Science program prepares the graduate to work as part of the health care delivery team in a non-profit clinical laboratory or research laboratory. Clinical practice is conducted in local health care facilities under the supervision of qualified registered professional personnel. Graduates are eligible for Florida State Licensure and Registry with the American Society of Clinical Pathologists and equivalent licensure.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 9.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester		Units: 11.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
PHI 2010	Introduction to Philosophy	3.0
Semester 3		Units: 9.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0
Semester 4 (In Program Milestone)		Units: 9.0

MLT 1040L	Fundamentals of Laboratory Operations	1.0
MLT 1210C	Clinical Urinalysis with Lab	2.0
MLT 1300	Clinical Hematology	2.0
MLT 1300L	Clinical Hematology Laboratory	2.0
MLT 1752	Quality Control Laboratory Mathematics	2.0

Semester 5 (In Program Milestone)	Units: 9.0
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MLT 1500	Clinical Immunology/Serology	2.0
MLT 1500L	Clinical Immunology/Serology Laboratory	1.0
MLT 1610	Clinical Chemistry 1	2.0
MLT 1610L	Clinical Chemistry 1 Laboratory	2.0
MLT 2440	Clinical Microbiology 1	1.0
MLT 2440L	Clinical Microbiology Lab 1	1.0

Semester 6 (In Program Milestone)	Units: 6.0
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MLT 1330	Clinical Coagulation	1.0
MLT 1300L	Clinical Hematology Laboratory	2.0
MLT 2525	Immunohematology	2.0
MLT 2525L	Immunohematology Laboratory	2.0

Semester 7 (In Program Milestone)	Units: 9.0
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MLT 2403	Clinical Microbiology 2	2.0
MLT 2403L	Clinical Microbiology Lab 2	2.0
MLT 2620	Clinical Chemistry 2	2.0
MLT 2620L	Clinical Chemistry 2 Laboratory	1.0
MLT 2624L	Special Techniques In Clinical Chemistry	2.0

Semester 8 (In Program Milestone)	Units: 14.0
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MLT 2807L	Hospital Practicum: Immunohematology	3.0
MLT 2809L	Hospital Practicum: Hematology	3.0
MLT 2810L	Hospital Practicum: Chemistry	3.0
MLT 2811L	Hospital Practicum: Microbiology	3.0
MLT 2930	Medical Laboratory Technology Seminar	2.0

Total: 76.0

ENC1101 English Composition 1 3.000 credits

This is a required general education core course in college-level writing. The students will learn to compose essays and other works using various methods of development. This course fulfills the Gordon Rule writing requirement and must be completed with a grade "C" or better. Pre-requisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Lecture Hours: 48.000

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

**BSC2085 Human Anatomy and Physiology 1
3.000 credits**

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

**BSC2085L Human Anatomy and Physiology 1
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

**CHM1033 Chemistry for Health Sciences 3.000
credits**

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions. Corequisites: CHM 1033L, MAT 1033 (3 hr. lecture)

Lecture Hours: 48.000

Corequisite: CHM1033L, Prerequisite: MAT1033

Course User Fee: \$15.00

**CHM1033L Chemistry for Health Sciences Lab
1.000 credits**

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite:

MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

**BSC2086 Human Anatomy & Physiology 2 3.000
credits**

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

**BSC2086L Human Anatomy & Physiology 2
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

MLT1040L Fundamentals of Laboratory Operations 1.000 credits

Collection of blood by venipuncture, skin puncture and donor room techniques. This includes handling of specimens, professional ethics, basic anatomy and physiology of the circulatory system, medical terminology and safety practices, including those for AIDS patients.

Lecture Hours: 32.000

Course User Fee: \$184.16

MLT1210C Clinical Urinalysis with Lab 2.000 credits

Theoretical concepts and practice in the collection and analysis of urine and other body fluids by combination of didactic and laboratory instruction. Performance of routine urinalysis procedures, including microscopy with identification of related disease states.

Lecture Hours: 48.000

Course User Fee: \$65.00

MLT1300 Clinical Hematology 2.000 credits

Didactic study of blood cells to include the origin, morphology, function and dysfunction of cells and related disease states of the blood. Theoretical concepts and principles of routine hematology procedures, quality control and instrumentation.

Lecture Hours: 32.000

Corequisite: MLT1300L

MLT1300L Clinical Hematology Laboratory 2.000 credits

Manual and automated procedures in hematology. This includes blood cell counts and other basic hematologic procedures in the simulated laboratory and in the clinical setting.

Lab Hours: 64.000

Corequisite: MLT1300

Course User Fee: \$90.00

MLT1752 Quality Control Laboratory Mathematics 2.000 credits

Emphasis on mathematical computations related to procedures in the clinical laboratory including dilutions, solutions, colorimetry, hematology math, enzymatic calculations, calculations relating to renal function tests, and mathematical principles related to ionic solutions. The student will also be given specific statistical tools necessary for quality control procedures as well as interpretations of Levy-Jennings charts and troubleshooting tools.

Lecture Hours: 32.000

MLT1500 Clinical Immunology/Serology 2.000 credits

Theoretical concepts of the human immune system in health and disease. Relationships to immunohematology, infection, and serological procedures are analyzed.

Lecture Hours: 32.000

Prerequisite: BSC 2085, BSC 2086, Corequisite: MLT1500L

MLT1500L Clinical Immunology/Serology Laboratory 1.000 credits

Performance of serological procedures that are identified in MLT 1500. The clinical significance of test results to disease states is included.

Lab Hours: 32.000

Prerequisite: BSC 2085L, BSC 2086L, Corequisite: MLT1500

Course User Fee: \$49.00

MLT1610 Clinical Chemistry 1 2.000 credits

Theoretical concepts and principles of carbohydrate, nonprotein nitrogen, and electrolyte chemistry analyses with emphasis on their relationships to various disease states. Analytical procedures to assess liver function and acid-base balance are also included.

Lecture Hours: 32.000

Prerequisite: CHM1025, Corequisite: MLT1610L

MLT1610L Clinical Chemistry 1 Laboratory 2.000 credits

Performance of chemistry procedures on body fluids with emphasis on manual and automated instrumentation.

Lab Hours: 64.000

Prerequisite: CHM1025L, Corequisite: MLT1610

Course User Fee: \$40.00

MLT2440 Clinical Microbiology 1 1.000 credits

This course will provide an overview of clinical mycology and parasitology. Topics will include both parasites and fungi and will cover life cycles, epidemiology, and etiology. Emphasis will be given to the most commonly encountered mycoses and parasitic infestations. This course should be taken concurrently with Clinical Microbiology 1 Lab.

Lecture Hours: 16.000

Corequisite: MLT 2400L

MLT2440L Clinical Microbiology Lab 1 1.000 credits

This course provides a practical overview of mycology and parasitology. Students will also obtain hands-on experience working with formalin preserve ova and parasites. They will also obtain the knowledge necessary to be able to identify at least the genus level of the most commonly encountered yeasts and fungi using microscopic and macroscopic techniques. This course should be taken concurrently with Clinical Microbiology.

Lab Hours: 32.000

Corequisite: MLT2440

Course User Fee: \$50.00

MLT1330 Clinical Coagulation 1.000 credits

Didactic study of hemostasis, various clotting mechanisms, and related disease states.

Lecture Hours: 16.000

Prerequisite: MLT1300, Corequisite: MLT 1330L

MLT1300L Clinical Hematology Laboratory 2.000 credits

Manual and automated procedures in hematology. This includes blood cell counts and other basic hematologic procedures in the simulated laboratory and in the clinical setting.

Lab Hours: 64.000

Corequisite: MLT1300

Course User Fee: \$90.00

MLT2525 Immunohematology 2.000 credits

Theoretical concepts involving blood group systems, hemolytic diseases, and blood bank procedures relating to transfusion and component therapy.

Lecture Hours: 32.000

Prerequisite: MLT1500, Corequisite: MLT2525L

MLT2525L Immunohematology Laboratory 2.000 credits

Performance of basic blood typing, blood bank assays on prepared specimens, and appropriate quality control procedures. Interpretation of results is included.

Lab Hours: 64.000

Prerequisite: MLT1500L, Corequisite: MLT2525

Course User Fee: \$45.00

MLT2403 Clinical Microbiology 2 2.000 credits

This course will provide a working knowledge of clinical bacteriology and should complement the Microbiology 2 Lab. The student will be exposed to some of the indigenous flora and the pathogenicity of microorganisms as they affect various body sites. Specimen transport, collection, laboratory identification techniques, and antimicrobial therapy also provides the knowledge base necessary for working in a clinical setting.

Lecture Hours: 32.000

Prerequisite: MLT2440, Corequisite: MLT2403L

MLT2403L Clinical Microbiology Lab 2 2.000 credits

This course is designed to complement the Microbiology 2 lecture and provide students with the necessary knowledge base and laboratory skills to effectively identify microorganisms associated with infectious diseases.

Lab Hours: 64.000

Prerequisite: MLT2440L, Corequisite: MLT2403

Course User Fee: \$50.00

MLT2620 Clinical Chemistry 2 2.000 credits

Theoretical concepts and principles of proteins, enzymes, and lipids with emphasis on their relationship to various disease states.

Lecture Hours: 32.000

Prerequisite: MLT1610, Corequisite: MLT2620L

MLT2620L Clinical Chemistry 2 Laboratory 1.000 credits

Performance on those analyses identified in MLT 2620 including electrophoresis and quality control.

Lab Hours: 32.000

Prerequisite: MLT1610L, Corequisite: MLT2620

Course User Fee: \$83.00

MLT2624L Special Techniques In Clinical Chemistry 2.000 credits

The principles and performance of radioimmunoassay, EMIT, ELISA, and toxicological techniques for thyroid function, hormones, and toxic substances.

Lab Hours: 64.000

Prerequisite: MLT 1610, MLT 1610L, Corequisite: MLT 2620, MLT 2620L

Course User Fee: \$62.00

MLT2807L Hospital Practicum: Immunohematology 3.000 credits

A supervised laboratory rotation in a clinical immunohematology facility. This provides the student an opportunity for the practice of skills previously learned and for the acquisition of new procedural skills. The development of interpersonal skills and the transition from student to professional are emphasized.

Clinical Hours: 144.000

Prerequisite: MLT 2525, MLT 2525L, Corequisite: MLT 2930

MLT2809L Hospital Practicum: Hematology 3.000 credits

A supervised laboratory rotation in a clinical hematology facility. This provides the student an opportunity for the practice of skills previously learned and for the acquisition of new procedural skills. The development of interpersonal skills and the transition from student to professional are emphasized.

Clinical Hours: 144.000

Prerequisite: MLT 1300, MLT 1300L, MLT 1330, MLT 1330L, Corequisite: MLT 2930

MLT2810L Hospital Practicum: Chemistry 3.000 credits

A supervised laboratory rotation in a clinical chemistry facility. The development of interpersonal skills the transition from student to professional are emphasized. This provides the student an opportunity for the practice of skills previously learned and for the acquisition of new procedural skills.

Clinical Hours: 144.000

Prerequisite: MLT 2620, MLT 2620L, MLT 2624L, Corequisite: MLT 2930

MLT2811L Hospital Practicum: Microbiology 3.000 credits

A supervised laboratory rotation in a clinical microbiology facility. This provides the student an opportunity for the practice of skills previously learned and for the acquisition of new procedural skills.

Clinical Hours: 144.000

Prerequisite: MLT 2403, MLT 2403L, Corequisite: MLT 2930

MLT2930 Medical Laboratory Technology Seminar 2.000 credits

Clinical correlations, professional issues, updates in Medical Laboratory Technology with student's reports on recent professional journal articles, and the use of microcomputers in the laboratory.

Lecture Hours: 32.000

Prerequisite: MLT 2807L, MLT 2809L, MLT 2810L, MLT 2811L

Course User Fee: \$35.00

Program Description

Computer Engineering Technology

Associate in Science | Code: 26052 | 68 credits

The Computer Engineering Technology program prepares students for employment as computer engineering technicians/technologists and in related occupations in electronics. It also provides supplemental training for persons currently or previously employed in these occupations. The program focuses on the understanding and applying of hardware and software concepts, in addition to troubleshooting techniques to digital, microprocessor or computer-based systems. Assembly, installation, operation, maintenance, calibration, troubleshooting, repairing and elementary designs of medical systems are taught using an integrated and theoretical approach.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

CET 1171	Introduction to Computer Service and Maintenance	3.0
EET 1082	Introduction to Electronics	3.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0

Semester 2

Units: 14.0

CET 1178C	A+ Computer Hardware Service	3.0
COP 2270	C for Engineers	4.0
EET 1015C	Direct Current Circuits	4.0
MAC 1114	Trigonometry	3.0

Semester 3

Units: 6.0

<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
CET 1487C	Network+	3.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT).

Semester 4

Units: 14.0

CET 1110C	Digital Circuits	4.0
EET 1025C	Alternating Current Circuits	4.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
GLY 1010	Physical Geology	3.0
OR		
PHY 2053	Physics (without Calculus) 1	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		

POS 2041	American Federal Government	3.0
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Semester 5		Units: 15.0
CET 2113C	Advanced Digital Circuits	4.0
CET 2123C	Microprocessors	4.0
CET 2588C	Server + Service and Maintenance	3.0
<i>Program Electives</i>		4.0
EET 1141C	Electronics 1	4.0

Semester 6		Units: 7.0
EET 2351C	Digital and Data Communications	4.0
<i>Program Electives*</i>		3.0

*Select 3 credits from the following course option: EET2101C, EET2323C, MAC1140, PHY2053.

Total: 68.0

CET1171 Introduction to Computer Service and Maintenance 3.000 credits

This course is designed as an introduction for students new to IT. Students will learn about the history, design, construction, and maintenance of microcomputers, including the proper handling and use of computer components and tools; how to assemble and disassemble computers; how to perform preventive maintenance; how to identify and upgrade components; how to interpret error messages, and how to perform basic troubleshooting.

Lecture Hours: 48.000

Course User Fee: \$63.00

EET1082 Introduction to Electronics 3.000 credits

Learn by building practical electronic circuits. Survey course suitable for both majors and non-majors. Instructors and tutors are available to assist in project completion. Topics include: schematics, pictorials, amplifiers, oscillators, burglar alarms, radios, digital circuits. Students will develop individual career plans and learn about employment opportunities within the field.

Lecture Hours: 48.000

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CET1178C A+ Computer Hardware Service 3.000 credits

This is an intermediate level course that prepares students for A+ hardware certification. Students will learn how to: install, configure, and upgrade components; diagnose and troubleshoot computer systems, identify, test, and troubleshoot motherboards, processors, memory, and printers, and connect network equipment.

Prerequisite: CET1171. Special fee. A.S. credit only.

Lecture Hours: 48.000

Course User Fee: \$63.00

COP2270 C for Engineers 4.000 credits

This course is intended for students majoring in Computer Engineering Technology, Electronics Engineering Technology, or any engineering discipline. Students will learn the "C" programming language, MATLAB, and the Engineering Problem Solving Method to analyze, design, code, compile and execute programs that solve engineering related problems. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105, Recommended Preparation: CGS1060 OR knowledge of computer skills

Course User Fee: \$63.00

EET1015C Direct Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn electrical components, inductance, capacitance, and elementary network analysis. Students will additionally learn to verify and apply basic theories and principles through hands-on, laboratory experiments utilizing modern testing equipment. Pre/corequisite: MAT 1033. Laboratory fee. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 48.000

Pre/Corequisite: MAT1033

Course User Fee: \$63.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

CET1487C Network+ 3.000 credits

This is an intermediate level course designed for students preparing for the hardware component of the Network+ certification. Students will learn how to install, configure, manage, troubleshoot and upgrade network devices including network interface cards, switches, hubs, wireless access points, routers, and patch panels. They will also learn about the construction, installation, testing and repair of the physical layer of the network, including wired cables, fiber optic media, wireless transmitters and antennas. Demonstrated knowledge of microcomputer fundamentals and system components is required. Recommended Preparation: CET1178C or A+ certification. Laboratory fee. A.S. degree credit only.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CET1110C Digital Circuits 4.000 credits

This is an introductory course in digital circuits for engineering and other technology majors. Students learn how to apply electronic principles to digital computer circuits and systems. Students will also learn how to simplify logic circuits, build digital circuits, and perform other laboratory activities. Pre/Co-requisite: MAC1105. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

EET1025C Alternating Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn inductance, capacitance, vector notation, AC circuits, impedance, phase shift, networks, transformers, and resonance. Students will apply and verify theories and principles through hands-on, laboratory experiments utilizing modern testing equipment.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1015C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view. Lecture Hours: 48.000

CET2113C Advanced Digital Circuits 4.000 credits

This is a second level course in digital circuits for students majoring in electronics and related engineering technologies that extends the application of sequential and combinational logic circuits and other digital applications. Students will learn to program, operate, and interface with a micro-computer and its elements. Prerequisites: CET1110C, COP2270. Laboratory fee. Lecture Hours: 32.000, Lab Hours: 64.000

CET2123C Microprocessors 4.000 credits

This course is intended for students majoring in Electronics or Computer Engineering Technology. Students will learn to apply digital principles to the understanding of microprocessor parameters and characteristics (addressing range and models, instruction set, architecture, input/output, interrupts, and programming). Students will build practical microprocessor and/or microcontroller based systems to perform a variety of engineering applications. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: CET1110C AND COP2270
Course User Fee: \$63.00

CET2588C Server + Service and Maintenance 3.000 credits

This course is designed for students preparing for the hardware component of the Server + certification. Students will learn how to install, configure, and upgrade workstations and servers, configure and test network and peripheral equipment, and diagnose and troubleshoot advanced computer systems. Laboratory fee. College readiness in reading and math required. Recommended Preparation: CET1178C or A+ certification. Special fee.

Lecture Hours: 48.000

Course User Fee: \$63.00

EET1141C Electronics 1 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology or related fields. Students will learn how to apply electronic principles to analog circuits and systems, including semiconductor diodes, applying the fundamental theory of transistors and other solid-state devices; analysis of amplifiers, oscillators, and other applications using a sinusoidal wave. Students also learn basic safety procedures to follow when working in an electronics laboratory and with electronic circuits and systems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1025C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

EET2351C Digital and Data Communications 4.000 credits

This course is intended for students majoring in Electronics or Computer Engineering Technology. It provides a theoretical and practical background in the basic concepts and applications of digital and data communications. Students will learn analog-to-digital (A/D) and digital-to-analog (D/A) conversions; data communications codes and standards; wired and wireless digital communications; modulation, transmission impairment, the telephone system, modems, multiplexers, and electrical interface standards.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2123C

Course User Fee: \$63.00

Program Description

Computer Information Technology

Associate in Science | Code: 25055 | 60 credits

The Computer Information Technology program is to provide an opportunity to establish a basic foundation in computer applications. Graduates are prepared for positions as microcomputer support specialists, user support specialists, applications system specialists and computer information managers to meet the demands of today's automated offices. In addition, program objectives are designed to assist students in their development of interpersonal and communication skills required by office professionals.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
		4.0
CTS 1134	Networking Technologies	4.0
OR		
CTS 1650	CCNA 1: Cisco Fundamentals	4.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Semester 2		Units: 12.0

CGS 1560	A+ Computer Operating Systems	4.0
CTS 1328	Supporting Microsoft Clients	4.0
Program Electives*		4.0

*Select 4 credits from the following course option: CAP1788, CAP2761C, CIS1949, CIS2010C, CIS2900, COP1047C, COP2041C, COP2800, CTS1111, CTS1120, CTS1145, CTS1651, CTS2143C, CTS2192C, CTS2375C, CTS2652, CTS2960, GIS1040, GIS2045.

Semester 3 Units: 9.0

<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4 Units: 12.0

CTS 1131	A+ Computer Essentials & Support	4.0
CTS 2148C	IT Project Management	4.0
CTS 2153	Supporting Windows Users & Applications	4.0

Semester 5 Units: 13.0

CGS 2108	Advanced Desktop Applications	4.0
Program Electives*		9.0

*Select 9 credits from the following course option: CAP1788, CAP2761C, CIS1949, CIS2010C, CIS2900, COP1047C, COP2041C, COP2800, CTS1111, CTS1120, CTS1145, CTS1651, CTS2143C, CTS2192C, CTS2375C, CTS2652, CTS2960, GIS1040, GIS2045.

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1650 CCNA 1: Cisco Fundamentals 4.000 credits

This is the first course of the four-course Cisco curriculum that will prepare students for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn networking concepts and practices, network terminology and protocols, the OSI reference model, cabling, cabling tools, routers, router and switch configurations, LAN/WAN topologies, IP addressing, and network standards.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Knowledge of computer operating systems and hardware is required.

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This is a required general education core course in college-level writing. The students will learn to compose essays and other works using various methods of development. This course fulfills the Gordon Rule writing requirement and must be completed with a grade "C" or better. Pre-requisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Lecture Hours: 48.000

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CGS1560 A+ Computer Operating Systems 4.000 credits

This is a comprehensive course in the use of operating systems for microcomputers suitable for students seeking preparation for A+ operating system certification. Students will learn how to install, configure, use, manage, and troubleshoot the Disk Operating System (DOS), Microsoft Windows, and other microcomputer operating systems. Recommended CGS 1060C or computer experience. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1328 Supporting Microsoft Clients 4.000 credits

This course is intended for students preparing for IT careers as desktop and network support specialists and server administrators, as well as candidates for industry certification. Students will learn how to implement and maintain a Microsoft client operating system.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Recommended: CGS1060C or equivalent computer experience, knowledge or skills.

Course User Fee: \$63.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CTS1131 A+ Computer Essentials & Support 4.000 credits

This is an intermediate level course designed for students preparing for A+ certification as a support technician. Students will learn how to install, configure, upgrade and replace computer system components; how to troubleshoot processors, memory, storage devices, adapter cards, peripherals and other system components; how to install, configure and troubleshoot operating systems, laptops, portable devices, printers, scanners, network devices, security measures and virtualization and cloud computing; and how to provide professional IT support and customer service. Prerequisite: CGS1560. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1560

Course User Fee: \$63.00

CTS2148C IT Project Management 4.000 credits

This course prepares students to effectively plan, implement, coordinate and manage small to medium size information technology-related projects. The concepts and tools covered in this course offers practical knowledge on managing project scope, planning and scheduling, risk assessment, and identification of project resources. This course also prepares students for the CompTIA Project+ examination, an industry standard IT certification. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2153 Supporting Windows Users & Applications 4.000 credits

This is an advanced course designed to help students prepare for the Microsoft Certified IT Professional Support Technician Certification. Students will learn how to install, configure and manage Windows applications in a networked Windows environment and how to support enterprise users. Students will also deploy Windows and applications using various methods, resolve installation and compatibility issues, establish group policies and user profiles, perform support functions, troubleshoot user and application issues, secure the desktop and network from unauthorized use, install software upgrades and updates, perform systems monitoring and documentation, and develop customer service skills.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1328

Course User Fee: \$63.00

CGS2108 Advanced Desktop Applications 4.000 credits

This is an advanced level course for major and non-major students who have completed CGS 1060C, Introduction to Computer Technology and Applications. Students will learn advanced computer skills using software applications, such as word processing, spreadsheets, database, presentation graphics, and communications and scheduling software. Students will also learn advanced file management techniques, deal with security issues, and troubleshoot hardware and software. Prerequisite: CGS1060C. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

Program Description

[Computer Programming and Analysis - Business Application Programming](#)

Associate in Science | Code: 25065 | 60 credits

The Computer Programming and Analysis program with a Business Application Programming concentration provides an opportunity to establish a foundation in computer programming in scientific, commercial, industrial and government data processing applications. Graduates are prepared for positions as entry-level application programmers, programmer specialists, computer programmers and programmer analysts.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 14.0

CGS 1060C	Introduction to Computer Technology and Applications	4.0
COP 1047C	Introduction to Python Programming	4.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0

Semester 2

Units: 13.0

CGS 1540C	Database Concepts Design	4.0
CIS 2331	Systems Analysis, Design and Implementation	5.0
COP 1334	Introduction to C++ Programming	4.0

Semester 3

Units: 7.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
COP 2800	Java Programming	4.0

Semester 4

Units: 13.0

		4.0
COP 2041C	Python Programming 2	4.0
OR		
COP 2335	C++ Programming 2	4.0
OR		
COP 2805C	Java Programming 2	4.0
GEB 1432	Applied Artificial Intelligence (AI) in Business	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
Program Elective *		3.0

*Select 3 credits from the following course option: CAI1001C, CAI1320C, CAI2100C, CAI2300C, CAI2820C, CAI2840C, CEN2211, CEN2212C, CGS2091, COP2041C, CTS1120, CTS1145, CTS2192C, CTS2375C, PHI2680.

Semester 5

Units: 13.0

		4.0
COP 2041C	Python Programming 2	4.0

OR		
COP 2335	C++ Programming 2	4.0
OR		
COP 2805C	Java Programming 2	4.0
Natural Science		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0
Program Electives *		6.0

* Select 6 credits from the following course option: CAI1001C, CAI1320C, CAI2100C, CAI2300C, CAI2820C, CAI2840C, CEN2211, CEN2212C, CGS2091, COP2041C, CTS1120, CTS1145, CTS2192C, CTS2375C, PHI2680.

Total: 60.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CIS2331 Systems Analysis, Design and Implementation 5.000 credits

This course is designed for computer and information technology majors. Students learn to investigate, analyze, plan, design, implement and document an information system using the traditional Systems Development Life Cycle (SDLC).

Lecture Hours: 64.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

COP2041C Python Programming 2 4.000 credits

This is an advanced level programming course using Python. Students will learn how to code, compile, and execute programs. Topics include problem solving, object-oriented programming, database programming, data visualization, and secure software development. Prerequisite: COP1047C; (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C

Course User Fee: \$63.00

COP2335 C++ Programming 2 4.000 credits

This course delves further into the application of C++ programming techniques including classes, object orientation, function and operator overloading, inheritance and dynamic polymorphism, templates, exception handling, standard template library, data structures, and other advanced C++ techniques. Prerequisite: COP 1334 (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1334

Course User Fee: \$63.00

COP2805C Java Programming 2 4.000 credits

This is an advanced level programming course using Java. Students will learn how to code, compile and execute programs. Topics include object serialization, Java Collection, sorting/searching algorithms, multithreading, and Java databases. Prerequisite: COP 2800. (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

GEB1432 Applied Artificial Intelligence (AI) in Business 3.000 credits

This course illustrates how Artificial Intelligence (AI) is currently applied in industries and across business functions to support business strategy formulation, implementation and evaluation. This course introduces the data-driven outcomes and ethical issues to be considered when implementing responsible AI in business and industry. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$35.20

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

COP2041C Python Programming 2 4.000 credits

This is an advanced level programming course using Python. Students will learn how to code, compile, and execute programs. Topics include problem solving, object-oriented programming, database programming, data visualization, and secure software development. Prerequisite: COP1047C; (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C

Course User Fee: \$63.00

COP2335 C++ Programming 2 4.000 credits

This course delves further into the application of C++ programming techniques including classes, object orientation, function and operator overloading, inheritance and dynamic polymorphism, templates, exception handling, standard template library, data structures, and other advanced C++ techniques. Prerequisite: COP 1334 (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1334

Course User Fee: \$63.00

COP2805C Java Programming 2 4.000 credits

This is an advanced level programming course using Java. Students will learn how to code, compile and execute programs. Topics include object serialization, Java Collection, sorting/searching algorithms, multithreading, and Java databases. Prerequisite: COP 2800. (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth’s connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth’s spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

Program Description

Computer Programming and Analysis - Internet of Things (IoT) Applications

Associate in Science | Code: 25076 | 60 credits

The Associate in Science in Computer Programming and Analysis with an IoT Application concentration provides students with a comprehensive foundation in the technologies driving today’s smart, connected world. IoT connects everyday devices to the internet to collect, exchange, and analyze data, enabling innovation across industries. This degree program emphasizes programming, database design, embedded systems, cloud platforms, and securing IoT systems. Students gain hands-on experience developing IoT solutions that integrate data analytics and artificial intelligence. Graduates are prepared to enter the workforce where IoT and emerging technologies converge.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 14.0

CGS 1060C	Introduction to Computer Technology and Applications	4.0
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COP 1334	Introduction to C++ Programming	4.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Semester 2		Units: 13.0
CEN 2211	C/C++ Programming for Embedded Devices	4.0
CIS 2010C	Introduction to Internet of Things Networking and Security	4.0
CIS 2331	Systems Analysis, Design and Implementation	5.0
Semester		Units: 7.0
COP 2800	Java Programming	4.0
		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 4		Units: 12.0
CEN 2211	C/C++ Programming for Embedded Devices	4.0
CEN 2212C	Introduction to Programming the Internet of Things (IoT)	4.0
CGS 1540C	Database Concepts Design	4.0
Semester 5		Units: 14.0
CIS 2012C	Introduction to Data Analytics and AI for the Internet of Things	4.0
CIS 2013C	Internet of Things Applications	4.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0

Total: 60.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CEN2211 C/C++ Programming for Embedded Devices 4.000 credits

This course teaches the principles of programming in the C/C++ languages for embedded devices. The student will learn how to create programs to control open-source hardware for building digital devices that can sense and control the physical world around them.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

CIS2010C Introduction to Internet of Things Networking and Security 4.000 credits

This course explores Internet of Things (IoT) networks' design, implementation, and security, providing a well-rounded understanding of protocols and standards governing the Internet and IoT. Students gain experience performing security assessments and vulnerability testing of IoT systems using industry-standard tools, focusing on identifying and mitigating risks to build secure and resilient IoT solutions.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS2331 Systems Analysis, Design and Implementation 5.000 credits

This course is designed for computer and information technology majors. Students learn to investigate, analyze, plan, design, implement and document an information system using the traditional Systems Development Life Cycle (SDLC).

Lecture Hours: 64.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view. Lecture Hours: 48.000

CEN2211 C/C++ Programming for Embedded Devices 4.000 credits

This course teaches the principles of programming in the C/C++ languages for embedded devices. The student will learn how to create programs to control open-source hardware for building digital devices that can sense and control the physical world around them.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

CEN2212C Introduction to Programming the Internet of Things (IoT) 4.000 credits

This course teaches the principles of programming Internet of Things devices using a computer language. The student will learn fundamental programming concepts and systematic design techniques. At the end of the course, the student will be able to write programs that control development boards, with sensors, connected to the Internet. Prerequisite: CEN2211.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CEN2211

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CIS2012C Introduction to Data Analytics and AI for the Internet of Things 4.000 credits

This course equips students with the knowledge and skills necessary to navigate the complex landscape of data analytics and artificial intelligence within the context of the Internet of Things (IoT), empowering them to contribute to the development of smart, data-driven IoT solutions. Prerequisites: CIS 2010C, CIS 2011C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS2010C AND CIS2011C

Course User Fee: \$63.00

CIS2013C Internet of Things Applications 4.000 credits

In this course, students learn about the practical applications of the Internet of Things (IoT) across various sectors and industries. They explore real-world use cases, ranging from smart homes and smart cities to the latest developments in the field. The course offers insights into how IoT solutions are designed, developed, and deployed, equipping students with the knowledge to create innovative IoT applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS2010C AND CIS2011C

Course User Fee: \$63.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

Program Description

Computer Programming and Analysis - Mobile Applications Development

Associate in Science | Code: 25070 | 60 credits

The Computer Programming and Analysis program provides an opportunity to establish a foundation in computer programming in scientific, commercial, industrial and government data processing applications. The Mobile Applications Development concentration additionally offers hands-on instruction with current technology for Apple and Android mobile device platforms. Graduates are prepared for positions as app developers.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 14.0

CGS 1060C	Introduction to Computer Technology and Applications	4.0
COP 1334	Introduction to C++ Programming	4.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0

Semester 2

Units: 13.0

CGS 1540C	Database Concepts Design	4.0
CIS 2331	Systems Analysis, Design and Implementation	5.0
COP 2800	Java Programming	4.0

Semester 3

Units: 9.0

<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4

Units: 12.0

COP 2664	iPhone Application Development 1	4.0
COP 2658	iPhone Application Development 2	4.0
Program Elective *		4.0

*Select 4 credits from the following course option: CAI1001C, CAI1320C, CAI2100C, CAI2300C, CAI2840C, CEN2211, CEN2212C, CGS2091, COP2805, CTS1120, CTS1145, CTS2192C, CTS2375C, GEB1432, PHI2680.

Semester 5

Units: 12.0

COP 2660	Android Application Development 1	4.0
COP 2662	Android Application Development 2	4.0
Program Elective *		4.0

*Select 4 credits from the following course option: CAI1001C, CAI1320C, CAI2100C, CAI2300C, CAI2840C, CEN2211, CEN2212C, CGS2091, COP2805, CTS1120, CTS1145, CTS2192C, CTS2375C, GEB1432, PHI2680.

Total: 60.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CIS2331 Systems Analysis, Design and Implementation 5.000 credits

This course is designed for computer and information technology majors. Students learn to investigate, analyze, plan, design, implement and document an information system using the traditional Systems Development Life Cycle (SDLC).

Lecture Hours: 64.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

COP2664 iPhone Application Development 1 4.000 credits

This is an introduction to iOS programming course using the Objective C computer language, recommended for Computer Science and Computer Information Systems majors. Students will learn to code, compile and execute mobile iOS applications while learning advanced programming concepts and object oriented programming design concepts and principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1332 OR COP1334

Course User Fee: \$63.00

COP2658 iPhone Application Development 2 4.000 credits

This intermediate iOS course teaches the principles of iPhone application development for majors in Computer Science, Computer Information Systems, and related disciplines. Students will learn how to create mobile applications that can be deployed to iPhone smartphones, tablets or simulators utilizing Cocoa and XCode for development. Emphasis will be placed on learning the underlying iPhone framework and components in order to create quality mobile applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2654

Course User Fee: \$63.00

COP2660 Android Application Development 1 4.000 credits

This course teaches the principles of Android application development for majors in Computer Science, Computer Information Systems, and related disciplines. Students will learn how to create mobile applications for deployment to Android smartphones, tablets or simulators utilizing open source software (Java, Eclipse IDE, Android Plug-In and Android SDK) for development. Emphasis will be placed on the underlying Android framework to create quality applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

COP2662 Android Application Development 2 4.000 credits

This course for majors in Computer Science, Computer Information Systems, and related disciplines teaches how to develop advanced Android applications. Students will learn how to create applications utilizing the advanced capabilities of Android smartphones, including interfacing the application to the device's content providers' databases, GPS and location based services, notifications, background threads, audio, video, SMS, motion sensors and network connectivity. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2660, COP2800

Course User Fee: \$58.00

Program Description

Crime Scene Technology - Crime Scene Investigation

Associate in Science | Code: 27026 | 60 credits

The Associate in Science Degree in Crime Scene Technology prepares students for employment in the field of criminalistics with a specialty in Crime Scene Investigation or Forensic Science. Graduates of this program may find employment as a Forensic Science Technician, Crime Scene Technician, Medical Examiner Investigator, Medical Investigator, Insurance Investigator, Legal Investigator, Forensic Paralegal, Crime Scene Investigator, and Laboratory Technician, among others. Students may also continue their formal education with the College and receive a Bachelor of Applied Science in Public Safety Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
CCJ 1020	Introduction to Criminal Justice	3.0
CJE 2671	Basic Fingerprinting	3.0
ENC 1101	English Composition 1	3.0
<i>Mathematics</i>		3.0
MAC 1105	College Algebra	3.0
OR		
STA 2023	Statistical Methods	3.0

Semester 2		Units: 16.0
CCJ 1191	Human Behavior in Criminal Justice	3.0
CHS 1522C	Forensic Science 1	4.0
CJE 1640	Crime Scene Technology 1	3.0
CJE 2600	Criminal Investigation	3.0
PHI 2010	Introduction to Philosophy	3.0

Semester 3		Units: 9.0
		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
CJE 1673	Crime Scene Photography 1	3.0
CJE 2644	Crime Scene Safety	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT).

Semester 4		Units: 9.0
CCJ 2358	Criminal Justice Reporting	3.0
PHI 2600	Introduction to Ethics	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0

OR		
POS 2041	American Federal Government	3.0

Semester 5		Units: 15.0
CCJ 2053	Criminal Justice Ethics and Professionalism	3.0
CJE 1642	Crime Scene Technology 2	3.0
CJE 1772	Crime Scene Photography 2	3.0
CJE 2672	Fingerprint Development	3.0
CJL 2610	Courtroom Presentation	3.0

Total: 61.0

CCJ1020 Introduction to Criminal Justice 3.000 credits

This course will explore and define the various components of the criminal justice system. It is designed for the entry level criminal justice practitioner and beginning student interested in entering the field.

Lecture Hours: 48.000

CJE2671 Basic Fingerprinting 3.000 credits

This course provides a foundation in basic fingerprinting. Students will learn topics which include classification, identification, filing and rolling of fingerprints, problems and practices associated with post mortem fingerprinting and proper presentation of fingerprint evidence. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$11.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CCJ1191 Human Behavior in Criminal Justice 3.000 credits

Human behavior and how it relates to the duties and responsibilities of the criminal justice practitioner.

Lecture Hours: 48.000

CHS1522C Forensic Science 1 4.000 credits

An introductory course in the principles and techniques of forensic science. Students will learn how forensic science pertains to crime scene investigation and crime laboratory analysis.

Lecture Hours: 48.000, Lab Hours: 32.000

CJE1640 Crime Scene Technology 1 3.000 credits

This is an introductory course in Crime Scene Technology. Students will learn the techniques, materials and instrumentation used in securing, searching, recording, collecting, and examining physical evidence. There will be special emphasis on the tools, instruments, and techniques used in the studies of crime scene reconstruction, fingerprints, firearms, tool marks, and blood stain pattern analysis.

Lecture Hours: 48.000

CJE2600 Criminal Investigation 3.000 credits

Fundamentals of criminal investigation, theory and practice, including crime scene search; preservation, collection and transportation of physical evidence; interviewing, interrogating; statement taking; and case preparation, with investigation of specific offenses; relationship with the police science laboratory.

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

CJE1673 Crime Scene Photography 1 3.000 credits

This is an introductory study of the history of photography including basic photography skills. Students will learn camera operations, exposure control, relational photographs and flash control for crime scene and evidentiary documentation. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

CJE2644 Crime Scene Safety 3.000 credits

This course provides the fundamentals of protecting and preserving the crime scene and identifies the essential techniques of properly handling physical evidence. Students will learn the understanding of various hazards and safety issues and provides basic techniques for preserving evidence as it relates to various hazardous chemical and biological materials. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$14.00

CCJ2358 Criminal Justice Reporting 3.000 credits

This course prepares students through instruction and practice to properly prepare written reports common to the criminal justice community. Students will learn a variety of criminal justice scenarios presented and students will be instructed as to proper report format and presentation. Prerequisite: ENC1101.

Lecture Hours: 48.000

Prerequisite: ENC1101

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CCJ2053 Criminal Justice Ethics and Professionalism 3.000 credits

This course will provide students with an overview of moral, ethical, and professional issues and dilemmas facing individuals and organizations within the criminal justice system. Students will learn to define and implement ethical and professional standards by examining what they will be confronted with and how to respond appropriately. Prerequisite: PHI 2600.

Lecture Hours: 48.000

Prerequisite: PHI2600

CJE1642 Crime Scene Technology 2 3.000 credits

This course covers advanced principles, theories and applications in crime scene technology. Students will learn collection procedures of weapons, arson, gunshot residue, blood spatter, and recovery of human remains. Data analysis, reporting and plan of action development are emphasized.

Lecture Hours: 48.000

Prerequisite: CJE1640

CJE1772 Crime Scene Photography 2 3.000 credits

This course expands upon concepts; knowledge and skills taught in Crime Scene Photography 1. Students will learn to include specialty light sources, darkroom techniques and procedures, filters and specialized equipment including black and white and color enlargers.

Lecture Hours: 48.000

Prerequisite: CJE1673

CJE2672 Fingerprint Development 3.000 credits

This course provides a continuation of Basic Fingerprinting. Students will learn different methods involved in detection, enhancement, and recovery of latent fingerprints. Techniques will involve chemical and mechanical methods on substrates and evaluation for proper application in both theory and practices.

Lecture Hours: 48.000

Prerequisite: CJE2671

CJL2610 Courtroom Presentation 3.000 credits

This course introduces students to proper courtroom presentation and procedures. Students will learn the appropriate techniques for proper attire, grooming, speaking, listening and stress control during courtroom proceedings, visual aid preparation, and presentations of all evidence (commonly referred to as "scientific evidence") collected at the crime scene are also included.

Lecture Hours: 48.000

Program Description

Crime Scene Technology - Forensic Science

Associate in Science | Code: 27027 | 60 credits

The Associate in Science Degree in Crime Scene Technology prepares students for employment in the field of criminalistics with a specialty in Crime Scene Investigation or Forensic Science. Graduates of this program may find employment as a Forensic Science Technician, Crime Scene Technician, Medical Examiner Investigator, Medical Investigator, Insurance Investigator, Legal Investigator, Forensic Paralegal, Crime Scene Investigator, and Laboratory Technician, among others. Students may also continue their formal education with the College and receive a Bachelor of Applied Science in Public Safety Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
CCJ 1020	Introduction to Criminal Justice	3.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
Semester 2		Units: 15.0
BSC 2426	Biotechnology Methods and Applications 1	3.0
BSC 2426L	Biotechnology Methods and Applications 1 Laboratory	2.0
CHS 1522C	Forensic Science 1	4.0
CJE 1640	Crime Scene Technology 1	3.0
CJE 2600	Criminal Investigation	3.0
Semester 3		Units: 14.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry and Qualitative Analysis Lab	2.0
CHS 2523	Forensic Science 2	3.0
CJE 2671	Basic Fingerprinting	3.0
PHI 2010	Introduction to Philosophy	3.0
Computer Competency Test *		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
 Departmental Recommendation: Computer Competency Test (CCT).

Semester 4 **Units: 11.0**

CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
CCJ 1191	Human Behavior in Criminal Justice	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 5

Units: 8.0

BSC 2427	Biotechnology Methods and Applications 2	3.0
BSC 2427L	Biotechnology Methods and Applications 2 Laboratory	2.0
SPC 1017	Introduction to Communication	3.0

Total: 60.0

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

CCJ1020 Introduction to Criminal Justice 3.000 credits

This course will explore and define the various components of the criminal justice system. It is designed for the entry level criminal justice practitioner and beginning student interested in entering the field.

Lecture Hours: 48.000

ENC1101 English Composition 1 3.000 credits

This is a required general education core course in college-level writing. The students will learn to compose essays and other works using various methods of development. This course fulfills the Gordon Rule writing requirement and must be completed with a grade "C" or better. Pre-requisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Lecture Hours: 48.000

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BSC2426 Biotechnology Methods and Applications 1 3.000 credits

This course addresses the basic principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical, biotechnology and/or research laboratory setting(s). Practical applications of biotechnology are explored.

Lecture Hours: 48.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426L

BSC2426L Biotechnology Methods and Applications 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2426 Biotechnology Methods and Applications 1. This is a hands-on course that emphasizes the basic laboratory principles, techniques, and instrumentation, necessary for effective work in pharmaceutical, biotechnology, and/or research laboratory settings(s).

Lab Hours: 64.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426

Course User Fee: \$50.00

CHS1522C Forensic Science 1 4.000 credits

An introductory course in the principles and techniques of forensic science. Students will learn how forensic science pertains to crime scene investigation and crime laboratory analysis.

Lecture Hours: 48.000, Lab Hours: 32.000

CJE1640 Crime Scene Technology 1 3.000 credits

This is an introductory course in Crime Scene Technology. Students will learn the techniques, materials and instrumentation used in securing, searching, recording, collecting, and examining physical evidence. There will be special emphasis on the tools, instruments, and techniques used in the studies of crime scene reconstruction, fingerprints, firearms, tool marks, and blood stain pattern analysis.

Lecture Hours: 48.000

CJE2600 Criminal Investigation 3.000 credits

Fundamentals of criminal investigation, theory and practice, including crime scene search; preservation, collection and transportation of physical evidence; interviewing, interrogating; statement taking; and case preparation, with investigation of specific offenses; relationship with the police science laboratory.

Lecture Hours: 48.000

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

CHM 1045 is the first semester of a two-semester general chemistry course for science, premedical science and engineering students. Students will learn stoichiometry, atomic structure, completing and balancing chemical reactions, nomenclature, bonding, acid-base theories, solutions, gas laws and beginning thermodynamics.

Special fee.

Lecture Hours: 48.000

Prerequisite: CHM1025or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: MAC1105, Corequisite: CHM1045, Prerequisite: CHM1025Or an approved ACS Proficiency Exam score.

Course User Fee: \$30.00

CHS2523 Forensic Science 2 3.000 credits

This is a continuation of Forensic Science 1. Students will learn topics which include but are not limited to: drug identification and toxicology; document analysis; death determination; soil examination methodology; forensic anthropology; tool marks and casts/impressions. Prerequisite: CHS1522C.

Lecture Hours: 48.000

CJE2671 Basic Fingerprinting 3.000 credits

This course provides a foundation in basic fingerprinting. Students will learn topics which include classification, identification, filing and rolling of fingerprints, problems and practices associated with post mortem fingerprinting and proper presentation of fingerprint evidence. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$11.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

CCJ1191 Human Behavior in Criminal Justice 3.000 credits

Human behavior and how it relates to the duties and responsibilities of the criminal justice practitioner.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view. Lecture Hours: 48.000

BSC2427 Biotechnology Methods and Applications 2 3.000 credits

This course addresses advanced principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical-biotechnology and/or research-laboratory setting(s). The following areas of contemporary biotechnology are covered: forensics, bioremediation, and animal-, plant-, and marine biotechnology.

Lecture Hours: 48.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427L

BSC2427L Biotechnology Methods and Applications 2 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2427 Biotechnology Methods and Applications 2. This is a hands-on course that emphasizes advanced laboratory principles, techniques, and instrumentation necessary for effective work in a pharmaceutical, biotechnology, and/or research-laboratory setting(s).

Lab Hours: 64.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427

Course User Fee: \$50.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

Program Description

Criminal Justice Technology

Associate in Science | Code: 27012 | 60 credits

Upon completion of the Criminal Justice Technology program, the student will be eligible for the Associate in Science degree. The A.S. degree opens up entry-level non-sworn positions in local, state and federal agencies, i.e., juvenile justice, private security, law enforcement, corrections, probation and parole, detention centers and community based intervention programs.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
CCJ 1020	Introduction to Criminal Justice	3.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
Semester 2		Units: 15.0
CCJ 1010	Introduction to Criminology	3.0
CJC 1000	Introduction to Corrections	3.0
CJC 1162	Parole and Probation	3.0
CJL 2062	Constitutional Law and Legal Procedure or Evidence	3.0
PHI 2010	Introduction to Philosophy	3.0
Semester 3		Units: 9.0
CCJ 1191	Human Behavior in Criminal Justice	3.0
CJE 2600	Criminal Investigation	3.0
CJL 1100	Criminal Law	3.0
Computer Competency Test *		
*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.		
Departmental Recommendation: Computer Competency Test (CCT).		
Semester 4		Units: 12.0
CCJ 2650	Narcotics and Drug Education	3.0
DSC 1006	Introduction to Homeland Security	3.0
PHI 2600	Introduction to Ethics	3.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 5

Units: 12.0

CCJ 2053	Criminal Justice Ethics and Professionalism	3.0
CJJ 2002	Juvenile Delinquency	3.0
DSC 2242	Transportation and Border Security	3.0
DSC 2590	Intelligence Analysis and Security Management	3.0

Total: 60.0

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

CCJ1020 Introduction to Criminal Justice 3.000 credits

This course will explore and define the various components of the criminal justice system. It is designed for the entry level criminal justice practitioner and beginning student interested in entering the field.

Lecture Hours: 48.000

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

CCJ1010 Introduction to Criminology 3.000 credits

Theories and causes of criminal and delinquent behavior, including its variations, ramifications, explanations and measures of prevention, control and treatment.

Lecture Hours: 48.000

CJC1000 Introduction to Corrections 3.000 credits

A comprehensive view of the historical and philosophical treatment programs and developments in the field of adult institutional corrections. Emphasis is on understanding the offender in the correctional system; an examination of the correctional client and staff, agencies and recidivism; (3 hr. lecture).

Lecture Hours: 48.000

CJ1162 Parole and Probation 3.000 credits

The history, current practices and the consideration of philosophical concepts in the areas of probation and parole; (3 hr. lecture)

Lecture Hours: 48.000

CJL2062 Constitutional Law and Legal Procedure or Evidence 3.000 credits

An examination of the United States and Florida Constitutions, with emphasis on leading cases dealing with arrest, search and seizure, confessions and the rules of evidence.

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CCJ1191 Human Behavior in Criminal Justice 3.000 credits

Human behavior and how it relates to the duties and responsibilities of the criminal justice practitioner.

Lecture Hours: 48.000

CJE2600 Criminal Investigation 3.000 credits

Fundamentals of criminal investigation, theory and practice, including crime scene search; preservation, collection and transportation of physical evidence; interviewing, interrogating; statement taking; and case preparation, with investigation of specific offenses; relationship with the police science laboratory.

Lecture Hours: 48.000

CJL1100 Criminal Law 3.000 credits

Historical background and foundations of American criminal law, including United States Constitutional requirements, Federal and State court organization and jurisdiction, criminal law basics, Florida statutes, rules of evidence and procedure.

Lecture Hours: 48.000

CCJ2650 Narcotics and Drug Education 3.000 credits

The general problems created by illegal use of narcotics and dangerous substances, with emphasis upon classification, description and history of drugs, etiology of addiction, extent of drug use and its relationship to criminal behavior and methods of control.

Lecture Hours: 48.000

DSC1006 Introduction to Homeland Security 3.000 credits

This course will introduce students to the vocabulary and important components of Homeland Security. Students will learn about the agencies associated with Homeland Security and their interrelated duties and relationships.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CCJ2053 Criminal Justice Ethics and Professionalism 3.000 credits

This course will provide students with an overview of moral, ethical, and professional issues and dilemmas facing individuals and organizations within the criminal justice system. Students will learn to define and implement ethical and professional standards by examining what they will be confronted with and how to respond appropriately. Prerequisite: PHI 2600.

Lecture Hours: 48.000

Prerequisite: PHI2600

CJJ2002 Juvenile Delinquency 3.000 credits

A history of the juvenile court system in the United States. An analysis of delinquency and the family structure. Delinquency control including the police function with juveniles and their families, preventative measures by adequate court and police enforcement, rehabilitative processes, and the philosophy and programs of schools for boys and girls and detention homes.

Lecture Hours: 48.000

DSC2242 Transportation and Border Security 3.000 credits

This course introduces students to global supply chains and intermodal transportation systems. Students will learn the threats to these systems, their vulnerabilities and potential for terrorist attacks, and the measures being undertaken to secure them.

Lecture Hours: 48.000

DSC2590 Intelligence Analysis and Security Management 3.000 credits

This course examines intelligence analysis and its indispensable relationship to the security management of terrorist attacks, man-made disasters and natural disasters. Students will learn substantive issues regarding intelligence support of Homeland Security measures implemented by the United States and explore how the intelligence community operates.

Lecture Hours: 48.000

Program Description

Culinary Arts Management

Associate in Science | Code: 22031 | 60 credits

The Associate in Science degree in Culinary Arts Management will groom individuals for careers in the widely varied areas of the culinary industry, including production line and supervisory positions. This competency based culinary program will provide the student with a unique combination of comprehensive theoretical knowledge and practical hands-on training. Students will master the fundamentals of culinary production in an environment that builds teamwork while gaining practical individualized experiences. The course content includes food preparation and service; identification, storage, selection and presentation of foods; training in communication, leadership, human relations, and employability skills; and sanitation and safe work practices. Earning a degree will

enable students to pursue further education at the university level or begin working in the field immediately upon graduation.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
FSS 1200	Culinary Terminology and Procedures	3.0
FSS 1202C	Food Production 1 - Foundational Skills	3.0
HFT 1212	Safety and Sanitation	3.0

Semester 2		Units: 12.0
FSS 1204C	Food Production 2 - American Regional Cuisine	3.0
FSS 1246C	Basic Baking - Foundational Skills	3.0
MGF 1130	Mathematical Thinking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 3		Units: 9.0
FSS 2248C	Garde Manger	3.0
HFT 1220	Supervisory Development	3.0
HUN 1201	Essentials of Human Nutrition	3.0
Computer Competency*		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT).

Semester 4		Units: 12.0
BSC 1005	General Education Biology	3.0
FSS 1801	Culinary Sustainability and Practices	3.0
FSS 2242C	International Cuisine	3.0
HFT 1454	Food and Beverage Cost Controls	3.0

Semester 5		Units: 8.0
ACG 1403	Excel for Business	1.0
FSS 2205C	Food Production 3 - Contemporary Cuisine and Modernist Methods	3.0
FSS 2381	MCI Café Management Internship	4.0

Semester 6		Units: 7.0
FSS 2940	Culinary Management Externship	4.0
PHI 2010	Introduction to Philosophy	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

FSS1200 Culinary Terminology and Procedures 3.000 credits

This is an introductory course in which students will learn to integrate industry vocabulary, terminology, knowledge, skills, and practices required for careers in the food service industry. The students will learn standard kitchen phrases, how to identify and describe equipment, recipe reading, costing, conversion formulas, product identification, measurements and basic cooking procedures. Corequisite: FSS1202C.

Lecture Hours: 48.000

Corequisite: FSS1202C

FSS1202C Food Production 1 - Foundational Skills 3.000 credits

This course is an introductory kitchen lecture/lab experience in which students will be provided hands-on orientation to tools, equipment, recipe production, measurements, knife cut techniques and basic cooking procedures. Students will practice classic cooking methods, product identification and the functions of the production kitchen in a "green" team environment. Corequisite: FSS1200. A.S. degree credit only.

Lecture Hours: 16.000, Practicum Hours: 96.000

Corequisite: FSS1200

Course User Fee: \$1981.97

HFT1212 Safety and Sanitation 3.000 credits

Students will learn industry standards on sanitation as it relates to food preparation and kitchen operations. Students will recognize proper sanitation techniques and explain how to implement sanitation programs in food service operations.

Lecture Hours: 48.000

Course User Fee: \$125.00

FSS1204C Food Production 2 - American Regional Cuisine 3.000 credits

The lecture/practicum course reinforces the knowledge and skills learned in the preceding classes and helps students build confidence in the techniques of basic cookery. The development of knife skills is accented. American Regional Cuisine explores the use of indigenous ingredients in the preparation of traditional and contemporary American specialties. The importance and basics of "sustainable agriculture" are introduced. The concepts of mise en place, timelines, plate presentation, and teamwork in a production setting are introduced and accentuated. Timing and organization skills are emphasized. Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200 AND FSS1202C

Course User Fee: \$1981.99

FSS1246C Basic Baking - Foundational Skills 3.000 credits

This is an introductory course in which students will learn the fundamentals of baking science, terminology, equipment, ingredients, weights, measures, formula conversion and storage. Students will learn the functions of various baking ingredients and execute baking recipes, including competencies relating to doughs, breads, cookies, pies, puff pastries, sweet & savory pastry fillings, quick breads, cakes and basic decorating techniques.

Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C

Course User Fee: \$1981.98

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

FSS2248C Garde Manger 3.000 credits

This is a foundational lecture/lab course in the preparation of cold food, hors d'oeuvres, display platters, charcuteries, smoking techniques, butchery and seafood. Students will prepare edible buffet centerpieces. Students will also have hands-on experiences in ice carving. Production methods and safe food handling techniques are re-emphasized. Prerequisites: FSS1200, FSS1202C, FSS1246C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C, FSS1204C

Course User Fee: \$1981.95

HFT1220 Supervisory Development 3.000 credits

This introductory course will assist students in learning basic supervisory skills. The students will learn how to conduct proper employee evaluations, as well as how to apply discipline. Effective communication techniques, use of delegation and diversity in the workforce are discussed. Issues concerning employee relations and current topics in management are addressed.

Lecture Hours: 48.000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

FSS1801 Culinary Sustainability and Practices 3.000 credits

This course will engage students in growing and producing food, processing, distribution, and end user/purchaser aspects of food systems. Students will explore how to reduce the carbon footprint of food service operations and understand the importance of bringing seasonal food to the table at its peak of freshness and height of nutritional value. With hands-on approach and a focus on biodiversity, sustainability, healthy food and animal welfare, students will develop critical thinking to understanding local and global food systems. The course includes on-site visits with farmers, food processors, and experts in our local food system to engage in health and sustainability practices related to food safety, water and waste systems, food marketing, and the heritage food movement. Prerequisite: FSS2248C.

Lecture Hours: 48.000

Prerequisite: FSS2248C

Course User Fee: \$583.82

FSS2242C International Cuisine 3.000 credits

This is an advanced course in which students will reinforce their knowledge of equipment, vocabulary and theories learned in prerequisite classes. Students will learn to use global ingredients, explore international cooking techniques and methods and be exposed to equipment specific to a myriad of world regions. Students will have the opportunity to develop international menus and prepare dishes from many countries.

Prerequisites: FSS1200, FSS1202C, FSS1204C, FSS1246C, FSS2248C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C, FSS1204C, FSS1246C, FSS2248C

Course User Fee: \$1981.97

HFT1454 Food and Beverage Cost Controls 3.000 credits

Covers the principles and procedures involved in an effective food and beverage control system, including standards determination, the operation budget, income and cost control, menu pricing, and computer applications.

Lecture Hours: 48.000

Course User Fee: \$15.00

ACG1403 Excel for Business 1.000 - 3.000 credits

This course will cover Excel topics relevant to the field of accounting and finance including, but not limited to, VLOOKUP, HLOOKUP, INDEX, MATCH, IF, AND, OR, Pivot Tables, Named Ranges, Array Formulas, Custom Number Formats, Conditional Formatting, Absolute References and Keyboard Shortcuts.

Lecture Hours: 16.000 - 48.000

Course User Fee: \$22.00

FSS2205C Food Production 3 - Contemporary Cuisine and Modernist Methods 3.000 credits

This lecture/lab course is an introduction to alternative methods of food production using science as the platform. Students will further develop their understanding of modernist and molecular cuisine through hands-on approach, utilizing state-of-art equipment and techniques. Prerequisites: FSS1200, FSS1202C, FSS1204C, FSS1246C, FSS2242C, FSS2248C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200 AND FSS1202C AND FSS1204C AND FSS1246C AND FSS2242C AND FSS2248C

Course User Fee: \$1981.96

FSS2381 MCI Café Management Internship 4.000 credits

This required internship is designed to provide hands-on culinary training through industry work experience. Students will be located in an approved site to reinforce their skills while being exposed to various stations in a food services operation. Students will learn to actively participate in various aspects of the operation including cooking, cost controls, and sanitation programs. Prerequisite: FSS2205C. .

Practicum Hours: 192.000

Prerequisite: FSS2205C

Course User Fee: \$1981.97

FSS2940 Culinary Management Externship 4.000 credits

This course is designed to provide advanced hands-on culinary training for culinary students through work experience in an approved commercial foodservice or hospitality establishment. Students on externship will receive feedback from their supervisor and keep an externship manual to record and reflect on their work experience.

Internship Hours: 192.000

Course User Fee: \$249.05

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

Program Description

Cybersecurity

Associate in Science | Code: 25079 | 60 credits

The Cybersecurity program prepares students to fill a critical and growing need for cybersecurity personnel through hands-on experience along with simulation training and group/team-based learning to simulate a

professional work environment. The program covers a wide range of topics including cybersecurity fundamentals, computer forensics and network security. Upon completion of the program, the student will have learned to evaluate security trends, recognize best practices, and understand IT security products and threats. Additionally, the curriculum is designed to aid students in preparing for many of the certification exams in the field.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
<i>Mathematics</i>		3.0
MAC 1105	College Algebra	3.0
OR		
STA 2023	Statistical Methods	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
CTS 1134	Networking Technologies	4.0
Semester 2		Units: 12.0
CGS 1700	Introduction to Operating Systems	4.0
CIS 1531	Introduction to Secure Scripting	4.0
CTS 1120	Cybersecurity Fundamentals	4.0
Semester 3		Units: 7.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
CTS 1111	Linux +	4.0
Semester 4		Units: 14.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
CET 2880C	Digital Forensics	4.0
CTS 2314	Network Defense and Countermeasures	4.0
Semester 5		Units: 13.0
CIS 2350	Cybersecurity Analysis (CySA+)	4.0
<i>Program Electives</i>		9.0

CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
OR		
CAI 1320C	Introduction to Prompt Engineering	3.0
OR		
CAI 2100C	Machine Learning Foundations	3.0
OR		
CAI 2300C	Introduction to Natural Language Processing	3.0
OR		
CAI 2820C	Artificial Intelligence Applications Solutions	3.0
OR		
CAI 2840C	Introduction to Computer Vision	3.0
OR		
CIS 1949	CO-Op Work Experience 1: CIS	1.0
OR		
CIS 2010C	Introduction to Internet of Things Networking and Security	4.0
OR		
CIS 2900	Directed Study in IT	1.0
OR		
CJE 1680	Introduction to Computer Crimes	3.0
OR		
CJL 2062	Constitutional Law and Legal Procedure or Evidence	3.0
OR		
COP 1047C	Introduction to Python Programming	4.0
OR		
COP 1334	Introduction to C++ Programming	4.0
OR		
COP 2041C	Python Programming 2	4.0
OR		
COP 2800	Java Programming	4.0
OR		
CTS 1145	Cloud Essentials	4.0
OR		
CTS 2143C	Server Administration	4.0
OR		
CTS 2148C	IT Project Management	4.0
OR		
CTS 2192C	Microsoft Azure Administration	4.0
OR		
CTS 2303	Windows Server Administration	4.0
OR		
CTS 2375C	Cloud Infrastructure and Services	4.0
OR		
CTS 2670	Check Point Security Administration	4.0
OR		
CTS 2671	Check Point Security Engineering	4.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1700 Introduction to Operating Systems 4.000 credits

This course examines the role of operating systems as the interface between the hardware, the software and the users of a computer system. It explores the concepts such as processes and threads, file systems, virtual memory, interrupt handling, virtualization and security.

Lecture Hours: 48.000, Lab Hours: 32.000

CIS1531 Introduction to Secure Scripting 4.000 credits

This course provides students with the knowledge and skills to: create secure scripts and programs using system shells and programming languages; implement and debug algorithms to solve problems; automate and perform administrative tasks; manage data handling, and backup and storage.

Lecture Hours: 48.000, Lab Hours: 32.000

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CTS1111 Linux + 4.000 - 32.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based work-stations and servers. Prerequisite: CGS 1060 or computer experience is required. (3 hr. lecture; 2 hr. lab)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

CET2880C Digital Forensics 4.000 credits

This is an introductory digital forensics course for students who are studying cybersecurity, electronics or computer engineering technologies. In this course, students will learn the setup and use of an investigator's laboratory, how to perform data acquisition, web forensics, email forensics, mobile forensics, network analysis, and file recovery.

Lecture Hours: 32.000, Lab Hours: 64.000

CTS2314 Network Defense and Countermeasures 4.000 credits

In this course, students will take an in-depth look at network defense concepts and techniques. Coverage includes network defensive concepts; policy development; problem-solving; and implementation of firewalls, DMZ, VPN, IDS, NAT and proxy servers.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120, CTS1134

CIS2350 Cybersecurity Analysis (CySA+) 4.000 credits

This course provides students an intermediate skills-level approach to cybersecurity analysis. Students learn to identify the phases of an attack, the motivations of the adversary, the resources and techniques they use, the intended effect, or end-game, and how to mitigate threats. Topics include intrusion detection and response, analytics and advanced threat visibility. Prerequisites: CTS1120, CTS1134. Laboratory Fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120, CTS1134

Course User Fee: \$63.00

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAI1320C Introduction to Prompt Engineering 3.000 credits

This course introduces students to prompt engineering, focusing on foundational concepts and practical applications across various tools. Students will learn to craft prompts that enhance creativity, precision, and efficiency, leveraging AI to optimize workflows and productivity.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAI2100C Machine Learning Foundations 3.000 credits

Students are introduced to machine learning concepts and Python applications, including data acquisition, supervised, unsupervised, and reinforced learning. In addition, students will develop and deploy artificial intelligence (AI) models utilizing classification algorithms. Prerequisites: CAI1001C, COP1047C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI1001C/COP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

CAI2300C Introduction to Natural Language Processing 3.000 credits

Students will learn the fundamental concepts of Natural Language Processing (NLP) and text processing. In addition, focus will be on knowledge and skills necessary to create a language recognition application. Pre/Corequisite: CAI2100C.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: CAI2100C

Course User Fee: \$63.00

CAI2820C Artificial Intelligence Applications Solutions 3.000 credits

A lower division course for students majoring in Applied Artificial Intelligence (AI). Students will demonstrate competence to scope, acquire/explore data, model, evaluate, and deploy an AI/machine learning solution in a team environment. Students will create and present a code or no-code AI solution. Must be taken during the last semester before graduation. Prerequisites: CAI2300C and CAI2840C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI2840C AND CAI2300C

Course User Fee: \$63.00

CAI2840C Introduction to Computer Vision 3.000 credits

Students will learn fundamental concepts in Computer Vision (CV) and image processing, including introduction to necessary proprietary and open-source Python libraries. Pre/Corequisite: CAI2100C.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: CAI2100C

Course User Fee: \$63.00

CIS1949 CO-Op Work Experience 1: CIS 1.000 - 4.000 credits

This course is designed as a work experience for students majoring in computer information systems programs. Students will learn to apply the skills and knowledge that they have acquired through their program of study in a real work environment.

Prerequisite: Successful completion of required program course work. Department approval required.

Lecture Hours: 16.000 - 64.000

Prerequisite: Successful completion of required program course work. Department approval required.

CIS2010C Introduction to Internet of Things Networking and Security 4.000 credits

This course explores Internet of Things (IoT) networks' design, implementation, and security, providing a well-rounded understanding of protocols and standards governing the Internet and IoT. Students gain experience performing security assessments and vulnerability testing of IoT systems using industry-standard tools, focusing on identifying and mitigating risks to build secure and resilient IoT solutions.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS2900 Directed Study in IT 1.000 credits

This course, for majors in Information Technology, provides students the opportunity to pursue projects under faculty supervision through self-directed research, experiences, and presentations.

Prerequisite: Successful completion of required program course work. Department approval required.

Lab Hours: 32.000

Recommended Preparation: Successful completion of required program course work. Department approval required

CJE1680 Introduction to Computer Crimes 3.000 credits

This course provides the student with an overview of crimes involving the use of computer technology and the internet. Students will learn how computer related crimes are committed and how they are investigated. Topics covered will include computer crime scene management and the legal issues involved in the prosecution of computer crimes and legislation enacted to protect the public.. (3 hr. lecture)

Lecture Hours: 48.000

CJL2062 Constitutional Law and Legal Procedure or Evidence 3.000 credits

An examination of the United States and Florida Constitutions, with emphasis on leading cases dealing with arrest, search and seizure, confessions and the rules of evidence.

Lecture Hours: 48.000

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2041C Python Programming 2 4.000 credits

This is an advanced level programming course using Python. Students will learn how to code, compile, and execute programs. Topics include problem solving, object-oriented programming, database programming, data visualization, and secure software development. Prerequisite: COP1047C; (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2143C Server Administration 4.000 credits

This course is designed for students preparing for IT careers as server administrators, and focuses on server hardware and operating systems, system management and administration, security and disaster recovery, infrastructure and network services, data storage, and troubleshooting. This course is intended to prepare students for the CompTIA Server+ certification.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Server operating systems experience.

Course User Fee: \$63.00

CTS2148C IT Project Management 4.000 credits

This course prepares students to effectively plan, implement, coordinate and manage small to medium size information technology-related projects. The concepts and tools covered in this course offers practical knowledge on managing project scope, planning and scheduling, risk assessment, and identification of project resources. This course also prepares students for the CompTIA Project+ examination, an industry standard IT certification. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2192C Microsoft Azure Administration 4.000 credits

This course is intended for students preparing for IT careers as cloud administrators, as well as candidates for industry certification. Students will learn the administration skills and knowledge required to implement, manage, and monitor identity, governance, storage, and compute virtual networks in the Microsoft Azure cloud environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS1145 or equivalent knowledge in Microsoft Azure Fundamentals

Course User Fee: \$63.00

CTS2303 Windows Server Administration 4.000 credits

This course is intended for students preparing for IT careers as network support specialists as well as server and cloud administrators. Students will learn the fundamental administration skills and knowledge required to deploy and support Windows Server core services.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CGS1060 and CTS1134 or equivalent knowledge

Course User Fee: \$63.00

CTS2375C Cloud Infrastructure and Services 4.000 credits

This course helps students develop technical expertise in Cloud computing and prepares them for industry certification. Students will learn the essentials of Cloud computing, business security and compliance considerations, migrating to the Cloud, architecting a Cloud server, and how to troubleshoot Cloud services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2145

Course User Fee: \$63.00

CTS2670 Check Point Security Administration 4.000 credits

This course, designed for students specializing in network security, prepares students for the Check Point Certified Security Administrator (CCSA) certification exam. Students will learn how to install security gateways; configure rules on servers; create a rule base; assign user permissions; schedule backups and upgrades; monitor and troubleshoot common network traffic.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120, CTS1134

CTS2671 Check Point Security Engineering 4.000 credits

This course, for students specializing in network security, prepares students for the Check Point Certified Security Expert (CCSE) certification examination. Students learn how to configure, build, modify, deploy and troubleshoot a secure network utilizing firewall technologies. Topics include clustering, software acceleration, advanced VPN concepts and implementation, and monitoring and reporting tools.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120, CTS1134, CTS2670

Course User Fee: \$63.00

Program Description

Database Technology - Oracle Database Administration

Associate in Science | Code: 25058 | 60 credits

The Database Technology program is designed to provide an opportunity to establish a basic foundation in the field of database administration for employment in commercial, industrial and government institutions. Graduates are prepared for positions as database administrators and database developers.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
COP 1334	Introduction to C++ Programming	4.0
Semester 2		Units: 12.0
CGS 1540C	Database Concepts Design	4.0
COP 2800	Java Programming	4.0
<i>Program Electives</i>		4.0
CAP 1788	Introduction to Data Analytics	4.0
OR		

CAP 2743C	Power BI - Data Visualization and Analysis	4.0
OR		
CAP 2761C	Intermediate Analytics	4.0
OR		
CAP 2791C	Power BI - Data Preparation and Modeling	4.0
OR		
CIS 1949	CO-Op Work Experience 1: CIS	1.0
OR		
CIS 2900	Directed Study in IT	1.0
OR		
COP 2805C	Java Programming 2	4.0
OR		
CTS 1111	Linux +	4.0
OR		
CTS 1120	Cybersecurity Fundamentals	4.0
OR		
CTS 1145	Cloud Essentials	4.0
OR		
CTS 2143C	Server Administration	4.0
OR		
CTS 2192C	Microsoft Azure Administration	4.0
OR		
CTS 2375C	Cloud Infrastructure and Services	4.0

Semester 3

Units: 6.0

<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 4

Units: 15.0

<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
CTS 2440	Introduction to Oracle: SQL And PL/SQL	4.0
CTS 2441	Introduction to Oracle Database Administration	4.0
Program Electives ¹⁾		4.0

¹⁾Select 4 credits from the following course options: CAP 1788, CAP 2743C, CAP 2761C, CAP 2791C, CIS 1949, CIS 2900, COP 2805C, CTS 1111, CTS 1120, CTS 1145, CTS 2143C, CTS 2192C, CTS 2375C

Semester 5

Units: 13.0

CTS 2442	Intermediate Oracle Database Administration	4.0
CTS 2444	Oracle Database Performance Tuning	4.0
Program Electives ¹⁾		5.0

¹⁾Select 5 credits from the following course options: CAP 1788, CAP 2743C, CAP 2761C, CAP 2791C, CIS 1949, CIS 2900, COP 2805C, CTS 1111, CTS 1120, CTS 1145, CTS 2143C, CTS 2192C, CTS 2375C

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.
Lecture Hours: 48.000, Lab Hours: 32.000
Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.
Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.
Lecture Hours: 48.000, Lab Hours: 32.000
Corequisite: COP1047C OR COP1334 OR COP2270
Course User Fee: \$63.00

CAP1788 Introduction to Data Analytics 4.000 credits

This course offers a broad introduction to data analytics and the role it plays in modern organizations. Students will use Excel and Tableau to import, clean, transform, and visualize data. Students will also learn how to effectively communicate findings to decision makers.
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

CAP2743C Power BI - Data Visualization and Analysis 4.000 credits

This is the second of two introductory courses aimed at preparing students for the Microsoft Power BI certification exam. Students will learn how to create informative data visualizations and leverage Power BI's analytic capabilities to provide meaningful business insights. Prerequisite: CAP2791C.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CAP2791C
Course User Fee: \$63.00

CAP2761C Intermediate Analytics 4.000 credits

This course offers an introduction to the SQL language and how it can be used by data analysts to query databases and gain valuable insights. Students will also gain extensive hands-on experience with Microsoft SQL Server, Azure, and GitHub.
Prerequisites: CAP1788, CGS1540C.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CGS1540C
Course User Fee: \$63.00

CAP2791C Power BI - Data Preparation and Modeling 4.000 credits

This is one of two introductory courses aimed at preparing students for the Microsoft Power BI certification exam. Students will learn how to get data from different sources, clean and transform the data, and design a data model that will be used in analysis and reports. Prerequisites: CAP1788 and CGS1540C.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CAP1788 AND CGS1540C
Course User Fee: \$63.00

CIS1949 CO-Op Work Experience 1: CIS 1.000 - 4.000 credits

This course is designed as a work experience for students majoring in computer information systems programs. Students will learn to apply the skills and knowledge that they have acquired through their program of study in a real work environment. Prerequisite: Successful completion of required program course work. Department approval required.
Lecture Hours: 16.000 - 64.000
Prerequisite: Successful completion of required program course work. Department approval required.

CIS2900 Directed Study in IT 1.000 credits

This course, for majors in Information Technology, provides students the opportunity to pursue projects under faculty supervision through self-directed research, experiences, and presentations.

Prerequisite: Successful completion of required program course work. Department approval required.

Lab Hours: 32.000

Recommended Preparation: Successful completion of required program course work. Department approval required

COP2805C Java Programming 2 4.000 credits

This is an advanced level programming course using Java. Students will learn how to code, compile and execute programs. Topics include object serialization, Java Collection, sorting/searching algorithms, multithreading and networking capabilities, and Java databases.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

CTS1111 Linux + 4.000 - 32.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based work-stations and servers. Prerequisite: CGS 1060 or computer experience is required. (3 hr. lecture; 2 hr. lab)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2143C Server Administration 4.000 credits

This course is designed for students preparing for IT careers as server administrators, and focuses on server hardware and operating systems, system management and administration, security and disaster recovery, infrastructure and network services, data storage, and troubleshooting. This course is intended to prepare students for the CompTIA Server+ certification.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Server operating systems experience.

Course User Fee: \$63.00

CTS2192C Microsoft Azure Administration 4.000 credits

This course is intended for students preparing for IT careers as cloud administrators, as well as candidates for industry certification. Students will learn the administration skills and knowledge required to implement, manage, and monitor identity, governance, storage, and compute virtual networks in the Microsoft Azure cloud environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS1145 or equivalent knowledge in Microsoft Azure Fundamentals

Course User Fee: \$63.00

CTS2375C Cloud Infrastructure and Services 4.000 credits

This course helps students develop technical expertise in Cloud computing and prepares them for industry certification. Students will learn the essentials of Cloud computing, business security and compliance considerations, migrating to the Cloud, architecting a Cloud server, and how to troubleshoot Cloud services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2145

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CTS2440 Introduction to Oracle: SQL And PL/SQL 4.000 credits

This is an introductory level course for students majoring in the Oracle Database Administrator and/or Solutions Developer programs. Students will learn the fundamentals of SQL and PL/SQL programming languages including the concepts of relational databases, how to create and maintain database objects, and how to store, retrieve, and manipulate data. Students will also learn to create PL/SQL blocks of application code that can be shared by multiple forms, reports, and data management applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1060C

Course User Fee: \$63.00

CTS2441 Introduction to Oracle Database Administration 4.000 credits

This course is designed to give students who are preparing to become Oracle database administrators (DBA) a firm foundation in basic administrative tasks. Students will learn through instructor-led learning, structured hands-on practices, and challenge-level exercise labs, the necessary knowledge and skills to set up, maintain and troubleshoot an Oracle database.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2440

Course User Fee: \$63.00

CTS2442 Intermediate Oracle Database Administration 4.000 credits

This is the second course in Oracle database administration. Students will learn basic network administration, including techniques to backup and to recover an Oracle database. The skills developed in this class will help prepare students for the Oracle database administrator (DBA) certification exam.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2441

Course User Fee: \$63.00

**CTS2444 Oracle Database Performance Tuning
4.000 credits**

This course teaches students tuning steps which can be used to improve database performance. Students will learn through a combination of demonstrations, lectures, and lab exercises, gaining practical experience tuning an Oracle database. Students will also learn how to recognize, troubleshoot and resolve common performance-related problems in administering an Oracle database.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2441

Course User Fee: \$63.00

Program Description

Dental Hygiene

Associate in Science | Code: 23021 | 88 credits

The Dental Hygienist is a licensed member of the dental health team dedicated to helping patients maintain good oral health and prevent dental disease and disorders. The dental hygienist performs dental cleaning, teaches patients proper oral care, takes x-rays and provides nutritional counseling for optimal oral health.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 15.0
ENC 1101	English Composition 1	3.0
CHM 1033	Chemistry for Health Sciences	3.0
MGF 1130	Mathematical Thinking	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
PSY 2012	Introduction to Psychology	3.0
Semester 2		Units: 7.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
PHI 2010	Introduction to Philosophy	3.0
Computer Competency *		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3		Units: 9.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0

MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0

Semester 4 (In Milestone)

Units: 13.0

DEH 1002	Pre-Clinical Dental Hygiene	2.0
DEH 1002L	Pre-Clinical Dental Hygiene Laboratory	2.0
DEH 1133	Dental Anatomy, Histology and Physiology	2.0
DEH 1710	Oral Health Literacy	1.0
DEH 1720	Preventative Dentistry	2.0
DES 1200	Dental Radiology	2.0
DES 1200L	Dental Radiology Laboratory	2.0

Semester 5 (In Milestone)

Units: 14.0

DEH 1400	General and Oral Pathology	3.0
DEH 1800	Dental Hygiene 1	2.0
DEH 1800L	Dental Hygiene 1 Clinic	3.0
DEH 1811	Professional Issues	2.0
DEH 2202	Nutrition and Dental Health	2.0
DES 1600	Dental Office Emergency	2.0

Semester 6 (In Milestone)

Units: 6.0

DEH 1802L	Dental Hygiene 2 Clinic	1.0
DEH 1804L	Dental Hygiene 3 Clinic	1.0
DEH 1840L	Advanced Radiographic & Clinical Assessment Tecniq	1.0
DEH 2300	Pharmacology and Pain Control	1.0
DEH 2300L	Pharmacology and Pain Control Laboratory	1.0
DEH 2602	Periodontology 1	1.0

Semester 7 (In Milestone)

Units: 12.0

DEH 2603	Periodontology 2	2.0
DEH 2603L	Periodontology 2 Laboratory	1.0
DEH 2701	Community Dental Health 1	3.0
DEH 2806	Oral Health for Special Populations	2.0
DEH 2806L	Dental Hygiene 4 Clinic	4.0

Semester 8 (In Milestone)

Units: 12.0

DEH 2702L	Community Dental Health 2 Clinic	2.0
DEH 2808	Dental Hygiene Practice Management and Leadership	2.0
DEH 2808L	Dental Hygiene 5 Clinic	4.0
DEH 2810L	Interprofessional Practice and Education Lab	1.0
DES 2100	Dental Materials	2.0
DES 2100L	Dental Materials Laboratory	1.0

Total: 88.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions.

Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence. Lecture Hours: 48.000

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Lab 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

DEH1002 Pre-Clinical Dental Hygiene 2.000 credits

Introduction to procedures relevant to the practice of dental hygiene.

Lecture Hours: 32.000

Corequisite: DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L

DEH1002L Pre-Clinical Dental Hygiene Laboratory 2.000 credits

Laboratory for DEH 1002.

Clinical Hours: 96.000

Corequisite: DEH 1002, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L

Course User Fee: \$120.00

DEH1133 Dental Anatomy, Histology and Physiology 2.000 credits

This course covers specific tissues of the oral cavity, head, neck and their embryonic development. The students will learn structure, morphology and function of the primary and permanent dentitions.

Lecture Hours: 32.000

Corequisite: DEH 1002, DEH 1002L, DEH 1710, DEH 1720, DES 1200, DES 1200L

DEH1710 Oral Health Literacy 1.000 credits

Students will learn the concepts of oral health literacy. Students will also identify how to improve patient's oral health literacy and the barriers that impede oral health care.

Lecture Hours: 16.000

Corequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1720, DES 1200, DES 1200L

DEH1720 Preventative Dentistry 2.000 credits

This is a foundation course in dental hygiene preventive care. Students will learn the concepts of oral health and how to prevent future disease. Students will become engaged in developing their own prevention strategies by selecting with a rationale, appropriate oral health devices used for self-care.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DES 1200, DES 1200L

DES1200 Dental Radiology 2.000 credits

Techniques and theory for the safe and effective use of radiographs as related to dentistry.

Lecture Hours: 32.000

Corequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200L

DES1200L Dental Radiology Laboratory 2.000 credits

Laboratory for DES 1200.

Lab Hours: 64.000

Corequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200

Course User Fee: \$90.00

DEH1400 General and Oral Pathology 3.000 credits

Processes of inflammation, necrosis, retrograde changes, diseases caused by bacteria, viruses, and other organisms. Emphasis will be placed on differentiating between normal and abnormal conditions of the oral cavity.

Lecture Hours: 48.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600

DEH1800 Dental Hygiene 1 2.000 credits

Theory of the removal of hard and soft deposits from the teeth, and other related postoperative and preventive procedures.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800L, DEH 1811, DEH 2202, DES 1600

DEH1800L Dental Hygiene 1 Clinic 3.000 credits

Clinic for DEH 1800.

Clinical Hours: 144.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800, DEH 1811, DEH 2202, DES 1600
Course User Fee: \$135.00

DEH1811 Professional Issues 2.000 credits

This course is designed to provide the dental hygiene student with an understanding of the political, social, environmental and professional issues that affect the practice of dental hygiene. These issues include: a) cultural diversity, b) legal and ethical responsibilities, c) sexual harassment, d) child abuse, e) problem solving, f) communication style.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 2202, DES 1600

DEH2202 Nutrition and Dental Health 2.000 credits

This course provides a study of nutrients, their nature, source, and utilization. Students will learn the relationship between diet and oral health care and oral manifestations of nutritional deficiencies.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DES 1600

DES1600 Dental Office Emergency 2.000 credits

This course is designed to instruct students in the fundamental patient assessment skills needed to identify and manage emergencies that may arise in the dental office.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202

DEH1802L Dental Hygiene 2 Clinic 1.000 credits

Continuation of clinical skills from DEH 1800L.

Clinical Hours: 48.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602
Course User Fee: \$75.00

DEH1804L Dental Hygiene 3 Clinic 1.000 credits

Designed to further student's knowledge and skills through clinical experiences more difficult than those experienced in DEH 1802L.

Clinical Hours: 48.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602

Course User Fee: \$75.00

DEH1840L Advanced Radiographic & Clinical Assessment Tecniq 1.000 credits

A laboratory course introducing advanced digital radiographic techniques, the intraoral camera, periodontal probing and dental charting software and other clinical assessment tools. These skills will enable the student to provide comprehensive patient treatment and enhance their ability to interpret intraoral conditions.

Clinical Hours: 48.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1804L, DEH 2300, DEH 2300L, DEH 2602

Course User Fee: \$60.00

DEH2300 Pharmacology and Pain Control 1.000 credits

This course introduces the student to a broad range of pharmacological concepts including drug categories, drug action, and adverse drug effects. Dental prescriptions such as the antibiotics, antifungals and antivirals will be studied. Students will learn the common medical conditions affecting dental hygiene care, such as cardiovascular disease, endocrine and neurological disorders as well as their drug management will be examined.

Lecture Hours: 16.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300L, DEH 2602

DEH2300L Pharmacology and Pain Control Laboratory 1.000 credits

This course is designed to prepare the dental hygiene student for the safe and effective administration of local anesthesia. Students will learn about the psychology of pain management, pharmacology of anesthetic agents, emergency precautions, and a review of anatomy and physiology as they relate to the administration of anesthetic agents. This course will include online and clinical instruction.

Lab Hours: 32.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2602

Course User Fee: \$100.00

DEH2602 Periodontology 1 1.000 credits

This course will introduce the student to the concepts of non-surgical periodontal therapy, risk factors in periodontal diseases, classifications of periodontal diseases, the components of the comprehensive periodontal assessment and care plan. Ultrasonic periodontal debridement will be studied. Furthermore, the course will include the study of behavior motivation, the dental hygiene human needs conceptual model, the phases of self-care education and the importance of case presentation in modifying client self-care.

Lecture Hours: 16.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L

DEH2603 Periodontology 2 2.000 credits

Etiology, classification, diagnosis, treatment and maintenance of the periodontal patient.

Lecture Hours: 32.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH 2603L, DEH 2701, DEH 2806, DEH 2806L

DEH2603L Periodontology 2 Laboratory 1.000 credits

Laboratory for DEH 2603.

Lab Hours: 32.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH 2603, DEH 2701, DEH 2806, DEH 2806L

Course User Fee: \$100.00

DEH2701 Community Dental Health 1 3.000 credits

Public Health Dentistry and the role of the dental hygienist.

Lecture Hours: 48.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH 2603, DEH 2603L, DEH 2806, DEH 2806L

DEH2806 Oral Health for Special Populations 2.000 credits

This course is a continuation of dental hygiene theory and practice. Students will learn the process and procedures for gingival curettage and root planning.

Lecture Hours: 32.000

Prerequisite: DEH1804L, Corequisite: DEH2806L

Course User Fee: \$362.24

DEH2806L Dental Hygiene 4 Clinic 4.000 credits

Clinic for DEH 2806.

Clinical Hours: 192.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806

Course User Fee: \$150.00

DEH2702L Community Dental Health 2 Clinic 2.000 credits

Provides the student an opportunity for application of the principles of public and community dentistry.
Lab Hours: 64.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2808, DEH 2808L, DEH 2810L, DES 2100, DES 2100L

DEH2808 Dental Hygiene Practice Management and Leadership 2.000 credits

Students will learn the basic dental and behavioral sciences within the practice of dental hygiene. Special emphasis is given to Florida laws governing dental hygiene practices.

Lecture Hours: 32.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808L, DEH 2810L, DES 2100, DES 2100L

DEH2808L Dental Hygiene 5 Clinic 4.000 credits

Ongoing experience in total dental hygiene care of the periodontally involved patient.

Clinical Hours: 192.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808, DEH 2810L, DES 2100, DES 2100L

Course User Fee: \$150.00

DEH2810L Interprofessional Practice and Education Lab 1.000 credits

Students will learn to examine the Dental Hygiene scope of practice through professional identity by way of interprofessional collaboration. Students will also develop knowledge and skills to serve on health professional teams to improve health outcomes.

Lab Hours: 32.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808, DEH 2808L, DES 2100, DES 2100L

DES2100 Dental Materials 2.000 credits

Physical properties of dental materials and their use in the oral cavity.

Lecture Hours: 32.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808, DEH 2808L, DEH 2810L, DES 2100L

DES2100L Dental Materials Laboratory 1.000 credits

Laboratory for DES 2100.

Lab Hours: 32.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808, DEH 2808L, DEH 2810L, DES 2100

Course User Fee: \$190.00

Program Description

Diagnostic Medical Sonography

Associate in Science | Code: 23038 | 77 credits

The Diagnostic Medical Sonography program prepares the student to become a Diagnostic Medical Sonographer. The Diagnostic Medical Sonographer provides patient services, using diagnostic ultrasound under the supervision of a doctor of medicine or osteopathy who is responsible for the use and interpretation of ultrasound procedures. The sonographer assists the physician in gathering sonographic data necessary to reach diagnostic decisions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2

Units: 10.0

BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
<i>Physics</i>		3.0
PHY 1020	General Education Physics	3.0
OR		
PHY 1025	Basic Physics	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 3		Units: 12.0
SON 1000L	Introduction to Sonography 1	1.0
SON 1006L	Professional Aspects of Sonography	1.0
SON 1100L	Principles of Protocols Of Imaging	2.0
SON 1111C	Abdominal Sonography 1	2.0
SON 1113L	Sonography Cross Sectional Anatomy	2.0
SON 1121C	Obstetrics/Gynecology Sonography 1	2.0
SON 1804	Clinic 1	2.0
Semester 4		Units: 11.0
SON 1001L	Introduction to Sonography 2	1.0
SON 1005L	Basic Sonography	2.0
SON 1112C	Abdominal Sonography 2	2.0
SON 1122C	Obstetrics/Gynecology Sonography 2	2.0
SON 1814	Clinic 2	2.0
SON 2400C	Echocardiography 1	2.0
Semester 5		Units: 5.0
SON 1141C	Small Parts Sonography	2.0
SON 1824	Clinic 3	3.0
Semester 6		Units: 11.0
SON 1115L	Duplex Abdominal Sonography	1.0
SON 1145L	Pediatric Sonography	1.0
SON 2139L	Cardiovascular Principles	1.0
SON 2151C	Neurosonography	2.0
SON 2401C	Echocardiography 2	2.0
SON 2614C	Acoustical Physics and Instrumentation 1	2.0
SON 2834	Clinic 4	2.0
Semester 7		Units: 11.0
PHI 2010	Introduction to Philosophy	3.0
SON 2618C	Acoustical Physics and Instrumentation 2	2.0
SON 2619C	Doppler Principles and Instrumentation	2.0
SON 2844	Clinic 5	3.0

SON 2935L	Diagnostic Ultrasound Imaging: Advanced Techniques and Case Analysis	1.0
Semester 8		Units: 7.0
SON 2854	Clinic 6	3.0
SON 2930L	Seminar in Sonography	1.0
<i>Elective</i>		3.0
DEP 2000	Human Growth and Development	3.0
OR		
SYG 2000	Introduction to Sociology	3.0

Total: 77.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SON1000L Introduction to Sonography 1 1.000 credits

An introduction to the physical principles of diagnostic ultrasound. Bases of imaging with ultra sound are discussed as well as clinical units in the various areas of specialization. In conjunction with the lectures, supervised laboratory classes are conducted to familiarize students with operations of the equipment in each of the clinical areas.

Lab Hours: 32.000

Corequisite: SON 1111C, SON 1121C

SON1006L Professional Aspects of Sonography 1.000 credits

An introduction to the professional aspects of sonography. Topics include: medical ethics and law, hospital administration, quality assurance/quality control and management. Laboratory experience includes actual phantom scanning conducting equipment QA protocols, and participation on a mock ethics board.

Lab Hours: 32.000

SON1100L Principles of Protocols Of Imaging 2.000 credits

An introduction to radiographic film, its handling & processing and the various radiographic specialties.

Laboratory experience includes: film composition and identification, rapid processing, photographic techniques, reading H&D curves, performing sensitometry and identifying film artifacts. During radiographic specialties, there will be an introduction to CT, MRI, and the areas of radiologic technology in order to discover how these modalities compliment sonography.

Lab Hours: 64.000

SON1111C Abdominal Sonography 1 2.000 credits

An in-depth course designed to cover all aspects of clinical abdominal ultrasound studies. Subject matter includes: review of normal anatomy (ultrasonic appearance), indications for ultrasound studies, clinical presentation and data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls.

Lecture Hours: 16.000, Lab Hours: 32.000

Corequisite: SON1000L

SON1113L Sonography Cross Sectional Anatomy 2.000 credits

A thorough course aimed at teaching the student to understand anatomical relationships and recognize structures on cross-sectional and sagittal diagrams, photographs of gross anatomy and sonography. The laboratory conducted in conjunction with the classroom lectures is designed to identify all normal anatomical landmarks in multiple planes in actual scanning situations.

Lab Hours: 64.000

SON1121C Obstetrics/Gynecology Sonography 1 2.000 credits

An in-depth course designed to present all aspects of clinical OB/GYN ultrasound studies. Subject matter includes: review of normal anatomy (ultrasound appearance), indications for ultrasonic studies, clinical presentation, clinical data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls.

Lecture Hours: 16.000, Lab Hours: 32.000

Corequisite: SON1000L

SON1804 Clinic 1 2.000 credits

This is the first in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded a hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 96.000

Corequisite: SON1000L

SON1001L Introduction to Sonography 2 1.000 credits

This second introductory course will cover the past present and future of sonography. After the historical landmarks are identified, the focus will be on the current diversity of applications of diagnostic medical sonography. Students will also discover future trends and developments on the technology horizon of the profession.

Lab Hours: 32.000

Prerequisite: SON1000L

SON1005L Basic Sonography 2.000 credits

This course is designed to cover the essential of the profession of Diagnostic Medical Sonography. Topics include: professionalism, medical ethics, hospital administration, sonographic terminology, quality assurance, photographic principles, related radiological specialties and scanning techniques. Laboratory experience will include equipment use and quality assurance techniques.

Lab Hours: 64.000

Prerequisite: SON1000L

Course User Fee: \$15.00

SON1112C Abdominal Sonography 2 2.000 credits

An in-depth course designed to cover all aspects of clinical abdominal ultrasound studies. Subject matter includes: review of normal anatomy (ultrasonic appearance), indications for ultrasound studies, clinical presentation and data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON1111C

Course User Fee: \$15.00

SON1122C Obstetrics/Gynecology Sonography 2 2.000 credits

An in-depth course designed to cover all aspects of clinical OB/GYN ultrasound studies. Subject matter includes: review of normal anatomy (ultrasound appearance), indications for ultrasonic studies, clinical presentation, clinical data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON1121C

SON1814 Clinic 2 2.000 credits

This is the second in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 96.000

Prerequisite: SON1804

SON2400C Echocardiography 1 2.000 credits

An in-depth course designed to present all aspects of clinical cardiovascular ultrasound studies. Topics discussed are: pathophysiological basis of diseases, clinical presentation and clinical data, Doppler and echocardiographic findings in disease, hemodynamic relationships, scanning pitfalls and differential diagnosis.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON1000L, SON1113L

SON1141C Small Parts Sonography 2.000 credits

An in-depth course designed to cover all aspects of clinical abdominal ultrasound studies. Subject matter includes: review of normal anatomy (ultrasonic appearance), indications for ultrasound studies, clinical presentation and data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls. Prerequisite: SON 1112C. (2 hr. lab.)

Lab Hours: 48.000

SON1824 Clinic 3 3.000 credits

This is the third in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 144.000

Prerequisite: SON1814

SON1115L Duplex Abdominal Sonography 1.000 credits

This course is designed to cover aspects of duplex abdominal sonography applications. Topics include: the aorta and its branches, the IVC and its tributaries, and the portal system. Subject matter includes: etiology, pathophysiology, clinical presentations, sonographic appearance and differential diagnosis of diseases.

Lab Hours: 32.000

Prerequisite: SON1112C

SON1145L Pediatric Sonography 1.000 credits

This course is designed to cover aspects of pediatric ultrasound examinations. Topics include: Liver, biliary, spleen, renal, adrenal, gastrointestinal, scrotum, and musculoskeletal structures. Subject matter includes: etiology, pathophysiology, clinical presentations, sonographic appearance and differential diagnosis.

Lab Hours: 32.000

Prerequisite: SON1141C

SON2139L Cardiovascular Principles 1.000 credits

An introductory course to techniques other than echocardiography utilized in the diagnosis of cardiovascular disease. Topics discussed include physical examination, electrocardiogram, Phonocardiogram, cardiac catheterization, and nuclear medicine cardiology.

Lab Hours: 32.000

Prerequisite: SON2400C, Corequisite: SON2401C

SON2151C Neurosonography 2.000 credits

A comprehensive course designed to examine sonographic imaging of the neonatal and infant brain, with an introduction to intraoperative brain and spinal cord imaging. Emphasis is placed on normal brain anatomy, congenital and malformations and acquired pathologic conditions.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON1113L, SON1141C

Course User Fee: \$15.00

SON2401C Echocardiography 2 2.000 credits

An in-depth course designed to cover all aspects of clinical cardiovascular ultrasound studies. Topics discussed are pathophysiological basis of diseases, clinical presentation and clinical data, Doppler and echocardiographic findings in disease, hemodynamic relationships, scanning pitfalls and differential diagnosis.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON2401C

SON2614C Acoustical Physics and Instrumentation 1 2.000 credits

The course will present a review of fundamental physics and in-depth study of the physical principles of diagnostic ultrasound. Topics discussed include: properties of sound waves, interaction of sound waves with matter, generation of ultrasound and principles of Doppler ultrasound.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON1005L

SON2834 Clinic 4 2.000 credits

This is the fourth in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 96.000

Prerequisite: SON1824

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

SON2618C Acoustical Physics and Instrumentation 2 2.000 credits

Physical principles of Ultrasound Instrumentation -A course designed to familiarize the student with the physical principles and modes of operation of diagnostic ultrasound equipment. Subject matter includes: transducers, display systems, component parts of a scanning system, real-time scanners, Doppler equipment, quality control, routine maintenance and recent developments.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON2614C

SON2619C Doppler Principles and Instrumentation 2.000 credits

This course presents a review of fundamental physics and an in-depth study of Doppler Physical Principles of Diagnostic Ultrasound. Topics also include Doppler Instrumentation, equipment, display systems, quality control, and hemodynamics of blood flow. Prerequisite: SON 2618C. Laboratory fee. (1 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000

Course User Fee: \$41.00

SON2844 Clinic 5 3.000 credits

This is the fifth in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 144.000

Prerequisite: SON2834

SON2935L Diagnostic Ultrasound Imaging: Advanced Techniques and Case Analysis 1.000 credits

An extensive laboratory aimed at teaching the student to recognize quality images, anatomy, patient positioning, pathology, and scanning technique errors as well as artifacts. For each class, the student will present a case from their current rotation of the teaching file. The presentation will include all technical and clinical information as well as the final interpretation by the supervising physician.

Lab Hours: 32.000

SON2854 Clinic 6 3.000 credits

This is the last in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 144.000

Prerequisite: SON2844

SON2930L Seminar in Sonography 1.000 credits

Students will participate in the various types of continuing education. This may include: society meetings, seminars, conferences and in-services.

Lab Hours: 32.000

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

Program Description

Electronics Engineering Technology

Associate in Science | Code: 26039 | 68 credits

The Electronics Engineering Technology program prepares students for work as technicians in various fields of electronics technology. No previous experience is required to enter. Courses offered cover basic and advanced electrical circuits, semi-conductors, integrated circuits, pulse circuits, digital computer circuits, electrical machinery, communication systems and industrial control. Theory and laboratory experience is provided. *Note: This program transfers to four-year institutions. See department advisor for information.*

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
COP 2270	C for Engineers	4.0
EET 1015C	Direct Current Circuits	4.0
Semester 2		Units: 13.0
MAC 1114	Trigonometry	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0

OR		
POS 2041	American Federal Government	3.0
CET 1110C	Digital Circuits	4.0
ETI 2670	Engineering Economic Analysis	3.0

Semester 3		Units: 8.0
EET 1025C	Alternating Current Circuits	4.0
PHY 2053	Physics (without Calculus) 1	3.0
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4		Units: 15.0
CET 2113C	Advanced Digital Circuits	4.0
EET 1141C	Electronics 1	4.0
ETS 2673C	Programmable Logic Controls	4.0
MAC 1140	Pre-Calculus Algebra	3.0

Semester 5		Units: 12.0
CET 2123C	Microprocessors	4.0
EET 2101C	Electronics 2	4.0
<i>Program Electives</i>		4.0
EET 1082	Introduction to Electronics	3.0
OR		
EET 2323C	Analog Communications	4.0
OR		
EET 2351C	Digital and Data Communications	4.0
OR		
EGN 2200	Computer Applications in Engineering	3.0

Semester 6		Units: 6.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
Program Electives ¹⁾		3.0

¹⁾Select 3 credits from the following course options: EET 1082, EET 2323C, EET 2351C, EGN 2200

Total: 68.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

COP2270 C for Engineers 4.000 credits

This course is intended for students majoring in Computer Engineering Technology, Electronics Engineering Technology, or any engineering discipline. Students will learn the "C" programming language, MATLAB, and the Engineering Problem Solving Method to analyze, design, code, compile and execute programs that solve engineering related problems. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105, Recommended

Preparation: CGS1060 OR knowledge of computer skills

Course User Fee: \$63.00

EET1015C Direct Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn electrical components, inductance, capacitance, and elementary network analysis. Students will additionally learn to verify and apply basic theories and principles through hands-on, laboratory experiments utilizing modern testing equipment. Pre/corequisite: MAT 1033. Laboratory fee. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 48.000

Pre/Corequisite: MAT1033

Course User Fee: \$63.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CET1110C Digital Circuits 4.000 credits

This is an introductory course in digital circuits for engineering and other technology majors. Students learn how to apply electronic principles to digital computer circuits and systems. Students will also learn how to simplify logic circuits, build digital circuits, and perform other laboratory activities. Pre/Co-requisite: MAC1105. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

ETI2670 Engineering Economic Analysis 3.000 credits

This course is designed for students who are majoring in any engineering discipline. Students will learn the basic methods of engineering cost analysis including equivalence, value measurement, interest relationships and decision support theory and techniques as applied to capital projects. Various problem solving methods will be used for decision making, multiple alternatives and uncertainty.

Lecture Hours: 48.000

Prerequisite: MAC1105

EET1025C Alternating Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn inductance, capacitance, vector notation, AC circuits, impedance, phase shift, networks, transformers, and resonance. Students will apply and verify theories and principles through hands-on, laboratory experiments utilizing modern testing equipment.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1015C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

CET2113C Advanced Digital Circuits 4.000 credits

This is a second level course in digital circuits for students majoring in electronics and related engineering technologies that extends the application of sequential and combinational logic circuits and other digital applications. Students will learn to program, operate, and interface with a micro-computer and its elements. Prerequisites: CET1110C, COP2270. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

EET1141C Electronics 1 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology or related fields. Students will learn how to apply electronic principles to analog circuits and systems, including semiconductor diodes, applying the fundamental theory of transistors and other solid-state devices; analysis of amplifiers, oscillators, and other applications using a sinusoidal wave. Students also learn basic safety procedures to follow when working in an electronics laboratory and with electronic circuits and systems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1025C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

ETS2673C Programmable Logic Controls 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology and Industrial Engineering Technology. Students will learn the principals of Programmable Logic Controls (PLC) including hardware, programming, and troubleshooting. Students will also develop advanced working programs, and troubleshoot hardware and software communication problems. Prerequisite: CET110C

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET1110C

Course User Fee: \$63.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

CET2123C Microprocessors 4.000 credits

This course is intended for students majoring in Electronics or Computer Engineering Technology. Students will learn to apply digital principles to the understanding of microprocessor parameters and characteristics (addressing range and models, instruction set, architecture, input/output, interrupts, and programming). Students will build practical microprocessor and/or microcontroller based systems to perform a variety of engineering applications. Prerequisites: CET1110C, COP 2270. Special Fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET1110C AND COP2270

Course User Fee: \$63.00

EET2101C Electronics 2 4.000 credits

This course is intended for students majoring in electronics or computer engineering technologies. Students will learn how to apply electronic principles to analog circuits including transistor amplifiers, feedback and frequency response of linear circuits, operational amplifiers, MOSFET and oscillators.

Lecture Hours: 32.000, Lab Hours: 48.000

Prerequisite: EET1141C

Course User Fee: \$63.00

EET1082 Introduction to Electronics 3.000 credits

Learn by building practical electronic circuits. Survey course suitable for both majors and non-majors. Instructors and tutors are available to assist in project completion. Topics include: schematics, pictorials, amplifiers, oscillators, burglar alarms, radios, digital circuits. Students will develop individual career plans and learn about employment opportunities within the field.

Lecture Hours: 48.000

Course User Fee: \$63.00

EET2323C Analog Communications 4.000 credits

This course is designed for students majoring in Electronics Engineering Technology, Telecommunications Engineering Technology, and related disciplines. Students will learn the principles of radio wave transmission and reception, including AM and FM transmitters, receivers, single sideband, television and digital data transmission lines, wave propagation antennas and microwaves. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1141C

Course User Fee: \$63.00

EET2351C Digital and Data Communications 4.000 credits

This course is intended for students majoring in Electronics or Computer Engineering Technology. It provides a theoretical and practical background in the basic concepts and applications of digital and data communications. Students will learn analog-to-digital (A/D) and digital-to-analog (D/A) conversions; data communications codes and standards; wired and wireless digital communications; modulation, transmission impairment, the telephone system, modems, multiplexers, and electrical interface standards.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2123C

Course User Fee: \$63.00

EGN2200 Computer Applications in Engineering 3.000 credits

An introduction to fundamental concepts and skills of mathematical programming and computer-aided design. This course explores the use of computer software to solve engineering problems and bring ideas from a concept to a model.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1114 OR MAC1147

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

Program Description

Emergency Medical Services

Associate in Science | Code: 23047 | 73 credits

The Emergency Medical Services program is designed according to national and state standards. Graduates will perform as advanced practitioners and as leaders in the technical supervisory and managerial aspects of advanced emergency care. Graduates will be prepared primarily for employment in agencies providing pre-hospital emergency medical care and secondarily, for jobs in emergency and other acute care areas of the hospital.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester		Units: 13.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy & Physiology Lab 1	1.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MGF 1130	Mathematical Thinking	3.0
Semester 2		Units: 10.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

EMS 1059	Emergency Medical Responder	1.0
EMS 1059L	Emergency Medical Responder Laboratory	1.0
MNA 1345	Effective Supervision	3.0
Computer Competency*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3 (In Milestone) Units: 10.0

<i>EMT</i>		10.0
<i>Group A</i>		10.0
EMS 1119	Emergency Medical Technician	4.0
EMS 1119L	Emergency Medical Technician Lab and Clinic	3.0
EMS 1431	EMT Hospital/Field Experience	3.0
OR		
<i>Group B</i>		10.0
EMS 9996	Emergency Medical Technician (EMT) - Basic	10.0

Semester 4 (In Milestone) Units: 15.0

EMS 2601	Paramedic Lecture 1	8.0
EMS 2601L	Paramedic Laboratory 1	4.0
EMS 2664	Paramedic Clinic 1	3.0

Semester 5 (In Milestone) Units: 15.0

EMS 2602	Paramedic Lecture 2	8.0
EMS 2602L	Paramedic Laboratory 2	4.0
EMS 2665	Paramedic Clinic 2	3.0

Semester 6 (In Milestone) Units: 10.0

EMS 2311	Emergency Medical Services Operations	2.0
EMS 2659	EMS-Field Internship and Conference	8.0

Total: 73.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy & Physiology Lab 1 1.000 credits

In this laboratory course, student will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy.

Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Lab 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

EMS1059 Emergency Medical Responder 1.000 credits

Provides training in emergency medical care for those who may be first to respond to an accident. The course meets the basic requirements of the U.S. Department of Transportation. Recommended for students who are not required to be certified EMTs.

Lecture Hours: 16.000

Corequisite: EMS1059L

EMS1059L Emergency Medical Responder Laboratory 1.000 credits

Provide training in emergency medical care for those who may be first to respond to an accident. The course meets the basic requirements of the U.S. Department of Transportation.

Lab Hours: 32.000

Corequisite: EMS1059

Course User Fee: \$35.00

MNA1345 Effective Supervision 3.000 credits

Students will learn to implement supervisory and management practices. Emphasis is placed on learning to communicate more effectively with employees, motivating employees, increasing one's leadership effectiveness, delegating, counseling problem employees, conducting performance reviews, maintaining a discrimination and harassment-free workplace, and managing time.

Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

EMS1119 Emergency Medical Technician 4.000 credits

A review of basic life support theory. Areas of emphasis include the prehospital environment, preparatory information, patient assessment, medical emergencies, behavioral emergencies, OB/GYN emergencies, trauma emergencies, pediatric emergencies and EMS operations.

Lecture Hours: 64.000

Corequisite: EMS1119L, EMS1431

EMS1119L Emergency Medical Technician Lab and Clinic 3.000 credits

Practical application of the content covered in EMS 1119 with an emphasis on cardiopulmonary resuscitation, splinting, bandaging, patient movement, and other skills as recommended by the U.S. Department of Transportation for the EMT-A level practitioner.

Lab Hours: 96.000

Corequisite: EMS1119, EMS1431

Course User Fee: \$55.00

EMS1431 EMT Hospital/Field Experience 3.000 credits

Practice in local emergency departments and rescue agencies under professional supervision. This course meets the skills recommended by the U.S.

Department of Transportation.

Clinical Hours: 144.000

Corequisite: EMS1119, EMS1119L

EMS9996 Emergency Medical Technician (EMT) - Basic 10.000 credits

This course provides academic credit for students who have successfully completed a State of Florida-approved Emergency Medical Technician (EMT-Basic) Applied Technology Diploma (ATD) program and are enrolled in the related Associate in Science degree within the School of Health Sciences. In accordance with TMOP 1100-816125, the course recognizes competencies attained through prior educational and certification experiences, including emergency patient assessment, airway management, trauma care, medical emergency response, patient transportation, and prehospital care procedures. Credit awarded through this course supports academic progression and acknowledges demonstrated achievement of industry-recognized emergency medical services competencies. Other Hours: 160.000

EMS2601 Paramedic Lecture 1 8.000 credits

EMS2601 is the first course in the sequence necessary for completion of the Paramedic Certificate program. The course is designed to reinforce concepts and clinical skills learned at the EMT level and to integrate this knowledge beginning with advanced life support concepts and skills. Emphasis is placed on EMS systems, illness and injury prevention, medical-legal issues, patient assessment, airway management and ventilation, pathophysiology, pharmacology, shock, decision-making, and the management of trauma-related injuries. This course includes Modules 1–4 of the 1998 DOT National Standard Curriculum for Paramedic Programs.

Lecture Hours: 128.000

Corequisite: EMS2601L

Course User Fee: \$96.00

EMS2601L Paramedic Laboratory 1 4.000 credits

A review of basic life support practice and an introduction to advanced life support practice. Areas of emphasis include patient assessment, trauma emergencies, obstetric emergencies, gynecological emergencies, pediatric emergencies and psychiatric emergencies. Students will be expected to master the techniques of patient assessment, intravenous techniques and endotracheal intubation.

Lab Hours: 128.000

Corequisite: EMS2601

Course User Fee: \$75.00

EMS2664 Paramedic Clinic 1 3.000 credits

EMS 2664 is designed to allow the students "hands-on" practice of the skills and theories learned in EMS 2601 and 2601L. Clinical experience will take place in many areas including the emergency department, operating room and medical examiner's office. All patient care experience will be practiced under the direct supervision of a medical professional (Paramedic, Nurse, Physician, etc.).

Clinical Hours: 144.000

Corequisite: EMS2601, EMS2601L

EMS2602 Paramedic Lecture 2 8.000 credits

EMS 2602 is the second course in the sequence necessary for the completion of the Paramedic Certificate Program. This course is designed to reinforce and expand upon the material and skills learned in Paramedic 1 level and to integrate prior learning with enhanced life support concepts and skills. Emphasis is placed on patient assessment and recognition of significant findings, pre-hospital diagnosis and differential diagnosis, treatment strategies, anatomy and physiology, pathophysiology, and the management of various emergencies, patients with special challenges, assessment based management, and EMS operations. This course includes Modules 5-8 of the 1998 DOT National Standard Curriculum for Paramedic Programs.

Lecture Hours: 128.000

Prerequisite: EMS2601L, EMS2601, EMS2664,

Corequisite: EMS2602L, EMS2665

EMS2602L Paramedic Laboratory 2 4.000 credits

Continuation of advanced life support practice. Areas of emphasis include patient assessment, trauma emergencies, obstetric emergencies, gynecological emergencies, pediatric emergencies and psychiatric emergencies. Students will be expected to master the techniques of patient assessment, intravenous techniques, endotracheal intubation, and advanced life support.

Lab Hours: 128.000

Corequisite: EMS2602, EMS2665

Course User Fee: \$75.00

EMS2665 Paramedic Clinic 2 3.000 credits

EMS 2665 is designed to allow the students "hands-on" practice of the skills and theories learned in EMS 2602 and 2602L. Clinical experience will take place in many areas including the emergency department, operating room and critical care unit. All patient care experience will be practiced under the direct supervision of a medical professional (paramedic, nurse, physician, etc.).

Clinical Hours: 144.000

Prerequisite: EMS2664, Corequisite: EMS2602, EMS2602L

**EMS2311 Emergency Medical Services Operations
2.000 credits**

Advanced theory of management operations currently used nationally by comprehensive emergency medical service systems. Legal issues as related to various aspects of the system, personnel policies, provider versus client roles, disaster planning, communications, budgeting and evaluation of the system will be discussed.

Lecture Hours: 32.000

Prerequisite: MNA1345

**EMS2659 EMS-Field Internship and Conference
8.000 credits**

A supervised clinical experience on an Advanced Life Support (ALS) vehicle. The student obtains increasing patient care responsibilities as a working member of the EMS team under the direct supervision of a designated preceptor.

Clinical Hours: 384.000

Prerequisite: EMS2602, EMS2602L, EMS2665

Program Description

Entrepreneurship

Associate in Science | Code: 20002 | 60 credits

The Associate in Science (AS) in Entrepreneurship prepares students to start-up and operate a business or social venture with a foundation in opportunity recognition, analysis, business model development and business plan creation. Graduates will bring a critically-informed perspective to their own start-up venture, family-owned business, or social venture.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
GEB 1011	Principles of Business	3.0
<i>Program Electives</i>		3.0
BUL 2241	Business Law 1	3.0
OR		
CGS 1060C	Introduction to Computer Technology and Applications	4.0
OR		
ENT 1501	Fundamentals of Changemaking and Social Innovation	3.0
OR		
ENT 2212	Entrepreneurial Leadership	3.0
OR		
MAC 1106	Integrated College and Precalculus Algebra	5.0
OR		
MAC 2233	Business Calculus	3.0
OR		
MAR 2704	Marketing Web Analytics	3.0
OR		
MAT 1033	Intermediate Algebra	3.0

Computer Competency Test*

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 12.0
MAC 1105	College Algebra	3.0
ECO 2013	Principles of Economics (Macro)	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
MAR 1720	Marketing in a Digital World	3.0

Semester 3		Units: 10.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
ECO 2023	Principles of Economics (Micro)	3.0

Semester 4		Units: 13.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
ENT 2201	Introduction to Lean Start-Up	3.0
Program Electives ¹⁾		6.0

¹⁾Select 7 credits from the following course options: BUL 2241, CGS 1060C, ENT 1501, ENT 2212, ENT 2502, ENT 2270, ENT 2421, MAC 1106, MAC 2233 *, MAR 2704, MAT 1033

Semester 5		Units: 13.0
<i>Accounting Option</i>		3.0
ACG 2450	Microcomputers in Accounting	3.0
OR		
TAX 2021	Taxation of Business Organizations	3.0
Program Electives ¹⁾		10.0

¹⁾Select 11 credits from the following course

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ENT1501 Fundamentals of Changemaking and Social Innovation 3.000 credits

This course introduces students to the work of changemaking and the field of social innovation. Students will explore principles of social innovation and social change, while developing the skills to analyze social issues, generate solutions to those issues, and become an effective social change agent.

Lecture Hours: 48.000

Course User Fee: \$43.00

ENT2212 Entrepreneurial Leadership 3.000 credits

The student will learn key skills and traits of successful entrepreneurs. Students will explore the notion of values-based business formation, personal strength and weakness identification, leadership for team building, project and personal time management, and story-telling.

Lecture Hours: 48.000

Course User Fee: \$100.00

MAC1106 Integrated College and Precalculus Algebra 5.000 credits

The student will learn to analyze linear, quadratic, polynomial, rational, radical, absolute value, composite, inverse, piecewise, exponential, and logarithmic functions, conic sections, systems of equations/inequalities, matrices and determinants, sequences & series, the Binomial Theorem, and applications of mathematical modeling including exponential growth and decay. Prerequisite: MAT1033 with a grade of "B" or better.

Computational Course.

Lecture Hours: 80.000

Prerequisite: MAT1033 or appropriate placement scores.

Course User Fee: \$10.00

MAC2233 Business Calculus 3.000 credits

This course introduces the basic concepts of differential and integral calculus for students majoring in business administration and related fields. Topics include limits, continuity, differentiation and integration of polynomials, logarithmic and exponential functions with applications to business, economics, and the life sciences. Prerequisite: MAC 1105 or MAC1106 with a grade of C or better or departmental permission. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

MAR2704 Marketing Web Analytics 3.000 credits

This course helps students to develop a deep understanding of the digital analytics landscape. The student will learn how to identify which metrics are best to measure web, mobile, social and marketing channels, and how organizations use web analytics to obtain higher profits, improved customer relationships, and measurable value. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$57.00

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033Any Mathematics level higher than MAT 1033 will satisfy this requirement.

MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011;

Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

ENT2201 Introduction to Lean Start-Up 3.000 credits

The student will learn how successful start-ups find a repeatable, scalable business model that creates value for themselves and customers. In this experientially driven course, student teams develop, validate and refine a business model by creating a hypothesis, designing products/services, testing the hypotheses, and reflecting on what has been learned.

Prerequisite: GEB2112.

Lecture Hours: 48.000

Prerequisite: GEB2112

Course User Fee: \$69.00

ACG2450 Microcomputers in Accounting 3.000 credits

Accounting application of electronic data processing including the preparation, interpretation and use of computer information in financial decision-making.

Pre/Corequisite: ACG2021. Special fee.

Lecture Hours: 48.000

Pre/Corequisite: ACG2021

Course User Fee: \$22.00

TAX2021 Taxation of Business Organizations 3.000 credits

This course will cover federal income taxation of corporations, S Corporations, limited liability companies and partnerships. Topics covered include determining the tax consequences of income, expenses, distributions, redemptions and liquidations for business entities. Practical application of the tax law will be emphasized along with analysis of tax procedures.

Lecture Hours: 48.000

Course User Fee: \$22.00

Program Description

Fashion Design

Associate in Science | Code: 22005 | 60 credits

The Associate in Science degree in Fashion Design offers students a comprehensive education of the fashion business. The program offers students instruction in process, skills, and designs from concept development through production for emergence into the fashion business or fashion design segments of the industry.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
CTE 1050	Introduction to Fashion Design and Related Industries	3.0
CTE 1743C	Patternmaking Level 1	3.0
CTE 2310C	Basic Clothing Construction Methods	3.0
CTE 2732	Fashion Illustration Technology	3.0
Semester 2		Units: 12.0
CTE 1401	Textiles	3.0
CTE 1721C	Fashion Design 1	3.0
CTE 2330C	Clothing Construction Methods Level 2	3.0
CTE 2745C	Patternmaking Level 2	3.0
Semester 3		Units: 12.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
PHI 2010	Introduction to Philosophy	3.0
Social Science		3.0

AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Computer Competency Test*

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4		Units: 12.0
CTE 1841C	Apparel Evaluation & Production	3.0
CTE 2342C	Clothing Construction Methods Level 3	3.0
CTE 2722C	Fashion Design 2	3.0
CTE 2749C	Patternmaking Level 3	3.0

Semester 5		Units: 12.0
CTE 2760C	Creative Design	3.0
CTE 1942	Fashion Industry Internship	3.0
EVR 1001	Introduction to Environmental Science	3.0
CTE 2120	Portfolio Collection Development	3.0

Total: 60.0

CTE1050 Introduction to Fashion Design and Related Industries 3.000 credits

In this course students will learn the basic design process, history, characteristics and global interrelationships of the fashion industry segments. The course explores how fashion, textile and apparel producers, retailers and related industries use merchandising and market their products within the industry and ultimately, to the consumer.
Lecture Hours: 48.000

CTE1743C Patternmaking Level 1 3.000 credits

This course will focus on the development of basic blocks, dart manipulation, bodice, skirt and sleeve styling methods in accordance with approved garment trade practices. Students will learn to draft and manipulate the various garment components and develop patterns and samples. Must be met with a grade of "C" or higher.
Lecture Hours: 16.000, Lab Hours: 64.000
Corequisite: CTE2310C
Course User Fee: \$241.00

CTE2310C Basic Clothing Construction Methods 3.000 credits

Students will learn the basic elements of sewing utilized and incorporated into all designs in the garment industry. These garment structures form the fundamentals of sewing and are integrated into the construction methods used by each company in the applications to a specific design. Must be met with a grade of "C" or higher.
Lecture Hours: 16.000, Lab Hours: 64.000
Corequisite: CTE1743C
Course User Fee: \$241.00

CTE2732 Fashion Illustration Technology 3.000 credits

This course introduces the basics of fashion illustration technology targeted specifically for the fashion industry. Using various computer software and tools, including but not limited to Photoshop, Fashion CAD, C-Design, Adobe Illustrator, and Digital Fashion Pro, students will learn the techniques to conceptualize, create, and manage fashion designs and collections, and create industry-standard presentations.
Lecture Hours: 48.000

CTE1401 Textiles 3.000 credits

This is a survey course designed for students majoring in fashion-related curriculum or with a general interest in textile materials. Students will learn basic elements of the transformation from fiber of textiles into finished goods. The course provides insights into textile manufactures with a primary focus on general textile applications relative to end-use consumer products. Students will learn the terminology needed for effective communication throughout the fashion supply chain, gain insight and appreciation for the relative value of textile products and the appropriateness of specific textile uses.

Lecture Hours: 48.000

CTE1721C Fashion Design 1 3.000 credits

This introductory course on flat patterns explores the two-dimensional method of creating basic slopers in order to execute designs. Students will learn the two methods of making a basic set of slopers.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE1401 AND CTE1401L

Course User Fee: \$241.00

CTE2330C Clothing Construction Methods Level 2 3.000 credits

This course focuses on the principles and techniques of applied clothing constructions. Students will learn to apply the principles and techniques of clothing construction using original student design and pattern work.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2310C, Corequisite: CTE2745C

Course User Fee: \$241.00

CTE2745C Patternmaking Level 2 3.000 credits

This course focuses on the integration of flat patterns and draping. Students will learn the foundations of Draping as a design development and execution technique. How to move from the form to the flat and on to the body. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE1743C, Corequisite: CTE2330C

Course User Fee: \$241.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CTE1841C Apparel Evaluation & Production 3.000 credits

This is an introductory course in the apparel development process. Students will learn to facilitate the communication and coordination of pre-product development tasks achieved through linking design, costing, and manufacturing technology in the production setup for each design. Students will learn how outsourcing affects the product development process in editing garment designs and the line development calendar.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: CTE1401 AND CTE1401L

CTE2342C Clothing Construction Methods Level 3 3.000 credits

This course advances students' knowledge about materials, core properties and construction techniques that inform their choices made for silhouette creation and hands-on prototype development. Students will explore more advanced principles and techniques of unstructured draping using soft fabric for the creative interpretation and artistic development of contemporary designs. Emphasis on proportion, balance and shape as related to design aesthetics. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2330C, Corequisite: CTE2749C

CTE2722C Fashion Design 2 3.000 credits

This course focuses on the use and development of basic slopers. Students will learn to proceed from basic applications and principles of patternmaking to the advanced process of design development. Combining the bodice and sleeve slopers to develop a kimono/dolman sloper is one example of the application of basic slopers to the comprehensive understanding of principles of pattern and design development.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE1721C

Course User Fee: \$241.00

CTE2749C Patternmaking Level 3 3.000 credits

This course reinforces the students' understanding of the art of draping and patternmaking and diverse methods the industry uses to create production patterns. Students will analyze draping and drafting techniques to create an awareness of which method is most applicable for a given situation. Analytical thinking and hands-on class experiences will strengthen students' skills, thus enabling them to expand on their creativity and provide a proper fit for their creations. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2745C, Corequisite: CTE2342C

CTE1942 Fashion Industry Internship 3.000 credits

This internship course provides eligible students with placement in premier fashion settings and with the professional and practical experiences needed to further their education in a variety of fashion-based positions. Students control the internship selection process and work with an internship coordinator in the revisions of their resumes and schedule of interviews within the network of fashion companies. Fashion internships are available in the areas of design, merchandising, buying, show-room, and fashion public relations. (144 hr. Internship)

Internship Hours: 144.000

Recommended Preparation: Departmental Approval & Completion of the Program Core Requirements

CTE2760C Creative Design 3.000 credits

In this capstone course, the student will focus on the acquisition of a multidisciplinary methodology needed to produce a collection. The student will also survey the various steps of building a collection, from conception and range planning through to design, execution and presentation. A minimum of 6 looks will be produced by the end of the semester.

Prerequisite: CTE2342C and CTE2722C and CTE2749C. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2722C AND CTE2342C AND CTE2749C

Course User Fee: \$241.00

CTE2120 Portfolio Collection Development 3.000 credits

In this course students will learn the basics for the development of a presentation portfolio collection. Students will research fabrics to use in the design of collection concepts for specific market areas. Various formats will be explored using digital and analog media. Retail market research and the internet may be applied as a basis for a design presentation.

Lecture Hours: 48.000

Prerequisite: CTE1401 AND CTE1401L

**EVR1001 Introduction to Environmental Science
3.000 credits**

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

Program Description

Fashion Merchandising

Associate in Science | Code: 22020 | 60 credits

The Associate in Science degree in Fashion Merchandising offers students a comprehensive education of the fashion business. The program offers students instruction in process, skills, and designs from concept development through production for emergence into the fashion business or fashion design segments of the industry.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
CTE 1050	Introduction to Fashion Design and Related Industries	3.0
CTE 2732	Fashion Illustration Technology	3.0
GEB 1011	Principles of Business	3.0
MAR 1011	Principles of Marketing	3.0
Semester 2		Units: 12.0
CTE 1801	Introduction to Fashion Merchandising	3.0
CTE 2301	Product Development	3.0
CTE 2610	Fashion Forecasting and Research	3.0
MAR 1720	Marketing in a Digital World	3.0
Semester 3		Units: 12.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
PHI 2010	Introduction to Philosophy	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0

OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Course Competency Test*

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4		Units: 12.0
CTE 2388	Principles of Contemporary Retailing	3.0
CTE 2802	Fashion Merchandising Strategies	3.0
CTE 2836	Global Merchandising	3.0
EVR 1001	Introduction to Environmental Science	3.0

Semester 5		Units: 12.0
CTE 1942	Fashion Industry Internship	3.0
CTE 2111C	Digital Fashion Portfolio	3.0
CTE 2800	Textile, Apparel & Retail Analysis	3.0
ENT 2212	Entrepreneurial Leadership	3.0

Total: 60.0

CTE1050 Introduction to Fashion Design and Related Industries 3.000 credits

In this course students will learn the basic design process, history, characteristics and global interrelationships of the fashion industry segments. The course explores how fashion, textile and apparel producers, retailers and related industries use merchandising and market their products within the industry and ultimately, to the consumer.
Lecture Hours: 48.000

CTE2732 Fashion Illustration Technology 3.000 credits

This course introduces the basics of fashion illustration technology targeted specifically for the fashion industry. Using various computer software and tools, including but not limited to Photoshop, Fashion CAD, C-Design, Adobe Illustrator, and Digital Fashion Pro, students will learn the techniques to conceptualize, create, and manage fashion designs and collections, and create industry-standard presentations.
Lecture Hours: 48.000

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.
Lecture Hours: 48.000
Course User Fee: \$20.00

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.
Lecture Hours: 48.000
Course User Fee: \$10.00

CTE1801 Introduction to Fashion Merchandising 3.000 credits

This introductory class provides an exposure to merchandising and terminology. Students will learn the entrepreneurs who influence the industry, career possibilities and an overview of the components of a manufacturer's or retailer's promotional techniques. Lecture Hours: 48.000

CTE2301 Product Development 3.000 credits

In this course students will learn the concepts and methods by which retailers create special, store-branded merchandise for targeted customer segments. The process of product development, from research to production to distribution, is studied.

Lecture Hours: 48.000

Prerequisite: CTE1401 AND CTE1401L

CTE2610 Fashion Forecasting and Research 3.000 credits

In this course students will learn to explore and apply forecast research methods in preparation for developing, planning, purchasing, or merchandising apparel lines and collections. Using the case study method, trend research is evaluated through the use of scholarly texts, articles, databases, and relevant websites to identify opportunities for growth and profitability in a fashion business.

Lecture Hours: 48.000

Prerequisite: CTE1050 AND MAR1011

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CTE2388 Principles of Contemporary Retailing 3.000 credits

In this course students will learn the operational segments of the fashion industry and their functions. The course focus is on the contributions employees add to sales productivity and customer satisfaction in retail establishments and on the exploration of new technologies and their impact on consumers' shopping experiences.

Lecture Hours: 48.000

Prerequisite: CTE1050

CTE2802 Fashion Merchandising Strategies 3.000 credits

In this course students will gain comprehensive knowledge of the merchandising environment, including the functions and objectives of the merchandising team, the principles and techniques of today's buyers, planners, product developers, and account executives.

Lecture Hours: 48.000

Prerequisite: CGS1060C

CTE2836 Global Merchandising 3.000 credits

In this course students will learn the merchandising practices used around the world in fashion apparel companies, both in retail and wholesale. American merchandising theory is used as a base of comparison in the consideration of various religions, cultures, legal systems, and other global systems.

Lecture Hours: 48.000

Corequisite: CTE2802

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

CTE1942 Fashion Industry Internship 3.000 credits

This internship course provides eligible students with placement in premier fashion settings and with the professional and practical experiences needed to further their education in a variety of fashion-based positions. Students control the internship selection process and work with an internship coordinator in the revisions of their resumes and schedule of interviews within the network of fashion companies.

Fashion internships are available in the areas of design, merchandising, buying, show-room, and fashion public relations. (144 hr. Internship)

Internship Hours: 144.000

Recommended Preparation: Departmental Approval & Completion of the Program Core Requirements

CTE2111C Digital Fashion Portfolio 3.000 credits

In this course students will learn to integrate computer aided design tools in the creative process of developing a digital fashion collection portfolio. Visual presentation and specific applications will be used.

Lecture Hours: 16.000, Lab Hours: 32.000

Recommended Preparation: CAD/Photoshop/ Illustrator

CTE2800 Textile, Apparel & Retail Analysis 3.000 credits

In this course students will learn about textile marketing of sustainable apparel and the textile value chains from product concept to the consumer. A variety of topics on global value chains, market analysis, product development, manufacturing, market and sourcing are explored. The global impact of trade and sourcing constraints are examined. Through readings, case studies and in-class industry presentations, students will explore a comprehensive array of contemporary issues, both social and regulatory, that help in understanding the complex value and supply chain.

Lecture Hours: 48.000

Prerequisite: CTE2388

ENT2212 Entrepreneurial Leadership 3.000 credits

The student will learn key skills and traits of successful entrepreneurs. Students will explore the notion of values-based business formation, personal strength and weakness identification, leadership for team building, project and personal time management, and story-telling.

Lecture Hours: 48.000

Course User Fee: \$100.00

Program Description

Associate in Science | Code: 26044 | 64 credits

The Film Production Technology program prepares students to learn all aspects of the film industry through hands-on, production-oriented classes both in the studio and on location. Students can receive training in cinematography, lighting, audio recording and editing. High-end equipment is used by students to shoot and edit sound 16mm film. Students are also exposed to video and the business aspects of the industry.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
FIL 1100	Screenwriting 1: Introduction to Story Structure	3.0
FIL 1420C	Film Production 1: Introduction to the Filmmaking	4.0
FIL 2552C	Editing Level 1: Introduction to Editing	3.0
VIC 1000	Visual Communications	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
ENC 1101	English Composition 1	3.0
FIL 1431C	Film Production 2: Cinematography and Sound	4.0
FIL 2131	Screenwriting 2: Character Development and Advanced Story Structure	3.0
FIL 2553C	Editing Level 2: Intermediate Editing and Visual Effects	3.0

Semester 3		Units: 12.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
FIL 2560C	Editing Level 3: Advanced Editing: Color Correction and Finishing	3.0
RTV 1240C	Sound Design	3.0

Semester 4		Units: 13.0
MGF 1130	Mathematical Thinking	3.0
FIL 1030	History of Film	3.0
FIL 2480C	Film Production 3: Directing	4.0
FIL 2611	Film Business Marketing Distribution Exhibition	3.0

Semester 5		Units: 13.0
<i>Natural Science</i>		3.0

BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
FIL 2515C	Film Production 4: Producing the Short Film	4.0
Program Electives		6.0
FIL 1055	American Independent Film	3.0
OR		
FIL 1060	Survey of Documentary Film	3.0
OR		
FIL 2413	Screenwriting 3	3.0
OR		
FIL 2945	Film Internship	3.0
OR		
FIL 2951	Film Festival Experience	3.0
OR		
TPP 1100	Acting 1	3.0
OR		
TPP 1110	Acting 1	3.0

Total: 64.0

FIL1100 Screenwriting 1: Introduction to Story Structure 3.000 credits

A workshop-style introductory class covering narrative script writing for film and television. Students will learn to develop a short-format screenplay incorporating three-act story structure, script elements, and standard industry formatting. Lecture Hours: 48.000

FIL1420C Film Production 1: Introduction to the Filmmaking 4.000 credits

An introductory overview of the art and technology of narrative motion picture production. Students will learn basic production techniques, from pre-production through production to final screening. Lecture Hours: 32.000, Lab Hours: 64.000

Corequisite: FIL2552C
Course User Fee: \$150.00

FIL2552C Editing Level 1: Introduction to Editing 3.000 credits

An introductory course in which students will learn the practice of editing digital media.

Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$35.00

VIC1000 Visual Communications 3.000 credits

An introductory course in which students will learn the visual aspects of film, video, photography and graphic arts, specifically dealing with design elements and principles.

Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$10.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

FIL1431C Film Production 2: Cinematography and Sound 4.000 credits

An introductory course in which students will learn cinematography and sync-sound motion picture production.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: FIL1420C AND Corequisite: FIL2553C
Course User Fee: \$150.00

FIL2131 Screenwriting 2: Character Development and Advanced Story Structure 3.000 credits

In this workshop-style intermediate level course students will learn about character development and various story structures for narrative motion picture screenplays.

Lecture Hours: 48.000

Prerequisite: FIL1100

FIL2553C Editing Level 2: Intermediate Editing and Visual Effects 3.000 credits

An intermediate course in which students will learn video editing with an emphasis on sound design and visual effects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2552C

Course User Fee: \$35.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

FIL2560C Editing Level 3: Advanced Editing: Color Correction and Finishing 3.000 credits

An advanced course in which students will learn the practice of color correcting and finishing fiction and non-fiction projects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2553C

Course User Fee: \$35.00

RTV1240C Sound Design 3.000 credits

In this introductory Sound Design course students will learn an overview of sound recording and audio post-production.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

FIL1030 History of Film 3.000 credits

In this introductory course students will learn about the history of motion pictures, with an emphasis on American and European films.

Lecture Hours: 48.000

FIL2480C Film Production 3: Directing 4.000 credits

An intermediate practicum in motion picture direction through the analysis of various directors and their cinematic styles. Students will learn the role of the director by interpreting dramatic material, effectively guiding acting performances, and communicating a story visually.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: FIL1431C AND FIL2553C, RTV1240C

Course User Fee: \$25.00

FIL2611 Film Business Marketing Distribution Exhibition 3.000 credits

Examination of the functional areas within marketing as well as the various distribution means (both current and projected) that are governing the sale of independent feature films or films financed outside of the studio system. Students learn to distribute their own selected films in this course. Prerequisite: FIL 1276. (3 hr. lecture)

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

FIL2515C Film Production 4: Producing the Short Film 4.000 credits

An advanced course in film production. Students will learn to apply the fundamentals of film production as introduced in Film Production 1, 2, & 3 to the production of a portfolio-quality narrative short film.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: FIL2480C

Course User Fee: \$250.00

FIL1055 American Independent Film 3.000 credits

In this introductory course, students will learn about the American independent film movement with an emphasis on American directors and producers.
Lecture Hours: 48.000

FIL1060 Survey of Documentary Film 3.000 credits

In this introductory course students will learn the history of nonfiction films, with an emphasis on American and European filmmakers.
Lecture Hours: 48.000

FIL2413 Screenwriting 3 3.000 credits

An advanced course in which the fundamentals of story structure and character development introduced in Screenwriting 1 and Screenwriting 2 are refined. The student will learn how to write an outline for a feature-length motion picture.
Lecture Hours: 48.000
Prerequisite: FIL2131

FIL2945 Film Internship 3.000 credits

Students will learn to apply the various skills gained throughout the program in a semester long immersion at an established film entertainment company.
Lecture Hours: 240.000
Prerequisite: FIL2480C AND Recommended Preparation: Professor Permission

FIL2951 Film Festival Experience 3.000 credits

An immersive experience in a film festival. Students engage directly with films and filmmakers through their attendance to festival screenings and professional panels and discussions. Students become acquainted with a festival's organization and develop their professional skills by participating as volunteers in a multi-cultural community event.
Lecture Hours: 48.000
Recommended Preparation: Registration with Professor Approval Only.

TPP1100 Acting 1 3.000 credits

The fundamentals of stage performance, stressing voice, movement, and the more formal and technical aspects of the actor's art. May be repeated for credit.
Lecture Hours: 48.000

TPP1110 Acting 1 3.000 credits

Continuation of TPP 1100.
Lecture Hours: 48.000
Prerequisite: TPP1100

Program Description

Financial Services - Banking

Associate in Science | Code: 22026 | 60 credits

The Financial Services program is designed to meet the needs of students who plan to seek employment with commercial banks, stock brokerage companies and related financial organizations. It is also planned for students who are currently employed and desire advancement to positions of greater responsibility with financial organizations. This program meets most of the requirements for the American Institute of Banking diploma/certificates. The Associate in Arts degree is also available to the student planning to transfer to a senior institution after graduation from MDC. Consult an advisor about which additional courses are included in that program.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
BAN 1004	Banking Fundamentals, Evolution, and Compliance	3.0
BAN 2511	Marketing for Financial Services	3.0

Semester 2

Units: 13.0

ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0

BAN 1240	Essentials for Retail Lending	3.0
BAN 2501	Money, Banking and Financial Markets	3.0
STA 2023	Statistical Methods	3.0
Course Competency *		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3	Units: 12.0
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<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
ACG 1403	Excel for Business	2.0
BAN 1231	Commercial Lending	3.0
BAN 2210	Analyzing Financial Statements	3.0
FIN 1930	Special Topic Seminar	1.0

Semester 4	Units: 13.0
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BAN 2211	Applied Financial Statement Analysis	3.0
ECO 2013	Principles of Economics (Macro)	3.0
MKA 1022	Relationship Selling	3.0
MNA 1130	Business for Financial Services	1.0
PHI 2010	Introduction to Philosophy	3.0

Semester 5	Units: 10.0
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<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
REE 2304	Commercial Real Estate	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BAN1004 Banking Fundamentals, Evolution, and Compliance 3.000 credits

In this course, students will learn the foundation, structure and function of financial systems. The course plan includes a review of the role of banks as businesses and their impact on the economy. The students will receive an overview of the main entities that comprise the financial system: financial intermediaries, investments and financial markets. The students will be able to describe the risk infrastructure of financial institutions, the regulatory environment, and the responsibilities inherent in complying with regulatory requirements associated with the safety and soundness of banks.

Lecture Hours: 48.000

BAN2511 Marketing for Financial Services 3.000 credits

Marketing of financial services is a specialized segment of marketing. It is highly competitive, making the process of selling for bankers a highly specialized and challenging endeavor. Changing market conditions, deregulation, the emergence of new competitors from within and external to the banking industry and the rapid integration of new technologies are some of the challenges bankers encounter. It covers the aspects of a consultative selling approach with emphasis on planning, implementing and fostering a long-term advisory relationship with clients. The students will learn to become proficient in how to prepare for a successful sales presentation and closing.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

BAN1240 Essentials for Retail Lending 3.000 credits

In this course, the student will learn a comprehensive approach to consumer lending, identifying financial risks and the regulatory environment that impact and promote safety and soundness in lending. The course centers around providing a practical approach to the fundamentals of consumer installment lending, including a review of different loan products, their life cycles, the credit application process and the essentials of the closing process.

Lecture Hours: 48.000

BAN2501 Money, Banking and Financial Markets 3.000 credits

The course is designed to provide students with an overview of the US financial system, including theory and practice of monetary policy and financial instruments. Students will learn about interest rate determination, the structure and role of banks, financial institutions in the intermediation process, the factors impacting inflation, and variables in the economy. The course plan is designed not only for students but for individuals working in the financial services industry, including junior officers to mid-management and entry level staff who are considering pursuing a career in the banking field.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

ACG1403 Excel for Business 1.000 - 3.000 credits

This course will cover Excel topics relevant to the field of accounting and finance including, but not limited to, VLOOKUP, HLOOKUP, INDEX, MATCH, IF, AND, OR, Pivot Tables, Named Ranges, Array Formulas, Custom Number Formats, Conditional Formatting, Absolute References and Keyboard Shortcuts.

Lecture Hours: 16.000 - 48.000

Course User Fee: \$22.00

BAN1231 Commercial Lending 3.000 credits

This course provides an increased awareness of the credit underwriting process. It provides a comprehensive foundation to the lending process including technical and interpretative analysis of financial information from liquidity, solvency, ratio and cash flow analysis to the basics for the completion of a credit approval memorandum. Students will learn the concepts of qualitative analysis including the assessment of industry risk, market risk and management risk. The course provides an understanding of the role of loan policy and the need to summarize the borrowers' various risks into an appropriate credit risk rating.

Prerequisite: BAN2210.

Lecture Hours: 48.000

Prerequisite: BAN2210

BAN2210 Analyzing Financial Statements 3.000 credits

In this course the students will learn the framework and resources available to analyze financial statements and to assess a company's operation and future performance. The course covers the critical objectives with the assessment of business performance such as: the relationship between the company's business and financial strategy, their financial statements, ratio, profitability, balance sheet and cash flow analysis. The course plan includes the use of financial and accounting information to make lending/credit and investment decisions. The course is targeted to a wide range of students including those pursuing careers in banking, general management, investment banking, financial analysis and consulting. Prerequisite: ACG2021.

Lecture Hours: 48.000

Prerequisite: ACG2021

FIN1930 Special Topic Seminar 1.000 credits

This course centers around topics of current interest or of special interest. Topics or focus may vary from semester to semester. (1 hr. lecture)

Lecture Hours: 16.000

Course User Fee: \$35.00

BAN2211 Applied Financial Statement Analysis 3.000 credits

This course provides a comprehensive analysis of business strategy, operating performance, financial condition and cash flow strength. The student will learn basic and advanced financial concepts impacting the viability of a business, including accounting rules, methods of credit analysis, assessment of financial ratios, historical financial analysis, cash flow and financial forecasting.

Prerequisite: BAN2210.

Lecture Hours: 48.000

Prerequisite: BAN2210

Course User Fee: \$20.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

MKA1022 Relationship Selling 3.000 credits

This course introduces the role that relationship building plays in sales. Students will explore the communication process and the importance of interpersonal skills in developing a relationship for an effective sales strategy. Students will develop a sales presentation based around consumer needs.

Lecture Hours: 48.000

Course User Fee: \$47.44

MNA1130 Business for Financial Services 1.000 credits

This course teaches business professionals a structured approach to create clear, effective, professional business writing, including e-mail, memos, letters, and reports. (1 hr. lecture)

Lecture Hours: 16.000

Course User Fee: \$45.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L
Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.
Lab Hours: 32.000
Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071
Course User Fee: \$32.00

REE2304 Commercial Real Estate 3.000 credits

In this course, the student will learn specific financial tools and techniques required to assess the feasibility of commercial real estate investment, funding and financing options, risks associated with real estate transactions and factors impacting real estate investment decisions. The student will learn how the commercial real estate process works, the different interested parties involved in the transaction, the essentials of risk and return, financing options and different phases in a commercial real estate transaction.

Lecture Hours: 48.000

Course User Fee: \$70.00

Program Description

Financial Services - Wealth Management

Associate in Science | Code: 22024 | 60 credits

The Financial Services program is designed to meet the needs of students who plan to seek employment with commercial banks, stock brokerage companies and related financial organizations. It is also planned for students who are currently employed and desire advancement to positions of greater responsibility with financial organizations. This program meets most of the requirements for the American Institute of Banking diploma/certificates. The Associate in Arts degree is also available to the student planning to transfer to a senior institution after graduation from MDC. Consult an advisor about which additional courses are included in that program.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
BAN 1004	Banking Fundamentals, Evolution, and Compliance	3.0
FIN 2000	Principles of Finance	3.0

Semester 2

Units: 14.0

ACG 1403	Excel for Business	2.0
FIN 2010	Investments in Stocks and Bonds	3.0
FIN 2032	Fundamentals of Wealth Management, Institutions, Markets and Products	3.0
FIN 2100	Personal Finance	3.0
STA 2023	Statistical Methods	3.0
Computer Competency*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3		Units: 12.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
FIN 1930	Special Topic Seminar	1.0
MNA 1130	Business for Financial Services	1.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
PHI 2010	Introduction to Philosophy	3.0
Semester 4		Units: 13.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
BAN 2210	Analyzing Financial Statements	3.0
ECO 2013	Principles of Economics (Macro)	3.0
MKA 1022	Relationship Selling	3.0
Semester 5		Units: 9.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
FIN 2031	Risk Management and Compliance	3.0
FIN 2642	Financial Analysis and Valuation	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BAN1004 Banking Fundamentals, Evolution, and Compliance 3.000 credits

In this course, students will learn the foundation, structure and function of financial systems. The course plan includes a review of the role of banks as businesses and their impact on the economy. The students will receive an overview of the main entities that comprise the financial system: financial intermediaries, investments and financial markets. The students will be able to describe the risk infrastructure of financial institutions, the regulatory environment, and the responsibilities inherent in complying with regulatory requirements associated with the safety and soundness of banks.

Lecture Hours: 48.000

FIN2000 Principles of Finance 3.000 credits

The creation, allocation, and utilization of money, and the effect of monetary policy upon individuals, business, national and international economics. This course provides a basis for further study of monetary theory, banking, finance and securities.

Lecture Hours: 48.000

ACG1403 Excel for Business 1.000 - 3.000 credits

This course will cover Excel topics relevant to the field of accounting and finance including, but not limited to, VLOOKUP, HLOOKUP, INDEX, MATCH, IF, AND, OR, Pivot Tables, Named Ranges, Array Formulas, Custom Number Formats, Conditional Formatting, Absolute References and Keyboard Shortcuts.

Lecture Hours: 16.000 - 48.000

Course User Fee: \$22.00

FIN2010 Investments in Stocks and Bonds 3.000 credits

The basic principles of the stock market as they affect the individual investor in stocks and bonds. Investment in these securities is studied from the standpoint of the short-term and long-term investors.

Lecture Hours: 48.000

Course User Fee: \$10.00

FIN2032 Fundamentals of Wealth Management, Institutions, Markets and Products 3.000 credits

This course is a standard introduction to the financial services profession, financial markets, and financial institutions. It touches on nearly every aspect of financial services. It assists professionals in understanding concepts, markets, products, regulations and the application of financial planning and development of wealth management skills.

Lecture Hours: 48.000

Prerequisite: ACG2021, ACG2021L

Course User Fee: \$70.00

FIN2100 Personal Finance 1.000 - 3.000 credits

A study of economic and personal goals including personal budgeting, credit budgeting, borrowing money, banking facilities, the nature of investments, life insurance, casualty insurance, home ownership, stocks and bonds, and retirement plans.

Lecture Hours: 16.000 - 48.000

Course User Fee: \$20.00

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

FIN1930 Special Topic Seminar 1.000 credits

This course centers around topics of current interest or of special interest. Topics or focus may vary from semester to semester. (1 hr. lecture)

Lecture Hours: 16.000

Course User Fee: \$35.00

MNA1130 Business for Financial Services 1.000 credits

This course teaches business professionals a structured approach to create clear, effective, professional business writing, including e-mail, memos, letters, and reports. (1 hr. lecture)

Lecture Hours: 16.000

Course User Fee: \$45.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision- making. Prerequisites:

ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

BAN2210 Analyzing Financial Statements 3.000 credits

In this course the students will learn the framework and resources available to analyze financial statements and to assess a company's operation and future performance. The course covers the critical objectives with the assessment of business performance such as: the relationship between the company's business and financial strategy, their financial statements, ratio, profitability, balance sheet and cash flow analysis. The course plan includes the use of financial and accounting information to make lending/credit and investment decisions. The course is targeted to a wide range of students including those pursuing careers in banking, general management, investment banking, financial analysis and consulting. Prerequisite: ACG2021.

Lecture Hours: 48.000

Prerequisite: ACG2021

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

MKA1022 Relationship Selling 3.000 credits

This course introduces the role that relationship building plays in sales. Students will explore the communication process and the importance of interpersonal skills in developing a relationship for an effective sales strategy. Students will develop a sales presentation based around consumer needs.

Lecture Hours: 48.000

Course User Fee: \$47.44

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

FIN2031 Risk Management and Compliance 3.000 credits

This course offers an analysis of the risks faced by investors and savers interacting through both financial institutions and financial markets. It will provide insight at the risks and opportunities of doing business in today's financial markets and the challenges presented by both regulators and market participants. The course will review the increasing integration of foreign and domestic financial markets. (3hr lecture)

Lecture Hours: 48.000

Course User Fee: \$70.00

FIN2642 Financial Analysis and Valuation 3.000 credits

This course will cover financial institutions, financial Investments, financial planning and analysis and international financial perspectives. It addresses core principles of value creation, merger, analysis of historical financial performance, cash flows and the identification of sources of value. (3 hr. lecture)

Lecture Hours: 48.000, Clinical Hours: 1.000

Course User Fee: \$70.00

Program Description

Financial Technology

Associate in Science | Code: 22035 | 60 credits

The Financial Technology Associate of Science is a comprehensive 60-credit degree program that combines finance expertise with advanced technology skills. Students learn financial big data modeling, machine learning, blockchain technology, and digital banking solutions while developing competencies in data analytics, cybersecurity, and regulatory compliance. The curriculum covers emerging fintech innovations including robo-advisors, mobile payments, and alternative lending platforms, alongside traditional banking principles and Federal Reserve oversight. Graduates are prepared for advanced careers in fintech companies, traditional banks adopting digital solutions, and other organizations integrating technology with financial services.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
BAN 1004	Banking Fundamentals, Evolution, and Compliance	3.0
FIN 2000	Principles of Finance	3.0

Semester 2

Units: 12.0

<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
ECO 2013	Principles of Economics (Macro)	3.0
FIT 2000	Introduction to Financial Technology	3.0
<i>Program Elective</i>		3.0
BUL 2241	Business Law 1	3.0
OR		
COP 1047C	Introduction to Python Programming	4.0
OR		
CTS 1120	Cybersecurity Fundamentals	4.0
OR		
CTS 1145	Cloud Essentials	4.0
OR		
ECO 2023	Principles of Economics (Micro)	3.0
OR		
FIN 2010	Investments in Stocks and Bonds	3.0
OR		
FIN 2051	International Financial Management	3.0
OR		
GEB 1432	Applied Artificial Intelligence (AI) in Business	3.0
OR		
MAR 1720	Marketing in a Digital World	3.0
OR		
MAR 1930	Introduction to Salesforce Marketing Cloud	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 3

Units: 10.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
FIN 2031	Risk Management and Compliance	3.0

Semester 4

Units: 13.0

<i>Natu</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		

EVR 1001	Introduction to Environmental Science	3.0
FIN 2032	Fundamentals of Wealth Management, Institutions, Markets and Products	3.0
Program Electives ¹⁾		7.0

¹⁾Select 7 credits from the following course options: BUL 2241, COP 1047C, CST 1120, CTS 1145, ECO 2023, FIN 2010, FIN 2051, GEB 1432, MAR 1720, MAR 1930

Semester 5		Units: 13.0
CGS 1540C	Database Concepts Design	4.0
GEB 2100	Introduction to Business Analytics	3.0
Program Electives ¹⁾		6.0

¹⁾Select 7 credits from the following course options: BUL 2241, COP 1047C, CST 1120, CTS 1145, ECO 2023, FIN 2010, FIN 2051, GEB 1432, MAR 1720, MAR 1930

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BAN1004 Banking Fundamentals, Evolution, and Compliance 3.000 credits

In this course, students will learn the foundation, structure and function of financial systems. The course plan includes a review of the role of banks as businesses and their impact on the economy. The students will receive an overview of the main entities that comprise the financial system: financial intermediaries, investments and financial markets. The students will be able to describe the risk infrastructure of financial institutions, the regulatory environment, and the responsibilities inherent in complying with regulatory requirements associated with the safety and soundness of banks.

Lecture Hours: 48.000

FIN2000 Principles of Finance 3.000 credits

The creation, allocation, and utilization of money, and the effect of monetary policy upon individuals, business, national and international economics. This course provides a basis for further study of monetary theory, banking, finance and securities.

Lecture Hours: 48.000

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

FIT2000 Introduction to Financial Technology 3.000 credits

This course introduces students to the foundations, innovations, and applications of Financial Technology (FinTech). Topics include the evolution of money, core banking systems, digital payments, blockchain, big data, artificial intelligence, cybersecurity, and FinTech regulations. Students will explore how technology is transforming banking, insurance, investment, and other sectors, while developing a framework for ethical and regulatory compliance in digital finance.

Lecture Hours: 48.000

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033 Any Mathematics level higher than MAT 1033 will satisfy this requirement.

MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

FIN2010 Investments in Stocks and Bonds 3.000 credits

The basic principles of the stock market as they affect the individual investor in stocks and bonds.

Investment in these securities is studied from the standpoint of the short-term and long-term investors.

Lecture Hours: 48.000

Course User Fee: \$10.00

FIN2051 International Financial Management 3.000 credits

The student will learn basic concepts and principles of international finance, with consideration of the financial environment, transactions, and flows.

Exchange rates, risks, and government policies affecting business are analyzed as well as management policies and decisions.

Lecture Hours: 48.000

Course User Fee: \$20.00

GEB1432 Applied Artificial Intelligence (AI) in Business 3.000 credits

This course illustrates how Artificial Intelligence (AI) is currently applied in industries and across business functions to support business strategy formulation, implementation and evaluation. This course introduces the data-driven outcomes and ethical issues to be considered when implementing responsible AI in business and industry. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$35.20

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR1930 Introduction to Salesforce Marketing Cloud 1.000 credits

This course introduces students on how to build and personalize a 1:1 marketing campaign that can help connect and deliver the right message, to the right person, at the right time. Students will learn how to manage the most utilized marketing channels in the industry and discover new marketing tools.

Lecture Hours: 16.000

Course User Fee: \$33.69

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

FIN2031 Risk Management and Compliance 3.000 credits

This course offers an analysis of the risks faced by investors and savers interacting through both financial institutions and financial markets. It will provide insight at the risks and opportunities of doing business in today's financial markets and the challenges presented by both regulators and market participants. The course will review the increasing integration of foreign and domestic financial markets. (3hr lecture)

Lecture Hours: 48.000

Course User Fee: \$70.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

FIN2032 Fundamentals of Wealth Management, Institutions, Markets and Products 3.000 credits

This course is a standard introduction to the financial services profession, financial markets, and financial institutions. It touches on nearly every aspect of financial services. It assists professionals in understanding concepts, markets, products, regulations and the application of financial planning and development of wealth management skills.

Lecture Hours: 48.000

Prerequisite: ACG2021, ACG2021L

Course User Fee: \$70.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement. Lecture Hours: 48.000, Lab Hours: 32.000
Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

GEB2100 Introduction to Business Analytics 3.000 credits

This course is for students majoring in business, marketing, business intelligence, computer science and other majors and introduces how information and technology are used in organizations to create market advantage. Students will learn about the operational units of an organization and the information requirements to support the organization.

Lecture Hours: 48.000

Program Description

Fire Science Technology

Associate in Science | Code: 27018 | 60 credits

The Fire Science Technology program prepares students for a wide variety of technical positions in the area of fire prevention and control. Students will learn about safety factors, building code requirements, national and local standards, hazardous materials, supervision and management skills, hydraulics, fire apparatus, tactics and strategy. The program has been designed to meet both the Florida Fire Fighters Pre-Officer Requirements and the NFPA 1021 Fire Officer Level Two Requirements.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 15.0

ENC 1101	English Composition 1	3.0
FFP 1000	Introduction to Fire Science	3.0
FFP 1301	Fire Service Hydraulics	3.0
FFP 1505	Fire Prevention Practices	3.0
FFP 1740	Fire Service Course Delivery	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2

Units: 15.0

ENC 1102	English Composition 2	3.0
MGF 1130	Mathematical Thinking	3.0
FFP 1540	Private Fire Protection Systems 1	3.0
FFP 1810	Firefighting Tactics & Strategy 1	3.0
FFP 2120	Building Construction for the Fire Service	3.0

Semester 3

Units: 15.0

STA 2023	Statistical Methods	3.0
<i>Social Science</i>		3.0

AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
<i>Program Electives</i>		9.0
<i>Fire Science Electives</i>		6.0
FFP 1302	Apparatus Operations	3.0
OR		
FFP 1510	Codes and Standards	3.0
OR		
FFP 2521	Construction Documents and Plans Review	3.0
OR		
FFP 2741	Fire Service Course Design	3.0
OR		
<i>Industry Certifications</i>		6.0
EMS 9995	Emergency Medical Responder ^{EMS}	2.0
OR		
EMS 9996	Emergency Medical Technician (EMT) - Basic	10.0
OR		
FFP 9996	Fire Fighting and Protection - Firefighter II	3.0
OR		
<i>Electives</i>		3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
OR		
MAT 1033	Intermediate Algebra	3.0
OR		
SPC 1017	Introduction to Communication	3.0

Semester 4

Units: 15.0

CHM 1020	General Education Chemistry	3.0
PHI 2010	Introduction to Philosophy	3.0
FFP 2720	Company Officer	3.0
Progra Electives ¹⁾		6.0

¹⁾BSC 2085, FFP 1302, FFP 1510, FFP 2521, FFP 2741, MAT 1033, SPC 1017

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

FFP1000 Introduction to Fire Science 3.000 credits

This course provides an overview of fire protection and emergency services; career opportunities in fire protection and related fields; culture and history of emergency services; fire loss analysis; organization and function of public and private fire protection services; fire departments as part of local government; laws and regulations affecting the fire service; fire service nomenclature; specific fire protection functions; basic fire chemistry and physics; introduction to fire protection systems; introduction to strategy and tactics; and life safety initiatives. (3 hr. lecture)

Lecture Hours: 48.000

FFP1301 Fire Service Hydraulics 3.000 credits

This course covers the relationship between flow, pressure, and mathematical hydraulic formulas. The course includes pump theory, pump rating, and pressure and vacuum gauges. (3 hr. lecture)

Lecture Hours: 48.000

FFP1505 Fire Prevention Practices 3.000 credits

The purpose of this course is to place emphasis on fire prevention through inspection, code enforcement, the use of model building and fire prevention codes, including detailed information on the legal, economic, and political aspects of the fire inspection process. (3 hr. lecture)

Lecture Hours: 48.000

FFP1740 Fire Service Course Delivery 3.000 credits

The curriculum draws from many recognized authorities in exploring the methods and mechanics of imparting information and adult learning principles. The course emphasizes techniques that have wide application in teaching situations, as well as devices for specific training areas. The course also stresses measuring teaching effectiveness, and the use of media and visual aids. (3 hr. lecture)

Lecture Hours: 48.000

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

FFP1540 Private Fire Protection Systems 1 3.000 credits

This is a study of private fire protection and detection systems such as sprinkler and standpipe systems, chemical extinguishing systems, and detection systems and devices. Each system is discussed as to its need, construction, preventative maintenance and individual use. (3 hr. lecture)
Lecture Hours: 48.000, Practicum Hours: 3.000

FFP1810 Firefighting Tactics & Strategy 1 3.000 credits

This course will provide students with a basic understanding of factors involved in coping with a fire emergency and determining the best use of available resources in protecting lives and property. The course emphasizes the changing nature of an emergency situation and the ways in which the fire officer can evaluate the effectiveness of their proposed incident action plan. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: , FFP1000

FFP2120 Building Construction for the Fire Service 3.000 credits

This course will provide the student with a basic understanding of building design and construction methods which will aid in decision making processes related to fire prevention and control. Students will gain an understanding of construction principles and practices related to fire loads and behavior to make more effective, safer, and timely decisions to protect people and property from potential and actual fires. (3 hr. lecture)
Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

FFP1302 Apparatus Operations 3.000 credits

This course covers the laws, rules and driving techniques for emergency vehicles as well as a review of hydraulics. Fireground evolutions and driving make up the practical part of the course. The evolution portion of the course includes the use of pre-connected lines, tandem pumping, drafting relays, and master streams. The student should have a basic understanding of fire stream hydraulics prior to entering this course. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: FFP1301

FFP1510 Codes and Standards 3.000 credits

The purpose of this course is to familiarize students or current inspectors with the Florida Fire Prevention Code which includes NFPA 101, NFPA 1, FSS 633 and FAC 69A. Students will learn how to research the codes and apply them to inspection activities. (3 hr. lecture)

Lecture Hours: 48.000

FFP2521 Construction Documents and Plans Review 3.000 credits

This curriculum is geared towards teaching the student how to assimilate information contained in working, drawing, and specifications as they relate to the fire inspector. The curriculum includes how to interpret conventional graphic communications. Accepted standards and conventions are introduced. Symbols, abbreviations, principles of technical projection, as well as a review of geometry are included. Related worksheets are used to allow for applied experience of finding and interpreting information from authentic drawings. (3 hr. lecture)

Lecture Hours: 48.000

FFP2741 Fire Service Course Design 3.000 credits

The curriculum draws from many recognized authorities in exploring the methods and mechanics of imparting information and adult learning principles. The course emphasizes techniques that have wide application in teaching situations, as well as devices for specific training areas. The course also stresses measuring teaching effectiveness, and the use of media and visual aids. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: FFP1740

EMS9995 Emergency Medical Responder 2.000 credits

This course is designed for students who have earned the Emergency Medical Responder (EMR) credential and provides academic recognition of the knowledge, skills, and competencies demonstrated through certification attainment. The course validates proficiency in emergency patient assessment, basic life support, trauma care, medical emergencies, patient stabilization, and emergency response procedures. Credit awarded through this course supports academic progression and recognizes documented achievement of industry-recognized competencies in prehospital emergency care. Clinical Hours: 96.000

EMS9996 Emergency Medical Technician (EMT) - Basic 10.000 credits

This course provides academic credit for students who have successfully completed a State of Florida-approved Emergency Medical Technician (EMT-Basic) Applied Technology Diploma (ATD) program and are enrolled in the related Associate in Science degree within the School of Health Sciences. In accordance with TMOP 1100-816125, the course recognizes competencies attained through prior educational and certification experiences, including emergency patient assessment, airway management, trauma care, medical emergency response, patient transportation, and prehospital care procedures. Credit awarded through this course supports academic progression and acknowledges demonstrated achievement of industry-recognized emergency medical services competencies. Other Hours: 160.000

FFP9996 Fire Fighting and Protection - Firefighter II 3.000 credits

This course provides academic credit for students who have successfully completed Firefighter II training through an approved fire science training provider. In accordance with TMOP 1100:815720, the course serves as a mechanism for awarding college credit to Miami Dade College (MDC) students based on documented completion of approved fire science training. The course recognizes competencies attained in fire suppression, rescue operations, hazardous materials response, fire prevention, emergency services, and firefighter safety. Credit awarded through this course supports academic progression and acknowledges industry-recognized skills and professional preparation in the field of fire science and emergency services. Lecture Hours: 48.000

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L. Lecture Hours: 48.000
Corequisite: BSC2085L
Course User Fee: \$8.00

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications. Lecture Hours: 48.000
Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.
Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture) Lecture Hours: 48.000
Course User Fee: \$7.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

FFP2720 Company Officer 3.000 credits

This course is designed to provide the student with an understanding of being a company officer through various areas of knowledge as well as solving the varied problems and situations they will be required to manage effectively in today's ever-changing fire service. The curriculum includes a review of fire department organization and administration, management theory, leadership, communications, motivation and group dynamics. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: FFP1000

Program Description

Funeral Service Education

Associate in Science | Code: 23049 | 72 credits

Students in the Funeral Service Education program are given a broad understanding of all phases of funeral home operations as well as the public health responsibilities of the funeral director and embalmer. The Funeral Service Education associate degree program at Miami Dade College is accredited by the American Board of Funeral Service Education (ABFSE), 992 Mantua Pike, Suite 108, Woodbury Heights, NJ 08097 (816) 233-3747. Web: www.abfse.org. All students must register to take the International Conference of Funeral Service Examining Boards, Inc. National Board Examination during their final semester. Students who plan Funeral Service licensure in other states must contract the respective state board to determine licensing requirements and the student is responsible for complying with all particular laws of that state. National Board Examination pass rates, graduation rates, and employment rates for this and other ABFSE-accredited programs are available at www.abfse.org. To request a printed copy of this program's rates, go to the Funeral Service Education Office (Building 3142) or by telephone 305-237-1244. Students who plan Funeral Service licensure in other states must contract the respective state board to determine licensing requirements and the student is responsible for complying with all particular laws of that state. National Board Examination pass rates, graduation rates, and employment rates for this and other ABFSE-accredited programs are available at www.abfse.org. To request a printed copy of this program's rates, go to the Funeral Service Education Office (Building 3142) or by telephone 305-237-1244.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

BSC 1084	Functional Human Anatomy	3.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
PHI 2010	Introduction to Philosophy	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2

Units: 12.0

<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
BSC 2020	Human Biology: Fundamentals of Anatomy/Physiology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
PHY 1020	General Education Physics	3.0
FSE 1000	History of Funeral Service	3.0
FSE 1105	Funeral Service Chemistry	3.0
<i>Business Law Option</i>		3.0
BUL 2131	Legal Environment	3.0
OR		
BUL 2241	Business Law 1	3.0

Semester 3

Units: 11.0

<i>Management Options</i>		3.0
FSE 2202	Funeral Service Business Management	3.0
OR		
GEB 1011	Principles of Business	3.0
FSE 2100	Embalming 1	3.0
FSE 2100L	Embalming 1 Lab	2.0
FSE 2106	Funeral Service Microbiology	3.0

Semester 4

Units: 14.0

FSE 2060	Funeral Directing	3.0
FSE 2061	Funeral Service Counseling and Ethics	3.0
FSE 2140	Embalming 2	3.0
FSE 2140L	Embalming 2 Laboratory	2.0
FSE 2160	Funeral Service Pathology	3.0

Semester 5

Units: 13.0

<i>Accounting Course Options¹⁾</i>		3.0
ACG 2001	Principles of Accounting I	3.0
OR		
ACG 2021	Financial Accounting	3.0
OR		

FSE 2200	Funeral Service Accounting	3.0
FSE 1080	Funeral Law	3.0
FSE 2120C	Restorative Art	4.0
FSE 2201	Funeral Home Operations	3.0

¹⁾Note: If a student selects ACG 2001 or ACG 2021, then the student must register for the corresponding laboratory.

Semester 6		Units: 10.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
FSE 2203	Funeral Home Applications	4.0
FSE 2930L	Funeral Service Professional Review	2.0
FSE 2932	Funeral Science Professional Review 2	1.0

Total: 72.0

BSC1084 Functional Human Anatomy 3.000 credits

This course covers basic human anatomy for students in allied health and mortuary science programs. It includes the dynamics of gross and functional anatomy, terminology, body orientation, and systematic relationships.

Lecture Hours: 48.000

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2020 Human Biology: Fundamentals of Anatomy/Physiology 3.000 credits

This course provides a basic understanding of the human body, its systems, and their functions. It includes the dynamics of physiology, terminology, and physiological relationships of the body systems.

Lecture Hours: 48.000

Course User Fee: \$10.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

FSE1000 History of Funeral Service 3.000 credits

This course traces the origins of funeral service practice from antiquity to modern day practice. Students will learn the evolution of the ethical obligations, fundamental requirements, skills, aptitudes, and qualifications of funeral service professionals. Requires a grade of C or better to pass the course.

Lecture Hours: 48.000

FSE1105 Funeral Service Chemistry 3.000 credits

A survey of the basic principles of chemistry as they relate to funeral service. Especially stressed are the chemical principles and precautions involved in sanitation, disinfection, public health and embalming practice.

Lecture Hours: 48.000

BUL2131 Legal Environment 3.000 credits

Law in relation to the proper conduct of business includes a consideration of the nature and sources of law, its legal environment and history. The topics of business torts, crimes, contracts, and forms of organizations are also covered.

Lecture Hours: 48.000

Course User Fee: \$10.00

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

FSE2202 Funeral Service Business Management 3.000 credits

The role and function of an effective manager is explored. Emphasis is placed on the management functions of planning, organizing, motivating, directing, and controlling. How to purchase a small business is also covered.

Lecture Hours: 48.000

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

FSE2100 Embalming 1 3.000 credits

Designed as an introductory course outlining the basic skills, aptitudes, and personal qualifications necessary for successful embalmers. The subject in embalming is taught so that the student is trained in all phases of the embalming process. Prerequisite: BSC1084 or FSE1001

Lecture Hours: 48.000, Lab Hours: 48.000

Pre/Corequisite: BSC1084 OR FSE1001

FSE2100L Embalming 1 Lab 2.000 credits

Laboratory for FSE 2100. Laboratory fee. Corequisite: FSE 2100. A.S. degree credit only. (4 hr. lab)

Lab Hours: 64.000

Pre/Corequisite: FSE2100

Course User Fee: \$490.00

FSE2106 Funeral Service Microbiology 3.000 credits

This course is a survey of the basic principles of microbiology as it relates to Funeral Science. It emphasizes the importance of sanitation, disinfection, and public health in the embalming practice.(3 hr. lecture)

Lecture Hours: 48.000

FSE2060 Funeral Directing 3.000 credits

Study of various religious, fraternal, military, traditional, nontraditional and humanistic variations of funeral ceremonies, including cultural, ethnic and geographic customs.

Lecture Hours: 48.000

FSE2061 Funeral Service Counseling and Ethics 3.000 credits

This course emphasizes the psychological and sociological dynamics of death, dying, and bereavement and the funeral director's role in counseling families through the process. A clear understanding of the grief process and its variations among individuals as influenced by psychological factors is addressed. Ethical and moral decisions involving end of life and the funeral arrangement process are discussed. The course also addresses the funeral director as a facilitating agent for effective mourning through personal interaction and the design and implementation of the funeral.

Lecture Hours: 48.000

FSE2140 Embalming 2 3.000 credits

This course is a continuation of Embalming 1 lecture and places emphasis on embalming considerations and procedures for pathogenesis and advanced decomposition. Students will learn to use specialized chemicals, treatment of post-mortem cases and advanced embalming techniques. Corequisite: FSE2140L. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: FSE2100, Recommended Preparation: Completed FSE2100L - Embalming 1 Lab - with a "C" or better.

FSE2140L Embalming 2 Laboratory 2.000 credits

This course is a continuation of FSE2100 and complements the lecture co-requisite FSE2140. Students will learn advanced procedures to enhance their foundational knowledge of embalming technique as learned in the prerequisite classes FSE2100/FSE2100L. Learning the fundamentals of general embalming technique as outlined by the American Board of Funeral Service Education Standards will enable students to work in a preparation room as embalming apprentices to licensed embalmers.

Lab Hours: 64.000

Prerequisite: FSE2100, FSE2100L, Corequisite: FSE2140

Course User Fee: \$490.00

FSE2160 Funeral Service Pathology 3.000 credits

General, systemic and forensic pathology with emphasis on analysis of pre-and post-mortem histology, cytology and etiology. Students will learn the causative factors relating to death and determination of cause of death.

Lecture Hours: 48.000

ACG2001 Principles of Accounting I 3.000 credits

An introduction to the basic principles of financial accounting with emphasis on basic accounting procedures such as recording of transactions and the preparation of financial statements. Other topics include inventory and cash. ACG2001 and ACG2011 can be substituted for ACG2021. Co-requisite: ACG2001L

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

FSE2200 Funeral Service Accounting 3.000 credits

An introduction to basic principles of accounting theory. This subject covers financial statements and their analysis, journalizing, receivables, payables, deferrals, and accruals. Inventory costing models, depreciation models and payroll accounting are included. Applications to funeral home operations are made throughout the subject material.

Lecture Hours: 48.000

FSE1080 Funeral Law 3.000 credits

Federal, state and municipal statutes, rules, regulations and ordinances pertaining to funeral service; torts, contract and administrative laws, and financial disclosures pertinent to funeral operations and management.

Lecture Hours: 48.000

FSE2120C Restorative Art 4.000 credits

Anatomical study of human features; familiarization with instruments, human proportions, special materials and techniques.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: FSE2100, FSE2100L

FSE2201 Funeral Home Operations 3.000 credits

Theoretical and practical training in all areas of funeral home operations; laboratory experience in merchandising and funeral arrangements.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.
Lecture Hours: 48.000

FSE2203 Funeral Home Applications 4.000 credits

Funeral Home Applications reinforces the academic and theory of prior classes with hands on, practical exercises. Students will learn procedures on taking first call, buying and selling of merchandise, funeral arranging, and conducting funerals. Must complete this course with a grade of "C" or better.
Lecture Hours: 64.000
Prerequisite: FSE2060, FSE2201

FSE2930L Funeral Service Professional Review 2.000 credits

This course is offered for the Funeral Service Education student who is in their final semester of study. The course is a comprehensive review of the assessed material on the National Board Exam to help prepare the student for the examination and professional licensure. A grade of "C" or better is required to successfully complete the class. Pre/Corequisite: FSE1000, FSE1080, FSE2060, FSE2061, FSE2100, FSE2106, FSE2120C, FSE2140, FSE2201, FSE2202, FSE2203, FSE2160.
Lab Hours: 64.000
Pre/Corequisite: FSE1000 AND FSE1080 AND FSE2060 AND FSE2061 AND FSE2100 AND FSE2106 AND FSE2120 AND FSE2140 AND FSE2160 AND FSE2201 AND FSE2203

FSE2932 Funeral Science Professional Review 2 1.000 credits

This course is for the funeral Science student who is graduating and taking the National Board Examination at the end of the semester that this course is being offered. The course is a review of the Arts section of the Funeral Science courses in order to help prepare the student for the National Board Examination.
Lecture Hours: 32.000
Prerequisite: Departmental Permission Required

Program Description

Game Development & Design

Associate in Science | Code: 25075 | 60 credits

This curriculum allows students to explore the entertainment technology landscape while still pursuing a broad-based education. With an emphasis on game programming, the program exposes students to the development and design processes. Students can further specialize in game design, production, engines and systems, graphics programming and animation, mobile, and more.

Program Modality: Both [In-Person (IP) and On-line]
Contracted: No

Semester 1

Units: 12.0

COP 1334 Introduction to C++ Programming 4.0

DIG 1710	Introduction to Game Development	4.0
DIG 1729C	Game Engines	4.0
Semester 2		Units: 14.0
MAC 1105	College Algebra	3.0
CAP 2047	User Interface Design	4.0
COP 2335	C++ Programming 2	4.0
DIG 1430	Storyboarding	3.0
Semester 3		Units: 6.0
ENC 1101	English Composition 1	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 4		Units: 14.0
<i>Hum</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
CAP 2048	Game Development Project I	3.0
DIG 1772C	Introduction to Virtual & Augmented Reality Technologies	4.0
DIG 2712	Level Building & Design	4.0
Semester 5		Units: 14.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
PHY 1020	General Education Physics	3.0
CAP 2920C	Game Development Project II	3.0
DIG 2626	Artificial Intelligence	4.0
DIG 2717C	Game Systems Design	4.0
		Total: 60.0

**COP1334 Introduction to C++ Programming
4.000 credits**

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

**DIG1710 Introduction to Game Development
4.000 credits**

This course is an introduction to the computer game design and development industry. Students will learn about game development careers, game development and design processes, marketing themes, copyright laws, game company structures, programming languages used by different types of games, the impact of video games on modern society, general programming concepts, how to create game design documentation, and how to use common game development environments.
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$97.00

DIG1729C Game Engines 4.000 credits

This course is an introduction to game engines and their uses. Students will learn the basic techniques for creating interactive applications and how this techniques can be used for Virtual Reality (VR) and Augmented Reality (AR) projects. (3 hr. lecture, 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$129.55

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.
Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

CAP2047 User Interface Design 4.000 credits

This course is for students majoring in game development. It covers designing and developing interfaces for games. Students will learn how to use different input/output hardware devices, how to create and use existing interfaces for different types of hardware, and the development process for different types of gaming systems. Prerequisites: DIG1710 and DIG1729C; Pre/Corequisite: COP2335.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: DIG1710, DIG1729C AND Pre/Corequisite: COP2335
Course User Fee: \$129.55

COP2335 C++ Programming 2 4.000 credits

This course delves further into the application of C++ programming techniques including classes, object orientation, function and operator overloading, inheritance and dynamic polymorphism, templates, exception handling, standard template library, data structures, and other advanced C++ techniques. Prerequisite: COP 1334 (3 hr. lecture; 2 hr. lab).
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: COP1334
Course User Fee: \$63.00

DIG1430 Storyboarding 3.000 credits

This course is for students majoring in Animation and Game Art. It introduces the necessary tasks in the storytelling phase of an animation project. Students will learn how to develop and design visual storyboards and how to sell their storyboard ideas. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$97.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CAP2048 Game Development Project I 3.000 credits

This capstone course is for students majoring in Game Development and Game Animation. Students will work in cross disciplinary teams to develop a working animated game or film. Students will learn how to apply the skills and knowledge they have acquired in a real-world working development environment. Prerequisites: CAP2047 or DIG1302, COP1334 or DIG1111, COP2335 or DIG1437, DIG1430, DIG1710 or DIG1132; Pre/Corequisite: DIG1712 or DIG2113.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAP2047, Corequisite: DIG2626, Pre/Corequisite: DIG1712

Course User Fee: \$129.55

DIG1772C Introduction to Virtual & Augmented Reality Technologies 4.000 credits

This course introduces students to basic concepts, history and tools commonly used for Virtual Reality (AR) and Augmented Reality (AR) technologies. Students will learn the history of VR/AR, its current role in the industry, developer tools, and physical principles. Prerequisite: DIG 1729C. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: DIG1729C

Course User Fee: \$129.55

DIG2712 Level Building & Design 4.000 credits

This course is intended for students majoring in Game Development and Design. Students will learn how to develop game environments in industry standard engines, how to create documentation to plan out effective game play experiences, and the requirements to create virtual worlds. Prerequisites: CAP 2047; COP 2335; DIG 1430. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP2047 AND COP2335 AND DIG1430

Course User Fee: \$129.55

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

CAP2920C Game Development Project II 3.000 credits

This capstone course is for students majoring in Game Development and Game Animation. Students will work in cross disciplinary teams to develop a working 5-10 min interactive game experience. Students will learn how to apply the skills and knowledge they have acquired in a real world working development environment. Prerequisites: CAP2048, DIG1712 or DIG 2113.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAP2048 AND DIG1430 AND

Corequisite: DIG1705

Course User Fee: \$129.55

DIG2626 Artificial Intelligence 4.000 credits

This course covers key aspects of Artificial Intelligence (AI) for students majoring in game development. Students will learn the origins and history of Artificial Intelligence, current and future uses of AI, AI methods algorithms such as: path planning, stimulus-response agents, agent architectures, decision-making systems, game trees, neural networks, and genetic algorithms. Students will create and modify existing games to include an AI system.

Lecture Hours: 48.000, Lab Hours: 32.000

Pre/Corequisite: COP1334

Course User Fee: \$97.00

DIG2717C Game Systems Design 4.000 credits

This is a core course for students majoring in game development and design. Students will learn how to develop game systems such as combat, economy, and social society. Students will also learn how to model and test systems before incorporating them into development, and how to use probability to create more interesting gameplay. Prerequisites: CAP 2047, DIG 1712, and MAC 1105. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000 - 58.000, Lab Hours: 32.000, Practicum Hours: 1.000

Prerequisite: CAP2047 AND DIG1712 AND MAC1105

Course User Fee: \$129.55

Program Description

Graphic Design Technology

Associate in Science | Code: 26031 | 64 credits

The Graphic Design Technology program is designed to give creative students a rewarding and challenging career in the artistic field of printing, publishing, electronic communication and advertising. An art aptitude is required or supplemental classes may be taken. Miami Dade College's graphic department offers one of the most extensive electronic publishing teaching facilities in the United States. This A.S. degree may transfer to upper- division universities with a Bachelor of Science degree in Graphic Design. This degree will give students employability skills for the printing, publishing, electronic communication, design or advertising industries.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1 **Units: 15.0**

GRA 1111C	Graphic Design Fundamentals	4.0
GRA 1280C	Digital Imaging Fundamentals	4.0
GRA 1750	Web Design Fundamentals	3.0
GRA 2117C	Digital Illustration Fundamentals	4.0

Semester 2 **Units: 14.0**

ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
GRA 1206C	Typography Fundamentals	4.0
GRA 2151C	Advanced Digital Illustration and Imaging	4.0

Semester 3 **Units: 6.0**

<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Computer Competency Test*

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4 **Units: 15.0**

GRA 1113C	Visual Identity & Branding Design	4.0
GRA 2121C	Publication Design	4.0
GRA 2545C	Package Design	4.0
Program Elective ¹⁾		3.0

¹⁾Select 3 credits from the following course options: ART 1300C, ART 1330C, GRA 1949, GRA 2305C, GRA 2949

Semester 5 **Units: 14.0**

<i>Natrual Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		

ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
GRA 2203C	Portfolio and Business Practices For Designers	3.0
GRA 2207C	Capstone Project	4.0
GRA 2546C	Advertising Design	4.0

Total: 64.0

GRA1111C Graphic Design Fundamentals 4.000 credits

The basics of graphic design comprise this introductory course. The student will analyze the principles and practices of graphic design, and evaluate their creative solutions to challenging design problems. Topics include design vocabulary, elements and principles, the design process, proportion and composition, creativity and visualization, as well as color and typography basics. Assignments are designed to emulate real-world projects, including brainstorming, design, execution, critique, revision and implementation.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$20.00

GRA1280C Digital Imaging Fundamentals 4.000 credits

The student will learn that leading industry-standard image editing software is used to prepare photos for print and digital media in this introductory course. Topics include: resolutions, compositing, masking, correcting and enhancing, as well as applying styles and effects. Assignments are designed to emulate real-world projects, including composite images for graphic design, photographic editing, and web pages.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$35.00

GRA1750 Web Design Fundamentals 3.000 credits

The basics of web design comprise this introductory course. An introduction to the history of the web, including current design trends, a survey of Internet architecture, as well as user experience fundamentals and usability will comprise this introductory-level course. The student will examine the structure and styling of web pages.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$118.00

GRA2117C Digital Illustration Fundamentals 4.000 credits

The student will create illustrations using industry-standard vector-based illustration software. Topics include creating vector-based illustrations, leading to an illustration portfolio. Emphasis will be on both representational as well as commercial art styles. Assignments are designed to emulate real-world projects, including illustration, design, and typography.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

GRA1206C Typography Fundamentals 4.000 credits

The student will design with type in this introductory course. The student will examine readability vs. legibility, the history of type, type anatomy and how to classify type into groups. The student will examine typographic layout design and how the principals of design apply. Studio projects will introduce the student to expressiveness through typographic creativity, symbolism and abstraction to express a given topic or design solution.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

GRA2151C Advanced Digital Illustration and Imaging 4.000 credits

This is an advanced course in digital illustration and imaging, building on existing skills to design and create portfolio-ready projects. Topics include: perspective illustration, advanced masking and compositing techniques, "comping" for layout presentations, outdoor advertisements or 3-D package designs. The student will explore optimizing graphics for web, animations and video.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1280C, GRA2117C

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

GRA1113C Visual Identity & Branding Design 4.000 credits

Studio projects involving visual identity and branding design will comprise this intermediate course. The student will examine color theory, branding, and best practices to develop unique and impactful brands, including sustainability and brand value. The student will design logos and style manuals. Trademarks and copyrights will be examined.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1111C, GRA1206C, GRA2117C

Course User Fee: \$20.00

GRA2121C Publication Design 4.000 credits

Studio projects focusing on periodical and catalog designs comprise this intermediate-level course. The student will apply knowledge and skills toward the design and creation of newsletters, magazines, catalogs, booklets, press media kits and interactive e-books. Emphasis will be on graphic design principles, composition, layout, pagination, style balance format and project planning.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1111C, GRA1206C, GRA2117C, Pre/
Corequisite: GRA1280C

Course User Fee: \$25.00

GRA2545C Package Design 4.000 credits

The student will create package designs for use in retail sales, merchandising, and point-of-purchase in this introductory-level course. Assignments are designed to emulate real-world projects, including package construction, design, illustration and typography, leading to portfolio-ready projects.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA2151C

Course User Fee: \$50.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GRA2203C Portfolio and Business Practices For Designers 3.000 credits

This advanced course enables students to create a portfolio for self-promotion to prospective employers and clients. The student will revise and assemble projects accomplished throughout their career tracks. The student will also analyze best practices involved in the profession: pricing freelance assignments, contracts, intellectual property rights, and other professional requirements. Course is taken in the semester prior to graduation.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: GRA2121C, GRA2151C

Course User Fee: \$3.00

GRA2207C Capstone Project 4.000 credits

The student will apply cumulative knowledge and integrative skills to the design and creation of a professional-level project. The graphic/web design assignment shall meet the requirements specified in a case study or real-world project. Course is taken in the semester prior to graduation.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA2151C

Course User Fee: \$50.00

GRA2546C Advertising Design 4.000 credits

The student will design and create effective advertising campaign concepts for print, outdoor and digital media in this intermediate-level course. Studio projects will emphasize originality, brainstorming, ad layouts, creative copywriting and persuasion, culminating in portfolio-ready projects.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1113C AND GRA2151C

Course User Fee: \$50.00

Program Description

Graphic Internet Technology

Associate in Science | Code: 26050 | 60 credits

The Graphic Internet Technology program is designed to prepare creative students for a rewarding and challenging career as a web designer. Students will develop a wide variety of internet communications skills and will learn to design, produce and distribute communications with the internet. Graduates will work in major corporations, web design studios, internet service providers, government departments and various types of communications organizations.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 15.0
GRA 1111C	Graphic Design Fundamentals	4.0
GRA 1280C	Digital Imaging Fundamentals	4.0
GRA 1750	Web Design Fundamentals	3.0
GRA 2117C	Digital Illustration Fundamentals	4.0
Semester 2		Units: 14.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
GRA 1206C	Typography Fundamentals	4.0
GRA 1751	Fixed-Layout Web Design	4.0
Semester 3		Units: 6.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0

OR		
POS 2041	American Federal Government	3.0

Course Competency Test¹⁾

¹⁾Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4		Units: 12.0
GRA 1752	Motion Graphics for Web Design	4.0
GRA 1754	Responsive Web Design	4.0
GRA 2156C	User Interface and Experience Design	4.0

Semester 5		Units: 13.0
		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
GRA 2203C	Portfolio and Business Practices For Designers	3.0
GRA 2727	Dynamic Web Design	4.0
Elective ¹		3.0

¹Select 3 credits from the following course options: GRA 1949, GRA 2151C, GRA 2207C, GRA 2755, GRA 2949

Total: 60.0

GRA1111C Graphic Design Fundamentals 4.000 credits

The basics of graphic design comprise this introductory course. The student will analyze the principles and practices of graphic design, and evaluate their creative solutions to challenging design problems. Topics include design vocabulary, elements and principles, the design process, proportion and composition, creativity and visualization, as well as color and typography basics. Assignments are designed to emulate real-world projects, including brainstorming, design, execution, critique, revision and implementation.
Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$20.00

GRA1280C Digital Imaging Fundamentals 4.000 credits

The student will learn that leading industry-standard image editing software is used to prepare photos for print and digital media in this introductory course. Topics include: resolutions, compositing, masking, correcting and enhancing, as well as applying styles and effects. Assignments are designed to emulate real-world projects, including composite images for graphic design, photographic editing, and web pages.
Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$35.00

GRA1750 Web Design Fundamentals 3.000 credits

The basics of web design comprise this introductory course. An introduction to the history of the web, including current design trends, a survey of Internet architecture, as well as user experience fundamentals and usability will comprise this introductory-level course. The student will examine the structure and styling of web pages.
Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$118.00

GRA2117C Digital Illustration Fundamentals 4.000 credits

The student will create illustrations using industry-standard vector-based illustration software. Topics include creating vector-based illustrations, leading to an illustration portfolio. Emphasis will be on both representational as well as commercial art styles. Assignments are designed to emulate real-world projects, including illustration, design, and typography.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

GRA1206C Typography Fundamentals 4.000 credits

The student will design with type in this introductory course. The student will examine readability vs. legibility, the history of type, type anatomy and how to classify type into groups. The student will examine typographic layout design and how the principals of design apply. Studio projects will introduce the student to expressiveness through typographic creativity, symbolism and abstraction to express a given topic or design solution.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

GRA1751 Fixed-Layout Web Design 4.000 credits

The student will design fixed-width layout web pages in this intermediate-level course. The student will apply knowledge and skills to the design and construction of single web pages and multipage websites using a visual editor. Topics include styling, linking, incorporating media, production workflows, and the creation of web forms. Assignments emulate real-world projects, including ideation, design, creation, and revision, leading to portfolio-ready projects.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1750

Course User Fee: \$118.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

GRA1752 Motion Graphics for Web Design 4.000 credits

This intermediate-level course examines web vector graphics and their ability to display illustrations and animations on a web page. An in-depth examination of web vector graphics which when combined with transitions and transformations, enables web designers to create dynamic online images and animations that are compatible across desktop and mobile web browsers. The student will create portfolio-ready web projects and animations.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1750, GRA2117C

Course User Fee: \$118.00

GRA1754 Responsive Web Design 4.000 credits

The student will design fluid-layout and responsive pages in this intermediate-level course. Responsive web pages adapt to various devices and user preferences, and examines the advantages of responsive design, as well as the latest industry research reporting on statistics for mobile device use globally. Topics include: fluid grids, responsive typography and images, CSS media queries, "mobile first" workflows, usability, tools and resources.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1751

Course User Fee: \$118.00

GRA2156C User Interface and Experience Design 4.000 credits

In this intermediate-level course, the student will examine the design and creation of mobile applications. The student will create visual designs for mobile apps, as well as analyze how to build and distribute them. The student will apply skills and knowledge toward creating portfolio-ready design projects. Prerequisites: GRA1751 and GRA1754. (2 hr. lecture; 4 hr. lab.)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1751, GRA1754

Course User Fee: \$50.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GRA2203C Portfolio and Business Practices For Designers 3.000 credits

This advanced course enables students to create a portfolio for self-promotion to prospective employers and clients. The student will revise and assemble projects accomplished throughout their career tracks. The student will also analyze best practices involved in the profession: pricing freelance assignments, contracts, intellectual property rights, and other professional requirements. Course is taken in the semester prior to graduation.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: GRA2121C, GRA2151C

Course User Fee: \$3.00

GRA2727 Dynamic Web Design 4.000 credits

The student will design and create dynamic web pages. This course presents PHP programming language to enable students to reduce website maintenance, change content depending on date and time, collect user input from an online form, and link to a MySQL database to display product information. Studio projects culminate in portfolio-ready artifacts.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1754, GRA2156C

Course User Fee: \$118.00

Program Description

Health Information Technology

Associate in Science | Code: 23052 | 70 credits

The Health Information Management program prepares the individual for employment as a Health Information Technician in a variety of health care facilities. The technician may function in various capacities, having responsibilities such as coding of diagnoses and procedures; processing of health information; storage and retrieval of health information and statistical reporting. Other aspects of the curriculum include medical legal aspects, quality assessment and supervision of the daily operations of a Health Information Department. Management of computerized health information is emphasized. Clinical experiences are provided under the

supervision of qualified professionals to enhance classroom instruction and demonstrate current advances in health information practice. A grade of "C" or better is required in all program courses.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
HIM 1000	Introduction to Health Information Technology	2.0
Semester 2		Units: 14.0
ENC 1102	English Composition 2	3.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0
HIM 1300	Health Care Facility and Delivery Systems	2.0
HIM 2012	Legal Aspects of Health Care	2.0
HIM 2472	Medical Terminology	3.0
Semester 3		Units: 12.0
STA 2023	Statistical Methods	3.0
HIM 1110	Health Information Technology and Data Collection	2.0
HIM 1110L	Health Information Management Data Collection Lab	3.0
HIM 1800	Professional Practice Experience 1	2.0
HIM 2253C	Current Procedural Terminology (CPT-4)	2.0
Semester 4		Units: 10.0
HIM 2222	ICD Coding Systems	2.0
HIM 2222L	ICD Coding Systems Laboratory	3.0
HIM 2433	Pathophysiology and Pharmacology	3.0
HIM 2512C	Supervision & Organization For Health Information Management	2.0
Semester 5		Units: 12.0
PHI 2010	Introduction to Philosophy	3.0
HIM 2211C	Health Information Technologies	2.0
HIM 2234	Advanced Coding & Reimbursement Systems	2.0
HIM 2234L	Advanced Coding & Reimbursement Systems Laboratory	1.0
HIM 2400C	Diversified Non-Hospital Health Records	2.0
HIM 2810	Professional Practice Experience 2	2.0
Semester 6		Units: 10.0
HIM 2214C	Health Statistics	2.0
HIM 2500	Data Management & Quality Assessment	2.0

HIM 2500L	Data Management & Quality Assessment Laboratory	1.0
HIM 2652C	Electronic Health Record	3.0
HIM 2820	Seminar and Professional Practice Experience 3	2.0

Total: 70.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

HIM1000 Introduction to Health Information Technology 2.000 credits

This course examines the role and functions of a health information technician. Students will learn about the structure, organization, and maintenance of the medical record. Students will also learn about the organization and function of various types of health care facilities, the responsibilities of national, state and local health agencies, and the organization and mission of the American Health Information Management Association.

Lecture Hours: 32.000

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Lab 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

HIM1300 Health Care Facility and Delivery Systems 2.000 credits

This course will examine healthcare complexities, the function of various types of health facilities, accreditation standards, Medicare law, and the American health delivery system. Students will learn the components of Medicare, Medicaid, Health Insurance Organizations (HMO's), and the federal laws that govern them.

Lecture Hours: 32.000

HIM2012 Legal Aspects of Health Care 2.000 credits

This course provides basic knowledge of the United States of America court system and the interconnection between the health care system and the federal government. The student will learn concepts relating to the Health Insurance Portability Accountability Act (HIPAA), ethics, release of health information, record retention, and the legalities of electronic health records.

Lecture Hours: 32.000

Prerequisite: ENC1101

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

HIM1110 Health Information Technology and Data Collection 2.000 credits

This course is designed to provide the skills necessary to function in a health information management department. Students will learn health record data collection and informatics. Students will also learn about the various components and approaches of the electronic health record.

Lecture Hours: 32.000

Prerequisite: HIM1000, HIM2472, Corequisite: HIM1110L

HIM1110L Health Information Management Data Collection Lab 3.000 credits

This course is designed to apply basic requirements imposed by regulatory agencies to health record data. Students will learn how clinical data repositories store health information. Concepts relating to confidentiality, ethics, and release of information will be applied.

Lab Hours: 96.000

Prerequisite: HIM1000, HIM2472, Corequisite: HIM1110

Course User Fee: \$25.00

HIM1800 Professional Practice Experience 1 2.000 credits

This course will provide the student with supervised professional practice experience in a health care setting, utilizing electronic health records and reports to manage health information data. Students will learn an in-depth knowledge of applying the minimum basic requirements for handling records imposed by regulatory agencies.

Clinical Hours: 96.000

Prerequisite: HIM1000, Corequisite: HIM1110, HIM1110L

Course User Fee: \$10.00

HIM2253C Current Procedural Terminology (CPT-4) 2.000 credits

This course provides an in-depth knowledge of coding and reporting using Current Procedural Terminology classification. Students will learn to read and interpret ambulatory health record documentation to classify and assign services and procure codes, and use of encoder and grouper software. HCPCS, APCs, and RBRVS will be discussed.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: BSC2085, BSC2085L, HIM2472

Course User Fee: \$15.00

HIM2222 ICD Coding Systems 2.000 credits

This course is designed to teach the current edition of International Classification of Diseases and Procedures codes. Students will learn code convention and format, usage of anatomy, physiology, and clinical disease processes. Pharmacology, present on admission indicators, correct code assignment and sequencing will be discussed.

Lecture Hours: 32.000

Prerequisite: BSC2085, BSC2085L, HIM2472,

Corequisite: HIM2222L

HIM2222L ICD Coding Systems Laboratory 3.000 credits

This course is designed to apply diagnoses and operations into numerical designations (codes) utilizing the International Classification of Diseases and Procedures. Students will learn correct sequencing, analysis of clinical disease processes, diagnosis and procedural terminology, and application of pharmacology in current coding systems. Encoding software is utilized.

Lab Hours: 96.000

Prerequisite: BSC2085, BSC2085L, HIM2472,

Corequisite: HIM2222

Course User Fee: \$25.00

HIM2433 Pathophysiology and Pharmacology 3.000 credits

This course provides an in-depth knowledge of disease, its etiology, medical complications, and pathophysiologic nature. Students will learn laboratory and other diagnostic tests used to confirm or rule out those diagnoses addressed. Current pharmacological treatments are explored with review and interpretation of health record data.

Lecture Hours: 48.000

Prerequisite: BSC2085, BSC2085L, HIM2472

HIM2512C Supervision & Organization For Health Information Management 2.000 credits

This course will review the basic principles of management and organizational life in a health information management department and the interrelationships within the health care organization. Emphasis will be placed on the supervisory role of the health information professional, including basic motivation and communication principles essential to the practice of health information management. The student will identify and use specific motivational and communication techniques in health information supervision.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: HIM 1000, HIM 1110, HIM 1110L, HIM 1800

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

HIM2211C Health Information Technologies 2.000 credits

This course is designed to examine computer technology used to collect and store health information. The student will learn a variety of applications used to maintain and secure health care data.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: HIM1110, HIM1110L, HIM1800

Course User Fee: \$85.00

HIM2234 Advanced Coding & Reimbursement Systems 2.000 credits

This course is designed to apply the fundamentals of the Prospective Payment Systems as it applies to coding and reimbursement. The student will learn documentation criteria, validation reports of coded data, health record for compliance, and optimum reimbursement under current payment methodologies.

Lecture Hours: 32.000

Prerequisite: HIM2222, HIM2222L, Corequisite: HIM2234L

HIM2234L Advanced Coding & Reimbursement Systems Laboratory 1.000 credits

This course is designed to apply and compute Prospective Payment Systems categories. Students will learn to apply health record documentation to identify and validate correct code and payment assignments. The focus is on the computation of MS-DRGs, APCs and a Case-Mix Index using encoder, grouper, and electronic billing software for reimbursement.

Lab Hours: 32.000

Prerequisite: HIM2222, HIM2222L, Corequisite: HIM2234

Course User Fee: \$15.00

HIM2400C Diversified Non-Hospital Health Records 2.000 credits

This course emphasizes the importance of quality record-keeping practices, data flow, and management of health information systems in a non-acute care setting. The student will learn the documentation requirements based on Federal and State statutes, accreditation standards, Medicare Conditions of Participation, payment systems, funding, Health Insurance Portability Accountability Act, and the evolution of the electronic health record. Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$15.00

HIM2810 Professional Practice Experience 2 2.000 credits

This course is a supervised professional practice experience in a health care setting utilizing health records and reports to perform medical coding functions. Students will learn to assign Medicare Severity-Diagnosis Related Groups/Ambulatory Payment Classifications using the organization's information systems, encoder and grouper software. Sequencing of International Classification of Disease (ICD) and Current Procedural Terminology (CPT) coding systems will be discussed.

Clinical Hours: 96.000

Prerequisite: HIM2234, HIM2234L

Course User Fee: \$15.00

HIM2214C Health Statistics 2.000 credits

This course will focus on the definitions for analysis, interpretation, and display of healthcare data. The student will learn the acceptable terminology and basic definitions for reporting health statistics.

Emphasis is placed on the use of the formulas necessary for computing standard rates, percentages, and averages from patient data.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: HIM1110, HIM1110L, Corequisite: HIM2512C

Course User Fee: \$20.00

HIM2500 Data Management & Quality Assessment 2.000 credits

This course examines the basic principles of quality assessment, quality improvement and utilization review. Students will learn about the accreditation process, risk management, managed care models, and the methodologies and relationships of these key areas within a health care facility. Prerequisites: HIM 1110, 1110L, 2211C. Corequisite: HIM 2500L.

Lecture Hours: 32.000

Course User Fee: \$75.00

HIM2500L Data Management & Quality Assessment Laboratory 1.000 credits

The application of the basic principles of quality assessment: quality improvement and utilization review. The student will learn to generate models for the evaluation of different types of medical care. Activities will center on the accreditation process, managed care, and risk management. The methodologies and relationships of these key areas within a health care facility are emphasized.

Lab Hours: 32.000

Prerequisite: HIM 1110, HIM 1110L, Corequisite: HIM 2500

HIM2652C Electronic Health Record 3.000 credits

This course reviews the current trends and preparation implementing the electronic health record and reviews documentation requirements for a variety of healthcare settings. Students will learn best practices, problem-solve associated issues, and directly participate in the transitions of electronic health records.

Lecture Hours: 16.000, Lab Hours: 32.000

Pre/Corequisite: HIM2211C

HIM2820 Seminar and Professional Practice Experience 3 2.000 credits

This course is designed to provide students with structured learning experiences necessary for them to enter the health information management field. Students will learn preparation skills for the national examination by analyzing major examination topics offered in curriculum. A professional practice experience will be a component of this course.

Clinical Hours: 96.000

Prerequisite: HIM2810, Corequisite: HIM 2500, 2500L

Course User Fee: \$10.00

Program Description

Health Science - Health Services Management

Associate in Science | Code: 23082 | 60 credits

The Associate in Science degree in Health Sciences: Health Service Management is designed to introduce graduates to the healthcare field and an in-depth background on health care management to prepare them for a health-related career or a graduate professional health program or other graduate programs. In addition, students will acquire extensive knowledge on legal and ethical responsibilities within the health care system, current issues in the health care system, health services management concepts, health care quality management, and managing organizations in a multicultural environment.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
Course Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
PHI 2010	Introduction to Philosophy	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
HIM 2472	Medical Terminology	3.0

Semester 3		Units: 13.0
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ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
GEB 1011	Principles of Business	3.0
MAN 2021	Principles of Management	3.0
HIM 2652C	Electronic Health Record	3.0

Semester 4

Units: 13.0

HIM 1300	Health Care Facility and Delivery Systems	2.0
HIM 2253C	Current Procedural Terminology (CPT-4)	2.0
HSA 1380	Health Care Quality Management	3.0
HSA 2181	Health Services Management Concepts	3.0
HSC 1149	General Pharmacology for Health Care Professions	3.0

Semester 5

Units: 11.0

HIM 2012	Legal Aspects of Health Care	2.0
HSA 1102	Current Issues in Health	3.0
HSA 2001	Interprofessional Team Based Health Care	2.0
HSC 2810	Professional Practice Experience	4.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

**BSC2085 Human Anatomy and Physiology 1
3.000 credits**

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

**BSC2085L Human Anatomy and Physiology 1
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

HIM2652C Electronic Health Record 3.000 credits

This course reviews the current trends and preparation implementing the electronic health record and reviews documentation requirements for a variety of healthcare settings. Students will learn best practices, problem-solve associated issues, and directly participate in the transitions of electronic health records.

Lecture Hours: 16.000, Lab Hours: 32.000

Pre/Corequisite: HIM2211C

HIM1300 Health Care Facility and Delivery Systems 2.000 credits

This course will examine healthcare complexities, the function of various types of health facilities, accreditation standards, Medicare law, and the American health delivery system. Students will learn the components of Medicare, Medicaid, Health Insurance Organizations (HMO's), and the federal laws that govern them.

Lecture Hours: 32.000

HIM2253C Current Procedural Terminology (CPT-4) 2.000 credits

This course provides an in-depth knowledge of coding and reporting using Current Procedural Terminology classification. Students will learn to read and interpret ambulatory health record documentation to classify and assign services and procure codes, and use of encoder and grouper software. HCPCS, APCs, and RBRVS will be discussed.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: BSC2085, BSC2085L, HIM2472

Course User Fee: \$15.00

HSA1380 Health Care Quality Management 3.000 credits

This course examines various efforts being introduced to improve the quality and efficiency of the health care system. Topics covered range from issues of medical error reduction, quality improvements in medical records, and utilization review, models for continuous quality improvement in physician-patient relations. (3 hr. lecture)

Lecture Hours: 48.000

HSA2181 Health Services Management Concepts 3.000 credits

This course provides preparation and fundamental skills required to enter the field of medical office management. Emphasis will be placed on health care delivery in a medical office setting. Topics include an introduction to health care management information systems, an introduction to essential policies and regulations, and an introduction to evaluation of performance and process outcomes.

Lecture Hours: 48.000

HSC1149 General Pharmacology for Health Care Professions 3.000 credits

This course will provide learners with a basic introduction to pharmacology concepts and principles associated with management of common diseases and treatment options. It presents information on major drug classifications, indications for use of common medications, and potential contraindications adversely affecting medical care outcomes.

Lecture Hours: 48.000

HIM2012 Legal Aspects of Health Care 2.000 credits

This course provides basic knowledge of the United States of America court system and the interconnection between the health care system and the federal government. The student will learn concepts relating to the Health Insurance Portability Accountability Act (HIPAA), ethics, release of health information, record retention, and the legalities of electronic health records.

Lecture Hours: 32.000

Prerequisite: ENC1101

HSA1102 Current Issues in Health 3.000 credits

This course covers current issues within the health care system. Emphasis focuses on varied areas of health care and is appropriate for persons directly or indirectly involved in the provision of health care or health education. (3 hr. lecture)

Lecture Hours: 48.000

HSA2001 Interprofessional Team Based Health Care 2.000 credits

This course is an introduction to interprofessional health care delivery. Reviewed are concepts of interprofessional collaboration practices in relation to team formation, ethical consideration with team management, effective communication approaches, and patient care processes. (3 hr. lecture)

Lecture Hours: 32.000

HSC 2810 Professional Practice Experience 4.000 credits

This course provides an opportunity for the student to observe basic principles and techniques of managers in a health care setting. Students apply knowledge from previous health service management courses in completing course assignments. This course has a service learning component at an area health care agency. (3 hr. lecture)

Lecture Hours: 64.000

Program Description

Health Sciences

Associate in Science | Code: 23081 | 60 credits

The Health Sciences program provides students an introduction to the healthcare field and an in-depth science background to prepare them for a health-related career or a graduate professional health program or other graduate program.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 15.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
Computer Competency Test ¹⁾		

¹⁾Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2

Units: 14.0

BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
CHM 1046	General Chemistry and Qualitative Analysis	3.0
CHM 1046L	General Chemistry and Qualitative Analysis Lab	2.0
MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0

Semester 3

Units: 10.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		

POS 2041	American Federal Government	3.0
HSA 2532	Medical Documentation in Health Care	1.0
PAS 1800C	Physical Diagnosis I	2.0
PAS 1803	Clinical Anatomy and Physiology	2.0
PAS 1831	Clinical Diagnostic Imaging	1.0
PAS 2936	Contemporary Issues for the PA	1.0

Semester 4

Units: 11.0

<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
PAS 1801C	Physical Diagnosis 2	2.0
PAS 1811C	Introduction to Medicine I for PAS	5.0
PAS 1812	Behavioral and Community Medicine	1.0

Semester 5

Units: 10.0

PAS 1813	Pathophysiological Basis of Disease I	2.0
PAS 1822C	Electrocardiography	1.0
PAS 1823	Pharmacology 1	4.0
PSY 2012	Introduction to Psychology	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

**BSC2085 Human Anatomy and Physiology 1
3.000 credits**

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

**BSC2085L Human Anatomy and Physiology 1
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

**CHM1045 General Chemistry and Qualitative
Analysis 3.000 credits**

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite:

BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites:

BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

HSA2532 Medical Documentation in Health Care 1.000 credits

Medical Documentation in Health Care will introduce the student to documentation in the written patient chart or electronic medical record. Through case discussions and in-class writing assignments, the student will acquire the necessary skills to document in the patient's medical record utilizing medico-legal principles and evaluation and management criteria. Patient confidentiality, billing, and coding will also be discussed.

Lecture Hours: 16.000

PAS1800C Physical Diagnosis I 2.000 credits

A course which provides the students with the critical basis for and clinical exposure to techniques used in the proper performance and recording of the physical examination of patients.

Lecture Hours: 16.000, Clinical Hours: 48.000

Prerequisite: BSC 2086, BSC 2086L, CHM 1046, CHM 1046L, MCB 2010, MCB 2010L

Course User Fee: \$560.00

PAS1803 Clinical Anatomy and Physiology 2.000 credits

This course is designed for students accepted into the Physician Assistant Program. The course will review basic Anatomy and Physiology principles, while integrating important clinical concepts.

Students will learn to transition from Anatomy and Physiology to Pathophysiology.

Lecture Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L AND BSC2086 AND BSC2086L AND CHM1045 AND CHM1045L AND CHM1046 AND CHM1046L AND MCB2010 AND MCB2010L

PAS1831 Clinical Diagnostic Imaging 1.000 credits

The course introduces the PA student to principles of diagnostic imaging and a familiarity with the many diagnostic techniques available, their values and limitations, and how they may be best used in the management of the patient in clinical practice.

Lecture Hours: 16.000

Prerequisite: BSC2085 AND BSC2085L AND BSC2086 AND BSC2086L AND CHM1045 AND CHM1045L AND CHM1046 AND CHM1046L AND MCB2010, MCB2010L

Course User Fee: \$280.00

PAS2936 Contemporary Issues for the PA 1.000 credits

In this course the student will examine current issues, challenges, and practices influencing the physician assistant profession. Historical perspectives will be examined and the contributions of physician assistants in the healthcare system will be researched. The student will learn to use evidenced based medicine to research topics including the changing nature of health care delivery system in the United States; demographic, economic, ethical, and political factors influencing the practice of the physician assistant.

Lecture Hours: 16.000

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PAS1801C Physical Diagnosis 2 2.000 credits

This course will build upon the skills learned in Physical Diagnosis I and will prepare the physician assistant student with the essential skills for entry into clinical practice. Students will learn to perform medical interviews and physical examinations on live and professional patients, and high-fidelity simulators. Prerequisite: PAS1800C , PAS1803, PAS1831 , PAS2936. (1 hr. lecture, 3 hr. clinical) Lecture Hours: 16.000, Clinical Hours: 48.000 Prerequisite: PAS1800C, PAS1803, PAS1831, PAS2936 Course User Fee: \$560.00

PAS1811C Introduction to Medicine I for PAS 5.000 credits

Students will learn the signs, symptoms, and pathophysiology of common diseases affecting pediatric, adult, and geriatric patients. In addition, diagnosis, therapeutic interventions, and follow-up will be learned. Patient education and preventative medicine will also included. 3 hr. lecture, 2 hr. clinical) Lecture Hours: 48.000, Clinical Hours: 96.000 Prerequisite: PAS 1800C, PAS 1803, PAS 1831, PAS 2936

Course User Fee: \$1400.00

PAS1812 Behavioral and Community Medicine 1.000 credits

This course utilizes a biopsychosocial system approach to identify social determinants within the health care delivery system. Students will participate in outreach projects to implement concepts learned in the classroom within a community environment. Prerequisites:

Lecture Hours: 16.000

Prerequisite: PAS1800C AND PAS1831 AND PAS2936, PAS 2YYY

Course User Fee: \$280.00

PAS1813 Pathophysiological Basis of Disease I 2.000 credits

The clinical pathophysiology course serves as a transition from the basic medical sciences to clinical medicine. Students will learn the study of disease, both congenital and acquired with an emphasis on providing understanding of pathologic physiology in conjunction with information regarding medical history and laboratory data to solve case based clinical problems. Students are taught how pathophysiology translates into patient signs, symptoms. Prerequisites: PAS1800C, 2936. (2 hr. clinical)

Clinical Hours: 96.000

Prerequisite: PAS1800C AND PAS2936

PAS1822C Electrocardiography 1.000 credits

Students will learn the basic knowledge to understand, interpret and utilize an electrocardiogram (EKG) as a diagnostic tool. This course provides the foundation for learning to interpret 12-lead EKG tracings and applying those principles to evaluate the EKG tracings of common cardiac disease.

Lab Hours: 32.000

Prerequisite: PAS 1800C, PAS 1803, PAS 1831, PAS 2936

Course User Fee: \$560.00

PAS1823 Pharmacology 1 4.000 credits

This course will focus on the preparation, uses and action of drugs. Students will learn about Pharmacokinetics and Pharmacodynamics. Students will be introduced to Clinical Pharmacology following an organ system approach, covering the major classes of therapeutically relevant drugs, and how they work at the molecular and cellular levels.

Prerequisites: PAS1800C, PAS1803, PAS1831, and PAS2936 (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: HSA2532, PAS1801C, PAS1803, PAS1831, PAS2936

Course User Fee: \$560.00

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

Program Description

Histologic Technology

Associate in Science | Code: 23064 | 76 credits

The Histologic Technology program prepares the student for employment in an unlimited choice of practice settings including: hospitals, clinics, clinical laboratories, veterinary pathology and forensic pathology. A Histotechnologist will be able to freeze, embed, and cut tissues, mount tissue samples on slides and stain them with dyes to make the cell details visible under the microscope. Graduates are eligible to sit for the Florida State Licensure and Registry with the American Society of Clinical Pathologists and equivalent licensure.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 9.0

ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Course Competency Test*

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 11.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
PHI 2010	Introduction to Philosophy	3.0
Semester 3		Units: 9.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0
MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0
Semester 4 (In Program Milestone)		Units: 12.0
MLT 1191	Histotechnology 1	3.0
MLT 1191	Histotechnology 1	3.0
MLT 1195C	Tissue Identification 1	3.0
MLT 1196	Laboratory Safety and Regulations	3.0
Semester 5 (In Program Milestone)		Units: 10.0
MLT 2192	Histotechnology 2	3.0
MLT 2192L	Histotechnology 2 Laboratory	3.0
MLT 2197C	Tissue Identification 2	4.0
Semester 6 (In Program Milestone)		Units: 9.0
MLT 2180C	Infectious Diseases & Control Practices	3.0
MLT 2198	Histochemistry	3.0
MLT 2198L	Histochemistry Laboratory	3.0
Semester 7 (In Program Milestone)		Units: 8.0
HIM 2472	Medical Terminology	3.0
MLT 1840L	Histotechnology Practicum 1	5.0
Semester 8 (In Program Milestone)		Units: 8.0
MLT 2841L	Histotechnology Practicum 2	5.0
MLT 2931	Histotechnology Seminar	3.0
		Total: 76.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions.

Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite:

MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy.

Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Lab 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

MLT1191 Histotechnology 1 3.000 credits

This course will introduce students to the fundamental principles of histologic technology. These include the principles of fixation, processing for paraffin-embedding, microtome sectioning, staining and cover-slipping and laboratory safety.

Lecture Hours: 48.000

MLT1191 Histotechnology 1 3.000 credits

This course will introduce students to the fundamental principles of histologic technology. These include the principles of fixation, processing for paraffin-embedding, microtome sectioning, staining and cover-slipping and laboratory safety.

Lecture Hours: 48.000

MLT1195C Tissue Identification 1 3.000 credits

This course will introduce students to the study of human organs and tissues for the purpose of developing histotechnological skills. It will include recognition, composition, and function of organs and tissues. Macroscopic and microscopic laboratory examination and evaluation of specimens will be included.

Lecture Hours: 32.000, Lab Hours: 32.000

MLT1196 Laboratory Safety and Regulations 3.000 credits

This course will introduce students to the rules and regulations governing safety in the histotechnology laboratory. It will also introduce students to the federal regulations pertaining to the histotechnology laboratory and methods of compliance. Prominent safety issues to be covered include the biological and chemical hazards in histology laboratory, formaldehyde standard, hazardous waste disposal and minimization.

Lecture Hours: 48.000

MLT2192 Histotechnology 2 3.000 credits

This course is a continuation of Histotechnology 1. Students will be introduced to advanced processing techniques of human tissue for anatomical pathology and concepts of instrumentation.

Lecture Hours: 48.000

Prerequisite: MLT1191

Course User Fee: \$100.00

MLT2192L Histotechnology 2 Laboratory 3.000 credits

This course is a continuation of Histotechnology Lab 1. Students will be introduced to more complex laboratory techniques in histotechnology.

Lab Hours: 96.000

Prerequisite: MLT1191, MLT1191L, Corequisite: MLT2197C

MLT2197C Tissue Identification 2 4.000 credits

This course will provide the students with the correlations between histotechnological procedures and diseases processes. Students will study the changes in tissue that are associated with various disease states, and will learn the usefulness of selected special stains and techniques in identifying disease processes.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: MLT1195C

MLT2180C Infectious Diseases & Control Practices 3.000 credits

This course will focus on the principles of transmission and control of diseases with an emphasis on infectious tissue specimens.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: MCB2010, MCB2010L

MLT2198 Histochemistry 3.000 credits

This course will introduce students to organic chemistry of stains and special stains, dyes, hydrocarbons; aromatics, alcohols, ethers, aldehydes, ketones, carbonyl compounds, amines and amides.

Lecture Hours: 48.000

Prerequisite: MLT2192, MLT2192L, Corequisite: MLT2198L

MLT2198L Histochemistry Laboratory 3.000 credits

This course will introduce students to biochemicals used in histology with emphasis on laboratory preparation and use of histochemical and immunohistochemical stains. Prerequisite: CHM 1033L; corequisite: MLT 2198. Laboratory fee. (4 hr. lab)

Lab Hours: 96.000

Prerequisite: CHM 1033, CHM 1033L, MLT 2192, MLT 2192L, Corequisite: MLT2198

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

MLT1840L Histotechnology Practicum 1 5.000 credits

This is a clinical experience in which students will learn the techniques of processing human tissue for histological purposes.

Practicum Hours: 240.000

Prerequisite: MLT 2180C, MLT 2198, MLT 2198L

MLT2841L Histotechnology Practicum 2 5.000 credits

This clinical experience will introduce the students to the basic techniques of microtomy, staining and preparation of human tissue for anatomical pathology.

Practicum Hours: 240.000

Prerequisite: MLT1840L, Corequisite: MLT2931

MLT2931 Histotechnology Seminar 3.000 credits

This course will prepare students for career entry. Emphasis will be placed on current topics in histotechnology, legal and ethical responsibilities of health care professionals, knowledge of the health care delivery system, including health policies and financing and employability skills.

Lecture Hours: 48.000

Prerequisite: MLT1840L, Corequisite: MLT2841L

Program Description

Hospitality & Tourism Management

Associate in Science | Code: 22016 | 60 credits

The Hospitality and Tourism Management program provides professional preparation for a career in the hospitality industry. Hospitality management is presented as a core curriculum with emphasis on hotel, cruise-line, resorts, conventions, and institutional management. An internship program is required to provide practical experience in the field of the student's choice. The Associate in Arts degree is also available to the student planning to transfer to a four-year institution after graduation from MDC. Consult an advisor about which additional courses are included in that program.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
HFT 1000	Introduction to Hospitality	3.0
HFT 1210	Human Resources	3.0

Semester 2		Units: 12.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
HFT 1949	Co-Op Work-Study Internships I	3.0
HFT 2449	E-Business For The Hospitality Industry	3.0
Program Electives*		3.0

Select 3 credits from the following course options: CGS 1060C, FSS 1100, FSS 1200*, FSS 1202C*, HFT 1212*, HFT 1220*, HFT 1300*, HFT 1454*, HFT 1611*, HFT 1631*, HFT 1841*, HFT 1841L, HFT 2223*, HFT 2241*, HFT 2261*, HFT 2410*, HFT 2500, HFT 2501*, HFT 2750*, HFT 2772*, HFT 2773*, HFT 2774*, HFT 2775*, HFT 2800*, HFT 2949*

Semester 3		Units: 12.0
BSC 1005	General Education Biology	3.0
Program Electives ¹⁾		9.0
Computer Competency Test ²⁾		

¹⁾Select 9 credits from the following course options: CGS 1060C, FSS 1100*, FSS 1200*, FSS 1202C*, HFT 1212*, HFT 1220*, HFT 1300*, HFT 1454*, HFT 1611*, HFT 1631*, HFT 1841*, HFT 1841L, HFT 2223*, HFT 2241*, HFT 2261*, HFT 2410*, HFT 2500, HFT 2501*, HFT 2750*, HFT 2772*, HFT 2773*, HFT 2774*, HFT 2775*, HFT 2800*, HFT 2949*

²⁾Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4		Units: 12.0
HFT 1852	Menu and Facilities Planning	3.0
HFT 2421	Managerial Accounting for Hospitality	3.0
Program Electives ¹⁾		6.0

¹⁾Select 6 credits from the following course options: CGS 1060C, FSS 1100*, FSS 1200*, FSS 1202C*, HFT 1212*, HFT 1220*, HFT 1300*, HFT 1454*, HFT 1611*, HFT 1631*, HFT 1841*, HFT 1841L, HFT 2223*, HFT 2241*, HFT 2261*, HFT 2410*, HFT 2500, HFT 2501*, HFT 2750*, HFT 2772*, HFT 2773*, HFT 2774*, HFT 2775*, HFT 2800*, HFT 2949*

Semester 5

Units: 12.0

PHI 2010	Introduction to Philosophy	3.0
Program Electives ¹⁾		9.0

¹⁾Select 9 credits from the following course options: CGS 1060C, FSS 1100*, FSS 1200*, FSS 1202C*, HFT 1212*, HFT 1220*, HFT 1300*, HFT 1454*, HFT 1611*, HFT 1631*, HFT 1841*, HFT 1841L, HFT 2223*, HFT 2241*, HFT 2261*, HFT 2410*, HFT 2500, HFT 2501*, HFT 2750*, HFT 2772*, HFT 2773*, HFT 2774*, HFT 2775*, HFT 2800*, HFT 2949*

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1210 Human Resources 3.000 credits

Provides information relating to the recruitment and selection of new staff, the handling of difficult employees, motivating employees and conducting performance evaluations. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

HFT1949 Co-Op Work-Study Internships I 3.000 credits

Provides a supervised cooperative education experience in the hospitality industry that combines academic learning with practical workplace training. Students gain hands-on experience in hospitality operations, customer service, professional communication, and industry best practices while working in approved hospitality settings. Emphasis is placed on skill development, workplace professionalism, and career readiness through structured experiential learning.

Lecture Hours: 48.000

Course User Fee: \$25.00

HFT2449 E-Business For The Hospitality Industry 3.000 credits

Prepares student to manage information systems within their organizations. Emphasis is on selecting the right computer systems technology and maximizing available technology in order to promote and sell services. Introduces the use of automation in the broad hospitality sector and exams technological applications ranging from distribution systems (GDS, CRS and Web based), Property Management Systems, and EPOS systems to developments in telecommunications, and assesses their effect on the tourism sector. While a broad interpretation of both technology and tourism will be used, particular emphasis will be placed on the hospitality sector (i.e. hotels and catering) and on distribution technology. A combination of lecture, case studies, seminars, visiting lecture and lab sessions are used.

Lecture Hours: 48.000

Prerequisite: HFT1000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

HFT1852 Menu and Facilities Planning 3.000 credits

This course is a study of the factors involved in planning effective menus for a variety of food service operations. Students will learn to develop a firm working knowledge of menu-writing techniques and participate in actual menu design and facilities layout of a food service establishment.

Lecture Hours: 48.000

Prerequisite: HFT1000

Course User Fee: \$40.00

HFT2421 Managerial Accounting for Hospitality 3.000 credits

Presents managerial accounting concepts and explains how they apply to specific operations within the hospitality industry.

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

Program Description

Interior Design Technology

Associate in Science | Code: 26030 | 75 credits

The Interior Design Technology program is planned to develop ability in the design of interiors, to encourage originality and to foster talent in this field. It includes theoretical and technical aspects of interior design. The program is open to those who study for pleasure and those preparing for a career.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
ARC 1115	Architectural Communications 1	2.0
ARC 1126C	Architectural Drawing 1	4.0
IND 1020	Interior Design 1	4.0

Semester 2		Units: 13.0
MAC 1105	College Algebra	3.0
ARC 2171C	Computer Aided Drafting 1	4.0
IND 1200	Interior Design 2	4.0
IND 1300	Interior Design Presentations 1	2.0

Semester 3		Units: 7.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Program Elective ¹⁾		4.0
Computer Competency Test ^{**}		

¹⁾ARC*, IND*, CGS 1060C

^{**}Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4		Units: 14.0
ARC 2461	Architectural Materials and Construction 1	4.0
IND 1100	History of Interiors 1	3.0
IND 2210	Interior Design 3	4.0
IND 2330	Interior Design Presentations 2	3.0

Semester 5		Units: 13.0
IND 1130	History of Interiors 2	3.0
IND 2220	Interior Design 4	4.0
IND 2430	Lighting Design	3.0
IND 2500	Professional Practices	3.0

Semester 6		Units: 6.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		

HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
Natural Science		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0

Semester 7

Units: 9.0

IND 2201	Design Principles for Kitchen and Bath	3.0
IND 2608	Sustainable Design	3.0
Program Electives*		3.0

Select 3 credits from the following course options: ARC, IND*, CGS 1060C

Total: 75.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARC1115 Architectural Communications 1 2.000 credits

Exercises in freehand drawing, sketching and linear perspective are designed to increase the student's awareness of the architectural environment. This is accomplished through a series of form studies of nature, architectural forms, and abstract elements of composition.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$25.00

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions. Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

IND1020 Interior Design 1 4.000 credits

Student's projects develop the ability to plan simple interior floor plans and elevations.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1115

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

ARC2171C Computer Aided Drafting 1 4.000 credits

Computer-aided drafting as it applies in the fields of architecture and interior design using office simulation. Emphasis is on the production of computer-aided drafting of working drawings involving different types of structure. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ARC1126C OR ARC2461
Course User Fee: \$50.00

IND1200 Interior Design 2 4.000 credits

Problems in room planning, correlation of color schemes and furnishings. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: IND1020
Course User Fee: \$25.00

IND1300 Interior Design Presentations 1 2.000 credits

An introductory course in the use of various media for presentation of plans, schemes, and interior perspective renderings. Lecture Hours: 16.000, Lab Hours: 32.000
Prerequisite: IND1020, Corequisite: IND1200
Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ARC2461 Architectural Materials and Construction 1 4.000 credits

An introduction to basic materials and methods of building construction. Emphasis is on wood, concrete, unit masonry, and light steel construction. Laboratory projects may include working drawings' interpretation, sketching construction details, or field trips to construction sites and fabricant plants. Designed primarily as the initial materials and methods course for architectural transfer students.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR BCN1251

Course User Fee: \$25.00

IND1100 History of Interiors 1 3.000 credits

Lecture Hours: 48.000

Course User Fee: \$10.00

IND2210 Interior Design 3 4.000 credits

Projects provide practice in planning traditional and contemporary interiors including working drawings and specifications.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: IND1200, Corequisite: IND2330

Course User Fee: \$25.00

IND2330 Interior Design Presentations 2 3.000 credits

Emphasis is on perfecting water color, casein and reproducible drawing techniques through the presentation of interior plans, elevations and perspectives. Projects also provide experience in assembling collages.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: IND1300, Corequisite: IND2210

Course User Fee: \$25.00

IND1130 History of Interiors 2 3.000 credits

Emphasis is on perfecting water color, casein and reproducible drawing techniques through the presentation of interior plans, elevations and perspectives. Projects also provide experience in assembling collages.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: IND1300, Corequisite: IND2210

Course User Fee: \$25.00

IND2220 Interior Design 4 4.000 credits

Emphasis is on perfecting watercolor, casein and reproducible drawing techniques through the presentation of interior plans, elevations and perspectives. Projects also provide experience in assembling collages.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: IND2210

Course User Fee: \$25.00

IND2430 Lighting Design 3.000 credits

A survey of utilitarian interior lighting and exterior architectural lighting including fundamentals and basic physic laws, practical applications to interior and exterior spaces and lighting design considering different levels of space utilization and fixture efficiency.

Lecture Hours: 48.000

Prerequisite: IND1200

Course User Fee: \$10.00

IND2500 Professional Practices 3.000 credits

Duties and responsibilities relative to employment and business practices.

Lecture Hours: 48.000

Prerequisite: Sophomore standing level or equivalent.

Course User Fee: \$10.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

**IND2201 Design Principles for Kitchen and Bath
3.000 credits**

A specialized design course for kitchen and baths. Students will learn the basic elements and principles of creating spaces for clients, blending architectural styles, colors and themes with function and purpose.

Lecture Hours: 48.000

Prerequisite: ARC2461, IND1200, IND1300

Course User Fee: \$10.00

IND2608 Sustainable Design 3.000 credits

Introduction to basic theories and practices of ecological design in interiors. Students will learn to take responsibility for their work by understanding the impact their designs have on our environment, and ways to mitigate this impact by gaining a practical understanding of the field of sustainable design.

Lecture Hours: 48.000

Prerequisite: ARC1126C, ARC2461, IND1200, IND1300

Course User Fee: \$10.00

Program Description

Landscape & Horticulture Technology

Associate in Science | Code: 21005 | 60 credits

The purpose of this program is to prepare students for employment in horticulture and landscape industries. The students will learn about plant growth, nutrition and fertilization, plant classification and identification, propagation, pest control, pruning, maintenance and drainage. Students will also gain business management and job skills. Students pursuing the Horticulture Specialization will obtain the skills necessary for protecting, processing, shipping and marketing of commercially viable plants. Students pursuing the Landscape Specialization will obtain the skills necessary for landscape design and installation (Note: The degree is awarded only once.).

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
HOS 1010	Horticulture 1	3.0
IPM 2112	Principles of Entomology	3.0
IPM 2301	Pesticide Applications	3.0
<i>Program Elective</i>		3.0
BOT 2150C	Native Plant Identification and Usage in South FL	3.0
OR		
ORH 1510	Landscape Plant Identification 1	3.0
Semester 2		Units: 15.0
ENC 1101	English Composition 1	3.0
HOS 1011	Horticulture 2	3.0
ORH 1251	Nursery Practices 1	3.0
ORH 2949	Landscape Technology Internship	3.0
SBM 1000	Small Business Management	3.0
Semester 3		Units: 13.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
<i>Natural Science</i>		3.0

CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0

Semester 4

Units: 12.0

<i>Mathematics</i>		3.0
MGF 1130	Mathematical Thinking	3.0
OR		
MGF 1131	Mathematics in Context	3.0
LDE 2310	Irrigation Design & Maintenance	3.0
Electives*		6.0

Select 6 credits from the following course options: ACG, BOT*, BSC*, CHM*, ECO*, ESC*, EVR*, FIN*, IPM*, LDE*, MAC*, MAN*, MAR*, MAT*, MKA*, ORH*, QMB*, STA*, TRA*, ZOO

Semester 5

Units: 8.0

Elective*	8.0
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Select 6 credits from the following course options: ACG, BOT*, BSC*, CHM*, ECO*, ESC*, EVR*, FIN*, IPM*, LDE*, MAC*, MAN*, MAR*, MAT*, MKA*, ORH*, QMB*, STA*, TRA*, ZOO

Total: 60.0

HOS1010 Horticulture 1 3.000 credits

This is an introductory course on the principles of horticulture. Students will learn plant structure and function, plant propagation, plant nutrients and fertilizers, potting media, soils, pruning, and plant pests. A survey of various fields in ornamental horticulture will also be covered.

Lecture Hours: 48.000

IPM2112 Principles of Entomology 3.000 credits

This is an introductory course on the principles of entomology. Students will learn to identify characteristics of arthropods, the insect orders, and the growth cycle of insects. Students will also address insect pests specific to South Florida and methods to responsibly manage plant pests.

Lecture Hours: 48.000

IPM2301 Pesticide Applications 3.000 credits

Students will learn government regulations with regards to pesticide mixing and application, and safety equipment. Preparation for the restricted use applicators license exam will be covered.

Lecture Hours: 48.000

BOT2150C Native Plant Identification and Usage in South FL 3.000 credits

Plants native to South Florida are identified and presented according to their typical ecological communities. Emphasis is placed on pineland, tropical hardwood hammock, mangrove and coastal, Everglades marsh, and cypress swamp ecosystems. Plants suitable for use in urban landscapes, as well as ecological restoration projects, are also covered. This course combines lecture and laboratory instruction.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$23.00

ORH1510 Landscape Plant Identification 1 3.000 credits

Students will learn the identification and usage of plants used in the horticultural trade in South Florida. Subject matter includes trees, shrubs, and flowering plants for both interior and outdoor use.

Lecture Hours: 48.000

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

HOS1011 Horticulture 2 3.000 credits

The student will learn the maintenance and management aspects of horticulture business (nursery facility or landscape maintenance and design) including irrigation systems, plant growing facilities, plant propagation equipment, and landscape maintenance equipment. Hands-on practice in programming of plant production crops and nursery design in our nursery.

Lecture Hours: 48.000

Prerequisite: HOS1010

ORH1251 Nursery Practices 1 3.000 credits

The student will learn the techniques and practices in commercial production of ornamental plants.

Emphasis on types of nurseries.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: HOS1010

ORH2949 Landscape Technology Internship 3.000 credits

The internship will provide students with hands-on work experience in horticulture. Landscape, or related technology industries. Students will learn employability skills, and the specific skills and safety requirements necessary for effective work in this environment.

Other Hours: 144.000

SBM1000 Small Business Management 3.000 credits

Students will learn that growing a business involves shifting from the search for and validation of a business model to executing and scaling the business model. The skill sets are different and more formal management is often required. The course reviews the context and complexity of scaling a business and executing a business model.

Lecture Hours: 48.000

Course User Fee: \$90.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

LDE2310 Irrigation Design & Maintenance 3.000 credits

The students will learn the design, maintenance, and installation of nursery and landscape irrigation systems. All types of nursery systems will be covered including field, shade house, and greenhouse. Both sprinkle and low volume drip systems will be surveyed for appropriateness in nursery and landscape uses. Includes occasional weekend hands-on activities.

Lecture Hours: 48.000

Program Description

Marketing

Associate in Science | Code: 25047 | 60 credits

The Associate in Science (AS) in Marketing is designed mainly for students who intend to seek immediate employment in the fields of marketing, international business and trade, or real estate; also for those desiring to work in a non-profit institution and those presently employed in marketing but seeking advancement. The Associate in Arts degree is also available to the student planning to transfer to a senior institution after graduating from Miami Dade College. Consult an advisor about additional courses for such plans.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 15.0

ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MAC 1105	College Algebra	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 2

Units: 15.0

<i>Accounting Selection</i>		4.0
		4.0
ACG 2011	Principles of Accounting II	3.0
ACG 2011L	Principles of Accounting 2 Lab	1.0
OR		
		4.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
ECO 2023		Principles of Economics (Micro)
Program Electives ¹		8.0
Computer Competency Test*		

¹Select 8 credits from the following course options: ACG*, BAN*, BRC*, BUL*, CGS*, CIS*, COP*, ECO*, ENC*, FIN*, GEB*, LIS*, MAC, MAN*, MAR*, MKA*, MNA*, MTB*, OST*, PUR*, QMB*, REE*, STA 2023, TRA*

*Computer Competency Test (CCT) ---OR--- CGS 1060C - Introduction to Computer Technology & Applications.
Departmental Recommendation: Computer Competency Test (CCT)

Semester 3

Units: 12.0

ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
Program Electives ¹		8.0

¹Select 8 credits from the following course options: ACG*, BAN*, BRC*, BUL*, CGS*, CIS*, COP*, ECO*, ENC*, FIN*, GEB*, LIS*, MAC, MAN*, MAR*, MKA*, MNA*, MTB*, OST*, PUR*, QMB*, REE*, STA 2023, TRA*

Semester 4

Units: 18.0

<i>Marketing Specialization</i>		18.0
<i>TRACK 1 – DIGITAL MARKETING</i>		18.0
MAR 1011	Principles of Marketing	3.0
MAR 1720	Marketing in a Digital World	3.0
MAR 2101	Social Media Marketing	3.0
MAR 2703	Marketing Content, Branding and Strategy	3.0
MAR 2704	Marketing Web Analytics	3.0
MAR 2952	Digital Marketing Capstone	3.0
OR		
<i>TRACK 2 – SALES AND CUSTOMER SERVICE</i>		18.0
MAR 1440	Fundamentals of Negotiations	3.0
MAR 1502	Sales and Consumer Behavior	3.0
MAR 2419	Technology in Sales	3.0
MKA 1022	Relationship Selling	3.0
MKA 1160	Customer Relationship Management	3.0
MKA 2024	Organizational Sales Management	3.0
OR		
<i>TRACK 3 – MARKETING¹</i>		15.0
OST 2335	Business Writing	3.0

¹Select 15 credits from the following course option: MAR*, MKA*

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method. Lecture Hours: 48.000
Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations. Lecture Hours: 48.000
Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues. Lecture Hours: 48.000
Course User Fee: \$10.00

**EVR1001 Introduction to Environmental Science
3.000 credits**

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ACG2011 Principles of Accounting II 3.000 credits

This is the second part of the introductory financial accounting series, which includes ACG2001 and ACG2011, with emphasis on accounting owner's equity and corporate financial statements. Other topics include receivables, fixed assets, intangible assets, current and long-term liabilities. Co-requisite: ACG2011L

Lecture Hours: 48.000

ACG2011L Principles of Accounting 2 Lab 1.000 credits

Students will use personal computer software and financial accounting applications to analyze accounting records, prepare financial statements, and compile EXCEL spreadsheets. This course will reinforce, with tutorial help and problem-solving, the concepts needed to achieve the objectives of ACG2011. Corequisite: ACG2011.

Lab Hours: 32.000

Corequisite: ACG2001

Course User Fee: \$22.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033 Any Mathematics level higher than MAT 1033 will satisfy this requirement.

MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR2101 Social Media Marketing 3.000 credits

This course addresses the changes in marketing as a result of social media. The student will explore in-depth the role of social media in marketing. The student will focus on developing an effective marketing plan utilizing social media, with an emphasis on the importance of building an online customer community and developing customer engagement and loyalty.

Lecture Hours: 48.000

MAR2703 Marketing Content, Branding and Strategy 3.000 credits

This course provides an introduction into branding and content strategy. Students will explore concepts, such as: integrated marketing communications, consumer psychology, buying motivations, appropriate content, branding, and marketing strategy as it relates to digital marketing. Students will have the opportunity to practice their writing and communications skills in the development of digital marketing content.

Lecture Hours: 48.000

Course User Fee: \$57.00

MAR2704 Marketing Web Analytics 3.000 credits

This course helps students to develop a deep understanding of the digital analytics landscape. The student will learn how to identify which metrics are best to measure web, mobile, social and marketing channels, and how organizations use web analytics to obtain higher profits, improved customer relationships, and measurable value. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$57.00

MAR2952 Digital Marketing Capstone 3.000 credits

This course provides a real-world approach to designing, implementing, managing and analyzing a digital marketing campaign or strategy. The student will engage in a comprehensive field experience designed to gain insights into the creative processes and demonstrate competence by applying the knowledge and concepts of marketing.

Lecture Hours: 48.000

MAR1440 Fundamentals of Negotiations 3.000 credits

This course introduces the negotiation process and identifying consumer motivations. Students will analyze the various positions of negotiations from gaining leverage to making considerations. Students will examine a long-term sales strategy and how a balanced negotiation plan impacts it. (3 hr. lecture)

Lecture Hours: 48.000, Lab Hours: 5.000

Course User Fee: \$47.44

MAR1502 Sales and Consumer Behavior 3.000 credits

This course introduces the process consumers use in making purchase decisions. Students will analyze how developing a sales personality factors into producing prospects, leads, and ultimately generating sales. Students will identify how both buyer and seller behaviors interact in the sales process. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$47.44

MAR2419 Technology in Sales 3.000 credits

This course introduces the various technological tools used in the sales process. Students will explore sales management software, data management, artificial intelligence sales technology, and automated sales functions. Students will analyze the benefits of utilizing technology and the competitive advantage it provides to a sales force. Prerequisites: MAR1440, MKA1160, MAR1502, MKA1022, MKA2024. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MKA1160

Course User Fee: \$47.44

MKA1022 Relationship Selling 3.000 credits

This course introduces the role that relationship building plays in sales. Students will explore the communication process and the importance of interpersonal skills in developing a relationship for an effective sales strategy. Students will develop a sales presentation based around consumer needs.

Lecture Hours: 48.000

Course User Fee: \$47.44

MKA1160 Customer Relationship Management 3.000 credits

This course provides an introduction into Customer Relationship Management (CRM). Students will be exposed to concepts ranging from identifying customer needs, communication channels, customer acquisition cost, demographics and target marketing. Students will learn the role CRM plays in a sales strategy. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$47.44

MKA2024 Organizational Sales Management 3.000 credits

This course introduces the role of a sales manager and sales team within an organization. Students will examine the management process of recruiting, leading, and setting sales targets and quotas for a sales team. Students will explore sales analytics that measure the effectiveness of an organizational sales strategy.

Lecture Hours: 48.000

Course User Fee: \$47.44

OST2335 Business Writing 3.000 credits

Covers the procedures for writing effective business letters and memorandums, a review of grammar, and the proper format of today's business correspondence. Students learn how to prepare inquiry letters, direct and indirect response letters, application letters and resumes, and short reports.

Lecture Hours: 48.000

Course User Fee: \$15.00

Program Description

Music Business - Management and Marketing

Associate in Science | Code: 25019 | 64 credits

The Associate of Science in Music Business with a concentration in Management and Marketing is tailored for students aspiring to build careers in the music and entertainment industries. This program blends core business skills with specialized knowledge and practices unique to the music industry. Graduates will gain the expertise needed to navigate key aspects of the industry, including artist management, event production, music publishing, licensing, administration, marketing, and entrepreneurship. Through hands-on learning and industry-focused coursework, students will be equipped to succeed as professionals in this dynamic field.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
GEB 1011	Principles of Business	3.0
MUM 2700	Music Business 1	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) --or-- CGS 1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 12.0
MGF 1130	Mathematical Thinking	3.0
MAN 2021	Principles of Management	3.0
MAR 1011	Principles of Marketing	3.0
MUM 2702	Music Business 2	3.0
Semester 3		Units: 9.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0

OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
<i>Program Electives</i>		3.0
Private Music Lessons: ¹⁾		
OR		
Television, Film Arts, and Graphic Design ²⁾		
OR		
Music Production/Performance/Theory ³⁾		
OR		
Business, Communications, and Technology ⁴⁾		

¹⁾MVB*, MVJ*, MVK*, MVO*, MVP*, MVS*, MVV*, MVW*, MVK 1111

²⁾FIL*, GRA*, RTV*

³⁾MUM*, MUN*, MUT*, MUC 2601, MUH 2111, MUH 2112, MUL 2380, MUT 1001, MUT 1003, MUS 2304

⁴⁾BUL 2242, CGS 1060C, GEB 2350, MAR*, PUR 2003, SBM 1000, SPC 2608

Semester 4

Units: 13.0

ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
MAR 1720	Marketing in a Digital World	3.0
MUM 2703	Music Business 3	3.0
		3.0
Private Music Lessons ¹⁾		
OR		
Television, Film Arts, and Graphic Design ²⁾		
OR		
Music Production/Performance/Theory ³⁾		
OR		
Business, Communications, and Technology ⁴⁾		

¹⁾MVB*, MVJ*, MVK*, MVK 1111, MVO*, MVP*, MVS*, MVV*, MVW*

²⁾FIL*, GRA*, RTV*

³⁾MUC 2601, MUH 2111, MUH 2112, MUL 2380, MUM*, MUN*, MUS 2304, MUT*, MUT 1001, MUT 1003,

⁴⁾BUL 2242, CGS 1060C, GEB 2350, MAR*, PUR 2003, SBM 1000, SPC 2608

Semester 5

Units: 12.0

BUL 2241	Business Law 1	3.0
MAR 2101	Social Media Marketing	3.0
MUM 2704	Music Business 4	3.0
<i>Program Electives</i>		3.0

Private Music Lessons¹⁾

OR

Television, Film Arts, and Graphic Design²⁾

OR

Music Production/Performance/Theory³⁾

OR

Business, Communications, and Technology⁴⁾

¹⁾MVB*, MVJ*, MVK*, MVK 1111, MVO*, MVP*, MVS*, MVV*, MVW*

²⁾FIL*, GRA*, RTV*

³⁾MUC 2601, MUH 2111, MUH 2112, MUL 2380, MUM*, MUN*, MUS 2304, MUT*, MUT 1001, MUT 1003,

⁴⁾BUL 2242, CGS 1060C, GEB 2350, MAR*, PUR 2003, SBM 1000, SPC 2608

Semester 6

Units: 6.0

MAR 2704	Marketing Web Analytics	3.0
MUM 1949	Music Industry Internship	3.0

Total: 64.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MUM2700 Music Business 1 3.000 credits

This course is an introduction to the foundational aspects of the music industry, focusing on the essential principles and applications of music copyright law, contracts, publishing, licensing, and management. Students will explore the roles of agencies and organizations in the industry, along with the processes of record production, marketing and distribution. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$20.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MUM2702 Music Business 2 3.000 credits

This course provides an in-depth exploration of entrepreneurship, employability, and management principles within key areas of the music industry, including music equipment sales, concert promotion, independent record labels, record production, music licensing, and publishing. Students will gain an understanding of the business models and strategic approaches that drive success in each sector, with an emphasis on career skill development such as business planning, marketing, financial management, and contract negotiation. Prerequisite MUM 2700. (3 hr. lecture).

Lecture Hours: 48.000

Prerequisite: MUM2700

Course User Fee: \$20.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

MUM2703 Music Business 3 3.000 credits

This course provides hands-on training in essential computer applications and online tools relevant to the music industry. Students will learn to use digital tools to create professional promotional materials, presentations, and documents typically used in music industry transactions. (6 hr. lab)

Lab Hours: 96.000

Course User Fee: \$25.00

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAR2101 Social Media Marketing 3.000 credits

This course addresses the changes in marketing as a result of social media. The student will explore in-depth the role of social media in marketing. The student will focus on developing an effective marketing plan utilizing social media, with an emphasis on the importance of building an online customer community and developing customer engagement and loyalty.

Lecture Hours: 48.000

MUM2704 Music Business 4 3.000 credits

This course provides hands-on experience with cross-platform software applications and online resources used within the Music Business environment. Students will create their own website for marketing their own Music Business enterprise.

Prerequisite: MUM 2703. (6 hr. lab)

Lab Hours: 96.000

Prerequisite: MUM2703

Course User Fee: \$25.00

MAR2704 Marketing Web Analytics 3.000 credits

This course helps students to develop a deep understanding of the digital analytics landscape. The student will learn how to identify which metrics are best to measure web, mobile, social and marketing channels, and how organizations use web analytics to obtain higher profits, improved customer relationships, and measurable value. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$57.00

MUM1949 Music Industry Internship 3.000 credits

This course is designed to provide students with practical, hands-on experience in their chosen field of study—such as Music Business, Music Production, or a related field—through on-the-job training and faculty guidance. Students are evaluated on the basis of documented learning outcomes as reported by both the student and the employer. Prerequisite: MUM2700, MUM 2702, and MUM 2703; or department approval.

Lecture Hours: 48.000

Prerequisite: MUM2700 AND MUM2702 AND MUM2703

Program Description

Music Business - Performance and Production

Associate in Science | Code: 25044 | 64 credits

The Associate of Science (A.S.) in Music Business - Performance and Production track offers a comprehensive program for students interested in performing, producing, and composing music. This track is designed to develop a well-rounded understanding of both musical artistry and the business aspects of the music and entertainment industries. Students in this program engage in coursework that includes business essentials, music business practices, music production, performance training, and music theory. The curriculum equips students with the skills needed to manage their own careers as performers and producers while building a strong foundation in the professional and creative aspects of the industry.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
MUM 2623C	MIDI Electronic Music 1	3.0
MUM 2700	Music Business 1	3.0
MVK 1111	Class Piano 1	1.0

Semester 2

Units: 14.0

MGF 1130	Mathematical Thinking	3.0
GEB 1011	Principles of Business	3.0

MUM 2702	Music Business 2	3.0
MUT 1111	Theory 1	3.0
MUT 1241	Sightsinging & Ear Training 1 Year	1.0
MVK 1112	Class Piano 2	1.0

Semester 3

Units: 9.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

<i>Program Elective</i>		6.0
Private Music Lessons ¹⁾		
OR		
Television, Film Arts, Graphic Design, and Marketing ²⁾		
OR		
Music Production/Performance/Theory ³⁾		

Computer Competenc^{*}

¹⁾MVB*, MVJ*, MVK*, MVO*, MVP*, MVS*, MVV*, MVW*

²⁾FIL*, GRA*, MAR*, RTV*

³⁾MUM*, MUN*, MUT*, MUC 2601, MUS 2304

*Computer Competency Test (CCT) --or-- CGS 1060C - Introduction to Computer Technology and Applications.
Departmental recommendation:

Semester 4

Units: 14.0

MAR 1011	Principles of Marketing	3.0
MUM 2600	Sound Recording 1	3.0
MUM 2600L	Sound Recording 1 Lab	1.0
MUM 2703	Music Business 3	3.0
<i>Program Elective</i>		4.0

Private Music Lessons ¹⁾		
OR		
Television, Film Arts, Graphic Design, and Marketing ²⁾		
OR		
Music Production/Performance/Theory ³⁾		

¹⁾MVB*, MVJ*, MVK*, MVO*, MVP*, MVS*, MVV*, MVW*

²⁾FIL*, GRA*, MAR*, RTV*

³⁾MUM*, MUN*, MUT*, MUC 2601, MUS 2304

Semester 5

Units: 14.0

<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0

OR		
EVR 1001	Introduction to Environmental Science	3.0
MUM 1949	Music Industry Internship	3.0
MUM 2704	Music Business 4	3.0
Music Ensemble*		3.0
<i>Elective</i>		2.0
Private Music Lessons ¹⁾		
Television, Film Arts, Graphic Design, and Marketing ²⁾		
Music Production/Performance/Theory ³⁾		

Select 3 credits from the following course options: MUM 2023 - Commercial Music Ensemble or MUN (Music Ensemble)

¹⁾MVB*, MVJ*, MVK*, MVO*, MVP*, MVS*, MVV*, MVW*

²⁾FIL*, GRA*, MAR*, RTV*

³⁾MUM*, MUN*, MUT* MUC 2601, MUS 2304

Total: 64.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

MUM2623C MIDI Electronic Music 1 3.000 credits

This course is designed to acquaint music students with basic applications of Musical Instrument Digital Interface (MIDI) for the purpose of composition and performance and learning pre-production concepts with multi-track recording studio. Emphasis will be placed on keyboards, outboard gear, drum machines, and computer-assisted operations. Special fee.

Lecture Hours: 32.000, Lab Hours: 16.000 - 32.000

Course User Fee: \$10.00

MUM2700 Music Business 1 3.000 credits

This course is an introduction to the foundational aspects of the music industry, focusing on the essential principles and applications of music copyright law, contracts, publishing, licensing, and management. Students will explore the roles of agencies and organizations in the industry, along with the processes of record production, marketing and distribution. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$20.00

MVK1111 Class Piano 1 1.000 credits

The secondary area of piano with emphasis on sight-reading, melody harmonization, and ensemble playing. Required of all music students except piano majors. May be repeated for credit.

Lab Hours: 32.000

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MUM2702 Music Business 2 3.000 credits

This course provides an in-depth exploration of entrepreneurship, employability, and management principles within key areas of the music industry, including music equipment sales, concert promotion, independent record labels, record production, music licensing, and publishing. Students will gain an understanding of the business models and strategic approaches that drive success in each sector, with an emphasis on career skill development such as business planning, marketing, financial management, and contract negotiation. Prerequisite MUM 2700. (3 hr. lecture).

Lecture Hours: 48.000

Prerequisite: MUM2700

Course User Fee: \$20.00

MUT1111 Theory 1 3.000 credits

The techniques of writing four-part chord progressions using root position and inversions of the primary and secondary triads and the dominant and supertonic seventh; also, non-harmonic tones, melodic writing, and an introduction modulation.

Lecture Hours: 48.000

Prerequisite: MUT1001or passing score on departmental placement exam., Corequisite: MUT1241

Course User Fee: \$15.00

MUT1241 Sightsinging & Ear Training 1 Year 1.000 credits

The development of aural skill by means of rhythmic and melodic dictation and sightsinging.

Lab Hours: 32.000

Corequisite: MUT1111, MUT1112

Course User Fee: \$10.00

MVK1112 Class Piano 2 1.000 credits

A continuation of MVK 1111.

Lab Hours: 32.000

Prerequisite: MVK1111or placement by exam.

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MUM2600 Sound Recording 1 3.000 credits

An introduction to techniques, practices and procedures in making eight-track recordings. The student will gain experience with acoustical balancing, editing and over-dubbing in a wide variety of sound situations.

Lecture Hours: 48.000

Corequisite: MUM2600L

MUM2600L Sound Recording 1 Lab 1.000 credits

Participation in MUM 2600L offers students directed "hands on" experience coinciding with lectures in MUM 2600.

Lab Hours: 32.000

Corequisite: MUM2600

Course User Fee: \$15.00

MUM2703 Music Business 3 3.000 credits

This course provides hands-on training in essential computer applications and online tools relevant to the music industry. Students will learn to use digital tools to create professional promotional materials, presentations, and documents typically used in music industry transactions. (6 hr. lab)

Lab Hours: 96.000

Course User Fee: \$25.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

**EVR1001 Introduction to Environmental Science
3.000 credits**

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

**MUM1949 Music Industry Internship 3.000
credits**

This course is designed to provide students with practical, hands-on experience in their chosen field of study—such as Music Business, Music Production, or a related field—through on-the-job training and faculty guidance. Students are evaluated on the basis of documented learning outcomes as reported by both the student and the employer. Prerequisite: MUM2700, MUM 2702, and MUM 2703; or department approval.

Lecture Hours: 48.000

Prerequisite: MUM2700 AND MUM2702 AND MUM2703

MUM2704 Music Business 4 3.000 credits

This course provides hands-on experience with cross-platform software applications and online resources used within the Music Business environment. Students will create their own website for marketing their own Music Business enterprise.

Prerequisite: MUM 2703. (6 hr. lab)

Lab Hours: 96.000

Prerequisite: MUM2703

Course User Fee: \$25.00

Program Description

Networking Services Technology - Enterprise Cloud Computing

Associate in Science | Code: 25077 | 60 credits

The Networking Services Technology - Enterprise Cloud Computing program is designed to provide an opportunity to establish foundation in architect scalable, highly available application solutions that leverage cloud computing services, utilizing best practices focusing on cloud security, cost, and reliability. Graduates will utilize core design patterns and infrastructure expertise to implement solutions to deploy and maintain workloads and applications.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 14.0**

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
CTS 1134	Networking Technologies	4.0

Semester 2**Units: 12.0**

CGS 1540C	Database Concepts Design	4.0
CTS 1120	Cybersecurity Fundamentals	4.0
CTS 1145	Cloud Essentials	4.0

Semester 3**Units: 7.0**

Social Science	3.0
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AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
CTS 1111	Linux +	4.0

Semester 4

Units: 15.0

<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
CTS 2303	Windows Server Administration	4.0
CTS 2375C	Cloud Infrastructure and Services	4.0
CTS 2960	Cloud Computing Capstone	4.0

Semester 5

Units: 12.0

<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1020	General Education Physics	3.0
CTS 2192C	Microsoft Azure Administration	4.0
<i>Program Elective</i>		5.0
CIS 1949	CO-Op Work Experience 1: CIS	1.0
OR		
CIS 2949	CO-OP WRK EXP 2: CIS	1.0
OR		
COP 1047C	Introduction to Python Programming	4.0
OR		
COP 2800	Java Programming	4.0
OR		
CTS 2143C	Server Administration	4.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement. Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional. Lecture Hours: 48.000, Lab Hours: 32.000
Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab). Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CTS1111 Linux + 4.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based workstations and servers. Students learn how to plan, install, maintain, document, and troubleshoot GNU/Linux operating system services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1060C

Course User Fee: \$63.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CTS2303 Windows Server Administration 4.000 credits

This course is intended for students preparing for IT careers as network support specialists as well as server and cloud administrators. Students will learn the fundamental administration skills and knowledge required to deploy and support Windows Server core services.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CGS1060 and CTS1134 or equivalent knowledge

Course User Fee: \$63.00

CTS2375C Cloud Infrastructure and Services 4.000 credits

This course helps students develop technical expertise in Cloud computing and prepares them for industry certification. Students will learn the essentials of Cloud computing, business security and compliance considerations, migrating to the Cloud, architecting a Cloud server, and how to troubleshoot Cloud services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2145

Course User Fee: \$63.00

CTS2960 Cloud Computing Capstone 4.000 credits

This course requires students to demonstrate their competence to analyze, design, develop, and test a cloud based complex system. Each student will create and present a cloud based solution proposal that includes: design documentation, implementation plan, cloud resources required, projected cost analysis, basic security plan and project test plan to create an operational cloud based system solution. Must be taken during the last semester before graduation and with a departmental permission. Prerequisite: Departmental Approval. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: Departmental Approval

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

CTS2192C Microsoft Azure Administration 4.000 credits

This course is intended for students preparing for IT careers as cloud administrators, as well as candidates for industry certification. Students will learn the administration skills and knowledge required to implement, manage, and monitor identity, governance, storage, and compute virtual networks in the Microsoft Azure cloud environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS1145 or equivalent knowledge in Microsoft Azure Fundamentals

Course User Fee: \$63.00

CIS1949 CO-Op Work Experience 1: CIS 1.000 - 4.000 credits

This course is designed as a work experience for students majoring in computer information systems programs. Students will learn to apply the skills and knowledge that they have acquired through their program of study in a real work environment. Prerequisite: Successful completion of required program course work. Department approval required.

Lecture Hours: 16.000 - 64.000

Prerequisite: Successful completion of required program course work. Department approval required.

CIS2949 CO-OP WRK EXP 2: CIS 1.000 - 4.000 credits

This course is designed as a second-level capstone for students majoring in computer information systems programs. Students apply advanced skills and knowledge that they have acquired through their first capstone course in a real work environment. Prerequisite: Successful completion of required program course work. Department approval required.

Lecture Hours: 16.000 - 64.000

Prerequisite: CIS1949

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

CTS2143C Server Administration 4.000 credits

This course is designed for students preparing for IT careers as server administrators, and focuses on server hardware and operating systems, system management and administration, security and disaster recovery, infrastructure and network services, data storage, and troubleshooting. This course is intended to prepare students for the CompTIA Server+ certification.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Server operating systems experience.

Course User Fee: \$63.00

Program Description

Networking Services Technology - Network Infrastructure

Associate in Science | Code: 25062 | 60 credits

The Networking Services Technology program provides an opportunity to establish a foundation in the field of network design and administration for employment in commercial, industrial and government institutions. The Network Infrastructure concentration offers the training needed to connect and secure multiple computing systems and software platforms. Students are additionally eligible to sit for the Cisco Certified Networking Associate (CCNA) certification exam. Graduates are prepared for positions as information technology specialists, help desk specialists, network specialists, entry level security specialists and network systems analysts. Note: There is only one A.S. program in Networking Services Technology with three (3) track specializations—Network Infrastructure, Network Security, and Enterprise Cloud Computing. Students may select from one of the three options, but the A.S. in Networking Services Technology will be awarded to the student only once.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 14.0**

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
CTS 1134	Networking Technologies	4.0

Semester 2**Units: 12.0**

CTS 1120	Cybersecurity Fundamentals	4.0
CTS 1650	CCNA 1: Cisco Fundamentals	4.0
CTS 1651	CCNA 2: Routing and Switching	4.0

Semester 3**Units: 10.0**

Humanities	3.0
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HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
CTS 1111	Linux +	4.0

Semester 4

Units: 12.0

CTS 2303	Windows Server Administration	4.0
CTS 2652	CCNA 3: Advanced Routing and Switching	4.0
<i>Electives*</i>		4.0
COP 1047C	Introduction to Python Programming	4.0
OR		
COP 2800	Java Programming	4.0
OR		
CTS 1145	Cloud Essentials	4.0
OR		
CTS 2375C	Cloud Infrastructure and Services	4.0

*Note: Listed courses are recommended options within the Elective block. While not required, CTS 1145 - Cloud Essentials is highly recommended in preparation for CTS 2143C and CTS 2192C.

Semester 5

Units: 12.0

<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1020	General Education Physics	3.0
CTS 2143C	Server Administration	4.0
CTS 2192C	Microsoft Azure Administration	4.0
<i>Elective</i>		1.0
CIS 1949	CO-Op Work Experience 1: CIS	1.0
OR		
CIS 2900	Directed Study in IT	1.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS1650 CCNA 1: Cisco Fundamentals 4.000 credits

This is the first course of the four-course Cisco curriculum that will prepare students for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn networking concepts and practices, network terminology and protocols, the OSI reference model, cabling, cabling tools, routers, router and switch configurations, LAN/WAN topologies, IP addressing, and network standards. Lecture Hours: 48.000, Lab Hours: 32.000
Recommended Preparation: Knowledge of computer operating systems and hardware is required.
Course User Fee: \$63.00

CTS1651 CCNA 2: Routing and Switching 4.000 credits

This is the second course of the four-course Cisco curriculum that will prepare the student for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn the architecture, components and operation of routers and switches, LAN (Local Area Networks) switch protocols and operations, VLANs (Virtual Local Area Networks), network routing protocols and concepts, static and dynamic routing, router and switch configuration and troubleshooting, and IP Address services. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS1650
Course User Fee: \$63.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

CTS1111 Linux + 4.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based workstations and servers. Students learn how to plan, install, maintain, document, and troubleshoot GNU/Linux operating system services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1060C

Course User Fee: \$63.00

CTS2303 Windows Server Administration 4.000 credits

This course is intended for students preparing for IT careers as network support specialists as well as server and cloud administrators. Students will learn the fundamental administration skills and knowledge required to deploy and support Windows Server core services.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CGS1060 and CTS1134 or equivalent knowledge

Course User Fee: \$63.00

CTS2652 CCNA 3: Advanced Routing and Switching 4.000 credits

This is the third course of the four-course Cisco curriculum that will prepare the student for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn how to create virtual local area networks (VLANs), configure inter VLAN routing, and implement wireless network access and VLAN security.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1651

Course User Fee: \$63.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2375C Cloud Infrastructure and Services 4.000 credits

This course helps students develop technical expertise in Cloud computing and prepares them for industry certification. Students will learn the essentials of Cloud computing, business security and compliance considerations, migrating to the Cloud, architecting a Cloud server, and how to troubleshoot Cloud services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2145

Course User Fee: \$63.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

CTS2143C Server Administration 4.000 credits

This course is designed for students preparing for IT careers as server administrators, and focuses on server hardware and operating systems, system management and administration, security and disaster recovery, infrastructure and network services, data storage, and troubleshooting. This course is intended to prepare students for the CompTIA Server+ certification.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Server operating systems experience.

Course User Fee: \$63.00

CTS2192C Microsoft Azure Administration 4.000 credits

This course is intended for students preparing for IT careers as cloud administrators, as well as candidates for industry certification. Students will learn the administration skills and knowledge required to implement, manage, and monitor identity, governance, storage, and compute virtual networks in the Microsoft Azure cloud environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS1145 or equivalent knowledge in Microsoft Azure Fundamentals

Course User Fee: \$63.00

CIS1949 CO-Op Work Experience 1: CIS 1.000 - 4.000 credits

This course is designed as a work experience for students majoring in computer information systems programs. Students will learn to apply the skills and knowledge that they have acquired through their program of study in a real work environment. Prerequisite: Successful completion of required program course work. Department approval required.

Lecture Hours: 16.000 - 64.000

Prerequisite: Successful completion of required program course work. Department approval required.

CIS2900 Directed Study in IT 1.000 credits

This course, for majors in Information Technology, provides students the opportunity to pursue projects under faculty supervision through self-directed research, experiences, and presentations.

Prerequisite: Successful completion of required program course work. Department approval required.

Lab Hours: 32.000

Recommended Preparation: Successful completion of required program course work. Department approval required

Program Description

Nuclear Medicine Technology

Associate in Science | Code: 23068 | 75 credits

The Nuclear Medicine Technology program is designed to prepare selected students to qualify as nuclear medicine technologists in hospitals, outpatient diagnostic imaging centers, and private physician offices. These contributing members of the allied health team prepare and administer the tracer radio pharmaceuticals to the patients and record the image using computerized detection systems for medical diagnosis. Successful completion of this two-year program qualifies graduates to apply for the American Registry for Radiologic Technologists examination in Nuclear Medicine and/or the Nuclear Medicine Technology Certification Board Examination leading to certification as a registered nuclear medicine technologist and gainful employment as such.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
MAC 1105	College Algebra	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 10.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
PHY 1004	Physics With Applications 1	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		

POS 2041	American Federal Government	3.0
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Semester 3 (In Program Milestone)		Units: 12.0
NMT 1002L	Introduction to Nuclear Medicine Laboratory	2.0
NMT 1312C	Radiation Protection and Radiobiology	2.0
NMT 1705C	Nuclear Medicine Pre-Clinical	2.0
NMT 2613	Nuclear Medicine Physics and Mathematical Applications	3.0
Elective*		3.0

*Select any appropriate transferable (Credit Type 1 or 2) courses (Check with Advisor).

Semester 4 (In Program Milestone)		Units: 12.0
PHI 2010	Introduction to Philosophy	3.0
NMT 1713C	Nuclear Medicine Methodology 1	2.0
NMT 2102	Nuclear Medicine Administration	1.0
NMT 2534C	Instrumentation, Quality Assurance and Control	3.0
NMT 2804C	Nuclear Medicine Clinic Practice & Conference 1	3.0

Semester 5 (In Program Milestone)		Units: 10.0
NMT 2130C	Nuclear Medicine Radiopharmacy	3.0
NMT 2723C	Nuclear Medicine Methodology 2	2.0
NMT 2814C	Nuclear Medicine Clinic Practice & Conference 2	5.0

Semester 6 (In Program Milestone)		Units: 9.0
NMT 2733C	Nuclear Medicine Methodology 3	2.0
NMT 2779C	Multi-Modalities and Cross-Sectional Anatomy	2.0
NMT 2824C	Nuclear Medicine Clinical Education 3	5.0

Semester 7 (In Program Milestone)		Units: 8.0
NMT 2834C	Nuclear Medicine Clinical Education 4	5.0
NMT 2932C	Nuclear Medicine Seminar	3.0

Total: 75.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions. Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite: MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

PHY1004 Physics With Applications 1 3.000 credits

Emphasizes the basic concepts and principles and their practical applications. Designed specifically for students in technical studies and for others wishing to strengthen their physics background before taking advanced courses.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: PHY1004L

Course User Fee: \$10.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

NMT1002L Introduction to Nuclear Medicine Laboratory 2.000 credits

Provides an overview of the field of nuclear medicine. Students will focus on medical terminology, the history of nuclear medicine, basic concepts of radiochemistry, radiation safety, medical law, patient care and positioning, and hospital administration. Field trips to nuclear medicine training facilities are included.

Prerequisites: CHM 1033, 1033L; Corequisites: NMT 1312C, NMT 1705C, NMT 2613. (4 hr. lab)

Lab Hours: 64.000

Prerequisite: CHM1033 AND CHM1033L AND Corequisite: NMT1312 AND NMT2613

NMT1312C Radiation Protection and Radiobiology 2.000 credits

This course covers all local, state and federal regulations related to Nuclear Medicine. Students will learn the appropriate protection procedures to limit exposure, the performance of area surveys and wipe tests, the proper decontamination procedures, the disposal of radioactive waste procedures, and personnel monitoring of radiation exposure. Corequisites: NMT 1002L, 2613. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000

Corequisite: NMT1002L AND NMT2613

NMT1705C Nuclear Medicine Pre-Clinical 2.000 credits

This nuclear medicine technology course prepares students to attend to patients, and evaluate data from patient records, make dose calculations, prepare radio-pharmaceuticals, perform in-vivo and in-vitro diagnostic studies, and perform quality control procedures. Prerequisites: BSC 2085/L, BSC 2086/L, CHM 1033/L; Corequisites: NMT 1002L, NMT 1312C, NMT 2613. (1 hr. lecture, 3 hr. clinical)

Lecture Hours: 16.000, Clinical Hours: 48.000

Course User Fee: \$293.31

NMT2613 Nuclear Medicine Physics and Mathematical Applications 3.000 credits

Students will learn the basic concepts of atomic, nuclear and radiation physics with an emphasis on the interactions of radiation with matter. Alpha, beta, and gamma sources are explained as well mathematical principles associated with the types of radiation sources. Prerequisites: MAC 1105 and PHY 1020; Corequisites: NMT 1002L, NMT 1312C. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MAC1105 AND PHY1020 AND Corequisite: NMT1002L AND NMT1312

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

NMT1713C Nuclear Medicine Methodology 1 2.000 credits

This course will include the imaging parameters necessary to obtain images for the basic procedures performed in a Nuclear Medicine department. Students will learn about imaging procedures related to the following systems: skeletal, central nervous, cardiovascular genitourinary, respiratory and gastrointestinal. Instrumentation necessary to produce the required images as well as patient management during the procedures will be addressed. Prerequisites: BSC 2085, 2085L, 2086, 2086L, CHM1033, 1033L; Co-requisites: NMT 2102, NMT 2534C, NMT 2804C. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000

NMT2102 Nuclear Medicine Administration 1.000 credits

The student will learn the administrative duties required of a nuclear medicine technologist. Areas covered include patient scheduling, radioisotope ordering; scheduling and testing; communication; patient and clinician satisfaction. Prerequisites: NMT 1002L, NMT 1312C, NMT 1705C, NMT 2613; Corequisites: NMT 1713C, NMT 2534C, NMT 2804C. (1 hr. lecture)

Lecture Hours: 16.000

NMT2534C Instrumentation, Quality Assurance and Control 3.000 credits

This course will integrate and correlate the principles of electrical and nuclear physics associated with the operation, calibration, quality assurance and control testing of imaging systems and radiation detection devices employed in nuclear medicine. The student will learn the various types of devices used to provide information obtained from diagnostic images. Prerequisites: NMT 1002L, NMT 1312C, NMT 2613 and PHY1020; Corequisites: NMT 1713C, 2102, 2804C. (1 hr. lecture, 4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

NMT2804C Nuclear Medicine Clinic Practice & Conference 1 3.000 credits

This course will introduce the student to the fundamentals of clinical nuclear medicine primarily through hospital involvement. The student will learn practical experience in a Nuclear Medicine department by performing the principles taught in class. Prerequisites: NMT 1002L, NMT 1312C, NMT 1705C, NMT 2613; Corequisites: NMT 1713C, NMT 2102, NMT 2534C. (144 hrs. clinic)

Clinical Hours: 144.000

Prerequisite: NMT 1002L, NMT 1312C, NMT 1705C, NMT 2613, Corequisite: NMT 1713C; NMT 2102, NMT 2534C

Course User Fee: \$25.00

NMT2130C Nuclear Medicine Radiopharmacy 3.000 credits

Students will learn how to maintain radiopharmaceutical laboratory records and materials, obtain a generator eluate, prepare radiopharmaceuticals and perform quality control tests, as well as dispose of radioactive waste appropriately. The ordering of pharmaceuticals in appropriate dosage and effective time frames will also be included. Prerequisites: NMT 1002L, NMT 1312C, NMT 1705C, NMT 2613; Corequisites: NMT 2130C, NMT 2723C, NMT 2814C. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

NMT2723C Nuclear Medicine Methodology 2 2.000 credits

A continuation of Nuclear Medicine Procedures 1, students will learn the imaging parameters necessary to obtain images as well as the use of instrumentation necessary to produce the required images performed in a nuclear medicine department. Exposure to patient management during the procedures will also be addressed. Prerequisites: NMT 1713C, NMT 2102, NMT 2534C, NMT 2804C; Corequisite: NMT 2130C, NMT 2814C. (1 hr. lecture, 3 hr. clinical)

Lecture Hours: 16.000, Clinical Hours: 48.000

Prerequisite: NMT1713 AND NMT2804C, Corequisite: NMT2573 AND NMT2814C

NMT2814C Nuclear Medicine Clinic Practice & Conference 2 5.000 credits

This course is a continuation of NMT 2804C Clinic 1 and will provide the student the opportunity to participate in the fundamentals of clinical nuclear medicine in the hospital involvement. The student will learn practical experience in a Nuclear Medicine department by performing the principles taught in class.

Clinical Hours: 240.000

Prerequisite: NMT2804C, NMT 2130, NMT 2534C, NMT 2613, NMT 2804C

Course User Fee: \$25.00

NMT2733C Nuclear Medicine Methodology 3 2.000 credits

A continuation of Nuclear Medicine Procedures 2, students will learn the imaging parameters necessary to obtain images as well as the use of instrumentation necessary to produce the required images performed in a nuclear medicine department. Exposure to patient management during the procedures will also be addressed. Prerequisites: BSC 2085/L, BSC 2086/L, NMT 1713C, NMT 2723C; Corequisites: NMT 2779C, NMT 2824C. (1 hr. lecture, 3 hr. clinical)

Lecture Hours: 16.000, Clinical Hours: 48.000

Course User Fee: \$239.31

NMT2779C Multi-Modalities and Cross-Sectional Anatomy 2.000 credits

This course educates the student upon proper recognition and interpretation of cross-sectional anatomy. The student will also compare and analyze images from complementary modalities. It is crucial for the nuclear medicine technologist to understand three-dimensional imaging in order to enhance patient care and be an asset to the facility. Prerequisites: NMT 2130C, NMT 2723C, NMT 2814C; Corequisites: NMT 2733C, NMT 2824C. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: NMT 2130C, NMT 2723C, NMT 2814C, Corequisite: NMT 2733C, 2824C

Course User Fee: \$239.38

NMT2824C Nuclear Medicine Clinical Education 3 5.000 credits

This is the final course in the series of three clinical courses. Students will learn to apply all didactic competencies in the Nuclear Medicine department setting, as well as perform all procedures from the two Nuclear Medicine Procedures courses with minimal supervision. The ARRT Competency Requirements must be completed in this course. Prerequisites: NMT 2814C.

Clinical Hours: 240.000

Prerequisite: NMT2814C

Course User Fee: \$25.00

NMT2834C Nuclear Medicine Clinical Education 4 5.000 credits

This is the final course in the series of four clinical courses. Students will learn to apply all didactic competencies in the Nuclear Medicine department setting, as well as perform all procedures from the two Nuclear Medicine Procedures courses with minimal supervision. The ARRT Competency Requirements must be completed in this course.

Clinical Hours: 240.000

Prerequisite: NMT2824C

Course User Fee: \$239.31

NMT2932C Nuclear Medicine Seminar 3.000 credits

The student will learn to incorporate all theory related to the production of a nuclear medicine image. The student will also learn about radiation protection, instrumentation, physics, pharmacology, and Quality Assurance/Quality Control.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: NMT1312NMT 2733C, NMT 2779C, NMT 2824C, Corequisite: NMT2824C

Course User Fee: \$25.00

Program Description

Nursing - R.N. (Accelerated)

Associate in Science | Code: 23031 | 72 credits

The Accelerated Option in Nursing is designed to prepare the student with a baccalaureate degree in other disciplines for a career as a Registered Nurse (RN). The content and clinical experiences are designed to meet the learning and professional socialization needs of this special category of student. The program is accredited by the National League for Nursing Accrediting Commission (NLNAC) (61 Broadway, New York, NY 10006, ph. 212-363-5555, www.nlnac.org) and approved by the Florida Board of Nursing (FBON). Graduates are eligible to apply to write the National Council Licensing Examination for Registered Nurses (NCLEX-RN). Selection is based on scores on pre-admission testing and completion of required pre-requisite courses. See a School of Nursing

Information Booklet for more specific details about admission requirements. Pre-Admission tests – Applicants must take the reading portion of the CPT and score 78 or above; the Test of Adult Basic Education (TABE) and score 11 or above in reading, language and total math and the Nurse Entrance Test (NET). The NET is given only on the Medical Campus. Scores on the NET and TABE are used for selection.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
SPC 2608	Introduction to Public Speaking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 2		Units: 10.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
MAC 1105	College Algebra	3.0
PHI 2010	Introduction to Philosophy	3.0

Semester 3		Units: 7.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
MCB 2010	Microbiology	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4 (In Program Milestone)		Units: 18.0
NUR 1025	Fundamentals of Nursing	3.0
NUR 1025C	Fundamentals of Nursing Skills Lab	1.0
NUR 1025L	Fundamentals of Nursing Clinical Lab	2.0
NUR 1060C	Adult Health Assessment	3.0
NUR 1141	Nursing Math & Pharmacology	3.0
NUR 1211	Medical-Surgical Nursing	3.0
NUR 1211L	Medical Surgical Nursing Clinical	2.0
NUR 1214C	Medical Surgical Nursing Skills Lab	1.0

Semester 5 (In Program Milestone)		Units: 11.0
NUR 2220	Medical Surgical Nursing II	3.0
NUR 2220L	Medical Surgical Nursing II Clinical	1.0
NUR 2310	Pediatric Nursing	2.0
NUR 2310L	Pediatric Nursing Clinical Lab	1.0

NUR 2420	Obstetrical Nursing	2.0
NUR 2420L	Obstetrical Nursing Clinical Lab	1.0
NUR 2680L	Community Health Nursing Lab	1.0

Semester 6 (In Program Milestone)

Units: 13.0

NUR 2212	Medical Surgical Nursing III	3.0
NUR 2212L	Medical Surgical Nursing III Clinical Lab	2.0
NUR 2520	Psychiatric Nursing	2.0
NUR 2520L	Psychiatric Nursing Clinical Lab	1.0
NUR 2811C	Professional Nursing Leadership	4.0
NUR 2960	Senior Seminar	1.0

Total: 72.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions. Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite: MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L. Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

NUR1025 Fundamentals of Nursing 3.000 credits

This course introduces students to the profession of Nursing. The student will learn the roles basic to the practice of nursing, the nursing process, and how nurses are involved in health promoting activities to meet client needs. Pre-Req: Program Admission; Co-Req: NUR 1025C, 1025L, 1060C, and 1141.

Lecture Hours: 48.000

Course User Fee: \$113.00

NUR1025C Fundamentals of Nursing Skills Lab 1.000 credits

Students will learn of the opportunities for the explanation, demonstration, and practice of care provider activities essential to the basic practice of nursing. Learning experiences are provided in the skills laboratory. Pre-Req: Program Admission; Co-Req: NUR 1025, 1025L, 1060C, 1141

Lab Hours: 32.000

Course User Fee: \$105.00

NUR1025L Fundamentals of Nursing Clinical Lab 2.000 credits

Students will learn of the opportunities for the explanation, demonstration, and practice of care provider activities essential to the basic practice of nursing. Learning experiences are provided in the skills laboratory. Pre-Req: Program Admission; Corequisite: NUR 1025, 1025C, 1141, 1060C.

Clinical Hours: 96.000

Course User Fee: \$8.00

NUR1060C Adult Health Assessment 3.000 credits

This course is designed to provide students with the necessary skills to perform an in-depth nursing history and a complete physical examination on an adult client. The focus will be on clients with minimal or no alterations in their health state. Students will be introduced to and will demonstrate the techniques used in physical examination. Prerequisite: Admission to the School of Nursing; Corequisites: NUR 1025, 1025C, 1025L, 1141.

Lecture Hours: 32.000 - 33.000, Lab Hours: 32.000, Practicum Hours: 2.000

Prerequisite: Admission to the program

Course User Fee: \$109.00

NUR1141 Nursing Math & Pharmacology 3.000 credits

Students will learn about medications and their effects on different body systems. The conceptual and mathematical operations necessary for safe and effective administration of intravenous medications, preparing medications that come in powdered form and adjusting medication administration based on medical protocols are discussed. Concepts of medications including history, nomenclature, sources of drug information, federal drug laws and standards, medication classifications, pharmacokinetics, pharmacodynamics, variables affecting medication actions and effects, and methods of delivery will be addressed. Co-Req: NUR 1025, 1025C, 1025L, 1060C.

Lecture Hours: 48.000, Practicum Hours: 1.000

Course User Fee: \$75.00

NUR1211 Medical-Surgical Nursing 3.000 credits

This course provides an introduction to the adult nursing care. Students will learn the nurse's role in meeting the short and long term needs of the client and community through preventive, therapeutic and palliative care are discussed. Pre-Req: NUR 1025, 1025C, 1060C, 1142; Co-Req: NUR 1211L, 1214C.

Lecture Hours: 48.000

Course User Fee: \$121.00

NUR1211L Medical Surgical Nursing Clinical 2.000 credits

Students will learn how to apply concepts of adult health nursing. Experiences in both in-patient and community settings will be provided. Students are encouraged to participate in projects emphasizing preventive aspects of nursing care. Prerequisites: NUR 1025, 1025C, 1025L, 1060C, and 1141

Clinical Hours: 96.000

Course User Fee: \$87.00

NUR1214C Medical Surgical Nursing Skills Lab 1.000 credits

Students will learn the opportunities for the explanation, demonstration, and practice of skills related to adult health nursing. Learning experiences are provided in the School of Nursing Skills Laboratory. Pre-Req: NUR 1025, 1025C, 1025L, 1060C, 1142; CO-REQ: NUR 1211, 1211L.

Lecture Hours: 8.000, Lab Hours: 16.000

Course User Fee: \$114.00

NUR2220 Medical Surgical Nursing II 3.000 credits

This course provides students with the opportunity to apply concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health.

Lecture Hours: 48.000

Prerequisite: NUR1025 AND NUR1025C AND NUR1025L AND NUR1060C AND NUR1141 AND NUR1211 AND NUR1211L AND NUR1214C AND Corequisite: NUR2220L

NUR2220L Medical Surgical Nursing II Clinical 1.000 credits

This course provides students with the opportunity to apply concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health.

Clinical Hours: 48.000

Prerequisite: NUR1025 AND NUR1025C AND NUR1025L AND NUR1060C AND NUR1141 AND NUR1211 AND NUR1211L AND NUR1214C AND Corequisite: NUR2220

NUR2310 Pediatric Nursing 2.000 credits

This course provides a family centered approach to the nursing care of pediatric clients and their families. Students will learn the nurse's role in meeting the short and long term needs of the pediatric client, family, and community through preventative, therapeutic and palliative care, with recognition for the multicultural aspects of client needs. Prerequisites: 1211, 1211L, 1214C or 1002, 1002L; 1142; corequisites: NUR 2310L, 2420, 2420L, 2680L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$126.00

NUR2310L Pediatric Nursing Clinical Lab 1.000 credits

This course allows the student to apply the nursing process to the care of clients in selected pediatric clinical settings. Students will learn to observe cultural diversity and implement care to the pediatric client, family, and community through preventive, therapeutic and palliative measures. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2420, 2420L, 2520, 2520L, 2680L. Laboratory fee.

Clinical Hours: 48.000

Course User Fee: \$86.00

NUR2420 Obstetrical Nursing 2.000 credits

This course provides a family centered approach to the nursing care of obstetrical clients and their families. Students will learn to assess the pregnant client, to implement caring behaviors for the laboring client, educate the postpartum client, manage the care of the newborn and collaboration of care for the high risk client. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142. Corequisites: NUR 2310, 2310L, 2420L, 2680L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

NUR2420L Obstetrical Nursing Clinical Lab 1.000 credits

This course provides an introduction to obstetrical nursing clinical practice. Students will learn to apply the nursing process to the care of clients in selected obstetrical clinical settings. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2310L, 2420, 2680L. Laboratory fee.

Clinical Hours: 48.000

Course User Fee: \$96.00

NUR2680L Community Health Nursing Lab 1.000 credits

This laboratory course assists the students in applying knowledge of community health resources. Students will learn to manage community health resources to support the delivery of care to the childbearing/child-rearing families. Special emphasis is placed on the understanding of cultural influences on the health practices and beliefs within the family. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; co-requisites: NUR 2310, 2310L, 2420, 2420L.0L. Special fee. (48 hr. clinical)

Clinical Hours: 48.000

Course User Fee: \$81.00

NUR2212 Medical Surgical Nursing III 3.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1008, 1002L; NUR 1141, 2310, 2310L, 2420 2420L, 2680L

Lecture Hours: 48.000

Course User Fee: \$450.00

NUR2212L Medical Surgical Nursing III Clinical Lab 2.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1002, 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L

Clinical Hours: 96.000

Course User Fee: \$150.00

NUR2520 Psychiatric Nursing 2.000 credits

This course introduces students to the basic concepts of psychiatric nursing. Students will learn to provide care in in-patient and community settings, focusing on the nurse's role in meeting the needs of the patient, family, and the community. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2520L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$62.00

NUR2520L Psychiatric Nursing Clinical Lab 1.000 credits

This course provides the student opportunities to apply concepts of psychiatric nursing. Students will learn psychiatric procedures for both in-patient and community settings, focusing on the nurse's role on meeting the needs of the client, family, and community. Students actively participate in projects assisting clients in preventative care and maintenance of mental health. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142

Clinical Hours: 48.000

Course User Fee: \$98.00

NUR2811C Professional Nursing Leadership 4.000 credits

This course provides the student with the theoretical and clinical knowledge necessary for actualization of the role of the registered professional nurse, with emphasis on delegation and supervision.

Prerequisites: NUR 2212, 2212L

Lecture Hours: 16.000, Clinical Hours: 144.000

Course User Fee: \$175.00

NUR2960 Senior Seminar 1.000 credits

This course provides a forum for the discussion, and preparation for the NCLEX-RN exam. Explain applicant requirements for the state board of nursing (and licensure eligibility requirements) and examination registration.

Lecture Hours: 16.000

Prerequisite: NUR2811C

Program Description

Nursing - R.N. (Generic Full-Time)

Associate in Science | Code: 23028 | 72 credits

The Generic Nursing Option is designed to prepare students without previous health care education for careers as Registered Nurses. This program is accredited by the National League for Nursing Accrediting Commission (NLNAC), (61 Broadway, New York, NY 10006, phone (212) 363-5555, www.nlnac.org) and approved by the Florida Board of Nursing (FBON). Graduates are eligible to apply to write the National Council Licensing Examination for Registered Nurses (NCLEX-RN). The part-time track is designed for individuals who must work while they attend school. Selection is based on scores on pre-admission testing and completion of required pre-requisite courses. See a School of Nursing Information Booklet for more specific details about admission requirements. Pre-Admission tests – Applicants must take the Test of Adult Basic Education (TABE) and score 11 or above in reading,

language and total math and the Nurse Entrance Test (NET).The NET is given only on the Medical Campus. Scores on the NET and TABE are used for selection.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 2		Units: 10.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
MAC 1105	College Algebra	3.0
PHI 2010	Introduction to Philosophy	3.0

Semester 3		Units: 10.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
MCB 2010	Microbiology	3.0
SPC 2608	Introduction to Public Speaking	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4 (In Milestone)		Units: 12.0
NUR 1025	Fundamentals of Nursing	3.0
NUR 1025C	Fundamentals of Nursing Skills Lab	1.0
NUR 1025L	Fundamentals of Nursing Clinical Lab	2.0
NUR 1060C	Adult Health Assessment	3.0
NUR 1141	Nursing Math & Pharmacology	3.0

Semester 5 (In Milestone)		Units: 12.0
NUR 1211	Medical-Surgical Nursing	3.0
NUR 1211L	Medical Surgical Nursing Clinical	2.0
NUR 1214C	Medical Surgical Nursing Skills Lab	1.0
NUR 2310	Pediatric Nursing	2.0
NUR 2310L	Pediatric Nursing Clinical Lab	1.0
NUR 2420	Obstetrical Nursing	2.0
NUR 2420L	Obstetrical Nursing Clinical Lab	1.0

Semester 6 (In Milestone)		Units: 8.0
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NUR 2220	Medical Surgical Nursing II	3.0
NUR 2220L	Medical Surgical Nursing II Clinical	1.0
NUR 2520	Psychiatric Nursing	2.0
NUR 2520L	Psychiatric Nursing Clinical Lab	1.0
NUR 2680L	Community Health Nursing Lab	1.0

Semester 7 (In Milestone)

Units: 10.0

NUR 2212	Medical Surgical Nursing III	3.0
NUR 2212L	Medical Surgical Nursing III Clinical Lab	2.0
NUR 2811C	Professional Nursing Leadership	4.0
NUR 2960	Senior Seminar	1.0

Total: 72.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions. Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite: MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L. Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

NUR1025 Fundamentals of Nursing 3.000 credits

This course introduces students to the profession of Nursing. The student will learn the roles basic to the practice of nursing, the nursing process, and how nurses are involved in health promoting activities to meet client needs. Pre-Req: Program Admission; Co-Req: NUR 1025C, 1025L, 1060C, and 1141.

Lecture Hours: 48.000

Course User Fee: \$113.00

NUR1025C Fundamentals of Nursing Skills Lab 1.000 credits

Students will learn of the opportunities for the explanation, demonstration, and practice of care provider activities essential to the basic practice of nursing. Learning experiences are provided in the skills laboratory. Pre-Req: Program Admission; Co-Req: NUR 1025, 1025L, 1060C, 1141

Lab Hours: 32.000

Course User Fee: \$105.00

NUR1025L Fundamentals of Nursing Clinical Lab 2.000 credits

Students will learn of the opportunities for the explanation, demonstration, and practice of care provider activities essential to the basic practice of nursing. Learning experiences are provided in the skills laboratory. Pre-Req: Program Admission; Corequisite: NUR 1025, 1025C, 1141, 1060C.

Clinical Hours: 96.000

Course User Fee: \$8.00

NUR1060C Adult Health Assessment 3.000 credits

This course is designed to provide students with the necessary skills to perform an in-depth nursing history and a complete physical examination on an adult client. The focus will be on clients with minimal or no alterations in their health state. Students will be introduced to and will demonstrate the techniques used in physical examination. Prerequisite: Admission to the School of Nursing; Corequisites: NUR 1025, 1025C, 1025L, 1141.

Lecture Hours: 32.000 - 33.000, Lab Hours: 32.000, Practicum Hours: 2.000

Prerequisite: Admission to the program

Course User Fee: \$109.00

NUR1141 Nursing Math & Pharmacology 3.000 credits

Students will learn about medications and their effects on different body systems. The conceptual and mathematical operations necessary for safe and effective administration of intravenous medications, preparing medications that come in powdered form and adjusting medication administration based on medical protocols are discussed. Concepts of medications including history, nomenclature, sources of drug information, federal drug laws and standards, medication classifications, pharmacokinetics, pharmacodynamics, variables affecting medication actions and effects, and methods of delivery will be addressed. Co-Req: NUR 1025, 1025C, 1025L, 1060C. Lecture Hours: 48.000, Practicum Hours: 1.000

Course User Fee: \$75.00

NUR1211 Medical-Surgical Nursing 3.000 credits

This course provides an introduction to the adult nursing care. Students will learn the nurse's role in meeting the short and long term needs of the client and community through preventive, therapeutic and palliative care are discussed. Pre-Req: NUR 1025, 1025C, 1060C, 1142; Co-Req: NUR 1211L, 1214C. Lecture Hours: 48.000

Course User Fee: \$121.00

NUR1211L Medical Surgical Nursing Clinical 2.000 credits

Students will learn how to apply concepts of adult health nursing. Experiences in both in-patient and community settings will be provided. Students are encouraged to participate in projects emphasizing preventive aspects of nursing care. Prerequisites: NUR 1025, 1025C, 1025L, 1060C, and 1141

Clinical Hours: 96.000

Course User Fee: \$87.00

NUR1214C Medical Surgical Nursing Skills Lab 1.000 credits

Students will learn the opportunities for the explanation, demonstration, and practice of skills related to adult health nursing. Learning experiences are provided in the School of Nursing Skills Laboratory. Pre-Req: NUR 1025, 1025C, 1025L, 1060C, 1142; CO-REQ: NUR 1211, 1211L. Lecture Hours: 8.000, Lab Hours: 16.000
Course User Fee: \$114.00

NUR2310 Pediatric Nursing 2.000 credits

This course provides a family centered approach to the nursing care of pediatric clients and their families. Students will learn the nurse's role in meeting the short and long term needs of the pediatric client, family, and community through preventative, therapeutic and palliative care, with recognition for the multicultural aspects of client needs. Prerequisites: 1211, 1211L, 1214C or 1002, 1002L; 1142; corequisites: NUR 2310L, 2420, 2420L, 2680L. Special fee. (2 hr. lecture)
Lecture Hours: 32.000
Course User Fee: \$126.00

NUR2310L Pediatric Nursing Clinical Lab 1.000 credits

This course allows the student to apply the nursing process to the care of clients in selected pediatric clinical settings. Students will learn to observe cultural diversity and implement care to the pediatric client, family, and community through preventive, therapeutic and palliative measures. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2420, 2420L, 2520, 2520L, 2680L. Laboratory fee.
Clinical Hours: 48.000
Course User Fee: \$86.00

NUR2420 Obstetrical Nursing 2.000 credits

This course provides a family centered approach to the nursing care of obstetrical clients and their families. Students will learn to assess the pregnant client, to implement caring behaviors for the laboring client, educate the postpartum client, manage the care of the newborn and collaboration of care for the high risk client. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142. Corequisites: NUR 2310, 2310L, 2420L, 2680L. Special fee. (2 hr. lecture)
Lecture Hours: 32.000

NUR2420L Obstetrical Nursing Clinical Lab 1.000 credits

This course provides an introduction to obstetrical nursing clinical practice. Students will learn to apply the nursing process to the care of clients in selected obstetrical clinical settings. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2310L, 2420, 2680L. Laboratory fee.
Clinical Hours: 48.000
Course User Fee: \$96.00

NUR2220 Medical Surgical Nursing II 3.000 credits

This course provides students with the opportunity to apply concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health.
Lecture Hours: 48.000
Prerequisite: NUR1025 AND NUR1025C AND NUR1025L AND NUR1060C AND NUR1141 AND NUR1211 AND NUR1211L AND NUR1214C AND
Corequisite: NUR2220L

NUR2220L Medical Surgical Nursing II Clinical 1.000 credits

This course provides students with the opportunity to apply concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health.
Clinical Hours: 48.000
Prerequisite: NUR1025 AND NUR1025C AND NUR1025L AND NUR1060C AND NUR1141 AND NUR1211 AND NUR1211L AND NUR1214C AND
Corequisite: NUR2220

NUR2520 Psychiatric Nursing 2.000 credits

This course introduces students to the basic concepts of psychiatric nursing. Students will learn to provide care in in-patient and community settings, focusing on the nurse's role in meeting the needs of the patient, family, and the community. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2520L. Special fee. (2 hr. lecture)
Lecture Hours: 32.000
Course User Fee: \$62.00

NUR2520L Psychiatric Nursing Clinical Lab 1.000 credits

This course provides the student opportunities to apply concepts of psychiatric nursing. Students will learn psychiatric procedures for both in-patient and community settings, focusing on the nurse's role on meeting the needs of the client, family, and community. Students actively participate in projects assisting clients in preventative care and maintenance of mental health. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142
Clinical Hours: 48.000
Course User Fee: \$98.00

NUR2680L Community Health Nursing Lab 1.000 credits

This laboratory course assists the students in applying knowledge of community health resources. Students will learn to manage community health resources to support the delivery of care to the childbearing/child-rearing families. Special emphasis is placed on the understanding of cultural influences on the health practices and beliefs within the family. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; co-requisites: NUR 2310, 2310L, 2420, 2420L.0L. Special fee. (48 hr. clinical)
Clinical Hours: 48.000
Course User Fee: \$81.00

NUR2212 Medical Surgical Nursing III 3.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L
Lecture Hours: 48.000
Course User Fee: \$450.00

NUR2212L Medical Surgical Nursing III Clinical Lab 2.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1002, 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L
Clinical Hours: 96.000
Course User Fee: \$150.00

NUR2811C Professional Nursing Leadership 4.000 credits

This course provides the student with the theoretical and clinical knowledge necessary for actualization of the role of the registered professional nurse, with emphasis on delegation and supervision. Prerequisites: NUR 2212, 2212L
Lecture Hours: 16.000, Clinical Hours: 144.000
Course User Fee: \$175.00

NUR2960 Senior Seminar 1.000 credits

This course provides a forum for the discussion, and preparation for the NCLEX-RN exam. Explain applicant requirements for the state board of nursing (and licensure eligibility requirements) and examination registration.
Lecture Hours: 16.000
Prerequisite: NUR2811C

Program Description

Nursing - R.N. (Generic Part-Time)

Associate in Science | Code: 23027 | 72 credits

The Generic Nursing Option is designed to prepare students without previous health care education for careers as Registered Nurses. This program is accredited by the National League for Nursing Accrediting Commission (NLNAC), (61 Broadway, New York, NY 10006, phone (212) 363-5555, www.nlnac.org) and approved by the Florida Board of Nursing (FBON). Graduates are eligible to apply to write the National Council Licensing Examination for Registered Nurses (NCLEX-RN). The part-time track is designed for individuals who must work while they attend school. Selection is based on scores on pre-admission testing and completion of required pre-requisite courses.

See a School of Nursing Information Booklet for more specific details about admission requirements. Pre-Admission tests – Applicants must take the Test of Adult Basic Education (TABE) and score 11 or above in reading, language and total math and the Nurse Entrance Test (NET). The NET is given only on the Medical Campus. Scores on the NET and TABE are used for selection.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 2		Units: 10.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
MAC 1105	College Algebra	3.0
PHI 2010	Introduction to Philosophy	3.0

Semester 3		Units: 10.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
MCB 2010	Microbiology	3.0
SPC 2608	Introduction to Public Speaking	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4 (In Milestone)		Units: 6.0
NUR 1025	Fundamentals of Nursing	3.0
NUR 1060C	Adult Health Assessment	3.0

Semester 5 (In Milestone)		Units: 6.0
NUR 1025C	Fundamentals of Nursing Skills Lab	1.0
NUR 1025L	Fundamentals of Nursing Clinical Lab	2.0
NUR 1141	Nursing Math & Pharmacology	3.0

Semester 6 (In Milestone)		Units: 6.0
NUR 1211	Medical-Surgical Nursing	3.0
NUR 1211L	Medical Surgical Nursing Clinical	2.0
NUR 1214C	Medical Surgical Nursing Skills Lab	1.0

Semester 7 (In Milestone)		Units: 6.0
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NUR 2310	Pediatric Nursing	2.0
NUR 2310L	Pediatric Nursing Clinical Lab	1.0
NUR 2420	Obstetrical Nursing	2.0
NUR 2420L	Obstetrical Nursing Clinical Lab	1.0

Semester 7 (In Milestone)

Units: 7.0

NUR 2520	Psychiatric Nursing	2.0
NUR 2520L	Psychiatric Nursing Clinical Lab	1.0
NUR 2220	Medical Surgical Nursing II	3.0
NUR 2220L	Medical Surgical Nursing II Clinical	1.0

Semester 9 (In Milestone)

Units: 6.0

NUR 2212	Medical Surgical Nursing III	3.0
NUR 2212L	Medical Surgical Nursing III Clinical Lab	2.0
NUR 2680L	Community Health Nursing Lab	1.0

Semester 10 (In Milestone)

Units: 5.0

NUR 2811C	Professional Nursing Leadership	4.0
NUR 2960	Senior Seminar	1.0

Total: 72.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy. Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view. Lecture Hours: 48.000

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions. Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite: MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L. Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

NUR1025 Fundamentals of Nursing 3.000 credits

This course introduces students to the profession of Nursing. The student will learn the roles basic to the practice of nursing, the nursing process, and how nurses are involved in health promoting activities to meet client needs. Pre-Req: Program Admission; Co-Req: NUR 1025C, 1025L, 1060C, and 1141.

Lecture Hours: 48.000

Course User Fee: \$113.00

NUR1060C Adult Health Assessment 3.000 credits

This course is designed to provide students with the necessary skills to perform an in-depth nursing history and a complete physical examination on an adult client. The focus will be on clients with minimal or no alterations in their health state. Students will be introduced to and will demonstrate the techniques used in physical examination. Prerequisite: Admission to the School of Nursing; Corequisites: NUR 1025, 1025C, 1025L, 1141.

Lecture Hours: 32.000 - 33.000, Lab Hours: 32.000, Practicum Hours: 2.000

Prerequisite: Admission to the program

Course User Fee: \$109.00

NUR1025C Fundamentals of Nursing Skills Lab 1.000 credits

Students will learn of the opportunities for the explanation, demonstration, and practice of care provider activities essential to the basic practice of nursing. Learning experiences are provided in the skills laboratory.

Pre-Req: Program Admission; Co-Req: NUR 1025, 1025L, 1060C, 1141

Lab Hours: 32.000

Course User Fee: \$105.00

NUR1025L Fundamentals of Nursing Clinical Lab 2.000 credits

Students will learn of the opportunities for the explanation, demonstration, and practice of care provider activities essential to the basic practice of nursing. Learning experiences are provided in the skills laboratory.

Pre-Req: Program Admission; Corequisite: NUR 1025, 1025C, 1141, 1060C.

Clinical Hours: 96.000

Course User Fee: \$8.00

NUR1141 Nursing Math & Pharmacology 3.000 credits

Students will learn about medications and their effects on different body systems. The conceptual and mathematical operations necessary for safe and effective administration of intravenous medications, preparing medications that come in powdered form and adjusting medication administration based on medical protocols are discussed. Concepts of medications including history, nomenclature, sources of drug information, federal drug laws and standards, medication classifications, pharmacokinetics, pharmacodynamics, variables affecting medication actions and effects, and methods of delivery will be addressed. Co-Req: NUR 1025, 1025C, 1025L, 1060C.

Lecture Hours: 48.000, Practicum Hours: 1.000

Course User Fee: \$75.00

NUR1211 Medical-Surgical Nursing 3.000 credits

This course provides an introduction to the adult nursing care. Students will learn the nurse's role in meeting the short and long term needs of the client and community through preventive, therapeutic and palliative care are discussed. Pre-Req: NUR 1025, 1025C, 1060C, 1142; Co-Req: NUR 1211L, 1214C.

Lecture Hours: 48.000

Course User Fee: \$121.00

NUR1211L Medical Surgical Nursing Clinical 2.000 credits

Students will learn how to apply concepts of adult health nursing. Experiences in both in-patient and community settings will be provided. Students are encouraged to participate in projects emphasizing preventive aspects of nursing care. Prerequisites: NUR 1025, 1025C, 1025L, 1060C, and 1141

Clinical Hours: 96.000

Course User Fee: \$87.00

NUR1214C Medical Surgical Nursing Skills Lab 1.000 credits

Students will learn the opportunities for the explanation, demonstration, and practice of skills related to adult health nursing. Learning experiences are provided in the School of Nursing Skills Laboratory. Pre-Req: NUR 1025, 1025C, 1025L, 1060C, 1142; CO-REQ: NUR 1211, 1211L.

Lecture Hours: 8.000, Lab Hours: 16.000

Course User Fee: \$114.00

NUR2310 Pediatric Nursing 2.000 credits

This course provides a family centered approach to the nursing care of pediatric clients and their families. Students will learn the nurse's role in meeting the short and long term needs of the pediatric client, family, and community through preventative, therapeutic and palliative care, with recognition for the multicultural aspects of client needs. Prerequisites: 1211, 1211L, 1214C or 1002, 1002L; 1142; corequisites: NUR 2310L, 2420, 2420L, 2680L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$126.00

NUR2310L Pediatric Nursing Clinical Lab 1.000 credits

This course allows the student to apply the nursing process to the care of clients in selected pediatric clinical settings. Students will learn to observe cultural diversity and implement care to the pediatric client, family, and community through preventive, therapeutic and palliative measures. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2420, 2420L, 2520, 2520L, 2680L. Laboratory fee.

Clinical Hours: 48.000

Course User Fee: \$86.00

NUR2420 Obstetrical Nursing 2.000 credits

This course provides a family centered approach to the nursing care of obstetrical clients and their families. Students will learn to assess the pregnant client, to implement caring behaviors for the laboring client, educate the postpartum client, manage the care of the newborn and collaboration of care for the high risk client.

Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142. Corequisites: NUR 2310, 2310L, 2420L, 2680L.

Special fee. (2 hr. lecture)

Lecture Hours: 32.000

NUR2420L Obstetrical Nursing Clinical Lab 1.000 credits

This course provides an introduction to obstetrical nursing clinical practice. Students will learn to apply the nursing process to the care of clients in selected obstetrical clinical settings. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2310L, 2420, 2680L. Laboratory fee.

Clinical Hours: 48.000

Course User Fee: \$96.00

NUR2520 Psychiatric Nursing 2.000 credits

This course introduces students to the basic concepts of psychiatric nursing. Students will learn to provide care in in-patient and community settings, focusing on the nurse's role in meeting the needs of the patient, family, and the community. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2520L.

Special fee. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$62.00

NUR2520L Psychiatric Nursing Clinical Lab 1.000 credits

This course provides the student opportunities to apply concepts of psychiatric nursing. Students will learn psychiatric procedures for both in-patient and community settings, focusing on the nurse's role on meeting the needs of the client, family, and community. Students actively participate in projects assisting clients in preventative care and maintenance of mental health. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142

Clinical Hours: 48.000

Course User Fee: \$98.00

NUR2220 Medical Surgical Nursing II 3.000 credits

This course provides students with the opportunity to apply concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health.

Lecture Hours: 48.000

Prerequisite: NUR1025 AND NUR1025C AND NUR1025L AND NUR1060C AND NUR1141 AND NUR1211 AND NUR1211L AND NUR1214C AND Corequisite: NUR2220L

NUR2220L Medical Surgical Nursing II Clinical 1.000 credits

This course provides students with the opportunity to apply concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health.

Clinical Hours: 48.000

Prerequisite: NUR1025 AND NUR1025C AND NUR1025L AND NUR1060C AND NUR1141 AND NUR1211 AND NUR1211L AND NUR1214C AND Corequisite: NUR2220

NUR2212 Medical Surgical Nursing III 3.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1008, 1002L; NUR 1141, 2310, 2310L, 2420 2420L, 2680L

Lecture Hours: 48.000

Course User Fee: \$450.00

NUR2212L Medical Surgical Nursing III Clinical Lab 2.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1002, 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L

Clinical Hours: 96.000

Course User Fee: \$150.00

NUR2680L Community Health Nursing Lab 1.000 credits

This laboratory course assists the students in applying knowledge of community health resources. Students will learn to manage community health resources to support the delivery of care to the childbearing/child-rearing families. Special emphasis is placed on the understanding of cultural influences on the health practices and beliefs within the family. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; co-requisites: NUR 2310, 2310L, 2420, 2420L.OL. Special fee. (48 hr. clinical)

Clinical Hours: 48.000

Course User Fee: \$81.00

NUR2811C Professional Nursing Leadership 4.000 credits

This course provides the student with the theoretical and clinical knowledge necessary for actualization of the role of the registered professional nurse, with emphasis on delegation and supervision. Prerequisites: NUR 2212, 2212L

Lecture Hours: 16.000, Clinical Hours: 144.000

Course User Fee: \$175.00

NUR2960 Senior Seminar 1.000 credits

This course provides a forum for the discussion, and preparation for the NCLEX-RN exam. Explain applicant requirements for the state board of nursing (and licensure eligibility requirements) and examination registration.

Lecture Hours: 16.000

Prerequisite: NUR2811C

Program Description

Nursing - R.N. (Transitional Full-Time)

Associate in Science | Code: 23025 | 72 credits

The Transitional Option in Nursing is designed to prepare Licensed Practical Nurses (LPN) and selected other individuals with National Licensure or Certification for practice as a Registered Nurse (RN). The content and clinical experiences are designed to meet the learning and professional socialization needs of this special category of student. The program is accredited by the National League for Nursing Accrediting Commission (NLNAC) (61 Broadway, New York, NY 10006, ph. 212-363-5555, www.nlnac.org) and approved by the Florida Board of Nursing (FBON). Graduates are eligible to apply to write the National Council Licensing Examination for Registered Nurses (NCLEX-RN). Selection is based on scores on pre-admission testing and completion of required pre-requisite courses. See a School of Nursing Information Booklet for more specific details about admission requirements. Pre-Admission tests – Applicants must take the Test of Adult Basic Education (TABE) and score 11 or above in reading, language and total math and the Nurse Entrance Test (NET). The NET is given only on the Medical Campus. Scores on the NET and TABE are used for selection.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 10.0**

ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 2**Units: 10.0**

BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
MAC 1105	College Algebra	3.0
PHI 2010	Introduction to Philosophy	3.0

Semester 3

Units: 10.0

CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
MCB 2010	Microbiology	3.0
SPC 2608	Introduction to Public Speaking	3.0
Computer Competency*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4 (In Milestone)

Units: 22.0

<i>Transitional Course</i>		2.0-5.0
NUR 1002	Transition to Registered Nursing	2.0
OR		
NUR 1006	Health Care Professions to RN Transition	5.0
NUR 1002L	Transition to Medical Surgical Nursing Clinical Lab	2.0
NUR 1008	Transition to Medical Surgical Nursing	5.0
NUR 1141	Nursing Math & Pharmacology	3.0
NUR 9995	Nursing Statewide Industry Certification	10.0

Semester 5 (In Milestone)

Units: 10.0

NUR 2310	Pediatric Nursing	2.0
NUR 2310L	Pediatric Nursing Clinical Lab	1.0
NUR 2420	Obstetrical Nursing	2.0
NUR 2420L	Obstetrical Nursing Clinical Lab	1.0
NUR 2520	Psychiatric Nursing	2.0
NUR 2520L	Psychiatric Nursing Clinical Lab	1.0
NUR 2680L	Community Health Nursing Lab	1.0

Semester 6 (In Milestone)

Units: 10.0

NUR 2212	Medical Surgical Nursing III	3.0
NUR 2212L	Medical Surgical Nursing III Clinical Lab	2.0
NUR 2811C	Professional Nursing Leadership	4.0
NUR 2960	Senior Seminar	1.0

Total: 72.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions.

Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite: MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L.

Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

NUR1002 Transition to Registered Nursing 2.000 credits

This course introduces the student with selected prior health care experience and education to the profession of nursing, the roles basic to nursing practice, nursing process and the implementation of health-promoting activities to meet patient needs.

Prerequisite: Admission to the School of Nursing. Co-requisites: NUR 1002L, 1141.

Lecture Hours: 32.000

Course User Fee: \$125.00

NUR1006 Health Care Professions to RN Transition 5.000 credits

This course is designed for Medical Assistants, Respiratory Therapist, and Paramedics entering into the Transition program and will introduce the framework of the nursing process to non-nurse health care providers desiring to transition into the role of the registered nurse. Fundamental nursing concepts related to role transition and the introduction of nursing practices will be emphasized along with essential nursing core competencies including the knowledge, skills, and abilities to provide safe nursing practice. This course also provides the opportunity for participants to observe, review and practice in nursing skills in a simulated laboratory setting. Critical Thinking and Clinical Analysis will be introduced along with the Nursing Process. (NUR 1XXX is required for Licensed Medical Assistant, Paramedic and/or Respiratory Therapist students only). 80.00 hr. lecture

Lecture Hours: 80.000

NUR1002L Transition to Medical Surgical Nursing Clinical Lab 2.000 credits

This lab introduces students with prior health care education to the nursing profession. Students will learn the nurse's role in meeting short and long term needs of the patient through preventive, therapeutic and palliative care. Students will also explore nursing care of the adults with moderate alterations in health within a body systems framework. Pre-Req: Program Admission; Co-Req: NUR 1002 and NUR 1141. (192 Hr. Clinical)

Clinical Hours: 96.000

Course User Fee: \$116.00

NUR1008 Transition to Medical Surgical Nursing 5.000 credits

This course introduces students with prior health care education to the nursing profession. Students will learn the nurse's role in meeting short- and long-term needs of the patient through preventive, therapeutic and palliative care. Students will also explore nursing care of the adults with moderate alterations in health within a body systems framework.

Lecture Hours: 80.000

NUR1141 Nursing Math & Pharmacology 3.000 credits

Students will learn about medications and their effects on different body systems. The conceptual and mathematical operations necessary for safe and effective administration of intravenous medications, preparing medications that come in powdered form and adjusting medication administration based on medical protocols are discussed. Concepts of medications including history, nomenclature, sources of drug information, federal drug laws and standards, medication classifications, pharmacokinetics, pharmacodynamics, variables affecting medication actions and effects, and methods of delivery will be addressed. Co-Req: NUR 1025, 1025C, 1025L, 1060C.

Lecture Hours: 48.000, Practicum Hours: 1.000

Course User Fee: \$75.00

NUR9995 Nursing Statewide Industry Certification 10.000 credits

This course is designed for students who have earned an approved Statewide Nursing Industry Certification and provides academic recognition of the knowledge, skills, and competencies demonstrated through credential attainment. The course validates proficiency in nursing principles, patient care, clinical decision-making, safety practices, and professional standards within healthcare settings. Credit awarded through this course supports academic progression and recognizes documented achievement of industry-recognized nursing competencies. See Advisor to apply Technical Manual of Procedure (TMOP): 1100-815848.

NUR2310 Pediatric Nursing 2.000 credits

This course provides a family centered approach to the nursing care of pediatric clients and their families. Students will learn the nurse's role in meeting the short and long term needs of the pediatric client, family, and community through preventative, therapeutic and palliative care, with recognition for the multicultural aspects of client needs. Prerequisites: 1211, 1211L, 1214C or 1002, 1002L; 1142; corequisites: NUR 2310L, 2420, 2420L, 2680L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$126.00

NUR2310L Pediatric Nursing Clinical Lab 1.000 credits

This course allows the student to apply the nursing process to the care of clients in selected pediatric clinical settings. Students will learn to observe cultural diversity and implement care to the pediatric client, family, and community through preventive, therapeutic and palliative measures. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2420, 2420L, 2680L. Laboratory fee.

Clinical Hours: 48.000

Course User Fee: \$112.00

NUR2420 Obstetrical Nursing 2.000 credits

This course provides a family centered approach to the nursing care of obstetrical clients and their families. Students will learn to assess the pregnant client, to implement caring behaviors for the laboring client, educate the postpartum client, manage the care of the newborn and collaboration of care for the high risk client. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142. Corequisites: NUR 2310, 2310L, 2420L, 2680L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

NUR2420L Obstetrical Nursing Clinical Lab 1.000 credits

This course provides an introduction to obstetrical nursing clinical practice. Students will learn to apply the nursing process to the care of clients in selected obstetrical clinical settings. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2310L, 2420, 2680L. Laboratory fee. Clinical Hours: 48.000
Course User Fee: \$96.00

NUR2520 Psychiatric Nursing 2.000 credits

This course introduces students to the basic concepts of psychiatric nursing. Students will learn to provide care in in-patient and community settings, focusing on the nurse's role in meeting the needs of the patient, family, and the community. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2520L. Special fee. (2 hr. lecture) Lecture Hours: 32.000
Course User Fee: \$62.00

NUR2520L Psychiatric Nursing Clinical Lab 1.000 credits

This course provides the student opportunities to apply concepts of psychiatric nursing. Students will learn psychiatric procedures for both in-patient and community settings, focusing on the nurse's role on meeting the needs of the client, family, and community. Students actively participate in projects assisting clients in preventative care and maintenance of mental health. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142 Clinical Hours: 48.000
Course User Fee: \$98.00

NUR2680L Community Health Nursing Lab 1.000 credits

This laboratory course assists the students in applying knowledge of community health resources. Students will learn to manage community health resources to support the delivery of care to the childbearing/child-rearing families. Special emphasis is placed on the understanding of cultural influences on the health practices and beliefs within the family. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; co-requisites: NUR 2310, 2310L, 2420, 2420L.0L. Special fee. (48 hr. clinical) Clinical Hours: 48.000
Course User Fee: \$81.00

NUR2960 Senior Seminar 1.000 credits

This course provides a forum for the discussion, and preparation for the NCLEX-RN exam. Explain applicant requirements for the state board of nursing (and licensure eligibility requirements) and examination registration. Lecture Hours: 16.000
Prerequisite: NUR2811C

NUR2811C Professional Nursing Leadership 4.000 credits

This course provides the student with the theoretical and clinical knowledge necessary for actualization of the role of the registered professional nurse, with emphasis on delegation and supervision. Prerequisites: NUR 2212, 2212L Lecture Hours: 16.000, Clinical Hours: 144.000
Course User Fee: \$175.00

NUR2212 Medical Surgical Nursing III 3.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L Lecture Hours: 48.000
Course User Fee: \$450.00

NUR2212L Medical Surgical Nursing III Clinical Lab 2.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1002, 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L Clinical Hours: 96.000
Course User Fee: \$150.00

Program Description

Nursing - R.N. (Transitional Part-Time)

Associate in Science | Code: 23026 | 72 credits

The Transitional Option in Nursing is designed to prepare Licensed Practical Nurses (LPN) and selected other individuals with National Licensure or Certification for practice as a Registered Nurse (RN). The content and clinical experiences are designed to meet the learning and professional socialization needs of this special category of student. The program is accredited by the National League for Nursing Accrediting Commission (NLNAC) (61 Broadway, New York, NY 10006, ph. 212-363-5555, www.nlnac.org) and approved by the Florida Board of Nursing (FBON). Graduates are eligible to apply to write the National Council Licensing Examination for Registered Nurses (NCLEX-RN). Selection is based on scores on pre-admission testing and completion of required pre-requisite courses. See a School of Nursing Information Booklet for more specific details about admission requirements. Pre-Admission tests – Applicants must take the Test of Adult Basic Education (TABE) and score 11 or above in reading, language and total math and the Nurse Entrance Test (NET). The NET is given only on the Medical Campus. Scores on the NET and TABE are used for selection.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Semester 2		Units: 10.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0
MAC 1105	College Algebra	3.0
PHI 2010	Introduction to Philosophy	3.0

Semester 3		Units: 10.0
CHM 1033	Chemistry for Health Sciences	3.0
CHM 1033L	Chemistry for Health Sciences Lab	1.0
MCB 2010	Microbiology	3.0
SPC 2608	Introduction to Public Speaking	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4 (In Program Milestone)		Units: 7.0
<i>Transitional Course</i>		2.0-5.0
NUR 1002	Transition to Registered Nursing	2.0
OR		
NUR 1006	Health Care Professions to RN Transition	5.0
NUR 1008	Transition to Medical Surgical Nursing	5.0

Semester 5 (In Program Milestone)		Units: 15.0
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NUR 1002L	Transition to Medical Surgical Nursing Clinical Lab	2.0
NUR 1141	Nursing Math & Pharmacology	3.0
NUR 9995	Nursing Statewide Industry Certification	10.0

Semester 6 (In Program Milestone)

Units: 6.0

NUR 2310	Pediatric Nursing	2.0
NUR 2310L	Pediatric Nursing Clinical Lab	1.0
NUR 2420	Obstetrical Nursing	2.0
NUR 2420L	Obstetrical Nursing Clinical Lab	1.0

Semester 7 (In Program Milestone)

Units: 4.0

NUR 2520	Psychiatric Nursing	2.0
NUR 2520L	Psychiatric Nursing Clinical Lab	1.0
NUR 2680L	Community Health Nursing Lab	1.0

Semester 8 (In Program Milestone)

Units: 10.0

NUR 2212	Medical Surgical Nursing III	3.0
NUR 2212L	Medical Surgical Nursing III Clinical Lab	2.0
NUR 2811C	Professional Nursing Leadership	4.0
NUR 2960	Senior Seminar	1.0

Total: 72.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Lab 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions. Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite: MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L. Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions.

Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

NUR1002 Transition to Registered Nursing 2.000 credits

This course introduces the student with selected prior health care experience and education to the profession of nursing, the roles basic to nursing practice, nursing process and the implementation of health-promoting activities to meet patient needs.

Prerequisite: Admission to the School of Nursing. Co-requisites: NUR 1002L, 1141.

Lecture Hours: 32.000

Course User Fee: \$125.00

NUR1006 Health Care Professions to RN Transition 5.000 credits

This course is designed for Medical Assistants, Respiratory Therapist, and Paramedics entering into the Transition program and will introduce the framework of the nursing process to non-nurse health care providers desiring to transition into the role of the registered nurse. Fundamental nursing concepts related to role transition and the introduction of nursing practices will be emphasized along with essential nursing core competencies including the knowledge, skills, and abilities to provide safe nursing practice. This course also provides the opportunity for participants to observe, review and practice in nursing skills in a simulated laboratory setting. Critical Thinking and Clinical Analysis will be introduced along with the Nursing Process. (NUR 1XXX is required for Licensed Medical Assistant, Paramedic and/or Respiratory Therapist students only). 80.00 hr. lecture

Lecture Hours: 80.000

NUR1008 Transition to Medical Surgical Nursing 5.000 credits

This course introduces students with prior health care education to the nursing profession. Students will learn the nurse's role in meeting short- and long-term needs of the patient through preventive, therapeutic and palliative care. Students will also explore nursing care of the adults with moderate alterations in health within a body systems framework.

Lecture Hours: 80.000

NUR1002L Transition to Medical Surgical Nursing Clinical Lab 2.000 credits

This lab introduces students with prior health care education to the nursing profession. Students will learn the nurse's role in meeting short and long term needs of the patient through preventive, therapeutic and palliative care. Students will also explore nursing care of the adults with moderate alterations in health within a body systems framework. Pre-Req: Program Admission; Co-Req: NUR 1002 and NUR 1141. (192 Hr. Clinical)

Clinical Hours: 96.000

Course User Fee: \$116.00

NUR1141 Nursing Math & Pharmacology 3.000 credits

Students will learn about medications and their effects on different body systems. The conceptual and mathematical operations necessary for safe and effective administration of intravenous medications, preparing medications that come in powdered form and adjusting medication administration based on medical protocols are discussed. Concepts of medications including history, nomenclature, sources of drug information, federal drug laws and standards, medication classifications, pharmacokinetics, pharmacodynamics, variables affecting medication actions and effects, and methods of delivery will be addressed. Co-Req: NUR 1025, 1025C, 1025L, 1060C.

Lecture Hours: 48.000, Practicum Hours: 1.000

Course User Fee: \$75.00

NUR9995 Nursing Statewide Industry Certification 10.000 credits

This course is designed for students who have earned an approved Statewide Nursing Industry Certification and provides academic recognition of the knowledge, skills, and competencies demonstrated through credential attainment. The course validates proficiency in nursing principles, patient care, clinical decision-making, safety practices, and professional standards within healthcare settings. Credit awarded through this course supports academic progression and recognizes documented achievement of industry-recognized nursing competencies. See Advisor to apply Technical Manual of Procedure (TMOP): 1100-815848.

NUR2310 Pediatric Nursing 2.000 credits

This course provides a family centered approach to the nursing care of pediatric clients and their families. Students will learn the nurse's role in meeting the short and long term needs of the pediatric client, family, and community through preventative, therapeutic and palliative care, with recognition for the multicultural aspects of client needs. Prerequisites: 1211, 1211L, 1214C or 1002, 1002L; 1142; corequisites: NUR 2310L, 2420, 2420L, 2680L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$126.00

NUR2310L Pediatric Nursing Clinical Lab 1.000 credits

This course allows the student to apply the nursing process to the care of clients in selected pediatric clinical settings. Students will learn to observe cultural diversity and implement care to the pediatric client, family, and community through preventive, therapeutic and palliative measures. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2420, 2420L, 2520, 2520L, 2680L. Laboratory fee.

Clinical Hours: 48.000

Course User Fee: \$86.00

NUR2420 Obstetrical Nursing 2.000 credits

This course provides a family centered approach to the nursing care of obstetrical clients and their families. Students will learn to assess the pregnant client, to implement caring behaviors for the laboring client, educate the postpartum client, manage the care of the newborn and collaboration of care for the high risk client. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142. Corequisites: NUR 2310, 2310L, 2420L, 2680L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

NUR2420L Obstetrical Nursing Clinical Lab 1.000 credits

This course provides an introduction to obstetrical nursing clinical practice. Students will learn to apply the nursing process to the care of clients in selected obstetrical clinical settings. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2310L, 2420, 2680L. Laboratory fee.

Clinical Hours: 48.000

Course User Fee: \$96.00

NUR2520 Psychiatric Nursing 2.000 credits

This course introduces students to the basic concepts of psychiatric nursing. Students will learn to provide care in in-patient and community settings, focusing on the nurse's role in meeting the needs of the patient, family, and the community. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2520L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$62.00

NUR2520L Psychiatric Nursing Clinical Lab 1.000 credits

This course provides the student opportunities to apply concepts of psychiatric nursing. Students will learn psychiatric procedures for both in-patient and community settings, focusing on the nurse's role on meeting the needs of the client, family, and community. Students actively participate in projects assisting clients in preventative care and maintenance of mental health. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142

Clinical Hours: 48.000

Course User Fee: \$98.00

NUR2680L Community Health Nursing Lab 1.000 credits

This laboratory course assists the students in applying knowledge of community health resources. Students will learn to manage community health resources to support the delivery of care to the childbearing/child-rearing families. Special emphasis is placed on the understanding of cultural influences on the health practices and beliefs within the family. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; co-requisites: NUR 2310, 2310L, 2420, 2420L. Special fee. (48 hr. clinical)

Clinical Hours: 48.000

Course User Fee: \$81.00

NUR2212 Medical Surgical Nursing III 3.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L

Lecture Hours: 48.000

Course User Fee: \$450.00

NUR2212L Medical Surgical Nursing III Clinical Lab 2.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1002, 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L

Clinical Hours: 96.000

Course User Fee: \$150.00

NUR2811C Professional Nursing Leadership 4.000 credits

This course provides the student with the theoretical and clinical knowledge necessary for actualization of the role of the registered professional nurse, with emphasis on delegation and supervision.

Prerequisites: NUR 2212, 2212L

Lecture Hours: 16.000, Clinical Hours: 144.000

Course User Fee: \$175.00

NUR2960 Senior Seminar 1.000 credits

This course provides a forum for the discussion, and preparation for the NCLEX-RN exam. Explain applicant requirements for the state board of nursing (and licensure eligibility requirements) and examination registration.

Lecture Hours: 16.000

Prerequisite: NUR2811C

Program Description

Opticianry

Associate in Science | Code: 23041 | 72 credits

The Opticianry program simultaneously prepares students for three ophthalmic health care careers: optician, optometric technician and ophthalmic medical assistant. A concentrated presentation of general education courses combined with career development and clinical experience accomplishes this multi-disciplinary approach. Among the marketable skills acquired are clinical data collection, ophthalmic fabrication and ophthalmic dispensing. The student begins working with patients during the third semester in clinics staffed by ophthalmologists, optometrists and opticians. A student must maintain a grade point average of 2.0 or better in each course with an "OPT" prefix in order to advance within the program. The successful completion of this program offers the graduate a challenging and rewarding career on an ophthalmic health care team. Graduates are eligible to sit for the Opticianry Licensure Examination and the Optometric Technician Registration Examination. After one year of work experience with an ophthalmologist, graduates may sit for the Ophthalmic Medical Assistant Certification Examination. The Opticianry program is approved by the Council on Optometric Education and the Commission on Opticianry Accreditation.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 15.0

ENC 1101

English Composition 1

3.0

MGF 1130	Mathematical Thinking	3.0
PHI 2010	Introduction to Philosophy	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
PHY 1020	General Education Physics	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Course Competency Text*

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2 (In Milestone)

Units: 11.0

OPT 1110	Physical & Geometrical Optics	4.0
OPT 1205	Ocular Anatomy, Physiology & Pathophysiology	3.0
OPT 1330	Clinical Data Collection 1	2.0
OPT 1450	Ophthalmic Dispensing Procedures 1	1.0
OPT 1450L	Ophthalmic Dispensing Procedures 1 Laboratory	1.0

Semester 3 (In Milestone)

Units: 11.0

OPT 1150	Ophthalmic Lenses	2.0
OPT 1331	Clinical Data Collection 2	2.0
OPT 1331L	Clinical Data Collection 2 Laboratory	1.0
OPT 2070L	Computers for Vision Care	1.0
OPT 2451	Ophthalmic Dispensing Procedures 2	1.0
OPT 2451L	Ophthalmic Dispensing Procedures Laboratory	1.0
OPT 2505	Contact Lenses 1	3.0

Semester 4 (In Milestone)

Units: 10.0

OPT 2375	Refractometry	1.0
OPT 2375L	Refractometry Laboratory 1	1.0
OPT 2420	Eyewear Fabrication 1	2.0
OPT 2420L	Eyewear Fabrication 1 Laboratory	1.0
OPT 2506	Contact Lenses 2	2.0
OPT 2506L	Contact Lenses 2 Lab	1.0
OPT 2800L	Vision Care Clinic 1	2.0

Semester 5 (In Milestone)

Units: 12.0

OPT 2376L	Refractometry Lab 2	1.0
OPT 2421C	Eyewear Fabrication 2	3.0
OPT 2801L	Vision Care Clinic 2	4.0

OPT 2830C	Contact Lenses Clinic 1	2.0
OPT 2875L	Ophthalmic Dispensing Practicum 1	2.0

Semester 6 (In Milestone)

Units: 13.0

OPT 2060	Ophthalmic Management Policy & Procedures	2.0
OPT 2377L	Refractometry 3 Laboratory	1.0
OPT 2422C	Eyewear Fabrication 3	3.0
OPT 2802L	Vision Care Clinic 3	4.0
OPT 2831L	Contact Lense Clinic 2	1.0
OPT 2876L	Ophthalmic Dispensing Practicum 2	2.0

Total: 72.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

OPT1110 Physical & Geometrical Optics 4.000 credits

Behavior of light energy as it passes through air, plastic, glass and water with emphasis on how light is modified by prism and curved lens surfaces. These principles relate to the effect these ophthalmic devices have in correcting the errors of human vision.

Lecture Hours: 64.000

Corequisite: OPT 1205, OPT 1330

OPT1205 Ocular Anatomy, Physiology & Pathophysiology 3.000 credits

The structure and function of the systems of the human body, emphasizing the anatomy, physiology and pathophysiology of the human eye. Visual recognition of common eye disorders is also discussed.

Lecture Hours: 48.000

Corequisite: OPT 1110, OPT 1330

OPT1330 Clinical Data Collection 1 2.000 credits

Techniques necessary in a clinical environment for the collection of patient case history, entrance visual acuity, basic visual skills of ocular mobility and accommodation, color discrimination, depth perception and binocular fusion. Emphasis is also placed on gaining familiarity with the medical terminology as it relates to the visual system.

Lecture Hours: 32.000

Corequisite: OPT 1110, OPT 1205

OPT1450 Ophthalmic Dispensing Procedures 1 1.000 credits

Basic procedures of ophthalmic dispensing such as frame selection, measurement and laboratory ordering. Emphasis will be placed on common ophthalmic frame materials; crown glass and CR-39 lenses; absorptive lenses; and frame alignment, adjustment and repair. The student will demonstrate skills necessary for entry level ophthalmic dispensing in Vision Care Clinic.

Lecture Hours: 16.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1450L

OPT1450L Ophthalmic Dispensing Procedures 1 Laboratory 1.000 credits

Laboratory for OPT 1450.

Lab Hours: 32.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1450

OPT1150 Ophthalmic Lenses 2.000 credits

Characteristic of unifocal and multifocal lens reference points for proper lens selection to meet visual needs of the patients. Emphasis is on accurate positioning of the optical centers and selected multi-focal addition design. ANSI and FDA standards; prescription ordering; verification procedures; absorptive lenses; and invisible and progressive multi-focals are presented.

Lecture Hours: 32.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1331, OPT 1331L, OPT 2505, OPT 2070L

OPT1331 Clinical Data Collection 2 2.000 credits

Techniques necessary in a clinical environment for the collection of subjective and objective patient diagnostic information including visual field plotting, tonometry, lensometry, keratometry, and sphygmomanometry.

Lecture Hours: 32.000

Prerequisite: OPT1330OPT 1110, OPT 1205, OPT 1330, Corequisite: OPT 1150, OPT 1331L, OPT 2505, OPT 2070L

OPT1331L Clinical Data Collection 2 Laboratory 1.000 credits

Laboratory for OPT1331 in which students are under the supervision of a licensed practitioner.

Lab Hours: 32.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1150, OPT 1331, OPT 2505, OPT 2070L

Course User Fee: \$10.00

OPT2070L Computers for Vision Care 1.000 credits

This course introduces students to the use of computers in ophthalmic practice. Students will learn computer basics and the use of application software for maintaining patient records and billing. Elements of ophthalmic coding are included.

Lab Hours: 32.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 2505

Course User Fee: \$15.00

OPT2451 Ophthalmic Dispensing Procedures 2 1.000 credits

Theory and terminology of advanced ophthalmic dispensing. Emphasis will be placed on new technology in ophthalmic frame materials; multifocal lenses including progressive power and blended bifocals; and high index lenses. The process of analyzing the patient's specific needs for the proper frame and lens selection is highlighted.

Lecture Hours: 16.000

Prerequisite: OPT1450OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070, Corequisite: OPT 2451L

OPT2451L Ophthalmic Dispensing Procedures Laboratory 1.000 credits

Laboratory for OPT 2451.

Lab Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2451

Course User Fee: \$10.00

OPT2505 Contact Lenses 1 3.000 credits

Basic principles of contact lens fitting, emphasizing soft lenses. Topics include lens-relate terminology, anatomy and physiology, patient examination, soft lens materials, design, parameters, handling, fitting and care. Includes introduction to rigid lenses.

Lecture Hours: 48.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 2070L

OPT2375 Refractometry 1.000 credits

Students will learn the basic principles of refractometry, theoretical aspects of retinoscopy, and the use of cross cylinders. Students will be able to describe various refractive problems and their solutions.

Lecture Hours: 16.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375L, OPT 2420, OPT 2420L, OPT 2506, OPT 2506L, OPT 2800L

OPT2375L Refractometry Laboratory 1 1.000 credits

Students will practice theoretical concepts of refractometry using a retinoscope, auto-refractor, and cross cylinders in a laboratory setting.

Lab Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2420, OPT 2420L, OPT 2506, OPT 2506L, OPT 2800L

Course User Fee: \$30.00

OPT2420 Eyewear Fabrication 1 2.000 credits

Theory of ophthalmic surfacing procedures. Students acquire knowledge to arrange single vision lenses; use lensometers and lens clock; operate project-o-makers for single vision lens layout; select or fabricate frame patterns; and utilize several systems for edging lenses for ophthalmic frames.

Lecture Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2375L, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L

OPT2420L Eyewear Fabrication 1 Laboratory 1.000 credits

Laboratory for OPT 2420.

Lab Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2506, OPT 2506L, OPT 2800L

Course User Fee: \$15.00

OPT2506 Contact Lenses 2 2.000 credits

Principles of contact lens fitting, emphasizing rigid lenses. Topics include materials, design parameters, verification, handling, fitting and care. Considers advanced and specialty design and ocular complications.

Lecture Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2506L, OPT 2800L

OPT2506L Contact Lenses 2 Lab 1.000 credits

Practical procedures designed to apply technical skills of contact lens application and removal, verification of the contact lens prescription, modification of hard and hard gas permeable contact lenses, and other skills discussed in previous lecture coursework.

Lab Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2506, OPT 2800L

Course User Fee: \$36.00

OPT2800L Vision Care Clinic 1 2.000 credits

Introductory clinic designed to apply technical skills acquired in previous course work. Recording of clinical data, administrative procedures and techniques in patient handling under close staff supervision.

Clinical Hours: 96.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2506, OPT 2506L

Course User Fee: \$25.00

OPT2376L Refractometry Lab 2 1.000 credits

This course is designed to provide the student with the hands on experience of hand neutralizing a pair of glasses, retinoscopy, and the use of the phoropter and the Snellen chart.

Lab Hours: 32.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L

Course User Fee: \$26.00

OPT2421C Eyewear Fabrication 2 3.000 credits

Advanced techniques in measurement, fabrication, and verification of unifocal and multifocal lenses. Students fabricate finished eyewear from written specifications ensuring that current ANSI and FDA standards are exceeded.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2376L, OPT 2801L, OPT 2830C, OPT 2875L

Course User Fee: \$25.00

OPT2801L Vision Care Clinic 2 4.000 credits

Development of skills in patient reception, clinical data collection, assisting clinician, and ophthalmic dispensing. This is an opportunity to follow the patient through the entire cycle of vision care under close supervision of the clinical staff.

Clinical Hours: 192.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2376L, OPT 2421C, OPT 2830C, OPT 2875L

Course User Fee: \$25.00

OPT2830C Contact Lenses Clinic 1 2.000 credits

Observe and assist an optometrist in the initial fitting and follow-up care of rigid and soft contact lenses for patients referred from the Vision Care Clinic when conventional eyewear will not suffice. Familiarization with the instructions for lens handling, cleaning, care and storage of contact lenses.

Lab Hours: 64.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2875L

Course User Fee: \$10.00

OPT2875L Ophthalmic Dispensing Practicum 1 2.000 credits

Externship in an approved finishing laboratory of a retail ophthalmic dispensing establishment. The student will gain a working knowledge of ophthalmic frame and lens stock, inventory system, layout and blocking, chemical and heat treating, edging, tinting assembly and alignment of eyewear according to the written prescription.

Practicum Hours: 96.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C

OPT2060 Ophthalmic Management Policy & Procedures 2.000 credits

Procedures and terminology used in the handling of patients, correspondence, legal and ethical principles, inter- and intra-professional relationships, and office management. Develop feasibility report of opening a retail ophthalmic dispensary. The history of opticianry, optometry and ophthalmology is traced. Special emphasis is on a comprehensive review of the curriculum.

Lecture Hours: 32.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2377L, OPT 2422C, OPT 2802L, OPT 2831L, OPT 2876L

OPT2377L Refractometry 3 Laboratory 1.000 credits

This laboratory course will continue to advance the skills already introduced in the previous laboratories 1 & 2. The improved skills will enhance the student's ability to determine the refractive status of the eye and be able to practice these skills on patients in the clinic.

Lab Hours: 32.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2422C, OPT 2802L, OPT 2831L, OPT 2876L

OPT2422C Eyewear Fabrication 3 3.000 credits

A continuation of OPT 2421. Theory of evaluation and analysis of eyewear for accuracy and quality. Advanced techniques in operation of automated lens analyzer and lens edgers, and maintenance of equipment.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2377L, OPT 2802L, OPT 2831L, OPT 2876L

Course User Fee: \$25.00

OPT2802L Vision Care Clinic 3 4.000 credits

Development of additional skills in visual fields, tonography, ocular photodocumentation, vision therapy/orthoptics, low vision, aseptic techniques, eye emergencies, and assisting in triage and laboratory diagnosis of external eye disease. On and off-campus clinics will be utilized under the close supervision of optometrists and ophthalmologists.

Clinical Hours: 192.000 - 19.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2377L, OPT 2422C, OPT 2831L, OPT 2876L

OPT2831L Contact Lens Clinic 2 1.000 credits

Use of the soft contact lens instruments to confirm all the parameters for replacement lens. Particular attention is devoted to the patient that is having problems with contact lenses after long-term wear due to corneal changes and sensitivity to solutions under direct supervision of an optometrist.

Clinical Hours: 48.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2377L, OPT 2422C, OPT 2802L, OPT 2876L

Course User Fee: \$10.00

OPT2876L Ophthalmic Dispensing Practicum 2 2.000 credits

Externship in an approved retail ophthalmic dispensing establishment in the area of frame styling, ordering of appropriately designed lenses, adjustment, repair and dispensing of eyewear. The student will gain a working knowledge of administrative management procedures of the practice.

Practicum Hours: 96.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2377L, OPT 2422C, OPT 2802L, OPT 2831L

Program Description

Paralegal Studies - ABA Approved

Associate in Science | Code: 27013 | 64 credits

The Paralegal program prepares students to obtain entry-level employment in law offices, government agencies, banks or business corporations. It also enables persons working in the field without a degree to upgrade their skills to become a qualified paralegal. The MDC Paralegal program is approved by the American Bar Association. The American Bar Association defines a paralegal or as "a person, qualified by education, training or work experience who is employed or retained by a lawyer, law office, corporation, governmental agency or other entity and who performs specifically delegated substantive legal work for which a lawyer is responsible." Paralegals cannot give legal advice, set fees, negotiate or represent clients in court as these activities involve unauthorized practice of law. Paralegals work under the supervision of attorneys and are not just "document preparers," working directly with the public. Additional Information: It is necessary to see an advisor prior to beginning the program and before registering each term. For more information, please contact the Paralegal Program at 305-237-7813 or visit <https://www.mdc.edu/paralegal/>.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
PHI 2010	Introduction to Philosophy	3.0
LCO 0999	Law Center Student Orientation	0.0
BUL 2241	Business Law 1	3.0
PLA 2003	Fundamentals of Law	3.0
PLA 2931	Legal Specialty Seminar	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 14.0
PLA 2120C	Legal Research and Analysis	4.0
PLA 2704	Professionalism & Civility in the Law	3.0
PLA 2890	Human Rights Law & Procedure	3.0
REE 2040	Real Estate Principles and Practices (P&P 1)	4.0

Semester 3		Units: 12.0
MGF 1130	Mathematical Thinking	3.0
BUL 2242	Business Law 2	3.0
PLA 2114	Legal Writing	3.0
PLA 2227	Civil Law & Litigation	3.0

Semester 4		Units: 13.0
<i>Specialization Selection</i>		13.0
<i>TRACK 1 – LITIGATION</i>		13.0
PLA 2273	Torts	3.0
PLA 2303	Criminal Law & Litigation	3.0
PLA 2736	Electronic Discovery, Investigations, and Evidence	3.0

PLA 2800	Family Law	3.0
<i>Legal Seminar</i>		1.0
PLA 2934	Legal Seminar: Immigration Law	1.0
OR		
PLA 2935	Legal Seminar: Intellectual Property Law	1.0
OR		
TRACK 2 – TRANSACTION		13.0
PLA 2631C	Real Estate Transactions and Settlement	4.0
PLA 2600	Wills, Trust, Estate	3.0
PLA 2610	Introduction to Real Property	4.0
PLA 2933	Legal Seminar: Real Estate Closing	1.0
Legal Seminar*		1.0
OR		
TRACK 3 – GENERAL LAW STUDIES**		13.0

*Select 1 credit from the following course options: PLA 2934 or PLA 2935

**Must take the following courses: PLA 2303 and PLA 2736 and PLA 2800 ---and--- Select 4 credits from the following course options: PLA 2631C, PLA 2273, PLA 2600, PLA 2610, PLA 2930, PLA 2933, PLA 2934, PLA 2935,

Semester 5		Units: 12.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
PLA 2763	Law Office Management	3.0
<i>Lea</i>		3.0
PLA 1949	Paralegal Studies Practicum	3.0
OR		
PLA 2940	Paralegal Studies Internship	3.0

Total: 64.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

LCO0999 Law Center Student Orientation 0.000 credits

Law Center Student Orientation

Lecture Hours: 6.000

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

PLA2003 Fundamentals of Law 3.000 credits

This course provides students with an overview of the American legal system. It explores the basic concepts of law in society including the different sources of law. The federal, state and county court systems are examined along with judicial interpretation of the law. The course also covers the distinctions between procedural and substantive law, civil versus criminal and a court of equity and a court of law. The roles of paralegals are discussed with an emphasis given in their professional relationships, functions, career opportunities and ethical obligations.

Lecture Hours: 48.000

Pre/Corequisite: ENC1101, Corequisite: LCO0999, PLA2931

Course User Fee: \$25.00

PLA2931 Legal Specialty Seminar 1.000 credits

Intensive practical and theoretical training is provided in a seminar format. The seminar topics cover current and timely legal issues and are addressed by practicing attorneys. The topics are announced at the beginning of the Fall and Winter semesters. Corequisite: PLA 2003. (1 hr. lecture)

Lecture Hours: 16.000

Corequisite: PLA2003

PLA2120C Legal Research and Analysis 4.000 credits

Legal Research and Analysis is an introduction to the legal system, the law library system, the ethical responsibilities and skills of a paralegal when conducting research for primary and secondary sources of law in the United States. Students must earn a minimum of a C to pass this course. Pre-requisite: ENC1101; Pre/Co-requisites: PLA2003 and PLA2931 with a grade of C or better, and Departmental approval for non-Paralegal Studies majors. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: ENC1101 AND Pre/Corequisite: PLA2003, PLA2931

Course User Fee: \$45.84

PLA2704 Professionalism & Civility in the Law 3.000 credits

This class will provide students guidance as they interact in a real-world setting as legal professionals while strengthening their communication skills. This course will cultivate the students' leadership capacities at the intra-personal, interpersonal, and industry levels. The coursework will proceed through experiential learning that is grounded in theory and includes reflection. The study material will develop students' knowledge in the context of the aspiration, relational, and practical aspects of professionalism and civility in the legal field (3 credits)

Lecture Hours: 48.000

Course User Fee: \$45.84

PLA2890 Human Rights Law & Procedure 3.000 credits

This course will address theoretical and practical questions the evolution and development of the laws addressing international human rights issues.

Emphasis will be placed on the core issues, debates, struggles, failures, and successes of legal claims. (3 hr. lecture)

Lecture Hours: 48.000

REE2040 Real Estate Principles and Practices (P&P 1) 4.000 credits

Topics include real property, liens, titles, contracts, tax factors, mortgages, property evaluation, real estate market, licensing requirements, legal aspects of the real estate business, and property management. Completion of this course is required by the Florida Real Estate Commission for approval to take the State Examination.

Lecture Hours: 64.000

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

BUL2242 Business Law 2 3.000 credits

Emphasis on the laws affecting agencies, the formation and operation of partnership and corporation, personal and real property, insurance, surety ship, estates and bankruptcy, and a general review of government regulations affecting usual business operations. Prerequisite: BUL2241. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

PLA2114 Legal Writing 3.000 credits

This course provides knowledge and understanding of how to present legal research and analysis in proper written format. As legal research is an integral part of legal writing, the course will reinforce the skills used in legal research. It will also cover basic writing skills, the process of legal analysis, methodology involved in drafting a Memorandum of law, practice in drafting pleadings, and various types of specific law office correspondence.

Lecture Hours: 48.000

Prerequisite: ENC 1101, PLA 2003, PLA 2120C, PLA 2931

Course User Fee: \$50.00

PLA2227 Civil Law & Litigation 3.000 credits

This course will introduce students to a comprehensive exploration of civil procedure, pleadings, evidence, motions, damages, appeals, and alternative dispute resolution. Students will gain skills in the application of law when preparing documents before, during and post trial. A minimum of a C is required to pass this course. Prerequisites: ENC 1101, PLA 2003, PLA 2120C, and PLA 2931. (3 hr. lecture).

Lecture Hours: 48.000

Prerequisite: ENC1101 AND PLA2003 AND PLA2931, PLA2120C

Course User Fee: \$45.84

PLA2273 Torts 3.000 credits

This course provides an examination of the theories governing tort law and the use of various pre-litigation tools. Topics covered include intentional torts, negligence and strict liability. The course also requires students to utilize the knowledge obtained to draft documents employed in practice.

Lecture Hours: 48.000

Prerequisite: ENC 1101, PLA 2203, PLA 2120C, PLA 2114, PLA 2227, PLA 2931

Course User Fee: \$50.00

PLA2303 Criminal Law & Litigation 3.000 credits

This course focuses on the substantive areas of criminal law including the offenses, elements, defenses and parties to a criminal law proceeding. It also emphasizes the role of the criminal justice system in adjudicating, enforcing and sentencing criminal defendants. It examines the Florida Rules of Criminal Procedure and provides practice in drafting documents required in the conduct of a criminal trial.

Lecture Hours: 48.000

Prerequisite: ENC 1101, PLA 2203, PLA 2120C, PLA 2114, PLA 2227, PLA 2931

Course User Fee: \$50.00

PLA2736 Electronic Discovery, Investigations, and Evidence 3.000 credits

This course will examine the electronic discovery (e-discovery) process primarily following the 9 stages of the Electronic Discovery Reference Model (EDRM). It will include analysis of e-discovery case law including past and current cases; the Federal Rules of Civil Procedure related to e-discovery; different types of e-discovery including litigation, government investigations, and internal audits; and ethical issues related to e-discovery. Additionally, it will have a practical aspect that will teach e-discovery skills related to e-discovery technology, including review platforms. Students must earn a grade of C or better to pass this course. Prerequisites: ENC1101, PLA2003, PLA2120C, PLA2227, and PLA2931.

Lecture Hours: 48.000

Prerequisite: PLA2003ENC 1101, PLA 2203, PLA 2120C, PLA 2114, PLA 2227, PLA 2931

Course User Fee: \$45.84

PLA2800 Family Law 3.000 credits

An examination of the legal aspects of domestic relations. This course focuses upon dissolution of marriage law with emphasis on pleadings, discovery, and property settlements. Other areas of family law such as adoption and annulment will be reviewed.

Lecture Hours: 48.000

Prerequisite: ENC 1101, PLA 2203, PLA 2120C, PLA 2114, PLA 2227, PLA 2931

Course User Fee: \$50.00

PLA2934 Legal Seminar: Immigration Law 1.000 credits

Intensive practical and theoretical training is provided in a seminar format. The seminar topics cover current and timely legal issues and are addressed by practicing attorneys. The topics are announced at the beginning of the Fall and Winter semesters.

Lecture Hours: 16.000

PLA2935 Legal Seminar: Intellectual Property Law 1.000 credits

Intensive practical and theoretical training is provided in a seminar format. The seminar topics cover current and timely legal issues and are addressed by practicing attorneys. The topics are announced at the beginning of the Fall and Winter semesters.

Lecture Hours: 16.000

PLA2631C Real Estate Transactions and Settlement 4.000 credits

This course is a combination of both class lecture and lab. The lecture segment includes an overview of intermediate real estate law topics with a specific emphasis on Florida real estate transactions. Students who complete this course will understand the fundamental concepts and sequences followed in a real estate transaction from initial contract review, through settlement (closing). The lab segment is transaction and problem-oriented. This segment trains students in the use of the industry settlement software to complete title examinations, updates, issuance of title policies & endorsements, and various federal and state regulations that affect real estate settlements. Students may also perform various duties typically assigned to a paralegal in a real estate transaction. Prerequisite: REE2040; Corequisite: PLA2610 (3 hr. lecture, 2 hr. lab). Lecture Hours: 48.000, Lab Hours: 32.000 Prerequisite: REE2040, Corequisite: PLA2610
Course User Fee: \$45.84

PLA2600 Wills, Trust, Estate 3.000 credits

Wills, Trusts, and Estates is a study of the laws governing wills and interstate succession. The course provides practice in drafting a simple will and trust. It also examines the procedures and rules involved in probate administrations and explains the ethical obligations of attorneys and paralegals who are involved in this area of practice. Lecture Hours: 48.000 Prerequisite: PLA 2114, REE 2040
Course User Fee: \$50.00

PLA2610 Introduction to Real Property 4.000 credits

This is an introductory course that provides an overview of the general concepts of real property law applied in most states, including substantive legal issues, ownership, terms and various legal forms, the mechanics of various residential, some commercial transactions, and mortgage foreclosures. Prerequisite: REE2040; Corequisite: PLA2631C (3 hr. lecture) Lecture Hours: 64.000 Prerequisite: REE2040, Corequisite: PLA2631C
Course User Fee: \$45.84

PLA2933 Legal Seminar: Real Estate Closing 1.000 credits

Intensive practical and theoretical training is provided in a seminar format. The seminar topics cover current and timely legal issues and are addressed by practicing attorneys. The topics are announced at the beginning of the Fall and Winter semesters. Prerequisite: REE 2040. (16 hr. Seminar) Lecture Hours: 16.000 Prerequisite: REE2040

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method. Lecture Hours: 48.000
Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues. Lecture Hours: 48.000
Course User Fee: \$10.00

**EVR1001 Introduction to Environmental Science
3.000 credits**

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PLA2763 Law Office Management 3.000 credits

A survey of economical and efficient law office practices and procedures including the proper use of law office equipment; business data processing law office management, personnel selection, training and management; employer/employee relationships; correct utilization of time and space; correct time keeping and billing procedures.

Lecture Hours: 48.000

Prerequisite: PLA2003

Course User Fee: \$50.00

PLA1949 Paralegal Studies Practicum 3.000 credits

This is a course designed to continue training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Co-operative Education Office to obtain registration approval.

Other Hours: 144.000

Prerequisite: Departmental Permission Required

PLA2940 Paralegal Studies Internship 3.000 credits

This course provides students with supervised practical experience in a law office, corporate legal department, government agency, court, or other legal setting. Students apply knowledge and skills acquired through paralegal studies coursework to real-world legal environments, gaining experience in legal research, document preparation, case management, client communication, and professional ethics. The internship is designed to enhance professional development, strengthen workplace competencies, and provide exposure to the responsibilities of a paralegal within the legal profession.

Other Hours: 144.000

Prerequisite: Departmental Permission Required

Program Description

Photographic Technology

Associate in Science | Code: 26032 | 64 credits

The Photographic Technology program is designed to meet individual students' needs for either further study or immediate employment in the field of commercial and industrial photography. Students develop a wide variety of photographic and art-related skills and the ability to use these skills to produce commercially viable photographs. Instruction covers portrait photography, still photography, fashion photography, illustrative photography as well as the business skills needed to manage a photographic enterprise. Various internships such as in biomedical and forensic technology are available to students.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 14.0**

ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
ART 1202C	Two-Dimensional Design	4.0
ART 1300C	Drawing	4.0
Course Competency Test*		

* Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2**Units: 14.0**

Humanities	3.0
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ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
PGY 2401C	Introduction to Photography	4.0
PGY 2404C	Intermediate Photography	4.0
Semester 3		Units: 12.0
PGY 2110C	Color Photography 1	4.0
PGY 2211	Portrait And Still Photography	4.0
PGY 2238	Illustrative Photography 1	4.0
Semester 4		Units: 12.0
PGY 2111C	Color Photography 2	4.0
PGY 2222	Fashion Photography	4.0
PGY 2239	Illustrative Photography 2	4.0
Semester 5		Units: 12.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
PHY 1020	General Education Physics	3.0
PGY 2470C	Photography Portfolio	4.0
Electives*		5.0

*Select 5 credits from the following course options: ART 1205C, ART 2600C, ART 2601C, ART 2602C, ART 2802C, GRA 1280C, GRA 2117C, GRA 2121C, GRA 2151C, GRA 2156C, GRA 2203C, PGY 2112C, PGY 2475, PGY 2800C

Total: 64.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

ART1202C Two-Dimensional Design 4.000 credits

This course is designed to give students an understanding of advanced concepts of two-dimensional design and to give hands on opportunity to transform visual and experiential information into two-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals.

Lecture Hours: 32.000, Lab Hours: 64.000

ART1300C Drawing 4.000 credits

Basic problems in freehand drawing, including perspective, still-life and landscape. Emphasis is on developing a sense of structure through line, form and texture.

Lecture Hours: 32.000, Lab Hours: 64.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PGY2401C Introduction to Photography 4.000 credits

Fundamentals of black and white photography as an art medium with emphasis on composition, design and processing. Students will supply their own camera, film and paper.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ART 1203C, ART 1300C, or Equivalent

Course User Fee: \$50.00

PGY2404C Intermediate Photography 4.000 credits

Emphasis on achieving more technical control of black and white photography with introduction to larger format photography utilization of studio aspects such as strobe, quartz lighting and view camera controls continued development of aesthetics.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

PGY2110C Color Photography 1 4.000 credits

An introductory course in the making of Type C photographic prints, including the darkroom techniques of developing color film, color filtering, color balance and density control. There will be an exploration of significant contributions to the aesthetics of color photography. Students must provide their own cameras, film and photographic paper.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2401C

Course User Fee: \$50.00

PGY2211 Portrait And Still Photography 4.000 credits

Fundamentals of portrait and still photography are presented. Basic and advanced exercises are taught in lighting, posing, make-up and camera angles. Composition, lighting and design functioning to describe people and objects for a variety of clients are explored.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2401C

PGY2238 Illustrative Photography 1 4.000 credits

The use of the camera to illustrate either an original concept or a concept provided by an art director for clients such as magazines, manufacturing concerns, advertising agents, newspapers, technical publications and schools. The creative approach is stressed in planning and production-effective color and black/white illustrations.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2401C

PGY2111C Color Photography 2 4.000 credits

Deals primarily with printing methods used in printing color negatives. Concentrated practice is given in light, color balancing, exposure and processing of color printing materials; the techniques of producing matched multi-size prints are demonstrated.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2110C

Course User Fee: \$50.00

PGY2222 Fashion Photography 4.000 credits

The production of commercially viable photographs illustrating clothes as desirable objects as well as recent trends in fashion industry are studied. An awareness of mood, make-up, and dramatic impact is stressed.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2239

Course User Fee: \$25.00

PGY2239 Illustrative Photography 2 4.000 credits

A sophisticated level of photographic illustration is reached and emphasis is given to conceptual and visual continuity. Concepts, methods and techniques necessary to produce slide presentations for variety of clients are stressed. Seminars and conferences prepare students for the business aspects of the illustration and advertising markets.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2238

Course User Fee: \$25.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

PGY2470C Photography Portfolio 4.000 credits

This course is the culmination of artistic and technical experiences acquired in previous photography courses. Students explore portfolio creation through a semester-long analysis, crafting an artist statement, and studio/critique experiences. The course addresses professional practices such as the creation of a resume/CV, a website, and a social media profile. Students gain insights into strategic approaches to pursue opportunities such as transfer applications, seeking exhibitions/grants, entrepreneurship, and job interview readiness.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY 2111C, PGY 2211, PGY 2222, PGY 2238, PGY 2239

Course User Fee: \$25.00

Program Description

Physical Therapist Assistant

Associate in Science | Code: 23034 | 74 credits

The Physical Therapist Assistant program prepares students for employment in hospitals, rehabilitation centers, nursing homes, private practices or other qualified health agencies. Graduates will work under the supervision of a physical therapist in the promotion of optimal human health and function through the application of scientific principles to prevent, identify, correct or alleviate acute or prolonged physical disability of anatomic or physiologic origin. Externship or clinical practice is conducted in local health care facilities under the supervision of qualified professional personnel. The program is accredited by the Commission on Accreditation in Physical Therapy Education. Graduates of the program are eligible to take the State Board Examination and receive an Associate in Science degree in Physical Therapist Assisting.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 18.0**

ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0

BSC 2085L	Human Anatomy & Physiology Lab 1	1.0
MAC 1105	College Algebra	3.0
PHY 1004	Physics With Applications 1	3.0
PHY 1004L	Physics with Applications 1 Lab	1.0
HIM 2472	Medical Terminology	3.0
<i>Program Electives</i>		1.0
IDS 1107	Tools for Success	1.0
OR		
LIS 2004	Digital Research Strategies	1.0
OR		
MAT 1033	Intermediate Algebra	3.0
OR		
SPC 1017	Introduction to Communication	3.0
OR		
PHI 2600	Introduction to Ethics	3.0
OR		
PSY 2012	Introduction to Psychology	3.0
Course Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2	Units: 11.0
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BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
PHT 1102C	Anatomy for the Physical Therapist Assistants	4.0

Semester 3	Units: 10.0
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PHI 2010	Introduction to Philosophy	3.0
PHT 1201	Introduction to Physical Therapy	2.0
PHT 1201L	Introduction to Physical Therapy Laboratory	1.0
PHT 1211	Disabilities and Therapeutic Procedures 1	2.0
PHT 1211L	Disabilities and Therapeutic Procedures 1 Lab	2.0

Semester 4	Units: 10.0
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PHT 2120	Applied Kinesiology	2.0
PHT 2120L	Applied Kinesiology Laboratory	1.0
PHT 2224	Disabilities and Therapeutic Procedures II	3.0
PHT 2224L	Disabilities and Therapeutic Procedures II Lab	2.0
PHT 2801C	Clinical Practice and Conference I	2.0

Semester 5	Units: 15.0
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PHT 2162	Survey of Neurological Deficits	3.0
PHT 2701	Rehabilitation Procedures	3.0
PHT 2701L	Rehabilitation Procedures Laboratory	2.0
PHT 2810	Clinical Practice and Conference II	7.0

Semester 6

Units: 10.0

PHT 2820	Clinical Practice and Conference III	7.0
PHT 2931	Seminar for Physical Therapist Assistants	3.0

Total: 74.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy & Physiology Lab 1 1.000 credits

In this laboratory course, student will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PHY1004 Physics With Applications 1 3.000 credits

Emphasizes the basic concepts and principles and their practical applications. Designed specifically for students in technical studies and for others wishing to strengthen their physics background before taking advanced courses.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: PHY1004L

Course User Fee: \$10.00

PHY1004L Physics with Applications 1 Lab 1.000 credits

Laboratory for PHY 1004.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: PHY1004

Course User Fee: \$20.00

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

IDS1107 Tools for Success 1.000 credits

This course is for students majoring in science, technology, engineering and mathematics fields (STEM). Students will learn writing, research, presentation, and technological skills necessary for success in STEM-related disciplines. Course topics include learning styles, collaborative skills, power study techniques and will use related technologies related to STEM.

Lecture Hours: 16.000

LIS2004 Digital Research Strategies 1.000 credits

LIS 2004, Digital Research Strategies, equips students with information literacy skills to locate, evaluate, and ethically use information in an evolving digital landscape. Students will apply critical thinking and research techniques to explore traditional and emerging research tools. (1 hr. lecture)

Lecture Hours: 16.000

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.

Course User Fee: \$18.00

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

PHT1102C Anatomy for the Physical Therapist Assistants 4.000 credits

Regional description of the musculoskeletal landmarks utilized in implementing and documenting assessment and treatment procedures in physical therapy. Prerequisite: Departmental Approval, Corequisites: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L. (3 hr. lecture and 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Departmental Approval, Corequisite: PHT1201 AND PHT1201L AND PHT1211 AND PHT1211L

Course User Fee: \$438.81

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

PHT1201 Introduction to Physical Therapy 2.000 credits

This course introduces students to the survey and history of the physical therapy profession. The role and responsibilities of the physical therapist assistant as they interact with patients and other health care workers are discussed. It incorporates discussion on legal and ethical issues, levels of supervision, and the medical record. An overview of common pathologies is presented. Prerequisites: BSC 2085, BSC 2085L, PHY 1004, PHY 1004L; Corequisites: PHT 1102C, PHT 1201L, PHT 1211, PHT 1211L. (2 hr. lecture)

Lecture Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, Corequisite: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L.

PHT1201L Introduction to Physical Therapy Laboratory 1.000 credits

Laboratory course for PHT1201 Introduction to Physical Therapy. This course includes laboratory practice for basic patient care skills and treatment procedures including body mechanics, vital signs, transfers, gait training, ROM, flexibility, manual resistance. Case scenarios to identify proper transfer techniques and gait training patterns are utilized. Principles of effective documentation and data collection are introduced.

Lab Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, Corequisite: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L.

Course User Fee: \$20.00

PHT1211 Disabilities and Therapeutic Procedures 1 2.000 credits

This course introduces theory and practical applications of physical therapy biophysical agents. The physiologic effects, indications, contraindications, and intervention parameters of patient care interventions including electrotherapeutic agents, compression therapies, cryotherapy, hydrotherapy, superficial and deep thermal agents, traction, and therapeutic massage are presented.

Lecture Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, Corequisite: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L.

PHT1211L Disabilities and Therapeutic Procedures 1 Lab 2.000 credits

Laboratory course for PHT1211 Disabilities and Therapeutic Procedures 1. This course includes laboratory practice of technical skills relating to practical applications of physical therapy biophysical agents including electrotherapeutic agents, compression therapies, cryotherapy, hydrotherapy, superficial and deep thermal agents, and traction. Students develop basic skills in therapeutic massage. Practical application of each intervention is emphasized.

Lab Hours: 64.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, Corequisite: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L.

Course User Fee: \$25.00

PHT2120 Applied Kinesiology 2.000 credits

Students learn concepts and principles of applied kinesiology and biomechanics with emphasis on the function of the musculoskeletal system as it pertains to human motion. Anatomical structures and movement concepts covered in PHT 1102C Anatomy for PTA are expanded upon. Goniometric measurement, manual muscle testing skills, plumb-line analysis, data collection skills, palpation skills, understanding of special tests, and concepts of reliability and validity as it relates to goniometry are emphasized.

Lecture Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L; , Corequisite: PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C

PHT2120L Applied Kinesiology Laboratory 1.000 credits

Laboratory course for PHT 2120 Applied Kinesiology. This course provides laboratory practice of technical skills relating to practical applications of foundational concepts and principles pertaining to Applied Kinesiology. Students learn goniometric measurement, manual muscle testing skills, plumb-line analysis, palpation skills, data collection, and develop an understanding of special tests.

Lab Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L , Corequisite: PHT 2120, PHT 2224, PHT 2224L, PHT 2801C

Course User Fee: \$25.00

PHT2224 Disabilities and Therapeutic Procedures II 3.000 credits

This course introduces foundational concepts and principles pertaining to therapeutic exercise prescription and interventions for the upper and lower extremity, spine, and Women's Health across the lifespan and continuum of care. Applied science of exercise and techniques, exercise interventions by body region, peripheral joint mobilization, and medical and surgical procedures and protocols are emphasized. Students engage in active and collaborative learning and utilize case scenarios for practical application.

Lecture Hours: 48.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L , Corequisite: PHT 2120, PHT 2120L, PHT 2224L, PHT 2801C

PHT2224L Disabilities and Therapeutic Procedures II Lab 2.000 credits

Laboratory course for PHT 2224 Disabilities and Therapeutic Procedures 2. This course is designed to provide students with technical skill practice and application of foundational concepts and principles pertaining to therapeutic exercise prescription and interventions for the upper and lower extremity, spine, and Women's Health across the lifespan and continuum of care. Students apply exercise and therapeutic interventions by body region, practice documentation, prescribe home exercise programs, and are introduced to peripheral joint mobilization for the upper and lower extremities.

Lab Hours: 64.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L, Corequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2801C

Course User Fee: \$25.00

PHT2801C Clinical Practice and Conference I 2.000 credits

The students will be exposed to clinical experiences in supervised patient care activities in a variety of clinical facilities including general hospitals and physical therapy clinics.

Clinical Hours: 96.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L, Corequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L

Course User Fee: \$389.50

PHT2162 Survey of Neurological Deficits 3.000 credits

This course introduces the etiology, pathophysiology, and clinical manifestations of common neurological diseases/conditions treated in physical therapy. Neuroanatomy of the central and peripheral nervous system is reviewed. Reflex integration as well as normal growth and development are discussed. Students have an end of course project-based topic assignment that pertains to commonly seen neurological conditions in physical therapy.

Lecture Hours: 48.000

Prerequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C , Corequisite: PHT 2701, PHT 2701L, PHT 2810

PHT2701 Rehabilitation Procedures 3.000 credits

This course presents treatment techniques related to physical therapy interventions and rehabilitation concepts and principles for neurological and other medical conditions across the lifespan and continuum of care.

Lecture Hours: 48.000

Prerequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C, Corequisite: PHT 2162, PHT 2701L, PHT 2810

PHT2701L Rehabilitation Procedures Laboratory 2.000 credits

Laboratory course for PHT 2701 Rehabilitation Procedures. This course provides laboratory practice of technical skills relating to physical therapy interventions and rehabilitation concepts and principles for neurological and other medical conditions across the lifespan and continuum of care.

Lab Hours: 64.000

Prerequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C, Corequisite: PHT 2162, PHT 2701L, PHT 2810

Course User Fee: \$25.00

PHT2810 Clinical Practice and Conference II 7.000 credits

Intermediate clinical experiences in selected patient care activities under the supervision of a licensed physical therapist.

Clinical Hours: 336.000

Prerequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C, Corequisite: PHT 2162, PHT 2701, PHT 2701L

PHT2820 Clinical Practice and Conference III 7.000 credits

Advanced clinical experiences in patient care activities under the direct supervision of a licensed physical therapist.

Clinical Hours: 336.000

Prerequisite: PHT 2162, PHT 2701, PHT 2701L, PHT 2810, Corequisite: PHT 2931

Course User Fee: \$117.00

PHT2931 Seminar for Physical Therapist Assistants 3.000 credits

Recognition of the expected current competency levels, and ethical and legal responsibilities of the physical therapist assistant in the health care system. A comprehensive course designed to enhance students' understanding of the curricular content. It provides a thorough review of the curriculum, ensuring that students consolidate their knowledge and skills. Special topics are presented. The role of the PTA including ethical and legal responsibilities are emphasized. The course prepares students for licensure requirements and preparation for the National Physical Therapy Examination (NPTE-PTA) and the Florida Laws and Rules examination.

Lecture Hours: 48.000

Prerequisite: PHT 2162, PHT 2701, PHT 2701L, PHT 2810, Corequisite: PHT 2820

Course User Fee: \$30.00

Program Description

Professional Pilot Technology

Associate in Science | Code: 26029 | 64 credits

The Professional Pilot Technology program is primarily developed to meet the challenging regional airline requirements for pilots; therefore, graduates of the program will earn the following Federal Aviation Administration (FAA) Certificates: Commercial Pilot with Single and Multi-Engine Ratings. In addition, these certificates can be applied toward a Certified Flight Instructor (CFI) Certificate. Additional Information: Students interested in this program must first pass an FAA Class I medical evaluation prior to beginning classes. Cost of flight training is in addition to normal tuition costs. Contact the Aviation Department at (305) 237-5950 for information and advisement.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: Yes

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
AST 1002	Descriptive Astronomy	3.0
ASC 1210	Aviation Meteorology	3.0
ATT 1100	Private Pilot Theory	3.0
Course Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 15.0
MAC 1105	College Algebra	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
ASC 1550	Aerodynamics	3.0
ASC 2320	Aviation Laws and Regulations	3.0
ATF 1100L	Private Pilot Flight Accelerated	3.0

Semester 3	Units: 12.0
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ASC 1610	Aircraft Engines and Structure Theory	3.0
ASC 2470	Physiology-Psychology of Flight	3.0
ATF 2305L	Instrument Pilot Flight Accelerated	3.0
ATT 2820	Air Traffic Control	3.0

Semester 4

Units: 12.0

ATF 2214L	Commercial Pilot Flight Accelerated	3.0
ATT 2110	Commercial Pilot Theory 3	3.0
ATT 2120	Instruments Pilot Theory	4.0
ATT 2133	Multi-Engine Pilot Theory	2.0

Semester 5

Units: 13.0

PHI 2010	Introduction to Philosophy	3.0
<i>Program Elective</i>		10.0
<i>Group A</i>		3.0
ATT 2821	Air Traffic Control (ATC) Radar	3.0
OR		
ATT 2822	VFR Tower Operations	3.0
OR		
ATT 2823	Air Traffic Control (ATC) Non-Radar	3.0
OR		
AVM 1949	Co-Op Work Experience 1: AVI	3.0
OR		
<i>Group B</i>		3.0-4.0
ASC 1560	Introduction to Unmanned Aerial Vehicles and Systems (UAVs)	3.0
OR		
ASC 1563	UAS Applications in Aerial Photography	3.0
OR		
ATT 2995	Occupational Practicum	4.0

Total: 64.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

ASC1210 Aviation Meteorology 3.000 credits

This is a core aviation course. The student will be prepared to understand weather and environmental issues in commercial aviation. Topics covered will be atmospheric phenomena relating to aircraft operations, the analysis and use of weather data as presented by the U.S. National Weather Service.

Prerequisite: ATT1100 or equivalent; Corequisite: ATT2110 or equivalent.

Lecture Hours: 48.000

ATT1100 Private Pilot Theory 3.000 credits

This course introduces basic subjects pertaining to pilot knowledge including basic aircraft systems, aircraft operation and performance, aerodynamic principles, human factors, and aeronautical decision making. This Course can be instructed as Part 61 or Part 141.

Lecture Hours: 48.000

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ASC1550 Aerodynamics 3.000 credits

This is a basic course in aerodynamics. Students will analyze the physics of flight and the application of basic aerodynamics to both airframe and power plant as preparation for the requirements of commercial aviation.

Lecture Hours: 48.000

ASC2320 Aviation Laws and Regulations 3.000 credits

Insight pertinent to federal governing bodies, and current local, federal and international laws forming the present structure of aviation law.

Lecture Hours: 48.000

ATF1100L Private Pilot Flight Accelerated 3.000 credits

This course provides flight training in the areas required to safely perform the duties of a private pilot. This course is to be completed in less than 16 weeks using "total immersion" approach to training to help lower the attrition rates for students. It fulfills the requirements for private pilot certification outlined in part 141 of the Federal Aviation Regulations as presented in the Jeppesen Sanderson Private Pilot Syllabus. Upon satisfactory completion of this course and the Federal Aviation Administration (FAA) knowledge and practical exams, the applicant will receive an FAA private pilot certificate. Special Fee. Note: This course is not approved for Veterans seeking to use Veterans Educational Benefits.

Lecture Hours: 48.000

Prerequisite: ATT1100, FAA First Class Medical Certificate.

Course User Fee: \$15094.00

ASC1610 Aircraft Engines and Structure Theory 3.000 credits

This is a foundation course in aircraft engines and structure. Students will learn the elements of aircraft engines, engine theory, construction, systems, operating procedures, performance diagnosis, and aircraft structures.

Lecture Hours: 48.000

ASC2470 Physiology-Psychology of Flight 3.000 credits

This is an introductory course in the physiology and psychology of flight. Students will learn aero-medical facts of significance to pilots, including causes, symptoms, prevention and emergency treatment of ailments common to the aviation environment through a basic understanding of a person's normal functioning. Cabin pressurization, communications, decompression sickness, hyperventilation, hypoxia, self-imposed stresses, spatial disorientation and vision are examined.

Lecture Hours: 48.000

ATF2305L Instrument Pilot Flight Accelerated 3.000 credits

This accelerated course provides the flight training required to safely conduct flights as an instrument rated pilot. This course is to be completed in less than 16-weeks. The training is conducted in accordance with Codes of Federal Regulations (CFR) Part 141 of the Federal Aviation Regulations as outlined in the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course and the Federal Aviation Administration (FAA) knowledge test and practical test, the applicant will receive an FAA Instrument Rating. Minimum approved FAA CFR Part 141 course hours include 35 hours of flight.
Lecture Hours: 48.000
Prerequisite: FAA Private Pilot Certificate, Corequisite: ATT2120and Current FAA 3rd Class Medical (1st Class preferred)
Course User Fee: \$12686.00

ATT2820 Air Traffic Control 3.000 credits

The basic elements of air traffic control operations, providing the necessary foundation for successful completion of the Air Traffic Control Basic Certification Examination. Prerequisite: sophomore standing in major program.
Lecture Hours: 48.000
Prerequisite: Sophomore standing in major program.

ATF2214L Commercial Pilot Flight Accelerated 3.000 credits

This accelerated course of instruction provides training required to allow the student to safely conduct flight as a Commercial Pilot. This course is expected to be completed in less than 4 months. The training will be conducted in accordance with the Codes of Federal Regulations (CFR) Part 141. This training will be completed utilizing the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course, the FAA knowledge test, and FAA practical test, the student will be awarded an FAA Commercial Pilot certificate. A 1st Class Medical Certificate with Instrument Rating is required. Minimum approved FAA CFR Part 141 course hours include 120 hours of flight. Special Fee.
Lecture Hours: 48.000
Recommended Preparation: Required: FAA Private Pilot Certificate and FAA 2nd Class Medical Certificate (1st Class preferred).
Course User Fee: \$23990.50

ATT2110 Commercial Pilot Theory 3 3.000 credits

This 35-hour course provides students with the aeronautical knowledge required to act as Commercial Pilot. Students will prepare for the FAA Commercial Written Exam. Private Pilot Certificate with Instrument Rating required.
Lecture Hours: 48.000
Prerequisite: ATF2305

ATT2120 Instruments Pilot Theory 4.000 credits

This 30-hour course introduces basic theories of instrument pilot operations to prepare students for the FAA Instrument Written Exam. Students will acquire aeronautical knowledge required to act as an Instrument Rated Pilot. It will prepare the students for the FAA Instrument Written Exam. Private Pilot Certificate required.
Lecture Hours: 64.000

ATT2133 Multi-Engine Pilot Theory 2.000 credits

This course introduces basic theories of multi-engine pilot operations to prepare students for the FAA Multi-Engine oral and practical exams. Students will acquire the aeronautical knowledge required to act as a multi-engine rated pilot.
Lecture Hours: 32.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.
Lecture Hours: 48.000

ATT2821 Air Traffic Control (ATC) Radar 3.000 credits

This course will provide the student with a fundamental knowledge of air traffic control practices, policies and procedures as they relate to the specifics of the controller function in an air traffic radar operating environment, with air traffic controllers utilizing the radar for traffic separation. The liberal use of the figures and example phraseology assist the student in achieving an overall use of understanding of the air traffic control system. A radar air traffic control simulator is utilized to provide realistic training exercises for the students.

Prerequisite: ASC1210.

Lecture Hours: 48.000

Prerequisite: ASC1210

Course User Fee: \$300.00

ATT2822 VFR Tower Operations 3.000 credits

This course expands the knowledge attained from ATT 2820 and is designed to further develop the aviation students' skills in the ATC environment. Emphasis is placed on the duties and responsibilities of operational positions in local, ground, flight data, and coordination. Students will also learn the FAA regulations which govern flight under visual conditions. Optimum use of the Hughes Virtual Tower incorporated into this course. Prerequisite: ATT2820. Special fee.

Lecture Hours: 48.000

Prerequisite: ATT2820

Course User Fee: \$300.00

ATT2823 Air Traffic Control (ATC) Non-Radar 3.000 credits

In this course, future air traffic controllers will acquire an understanding of air traffic control practices, policies and procedures and their application in a non-radar air traffic environment. Throughout this course, (Non-Radar Procedures) appropriate real-life examples are used to illustrate the reasoning behind procedures used by air traffic controllers utilizing the non-radar methods. The liberal use of figures and example phraseology is used to assist the student in achieving an overall understanding of the air traffic control system. Prerequisites: ATT2820, ASC1210.

Special fee.

Lecture Hours: 48.000

Prerequisite: ASC1210, ATT2820

Course User Fee: \$20.00

AVM1949 Co-Op Work Experience 1: AVI 3.000 credits

This is a course designed to provide training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Co-operative Education Office to obtain registration approval. Prerequisite: 2.0 minimum GPA, approval of Co-op Program Director, minimum of 6 credits in field or work approved experience.

Lecture Hours: 48.000

ASC1560 Introduction to Unmanned Aerial Vehicles and Systems (UAVs) 3.000 credits

This course introduces the fundamentals of unmanned aerial vehicles and systems, covering UAV types, components, principles of flight, regulations, and safe operational practices. Students will explore industry applications and gain foundational knowledge for further study in UAV technology.

Lecture Hours: 48.000

Course User Fee: \$59.33

ASC1563 UAS Applications in Aerial Photography 3.000 credits

Students learn techniques and best practices for capturing aerial images using unmanned aerial systems. The course covers camera operation, flight planning for photography missions, image processing, and practical applications in various industries.

Lecture Hours: 48.000

Course User Fee: \$59.33

ATT2995 Occupational Practicum 4.000 credits

This course provides supervised, work-based learning experience in an aviation setting that enables students to apply classroom knowledge in a professional environment. Students gain practical experience in aviation operations, safety procedures, regulatory compliance, communication, and workplace professionalism while developing career readiness and industry-specific skills through guided experiential learning.

Lecture Hours: 64.000

Program Description

Radiation Therapy

Associate in Science | Code: 23058 | 77 credits

The Radiation Therapy program is a comprehensive educational program designed to prepare students for a career as a skilled and knowledgeable Radiation Therapist. As a key member of the healthcare team, Radiation Therapists use various forms of radiation, such as high-energy X-rays and electron beams, to treat cancer patients. The program provides students with extensive knowledge of radiation therapy procedures, treatment planning, and imaging technologies. In addition, students develop cognitive skills such as critical thinking, problem-solving, and effective communication with patients, their families, and other healthcare professionals.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
PHI 2010	Introduction to Philosophy	3.0
Semester 2		Units: 13.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
MAC 1105	College Algebra	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
HIM 2472	Medical Terminology	3.0
Semester 3 (In-Milestone)		Units: 13.0
RAT 1001	Introduction to Radiation Oncology	2.0
RAT 1615	Radiation Therapy Medical Imaging	3.0
RAT 1804	Clinical Education I	5.0
RAT 2123	Patient Care in Radiation Therapy	3.0
Semester 4 (In-Milestone)		Units: 12.0
RAT 1021C	Principles and Practices of Radiation Therapy I	2.0
RAT 1242	Clinical Oncology & Pathology	2.0
RAT 1614C	Radiation Therapy Physics 1	2.0
RAT 1814	Clinical Education II	5.0
RAT 2240C	Radiation Oncology Sectional Anatomy	1.0
Semester 5 (In-Milestone)		Units: 16.0
RAT 1657	Radiation Protection/Quality Assurance	1.0
RAT 1824L	Clinical Education III	8.0
RAT 2022C	Principles & Practice of Radiation Therapy 2	2.0
RAT 2243C	Clinical Oncology and Neoplasms	3.0
RAT 2618C	Radiation Therapy Physics II	2.0
Semester 6 (In-Milestone)		Units: 13.0
RAT 1619	Elements of Treatment Planning	2.0

RAT 2061	Radiation Therapy Seminar	3.0
RAT 2241	Radiobiology	2.0
RAT 2834L	Clinical Education IV	6.0

Total: 77.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

RAT1001 Introduction to Radiation Oncology 2.000 credits

Introduction to the clinical setting in a radiation therapy department. The course includes radiation protection, mathematical concepts in radiation oncology, and medical terminology in the treatment of patients in a radiation oncology setting.

Lecture Hours: 32.000

RAT1615 Radiation Therapy Medical Imaging 3.000 credits

This course is an introductory study to radiographic processes which includes the processes behind computed tomography, magnetic resonance imaging, nuclear medicine, positron emitting tomography, and ultrasound as it pertains to simulation, detection, and diagnosis of cancer.

Lecture Hours: 48.000

RAT1804 Clinical Education I 5.000 credits

Students will learn radiation therapy procedures in a local radiation therapy department. Students are closely supervised by certified radiation therapy technologists as they are introduced to record-keeping and treatment units.

Clinical Hours: 240.000

Course User Fee: \$10.50

RAT2123 Patient Care in Radiation Therapy 3.000 credits

This course presents patient care procedures relevant to the radio logic health science student. Content includes specific knowledge and skills required for patient care such as pharmacology, vital sign monitoring, recognizing side effects of treatment, nutritional needs of patients and psychological needs of patients.

Lecture Hours: 48.000

RAT1021C Principles and Practices of Radiation Therapy I 2.000 credits

A study of all major radiotherapy equipment such as linear accelerators and superficial ortho- and megavoltage units. Auxiliary equipment such as simulators, immobilization devices, beam directors and modifiers will also be discussed. Patient positioning, treatment planning, patient flow, and quality assurance will be presented in detail.

Lecture Hours: 16.000, Clinical Hours: 48.000

RAT1242 Clinical Oncology & Pathology 2.000 credits

Malignant conditions, etiology, and methods of treatment. Patient management, treatment planning, patient prognosis, treatment results, and the use and effect of combined therapies will be discussed. Contributing factors, growth and biologic behavior of neoplastic diseases as well as specific types of tumors and tumor sites will also be discussed.

Lecture Hours: 32.000

RAT1614C Radiation Therapy Physics 1 2.000 credits

A basic radiation physics course containing fundamental principles and concepts. The course includes radiation production, properties, and characteristics as well as structure of the atom and matter, electrostatics, magnetism, electrodynamics, and the electromagnetic spectrum.

Lecture Hours: 16.000, Clinical Hours: 48.000, Other Hours: 48.000

RAT1814 Clinical Education II 5.000 credits

Continued patient treatment assignments. The responsibilities of the students increase as more complex competencies in patient treatment are mastered under direct supervision

Clinical Hours: 240.000

Prerequisite: RAT1804

Course User Fee: \$10.50

RAT2240C Radiation Oncology Sectional Anatomy 1.000 credits

An introduction to the concept of disease and general pathology. The types of growth, causative factors and biological behavior of neoplastic diseases are stressed. Pharmacology with emphasis on the radiation therapy patient is included in this course. Clinical Hours: 48.000

RAT1657 Radiation Protection/Quality Assurance 1.000 credits

The student will learn to present basic principles of radiation protection and safety in radiation therapy. Radiation health and safety requirements of federal and state regulatory agencies, accreditation agencies and health care organizations are included. Specific responsibilities of the radiation therapist are discussed, examined and evaluated.

Lecture Hours: 16.000

RAT1824L Clinical Education III 8.000 credits

Continuation of advanced patient treatment competencies under the supervision of an ARRT Certified Radiation Therapy Technologist.

Clinical Hours: 384.000

Prerequisite: RAT1814

RAT2022C Principles & Practice of Radiation Therapy 2 2.000 credits

Continued application of radiation therapy and its effectiveness in treatments. Advanced patient positioning, planning and flow, and quality assurance will be discussed.

Lecture Hours: 16.000, Clinical Hours: 48.000

Prerequisite: RAT1021C

RAT2243C Clinical Oncology and Neoplasms 3.000 credits

A continuation of medical oncology and pathology
Lecture Hours: 1.000 - 16.000, Clinical Hours: 96.000

RAT2618C Radiation Therapy Physics II 2.000 credits

Specifics of ionizing radiation such as details of production, interactions, and types of radiation and their application to the patient treatment. Properties of production, photon interactions, beam characteristics, and particle irradiation will be discussed.

Lecture Hours: 16.000, Clinical Hours: 48.000

RAT1619 Elements of Treatment Planning 2.000 credits

Determination of radiation doses in treatment planning using computerized methodology.
Lecture Hours: 32.000

RAT2061 Radiation Therapy Seminar 3.000 credits

This course will provide the opportunity for the radiation therapy student to evaluate their cumulative knowledge through comprehensive testing, refinement of accumulated knowledge, and retention of all aspects of radiation therapy. The course challenges the student to be prepared for the American Registry of Radiologic Technologist (ARRT) comprehensive national examination upon completion of graduation.

Lecture Hours: 48.000

RAT2241 Radiobiology 2.000 credits

Principles of cell response to radiation. Factors influencing the effects of radiation, tissue sensitivity, and environmental factors are discussed.

Lecture Hours: 32.000

RAT2834L Clinical Education IV 6.000 credits

This course includes clinical rotations through the radiation therapy department. Students will be provided the opportunity to apply theory learned from the previous semester in the various areas of the treatment process.

Clinical Hours: 288.000

Prerequisite: RAT1824L

Program Description

Radio & Television Broadcast Programming

Associate in Science | Code: 26043 | 64 credits

The Radio and Television Broadcasting Programming program is designed for students who intend to seek employment in radio, television and production companies, as well as allied fields such as in-house educational and industrial studios. The curriculum provides introductory and advanced courses essential to the professional program. It stresses hands-on equipment use in both the radio and TV laboratories. Students will have access to high-end cameras, editing suites and video graphics animation facilities and will complete portfolio-quality productions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
FIL 2552C	Editing Level 1: Introduction to Editing	3.0
RTV 1241C	Television Production 1	4.0
RTV 2300	Broadcast Writing	3.0
VIC 1000	Visual Communications	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
ENC 1101	English Composition 1	3.0
FIL 2553C	Editing Level 2: Intermediate Editing and Visual Effects	3.0
RTV 1000	Fundamentals of Broadcasting	3.0
RTV 1242C	Television Production 2	4.0

Semester 3		Units: 12.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
THE 2000	Theatre Appreciation	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
FIL 2560C	Editing Level 3: Advanced Editing: Color Correction and Finishing	3.0
RTV 1240C	Sound Design	3.0

Semester 4		Units: 13.0
MGF 1130	Mathematical Thinking	3.0
MMC 2000	Introduction to Mass Communications	3.0
RTV 2243C	Directing	3.0
RTV 2245C	Electronic Field Production 1	4.0

Semester 5**Units: 10.0**

<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
RTV 2246C	Electronic Field Production 2	4.0
<i>Program Electives</i>		3.0
FIL 1030	History of Film	3.0
OR		
FIL 1100	Screenwriting 1: Introduction to Story Structure	3.0
OR		
RTV 2940	Television Internship	3.0
OR		
RTV 2942	Spring Television Practicum	3.0

Total: 61.0

FIL2552C Editing Level 1: Introduction to Editing 3.000 credits

An introductory course in which students will learn the practice of editing digital media.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$35.00

RTV1241C Television Production 1 4.000 credits

An introductory overview of the practices and procedures used in a television studio. Students will learn basic operation of studio and control room equipment and work towards the completion of broadcast standard projects.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

RTV2300 Broadcast Writing 3.000 credits

In this intermediate workshop style class students will learn broadcast writing emphasizing news, documentary, commercials, and long-form programming.

Lecture Hours: 16.000, Lab Hours: 32.000

VIC1000 Visual Communications 3.000 credits

An introductory course in which students will learn the visual aspects of film, video, photography and graphic arts, specifically dealing with design elements and principles.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$10.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

FIL2553C Editing Level 2: Intermediate Editing and Visual Effects 3.000 credits

An intermediate course in which students will learn video editing with an emphasis on sound design and visual effects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2552C

Course User Fee: \$35.00

RTV1000 Fundamentals of Broadcasting 3.000 credits

In this introductory course for television and radio broadcasting, students will learn about the foundations of the the American broadcast system.
Lecture Hours: 48.000

RTV1242C Television Production 2 4.000 credits

An intermediate course in television studio production. Students will learn to implement the fundamentals as introduced in Television Production 1 to the production of scripted programs.
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: RTV1241C
Course User Fee: \$25.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.
Lecture Hours: 48.000
Course User Fee: \$10.00

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.
Lecture Hours: 48.000
Course User Fee: \$15.00

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.
Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.
Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

FIL2560C Editing Level 3: Advanced Editing: Color Correction and Finishing 3.000 credits

An advanced course in which students will learn the practice of color correcting and finishing fiction and non-fiction projects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2553C

Course User Fee: \$35.00

RTV1240C Sound Design 3.000 credits

In this introductory Sound Design course students will learn an overview of sound recording and audio post-production.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MMC2000 Introduction to Mass Communications 3.000 credits

Development of a critical perception of the mass communications process and its results in both printed and electronic media. Applications of the ethics and codes of journalism to the changing roles and forms of journalistic media. MMC 2000 will transfer for mass communications majors to various universities within the Florida State System.

Lecture Hours: 48.000

RTV2243C Directing 3.000 credits

An introductory overview of television directing. Students will learn the responsibilities of the television director in coordinating production elements and television control room techniques. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: RTV1242C

RTV2245C Electronic Field Production 1 4.000 credits

An introductory course in which students will learn single-camera field production and electronic news-gathering.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: RTV1242C

Course User Fee: \$40.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

RTV2246C Electronic Field Production 2 4.000 credits

In this intermediate workshop style class students will learn about documentary production.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: RTV2245C

Course User Fee: \$40.00

FIL1030 History of Film 3.000 credits

In this introductory course students will learn about the history of motion pictures, with an emphasis on American and European films.

Lecture Hours: 48.000

FIL1100 Screenwriting 1: Introduction to Story Structure 3.000 credits

A workshop-style introductory class covering narrative script writing for film and television. Students will learn to develop a short-format screenplay incorporating three-act story structure, script elements, and standard industry formatting. Lecture Hours: 48.000

RTV2940 Television Internship 3.000 credits

In this lower division internship designed for AS students in Television Production, students will learn to apply skills and knowledge learn in the program at an established television entertainment company (144 hr. Internship).

Other Hours: 144.000

Prerequisite: RTV1242C

RTV2942 Spring Television Practicum 3.000 credits

In this semester long on-site training with MDC Television students will learn from Television Professionals by working on current MDC TV programming (144 hr. Practicum).
Other Hours: 144.000
Prerequisite: RTV1242C

Program Description

Radiography

Associate in Science | Code: 23037 | 77 credits

The Radiography program is an Associate in Applied Science degree, which provides a broad base of education and performance-based clinical experience in all technical aspects of work as a Radiographer. Experience is provided in all routine general and fluoroscopic procedures, special procedures and in the use of the specialized equipment and techniques available in the affiliated clinical education centers. The graduate is eligible to apply to take the Registry Examination of the American Registry of Radiologic Technologists. The application deadline is February 15th for the class beginning the following Summer term. Additional Information: All applicants must attend an Information Session before acceptance into the Radiography program. Applicants must pass a physical, meet physical requirements, must complete an approved CPR course and an approved HIV/AIDS course before beginning the Radiography program. Due to the limited number of students that can be accepted into the Radiography program, it is important that applicants be properly informed. For information, advisement, application forms and deadline dates, interested students should contact the Department of Radiologic Sciences, Medical Center Campus.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
<i>Humanities</i>		3.0
HUM 1020	Humanities	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2	Units: 12.0
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BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
MGF 1130	Mathematical Thinking	3.0
PHI 2600	Introduction to Ethics	3.0
RTE 1000	Orientation to The Imaging Sciences	2.0
Semester 3 (In Milestone)		Units: 11.0
RTE 1503	Radiographic Positioning 1	3.0
RTE 1503L	Radiographic Positioning Laboratory 1	1.0
RTE 1613	Radiologic Physics	2.0
RTE 1804	Radiographic Clinic 1	5.0
Semester 4 (In Milestone)		Units: 12.0
RTE 1418	Radiographic Technology 1	3.0
RTE 1513	Radiographic Positioning 2	3.0
RTE 1513L	Radiographic Positioning Laboratory 2	1.0
RTE 1814	Radiographic Clinic 2	5.0
Semester 5 (In Milestone)		Units: 5.0
RTE 1814	Radiographic Clinic 2	5.0
Semester 6 (In Milestone)		Units: 10.0
RTE 2457	Radiologic Technology 2	3.0
RTE 2563	Radiographic Positioning 3	2.0
RTE 2834	Radiographic Clinic 4	5.0
Semester 7 (In Milestone)		Units: 11.0
RTE 2010	New Imaging Modalities In Radiology	1.0
RTE 2385	Radiation Biology	2.0
RTE 2844	Radiographic Clinic 5	8.0
Semester 8 (In Milestone)		Units: 3.0
RTE 2854	Radiographic Clinic 6	3.0
		Total: 77.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

RTE1000 Orientation to The Imaging Sciences 2.000 credits

This course is an introduction to the overall field of Imaging Sciences, radiography in particular. Students will learn the basic principles of radiation protection as it applies to the various modalities in imaging and treatment, a history of the imaging sciences, ethical/legal issues, professional behavior, medical terminology, and math/physics.

Lecture Hours: 32.000

RTE1503 Radiographic Positioning 1 3.000 credits

Basic routine positioning of the chest, abdomen, upper and lower extremities, digestive and urinary systems.

Lecture Hours: 48.000

Corequisite: RTE1503L

RTE1503L Radiographic Positioning Laboratory 1 1.000 credits

Laboratory for RTE 1503

Lab Hours: 32.000

Corequisite: RTE1503

RTE1613 Radiologic Physics 2.000 credits

Basic principles of physics involving x-radiation equipment, production and control.

Lecture Hours: 32.000

RTE1804 Radiographic Clinic 1 5.000 credits

The first in a series of six clinical courses. Under direct supervision of faculty and clinical staff, performance of basic diagnostic radiographic procedures is carried out.

Clinical Hours: 240.000

Corequisite: RTE1503, RTE1503L

Course User Fee: \$40.00

RTE1418 Radiographic Technology 1 3.000 credits

Introduction to radiographic imaging including the relation of technical factors and accessories. The chemistry of manual and automatic film processing is included.

Lecture Hours: 48.000

Prerequisite: RTE1503, RTE1503L

RTE1513 Radiographic Positioning 2 3.000 credits

Positioning of the bony pelvis, shoulder girdle, bony thorax, spinal column, skull and facial bones.

Lecture Hours: 48.000

Prerequisite: RTE1503, RTE1503L, Corequisite: RTE1513L

RTE1513L Radiographic Positioning Laboratory 2 1.000 credits

Laboratory for RTE 1513

Lab Hours: 32.000

Prerequisite: RTE1503, RTE1503L, Corequisite: RTE1513

Course User Fee: \$25.00

RTE1814 Radiographic Clinic 2 5.000 credits

The student will be evaluated on competency performances in routine fluoroscopic, and in urographic procedures. This is the second of six clinical education courses.

Clinical Hours: 240.000

Prerequisite: RTE1503, RTE1503L, RTE1804

Course User Fee: \$25.00

RTE1814 Radiographic Clinic 2 5.000 credits

The student will be evaluated on competency performances in routine fluoroscopic, and in urographic procedures. This is the second of six clinical education courses.

Clinical Hours: 240.000

Prerequisite: RTE1503, RTE1503L, RTE1804

Course User Fee: \$25.00

RTE2457 Radiologic Technology 2 3.000 credits

A more in-depth study of radiographic exposure factors as they relate to specialized procedures and equipment.

Lecture Hours: 48.000

RTE2563 Radiographic Positioning 3 2.000 credits

Radiographic procedures which utilize contrast media, sterile techniques, and/or specialized equipment and accessories.

Lecture Hours: 32.000

Prerequisite: RTE1513, RTE1513L

RTE2834 Radiographic Clinic 4 5.000 credits

Performance of procedures of increasing levels of complexity and responsibility including specialized diagnostic procedures. At this level the program faculty and clinical supervisor will determine if the student can perform procedures with less supervision.

Clinical Hours: 240.000

Prerequisite: RTE1824

Course User Fee: \$66.00

RTE2010 New Imaging Modalities In Radiology 1.000 credits

This course will enable the students to compare and contrast the current imaging modalities with the emerging technologies available in Radiology departments. Included in this course will be pictorial archiving and communications systems (PACs), digital imaging, and fusion imaging.

Lecture Hours: 16.000

Course User Fee: \$83.00

RTE2385 Radiation Biology 2.000 credits

The biologic effects of the interaction of ionizing radiation with living matter.

Lecture Hours: 32.000

RTE2844 Radiographic Clinic 5 8.000 credits

The fifth in a series of six clinical education courses.

During this clinical course the student will perform standard quality assurance tests on radiographic equipment and accessories. In addition, the student will have competency evaluations to include a gastrointestinal series and either paranasal sinuses or facial bone studies.

Clinical Hours: 384.000

Prerequisite: RTE2834

Course User Fee: \$25.00

RTE2854 Radiographic Clinic 6 3.000 credits

The student will complete the competencies required by the American Registry of Radiologic Technologists to become eligible to apply to sit for the certification exam. The student will socialize into radiography practice by beginning to work more independently of a radiographer. The student will use organizational skills to provide care to patient clients assigned to them during radiographic exams. During this course the student will be assigned to one rotation during hours other than the normal working hours of the radiology department to gain competency in procedures not usually available during the day.

Clinical Hours: 144.000

Prerequisite: RTE2844

Program Description

Respiratory Care

Associate in Science | Code: 23046 | 76 credits

The Respiratory Care program prepares the successful graduates for employment in health agencies where they will work with the physician and other professionals in treating patients with respiratory ailments or injuries affecting the respiratory function. Emphasis will be placed on supervised clinical instruction and practice in local health care facilities. Completion of this two-year accredited program enables the graduate to apply for entry into the Examination Process of the N.B.R.C.A grade of "C" or better is required in each course.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Course Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 10.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
BSC 2086	Human Anatomy & Physiology 2	3.0
CHM 1025	Introductory Chemistry	3.0
MAC 1105	College Algebra	3.0

Semester 3 (In Milestone)		Units: 11.0
PHI 2010	Introduction to Philosophy	3.0
RET 1024C	Fundamentals of Respiratory Care	3.0
RET 1484	Respiratory Care Pathophysiology 1	2.0
RET 2274C	Respiratory Care Equipment & Procedures	3.0

Semester 4 (In Milestone)		Units: 13.0
RET 2350	Respiratory Care Pharmacology	2.0
RET 2275C	Advanced Respiratory Care Equipment & Procedures	3.0
RET 2832	Respiratory Care Clinic 1	2.0
RET 2414	Pulmonary Studies	2.0
RET 2414L	Pulmonary Studies Laboratory	1.0
RET 2503	Respiratory Care Pathophysiology 2	3.0

Semester 5 (In Milestone)		Units: 9.0
RET 2284	Principles of Mechanical Ventilation	2.0
RET 2284L	Principles of Mechanical Ventilation Laboratory	2.0
RET 2833	Respiratory Care Clinic 2	5.0

Semester 6 (In Milestone)		Units: 13.0
RET 2264	Advanced Modalities and Monitoring	2.0
RET 2714	Perinatal and Pediatric Respiratory Care	2.0
RET 2714L	Perinatal & Pediatric Respiratory Care Laboratory	1.0
RET 2834	Respiratory Care Clinic 3	8.0

Semester 7 (In Milestone)		Units: 10.0
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RET 2601	Respiratory Care Seminar	2.0
RET 2835	Respiratory Care Clinic 4	8.0

Total: 76.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

RET1024C Fundamentals of Respiratory Care 3.000 credits

This is an introductory course to the Respiratory Care discipline. Students will learn the history of the profession, essential medical terminology, hospital and patient safety, infection control, patient assessment, accessing and utilizing the patient's medical record, critical thinking, Respiratory Care protocols, and patient education. Lecture Hours: 32.000, Clinical Hours: 48.000
Prerequisite: ENC1101, Corequisite: RET2274C

RET1484 Respiratory Care Pathophysiology 1 2.000 credits

This is an introductory course in the study of pulmonary and cardiovascular anatomy, physiology and pathology. Students will learn terminology, disease classification, diagnostic techniques and related physiological concepts.

Lecture Hours: 32.000

Prerequisite: ENC1101, Corequisite: RET1024C

RET2274C Respiratory Care Equipment & Procedures 3.000 credits

An overview of oxygen and aerosol therapy equipment and procedures normally used for respiratory therapy. Especially emphasized are methods used in medical, surgical, and pediatric patients and their cardiopulmonary physiology as it relates to therapeutic oxygen techniques.

Lecture Hours: 32.000, Clinical Hours: 48.000

Corequisite: RET1024C

RET2350 Respiratory Care Pharmacology 2.000 credits

This course is designed to provide training in the basic principles of the administration of medications including dosage and solutions. The drugs administered by respiratory therapists are covered in-depth, along with an introduction to the general pharmacological classifications of other drugs that may be administered to pulmonary patients. Lecture Hours: 32.000

RET2275C Advanced Respiratory Care Equipment & Procedures 3.000 credits

This course reviews airway equipment and procedures normally used for respiratory therapy. Especially emphasized are methods used in airway management, bronchial hygiene therapies, hyperventilation therapies, and airway clearance techniques.

Lecture Hours: 32.000, Clinical Hours: 48.000

Prerequisite: RET2274C

RET2832 Respiratory Care Clinic 1 2.000 credits

This is an introductory clinical practice course. The student will learn psychomotor skills related to basic respiratory care and patient care procedures including patient charting, vital signs, infection control and non-pressurized oxygen adjuncts. Clinical Hours: 96.000

RET2414 Pulmonary Studies 2.000 credits

In-depth study of diagnostic techniques in the field of pulmonary medicine which includes lung volumes, static and dynamic mechanics of breathing, ventilation, distribution of gases, diffusion and arterial blood gas sampling and handling.

Corequisite: RET 2414L (2 hr. lecture)

Lecture Hours: 32.000

Corequisite: RET2414L

RET2414L Pulmonary Studies Laboratory 1.000 credits

Laboratory for RET 2414. Simulated clinical settings of diagnostic techniques used to evaluate pulmonary functions.

Lab Hours: 32.000

Prerequisite: RET2414

Course User Fee: \$20.00

RET2503 Respiratory Care Pathophysiology 2 3.000 credits

This is a foundation course on cardiopulmonary disease. The student will learn the pathogenesis, diagnosis, treatment and rehabilitation of the diseases included in the course.

Lecture Hours: 48.000

RET2284 Principles of Mechanical Ventilation 2.000 credits

A continuation of RET 2275. A concentrated course of study which focuses on the theoretical operation, application and procedures related to critical care and mechanical ventilation.

Lecture Hours: 32.000

Prerequisite: RET2275C, RET2284L

RET2284L Principles of Mechanical Ventilation Laboratory 2.000 credits

Laboratory for RET 2284. This course will provide an in depth study of the operation of mechanical ventilation devices and associated monitors. Patient safety, troubleshooting and application are stressed.

Lab Hours: 64.000

Corequisite: RET2284

Course User Fee: \$35.00

RET2833 Respiratory Care Clinic 2 5.000 credits

In conjunction with RET 2274, 2274L and RET 1024, 1024L, RET 2832 is designed to allow the student to develop psychomotor skills related to basic respiratory care and patient care procedures (patient charting, vital signs, infection control and non-pressurized oxygen adjuncts). During the rotation, the student is provided with the opportunity to apply and discuss the theory and techniques as presented in corequisite courses.

Clinical Hours: 240.000

RET2264 Advanced Modalities and Monitoring 2.000 credits

This is an advanced course relating to critical care. Students will learn advanced techniques in invasive and non-invasive monitoring, electrocardiographic monitoring and interpretation, alternatives to conventional ventilation and advanced cardiovascular support systems.

Lecture Hours: 32.000

Prerequisite: RET2284, Corequisite: RET2714, RET2714L

RET2714 Perinatal and Pediatric Respiratory Care 2.000 credits

This course is designed to provide training in perinatal and pediatric respiratory care. Students will learn assessment and therapeutic techniques related to critical care.

Lecture Hours: 32.000

Corequisite: RET2264, RET2714L

RET2714L Perinatal & Pediatric Respiratory Care Laboratory 1.000 credits

This is an introductory laboratory course that will explore assessment and Respiratory Care therapeutics of the perinatal and pediatric patient populations. The student will learn to apply physical assessment techniques, oxygen aerosol and humidity therapies, therapeutic procedures, airway management, resuscitation and management of mechanical ventilation.

Lab Hours: 32.000

Corequisite: RET2264, RET2714L

RET2834 Respiratory Care Clinic 3 8.000 credits

This course is a continuation of RET 2833. Training will be provided on the clinical application of procedures and techniques relating to respiratory critical care.

Clinical Hours: 384.000

Prerequisite: RET2284, RET2284L, Corequisite: RET2714, RET2714L

RET2601 Respiratory Care Seminar 2.000 credits

This is an advanced course focuses on clinical and nonclinical issues. Students will learn concepts including, but not limited to, clinical research, legal and ethical concerns, home care, extended care, rehabilitation and management. ACLS and PALS certification obtained.

Lecture Hours: 32.000

Course User Fee: \$90.00

RET2835 Respiratory Care Clinic 4 8.000 credits

This course is designed to provide the student with the clinical application of adult, pediatric, and neonatal intensive respiratory care. Procedures and techniques presented in RET 2280, 2714, 2264 as it relates to their clinical application will be emphasized.

Clinical Hours: 384.000

Prerequisite: RET2834, Corequisite: RET2601

Program Description

Respiratory Care (Accelerated) - CRT to RRT

Associate in Science | Code: 23067 | 76 credits

The Respiratory Care CRT to RRT Completion program prepares the practicing Certified Respiratory Therapist (CRT) to acquire the Registered Respiratory Therapist (RRT) credential, by completing the program requirements in order to meet eligibility for the National Board for Respiratory Care (NBRC) RRT examination. The emphasis of the accelerated CRT to RRT completion program is on teaching the didactic, laboratory, and clinical competencies required of a registered respiratory therapist (RRT). CRT to RRT students will be classified as Advanced Placement in the Respiratory Care Program.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
Social Sc		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Course Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
CHM 1025	Introductory Chemistry	3.0
MAC 1105	College Algebra	3.0
PHI 2010	Introduction to Philosophy	3.0
Semester 3 (In Milestone)		Units: 27.0
RET 2284	Principles of Mechanical Ventilation	2.0
RET 2284L	Principles of Mechanical Ventilation Laboratory	2.0

RET 9995	Respiratory Care	23.0
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Semester 4 (In Milestone)

Units: 13.0

RET 2264	Advanced Modalities and Monitoring	2.0
RET 2714	Perinatal and Pediatric Respiratory Care	2.0
RET 2714L	Perinatal & Pediatric Respiratory Care Laboratory	1.0
RET 2834	Respiratory Care Clinic 3	8.0

Semester 5 (In Milestone)

Units: 13.0

RET 2414	Pulmonary Studies	2.0
RET 2414L	Pulmonary Studies Laboratory	1.0
RET 2601	Respiratory Care Seminar	2.0
RET 2835	Respiratory Care Clinic 4	8.0

Total: 76.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

RET2284 Principles of Mechanical Ventilation 2.000 credits

A continuation of RET 2275. A concentrated course of study which focuses on the theoretical operation, application and procedures related to critical care and mechanical ventilation.

Lecture Hours: 32.000

Prerequisite: RET2275C, RET2284L

RET2284L Principles of Mechanical Ventilation Laboratory 2.000 credits

Laboratory for RET 2284. This course will provide an in depth study of the operation of mechanical ventilation devices and associated monitors. Patient safety, troubleshooting and application are stressed.

Lab Hours: 64.000

Corequisite: RET2284

Course User Fee: \$35.00

RET9995 Respiratory Care 23.000 - 232.000 credits

This course is designed for students who currently hold the Certified Respiratory Therapist (CRT) Credential (TMOP 1100:815732) and provides academic recognition of the knowledge, skills, and competencies acquired through professional credential attainment. The course validates expertise in respiratory care, including patient assessment, airway management, oxygen therapy, mechanical ventilation, diagnostic testing, and therapeutic interventions. Credit awarded through this course supports academic progression and recognizes demonstrated competency in the respiratory care profession.

Lecture Hours: 640.000

RET2264 Advanced Modalities and Monitoring 2.000 credits

This is an advanced course relating to critical care. Students will learn advanced techniques in invasive and non-invasive monitoring, electrocardiographic monitoring and interpretation, alternatives to conventional ventilation and advanced cardiovascular support systems.

Lecture Hours: 32.000

Prerequisite: RET2284, Corequisite: RET2714, RET2714L

RET2714 Perinatal and Pediatric Respiratory Care 2.000 credits

This course is designed to provide training in perinatal and pediatric respiratory care. Students will learn assessment and therapeutic techniques related to critical care.

Lecture Hours: 32.000

Corequisite: RET2264, RET2714L

RET2714L Perinatal & Pediatric Respiratory Care Laboratory 1.000 credits

This is an introductory laboratory course that will explore assessment and Respiratory Care therapeutics of the perinatal and pediatric patient populations. The student will learn to apply physical assessment techniques, oxygen aerosol and humidity therapies, therapeutic procedures, airway management, resuscitation and management of mechanical ventilation.

Lab Hours: 32.000

Corequisite: RET2264, RET2714L

RET2834 Respiratory Care Clinic 3 8.000 credits

This course is a continuation of RET 2833. Training will be provided on the clinical application of procedures and techniques relating to respiratory critical care.

Clinical Hours: 384.000

Prerequisite: RET2284, RET2284L, Corequisite: RET2714, RET2714L

RET2414 Pulmonary Studies 2.000 credits

In-depth study of diagnostic techniques in the field of pulmonary medicine which includes lung volumes, static and dynamic mechanics of breathing, ventilation, distribution of gases, diffusion and arterial blood gas sampling and handling.

Corequisite: RET 2414L. (2 hr. lecture)

Lecture Hours: 32.000

Corequisite: RET2414L

RET2414L Pulmonary Studies Laboratory 1.000 credits

Laboratory for RET 2414. Simulated clinical settings of diagnostic techniques used to evaluate pulmonary functions.

Lab Hours: 32.000

Prerequisite: RET2414

Course User Fee: \$20.00

RET2601 Respiratory Care Seminar 2.000 credits

This is an advanced course focuses on clinical and nonclinical issues. Students will learn concepts including, but not limited to, clinical research, legal and ethical concerns, home care, extended care, rehabilitation and management. ACLS and PALS certification obtained.

Lecture Hours: 32.000

Course User Fee: \$90.00

RET2835 Respiratory Care Clinic 4 8.000 credits

This course is designed to provide the student with the clinical application of adult, pediatric, and neonatal intensive respiratory care. Procedures and techniques presented in RET 2280, 2714, 2264 as it relates to their clinical application will be emphasized.

Clinical Hours: 384.000

Prerequisite: RET2834, Corequisite: RET2601

Program Description

Semiconductor Technology and Nanomanufacturing

Associate in Science | Code: 25085 | 60 credits

The Associate of Science in Semiconductor Technology and Nanomanufacturing program equips students with the skills and knowledge needed to excel in nanomanufacturing, a critical field that drives innovation across diverse industries such as semiconductors, biotechnology, and energy. Combining hands-on training with foundational theory, students gain in-depth knowledge of semiconductor circuits, cleanroom protocols, patterning techniques, and metrology. Graduates are well-prepared to excel in the dynamic field of manufacturing nanoscale materials, meeting the evolving demands of the semiconductor industry.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1 Units: 9.0

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
CHM 1025	Introductory Chemistry	3.0

Semester 2 Units: 13.0

MAC 1114	Trigonometry	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
EET 1015C	Direct Current Circuits	4.0
ETS 2163C	Cleanroom Safety and Introduction to Cleanroom Processing	3.0

Semester 3 Units: 4.0

PHY 2053	Physics (without Calculus) 1	3.0
PHY 2053L	Physics (Without Calculus) 1 Lab	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 4 Units: 12.0

CET 1110C	Digital Circuits	4.0
EET 1025C	Alternating Current Circuits	4.0
ETS 2114C	Lithography Methods in Semiconductor Manufacturing	4.0

Semester 5 Units: 12.0

EET 1141C	Electronics 1	4.0
ETS 2168C	Semiconductor Plasma Techniques	4.0
ETS 2673C	Programmable Logic Controls	4.0

Semester 6**Units: 7.0**

<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
ETS 2165C	Semiconductor Characterization and Packaging	4.0

Total: 57.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

EET1015C Direct Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn electrical components, inductance, capacitance, and elementary network analysis. Students will additionally learn to verify and apply basic theories and principles through hands-on, laboratory experiments utilizing modern testing equipment. Pre/corequisite: MAT 1033. Laboratory fee. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 48.000

Pre/Corequisite: MAT1033

Course User Fee: \$63.00

ETS2163C Cleanroom Safety and Introduction to Cleanroom Processing 3.000 credits

Students learn cleanroom protocols focused on safety and contamination control. Semiconductor manufacturing processes are introduced, including photolithography, etching, deposition, and doping. Students explore characterization techniques such as microscopy and ellipsometry through hands-on cleanroom experience, emphasizing process optimization and compliance with industry standards. Recommended: Students are recommended to have completed CHM 1025. (1 hr. lecture 4 hr. lab).

Lecture Hours: 16.000, Lab Hours: 64.000

Recommended Preparation:

CHM1025 Recommended: Students are recommended to have completed CHM 1025.

Course User Fee: \$98.00

PHY2053 Physics (without Calculus) 1 3.000 credits

This course is the first in a two-part series intended for non-physics majors, offering an algebra and trigonometry approach to topics such as kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. The course fosters analytical and critical thinking skills to promote a scientific understanding of the real world. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion using algebra and trigonometry; students will apply Newton's laws, and conservation laws by using algebra and trigonometry to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053L

Course User Fee: \$10.00

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

CET1110C Digital Circuits 4.000 credits

This is an introductory course in digital circuits for engineering and other technology majors. Students learn how to apply electronic principles to digital computer circuits and systems. Students will also learn how to simplify logic circuits, build digital circuits, and perform other laboratory activities. Pre/Co-requisite: MAC1105. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

EET1025C Alternating Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn inductance, capacitance, vector notation, AC circuits, impedance, phase shift, networks, transformers, and resonance. Students will apply and verify theories and principles through hands-on, laboratory experiments utilizing modern testing equipment.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1015C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

ETS2114C Lithography Methods in Semiconductor Manufacturing 4.000 credits

This course covers photolithography in semiconductor fabrication, focusing on pattern generation, statistical process control, and process fundamentals. Students gain hands-on experience with photomasks, spin-coating, exposure systems, and chemical development, while exploring various lithography methods and evaluating key process variables for successful pattern transfer.

Prerequisites: ETS 2160C, ETS 2163C. (2 hr. lecture 4 hr. lab).

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETS2160C AND ETS2163C

Course User Fee: \$98.00

EET1141C Electronics 1 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology or related fields. Students will learn how to apply electronic principles to analog circuits and systems, including semiconductor diodes, applying the fundamental theory of transistors and other solid-state devices; analysis of amplifiers, oscillators, and other applications using a sinusoidal wave. Students also learn basic safety procedures to follow when working in an electronics laboratory and with electronic circuits and systems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1025C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

ETS2168C Semiconductor Plasma Techniques 4.000 credits

This course introduces students to the principles and applications of plasma in semiconductor manufacturing, with a focus on etching and deposition processes. Students will explore wet and dry etching methods, including reactive ion etching (RIE) and inductively coupled plasma (ICP), while gaining hands-on experience in plasma characterization, process optimization, and plasma-enhanced chemical vapor deposition (PECVD).

Prerequisites: ETS 2160C, ETS 2163C. (2 hr. lecture 4 hr. lab).

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETS2160C AND ETS2163C

Course User Fee: \$98.00

ETS2673C Programmable Logic Controls 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology and Industrial Engineering Technology. Students will learn the principals of Programmable Logic Controls (PLC) including hardware, programming, and troubleshooting. Students will also develop advanced working programs, and troubleshoot hardware and software communication problems. Prerequisite: CET110C

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET1110C

Course User Fee: \$63.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.
Lecture Hours: 48.000
Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.
Lecture Hours: 48.000
Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.
Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.
Lecture Hours: 48.000

ETS2165C Semiconductor Characterization and Packaging 4.000 credits

This course introduces advanced semiconductor metrology and packaging processes. Students will gain hands-on experience with tools like scanning electron microscope, ellipsometers, and profilometers, focusing on thin-film measurement, wafer probing, and packaging techniques while addressing safety, quality standards, and emerging trends in nanomanufacturing. Prerequisites: ETS 2160C, ETS 2163C. (2 hr. lecture 4 hr. lab).
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ETS2160C AND ETS2163C
Course User Fee: \$98.00

Program Description

Sign Language Interpretation

Associate in Science | Code: 23033 | 66 credits

The Sign Language Interpretation program is designed to develop the skills necessary to interpret the communications between deaf or hard of hearing persons and hearing individuals in an accurate and effective

manner. Also developed is a practical understanding of aspects of deaf studies and deaf culture and community. Graduates should be able to interpret at a basic level, and to achieve a minimum of Level 1 on the Quality Assurance (QA) Screening of the Florida Registry of Interpreters for the Deaf, which is traditionally required for employment as an interpreter in the state. In addition, the program will provide a foundation, especially with an accompanying Associate in Arts degree, for those persons who wish to pursue advanced degrees in preparation for careers in special education, vocational rehabilitation or other human service fields.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
ASL 1140C	American Sign Language 1	4.0
Semester 2		Units: 13.0
PHI 2010	Introduction to Philosophy	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
ASL 1150C	American Sign Language 2	4.0
ASL 2510	Deaf Culture And Community	3.0
Semester 3		Units: 13.0
ASL 2160C	American Sign Language 3	4.0
ASL 2210	ASL Conversational Skills	3.0
ASL 2220	Receptive Skills Development	3.0
INT 1000	Interpreting Ethics and Professionalism	3.0
Semester 4		Units: 13.0
ASL 2200C	American Sign Language 4	4.0
ASL 2400	Linguistics of American Sign Language	3.0
ASL 2430	Manual Alphabet Skills Development	3.0
INT 1400	Educational Interpreting	3.0
Semester 5		Units: 9.0
INT 1202	Sign To Voice Interpreting	3.0
INT 1240	Voice to Sign Interpreting	3.0
INT 1480	Interpreting: Special Settings & Populations	3.0
Semester 6		Units: 5.0

*Computer Competency Test (CCT) ---or--- CGS 1060C. Departmental Recommendation: Computer Competency Test (CCT)

Total: 66.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ASL1140C American Sign Language 1 4.000 credits

This course provides introductory information on the linguistics of American Sign Language and approximately 500 sign concepts. The course includes lecture, discussion and lab practice.

Lecture Hours: 64.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

ASL1150C American Sign Language 2 4.000 credits

This course provides continued instruction in the linguistic principles of American Sign Language and an additional 500 sign concepts. The course includes lecture, discussion and lab practice which is conducted in ASL. Prerequisite: ASL1140C.

Lecture Hours: 64.000

ASL2510 Deaf Culture And Community 3.000 credits

The course provides an in-depth study of the lives and experiences of deaf and hard of hearing persons and examines perceptions of belonging to a unique cultural group. Cultural characteristics are examined alongside the impact of hearing loss on one's family, friends and employment. Multicultural issues and the impact of hearing loss are examined within various ethnic groups. Societal attitudes regarding disability in general, and hearing loss and communication difficulties in particular, are addressed.

Lecture Hours: 48.000

ASL2160C American Sign Language 3 4.000 credits

This course provides linguistic principles of American Sign Language at the intermediate level and an additional 500 sign concepts. Lecture, discussion and lab practice are included. Students have increased opportunities for interaction with members of the deaf community. Increasingly, class sessions are conducted in ASL. Prerequisite: ASL1150C; Pre/Corequisite: ASL2210.

Lecture Hours: 64.000

ASL2210 ASL Conversational Skills 3.000 credits

The course provides opportunities to utilize all previously acquired knowledge and skill in ASL, including vocabulary and all components and linguistics principles of the language. Emphasis is placed on the more difficult aspects of ASL, such as role shifting for relating stories and the use of classifier predicates for providing visual descriptions. Class sessions are conducted entirely without voice. Prerequisites: ASL1150C, ASL2160C. Lecture Hours: 48.000

ASL2220 Receptive Skills Development 3.000 credits

The course will focus on increasing the student's receptive understanding of signed communications. Examples of American Sign Language (ASL) will be presented via videotapes and live interactions with deaf persons. Students will identify all the components and linguistic features of ASL and will provide appropriate English translations either in speech (paraphrasing) or in written form. Prerequisite: ASL 1150C. Lecture Hours: 48.000

INT1000 Interpreting Ethics and Professionalism 3.000 credits

The course provides an overview of the career of sign language interpreter. Included are the interpreter's role and responsibilities, Code of Ethics issues, evaluation systems for determining competency and logistical considerations. Various statutes will be examined with regard to their implications for interpreting and related services. These include The American with Disabilities Act (ADA), the education for all Handicapped Children Act and the Rehabilitation Act. Lecture Hours: 48.000 Prerequisite: ASL1150C

ASL2200C American Sign Language 4 4.000 credits

This course provides linguistic principles of American Sign Language at the advanced level and an additional 500 sign concepts including idioms used in ASL. Lecture, discussion and lab practice are included. Class sessions are conducted predominately in ASL. Prerequisite: ASL 2160C. Lecture Hours: 64.000

ASL2400 Linguistics of American Sign Language 3.000 credits

The course is designed for persons who already have an understanding of ASL principles. Provides an overview of the various systems of manual communication used in the U.S. including PSE, Cued Speech and signed English. Prerequisite: ASL 2160C. Lecture Hours: 48.000

ASL2430 Manual Alphabet Skills Development 3.000 credits

The course content focuses on acquiring both expressive and receptive skill in the manual alphabet of American Sign Language, commonly known as fingerspelling. A performance test is given at the beginning of the course to determine existing competency. Prerequisites: ASL 1140C, ASL1150C. Lecture Hours: 48.000 Prerequisite: ASL1150C, ASL1140C

INT1400 Educational Interpreting 3.000 credits

Provides an overview of the field, including the role and responsibilities of educational interpreters, their working conditions and related issues. Also covered are evaluation systems for educational interpreters and the Florida Educational Code of Ethics. Opportunities for skill building will be included with emphasis placed on signing with conceptual accuracy, mastering various sign systems and developing expertise in the use of technical signs. Lecture Hours: 48.000 Prerequisite: INT1000

INT1202 Sign To Voice Interpreting 3.000 credits

In-depth discussion and application of techniques and principles for interpreting legal, medical, oral and deaf/blind situations. Lecture Hours: 48.000 Prerequisite: ASL2160C, INT1240

INT1240 Voice to Sign Interpreting 3.000 credits

In-depth discussion and application of techniques and principles for interpreting situations in educational, social service, free-lance interpreting and the business aspects of interpreting. Lecture Hours: 48.000 Prerequisite: ASL2160C, INT1000

INT1480 Interpreting: Special Settings & Populations 3.000 credits

The course examines various settings in which interpreters work. These include social service and rehabilitation, employment-related, mental health and substance abuse treatment, religious, performing arts, legal and other settings. Also considered are specific deaf and hard of hearing consumers who present unique challenges for interpreters such as oral deaf persons, people who are both deaf and blind and those who would be classified as having minimal language skills (MLS). The course includes lecture and skill building opportunities. Lecture Hours: 48.000 Prerequisite: ASL2160C, INT1000

INT1941 Interpreting Internship 5.000 credits

The course includes field observation and supervised practical interpreting experience in a one-to-one interpreting situation in the community. The student is assigned to a practicing interpreter who acts as a mentor for the duration of the internship. A minimum of 240 hours is spent in the internship experience. This includes meetings with college staff and the interpreter/mentor. Prerequisites: All courses in the subject major must have been completed prior to enrolling in this course.

Other Hours: 240.000

Prerequisite: INT1000, INT1202, INT1240, INT1400, INT1480

Program Description

Social and Human Services - Addictions Studies

Associate in Science | Code: 25067 | 60 credits

The Social and Human Services program with a specialization in Addiction Studies is designed to prepare students for employment as human services specialists, human services practitioners, chemical dependency practitioners, addiction specialists, mental health and social service practitioners, or to provide supplemental training for persons previously or currently employed in these occupations. The program is also designed to provide most of the general academic and addiction specific requirements of the Certification Board for Addiction Professionals of Florida.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
PSY 2012	Introduction to Psychology	3.0
HUS 1001	Introduction to Human Services	3.0
Semester 2		Units: 12.0
PHI 2010	Introduction to Philosophy	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
HUS 1302	Basic Counseling Skills	3.0
PSB 2442	Psychology of Addiction	3.0
Semester 3		Units: 12.0
EVR 1001	Introduction to Environmental Science	3.0
HUS 1318	Domestic Abuse and Family Violence	3.0
HUS 1428	Addiction Treatment Delivery Systems	3.0

HUS 2303	Counseling Techniques	3.0
Semester 4		Units: 12.0
HUS 1423	Group Counseling in Substance Abuse	3.0
HUS 1480	HIV/AIDS and the Substance Abuser	3.0
HUS 2493	Addiction Counseling	3.0
HUS 2500	Issues & Ethics in Human Services	3.0
Semester 5		Units: 12.0
HUS 1421	Assessment and Treatment Planning in Addictions	3.0
HUS 1440	Family Issues in Chemical Dependency	3.0
HUS 1475	Addiction Counseling and the Law	3.0
HUS 2820	Field Experience in Human Service	3.0
		Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

HUS1001 Introduction to Human Services 3.000 credits

An introduction to an overview of the field of Human Services, including the role of the human services worker as it relates to various agencies, counseling, interviewing and managing.

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics.

Writing Intensive Course.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

HUS1302 Basic Counseling Skills 3.000 credits

Development of the skills of observation, recording, reporting, interviewing and counseling. These skills are presented in the context of general counseling theory.

Lecture Hours: 48.000

PSB2442 Psychology of Addiction 3.000 credits

This course will examine psychological, medical, pharmacological, legal, economic and sociological aspects of addiction to and use of various chemicals. The course will take an in-depth look at narcotic sedatives, and stimulants including alcohol, cocaine, heroin, cannabis, caffeine and tobacco.

Lecture Hours: 48.000

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUS1318 Domestic Abuse and Family Violence 3.000 credits

This course is designed to educate human services workers for the evaluation, counseling and outreach skills necessary for working with victims of domestic violence. The dynamics of partner violence, child abuse, and elder abuse will all be explored.

Lecture Hours: 48.000

HUS1428 Addiction Treatment Delivery Systems 3.000 credits

This course is designed to survey the modalities of addiction treatment. The course will study federal and state systems as well as private-not-for-profit and private-for-profit programs. All of these will be described using examples drawn from local agencies, the diverse populations they serve, and the politics and economics of the systems. This course will also present a critical exploration of the history and theory defining problems of addiction treatment and the characteristics and career issues of an addiction treatment services worker.

Lecture Hours: 48.000

HUS2303 Counseling Techniques 3.000 credits

Specific counseling techniques are introduced within the various counseling theories. Work involves both group and individual techniques.

Lecture Hours: 48.000

HUS1423 Group Counseling in Substance Abuse 3.000 credits

This course stresses development of effective group counseling leadership skills including organizing, implementing, and evaluating group counseling programs. The course includes actual group experiences.

Prerequisite: PSB2442.

Lecture Hours: 48.000

Prerequisite: PSB2442

HUS1480 HIV/AIDS and the Substance Abuser 3.000 credits

This course is designed to educate prospective addiction counselors for the evaluation, counseling and outreach skills necessary for working with HIV disease and AIDS. The course will explore not only how this disease affects one personally, but also how this pandemic has affected many psychosocial aspects of society.

Lecture Hours: 48.000

HUS2493 Addiction Counseling 3.000 credits

This course is designed to enable students to master the TAP 21 competencies: clinical evaluation, treatment planning, referral, service coordination, counseling, client, family, and community education, documentation and professional and ethical responsibilities. Additionally, the course will teach the student the process of identifying problems, establishing goals and deciding on a client treatment plan. Students will learn how to respond to an individual's needs during acute emotional and physical distress. Prerequisite: PSB2442

Lecture Hours: 48.000

Prerequisite: PSB2442

HUS2500 Issues & Ethics in Human Services 3.000 credits

This course is designed to familiarize students with the ethical problems that emerge from counseling the chemically dependent client. Emphasis will be placed on the following: the history and theory of ethics in health care; professionals' and patients' rights and responsibilities; the relationship between ethics and law; confidentiality and truth-telling in clinical relationships; technology; diagnostic testing and treatment; treatment of terminal illness; distribution of scarce medical resources and access to health care and systems payment. Prerequisite: PSB2442.

Lecture Hours: 48.000

HUS1421 Assessment and Treatment Planning in Addictions 3.000 credits

This course is designed to familiarize students with the core functions of Assessment and Treatment Planning for the chemically dependent client. Emphasis on treatment planning will be accomplished drawing from the Florida Certification Board for addiction professionals and the Department of Children's and Family Services guidelines. Prerequisites: HUS2493, PSB2442.

Lecture Hours: 48.000

Prerequisite: HUS2493, PSB2442

HUS1440 Family Issues in Chemical Dependency 3.000 credits

This course is designed to analyze the effects of chemical abuse on the family system. Emphasis will be placed on family roles and dynamics; characteristics of children (including adult children) of chemical abusers; theories of codependence; and adaptations made individually and socially by family members. Critical issues and strategies in family treatment will be explored.

Lecture Hours: 48.000

HUS1475 Addiction Counseling and the Law 3.000 credits

This course is designed to introduce addiction counseling students to the vocabulary, agencies and processes required to work with clients involved in both the criminal and civil justice systems. This course focuses on the relationship between the law and Human Services institutions, patterns of lawmaking and law-breaking, the legal structures and processes, and law as an instrument of public policy, social control and social change. The roles and functions of police, courts and correctional services will be surveyed. Common civil issues that affect clients in recovery will be explored. In addition, this course will enable students to explain the legal basis for alcohol and other drug services in Florida. State statutes pertaining to alcohol and drugs and their administrative rules will be reviewed. Confidentiality requirements, compliance standards, and professional ethics will be presented. Prerequisite: PSB2442.

Lecture Hours: 48.000

Prerequisite: PSB2442

HUS2820 Field Experience in Human Service 3.000 credits

Volunteer work as counseling paraprofessionals in a community agency under supervision. Students meet regularly with the Field Coordinator. Prerequisites: HUS1001, HUS1302, HUS2303. (120 hrs. per term)

Other Hours: 144.000

Prerequisite: HUS1001, HUS1302, HUS2303

Program Description

[Social and Human Services - Generalist](#)

Associate in Science | Code: 25026 | 60 credits

The Generalist Human Services Associate in Science degree prepares the students for employment in the network of programs and agencies which provide a vast array of human needs. These include areas such as child care, criminal justice, education, health, housing, income maintenance, mental health and retardation, among others. These needs are provided for a variety of settings, such as clinics, hospitals, nursing homes, rehabilitation centers and social agencies.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

ENC 1101	English Composition 1	3.0
<i>Mathematics</i>		3.0
MGF 1130	Mathematical Thinking	3.0
OR		
MGF 1131	Mathematics in Context	3.0
PSY 1212	Introduction to Psychology	3.0
HUS 1001	Introduction to Human Services	3.0
Semester 2		Units: 12.0
HUM 1020	Humanities	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
HUS 1302	Basic Counseling Skills	3.0
PSB 2442	Psychology of Addiction	3.0
Semester 3		Units: 12.0
EVR 1001	Introduction to Environmental Science	3.0
HUS 1318	Domestic Abuse and Family Violence	3.0
HUS 2303	Counseling Techniques	3.0
SYG 2000	Introduction to Sociology	3.0
Semester 4		Units: 12.0
HUS 1480	HIV/AIDS and the Substance Abuser	3.0
HUS 2500	Issues & Ethics in Human Services	3.0
SOP 2002	Social Psychology	3.0
SYG 2010	Social Problems	3.0
Semester 5		Units: 12.0
CLP 2140	Psychopathology	3.0
DEP 2000	Human Growth and Development	3.0
HUS 2820	Field Experience in Human Service	3.0
SYG 2430	Marriage and The Family	3.0
		Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

HUS1001 Introduction to Human Services 3.000 credits

An introduction to an overview of the field of Human Services, including the role of the human services worker as it relates to various agencies, counseling, interviewing and managing.

Lecture Hours: 48.000

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000

Course User Fee: \$5.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

HUS1302 Basic Counseling Skills 3.000 credits

Development of the skills of observation, recording, reporting, interviewing and counseling. These skills are presented in the context of general counseling theory.

Lecture Hours: 48.000

PSB2442 Psychology of Addiction 3.000 credits

This course will examine psychological, medical, pharmacological, legal, economic and sociological aspects of addiction to and use of various chemicals. The course will take an in-depth look at narcotic sedatives, and stimulants including alcohol, cocaine, heroin, cannabis, caffeine and tobacco.

Lecture Hours: 48.000

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

HUS1318 Domestic Abuse and Family Violence 3.000 credits

This course is designed to educate human services workers for the evaluation, counseling and outreach skills necessary for working with victims of domestic violence. The dynamics of partner violence, child abuse, and elder abuse will all be explored.

Lecture Hours: 48.000

HUS2303 Counseling Techniques 3.000 credits

Specific counseling techniques are introduced within the various counseling theories. Work involves both group and individual techniques.

Lecture Hours: 48.000

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

HUS1480 HIV/AIDS and the Substance Abuser 3.000 credits

This course is designed to educate prospective addiction counselors for the evaluation, counseling and outreach skills necessary for working with HIV disease and AIDS. The course will explore not only how this disease affects one personally, but also how this pandemic has affected many psychosocial aspects of society.

Lecture Hours: 48.000

HUS2500 Issues & Ethics in Human Services 3.000 credits

This course is designed to familiarize students with the ethical problems that emerge from counseling the chemically dependent client. Emphasis will be placed on the following: the history and theory of ethics in health care; professionals' and patients' rights and responsibilities; the relationship between ethics and law; confidentiality and truth-telling in clinical relationships; technology; diagnostic testing and treatment; treatment of terminal illness; distribution of scarce medical resources and access to health care and systems payment. Prerequisite: PSB2442.

Lecture Hours: 48.000

SOP2002 Social Psychology 3.000 credits

Combines a knowledge of psychology and sociology, in an interdisciplinary approach to the study of human interaction. Main themes deal with the nature of attitudes, how attitudes may be changed, the processes of interaction and the nature of group structures.

Lecture Hours: 48.000

SYG2010 Social Problems 3.000 credits

An analysis of the major contemporary and recurring social problems, emphasizing scientific search for variables involved and exploring alternative solutions.

Lecture Hours: 48.000

CLP2140 Psychopathology 3.000 credits

This course examines the major categories of mental disorders. Diagnostic criteria, treatment methods, cultural factors, public attitudes, community resources, ethical issues and legislation applicable to individuals with mental disorders are studied. The impact of mental disorders on individuals, families and society are discussed.

Lecture Hours: 48.000

Prerequisite: PSY2012

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

HUS2820 Field Experience in Human Service 3.000 credits

Volunteer work as counseling paraprofessionals in a community agency under supervision. Students meet regularly with the Field Coordinator. Prerequisites: HUS1001, HUS1302, HUS2303. (120 hrs. per term)

Other Hours: 144.000

Prerequisite: HUS1001, HUS1302, HUS2303

SYG2430 Marriage and The Family 3.000 credits

The family as a social institution — its origin and development, its forms and functions, its interrelation with other social institutions, and its role in contemporary civilization. Areas of study include factors contributing to or acting against successful, stable marriage.

Lecture Hours: 48.000

Program Description

Sport Management

Associate in Science | Code: 28010 | 60 credits

The Associate in Science (AS) in Sport Management focuses on learning through practical, hands-on experiences, and bolsters the students' learned knowledge with theory, industry insight gained from professionals with years of experience within sport. Students will experience sporting events through the lens of a manager to better understand the roles and responsibilities that are necessary to operate sports at an amateur and professional level. In addition, AS will help students understand the best practices of business, professional networking, problem-solving, and project and event management. Graduates will be able to gain employment in various athletic, entertainment, and institutional settings.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 12.0**

ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
GEB 1011	Principles of Business	3.0
<i>Program Elective</i>		3.0
PET 1170	Fundamentals of Athletic Coaching and Management	3.0
OR		
SPM 1000	Introduction to Sports Management	3.0

Semester 2**Units: 14.0**

ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
ECO 2013	Principles of Economics (Macro)	3.0
<i>Program Elective</i>		3.0

PET 1173	Strategies of Coaching and Competition	3.0
OR		
SPM 2105	Sports Events and Facility Management	3.0
Electives*		4.0

Select 4 credits from the following course options: ACG, BAN*, BUL*, CGS 1060C, FIN*, GEB*, HFT*, MAC 2233, MAN*, MAR*, MKA*, MNA*, MTB*, OST*, PUR*, QMB*, TRA*

Semester 3

Units: 13.0

<i>Natural Science</i>		3.0
AST 1002	Descriptive Astronomy	3.0
OR		
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
GLY 1010	Physical Geology	3.0
OR		
OCE 1001	Introduction to Oceanography	3.0
OR		
PHY 1020	General Education Physics	3.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
ECO 2023	Principles of Economics (Micro)	3.0
MAN 2021	Principles of Management	3.0

Semester 4

Units: 12.0

<i>Hu</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
QMB 2100	Basic Business Statistics	3.0
GEB 1000	Business Career Strategies	3.0
<i>Program Elective</i>		3.0
SPM 2151	The Business of Sports	3.0
OR		

SPM 2201	Sports Ethics and Leadership	3.0
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Semester 5

Units: 9.0

<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
GEB 1949	General Business Internship 1	3.0
Elective*		3.0

Select 3 credits from the following course options: ACG, BAN*, BUL*, CGS 1060C, FIN*, GEB*, HFT*, MAC 2233, MAN*, MAR*, MKA*, MNA*, MTB*, OST*, PUR*, QMB*, TRA*

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law. Lecture Hours: 48.000

Course User Fee: \$20.00

PET1170 Fundamentals of Athletic Coaching and Management 3.000 credits

This course will introduce students to the basic principles of athletic coaching and team management. Students will discuss the fundamentals of developing an individual athlete and team, effective athlete training programming, and factors which impact the operations of athletic organizations. Students will develop an understanding of how policy and coaching decisions influence an organization at all levels (Credit hours 3.00).

Lecture Hours: 48.000

Course User Fee: \$55.83

SPM1000 Introduction to Sports Management 3.000 credits

This course will provide students with the fundamental knowledge of the field of sports management. The course will provide an overview of sports across different levels, key events within the sports industry, the stakeholders involved, and jobs within sports. Students will learn skills and competencies that will lend towards a career in sports management.

Lecture Hours: 48.000

Course User Fee: \$64.65

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

PET1173 Strategies of Coaching and Competition 3.000 credits

This course will provide students with the tools and resources necessary to effectively evaluate team and athlete performance. In addition, the utilization of strategic thinking, game strategy, and available resources will be applied to coaching scenarios. Students will leverage their aggregate knowledge of coaching and competition to develop and adapt coaching and management plans to fulfill the needs of participants (Credit hours: 3.00).

Course User Fee: \$55.83

SPM2105 Sports Events and Facility Management 3.000 credits

This course examines the different kinds of sporting events and the facilities in which they are hosted. Through the examination of events and facilities, students will learn the basics of event and facility management and apply them in a sports-specific environment. The course will prepare students for event planning, facility and event operations, event management, and other essential competencies for career readiness.

Lecture Hours: 48.000

Course User Fee: \$64.67

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

GLY1010 Physical Geology 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites:

ACG2001, ACG2021L, ACG2021, ACG2011;

Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite:

MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033 Any Mathematics level higher than MAT 1033 will satisfy this requirement.

MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon.

Written analysis of literary works may be required.

Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

GEB1000 Business Career Strategies 3.000 credits

This course is designed to enable students to thrive in a competitive business environment. Students will learn business etiquette, alternative career pathways, personal financial management, and budgets. Additionally, students will learn interviewing, networking, and career development skills.

Lecture Hours: 48.000

Course User Fee: \$24.00

SPM2151 The Business of Sports 3.000 credits

This course is designed to provide an overview of the financial, management, and business components of sports. Students will learn how concepts such as ticket pricing, sales, marketing, media, and sponsorship affect amateur and professional sporting events. By examining past events in sports, best business practices will be identified and discussed.

Lecture Hours: 48.000

Course User Fee: \$64.67

SPM2201 Sports Ethics and Leadership 3.000 credits

This course will provide students with an understanding of effective leadership and help students begin to develop their own ethical framework. Students will examine ethical issues in sport such as the use of performance-enhancing drugs, cheating, match-fixing, etc. Additionally, students will learn how to apply different leadership styles and theories to develop a cohesive team culture and effectively navigate contemporary ethical issues within sport (Credit hours: 3.00).

Lecture Hours: 48.000

Course User Fee: \$55.83

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

GEB1949 General Business Internship 1 3.000 credits

This internship provides students with an opportunity to gain business experience while receiving academic credit. Students will learn to make connections between their internship experiences, academic coursework, and career goals. Students are required to complete 144 hours of internship in an organizational setting.

Other Hours: 144.000

Program Description

Surgical Technology

Associate in Science | Code: 23077 | 64 credits

The Associate in Science degree in Surgical Technology is designed to educate and prepare graduates with skills required for entry-level employment as a member of the healthcare team. In addition, students will acquire extensive knowledge of legal and ethical responsibilities, anatomy, physiology, pathophysiology, microbiology, aseptic techniques, patient care procedures, surgical technology procedures, patient safety, and use and care of equipment and supplies.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 10.0
ENC 1101	English Composition 1	3.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 2		Units: 10.0
MGF 1130	Mathematical Thinking	3.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Laboratory	1.0
PHI 2010	Introduction to Philosophy	3.0
Semester 3 (In Milestone)		Units: 11.0
HIM 2472	Medical Terminology	3.0
STS 1302	Introduction to Surgical Technology	2.0
STS 1303	Fundamentals of Surgical Technology	3.0
STS 1307	Surgical Equipment and Instrumentation	3.0
Semester 4 (In Milestone)		Units: 13.0
STS 1308	Perioperative Patient Care Concepts	2.0
STS 1323	Surgical Procedures I	3.0
STS 1327L	Principles and Practices of Surgical Technology Laboratory	2.0
STS 2179	Surgical Biomedical Fundamentals	3.0
STS 2944	Surgical Clinical I	3.0
Semester 5 (In Milestone)		Units: 12.0
STS 1304L	Operating Room Techniques Laboratory	3.0
STS 2324	Surgical Procedures II	3.0
STS 2340	Surgical Pharmacology	3.0
STS 2945	Surgical Clinical II	3.0
Semester 6 (In Milestone)		Units: 8.0
STS 1931	Surgical Technology Special Topics Seminar	3.0
STS 2360	Professional Skills for the Surgical Technologist	2.0
STS 2946	Surgical Clinical III	3.0
		Total: 64.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy and Physiology 1 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

BSC2086L Human Anatomy & Physiology 2 Laboratory 1.000 credits

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach. Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

STS1302 Introduction to Surgical Technology 2.000 credits

This course is intended to teach the role of the surgical technologist and central sterile supply technician in the operating room, delivery room and related areas will be covered. The student will learn the basic knowledge of equipment, supplies and instrumentation including the physical environment of the surgical suite. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$28.65

STS1303 Fundamentals of Surgical Technology 3.000 credits

This course introduces the discipline of surgical technology and the role of the surgical technologist in preventing perioperative disease transmission and microbiology to include the characteristics and activities of microorganisms. It surveys the various microbial groups, especially the bacteria, viruses and fungi with emphasis on pathogenic forms. Various significant aspects of infectious disease that occur in humans are also covered. The course will also include an introduction to principles of perioperative care, asepsis/infection control, proper disinfection, assembly, and sterilization processes for instrumentation following surgical procedures. (2 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$28.65

STS1307 Surgical Equipment and Instrumentation 3.000 credits

This course prepares the student for the scrub role during surgical procedures, identification of basic surgical instrumentation, passing of the instruments, proper care, reprocessing of instruments following a surgical procedure, and utilizing equipment in the operating room. Students will also learn about surgical wounds, proper tissue handling techniques, wound closure, wound healing, suture material, and stapling devices.

Lecture Hours: 48.000

Course User Fee: \$46.83

STS1308 Perioperative Patient Care Concepts 2.000 credits

This course will introduce the student to the needs of the surgical patient, with a focus on the special needs patient care concept, which includes adult, pediatric, geriatric, and bariatric patients. Discussion will also include types of consents, transportation and transfer of the surgical patient, preoperative patient routines, positioning, safety issues, and death and dying will be included. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$28.65

STS1323 Surgical Procedures I 3.000 credits

This course is designed to prepare students for surgical procedures by providing instruction on diagnostic procedures and relevant equipment, supplies, and techniques. The course will also include the review of surgical anatomy, physiology, and pathophysiology in relation to general surgery, endoscopic surgery, gynecological and obstetrical surgery, genitourinary surgery, orthopedic surgery, and ophthalmic surgery.

Lecture Hours: 48.000

Course User Fee: \$46.83

STS1327L Principles and Practices of Surgical Technology Laboratory 2.000 credits

This course is an introduction to surgical instrumentation, to include identification, classifications, selection, passing, proper care, and handling equipment and supplies. The course will include proper care, disinfection, assembly, and sterilization processes for instrumentation following surgical procedures. The decontamination of the operating room will be covered. Hands-on-experience in the Sterile Processing Department at local hospitals will be included in the course.

Lab Hours: 64.000, Clinical Hours: 4.000

Course User Fee: \$46.83

STS2179 Surgical Biomedical Fundamentals 3.000 credits

This course teaches the skills necessary to function as a surgical technologist in the operating room, including principles of aseptic technique, and a basic understanding of robotics and their use in the operating room setting. In addition, an understanding of the principles of physics and electricity as it relates to the operating room environment will be covered.

Lecture Hours: 48.000

Course User Fee: \$28.65

STS2944 Surgical Clinical I 3.000 credits

This course is the first in a series of three clinical courses. The course will introduce the student to the surgical clinical environment, provide the ability to begin utilizing skills acquired during previous lab courses, and implement the principles of aseptic technique while participating in the role of a surgical technologist.

Clinical Hours: 144.000

Course User Fee: \$46.83

STS1304L Operating Room Techniques Laboratory 3.000 credits

This course will provide the student with information on operating room furniture, equipment, and supplies used during surgery. The preparation of the operating room, surgical scrub, gowning and gloving, development of the sterile field, patient positioning, surgical skin prep, and draping will be included and will be part of the lab practical exam. Students will be required to simulate a surgical procedure from start to finish. Students must complete this course with a grade of "C" or higher in order to start the clinical rotations.

Lab Hours: 96.000

Course User Fee: \$46.83

STS2324 Surgical Procedures II 3.000 credits

This course is designed to prepare students for specialty surgical procedures including the review of surgical anatomy, physiology, pathophysiology, relevant equipment, supplies, and techniques regarding otorhinolaryngological surgery (ear, nose, and throat/ENT), oral and maxillofacial surgery, plastic and reconstructive surgery, cardiothoracic surgery, peripheral vascular surgery, neurosurgery, transplant and trauma surgery. The depth of coverage is determined by the current edition of the core curriculum for surgical technologists published by the Association of Surgical Technologists.

Lecture Hours: 48.000

Course User Fee: \$46.83

STS2340 Surgical Pharmacology 3.000 credits

This course introduces general pharmacological concepts and principles in the management of patient care. Effective administration of therapeutic drugs, indications, and contraindications are discussed, including effects of medication on body systems, drug classifications and their principle action. Correct drug and dose identification is emphasized, including medications handled by the surgical technologist on the sterile field.

Lecture Hours: 48.000

Course User Fee: \$28.65

STS2945 Surgical Clinical II 3.000 credits

This course is the second in a series of three clinical courses. In this course students will concentrate their clinical experience on circulating and scrubbing in on all phases of specialty surgical cases, assisting in the use of special instruments, procedures, and surgical equipment. The student will also focus on obtaining the required surgical cases in the appropriate role as a surgical technologist. (96 hr. clinical)

Clinical Hours: 144.000

Course User Fee: \$46.83

STS1931 Surgical Technology Special Topics Seminar 3.000 credits

This course provides a review of all materials covered throughout the five-semester program. Certification examination process are covered. Employability skills are also covered.

Lecture Hours: 48.000

Course User Fee: \$332.05

STS2360 Professional Skills for the Surgical Technologist 2.000 credits

This course will cover professional management, communication skills and teamwork, ethical and moral issues in the healthcare setting, and legal issues and risk management. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$46.83

STS2946 Surgical Clinical III 3.000 credits

This course is the third in a series of three clinical courses. In this course students will concentrate their clinical experience on scrubbing in on general and specialty surgical cases, assisting in the use of special instruments, procedures, and surgical equipment. The student will also focus on obtaining the required surgical cases in the appropriate role as a surgical technologist.

Clinical Hours: 144.000

Course User Fee: \$46.83

Program Description

Teaching and Learning: Early Childhood Education

Associate in Science | Code: 27014 | 60 credits

The Associate in Science (A.S.) in Teaching and Learning: Early Childhood Education prepares students for careers working with young children from birth through age eight in educational and childcare settings. The program provides a strong foundation in child growth and development, curriculum planning, classroom management, literacy development, family engagement, and developmentally appropriate teaching practices. Through a combination of academic coursework and field-based experiences, students develop the knowledge and skills needed to support the learning, development, and well-being of young children. Graduates are prepared for employment in early childhood education settings and may pursue further education in related teaching and education degree programs.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ENC 1101	English Composition 1	3.0
MGF 1130	Mathematical Thinking	3.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
EDF 1005	Introduction to the Education Profession	3.0
Semester 2		Units: 12.0
ENC 1102	English Composition 2	3.0
MGF 1131	Mathematics in Context	3.0
PHI 2680	Artificial Intelligence and Ethics	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
Semester 3		Units: 12.0
SPC 2608	Introduction to Public Speaking	3.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0

OR		
THE 2000	Theatre Appreciation	3.0
<i>Natural Science</i>		3.0
HUN 1201	Essentials of Human Nutrition	3.0
OR		
PSC 1515	Energy in the Natural Environment	3.0
Computer Competency Test*		3.0

*Computer Competency Test (CCT) ---OR--- CGS 1060C. Departmental Recommendation: CAI 1001C - Artificial Intelligence (AI) Thinking. Note: Check with Academic Advisor and School of Engineering, Technology, and Design for confirmation.

Semester 4

Units: 12.0

<i>Specialization Option</i>		12.0
<i>Track 1: Birth through Age</i>		12.0
EEC 1000	Introduction to Early Childhood Education	3.0
EEC 1200	Early Childhood Curriculum 1 (Birth - Age 5)	3.0
EEC 1311	Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5)	3.0
EEC 2407	Foundations of Learning and Executive Function in Early Childhood (Birth - 5)	3.0
OR		
<i>Track 2: Administrator</i>		12.0
EEC 2002	Operation of an Early Childhood Facility	3.0
EEC 2520	Early Childhood Organization Leadership	3.0
EEC 2523	Programming & Management for Early Childhood Admin	3.0
EEC 2527	Legal and Financial Issues in Childcare	3.0

Semester 5

Units: 12.0

<i>Educational Core</i>		6.0
EDF 2144	Maximizing Student Potential in the School Context	3.0
OR		
EDG 2943	Educational Service Field Work	1.0
OR		
EME 2040	Creativity, Innovation, and Technology	3.0
DEP 2000	Human Growth and Development	3.0
ECW 2010	Teaching with AI: Designing Human-Centered Classrooms	3.0

Total: 60.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000

Course User Fee: \$18.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

HUN1201 Essentials of Human Nutrition 3.000 credits

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

EEC1000 Introduction to Early Childhood Education 3.000 credits

Students will be introduced to the field of early childhood education with a focus on infants, toddlers, and preschool-aged children from birth through age five, highlighting play as a developmentally appropriate and research-supported approach to early learning. The course examines the purpose and scope of early childhood education and the roles of children, families, and educators. Students gain an overview of foundational theories and philosophies, family and societal influences, learning environments, classroom routines, and guidance practices that support development across domains. Common early childhood program types and curriculum models serving birth-five settings are also explored. Ten hours of service learning in an early childhood setting are required. Fingerprinting through MDC's school of education must be completed before starting this course. See the MDC SOE Work-Force handbook.

Lecture Hours: 48.000

EEC1200 Early Childhood Curriculum 1 (Birth - Age 5) 3.000 credits

Students will design and implement developmentally appropriate curriculum for children from birth to age five using the NAEYC Professional Standards as a guiding framework. Emphasis is placed on foundational curriculum planning grounded in children's interests, with focus on development, social studies, language and literacy, and creative expression through the arts. Course content includes observation-based planning, learning centers, classroom and outdoor environments, daily routines, and play-based experiences. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. This course is paired with EEC1311, which extends curriculum planning through integrated and inquiry-based applications. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000

EEC1311 Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5) 3.000 credits

Students will design and implement integrated, developmentally appropriate curriculum for children from birth to age five, building on foundational curriculum planning introduced in EEC 1200. Emphasis is placed on arts-integrated STEM learning through inquiry-based curriculum planning, with focus on scientific inquiry, mathematical thinking, building and engineering, technology, and maker spaces for creative and artistic expression. Course content includes inquiry-based planning, learning centers, enriching classroom and outdoor environments, and play-based investigations that extend children's thinking across domains. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000 AND EEC1200

Course User Fee: \$10.00

EEC2407 Foundations of Learning and Executive Function in Early Childhood (Birth - 5) 3.000 credits

Students will build foundational knowledge of executive-function life skills, including focus and self-control, perspective taking, communication, making connections, critical thinking, taking on challenges, and self-directed learning, as the basis for learning in early childhood (Birth–5). The course examines how these skills interact with interpersonal development and cognitive growth, drawing on neuroscience and research on child development. Reciprocal family partnerships, observation-based reflection, and ethical considerations guide the analysis of young children's interactions and abilities. Emphasis is placed on research-based inhibitory control strategies that support neurocognitive development. Students will connect these executive-function skills and regulation strategies to developmentally appropriate teaching methods across home and early education settings. Ten hours of service learning in an approved early childhood setting are required. See the School of Education workforce handbook.

Lecture Hours: 48.000

EEC2002 Operation of an Early Childhood Facility 3.000 credits

This introductory course provides an overview for early childhood administrators to develop and enhance their leadership role in designing and implementing quality early care and education programs. The student will study the following topics: organizational leadership and management, programming, and financial and legal issues. This course meets the requirements for the Education Program Administrator Foundational Level Credential.

Lecture Hours: 48.000

EEC2520 Early Childhood Organization Leadership 3.000 credits

The student will develop skills needed by early childhood program administrators to manage a high-quality center. The student will learn about organizational structure of a center; ethics and professionalism; leadership strategies, skills, and competencies; self-reflection and work environment; quality improvement; staff recruitment, evaluation, and retention. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.)

Lecture Hours: 48.000

Prerequisite: EEC2002

Course User Fee: \$10.00

EEC2523 Programming & Management for Early Childhood Admin 3.000 credits

The student will learn about developmentally and culturally appropriate environments, curriculum, and professional standards for early childhood care program administrators. The student will develop an understanding of child observation, assessment, documentation, and referrals and their importance. The student will learn about program evaluation, health, safety, healthy nutrition practices, and the importance of partnerships with families. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.).

Lecture Hours: 48.000

Course User Fee: \$10.00

EEC2527 Legal and Financial Issues in Childcare 3.000 credits

The student will learn about the legal and financial issues related to operating a successful early childhood center. The student learns about financial planning, personnel cost and allocation, budgeting and accounting. The student will develop knowledge about compensation and benefit, risk management, marketing and public relations, regulatory requirements, and legal issues related to childcare management. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential.

Lecture Hours: 48.000

Prerequisite: EEC2002

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.

Lecture Hours: 48.000

EDG2943 Educational Service Field Work 1.000 - 3.000 credits

The student will learn to compile the necessary documents and complete the process of obtaining a state and/or national early childhood credential. The student must complete the four courses in either the Infant/Toddler Specialization or the Preschool Specialization in order to take this course.

Lecture Hours: 16.000 - 48.000

Prerequisite: EEC1000, EEC1200, EEC1311, EEC2202, or EEC2002, EEC2520, EEC2523, EEC2527

EME2040 Creativity, Innovation, and Technology 3.000 credits

The student will learn to manage a productive & safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Prerequisite: CGS1060 OR Passing Score on the Computer Competency Exam

Course User Fee: \$190.03

DEP2000 Human Growth and Development 3.000 credits

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

ECW2010 Teaching with AI: Designing Human-Centered Classrooms 3.000 credits

This course invites learners to explore the integration of artificial intelligence through a pedagogical lens, where technology serves as a tool for reimagining teaching and learning. Learners will critically engage with AI tools by grounding their use in learning theory and evidence-based design practices.

Through inquiry, collaboration, and hands-on experimentation, learners will examine how AI can amplify creativity, personalize learning, and promote human-centeredness.

Lecture Hours: 48.000

Program Description

[Translation/Interpretation Studies](#)

Associate in Science | Code: 24050 | 60 credits

This program is designed to provide bilingual students with the knowledge and skills necessary to carry out the work associated with areas of translation (written) and interpretation (oral) in the workplace. Graduates are prepared for positions as court translators/ interpreters, in-house translators/interpreters for the private sector (including translation/interpretation agencies), translators for government agencies, hospital interpreters/ translators, freelance translators/interpreters and telephone interpreters. Graduates will have the basic foundation to establish their own translation/interpretation business.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
FOT 2802	Introduction to Translation	3.0
FOT 2821	Introduction to Interpretation	3.0
SPT 2842	Contrastive Analysis Spanish/English	3.0
Semester 2		Units: 12.0
ENC 1101	English Composition 1	3.0
FOT 2823	Consecutive Interpretation	3.0
FOT 2825	Computer Assisted Translation 1	3.0
LIN 2011	Introduction to Linguistics	3.0
Semester 3		Units: 6.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
MGF 1130	Mathematical Thinking	3.0
Semester 4		Units: 12.0
FOT 2220	Localization and Project Management	3.0
FOT 2822	Court Interpreting Skills	3.0
FOT 2824	Simultaneous Interpretation Strategies	3.0
FOT 2828	Medical Interpretation	3.0
Semester 5		Units: 12.0
FOT 2701	Simultaneous Conference Interpretation	3.0
FOT 2826	Legal Translation	3.0
FOT 2827	Medical Translation	3.0
FOT 2829	Financial and Business Translation	3.0
Semester 6		Units: 6.0

Natural Science		3.0
AST 1002	Descriptive Astronomy	3.0
OR		
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0
OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1020	General Education Physics	3.0
Social Science		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0

Total: 61.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

FOT2802 Introduction to Translation 3.000 credits

Develops the ability to do accurate written translations in general. Includes the application of contrastive structures and grammar rules of source and target languages; translation of idiomatic expressions and an introduction to legal and technical vocabulary; the use of bilingual dictionaries and glossaries. The demands of translation as a profession and its code of ethics are stressed.

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2821 Introduction to Interpretation 3.000 credits

The acquisition and development of skills to sight translate a text from the source to the target language. Training in the conceptualization and analysis of an oral message to transfer its content to another language consecutively. Preparatory exercises for the development of simultaneous interpretation skills.

Lecture Hours: 48.000

Course User Fee: \$15.00

SPT2842 Contrastive Analysis Spanish/English 3.000 credits

Comparison/Contrastive study of the phonology, morphology and syntax of Spanish and English. Recommended for students of translation and interpretation.

Lecture Hours: 48.000

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

FOT2823 Consecutive Interpretation 3.000 credits

This course builds on the foundation established in Introduction to Interpretation (FOT2810) and acquaints the students with the practice and application of consecutive interpretation (English/Spanish). Development of active listening, concentration and retention skills as well as the ability to perceive essential meaning for subsequent recall is emphasized. This course also explores basic note taking techniques and provides practice in monolateral and bilateral consecutive interpretation.

Lecture Hours: 48.000

Prerequisite: FOT2821

Course User Fee: \$15.00

FOT2825 Computer Assisted Translation 1 3.000 credits

Examines the types of translation software currently used in the translation/interpretation profession as well as the commercial use and business application of these. Description and application of tools such as translation memory, electronic dictionaries, desktop-publishing systems, and website translation technologies are covered.

Lecture Hours: 48.000

Prerequisite: FOT2802

Course User Fee: \$25.00

LIN2011 Introduction to Linguistics 3.000 credits

The course provides an exploration of basic linguistic concepts including phonology, morphology, syntax, pragmatics and semantics, how a language is organized and functions, and will establish a connection between Linguistics and Translation Theory. The course will also introduce students to the study of how cultural and social factors affect human communication.

Lecture Hours: 48.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

**FOT2220 Localization and Project Management
3.000 credits**

Software and website localization is a new and growing area of translation and is closely linked to the concepts of globalization and internationalization in electronic communications. In this introductory course students will learn how to adapt software for international markets as well as to how to translate websites. Students will also be introduced to basic concepts in localization project management.

Lecture Hours: 48.000

Prerequisite: FOT2802 AND FOT2825

Course User Fee: \$25.00

FOT2822 Court Interpreting Skills 3.000 credits

Continuation of FOT 2821 including deepening and broadening the type of exercise of FOT 2821 and gradual introduction to simultaneous interpretation. Oral translation with notes and conversations, ratio or tape passages. Extensive practice in the process of hearing, understanding, remembering and speaking for simultaneous oral interpretation. Participation in an internship or practical training program.

Lecture Hours: 48.000

**FOT2824 Simultaneous Interpretation Strategies
3.000 credits**

This course builds on the foundation established in previous interpretation courses while introducing the students to simultaneous interpretation (English/Spanish) by providing preparatory exercises such as shadowing, lagging, paraphrasing etc. Through a variety of recorded materials, students practice the simultaneous interpretation mode so as to acquire smooth delivery techniques while forming good professional habits.

Lecture Hours: 48.000

Prerequisite: FOT2821, FOT2823

Course User Fee: \$15.00

FOT2828 Medical Interpretation 3.000 credits

This course develops the techniques, practices and knowledge needed to function as interpreters in a medical environment. Interpreting models such as sight, consecutive and simultaneous — as they apply to the medical setting — are revisited. Medical vocabulary/terminology in English and foreign language as well as code of ethics will also be introduced.

Lecture Hours: 48.000

Prerequisite: FOT2821, FOT2823, FOT2824

**FOT2701 Simultaneous Conference Interpretation
3.000 credits**

This course builds on the foundation established in the previous Simultaneous Interpretation Strategies course. Students will continue developing their simultaneous interpretation skills through exercises such as shadowing, decálage, paraphrasing, etc. Through a variety of authentic recording materials, students will practice the simultaneous interpretation mode in the context of international organizations and conferences so as to acquire smooth delivery techniques while forming professional habits both in conference booths. Extensive practice in simultaneous interpretation will be provided both in class and online.

Lecture Hours: 48.000

Prerequisite: FOT2824

Course User Fee: \$25.00

FOT2826 Legal Translation 3.000 credits

Continuation of FOT 2802. Written translations of multi-page documents and/or articles containing legal, technical and other specialized vocabulary from the source language into the target language. Firsthand translation experience by participating in a "translator's bureau," or an "internship" or practical training program.

Lecture Hours: 48.000

FOT2827 Medical Translation 3.000 credits

This course further develops translation strategies while familiarizing the student with the characteristics of medical and health-related discourse in both English and Spanish. Included is the acquisition of medical and hospital/clinic terminology and the analysis of related linguistic structures so students can engage in translating texts from English into foreign language and vice versa.

Lecture Hours: 48.000

Prerequisite: FOT2802

FOT2829 Financial and Business Translation 3.000 credits

This course further develops translation strategies while familiarizing the students with the characteristics of financial and business discourse in both English and Spanish. Included is the learning of special terminology and related linguistic structures so students can engage in the translation of texts containing financial/business or economic discourse from English into Spanish and vice versa. As in legal translation, students engage in terminology research and glossary development through the use of specialized bilingual financial and business dictionaries and other pertinent sources.

Prerequisites: FOT 2802, 2803. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

AST1002 Descriptive Astronomy 3.000 credits

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

Program Description

Transportation and Logistics

Associate in Science | Code: 28000 | 64 credits

This Transportation and Logistics A.S. degree program provides core courses in transportation policy, law, safety, security, management and marketing, and an integrated understanding of the intermodal relationship between the various modes of transportation. In addition, college credit certificates will be offered in Intermodal Freight Transportation and Surface Transportation.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
ENC 1101	English Composition 1	3.0
MAC 1105	College Algebra	3.0
TRA 1154	Introduction to Supply Chain Management	3.0
Program Elective*		4.0
Computer Competency Test**		

Select 4 credits from the following course options: ACG, BUL*, CGS 1060C, ECO*, FIN1*, FIN2*, GEB*, HFT*, MAN1*, MAN2*, MAR*, OST*, PET*, PUR*, QMB*, SBM*, SCM*, SPM*, TAX*, TRA*

**Computer Competency Test (CCT) ---OR--- CGS1060C. Departmental Recommendation: Computer Competency Test (CCT)

Semester 2		Units: 13.0
<i>Humanities</i>		3.0
ARH 1000	Art Appreciation	3.0
OR		
HUM 1020	Humanities	3.0
OR		
LIT 2000	Introduction to Literature	3.0
OR		
MUL 1010	Music Appreciation	3.0
OR		
PHI 2010	Introduction to Philosophy	3.0
OR		
THE 2000	Theatre Appreciation	3.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
ECO 2013	Principles of Economics (Macro)	3.0
TRA 2010	Introduction to Transportation and Logistics	3.0

Semester 3		Units: 13.0
<i>Natural Science</i>		3.0
BSC 1005	General Education Biology	3.0
OR		
CHM 1020	General Education Chemistry	3.0
OR		
ESC 1000	General Education Earth Science	3.0

OR		
EVR 1001	Introduction to Environmental Science	3.0
OR		
PHY 1020	General Education Physics	3.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
ECO 2023	Principles of Economics (Micro)	3.0
TRA 2156	Operations Management for Transportation	3.0

Semester 4	Units: 12.0
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<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		
POS 2041	American Federal Government	3.0
MAN 2021	Principles of Management	3.0
QMB 2100	Basic Business Statistics	3.0
TRA 2321	Transportation Public Policy, Law, and Regulations	3.0

Semester 5	Units: 13.0
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SCM 1114	Applied Advanced Technologies & AI in Supply Chain Management	3.0
TRA 2402	Intermodal Transportation Operations and Project Management	3.0
TRA 2702	International Logistics and Transportation	3.0
Program Elective		4.0

Total: 64.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course. Lecture Hours: 48.000
Prerequisite: MAT1033
Course User Fee: \$18.00

TRA1154 Introduction to Supply Chain Management 3.000 credits

This course is an introduction to the concepts, principles and techniques in the field of supply chain management (SCM) with particular emphasis on the economic significance of distribution to business and the U.S. economy. Students will learn the interrelationship between logistics and other areas of business, noting how the SCM pipeline can significantly impact customer loyalty by adding value. Lecture Hours: 48.000

ARH1000 Art Appreciation 3.000 credits

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context. Lecture Hours: 48.000
Course User Fee: \$10.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas. Lecture Hours: 48.000
Course User Fee: \$5.00

LIT2000 Introduction to Literature 3.000 credits

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course. Lecture Hours: 48.000
Prerequisite: ENC1101

MUL1010 Music Appreciation 3.000 credits

In this course, students will survey the history of classical music from antiquity to the modern period, focusing on western music. The curriculum may also integrate a variety of popular and global styles where appropriate. Student learning outcomes: students will discuss and analyze music using terminology appropriate for the course; students will demonstrate fundamental knowledge of the works of significant composers; students will identify connections between music and the other arts; and students will identify historical styles and periods based on instruments and performance practices utilized.

Lecture Hours: 48.000

Course User Fee: \$15.00

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course.

Lecture Hours: 48.000

THE2000 Theatre Appreciation 3.000 credits

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

ECO2013 Principles of Economics (Macro) 3.000 credits

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$15.00

TRA2010 Introduction to Transportation and Logistics 3.000 credits

This course surveys the organization and operations of the commercial transportation industry and its impact on the bottom line of today's modern businesses. Students will learn to review regulations and processes affecting transportation and logistics functions as well as explore the industry job market and look at technologies and current issues shaping transportation and logistics. A.S. degree only.

Lecture Hours: 48.000

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

EVR1001 Introduction to Environmental Science 3.000 credits

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems.

Lecture Hours: 48.000

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033 Any Mathematics level higher than MAT 1033 will satisfy this requirement.

MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

TRA2156 Operations Management for Transportation 3.000 credits

This course includes the skills necessary for a supervisory role in logistics. Students will learn the roles and responsibilities in managing different types of transportation operations. Topics include human resources, design and management of production operations, productivity, capacity planning, resource management, just-in-time systems, hazardous materials management, planning and project management. Prerequisite: MAN2021.

Lecture Hours: 48.000

Prerequisite: MAN2021

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

TRA2321 Transportation Public Policy, Law, and Regulations 3.000 credits

Students will learn the transportation regulatory environment, including the various levels of government regulations. Review of security, environmental requirements, regulatory research and labor laws are also covered. Prerequisite: TRA 2010.

Lecture Hours: 48.000

Prerequisite: TRA2010

SCM1114 Applied Advanced Technologies & AI in Supply Chain Management 3.000 credits

This course provides an in-depth exploration of Industry 4.0 and Artificial Intelligence (AI) applications within the supply chain industry, focusing on strategic support, optimization, data-driven decision-making, and organizational readiness. Students will discover how AI technologies enhance operational efficiency, reduce costs, and improve overall supply chain performance while ensuring responsible and ethical implementation. Through case studies and practical exercises, learners will gain the skills needed to leverage AI for innovative solutions and competitive advantage in supply chain management.

Lecture Hours: 48.000

TRA2402 Intermodal Transportation Operations and Project Management 3.000 credits

Students will learn the fundamental elements necessary to plan, implement and control efficient and market-responsive integrated transportation systems. Topics include strategic, operational, and project management roles of transportation in supply chains. Emphasis is placed on services pricing, carrier selection, equipment and shipment planning, intermodal operations, financial/budgetary constraints, security and distribution services.

Prerequisite: AVM2120, TRA2010; Corequisites: TRA1410, TRA1420, TRA1430.

Lecture Hours: 48.000

Prerequisite: AVM2120 AND TRA2010, Corequisite: TRA1430 AND TRA1410 AND TRA1420

TRA2702 International Logistics and Transportation 3.000 credits

International logistics concerns the flow of materials into, through and out of an international corporation as it relates to materials management, storage, inventory locations, physical distribution and documentation. This course will emphasize international transportation infrastructure and modes such as ocean, airfreight, intermodal movement, truck and rail. Choices among these modes will be explored considering such factors as transit time, packaging, risks, predictability and cost. The roles of freight forwarders and custom brokers in moving international cargo and operation of foreign trade zones will be discussed. Prerequisites: TRA2010, TRA2321.

Lecture Hours: 48.000

Prerequisite: TRA2010, TRA2321

Program Description

Veterinary Technology

Associate in Science | Code: 23065 | 73 credits

The Veterinary Technology program prepares students to assist veterinarians in their daily practice, working with all types of animals and in various disciplines within the realm of veterinary medicine. Tasks include providing total nursing care to the sick or injured patient, handling and restraint, assisting during examinations and surgical procedures, performing dental hygiene and radiographic exams and collection and analysis of diagnostic specimens. Graduates are eligible to apply to take the Veterinary Technician National Examination (VTNE) and the Florida Practical Exam (FPE).

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 10.0

ENC 1101	English Composition 1	3.0
HUM 1020	Humanities	3.0
BSC 1005	General Education Biology	3.0
BSC 1005L	General Education Biology Laboratory	1.0
Computer Competency Test*		

*Computer Competency Test (CCT) ---OR--- CGS1060C - Introduction to Computer Technology and Applications.
Departmental Recommendation: Computer Competency Test (CCT)

Semester 2

Units: 6.0

MGF 1130	Mathematical Thinking	3.0
<i>Social Science</i>		3.0
AMH 2010	History of the US to 1877	3.0
OR		
AMH 2020	History of the US since 1877	3.0
OR		

POS 2041	American Federal Government	3.0
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Semester 3 (In Milestone)	Units: 8.0
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ATE 1110	Animal Anatomy	3.0
ATE 1110L	Animal Anatomy and Physiology Lab	1.0
ATE 1211	Animal Physiology	3.0
ATE 1650L	Introduction to Clinical Practice 1	1.0

Semester 4 (In Milestone)	Units: 11.0
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ATE 2636	Large Animal Clinic anNursing Skills	2.0
ATE 2636L	Large Animal Clinic and Nursing Skills Laboratory	1.0
ATE 2638	Animal Lab Procedures 1	3.0
ATE 2638L	Animal Lab Procedures 1 Laboratory	2.0
ATE 2652L	Introduction to Clinical Practice 2	1.0
ATE 2671C	Lab Animal Medicine	2.0

Semester 5 (In Milestone)	Units: 9.0
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ATE 1630	Pharmacology for Veterinary Technicians	2.0
ATE 2639	Animal Lab Procedures 2	3.0
ATE 2639L	Animal Lab Procedures 2 Laboratory	2.0
ATE 2722C	Avian and Exotic Pet Medicine	2.0

Semester 6 (In Milestone)	Units: 12.0
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ATE 1940	Veterinary Clinical Experience 1	3.0
ATE 2611	Animal Medicine 1	3.0
ATE 2631	Small Animal Nursing 1	3.0
ATE 2655L	Animal Nursing and Medicine Laboratory 1	2.0
ATE 2661	Large Animal Diseases	1.0

Semester 7 (In Milestone)	Units: 13.0
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ATE 1941	Veterinary Clinical Experience 2	3.0
ATE 2050L	Animal Nursing and Medicine Laboratory 2	2.0
ATE 2612	Small Animal Nursing	3.0
ATE 2614	Animal Medicine 2	3.0
ATE 2710	Animal Emergency Medicine	2.0

Semester 8 (In Milestone)	Units: 4.0
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ATE 2942	Veterinary Clinical Experience 3	4.0
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Total: 73.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

HUM1020 Humanities 3.000 credits

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe.

Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

BSC1005 General Education Biology 3.000 credits

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC1005L General Education Biology Laboratory 1.000 credits

This is an optional one-credit lab to provide students with experience in the scientific process. This course applies the scientific method to critically examine and explain the natural world, including but not limited to cells, organisms, genetics, evolution, ecology, and behavior.

Lab Hours: 32.000

Pre/Corequisite: BSC1005

Course User Fee: \$25.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

POS2041 American Federal Government 3.000 credits

In this course, students will investigate how the national government is structured and how the American constitutional republic operates. It covers the philosophical and historical foundations of American government, including but not limited to the declaration of independence, the United States Constitution and all its amendments, and the federalist papers. The course examines the branches of government and the government's laws, policies, and programs. It also examines the ways in which citizens participate in their government and ways their government responds to citizens. Student learning outcomes: students will demonstrate an understanding of the basic principles and practices of America's constitutional republic; students will demonstrate knowledge of the nation's founding documents, including the declaration of independence, the U.S. Constitution and its amendments, and the Federalist Papers; students will demonstrate knowledge of landmark U.S. Supreme court cases, landmark legislation, and landmark executive actions; students will demonstrate knowledge of the history and development of the American federal government and its impact on law and society; students will demonstrate an ability to apply course material to contemporary political issues and debates; and students will demonstrate the ability to engage in discussion and civil debate on American politics that are associated with multiple points of view.

Lecture Hours: 48.000

ATE1110 Animal Anatomy 3.000 credits

This course explores the physical and functional phenomena that interact to sustain life in animals. The student will learn the relationships of all of the systems in domestic animals, such as the osseous apparatus, the respiratory, digestive, genitourinary, endocrine, and nervous systems. The student will also be introduced to the descriptive and topographical terms needed to communicate with the professional staff.

Lecture Hours: 48.000

Prerequisite: BSC1005, BSC1005L, ENC1101,

Corequisite: ATE1110L, ATE1211, ATE1650L

ATE1110L Animal Anatomy and Physiology Lab 1.000 credits

This course will complete the coverage and understanding of the physiological and anatomical relationships required for further development as a veterinary technician. This course will correlate with lecture material learned in the Animal anatomy and Animal Physiology lecture courses. Anatomical dissection, necropsy, examination of live animals will be used as well as the study of radiographs, skeleton models and histological sections.

Lab Hours: 32.000

Corequisite: ATE1110, ATE1211, ATE1650L

Course User Fee: \$50.00

ATE1211 Animal Physiology 3.000 credits

This course is designed to explore the terminology related to animal physiology, in addition to all aspects of the functions of systems in small and large animals.

Lecture Hours: 48.000

Corequisite: ATE1110, ATE1110L, ATE1650L

ATE1650L Introduction to Clinical Practice 1 1.000 credits

This introductory course is designed to acquaint the student with skills associated with veterinary clinical practice. Students will learn basic office, laboratory and nursing skills, including hospital/office management, restraint, history taking, examination room techniques, administration of medication, basic parasitology, and basic clinical pathology procedures.

Lab Hours: 32.000

Corequisite: ATE1110, ATE1110L, ATE1211

ATE2636 Large Animal Clinic and Nursing Skills 2.000 credits

This course is designed to acquaint the student with the fundamentals of large animal herd management, reproductive physiology and lactation physiology. Aspects of equine, bovine, ovine and porcine husbandry will be included.

Lecture Hours: 32.000

Prerequisite: ATE1110, ATE1110L, ATE1211,

Corequisite: ATE2636L

ATE2636L Large Animal Clinic and Nursing Skills Laboratory 1.000 credits

This course is designed to acquaint the student with the fundamentals of large animal husbandry, herd health management, preventive medicine, animal restraint and nutrition as it relates to the bovine, equine, porcine and caprine species. Techniques discussed in the Large Animal Clinic and Nursing skills course such as venipuncture, injections and administration of other oral medications will be reviewed and demonstrated. One laboratory session will be devoted to poultry science.

Lab Hours: 32.000

Corequisite: ATE2636

Course User Fee: \$50.00

ATE2638 Animal Lab Procedures 1 3.000 credits

This course is designed to introduce the veterinary technician to common parasites and their life cycles seen in routine veterinary practice. Also, hematology and the kinetics of the hematopoietic system are discussed with emphasis on normal blood smears and common changes seen during disease stages of domestic animals.

Lecture Hours: 48.000

Prerequisite: ATE1110, ATE1110L, ATE1211,

Corequisite: ATE2638L

ATE2638L Animal Lab Procedures 1 Laboratory 2.000 credits

This course is designed to acquaint the student with clinical laboratory procedures covered in the Animal Laboratory Procedures 1 course. Areas of emphasis include hematology, coagulation and parasitology as well as general laboratory etiquette.

Lab Hours: 64.000

Corequisite: ATE2638

Course User Fee: \$100.00

ATE2652L Introduction to Clinical Practice 2 1.000 credits

The clinical application of basic veterinary radiology and surgical nursing skills will be the primary focus of this practicum. The student will demonstrate skills under supervised instruction.

Lab Hours: 32.000

Prerequisite: ATE1110, ATE1110L, ATE1650L

ATE2671C Lab Animal Medicine 2.000 credits

This foundation course provides instruction on laboratory animal care. Students will learn the technical aspects of laboratory animal care, including restraint and handling, common diseases and nutrition. The animals studied include rabbits, rats, mice, guinea pigs, hamsters and primates.

Lecture Hours: 16.000, Lab Hours: 32.000

ATE1630 Pharmacology for Veterinary Technicians 2.000 credits

This introductory course reviews drug classification and office procedures/management. Students will learn methods of calculating appropriate drug dosage, routes of administration, and evaluation of drug efficacy as well as office procedures used in veterinary hospital management.

Lecture Hours: 32.000

ATE2639 Animal Lab Procedures 2 3.000 credits

This course serves as a continuation of Animal Laboratory Procedures 1 and covers immunology, liver function and diagnostic testing for liver abnormalities, kidney function and testing used in disease states, urinalysis, pancreatic evaluation; normal and abnormal exfoliative cytology; and the evaluation of endocrine disorders. It also will include principles of serological testing and microbiological methods and protocols. Prerequisites: ATE2638, ATE2638L; Corequisite: ATE2639L.

Lecture Hours: 48.000

Prerequisite: ATE2638, ATE2638L, Corequisite: ATE2639L

ATE2639L Animal Lab Procedures 2 Laboratory 2.000 credits

This course provides experience in the practical applications discussed in Animal Laboratory procedures 2. It will also include principles of serological testing and microbiological methods and protocols as well as dentistry for the veterinary technician. Prerequisites: ATE2638, ATE2638L; Corequisite: ATE2639.

Lab Hours: 64.000

Prerequisite: ATE2638, ATE2638L, Corequisite: ATE2639

Course User Fee: \$75.00

ATE2722C Avian and Exotic Pet Medicine 2.000 credits

This course is designed to acquaint students with the medical care associated with exotic animals and avians. Students will learn types of species that may be encountered in a practice and their associated care techniques.

Lecture Hours: 16.000, Lab Hours: 32.000

ATE1940 Veterinary Clinical Experience 1 3.000 credits

This entry clinical course provides supervised clinical experience in a veterinary facility. Students will learn and reinforce competencies in clinical laboratory procedures, venipuncture techniques, physical examination of patients, administration of intramuscular and subcutaneous injections and exam room protocol.

Clinical Hours: 144.000

Course User Fee: \$28.00

ATE2611 Animal Medicine 1 3.000 credits

This course is designed to acquaint the student with anesthesiology, asepsis and general surgical nursing care, essentials in pharmacy and pharmacology, and concepts in microbiology, virology and immunology.

Prerequisites: ATE1110, ATE1211; Corequisites:

ATE2661, ATE2942, ATE2631, ATE2655L.

Lecture Hours: 48.000

ATE2631 Small Animal Nursing 1 3.000 credits

The student will master the technical skills of medicating animals and the taking and processing of radiographs. This course also covers general care, including grooming and bathing, feeding and watering, nail trimming, ear cleaning, anal sac expression, and determination of vital signs.

Prerequisites: ATE1110, ATE1211; Corequisites:

ATE2611, ATE2655L.

Lecture Hours: 48.000

Prerequisite: ATE1110, ATE1211, Corequisite:

ATE2611, ATE2655L

ATE2655L Animal Nursing and Medicine Laboratory 1 2.000 credits

This course is designed to acquaint the student with exam room and restraining techniques, anesthesia and surgical protocols and diagnostic imaging procedures used in veterinary hospitals.

Lab Hours: 64.000

Course User Fee: \$75.00

ATE2661 Large Animal Diseases 1.000 credits

This course is designed to acquaint the student with the fundamentals of preventative medicine and with the common disease seen in the large animal species. Aspects of equine, bovine, ovine and porcine diseases and common treatments will be emphasized. Prerequisites: ATE1110, ATE2636, ATE2636L; Corequisite: ATE2611.

Lecture Hours: 16.000

Prerequisite: ATE1110, ATE2636, ATE2636L,

Corequisite: ATE2611

ATE1941 Veterinary Clinical Experience 2 3.000 credits

This course consists of supervised clinical experience in the veterinary workplace. Students will learn to enhance the competencies from ATE 1940 Veterinary Clinical Experience 1 while adding application of classroom knowledge in pharmacology, clinical laboratory procedures, and radiology.

Clinical Hours: 144.000

ATE2050L Animal Nursing and Medicine Laboratory 2 2.000 credits

The student will practice training a dog, and applying corrections for common behavioral problems. Clinical training in a small animal necropsy is also presented.

Prerequisites: ATE1110, ATE2631, ATE2655L;

Corequisite: ATE2612.

Lab Hours: 64.000

Prerequisite: ATE1110, ATE2631, ATE2655L,

Corequisite: ATE2612

Course User Fee: \$50.00

ATE2612 Small Animal Nursing 3.000 credits

A study of the basic concepts of nutrition, obstetrics, and pediatric care, as well as the important aspects regarding zoonotic diseases, public health, and animal behavior. The student will also be introduced to alternative medicine, including holistic concepts, homeopathic, acupuncture, chiropractic and other emerging specialties. Prerequisites: ATE1110, ATE2611, ATE2631, ATE2655L; Corequisite: ATE2050L.

Lecture Hours: 48.000

Prerequisite: ATE1110, ATE2611, ATE2631, ATE2655L,

Corequisite: ATE2050L

ATE2614 Animal Medicine 2 3.000 credits

This course will explore general pathology, causes and nature of disease, toxicology, and an overview of the pathologies of the major systems, as well as immunity, disease prevention, common vaccinations and diseases relating to small animals. Prerequisites: ATE1110, ATE2611.

Lecture Hours: 48.000

Prerequisite: ATE1110, ATE2611

ATE2710 Animal Emergency Medicine 2.000 credits

This course is designed to acquaint the student with fundamentals of emergency veterinary medicine, including veterinary first aid, toxicology and specialized medical techniques and procedures.

Prerequisites: ATE1110, ATE1211; Corequisites:

ATE2611, ATE2631, ATE2655L.

Lecture Hours: 32.000

Prerequisite: ATE1110, ATE1211, Corequisite:

ATE2611, ATE2631, ATE2655L

ATE2942 Veterinary Clinical Experience 3 4.000 credits

This course provides clinical experience to the student, under the supervision of a veterinarian. Students will enhance the competencies learned in ATE1940 - Clinical Experience 1 and ATE1941 - Veterinary Clinical Experience 2 and master skills associated with advanced veterinary technology practice. Prerequisite ATE1941.

Clinical Hours: 192.000

Prerequisite: ATE1941

Course User Fee: \$35.00

Advanced Technical Certificate (ATC)

Program Description

Biotechnology

Advanced Technical Certificate | Code: C6028 | 33 credits

A fast-track certification program in Biotechnology is available for students with bachelor's degrees, or having at least an associate degree and a strong background in college-level math and science courses. This background, combined with selected courses in the biotechnology program, will provide students with the necessary skills to seek employment in Biotechnology and related industries.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

BSC 2426	Biotechnology Methods and Applications 1	3.0
BSC 2426L	Biotechnology Methods and Applications 1 Laboratory	2.0
BSC 2427	Biotechnology Methods and Applications 2	3.0
BSC 2427L	Biotechnology Methods and Applications 2 Laboratory	2.0
BSC 2943L	BIOSCIENCE INTRNSHIP	3.0

Semester 2

Units: 19.0-21.0

CHM 2200	Survey of Organic Chemistry	3.0
CHM 2200L	Survey of Organic Chemistry Laboratory	1.0
EGS 1010	Applied Research Methods	1.0-3.0
ETI 1040	Introduction to Bioscience Manufacturing	3.0
ETI 1172	Introduction to Quality Assurance	3.0
ETI 1622	Concepts of Lean and Six Sigma	3.0
MCB 2010	Microbiology	3.0
MCB 2010L	Microbiology Laboratory	2.0

Total: 32.0-34.0

BSC2426 Biotechnology Methods and Applications 1 3.000 credits

This course addresses the basic principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical, biotechnology and/or research laboratory setting(s). Practical applications of biotechnology are explored.

Lecture Hours: 48.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426L

BSC2426L Biotechnology Methods and Applications 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2426 Biotechnology Methods and Applications 1. This is a hands-on course that emphasizes the basic laboratory principles, techniques, and instrumentation, necessary for effective work in pharmaceutical, biotechnology, and/or research laboratory settings(s).

Lab Hours: 64.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426

Course User Fee: \$50.00

BSC2427 Biotechnology Methods and Applications 2 3.000 credits

This course addresses advanced principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical-biotechnology and/or research-laboratory setting(s). The following areas of contemporary biotechnology are covered: forensics, bioremediation, and animal-, plant-, and marine biotechnology.

Lecture Hours: 48.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427L

BSC2427L Biotechnology Methods and Applications 2 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2427 Biotechnology Methods and Applications 2. This is a hands-on course that emphasizes advanced laboratory principles, techniques, and instrumentation necessary for effective work in a pharmaceutical, biotechnology, and/or research-laboratory setting(s).

Lab Hours: 64.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427

Course User Fee: \$50.00

BSC2943L BIOSCIENCE INTRNSHIP 3.000 - 6.000 credits

Lab Hours: 144.000 - 288.000

Course User Fee: \$21.00

CHM2200 Survey of Organic Chemistry 3.000 credits

XXXX This one-semester course briefly examines the structure, synthesis, nomenclature and reactivity of selected mono-and-poly-functional organic compounds. Theories that relate the structure of organic molecules to their chemical reactivity will be presented as a unifying principle. Prerequisite: CHM 1046 and CHM 1046L; Corequisite: CHM 2200L. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2200L

CHM2200L Survey of Organic Chemistry Laboratory 1.000 credits

Experiments and exercises will be conducted to introduce students to the basic laboratory techniques that are used in organic chemistry and that re-enforce and illustrate several important topics in organic chemistry.

Prerequisite: CHM 1046 and CHM 1046L; Corequisite: CHM 2200. Special fee. (2 hr. lab)

Lab Hours: 32.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2200

Course User Fee: \$46.00

EGS1010 Applied Research Methods 1.000 - 3.000 credits

This course is designed for STEM majors. Students will learn basic research practices: research methods, experimentation, validation, technical writing, and presentations. Using the Affinity Research Group model, students will work in groups to conduct theory-based STEM research, develop poster presentations, and write conference and journal publications.

Lecture Hours: 16.000 - 48.000

ETI1040 Introduction to Bioscience Manufacturing 3.000 credits

This course introduces students to the field of bioscience manufacturing. Topics will include basic principles of the industry, large-scale process development and the future of the bioscience industry. Current Good Manufacturing Practices (cGMPs), and the nature and delivery system of products will also be discussed.

Lecture Hours: 48.000

ETI1172 Introduction to Quality Assurance 3.000 credits

This course describes the role and aspects of quality systems and Regulatory affairs in research laboratories, regulated companies, and firms that comply with voluntary standards. Topics include stages in development and submission of drugs and medical devices, patents legislation, and quality systems such as auditing, standard procedures, good manufacturing and laboratory practices.

Lecture Hours: 48.000

ETI1622 Concepts of Lean and Six Sigma 3.000 credits

This course is designed for students who are preparing for careers in the manufacturing industry. Students will learn the basic concepts, frameworks, and techniques used in six sigma, including total quality philosophies, the calculation of six sigma and other vital statistics, tools of lean six sigma, and knowledge of various methodologies.

Lecture Hours: 48.000

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L. Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L
Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

Program Description

Biotechnology - Bioinformatics

Advanced Technical Certificate | Code: C6029 | 33 credits

Semester 1

Units: 13.0

BSC 2426	Biotechnology Methods and Applications 1	3.0
BSC 2426L	Biotechnology Methods and Applications 1 Laboratory	2.0
BSC 2427	Biotechnology Methods and Applications 2	3.0
BSC 2427L	Biotechnology Methods and Applications 2 Laboratory	2.0
BSC 2943L	Bioscience Internship	3.0

Semester 2

Units: 12.0

CGS 1540C	Database Concepts Design	
CGS 1700	Introduction to Operating Systems	4.0
CIS 1321	INTRO SYS ANAL&DESGN	4.0
CIS 1531	Introduction to Secure Scripting	
COP 1334	Introduction to C++ Programming	4.0

Total: 25.0

BSC2426 Biotechnology Methods and Applications 1 3.000 credits

This course addresses the basic principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical, biotechnology and/or research laboratory setting(s). Practical applications of biotechnology are explored.

Lecture Hours: 48.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426L

BSC2426L Biotechnology Methods and Applications 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2426 Biotechnology Methods and Applications 1. This is a hands-on course that emphasizes the basic laboratory principles, techniques, and instrumentation, necessary for effective work in pharmaceutical, biotechnology, and/or research laboratory settings(s).

Lab Hours: 64.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426

Course User Fee: \$50.00

BSC2427 Biotechnology Methods and Applications 2 3.000 credits

This course addresses advanced principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical-biotechnology and/or research-laboratory setting(s). The following areas of contemporary biotechnology are covered: forensics, bioremediation, and animal-, plant-, and marine biotechnology.

Lecture Hours: 48.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427L

BSC2427L Biotechnology Methods and Applications 2 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2427 Biotechnology Methods and Applications 2. This is a hands-on course that emphasizes advanced laboratory principles, techniques, and instrumentation necessary for effective work in a pharmaceutical, biotechnology, and/or research-laboratory setting(s).

Lab Hours: 64.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427

Course User Fee: \$50.00

BSC2943L Bioscience Internship 3.000 - 6.000 credits

This internship course is a capstone for students majoring in bioscience and related programs. Students will learn to apply acquired knowledge and skills to gain experience in the bioscience workplace.

Other Hours: 144.000 - 288.000

Course User Fee: \$21.00

CGS1540C Database Concepts Design credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

**CGS1700 Introduction to Operating Systems
4.000 credits**

This course examines the role of operating systems as the interface between the hardware, the software and the users of a computer system. It explores the concepts such as processes and threads, file systems, virtual memory, interrupt handling, virtualization and security.

Lecture Hours: 48.000, Lab Hours: 32.000

CIS1321 INTRO SYS ANAL&DESGN 4.000 credits

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS1531 Introduction to Secure Scripting credits

This course provides students with the knowledge and skills to: create secure scripts and programs using system shells and programming languages; implement and debug algorithms to solve problems; automate and perform administrative tasks; manage data handling, and backup and storage. (3 hr. lecture; 2 hr. lab)

Course User Fee: \$63.00

**COP1334 Introduction to C++ Programming
4.000 credits**

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

Program Description

Certified Flight Instructor

Advanced Technical Certificate | Code: C6027 | 13 credits

Semester 1

Units: 13.0

ATF 2501	Flight Instructor-Flight Training	3.0
ATF 2501L	Flight Instructor Laboratory	1.0
ATT 2131	Flight Instructor Theory	3.0
AVM 1949	Co-Op Work Experience 1: AVI	3.0
AVM 2949	Co-Op Work Experience 2: AVI	3.0

Total: 13.0

ATF2501 Flight Instructor-Flight Training 3.000 credits

This course provides flight training for the student to develop the ability to analyze the performance of private and commercial flight maneuvers from the right seat of a training aircraft, in compliance with the Federal Aviation Administration Certified Flight Instructor Certificate. Prerequisite: ATF2300; Corequisites: ATT2131, ATF2501L. Special fee.

Lecture Hours: 48.000

Prerequisite: FAA Commercial Certificate with an Instrument Rating or ATP, ATT2133

ATF2501L Flight Instructor Laboratory 1.000 credits

Provides the student with internship teaching experience based upon the principles of flight instruction learned in ATT2131 and ATF 2501. Students will learn to develop lesson plans and how to communicate effectively using instructional materials Prerequisite: ATF2300; Corequisites: ATT2131, ATF2501.

Lab Hours: 32.000

Prerequisite: ATF2300 AND Corequisite: ATF2501 AND ATT2131

Course User Fee: \$8703.50

ATT2131 Flight Instructor Theory 3.000 credits

Provides the student ground instruction to obtain the necessary aeronautical knowledge, to meet the FAA written standards for the Certified Flight Instructors Certificate. Preparation for the written exam is included in the course content. Prerequisite: ATF2305.

Lecture Hours: 48.000

AVM1949 Co-Op Work Experience 1: AVI 3.000 credits

This is a course designed to provide training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Co-operative Education Office to obtain registration approval. Prerequisite: 2.0 minimum GPA, approval of Co-op Program Director, minimum of 6 credits in field or work approved experience.

Lecture Hours: 48.000

AVM2949 Co-Op Work Experience 2: AVI 3.000 credits

This is a course designed to continue training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Co-operative Education Office to obtain registration approval. Prerequisite: 2.0 minimum GPA, approval of Co-Op Program Director, completion of AVM1949.

Lecture Hours: 48.000

Program Description

Compensation and Benefits

Advanced Technical Certificate | Code: C6002 | 18 credits

Semester 1		Units: 6.0
MAN 3025	Organization Management	3.0
MAN 3301	Human Resource Management	3.0
Semester 2		Units: 6.0
MAN 4330	Compensation Management	3.0
MAN 4335	Employee Benefit Planning	3.0
Semester 3		Units: 6.0
MAN 4361	Organizational Staffing	3.0
MAN 4402	Employment Law and Regulation	3.0

Total: 18.0

MAN3025 Organization Management 3.000 credits

The student will learn the major functions of supervision and management including the functions of planning, staffing, directing and controlling. Emphasis is placed on learning how to communicate more effectively with employees and how to increase leadership effectiveness. Major topics include goal setting and attainment, organizational structure, decision-making, strategic planning, managing stress, and ethical behavior and ethical role modeling. Cases that present the student with opportunities to make supervisory and management decisions, along with timely feedback on their effectiveness, will be used. Must pass this course with a grade of C or higher.

Lecture Hours: 48.000

Prerequisite: MAN2021

Course User Fee: \$60.00

MAN3301 Human Resource Management 3.000 credits

The student will learn the functions of Human Resource Management including human resource planning, strategic development of human resources, recruitment techniques, selection and hiring processes, compensation systems, development of policy and procedures for effective and ethical human resource management, performance review and evaluation systems, working effectively with organized labor, retention of employees, and current issues in human resource management. The course will also include an exploration of human resources within the global business environment of a boundless organization. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: MAN2021

Course User Fee: \$60.00

MAN4330 Compensation Management 3.000 credits

This course is a study of the strategic use of compensation system for the purposes of attracting, retaining, and motivating a competitive workforce. The student will address topics such as designing compensation systems, bases for pay, employee benefit programs, laws affecting compensation practices, and compensation challenges for various employee groups. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4335 Employee Benefit Planning 3.000 credits

This course focuses on the both wage and nonwage related benefits made available to employees by the firm and various related social and governmental programs. The student will examine various state and federal laws regarding employee benefits planning. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4361 Organizational Staffing 3.000 credits

This course provides the student with an overview of the staffing function in organizations, including the topics of job analysis, forecasting, recruitment, selection, retention and turnover. It serves as an introductory course for the prospective human resources manager and as a survey of responsibility and activities of any manager with supervisory responsibilities. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4402 Employment Law and Regulation 3.000 credits

This course analyzes the federal and state regulation of the employment relationship, including wage and hour laws, EEO, and Affirmative Action programs. The student will address human resource issues such as employee benefits, insurance, workers' compensation, safety, health, employees' personal rights and collective bargaining legislation. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

Program Description

Computed Tomography

Advanced Technical Certificate | Code: C6037 | 9 credits

Semester 1		Units: 9.0
RTE 2571	Principles of Computed Tomography	3.0
RTE 2762	Cross Sectional Anatomy	3.0
RTE 2940C	Computed Tomography Clinical Education	3.0
		Total: 9.0

RTE2571 Principles of Computed Tomography 3.000 credits

This course provides the radiologic technologist advanced imaging techniques of computed tomography. This introduction to the CT scanning technology will include history and development, equipment, terminology, patient preparation and care, and the principles of image formation, acquisition, and production. Corequisite: RTE2762.

Lecture Hours: 48.000

Corequisite: RTE2762

RTE2762 Cross Sectional Anatomy 3.000 credits

This course is designed to provide the students a thorough knowledge of Axial, coronal, and sagittal imaging planes of our body. Students will also learn to identify various anatomical structures in axial, coronal, and sagittal images of our body.

Lecture Hours: 48.000

RTE2940C Computed Tomography Clinical Education 3.000 credits

The course provides the radiologic technologist practical, first-hand experience of scanning procedures and techniques at a supervised clinical site; theories learned in RTE 2571 will be applied. Students will observe, assist, and perform Computed Tomography under the supervision and guidance of a qualified CT Technologist. Prerequisites: RTE2571, RTE2762.

Clinical Hours: 144.000

Prerequisite: RTE2571 AND RTE2762

Course User Fee: \$29.36

Program Description

Hospitality Management

Advanced Technical Certificate | Code: C6000 | 18 credits

Semester 1		Units: 9.0
HFT 3263	Restaurant Management	3.0
HFT 3603	Hospitality Law	3.0
HFT 4064	Bar and Beverage Management	3.0
Semester 2		Units: 9.0
HFT 4253	Hotel Management	3.0
HFT 4809	Food Service Management	3.0
HFT 4468	Revenue Management in the Hospitality Industry	3.0
		Total: 18.0

HFT3263 Restaurant Management 3.000 credits

Students will examine restaurant management and operations. Topics include operational issues, customer satisfaction, restaurant trends and challenges, new technologies, financial accountability, cost controls, service issues, regulatory and safety compliance, leadership, training and staff development. Upon completion, students will be able to articulate knowledge of all aspects of a restaurant operation.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT3603 Hospitality Law 3.000 credits

Students will examine the legal rights and responsibilities related to the hospitality industry. Topics include federal and state regulations, historical and current practices, safety and security, risk management, loss prevention, liability, relevant torts and contracts. Upon completion, students will be able to articulate knowledge of the legal issues in the hospitality industry, and methods of preventing or minimizing organizational liability.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT4064 Bar and Beverage Management 3.000 credits

Students will examine the management of beverages served in hospitality operations. Topics include history and trends; service, procurement and storage; knowledge and control of wines and fermented/distilled beverages; and non-alcoholic beverages, coffees, and teas. Upon completion, students will be able to articulate knowledge of responsible alcohol service and knowledge of beverages consumed in a hospitality operation.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT4253 Hotel Management 3.000 credits

Students will examine the working components of a hotel and their interrelationships. Topics include hotel organization, management and leadership, human resources, food and beverage service, meeting industry, marketing and selling, communications, management companies, and ethics in hospitality management. Upon completion, students will be able to articulate knowledge of the responsibilities of each hotel department and how they interact to provide the overall guest experience.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT4809 Food Service Management 3.000 credits

Students will examine food service management and operations. Topics include the management process, marketing, sales techniques, cost and pricing strategies, food purchase controls, production and service, sanitation and safety, management of bar and beverage design and financial management. Upon completion, students will be able to articulate knowledge of the financial relationship of the food and beverage aspect of the hotel industry.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT4468 Revenue Management in the Hospitality Industry 3.000 credits

Students will examine the basics of revenue management in the hotel industry. Topics include pricing and length-of-stay tools, revenue management performance measures, capacity management, demand and revenue forecasting, discounting, reservation systems, and pricing execution. Upon completion, students will be able to articulate knowledge of how to accurately forecast hotel guest arrivals, examine pricing models in accordance with revenue management principles, and how to manage overbooking. Prerequisite: ACG2021 , ACG2021L, or HFT2421.

Lecture Hours: 48.000

Pre/Corequisite: HFT2421, Prerequisite: ACG2021, ACG2021L

Course User Fee: \$35.00

Program Description

Semester 1		Units: 6.0
MAN 3025	Organization Management	3.0
MAN 3301	Human Resource Management	3.0
Semester 2		Units: 9.0
MAN 3322	Human Resources Information Systems	3.0
MAN 4330	Compensation Management	3.0
MAN 4335	Employee Benefit Planning	3.0
Semester 3		Units: 9.0
MAN 4352	Effective Employee Training	3.0
MAN 4361	Organizational Staffing	3.0
MAN 4402	Employment Law and Regulation	3.0
		Total: 24.0

MAN3025 Organization Management 3.000 credits

The student will learn the major functions of supervision and management including the functions of planning, staffing, directing and controlling. Emphasis is placed on learning how to communicate more effectively with employees and how to increase leadership effectiveness. Major topics include goal setting and attainment, organizational structure, decision-making, strategic planning, managing stress, and ethical behavior and ethical role modeling. Cases that present the student with opportunities to make supervisory and management decisions, along with timely feedback on their effectiveness, will be used. Must pass this course with a grade of C or higher.

Lecture Hours: 48.000

Prerequisite: MAN2021

Course User Fee: \$60.00

MAN3301 Human Resource Management 3.000 credits

The student will learn the functions of Human Resource Management including human resource planning, strategic development of human resources, recruitment techniques, selection and hiring processes, compensation systems, development of policy and procedures for effective and ethical human resource management, performance review and evaluation systems, working effectively with organized labor, retention of employees, and current issues in human resource management. The course will also include an exploration of human resources within the global business environment of a boundless organization. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: MAN2021

Course User Fee: \$60.00

MAN3322 Human Resources Information Systems 3.000 credits

This course examines the role of human resources information system (HRIS) in today's organizations and human resources departments. The student will address topics such as human resource information systems design, acquisition, and implementation. The role of these systems in talent acquisition and management is also examined. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4330 Compensation Management 3.000 credits

This course is a study of the strategic use of compensation system for the purposes of attracting, retaining, and motivating a competitive workforce. The student will address topics such as designing compensation systems, bases for pay, employee benefit programs, laws affecting compensation practices, and compensation challenges for various employee groups. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4335 Employee Benefit Planning 3.000 credits

This course focuses on the both wage and nonwage related benefits made available to employees by the firm and various related social and governmental programs. The student will examine various state and federal laws regarding employee benefits planning. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4352 Effective Employee Training 3.000 credits

This course focuses on professional development activities as performed by human resources specialists or organizational specialists. The student will analyze the benefits of employee training, establish employee development programs, and address the theories, issues, practices and problems regarding employee training and development. Prerequisites: MAN3025, MAN3301

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4361 Organizational Staffing 3.000 credits

This course provides the student with an overview of the staffing function in organizations, including the topics of job analysis, forecasting, recruitment, selection, retention and turnover. It serves as an introductory course for the prospective human resources manager and as a survey of responsibility and activities of any manager with supervisory responsibilities. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4402 Employment Law and Regulation 3.000 credits

This course analyzes the federal and state regulation of the employment relationship, including wage and hour laws, EEO, and Affirmative Action programs. The student will address human resource issues such as employee benefits, insurance, workers' compensation, safety, health, employees' personal rights and collective bargaining legislation. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

Program Description

Magnetic Resonance Imaging

Advanced Technical Certificate | Code: C6036 | 15 credits

Semester 1

Units: 15.0

RTE 2762	Cross Sectional Anatomy	3.0
RTE 2575	MRI Principles and MR Safety	3.0
RTE 2576	MRI Procedures & Patient Care	3.0
RTE 2577C	Magnetic Resonance Imaging (MRI) Practicum	6.0

Total: 15.0

RTE2762 Cross Sectional Anatomy 3.000 credits

This course is designed to provide the students a thorough knowledge of Axial, coronal, and sagittal imaging planes of our body. Students will also learn to identify various anatomical structures in axial, coronal, and sagittal images of our body.

Lecture Hours: 48.000

RTE2575 MRI Principles and MR Safety 3.000 credits

This course is an introduction to the basic principles of MRI, including signal production, basic concepts in image formation and image contrast and resolution. It is intended to provide a solid foundation in MRI technology as well as an understanding of the MRI equipment and its appropriate and safe use. A practical approach to using the correct imaging sequences and parameters will be discussed. Emphasis will be placed in MRI Screening and MRI Safety. Corequisites: RTE2762, RTE2576.

Lecture Hours: 48.000

Corequisite: RTE2762, RTE2576

RTE2576 MRI Procedures & Patient Care 3.000 credits

This course is designed to introduce the student to the different image acquisition techniques and common scan sequences used in MRI imaging to describe normal and abnormal anatomy. Emphasis will be placed on basic patient care skills and safe use of equipment used in MRI. Corequisites: RTE2762, RTE2575.

Lecture Hours: 48.000

Corequisite: RTE2762, RTE2575

RTE2577C Magnetic Resonance Imaging (MRI) Practicum 6.000 credits

This course is designed to provide the students with hands-on experience for body imaging. Students will learn to position the patients correctly inside the Magnetic Resonance Imaging (MRI) machine and program the machine to create axial, coronal, and sagittal images of the brain and spinal column. Prerequisites: RTE2762, RTE2575, RTE 2576.

Practicum Hours: 288.000

Prerequisite: RTE2762 AND RTE2575 AND RTE2576

Program Description

Museum Studies

Advanced Technical Certificate | Code: C6050 | 16 credits

The Advanced Technical Certificate (ATC) in Museum Studies offers a dynamic and interdisciplinary education that prepares students for impactful careers at the intersection of museum work and interactive media production. This innovative program equips learners with critical skills enabling them to craft engaging traditional and interactive experiences. Simultaneously, students will explore the field of museum studies, gaining in-depth training in the analysis, interpretation, and curation of museum artifacts, while developing a profound appreciation for their historical and cultural contexts. Through hands-on learning opportunities, participants will apply ethical standards and professional practices in the responsible management and preservation of collections. With a strong focus on research, documentation, and exhibition design, graduates will emerge as versatile professionals prepared to navigate the evolving media and museum landscapes, effectively engaging various audiences and fostering a deeper understanding of cultural heritage and storytelling across multiple platforms.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 9.0

ENC 1101	English Composition 1	3.0
ARH 2051	Art History 2	3.0
ARH 2857	Introduction to Museum Studies	3.0

Semester 2

Units: 7.0

ARH 2833	Museum and Gallery Exhibition Design	3.0
ARH 2990	Museum and Gallery Co-Op Work Experience	3.0
LIS 2004	Digital Research Strategies	1.0

Total: 16.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ARH2051 Art History 2 3.000 credits

In this course, students will embark on a world survey of the visual arts from 800 to 1850 A.D., exploring various artistic movements, styles, and cultural contexts. The course will delve into the rich and diverse artistic traditions from different regions, including works from the Western canon. Writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$5.00

ARH2857 Introduction to Museum Studies 3.000 credits

This is a foundation course that offers a practical introduction to the nature, context, and operations of museums with the intention of providing practical examples and indicators of good practices. This course introduces students to the history and theory of museums and museum practices, museum administration, exhibition planning, museum education, and museum careers. In addition to the comprehensive overview of museum studies, this course also delves into the influence of the Western canon on museum curation and interpretation, providing students with a well-rounded understanding of the cultural and historical significance of the artifacts and artworks displayed in museums.

Lecture Hours: 48.000

ARH2833 Museum and Gallery Exhibition Design 3.000 credits

The course examines the historical context of museum exhibitions, tracing their evolution from traditional displays to contemporary immersive experiences. Students will explore the key milestones and influential figures that have shaped the field, gaining insights into the changing trends and philosophies that guide exhibition practices today.

Lecture Hours: 32.000, Lab Hours: 32.000

ARH2990 Museum and Gallery Co-Op Work Experience 3.000 credits

This course offers an exceptional opportunity for students to further enhance their skills and knowledge in the field of Art and related studies through practical work experience. The primary focus of this course is to provide students with hands-on training that complements their academic studies.

Internship Hours: 144.000

LIS2004 Digital Research Strategies 1.000 credits

LIS 2004, Digital Research Strategies, equips students with information literacy skills to locate, evaluate, and ethically use information in an evolving digital landscape. Students will apply critical thinking and research techniques to explore traditional and emerging research tools. (1 hr. lecture)
Lecture Hours: 16.000

Program Description

Procurement Management

Advanced Technical Certificate | Code: C6003 | 18 credits

Semester 1		Units: 9.0
QMB 2100	Basic Business Statistics	3.0
OR		3.0
STA 2023	Statistical Methods	3.0
MAN 3562	Purchasing, Inventory and Warehouse Management	3.0
MAN 3577	Procurement for Major Projects	3.0
Semester 2		Units: 9.0
MAN 3506	Operations Management	3.0
MAN 3578	Global Procurement Management	3.0
MAN 4570	Purchasing for Industry	3.0
		Total: 18.0

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

MAN3562 Purchasing, Inventory and Warehouse Management 3.000 credits

This course explores the integration of production operations and logistics management. It will also examine purchasing, Inventory and Warehouse Management concepts and theory. Special emphasis is placed on the relationships that exist between operations and the supply chain to include operations and supply chain strategies, business processes, project management, product design and development, inventory management and effective warehouse and stores management.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3577 Procurement for Major Projects 3.000 credits

Students will learn the principles of procurement for major projects concepts and theory, including advanced purchasing and cost savings techniques, contracts administration from award to completion, strategic procurement management.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3506 Operations Management 3.000 credits

The course emphasizes the application of operational decision-making techniques to improve process, productivity, and the effective utilization of resources within organizations. Students will learn to recognize the trade-offs associated with operations management decisions and their effect on resource allocation. Topics include production processes, operations strategies, quantitative techniques, quality, performance, capacity planning, efficiency, forecasting, resource management, statistical process control, project management, and supply chains. Prerequisite: QMB2100 or STA2023.

Lecture Hours: 48.000

Prerequisite: QMB2100 OR STA2023

MAN3578 Global Procurement Management 3.000 credits

Students will learn Global Procurement Management concepts and theory, including a good understanding in the differences in cultures, law, currency, communications and buying from foreign suppliers, international procurement is a challenge, buying internationally and to help support international procurement efforts.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4570 Purchasing for Industry 3.000 credits

Students will learn purchasing for industry concepts and theory, including negotiation skills: mastering the art of deal making strategies, contracts: bidding, evaluation, negotiation and award, purchasing management, tendering and supplier selection, vendor qualification: managing performance and contract compliance.

Lecture Hours: 48.000

Course User Fee: \$62.76

Program Description

Project Management

Advanced Technical Certificate | Code: C6004 | 18 credits

Semester 1

Units: 9.0

		3.0
QMB 2100	Basic Business Statistics	3.0
OR		
STA 2023	Statistical Methods	3.0
MAN 3583	Project Management	3.0
MAN 3731	Assessing and Managing Project Risk	3.0

Semester 2

Units: 9.0

MAN 3506	Operations Management	3.0
MAN 3888	Project Leadership	3.0
MAN 4887	Project Planning and Control Systems for Supply Chain Management	3.0

Total: 18.0

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

MAN3583 Project Management 3.000 credits

Students will learn project management concepts and theory, including attributes for a project life cycle, global project management, benefits of project management, project management environments, planning and managing risk, project planning controls, and terminating a project.

Lecture Hours: 48.000

MAN3731 Assessing and Managing Project Risk 3.000 credits

This course explores project uncertainty, and how to manage risk to keep the project on track and meet project goals. Project risk management is a vital part of all projects, which requires a purposeful strategy to avoid any setbacks. In this course there will be an examination of what the risk management process involves through the identification, assessment, and response to project risks.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3506 Operations Management 3.000 credits

The course emphasizes the application of operational decision-making techniques to improve process, productivity, and the effective utilization of resources within organizations. Students will learn to recognize the trade-offs associated with operations management decisions and their effect on resource allocation. Topics include production processes, operations strategies, quantitative techniques, quality, performance, capacity planning, efficiency, forecasting, resource management, statistical process control, project management, and supply chains. Prerequisite: QMB2100 or STA2023.

Lecture Hours: 48.000

Prerequisite: QMB2100 OR STA2023

MAN3888 Project Leadership 3.000 credits

Students will learn the fundamentals of management vs. leadership, monitoring/controlling, delegation, personal traits, communication, and people management when in a leadership role. The specifics include techniques and strategies using the company's vision and project charter needed to effectively manage, lead and direct a team to ensure reaching project objectives. As part of this process students will determine their personal leadership traits and approach.

Lecture Hours: 48.000, Lab Hours: 3.000

Course User Fee: \$62.76

MAN4887 Project Planning and Control Systems for Supply Chain Management 3.000 credits

Students will learn about the complex supply-chain transformation that is required to manage resources from many different departments. Additionally, an exploration of how supply-chain project managers ensure internal and external stakeholder alignment, mitigate large amounts of risk, and implement communication, risk mitigation, and change management plans to ensure a successful project. Finally, the student will gain an integrated view of supply-chain transformation that incorporates elements of change management, test plan development, project management techniques, and establishing effective project management teams.

Lecture Hours: 48.000

Course User Fee: \$62.76

Program Description

Supply Chain Analytics

Advanced Technical Certificate | Code: C6005 | 18 credits

Semester 1		Units: 9.0
QMB 2100	Basic Business Statistics	3.0
OR		3.0
STA 2023	Statistical Methods	3.0
MAN 4523	Production Information Systems	3.0
MAN 4552	Supply Chain Analytics & Decision Making	3.0
Semester 2		Units: 9.0
MAN 3506	Operations Management	3.0
MAN 4719	Challenges in the Digital Supply Chain	3.0
MAN 4732	Business Intelligence for Supply Chain	3.0
		Total: 18.0

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression.

Prerequisite: Satisfactory score/grade on an approved placement method.

Lecture Hours: 48.000

Recommended Preparation: Completion of MGF1131 prior to registration.

Course User Fee: \$18.00

MAN4523 Production Information Systems 3.000 credits

This course presents the fundamental aspects of computer technology required by the systems that provide data to, and derive information from, production in manufacturing. Students will learn the techniques to organize, store, manipulate data, report, derive and analyze production information, basics networking used in production, as well as various forms of information systems.

Lecture Hours: 48.000

MAN4552 Supply Chain Analytics & Decision Making 3.000 credits

Through this course students will focus on the science of learning from data to support decision making. The objective of this course is to explore and apply basic statistical concepts and procedures that are used to collect, analyze, summarize, and report data in a supply chain management role. Course topics covered include methods for: collecting, analyzing, summarizing data; making statistical inferences about populations and exploring the relationship between variables. The course introduces analytical tools and techniques that provide a cause and effect understanding linking operational plans with corporate objectives.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3506 Operations Management 3.000 credits

The course emphasizes the application of operational decision-making techniques to improve process, productivity, and the effective utilization of resources within organizations. Students will learn to recognize the trade-offs associated with operations management decisions and their effect on resource allocation. Topics include production processes, operations strategies, quantitative techniques, quality, performance, capacity planning, efficiency, forecasting, resource management, statistical process control, project management, and supply chains. Prerequisite: QMB2100 or STA2023.

Lecture Hours: 48.000

Prerequisite: QMB2100 OR STA2023

MAN4719 Challenges in the Digital Supply Chain 3.000 credits

Through this course, students will analyze the social, ethical, and legal issues in supply chain management. Students will examine the importance of fairness, accountability, and transparency of data policies in various supply chain operations. Various issues governing the collection, storage, access, use, and sharing of data will also be explored. Prerequisites: MAN3506, QMB2100 or STA2023.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4732 Business Intelligence for Supply Chain 3.000 credits

This course will enable students to synthesize their learning from program coursework. The global context within which firms operate will provide the foundation for understanding global supply chain strategy, from market entry through to the integration of demand management and supply management processes to meet corporate objectives. The course will utilize exercises and a simulation to allow students to analyze and synthesize program coursework. The course will help students understand how to best use program knowledge to maximize value creation for employers and in their own career.

Lecture Hours: 48.000

Course User Fee: \$62.76

College Credit Certificate (CCC)

Program Description

Accounting and Budgeting

College Credit Certificate | Code: 65075 | 18 credits

Semester 1		Units: 8.0
ACG 1403	Excel for Business	1.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
TAX 2000	Income Tax	3.0
Semester 2		Units: 10.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
ACG 2450	Microcomputers in Accounting	3.0
TAX 2021	Taxation of Business Organizations	3.0
		Total: 18.0

ACG1403 Excel for Business 1.000 - 3.000 credits

This course will cover Excel topics relevant to the field of accounting and finance including, but not limited to, VLOOKUP, HLOOKUP, INDEX, MATCH, IF, AND, OR, Pivot Tables, Named Ranges, Array Formulas, Custom Number Formats, Conditional Formatting, Absolute References and Keyboard Shortcuts.

Lecture Hours: 16.000 - 48.000

Course User Fee: \$22.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

TAX2000 Income Tax 3.000 credits

Federal income tax fundamentals with emphasis on individual returns. Topics considered include gross income, capital gains and losses, deductions and exemptions, and tax credits.

Lecture Hours: 48.000

Course User Fee: \$15.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision- making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

ACG2450 Microcomputers in Accounting 3.000 credits

Accounting application of electronic data processing including the preparation, interpretation and use of computer information in financial decision-making. Pre/Corequisite: ACG2021. Special fee.

Lecture Hours: 48.000

Pre/Corequisite: ACG2021

Course User Fee: \$22.00

TAX2021 Taxation of Business Organizations 3.000 credits

This course will cover federal income taxation of corporations, S Corporations, limited liability companies and partnerships. Topics covered include determining the tax consequences of income, expenses, distributions, redemptions and liquidations for business entities. Practical application of the tax law will be emphasized along with analysis of tax procedures.

Lecture Hours: 48.000

Course User Fee: \$22.00

Program Description

Accounting Technology Management

College Credit Certificate | Code: 65077 | 27 credits |

Semester 1		Units: 13.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
GEB 1011	Principles of Business	3.0
MTB 1103	Business Mathematics	3.0
OST 2335	Business Writing	3.0
Semester		Units: 14.0
ACG 2071	Managerial Accounting	3.0
ACG 2071L	Managerial Accounting Lab	1.0
ACG 2450	Microcomputers in Accounting	3.0
TAX 2000	Income Tax	3.0
TAX 2021	Taxation of Business Organizations	3.0
Major Course Elective	ACG*, BAN*, FIN*, GEB*, QMB*, TAX*	1.0
		Total: 27.0

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MTB1103 Business Mathematics 3.000 credits

Students will learn the basic arithmetic processes related to mathematics and computations used in businesses, to include cash and trade discounts, commissions, markups and markdowns, depreciation, simple and compound interest, bank discounts, payroll records, payroll taxes, charts and graphs, and consumer applications.

Lecture Hours: 48.000

Course User Fee: \$10.00

OST2335 Business Writing 3.000 credits

Covers the procedures for writing effective business letters and memorandums, a review of grammar, and the proper format of today's business correspondence. Students learn how to prepare inquiry letters, direct and indirect response letters, application letters and resumes, and short reports.

Lecture Hours: 48.000

Course User Fee: \$15.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field. Additional support will be provided to students in order to achieve the objectives of ACG2071. Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

ACG2450 Microcomputers in Accounting 3.000 credits

Accounting application of electronic data processing including the preparation, interpretation and use of computer information in financial decision-making. Pre/Corequisite: ACG2021. Special fee.

Lecture Hours: 48.000

Pre/Corequisite: ACG2021

Course User Fee: \$22.00

TAX2000 Income Tax 3.000 credits

Federal income tax fundamentals with emphasis on individual returns. Topics considered include gross income, capital gains and losses, deductions and exemptions, and tax credits.

Lecture Hours: 48.000

Course User Fee: \$15.00

TAX2021 Taxation of Business Organizations 3.000 credits

This course will cover federal income taxation of corporations, S Corporations, limited liability companies and partnerships. Topics covered include determining the tax consequences of income, expenses, distributions, redemptions and liquidations for business entities. Practical application of the tax law will be emphasized along with analysis of tax procedures.

Lecture Hours: 48.000

Course User Fee: \$22.00

Program Description

Addiction Studies

College Credit Certificate | Code: 65078 | 24 credits

Semester 1		Units: 12.0
HUS 1302	Basic Counseling Skills	3.0
HUS 1440	Family Issues in Chemical Dependency	3.0
HUS 1475	Addiction Counseling and the Law	3.0
PSB 2442	Psychology of Addiction	3.0
Semester 2		Units: 12.0
HUS 1421	Assessment and Treatment Planning in Addictions	3.0
HUS 1423	Group Counseling in Substance Abuse	3.0
HUS 1428	Addiction Treatment Delivery Systems	3.0
HUS 2303	Counseling Techniques	3.0
		Total: 24.0

HUS1302 Basic Counseling Skills 3.000 credits

Development of the skills of observation, recording, reporting, interviewing and counseling. These skills are presented in the context of general counseling theory.

Lecture Hours: 48.000

HUS1440 Family Issues in Chemical Dependency 3.000 credits

This course is designed to analyze the effects of chemical abuse on the family system. Emphasis will be placed on family roles and dynamics; characteristics of children (including adult children) of chemical abusers; theories of codependence; and adaptations made individually and socially by family members. Critical issues and strategies in family treatment will be explored.

Lecture Hours: 48.000

HUS1475 Addiction Counseling and the Law 3.000 credits

This course is designed to introduce addiction counseling students to the vocabulary, agencies and processes required to work with clients involved in both the criminal and civil justice systems. This course focuses on the relationship between the law and Human Services institutions, patterns of lawmaking and law-breaking, the legal structures and processes, and law as an instrument of public policy, social control and social change. The roles and functions of police, courts and correctional services will be surveyed. Common civil issues that affect clients in recovery will be explored. In addition, this course will enable students to explain the legal basis for alcohol and other drug services in Florida. State statutes pertaining to alcohol and drugs and their administrative rules will be reviewed. Confidentiality requirements, compliance standards, and professional ethics will be presented. Prerequisite: PSB2442.

Lecture Hours: 48.000

Prerequisite: PSB2442

PSB2442 Psychology of Addiction 3.000 credits

This course will examine psychological, medical, pharmacological, legal, economic and sociological aspects of addiction to and use of various chemicals. The course will take an in-depth look at narcotic sedatives, and stimulants including alcohol, cocaine, heroin, cannabis, caffeine and tobacco.

Lecture Hours: 48.000

HUS1421 Assessment and Treatment Planning in Addictions 3.000 credits

This course is designed to familiarize students with the core functions of Assessment and Treatment Planning for the chemically dependent client. Emphasis on treatment planning will be accomplished drawing from the Florida Certification Board for addiction professionals and the Department of Children's and Family Services guidelines. Prerequisites: HUS2493, PSB2442.

Lecture Hours: 48.000

Prerequisite: HUS2493, PSB2442

HUS1423 Group Counseling in Substance Abuse 3.000 credits

This course stresses development of effective group counseling leadership skills including organizing, implementing, and evaluating group counseling programs. The course includes actual group experiences.

Prerequisite: PSB2442.

Lecture Hours: 48.000

Prerequisite: PSB2442

HUS1428 Addiction Treatment Delivery Systems 3.000 credits

This course is designed to survey the modalities of addiction treatment. The course will study federal and state systems as well as private-not-for-profit and private-for-profit programs. All of these will be described using examples drawn from local agencies, the diverse populations they serve, and the politics and economics of the systems. This course will also present a critical exploration of the history and theory defining problems of addiction treatment and the characteristics and career issues of an addiction treatment services worker.

Lecture Hours: 48.000

HUS2303 Counseling Techniques 3.000 credits

Specific counseling techniques are introduced within the various counseling theories. Work involves both group and individual techniques.

Lecture Hours: 48.000

Program Description

Air Cargo Management

College Credit Certificate | Code: 66030 | 16 credits

The Air Cargo Management College Credit Certificate program is designed to give students the skills required to gain employment as an air cargo agent. The program can be completed in one or two semesters with classes offered during the day or evening hours. All of the credits earned can be applied towards an A.S. degree in Aviation Administration. Contact the Aviation Department at 305-237-5950 for information and advisement.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 16.0
AVM 1010	Aviation Industry Operation	3.0
AVM 1022	Flight Operations	3.0
AVM 1062	Aviation Career Planning	1.0
AVM 1121	Hazardous Materials/Dangerous Goods	3.0
AVM 2120	Air Cargo	3.0
AVM 2431	Customer Service Agent	3.0

Total: 16.0

AVM1010 Aviation Industry Operation 3.000 credits

The course provides insight into the development and present status of aircraft and air transportation, governmental organizations, controls and regulations, and career opportunities in the field.

Lecture Hours: 48.000

AVM1022 Flight Operations 3.000 credits

An investigation of the occupational duties, responsibilities, and physical facilities required by the positions of pilot, co-pilot, flight engineer, dispatcher and flight attendant.

Lecture Hours: 48.000

AVM1062 Aviation Career Planning 1.000 credits

This course provides direction and guidance in career planning for all aviation students. Topics of discussion will include the job search education and training requirements, resume writing, business etiquette, interview skills and follow-up techniques

Lecture Hours: 16.000

AVM1121 Hazardous Materials/Dangerous Goods 3.000 credits

This course is designed to provide the student with knowledge of dangerous goods/ hazardous materials and their effect in air transportation and logistics. The students will be conversant in hazardous material regulations for cargo and passenger transportation. The course will encompass the identification, labeling, packaging and handling of 9 types of dangerous goods in air transportation and general logistics. Prerequisite: AVM2120.

Special fee.

Lecture Hours: 48.000

Prerequisite: AVM2120

Course User Fee: \$25.00

AVM2120 Air Cargo 3.000 credits

The course develops a comprehensive grasp of the characteristics and evolution of air cargo, its impact on United States industry, inherent problems and future development.

Lecture Hours: 48.000

AVM2431 Customer Service Agent 3.000 credits

This course covers the generic skills needed for any airline position involving regular contact with the traveling public. It includes human relations, personal appearance enhancement, etiquette, conflict management, speech skills, and the acquisition of attributes that would promote a proper professional image.

Lecture Hours: 48.000

Program Description

Airline Maintenance Procedures and Records Management

College Credit Certificate | Code: 63018 | 18 credits

The College Credit Certificate (CCC) in Airline Maintenance Procedures and Records Management will fill a growing need to train personnel in how to organize, review and classify aircraft records to comply with FAA regulations, as well as successfully manage records at aircraft manufacturers, airlines and maintenance repair operators. Students will receive a wide-range of classroom and hands-on training that will provide the student with the education and knowledge to successfully manage the aircraft records department in an aviation company.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 18.0

ASC 1610	Aircraft Engines and Structure Theory	3.0
AVM 1160	Aviation Maintenance Programs and Inspections	3.0
AVM 1161	Aircraft Performance Measures and Maintenance Requirements	3.0

AVM 1162	Maintenance Repair and Overhaul (MRO) Interactions with Commercial Airline Operations	3.0
AVM 1163	Policies and Procedures for Commercial Airlines Maintenance Programs	3.0
AVM 1164	Logistics and Maintenance Programs for Commercial Airlines	3.0

Total: 18.0

ASC1610 Aircraft Engines and Structure Theory 3.000 credits

This is a foundation course in aircraft engines and structure. Students will learn the elements of aircraft engines, engine theory, construction, systems, operating procedures, performance diagnosis, and aircraft structures.

Lecture Hours: 48.000

AVM1160 Aviation Maintenance Programs and Inspections 3.000 credits

This course provides an in-depth study of aircraft inspection programs and maintenance scheduling procedures. Students will learn national and international regulations governing aircraft inspection, maintenance evaluation, and the required procedures to update airline and governmental maintenance technical manuals.

Lecture Hours: 48.000

AVM1161 Aircraft Performance Measures and Maintenance Requirements 3.000 credits

Students will learn aircraft performance measures and maintenance requirements for airplanes powered by reciprocating, turboprop, and/or jet turbine and turbofan engines. Topics include stability and control, weight and balance, performance charts and graphs, takeoff and cruise control, airplane performance characteristics, from which they will extract data that maximizes performance.

Lecture Hours: 48.000

AVM1162 Maintenance Repair and Overhaul (MRO) Interactions with Commercial Airline Operations 3.000 credits

Students will learn the Maintenance Repair and Overhaul (MRO) Procedures in Commercial Airline Operations. Topics include airline maintenance operations, engineering, maintenance, repair of structures, systems, and aircraft components. In addition, students will explore MRO financing, domestic and offshore operations, regulatory requirements, logistics, supply chain support, human resources and industry oversight.

Lecture Hours: 48.000

AVM1163 Policies and Procedures for Commercial Airlines Maintenance Programs 3.000 credits

Students will learn the maintenance policies and programs for commercial airlines. Concepts of Maintenance Steering Group (MSG) and Reliability Centered Maintenance (RCM) programs, Maintenance Control by Reliability Methods (MCEM) program, and Operational Availability (OA) for Commercial Aircraft will be discussed.

Lecture Hours: 48.000

AVM1164 Logistics and Maintenance Programs for Commercial Airlines 3.000 credits

This course focuses on the concepts and application of logistics and supply chain management utilized within the aviation maintenance industry to increase efficiency in production and maintenance. Students will learn the logistics support from Maintenance Repair Operators and Original Equipment Manufacturers in aviation maintenance operations.

Lecture Hours: 48.000

Program Description

[Airport Management](#)

College Credit Certificate | Code: 63017 | 16 credits

The Airport Management College credit certificate program provides the student with the skills required to advance to management positions at airport (city & government) and/or airline terminal operations. Students will understand the cost centers, design processes and financial considerations required to be an effective manager in

the aviation industry. Additional Information: Contact the Aviation Department at (305) 237-5950 for information and advisement.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 16.0

ASC 2320	Aviation Laws and Regulations	3.0
AVM 1010	Aviation Industry Operation	3.0
AVM 1062	Aviation Career Planning	1.0
AVM 1440	Aviation/Airport Security	3.0
AVM 2410	Principles of Airport Management	3.0
AVM 2450	Airport Facilities/Financial Planning	3.0

Total: 16.0

ASC2320 Aviation Laws and Regulations 3.000 credits

Insight pertinent to federal governing bodies, and current local, federal and international laws forming the present structure of aviation law.

Lecture Hours: 48.000

AVM1010 Aviation Industry Operation 3.000 credits

The course provides insight into the development and present status of aircraft and air transportation, governmental organizations, controls and regulations, and career opportunities in the field.

Lecture Hours: 48.000

AVM1062 Aviation Career Planning 1.000 credits

This course provides direction and guidance in career planning for all aviation students. Topics of discussion will include the job search education and training requirements, resume writing, business etiquette, interview skills and follow-up techniques

Lecture Hours: 16.000

AVM1440 Aviation/Airport Security 3.000 credits

This course will provide the student with knowledge of the issues and strategies that are used to protect the national airspace system, airports and airlines from security threats. The various types of threats and responses to those threats will be covered. In addition, the legal requirements, planning issues, physical equipment and facility requirements and personnel issues will also be discussed.

Lecture Hours: 48.000

Course User Fee: \$50.00

AVM2410 Principles of Airport Management 3.000 credits

This course provides the student with a broad background in the Principles of Airport Management. This includes the airport system and its history, planning, land use, community relation issues, financial issues, capacity and growth, operations, organization and administration. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$25.00

AVM2450 Airport Facilities/Financial Planning 3.000 credits

This course provides the student with an in-depth knowledge of the techniques and strategies of the airport master plan in planning airport facilities and financial resources. Forecasting, demand analysis, sources of funding, planning requirements, environmental issues and requirements and compliance issues will be discussed. Also, implementation and control issues, financial management, budgets, costs and revenues as well as airport economics will be discussed. Prerequisite: AVM2410. Special fee.

Lecture Hours: 48.000

Prerequisite: AVM2410

Course User Fee: \$25.00

Program Description

American Sign Language

College Credit Certificate | Code: 64053 | 22 credits

The American Sign Language College Credit Certificate program equips students with basic American Sign Language skills to communicate with Deaf and Hard of Hearing individuals using expressive fingerspelling, signing, and conversational behaviors. Learning basic American Sign Language and understanding Deaf Culture benefits all students as it diversifies their communications skills, particularly those planning to enter or are currently employed in the highly competitive human services and customer relations job markets. While this program does not prepare students to become interpreters, students who earn this certificate may choose to continue their studies in the A.S. in Sign Language Interpretation program.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 7.0
ASL 1140C	American Sign Language 1	4.0
ASL 2510	Deaf Culture And Community	3.0
Semester 2		Units: 4.0
ASL 1150C	American Sign Language 2	4.0
Semester 3		Units: 7.0
ASL 2160C	American Sign Language 3	4.0
ASL 2210	ASL Conversational Skills	3.0
Semester 4		Units: 4.0
ASL 2200C	American Sign Language 4	4.0
		Total: 22.0

ASL1140C American Sign Language 1 4.000 credits

This course provides introductory information on the linguistics of American Sign Language and approximately 500 sign concepts. The course includes lecture, discussion and lab practice.

Lecture Hours: 64.000

ASL2510 Deaf Culture And Community 3.000 credits

The course provides an in-depth study of the lives and experiences of deaf and hard of hearing persons and examines perceptions of belonging to a unique cultural group. Cultural characteristics are examined alongside the impact of hearing loss on one's family, friends and employment. Multicultural issues and the impact of hearing loss are examined within various ethnic groups. Societal attitudes regarding disability in general, and hearing loss and communication difficulties in particular, are addressed.

Lecture Hours: 48.000

ASL1150C American Sign Language 2 4.000 credits

This course provides continued instruction in the linguistic principles of American Sign Language and an additional 500 sign concepts. The course includes lecture, discussion and lab practice which is conducted in ASL. Prerequisite: ASL1140C.

Lecture Hours: 64.000

ASL2160C American Sign Language 3 4.000 credits

This course provides linguistic principles of American Sign Language at the intermediate level and an additional 500 sign concepts. Lecture, discussion and lab practice are included. Students have increased opportunities for interaction with members of the deaf community. Increasingly, class sessions are conducted in ASL. Prerequisite: ASL1150C; Pre/Corequisite: ASL2210.

Lecture Hours: 64.000

ASL2210 ASL Conversational Skills 3.000 credits

The course provides opportunities to utilize all previously acquired knowledge and skill in ASL, including vocabulary and all components and linguistics principles of the language. Emphasis is placed on the more difficult aspects of ASL, such as role shifting for relating stories and the use of classifier predicates for providing visual descriptions. Class sessions are conducted entirely without voice. Prerequisites: ASL1150C, ASL2160C.

Lecture Hours: 48.000

ASL2200C American Sign Language 4 4.000 credits

This course provides linguistic principles of American Sign Language at the advanced level and an additional 500 sign concepts including idioms used in ASL. Lecture, discussion and lab practice are included. Class sessions are conducted predominately in ASL. Prerequisite: ASL 2160C.

Lecture Hours: 64.000

Artificial Intelligence Awareness

College Credit Certificate | Code: 66040 | 9 credits

As the impact of artificial intelligence (AI) expands to every industry, foundational knowledge of the applications of these systems is something employers are exceedingly looking for. The College Credit Certificate in Artificial Intelligence Awareness is suited for students in any field interested in learning about the applications of AI across a variety of sectors, stages involved in a typical AI project, and the ethical considerations when adopting these technologies.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1**Units: 9.0-10.0**

CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
PHI 2680	Artificial Intelligence and Ethics	3.0
<i>Discipline Specific Elective**</i>		3.0-4.0
CAI 1320C	Introduction to Prompt Engineering	3.0
OR		
COP 1047C	Introduction to Python Programming	4.0
OR		
CCJ 2840	Introduction to Criminal Justice Research with AI	3.0
OR		
ECW 2010	Teaching with AI: Designing Human-Centered Classrooms	3.0
OR		
ETS 1603C	Introduction to Robotics	4.0
OR		
GEB 1432	Applied Artificial Intelligence (AI) in Business	3.0
OR		
HSC 2060	Artificial Intelligence Applications in Healthcare	3.0
OR		
PLA 2872	Artificial Intelligence (AI) and the Law	3.0
OR		
SCM 1114	Applied Advanced Technologies & AI in Supply Chain Management	3.0

*(Select one (1) course from the following options)

Total: 9.0-10.0

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

CAI1320C Introduction to Prompt Engineering 3.000 credits

This course introduces students to prompt engineering, focusing on foundational concepts and practical applications across various tools. Students will learn to craft prompts that enhance creativity, precision, and efficiency, leveraging AI to optimize workflows and productivity.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CCJ2840 Introduction to Criminal Justice Research with AI 3.000 credits

This course provides an introduction to core research methods utilized in the field of Criminology and Criminal Justice. Students will learn foundational research concepts, including research design, data collection methods, and data analysis techniques. A key focus will be on the ethical and effective use of Artificial Intelligence (AI) tools to assist in various stages of the research process, such as data cleaning, analysis, and interpretation. Students will gain practical experience in applying these methods through hands-on projects and develop critical thinking skills necessary for evaluating research findings and understanding the social implications of criminal justice research. Recommended: Students are encouraged to take ENC1101 or CCJ1020 prior to registration.

Lecture Hours: 48.000

Recommended Preparation: ENC1101 OR CCJ1020

ECW2010 Teaching with AI: Designing Human-Centered Classrooms 3.000 credits

This course invites learners to explore the integration of artificial intelligence through a pedagogical lens, where technology serves as a tool for reimagining teaching and learning. Learners will critically engage with AI tools by grounding their use in learning theory and evidence-based design practices. Through inquiry, collaboration, and hands-on experimentation, learners will examine how AI can amplify creativity, personalize learning, and promote human-centeredness.

Lecture Hours: 48.000

ETS1603C Introduction to Robotics 4.000 credits

This is an introductory level course designed as an introduction to robotics and robotic applications. Students learn topics related to robotic design including robotic terminology, robotic programming, sensing and sensors, actuators, modeling and sensing, robotic platforms, and the application of artificial intelligence in robotics. Laboratory activities provide hands-on application of concepts and theories.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

GEB1432 Applied Artificial Intelligence (AI) in Business 3.000 credits

This course illustrates how Artificial Intelligence (AI) is currently applied in industries and across business functions to support business strategy formulation, implementation and evaluation. This course introduces the data-driven outcomes and ethical issues to be considered when implementing responsible AI in business and industry. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$35.20

HSC2060 Artificial Intelligence Applications in Healthcare 3.000 credits

An introduction to the health care environment and delivery systems. In this course, students will explore how Artificial Intelligence (AI) is impacting healthcare professions. Students will gain a fundamental understanding of ethical implications, legal principles, and applications of AI in various health settings.

Lecture Hours: 48.000

Course User Fee: \$28.02

PLA2872 Artificial Intelligence (AI) and the Law 3.000 credits

Introduction to the intersection of artificial intelligence and the law, focusing on its challenges, opportunities, ethics, and impact on the law, the legal industry, and society. (3 credits)

Lecture Hours: 48.000

Recommended Preparation: Students are encouraged to enroll in ENC1101 prior to registering for PLA2872.

SCM1114 Applied Advanced Technologies & AI in Supply Chain Management 3.000 credits

This course provides an in-depth exploration of Industry 4.0 and Artificial Intelligence (AI) applications within the supply chain industry, focusing on strategic support, optimization, data-driven decision-making, and organizational readiness. Students will discover how AI technologies enhance operational efficiency, reduce costs, and improve overall supply chain performance while ensuring responsible and ethical implementation. Through case studies and practical exercises, learners will gain the skills needed to leverage AI for innovative solutions and competitive advantage in supply chain management.

Lecture Hours: 48.000

Program Description

Artificial Intelligence Practitioner

College Credit Certificate | Code: 66041 | 18 credits

As artificial intelligence (AI) continues to expand in every industry, so will the demand for skilled professionals in applied AI. The College Credit Certificate in Artificial Intelligence Practitioner is for individuals with or without a technical background who are interested in entering this growing field. Learners will be introduced to core AI applications and the ethical considerations in their design and execution. Case studies and real-world applications will support learning in the fundamentals of machine learning, the functions of AI virtual assistants, and techniques used in computer vision.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 7.0
CAI 1001C	Artificial Intelligence (AI) Thinking	3.0
COP 1047C	Introduction to Python Programming	4.0
Semester 2		Units: 6.0
CAI 2100C	Machine Learning Foundations	3.0
PHI 2680	Artificial Intelligence and Ethics	3.0
Semester 3		Units: 6.0

CAI 2300C	Introduction to Natural Language Processing	3.0
CAI 2840C	Introduction to Computer Vision	3.0

Total: 19.0

CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAI2100C Machine Learning Foundations 3.000 credits

Students are introduced to machine learning concepts and Python applications, including data acquisition, supervised, unsupervised, and reinforced learning. In addition, students will develop and deploy artificial intelligence (AI) models utilizing classification algorithms. Prerequisites: CAI1001C, COP1047C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI1001CCOP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

CAI2300C Introduction to Natural Language Processing 3.000 credits

Students will learn the fundamental concepts of Natural Language Processing (NLP) and text processing. In addition, focus will be on knowledge and skills necessary to create a language recognition application. Pre/Corequisite: CAI2100C.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: CAI2100C

Course User Fee: \$63.00

CAI2840C Introduction to Computer Vision 3.000 credits

Students will learn fundamental concepts in Computer Vision (CV) and image processing, including introduction to necessary proprietary and open-source Python libraries. Pre/Corequisite: CAI2100C.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: CAI2100C

Course User Fee: \$63.00

Program Description

[Audio Technology](#)

College Credit Certificate | Code: 61000 | 15 credits

The Audio Technology College Credit Certificate (CCC) is designed for students who intend to seek employment within the music business industry as an alternative to the strictly traditional Music degree program. The Audio Technology college credit certificate combines a traditional music curriculum with industry-related courses and experiences. The curriculum stresses hands-on equipment use in sound engineering and recording, midi music creation, sound reinforcement methods and technologies, and computer applications.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 7.0
MUM 2600	Sound Recording 1	3.0
MUM 2600L	SOUND RECORDNG 1 LAB	1.0
MUM 2623C	MIDI Electronic Music 1	3.0
Semester 2		Units: 8.0
MUM 2601	Sound Recording 2	3.0
MUM 2601L	SOUND RECORDNG 2 LAB	1.0
MUM 1662	SOUND REINFORCE FUND	3.0
MUM 1622L	SOUND REINF/FUND LAB	1.0
		Total: 15.0

MUM2600 Sound Recording 1 3.000 credits

An introduction to techniques, practices and procedures in making eight-track recordings. The student will gain experience with acoustical balancing, editing and over-dubbing in a wide variety of sound situations.

Lecture Hours: 48.000

Corequisite: MUM2600L

MUM2600L SOUND RECORDNG 1 LAB 1.000 credits

Participation in MUM 2600L offers students directed "hands on" experience coinciding with lectures in MUM 2600. Corequisite: MUM 2600. Special fee. (2 hr lab)

Lab Hours: 32.000

Course User Fee: \$15.00

MUM2623C MIDI Electronic Music 1 3.000 credits

This course is designed to acquaint music students with basic applications of Musical Instrument Digital Interface (MIDI) for the purpose of composition and performance and learning pre-production concepts with multi-track recording studio. Emphasis will be placed on keyboards, outboard gear, drum machines, and computer-assisted operations. Special fee.

Lecture Hours: 32.000, Lab Hours: 16.000 - 32.000

Course User Fee: \$10.00

MUM2601 Sound Recording 2 3.000 credits

This course explores advanced multi-track recording skills and audio production techniques. Emphasis is on mixing board skills, microphone techniques, use of outboard equipment and live 2 track recording.

Prerequisites: MUM 2600 and MUM 2600L; Corequisite: MUM 2601L. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MUM2600 AND MUM2600L AND Corequisite: MUM2601L

MUM2601L SOUND RECORDNG 2 LAB 1.000 credits

Corequisite for MUM 2601. Advanced Sound Recording. Participation in MUM 2601L offers students directed "hands on" experience parallel- ling lectures in MUM 2601. Special fee. (2 hr lab)

Lab Hours: 32.000

Course User Fee: \$15.00

MUM1662 SOUND REINFORCE FUND 3.000 credits

Sound reinforcement fundamentals is a course designed to provide students with background in live sound reinforcement, concert sound practices, and general PA work associated with sound engineering. Corequisite: MUM 1622L. (3 hr. lecture)
Lecture Hours: 48.000

MUM1622L SOUND REINF/FUND LAB 1.000 credits

This course is designed to provide students with "hands on" experience in conjunction with music school performance activities. Corequisite: MUM 1662. (2 hr. lab)
Lab Hours: 32.000

Program Description

Automation

College Credit Certificate | Code: 68000 | 15 credits

This certificate prepares students for initial employment with an occupational title as an Automation or Applied Automation Specialist in various specialized areas. It can also provide supplemental training for persons previously or currently employed in these occupations. These courses can be applied toward the Associate of Science in Advanced Manufacturing.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 9.0

MAC 1105	College Algebra	3.0
ETI 1040	Introduction to Bioscience Manufacturing	3.0
ETI 1701	Industrial Safety	3.0

Semester 2

Units: 6.0

ETI 2404	Advanced Manufacturing Technology	3.0
ETM 1315C	Applied Pneumatics and Hydraulics	3.0

Total: 15.0

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

ETI1040 Introduction to Bioscience Manufacturing 3.000 credits

This course introduces students to the field of bioscience manufacturing. Topics will include basic principles of the industry, large-scale process development and the future of the bioscience industry. Current Good Manufacturing Practices (cGMPs), and the nature and delivery system of products will also be discussed.

Lecture Hours: 48.000

ETI1701 Industrial Safety 3.000 credits

This course provides the student with the knowledge and skills to recognize hazardous situations in industrial plants and the precautions to be observed and practiced to perform work activities safely. Among the topics covered are industrial safety hazards, electrical safety, working with chemicals, gases, and solvents, protective equipment, and safe working conditions.

Lecture Hours: 48.000

ETI2404 Advanced Manufacturing Technology 3.000 credits

This is a course for students intending to work in manufacturing environments. Students will learn the basic concepts about advanced manufacturing operations and processes, including sourcing materials, production planning and process monitoring, and control to distribution activities. Students also review the facility and regulatory requirements needed to support manufacturing operations. Activities may include facility tours and site visits.

Lecture Hours: 48.000

ETM1315C Applied Pneumatics and Hydraulics 3.000 credits

This course prepares students to perform mechanical maintenance on industrial equipment and devices. Students learn the theory and application of fluid mechanics, how to calibrate metering devices, and conduct elementary hydraulic tests. Pre/Corequisite: MAC1105.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

Program Description

Biotechnology

College Credit Certificate | Code: 60002 | 19 credits

The College Credit Certificate in Biotechnology seeks to prepare students for immediate entry-level employment in the biotechnology-, pharmaceutical-, or medical device manufacturing- industry. The program prepares individuals in the principles, procedures, and practices used in the bioscience industries. The college credits granted in this program will apply toward an Associate in Science degree in Biotechnology.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 11.0

BSC 2426	Biotechnology Methods and Applications 1	3.0
BSC 2426L	Biotechnology Methods and Applications 1 Laboratory	2.0
MAC 1105	College Algebra	3.0
ETI 1040	Introduction to Bioscience Manufacturing	3.0

Semester 2

Units: 8.0

CHM 1045	General Chemistry and Qualitative Analysis	3.0
CHM 1045L	General Chemistry Lab	2.0
ETI 1172	Introduction to Quality Assurance	3.0

Total: 19.0

BSC2426 Biotechnology Methods and Applications 1 3.000 credits

This course addresses the basic principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical, biotechnology and/or research laboratory setting(s). Practical applications of biotechnology are explored.

Lecture Hours: 48.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426L

BSC2426L Biotechnology Methods and Applications 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2426 Biotechnology Methods and Applications 1. This is a hands-on course that emphasizes the basic laboratory principles, techniques, and instrumentation, necessary for effective work in pharmaceutical, biotechnology, and/or research laboratory settings(s).

Lab Hours: 64.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426

Course User Fee: \$50.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

ETI1040 Introduction to Bioscience Manufacturing 3.000 credits

This course introduces students to the field of bioscience manufacturing. Topics will include basic principles of the industry, large-scale process development and the future of the bioscience industry. Current Good Manufacturing Practices (cGMPs), and the nature and delivery system of products will also be discussed.

Lecture Hours: 48.000

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045

Course User Fee: \$30.00

ETI1172 Introduction to Quality Assurance 3.000 credits

This course describes the role and aspects of quality systems and Regulatory affairs in research laboratories, regulated companies, and firms that comply with voluntary standards. Topics include stages in development and submission of drugs and medical devices, patents legislation, and quality systems such as auditing, standard procedures, good manufacturing and laboratory practices.

Lecture Hours: 48.000

Program Description

Building Construction Specialist

College Credit Certificate | Code: 66025 | 18 credits

The Building Construction Specialist college credit certificate prepares students for entry in the construction field through acquisition of a basic understanding of drawings and construction estimating through curriculum and practical experiences. Following successful completion, students can obtain employment assisting in construction offices or job sites.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 10.0
ARC 1126C	Architectural Drawing 1	4.0
BCN 1272	Building Construction Plans Interpretation 1	3.0
BCN 1275	Building Construction Plans Interpretation 2	3.0
Semester 2		Units: 8.0
ARC 1949	Co-op Work Experience 1: ARC	3.0
BCT 1770	Building Construction Estimating Fundamentals	3.0
BCT 1771	Building Construction Advanced Estimating	3.0

Total: 18.0

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions. Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

BCN1272 Building Construction Plans Interpretation 1 3.000 credits

Develops the students' ability to interpret working drawings. Students will learn the conventions of graphic and symbolic language used by construction professionals to communicate information on drawings. Emphasis is on architectural and structural details with limited coverage on mechanical and electrical aspects. Develops the students' ability to interpret working drawings. Students will learn the conventions of graphic and symbolic language used by construction professionals to communicate information on drawings. Emphasis is on architectural and structural details with limited coverage on mechanical and electrical aspects.

Lecture Hours: 48.000

Course User Fee: \$10.00

BCN1275 Building Construction Plans Interpretation 2 3.000 credits

In this course, students will plan interpretation of complex working drawings for multi-story residential and commercial buildings. Students will learn to read and understand construction working drawings, identify structural systems and their details. Familiarity with all aspects of complex working drawings will be addressed.

Prerequisite: BCN1272.

Lecture Hours: 48.000

Prerequisite: BCN1272

Course User Fee: \$10.00

ARC1949 Co-op Work Experience 1: ARC 3.000 credits

This course is designed to provide training in the students' field of study through work experience. Students will learn to make connections between their internship experiences, academic coursework, and career goals. Students are graded on the basis of documentation of learning and goal achievement as reported by both student and employer. Prerequisite: departmental approval.

Other Hours: 48.000

Recommended Preparation: 2.0 GPA, minimum of 6 credits in major or approved work experience, and departmental approval required.

BCT1770 Building Construction Estimating Fundamentals 3.000 credits

An analysis and calculation of building construction costs. Students will learn the classification of materials, labor, and subcontracted work into the smallest manageable units. Students will develop a simple estimate for a residential structure. Laboratory fee.

Lecture Hours: 48.000

Prerequisite: BCN1272

Course User Fee: \$10.00

BCT1771 Building Construction Advanced Estimating 3.000 credits

Estimating more advanced elements of building construction. Students will learn to calculate direct, indirect, and overhead costs, as well as prepare bid proposals and related documents for commercial buildings. Laboratory fee.

Lecture Hours: 48.000

Prerequisite: BCN1275, BCT1770

Course User Fee: \$10.00

Program Description

Entrepreneurial Operations and Management

College Credit Certificate | Code: 65110 | 18 credits

Semester 1		Units: 12.0
ENT 2421	Funding Your Venture	3.0
ENT 2820	Entrepreneurship Capstone: Entrepreneurship in Contemporary Society	3.0
GEB 2112	Introduction to Entrepreneurship	3.0
MAR 1720	Marketing in a Digital World	3.0
Semester 2		Units: 6.0
Program Electives*		

Select 6 credits from the following course options: ACG2450, ENT1501**, ENT2201*, ENT2502**, ENT2511**, ENT2212**, GEB2YYY*, SMB1000*, TAX2021*. Note: • All courses must be met with a grade of "C" or higher. •

*Courses recommended for students pursuing careers in family-owned or legacy businesses. • **Courses recommended for students focused on social entrepreneurship and impact-driven ventures.

Total: 18.0

ENT2421 Funding Your Venture 3.000 credits

This course focuses on critical skills necessary to develop appropriate funding strategies for new venture creation and growth. Students will explore a variety of ways to raise capital and gain an understanding of investors' expectations and how to evaluate the advantages and pitfalls of various sources of capital. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$43.00

ENT2820 Entrepreneurship Capstone: Entrepreneurship in Contemporary Society 3.000 credits

This course explores how entrepreneurship transforms ideas into practical solutions within contemporary society. Students will apply the problem-opportunity venture operations (POVO) model to identify real world problems, recognize opportunities and develop evidence-based venture concepts and solutions. The course emphasizes opportunity assessment, problem-solving and professional entrepreneurial communication, culminating in a final project and/or presentation and evaluation. Learning takes place through practicum-based experiences, which may include real small business consulting or applied case projects. Prerequisites: ENT2421, GEB2112.

Lecture Hours: 48.000

Prerequisite: ENT2421 AND GEB2112

GEB2112 Introduction to Entrepreneurship 3.000 credits

Students will learn that start-ups are not smaller versions of big businesses. They are unique. This foundation course in business entrepreneurship covers the attributes of successful entrepreneurs, opportunity identification, opportunity analysis and development, as well as an overview of the key activities and functions which start-ups must address.

Lecture Hours: 48.000

Course User Fee: \$110.00

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

Program Description

Business Industry Operations

College Credit Certificate | Code: 65061 | 9 credits

The 9-credit-hour certificate program in Business Industry Operations, includes a series of courses that offer students the technical knowledge and skills needed to prepare for further education and careers in a variety of industries aligned to a student's pathway.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 9.0**

ENC 1101	English Composition 1	3.0
GEB 1011	Principles of Business	3.0
Program Elective*		3.0

*Select 3 credits from the following course options: ACG1403, CAP1788, ECO2013, EEC2002, FSE1000, FSE2202, GEB1000, GEB2112, HFT1000, HLP1081, HOS1010, HUN1201, MAN2021, MAN2300, MAR1011, TAX2000, TRA1154.

Total: 9.0

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

Program Description

[Business Intelligence Professional](#)

College Credit Certificate | Code: 66038 | 20 credits

The Business Intelligence Professional College Credit Certificate prepares students for employment in various business intelligence positions and/or for continued studies in the Business Intelligence Specialist AS degree. Students will learn to work with a variety of datasets and transform them into valuable insights via visualizations and reports.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 8.0
CAP 1788	Introduction to Data Analytics	4.0
CGS 1540C	Database Concepts Design	4.0
Semester 2		Units: 12.0
CAP 2743C	Power BI - Data Visualization and Analysis	4.0
CAP 2761C	Intermediate Analytics	4.0
CAP 2791C	Power BI - Data Preparation and Modeling	4.0
		Total: 20.0

CAP1788 Introduction to Data Analytics 4.000 credits

This course offers a broad introduction to data analytics and the role it plays in modern organizations. Students will use Excel and Tableau to import, clean, transform, and visualize data. Students will also learn how to effectively communicate findings to decision makers.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CAP2743C Power BI - Data Visualization and Analysis 4.000 credits

This is the second of two introductory courses aimed at preparing students for the Microsoft Power BI certification exam. Students will learn how to create informative data visualizations and leverage Power BI's analytic capabilities to provide meaningful business insights. Prerequisite: CAP2791C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP2791C

Course User Fee: \$63.00

CAP2761C Intermediate Analytics 4.000 credits

This course offers an introduction to the SQL language and how it can be used by data analysts to query databases and gain valuable insights. Students will also gain extensive hands-on experience with Microsoft SQL Server, Azure, and GitHub. Prerequisites: CAP1788, CGS1540C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1540C

Course User Fee: \$63.00

CAP2791C Power BI - Data Preparation and Modeling 4.000 credits

This is one of two introductory courses aimed at preparing students for the Microsoft Power BI certification exam. Students will learn how to get data from different sources, clean and transform the data, and design a data model that will be used in analysis and reports. Prerequisites: CAP1788 and CGS1540C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND CGS1540C

Course User Fee: \$63.00

Program Description

Business Operations - Business Management

College Credit Certificate | Code: 65022 | 18 credits

The Business Operations College Credit Certificate program is the second in a series of three College Credit Certificate programs designed to prepare students for employment and advancement in the following areas: accounting/budgeting, business/ management, customer service, finance, human resources, international business, marketing, nonprofit, real estate, retail and small business. There is only one College Credit Certificate in Business Operations. Students may select one of the 3 options, but the certificate is awarded only once.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
GEB 1011	Principles of Business	3.0
MAN 2021	Principles of Management	3.0
MAN 2604	Managing in a Multi-Cultural Environment	3.0
MAR 1011	Principles of Marketing	3.0
Semester 2		Units: 6.0

MAR 2704	Marketing Web Analytics	3.0
SBM 1000	Small Business Management	3.0

Total: 18.0

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

MAN2604 Managing in a Multi-Cultural Environment 3.000 credits

This course will introduce opportunities and problems encountered by managers operating in a diverse environment either within or outside their home country's borders. Discussions will cover the environment of multinational management as well as planning, organizing, staffing, leading and controlling in both domestic and multinational companies. Current events and cultural issues that significantly affect international business will also be examined.

Lecture Hours: 48.000

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAR2704 Marketing Web Analytics 3.000 credits

This course helps students to develop a deep understanding of the digital analytics landscape. The student will learn how to identify which metrics are best to measure web, mobile, social and marketing channels, and how organizations use web analytics to obtain higher profits, improved customer relationships, and measurable value. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$57.00

SBM1000 Small Business Management 3.000 credits

Students will learn that growing a business involves shifting from the search for and validation of a business model to executing and scaling the business model. The skill sets are different and more formal management is often required. The course reviews the context and complexity of scaling a business and executing a business model.

Lecture Hours: 48.000

Course User Fee: \$90.00

Program Description

College Credit Certificate | Code: 65025 | 18 credits

Semester 1		Units: 12.0
GEB 2112	Introduction to Entrepreneurship	3.0
MAN 2021	Principles of Management	3.0
MAN 2300	Human Resources Management	3.0
MNA 1345	Effective Supervision	3.0
Semester 2		Units: 6.0
MAR 2101	Social Media Marketing	3.0
MNA 2120	Human Relations in Business	3.0
		Total: 18.0

Program Description

Business Operations - Marketing

College Credit Certificate | Code: 65027 | 18 credits

The Business Operations College Credit Certificate program is the second in a series of three College Credit Certificate programs designed to prepare students for employment and advancement in the following areas: accounting/budgeting, business/ management, customer service, finance, human resources, international business, marketing, nonprofit, real estate, retail and small business. There is only one College Credit Certificate in Business Operations. Students may select one of the 3 options, but the certificate is awarded only once.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
MAR 1011	Principles of Marketing	3.0
MAR 1720	Marketing in a Digital World	3.0
MAR 2101	Social Media Marketing	3.0
MAR 2150	International Marketing	3.0
Semester 2		Units: 6.0
MAR 2520	Hispanic Marketing Communications	3.0
MKA 1511	Principles of Advertising and Copywriting	3.0
		Total: 18.0

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR2101 Social Media Marketing 3.000 credits

This course addresses the changes in marketing as a result of social media. The student will explore in-depth the role of social media in marketing. The student will focus on developing an effective marketing plan utilizing social media, with an emphasis on the importance of building an online customer community and developing customer engagement and loyalty.

Lecture Hours: 48.000

MAR2150 International Marketing 3.000 credits

Students will learn the four P's of product, price, place (distribution), and promotion as they relate to a global marketing strategy. The concepts are introduced within the international trade framework, as well as the cultural, social, economic, regulatory, and political environments affecting global marketing efforts.

Lecture Hours: 48.000

MAR2520 Hispanic Marketing Communications 3.000 credits

Students will learn how firms can formulate strategies to attract the U.S. Hispanic market and its various sub-segments. Emphasis is on strategy formulation and integrated marketing communications, including the use of case studies.

Lecture Hours: 48.000

MKA1511 Principles of Advertising and Copywriting 3.000 credits

Techniques and behavioral factors used in advertising and copyrighting which best motivate the consumer. Principles are applied in clear, concise written expression of various appeals used in selling goods and services. (3 hr. lecture)

Lecture Hours: 48.000

Program Description

Business Specialist - Business Management

College Credit Certificate | Code: 65011 | 12 credits

The Business Specialist College Credit Certificate program is the first in a series of three College Credit Certificate programs designed to prepare students for employment in entry-level positions in the following areas: accounting/budgeting, business management, finance, human resources, international business, marketing, small business. There is only one College Credit Certificate in Business Specialist. Students may select one of the 5 options, but the certificate is awarded only once.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

GEB 1011	Principles of Business	3.0
MAR 1011	Principles of Marketing	3.0
MNA 1345	Effective Supervision	3.0
SBM 1000	Small Business Management	3.0

Total: 12.0

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MNA1345 Effective Supervision 3.000 credits

Students will learn to implement supervisory and management practices. Emphasis is placed on learning to communicate more effectively with employees, motivating employees, increasing one's leadership effectiveness, delegating, counseling problem employees, conducting performance reviews, maintaining a discrimination and harassment-free workplace, and managing time. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

SBM1000 Small Business Management 3.000 credits

Students will learn that growing a business involves shifting from the search for and validation of a business model to executing and scaling the business model. The skill sets are different and more formal management is often required. The course reviews the context and complexity of scaling a business and executing a business model.

Lecture Hours: 48.000

Course User Fee: \$90.00

Program Description

Business Specialist - General Business

College Credit Certificate | Code: 65100 | 12 credits

The Business Specialist College Credit Certificate is the first in a series of three College Credit Certificate programs designed to prepare students for employment in entry-level positions in the following areas: accounting/budgeting, business management, finance, general business, human resources, international business, marketing, and small business. Students may select one of the 5 options, but the certificate is awarded only once.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

GEB 1011	Principles of Business	3.0
ENC 1101	English Composition 1	3.0
Group B - Business Course*		3.0
Group C - Career Area of Interest Course**		3.0

Select 3 credits from the following course option: ACG, BUL*, ECO*, FIN1*, FIN2*, MAN1*, MAN2*, MAR*, MAR1011, MAR2150, OST*, PUR*, QMB*, SBM*, TAX*

**Select 3 credits from any transferable Credit Type 1 or Credit Type 2 course.

Total: 12.0

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

Program Description

Business Specialist - Human Resources

College Credit Certificate | Code: 65014 | 12 credits

The Business Specialist College Credit Certificate program is the first in a series of three College Credit Certificate programs designed to prepare students for employment in entry-level positions in the following areas: accounting/budgeting, business management, finance, human resources, international business, marketing, small business. There is only one College Credit Certificate in Business Specialist. Students may select one of the 5 options, but the certificate is awarded only once.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

MAN 2021	Principles of Management	3.0
MAN 2300	Human Resources Management	3.0
MNA 1345	Effective Supervision	3.0
MNA 2120	Human Relations in Business	3.0

Total: 12.0

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

MAN2300 Human Resources Management 3.000 credits

Reviews how the personnel/human resources department contributes to overall planning and profitability of an organization. Major topics include typical personnel functions: recruitment and selection, training, performance appraisal, job analysis, and compensation and benefits administration. Class discussions will focus on changing value systems in the work force and the resulting challenges for managers.

Lecture Hours: 48.000

Course User Fee: \$20.00

MNA1345 Effective Supervision 3.000 credits

Students will learn to implement supervisory and management practices. Emphasis is placed on learning to communicate more effectively with employees, motivating employees, increasing one's leadership effectiveness, delegating, counseling problem employees, conducting performance reviews, maintaining a discrimination and harassment-free workplace, and managing time. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

MNA2120 Human Relations in Business 3.000 credits

Students will learn to implement human relations and communication skills necessary for superior performance and career advancement in the business profession. Emphasis is placed on learning and practicing effective interpersonal communication skills, giving criticism tactfully, expressing feelings constructively, being more sensitive to body language messages, and active listening. Other major topics emphasized are building self-esteem, learning how values and attitudes influence job performance and work relationships, assertion skills, group dynamics and team building, managing conflict, dealing with difficult people, and the challenges and opportunities of getting along in a culturally diverse workplace. Special fee.

Lecture Hours: 48.000

Course User Fee: \$20.00

Program Description

Business Specialist - International Business

College Credit Certificate | Code: 65015 | 12 credits

The Business Specialist College Credit Certificate program is the first in a series of three College Credit Certificate programs designed to prepare students for employment in entry-level positions in the following areas: accounting/budgeting, business management, finance, human resources, international business, marketing, small business. There is only one College Credit Certificate in Business Specialist. Students may select one of the 5 options, but the certificate is awarded only once.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

GEB 1011	Principles of Business	3.0
GEB 2350	Introduction to International Business	3.0
MAN 2604	Managing in a Multi-Cultural Environment	3.0
MAR 2150	International Marketing	3.0

Total: 12.0

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

GEB2350 Introduction to International Business 3.000 credits

Provides an overview of the cultural environment of international business and the institution which affects business today. International economic, political, and trade issues are analyzed in the context of socio-economic goals and policies of the nations involved.

Lecture Hours: 48.000

MAN2604 Managing in a Multi-Cultural Environment 3.000 credits

This course will introduce opportunities and problems encountered by managers operating in a diverse environment either within or outside their home country's borders. Discussions will cover the environment of multinational management as well as planning, organizing, staffing, leading and controlling in both domestic and multinational companies. Current events and cultural issues that significantly affect international business will also be examined.

Lecture Hours: 48.000

MAR2150 International Marketing 3.000 credits

Students will learn the four P's of product, price, place (distribution), and promotion as they relate to a global marketing strategy. The concepts are introduced within the international trade framework, as well as the cultural, social, economic, regulatory, and political environments affecting global marketing efforts.

Lecture Hours: 48.000

Program Description

Business Specialist - Marketing

College Credit Certificate | Code: 65016 | 12 credits

The Business Specialist College Credit Certificate program is the first in a series of three College Credit Certificate programs designed to prepare students for employment in entry-level positions in the following areas: accounting/budgeting, business management, finance, human resources, international business, marketing, small business. There is only one College Credit Certificate in Business Specialist. Students may select one of the 5 options, but the certificate is awarded only once.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

MAR 1011	Principles of Marketing	3.0
MAR 1720	Marketing in a Digital World	3.0
MAR 2101	Social Media Marketing	3.0
MKA 1511	Principles of Advertising and Copywriting	3.0

Total: 12.0

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR2101 Social Media Marketing 3.000 credits

This course addresses the changes in marketing as a result of social media. The student will explore in-depth the role of social media in marketing. The student will focus on developing an effective marketing plan utilizing social media, with an emphasis on the importance of building an online customer community and developing customer engagement and loyalty.

Lecture Hours: 48.000

MKA1511 Principles of Advertising and Copywriting 3.000 credits

Techniques and behavioral factors used in advertising and copywriting which best motivate the consumer. Principles are applied in clear, concise written expression of various appeals used in selling goods and services. (3 hr. lecture)

Lecture Hours: 48.000

Program Description

Chef Apprentice

College Credit Certificate | Code: 65059 | 12 credits

The Chef Apprentice Certificate is designed to prepare students with a theoretical and practical foundation for a successful career in the culinary industry. Credits earned can be applied to Associate in Science degree in Culinary Arts Management, which may be transferrable to upper division public institutions. *Students will be given opportunity to take the Food Safety exam for ServSafe Florida State Certification (State mandated for food handlers in Florida).

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 6.0
FSS 1200	Culinary Terminology and Procedures	3.0
FSS 1202C	Food Production 1 - Foundational Skills	3.0
Semester 2		Units: 6.0
FSS 1204C	Food Production 2 - American Regional Cuisine	3.0
FSS 2248C	Garde Manger	3.0
		Total: 12.0

FSS1200 Culinary Terminology and Procedures 3.000 credits

This is an introductory course in which students will learn to integrate industry vocabulary, terminology, knowledge, skills, and practices required for careers in the food service industry. The students will learn standard kitchen phrases, how to identify and describe equipment, recipe reading, costing, conversion formulas, product identification, measurements and basic cooking procedures. Corequisite: FSS1202C.

Lecture Hours: 48.000

Corequisite: FSS1202C

FSS1202C Food Production 1 - Foundational Skills 3.000 credits

This course is an introductory kitchen lecture/lab experience in which students will be provided hands-on orientation to tools, equipment, recipe production, measurements, knife cut techniques and basic cooking procedures. Students will practice classic cooking methods, product identification and the functions of the production kitchen in a "green" team environment. Corequisite: FSS1200. A.S. degree credit only.

Lecture Hours: 16.000, Practicum Hours: 96.000

Corequisite: FSS1200

Course User Fee: \$1981.97

FSS1204C Food Production 2 - American Regional Cuisine 3.000 credits

The lecture/practicum course reinforces the knowledge and skills learned in the preceding classes and helps students build confidence in the techniques of basic cookery. The development of knife skills is accented. American Regional Cuisine explores the use of indigenous ingredients in the preparation of traditional and contemporary American specialties. The importance and basics of "sustainable agriculture" are introduced. The concepts of mise en place, timelines, plate presentation, and teamwork in a production setting are introduced and accentuated. Timing and organization skills are emphasized. Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200 AND FSS1202C

Course User Fee: \$1981.99

FSS2248C Garde Manger 3.000 credits

This is a foundational lecture/lab course in the preparation of cold food, hors d'oeuvres, display platters, charcuteries, smoking techniques, butchery and seafood. Students will prepare edible buffet centerpieces. Students will also have hands-on experiences in ice carving. Production methods and safe food handling techniques are re-emphasized. Prerequisites: FSS1200, FSS1202C, FSS1246C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C, FSS1204C

Course User Fee: \$1981.95

Program Description

Cisco Certified Network Associate (CCNA)

College Credit Certificate | Code: 66050 | 16 credits

The Cisco Certified Network Associate College Credit Certificate program is designed to provide you with an opportunity to establish a foundation in the field of networking technologies and security, with a concentration in Cisco network design and implementation. Program coursework will additionally help you prepare for a variety of industry-recognized certifications, including: CompTIA Network+, CompTIA Security+, and Cisco Certified Network Associate (CCNA).

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
CTS 1134	Networking Technologies	4.0
CTS 1650	CCNA 1: Cisco Fundamentals	4.0
CTS 1651	CCNA 2: Routing and Switching	4.0
Semester 2		Units: 9.0
CIS 1949	CO-Op Work Experience 1: CIS	1.0
OR		1.0
CIS 2900	Directed Study in IT	1.0
CTS 1120	Cybersecurity Fundamentals	4.0
CTS 2652	CCNA 3: Advanced Routing and Switching	4.0
		Total: 21.0

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1650 CCNA 1: Cisco Fundamentals 4.000 credits

This is the first course of the four-course Cisco curriculum that will prepare students for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn networking concepts and practices, network terminology and protocols, the OSI reference model, cabling, cabling tools, routers, router and switch configurations, LAN/WAN topologies, IP addressing, and network standards.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Knowledge of computer operating systems and hardware is required.

Course User Fee: \$63.00

CTS1651 CCNA 2: Routing and Switching 4.000 credits

This is the second course of the four-course Cisco curriculum that will prepare the student for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn the architecture, components and operation of routers and switches, LAN (Local Area Networks) switch protocols and operations, VLANs (Virtual Local Area Networks), network routing protocols and concepts, static and dynamic routing, router and switch configuration and troubleshooting, and IP Address services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1650

Course User Fee: \$63.00

CIS1949 CO-Op Work Experience 1: CIS 1.000 - 4.000 credits

This course is designed as a work experience for students majoring in computer information systems programs. Students will learn to apply the skills and knowledge that they have acquired through their program of study in a real work environment. Prerequisite: Successful completion of required program course work.

Department approval required.

Lecture Hours: 16.000 - 64.000

Prerequisite: Successful completion of required program course work. Department approval required.

CIS2900 Directed Study in IT 1.000 credits

This course, for majors in Information Technology, provides students the opportunity to pursue projects under faculty supervision through self-directed research, experiences, and presentations. Prerequisite: Successful completion of required program course work. Department approval required.

Lab Hours: 32.000

Recommended Preparation: Successful completion of required program course work. Department approval required

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS2652 CCNA 3: Advanced Routing and Switching 4.000 credits

This is the third course of the four-course Cisco curriculum that will prepare the student for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn how to create virtual local area networks (VLANs), configure inter VLAN routing, and implement wireless network access and VLAN security.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1651

Course User Fee: \$63.00

Program Description

Commercial Transport Pilot

College Credit Certificate | Code: 68012 | 24 credits

The College Credit Certificate (CCC) in Commercial Transport Pilot will fill a growing need to train personnel in how to organize, review and classify aircraft records to comply with FAA regulations. Students will receive a wide-range of classroom and hands-on training that will provide the student with the education and knowledge to successfully manage the aircraft records department in an aviation company.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: Yes

Semester 1		Units: 13.0
ASC 1210	Aviation Meteorology	3.0
ATT 1100	Private Pilot Theory	3.0
ATF 1100L	Private Pilot Flight Accelerated	3.0
ATT 2120	Instruments Pilot Theory	4.0
Semester 2		Units: 11.0
ATF 2214L	Commercial Pilot Flight Accelerated	3.0
ATF 2310L	Instrument Pilot Flight Accelerated	3.0
ATT 2110	Commercial Pilot Theory 3	3.0
ATT 2133	Multi-Engine Pilot Theory	2.0
		Total: 24.0

ASC1210 Aviation Meteorology 3.000 credits

This is a core aviation course. The student will be prepared to understand weather and environmental issues in commercial aviation. Topics covered will be atmospheric phenomena relating to aircraft operations, the analysis and use of weather data as presented by the U.S. National Weather Service. Prerequisite: ATT1100 or equivalent; Corequisite: ATT2110 or equivalent.

Lecture Hours: 48.000

ATT1100 Private Pilot Theory 3.000 credits

This course introduces basic subjects pertaining to pilot knowledge including basic aircraft systems, aircraft operation and performance, aerodynamic principles, human factors, and aeronautical decision making. This Course can be instructed as Part 61 or Part 141.

Lecture Hours: 48.000

ATF1100L Private Pilot Flight Accelerated 3.000 credits

This course provides flight training in the areas required to safely perform the duties of a private pilot. This course is to be completed in less than 16 weeks using "total immersion" approach to training to help lower the attrition rates for students. It fulfills the requirements for private pilot certification outlined in part 141 of the Federal Aviation Regulations as presented in the Jeppesen Sanderson Private Pilot Syllabus. Upon satisfactory completion of this course and the Federal Aviation Administration (FAA) knowledge and practical exams, the applicant will receive an FAA private pilot certificate. Special Fee. Note: This course is not approved for Veterans seeking to use Veterans Educational Benefits.

Lecture Hours: 48.000

Prerequisite: ATT1100, FAA First Class Medical Certificate.

Course User Fee: \$15094.00

ATT2120 Instruments Pilot Theory 4.000 credits

This 30-hour course introduces basic theories of instrument pilot operations to prepare students for the FAA Instrument Written Exam. Students will acquire aeronautical knowledge required to act as an Instrument Rated Pilot. It will prepare the students for the FAA Instrument Written Exam. Private Pilot Certificate required.

Lecture Hours: 64.000

ATF2214L Commercial Pilot Flight Accelerated 3.000 credits

This accelerated course of instruction provides training required to allow the student to safely conduct flight as a Commercial Pilot. This course is expected to be completed in less than 4 months. The training will be conducted in accordance with the Codes of Federal Regulations (CFR) Part 141. This training will be completed utilizing the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course, the FAA knowledge test, and FAA practical test, the student will be awarded an FAA Commercial Pilot certificate. A 1st Class Medical Certificate with Instrument Rating is required. Minimum approved FAA CFR Part 141 course hours include 120 hours of flight. Special Fee.

Lecture Hours: 48.000

Recommended Preparation: Required: FAA Private Pilot Certificate and FAA 2nd Class Medical Certificate (1st Class preferred).

Course User Fee: \$23990.50

ATF2310L Instrument Pilot Flight Accelerated 3.000 credits

This accelerated course provides the flight training required to safely conduct flights as an instrument rated pilot. This course is to be completed in less than 16-weeks. The training is conducted in accordance with Codes of Federal Regulations (CFR) Part 141 of the Federal Aviation Regulations as outlined in the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course and the Federal Aviation Administration (FAA) knowledge test and practical test, the applicant will receive an FAA Instrument Rating. Minimum approved FAA CFR Part 141 course hours include 35 hours of flight. Special Fee.

Lecture Hours: 48.000

Prerequisite: Required: FAA Private Pilot Certificate, Corequisite: ATT2120, Current FAA 3rd Class Medical (1st Class preferred)

Course User Fee: \$12686.00

ATT2110 Commercial Pilot Theory 3 3.000 credits

This 35-hour course provides students with the aeronautical knowledge required to act as Commercial Pilot. Students will prepare for the FAA Commercial Written Exam. Private Pilot Certificate with Instrument Rating required.

Lecture Hours: 48.000

Prerequisite: ATF2305

ATT2133 Multi-Engine Pilot Theory 2.000 credits

This course introduces basic theories of multi-engine pilot operations to prepare students for the FAA Multi-Engine oral and practical exams. Students will acquire the aeronautical knowledge required to act as a multi-engine rated pilot.

Lecture Hours: 32.000

Program Description

Computer Aided Design Assistant

College Credit Certificate | Code: 66070 | 14 credits

The Computer-Aided Design Assistant College Credit Certificate program is designed to prepare students to work as CAD assistants in an architectural office by acquiring a basic understanding of the architectural graphic skills needed to produce working and presentation drawings.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 6.0
ARC 1115	Architectural Communications 1	2.0
ARC 1126C	Architectural Drawing 1	4.0
Semester 2		Units: 4.0
ARC 2171C	Computer Aided Drafting 1	4.0
Semester 3		Units: 4.0
ARC 2172	Computer Aided Drafting 2	4.0

Total: 14.0

ARC1115 Architectural Communications 1 2.000 credits

Exercises in freehand drawing, sketching and linear perspective are designed to increase the student's awareness of the architectural environment. This is accomplished through a series of form studies of nature, architectural forms, and abstract elements of composition.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$25.00

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions. Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC2171C Computer Aided Drafting 1 4.000 credits

Computer-aided drafting as it applies in the fields of architecture and interior design using office simulation. Emphasis is on the production of computer-aided drafting of working drawings involving different types of structure.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR ARC2461

Course User Fee: \$50.00

ARC2172 Computer Aided Drafting 2 4.000 credits

This course is designed for students with previous computer-aided design knowledge. Students will use both 2-dimensional and 3-dimensional CAD software to further develop their abilities to apply CAD techniques to the solution of architectural, engineering, and interior design problems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC2171C

Course User Fee: \$50.00

Program Description

Computer Aided Design Operator

College Credit Certificate | Code: 66071 | 22 credits

The Computer-Aided Design Operator College Credit Certificate program is designed to prepare students in an architectural office by obtaining intermediate skills in architectural graphics needed to produce working and presentation drawings. After successfully completing the following courses, students can obtain employment assisting architects and drafters with computer-aided drawings and design presentations.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 6.0
ARC 1115	Architectural Communications 1	2.0
ARC 1126C	Architectural Drawing 1	4.0
Semester 2		Units: 8.0
ARC 2171C	Computer Aided Drafting 1	4.0
ARC 2461	Architectural Materials and Construction 1	4.0
Semester 3		Units: 8.0
ARC 2056	Computer Aided Architectural Presentation	4.0
ARC 2172	Computer Aided Drafting 2	4.0
		Total: 22.0

ARC1115 Architectural Communications 1 2.000 credits

Exercises in freehand drawing, sketching and linear perspective are designed to increase the student's awareness of the architectural environment. This is accomplished through a series of form studies of nature, architectural forms, and abstract elements of composition.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$25.00

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions. Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC2171C Computer Aided Drafting 1 4.000 credits

Computer-aided drafting as it applies in the fields of architecture and interior design using office simulation. Emphasis is on the production of computer-aided drafting of working drawings involving different types of structure.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR ARC2461

Course User Fee: \$50.00

ARC2461 Architectural Materials and Construction 1 4.000 credits

An introduction to basic materials and methods of building construction. Emphasis is on wood, concrete, unit masonry, and light steel construction. Laboratory projects may include working drawings' interpretation, sketching construction details, or field trips to construction sites and fabricant plants. Designed primarily as the initial materials and methods course for architectural transfer students.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR BCN1251

Course User Fee: \$25.00

ARC2056 Computer Aided Architectural Presentation 4.000 credits

This course is designed to introduce the student to the concept of three-dimensional modeling and rendering for the purpose of producing an animated architectural presentation. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

ARC2172 Computer Aided Drafting 2 4.000 credits

This course is designed for students with previous computer-aided design knowledge. Students will use both 2-dimensional and 3-dimensional CAD software to further develop their abilities to apply CAD techniques to the solution of architectural, engineering, and interior design problems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC2171C

Course User Fee: \$50.00

Program Description

Computer Programmer - Business Applications

College Credit Certificate | Code: 66045 | 36 credits

The College Credit Certificate program in Computer Programmer - Business Applications is designed to provide an opportunity to establish a foundation in computer programming for employment in scientific, commercial, industrial and government data processing applications. Graduates are prepared for positions as entry-level programmers, programmer specialists, and software developers for both applications and systems software.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
CGS 1540C	Database Concepts Design	4.0
COP 1334	Introduction to C++ Programming	4.0
Semester 2		Units: 13.0
CIS 2331	Systems Analysis, Design and Implementation	5.0
COP 1047C	Introduction to Python Programming	4.0
COP 2800	Java Programming	4.0
Semester 3		Units: 11.0
GEB 1432	Applied Artificial Intelligence (AI) in Business	3.0
Program Elective		8.0
COP 2041C	Python Programming 2	4.0
OR		
COP 2335	C++ Programming 2	4.0
OR		
COP 2805C	Java Programming 2	4.0
		Total: 36.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS2331 Systems Analysis, Design and Implementation 5.000 credits

This course is designed for computer and information technology majors. Students learn to investigate, analyze, plan, design, implement and document an information system using the traditional Systems Development Life Cycle (SDLC).

Lecture Hours: 64.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1047C Introduction to Python Programming 4.000 credits

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

GEB1432 Applied Artificial Intelligence (AI) in Business 3.000 credits

This course illustrates how Artificial Intelligence (AI) is currently applied in industries and across business functions to support business strategy formulation, implementation and evaluation. This course introduces the data-driven outcomes and ethical issues to be considered when implementing responsible AI in business and industry. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$35.20

COP2041C Python Programming 2 4.000 credits

This is an advanced level programming course using Python. Students will learn how to code, compile, and execute programs. Topics include problem solving, object-oriented programming, database programming, data visualization, and secure software development. Prerequisite: COP1047C; (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C

Course User Fee: \$63.00

COP2335 C++ Programming 2 4.000 credits

This course delves further into the application of C++ programming techniques including classes, object orientation, function and operator overloading, inheritance and dynamic polymorphism, templates, exception handling, standard template library, data structures, and other advanced C++ techniques. Prerequisite: COP 1334 (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1334

Course User Fee: \$63.00

COP2805C Java Programming 2 4.000 credits

This is an advanced level programming course using Java. Students will learn how to code, compile and execute programs. Topics include object serialization, Java Collection, sorting/searching algorithms, multithreading and networking capabilities, and Java databases.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

Program Description

Computer Programmer - Mobile Applications Development

College Credit Certificate | Code: 66036 | 36 credits

The College Credit Certificate program is designed to provide an opportunity to establish a basic foundation in computer programming for employment in scientific, commercial, industrial and government data processing applications. The Mobile Applications Development concentration additionally offers hands-on instruction with current technology for Apple and Android mobile device platforms.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
CGS 1540C	Database Concepts Design	4.0
COP 1334	Introduction to C++ Programming	4.0
Semester 2		Units: 12.0
COP 2800	Java Programming	4.0
COP 2664	iPhone Application Development 1	4.0
COP 2658	iPhone Application Development 2	4.0
Semester 3		Units: 12.0
CIS 2331	Systems Analysis, Design and Implementation	5.0
COP 2660	Android Application Development 1	4.0
COP 2662	Android Application Development 2	4.0
		Total: 36.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement. Lecture Hours: 48.000, Lab Hours: 32.000
Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles. Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

COP2664 iPhone Application Development 1 4.000 credits

This is an introduction to iOS programming course using the Objective C computer language, recommended for Computer Science and Computer Information Systems majors. Students will learn to code, compile and execute mobile iOS applications while learning advanced programming concepts and object oriented programming design concepts and principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1332 OR COP1334

Course User Fee: \$63.00

COP2658 iPhone Application Development 2 4.000 credits

This intermediate iOS course teaches the principles of iPhone application development for majors in Computer Science, Computer Information Systems, and related disciplines. Students will learn how to create mobile applications that can be deployed to iPhone smartphones, tablets or simulators utilizing Cocoa and XCode for development. Emphasis will be placed on learning the underlying iPhone framework and components in order to create quality mobile applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2664

Course User Fee: \$63.00

CIS2331 Systems Analysis, Design and Implementation 5.000 credits

This course is designed for computer and information technology majors. Students learn to investigate, analyze, plan, design, implement and document an information system using the traditional Systems Development Life Cycle (SDLC). Lecture Hours: 64.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2660 Android Application Development 1 4.000 credits

This course teaches the principles of Android application development for majors in Computer Science, Computer Information Systems, and related disciplines. Students will learn how to create mobile applications for deployment to Android smartphones, tablets or simulators utilizing open source software (Java, Eclipse IDE, Android Plug-In and Android SDK) for development. Emphasis will be placed on the underlying Android framework to create quality applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

COP2662 Android Application Development 2 **4.000 credits**

This course for majors in Computer Science, Computer Information Systems, and related disciplines teaches how to develop advanced Android applications. Students will learn how to create applications utilizing the advanced capabilities of Android smartphones, including interfacing the application to the device's content providers' databases, GPS and location based services, notifications, background threads, audio, video, SMS, motion sensors and network connectivity. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2660, COP2800

Course User Fee: \$58.00

Program Description

[Crime Scene Technician](#)

College Credit Certificate | Code: 66072 | 28 credits

The College Credit Certificate in Crime Scene Technician will prepare students for employment in the field of criminalistics with a specialty in Crime Scene Investigation or Forensic Science. The student can serve as, but is not limited to, positions of Forensic Science Technician (SOC 194092), Crime Scene Technician, Medical Examiner Investigator, Medical Investigator, Insurance Investigator, Legal Investigator, Forensic Paralegal, Crime Scene Investigator, and Laboratory Technician. Students may also continue their formal education with the College and AS in Crime Scene Technology.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 15.0
BSC 1084	Functional Human Anatomy	3.0
CCJ 1020	Introduction to Criminal Justice	3.0
CCJ 1191	Human Behavior in Criminal Justice	3.0
PSY 2012	Introduction to Psychology	3.0
SPC 1017	Introduction to Communication	3.0
Semester 2		Units: 13.0
CHS 1522C	Forensic Science 1	4.0
CJE 1640	Crime Scene Technology 1	3.0
CJE 2600	Criminal Investigation	3.0
CJE 2671	Basic Fingerprinting	3.0
		Total: 28.0

BSC1084 Functional Human Anatomy 3.000 credits

This course covers basic human anatomy for students in allied health and mortuary science programs. It includes the dynamics of gross and functional anatomy, terminology, body orientation, and systematic relationships.

Lecture Hours: 48.000

CCJ1020 Introduction to Criminal Justice 3.000 credits

This course will explore and define the various components of the criminal justice system. It is designed for the entry level criminal justice practitioner and beginning student interested in entering the field.

Lecture Hours: 48.000

CCJ1191 Human Behavior in Criminal Justice 3.000 credits

Human behavior and how it relates to the duties and responsibilities of the criminal justice practitioner.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

CHS1522C Forensic Science 1 4.000 credits

An introductory course in the principles and techniques of forensic science. Students will learn how forensic science pertains to crime scene investigation and crime laboratory analysis.

Lecture Hours: 48.000, Lab Hours: 32.000

CJE1640 Crime Scene Technology 1 3.000 credits

This is an introductory course in Crime Scene Technology. Students will learn the techniques, materials and instrumentation used in securing, searching, recording, collecting, and examining physical evidence. There will be special emphasis on the tools, instruments, and techniques used in the studies of crime scene reconstruction, fingerprints, firearms, tool marks, and blood stain pattern analysis.

Lecture Hours: 48.000

CJE2600 Criminal Investigation 3.000 credits

Fundamentals of criminal investigation, theory and practice, including crime scene search; preservation, collection and transportation of physical evidence; interviewing, interrogating; statement taking; and case preparation, with investigation of specific offenses; relationship with the police science laboratory.

Lecture Hours: 48.000

CJE2671 Basic Fingerprinting 3.000 credits

This course provides a foundation in basic fingerprinting. Students will learn topics which include classification, identification, filing and rolling of fingerprints, problems and practices associated with post mortem fingerprinting and proper presentation of fingerprint evidence. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$11.00

Program Description

[Cruise Line Operations](#)

College Credit Certificate | Code: 65053 | 18 credits

All departments unrelated to navigation engine and entertainment, fall under the hotel division. This includes all concessionaires, housekeeping, food and beverage and pursers. Jobs in the hotel department on a cruise ship are very similar to jobs you will find in a hotel or resort. The hotel manager oversees all shipboard services and is responsible for supervising all staff and crew in these departments.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 9.0
HFT 1000	Introduction to Hospitality	3.0
HFT 2241	Leadership and Quality Assurance Management	3.0
HFT 2772	Introduction to Cruise Line Industry	3.0
Semester 2		Units: 9.0
HFT 2773	Cruise Line Sales and Marketing	3.0
HFT 2774	Shipboard Operations	3.0
HFT 2775	Shoreside Operations	3.0

Total: 18.0

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT2241 Leadership and Quality Assurance Management 3.000 credits

Provides an analysis of management issues related to the "personal touch" in customer service and quality assurance in the hospitality industry. Emphasis is placed on the importance of contemporary management and business practices to keep up with the demands of an ever-changing industry.

Lecture Hours: 48.000

HFT2772 Introduction to Cruise Line Industry 3.000 credits

Provides students with an introduction to the cruise line industry, its evolution and relationship to other segments of tourism and hospitality, sales and marketing methods, management, and strategic planning are major topics. Corequisite: HFT 1000. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: HFT 2000

HFT2773 Cruise Line Sales and Marketing 3.000 credits

Provides an introduction to the sales and marketing functions of the cruise industry. Students will gain an understanding of how cruise lines position themselves in the competitive business environment and the sales and marketing techniques used to attract customers and group business. Students will gain an understanding of yield management and the issues surrounding travel agents during the sales process.

Lecture Hours: 48.000

Prerequisite: HFT2772, Corequisite: HFT2774, HFT2775

HFT2774 Shipboard Operations 3.000 credits

Provides an understanding of shipboard operations on cruise ships and their relationship to the shoreside headquarters office. Students will gain knowledge of the history of cruise ships and the activities and facilities that make cruise line operations complementary both to the industry and the guest. This course will focus on the ship as a hotel for passengers with the wining and dining aspects of service, as well as, casino practices on board.

Lecture Hours: 48.000

Prerequisite: HFT2772, Corequisite: HFT2773, HFT2775

HFT2775 Shoreside Operations 3.000 credits

Provides a basic understanding of the shoreside office operations and sales procedures of the cruise line and how they relate to the general operations of the cruise ship itself. Students will acquire knowledge of pier, airport, ground services and hotel operations and create elements for cruise line sales.

Lecture Hours: 48.000

Prerequisite: HFT2772, Corequisite: HFT2773, HFT2774

Program Description

Culinary Arts Management Operations

College Credit Certificate | Code: 65060 | 18 credits

The Culinary Arts Management Operations College Credit Certificate is designed to prepare students with an in-depth study of food production, and a practical foundation in international cuisine for a successful career in the culinary industry. Credits earned can be applied to an Associate in Arts degree or an Associate in Science degree in Culinary Arts Management, which may be transferrable to upper division public institutions. *Students will be given opportunity to take the Food Safety exam for ServSafe Florida State Certification (State mandated for food handlers in Florida).

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 6.0
FSS 1200	Culinary Terminology and Procedures	3.0
FSS 1202C	Food Production 1 - Foundational Skills	3.0
Semester 2		Units: 6.0
FSS 1204C	Food Production 2 - American Regional Cuisine	3.0
FSS 1246C	Basic Baking - Foundational Skills	3.0
Semester 3		Units: 6.0
FSS 2242C	International Cuisine	3.0
FSS 2248C	Garde Manger	3.0
		Total: 18.0

FSS1200 Culinary Terminology and Procedures 3.000 credits

This is an introductory course in which students will learn to integrate industry vocabulary, terminology, knowledge, skills, and practices required for careers in the food service industry. The students will learn standard kitchen phrases, how to identify and describe equipment, recipe reading, costing, conversion formulas, product identification, measurements and basic cooking procedures. Corequisite: FSS1202C.

Lecture Hours: 48.000

Corequisite: FSS1202C

FSS1202C Food Production 1 - Foundational Skills 3.000 credits

This course is an introductory kitchen lecture/lab experience in which students will be provided hands-on orientation to tools, equipment, recipe production, measurements, knife cut techniques and basic cooking procedures. Students will practice classic cooking methods, product identification and the functions of the production kitchen in a "green" team environment. Corequisite: FSS1200. A.S. degree credit only.

Lecture Hours: 16.000, Practicum Hours: 96.000

Corequisite: FSS1200

Course User Fee: \$1981.97

FSS1204C Food Production 2 - American Regional Cuisine 3.000 credits

The lecture/practicum course reinforces the knowledge and skills learned in the preceding classes and helps students build confidence in the techniques of basic cookery. The development of knife skills is accented. American Regional Cuisine explores the use of indigenous ingredients in the preparation of traditional and contemporary American specialties. The importance and basics of "sustainable agriculture" are introduced. The concepts of mise en place, timelines, plate presentation, and teamwork in a production setting are introduced and accentuated. Timing and organization skills are emphasized. Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200 AND FSS1202C

Course User Fee: \$1981.99

FSS1246C Basic Baking - Foundational Skills 3.000 credits

This is an introductory course in which students will learn the fundamentals of baking science, terminology, equipment, ingredients, weights, measures, formula conversion and storage. Students will learn the functions of various baking ingredients and execute baking recipes, including competencies relating to doughs, breads, cookies, pies, puff pastries, sweet & savory pastry fillings, quick breads, cakes and basic decorating techniques.

Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C

Course User Fee: \$1981.98

FSS2242C International Cuisine 3.000 credits

This is an advanced course in which students will reinforce their knowledge of equipment, vocabulary and theories learned in prerequisite classes. Students will learn to use global ingredients, explore international cooking techniques and methods and be exposed to equipment specific to a myriad of world regions. Students will have the opportunity to develop international menus and prepare dishes from many countries.

Prerequisites: FSS1200, FSS1202C, FSS1204C, FSS1246C, FSS2248C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C, FSS1204C, FSS1246C, FSS2248C

Course User Fee: \$1981.97

FSS2248C Garde Manger 3.000 credits

This is a foundational lecture/lab course in the preparation of cold food, hors d'oeuvres, display platters, charcuteries, smoking techniques, butchery and seafood. Students will prepare edible buffet centerpieces. Students will also have hands-on experiences in ice carving. Production methods and safe food handling techniques are re-emphasized. Prerequisites: FSS1200, FSS1202C, FSS1246C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C, FSS1204C

Course User Fee: \$1981.95

Program Description

Cybersecurity Analyst

College Credit Certificate | Code: 66061 | 24 credits

The College Credit Certificate (CCC) in Cybersecurity offers essential cybersecurity skills and knowledge needed to secure networks, infrastructure, and use strategies to reduce the risk of incidents. The program curriculum offers

hands-on learning to defend against cyber-attacks and the opportunity to prepare for multiple in-demand industry certifications.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
CTS 1111	Linux +	4.0
CTS 1120	Cybersecurity Fundamentals	4.0
CTS 1134	Networking Technologies	4.0
Semester 2		Units: 12.0
CIS 1531	Introduction to Secure Scripting	4.0
CIS 2350	Cybersecurity Analysis (CySA+)	4.0
CTS 2314	Network Defense and Countermeasures	4.0

Total: 24.0

CTS1111 Linux + 4.000 - 32.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based work-stations and servers. Prerequisite: CGS 1060 or computer experience is required. (3 hr. lecture; 2 hr. lab)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS1531 Introduction to Secure Scripting 4.000 credits

This course provides students with the knowledge and skills to: create secure scripts and programs using system shells and programming languages; implement and debug algorithms to solve problems; automate and perform administrative tasks; manage data handling, and backup and storage.

Lecture Hours: 48.000, Lab Hours: 32.000

CIS2350 Cybersecurity Analysis (CySA+) 4.000 credits

This course provides students an intermediate skills-level approach to cybersecurity analysis. Students learn to identify the phases of an attack, the motivations of the adversary, the resources and techniques they use, the intended effect, or end-game, and how to mitigate threats. Topics include intrusion detection and response, analytics and advanced threat visibility. Prerequisites: CTS1120, CTS1134. Laboratory Fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120, CTS1134

Course User Fee: \$63.00

CTS2314 Network Defense and Countermeasures 4.000 credits

In this course, students will take an in-depth look at network defense concepts and techniques. Coverage includes network defensive concepts; policy development; problem-solving; and implementation of firewalls, DMZ, VPN, IDS, NAT and proxy servers.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120, CTS1134

Program Description

Digital Marketing Specialist

College Credit Certificate | Code: 65096 | 12 credits

The Digital Marketing Specialist Certificate is designed to prepare students to build and execute a digital marketing plan using digital tools to design, distribute, promote, and price a product or service. Graduates will understand how to connect with consumers using multiple digital platforms to create effective targeted promotional campaigns.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

MAR 1011	Principles of Marketing	3.0
MAR 1720	Marketing in a Digital World	3.0
MAR 2101	Social Media Marketing	3.0
MAR 2704	Marketing Web Analytics	3.0

Total: 12.0

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR2101 Social Media Marketing 3.000 credits

This course addresses the changes in marketing as a result of social media. The student will explore in-depth the role of social media in marketing. The student will focus on developing an effective marketing plan utilizing social media, with an emphasis on the importance of building an online customer community and developing customer engagement and loyalty.

Lecture Hours: 48.000

MAR2704 Marketing Web Analytics 3.000 credits

This course helps students to develop a deep understanding of the digital analytics landscape. The student will learn how to identify which metrics are best to measure web, mobile, social and marketing channels, and how organizations use web analytics to obtain higher profits, improved customer relationships, and measurable value. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$57.00

Program Description

Digital Marketing Strategy

College Credit Certificate | Code: 65102 | 18 credits

The Digital Marketing Strategy College Credit Certificate is designed to prepare students to design, implement, manage and analyze digital marketing strategies and campaigns. Graduates will understand how to connect with consumers using multiple digital platforms to create effective customer-focused promotional campaigns.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 9.0
MAR 1011	Principles of Marketing	3.0
MAR 1720	Marketing in a Digital World	3.0
MAR 2101	Social Media Marketing	3.0
Semester 2		Units: 9.0
MAR 2703	Marketing Content, Branding and Strategy	3.0
MAR 2704	Marketing Web Analytics	3.0
MAR 2952	Digital Marketing Capstone	3.0
		Total: 18.0

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR2101 Social Media Marketing 3.000 credits

This course addresses the changes in marketing as a result of social media. The student will explore in-depth the role of social media in marketing. The student will focus on developing an effective marketing plan utilizing social media, with an emphasis on the importance of building an online customer community and developing customer engagement and loyalty.

Lecture Hours: 48.000

MAR2703 Marketing Content, Branding and Strategy 3.000 credits

This course provides an introduction into branding and content strategy. Students will explore concepts, such as: integrated marketing communications, consumer psychology, buying motivations, appropriate content, branding, and marketing strategy as it relates to digital marketing. Students will have the opportunity to practice their writing and communications skills in the development of digital marketing content.

Lecture Hours: 48.000

Course User Fee: \$57.00

MAR2704 Marketing Web Analytics 3.000 credits

This course helps students to develop a deep understanding of the digital analytics landscape. The student will learn how to identify which metrics are best to measure web, mobile, social and marketing channels, and how organizations use web analytics to obtain higher profits, improved customer relationships, and measurable value. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$57.00

MAR2952 Digital Marketing Capstone 3.000 credits

This course provides a real-world approach to designing, implementing, managing and analyzing a digital marketing campaign or strategy. The student will engage in a comprehensive field experience designed to gain insights into the creative processes and demonstrate competence by applying the knowledge and concepts of marketing.

Lecture Hours: 48.000

Program Description

Early Childhood Education - Administrator

College Credit Certificate | Code: 60004 | 12 credits

This is a College Credit Certificate with a specialization in Child Care Management. The purpose of this program is to prepare students as early childhood education administrators with the knowledge and skills to effectively manage a quality childcare program or to provide supplementary training for persons previously or currently employed in these occupations. This CCC will allow its holder to apply for a Advanced Director's Credential or continue their education.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

EEC 2523	Programming & Management for Early Childhood Admin	3.0
EEC 2002	Operation of an Early Childhood Facility	3.0
EEC 2527	Legal and Financial Issues in Childcare	3.0
EEC 2520	Early Childhood Organization Leadership	3.0

Total: 12.0

EEC2002 Operation of an Early Childhood Facility 3.000 credits

This introductory course provides an overview for early childhood administrators to develop and enhance their leadership role in designing and implementing quality early care and education programs. The student will study the following topics: organizational leadership and management, programming, and financial and legal issues. This course meets the requirements for the Education Program Administrator Foundational Level Credential.

Lecture Hours: 48.000

EEC2523 Programming & Management for Early Childhood Admin 3.000 credits

The student will learn about developmentally and culturally appropriate environments, curriculum, and professional standards for early childhood care program administrators. The student will develop an understanding of child observation, assessment, documentation, and referrals and their importance. The student will learn about program evaluation, health, safety, healthy nutrition practices, and the importance of partnerships with families. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.).

Lecture Hours: 48.000

Course User Fee: \$10.00

**EEC2527 Legal and Financial Issues in Childcare
3.000 credits**

The student will learn about the legal and financial issues related to operating a successful early childhood center. The student learns about financial planning, personnel cost and allocation, budgeting and accounting. The student will develop knowledge about compensation and benefit, risk management, marketing and public relations, regulatory requirements, and legal issues related to childcare management. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential.

Lecture Hours: 48.000

Prerequisite: EEC2002

**EEC2520 Early Childhood Organization Leadership
3.000 credits**

The student will develop skills needed by early childhood program administrators to manage a high-quality center. The student will learn about organizational structure of a center; ethics and professionalism; leadership strategies, skills, and competencies; self-reflection and work environment; quality improvement; staff recruitment, evaluation, and retention. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.)

Lecture Hours: 48.000

Prerequisite: EEC2002

Course User Fee: \$10.00

Program Description

Early Childhood Education - Birth through Age Five

College Credit Certificate | Code: 60003 | 12 credits

This is a College Credit Certificate in early childhood education with a Birth through Age Five specialization. The purpose of this program is to prepare students as early childhood education caregivers with a birth through age five specialization or to provide supplementary training for persons previously or currently employed in these occupations. This CCC will allow its holder to apply for a National Child Development Associate credential, enabling this student to pursue work as a childcare provider nationally or continue their education.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 12.0**

EEC 1000	Introduction to Early Childhood Education	3.0
EEC 1200	Early Childhood Curriculum 1 (Birth - Age 5)	3.0
EEC 1311	Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5)	3.0
EEC 2407	Foundations of Learning and Executive Function in Early Childhood (Birth - 5)	3.0

Total: 12.0

EEC1000 Introduction to Early Childhood Education 3.000 credits

Students will be introduced to the field of early childhood education with a focus on infants, toddlers, and preschool-aged children from birth through age five, highlighting play as a developmentally appropriate and research-supported approach to early learning. The course examines the purpose and scope of early childhood education and the roles of children, families, and educators. Students gain an overview of foundational theories and philosophies, family and societal influences, learning environments, classroom routines, and guidance practices that support development across domains. Common early childhood program types and curriculum models serving birth–five settings are also explored. Ten hours of service learning in an early childhood setting are required. Fingerprinting through MDC's school of education must be completed before starting this course. See the MDC SOE Work-Force handbook.

Lecture Hours: 48.000

EEC1200 Early Childhood Curriculum 1 (Birth - Age 5) 3.000 credits

Students will design and implement developmentally appropriate curriculum for children from birth to age five using the NAEYC Professional Standards as a guiding framework. Emphasis is placed on foundational curriculum planning grounded in children's interests, with focus on development, social studies, language and literacy, and creative expression through the arts. Course content includes observation-based planning, learning centers, classroom and outdoor environments, daily routines, and play-based experiences. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. This course is paired with EEC1311, which extends curriculum planning through integrated and inquiry-based applications. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000

EEC1311 Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5) 3.000 credits

Students will design and implement integrated, developmentally appropriate curriculum for children from birth to age five, building on foundational curriculum planning introduced in EEC 1200. Emphasis is placed on arts-integrated STEM learning through inquiry-based curriculum planning, with focus on scientific inquiry, mathematical thinking, building and engineering, technology, and maker spaces for creative and artistic expression. Course content includes inquiry-based planning, learning centers, enriching classroom and outdoor environments, and play-based investigations that extend children's thinking across domains. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000 AND EEC1200

Course User Fee: \$10.00

EEC2407 Foundations of Learning and Executive Function in Early Childhood (Birth - 5) 3.000 credits

Students will build foundational knowledge of executive-function life skills, including focus and self-control, perspective taking, communication, making connections, critical thinking, taking on challenges, and self-directed learning, as the basis for learning in early childhood (Birth–5). The course examines how these skills interact with interpersonal development and cognitive growth, drawing on neuroscience and research on child development. Reciprocal family partnerships, observation-based reflection, and ethical considerations guide the analysis of young children's interactions and abilities. Emphasis is placed on research-based inhibitory control strategies that support neurocognitive development. Students will connect these executive-function skills and regulation strategies to developmentally appropriate teaching methods across home and early education settings. Ten hours of service learning in an approved early childhood setting are required. See the School of Education workforce handbook.

Lecture Hours: 48.000

Program Description

[Emergency Medical Technician](#)

College Credit Certificate | Code: 63013 | 12 credits

The Emergency Medical Technician – Basic College Credit Certificate is a one-semester program, which prepares students to function in the hospital and pre-hospital environment. Graduates of this program can perform clinical data collection, patient assessment and provide immediate care and safe relocation of the acutely ill. Satisfactory completion of this program will qualify the graduate to sit for the State and/or National EMT Certification Examination. This program is approved by the State of Florida, Department of Health and Rehabilitative Services.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 2.0
EMS 1059	Emergency Medical Responder	1.0
EMS 1059L	Emergency Medical Responder Laboratory	1.0
Semester 2		Units: 10.0
EMS 1119	Emergency Medical Technician	4.0
EMS 1119L	Emergency Medical Technician Lab and Clinic	3.0
EMS 1431	EMT Hospital/Field Experience	3.0
		Total: 12.0

EMS1059 Emergency Medical Responder 1.000 credits

Provides training in emergency medical care for those who may be first to respond to an accident. The course meets the basic requirements of the U.S. Department of Transportation. Recommended for students who are not required to be certified EMTs.

Lecture Hours: 16.000

Corequisite: EMS1059L

EMS1059L Emergency Medical Responder Laboratory 1.000 credits

Provide training in emergency medical care for those who may be first to respond to an accident. The course meets the basic requirements of the U.S. Department of Transportation.

Lab Hours: 32.000

Corequisite: EMS1059

Course User Fee: \$35.00

EMS1119 Emergency Medical Technician 4.000 credits

A review of basic life support theory. Areas of emphasis include the prehospital environment, preparatory information, patient assessment, medical emergencies, behavioral emergencies, OB/GYN emergencies, trauma emergencies, pediatric emergencies and EMS operations.

Lecture Hours: 64.000

Corequisite: EMS1119L, EMS1431

EMS1119L Emergency Medical Technician Lab and Clinic 3.000 credits

Practical application of the content covered in EMS 1119 with an emphasis on cardiopulmonary resuscitation, splinting, bandaging, patient movement, and other skills as recommended by the U.S. Department of Transportation for the EMT-A level practitioner.

Lab Hours: 96.000

Corequisite: EMS1119, EMS1431

Course User Fee: \$55.00

EMS1431 EMT Hospital/Field Experience 3.000 credits

Practice in local emergency departments and rescue agencies under professional supervision. This course meets the skills recommended by the U.S.

Department of Transportation.

Clinical Hours: 144.000

Corequisite: EMS1119, EMS1119L

Program Description

Endoscopic Technician

College Credit Certificate | Code: 63030 | 24 credits

The College Credit Certificate in Endoscopic Technician is designed to educate and prepare graduates with skills required for entry-level employment as a member of the healthcare team. In addition, students will acquire extensive knowledge of legal and ethical responsibilities, anatomy, physiology, pathophysiology, microbiology, aseptic techniques, patient care procedures, patient safety, and use and care of equipment and supplies.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 11.0

HIM 2472	Medical Terminology	3.0
STS 1307	Surgical Equipment and Instrumentation	3.0
STS 1323	Surgical Procedures I	3.0
STS 1925C	Endoscopy Technician Theory 1 and Lab	2.0

Semester 2

Units: 8.0

STS 1304L	Operating Room Techniques Laboratory	2.0
STS 1926C	Endoscopy Technician Theory 2 and Lab	2.0
STS 2179	Surgical Biomedical Fundamentals	2.0
STS 2944	Surgical Clinical I	2.0

Total: 19.0

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

STS1307 Surgical Equipment and Instrumentation 3.000 credits

This course prepares the student for the scrub role during surgical procedures, identification of basic surgical instrumentation, passing of the instruments, proper care, reprocessing of instruments following a surgical procedure, and utilizing equipment in the operating room. Students will also learn about surgical wounds, proper tissue handling techniques, wound closure, wound healing, suture material, and stapling devices.

Lecture Hours: 48.000

Course User Fee: \$46.83

STS1323 Surgical Procedures I 3.000 credits

This course is designed to prepare students for surgical procedures by providing instruction on diagnostic procedures and relevant equipment, supplies, and techniques. The course will also include the review of surgical anatomy, physiology, and pathophysiology in relation to general surgery, endoscopic surgery, gynecological and obstetrical surgery, genitourinary surgery, orthopedic surgery, and ophthalmic surgery.

Lecture Hours: 48.000

Course User Fee: \$46.83

STS1925C Endoscopy Technician Theory 1 and Lab 3.000 credits

Upon completion of the course, the student will have a basic understanding of the care and handling of endoscopic equipment, including storage, testing, decontamination, disinfection and preparation for use. Modules include anatomy and physiology review relevant to endoscopy procedures, essential communication, roles and responsibilities of the endoscopy team, equipment, risk management, safety, reprocessing/infection control, and emergency preparedness.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$39.45

STS1304L Operating Room Techniques Laboratory 3.000 credits

This course will provide the student with information on operating room furniture, equipment, and supplies used during surgery. The preparation of the operating room, surgical scrub, gowning and gloving, development of the sterile field, patient positioning, surgical skin prep, and draping will be included and will be part of the lab practical exam. Students will be required to simulate a surgical procedure from start to finish. Students must complete this course with a grade of "C" or higher in order to start the clinical rotations.

Lab Hours: 96.000

Course User Fee: \$46.83

STS1926C Endoscopy Technician Theory 2 and Lab 3.000 credits

This course is designed to prepare students for the role of Gastrointestinal (GI) technician regarding flexible endoscopic procedures, including the specialties of gastroenterology and pulmonology.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: STS1925C

Course User Fee: \$40.45

STS2179 Surgical Biomedical Fundamentals 3.000 credits

This course teaches the skills necessary to function as a surgical technologist in the operating room, including principles of aseptic technique, and a basic understanding of robotics and their use in the operating room setting. In addition, an understanding of the principles of physics and electricity as it relates to the operating room environment will be covered.

Lecture Hours: 48.000

Course User Fee: \$28.65

STS2944 Surgical Clinical I 3.000 credits

This course is the first in a series of three clinical courses. The course will introduce the student to the surgical clinical environment, provide the ability to begin utilizing skills acquired during previous lab courses, and implement the principles of aseptic technique while participating in the role of a surgical technologist.

Clinical Hours: 144.000

Course User Fee: \$46.83

Program Description

Engineering Technology Support Specialist

College Credit Certificate | Code: 66052 | 18 credits

This program offers a sequence of courses that provides students with the relevant technical knowledge and skills needed to prepare for further education and careers in the manufacturing career cluster.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester1**Units: 9.0**

ETD 1340	Computer Aided Drawing & Design	3.0
EET 1082	Introduction to Electronics	3.0
ETI 1172	Introduction to Quality Assurance	3.0

Semester 2**Units: 9.0**

ETI 1152C	Mechanical Measurement and Instrumentation	3.0
ETI 1701	Industrial Safety	3.0
ETI 2404	Advanced Manufacturing Technology	3.0

Total: 18.0

**ETD1340 Computer Aided Drawing & Design
3.000 credits**

This course is recommended for all engineering students as an introduction to the basic concepts of drafting and designing using a computer. Students will learn industry standard drafting and design practices using AutoCAD in a laboratory environment. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

EET1082 Introduction to Electronics 3.000 credits

Learn by building practical electronic circuits. Survey course suitable for both majors and non-majors. Instructors and tutors are available to assist in project completion. Topics include: schematics, pictorials, amplifiers, oscillators, burglar alarms, radios, digital circuits. Students will develop individual career plans and learn about employment opportunities within the field.

Lecture Hours: 48.000

Course User Fee: \$63.00

ETI1172 Introduction to Quality Assurance 3.000 credits

This course describes the role and aspects of quality systems and Regulatory affairs in research laboratories, regulated companies, and firms that comply with voluntary standards. Topics include stages in development and submission of drugs and medical devices, patents legislation, and quality systems such as auditing, standard procedures, good manufacturing and laboratory practices.

Lecture Hours: 48.000

ETI1152C Mechanical Measurement and Instrumentation 3.000 credits

This course provides the basic foundation for mechanical measurement techniques used in manufacturing environments. The course will integrate the concepts, principles, and techniques of mechanical measurement with the use of various types of instruments including micrometers, verniers, calipers, gages, and other types of measuring equipment.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

ETI1701 Industrial Safety 3.000 credits

This course provides the student with the knowledge and skills to recognize hazardous situations in industrial plants and the precautions to be observed and practiced to perform work activities safely. Among the topics covered are industrial safety hazards, electrical safety, working with chemicals, gases, and solvents, protective equipment, and safe working conditions.

Lecture Hours: 48.000

ETI2404 Advanced Manufacturing Technology 3.000 credits

This is a course for students intending to work in manufacturing environments. Students will learn the basic concepts about advanced manufacturing operations and processes, including sourcing materials, production planning and process monitoring, and control to distribution activities. Students also review the facility and regulatory requirements needed to support manufacturing operations. Activities may include facility tours and site visits.

Lecture Hours: 48.000

Program Description

Enterprise Cloud Computing

College Credit Certificate | Code: 66058 | 24 credits

The Enterprise Cloud Computing Credit Certificate program is designed to provide an opportunity to establish foundation in architect scalable, highly available application solutions that leverage cloud computing services utilizing best practices focusing on cloud security, cost, and reliability. Graduates will utilize core design patterns and infrastructure expertise to implement solutions to deploy and maintain workloads and applications.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 12.0**

CGS 1540C	Database Concepts Design	4.0
CTS 1134	Networking Technologies	4.0
CTS 1145	Cloud Essentials	4.0

Semester 2**Units: 12.0**

CTS 1111	Linux +	4.0
CTS 2375C	Cloud Infrastructure and Services	4.0
CTS 2960	Cloud Computing Capstone	4.0

Total: 24.0

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1111 Linux + 4.000 - 32.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based work-stations and servers. Prerequisite: CGS 1060 or computer experience is required. (3 hr. lecture; 2 hr. lab)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CTS2375C Cloud Infrastructure and Services 4.000 credits

This course helps students develop technical expertise in Cloud computing and prepares them for industry certification. Students will learn the essentials of Cloud computing, business security and compliance considerations, migrating to the Cloud, architecting a Cloud server, and how to troubleshoot Cloud services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2145

Course User Fee: \$63.00

CTS2960 Cloud Computing Capstone 4.000 credits

This course requires students to demonstrate their competence to analyze, design, develop, and test a cloud based complex system. Each student will create and present a cloud based solution proposal that includes: design documentation, implementation plan, cloud resources required, projected cost analysis, basic security plan and project test plan to create an operational cloud based system solution. Must be taken during the last semester before graduation and with a departmental permission. Prerequisite: Departmental Approval. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: Departmental Approval

Course User Fee: \$63.00

Program Description

Film Production Fundamentals

College Credit Certificate | Code: 61001 | 24 credits

The Film Production Fundamentals College Credit Certificate (CCC) is designed to prepare students for entry-level employment in the motion picture industry. Students will understand the fundamentals in the following skills: lighting, grip, camera, audio recording, and editing.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 14.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
FIL 1100	Screenwriting 1: Introduction to Story Structure	3.0
FIL 1420C	Film Production 1: Introduction to the Filmmaking	4.0
FIL 2552C	Editing Level 1: Introduction to Editing	3.0
Semester 2		Units: 10.0
FIL 1431C	Film Production 2: Cinematography and Sound	4.0
FIL 2553C	Editing Level 2: Intermediate Editing and Visual Effects	3.0
VIC 1000	Visual Communications	3.0
		Total: 24.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.
Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

FIL1100 Screenwriting 1: Introduction to Story Structure 3.000 credits

A workshop-style introductory class covering narrative script writing for film and television. Students will learn to develop a short-format screenplay incorporating three-act story structure, script elements, and standard industry formatting.
Lecture Hours: 48.000

FIL1420C Film Production 1: Introduction to the Filmmaking 4.000 credits

An introductory overview of the art and technology of narrative motion picture production. Students will learn basic production techniques, from pre-production through production to final screening.
Lecture Hours: 32.000, Lab Hours: 64.000
Corequisite: FIL2552C
Course User Fee: \$150.00

FIL2552C Editing Level 1: Introduction to Editing 3.000 credits

An introductory course in which students will learn the practice of editing digital media.
Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$35.00

FIL1431C Film Production 2: Cinematography and Sound 4.000 credits

An introductory course in which students will learn cinematography and sync-sound motion picture production.
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: FIL1420C AND Corequisite: FIL2553C
Course User Fee: \$150.00

FIL2553C Editing Level 2: Intermediate Editing and Visual Effects 3.000 credits

An intermediate course in which students will learn video editing with an emphasis on sound design and visual effects.
Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: FIL2552C
Course User Fee: \$35.00

VIC1000 Visual Communications 3.000 credits

An introductory course in which students will learn the visual aspects of film, video, photography and graphic arts, specifically dealing with design elements and principles.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$10.00

Program Description

Florida Funeral Director

College Credit Certificate | Code: 63024 | 31 credits

The Florida Funeral Director certificate program is intended for individuals who have obtained a prior degree from an accredited institution and intend to transition into a new career path. Funeral Directors arrange the details and handle the logistics of funerals, assist the family in establishing the location, date, and time of wakes, memorial services, and burials, and determine whether the body should be buried, entombed, or cremated depending on the family's cultural and/or religious practices. The Florida Funeral Director certificate program will allow students to be eligible for the Florida Funeral Directors State Licensing Exam.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 18.0
BUL 2131	Legal Environment	3.0
ENC 1101	English Composition 1	3.0
FSE 1000	History of Funeral Service	3.0
FSE 1080	Funeral Law	3.0
FSE 2060	Funeral Directing	3.0
PHI 2600	Introduction to Ethics	3.0
Semester 2		Units: 13.0
FSE 2061	Funeral Service Counseling and Ethics	3.0
FSE 2200	Funeral Service Accounting	3.0
FSE 2201	Funeral Home Operations	3.0
FSE 2202	Funeral Service Business Management	3.0
FSE 2932	Funeral Science Professional Review 2	1.0
		Total: 31.0

BUL2131 Legal Environment 3.000 credits

Law in relation to the proper conduct of business includes a consideration of the nature and sources of law, its legal environment and history. The topics of business torts, crimes, contracts, and forms of organizations are also covered.

Lecture Hours: 48.000

Course User Fee: \$10.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

FSE1000 History of Funeral Service 3.000 credits

This course traces the origins of funeral service practice from antiquity to modern day practice. Students will learn the evolution of the ethical obligations, fundamental requirements, skills, aptitudes, and qualifications of funeral service professionals. Requires a grade of C or better to pass the course.

Lecture Hours: 48.000

FSE1080 Funeral Law 3.000 credits

Federal, state and municipal statutes, rules, regulations and ordinances pertaining to funeral service; torts, contract and administrative laws, and financial disclosures pertinent to funeral operations and management.

Lecture Hours: 48.000

FSE2060 Funeral Directing 3.000 credits

Study of various religious, fraternal, military, traditional, nontraditional and humanistic variations of funeral ceremonies, including cultural, ethnic and geographic customs.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

FSE2061 Funeral Service Counseling and Ethics 3.000 credits

This course emphasizes the psychological and sociological dynamics of death, dying, and bereavement and the funeral director's role in counseling families through the process. A clear understanding of the grief process and its variations among individuals as influenced by psychological factors is addressed. Ethical and moral decisions involving end of life and the funeral arrangement process are discussed. The course also addresses the funeral director as a facilitating agent for effective mourning through personal interaction and the design and implementation of the funeral.

Lecture Hours: 48.000

FSE2200 Funeral Service Accounting 3.000 credits

An introduction to basic principles of accounting theory. This subject covers financial statements and their analysis, journalizing, receivables, payables, deferrals, and accruals. Inventory costing models, depreciation models and payroll accounting are included. Applications to funeral home operations are made throughout the subject material.

Lecture Hours: 48.000

FSE2201 Funeral Home Operations 3.000 credits

Theoretical and practical training in all areas of funeral home operations; laboratory experience in merchandising and funeral arrangements.

Lecture Hours: 48.000

FSE2202 Funeral Service Business Management 3.000 credits

The role and function of an effective manager is explored. Emphasis is placed on the management functions of planning, organizing, motivating, directing, and controlling. How to purchase a small business is also covered.
Lecture Hours: 48.000

FSE2932 Funeral Science Professional Review 2 1.000 credits

This course is for the funeral Science student who is graduating and taking the National Board Examination at the end of the semester that this course is being offered. The course is a review of the Arts section of the Funeral Science courses in order to help prepare the student for the National Board Examination.

Lecture Hours: 32.000

Prerequisite: Departmental Permission Required

Program Description

Food and Beverage Management

College Credit Certificate | Code: 65051 | 30 credits

The Food Service Management College Credit Certificate program is designed to prepare students with a theoretical and practical foundation for a successful career in the food and beverage industry. Students enrolled in this program are prepared for positions such as Catering/Banquet Manager, Food & Beverage Manager, Restaurant Manager and Bar/Lounge Manager. Credits earned can be applied to an Associate in Science degree in Hospitality Management, which is fully transferable to public universities within the State of Florida.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
HFT 1000	Introduction to Hospitality	3.0
HFT 1212	Safety and Sanitation	3.0
HFT 1841	Dining Room Service	3.0
HFT 1852	Menu and Facilities Planning	3.0

Semester 2		Units: 9.0
HFT 2261	Restaurant Management	3.0
HFT 2800	Food and Beverage Management	3.0
<i>Program Elective</i>		3.0
FSS 1100	Foodservice Purchasing	3.0
OR		
FSS 1200	Culinary Terminology and Procedures	3.0
OR		
FSS 1202C	Food Production 1 - Foundational Skills	3.0
OR		
HFT 1210	Human Resources	3.0
OR		
HFT 1454	Food and Beverage Cost Controls	3.0
OR		
HFT 1631	Risk Management and Security	3.0
OR		
HFT 2241	Leadership and Quality Assurance Management	3.0
OR		
HFT 2421	Managerial Accounting for Hospitality	3.0
OR		

HFT 2449	E-Business For The Hospitality Industry	3.0
OR		
HFT 2949	Co-Op Work-Study Internships 2	3.0

Semester 3

Units: 9.0

Program Electives*	9.0
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*Select 9 credits from the following course options: FSS 1100, FSS 1200, FSS 1202C, HFT 1210, HFT 1454, HFT 1631, HFT 2241, HFT 2421, HFT 2449, HFT 2949

Total: 30.0

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1212 Safety and Sanitation 3.000 credits

Students will learn industry standards on sanitation as it relates to food preparation and kitchen operations. Students will recognize proper sanitation techniques and explain how to implement sanitation programs in food service operations.

Lecture Hours: 48.000

Course User Fee: \$125.00

HFT1841 Dining Room Service 3.000 credits

Provides students with the opportunity to acquire knowledge of advanced service techniques, including guest satisfaction, food, wine and beverage serving, types of menus, table service techniques, tableside cooking, napkin folding, table setting, safety, sanitation, emergency procedures, restaurant technology, and service styles. Students will gain experience in cash and non-cash handling, forecasting sales, and merchandising techniques. Corequisite: HFT 1000. (3 hr. lecture)

Lecture Hours: 48.000

HFT1852 Menu and Facilities Planning 3.000 credits

This course is a study of the factors involved in planning effective menus for a variety of food service operations. Students will learn to develop a firm working knowledge of menu-writing techniques and participate in actual menu design and facilities layout of a food service establishment.

Lecture Hours: 48.000

Prerequisite: HFT1000

Course User Fee: \$40.00

HFT2261 Restaurant Management 3.000 credits

Familiarizes students with the general principles of food production management, work scheduling, and preparation supervision. Emphasis is placed on purchasing and financing, planning and equipping a kitchen, pricing and menu design, and marketing and promoting restaurants.

Lecture Hours: 48.000

Prerequisite: HFT1000

HFT2800 Food and Beverage Management 3.000 credits

Provides a basic understanding of the principles of food production and service management, menu planning, serving, purchasing, labor, food/bar service and costs, storage, beverage management, sales promotions, entertainment, and liability laws.

Lecture Hours: 48.000

Course User Fee: \$15.00

FSS1100 Foodservice Purchasing 3.000 credits

This is an introductory course where students will learn the principals of menu planning for various types of facilities and service as well as menu layout, selection, development and pricing structures. Students will learn the principals and practices concerned with the purchase and receipt of food, supplies and equipment for various food service operations. Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 48.000

Prerequisite: FSS1200 AND FSS1202C

FSS1200 Culinary Terminology and Procedures 3.000 credits

This is an introductory course in which students will learn to integrate industry vocabulary, terminology, knowledge, skills, and practices required for careers in the food service industry. The students will learn standard kitchen phrases, how to identify and describe equipment, recipe reading, costing, conversion formulas, product identification, measurements and basic cooking procedures.

Corequisite: FSS1202C.

Lecture Hours: 48.000

Corequisite: FSS1202C

FSS1202C Food Production 1 - Foundational Skills 3.000 credits

This course is an introductory kitchen lecture/lab experience in which students will be provided hands-on orientation to tools, equipment, recipe production, measurements, knife cut techniques and basic cooking procedures. Students will practice classic cooking methods, product identification and the functions of the production kitchen in a "green" team environment. Corequisite: FSS1200. A.S. degree credit only.

Lecture Hours: 16.000, Practicum Hours: 96.000

Corequisite: FSS1200

Course User Fee: \$1981.97

HFT1210 Human Resources 3.000 credits

Provides information relating to the recruitment and selection of new staff, the handling of difficult employees, motivating employees and conducting performance evaluations. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1454 Food and Beverage Cost Controls 3.000 credits

Covers the principles and procedures involved in an effective food and beverage control system, including standards determination, the operation budget, income and cost control, menu pricing, and computer applications.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1631 Risk Management and Security 3.000 credits

Provides the opportunity to examine issues surrounding the need for individualized security and surveillance programs, risk management and review systems. The student will examine a wide variety of security and safety equipment procedures and discuss guest protection, internal security for asset protection, and OSHA regulations that apply to lodging properties.

Lecture Hours: 48.000

HFT2241 Leadership and Quality Assurance Management 3.000 credits

Provides an analysis of management issues related to the "personal touch" in customer service and quality assurance in the hospitality industry. Emphasis is placed on the importance of contemporary management and business practices to keep up with the demands of an ever-changing industry.

Lecture Hours: 48.000

HFT2421 Managerial Accounting for Hospitality 3.000 credits

Presents managerial accounting concepts and explains how they apply to specific operations within the hospitality industry.

Lecture Hours: 48.000

HFT2449 E-Business For The Hospitality Industry 3.000 credits

Prepares student to manage information systems within their organizations. Emphasis is on selecting the right computer systems technology and maximizing available technology in order to promote and sell services. Introduces the use of automation in the broad hospitality sector and exams technological applications ranging from distribution systems (GDS, CRS and Web based), Property Management Systems, and EPOS systems to developments in telecommunications, and assesses their effect on the tourism sector. While a broad interpretation of both technology and tourism will be used, particular emphasis will be placed on the hospitality sector (i.e. hotels and catering) and on distribution technology. A combination of lecture, case studies, seminars, visiting lecture and lab sessions are used.

Lecture Hours: 48.000

Prerequisite: HFT1000

HFT2949 Co-Op Work-Study Internships 2 3.000 credits

Provides an advanced supervised cooperative education experience in the hospitality industry that builds on foundational workplace skills and knowledge. Students apply hospitality management principles in professional settings while enhancing competencies in operations, customer service, leadership, communication, and problem-solving. Emphasis is placed on increased responsibility, professional growth, and career development through structured experiential learning in approved hospitality organizations.

Other Hours: 48.000

Course User Fee: \$25.00

Program Description

Food and Beverage Operations

College Credit Certificate | Code: 65058 | 18 credits

The Food and Beverage Operations College Credit Certificate is designed to prepare students with an in-depth and practical foundation in management for a successful career in the food and beverage industry. Students enrolled in this certificate are prepared for positions such as Shift Supervisor, Restaurant Supervisor, or Bar/Lounge Supervisor. Credits earned can be applied to an Associate in Science degree in Hospitality Management, which is fully transferable to public universities within the state of Florida.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 9.0
HFT 1000	Introduction to Hospitality	3.0
HFT 1212	Safety and Sanitation	3.0
HFT 1454	Food and Beverage Cost Controls	3.0
Semester 2		Units: 9.0
HFT 2241	Leadership and Quality Assurance Management	3.0
HFT 2800	Food and Beverage Management	3.0
Program E		3.0
HFT 1852	Menu and Facilities Planning	3.0
OR		
HFT 2261	Restaurant Management	3.0

Total: 18.0

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1212 Safety and Sanitation 3.000 credits

Students will learn industry standards on sanitation as it relates to food preparation and kitchen operations. Students will recognize proper sanitation techniques and explain how to implement sanitation programs in food service operations.

Lecture Hours: 48.000

Course User Fee: \$125.00

HFT1454 Food and Beverage Cost Controls 3.000 credits

Covers the principles and procedures involved in an effective food and beverage control system, including standards determination, the operation budget, income and cost control, menu pricing, and computer applications.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT2241 Leadership and Quality Assurance Management 3.000 credits

Provides an analysis of management issues related to the "personal touch" in customer service and quality assurance in the hospitality industry. Emphasis is placed on the importance of contemporary management and business practices to keep up with the demands of an ever-changing industry.

Lecture Hours: 48.000

HFT2800 Food and Beverage Management 3.000 credits

Provides a basic understanding of the principles of food production and service management, menu planning, serving, purchasing, labor, food/bar service and costs, storage, beverage management, sales promotions, entertainment, and liability laws.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1852 Menu and Facilities Planning 3.000 credits

This course is a study of the factors involved in planning effective menus for a variety of food service operations. Students will learn to develop a firm working knowledge of menu-writing techniques and participate in actual menu design and facilities layout of a food service establishment.

Lecture Hours: 48.000

Prerequisite: HFT1000

Course User Fee: \$40.00

HFT2261 Restaurant Management 3.000 credits

Familiarizes students with the general principles of food production management, work scheduling, and preparation supervision. Emphasis is placed on purchasing and financing, planning and equipping a kitchen, pricing and menu design, and marketing and promoting restaurants.

Lecture Hours: 48.000

Prerequisite: HFT1000

Program Description

Food and Beverage Specialist

College Credit Certificate | Code: 65057 | 12 credits

The Food and Beverage Specialist College Credit Certificate is designed to prepare students with a theoretical and practical foundation for a successful career in the food and beverage industry. Students enrolled in this certificate are prepared for positions such as Restaurant Server, Room Service Attendant, or Banquet Set-Up Staff. Credits earned can be applied to an Associate in Science degree in Hospitality Management, which is fully transferable to public universities within the state of Florida.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

HFT 1000	Introduction to Hospitality	3.0
HFT 1212	Safety and Sanitation	3.0
HFT 1454	Food and Beverage Cost Controls	3.0
<i>Program Elective</i>		3.0
HFT 2449	E-Business For The Hospitality Industry	3.0
OR		
HFT 2949	Co-Op Work-Study Internships 2	3.0

Total: 12.0

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1212 Safety and Sanitation 3.000 credits

Students will learn industry standards on sanitation as it relates to food preparation and kitchen operations. Students will recognize proper sanitation techniques and explain how to implement sanitation programs in food service operations.

Lecture Hours: 48.000

Course User Fee: \$125.00

HFT1454 Food and Beverage Cost Controls 3.000 credits

Covers the principles and procedures involved in an effective food and beverage control system, including standards determination, the operation budget, income and cost control, menu pricing, and computer applications.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT2449 E-Business For The Hospitality Industry 3.000 credits

Prepares student to manage information systems within their organizations. Emphasis is on selecting the right computer systems technology and maximizing available technology in order to promote and sell services. Introduces the use of automation in the broad hospitality sector and exams technological applications ranging from distribution systems (GDS, CRS and Web based), Property Management Systems, and EPOS systems to developments in telecommunications, and assesses their effect on the tourism sector. While a broad interpretation of both technology and tourism will be used, particular emphasis will be placed on the hospitality sector (i.e. hotels and catering) and on distribution technology. A combination of lecture, case studies, seminars, visiting lecture and lab sessions are used.

Lecture Hours: 48.000

Prerequisite: HFT1000

HFT2949 Co-Op Work-Study Internships 2 3.000 credits

Provides an advanced supervised cooperative education experience in the hospitality industry that builds on foundational workplace skills and knowledge. Students apply hospitality management principles in professional settings while enhancing competencies in operations, customer service, leadership, communication, and problem-solving. Emphasis is placed on increased responsibility, professional growth, and career development through structured experiential learning in approved hospitality organizations.

Other Hours: 48.000

Course User Fee: \$25.00

Program Description

Geographic Information Systems

College Credit Certificate | Code: 66080 | 21 credits

Geographic Information Systems (GIS) is a technological field that promotes cooperation within and across organizations by enabling the integration of several information platforms to facilitate mapping, analysis and planning strategies. The College Credit Certificate program in Geographic Information Systems Technology prepares students with professional and technical training in geographic information systems (GIS). Graduates of the CCC acquire a skill-set that equips them to create, modify and enhance GIS for analysis, prediction, decision making and planning in a variety of fields.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 8.0

CGS 1540C

Database Concepts Design

4.0

GIS 1040	Introduction to GIS Technology	4.0
Semester 2		Units: 4.0
GIS 2045	Intermediate GIS Technology	4.0
Semester 3		Units: 9.0
GIS 2046	Advanced GIS Technology	4.0
GIS 2047	Applications of GIS Technology	4.0
CIS 2900	Directed Study in IT	1.0

Total: 21.0

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement. Lecture Hours: 48.000, Lab Hours: 32.000
Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

GIS1040 Introduction to GIS Technology 4.000 credits

This course teaches the principles of Geographic Information Systems (GIS). The student will learn fundamental GIS concepts, GIS models and formats, as well as map creation and analysis. At the end of the course, the student will be able to perform basic operations in GIS software. (3 hr. lecture, 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

GIS2045 Intermediate GIS Technology 4.000 credits

This course teaches intermediate-level concepts of Geographic Information Systems (GIS). The student will acquire an understanding of discrete geocoding and georeferencing, data input, working with spatial data-bases, and data creation. At the end of the course, the student will be able to perform intermediate-level operations in GIS software.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: GIS1040

Course User Fee: \$63.00

GIS2046 Advanced GIS Technology 4.000 credits

This course teaches advanced concepts of Geographic Information Systems (GIS). The student will learn how to use vector, raster and 3D data, geospatial structures, and write GIS functions using a programming language. At the end of the course, the student will be able to perform advanced operations in GIS software.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: GIS2045

Course User Fee: \$63.00

GIS2047 Applications of GIS Technology 4.000 credits

This course teaches the principles of urban analytics and disaster management through what-if scenario modeling in which risks are evaluated and managed in order to support better decision-making. By the end of the course, the student will apply a remote sensing technique to generate GIS data.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: GIS2045

Course User Fee: \$63.00

CIS2900 Directed Study in IT 1.000 credits

This course, for majors in Information Technology, provides students the opportunity to pursue projects under faculty supervision through self-directed research, experiences, and presentations.

Prerequisite: Successful completion of required program course work. Department approval required.

Lab Hours: 32.000

Recommended Preparation: Successful completion of required program course work. Department approval required

Program Description

College Credit Certificate | Code: 61002 | 15 credits

The Graphic Design Support College Credit Certificate (CCC) is designed to prepare students for initial employment as a graphic design assistant, graphic production artist, or to provide supplemental training for persons previously or currently employed in these occupations.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 15.0

GRA 1111C	Graphic Design Fundamentals	4.0
GRA 1280C	Digital Imaging Fundamentals	4.0
GRA 1750	Web Design Fundamentals	3.0
GRA 2117C	Digital Illustration Fundamentals	4.0

Total: 15.0

GRA1111C Graphic Design Fundamentals 4.000 credits

The basics of graphic design comprise this introductory course. The student will analyze the principles and practices of graphic design, and evaluate their creative solutions to challenging design problems. Topics include design vocabulary, elements and principles, the design process, proportion and composition, creativity and visualization, as well as color and typography basics. Assignments are designed to emulate real-world projects, including brainstorming, design, execution, critique, revision and implementation.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$20.00

GRA1280C Digital Imaging Fundamentals 4.000 credits

The student will learn that leading industry-standard image editing software is used to prepare photos for print and digital media in this introductory course. Topics include: resolutions, compositing, masking, correcting and enhancing, as well as applying styles and effects. Assignments are designed to emulate real-world projects, including composite images for graphic design, photographic editing, and web pages.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$35.00

GRA1750 Web Design Fundamentals 3.000 credits

The basics of web design comprise this introductory course. An introduction to the history of the web, including current design trends, a survey of Internet architecture, as well as user experience fundamentals and usability will comprise this introductory-level course. The student will examine the structure and styling of web pages.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$118.00

GRA2117C Digital Illustration Fundamentals 4.000 credits

The student will create illustrations using industry-standard vector-based illustration software. Topics include creating vector-based illustrations, leading to an illustration portfolio. Emphasis will be on both representational as well as commercial art styles. Assignments are designed to emulate real-world projects, including illustration, design, and typography.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

Program Description

Health Care Services Specialist

College Credit Certificate | Code: 63021 | 27 credits

The 27-credit hour certificate program includes a series of courses that prepare students to become proficient on the latest software used in medical offices. A medical office manager plays an integral part in the day-to-day operations of a medical practice. Medical office manager responsibilities include, but are not limited to, monitoring the office budget, ordering medical supplies and implementing office policies and procedures.

Graduates will also be trained to understand the complexities of healthcare insurance regulations and payment systems. This certificate is upward compatible with the A.S. degree, Health Services Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 14.0

BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy & Physiology Lab 1	1.0
GEB 1011	Principles of Business	3.0
HIM 1300	Health Care Facility and Delivery Systems	2.0
HIM 2253C	Current Procedural Terminology (CPT-4)	2.0
HIM 2472	Medical Terminology	3.0

Semester 2

Units: 13.0

HSA 1380	Health Care Quality Management	3.0
HSA 2181	Health Services Management Concepts	3.0
HSC 2810	Professional Practice Experience	4.0
MAN 2021	Principles of Management	3.0

Total: 27.0

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

BSC2085L Human Anatomy & Physiology Lab 1 1.000 credits

In this laboratory course, student will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

HIM1300 Health Care Facility and Delivery Systems 2.000 credits

This course will examine healthcare complexities, the function of various types of health facilities, accreditation standards, Medicare law, and the American health delivery system. Students will learn the components of Medicare, Medicaid, Health Insurance Organizations (HMO's), and the federal laws that govern them.

Lecture Hours: 32.000

HIM2253C Current Procedural Terminology (CPT-4) 2.000 credits

This course provides an in-depth knowledge of coding and reporting using Current Procedural Terminology classification. Students will learn to read and interpret ambulatory health record documentation to classify and assign services and procure codes, and use of encoder and grouper software. HCPCS, APCs, and RBRVS will be discussed. Lecture Hours: 16.000, Lab Hours: 32.000
Prerequisite: BSC2085, BSC2085L, HIM2472
Course User Fee: \$15.00

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.
Lecture Hours: 48.000

HSA1380 Health Care Quality Management 3.000 credits

This course examines various efforts being introduced to improve the quality and efficiency of the health care system. Topics covered range from issues of medical error reduction, quality improvements in medical records, and utilization review, models for continuous quality improvement in physician-patient relations. (3 hr. lecture)
Lecture Hours: 48.000

HSA2181 Health Services Management Concepts 3.000 credits

This course provides preparation and fundamental skills required to enter the field of medical office management. Emphasis will be placed on health care delivery in a medical office setting. Topics include an introduction to health care management information systems, an introduction to essential policies and regulations, and an introduction to evaluation of performance and process outcomes.
Lecture Hours: 48.000

HSC 2810 Professional Practice Experience 4.000 credits

This course provides an opportunity for the student to observe basic principles and techniques of managers in a health care setting. Students apply knowledge from previous health service management courses in completing course assignments. This course has a service learning component at an area health care agency. (3 hr. lecture)
Lecture Hours: 64.000

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.
Lecture Hours: 48.000
Course User Fee: \$15.00

Program Description

Healthcare Informatics Specialist

College Credit Certificate | Code: 63014 | 24 credits

This program is designed to prepare students for employment in various healthcare settings where the Electronic Health Record is being implemented or maintained. Students will learn the concepts of collection of health information, integration of technology into the management of healthcare records, basic concepts in health data management, and database management in a healthcare setting.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 6.0**

HIM 2472	Medical Terminology	3.0
SPC 1017	Introduction to Communication	3.0

Semester 2**Units: 11.0**

HIM 1000	Introduction to Health Information Technology	2.0
HIM 1110	Health Information Technology and Data Collection	2.0
HIM 1110L	Health Information Management Data Collection Lab	3.0
HIM 1800	Professional Practice Experience 1	2.0
HIM 2012	Legal Aspects of Health Care	2.0

Semester 3

Units: 7.0

HIM 2211C	Health Information Technologies	2.0
HIM 2400C	Diversified Non-Hospital Health Records	2.0
HIM 2652C	Electronic Health Record	3.0

Total: 24.0

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

HIM1000 Introduction to Health Information Technology 2.000 credits

This course examines the role and functions of a health information technician. Students will learn about the structure, organization, and maintenance of the medical record. Students will also learn about the organization and function of various types of health care facilities, the responsibilities of national, state and local health agencies, and the organization and mission of the American Health Information Management Association.

Lecture Hours: 32.000

HIM1110 Health Information Technology and Data Collection 2.000 credits

This course is designed to provide the skills necessary to function in a health information management department. Students will learn health record data collection and informatics. Students will also learn about the various components and approaches of the electronic health record.

Lecture Hours: 32.000

Prerequisite: HIM1000, HIM2472, Corequisite: HIM1110L

HIM1110L Health Information Management Data Collection Lab 3.000 credits

This course is designed to apply basic requirements imposed by regulatory agencies to health record data. Students will learn how clinical data repositories store health information. Concepts relating to confidentiality, ethics, and release of information will be applied.

Lab Hours: 96.000

Prerequisite: HIM1000, HIM2472, Corequisite: HIM1110

Course User Fee: \$25.00

HIM1800 Professional Practice Experience 1 2.000 credits

This course will provide the student with supervised professional practice experience in a health care setting, utilizing electronic health records and reports to manage health information data. Students will learn an in-depth knowledge of applying the minimum basic requirements for handling records imposed by regulatory agencies.

Clinical Hours: 96.000

Prerequisite: HIM1000, Corequisite: HIM1110, HIM1110L

Course User Fee: \$10.00

HIM2012 Legal Aspects of Health Care 2.000 credits

This course provides basic knowledge of the United States of America court system and the interconnection between the health care system and the federal government. The student will learn concepts relating to the Health Insurance Portability Accountability Act (HIPAA), ethics, release of health information, record retention, and the legalities of electronic health records.

Lecture Hours: 32.000

Prerequisite: ENC1101

HIM2211C Health Information Technologies 2.000 credits

This course is designed to examine computer technology used to collect and store health information. The student will learn a variety of applications used to maintain and secure health care data.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: HIM1110, HIM1110L, HIM1800

Course User Fee: \$85.00

HIM2400C Diversified Non-Hospital Health Records 2.000 credits

This course emphasizes the importance of quality record-keeping practices, data flow, and management of health information systems in a non-acute care setting. The student will learn the documentation requirements based on Federal and State statutes, accreditation standards, Medicare Conditions of Participation, payment systems, funding, Health Insurance Portability Accountability Act, and the evolution of the electronic health record.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$15.00

HIM2652C Electronic Health Record 3.000 credits

This course reviews the current trends and preparation implementing the electronic health record and reviews documentation requirements for a variety of healthcare settings. Students will learn best practices, problem-solve associated issues, and directly participate in the transitions of electronic health records.

Lecture Hours: 16.000, Lab Hours: 32.000

Pre/Corequisite: HIM2211C

Program Description

Help Desk Support Technician

College Credit Certificate | Code: 66037 | 16 credits

The Help Desk Support Technician College Credit Certificate is designed to prepare students with the technical knowledge and skills for employment as entry-level computer help desk and support technicians in commercial, industrial and government institutions. Graduates are also prepared for CompTIA A+ and Network+ industry certifications.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 16.0**

CGS 1560	A+ Computer Operating Systems	4.0
CTS 1131	A+ Computer Essentials & Support	4.0
CTS 1134	Networking Technologies	4.0
CTS 1328	Supporting Microsoft Clients	4.0

Total: 16.0

CGS1560 A+ Computer Operating Systems 4.000 credits

This is a comprehensive course in the use of operating systems for microcomputers suitable for students seeking preparation for A+ operating system certification. Students will learn how to install, configure, use, manage, and troubleshoot the Disk Operating System (DOS), Microsoft Windows, and other microcomputer operating systems. Recommended CGS 1060C or computer experience. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1131 A+ Computer Essentials & Support 4.000 credits

This is an intermediate level course designed for students preparing for A+ certification as a support technician. Students will learn how to install, configure, upgrade and replace computer system components; how to troubleshoot processors, memory, storage devices, adapter cards, peripherals and other system components; how to install, configure and troubleshoot operating systems, laptops, portable devices, printers, scanners, network devices, security measures and virtualization and cloud computing; and how to provide professional IT support and customer service. Prerequisite: CGS1560. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1560

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1328 Supporting Microsoft Clients 4.000 credits

This course is intended for students preparing for IT careers as desktop and network support specialists and server administrators, as well as candidates for industry certification. Students will learn how to implement and maintain a Microsoft client operating system.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Recommended: CGS1060C or equivalent computer experience, knowledge or skills.

Course User Fee: \$63.00

Program Description

Homeland Security

College Credit Certificate | Code: 66074 | 15 credits

This is a college credit certificate in Homeland Security. This certificate is designed to enhance student knowledge in the areas of counter-terrorism/intelligence, national security, or governmental security. Students may apply these courses to the associated A.S. degree in Criminal Justice Technology.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 15.0

CCJ 1191	Human Behavior in Criminal Justice	3.0
CJL 2062	Constitutional Law and Legal Procedure or Evidence	3.0
DSC 1006	Introduction to Homeland Security	3.0
DSC 2242	Transportation and Border Security	3.0
DSC 2590	Intelligence Analysis and Security Management	3.0

Total: 15.0

**CCJ1191 Human Behavior in Criminal Justice
3.000 credits**

Human behavior and how it relates to the duties and responsibilities of the criminal justice practitioner.
Lecture Hours: 48.000

**CJL2062 Constitutional Law and Legal Procedure
or Evidence 3.000 credits**

An examination of the United States and Florida Constitutions, with emphasis on leading cases dealing with arrest, search and seizure, confessions and the rules of evidence.
Lecture Hours: 48.000

**DSC1006 Introduction to Homeland Security
3.000 credits**

This course will introduce students to the vocabulary and important components of Homeland Security. Students will learn about the agencies associated with Homeland Security and their interrelated duties and relationships.
Lecture Hours: 48.000

**DSC2242 Transportation and Border Security
3.000 credits**

This course introduces students to global supply chains and intermodal transportation systems. Students will learn the threats to these systems, their vulnerabilities and potential for terrorist attacks, and the measures being undertaken to secure them.
Lecture Hours: 48.000

**DSC2590 Intelligence Analysis and Security
Management 3.000 credits**

This course examines intelligence analysis and its indispensable relationship to the security management of terrorist attacks, man-made disasters and natural disasters. Students will learn substantive issues regarding intelligence support of Homeland Security measures implemented by the United States and explore how the intelligence community operates.
Lecture Hours: 48.000

Program Description

Horticulture Professional

College Credit Certificate | Code: 63025 | 18 credits

The College Credit Certificate in Agriscience for the Horticulture Professional is an advanced certificate for managerial positions in nursery and landscape technology industries. The certificate will prepare students for employment in horticulture and landscape industries as nursery managers, landscape and grounds keeping managers, nursery supervisors, landscape gardeners, and parks workers. Students will learn concepts of plant physiology and growth, plant classification, plant identification and plant care and maintenance to satisfy the growing needs of the nursery industry with an additional emphasis on management skills. If a student should choose to continue their education in Agriscience, the college credits granted in this program will apply toward an A.S. degree in Landscape and Horticulture Technology.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 9.0
ORH 1510	Landscape Plant Identification 1	3.0
HOS 1010	Horticulture 1	3.0
IPM 2112	Principles of Entomology	3.0
Semester 2		Units: 9.0
ORH 1251	Nursery Practices 1	3.0
ORH 2949	Landscape Technology Internship	3.0
SBM 1000	Small Business Management	3.0
		Total: 18.0

ORH1510 Landscape Plant Identification 1 3.000 credits

Students will learn the identification and usage of plants used in the horticultural trade in South Florida. Subject matter includes trees, shrubs, and flowering plants for both interior and outdoor use.
Lecture Hours: 48.000

HOS1010 Horticulture 1 3.000 credits

This is an introductory course on the principles of horticulture. Students will learn plant structure and function, plant propagation, plant nutrients and fertilizers, potting media, soils, pruning, and plant pests. A survey of various fields in ornamental horticulture will also be covered.
Lecture Hours: 48.000

IPM2112 Principles of Entomology 3.000 credits

This is an introductory course on the principles of entomology. Students will learn to identify characteristics of arthropods, the insect orders, and the growth cycle of insects. Students will also address insect pests specific to South Florida and methods to responsibly manage plant pests.
Lecture Hours: 48.000

ORH1251 Nursery Practices 1 3.000 credits

The student will learn the techniques and practices in commercial production of ornamental plants. Emphasis on types of nurseries.
Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: HOS1010

ORH2949 Landscape Technology Internship 3.000 credits

The internship will provide students with hands-on work experience in horticulture. Landscape, or related technology industries. Students will learn employability skills, and the specific skills and safety requirements necessary for effective work in this environment.
Other Hours: 144.000

SBM1000 Small Business Management 3.000 credits

Students will learn that growing a business involves shifting from the search for and validation of a business model to executing and scaling the business model. The skill sets are different and more formal management is often required. The course reviews the context and complexity of scaling a business and executing a business model.
Lecture Hours: 48.000
Course User Fee: \$90.00

Program Description

Horticulture Specialist

College Credit Certificate | Code: 63026 | 12 credits

The College Credit Certificate in Agriscience for the Horticulture Specialist is an introductory certificate designed to prepare students for positions in the nursery and landscape industries at the entry level. The certificate will prepare students for employment as supervisors in grounds keeping, nursery and greenhouse production, landscape gardeners, and parks workers. Students will learn plant physiology and growth, plant classification, plant identification and plant care and maintenance to satisfy the growing needs of the nursery industry. If a student should choose to continue their education in Agriscience, the college credits granted in this program will apply toward an A.S. degree in Landscape and Horticulture Technology.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

HOS 1010	Horticulture 1	3.0
IPM 2112	Principles of Entomology	3.0
ORH 1251	Nursery Practices 1	3.0
ORH 1510	Landscape Plant Identification 1	3.0

Total: 12.0

HOS1010 Horticulture 1 3.000 credits

This is an introductory course on the principles of horticulture. Students will learn plant structure and function, plant propagation, plant nutrients and fertilizers, potting media, soils, pruning, and plant pests. A survey of various fields in ornamental horticulture will also be covered.

Lecture Hours: 48.000

IPM2112 Principles of Entomology 3.000 credits

This is an introductory course on the principles of entomology. Students will learn to identify characteristics of arthropods, the insect orders, and the growth cycle of insects. Students will also address insect pests specific to South Florida and methods to responsibly manage plant pests.

Lecture Hours: 48.000

ORH1251 Nursery Practices 1 3.000 credits

The student will learn the techniques and practices in commercial production of ornamental plants.

Emphasis on types of nurseries.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: HOS1010

ORH1510 Landscape Plant Identification 1 3.000 credits

Students will learn the identification and usage of plants used in the horticultural trade in South Florida. Subject matter includes trees, shrubs, and flowering plants for both interior and outdoor use.

Lecture Hours: 48.000

Program Description

Information Technology Support

College Credit Certificate | Code: 66044 | 28 credits

The Information Technology Support College Credit Certificate program is designed to provide an opportunity to establish a basic foundation in computer applications for employment in scientific, commercial, industrial and government institutions. Graduates are prepared for positions as data-entry specialists, software applications specialists and office systems specialists to meet the demands of today's automated offices. Additional Information: Certificate Pre-Requisite: CGS1060 or a working knowledge of the Microsoft Operating System and Microsoft Office Application Suite. Operational understanding of the following microcomputer topics: Operating systems, memory, hard disks, types of central processing units (CPUs), communications ports, printer ports, display adapters and pointing devices.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
CGS 1060C	Introduction to Computer Technology and Applications	4.0
CGS 1560	A+ Computer Operating Systems	4.0
CTS 1134	Networking Technologies	4.0
Semester 2		Units: 12.0
CTS 1131	A+ Computer Essentials & Support	4.0
CTS 1328	Supporting Microsoft Clients	4.0
CTS 2148C	IT Project Management	4.0
Semester 3		Units: 4.0
CTS 2153	Supporting Windows Users & Applications	4.0
		Total: 28.0

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1560 A+ Computer Operating Systems 4.000 credits

This is a comprehensive course in the use of operating systems for microcomputers suitable for students seeking preparation for A+ operating system certification. Students will learn how to install, configure, use, manage, and troubleshoot the Disk Operating System (DOS), Microsoft Windows, and other microcomputer operating systems. Recommended CGS 1060C or computer experience. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1131 A+ Computer Essentials & Support 4.000 credits

This is an intermediate level course designed for students preparing for A+ certification as a support technician. Students will learn how to install, configure, upgrade and replace computer system components; how to troubleshoot processors, memory, storage devices, adapter cards, peripherals and other system components; how to install, configure and troubleshoot operating systems, laptops, portable devices, printers, scanners, network devices, security measures and virtualization and cloud computing; and how to provide professional IT support and customer service. Prerequisite: CGS1560. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1560

Course User Fee: \$63.00

CTS1328 Supporting Microsoft Clients 4.000 credits

This course is intended for students preparing for IT careers as desktop and network support specialists and server administrators, as well as candidates for industry certification. Students will learn how to implement and maintain a Microsoft client operating system.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Recommended: CGS1060C or equivalent computer experience, knowledge or skills.

Course User Fee: \$63.00

CTS2148C IT Project Management 4.000 credits

This course prepares students to effectively plan, implement, coordinate and manage small to medium size information technology-related projects. The concepts and tools covered in this course offers practical knowledge on managing project scope, planning and scheduling, risk assessment, and identification of project resources. This course also prepares students for the CompTIA Project+ examination, an industry standard IT certification. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2153 Supporting Windows Users & Applications 4.000 credits

This is an advanced course designed to help students prepare for the Microsoft Certified IT Professional Support Technician Certification. Students will learn how to install, configure and manage Windows applications in a networked Windows environment and how to support enterprise users. Students will also deploy Windows and applications using various methods, resolve installation and compatibility issues, establish group policies and user profiles, perform support functions, troubleshoot user and application issues, secure the desktop and network from unauthorized use, install software upgrades and updates, perform systems monitoring and documentation, and develop customer service skills.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1328

Course User Fee: \$63.00

Program Description

[Intelligence Studies](#)

College Credit Certificate | Code: 66075 | 16 credits

In the College Credit Certificate (CCC) in Intelligence Studies students will learn what intelligence is and how it is used retroactively and proactively against computer crimes, homeland security critical issues and natural disasters. Students will also review the history of intelligence as practiced in the United States.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 16.0

CET 2880C	Digital Forensics	4.0
DSC 1002	Domestic and International Terrorism	3.0
DSC 1590	Introduction to Intelligence Studies	3.0
DSC 2501	Effective Communication Skills for Security Professionals	3.0
DSC 2590	Intelligence Analysis and Security Management	3.0

Total: 16.0

CET2880C Digital Forensics 4.000 credits

This is an introductory digital forensics course for students who are studying cybersecurity, electronics or computer engineering technologies. In this course, students will learn the setup and use of an investigator's laboratory, how to perform data acquisition, web forensics, email forensics, mobile forensics, network analysis, and file recovery.

Lecture Hours: 32.000, Lab Hours: 64.000

DSC1002 Domestic and International Terrorism 3.000 credits

This course is a study of domestic and international terrorism as it relates to domestic security. Through focused topics, students will learn about terrorist organizations and motivations, investigating terrorism threats, conducting vulnerability assessments of potential terrorist targets, and the role of government agencies in response to a terrorist incident and recovery afterwards.

Lecture Hours: 48.000

DSC1590 Introduction to Intelligence Studies 3.000 credits

This course will provide a comprehensive overview of intelligence for the purpose of national security for entry-level intelligence practitioners and beginning students. The student will learn security issues, define critical terms and review the history of intelligence as practiced in the United States.

Lecture Hours: 48.000

DSC2501 Effective Communication Skills for Security Professionals 3.000 credits

This course will provide a focused review and practice of the basic elements necessary for effective writing in any situation or any type of report specifically within the intelligence community. It is designed for the entry-level criminal justice practitioner and beginning student interested in entering the field. Pre-requisite: ENC1101

Lecture Hours: 48.000

Prerequisite: ENC1101

DSC2590 Intelligence Analysis and Security Management 3.000 credits

This course examines intelligence analysis and its indispensable relationship to the security management of terrorist attacks, man-made disasters and natural disasters. Students will learn substantive issues regarding intelligence support of Homeland Security measures implemented by the United States and explore how the intelligence community operates.

Lecture Hours: 48.000

Program Description

Intermodal Freight Transportation

College Credit Certificate | Code: 68011 | 18 credit

This certificate provides specialized knowledge in the intermodal transportation of goods. It covers technical skills, procedures, processes and occupation-specific skills that will enhance employability. This certificate is part of the Associate in Science degree in Transportation and Logistics.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 9.0
AVM 2120	Air Cargo	3.0
TRA 2010	Introduction to Transportation and Logistics	3.0
TRA 2402	Intermodal Transportation Operations and Project Management	3.0
Semester 2		Units: 9.0
TRA 1410	Introduction to Rail Freight Operations	3.0
TRA 1420	Introduction to Trucking Operations	3.0
TRA 1430	Introduction to Port Freight Operations	3.0
		Total: 18.0

AVM2120 Air Cargo 3.000 credits

The course develops a comprehensive grasp of the characteristics and evolution of air cargo, its impact on United States industry, inherent problems and future development.

Lecture Hours: 48.000

TRA2010 Introduction to Transportation and Logistics 3.000 credits

This course surveys the organization and operations of the commercial transportation industry and its impact on the bottom line of today's modern businesses. Students will learn to review regulations and processes affecting transportation and logistics functions as well as explore the industry job market and look at technologies and current issues shaping transportation and logistics. A.S. degree only.

Lecture Hours: 48.000

TRA2402 Intermodal Transportation Operations and Project Management 3.000 credits

Students will learn the fundamental elements necessary to plan, implement and control efficient and market-responsive integrated transportation systems. Topics include strategic, operational, and project management roles of transportation in supply chains. Emphasis is placed on services pricing, carrier selection, equipment and shipment planning, intermodal operations, financial/budgetary constraints, security and distribution services. Prerequisite: AVM2120, TRA2010; Corequisites: TRA1410, TRA1420, TRA1430.

Lecture Hours: 48.000

Prerequisite: AVM2120 AND TRA2010, Corequisite: TRA1430 AND TRA1410 AND TRA1420

TRA1410 Introduction to Rail Freight Operations 3.000 credits

This is an introductory course in Rail Freight Operations. Students will learn the advantages and disadvantages of freight movement by rail, how rail lines are organized and operate, including the use of intermodal transfer facilities and on-dock rail. Topics include shipping documents, shipment tracking, management of human resources and equipment, and an overview of hazardous materials shipments and security issues. Corequisites: TRA1420, TRA1430.

Lecture Hours: 48.000

Corequisite: TRA1430 AND TRA1420

TRA1420 Introduction to Trucking Operations 3.000 credits

This is an introductory course in Trucking Operations and the movement of goods via highways and roadways. Students will learn U.S. Department of Transportation requirements, documents for shipping, vehicle and shipment tracking, scheduling, management of human resources and equipment, just-in-time implications and integration with other transportation modes. Hazardous materials shipments and security issues will be discussed. Corequisites: TRA1410, TRA1430.

Lecture Hours: 48.000

Corequisite: TRA1430 AND TRA1410

TRA1430 Introduction to Port Freight Operations 3.000 credits

This is an introductory course in Port Freight Operations. Students will learn how Seaports and Inland Ports are organized and operate, how freight is moved domestically and internationally, including the integration of port operations with other modes of transportation. Topics include break-bulk handling during loading, discharging, in-transit carriage, on-dock rail, harbor drayage, equipment and cargo management, and an overview of hazardous materials shipments and security. Corequisites: TRA1410, TRA1420.

Lecture Hours: 48.000

Corequisite: TRA1410 AND TRA1420

Program Description**International Freight Transportation**

College Credit Certificate | Code: 68010 | 15 credit

This certificate provides specialized knowledge in the area international movement of freight. It covers the regulations and policies governing international shipments and border security, the intermodal transportation of international freight, the regulatory agencies and the documents and processes involved. This certificate is part of the Associate of Science Degree in Transportation and Logistics.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 15.0**

DSC 2242	Transportation and Border Security	3.0
TRA 1154	Introduction to Supply Chain Management	3.0
TRA 2010	Introduction to Transportation and Logistics	3.0
TRA 2321	Transportation Public Policy, Law, and Regulations	3.0
TRA 2702	International Logistics and Transportation	3.0

Total: 15.0

**DSC2242 Transportation and Border Security
3.000 credits**

This course introduces students to global supply chains and intermodal transportation systems. Students will learn the threats to these systems, their vulnerabilities and potential for terrorist attacks, and the measures being undertaken to secure them.
Lecture Hours: 48.000

TRA1154 Introduction to Supply Chain Management 3.000 credits

This course is an introduction to the concepts, principles and techniques in the field of supply chain management (SCM) with particular emphasis on the economic significance of distribution to business and the U.S. economy. Students will learn the interrelationship between logistics and other areas of business, noting how the SCM pipeline can significantly impact customer loyalty by adding value.
Lecture Hours: 48.000

TRA2010 Introduction to Transportation and Logistics 3.000 credits

This course surveys the organization and operations of the commercial transportation industry and its impact on the bottom line of today's modern businesses. Students will learn to review regulations and processes affecting transportation and logistics functions as well as explore the industry job market and look at technologies and current issues shaping transportation and logistics. A.S. degree only.
Lecture Hours: 48.000

TRA2321 Transportation Public Policy, Law, and Regulations 3.000 credits

Students will learn the transportation regulatory environment, including the various levels of government regulations. Review of security, environmental requirements, regulatory research and labor laws are also covered. Prerequisite: TRA 2010.
Lecture Hours: 48.000
Prerequisite: TRA2010

TRA2702 International Logistics and Transportation 3.000 credits

International logistics concerns the flow of materials into, through and out of an international corporation as it relates to materials management, storage, inventory locations, physical distribution and documentation. This course will emphasize international transportation infrastructure and modes such as ocean, airfreight, intermodal movement, truck and rail. Choices among these modes will be explored considering such factors as transit time, packaging, risks, predictability and cost. The roles of freight forwarders and custom brokers in moving international cargo and operation of foreign trade zones will be discussed. Prerequisites: TRA2010, TRA2321.
Lecture Hours: 48.000
Prerequisite: TRA2010, TRA2321

Program Description

Internet of Things (IoT) Applications

College Credit Certificate | Code: 66057 | 24 credit

The Internet of Things (IoT) Applications College Credit Certificate prepares students with multidisciplinary workforce skills and provides an accelerated credential that is useful for immediate employment and career experience. Graduates of the CCC in the IoT Applications acquire a skill-set that leads to producing connected devices by developing applications that can run on microcontroller development boards, simulating the functioning of the devices, and building physical prototypes. Upon completion of the program, the student will have learned how to program in the dominant programming languages used in IoT, completed projects that they can include in their portfolio, and configured different single board computers.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 8.0**

CEN 2211	C/C++ Programming for Embedded Devices	4.0
CIS 2010C	Introduction to Internet of Things Networking and Security	4.0

Semester 2**Units: 8.0**

CEN 2212C	Introduction to Programming the Internet of Things (IoT)	4.0
CIS 2011C	Introduction to Cloud Computing for Internet of Things	4.0

Semester 3

Units: 8.0

CIS 2012C	Introduction to Data Analytics and AI for the Internet of Things	4.0
CIS 2013C	Internet of Things Applications	4.0

Total: 24.0

CEN2211 C/C++ Programming for Embedded Devices 4.000 credits

This course teaches the principles of programming in the C/C++ languages for embedded devices. The student will learn how to create programs to control open-source hardware for building digital devices that can sense and control the physical world around them.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

CIS2010C Introduction to Internet of Things Networking and Security 4.000 credits

This course explores Internet of Things (IoT) networks' design, implementation, and security, providing a well-rounded understanding of protocols and standards governing the Internet and IoT. Students gain experience performing security assessments and vulnerability testing of IoT systems using industry-standard tools, focusing on identifying and mitigating risks to build secure and resilient IoT solutions.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CEN2212C Introduction to Programming the Internet of Things (IoT) 4.000 credits

This course teaches the principles of programming Internet of Things devices using a computer language. The student will learn fundamental programming concepts and systematic design techniques. At the end of the course, the student will be able to write programs that control development boards, with sensors, connected to the Internet. Prerequisite: CEN2211.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CEN2211

Course User Fee: \$63.00

CIS2011C Introduction to Cloud Computing for Internet of Things 4.000 credits

This course explores the intersection of cloud computing and the Internet of Things (IoT). Students learn how cloud platforms support storing, analyzing, and managing vast amounts of data created by IoT devices. From understanding cloud infrastructure to implementing IoT solutions using cloud services, the course provides practical insights into optimizing IoT performance using the power of cloud computing.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CEN2211

Course User Fee: \$63.00

CIS2012C Introduction to Data Analytics and AI for the Internet of Things 4.000 credits

This course equips students with the knowledge and skills necessary to navigate the complex landscape of data analytics and artificial intelligence within the context of the Internet of Things (IoT), empowering them to contribute to the development of smart, data-driven IoT solutions. Prerequisites: CIS 2010C, CIS 2011C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS2010C AND CIS2011C

Course User Fee: \$63.00

CIS2013C Internet of Things Applications 4.000 credits

In this course, students learn about the practical applications of the Internet of Things (IoT) across various sectors and industries. They explore real-world use cases, ranging from smart homes and smart cities to the latest developments in the field. The course offers insights into how IoT solutions are designed, developed, and deployed, equipping students with the knowledge to create innovative IoT applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS2010C AND CIS2011C

Course User Fee: \$63.00

Program Description

Lean Manufacturing

College Credit Certificate | Code: 68001 | 12 credit

This certificate prepares students for initial employment with an occupational title as a Quality Specialist or Lean Specialist in various specialized areas. It also can provide supplemental training for persons previously or currently employed in these occupations. These courses can be applied toward the Associate of Science in Advanced Manufacturing.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
ETI 1172	Introduction to Quality Assurance	3.0
ETI 1622	Concepts of Lean and Six Sigma	3.0
ETI 1644	Advanced Manufacturing Supply Chain	3.0
ETI 2404	Advanced Manufacturing Technology	3.0

Total: 12.0

ETI1172 Introduction to Quality Assurance 3.000 credits

This course describes the role and aspects of quality systems and Regulatory affairs in research laboratories, regulated companies, and firms that comply with voluntary standards. Topics include stages in development and submission of drugs and medical devices, patents legislation, and quality systems such as auditing, standard procedures, good manufacturing and laboratory practices.

Lecture Hours: 48.000

ETI1622 Concepts of Lean and Six Sigma 3.000 credits

This course is designed for students who are preparing for careers in the manufacturing industry. Students will learn the basic concepts, frameworks, and techniques used in six sigma, including total quality philosophies, the calculation of six sigma and other vital statistics, tools of lean six sigma, and knowledge of various methodologies.

Lecture Hours: 48.000

ETI1644 Advanced Manufacturing Supply Chain 3.000 credits

This course is designed to provide students who are preparing to be manufacturing support technologists with the fundamental concepts of advanced manufacturing supply chain management (SCM) principles. Students will learn how to use manufacturing planning and control systems to coordinate material, labor, capacity and other resources to optimize manufacturing operations. Students also learn the key features of automated systems that can be used to manage the supply chain process.

Lecture Hours: 48.000

ETI2404 Advanced Manufacturing Technology 3.000 credits

This is a course for students intending to work in manufacturing environments. Students will learn the basic concepts about advanced manufacturing operations and processes, including sourcing materials, production planning and process monitoring, and control to distribution activities. Students also review the facility and regulatory requirements needed to support manufacturing operations. Activities may include facility tours and site visits.

Lecture Hours: 48.000

Program Description

Lending

College Credit Certificate | Code: 65080 | 32 credit

The primary vision of the Lending Certificate is to develop in the students the ability to identify risks by providing a comprehensive foundation in financial analysis and credit underwriting fundamentals. The student will learn how to recognize first signals of credit deterioration and how lending officers manage problematic credit risk situations and implement strategies to minimize losses. The Program content includes identification, analysis of key qualitative risk factors, assessment, calculation, and interpretations of key ratios and elements of financial accounting.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 16.0
ACG 2021	Financial Accounting	3.0
ACG 2021L	Financial Accounting Lab	1.0
BAN 1004	Banking Fundamentals, Evolution, and Compliance	3.0
BAN 1240	Essentials for Retail Lending	3.0
BAN 2501	Money, Banking and Financial Markets	3.0
BAN 2511	Marketing for Financial Services	3.0

Semester 2		Units: 16.0
BAN 1231	Commercial Lending	3.0
BAN 2210	Analyzing Financial Statements	3.0
BAN 2211	Applied Financial Statement Analysis	3.0
FIN 1930	Special Topic Seminar	1.0
MKA 1022	Relationship Selling	3.0
REE 2304	Commercial Real Estate	3.0

Total: 32.0

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the financial field. Additional support will be provided to the students in order to achieve the objectives of ACG2021. May be repeated for credit.

Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

BAN1004 Banking Fundamentals, Evolution, and Compliance 3.000 credits

In this course, students will learn the foundation, structure and function of financial systems. The course plan includes a review of the role of banks as businesses and their impact on the economy. The students will receive an overview of the main entities that comprise the financial system: financial intermediaries, investments and financial markets. The students will be able to describe the risk infrastructure of financial institutions, the regulatory environment, and the responsibilities inherent in complying with regulatory requirements associated with the safety and soundness of banks.

Lecture Hours: 48.000

BAN1240 Essentials for Retail Lending 3.000 credits

In this course, the student will learn a comprehensive approach to consumer lending, identifying financial risks and the regulatory environment that impact and promote safety and soundness in lending. The course centers around providing a practical approach to the fundamentals of consumer installment lending, including a review of different loan products, their life cycles, the credit application process and the essentials of the closing process.

Lecture Hours: 48.000

BAN2501 Money, Banking and Financial Markets 3.000 credits

The course is designed to provide students with an overview of the US financial system, including theory and practice of monetary policy and financial instruments. Students will learn about interest rate determination, the structure and role of banks, financial institutions in the intermediation process, the factors impacting inflation, and variables in the economy. The course plan is designed not only for students but for individuals working in the financial services industry, including junior officers to mid-management and entry level staff who are considering pursuing a career in the banking field.

Lecture Hours: 48.000

BAN2511 Marketing for Financial Services 3.000 credits

Marketing of financial services is a specialized segment of marketing. It is highly competitive, making the process of selling for bankers a highly specialized and challenging endeavor. Changing market conditions, deregulation, the emergence of new competitors from within and external to the banking industry and the rapid integration of new technologies are some of the challenges bankers encounter. It covers the aspects of a consultative selling approach with emphasis on planning, implementing and fostering a long-term advisory relationship with clients. The students will learn to become proficient in how to prepare for a successful sales presentation and closing.

Lecture Hours: 48.000

BAN1231 Commercial Lending 3.000 credits

This course provides an increased awareness of the credit underwriting process. It provides a comprehensive foundation to the lending process including technical and interpretative analysis of financial information from liquidity, solvency, ratio and cash flow analysis to the basics for the completion of a credit approval memorandum. Students will learn the concepts of qualitative analysis including the assessment of industry risk, market risk and management risk. The course provides an understanding of the role of loan policy and the need to summarize the borrowers' various risks into an appropriate credit risk rating.

Prerequisite: BAN2210.

Lecture Hours: 48.000

Prerequisite: BAN2210

BAN2210 Analyzing Financial Statements 3.000 credits

In this course the students will learn the framework and resources available to analyze financial statements and to assess a company's operation and future performance. The course covers the critical objectives with the assessment of business performance such as: the relationship between the company's business and financial strategy, their financial statements, ratio, profitability, balance sheet and cash flow analysis. The course plan includes the use of financial and accounting information to make lending/credit and investment decisions. The course is targeted to a wide range of students including those pursuing careers in banking, general management, investment banking, financial analysis and consulting. Prerequisite: ACG2021.

Lecture Hours: 48.000

Prerequisite: ACG2021

BAN2211 Applied Financial Statement Analysis 3.000 credits

This course provides a comprehensive analysis of business strategy, operating performance, financial condition and cash flow strength. The student will learn basic and advanced financial concepts impacting the viability of a business, including accounting rules, methods of credit analysis, assessment of financial ratios, historical financial analysis, cash flow and financial forecasting.

Prerequisite: BAN2210.

Lecture Hours: 48.000

Prerequisite: BAN2210

Course User Fee: \$20.00

FIN1930 Special Topic Seminar 1.000 credits

This course centers around topics of current interest or of special interest. Topics or focus may vary from semester to semester. (1 hr. lecture)

Lecture Hours: 16.000

Course User Fee: \$35.00

MKA1022 Relationship Selling 3.000 credits

This course introduces the role that relationship building plays in sales. Students will explore the communication process and the importance of interpersonal skills in developing a relationship for an effective sales strategy. Students will develop a sales presentation based around consumer needs.

Lecture Hours: 48.000

Course User Fee: \$47.44

REE2304 Commercial Real Estate 3.000 credits

In this course, the student will learn specific financial tools and techniques required to assess the feasibility of commercial real estate investment, funding and financing options, risks associated with real estate transactions and factors impacting real estate investment decisions. The student will learn how the commercial real estate process works, the different interested parties involved in the transaction, the essentials of risk and return, financing options and different phases in a commercial real estate transaction.

Lecture Hours: 48.000

Course User Fee: \$70.00

Program Description

Logistics and Transportation Specialist

College Credit Certificate | Code: 63019 | 18 credit

The Logistics and Transportation Specialist College Credit Certificate (CCC) will prepare students for further education and employment in the Transportation, Distribution and Logistics career cluster. The program content is broad-based to reflect the cross-functional relationships prevalent in supply chain management. Students are exposed to related business practices such as standard operating procedures, negotiation techniques, planning, organizing, and accounting concepts, purchasing, sustainability, warehousing, project management, quality control, import/export, and asset management theory. Emphasis is placed on understanding the planning, acquisition, flow, and distribution of goods and services while managing the complexity of operational linkages in a fast-paced global supply chain.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester		Units: 15.0
TRA 1154	Introduction to Supply Chain Management	3.0
TRA 2010	Introduction to Transportation and Logistics	3.0
TRA 2156	Operations Management for Transportation	3.0
TRA 2402	Intermodal Transportation Operations and Project Management	3.0
TRA 2702	International Logistics and Transportation	3.0
Semester 2		Units: 3.0
Program Elective		3.0
TRA 1410	Introduction to Rail Freight Operations	3.0
OR		
TRA 1420	Introduction to Trucking Operations	3.0
OR		
TRA 1430	Introduction to Port Freight Operations	3.0
		Total: 18.0

TRA1154 Introduction to Supply Chain Management 3.000 credits

This course is an introduction to the concepts, principles and techniques in the field of supply chain management (SCM) with particular emphasis on the economic significance of distribution to business and the U.S. economy. Students will learn the interrelationship between logistics and other areas of business, noting how the SCM pipeline can significantly impact customer loyalty by adding value.

Lecture Hours: 48.000

TRA2010 Introduction to Transportation and Logistics 3.000 credits

This course surveys the organization and operations of the commercial transportation industry and its impact on the bottom line of today's modern businesses. Students will learn to review regulations and processes affecting transportation and logistics functions as well as explore the industry job market and look at technologies and current issues shaping transportation and logistics. A.S. degree only.

Lecture Hours: 48.000

TRA2156 Operations Management for Transportation 3.000 credits

This course includes the skills necessary for a supervisory role in logistics. Students will learn the roles and responsibilities in managing different types of transportation operations. Topics include human resources, design and management of production operations, productivity, capacity planning, resource management, just-in-time systems, hazardous materials management, planning and project management. Prerequisite: MAN2021.

Lecture Hours: 48.000

Prerequisite: MAN2021

TRA2402 Intermodal Transportation Operations and Project Management 3.000 credits

Students will learn the fundamental elements necessary to plan, implement and control efficient and market-responsive integrated transportation systems. Topics include strategic, operational, and project management roles of transportation in supply chains. Emphasis is placed on services pricing, carrier selection, equipment and shipment planning, intermodal operations, financial/budgetary constraints, security and distribution services.

Prerequisite: AVM2120, TRA2010; Corequisites:

TRA1410, TRA1420, TRA1430.

Lecture Hours: 48.000

Prerequisite: AVM2120 AND TRA2010, Corequisite:

TRA1430 AND TRA1410 AND TRA1420

TRA2702 International Logistics and Transportation 3.000 credits

International logistics concerns the flow of materials into, through and out of an international corporation as it relates to materials management, storage, inventory locations, physical distribution and documentation. This course will emphasize international transportation infrastructure and modes such as ocean, airfreight, intermodal movement, truck and rail. Choices among these modes will be explored considering such factors as transit time, packaging, risks, predictability and cost. The roles of freight forwarders and custom brokers in moving international cargo and operation of foreign trade zones will be discussed. Prerequisites: TRA2010, TRA2321.

Lecture Hours: 48.000

Prerequisite: TRA2010, TRA2321

TRA1410 Introduction to Rail Freight Operations 3.000 credits

This is an introductory course in Rail Freight Operations. Students will learn the advantages and disadvantages of freight movement by rail, how rail lines are organized and operate, including the use of intermodal transfer facilities and on-dock rail. Topics include shipping documents, shipment tracking, management of human resources and equipment, and an overview of hazardous materials shipments and security issues. Corequisites: TRA1420, TRA1430.

Lecture Hours: 48.000

Corequisite: TRA1430 AND TRA1420

TRA1420 Introduction to Trucking Operations 3.000 credits

This is an introductory course in Trucking Operations and the movement of goods via highways and roadways. Students will learn U.S. Department of Transportation requirements, documents for shipping, vehicle and shipment tracking, scheduling, management of human resources and equipment, just-in-time implications and integration with other transportation modes. Hazardous materials shipments and security issues will be discussed.

Corequisites: TRA1410, TRA1430.

Lecture Hours: 48.000

Corequisite: TRA1430 AND TRA1410

TRA1430 Introduction to Port Freight Operations**3.000 credits**

This is an introductory course in Port Freight Operations. Students will learn how Seaports and Inland Ports are organized and operate, how freight is moved domestically and internationally, including the integration of port operations with other modes of transportation. Topics include break-bulk handling during loading, discharging, in-transit carriage, on-dock rail, harbor drayage, equipment and cargo management, and an overview of hazardous materials shipments and security. Corequisites: TRA1410, TRA1420.

Lecture Hours: 48.000

Corequisite: TRA1410 AND TRA1420

Program Description

Mechatronics

College Credit Certificate | Code: 66053 | 30 credit

This program offers students instruction in maintenance techniques, computer aided drafting/design skills, technical communications, maintenance and operation of various industrial components, quality control and testing, material handling protocols, and proper usage of tools and instrumentation.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 11.0
CET 1110C	Digital Circuits	4.0
COP 2270	C for Engineers	4.0
ETI 1701	Industrial Safety	3.0
Semester 2		Units: 11.0
EET 1015C	Direct Current Circuits	4.0
EGN 2200	Computer Applications in Engineering	3.0
ETS 2673C	Programmable Logic Controls	4.0
Semester 3		Units: 9.0
ETI 1152C	Mechanical Measurement and Instrumentation	3.0
ETI 2404	Advanced Manufacturing Technology	3.0
ETM 1315C	Applied Pneumatics and Hydraulics	3.0
		Total: 31.0

CET1110C Digital Circuits 4.000 credits

This is an introductory course in digital circuits for engineering and other technology majors. Students learn how to apply electronic principles to digital computer circuits and systems. Students will also learn how to simplify logic circuits, build digital circuits, and perform other laboratory activities. Pre/Co-requisite: MAC1105. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

COP2270 C for Engineers 4.000 credits

This course is intended for students majoring in Computer Engineering Technology, Electronics Engineering Technology, or any engineering discipline. Students will learn the "C" programming language, MATLAB, and the Engineering Problem Solving Method to analyze, design, code, compile and execute programs that solve engineering related problems. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105, Recommended

Preparation: CGS1060 OR knowledge of computer skills

Course User Fee: \$63.00

ETI1701 Industrial Safety 3.000 credits

This course provides the student with the knowledge and skills to recognize hazardous situations in industrial plants and the precautions to be observed and practiced to perform work activities safely.

Among the topics covered are industrial safety hazards, electrical safety, working with chemicals, gases, and solvents, protective equipment, and safe working conditions.

Lecture Hours: 48.000

EET1015C Direct Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn electrical components, inductance, capacitance, and elementary network analysis. Students will additionally learn to verify and apply basic theories and principles through hands-on, laboratory experiments utilizing modern testing equipment. Pre/corequisite: MAT 1033. Laboratory fee. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 48.000

Pre/Corequisite: MAT1033

Course User Fee: \$63.00

EGN2200 Computer Applications in Engineering 3.000 credits

An introduction to fundamental concepts and skills of mathematical programming and computer-aided design. This course explores the use of computer software to solve engineering problems and bring ideas from a concept to a model.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1114 OR MAC1147

ETS2673C Programmable Logic Controls 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology and Industrial Engineering Technology. Students will learn the principals of Programmable Logic Controls (PLC) including hardware, programming, and troubleshooting. Students will also develop advanced working programs, and troubleshoot hardware and software communication problems. Prerequisite: CET110C

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET110C

Course User Fee: \$63.00

ETI1152C Mechanical Measurement and Instrumentation 3.000 credits

This course provides the basic foundation for mechanical measurement techniques used in manufacturing environments. The course will integrate the concepts, principles, and techniques of mechanical measurement with the use of various types of instruments including micrometers, verniers, calipers, gages, and other types of measuring equipment.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

ETI2404 Advanced Manufacturing Technology 3.000 credits

This is a course for students intending to work in manufacturing environments. Students will learn the basic concepts about advanced manufacturing operations and processes, including sourcing materials, production planning and process monitoring, and control to distribution activities. Students also review the facility and regulatory requirements needed to support manufacturing operations. Activities may include facility tours and site visits.

Lecture Hours: 48.000

ETM1315C Applied Pneumatics and Hydraulics
3.000 credits

This course prepares students to perform mechanical maintenance on industrial equipment and devices. Students learn the theory and application of fluid mechanics, how to calibrate metering devices, and conduct elementary hydraulic tests. Pre/Corequisite: MAC1105.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

Program Description

Medical Coder/Biller

College Credit Certificate - Health Sciences | Code: 63020 | 37 credit

The Medical Coder/Biller program prepares individuals for employment as Medical Coder/Billers. The student will learn to translate diagnoses and procedures into numerical designation (coding) using the International Classification of Diseases and Procedures (ICD-10-CM and PCS) and Current Procedural Terminology (CPT-4). The program involves coding, classifying and indexing diagnoses and procedures for purposes of standardization, retrieval and statistical analysis. The student will also be trained to prepare and file medical insurance claim forms for reimbursement. Electronic claims transmission is included. There is special emphasis on ethical and legal responsibilities, data quality, financial reimbursement, Diagnosis Related Groups (DRGs) and Ambulatory Patient Classification (APCs).

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 11.0
BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy and Physiology 1 Laboratory	1.0
HIM 1000	Introduction to Health Information Technology	2.0
HIM 1300	Health Care Facility and Delivery Systems	2.0
HIM 2472	Medical Terminology	3.0
Semester 2		Units: 16.0
BSC 2086	Human Anatomy & Physiology 2	3.0
BSC 2086L	Human Anatomy & Physiology 2 Lab	1.0
HIM 1110	Health Information Technology and Data Collection	2.0
HIM 1110L	Health Information Management Data Collection Lab	3.0
HIM 2222	ICD Coding Systems	2.0
HIM 2222L	ICD Coding Systems Laboratory	3.0
HIM 2253C	Current Procedural Terminology (CPT-4)	2.0
Semester 3		Units: 10.0
HIM 2211C	Health Information Technologies	2.0
HIM 2234	Advanced Coding & Reimbursement Systems	2.0
HIM 2234L	Advanced Coding & Reimbursement Systems Laboratory	1.0
HIM 2433	Pathophysiology and Pharmacology	3.0
HIM 2813	Professional Practice Experience	2.0
		Total: 37.0

**BSC2085 Human Anatomy and Physiology 1
3.000 credits**

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

**BSC2085L Human Anatomy and Physiology 1
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

**HIM1000 Introduction to Health Information
Technology 2.000 credits**

This course examines the role and functions of a health information technician. Students will learn about the structure, organization, and maintenance of the medical record. Students will also learn about the organization and function of various types of health care facilities, the responsibilities of national, state and local health agencies, and the organization and mission of the American Health Information Management Association.

Lecture Hours: 32.000

**HIM1300 Health Care Facility and Delivery
Systems 2.000 credits**

This course will examine healthcare complexities, the function of various types of health facilities, accreditation standards, Medicare law, and the American health delivery system. Students will learn the components of Medicare, Medicaid, Health Insurance Organizations (HMO's), and the federal laws that govern them.

Lecture Hours: 32.000

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

**BSC2086 Human Anatomy & Physiology 2 3.000
credits**

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems. Prerequisite: BSC2085; Corequisite: BSC2086L

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$10.00

**BSC2086L Human Anatomy & Physiology 2 Lab
1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development, from an experiential approach.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

**HIM1110 Health Information Technology and
Data Collection 2.000 credits**

This course is designed to provide the skills necessary to function in a health information management department. Students will learn health record data collection and informatics. Students will also learn about the various components and approaches of the electronic health record.

Lecture Hours: 32.000

Prerequisite: HIM1000, HIM2472, Corequisite: HIM1110L

HIM1110L Health Information Management Data Collection Lab 3.000 credits

This course is designed to apply basic requirements imposed by regulatory agencies to health record data. Students will learn how clinical data repositories store health information. Concepts relating to confidentiality, ethics, and release of information will be applied.

Lab Hours: 96.000

Prerequisite: HIM1000, HIM2472, Corequisite: HIM1110

Course User Fee: \$25.00

HIM2222 ICD Coding Systems 2.000 credits

This course is designed to teach the current edition of International Classification of Diseases and Procedures codes. Students will learn code convention and format, usage of anatomy, physiology, and clinical disease processes. Pharmacology, present on admission indicators, correct code assignment and sequencing will be discussed.

Lecture Hours: 32.000

Prerequisite: BSC2085, BSC2085L, HIM2472, Corequisite: HIM2222L

HIM2222L ICD Coding Systems Laboratory 3.000 credits

This course is designed to apply diagnoses and operations into numerical designations (codes) utilizing the International Classification of Diseases and Procedures. Students will learn correct sequencing, analysis of clinical disease processes, diagnosis and procedural terminology, and application of pharmacology in current coding systems. Encoding software is utilized.

Lab Hours: 96.000

Prerequisite: BSC2085, BSC2085L, HIM2472, Corequisite: HIM2222

Course User Fee: \$25.00

HIM2253C Current Procedural Terminology (CPT-4) 2.000 credits

This course provides an in-depth knowledge of coding and reporting using Current Procedural Terminology classification. Students will learn to read and interpret ambulatory health record documentation to classify and assign services and procure codes, and use of encoder and grouper software. HCPCS, APCs, and RBRVS will be discussed.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: BSC2085, BSC2085L, HIM2472

Course User Fee: \$15.00

HIM2211C Health Information Technologies 2.000 credits

This course is designed to examine computer technology used to collect and store health information. The student will learn a variety of applications used to maintain and secure health care data.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: HIM1110, HIM1110L, HIM1800

Course User Fee: \$85.00

HIM2234 Advanced Coding & Reimbursement Systems 2.000 credits

This course is designed to apply the fundamentals of the Prospective Payment Systems as it applies to coding and reimbursement. The student will learn documentation criteria, validation reports of coded data, health record for compliance, and optimum reimbursement under current payment methodologies.

Lecture Hours: 32.000

Prerequisite: HIM2222, HIM2222L, Corequisite: HIM2234L

HIM2234L Advanced Coding & Reimbursement Systems Laboratory 1.000 credits

This course is designed to apply and compute Prospective Payment Systems categories. Students will learn to apply health record documentation to identify and validate correct code and payment assignments. The focus is on the computation of MS-DRGs, APCs and a Case-Mix Index using encoder, grouper, and electronic billing software for reimbursement.

Lab Hours: 32.000

Prerequisite: HIM2222, HIM2222L, Corequisite: HIM2234

Course User Fee: \$15.00

HIM2433 Pathophysiology and Pharmacology 3.000 credits

This course provides an in-depth knowledge of disease, its etiology, medical complications, and pathophysiologic nature. Students will learn laboratory and other diagnostic tests used to confirm or rule out those diagnoses addressed. Current pharmacological treatments are explored with review and interpretation of health record data.

Lecture Hours: 48.000

Prerequisite: BSC2085, BSC2085L, HIM2472

HIM2813 Professional Practice Experience 2.000 credits

This course is an advanced coding/billing professional practice. Students will learn advanced coding and abstracting of actual inpatient and outpatient health records, with an emphasis on compliance and improving accuracy and productivity.

Clinical Hours: 96.000

Prerequisite: HIM2222 AND HIM2222L AND HIM2234 AND HIM2234L AND HIM2253C

Program Description

Mental Health - Neuroscience and Aging

College Credit Certificate | Code: 65079 | 18 credit

This College Credit Certificate (CCC) in Mental Health - Neuroscience and Aging is designed to prepare students to meet community needs and industry demands in the field of mental health services with a diverse aging population.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 9.0

INP 2390	Psychology of Work	3.0
PSB 2041	Behavioral Neuroscience	3.0
PSY 2012	Introduction to Psychology	3.0

Semester 2

Units: 9.0

CLP 2140	Psychopathology	3.0
DEP 2402	Psychology of Adulthood and Aging	3.0
PSY 2940C	Mental Health - Aging Practicum	3.0

Total: 18.0

INP2390 Psychology of Work 3.000 credits

Applies the understanding of effective human relations to work situations. Personal dynamics for success are also considered. Students will be taught how to influence behavior on the job as they apply their knowledge and interpersonal skills to specific experiences in the workplace.

Lecture Hours: 48.000

PSB2041 Behavioral Neuroscience 3.000 credits

This course presents an approach to the study of the biology of human behavior and aims to introduce the field of Biopsychology. It is designed to provide each student with comprehensive exposure to the nervous system and how it governs various behaviors. This is an introduction course that explores how our brains develop, how they work, and how they are changed by life experiences. Topics include neural communication, localization of brain function, neural systems, and control of behavior. Upon completion of the course, the student will have a solid foundation regarding the biological basis of behavior upon which to build on in more advanced courses of study.

Lecture Hours: 48.000

PSY2012 Introduction to Psychology 3.000 credits

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence.
Lecture Hours: 48.000

CLP2140 Psychopathology 3.000 credits

This course examines the major categories of mental disorders. Diagnostic criteria, treatment methods, cultural factors, public attitudes, community resources, ethical issues and legislation applicable to individuals with mental disorders are studied. The impact of mental disorders on individuals, families and society are discussed.
Lecture Hours: 48.000
Prerequisite: PSY2012

DEP2402 Psychology of Adulthood and Aging 3.000 credits

This course will examine adulthood and aging from a multidisciplinary and diversity perspective. This course will explore the biological, physical, psychological, cognitive, and cultural changes during adulthood and aging including contemporary aspects of work-family life, and retirement. Aspects of death and dying will also be discussed from a multidisciplinary perspective. (3 hr. lecture)
Lecture Hours: 48.000

PSY2940C Mental Health - Aging Practicum 3.000 credits

This Capstone course will provide students with field experience on neuroscience and aging through a practicum (non-paid), in a for profit or not for profit setting that serves older adult populations. The students will have the opportunity to apply the knowledge, skills and abilities gained through the Neuroscience and Aging CCC. Departmental Permission Required. (144 hours practicum).
Practicum Hours: 144.000

Program Description

Microcomputer Repairer/Installer

College Credit Certificate | Code: 66032 | 15 credit

The Microcomputer Repairer/Installer College Credit Certificate program is designed to prepare students to work as Computer Repair Assistants in a computer repair shop or the computer maintenance division of a corporation, by acquiring a basic understanding of computer internal architecture and operations.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 6.0
CET 1171	Introduction to Computer Service and Maintenance	3.0
EET 1082	Introduction to Electronics	3.0
Semester 2		Units: 9.0
CET 1178C	A+ Computer Hardware Service	3.0
CET 1487C	Network+	3.0
CET 2588C	Server + Service and Maintenance	3.0
		Total: 15.0

CET1171 Introduction to Computer Service and Maintenance 3.000 credits

This course is designed as an introduction for students new to IT. Students will learn about the history, design, construction, and maintenance of microcomputers, including the proper handling and use of computer components and tools; how to assemble and disassemble computers; how to perform preventive maintenance; how to identify and upgrade components; how to interpret error messages, and how to perform basic troubleshooting.

Lecture Hours: 48.000

Course User Fee: \$63.00

EET1082 Introduction to Electronics 3.000 credits

Learn by building practical electronic circuits. Survey course suitable for both majors and non-majors. Instructors and tutors are available to assist in project completion. Topics include: schematics, pictorials, amplifiers, oscillators, burglar alarms, radios, digital circuits. Students will develop individual career plans and learn about employment opportunities within the field.

Lecture Hours: 48.000

Course User Fee: \$63.00

CET1178C A+ Computer Hardware Service 3.000 credits

This is an intermediate level course that prepares students for A+ hardware certification. Students will learn how to: install, configure, and upgrade components; diagnose and troubleshoot computer systems, identify, test, and troubleshoot motherboards, processors, memory, and printers, and connect network equipment. Prerequisite: CET1171. Special fee. A.S. credit only.

Lecture Hours: 48.000

Course User Fee: \$63.00

CET1487C Network+ 3.000 credits

This is an intermediate level course designed for students preparing for the hardware component of the Network+ certification. Students will learn how to install, configure, manage, troubleshoot and upgrade network devices including network interface cards, switches, hubs, wireless access points, routers, and patch panels. They will also learn about the construction, installation, testing and repair of the physical layer of the network, including wired cables, fiber optic media, wireless transmitters and antennas. Demonstrated knowledge of microcomputer fundamentals and system components is required. Recommended Preparation: CET1178C or A+ certification. Laboratory fee. A.S. degree credit only.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CET2588C Server + Service and Maintenance 3.000 credits

This course is designed for students preparing for the hardware component of the Server + certification. Students will learn how to install, configure, and upgrade workstations and servers, configure and test network and peripheral equipment, and diagnose and troubleshoot advanced computer systems. Laboratory fee. College readiness in reading and math required. Recommended Preparation: CET1178C or A+ certification. Special fee.

Lecture Hours: 48.000

Course User Fee: \$63.00

Program Description

Network Server Administration

College Credit Certificate | Code: 66035 | 24 credit

The Network Server Administration College Credit Certificate is designed to prepare students with a foundation in how to implement, support, and maintain Microsoft-based servers and clients. Graduates are prepared for employment as entry-level network administrators, network technicians, information technology support specialists, network coordinators and as candidates for industry certifications.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

CTS 1111 Linux + 4.0

CTS 1120	Cybersecurity Fundamentals	4.0
CTS 1134	Networking Technologies	4.0

Semester 2

Units: 12.0

CTS 2143C	Server Administration	4.0
CTS 2192C	Microsoft Azure Administration	4.0
CTS 2303	Windows Server Administration	4.0

Total: 24.0

CTS1111 Linux + 4.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based workstations and servers. Students learn how to plan, install, maintain, document, and troubleshoot GNU/Linux operating system services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1060C

Course User Fee: \$63.00

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2143C Server Administration 4.000 credits

This course is designed for students preparing for IT careers as server administrators, and focuses on server hardware and operating systems, system management and administration, security and disaster recovery, infrastructure and network services, data storage, and troubleshooting. This course is intended to prepare students for the CompTIA Server+ certification.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Server operating systems experience.

Course User Fee: \$63.00

CTS2192C Microsoft Azure Administration 4.000 credits

This course is intended for students preparing for IT careers as cloud administrators, as well as candidates for industry certification. Students will learn the administration skills and knowledge required to implement, manage, and monitor identity, governance, storage, and compute virtual networks in the Microsoft Azure cloud environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS1145 or equivalent knowledge in Microsoft Azure Fundamentals

Course User Fee: \$63.00

CTS2303 Windows Server Administration 4.000 credits

This course is intended for students preparing for IT careers as network support specialists as well as server and cloud administrators. Students will learn the fundamental administration skills and knowledge required to deploy and support Windows Server core services.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CGS1060 and CTS1134 or equivalent knowledge

Course User Fee: \$63.00

Program Description

Network Systems Developer

College Credit Certificate | Code: 66034 | 41 credit

The Network Systems Developer College Credit Certificate is designed to prepare students to work as Computer Repair Technicians in a computer repair shop or the computer maintenance division of a corporation, by acquiring an indepth understanding of computer internal architecture, operations and digital systems design operations.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
CET 1110C	Digital Circuits	4.0
CET 1171	Introduction to Computer Service and Maintenance	3.0
EET 1082	Introduction to Electronics	3.0
MAC 1105	College Algebra	3.0
Semester		Units: 14.0
CET 1178C	A+ Computer Hardware Service	3.0
CET 1487C	Network+	3.0
CET 2113C	Advanced Digital Circuits	4.0
EET 1015C	Direct Current Circuits	4.0
Semester 3		Units: 4.0
CET 2123C	Microprocessors	4.0
Semester 4		Units: 10.0
CET 2588C	Server + Service and Maintenance	3.0
EET 2351C	Digital and Data Communications	4.0
MTB 1322	Technical Mathematics 2	3.0
		Total: 41.0

CET1110C Digital Circuits 4.000 credits

This is an introductory course in digital circuits for engineering and other technology majors. Students learn how to apply electronic principles to digital computer circuits and systems. Students will also learn how to simplify logic circuits, build digital circuits, and perform other laboratory activities. Pre/Co-requisite: MAC1105. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

CET1171 Introduction to Computer Service and Maintenance 3.000 credits

This course is designed as an introduction for students new to IT. Students will learn about the history, design, construction, and maintenance of microcomputers, including the proper handling and use of computer components and tools; how to assemble and disassemble computers; how to perform preventive maintenance; how to identify and upgrade components; how to interpret error messages, and how to perform basic troubleshooting.

Lecture Hours: 48.000

Course User Fee: \$63.00

EET1082 Introduction to Electronics 3.000 credits

Learn by building practical electronic circuits. Survey course suitable for both majors and non-majors. Instructors and tutors are available to assist in project completion. Topics include: schematics, pictorials, amplifiers, oscillators, burglar alarms, radios, digital circuits. Students will develop individual career plans and learn about employment opportunities within the field.

Lecture Hours: 48.000

Course User Fee: \$63.00

MAC1105 College Algebra 3.000 credits

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

CET1178C A+ Computer Hardware Service 3.000 credits

This is an intermediate level course that prepares students for A+ hardware certification. Students will learn how to: install, configure, and upgrade components; diagnose and troubleshoot computer systems, identify, test, and troubleshoot motherboards, processors, memory, and printers, and connect network equipment.

Prerequisite: CET1171. Special fee. A.S. credit only.

Lecture Hours: 48.000

Course User Fee: \$63.00

CET1487C Network+ 3.000 credits

This is an intermediate level course designed for students preparing for the hardware component of the Network+ certification. Students will learn how to install, configure, manage, troubleshoot and upgrade network devices including network interface cards, switches, hubs, wireless access points, routers, and patch panels. They will also learn about the construction, installation, testing and repair of the physical layer of the network, including wired cables, fiber optic media, wireless transmitters and antennas. Demonstrated knowledge of microcomputer fundamentals and system components is required. Recommended Preparation: CET1178C or A+ certification. Laboratory fee. A.S. degree credit only.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CET2113C Advanced Digital Circuits 4.000 credits

This is a second level course in digital circuits for students majoring in electronics and related engineering technologies that extends the application of sequential and combinational logic circuits and other digital applications. Students will learn to program, operate, and interface with a micro-computer and its elements.

Prerequisites: CET1110C, COP2270. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

EET1015C Direct Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn electrical components, inductance, capacitance, and elementary network analysis. Students will additionally learn to verify and apply basic theories and principles through hands-on, laboratory experiments utilizing modern testing equipment. Pre/corequisite: MAT 1033. Laboratory fee. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 48.000

Pre/Corequisite: MAT1033

Course User Fee: \$63.00

CET2123C Microprocessors 4.000 credits

This course is intended for students majoring in Electronics or Computer Engineering Technology. Students will learn to apply digital principles to the understanding of microprocessor parameters and characteristics (addressing range and models, instruction set, architecture, input/output, interrupts, and programming). Students will build practical microprocessor and/or microcontroller based systems to perform a variety of engineering applications. Prerequisites: CET1110C, COP 2270. Special Fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET1110C AND COP2270

Course User Fee: \$63.00

CET2588C Server + Service and Maintenance 3.000 credits

This course is designed for students preparing for the hardware component of the Server + certification. Students will learn how to install, configure, and upgrade workstations and servers, configure and test network and peripheral equipment, and diagnose and troubleshoot advanced computer systems. Laboratory fee. College readiness in reading and math required. Recommended Preparation: CET1178C or A+ certification. Special fee.

Lecture Hours: 48.000

Course User Fee: \$63.00

EET2351C Digital and Data Communications 4.000 credits

This course is intended for students majoring in Electronics or Computer Engineering Technology. It provides a theoretical and practical background in the basic concepts and applications of digital and data communications. Students will learn analog-to-digital (A/D) and digital-to-analog (D/A) conversions; data communications codes and standards; wired and wireless digital communications; modulation, transmission impairment, the telephone system, modems, multiplexers, and electrical interface standards.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2123C

Course User Fee: \$63.00

MTB1322 Technical Mathematics 2 3.000 credits

Applications of algebra, trigonometry, and analytic geometry are needed in technical programs.

Lecture Hours: 48.000

Program Description

Oracle Certified Database Administrator

College Credit Certificate | Code: 66048 | 16 credit

The Oracle Certified Database Administrator College Credit Certificate program is designed to provide an opportunity to establish a basic foundation in the field of database administration for employment in commercial, industrial and government institutions. Graduates are prepared for the position of Oracle Database Administrator.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 8.0
CTS 2440	Introduction to Oracle: SQL And PL/SQL	4.0
CTS 2441	Introduction to Oracle Database Administration	4.0
Semester		Units: 8.0
CTS 2442	Intermediate Oracle Database Administration	4.0
CTS 2444	Oracle Database Performance Tuning	4.0
		Total: 16.0

**CTS2440 Introduction to Oracle: SQL And PL/SQL
4.000 credits**

This is an introductory level course for students majoring in the Oracle Database Administrator and/or Solutions Developer programs. Students will learn the fundamentals of SQL and PL/SQL programming languages including the concepts of relational databases, how to create and maintain database objects, and how to store, retrieve, and manipulate data. Students will also learn to create PL/SQL blocks of application code that can be shared by multiple forms, reports, and data management applications. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CGS1060C
Course User Fee: \$63.00

**CTS2441 Introduction to Oracle Database
Administration 4.000 credits**

This course is designed to give students who are preparing to become Oracle database administrators (DBA) a firm foundation in basic administrative tasks. Students will learn through instructor-led learning, structured hands-on practices, and challenge-level exercise labs, the necessary knowledge and skills to set up, maintain and troubleshoot an Oracle database. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS2440
Course User Fee: \$63.00

**CTS2442 Intermediate Oracle Database
Administration 4.000 credits**

This is the second course in Oracle database administration. Students will learn basic network administration, including techniques to backup and to recover an Oracle database. The skills developed in this class will help prepare students for the Oracle database administrator (DBA) certification exam. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS2441
Course User Fee: \$63.00

**CTS2444 Oracle Database Performance Tuning
4.000 credits**

This course teaches students tuning steps which can be used to improve database performance. Students will learn through a combination of demonstrations, lectures, and lab exercises, gaining practical experience tuning an Oracle database. Students will also learn how to recognize, troubleshoot and resolve common performance-related problems in administering an Oracle database. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS2441
Course User Fee: \$63.00

Program Description

Paramedic

College Credit Certificate | Code: 63008 | 42 credit

The Paramedic College Credit Certificate provides students with the advanced skills, knowledge, and clinical experience required to provide safe and effective hospital and pre-hospital care to the sick and injured. Satisfactory completion of the program will qualify the graduate to sit for the State and/or National Paramedic Certification Examination. This program is accredited by the Commission on Accreditation of Allied Health Education Programs (www.caahep.org) upon the recommendation of the Committee on Accreditation of Educational Programs for the Emergency Medical Services Professions (CoAEMSP) and approved by the State of Florida, Department of Health and Rehabilitative Services

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

General Education Core Requirements - Science Courses**Units: 4.0**

BSC 2085	Human Anatomy and Physiology 1	3.0
BSC 2085L	Human Anatomy & Physiology Lab 1	1.0

Semester 1**Units: 15.0**

EMS 2601	Paramedic Lecture 1	8.0
EMS 2601L	Paramedic Laboratory 1	4.0
EMS 2664	Paramedic Clinic 1	3.0

Semester 2		Units: 15.0
EMS 2602	Paramedic Lecture 2	8.0
EMS 2602L	Paramedic Laboratory 2	4.0
EMS 2665	Paramedic Clinic 2	3.0

Semester 3		Units: 8.0
EMS 2659	EMS-Field Internship and Conference	8.0

Total: 42.0

BSC2085 Human Anatomy and Physiology 1 3.000 credits

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data. Corequisite: BSC2085L.

Lecture Hours: 48.000
Corequisite: BSC2085L
Course User Fee: \$8.00

BSC2085L Human Anatomy & Physiology Lab 1 1.000 credits

In this laboratory course, student will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach.

Lab Hours: 32.000
Corequisite: BSC2085
Course User Fee: \$25.00

EMS2601 Paramedic Lecture 1 8.000 credits

EMS2601 is the first course in the sequence necessary for completion of the Paramedic Certificate program. The course is designed to reinforce concepts and clinical skills learned at the EMT level and to integrate this knowledge beginning with advanced life support concepts and skills. Emphasis is placed on EMS systems, illness and injury prevention, medical-legal issues, patient assessment, airway management and ventilation, pathophysiology, pharmacology, shock, decision-making, and the management of trauma-related injuries. This course includes Modules 1–4 of the 1998 DOT National Standard Curriculum for Paramedic Programs.

Lecture Hours: 128.000
Corequisite: EMS2601L
Course User Fee: \$96.00

EMS2601L Paramedic Laboratory 1 4.000 credits

A review of basic life support practice and an introduction to advanced life support practice. Areas of emphasis include patient assessment, trauma emergencies, obstetric emergencies, gynecological emergencies, pediatric emergencies and psychiatric emergencies. Students will be expected to master the techniques of patient assessment, intravenous techniques and endotracheal intubation.

Lab Hours: 128.000
Corequisite: EMS2601
Course User Fee: \$75.00

EMS2664 Paramedic Clinic 1 3.000 credits

EMS 2664 is designed to allow the students "hands-on" practice of the skills and theories learned in EMS 2601 and 2601L. Clinical experience will take place in many areas including the emergency department, operating room and medical examiner's office. All patient care experience will be practiced under the direct supervision of a medical professional (Paramedic, Nurse, Physician, etc.).

Clinical Hours: 144.000
Corequisite: EMS2601, EMS2601L

EMS2602 Paramedic Lecture 2 8.000 credits

EMS 2602 is the second course in the sequence necessary for the completion of the Paramedic Certificate Program. This course is designed to reinforce and expand upon the material and skills learned in Paramedic 1 level and to integrate prior learning with enhanced life support concepts and skills. Emphasis is placed on patient assessment and recognition of significant findings, pre-hospital diagnosis and differential diagnosis, treatment strategies, anatomy and physiology, pathophysiology, and the management of various emergencies, patients with special challenges, assessment based management, and EMS operations. This course includes Modules 5-8 of the 1998 DOT National Standard Curriculum for Paramedic Programs.

Lecture Hours: 128.000

Prerequisite: EMS2601L, EMS2601, EMS2664,

Corequisite: EMS2602L, EMS2665

EMS2602L Paramedic Laboratory 2 4.000 credits

Continuation of advanced life support practice. Areas of emphasis include patient assessment, trauma emergencies, obstetric emergencies, gynecological emergencies, pediatric emergencies and psychiatric emergencies. Students will be expected to master the techniques of patient assessment, intravenous techniques, endotracheal intubation, and advanced life support.

Lab Hours: 128.000

Corequisite: EMS2602, EMS2665

Course User Fee: \$75.00

EMS2665 Paramedic Clinic 2 3.000 credits

EMS 2665 is designed to allow the students "hands-on" practice of the skills and theories learned in EMS 2602 and 2602L. Clinical experience will take place in many areas including the emergency department, operating room and critical care unit. All patient care experience will be practiced under the direct supervision of a medical professional (paramedic, nurse, physician, etc.).

Clinical Hours: 144.000

Prerequisite: EMS2664, Corequisite: EMS2602, EMS2602L

EMS2659 EMS-Field Internship and Conference 8.000 credits

A supervised clinical experience on an Advanced Life Support (ALS) vehicle. The student obtains increasing patient care responsibilities as a working member of the EMS team under the direct supervision of a designated preceptor.

Clinical Hours: 384.000

Prerequisite: EMS2602, EMS2602L, EMS2665

Program Description

Passenger Service Agent

College Credit Certificate | Code: 66028 | 16 credits

The Passenger Service Agent College Credit Certificate program is designed to give students the skills required to gain employment as a passenger service agent, including gate and ramp responsibilities. Students will be required to do an internship with a commuter or major airline. Additional Information: Contact the Aviation Department at (305) 237-5950 for more information and advisement.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 10.0

AVM 1022	Flight Operations	3.0
AVM 1062	Aviation Career Planning	1.0
AVM 1440	Aviation/Airport Security	3.0
AVM 2431	Customer Service Agent	3.0

Total: 10.0

AVM1022 Flight Operations 3.000 credits

An investigation of the occupational duties, responsibilities, and physical facilities required by the positions of pilot, co-pilot, flight engineer, dispatcher and flight attendant.

Lecture Hours: 48.000

AVM1062 Aviation Career Planning 1.000 credits

This course provides direction and guidance in career planning for all aviation students. Topics of discussion will include the job search education and training requirements, resume writing, business etiquette, interview skills and follow-up techniques

Lecture Hours: 16.000

AVM1440 Aviation/Airport Security 3.000 credits

This course will provide the student with knowledge of the issues and strategies that are used to protect the national airspace system, airports and airlines from security threats. The various types of threats and responses to those threats will be covered. In addition, the legal requirements, planning issues, physical equipment and facility requirements and personnel issues will also be discussed.

Lecture Hours: 48.000

Course User Fee: \$50.00

AVM2431 Customer Service Agent 3.000 credits

This course covers the generic skills needed for any airline position involving regular contact with the traveling public. It includes human relations, personal appearance enhancement, etiquette, conflict management, speech skills, and the acquisition of attributes that would promote a proper professional image.

Lecture Hours: 48.000

Program Description

Patternmaking and Construction

College Credit Certificate | Code: 64002 | 21 credit

The College Credit Certificate Program in Patternmaking and Construction prepares students to fulfill the essential roles of machinists, pattern makers, and pattern cutters in the growing field of fashion. Classes focus on the technical skills and basic textile knowledge required for entry level positions in the industries of textile, apparel, and furnishings.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 9.0
CTE 1401	Textiles	3.0
CTE 1743C	Patternmaking Level 1	3.0
CTE 2310C	Basic Clothing Construction Methods	3.0
Semester 2		Units: 6.0
CTE 2330C	Clothing Construction Methods Level 2	3.0
CTE 2745C	Patternmaking Level 2	3.0
Semester 3		Units: 6.0
CTE 2342C	Clothing Construction Methods Level 3	3.0
CTE 2749C	Patternmaking Level 3	3.0
		Total: 21.0

CTE1401 Textiles 3.000 credits

This is a survey course designed for students majoring in fashion-related curriculum or with a general interest in textile materials. Students will learn basic elements of the transformation from fiber of textiles into finished goods. The course provides insights into textile manufactures with a primary focus on general textile applications relative to end-use consumer products. Students will learn the terminology needed for effective communication throughout the fashion supply chain, gain insight and appreciation for the relative value of textile products and the appropriateness of specific textile uses.

Lecture Hours: 48.000

CTE1743C Patternmaking Level 1 3.000 credits

This course will focus on the development of basic blocks, dart manipulation, bodice, skirt and sleeve styling methods in accordance with approved garment trade practices. Students will learn to draft and manipulate the various garment components and develop patterns and samples. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Corequisite: CTE2310C

Course User Fee: \$241.00

CTE2310C Basic Clothing Construction Methods 3.000 credits

Students will learn the basic elements of sewing utilized and incorporated into all designs in the garment industry. These garment structures form the fundamentals of sewing and are integrated into the construction methods used by each company in the applications to a specific design. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Corequisite: CTE1743C

Course User Fee: \$241.00

CTE2330C Clothing Construction Methods Level 2 3.000 credits

This course focuses on the principles and techniques of applied clothing constructions. Students will learn to apply the principles and techniques of clothing construction using original student design and pattern work.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2310C, Corequisite: CTE2745C

Course User Fee: \$241.00

CTE2745C Patternmaking Level 2 3.000 credits

This course focuses on the integration of flat patterns and draping. Students will learn the foundations of Draping as a design development and execution technique. How to move from the form to the flat and on to the body. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE1743C, Corequisite: CTE2330C

Course User Fee: \$241.00

CTE2342C Clothing Construction Methods Level 3 3.000 credits

This course advances students' knowledge about materials, core properties and construction techniques that inform their choices made for silhouette creation and hands-on prototype development. Students will explore more advanced principles and techniques of unstructured draping using soft fabric for the creative interpretation and artistic development of contemporary designs. Emphasis on proportion, balance and shape as related to design aesthetics. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2330C, Corequisite: CTE2749C

CTE2749C Patternmaking Level 3 3.000 credits

This course reinforces the students' understanding of the art of draping and patternmaking and diverse methods the industry uses to create production patterns. Students will analyze draping and drafting techniques to create an awareness of which method is most applicable for a given situation. Analytical thinking and hands-on class experiences will strengthen students' skills, thus enabling them to expand on their creativity and provide a proper fit for their creations. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2745C, Corequisite: CTE2342C

Program Description

Rapid Prototyping Specialist

College Credit Certificate | Code: 66054 | 12 credit

This program provides students with the technical skills proficiency and competency-based applied learning that contributes to the academic and career knowledge for students entering the advanced manufacturing industry.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 6.0
ETD 1340	Computer Aided Drawing & Design	3.0
ETI 2404	Advanced Manufacturing Technology	3.0

Semester 2		Units: 6.0
EGN 2200	Computer Applications in Engineering	3.0
ETS 2632C	Introduction to Computer Aided Manufacturing	3.0

Total: 12.0

ETD1340 Computer Aided Drawing & Design 3.000 credits

This course is recommended for all engineering students as an introduction to the basic concepts of drafting and designing using a computer. Students will learn industry standard drafting and design practices using AutoCAD in a laboratory environment. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

ETI2404 Advanced Manufacturing Technology 3.000 credits

This is a course for students intending to work in manufacturing environments. Students will learn the basic concepts about advanced manufacturing operations and processes, including sourcing materials, production planning and process monitoring, and control to distribution activities. Students also review the facility and regulatory requirements needed to support manufacturing operations. Activities may include facility tours and site visits.

Lecture Hours: 48.000

EGN2200 Computer Applications in Engineering 3.000 credits

An introduction to fundamental concepts and skills of mathematical programming and computer-aided design. This course explores the use of computer software to solve engineering problems and bring ideas from a concept to a model.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1114 OR MAC1147

ETS2632C Introduction to Computer Aided Manufacturing 3.000 credits

An introduction into the fundamentals of Computer Integrated Manufacturing as it relates to theory, operation, setup, safety, and practices. Students will learn the application of Computer Aided Drawing (CAD) and Computer Aided Manufacturing (CAM) software to develop prototypes. Prerequisite: MAC1105.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

Program Description

[Real Estate Legal Specialization Certificate](#)

College Credit Certificate | Code: 67015 | 12 credit

The College Credit Certificate (CCC) in Real Estate Legal Specialization Certificate program equips students with the necessary skills to pursue entry-level positions in law offices, lending institutions, title companies, real estate brokerage offices, real estate settlement offices, and government agencies. Additionally, it provides an opportunity for individuals already working in the field without real estate expertise to enhance their skills, thereby increasing their marketability and potentially advancing their careers in the legal and real estate sectors.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1**Units: 12.0**

LCO 0999	Law Center Student Orientation	0.0
PLA 2610	Introduction to Real Property	4.0
PLA 2631C	Real Estate Transactions and Settlement	4.0
PLA 2704	Professionalism & Civility in the Law	3.0
PLA 2930	Special Topics in Law	1.0

Total: 12.0

LCO0999 Law Center Student Orientation 0.000 credits

Law Center Student Orientation

Lecture Hours: 6.000

PLA2610 Introduction to Real Property 4.000 credits

This is an introductory course that provides an overview of the general concepts of real property law applied in most states, including substantive legal issues, ownership, terms and various legal forms, the mechanics of various residential, some commercial transactions, and mortgage foreclosures.

Prerequisite: REE2040; Corequisite: PLA2631C (3 hr. lecture)

Lecture Hours: 64.000

Prerequisite: REE2040, Corequisite: PLA2631C

Course User Fee: \$45.84

PLA2631C Real Estate Transactions and Settlement 4.000 credits

This course is a combination of both class lecture and lab. The lecture segment includes an overview of intermediate real estate law topics with a specific emphasis on Florida real estate transactions.

Students who complete this course will understand the fundamental concepts and sequences followed in a real estate transaction from initial contract review, through settlement (closing). The lab segment is transaction and problem-oriented. This segment trains students in the use of the industry settlement software to complete title examinations, updates, issuance of title policies & endorsements, and various federal and state regulations that affect real estate settlements. Students may also perform various duties typically assigned to a paralegal in a real estate transaction. Prerequisite: REE2040;

Corequisite: PLA2610 (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: REE2040, Corequisite: PLA2610

Course User Fee: \$45.84

PLA2704 Professionalism & Civility in the Law 3.000 credits

This class will provide students guidance as they interact in a real-world setting as legal professionals while strengthening their communication skills. This course will cultivate the students' leadership capacities at the intra-personal, interpersonal, and industry levels. The coursework will proceed through experiential learning that is grounded in theory and includes reflection. The study material will develop students' knowledge in the context of the aspiration, relational, and practical aspects of professionalism and civility in the legal field (3 credits)

Lecture Hours: 48.000

Course User Fee: \$45.84

PLA2930 Special Topics in Law 1.000 - 3.000 credits

This course will explore a selection of emerging trends in the law. May be repeated for credit.

Lecture Hours: 16.000

Program Description[Rooms Division Management](#)

College Credit Certificate | Code: 65048 | 30 credit

The Rooms Division Management College Credit Certificate program is designed to prepare students with a theoretical and practical foundation for a successful career in the hotel sales and marketing industry. Students enrolled in this certificate are prepared for positions such as Front Desk Manager, and Guest Relations Manager. Credits earned can be applied to an Associate in Applied Science degree in Hospitality and Tourism Management.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 12.0
HFT 1000	Introduction to Hospitality	3.0
HFT 1210	Human Resources	3.0
HFT 1631	Risk Management and Security	3.0
HFT 2421	Managerial Accounting for Hospitality	3.0
Semester 2		Units: 9.0
HFT 2449	E-Business For The Hospitality Industry	3.0
<i>Elect</i>		6.0
HFT 1220	Supervisory Development	3.0
OR		
HFT 1300	Executive Housekeeping	3.0
OR		
HFT 2223	Training Skills And Development	3.0
OR		
HFT 2241	Leadership and Quality Assurance Management	3.0
OR		
HFT 2410	Front Office Procedures and Lodging Operations	4.0
OR		
HFT 2501	Hospitality Sales	3.0
OR		
HFT 2750	Event and Meeting Management	3.0
OR		
HFT 2949	Co-Op Work-Study Internships 2	3.0
Semester 3		Units: 9.0
Elective*		9.0

*Select 9 credits from the following course options: HFT 1220, HFT 1300, HFT 2223, HFT 2241, HFT 2410, HFT 2501, HFT 2750, HFT 2949

Total: 30.0

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1210 Human Resources 3.000 credits

Provides information relating to the recruitment and selection of new staff, the handling of difficult employees, motivating employees and conducting performance evaluations. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1631 Risk Management and Security 3.000 credits

Provides the opportunity to examine issues surrounding the need for individualized security and surveillance programs, risk management and review systems. The student will examine a wide variety of security and safety equipment procedures and discuss guest protection, internal security for asset protection, and OSHA regulations that apply to lodging properties.

Lecture Hours: 48.000

HFT2421 Managerial Accounting for Hospitality 3.000 credits

Presents managerial accounting concepts and explains how they apply to specific operations within the hospitality industry.

Lecture Hours: 48.000

HFT2449 E-Business For The Hospitality Industry 3.000 credits

Prepares student to manage information systems within their organizations. Emphasis is on selecting the right computer systems technology and maximizing available technology in order to promote and sell services. Introduces the use of automation in the broad hospitality sector and exams technological applications ranging from distribution systems (GDS, CRS and Web based), Property Management Systems, and EPOS systems to developments in telecommunications, and assesses their effect on the tourism sector. While a broad interpretation of both technology and tourism will be used, particular emphasis will be placed on the hospitality sector (i.e. hotels and catering) and on distribution technology. A combination of lecture, case studies, seminars, visiting lecture and lab sessions are used.

Lecture Hours: 48.000

Prerequisite: HFT1000

HFT1220 Supervisory Development 3.000 credits

This introductory course will assist students in learning basic supervisory skills. The students will learn how to conduct proper employee evaluations, as well as how to apply discipline. Effective communication techniques, use of delegation and diversity in the workforce are discussed. Issues concerning employee relations and current topics in management are addressed.

Lecture Hours: 48.000

HFT1300 Executive Housekeeping 3.000 credits

In this introductory course, the student will learn the fundamentals of housekeeping management. This course describes the management functions, tools and practices required in today's lodging and institutional housekeeping departments. Special attention to environmental and safety implementations. Design and architectural elements and their relation to housekeeping will be discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT2223 Training Skills And Development 3.000 credits

In this intermediary course, the student will learn effective training processes. The course considers the assessment and analysis of training needs, the systematic design of instruction, the evaluation of training programs, and management of the training programs, and functions. The student will have the necessary skills in order to develop and conduct training sessions.

Lecture Hours: 48.000

HFT2241 Leadership and Quality Assurance Management 3.000 credits

Provides an analysis of management issues related to the "personal touch" in customer service and quality assurance in the hospitality industry. Emphasis is placed on the importance of contemporary management and business practices to keep up with the demands of an ever-changing industry.

Lecture Hours: 48.000

HFT2410 Front Office Procedures and Lodging Operations 4.000 credits

In this intermediary course, the student will learn about various positions in the rooms division. Focus on hotel front office procedures involved in reservations, registering and checking out guests. Guest services, revenue management, accounting procedures and the operation of a PMS (Property Management System) are discussed. Special attention to hotel operations including security, housekeeping and engineering.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$15.00

HFT2501 Hospitality Sales 3.000 credits

In this intermediary course, the student will learn sales presentation skills and the importance of sales in an organization. Topics discussed are the organization of the sales department, including the responsibilities of all involved in the sales process. Emphasis on sales presentations, looking for leads, negotiating skills, building rapport, and closing the sale. Role-playing and other activities will be used to enhance learning.

Lecture Hours: 48.000

HFT2750 Event and Meeting Management 3.000 credits

In this intermediary course, the student will learn the complete set of skills necessary to adequately perform as an event planner, hotel banquet manager and convention and meeting planner. Actual events will be used to reinforce the general rules of table service, booking functions, staffing banquets/conventions, and responsibilities of a host venue as they apply to buffets and banquets. Prepares students in trade show administration, meeting management, and legal issues associated with banquets and conventions.

Lecture Hours: 48.000

HFT2949 Co-Op Work-Study Internships 2 3.000 credits

Provides an advanced supervised cooperative education experience in the hospitality industry that builds on foundational workplace skills and knowledge. Students apply hospitality management principles in professional settings while enhancing competencies in operations, customer service, leadership, communication, and problem-solving. Emphasis is placed on increased responsibility, professional growth, and career development through structured experiential learning in approved hospitality organizations.

Other Hours: 48.000

Course User Fee: \$25.00

Program Description

Rooms Division Operations

College Credit Certificate | Code: 65056 | 19 credits

The Rooms Divisions Operations College Credit Certificate is designed to prepare students with a theoretical and practical foundation for a successful career in the lodging industry. Students enrolled in this certificate are prepared for positions such as Front Desk Supervisor or Guest Relations Supervisor. Credits earned can be applied to an Associate in Science degree in Hospitality Management, which is fully transferable to public universities within the state of Florida.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 9.0
HFT 1000	Introduction to Hospitality	3.0
HFT 1212	Safety and Sanitation	3.0
HFT 1300	Executive Housekeeping	3.0
Semester		Units: 10.0
HFT 2410	Front Office Procedures and Lodging Operations	4.0
HFT 2449	E-Business For The Hospitality Industry	3.0
<i>Elective*</i>		3.0
HFT 1631	Risk Management and Security	3.0
OR		
HFT 1949	Co-Op Work-Study Internships I	3.0
OR		
HFT 2241	Leadership and Quality Assurance Management	3.0
OR		
HFT 2750	Event and Meeting Management	3.0

*Departmental Recommendation: HFT 2241

Total: 19.0

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1212 Safety and Sanitation 3.000 credits

Students will learn industry standards on sanitation as it relates to food preparation and kitchen operations. Students will recognize proper sanitation techniques and explain how to implement sanitation programs in food service operations.

Lecture Hours: 48.000

Course User Fee: \$125.00

HFT1300 Executive Housekeeping 3.000 credits

In this introductory course, the student will learn the fundamentals of housekeeping management. This course describes the management functions, tools and practices required in today's lodging and institutional housekeeping departments. Special attention to environmental and safety implementations. Design and architectural elements and their relation to housekeeping will be discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT2410 Front Office Procedures and Lodging Operations 4.000 credits

In this intermediary course, the student will learn about various positions in the rooms division. Focus on hotel front office procedures involved in reservations, registering and checking out guests. Guest services, revenue management, accounting procedures and the operation of a PMS (Property Management System) are discussed. Special attention to hotel operations including security, housekeeping and engineering.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$15.00

HFT2449 E-Business For The Hospitality Industry 3.000 credits

Prepares student to manage information systems within their organizations. Emphasis is on selecting the right computer systems technology and maximizing available technology in order to promote and sell services. Introduces the use of automation in the broad hospitality sector and exams technological applications ranging from distribution systems (GDS, CRS and Web based), Property Management Systems, and EPOS systems to developments in telecommunications, and assesses their effect on the tourism sector. While a broad interpretation of both technology and tourism will be used, particular emphasis will be placed on the hospitality sector (i.e. hotels and catering) and on distribution technology. A combination of lecture, case studies, seminars, visiting lecture and lab sessions are used.

Lecture Hours: 48.000

Prerequisite: HFT1000

HFT1631 Risk Management and Security 3.000 credits

Provides the opportunity to examine issues surrounding the need for individualized security and surveillance programs, risk management and review systems. The student will examine a wide variety of security and safety equipment procedures and discuss guest protection, internal security for asset protection, and OSHA regulations that apply to lodging properties.

Lecture Hours: 48.000

HFT1949 Co-Op Work-Study Internships I 3.000 credits

Provides a supervised cooperative education experience in the hospitality industry that combines academic learning with practical workplace training. Students gain hands-on experience in hospitality operations, customer service, professional communication, and industry best practices while working in approved hospitality settings. Emphasis is placed on skill development, workplace professionalism, and career readiness through structured experiential learning.

Lecture Hours: 48.000

Course User Fee: \$25.00

HFT2241 Leadership and Quality Assurance Management 3.000 credits

Provides an analysis of management issues related to the "personal touch" in customer service and quality assurance in the hospitality industry. Emphasis is placed on the importance of contemporary management and business practices to keep up with the demands of an ever-changing industry.

Lecture Hours: 48.000

HFT2750 Event and Meeting Management 3.000 credits

In this intermediary course, the student will learn the complete set of skills necessary to adequately perform as an event planner, hotel banquet manager and convention and meeting planner. Actual events will be used to reinforce the general rules of table service, booking functions, staffing banquets/conventions, and responsibilities of a host venue as they apply to buffets and banquets. Prepares students in trade show administration, meeting management, and legal issues associated with banquets and conventions.

Lecture Hours: 48.000

Program Description

Rooms Division Specialist

College Credit Certificate | Code: 65055 | 13 credit

The Rooms Divisions Specialist College Credit Certificate is designed to prepare students with a theoretical and practical foundation for a successful career in the lodging industry. Students enrolled in this certificate are prepared for positions such as Front Desk Agent, Guest Relations Agent, or Reservation Clerk. Credits earned can be applied to an Associate in Science degree in Hospitality Management, which is fully transferable to public universities within the state of Florida.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 13.0
HFT 1000	Introduction to Hospitality	3.0
HFT 1300	Executive Housekeeping	3.0
HFT 2410	Front Office Procedures and Lodging Operations	4.0
<i>Elective*</i>		3.0
HFT 1631	Risk Management and Security	3.0
OR		
HFT 1949	Co-Op Work-Study Internships I	3.0
OR		
HFT 2241	Leadership and Quality Assurance Management	3.0

*Departmental Recommendation: HFT 2241

Total: 13.0

HFT1000 Introduction to Hospitality 3.000 credits

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1300 Executive Housekeeping 3.000 credits

In this introductory course, the student will learn the fundamentals of housekeeping management. This course describes the management functions, tools and practices required in today's lodging and institutional housekeeping departments. Special attention to environmental and safety implementations. Design and architectural elements and their relation to housekeeping will be discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT2410 Front Office Procedures and Lodging Operations 4.000 credits

In this intermediary course, the student will learn about various positions in the rooms division. Focus on hotel front office procedures involved in reservations, registering and checking out guests. Guest services, revenue management, accounting procedures and the operation of a PMS (Property Management System) are discussed. Special attention to hotel operations including security, housekeeping and engineering.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$15.00

HFT1631 Risk Management and Security 3.000 credits

Provides the opportunity to examine issues surrounding the need for individualized security and surveillance programs, risk management and review systems. The student will examine a wide variety of security and safety equipment procedures and discuss guest protection, internal security for asset protection, and OSHA regulations that apply to lodging properties.

Lecture Hours: 48.000

HFT1949 Co-Op Work-Study Internships I 3.000 credits

Provides a supervised cooperative education experience in the hospitality industry that combines academic learning with practical workplace training. Students gain hands-on experience in hospitality operations, customer service, professional communication, and industry best practices while working in approved hospitality settings. Emphasis is placed on skill development, workplace professionalism, and career readiness through structured experiential learning.

Lecture Hours: 48.000

Course User Fee: \$25.00

HFT2241 Leadership and Quality Assurance Management 3.000 credits

Provides an analysis of management issues related to the "personal touch" in customer service and quality assurance in the hospitality industry. Emphasis is placed on the importance of contemporary management and business practices to keep up with the demands of an ever-changing industry.

Lecture Hours: 48.000

Program Description

Sales and Customer Management

College Credit Certificate | Code: 65107 | 12 credit

The College Credit Certificate in Sales and Customer Management is designed to enhance knowledge and practical skills for a successful career in the sales profession. Students will be able to use customer relationship management (CRM) and selling techniques for reaching target markets to sell a product or service to sustain profitability and competitiveness.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

MAR 1440	Fundamentals of Negotiations	3.0
MAR 1502	Sales and Consumer Behavior	3.0

MKA 1022	Relationship Selling	3.0
MKA 1160	Customer Relationship Management	3.0

Total: 12.0

MAR1440 Fundamentals of Negotiations 3.000 credits

This course introduces the negotiation process and identifying consumer motivations. Students will analyze the various positions of negotiations from gaining leverage to making considerations. Students will examine a long-term sales strategy and how a balanced negotiation plan impacts it. (3 hr. lecture)
Lecture Hours: 48.000, Lab Hours: 5.000

Course User Fee: \$47.44

MAR1502 Sales and Consumer Behavior 3.000 credits

This course introduces the process consumers use in making purchase decisions. Students will analyze how developing a sales personality factors into producing prospects, leads, and ultimately generating sales. Students will identify how both buyer and seller behaviors interact in the sales process. (3 hr. lecture)
Lecture Hours: 48.000

Course User Fee: \$47.44

MKA1022 Relationship Selling 3.000 credits

This course introduces the role that relationship building plays in sales. Students will explore the communication process and the importance of interpersonal skills in developing a relationship for an effective sales strategy. Students will develop a sales presentation based around consumer needs.

Lecture Hours: 48.000

Course User Fee: \$47.44

MKA1160 Customer Relationship Management 3.000 credits

This course provides an introduction into Customer Relationship Management (CRM). Students will be exposed to concepts ranging from identifying customer needs, communication channels, customer acquisition cost, demographics and target marketing. Students will learn the role CRM plays in a sales strategy. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$47.44

Program Description

[Solar Energy Systems Specialist](#)

College Credit Certificate | Code: 66056 | 18 credit

The College Credit Certificate (CCC) in Solar Energy Systems Specialist will fill a growing need to train personnel in how to build and install Solar Photo-Voltaic panels in both commercial and private structures. Students will receive a wide-range of classroom and hands-on training that will provide the student with the education and knowledge of the following grid connect solar power systems, off grid solar-remote power systems, commercial and medium scale solar, solar hot water, solar hydronics/thermal, and solar water pumping.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 7.0
EET 1033C	Electrical Fundamentals	4.0
ETI 1701	Industrial Safety	3.0

Semester 2		Units: 11.0
ETP 2040C	Electric Power Distribution	4.0
ETP 2410C	Design, Installation and Operation of Solar PV Systems	4.0
ETP 2501C	Introduction to Alternative and Renewable Energy	3.0

Total: 18.0

EET1033C Electrical Fundamentals 4.000 credits

This course is designed for students obtaining a CCC or AS degree in Engineering Technology, and related disciplines. The student will learn the basic concepts of electronics principles including Direct Current, Alternating Current, Series and Parallel circuits topologies, Basic electronics components, electronics measure tools and software simulation tools.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$103.00

ETI1701 Industrial Safety 3.000 credits

This course provides the student with the knowledge and skills to recognize hazardous situations in industrial plants and the precautions to be observed and practiced to perform work activities safely.

Among the topics covered are industrial safety hazards, electrical safety, working with chemicals, gases, and solvents, protective equipment, and safe working conditions.

Lecture Hours: 48.000

ETP2040C Electric Power Distribution 4.000 credits

This is a required course for all students pursuing a certificate in solar photovoltaic energy generation. The student will acquire an understanding of multi-phase power transmission, how to connect to the electric grid and the major components used in electric power distribution including: power transformers, circuit breakers, transmission lines, reclosures, relay coordination, fuses, motors and generators. The student will learn about the maintenance, troubleshooting and protection of these devices against lightning and other abnormal conditions through hands-on laboratory experiments utilizing modern testing and simulation equipment. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1033C

Course User Fee: \$103.00

ETP2410C Design, Installation and Operation of Solar PV Systems 4.000 credits

This is a required course for all students pursuing a certificate in solar photovoltaic energy generation. Students will learn about solar photovoltaic (PV) systems and the different components comprising the system including: PV panels, controllers and the batteries used with PV systems. Students will learn to size, install, maintain, troubleshoot and repair the PV system through hands-on laboratory experiments utilizing modern testing and simulation equipment.

Prerequisite: EET1033C.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1033C

Course User Fee: \$103.00

ETP2501C Introduction to Alternative and Renewable Energy 3.000 credits

This course is an introductory course to the concepts of renewable energy. It is designed to prepare students to enter the field of renewable energy and green technology. This course focuses on Solar photo-voltaics, solar power and tracking systems, charge controllers and inverters, Wind power systems, Biomass and geothermal power generation.

Prerequisite: EET1033C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: EET1033C

Course User Fee: \$103.00

Program Description

Sport Management Operations

College Credit Certificate | Code: 65120 | 18 credi

The College Credit Certificate in Sport Management Operations is designed to provide instruction to individuals in the areas of sports management, competitive gaming, collect and distribution of data, directing and controlling of a business, with emphasis on selected theories of management and decision-making and the knowledge and understanding necessary for managing people and functions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 9.0

MAN 2021	Principles of Management	3.0
SPM 1000	Introduction to Sports Management	3.0
SPM 2151	The Business of Sports	3.0

Semester 2

Units: 9.0

GEB 1000	Business Career Strategies	3.0
GEB 1949	General Business Internship 1	3.0
SPM 2105	Sports Events and Facility Management	3.0

Total: 18.0

MAN2021 Principles of Management 3.000 credits

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

SPM1000 Introduction to Sports Management 3.000 credits

This course will provide students with the fundamental knowledge of the field of sports management. The course will provide an overview of sports across different levels, key events within the sports industry, the stakeholders involved, and jobs within sports. Students will learn skills and competencies that will lend towards a career in sports management.

Lecture Hours: 48.000

Course User Fee: \$64.65

SPM2151 The Business of Sports 3.000 credits

This course is designed to provide an overview of the financial, management, and business components of sports. Students will learn how concepts such as ticket pricing, sales, marketing, media, and sponsorship affect amateur and professional sporting events. By examining past events in sports, best business practices will be identified and discussed.

Lecture Hours: 48.000

Course User Fee: \$64.67

GEB1000 Business Career Strategies 3.000 credits

This course is designed to enable students to thrive in a competitive business environment. Students will learn business etiquette, alternative career pathways, personal financial management, and budgets. Additionally, students will learn interviewing, networking, and career development skills.

Lecture Hours: 48.000

Course User Fee: \$24.00

GEB1949 General Business Internship 1 3.000 credits

This internship provides students with an opportunity to gain business experience while receiving academic credit. Students will learn to make connections between their internship experiences, academic coursework, and career goals. Students are required to complete 144 hours of internship in an organizational setting.

Other Hours: 144.000

SPM2105 Sports Events and Facility Management 3.000 credits

This course examines the different kinds of sporting events and the facilities in which they are hosted. Through the examination of events and facilities, students will learn the basics of event and facility management and apply them in a sports-specific environment. The course will prepare students for event planning, facility and event operations, event management, and other essential competencies for career readiness.

Lecture Hours: 48.000

Course User Fee: \$64.67

Program Description

[Tax Specialist](#)

College Credit Certificate | Code: 65076 | 12 credit

The College Credit Certificate in Tax Specialist will prepare students to fill out the necessary forms for their clients or for the businesses in which they work. In addition, students will learn how to reduce taxable income and discover the current income tax regulations and their impact on individuals, couples, families, and business owners. In addition, students will gain a working knowledge about business income, tax credits, itemized deductions, LLCs and S Corps, retirement plans, and home businesses.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester1

Units: 12.0

TAX 2000	Income Tax	3.0
TAX 2002	Taxation Practices and Procedures	3.0
TAX 2021	Taxation of Business Organizations	3.0
TAX 2401	Tax of Estates, Gifts and Trusts	3.0

Total: 12.0

TAX2000 Income Tax 3.000 credits

Federal income tax fundamentals with emphasis on individual returns. Topics considered include gross income, capital gains and losses, deductions and exemptions, and tax credits.

Lecture Hours: 48.000

Course User Fee: \$15.00

TAX2002 Taxation Practices and Procedures 3.000 credits

This course will cover Internal Revenue Service taxation practices and procedures. Topics covered will convey knowledge of IRS rules and penalties, rules for representing taxpayers before the IRS and in the courts, rules and requirements associated with the tax return filing process and records maintenance rules and basic tax research skills.

Lecture Hours: 48.000

Course User Fee: \$22.00

TAX2021 Taxation of Business Organizations 3.000 credits

This course will cover federal income taxation of corporations, S Corporations, limited liability companies and partnerships. Topics covered include determining the tax consequences of income, expenses, distributions, redemptions and liquidations for business entities. Practical application of the tax law will be emphasized along with analysis of tax procedures.

Lecture Hours: 48.000

Course User Fee: \$22.00

TAX2401 Tax of Estates, Gifts and Trusts 3.000 credits

This course covers definitions and operations of various fiduciary forms of wealth transfer including but not limited to fiduciary accounting principles and concepts; record keeping requirements; and various tax reporting requirements, forms and calculations.

Lecture Hours: 48.000

Course User Fee: \$22.00

Program Description

Television Studio Production

College Credit Certificate | Code: 61003 | 12 credit

The Television Studio College Credit Certificate (CCC) is designed for students who intend to seek employment in radio, television and production companies, as well as allied fields such as in-house educational and industrial studios. The curriculum stresses hands-on equipment use in TV laboratories. Students will have access to high-end cameras, editing suites and video graphics animation facilities and will complete portfolio-quality productions.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

RTV 1241C	Television Production 1	4.0
RTV 1242C	Television Production 2	4.0
RTV 2245C	Electronic Field Production 1	4.0

Total: 12.0

RTV1241C Television Production 1 4.000 credits

An introductory overview of the practices and procedures used in a television studio. Students will learn basic operation of studio and control room equipment and work towards the completion of broadcast standard projects.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

RTV1242C Television Production 2 4.000 credits

An intermediate course in television studio production. Students will learn to implement the fundamentals as introduced in Television Production 1 to the production of scripted programs.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: RTV1241C

Course User Fee: \$25.00

RTV2245C Electronic Field Production 1 4.000 credits

An introductory course in which students will learn single-camera field production and electronic news-gathering.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: RTV1242C

Course User Fee: \$40.00

Program Description

Translation and Interpretation

College Credit Certificate | Code: 64052 | 18 credit

This College Credit Certificate (CCC) focuses on the basic preparation needed for a career in Translation and Interpretation. Students learn how language functions and apply this knowledge to the development of Sight and Consecutive Interpretation skills. The development of basic translation competencies is also covered with emphasis on the use of Computer Assisted Translation software.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 12.0

LIN 2011	Introduction to Linguistics	3.0
FOT 2802	Introduction to Translation	3.0
FOT 2821	Introduction to Interpretation	3.0
SPT 2842	Contrastive Analysis Spanish/English	3.0

Semester 2

Units: 6.0

FOT 2823	Consecutive Interpretation	3.0
FOT 2825	Computer Assisted Translation 1	3.0

Total: 18.0

LIN2011 Introduction to Linguistics 3.000 credits

The course provides an exploration of basic linguistic concepts including phonology, morphology, syntax, pragmatics and semantics, how a language is organized and functions, and will establish a connection between Linguistics and Translation Theory. The course will also introduce students to the study of how cultural and social factors affect human communication.

Lecture Hours: 48.000

FOT2802 Introduction to Translation 3.000 credits

Develops the ability to do accurate written translations in general. Includes the application of contrastive structures and grammar rules of source and target languages; translation of idiomatic expressions and an introduction to legal and technical vocabulary; the use of bilingual dictionaries and glossaries. The demands of translation as a profession and its code of ethics are stressed.

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2821 Introduction to Interpretation 3.000 credits

The acquisition and development of skills to sight translate a text from the source to the target language. Training in the conceptualization and analysis of an oral message to transfer its content to another language consecutively. Preparatory exercises for the development of simultaneous interpretation skills.

Lecture Hours: 48.000

Course User Fee: \$15.00

SPT2842 Contrastive Analysis Spanish/English 3.000 credits

Comparison/Contrastive study of the phonology, morphology and syntax of Spanish and English. Recommended for students of translation and interpretation.

Lecture Hours: 48.000

FOT2823 Consecutive Interpretation 3.000 credits

This course builds on the foundation established in Introduction to Interpretation (FOT2810) and acquaints the students with the practice and application of consecutive interpretation (English/Spanish). Development of active listening, concentration and retention skills as well as the ability to perceive essential meaning for subsequent recall is emphasized. This course also explores basic note taking techniques and provides practice in monolateral and bilateral consecutive interpretation.

Lecture Hours: 48.000

Prerequisite: FOT2821

Course User Fee: \$15.00

FOT2825 Computer Assisted Translation 1 3.000 credits

Examines the types of translation software currently used in the translation/interpretation profession as well as the commercial use and business application of these. Description and application of tools such as translation memory, electronic dictionaries, desktop-publishing systems, and website translation technologies are covered.

Lecture Hours: 48.000

Prerequisite: FOT2802

Course User Fee: \$25.00

Program Description

UAV/Aviation Operations Management

College Credit Certificate | Code: 63012 | 16 credit

The UAV and Aviation Operations Management College Credit Certificate program will provide students with comprehensive management skills in both traditional aviation and unmanned aerial systems in an accelerated time frame. These include areas such as UAV operations and regulations, aviation industry knowledge, management skills, marketing, law, and flight operations. Students will learn how to take industry concepts and apply them both individually and as a team across manned and unmanned aviation sectors. They will be able to gain insight into the actual issues involved in managing aviation operations at both the micro and macro levels, from a leadership perspective. Students will also acquire up-to-date knowledge about emerging UAV technologies, aviation systems, regulatory compliance (including FAA Part 107), and the latest management concepts with practical application of theories to real-world aviation and UAS scenarios.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 16.0

Major Course Requirements		16.0
Group A: Aviation Management		16.0
ASC 2320	Aviation Laws and Regulations	3.0
AVM 1010	Aviation Industry Operation	3.0
AVM 1022	Flight Operations	3.0
AVM 1062	Aviation Career Planning	1.0
AVM 2510	Airline Management	3.0
AVM 2515	Airline Marketing	3.0
OR		
Group B: Unmanned Aerial Vehicles (UAV) Operations		16.0
ASC 1210	Aviation Meteorology	3.0
ASC 1560	Introduction to Unmanned Aerial Vehicles and Systems (UAVs)	3.0
ASC 1563	UAS Applications in Aerial Photography	3.0
ATT 1100	Private Pilot Theory	3.0
ATT 2995	Occupational Practicum	4.0

Total: 16.0

ASC2320 Aviation Laws and Regulations 3.000 credits

Insight pertinent to federal governing bodies, and current local, federal and international laws forming the present structure of aviation law.

Lecture Hours: 48.000

AVM1010 Aviation Industry Operation 3.000 credits

The course provides insight into the development and present status of aircraft and air transportation, governmental organizations, controls and regulations, and career opportunities in the field.

Lecture Hours: 48.000

AVM1022 Flight Operations 3.000 credits

An investigation of the occupational duties, responsibilities, and physical facilities required by the positions of pilot, co-pilot, flight engineer, dispatcher and flight attendant.

Lecture Hours: 48.000

AVM1062 Aviation Career Planning 1.000 credits

This course provides direction and guidance in career planning for all aviation students. Topics of discussion will include the job search education and training requirements, resume writing, business etiquette, interview skills and follow-up techniques

Lecture Hours: 16.000

AVM2510 Airline Management 3.000 credits

This course offers an insight relative to the business policies and the functions of management in airline operations. The course involves various internal managerial facets and the impact of external regulatory and economic implications.

Lecture Hours: 48.000

AVM2515 Airline Marketing 3.000 credits

A differentiation of the functions of marketing in airline operations; market research, demand analysis, advertising and promotion, sales, traffic, and the theory of price determination.

Lecture Hours: 48.000

ASC1210 Aviation Meteorology 3.000 credits

This is a core aviation course. The student will be prepared to understand weather and environmental issues in commercial aviation. Topics covered will be atmospheric phenomena relating to aircraft operations, the analysis and use of weather data as presented by the U.S. National Weather Service.

Prerequisite: ATT1100 or equivalent; Corequisite: ATT2110 or equivalent.

Lecture Hours: 48.000

ASC1560 Introduction to Unmanned Aerial Vehicles and Systems (UAVs) 3.000 credits

This course introduces the fundamentals of unmanned aerial vehicles and systems, covering UAV types, components, principles of flight, regulations, and safe operational practices. Students will explore industry applications and gain foundational knowledge for further study in UAV technology.

Lecture Hours: 48.000

Course User Fee: \$59.33

ASC1563 UAS Applications in Aerial Photography 3.000 credits

Students learn techniques and best practices for capturing aerial images using unmanned aerial systems. The course covers camera operation, flight planning for photography missions, image processing, and practical applications in various industries.

Lecture Hours: 48.000

Course User Fee: \$59.33

ATT1100 Private Pilot Theory 3.000 credits

This course introduces basic subjects pertaining to pilot knowledge including basic aircraft systems, aircraft operation and performance, aerodynamic principles, human factors, and aeronautical decision making. This Course can be instructed as Part 61 or Part 141.

Lecture Hours: 48.000

ATT2995 Occupational Practicum 4.000 credits

This course provides supervised, work-based learning experience in an aviation setting that enables students to apply classroom knowledge in a professional environment. Students gain practical experience in aviation operations, safety procedures, regulatory compliance, communication, and workplace professionalism while developing career readiness and industry-specific skills through guided experiential learning.

Lecture Hours: 64.000

Program Description

Virtual and Augmented Reality Technologies

College Credit Certificate | Code: 66060 | 19 credit

The program will teach students the fundamentals of Virtual and Augmented Reality. Students will learn basic concepts, history and tools commonly used for stereoscopic image acquisition and immersive technologies. Students will also learn the origins of Virtual Reality (VR) and its current role in the industry, its applications and opportunities and how to generate and manipulate VR imagery.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 7.0
DIG 1430	Storyboarding	3.0
DIG 1729C	Game Engines	4.0
Semester 2		Units: 4.0
DIG 1772C	Introduction to Virtual & Augmented Reality Technologies	4.0
Semester 3		Units: 8.0
DIG 2776C	Virtual Reality Platform Development	4.0
DIG 2777C	Augmented Reality Platform Development	4.0
		Total: 19.0

DIG1430 Storyboarding 3.000 credits

This course is for students majoring in Animation and Game Art. It introduces the necessary tasks in the storytelling phase of an animation project. Students will learn how to develop and design visual storyboards and how to sell their storyboard ideas. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$97.00

DIG1729C Game Engines 4.000 credits

This course is an introduction to game engines and their uses. Students will learn the basic techniques for creating interactive applications and how this techniques can be used for Virtual Reality (VR) and Augmented Reality (AR) projects. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$129.55

DIG1772C Introduction to Virtual & Augmented Reality Technologies 4.000 credits

This course introduces students to basic concepts, history and tools commonly used for Virtual Reality (AR) and Augmented Reality (AR) technologies.

Students will learn the history of VR/AR, its current role in the industry, developer tools, and physical principles. Prerequisite: DIG 1729C. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: DIG1729C

Course User Fee: \$129.55

DIG2776C Virtual Reality Platform Development 4.000 credits

Students will learn the fundamentals of Virtual Reality (VR) by gaining practical experience using state-of-the-art technology. This course mixes together knowledge from a variety of correlated topics, including computer graphics, tracking systems, and perceptual psychology.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: DIG1729C AND DIG1772C

DIG2777C Augmented Reality Platform Development 4.000 credits

This course provides a comprehensive curriculum that targets the key areas of augmented reality (AR). Students will learn how to enhance real-life objects and environments with digitally generated image overlays.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: DIG1729C AND DIG1772C

Career Technical Certificate (CTC)

Program Description

Advanced Automotive Service Technology - Tesla Technician

Career Technical Certificate | Code: 51000 | 26.66 credit

The Advanced Automotive Service Technology – Tesla Technician Career Technical Certificate prepares students to become entry-level technicians to the TESLA Motors technology. The coursework includes safety when working in or around high voltage, maintenance, regeneration braking, inverter power transfer, battery technologies, battery management systems, High-Voltage Bus & Charging, Pack Connector & Penthouse controls, and Autonomous technology. Electric vehicle applications and their integrated systems from TESLA Motors will be used to discuss physics of battery storage and material composition and High-Voltage generation systems. This program is suitable for student's already working in the battery electric vehicle/Hybrid vehicle and energy technology field. This program is a requirement for TESLA Motors Technician. Technicians successfully completing this training can enter gainful employment as a Level I, Level II, or Level III technician, Mobile Service Technician or a Field Service Technician.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 26.7

AER 0605	Tesla Electronic System Technician	13.3
AER 0606	Tesla Maintenance Technician	13.3

Total: 26.7

AER0605 Tesla Electronic System Technician 13.330 credits

This course prepares the student to apply technical knowledge and skills to repair, service, and maintain Tesla vehicles. At the end of the course, the student will be able to diagnose malfunctions in and repair electrical, charging, penthouse, driver assist and infotainment systems and components. Corequisite: AER0606.

Clock Hours: 400.000

AER0606 Tesla Maintenance Technician 13.330 credits

This course prepares the student to apply technical knowledge and skills to repair, service, and maintain Tesla vehicles. The student will learn Tesla's products and diagnostic tools, as well as safety equipment and procedures. At the end of the course, the student will be able to diagnose malfunctions in and repair HVAC, brake, and chassis systems and components.

Corequisite: AER0605.

Clock Hours: 400.000

Program Description

[Aircraft Structural Assembly and Fabrication Apprenticeship](#)

Career Technical Certificate | Code: 59005 | 143.20 credits

The Career Technical Certificate (CTC) in Aircraft Structural Assembly and Fabrication Apprenticeship will prepare the student to perform scheduled and non-scheduled maintenance and repairs on aircraft. The student will assist aviation maintenance technicians in commercial jet modifications, minor and major maintenance checks and corrosion control and preventions, and line services for a wide array of commercial and military aircraft.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: Yes

Semester		Units: 24.9
AMT 0044	Tools, Materials, and Processes 1	1.3
AMT 0045	Tools, Materials, and Process 2	1.3
AMT 0949	On the Job Training (OJT)	22.2
Semester 2		Units: 29.0
AMT 0046	Aircraft Materials, Hardware & Processes	2.3
AMT 0047	Applying the Design Process	1.8
AMT 0219	Aircraft Hydraulics & Aviation Mathematics	0.6
AMT 0269	Aircraft Electrical Systems & Quality Control	2.1
AMT 0949	On the Job Training (OJT)	22.2
Semester 3		Units: 22.6
AMT 0509	Composites and Capstone Project	0.4
AMT 0949	On the Job Training (OJT)	22.2

Total: 76.5

AMT0044 Tools, Materials, and Processes 1 1.330 credits

This course provides an introduction to the tools, hardware and material used in aircraft maintenance, repair, and safety. The student will be able to understand the information found in aircraft drawings, blueprints, charts and graphs Including stress materials' internal resistance and counterforce that opposes deformation.

Clock Hours: 40.000

AMT0045 Tools, Materials, and Process 2 1.330 credits

In this course, the student will learn the principles of corrosion and how to control it are studied and applied. This course provides experience in detecting, identifying, removal, and treatment of the various types of corrosion found on ferrous and non-ferrous metals. In addition, the student will learn about various forms of truss-types and how beams, struts, and bars resist deformation by applied loads.

Clock Hours: 40.000

AMT0949 On the Job Training (OJT) 22.220 credits

In this training course, the student will learn to perform the required tasks at the workplace. This course adds diversified experience and abilities while enriching the self-acquired proficiency of the student. Training is designed to minimize the margin of errors at the workplace.

Clock Hours: 666.670

AMT0046 Aircraft Materials, Hardware & Processes 2.260 credits

In this course, the student will learn about chemical processes, hazards of aviation solvents, lubricants, effects of corrosion on metals and aluminum, and measuring and sheet metal layout. Prerequisites: AMT0044 and AMT0045.

Clock Hours: 68.000

Prerequisite: AMT0044 AND AMT0045

AMT0047 Applying the Design Process 1.760 credits

In this course, the student will learn to create aircraft drawings, blueprints, charts and graphs; test equipment, including non-destructive inspection and testing; and determine best practices with respect to sealants and epoxies. Prerequisites: AMT0044 and AMT0045.

Clock Hours: 53.000

Prerequisite: AMT0044 AND AMT0045

AMT0219 Aircraft Hydraulics & Aviation Mathematics 0.630 credits

In this course, the student will learn about aircraft hydraulic and pneumatic systems and advanced aviation mathematics. Prerequisites: AMT0044 and AMT0045.

Clock Hours: 19.000

Prerequisite: AMT0044 AND AMT0045

AMT0269 Aircraft Electrical Systems & Quality Control 2.130 credits

In this course, the student will learn the principles of basic electricity as it relates to aviation electrical systems; work with data, including computer-generated flow diagrams and spreadsheets; working with wiring & fiber optics; advanced blueprint reading with respect to performing actual tasks; and key elements of quality control. Prerequisites: AMT0044 and AMT0045.

Clock Hours: 64.000

Prerequisite: AMT0044 AND AMT0045

AMT0949 On the Job Training (OJT) 22.220 credits

In this training course, the student will learn to perform the required tasks at the workplace. This course adds diversified experience and abilities while enriching the self-acquired proficiency of the student. Training is designed to minimize the margin of errors at the workplace.

Clock Hours: 666.670

AMT0509 Composites and Capstone Project 0.400 credits

In this course, the student will learn about composite materials. In addition, the student will demonstrate their competence to analyze, design, develop, and test Aircraft Structural Assembly and Fabrication techniques.

Prerequisites: AMT0044, 0045, 0046, 0047, 0219, 0269.

Clock Hours: 12.000

Prerequisite: AMT0044 AND AMT0045 AND AMT0046 AND AMT0047 AND AMT0269 AND AMT0219

AMT0949 On the Job Training (OJT) 22.220 credits

In this training course, the student will learn to perform the required tasks at the workplace. This course adds diversified experience and abilities while enriching the self-acquired proficiency of the student. Training is designed to minimize the margin of errors at the workplace.

Clock Hours: 666.670

Program Description

Central Sterile Processing Technician

Career Technical Certificate | Code: 53014 | 21.66 credits

The Career Technical Certificate in Central Sterile Processing Technology is designed to educate and prepare graduates with skills required for entry-level employment as a member of the healthcare team. In addition, students will acquire extensive knowledge of legal and ethical responsibilities, anatomy, physiology, pathophysiology, microbiology, aseptic techniques, patient care procedures, surgical technology procedures, patient safety, and use and care of equipment and supplies.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1 **Units: 8.0**

HSC 0003	Introduction to Healthcare	2.5
HSC 0003L	Introduction to Healthcare Lab	0.5
STS 0019	Central Sterile Service Material Management	5.0

Semester 2 **Units: 13.7**

STS 0013	Central Sterile Processing Technician	13.7
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Total: 21.7

HSC0003 Introduction to Healthcare 2.500 credits

An introduction to the health care environment, this course focuses on the health care team and delivery systems. Students will learn about legal responsibilities, ethical issues, safety, infection control, communication, interpersonal behaviors, wellness, and disease. Corequisite: HSC0003L

Clock Hours: 75.000

Corequisite: HSC0003L

Course User Fee: \$20.00

HSC0003L Introduction to Healthcare Lab 0.500 credits

This course focuses on the performance of basic health care skills. Students will apply body mechanics and ergonomics, standard precautions used in infection control procedures and perform and record vital signs. Corequisite: HSC0003

STS0019 Central Sterile Service Material Management 5.000 credits

This course is intended to teach the role of the central sterile processing technician in the sterile processing department and related areas will be covered. The course introduces anatomy and physiology to enable the SPD student to better understand each job function that they will perform. As well as microbiology to include the characteristics and activities of microorganisms and various significant aspects of infectious disease that occur in humans and infection control. This course will also cover professional management, communication skills and teamwork, ethical and moral issues in the healthcare setting, and legal issues and risk management. Clock Hours: 150.000

STS0013 Central Sterile Processing Technician 13.660 credits

This course focuses on the application of central sterile processing including proper care, disinfection, assembly, sterilization processes, and sterile storage for instrumentation following surgical procedures. The course will also include the review of basic anatomy, physiology, microbiology, relevant equipment, supplies, and techniques regarding different surgical procedures. Hands-on-experience in the Sterile Processing Department at local hospitals will be included in the course.

Clock Hours: 410.000

Course User Fee: \$49.50

Program Description

Commercial Class B Driving

Career Technical Certificate | Code: 56016 | 5 credits

The Career Technical Certificate in Commercial Class B Driving is designed to educate and prepare students for a Class B Commercial Driver License. The Truck Driver Heavy Florida Class B course prepares students for entry into the trucking and logistics industry. Students explore career opportunities and requirements of a professional class B truck driver. Students study vehicle safety, accident prevention, operating regulations, cargo handling, documentation procedures, pre-trip preparation, vehicle inspection, maintenance, service, control procedures, backing, maneuvering, road and hazardous driving skills, and licensing requirements.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1		Units: 5.0
TRA 0084	Truck Driver Heavy Florida Class "B"	4.6
TRA 0084L	On-the-Road Practice and Driving Skills - Class B	0.4
		Total: 5.0

TRA0084 Truck Driver Heavy Florida Class "B" 4.600 credits

The Truck Driver Heavy Florida Class "B" course prepares students for entry into the trucking and logistics industry. Students explore career opportunities and requirements of a professional class "B" truck driver. Students study vehicle safety, accident prevention, operating regulations, cargo handling, documentation procedures, pre-trip preparation, vehicle inspection, maintenance, service, control procedures, backing, maneuvering, road and hazardous driving skills, and licensing requirements. Prerequisites/Corequisite Information: Must have a valid Florida Driver's License and good driving record, must have Commercial Learner's Permit, must pass DOT Health Examination, must pass DOT Drug Test.

Clock Hours: 138.000

Pre/Corequisite: Prerequisites/Corequisite Information: • Must have a valid Florida Driver's License and good driving record • Must have a Commercial Learner's Permit • Must pass DOT Health Examination • Must pass DOT Drug Test

TRA0084L On-the-Road Practice and Driving Skills - Class B 0.400 credits

This course focuses on on-the-road practice for the Truck Driver Heavy Florida Class "B" License. The course prepares students to study vehicle safety, accident prevention, operating regulations, cargo handling, documentation procedures, pre-trip preparation, vehicle inspection, maintenance, service, control procedures, backing, maneuvering, road and hazardous driving skills, and licensing requirements. Prerequisites/Co-requisite Information: Must have a valid Florida Driver's License and good driving record, must have Commercial Learner's Permit, must pass DOT Health Examination, must pass DOT Drug Test.

Clock Hours: 12.000

Program Description

Commercial Vehicle Driving

Career Technical Certificate | Code: 56015 | 10.67 credits

Semester 1	Units: 10.7
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TRA 0080	Tractor Trailer Truck Driver	10.1
TRA 0080L	On-the-Road Practice and Driving Skills - Class A	0.5

Total: 10.7

TRA0080 Tractor Trailer Truck Driver 10.130 credits

The Tractor Trailer Truck Driver course prepares students for entry into the trucking and logistics industry. Students explore career opportunities and requirements of a professional tractor trailer driver. Students study vehicle safety, accident prevention, operating regulations, cargo handling, documentation procedures, pre-trip preparation, vehicle inspection, maintenance, service, control procedures, backing, coupling, uncoupling, maneuvering, road and hazardous driving skills, and licensing requirements. Pre/Corequisite: Must have a valid Florida Driver's License and good driving record; Must have Commercial Learner's Permit; Must pass DOT Health Examination; Must pass DOT Drug Test.

Clock Hours: 320.000

Pre/Corequisite: Must have a valid Florida Driver's License and good driving record; Must have Commercial Learner's Permit; Must pass DOT Health Examination; Must pass DOT Drug Test.

TRA0080L On-the-Road Practice and Driving Skills - Class A 0.530 credits

This course focuses on on-the-road practice, honing driving skills crucial for Tractor Trailer Truck Drivers. Students will engage in practical exercises involving backing, coupling, uncoupling, maneuvering, and road driving. Special attention will be given to hazardous driving skills, ensuring students are adept at handling various challenging situations on the road. This is a hands-on training course conducted on the road, providing practical experience for students pursuing a Class 'A' Commercial Driver's License.

Clock Hours: 16.000

Program Description

Correctional Officer - State

Career Technical Certificate | Code: 57021 | 14.83 credits

The Correctional Officer – State program prepares students for certification as Correctional Officers in the State of Florida. All criminal justice standards and training, Department of Education, and local standards will be met. Graduates are eligible for employment with any correctional agency in the state upon graduation from the program and successful completion of the State Certification Exam. Topics include human behavior, law, communications, facility operations, first aid and other related topics. There is emphasis on practical applications and competency-based performance. This program is offered at the School of Justice. Students seeking entrance into the MDC School of Justice basic recruit training programs for a career in corrections are required to pass a physical screening, physical agility, Voice Stress Analysis Test, psychological test, fingerprinting and background check and the Florida Basic Abilities Test (FBAT). For more information please contact the School of Justice, FBAT Department and/or visit the FBAT Web site, at www.mdc.edu.north.fbat.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 14.8

CJK 0301	Introduction to Corrections	1.1
CJK 0355	Legal for Correctional Officers	0.7
CJK 0306	Communications for Correctional Officers	1.1
CJK 0111	Interviewing and Report Writing in Corrections	0.5
CJK 0327	Shift Management and Safety	0.7
CJK 0321	Intake and Release	0.5
CJK 0324	Supervision in a Correctional Facility	1.1
CJK 0326	Supervising Correctional Populations	0.8

CJK 0336	Incidents and Emergencies in Correctional Facilities	0.7
CJK 0031	First Aid for Criminal Justice Officers	1.3
CJK 0040	Criminal Justice Firearms	2.7
CJK 0051	Criminal Justice Defensive Tactics	2.7
CJK 0340	Criminal Justice Officer Physical Fitness Training	1.0

Total: 14.8

CJK0301 Introduction to Corrections 1.066 credits

The student will learn the responsibilities of a correctional officer to provide safety for him/herself, public, staff, and inmates. The student will learn an overview on safety and security concerns, identification, manipulation and deception, contraband, and searches in a correctional setting. This course is for School of Justice students only.

CJK0355 Legal for Correctional Officers 0.730 credits

Students will study the legal framework governing correctional operations, including constitutional rights of inmates, statutory requirements, liability issues, and court decisions that impact daily correctional practices. The course covers search and seizure procedures, disciplinary processes, and legal standards for institutional operations.

Clock Hours: 22.000

CJK0306 Communications for Correctional Officers 1.070 credits

This course develops essential communication skills needed in correctional environments. Students learn effective verbal and non-verbal communication techniques, conflict resolution strategies, professional documentation methods, and appropriate interaction protocols with inmates, colleagues, and external agencies.

Clock Hours: 32.000

CJK0111 Interviewing and Report Writing in Corrections 0.530 credits

Students will master the fundamentals of conducting interviews and preparing accurate, detailed reports in correctional settings. The course covers investigative interviewing techniques, documentation standards, evidence collection, and the preparation of incident reports, disciplinary reports, and other official documentation.

Clock Hours: 16.000

CJK0327 Shift Management and Safety 0.670 credits

This course focuses on the principles of managing correctional facility operations during assigned shifts. Students learn security procedures, safety protocols, routine operational checks, emergency preparedness, and effective coordination with other shifts and departments to maintain institutional security and order.

Clock Hours: 20.000

CJK0321 Intake and Release 0.530 credits

Students will study the processes and procedures involved in receiving new inmates and processing releases. The course covers booking procedures, classification systems, orientation processes, property management, record-keeping requirements, and coordination with courts and other agencies during intake and release operations.

Clock Hours: 16.000

CJK0324 Supervision in a Correctional Facility 1.070 credits

This course examines the principles and practices of supervising inmates within institutional settings. Students learn observation techniques, behavioral management strategies, rule enforcement, recreational supervision, work program oversight, and methods for maintaining order while promoting positive outcomes.

Clock Hours: 32.000

CJK0326 Supervising Correctional Populations 0.830 credits

Students will explore advanced supervision techniques for managing diverse correctional populations. The course addresses special population needs, risk assessment methods, case management principles, program coordination, and strategies for working with individuals who have varying security levels and behavioral challenges.

Clock Hours: 25.000

CJK0336 Incidents and Emergencies in Correctional Facilities 0.670 credits

This course prepares students to respond effectively to various incidents and emergency situations within correctional facilities. Students learn emergency response protocols, crisis management techniques, evacuation procedures, medical emergencies, disturbance control, and coordination with emergency services and administration.

Clock Hours: 20.000

CJK0031 First Aid for Criminal Justice Officers 1.330 credits

Students will learn to initiate treatment for a variety of medical emergencies, understand and perform CPR, and know when to activate EMS and perform basic life support until help arrives. CPR and First Responder certification cards are issued upon successful completion. This course prepares criminal justice recruits for a variety of medical emergencies with minimal medical supplies. Basic training for School of Justice students only.

Course User Fee: \$237.00

CJK0040 Criminal Justice Firearms 2.670 credits

Students will learn how to use both handguns and shotguns. Students must qualify with both weapons under both daylight and night conditions. Students must also demonstrate ability for both accuracy and decision-making. Students are also introduced to chemical weapons and their effects.

Course User Fee: \$1574.00

CJK0051 Criminal Justice Defensive Tactics 2.670 credits

Students will learn how to physically defend themselves, physically control persons under arrest, and know what level of force is appropriate under differing circumstances. Students will learn escalate, de-escalate, or disengage techniques and when to use them. Additionally, a physical conditioning program is part of this course. For School of Justice students only.

Clock Hours: 80.000

Course User Fee: \$335.00

CJK0340 Criminal Justice Officer Physical Fitness Training 1.000 credits

Students will learn the benefits of maintaining physical fitness to include nutrition and diet. Students will learn of the effects of stress and how to deal with it; how to build up muscular and cardiovascular endurance and perform the exercises as required. For School of Justice students only. Special fee.

Clock Hours: 30.000

Course User Fee: \$57.00

Program Description

Dental Assisting Technology and Management

Career Technical Certificate | Code: 53015 | 41 credits

Semester 1		Units: 19.0
DEA 0020	Pre-Clinical Procedures Theory	3.0
DEA 0020L	Pre-Clinical Practice Lab	3.0
DEA 0031	Oral Pathology	1.0
DEA 0132	Dental Nutrition	1.0
DES 0103	Dental Materials	2.0
DES 0103L	Dental Materials Lab	3.0
DES 0021	Dental Anatomy	3.0
HSC 0003	Introduction to Healthcare	2.5
HSC 0003L	Introduction to Healthcare Lab	0.5
Semester 2		Units: 11.0

DES 0205	Dental Radiology	2.0
DES 0205L	Dental Radiology Lab	2.0
DES 0206	Dental Radiology II	1.0
DES 0206L	Dental Radiology II Lab	1.0
DES 0501	Dental Practice Management	1.0
DES 0602	Pharmacology / Dental Office Emergencies	1.0
DES 0830	Expanded Functions Theory	1.0
DES 0830L	Expanded Functions Lab	1.0
DEA 0930	Dental Assisting Seminar	1.0

Semester 3

Units: 11.0

DEA 0800L	Clinical Practice 1	5.0
DEA 0804L	Clinical Practice 2	6.0

Total: 41.0

DEA0020 Pre-Clinical Procedures Theory 3.000 credits

This course is an introduction to dentistry and the dental assisting profession including duties and responsibilities, orientation to the dental office, basic concepts of microbiology, introduction to the basic skills in dental assisting including, but not limited to infection control procedures, instrument identification and four-handed dentistry.

Clock Hours: 90.000

DEA0020L Pre-Clinical Practice Lab 3.000 credits

This course introduces students to the basic clinical dental assisting skills as theorized in DEA 0020.

Concentration of study will include areas such as reception and dismissal of patients, patient management, establishing and maintaining records, obtaining, and recording medical/dental histories and vital signs, charting, planning appointments, assisting with, or performing various dental procedures, and utilizing practice management system.

Clock Hours: 90.000

Course User Fee: \$76.53

DEA0031 Oral Pathology 1.000 credits

This course introduces principles of general pathology with an emphasis on conditions related to the oral cavity. There is a concentration on recognition of normal and abnormal conditions of the oral cavity and surrounding tissues.

Clock Hours: 30.000

DEA0132 Dental Nutrition 1.000 credits

This course is designed to integrate nutrition into the diagnosis, care, and treatment of dental patients demonstrating the relationship between diseases, diets, and oral health. An emphasis on dietary assessment methods in relation to dental health is explored.

Clock Hours: 30.000

DES0103 Dental Materials 2.000 credits

This course provides a study of the properties, manipulation, and care of materials used in the prevention and treatment of oral disease. It will include the study of physical, mechanical, chemical, and biological characteristics of materials.

Clock Hours: 60.000

DES0103L Dental Materials Lab 3.000 credits

This course provides a study of the properties, manipulation, and care of materials used in the prevention and treatment of oral disease. It will include the study of physical, mechanical, chemical, and biological characteristics of materials.

Clock Hours: 90.000

Course User Fee: \$50.53

DES0021 Dental Anatomy 3.000 credits

This course provides a study of the anatomy of the human dentition, its development and function. A detailed study of bony, muscular, circulatory, and nervous systems of the head and neck is included. An introduction to oral histology will also be offered.

Clock Hours: 90.000

HSC0003 Introduction to Healthcare 2.500 credits

An introduction to the health care environment, this course focuses on the health care team and delivery systems. Students will learn about legal responsibilities, ethical issues, safety, infection control, communication, interpersonal behaviors, wellness, and disease. Corequisite: HSC0003L

Clock Hours: 75.000

Corequisite: HSC0003L

Course User Fee: \$20.00

HSC0003L Introduction to Healthcare Lab 0.500 credits

This course focuses on the performance of basic health care skills. Students will apply body mechanics and ergonomics, standard precautions used in infection control procedures and perform and record vital signs. Corequisite: HSC0003

DES0205 Dental Radiology 2.000 credits

This course is designed to provide students with a study of the principles of radiology, theory, and operation of X-ray equipment, darkroom procedures, film storage, and the hazards and precautions involved in dental radiology.

Lab Hours: 3.000, Clock Hours: 6.000 - 60.000

DES0205L Dental Radiology Lab 2.000 credits

This course is designed to provide students with a study of the principles of radiology, theory, and operation of X-ray equipment, darkroom procedures, film storage, and the hazards and precautions involved in dental radiology.

Clock Hours: 60.000

Course User Fee: \$88.60

DES0206 Dental Radiology II 1.000 credits

This course is a continuation of DES 0205. This course focuses on identification and interpretation of normal and abnormal dental pathology, accessory techniques and tomographic imaging systems.

Clock Hours: 30.000

DES0206L Dental Radiology II Lab 1.000 credits

This course is a continuation of DES 0205L. This course focuses on identification and interpretation of normal and abnormal dental pathology, accessory techniques and tomographic imaging systems. Prerequisite: DES0205L.

Clock Hours: 30.000

Course User Fee: \$88.60

DES0501 Dental Practice Management 1.000 credits

This course introduces students to the principles of dental office management. Areas of study includes reception procedures, telephone techniques, appointment book control, inventory control, purchasing, filing systems, bookkeeping, and banking procedures, tax records, legal and ethical aspects of dentistry.

Clock Hours: 30.000

DES0602 Pharmacology / Dental Office Emergencies 1.000 credits

This course is designed to provide students with the basic concepts and considerations regarding pharmacology and pharmaceutical preparations used in dentistry and by patients; chemical and physical properties, actions, dose, methods of administration, therapeutic usage, side effects, indications and contraindications will be explored. In addition, this course also examines the anticipation and prevention of emergency situations in the dental office. The relationship of health and medical histories will be emphasized as well as recognizing and assisting with emergency care of dental patients.

Clock Hours: 30.000

DES0830 Expanded Functions Theory 1.000 credits

This course is designed to provide students with the foundational theory prior to gaining laboratory and clinical proficiency in the expanded functions legally allowable in the state of Florida.

Clock Hours: 30.000

DES0830L Expanded Functions Lab 1.000 credits

This course is designed to provide students with the foundational theory prior to gaining laboratory and clinical proficiency in the expanded functions legally allowable in the state of Florida

Clock Hours: 30.000

Course User Fee: \$50.53

DEA0930 Dental Assisting Seminar 1.000 credits

This course provides learners with a review of materials commonly covered on the Dental Assisting National Board (DANB). Topics include but are not limited to infection control, general chair side procedures, and radiographic techniques. The seminar will provide a mock board examination, and opportunities for review, problem solving and strengthening skills.

Clock Hours: 30.000

Course User Fee: \$16.05

DEA0800L Clinical Practice 1 5.000 credits

This course introduces students to the basic clinical dental assisting skills as theorized in DEA 0020. Students will be required to participate and observe clinical and administrative activities in the campus dental clinic and rotate through other community clinics/dental offices where they will get an opportunity to work with a varied patient population. Concentration of study will include reception and dismissal of patients, patient management, establishing and maintaining records, obtaining and recording medical/dental histories and vital signs, charting, planning appointments, assisting with or performing various dental procedures, and utilizing practice management system.

Clock Hours: 150.000

DEA0804L Clinical Practice 2 6.000 credits

This course is a continuation of DEA 0800L. Students will be required to participate and observe clinical and administrative activities in the campus dental clinic and rotate through other community clinics/dental offices where they will get an opportunity to work with a varied patient population as they work on their competence in performing chairside assisting, expanded functions, laboratory, and basic business office procedures.

Clock Hours: 180.000

Program Description

[Fire Fighter Minimum Standards](#)

Career Technical Certificate | Code: 57004 | 16.40 credits

This program offers a broad foundation of knowledge and skills to prepare the student for employment as a frontline, entry-level firefighter. The program content includes, but is not limited to, orientation to the fire service, fire alarms and communication, vehicles, apparatus and equipment, fire behavior, portable extinguishers, fire streams, fundamentals of extinguishment, ladders, hoses, tools and equipment, forcible entry, salvage, overhaul, ventilation, rescue, protective breathing equipment, first responder emergency medical techniques, water supplies, principles of in-service inspections, safety, controlled burning, and employability skills. Students who successfully complete the program will be required to pass a written and practical exam to obtain certification as a Firefighter 2 from the Florida Bureau of Fire Standards and Training. Students must be a minimum of 18 years old to participate. Fire Academy Students Only. TABE Requirement: Computation (Mathematics): 10 and Communications (Reading Language Arts): 10.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 16.4

Total: 16.4

FFP0021 Fire Fighter Minimum Standards 16.400 credits

This program offers a broad foundation of knowledge and skills to prepare the student for employment as a frontline, entry-level firefighter. The program content includes, but is not limited to, orientation to the fire service, fire alarms and communication, vehicles, apparatus and equipment, fire behavior, portable extinguishers, fire streams, fundamentals of extinguishment, ladders, hoses, tools and equipment, forcible entry, salvage, overhaul, ventilation, rescue, protective breathing equipment, first responder emergency medical techniques, water supplies, principles of in-service inspections, safety, controlled burning, and employability skills. Students who successfully complete the program will be required to pass a written and practical exam to obtain certification as a Firefighter 2 from the Florida Bureau of Fire Standards and Training. Students must be a minimum of 18 years old to participate. Fire Academy Students Only.

Lecture Hours: 492.000

Course User Fee: \$653.00

Program Description

Fire Fighter/Emergency Medical Technician - Combined

Career Technical Certificate | Code: 58000 | 26.40 credits

Semester 1**Units: 16.4**

FFP 0021

Fire Fighter Minimum Standards

16.4

Semester 2**Units: 10.0**

EMS 0110

Emergency Medical Technician

10.0

Total: 26.4

FFP0021 Fire Fighter Minimum Standards 16.400 credits

This program offers a broad foundation of knowledge and skills to prepare the student for employment as a frontline, entry-level firefighter. The program content includes, but is not limited to, orientation to the fire service, fire alarms and communication, vehicles, apparatus and equipment, fire behavior, portable extinguishers, fire streams, fundamentals of extinguishment, ladders, hoses, tools and equipment, forcible entry, salvage, overhaul, ventilation, rescue, protective breathing equipment, first responder emergency medical techniques, water supplies, principles of in-service inspections, safety, controlled burning, and employability skills. Students who successfully complete the program will be required to pass a written and practical exam to obtain certification as a Firefighter 2 from the Florida Bureau of Fire Standards and Training. Students must be a minimum of 18 years old to participate. Fire Academy Students Only.

Lecture Hours: 492.000

Course User Fee: \$653.00

EMS0110 Emergency Medical Technician 10.000 credits

The Emergency Medical Technician (EMT) course prepares students to provide pre-hospital assessment and care for patients of all ages with a variety of medical conditions and traumatic injuries. Areas of study include an introduction to emergency medical services systems, roles and responsibilities of EMTs, anatomy and physiology, medical emergencies, trauma, special considerations for working in the pre-hospital setting, and providing patient transportation.

Lecture Hours: 300.000

Program Description

Florida Law Enforcement Academy

Career Technical Certificate | Code: 57022 | 25.66 credit

The Florida Law Enforcement Academy program prepares students for certification as Police Officers in the State of Florida. All criminal justice standards and training, Department of Education, and local standards will be met. Graduates are eligible for employment with any law enforcement agency in the state upon graduation from the program and successful completion of the State Certification Exam. Topics include law, human issues, patrol, traffic, investigations and communications and more. There is an emphasis on practical applications and competency-based performance. This program is offered at the School of Justice. Students seeking entrance into the MDC School of Justice basic recruit training programs for a career in law enforcement are required to pass a physical screening, physical agility, Voice Stress Analysis Test, psychological test, fingerprinting and background check and the Florida Basic Abilities Test (FBAT). For more information, please contact the School of Justice, Assessment Center and/or visit the Assessment Center website, <https://www.mdc.edu/justice/assessment-center.aspx>.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 25.5
CJK 0002	Introduction to Law Enforcement	0.4
CJK 0016	Communications	0.8
CJK 0018	Legal	2.1
CJK 0019	Interviewing and Report Writing	1.8
CJK 0020	Law Enforcement Vehicle Operations	1.6
CJK 0021	Serving Your Community	1.1
CJK 0031	First Aid for Criminal Justice Officers	1.3
CJK 0040	Criminal Justice Firearms	2.6
CJK 0051	Criminal Justice Defensive Tactics	2.6
CJK 0063	Fundamentals of Patrol	1.3
CJK 0072	Crimes Against Persons	1.6
CJK 0073	Crimes Involving Property and Society	0.4
CJK 0079	Crime Scene Follow-up Investigations	1.1
CJK 0093	Critical Incidents	1.5
CJK 0096	Criminal Justice Officer Physical Fitness Training	2.0
CJK 0400	Traffic Incidents	0.4
CJK 0401	Traffic Stops	0.8
CJK 0402	Traffic Crash Investigations	1.0
CJK 0403	DUI Traffic Stops	0.8
CJK 0421	Dart-Firing Stun Gun	0.1

Total: 25.5

CJK0002 Introduction to Law Enforcement 0.400 credits

In this course, students will learn to communicate effectively using empathy, courtesy, and professionalism while serving the community. The student will be able to understand the importance of self-talk and self-awareness in preparation for using the core communication competencies. For School of Justice students only.

Clock Hours: 12.000

Course User Fee: \$91.00

CJK0016 Communications 0.800 credits

In this course, students will learn to communicate effectively using empathy, courtesy, and professionalism while serving the community. The student will be able to understand the importance of self-talk and self-awareness in preparation for using the core communication competencies. For School of Justice students only.

Clock Hours: 24.000

CJK0018 Legal 2.130 credits

In this course, students will learn legal rules and concepts, amendments, and law. The student will learn the basics of law, civil and criminal liability. For School of Justice students only.

Clock Hours: 64.000

CJK0019 Interviewing and Report Writing 1.800 - 1.870 credits

The student will learn note-taking, interviewing, elements and principles of effective report writing. The student will also learn to take statements from victims, witnesses, and suspects; write clear, concise and accurate incident and arrest reports. For School of Justice students only.

Clock Hours: 56.000

CJK0020 Law Enforcement Vehicle Operations 1.600 credits

Students will learn the physiological and psychological factors affecting vehicle operations. This course stresses the importance of vehicle maintenance, environmental conditions affecting driving, and elements of basic driving skills including skids and other causes of accidents. Students will demonstrate hands-on basic driving skills. For School of Justice students only.

Clock Hours: 48.000

Course User Fee: \$623.00

CJK0021 Serving Your Community 1.130 credits

The student will learn a community's relationship with law enforcement can greatly impact how law enforcement officers do their jobs. Students will also learn how to respond to people in crisis and high risk groups. For School of Justice students only.

Clock Hours: 34.000

CJK0031 First Aid for Criminal Justice Officers 1.330 credits

Students will learn to initiate treatment for a variety of medical emergencies, understand and perform CPR, and know when to activate EMS and perform basic life support until help arrives. CPR and First Responder certification cards are issued upon successful completion. This course prepares criminal justice recruits for a variety of medical emergencies with minimal medical supplies. Basic training for School of Justice students only.

Course User Fee: \$237.00

CJK0040 Criminal Justice Firearms 2.670 credits

Students will learn how to use both handguns and shotguns. Students must qualify with both weapons under both daylight and night conditions. Students must also demonstrate ability for both accuracy and decision-making. Students are also introduced to chemical weapons and their effects.

Course User Fee: \$1574.00

CJK0051 Criminal Justice Defensive Tactics 2.670 credits

Students will learn how to physically defend themselves, physically control persons under arrest, and know what level of force is appropriate under differing circumstances. Students will learn escalate, de-escalate, or disengage techniques and when to use them. Additionally, a physical conditioning program is part of this course. For School of Justice students only.

Clock Hours: 80.000

Course User Fee: \$335.00

CJK0063 Fundamentals of Patrol 1.330 credits

The student will learn law enforcement techniques and tactics that you will use while on patrol. The student will also learn how to respond to non-criminal calls and conduct structure and area searches, and provide resources that officers use while on patrol. For School of Justice students only.

Clock Hours: 40.000

CJK0072 Crimes Against Persons 1.600 credits

The student will learn to conduct an investigation with attention to accuracy, detail, and professionalism. The student will learn to identify and respond to different types of crimes against persons. For School of Justice students only.

Clock Hours: 48.000

CJK0073 Crimes Involving Property and Society 0.400 credits

The student will learn the different types of crimes such as theft, burglary, white collar crimes and animal cruelty. The student will also learn about loitering, disorderly behavior, illicit drugs and vice crimes. For School of Justice students only.

Clock Hours: 12.000

CJK0079 Crime Scene Follow-up Investigations 1.130 credits

The student will learn evidence rules and concepts, securing the crime scene and documentation and handling procedures. The student will also learn to follow-up and review of initial information procedures and how to prepare for court. For School of Justice students only.

Clock Hours: 34.000

CJK0093 Critical Incidents 1.470 credits

The student will learn about events that can put lives at risk and cause major damage to property and the environment. The student will also learn about the causes of these events and how an effective response can affect a community's ability to recover from an event. For School of Justice students only.

Clock Hours: 44.000

CJK0096 Criminal Justice Officer Physical Fitness Training 2.000 credits

The student will learn the benefits of maintaining physical fitness to include nutrition and diet. The student will also learn the effects of and how to deal with stress, how to build up muscular and cardiovascular endurance and perform the exercises as required. For School of Justice students only.

Clock Hours: 60.000

Course User Fee: \$65.00

CJK0400 Traffic Incidents 0.400 credits

The student will learn the necessary information about traffic statutes and procedures and will lay the foundations for the practice of excellent traffic enforcement. The student will learn all aspects of law enforcement related to traffic enforcement, vehicles, roadways, and pedestrians. For School of Justice students only.

Clock Hours: 12.000

CJK0401 Traffic Stops 0.800 credits

The student will learn the goal of every traffic stop is to promote driver education and safer roadways. The student will also learn to exercise procedural justice and remain professional at all times when issuing a uniform traffic citation or warning, making an arrest, or finding that someone needs help. For School of Justice students only.

Clock Hours: 24.000 - 247.000

Course User Fee: \$44.00

CJK0402 Traffic Crash Investigations 1.000 credits

The student will learn about conducting a traffic crash investigations using a systematic approach. The student will learn about responding to, assessing, and protecting the scene; gathering information and evidence, documenting a crash. For School of Justice students only.

Clock Hours: 30.000

CJK0403 DUI Traffic Stops 0.800 credits

The student will learn how a driver that is under the influence of drugs or alcohol poses a hazard to themselves and others. The student will also learn how a law enforcement officer detects impaired driving, administers field sobriety tests, makes arrests when appropriate, and records the evidence of a DUI (driving under the influence) offense. For School of Justice students only.

Clock Hours: 24.000

Course User Fee: \$4.00

CJK0421 Dart-Firing Stun Gun 0.130 credits

The student will learn the basics of the conducted electrical weapon (CEW), particularly the dart-firing stun gun. The student will learn information about its practical use based on statutory requirements. For School of Justice students only.

Clock Hours: 4.000

Program Description

Licensed Practical Nursing (LPN)

Career Technical Certificate | Code: 53020 | 45 credits

Semester 1		Units: 11.5
PRN 0005C	Fundamentals of Nursing	5.0
PRN 0021C	Anatomy and Physiology and Medical Terminology 1	3.5
PRN 0035C	Pharmacology	3.0
Semester 2		Units: 18.5
PRN 0130C	Special Populations	5.0
PRN 0210C	Adult Health I	10.0
PRN 0082C	Anatomy and Physiology and Medical Terminology	3.5
Semester 3		Units: 15.0
PRN 0211C	Adult Health II	10.0
PRN 0006C	Professional Role Transition	5.0
		Total: 45.0

PRN0005C Fundamentals of Nursing 5.000 credits

This course provides an introduction to nursing and roles of the practical nurse in health care settings; as well as profession related and client-care concepts. Emphasis is placed on the knowledge and skills needed to provide safe, quality care. The theoretical foundation for basic assessment and nursing skills is presented, and the student is given an opportunity to demonstrate these skills in a laboratory setting. An introduction to the nursing process provides the student with a beginning framework for clinical decision making.

Lecture Hours: 60.000, Clinical Hours: 90.000

Course User Fee: \$451.06

PRN0021C Anatomy and Physiology and Medical Terminology 1 3.500 credits

This course is designed to provide the student with basic knowledge of normal human body structure and function. The student will learn major systems, organs and terminology necessary for the provision of safe and effective nursing care.

Lecture Hours: 80.000, Lab Hours: 25.000

Course User Fee: \$451.06

PRN0035C Pharmacology 3.000 credits

The purpose of this course is to examine pharmacotherapeutic agents used in the treatment of illness and the promotion, maintenance, and restoration of wellness in diverse individuals. It focuses on drug classification, concepts, and principles of pharmacology, with special consideration for the nursing role in developing a comprehensive approach to the clinical application of drug therapy through the use of the nursing process. Nursing implications relative to the utilization of drug therapy are examined. Safety and legal implications of drug administration are discussed.

Lecture Hours: 90.000

Course User Fee: \$451.06

PRN0130C Special Populations 5.000 credits

This course examines the application of nursing to the care of special populations throughout the life cycle. Emphasis is placed on nursing care of clients with psychosocial disorders, childbearing women, newborns, and children to promote independence, wellness, and maximizing quality of life.

Lecture Hours: 75.000, Clinical Hours: 75.000

Prerequisite: PRN0082C, PRN0210C

Course User Fee: \$451.06

PRN0210C Adult Health I 10.000 credits

This course focuses on the care of adult/older adult patients with health alterations that require medical and/or surgical intervention. Emphasis is placed on the care of patients with alterations in selected body functions. Concepts of client-centered care, informatics, safe practice, and professionalism are emphasized throughout the course. Clinical experiences provide the student an opportunity to apply theoretical concepts and implement safe patient care to adults in a variety of settings.

Lecture Hours: 150.000, Clinical Hours: 150.000

Prerequisite: PRN0005C AND PRN0021C, PRN0035C

Course User Fee: \$451.06

PRN0082C Anatomy and Physiology and Medical Terminology 3.500 credits

This course is designed to provide the student with an increasingly complex knowledge of normal human body structure and function. The student will learn major systems, organs and terminology necessary for the provision of safe and effective nursing care.

Lecture Hours: 80.000, Lab Hours: 25.000

Prerequisite: PRN0005C, PRN0021C, PRN0035C

Course User Fee: \$451.06

PRN0211C Adult Health II 10.000 credits

This course focuses on the care of adult/older adult patients with health alterations that require medical and/or surgical intervention. Emphasis is placed on the care of patients with alterations in selected body functions. Concepts of client-centered care, informatics, safe practice, and professionalism are emphasized throughout the course. Clinical experiences provide the student an opportunity to apply theoretical concepts and implement safe patient care to adults in a variety of settings.

Lecture Hours: 150.000, Clinical Hours: 150.000

Prerequisite: PRN0210C, PRN0082C

Course User Fee: \$451.06

PRN0006C Professional Role Transition 5.000 credits

This course facilitates the transition of the student to the role of PN. Emphasis is placed on issues related to nursing and healthcare, as well as skills necessary to provide care to multiple patients and assign tasks to other PN and unlicensed personnel. Concepts related to leadership and management are presented, as well as career development options that enhance career mobility. Standards of practice and the importance of practicing according to state regulations and statutes are examined. Clinical experiences provide the student the opportunity to apply theoretical concepts.

Lecture Hours: 25.000, Clinical Hours: 125.000

Prerequisite: PRN0210C, PRN0082C

Course User Fee: \$451.06

Program Description

Low Voltage Technician

Career Technical Certificate | Code: 56010 | 25 credits

The Low Voltage Technician certificate program is designed to prepare students for employment or advanced training in a variety of occupations in the Electronics Technology Integration industry. The program focuses on technical and product skills, underlying principles of technology, planning, management, finance, labor issues, health, safety and environmental issues.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 25.0

EEV 0162	Low Voltage Technician 1	5.0
EEV 0163	Low Voltage Technician 2	5.0
EEV 0164	Low Voltage Technician 3	5.0
EEV 0165	Low Voltage Technician 4	5.0
EEV 0166	Low Voltage Technician 5	5.0

TABE REQUIREMENT

A student enrolling in a Career Technical Education (CTE) program of 450 or more clock hours is required to complete a basic skills test within the first six weeks of admission into the program. The Tests of Adult Basic Education (TABE) test provided by MDC satisfies this requirement. The minimum passing scores vary among the CTE programs. Students should speak with their program advisor to validate score expectations and testing exemptions. Students who complete the CTE program must meet basic skills competencies before a certificate of completion is awarded.

(Provide appropriate TABE Scores if certificate is 450 hours or higher. If not, put N/A.)

- Mathematics: 10

- Language: 10
- Reading: 10

Total: 25.0

EEV0162 Low Voltage Technician 1 5.000 credits

This course is an overview of the trade with an introduction to construction safety and hand and power tools used in the construction industry. Students will learn basic written and verbal communication skills, basic mathematics, construction methods and techniques, building codes, and how to read construction drawings. Clock Hours: 150.000

Prerequisite: TABE REQUIREMENT A student enrolling in a Career Technical Education (CTE) program of 450 or more clock hours is required to complete a basic skills test within the first six weeks of admission into the program. The Tests of Adult Basic Education (TABE) test provided by MDC satisfies this requirement. The minimum passing scores vary among the CTE programs. Students should speak with their program advisor to validate score expectations and testing exemptions. Students who complete the CTE program must meet basic skills competencies before a certificate of completion is awarded. TABE Scores: Mathematics:10 Language:10 Reading: 10

EEV0163 Low Voltage Technician 2 5.000 credits

This course focuses on alternating current (AC) and direct current (DC) circuits and electronic devices, including an overview of applicable test equipment, cable and terminations used in the installation of low-voltage systems. Further instruction on specific trade-related drawings, codes and standards will be discussed. OSHA 10 or OSHA 30 Construction Industry training will be provided.

Clock Hours: 150.000

Prerequisite: EEV0162

EEV0164 Low Voltage Technician 3 5.000 credits

This course is an overview of hard-wired cabling, fiber, wireless networks, maintenance, and repair of low-voltage systems, plus project management.

Clock Hours: 150.000

Pre/Corequisite: EEV0163

EEV0165 Low Voltage Technician 4 5.000 credits

This course is an overview of audio, video, broadband, media management, telecommunication systems, and residential/commercial building networks.

Clock Hours: 150.000

Pre/Corequisite: EEV0164

EEV0166 Low Voltage Technician 5 5.000 credits

This course is an overview of intrusion detection, fire alarm systems, fiber optics, nurse call, CCTV, and access control systems.

Clock Hours: 150.000

Pre/Corequisite: EEV0165

Program Description

[Massage Therapy - Accelerated Option](#)

Career Technical Certificate | Code: 53022 | 25 credit

The two-semester program prepares individuals to provide various techniques of massage of the back, head, and feet, including reflexology, rolling, and trigger point therapy. There is an emphasis on the therapist/client relationship and record management for clients and payment. Upon successful completion of this program, the graduate is eligible to sit for the State of Florida Massage Therapy licensure examination. Test of Adult Basic Education (TABE) is required.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 25.0

MSS 0215	History and Standards for Massage Therapy	1.0
MSS 0250	Introduction to Massage Therapy	1.0
MSS 0250L	Introduction to Massage Therapy Laboratory	6.0
MSS 0281	Allied Modalities	3.5
MSS 0995	Massage Therapy - Accelerated	13.5

Total: 25.0

MSS0215 History and Standards for Massage Therapy 1.000 credits

This course examines the history and development of massage therapy, basic legal concepts related to health care employment, and legal requirements for practice as a Massage Therapist in the State of Florida.

Course User Fee: \$25.00

MSS0250 Introduction to Massage Therapy 1.000 credits

This course focuses on the theories and principles of therapeutic massage. The Massage Therapist/Client Relationship, the effects of massage on the systems of the body, massage facilities, equipment/supplies, and furniture requirements will be discussed.

Course User Fee: \$25.00

MSS0250L Introduction to Massage Therapy Laboratory 6.000 credits

Laboratory for MSS 0250. This course provides opportunities for the practical application of the theories and principles of therapeutic massage.

Course User Fee: \$150.00

MSS0281 Allied Modalities 3.500 credits

A study of the advanced theories and techniques for massage therapy. Content includes: oriental bodywork, reflexology, trager approach, rolfing, craniosacral therapy, infant massage, pregnancy massage and aromatherapy.

Course User Fee: \$25.00

MSS0995 Massage Therapy - Accelerated 13.500 - 15.000 credits

This course is designed to provide PSAV credit for students with training and State of Florida licensure as a Physical Therapist or Physical Therapist Assistant. Students must provide documentation of a current state license and be a graduate of an accredited program. This course requires special permission and students must contact the program coordinator for registration approval.

Program Description

Massage Therapy - Generic Option

Career Technical Certificate | Code: 53021 | 25 credit

The two-semester program prepares individuals to provide various techniques of massage of the back, head, and feet, including reflexology, rolling, and trigger point therapy. There is an emphasis on the therapist/client relationship and record management for clients and payment. Upon successful completion of this program, the graduate is eligible to sit for the State of Florida Massage Therapy licensure examination. Test of Adult Basic Education (TABE) is required.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 13.0

MSS 0156	Anatomy & Physiology for Massage Therapy	2.5
MSS 0156L	Anatomy & Physiology for Massage Therapy Lab	2.5
MSS 0215	History and Standards for Massage Therapy	1.0
MSS 0250	Introduction to Massage Therapy	1.0

MSS 0250L	Introduction to Massage Therapy Laboratory	6.0
Semester 2		Units: 12.0
HSC 0003L	Introduction to Healthcare Lab	0.5
HSC 0003	Introduction to Healthcare	2.5
MSS 0281	Allied Modalities	3.5
MSS 0300	Hydrotherapy Modalities	1.0
MSS 0300L	Hydrotherapy Modalities Laboratory	1.5
MSS 0803C	Massage Therapy Clinical Practicum	3.0

Total: 25.0

MSS0156 Anatomy & Physiology for Massage Therapy 2.500 credits

This course will focus on the relationship between the anatomical and physiological effects of massage therapy on the body. Students will focus on the structure of organs, muscles, bones and tissues. Primary focus will center on the musculoskeletal system and innervations.

Course User Fee: \$62.50

MSS0156L Anatomy & Physiology for Massage Therapy Lab 2.500 credits

This course will examine the practical application and physiological effects of massage therapy on the body. Students will focus on the structure of organs, muscles, bones and tissues. Primary focus will center on the musculoskeletal systems and innervations as well as clinical pathologies related to those systems.

Course User Fee: \$62.50

MSS0215 History and Standards for Massage Therapy 1.000 credits

This course examines the history and development of massage therapy, basic legal concepts related to health care employment, and legal requirements for practice as a Massage Therapist in the State of Florida.

Course User Fee: \$25.00

MSS0250 Introduction to Massage Therapy 1.000 credits

This course focuses on the theories and principles of therapeutic massage. The Massage Therapist/Client Relationship, the effects of massage on the systems of the body, massage facilities, equipment/supplies, and furniture requirements will be discussed.

Course User Fee: \$25.00

MSS0250L Introduction to Massage Therapy Laboratory 6.000 credits

Laboratory for MSS 0250. This course provides opportunities for the practical application of the theories and principles of therapeutic massage.

Course User Fee: \$150.00

HSC0003L Introduction to Healthcare Lab 0.500 credits

This course focuses on the performance of basic health care skills. Students will apply body mechanics and ergonomics, standard precautions used in infection control procedures and perform and record vital signs.

Corequisite: HSC0003

HSC0003 Introduction to Healthcare 2.500 credits

An introduction to the health care environment, this course focuses on the health care team and delivery systems. Students will learn about legal responsibilities, ethical issues, safety, infection control, communication, interpersonal behaviors, wellness, and disease. Corequisite: HSC0003L

Clock Hours: 75.000

Corequisite: HSC0003L

Course User Fee: \$20.00

MSS0281 Allied Modalities 3.500 credits

A study of the advanced theories and techniques for massage therapy. Content includes: oriental bodywork, reflexology, trager approach, rolfing, craniosacral therapy, infant massage, pregnancy massage and aromatherapy.

Course User Fee: \$25.00

MSS0300 Hydrotherapy Modalities 1.000 credits

This course focuses on the history and development of hydrotherapy, application of equipment used, and the associated standards.

Course User Fee: \$25.00

MSS0300L Hydrotherapy Modalities Laboratory 1.500 credits

This course presents an opportunity for the students to safely and effectively apply various types of hydrotherapy and evaluate their effectiveness.

Course User Fee: \$62.50

MSS0803C Massage Therapy Clinical Practicum 3.000 credits

This course provides the student with the opportunity to practice and further develop an understanding of various massage techniques in a clinical placement setting under supervision of a licensed Massage Therapist.

Course User Fee: \$75.00

Program Description

Massage Therapy - Transitional Option

Career Technical Certificate | Code: 53023 | 25 credit

The two-semester program prepares individuals to provide various techniques of massage of the back, head, and feet, including reflexology, rolling, and trigger point therapy. There is an emphasis on the therapist/client relationship and record management for clients and payment. Upon successful completion of this program, the graduate is eligible to sit for the State of Florida Massage Therapy licensure examination. Test of Adult Basic Education (TABE) is required.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 8.0
MSS 0215	History and Standards for Massage Therapy	1.0
MSS 0250	Introduction to Massage Therapy	1.0
MSS 0250L	Introduction to Massage Therapy Laboratory	6.0
Semester 2		Units: 17.0
MSS 0281	Allied Modalities	3.5
MSS 0300	Hydrotherapy Modalities	1.0
MSS 0300L	Hydrotherapy Modalities Laboratory	1.5
MSS 0803C	Massage Therapy Clinical Practicum	3.0
MSS 0996	Massage Therapy - Transitional	8.0
		Total: 25.0

MSS0215 History and Standards for Massage Therapy 1.000 credits

This course examines the history and development of massage therapy, basic legal concepts related to health care employment, and legal requirements for practice as a Massage Therapist in the State of Florida.

Course User Fee: \$25.00

MSS0250 Introduction to Massage Therapy 1.000 credits

This course focuses on the theories and principles of therapeutic massage. The Massage Therapist/Client Relationship, the effects of massage on the systems of the body, massage facilities, equipment/supplies, and furniture requirements will be discussed.

Course User Fee: \$25.00

MSS0250L Introduction to Massage Therapy Laboratory 6.000 credits

Laboratory for MSS 0250. This course provides opportunities for the practical application of the theories and principles of therapeutic massage.

Course User Fee: \$150.00

MSS0281 Allied Modalities 3.500 credits

A study of the advanced theories and techniques for massage therapy. Content includes: oriental bodywork, reflexology, trager approach, rolfing, craniosacral therapy, infant massage, pregnancy massage and aromatherapy.

Course User Fee: \$25.00

MSS0300 Hydrotherapy Modalities 1.000 credits

This course focuses on the history and development of hydrotherapy, application of equipment used, and the associated standards.

Course User Fee: \$25.00

MSS0300L Hydrotherapy Modalities Laboratory 1.500 credits

This course presents an opportunity for the students to safely and effectively apply various types of hydrotherapy and evaluate their effectiveness.

Course User Fee: \$62.50

MSS0803C Massage Therapy Clinical Practicum 3.000 credits

This course provides the student with the opportunity to practice and further develop an understanding of various massage techniques in a clinical placement setting under supervision of a licensed Massage Therapist.

Course User Fee: \$75.00

MSS0996 Massage Therapy - Transitional 8.000 credits

This course is designed to provide PSAV credit for students with training and State of Florida licensure as an Allied Health Professional or Registered Nurse. Students must provide documentation of a current state license and be a graduate of an approved Associate Degree program. This course requires special permission and students must contact the program coordinator for registration approval.

Program Description

Medical Assisting

Career Technical Certificate | Code: 53007 | 43.30 credit

The Medical Assisting program, which is 1 year (3 semesters) in length, prepares individuals to provide health services in ambulatory out-patient facilities, including medical offices and clinics. Medical Assistants participate in diagnostic, clinical, and administrative functions. Diagnostic functions include drawing blood, performing basic laboratory tests, and taking EKG's and X-Rays. Clinical functions include obtaining vital signs, preparing patients for and assisting with examinations and procedures, administering medications and performing treatments. Administrative functions include serving as receptionists, scheduling appointments and diagnostic procedures, managing records, completing insurance coding, and providing for billing and collecting. Medical Assistants use computer technology to manage records, billing and other aspects of a medical office or clinic. Students participate in an externship each semester to gain experience in every aspect of the medical assistant's practice. Test of Adult Basic Education (TABE) is required.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester		Units: 3.0
HSC 0003	Introduction to Healthcare	2.5
HSC 0003L	Introduction to Healthcare Lab	0.5
Semester 2		Units: 15.3
MEA 0204	Theoretical Aspects of Clinical Skills	1.0
MEA 0204L	Application of Clinical Skills	2.0
MEA 0231	Anatomy & Physiology and Medical Terminology	2.3
MEA 0234	Pathophysiology & Disease for Medical Assistants	4.0
MEA 0242	Pharmacology for the Medical Assistant	3.0
MEA 0802	Clinical Externship for the Medical Assistant	3.0
Semester 3		Units: 13.0

MEA 0322	Office Mgmt. & Professional Issues for the MA	3.0
MEA 0334C	Medical Coding/Insurance Billing with Collections	4.0
MEA 0343	Computers in the Medical Office	3.0
MEA 0810	Administrative Externship for the MA	3.0

Semester 4

Units: 12.0

MEA 0254	Physician Office Laboratory Procedures	2.0
MEA 0254L	Physician Office Laboratory Procedure Applications	2.0
MEA 0258	Radiology for the Medical Assistant	3.0
MEA 0540	Electrocardiography/Emergency Procedures	2.0
MEA 0832	Diagnostic Externship in Medical Assistant	3.0

Total: 43.3

HSC0003 Introduction to Healthcare 2.500 credits

An introduction to the health care environment, this course focuses on the health care team and delivery systems. Students will learn about legal responsibilities, ethical issues, safety, infection control, communication, interpersonal behaviors, wellness, and disease. Corequisite: HSC0003L
Clock Hours: 75.000
Corequisite: HSC0003L
Course User Fee: \$20.00

HSC0003L Introduction to Healthcare Lab 0.500 credits

This course focuses on the performance of basic health care skills. Students will apply body mechanics and ergonomics, standard precautions used in infection control procedures and perform and record vital signs.
Corequisite: HSC0003

MEA0204 Theoretical Aspects of Clinical Skills 1.000 credits

This course is designed to develop and further support students' knowledge and ability to organize and work efficiently and effectively in both performing and assisting with clinical procedures performed in medical offices. Emphasis will be on the role and responsibility of the Medical Assistant.
Clock Hours: 30.000

MEA0204L Application of Clinical Skills 2.000 credits

This course is designed to develop and support students' ability to perform and assist in basic clinical skills. Emphasis will be on the role and responsibility of the medical assistant in performing sterile techniques and the use of organization and efficiency in performing and assisting with patient examination, sterile procedures, and diagnostic procedures and treatment performed in medical offices.
Clock Hours: 60.000
Course User Fee: \$3.00

MEA0231 Anatomy & Physiology and Medical Terminology 2.300 credits

This course is designed to introduce the student to basic anatomy and physiology and to develop the ability to communicate verbally and in writing within the medical field. Special fee.
Clock Hours: 69.000
Course User Fee: \$10.00

MEA0234 Pathophysiology & Disease for Medical Assistants 4.000 credits

This course is designed to introduce students to common diseases and medical conditions which affect patients who present themselves to medical offices for diagnosis and treatment. Emphasis will be on the role and responsibility of the Medical Assistant in prevention, diagnosis, and treatment.
Clock Hours: 120.000

MEA0242 Pharmacology for the Medical Assistant 3.000 credits

This course is designed to introduce students to principles of pharmacology and provide a basis to comprehend the role and responsibility of Medical Assistants in administering medication. Emphasis will be placed on calculation of dosages, frequently used drugs, and classification of drugs as they relate to the body systems.
Course User Fee: \$10.00

MEA0802 Clinical Externship for the Medical Assistant 3.000 credits

This course is designed to provide students with experience in the practice of the clinical aspect of medical assisting. Students will be assigned to a physician's office or clinics where they will provide direct patient care under the guidance of an experienced Medical Assistant.
Clock Hours: 90.000
Course User Fee: \$10.00

MEA0322 Office Mgmt. & Professional Issues for the MA 3.000 credits

Office management procedures, including planning and organization; financial and medical record keeping procedures; billing and collection; processing insurance claims using procedural and diagnostic coding. Legal and ethical responsibilities, credentialing and other professional issues of Medical Assisting.

Course User Fee: \$15.00

MEA0334C Medical Coding/Insurance Billing with Collections 4.000 credits

Processing health insurance claims using procedural and diagnostic coding. The student will learn and apply current government regulations affecting third-party reimbursement. Billing, electronic claims transmission, and collection systems are emphasized. Clock Hours: 120.000

MEA0343 Computers in the Medical Office 3.000 credits

The application of computer concepts to medical office practices. The student will keyboard documents using word processing software. Emphasis will be on operating transcription equipment and transcribing medical records. The student will also be introduced to electronic spreadsheet and database applications. Clock Hours: 90.000

Course User Fee: \$25.00

MEA0810 Administrative Externship for the MA 3.000 credits

The student is assigned to a physician's office, clinic, laboratory, or other community health care facility. Emphasis is on integrating basic administrative skills demonstrated in previous courses.

MEA0254 Physician Office Laboratory Procedures 2.000 credits

Theoretical concepts of specimen collection and processing. This course focuses on the fundamentals of diagnostic tests, including urinalysis, basic office bacteriology, hematology, and blood chemistry. The principles of aseptic techniques, infection control, and safety procedures are discussed. Compliance with quality assurance practices are emphasized. Clock Hours: 60.000

MEA0254L Physician Office Laboratory Procedure Applications 2.000 credits

A clinical laboratory course designed for the Medical Assistant student to practice specimen collection, microscopy and urinalysis. Includes basic office bacteriology, hematology, and blood chemistry. The student will apply principles of aseptic techniques and infection control.

Clock Hours: 60.000

Course User Fee: \$80.00

MEA0258 Radiology for the Medical Assistant 3.000 credits

This course focuses on the basic principles of x-ray, film handling and processing, radiographic technique, and radiation biology. The course prepares the student to take the examination given by the Florida Department of Professional Regulations (DPR) for the Basic Radiographer License.

Clock Hours: 90.000

Course User Fee: \$15.00

MEA0540 Electrocardiography/Emergency Procedures 2.000 credits

The nature and purpose of the electrocardiograph (EKG); maintenance of equipment and materials needed; preparation of the patient and the procedure for taking and mounting the EKG record and monitoring the record for abnormal or erratic tracings. The maintenance of emergency equipment and implementing emergency procedures in the medical office.

Clock Hours: 60.000

Course User Fee: \$10.00

MEA0832 Diagnostic Externship in Medical Assistant 3.000 credits

This course is designed to provide students with experience in the diagnostic aspect of Medical Assisting. Students will be assigned to a physician's office or clinics where they will perform diagnostic clinical laboratory procedures, electrocardiographic and basic x-ray procedures under the guidance of an experienced Medical Assistant.

Clock Hours: 90.000

Program Description

Pharmacy Technician

Career Technical Certificate | Code: 53004 | 35 credit

The Pharmacy Technician program prepares individuals for employment as Pharmacy Technicians. The Pharmacy Technician works primarily in retail and hospital pharmacies under the supervision of a registered pharmacist in the packaging and distribution of medication. Test of Adult Basic Education (TABE) is required.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1		Units: 3.0
HSC 0003	Introduction to Healthcare	2.5
HSC 0003L	Introduction to Healthcare Lab	0.5
Semester 2		Units: 32.0
PTN 0003	Intro to Pharmacy Practice & Medical Terminology	3.0
PTN 0004	Pharmacy Practitioner Applications	3.0
PTN 0006	Pharmacy Calculations	3.0
PTN 0021	Drug Classifications	3.0
PTN 0041	Pharmacy Technician Hospital Field Experience	10.0
PTN 0049	Pharmacy Technician Retail Store Field Experience	10.0
		Total: 35.0

HSC0003 Introduction to Healthcare 2.500 credits

An introduction to the health care environment, this course focuses on the health care team and delivery systems. Students will learn about legal responsibilities, ethical issues, safety, infection control, communication, interpersonal behaviors, wellness, and disease. Corequisite: HSC0003L

Clock Hours: 75.000

Corequisite: HSC0003L

Course User Fee: \$20.00

HSC0003L Introduction to Healthcare Lab 0.500 credits

This course focuses on the performance of basic health care skills. Students will apply body mechanics and ergonomics, standard precautions used in infection control procedures and perform and record vital signs. Corequisite: HSC0003

PTN0003 Intro to Pharmacy Practice & Medical Terminology 3.000 credits

This course is an orientation to the overall functions and services of a hospital pharmacy. Students will learn medical abbreviations, terminology, chemical symbols, formulas, and incompatibilities.

Prerequisite: HSC0003, HSC0003L, Corequisite: PTN0006

Course User Fee: \$10.00

PTN0004 Pharmacy Practitioner Applications 3.000 credits

This course focuses on pharmacy practitioner applications. Students will learn to develop skills relating to the specific, technical, manipulative and clerical tasks involved with the preparation and distribution of medications under the supervision of Licensed Pharmacists.

Clock Hours: 90.000

Prerequisite: HSC0003, HSC0003L, Corequisite: PTN0021

PTN0006 Pharmacy Calculations 3.000 credits

This is a course in Pharmacy Calculations. Students will learn to define systems of measurement, convert from one system to another, and calculate pharmacology problems.

Clock Hours: 90.000

Prerequisite: HSC0003, HSC0003L, Corequisite: PTN0003

PTN0021 Drug Classifications 3.000 credits

This course covers the major classifications of pharmaceuticals, standards for quality and purity of drugs, and authoritative information on dosage and administration. Students will learn about poisons, placebos, and the sources from which medications are produced.

Prerequisite: HSC0003, HSC0003L, Corequisite: PTN0004

Course User Fee: \$15.00

PTN0041 Pharmacy Technician Hospital Field Experience 10.000 credits

This course covers clinical hospital training to develop the student's knowledge and skills on the job. Students will learn how to properly prepare doses of medications and intravenous admixtures.

Prerequisite: HSC0003, HSC0003L, PTN0003, PTN0004, PTN0006, PTN0021, Corequisite: PTN0049

PTN0049 Pharmacy Technician Retail Store Field Experience 10.000 credits

This course covers the clinical field experiences in a retail establishment. Students will learn about pharmaceutical chemistry, proper medication, and how to deliver medications correctly.

Prerequisite: HSC0003, HSC0003L, PTN0003, PTN0004, PTN0006, PTN0021, Corequisite: PTN0041

Course User Fee: \$10.00

Program Description

Phlebotomy Technician

Career Technical Certificate | Code: 53003 | 5.50 credit

The Phlebotomy Technician program is designed to prepare students for employment in a hospital laboratory, blood center, or other health care facility to draw blood by venipuncture and capillary puncture. Students are taught safe and efficient work practices in obtaining adequate and correct blood specimens, labeling specimens, and transporting specimens correctly to the appropriate laboratory sections. The Center for Disease Control (CDC) guidelines for HIV/AIDS, Hepatitis B and other diseases are stressed.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 5.5

HSC 0003	Introduction to Healthcare	2.5
HSC 0003L	Introduction to Healthcare Lab	0.5
MLT 0041	Phlebotomy Theory	0.5
MLT 0048	Phlebotomy Practicum	1.5
MLT 0061	Practical Aspects of Phlebotomy	0.5

Total: 5.5

HSC0003 Introduction to Healthcare 2.500 credits

An introduction to the health care environment, this course focuses on the health care team and delivery systems. Students will learn about legal responsibilities, ethical issues, safety, infection control, communication, interpersonal behaviors, wellness, and disease. Corequisite: HSC0003L

Clock Hours: 75.000

Corequisite: HSC0003L

Course User Fee: \$20.00

HSC0003L Introduction to Healthcare Lab 0.500 credits

This course focuses on the performance of basic health care skills. Students will apply body mechanics and ergonomics, standard precautions used in infection control procedures and perform and record vital signs.

Corequisite: HSC0003

MLT0041 Phlebotomy Theory 0.500 credits

This course covers the theory of phlebotomy techniques by venipuncture and skin puncture. This includes basic anatomy and physiology of the circulatory system, types of tubes to select for various blood tests, possible interfering substances, hospital hierarchy, professionalism, risk factors for Hepatitis, AIDS, and all sexually transmitted diseases, infection control guidelines, and employability skills. Special fee.

Course User Fee: \$10.00

MLT0048 Phlebotomy Practicum 1.500 credits

This course is designed to prepare students to draw blood by venipuncture and capillary puncture and to prepare them for employment in a hospital laboratory, blood center, or other health care facility. Students are taught safe and efficient work practices in obtaining adequate and correct blood specimens, labeling specimens, and transporting specimens correctly to the appropriate laboratory sections. The Center for Disease Control (CDC) guidelines for HIV/AIDS, Hepatitis B and other diseases are stressed. Clock Hours: 45.000

MLT0061 Practical Aspects of Phlebotomy 0.500 credits

This course covers the collection of blood by venipuncture, skin puncture and donor room techniques. This includes the handling, labeling, transporting, and logging-in of specimens as well as the demonstration of correct infection control techniques. Special fee. Lab Hours: 15.000

Program Description

Private Investigator Intern

Career Technical Certificate | Code: 57023 | 1.33 credit

The purpose of this program is to prepare students for employment as Private Investigator Interns as required by Section 493.6203(6)(b) Florida Statutes, offered in a 24-contact hour course and a 16 contact hour course.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 1.3

SCY 0051	Private Investigator Intern Course A	0.8
SCY 0052	Private Investigator Intern Course B	0.5

Total: 1.3

SCY0051 Private Investigator Intern Course A 0.800 credits

This course requires twenty-four hours of training as required by Section 493.6203(b) F.S. for Private Investigator Interns. Students will learn topics which include Florida Statutes and Florida Administrative Code, the Intern/Sponsor Relationship, Ethics, Liability, Surveillance, Report Writing, Equipment, Interviewing, Sources of Information, the Computer and Investigations, and Restrictions on Records.

Course User Fee: \$35.00

SCY0052 Private Investigator Intern Course B 0.530 credits

This course requires sixteen hours of training as required by Section 493.6203(b) F.S. for Private Investigator Interns. Students will learn topics which include locating people and performing background investigations, evidence, executive protection, anti-terrorism, courtroom and formal hearing demeanor, pretrial responsibilities, and the investigator as a witness.

Prerequisite: SCY0051

Course User Fee: \$35.00

Program Description

Private Security Officer

Career Technical Certificate | Code: 57006 | 2.26 credit

The Private Security Officer program offers a comprehensive curriculum designed to meet Florida's requirements for both Class D and Class G Security Licenses. Students will engage in rigorous training that covers security principles, legal regulations, firearms proficiency, emergency response procedures, and effective communication skills, ensuring they are well-prepared for diverse roles in the private security industry. By completing this program, participants will be equipped to apply for the Florida Class D Security License for unarmed positions and

the Class G Security License for armed roles, broadening their career opportunities. This program not only lays a strong foundation for entry-level security positions but also provides advanced training for those aiming to excel in higher-stakes security environments.

Program Modality: Both [In-Person (IP) and On-line]

Contracted: No

Semester 1

Units: 2.3

CJK 0132	Basic Security Officer Training	1.3
CJT 0801	Private Security Guard Training 2: Class "G" Licens	0.9

Total: 2.3

CJK0132 Basic Security Officer Training 1.330 credits

This course is designed to prepare students to apply for Class "D" Private Security Officer license. Students will learn regulatory compliance, emergency procedures, ethics and entrepreneurship, courtroom procedures, traffic direction and crowd control, and more. Special Fee.

Course User Fee: \$25.00

CJT0801 Private Security Guard Training 2: Class "G" Licens 0.940 credits

This course is necessary for compliance with the state minimum training standard for a Class "G" (Armed Security Guard License).

Course User Fee: \$142.00

Program Description

Telecommunications and Fiber Optics Technician

Career Technical Certificate | Code: 59012 | 15.34 credits |

The Telecommunications and Fiber Optics Technician program provides students with the technical skills and knowledge needed for employment or further training in telecommunications. It covers installation, maintenance, and troubleshooting of copper, fiber optic, and wireless networks, as well as advanced technologies like hybrid fiber-coaxial (HFC) systems and digital signal processing. The curriculum also prepares students for industry certifications, emphasizing safety, regulatory compliance, and customer service, ensuring they are well-equipped to succeed in the telecommunications industry.

Program Modality: Resident [In-Person (IP)]

Contracted: No

Semester 1

Units: 15.3

TCN 0300	Telecommunications Fundamentals	3.8
TCN 0310	Network Cabling and Installation	3.8
TCN 0312	Fiber Optics and Splicing Techniques	3.8
TCN 0320	Advanced Telecommunications Systems	3.8

Total: 15.3

TCN0300 Telecommunications Fundamentals 3.830 credits

This course introduces students to basic concepts in telecommunications, including safety practices, electrical fundamentals, and industry standards. Students will learn about telecommunications systems architecture, cable types, and basic troubleshooting techniques. The course also covers customer service skills and professional conduct in the telecommunications industry.

Course User Fee: \$96.92

TCN0310 Network Cabling and Installation 3.830 credits

This course provides a comprehensive overview of network cabling and installation techniques essential for building robust telecommunications infrastructures. Students will explore various types of cabling, including twisted pair, coaxial, and fiber optic, while gaining hands-on experience in installation practices, cable termination, and testing procedures.

Course User Fee: \$96.92

TCN0312 Fiber Optics and Splicing Techniques 3.830 credits

This hands-on course provides students with a comprehensive understanding of fiber optic technology and essential splicing techniques crucial for modern telecommunications networks. Students will explore the fundamental principles of fiber optic transmission, including light propagation, cable types, and optical components.

Course User Fee: \$96.92

TCN0320 Advanced Telecommunications Systems 3.840 credits

This comprehensive course delves into cutting-edge telecommunications technologies and advanced network architectures. Students will explore the intricacies of modern communication systems, including 5G networks, Internet of Things (IoT) infrastructures, and software-defined networking (SDN).

Clock Hours: 115.000

Course User Fee: \$96.92

COURSE DESCRIPTIONS

ACG

ACG1403 Excel for Business 1.000 - 3.000 credits

This course will cover Excel topics relevant to the field of accounting and finance including, but not limited to, VLOOKUP, HLOOKUP, INDEX, MATCH, IF, AND, OR, Pivot Tables, Named Ranges, Array Formulas, Custom Number Formats, Conditional Formatting, Absolute References and Keyboard Shortcuts.

Lecture Hours: 16.000 - 48.000

Course User Fee: \$22.00

ACG2001 Principles of Accounting 1 3.000 credits

An introduction to the basic principles of financial accounting with emphasis on basic accounting procedures such as the recording of transactions and the preparation of financial statements. Other topics include inventories, receivables, and cash.

ACG2001 and ACG2011 can be substituted for ACG2021. Corequisite: ACG2001L.

Lecture Hours: 48.000

Corequisite: ACG2001L

ACG2001L Principles of Accounting 1 Lab 1.000 credits

The student will use computer software and financial accounting applications to analyze accounting records and prepare financial statements. The student will learn, with tutorial help and problem-solving, the concepts needed to achieve the objectives of ACG2021.

Lab Hours: 32.000

Course User Fee: \$22.00

ACG2011 Principles of Accounting 2 3.000 credits

Accounting for owners' equity with emphasis on corporate financial statements. Other topics include plant assets, intangible assets, current and long-term liabilities. ACG 2001 and 2011 can be substituted for ACG 2021. Prerequisite: ACG 2001.

Corequisite: ACG 2011L.

Lecture Hours: 48.000

Prerequisite: ACG2001, Corequisite: ACG2011L

ACG2011L Principles of Accounting 2 Lab 1.000 credits

Students will use personal computer software and financial accounting applications to analyze accounting records, prepare financial statements, and compile EXCEL spreadsheets. This course will reinforce, with tutorial help and problem-solving, the concepts needed to achieve the objectives of ACG2011. Corequisite: ACG2011.

Lab Hours: 32.000

Corequisite: ACG2001

Course User Fee: \$22.00

ACG2021 Financial Accounting 3.000 credits

An introduction to financial accounting concepts and analysis with emphasis on corporate financial statements and determination of income.

Lecture Hours: 48.000

Corequisite: ACG2021L

ACG2021L Financial Accounting Lab 1.000 credits

The student will use computer software and financial accounting applications to analyze accounting records and prepare financial statements. The student will learn, with tutorial help and problem-solving, the concepts needed to achieve the objectives of ACG2021. May be repeated for credit. Laboratory fee.

Lab Hours: 32.000

Corequisite: ACG2021

Course User Fee: \$32.00

ACG2031 Accounting Theory 3.000 credits

Designed primarily for the transferring accounting major, the course covers current topics in both financial and managerial accounting. It exposes the student to a computerized accounting system. It also familiarizes the student with current accounting literature and includes a review of the preparation and analysis of financial statements. Prerequisite: ACG2071.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$27.00

ACG2071 Managerial Accounting 3.000 credits

Managerial Accounting focuses on the accounting information needs of the various levels of internal management within an organization. Internal responsibility is directed at three major areas of management responsibility: cost determination, planning and control, and long-term decision-making. Prerequisites: ACG2011 and ACG2001 or ACG2021; Corequisite: ACG2071L

Lecture Hours: 48.000

ACG2071L Managerial Accounting Lab 1.000 credits

Students will learn to interpret and solve problems related to the managerial accounting field.

Additional support will be provided to students in order to achieve the objectives of ACG2071.

Prerequisites: ACG2001, ACG2021L, ACG2021, ACG2011; Corequisite: ACG2071.

Lab Hours: 32.000

Prerequisite: ACG2021, ACG2021L, Corequisite: ACG2071

Course User Fee: \$32.00

ACG2450 Microcomputers in Accounting 3.000 credits

Accounting application of electronic data processing including the preparation, interpretation and use of computer information in financial decision-making. Pre/Corequisite: ACG2021. Special fee.

Lecture Hours: 48.000

Pre/Corequisite: ACG2021

Course User Fee: \$22.00

ACG2500 FIN MGT NON-PRFT ORG 3.000 credits

This course will be deleted.

Lecture Hours: 48.000

ACG2630 Auditing 3.000 credits

Fundamental principles of audit practice and procedure including the verification of balance sheets and income statement items, the preparation of audit working papers, and the compilation of audit reports. The course includes short problems and audit of accounting records.

Lecture Hours: 48.000

Prerequisite: ACG2011

Course User Fee: \$27.00

ACG3103 Intermediate Accounting I 3.000 credits

Theory and methodology underlying financial reporting, including the FASB's conceptual framework, the accounting process, financial statements, accounting changes, present value applications, and current assets. Prerequisites: ACG2071, MAC2233, QMB2100

Lecture Hours: 48.000

Course User Fee: \$98.00

ACG3113 Intermediate Financial Accounting II 3.000 credits

Continuation of ACG3103. Particular emphasis on analysis of balance sheet accounts through problem solving. Provides students with a more in-depth knowledge of Generally Accepted Accounting Principles (GAAP), including the advanced study of long-term assets, current and long-term liabilities and equities. Prerequisite: ACG3103.

Lecture Hours: 48.000

Prerequisite: ACG3103

ACG3343 Cost Accounting and Controls 3.000 credits

This course covers the preparation of accounting information for use in management decision making process. It contains information on budgeting, standard costing, direct costing, performance evaluation, and use of accounting information. Prerequisite: ACG2701

Lecture Hours: 48.000

Prerequisite: ACG2071

Course User Fee: \$98.00

ACG4401 Accounting Information Systems 3.000 credits

This is an information systems course for accounting students, and not a traditional accounting course, nor a traditional MIS course. The course aims to provide students with a broad conceptual knowledge of accounting information systems, the need for and control of accounting information systems, current topics in accounting information systems, and discussions of specific transaction-cycle based accounting information systems. The course seeks to examine the linkages between information systems and accounting, and it also provides an overview of how to store, retrieve, analyze, and control data using information systems. Students must pass the course with a grade of "C" or higher. Prerequisites: ACG3113 and ACG3343.

Lecture Hours: 48.000

Prerequisite: ACG3113 AND ACG3343

ACG4632 Auditing 3.000 credits

This course provides a sound conceptual foundation of basic auditing processes from the perspective of the public accounting profession. Professional standards, ethics, legal responsibilities, and utilization of technology are addressed. Principles and procedures of internal and public auditing are discussed, including professional standards, ethics, legal responsibilities, and the utilization of technology.

Lecture Hours: 48.000

Prerequisite: ACG3113

Course User Fee: \$98.00

**AER0605 Tesla Electronic System Technician
13.330 credits**

This course prepares the student to apply technical knowledge and skills to repair, service, and maintain Tesla vehicles. At the end of the course, the student will be able to diagnose malfunctions in and repair electrical, charging, penthouse, driver assist and infotainment systems and components. Corequisite: AER0606.

Clock Hours: 400.000

AER0606 Tesla Maintenance Technician 13.330 credits

This course prepares the student to apply technical knowledge and skills to repair, service, and maintain Tesla vehicles. The student will learn Tesla's products and diagnostic tools, as well as safety equipment and procedures. At the end of the course, the student will be able to diagnose malfunctions in and repair HVAC, brake, and chassis systems and components. Corequisite: AER0605.

Clock Hours: 400.000

AFR

AFR1101 The Foundation of the United States Air Force - Part 1 1.000 credits

This course is designed to show the potential Air Force officer, what role today's Air Force plays in defense of our nation, what role they can fill in today's Air Force, and finally what the Air Force offers them both today and AFROTC and later should they choose the Air Force as a profession after AFROTC.

Lecture Hours: 16.000

AFR1111 Introduction to the United States Air Force - Part 3 Sem Basic Air Force ROTC 1.000 credits

This course is designed to examine general aspects of air and space power through a historical perspective. We will cover the time period from the first balloons and dirigibles to the space-age global positioning systems to the Persian Gulf War.

Historical examples will be provided to extrapolate the development of Air Force capabilities and missions to demonstrate the evolution of what has become today's U.S. Air Force air and space power.

Lecture Hours: 32.000

AFR2130 The Foundation of the United States Air Force - Part 2 1.000 credits

This course is designed to show the potential Air Force officer, what role today's Air Force plays in defense of our nation, what role they can fill into today's Air Force, and finally what the Air Force offers them both today and AFROTC and later should they choose the Air Force as a profession after AFROTC.

Lecture Hours: 16.000

AFR2131 Introduction to the United States Air Force - Part 2 1.000 credits

This course is designed to examine general aspects of air and space power through a historical perspective. We will cover the time period from the first balloons and dirigibles to the space-age global positioning systems to the Persian Gulf War.

Historical examples will be provided to extrapolate the development of Air Force capabilities and missions to demonstrate the evolution of what has become today's U.S. Air Force air and space power.

Lecture Hours: 32.000

AMH

AMH2010 History of the US to 1877 3.000 credits

In this course, students will examine United States History from before European contact to 1877. Topics will include but are not limited to indigenous peoples, the European background, the colonial period, the American Revolution, the Articles of Confederation, the Constitution, issues within the new republic, sectionalism, manifest destiny, slavery, the American Civil War, and Reconstruction. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history from before European contact to 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas; values, and perceptions that have shaped United States history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2020 History of the US since 1877 3.000 credits

In this course, students will trace the history of the United States from the end of the reconstruction era to the contemporary era. Topics will include but are not limited to the rise of industrialization, the United States' emergence as an actor on the world stage, constitutional amendments and their impact, the progressive era, World War I, the great depression and new deal, World War II, the civil rights era, the cold war, and the United States since 1989. Student Learning Outcomes: students will describe the factual details of the substantive historical episodes under study; students will identify and analyze foundational developments that shaped American history since 1877 using critical thinking skills; students will demonstrate an understanding of the primary ideas, values, and perceptions that have shaped American history; and students will demonstrate competency in civic literacy.

Lecture Hours: 48.000

AMH2035 Recent American History Since 1945 3.000 credits

This course covers the internal development of the United States and the role of the U.S. in world affairs since World War II.

Lecture Hours: 48.000

AMH2079 History of South Florida 3.000 credits

A study of the history of South Florida (Lake Okeechobee south to Key West) including geological foundations exploration, settlement and contemporary cultural trends.

Lecture Hours: 48.000

AMH2090 History of Women in the United States 3.000 credits

This course will explore the history of women from precolonial times to the present, by examining how gender roles, class, and ethnicity defined women's experiences in the social, political, and economic contexts. It will also analyze how women shaped and contributed to historical events and movements within the United States.

Lecture Hours: 48.000

AMH2091 Afro-American History 3.000 credits

A survey, including the African background, of the Afro-Americans in the United States history, with emphasis on their economic, political, and cultural development.

Lecture Hours: 48.000

AML**AML2010 American Literature 1 3.000 credits**

American Literature from Colonial times to the Civil War. Prerequisites: ENC1101, ENC1102.

Lecture Hours: 48.000

Prerequisite: ENC1101

AML2020 American Literature 2 3.000 credits

American literature from the Civil War to the present. Prerequisites: ENC1101, ENC1102.

Lecture Hours: 48.000

Prerequisite: ENC1101

AML2600 African American Literature 3.000 credits

A study of African American historical and contemporary literature from various genres such as poetry, fiction, narratives, speeches, films, and drama. Prerequisite of ENC1101 with a Grade of "C" or higher, or equivalent.

Lecture Hours: 48.000

Prerequisite: ENC1101

AMT**AMT0044 Tools, Materials, and Processes 1 1.330 credits**

This course provides an introduction to the tools, hardware and material used in aircraft maintenance, repair, and safety. The student will be able to understand the information found in aircraft drawings, blueprints, charts and graphs Including stress materials' internal resistance and counterforce that opposes deformation.

Clock Hours: 40.000

AMT0045 Tools, Materials, and Process 2 1.330 credits

In this course, the student will learn the principles of corrosion and how to control it are studied and applied. This course provides experience in detecting, identifying, removal, and treatment of the various types of corrosion found on ferrous and non-ferrous metals. In addition, the student will learn about various forms of truss-types and how beams, struts, and bars resist deformation by applied loads.

Clock Hours: 40.000

AMT0046 Aircraft Materials, Hardware & Processes 2.260 credits

In this course, the student will learn about chemical processes, hazards of aviation solvents, lubricants, effects of corrosion on metals and aluminum, and measuring and sheet metal layout. Prerequisites:

AMT0044 and AMT0045.

Clock Hours: 68.000

Prerequisite: AMT0044 AND AMT0045

AMT0047 Applying the Design Process 1.760 credits

In this course, the student will learn to create aircraft drawings, blueprints, charts and graphs; test equipment, including non-destructive inspection and testing; and determine best practices with respect to sealants and epoxies. Prerequisites: AMT0044 and AMT0045.

Clock Hours: 53.000

Prerequisite: AMT0044 AND AMT0045

AMT0219 Aircraft Hydraulics & Aviation Mathematics 0.630 credits

In this course, the student will learn about aircraft hydraulic and pneumatic systems and advanced aviation mathematics. Prerequisites: AMT0044 and AMT0045.

Clock Hours: 19.000

Prerequisite: AMT0044 AND AMT0045

AMT0269 Aircraft Electrical Systems & Quality Control 2.130 credits

In this course, the student will learn the principles of basic electricity as it relates to aviation electrical systems; work with data, including computer-generated flow diagrams and spreadsheets; working with wiring & fiber optics; advanced blueprint reading with respect to performing actual tasks; and key elements of quality control. Prerequisites: AMT0044 and AMT0045.

Clock Hours: 64.000

Prerequisite: AMT0044 AND AMT0045

AMT0509 Composites and Capstone Project 0.400 credits

In this course, the student will learn about composite materials. In addition, the student will demonstrate their competence to analyze, design, develop, and test Aircraft Structural Assembly and Fabrication techniques. Prerequisites: AMT0044, 0045, 0046, 0047, 0219, 0269.

Clock Hours: 12.000

Prerequisite: AMT0044 AND AMT0045 AND AMT0046 AND AMT0047 AND AMT0269 AND AMT0219

AMT0949 On the Job Training (OJT) 22.220 credits

In this training course, the student will learn to perform the required tasks at the workplace. This course adds diversified experience and abilities while enriching the self-acquired proficiency of the student. Training is designed to minimize the margin of errors at the workplace.

Clock Hours: 666.670

ANT**ANT2000 Introduction to Anthropology 3.000 credits**

In this course, students will learn the foundations of anthropology as the study of human variation in its biological, social, and cultural dimensions. Students will learn about anthropological concepts, principles, and methodologies to understand and explore past and present human behavior. They will apply the anthropological approach to analyze issues pertaining to past and contemporary cultures, and develop intellectual skills and habits to understand behavioral, social, and cultural issues from multiple disciplinary perspectives. Student learning outcomes: students will explain scientific approaches to the study of human variation and human origins, including primatology, extinct and extant human cultures, language, and ethnicity; students will explain the origins of anthropology as a foundation discipline in the social sciences that examines the nature and definition of culture; students will apply anthropological concepts, principles, and methods to the scientific study of past and present human behavior; students will explain how anthropology incorporates multidisciplinary knowledge and perspectives; students will describe contemporary anthropological contributions.

Lecture Hours: 48.000

ANT2410 Introduction to Cultural Anthropology 3.000 credits

This course focuses on the study of human organization and culture by exploring similarities and differences among a variety of cultures. Students will apply theories and methods developed by anthropologists to understand complex human behaviors revealed in the cultural expression of race, ethnicity, language, economic systems, political systems, families and kinship, marriage, gender, religion, ecology, art, transnational migration, and the globalization of the world system.

Lecture Hours: 48.000

ANT2511 Introduction to Physical Anthropology 3.000 credits

This course introduces students to the origins of humanity from early primates, extinct hominins, and to modern people. Students in this course will learn the basics of evolutionary theory and genetics, investigate human evolutionary history through the fossil record, observe contemporary non-human primates, and apply this knowledge to a biocultural understanding of human variation, past and present.

Lecture Hours: 48.000

ARC

ARC1115 Architectural Communications 1 2.000 credits

Exercises in freehand drawing, sketching and linear perspective are designed to increase the student's awareness of the architectural environment. This is accomplished through a series of form studies of nature, architectural forms, and abstract elements of composition.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$25.00

ARC1126C Architectural Drawing 1 4.000 credits

This course exercises the visualization and drafting of architectural objects and construction conditions. Students will learn to draw orthographic projections, isometric and sectional drawings as an expression of architectural communication. Topics include drawing plans, elevations, details, schedules, and sections of wood frame and masonry structures.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC1128 Architecture Drawing 2 4.000 credits

A simulation of an actual architectural drafting room. The instructor issues preliminary design drawings from which the student prepares working drawings. The problems presented have varied materials and structural systems, differing occupancies, etc., offering a series of new experiences in architectural drawing. Prerequisite: ARC1126.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C AND ARC2171C

Course User Fee: \$50.00

ARC1301 Architectural Design 1 4.000 credits

Introductory course to architectural design, its scope, methods and vocabulary interfacing graphics and design as a means towards an awareness and understanding of basic organizational principles. Design concepts analyzed through graphical representation and modeling. Pre/Corequisite: ARC1115.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC1302 Architectural Design 2 4.000 credits

A continuation of ARC1301, emphasizing the application of ordering concepts, and aspects and determinants of form and space. An individual design process is developed by the student. Prerequisite: ARC1301; Pre/Corequisites: ARC1126, ARC2701.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

ARC1949 Co-op Work Experience 1: ARC 3.000 credits

This course is designed to provide training in the students' field of study through work experience. Students will learn to make connections between their internship experiences, academic coursework, and career goals. Students are graded on the basis of documentation of learning and goal achievement as reported by both student and employer.

Prerequisite: departmental approval.

Other Hours: 48.000

Recommended Preparation: 2.0 GPA, minimum of 6 credits in major or approved work experience, and departmental approval required.

ARC2053 Architectural Computer Applications 4.000 credits

Applications of software and computer languages in the fields of architecture, building construction and interior design.

Lecture Hours: 32.000, Lab Hours: 64.000

Corequisite:

Course User Fee: \$50.00

ARC2056 Computer Aided Architectural Presentation 4.000 credits

This course is designed to introduce the student to the concept of three-dimensional modeling and rendering for the purpose of producing an animated architectural presentation. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

ARC2171C Computer Aided Drafting 1 4.000 credits

Computer-aided drafting as it applies in the fields of architecture and interior design using office simulation. Emphasis is on the production of computer-aided drafting of working drawings involving different types of structure.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR ARC2461

Course User Fee: \$50.00

ARC2172 Computer Aided Drafting 2 4.000 credits

This course is designed for students with previous computer-aided design knowledge. Students will use both 2-dimensional and 3-dimensional CAD software to further develop their abilities to apply CAD techniques to the solution of architectural, engineering, and interior design problems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC2171C

Course User Fee: \$50.00

ARC2180C Introduction to 3D Building Modeling 4.000 credits

An introduction to 3D building modeling and generative drafting as it applies to the fields of architecture and interior design. Students will learn current practices in 3D building design by emphasizing the manipulation of commands used for modeling, drawing, editing, dimensioning, basic drawing management, and drawing output.

Prerequisites: ARC2172, CGS1060, MAC1105.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

ARC2201 Theory of Architecture 3.000 credits

An introduction to the meaning of Architecture to society, the foundation theories of architecture and an exposure to the ways and means of the creative process. Prerequisite: ARC1115.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2303 Architectural Design 3 5.000 credits

Integration of the natural and built environment with physiological, functional, organizational, spatial and environmental forces. Prerequisites: ARC1302 and 2461. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 96.000

Course User Fee: \$25.00

ARC2304 Architectural Design 4 5.000 credits

A continuation of ARC2303. Introduction to programming and design methods in architecture. Applications of building technology in the design process. Overview of computer applications in design. Prerequisite: ARC2303; Pre/Corequisites: ARC2053, ARC2681. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 96.000

Course User Fee: \$25.00

ARC2312C Introduction to Building Assembly Modeling 4.000 credits

Introduction to the principles of Building Assembly Modeling. Students will learn to explore a building as an assembly of architectural objects and subassemblies using virtual design and construction software. In addition, students will learn part modeling, assembly modeling, generative drafting, and general representational and modeling techniques. Prerequisites: ARC2172, CGS1060, and MAC1105.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

ARC2461 Architectural Materials and Construction 1 4.000 credits

An introduction to basic materials and methods of building construction. Emphasis is on wood, concrete, unit masonry, and light steel construction. Laboratory projects may include working drawings' interpretation, sketching construction details, or field trips to construction sites and fabricant plants. Designed primarily as the initial materials and methods course for architectural transfer students.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1126C OR BCN1251

Course User Fee: \$25.00

ARC2580 Architectural Structures 1 4.000 credits

A basic structural course designed primarily for Architectural and Construction majors, covering the fundamentals of statics. Timber design emphasized. Prerequisite: MAC1114; Pre/Corequisites: ARC1126, ARC2461, PHY2053, PHY2053L. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$25.00

ARC2681 Environmental Technology 3.000 credits

An introduction to technological aspects of building design, which relates to human comfort, safety, and building performance. Includes a survey of the fundamentals of water supply, waste lines, plumbing equipment, heat and air conditioning, solar applications, and electrical components and equipment in the design and construction of buildings. Prerequisite: ARC1126C.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2701 History of Architecture 1 3.000 credits

This course offers a survey of architecture from ancient times through the 18th century and includes works from the Western canon. Students will examine the integration of art forms, structural elements, and ornamental designs across world civilizations, analyzing their connections to broader aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societies throughout historical periods.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2702 History of Architecture 2 3.000 credits

This course provides a survey of architecture from the 19th century to the present and includes works from the Western canon and examining architectural developments. Students will analyze the integration of art forms, structural elements, and ornamental designs across civilizations, exploring their connections to human development, including art and history. Through this study, students will develop critical thinking skills and gain understanding of how architecture reflects and influences societal changes and cultural expressions throughout modern and contemporary periods. This is a writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARC2949 Co-Op Work Experience 2: ARC 3.000 credits

This is a course designed to continue training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Co-operative Education Office to obtain registration approval. Prerequisite: 2.0 minimum GPA, approval of Co-op Program Director, and completion of ARC1949.

Lecture Hours: 48.000

ARH**ARH1000 Art Appreciation 3.000 credits**

In this course, students will develop an appreciation of and the ability to think critically about culture and be provided with the tools to understand, analyze, and discuss works of visual art and material culture. Student learning outcomes: students will identify and describe terms, concepts, and methods used in the discipline of art history; students will apply terms, concepts, and methods used in the discipline of art history to works of visual art and material culture; students will identify and describe works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; students will analyze works of visual art and material culture in the works' cultural context, including works from or inspired by the western canon and other cultural traditions; and students will generate an analytical response to works of visual art and material culture in the works' cultural context.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARH2050 Art History 1 3.000 credits

This course offers a survey of visual arts from prehistory to 800 C.E. and includes works from the western canon. Students will examine artistic expressions across civilizations, analyzing their connections to historical contexts and aspects of human development. Through this exploration, students will develop critical thinking skills and gain understanding of how visual arts reflect and influence societal developments, ideas, and values throughout early history.

Lecture Hours: 48.000

Course User Fee: \$10.00

ARH2051 Art History 2 3.000 credits

In this course, students will embark on a world survey of the visual arts from 800 to 1850 A.D., exploring various artistic movements, styles, and cultural contexts. The course will delve into the rich and diverse artistic traditions from different regions, including works from the Western canon. Writing intensive course.

Lecture Hours: 48.000

Course User Fee: \$5.00

ARH2740 Cinema Appreciation 3.000 credits

An analysis of the cinema as an important social force and an artistic medium. Significant American, British, and foreign language films will be shown and discussed. Writing intensive course.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: HUM1020

Course User Fee: \$5.00

ARH2833 Museum and Gallery Exhibition Design 3.000 credits

The course examines the historical context of museum exhibitions, tracing their evolution from traditional displays to contemporary immersive experiences. Students will explore the key milestones and influential figures that have shaped the field, gaining insights into the changing trends and philosophies that guide exhibition practices today.

Lecture Hours: 32.000, Lab Hours: 32.000

ARH2857 Introduction to Museum Studies 3.000 credits

This is a foundation course that offers a practical introduction to the nature, context, and operations of museums with the intention of providing practical examples and indicators of good practices. This course introduces students to the history and theory of museums and museum practices, museum administration, exhibition planning, museum education, and museum careers. In addition to the comprehensive overview of museum studies, this course also delves into the influence of the Western canon on museum curation and interpretation, providing students with a well-rounded understanding of the cultural and historical significance of the artifacts and artworks displayed in museums.

Lecture Hours: 48.000

ARH2990 Museum and Gallery Co-Op Work Experience 3.000 credits

This course offers an exceptional opportunity for students to further enhance their skills and knowledge in the field of Art and related studies through practical work experience. The primary focus of this course is to provide students with hands-on training that complements their academic studies.

Internship Hours: 144.000

ART**ART1201C Basic Design 3.000 - 4.000 credits**

This introductory course is designed to familiarize students with the basic elements and principles of design and to give hands-on opportunity to transform visual and experiential information into basic forms. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals.

Lecture Hours: 96.000

ART1202C Two-Dimensional Design 3.000 - 4.000 credits

This course is designed to give students an understanding of advanced concepts of two-dimensional design and to give hands on opportunity to transform visual and experiential information into two-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals.

Lecture Hours: 96.000

ART1203C Three-Dimensional Design 4.000 credits

This course is designed to give students an understanding of the concepts of three-dimensional design and to provide hands-on opportunity to transform visual and experiential information into three-dimensional form. Creative individual thinking and image making and successful problem solving both aesthetically and technically are ultimate goals. Self-evaluation and safety skills will also figure prominently. Prerequisite: ART1202C.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ART1202C

Course User Fee: \$40.00

ART1205C Color and Composition 1 3.000 - 4.000 credits

This course is a studio art course that is focused on learning the theory and practice of color mixing and compositional arrangement. The course will examine the various interactions of color and their creative application so that the student may use color more effectively in fine arts and applied design.

Lecture Hours: 16.000 - 32.000, Lab Hours: 64.000

ART1300C Drawing 4.000 credits

Basic problems in freehand drawing, including perspective, still-life and landscape. Emphasis is on developing a sense of structure through line, form and texture.

Lecture Hours: 32.000, Lab Hours: 64.000

ART1330C Figure Drawing 4.000 credits

Drawing and painting from the live model with emphasis on structure, movement and expression.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$80.00

ART1803C Workshop for ART Research and Practice: Studio 6.000 credits

Small enrollment sections. Interdisciplinary, team teaching, introductory studio experience in a wide variety of media. In-depth exploration of creative processes, principles of artistic integrity, and the nature of artistic meaning. Concepts in two-dimensional and three-dimensional design will be explored through studio experience.

Lab Hours: 192.000

ART2301C Drawing 2 3.000 - 4.000 credits

In this course students will execute drawings in various media, working with the figure or from various assigned drawing problems which are more complex and incorporate other design possibilities. Assignments in drawing will go beyond the realistic or literal and will incorporate media not usually used such as painting, collage, mixed media, and found objects.

Lecture Hours: 16.000, Lab Hours: 64.000

ART2302C Advanced Drawing 3.000 - 4.000 credits

An exploration of varied approaches to drawing through studio problems. May be repeated for credit. Prerequisites: ART1300C, ART1330C.

Lecture Hours: 16.000, Lab Hours: 64.000

ART2400C Printmaking 1 3.000 - 4.000 credits

Basic techniques of printmaking including relief prints (woodcut and wood engraving), intaglio (dry point and etching) and lithography. Prerequisite: ART1202C or ART1300C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$20.00

ART2401C Printmaking 2 3.000 - 4.000 credits

Advanced techniques in printmaking. Prerequisite: ART2400C

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$20.00

ART2406C Advanced Printmaking 3.000 - 4.000 credits

Individualized instruction on printmaking concepts specifically oriented toward the student's aesthetic concerns. May be repeated for credit. Prerequisites: ART2400C, ART2401C.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: ART2400C, ART2401C

Course User Fee: \$40.00

ART2500C Painting 1 3.000 - 4.000 credits

Study problems in painting involving contemporary styles, techniques and materials. Prerequisite: ART1202C or 1300C.

Lab Hours: 96.000

ART2501C Painting 2 3.000 - 4.000 credits

Advanced techniques in painting. Prerequisite: ART2500C.

Lecture Hours: 16.000 - 32.000, Lab Hours: 64.000

ART2502C Advanced Painting 3.000 - 4.000 credits

Individualized instruction in painting concepts specifically oriented to the student's aesthetic concerns. May be repeated for credit. Prerequisites: ART2500C, ART2501C.

Lecture Hours: 16.000, Lab Hours: 64.000

ART2600C Computer Art 3.000 - 4.000 credits

This course is an introduction to basic theory and skill techniques of visual communications using computers. It gives students a basic understanding of technical devices for the electronic production of visual images. Prerequisites: ART1201C, ART1300C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$70.00

ART2601C Intermediate Computer Art 3.000 - 4.000 credits

An intermediate computer art course focusing on the integration of computer technology with traditional design and fine art media such as illustration, painting, printmaking, and photography. Prerequisite: ART2600C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$70.00

ART2602C Advanced Computer Art 4.000 credits

An advanced computer art class that focuses on new and emerging computer technology utilizing multiple platforms to produce advanced computer art portfolio assignments in illustration, fine art, 2D animation, and digital photography.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$70.00

ART2701C Sculpture 1 3.000 - 4.000 credits

An introduction to sculpting techniques and materials. Prerequisite: ART1202C or ART1300C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$50.00

ART2702C Sculpture 2 3.000 - 4.000 credits

Advanced sculpturing techniques. Prerequisite: ART2701C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$50.00

ART2703C Advanced Sculpture 3.000 - 4.000 credits

Individualized instruction in sculptural concepts specifically oriented to the student's aesthetic concerns. May be repeated for credit. Prerequisites: ART2701C, ART2702C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$50.00

ART2750C Ceramics 1 3.000 - 4.000 credits

Basic techniques in pottery design - forming, decorating, glazing and firing. Prerequisite: ART1202C or ART1300C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$50.00

ART2751C Ceramics 2 3.000 - 4.000 credits

Advanced techniques in painting. Prerequisite: ART2500C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$50.00

ART2771C Advanced Ceramics 3.000 - 4.000 credits

Advanced work in ceramics. Emphasis placed on individual concepts and their application in ceramics. May be repeated for credit. Prerequisites: ART2750C, ART2751C.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$50.00

ART2802C Visual Arts Workshop 1.000 - 4.000 credits

Special Studio Topics including methods, materials and theory related to specific studio processes. Permission required from department chairperson. May be repeated for credit.

Lecture Hours: 16.000, Lab Hours: 32.000

ART2950 Portfolio Preparation - ART 3.000 credits

Provides students with knowledge and skills to compile a portfolio which prepares them for a college or professional career. Course content focuses on individual development through the use of varied media and styles. Emphasis is placed on selection, evaluation, and presentation. May be repeated for credit.

Lab Hours: 96.000

ASC**ASC1210 Aviation Meteorology 3.000 credits**

This is a core aviation course. The student will be prepared to understand weather and environmental issues in commercial aviation. Topics covered will be atmospheric phenomena relating to aircraft operations, the analysis and use of weather data as presented by the U.S. National Weather Service. Prerequisite: ATT1100 or equivalent; Corequisite: ATT2110 or equivalent.

Lecture Hours: 48.000

ASC1550 Aerodynamics 3.000 credits

This is a basic course in aerodynamics. Students will analyze the physics of flight and the application of basic aerodynamics to both airframe and power plant as preparation for the requirements of commercial aviation.

Lecture Hours: 48.000

ASC1560 Introduction to Unmanned Aerial Vehicles and Systems (UAVs) 3.000 credits

This course introduces the fundamentals of unmanned aerial vehicles and systems, covering UAV types, components, principles of flight, regulations, and safe operational practices. Students will explore industry applications and gain foundational knowledge for further study in UAV technology.

Lecture Hours: 48.000

Course User Fee: \$59.33

ASC1563 UAS Applications in Aerial Photography 3.000 credits

Students learn techniques and best practices for capturing aerial images using unmanned aerial systems. The course covers camera operation, flight planning for photography missions, image processing, and practical applications in various industries.

Lecture Hours: 48.000

Course User Fee: \$59.33

ASC1610 Aircraft Engines and Structure Theory 3.000 credits

This is a foundation course in aircraft engines and structure. Students will learn the elements of aircraft engines, engine theory, construction, systems, operating procedures, performance diagnosis, and aircraft structures.

Lecture Hours: 48.000

ASC2320 Aviation Laws and Regulations 3.000 credits

Insight pertinent to federal governing bodies, and current local, federal and international laws forming the present structure of aviation law.

Lecture Hours: 48.000

ASC2470 Physiology-Psychology of Flight 3.000 credits

This is an introductory course in the physiology and psychology of flight. Students will learn aero-medical facts of significance to pilots, including causes, symptoms, prevention and emergency treatment of ailments common to the aviation environment through a basic understanding of a person's normal functioning. Cabin pressurization, communications, decompression sickness, hyperventilation, hypoxia, self-imposed stresses, spatial disorientation and vision are examined.

Lecture Hours: 48.000

ASC2670 Aircraft Systems 3.000 credits

As preparation for commercial aviation requirements, this course is concerned with a detailed study of aircraft systems, their various sources of basic power and the functional application of mechanisms operated by these systems. Prerequisite: ASC1610.
Lecture Hours: 48.000

ASL**ASL1140C American Sign Language 1 4.000 credits**

This course provides introductory information on the linguistics of American Sign Language and approximately 500 sign concepts. The course includes lecture, discussion and lab practice.
Lecture Hours: 64.000

ASL1150C American Sign Language 2 4.000 credits

This course provides continued instruction in the linguistic principles of American Sign Language and an additional 500 sign concepts. The course includes lecture, discussion and lab practice which is conducted in ASL. Prerequisite: ASL1140C.
Lecture Hours: 64.000

ASL2160C American Sign Language 3 4.000 credits

This course provides linguistic principles of American Sign Language at the intermediate level and an additional 500 sign concepts. Lecture, discussion and lab practice are included. Students have increased opportunities for interaction with members of the deaf community. Increasingly, class sessions are conducted in ASL. Prerequisite: ASL1150C; Pre/Corequisite: ASL2210.
Lecture Hours: 64.000

ASL2200C American Sign Language 4 4.000 credits

This course provides linguistic principles of American Sign Language at the advanced level and an additional 500 sign concepts including idioms used in ASL. Lecture, discussion and lab practice are included. Class sessions are conducted predominately in ASL. Prerequisite: ASL 2160C.
Lecture Hours: 64.000

ASL2210 ASL Conversational Skills 3.000 credits

The course provides opportunities to utilize all previously acquired knowledge and skill in ASL, including vocabulary and all components and linguistics principles of the language. Emphasis is placed on the more difficult aspects of ASL, such as role shifting for relating stories and the use of classifier predicates for providing visual descriptions. Class sessions are conducted entirely without voice. Prerequisites: ASL1150C, ASL2160C.
Lecture Hours: 48.000

ASL2220 Receptive Skills Development 3.000 credits

The course will focus on increasing the student's receptive understanding of signed communications. Examples of American Sign Language (ASL) will be presented via videotapes and live interactions with deaf persons. Students will identify all the components and linguistic features of ASL and will provide appropriate English translations either in speech (paraphrasing) or in written form. Prerequisite: ASL 1150C.
Lecture Hours: 48.000
Prerequisite: ASL1150C

ASL2400 Linguistics of American Sign Language 3.000 credits

The course is designed for persons who already have an understanding of ASL principles. Provides an overview of the various systems of manual communication used in the U.S. including PSE, Cued Speech and signed English. Prerequisite: ASL 2160C.
Lecture Hours: 48.000

ASL2430 Manual Alphabet Skills Development 3.000 credits

The course content focuses on acquiring both expressive and receptive skill in the manual alphabet of American Sign Language, commonly known as fingerspelling. A performance test is given at the beginning of the course to determine existing competency. Prerequisites: ASL 1140C, 1150C.
Lecture Hours: 48.000
Prerequisite: ASL1140C, ASL1150C

ASL2510 Deaf Culture And Community 3.000 credits

The course provides an in-depth study of the lives and experiences of deaf and hard of hearing persons and examines perceptions of belonging to a unique cultural group. Cultural characteristics are examined alongside the impact of hearing loss on one's family, friends and employment. Multicultural issues and the impact of hearing loss are examined within various ethnic groups. Societal attitudes regarding disability in general, and hearing loss and communication difficulties in particular, are addressed.

Lecture Hours: 48.000

AST**AST1002 Descriptive Astronomy 3.000 credits**

This course provides a comprehensive look at modern astronomy, emphasizing the use of the scientific method and the application of physical laws to understand the universe including earth and its environment. Throughout this course, students will develop the ability to discern scientific knowledge from non-scientific information by using critical thinking. Student learning outcomes: students will define terms used to measure and describe the universe; students will explain the processes involved in the formation and evolution of celestial bodies over astronomical time according to different models and theories; students will describe how scientific theories evolve in response to new observations and critically evaluate their impact on society; students will formulate empirically testable hypotheses derived from the study of physical processes and phenomena; students will apply logical reasoning skills through scientific criticism and argument to separate science from non-science; and students will gather and analyze astronomical data and communicate results in graphic and written forms.

Lecture Hours: 48.000

Course User Fee: \$10.00

AST1002L Descriptive Astronomy Laboratory 1.000 credits

This is a laboratory course available to students taking the introductory Astronomy course AST 1002. Students will learn to obtain astronomically relevant scientific information by performing experiments, exercises or observations. They will learn to measure, collect, and analyze scientific data, to do calculations with the data, and to report their results.

Lab Hours: 32.000

Corequisite: AST1002

Course User Fee: \$20.00

ATE**ATE1110 Animal Anatomy 3.000 credits**

This course explores the physical and functional phenomena that interact to sustain life in animals. The student will learn the relationships of all of the systems in domestic animals, such as the osseous apparatus, the respiratory, digestive, genitourinary, endocrine, and nervous systems. The student will also be introduced to the descriptive and topographical terms needed to communicate with the professional staff.

Lecture Hours: 48.000

Prerequisite: BSC1005, BSC1005L, ENC1101,

Corequisite: ATE1110L, ATE1211, ATE1650L

ATE1110L Animal Anatomy and Physiology Lab 1.000 credits

This course will complete the coverage and understanding of the physiological and anatomical relationships required for further development as a veterinary technician. This course will correlate with lecture material learned in the Animal anatomy and Animal Physiology lecture courses. Anatomical dissection, necropsy, examination of live animals will be used as well as the study of radiographs, skeleton models and histological sections.

Lab Hours: 32.000

Corequisite: ATE1110, ATE1211, ATE1650L

Course User Fee: \$50.00

ATE1211 Animal Physiology 3.000 credits

This course is designed to explore the terminology related to animal physiology, in addition to all aspects of the functions of systems in small and large animals.

Lecture Hours: 48.000

Corequisite: ATE1110, ATE1110L, ATE1650L

ATE1630 Pharmacology for Veterinary Technicians 2.000 credits

This introductory course reviews drug classification and office procedures/management. Students will learn methods of calculating appropriate drug dosage, routes of administration, and evaluation of drug efficacy as well as office procedures used in veterinary hospital management.

Lecture Hours: 32.000

ATE1650L Introduction to Clinical Practice 1 1.000 credits

This introductory course is designed to acquaint the student with skills associated with veterinary clinical practice. Students will learn basic office, laboratory and nursing skills, including hospital/office management, restraint, history taking, examination room techniques, administration of medication, basic parasitology, and basic clinical pathology procedures.

Lab Hours: 32.000

Corequisite: ATE1110, ATE1110L, ATE1211

ATE1940 Veterinary Clinical Experience 1 3.000 credits

This entry clinical course provides supervised clinical experience in a veterinary facility. Students will learn and reinforce competencies in clinical laboratory procedures, venipuncture techniques, physical examination of patients, administration of intramuscular and subcutaneous injections and exam room protocol.

Clinical Hours: 144.000

Course User Fee: \$28.00

ATE1941 Veterinary Clinical Experience 2 3.000 credits

This course consists of supervised clinical experience in the veterinary workplace. Students will learn to enhance the competencies from ATE 1940 Veterinary Clinical Experience 1 while adding application of classroom knowledge in pharmacology, clinical laboratory procedures, and radiology.

Clinical Hours: 144.000

ATE2050L Animal Nursing and Medicine Laboratory 2 2.000 credits

The student will practice training a dog, and applying corrections for common behavioral problems. Clinical training in a small animal necropsy is also presented. Prerequisites: ATE1110, ATE2631, ATE2655L; Corequisite: ATE2612.

Lab Hours: 64.000

Prerequisite: ATE1110, ATE2631, ATE2655L,

Corequisite: ATE2612

Course User Fee: \$50.00

ATE2611 Animal Medicine 1 3.000 credits

This course is designed to acquaint the student with anesthesiology, sepsis and general surgical nursing care, essentials in pharmacy and pharmacology, and concepts in microbiology, virology and immunology. Prerequisites: ATE1110, ATE1211; Corequisites: ATE2661, ATE2942, ATE2631, ATE2655L.

Lecture Hours: 48.000

ATE2612 Small Animal Nursing 3.000 credits

A study of the basic concepts of nutrition, obstetrics, and pediatric care, as well as the important aspects regarding zoonotic diseases, public health, and animal behavior. The student will also be introduced to alternative medicine, including holistic concepts, homeopathic, acupuncture, chiropractic and other emerging specialties. Prerequisites: ATE1110, ATE2611, ATE2631, ATE2655L; Corequisite: ATE2050L.

Lecture Hours: 48.000

Prerequisite: ATE1110, ATE2611, ATE2631, ATE2655L, Corequisite: ATE2050L

ATE2614 Animal Medicine 2 3.000 credits

This course will explore general pathology, causes and nature of disease, toxicology, and an overview of the pathologies of the major systems, as well as immunity, disease prevention, common vaccinations and diseases relating to small animals. Prerequisites: ATE1110, ATE2611.

Lecture Hours: 48.000

Prerequisite: ATE1110, ATE2611

ATE2631 Small Animal Nursing 1 3.000 credits

The student will master the technical skills of medicating animals and the taking and processing of radiographs. This course also covers general care, including grooming and bathing, feeding and watering, nail trimming, ear cleaning, anal sac expression, and determination of vital signs.

Prerequisites: ATE1110, ATE1211; Corequisites: ATE2611, ATE2655L.

Lecture Hours: 48.000

Prerequisite: ATE1110, ATE1211, Corequisite: ATE2611, ATE2655L

ATE2636 Large Animal Clinic and Nursing Skills 2.000 credits

This course is designed to acquaint the student with the fundamentals of large animal herd management, reproductive physiology and lactation physiology. Aspects of equine, bovine, ovine and porcine husbandry will be included.

Lecture Hours: 32.000

Prerequisite: ATE1110, ATE1110L, ATE1211, Corequisite: ATE2636L

ATE2636L Large Animal Clinic and Nursing Skills Laboratory 1.000 credits

This course is designed to acquaint the student with the fundamentals of large animal husbandry, herd health management, preventive medicine, animal restraint and nutrition as it relates to the bovine, equine, porcine and caprine species. Techniques discussed in the Large Animal Clinic and Nursing skills course such as venipuncture, injections and administration of other oral medications will be reviewed and demonstrated. One laboratory session will be devoted to poultry science.

Lab Hours: 32.000

Corequisite: ATE2636

Course User Fee: \$50.00

ATE2638 Animal Lab Procedures 1 3.000 credits

This course is designed to introduce the veterinary technician to common parasites and their life cycles seen in routine veterinary practice. Also, hematology and the kinetics of the hematopoietic system are discussed with emphasis on normal blood smears and common changes seen during disease stages of domestic animals.

Lecture Hours: 48.000

Prerequisite: ATE1110, ATE1110L, ATE1211,

Corequisite: ATE2638L

ATE2638L Animal Lab Procedures 1 Laboratory 2.000 credits

This course is designed to acquaint the student with clinical laboratory procedures covered in the Animal Laboratory Procedures 1 course. Areas of emphasis include hematology, coagulation and parasitology as well as general laboratory etiquette.

Lab Hours: 64.000

Corequisite: ATE2638

Course User Fee: \$100.00

ATE2639 Animal Lab Procedures 2 3.000 credits

This course serves as a continuation of Animal Laboratory Procedures 1 and covers immunology, liver function and diagnostic testing for liver abnormalities, kidney function and testing used in disease states, urinalysis, pancreatic evaluation; normal and abnormal exfoliative cytology; and the evaluation of endocrine disorders. It also will include principles of serological testing and microbiological methods and protocols.

Prerequisites: ATE2638, ATE2638L; Corequisite: ATE2639L.

Lecture Hours: 48.000

Prerequisite: ATE2638, ATE2638L, Corequisite: ATE2639L

ATE2639L Animal Lab Procedures 2 Laboratory 2.000 credits

This course provides experience in the practical applications discussed in Animal Laboratory procedures 2. It will also include principles of serological testing and microbiological methods and protocols as well as dentistry for the veterinary technician. Prerequisites: ATE2638, ATE2638L; Corequisite: ATE2639.

Lab Hours: 64.000

Prerequisite: ATE2638, ATE2638L, Corequisite: ATE2639

Course User Fee: \$75.00

ATE2652L Introduction to Clinical Practice 2 1.000 credits

The clinical application of basic veterinary radiology and surgical nursing skills will be the primary focus of this practicum. The student will demonstrate skills under supervised instruction.

Lab Hours: 32.000

Prerequisite: ATE1110, ATE1110L, ATE1650L

ATE2655L Animal Nursing and Medicine Laboratory 1 2.000 credits

This course is designed to acquaint the student with exam room and restraining techniques, anesthesia and surgical protocols and diagnostic imaging procedures used in veterinary hospitals.

Lab Hours: 64.000

Course User Fee: \$75.00

ATE2661 Large Animal Diseases 1.000 credits

This course is designed to acquaint the student with the fundamentals of preventative medicine and with the common disease seen in the large animal species. Aspects of equine, bovine, ovine and porcine diseases and common treatments will be emphasized. Prerequisites: ATE1110, ATE2636, ATE2636L; Corequisite: ATE2611.

Lecture Hours: 16.000

Prerequisite: ATE1110, ATE2636, ATE2636L,

Corequisite: ATE2611

ATE2671C Lab Animal Medicine 2.000 credits

This foundation course provides instruction on laboratory animal care. Students will learn the technical aspects of laboratory animal care, including restraint and handling, common diseases and nutrition. The animals studied include rabbits, rats, mice, guinea pigs, hamsters and primates.

Lecture Hours: 16.000, Lab Hours: 32.000

ATE2710 Animal Emergency Medicine 2.000 credits

This course is designed to acquaint the student with fundamentals of emergency veterinary medicine, including veterinary first aid, toxicology and specialized medical techniques and procedures.

Prerequisites: ATE1110, ATE1211; Corequisites: ATE2611, ATE2631, ATE2655L.

Lecture Hours: 32.000

Prerequisite: ATE1110, ATE1211, Corequisite: ATE2611, ATE2631, ATE2655L

ATE2722C Avian and Exotic Pet Medicine 2.000 credits

This course is designed to acquaint students with the medical care associated with exotic animals and avians. Students will learn types of species that may be encountered in a practice and their associated care techniques.

Lecture Hours: 16.000, Lab Hours: 32.000

ATE2942 Veterinary Clinical Experience 3 4.000 credits

This course provides clinical experience to the student, under the supervision of a veterinarian. Students will enhance the competencies learned in ATE1940 - Clinical Experience 1 and ATE1941 - Veterinary Clinical Experience 2 and master skills associated with advanced veterinary technology practice. Prerequisite: ATE1941.

Clinical Hours: 192.000

Prerequisite: ATE1941

Course User Fee: \$35.00

ATF**ATF1100 Private Pilot Flight 3.000 credits**

This course provides flight training in the areas required to safely perform the duties of a Private Pilot. This will fulfill the requirements as outlined in FAR part 141 and as presented in the Jeppesen Sanderson Private Pilot syllabus. Upon satisfactory completion of this course, the FAA written exam and the practical exam, the applicant will receive an FAA Private Pilot Certificate. A Class 1 FAA Medical Certificate is required. Special fee.

Lecture Hours: 48.000

Corequisite: ASC1210, ATT1100

Course User Fee: \$15094.00

ATF1100L Private Pilot Flight Accelerated 3.000 credits

This course provides flight training in the areas required to safely perform the duties of a private pilot. This course is to be completed in less than 16 weeks using "total immersion" approach to training to help lower the attrition rates for students. It fulfills the requirements for private pilot certification outlined in part 141 of the Federal Aviation Regulations as presented in the Jeppesen Sanderson Private Pilot Syllabus. Upon satisfactory completion of this course and the Federal Aviation Administration (FAA) knowledge and practical exams, the applicant will receive an FAA private pilot certificate. Special Fee. Note: This course is not approved for Veterans seeking to use Veterans Educational Benefits.

Lecture Hours: 48.000

Prerequisite: ATT1100, FAA First Class Medical Certificate.

Course User Fee: \$15094.00

ATF1601L Flight Orientation Simulator Lab 1.000 credits

This course will provide the student with an introduction to the environment of operating an aircraft from a pilot's point of view. It is designed to provide this knowledge to those students such as Air Traffic Controllers and Aviation Administration Students who have no piloting experience. Special fee.

Lab Hours: 32.000

Course User Fee: \$200.00

ATF2210 Commercial Pilot Flight 3.000 credits

This course provides pilot training required to allow the student to safely conduct a flight as a Commercial Pilot. The training will be conducted in accordance with FAR Part 141 and in concert with stages 5 and 6 of the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course, the FAA written exam, and FAA practical exam, the student will receive an FAA Commercial Rating. Minimum approved FAA Part 141 course hours include 120 hours of flight, any additional training required beyond the FAA minimum is the financial responsibility of the student. Prerequisite: FAA Instrument Rating; Corequisite: ATT2110.

Lecture Hours: 48.000

ATF2214L Commercial Pilot Flight Accelerated 3.000 credits

This accelerated course of instruction provides training required to allow the student to safely conduct flight as a Commercial Pilot. This course is expected to be completed in less than 4 months. The training will be conducted in accordance with the Codes of Federal Regulations (CFR) Part 141. This training will be completed utilizing the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course, the FAA knowledge test, and FAA practical test, the student will be awarded an FAA Commercial Pilot certificate. A 1st Class Medical Certificate with Instrument Rating is required. Minimum approved FAA CFR Part 141 course hours include 120 hours of flight. Special Fee.

Lecture Hours: 48.000

Recommended Preparation: Required: FAA Private Pilot Certificate and FAA 2nd Class Medical Certificate (1st Class preferred).

Course User Fee: \$23990.50

ATF2305 Instrument Pilot Flight 3.000 credits

This course provides the flight training required to safely conduct flights as an instrument-rated pilot. The training is conducted in accordance with Part 141 of the Federal Aviation Regulations as outlined in stages 1 through 4 of the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course and the Federal Aviation Administration (FAA) knowledge and practical exams, the applicant will receive an FAA instrument rating. Prerequisites: ATF 1100; FAA Private Pilot Certificate; Corequisites: ATT 2120; current FAA Medical Certificate. Special fee.

Lecture Hours: 48.000

Prerequisite: ATF1100and FAA Private Pilot Certificate, and Current FAA Medical Certificate.,
Corequisite: ATT2120

ATF2305L Instrument Pilot Flight Accelerated 3.000 credits

This accelerated course provides the flight training required to safely conduct flights as an instrument rated pilot. This course is to be completed in less than 16-weeks. The training is conducted in accordance with Codes of Federal Regulations (CFR) Part 141 of the Federal Aviation Regulations as outlined in the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course and the Federal Aviation Administration (FAA) knowledge test and practical test, the applicant will receive an FAA Instrument Rating. Minimum approved FAA CFR Part 141 course hours include 35 hours of flight.

Lecture Hours: 48.000

Prerequisite: FAA Private Pilot Certificate,
Corequisite: ATT2120and Current FAA 3rd Class Medical (1st Class preferred)

Course User Fee: \$12686.00

ATF2310L Instrument Pilot Flight Accelerated 3.000 credits

This accelerated course provides the flight training required to safely conduct flights as an instrument rated pilot. This course is to be completed in less than 16-weeks. The training is conducted in accordance with Codes of Federal Regulations (CFR) Part 141 of the Federal Aviation Regulations as outlined in the Jeppesen Sanderson Instrument/Commercial Syllabus. Upon satisfactory completion of this course and the Federal Aviation Administration (FAA) knowledge test and practical test, the applicant will receive an FAA Instrument Rating. Minimum approved FAA CFR Part 141 course hours include 35 hours of flight.Special Fee.

Lecture Hours: 48.000

Prerequisite: Required: FAA Private Pilot Certificate,
Corequisite: ATT2120, Current FAA 3rd Class Medical (1st Class preferred)

Course User Fee: \$12686.00

ATF2400 Multi-Engine Pilot Flight 3.000 credits

This course provides the flight training required to prepare the student to safely conduct flight as a Multi-Engine Pilot. Upon satisfactory completion of this course, and the FAA oral and practical exams, the student will receive an FAA Multi-Engine Rating. Minimum approved FAA Part 141 course hours include 20 hours of flight. Cost per hour Wet/Dual at Wayman Flight School \$300/\$359. Prerequisites: FAA Private Pilot Certificate; Corequisite: ATT2133.

Lecture Hours: 48.000

Recommended Preparation: Required: FAA Private Pilot or Commercial Pilot Certificate. AND
Corequisite: ATT2133

Course User Fee: \$8945.75

ATF2501 Flight Instructor-Flight Training 3.000 credits

This course provides flight training for the student to develop the ability to analyze the performance of private and commercial flight maneuvers from the right seat of a training aircraft, in compliance with the Federal Aviation Administration Certified Flight Instructor Certificate. Prerequisite: ATF2300; Corequisites: ATT2131, ATF2501L. Special fee. Lecture Hours: 48.000
Prerequisite: FAA Commercial Certificate with an Instrument Rating or ATP, ATT2133

ATF2501L Flight Instructor Laboratory 1.000 credits

Provides the student with internship teaching experience based upon the principles of flight instruction learned in ATT2131 and ATF 2501. Students will learn to develop lesson plans and how to communicate effectively using instructional materials Prerequisite: ATF2300; Corequisites: ATT2131, ATF2501.
Lab Hours: 32.000
Prerequisite: ATF2300 AND Corequisite: ATF2501 AND ATT2131
Course User Fee: \$8703.50

ATT**ATT1100 Private Pilot Theory 3.000 credits**

This course introduces basic subjects pertaining to pilot knowledge including basic aircraft systems, aircraft operation and performance, aerodynamic principles, human factors, and aeronautical decision making. This Course can be instructed as Part 61 or Part 141.
Lecture Hours: 48.000

ATT2110 Commercial Pilot Theory 3 3.000 credits

This 35-hour course provides students with the aeronautical knowledge required to act as Commercial Pilot. Students will prepare for the FAA Commercial Written Exam. Private Pilot Certificate with Instrument Rating required.
Lecture Hours: 48.000
Prerequisite: ATF2305

ATT2120 Instruments Pilot Theory 4.000 credits

This 30-hour course introduces basic theories of instrument pilot operations to prepare students for the FAA Instrument Written Exam. Students will acquire aeronautical knowledge required to act as an Instrument Rated Pilot. It will prepare the students for the FAA Instrument Written Exam. Private Pilot Certificate required.
Lecture Hours: 64.000

ATT2131 Flight Instructor Theory 3.000 credits

Provides the student ground instruction to obtain the necessary aeronautical knowledge, to meet the FAA written standards for the Certified Flight Instructors Certificate. Preparation for the written exam is included in the course content. Prerequisite: ATF2305.
Lecture Hours: 48.000

ATT2133 Multi-Engine Pilot Theory 2.000 credits

This course introduces basic theories of multi-engine pilot operations to prepare students for the FAA Multi-Engine oral and practical exams. Students will acquire the aeronautical knowledge required to act as a multi-engine rated pilot.
Lecture Hours: 32.000

ATT2660 Regional Airline Operations 3.000 credits

This course provides theoretical instruction and practical experience in flight planning inclusive of navigation, weather, fuel management, flight and communication procedures, aircraft performance, crew coordination and simulator procedures. Utilizing flight systems and automated panels, the course additionally provides practical instruction in the operation of aircraft systems. Prerequisites: ASC1610, ATT2110, ATT2120
Lecture Hours: 48.000
Prerequisite: ASC1610, ATT2110, ATT2120

ATT2820 Air Traffic Control 3.000 credits

The basic elements of air traffic control operations, providing the necessary foundation for successful completion of the Air Traffic Control Basic Certification Examination. Prerequisite: sophomore standing in major program.
Lecture Hours: 48.000
Prerequisite: Sophomore standing in major program.

ATT2821 Air Traffic Control (ATC) Radar 3.000 credits

This course will provide the student with a fundamental knowledge of air traffic control practices, policies and procedures as they relate to the specifics of the controller function in an air traffic radar operating environment, with air traffic controllers utilizing the radar for traffic separation. The liberal use of the figures and example phraseology assist the student in achieving an overall use of understanding of the air traffic control system. A radar air traffic control simulator is utilized to provide realistic training exercises for the students. Prerequisite: ASC1210.
Lecture Hours: 48.000
Prerequisite: ASC1210
Course User Fee: \$300.00

ATT2822 VFR Tower Operations 3.000 credits

This course expands the knowledge attained from ATT 2820 and is designed to further develop the aviation students' skills in the ATC environment. Emphasis is placed on the duties and responsibilities of operational positions in local, ground, flight data, and coordination. Students will also learn the FAA regulations which govern flight under visual conditions. Optimum use of the Hughes Virtual Tower incorporated into this course. Prerequisite: ATT2820. Special fee.

Lecture Hours: 48.000

Prerequisite: ATT2820

Course User Fee: \$300.00

ATT2823 Air Traffic Control (ATC) Non-Radar 3.000 credits

In this course, future air traffic controllers will acquire an understanding of air traffic control practices, policies and procedures and their application in a non-radar air traffic environment. Throughout this course, (Non-Radar Procedures) appropriate real-life examples are used to illustrate the reasoning behind procedures used by air traffic controllers utilizing the non-radar methods. The liberal use of figures and example phraseology is used to assist the student in achieving an overall understanding of the air traffic control system.

Prerequisites: ATT2820, ASC1210. Special fee.

Lecture Hours: 48.000

Prerequisite: ASC1210, ATT2820

Course User Fee: \$20.00

ATT2995 Occupational Practicum 4.000 credits

This course provides supervised, work-based learning experience in an aviation setting that enables students to apply classroom knowledge in a professional environment. Students gain practical experience in aviation operations, safety procedures, regulatory compliance, communication, and workplace professionalism while developing career readiness and industry-specific skills through guided experiential learning.

Lecture Hours: 64.000

AVM**AVM1010 Aviation Industry Operation 3.000 credits**

The course provides insight into the development and present status of aircraft and air transportation, governmental organizations, controls and regulations, and career opportunities in the field.

Lecture Hours: 48.000

AVM1022 Flight Operations 3.000 credits

An investigation of the occupational duties, responsibilities, and physical facilities required by the positions of pilot, co-pilot, flight engineer, dispatcher and flight attendant.

Lecture Hours: 48.000

AVM1062 Aviation Career Planning 1.000 credits

This course provides direction and guidance in career planning for all aviation students. Topics of discussion will include the job search education and training requirements, resume writing, business etiquette, interview skills and follow-up techniques

Lecture Hours: 16.000

AVM1121 Hazardous Materials/Dangerous Goods 3.000 credits

This course is designed to provide the student with knowledge of dangerous goods/ hazardous materials and their effect in air transportation and logistics. The students will be conversant in hazardous material regulations for cargo and passenger transportation. The course will encompass the identification, labeling, packaging and handling of 9 types of dangerous goods in air transportation and general logistics. Prerequisite: AVM2120. Special fee.

Lecture Hours: 48.000

Prerequisite: AVM2120

Course User Fee: \$25.00

AVM1160 Aviation Maintenance Programs and Inspections 3.000 credits

This course provides an in-depth study of aircraft inspection programs and maintenance scheduling procedures. Students will learn national and international regulations governing aircraft inspection, maintenance evaluation, and the required procedures to update airline and governmental maintenance technical manuals.

Lecture Hours: 48.000

AVM1161 Aircraft Performance Measures and Maintenance Requirements 3.000 credits

Students will learn aircraft performance measures and maintenance requirements for airplanes powered by reciprocating, turboprop, and/or jet turbine and turbofan engines. Topics include stability and control, weight and balance, performance charts and graphs, takeoff and cruise control, airplane performance characteristics, from which they will extract data that maximizes performance.

Lecture Hours: 48.000

AVM1162 Maintenance Repair and Overhaul (MRO) Interactions with Commercial Airline Operations 3.000 credits

Students will learn the Maintenance Repair and Overhaul (MRO) Procedures in Commercial Airline Operations. Topics include airline maintenance operations, engineering, maintenance, repair of structures, systems, and aircraft components. In addition, students will explore MRO financing, domestic and offshore operations, regulatory requirements, logistics, supply chain support, human resources and industry oversight.
Lecture Hours: 48.000

AVM1163 Policies and Procedures for Commercial Airlines Maintenance Programs 3.000 credits

Students will learn the maintenance policies and programs for commercial airlines. Concepts of Maintenance Steering Group (MSG) and Reliability Centered Maintenance (RCM) programs, Maintenance Control by Reliability Methods (MCEM) program, and Operational Availability (OA) for Commercial Aircraft will be discussed.
Lecture Hours: 48.000

AVM1164 Logistics and Maintenance Programs for Commercial Airlines 3.000 credits

This course focuses on the concepts and application of logistics and supply chain management utilized within the aviation maintenance industry to increase efficiency in production and maintenance. Students will learn the logistics support from Maintenance Repair Operators and Original Equipment Manufacturers in aviation maintenance operations.
Lecture Hours: 48.000

AVM1440 Aviation/Airport Security 3.000 credits

This course will provide the student with knowledge of the issues and strategies that are used to protect the national airspace system, airports and airlines from security threats. The various types of threats and responses to those threats will be covered. In addition, the legal requirements, planning issues, physical equipment and facility requirements and personnel issues will also be discussed.
Lecture Hours: 48.000

Course User Fee: \$50.00

AVM1949 Co-Op Work Experience 1: AVI 3.000 credits

This is a course designed to provide training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Co-operative Education Office to obtain registration approval. Prerequisite: 2.0 minimum GPA, approval of Co-op Program Director, minimum of 6 credits in field or work approved experience.
Lecture Hours: 48.000

AVM2120 Air Cargo 3.000 credits

The course develops a comprehensive grasp of the characteristics and evolution of air cargo, its impact on United States industry, inherent problems and future development.
Lecture Hours: 48.000

AVM2410 AIRPORT*MANAGEMENT 3.000 credits

Duplicate course. It will be removed.
Lecture Hours: 48.000

AVM2410 Principles of Airport Management 3.000 credits

This course provides the student with a broad background in the Principles of Airport Management. This includes the airport system and its history, planning, land use, community relation issues, financial issues, capacity and growth, operations, organization and administration. Special Fee.
Lecture Hours: 48.000
Course User Fee: \$25.00

AVM2431 Customer Service Agent 3.000 credits

This course covers the generic skills needed for any airline position involving regular contact with the traveling public. It includes human relations, personal appearance enhancement, etiquette, conflict management, speech skills, and the acquisition of attributes that would promote a proper professional image.
Lecture Hours: 48.000

AVM2441 Aviation Safety and Human Factors 3.000 credits

This course will provide the student with an understanding of human factors and safety concepts as they apply to aviation. There will be an evaluation of aircraft accidents and their causal factors. Accident prevention measures are stressed as integral parts of an aviation safety program.
Lecture Hours: 48.000
Course User Fee: \$25.00

AVM2450 airport fac/fin plan 0.000 credits

Duplicate course. It will be removed

AVM2450 Airport Facilities/Financial Planning 3.000 credits

This course provides the student with an in-depth knowledge of the techniques and strategies of the airport master plan in planning airport facilities and financial resources. Forecasting, demand analysis, sources of funding, planning requirements, environmental issues and requirements and compliance issues will be discussed. Also, implementation and control issues, financial management, budgets, costs and revenues as well as airport economics will be discussed. Prerequisite: AVM2410. Special fee.

Lecture Hours: 48.000

Prerequisite: AVM2410

Course User Fee: \$25.00

AVM2510 Airline Management 3.000 credits

This course offers an insight relative to the business policies and the functions of management in airline operations. The course involves various internal managerial facets and the impact of external regulatory and economic implications.

Lecture Hours: 48.000

AVM2515 Airline Marketing 3.000 credits

A differentiation of the functions of marketing in airline operations; market research, demand analysis, advertising and promotion, sales, traffic, and the theory of price determination.

Lecture Hours: 48.000

AVM2949 Co-Op Work Experience 2: AVI 3.000 credits

This is a course designed to continue training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Co-operative Education Office to obtain registration approval. Prerequisite: 2.0 minimum GPA, approval of Co-Op Program Director, completion of AVM1949.

Lecture Hours: 48.000

AVM9996 FAA Certification 45.000 credits

The Federal Aviation Administration (FAA) Certification placeholder awards equivalent college credit for non-collegiate training based on Technical Manual of Procedure Number 1100:815701(A) and 1100:815701B. Students must provide original FAA certificate in order to receive credit.

Lecture Hours: 720.000

BAN**BAN1004 Banking Fundamentals, Evolution, and Compliance 3.000 credits**

In this course, students will learn the foundation, structure and function of financial systems. The course plan includes a review of the role of banks as businesses and their impact on the economy. The students will receive an overview of the main entities that comprise the financial system: financial intermediaries, investments and financial markets. The students will be able to describe the risk infrastructure of financial institutions, the regulatory environment, and the responsibilities inherent in complying with regulatory requirements associated with the safety and soundness of banks.

Lecture Hours: 48.000

BAN1231 Commercial Lending 3.000 credits

This course provides an increased awareness of the credit underwriting process. It provides a comprehensive foundation to the lending process including technical and interpretative analysis of financial information from liquidity, solvency, ratio and cash flow analysis to the basics for the completion of a credit approval memorandum. Students will learn the concepts of qualitative analysis including the assessment of industry risk, market risk and management risk. The course provides an understanding of the role of loan policy and the need to summarize the borrowers' various risks into an appropriate credit risk rating.

Prerequisite: BAN2210.

Lecture Hours: 48.000

Prerequisite: BAN2210

BAN1240 Essentials for Retail Lending 3.000 credits

In this course, the student will learn a comprehensive approach to consumer lending, identifying financial risks and the regulatory environment that impact and promote safety and soundness in lending. The course centers around providing a practical approach to the fundamentals of consumer installment lending, including a review of different loan products, their life cycles, the credit application process and the essentials of the closing process.

Lecture Hours: 48.000

BAN2210 Analyzing Financial Statements 3.000 credits

In this course the students will learn the framework and resources available to analyze financial statements and to assess a company's operation and future performance. The course covers the critical objectives with the assessment of business performance such as: the relationship between the company's business and financial strategy, their financial statements, ratio, profitability, balance sheet and cash flow analysis. The course plan includes the use of financial and accounting information to make lending/credit and investment decisions. The course is targeted to a wide range of students including those pursuing careers in banking, general management, investment banking, financial analysis and consulting. Prerequisite: ACG2021.

Lecture Hours: 48.000

Prerequisite: ACG2021

BAN2211 Applied Financial Statement Analysis 3.000 credits

This course provides a comprehensive analysis of business strategy, operating performance, financial condition and cash flow strength. The student will learn basic and advanced financial concepts impacting the viability of a business, including accounting rules, methods of credit analysis, assessment of financial ratios, historical financial analysis, cash flow and financial forecasting.

Prerequisite: BAN2210.

Lecture Hours: 48.000

Prerequisite: BAN2210

Course User Fee: \$20.00

BAN2253 Residential Mortgage Lending 3.000 credits

This course covers all phases of the residential mortgage lending process, including a foundation to the underwriting, credit analysis and servicing of residential mortgage loans. The student will learn the aspects of construction and permanent financing for residential property, real estate and mortgage law, financing options in real estate, appraisals, servicing, regulatory requirements and secondary mortgage markets. The students will become familiar with the critical issues necessary for any individual to operate successfully in the residential mortgage loan market. Target audience for this course includes loan officers, underwriters, loan processors and individuals interested in pursuing a career in the mortgage lending business. Lecture Hours: 48.000

BAN2501 Money, Banking and Financial Markets 3.000 credits

The course is designed to provide students with an overview of the US financial system, including theory and practice of monetary policy and financial instruments. Students will learn about interest rate determination, the structure and role of banks, financial institutions in the intermediation process, the factors impacting inflation, and variables in the economy. The course plan is designed not only for students but for individuals working in the financial services industry, including junior officers to mid-management and entry level staff who are considering pursuing a career in the banking field. Lecture Hours: 48.000

BAN2511 Marketing for Financial Services 3.000 credits

Marketing of financial services is a specialized segment of marketing. It is highly competitive, making the process of selling for bankers a highly specialized and challenging endeavor. Changing market conditions, deregulation, the emergence of new competitors from within and external to the banking industry and the rapid integration of new technologies are some of the challenges bankers encounter. It covers the aspects of a consultative selling approach with emphasis on planning, implementing and fostering a long-term advisory relationship with clients. The students will learn to become proficient in how to prepare for a successful sales presentation and closing.

Lecture Hours: 48.000

BCH**BCH3023 Introductory Biochemistry 3.000 credits**

This course surveys the fundamental components of biochemistry. In this course, students will learn concepts such as the structure and function of amino acids, proteins, carbohydrates, lipids, and nucleic acids, together with discussions of oxidative metabolism and regulation. Prerequisites: BSC2010, BSC2010L, BSC2011, BSC2011L, CHM2200 or CHM2211, CHM2211L; Corequisite: BCH3023L.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BCH3023L

Course User Fee: \$25.00

**BCH3023L Introductory Biochemistry Laboratory
2.000 credits**

This laboratory course complements the lecture corequisite BCH 3023, which involves the study of the fundamental components of biochemistry. In this laboratory course, students will learn and will be provided with hands-on experiences with the concepts addressed in the lecture course.

Prerequisites: BSC2010, BSC2010L, BSC2011, BSC2011L, CHM2200 or CHM2211, CHM2211L;

Corequisite: BCH3023.

Lab Hours: 64.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BCH3023L

Course User Fee: \$110.00

BCN**BCN1251 Construction Drafting 3.000 credits**

An introductory drafting course specifically oriented towards students entering the careers of air conditioning, architecture, building construction and interior design with limited drafting skills (without high school drafting).

Lecture Hours: 32.000, Lab Hours: 32.000

**BCN1272 Building Construction Plans
Interpretation 1 3.000 credits**

Develops the students' ability to interpret working drawings. Students will learn the conventions of graphic and symbolic language used by construction professionals to communicate information on drawings. Emphasis is on architectural and structural details with limited coverage on mechanical and electrical aspects. Develops the students' ability to interpret working drawings. Students will learn the conventions of graphic and symbolic language used by construction professionals to communicate information on drawings. Emphasis is on architectural and structural details with limited coverage on mechanical and electrical aspects.

Lecture Hours: 48.000

Course User Fee: \$10.00

**BCN1275 Building Construction Plans
Interpretation 2 3.000 credits**

In this course, students will plan interpretation of complex working drawings for multi-story residential and commercial buildings. Students will learn to read and understand construction working drawings, identify structural systems and their details. Familiarity with all aspects of complex working drawings will be addressed. Prerequisite: BCN1272.

Lecture Hours: 48.000

Prerequisite: BCN1272

Course User Fee: \$10.00

BCT**BCT1743 Building Construction Law 3.000
credits**

In this course, students will learn the legal aspects of construction contracts and the responsibilities arising particularly from the field operations. This course also includes the relationship of the general contractor to owner, architect, and subcontractor; material men and mechanics lien law; bonds; labor law; and other statutes and ordinances regulating contractors.

Lecture Hours: 48.000

Course User Fee: \$10.00

**BCT1750 Building Construction Financing 3.000
credits**

This course is a study of building construction financing and related contract requirements. Topics include construction loans, permanent building mortgages, construction bids and contracts, penalty and incentive provisions, progress payments and retention, escalation, escalation provisions, costs extras, performance and bid bonds, company profits, cash flow, and business loans.

Lecture Hours: 48.000

Course User Fee: \$10.00

**BCT1770 Building Construction Estimating
Fundamentals 3.000 credits**

This course is an analysis and calculation of building construction costs. Students will learn the classification of materials, labor, and subcontracted work into the smallest manageable units. Students will also develop a simple estimate for a residential structure. Prerequisite: BCN1272.

Lecture Hours: 48.000

Prerequisite: BCN1272

Course User Fee: \$10.00

**BCT1771 Building Construction Advanced
Estimating 3.000 credits**

Estimating more advanced elements of building construction. Students will learn to calculate direct, indirect, and overhead costs, as well as prepare bid proposals and related documents for commercial buildings. Prerequisites: BCN1275, BCT1770.

Lecture Hours: 48.000

Prerequisite: BCN1275, BCT1770

Course User Fee: \$10.00

**BCT2760 Building Code Regulations 3.000
credits**

In this course, students will learn the restrictions and limitations of the various agencies concerned with the building industry. Provisions of the South Florida Building Code are stressed.

Lecture Hours: 48.000

Course User Fee: \$10.00

BCT2990 CBE Building Construction 1.000 - 18.000 credits

This course is designed to assess learner mastery of the competencies and skills necessary for a successful career in the Construction Industry. It provides a foundation in pursuing a career in building inspection and quality control. The course offers a sequence of coherent and rigorous content aligned with challenging academic standards and relevant technical knowledge and skills needed for further education and careers; provides technical knowledge and skills proficiency, and includes competency-based applied learning that contributes to the academic knowledge, occupation-specific skills, and knowledge of all aspects of the Architecture and Construction career cluster.

Lab Hours: 16.000 - 288.000

BOT**BOT1010 Botany 3.000 credits**

This course is a survey of the plant kingdom based on a detailed study of the morphology, anatomy and physiology of selected representative specimens. Corequisite: BOT1010L

Lecture Hours: 48.000

Corequisite: BOT1010L

Course User Fee: \$8.00

BOT1010L Botany Laboratory 1.000 credits

Laboratory for BOT 1010. Corequisite: BOT 1010. Laboratory fee.

Lab Hours: 32.000

Course User Fee: \$38.00

BOT2150C Native Plant Identification and Usage in South FL 3.000 credits

Plants native to South Florida are identified and presented according to their typical ecological communities. Emphasis is placed on pineland, tropical hardwood hammock, mangrove and coastal, Everglades marsh, and cypress swamp ecosystems. Plants suitable for use in urban landscapes, as well as ecological restoration projects, are also covered. This course combines lecture and laboratory instruction.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$23.00

BOT3015 Survey of Plant Diversity 3.000 credits

This course explores the plant kingdom and emphasizes structure, function and genetics of plants. Students will learn the evolutionary relationships, natural history, ecological adaptations, physiology, morphology and reproductive biology of gymnosperms and angiosperms.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BOT3015L

Course User Fee: \$10.00

BOT3015L Survey of Plant Diversity Laboratory 1.000 credits

This course is designed to provide the necessary laboratory experiments and dissection exercises to supplement/accompany the BOT3015 Survey of Plant Diversity lecture course. Students will learn about the plant kingdom and with emphasis on structure, function and genetics of plants.

Dissections and laboratory exercises are designed to explore the fundamental cell and tissue structures of both vascular and non-vascular plants.

Lab Hours: 32.000

Prerequisite: BSC2011, BSC2011L, Corequisite: BOT3015

Course User Fee: \$40.00

BSC**BSC1005 General Education Biology 3.000 credits**

This course applies the scientific method to critically examine and explain the natural world including but not limited to cells, organisms, genetics, evolution, ecology, and behavior. Student learning outcomes: students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will describe the natural world; and students will articulate and practice the scientific method.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC1005L General Education Biology Laboratory 1.000 credits

This is an optional one-credit lab to provide students with experience in the scientific process. This course applies the scientific method to critically examine and explain the natural world, including but not limited to cells, organisms, genetics, evolution, ecology, and behavior.

Lab Hours: 32.000

Pre/Corequisite: BSC1005

Course User Fee: \$25.00

BSC1084 Functional Human Anatomy 3.000 credits

This course covers basic human anatomy for students in allied health and mortuary science programs. It includes the dynamics of gross and functional anatomy, terminology, body orientation, and systematic relationships.

Lecture Hours: 48.000

BSC1949 Co-op Work Experience 1 - BIO 3.000 credits

This is a course designed to provide training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Prerequisite: Co-Op Department approval. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Cooperative Education Office to obtain registration approval. (3 hr. lecture).

Lecture Hours: 48.000

BSC1949 Co-op Work Experience 1 - BIO 3.000 credits

This is a course designed to provide training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Prerequisite: Co-Op Department approval. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Cooperative Education Office to obtain registration approval. (3 hr. lecture).

Lecture Hours: 48.000

BSC2010 Principles of Biology 3.000 credits

In this course students will apply the scientific method to critically examine and explain the natural world. This course will cover molecular biology, cellular biology, genetics, metabolism, and replication. Student learning outcomes: students will demonstrate scientific literacy by articulating and practicing the scientific method; students will evaluate data regarding validity; students will read and interpret a variety of scientific data; students will identify major macromolecules and state their importance to living organisms; students will explain metabolism; students will compare and contrast prokaryotic and eukaryotic structures and processes of cell division and replication; and students will explain gene expression; students will solve problems in transmission genetics. Pre/Corequisites: BSC2010L, CHM1045. Special Fee.

Lecture Hours: 48.000

Course User Fee: \$8.00

BSC2010L Principles of Biology 1 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2010, Principles of Biology 1. It covers the nature of scientific investigation, the chemistry of life, microscopy, cell structure and function, metabolism, and the continuity of life. Corequisite: BSC2010, Special Fee.

Lab Hours: 64.000

Prerequisite: CHM1045 AND CHM1045L,

Corequisite: BSC2010

Course User Fee: \$40.00

BSC2011 Principles of Biology 2 3.000 credits

This is the second in a sequence of two courses that deals with the principles of modern biology. It covers organic evolution, phylogeny, biological diversity, overviews of plant and animal form and function, behavior, as well as population, community, and ecosystem ecology.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011L

Course User Fee: \$8.00

BSC2011L Principles of Biology 2 Laboratory 2.000 credits

This course is intended for students pursuing a major and complements the lecture course BSC2011. As such, it functions to provide them with hands-on experience with laboratory exercises designed to complement the presentation of the principles of biology as they relate to evolution, biological diversity, form and function in plants and animals, ethnology, ecology and conservation biology.

Lab Hours: 64.000

Prerequisite: BSC2010, BSC2010L, Corequisite: BSC2011

Course User Fee: \$40.00

BSC2020 Human Biology: Fundamentals of Anatomy/Physiology 3.000 credits

This course provides a basic understanding of the human body, its systems, and their functions. It includes the dynamics of physiology, terminology, and physiological relationships of the body systems.

Lecture Hours: 48.000

Course User Fee: \$10.00

**BSC2085 Human Anatomy and Physiology 1
3.000 credits**

This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems. Student learning outcomes: students will identify cell structures and describe their functions; students will distinguish tissues by structure, location in the body, and contrast their normal physiology; students will demonstrate an understanding of anatomical structure, organization of the body, cavities, planes, and directional terms; students will identify and describe structures of integumentary, skeletal, muscular, and nervous systems; students will interpret the functions of the integumentary, skeletal, muscular, and nervous systems; students will explain how the components of the human body maintain homeostasis; and students will analyze and interpret physiological data.

Corequisite: BSC2085L.

Lecture Hours: 48.000

Corequisite: BSC2085L

Course User Fee: \$8.00

**BSC2085L Human Anatomy and Physiology 1
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2085 as it pertains to structure and function of the human body from an experiential approach. This course is the first part of a two-semester sequence in which students examine human anatomy and physiology through a systems approach based on the interaction between form and function, from the microscopic components of cells and tissues to the organismal level. Emphasis is placed on histology and the integumentary, skeletal, muscular, and nervous systems.

Lab Hours: 32.000

Corequisite: BSC2085

Course User Fee: \$25.00

BSC2086 Human Anatomy & Physiology 2 3.000 credits

Building on concepts learned in BSC2085, students will learn the structure, function, and physiology of the human body, with an emphasis on the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems.

Prerequisite: BSC2085; Corequisite: BSC2086L.

Lecture Hours: 48.000

Prerequisite: BSC2085, Corequisite: BSC2086L

Course User Fee: \$9.00

**BSC2086L Human Anatomy & Physiology 2
Laboratory 1.000 credits**

In this laboratory course, students will learn to apply the concepts covered in BSC2086, which include the structure and function of the Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive Systems and development from an experiential approach.

Prerequisites: BSC2085, BSC2085L; Corequisite: BSC2086.

Lab Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L, Corequisite: BSC2086

Course User Fee: \$25.00

**BSC2426 Biotechnology Methods and
Applications 1 3.000 credits**

This course addresses the basic principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical, biotechnology and/or research laboratory setting(s). Practical applications of biotechnology are explored.

Lecture Hours: 48.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426L

**BSC2426L Biotechnology Methods and
Applications 1 Laboratory 2.000 credits**

This laboratory course is designed to complement BSC2426 Biotechnology Methods and Applications 1. This is a hands-on course that emphasizes the basic laboratory principles, techniques, and instrumentation, necessary for effective work in pharmaceutical, biotechnology, and/or research laboratory settings(s).

Lab Hours: 64.000

Prerequisite: Previous knowledge of chemistry and biology strongly recommended, Corequisite: BSC2426

Course User Fee: \$50.00

**BSC2427 Biotechnology Methods and
Applications 2 3.000 credits**

This course addresses advanced principles, concepts and techniques of biotechnology necessary for an understanding of the field, and effective work in a pharmaceutical-biotechnology and/or research-laboratory setting(s). The following areas of contemporary biotechnology are covered: forensics, bioremediation, and animal-, plant-, and marine biotechnology.

Lecture Hours: 48.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427L

BSC2427L Biotechnology Methods and Applications 2 Laboratory 2.000 credits

This laboratory course is designed to complement BSC2427 Biotechnology Methods and Applications 2. This is a hands-on course that emphasizes advanced laboratory principles, techniques, and instrumentation necessary for effective work in a pharmaceutical, biotechnology, and/or research-laboratory setting(s).

Lab Hours: 64.000

Prerequisite: BSC2426, BSC2426L, Corequisite: BSC2427

Course User Fee: \$50.00

BSC2943L Bioscience Internship 3.000 - 6.000 credits

This internship course is a capstone for students majoring in bioscience and related programs. Students will learn to apply acquired knowledge and skills to gain experience in the bioscience workplace.

Other Hours: 144.000 - 288.000

Course User Fee: \$21.00

BSC2949 Co-op Work Experience 2: BSC 3.000 credits

Course Description: This is a course designed to continue training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer

Lecture Hours: 48.000

BSC2949 Co-op Work Experience 2: BSC 3.000 credits

Course Description: This is a course designed to continue training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer

Lecture Hours: 48.000

BSC2949 Co-op Work Experience 2: BSC 3.000 credits

Course Description: This is a course designed to continue training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer

Lecture Hours: 48.000

BSC3930 Biological Sciences Seminar 1.000 credits

This course is designed for biological science majors. Students will gain an understanding of the broad range of career options within the biological sciences. Additionally, students will learn how to read, interpret, discuss, and cite selected examples of scientific literature in different areas of biology.

Lecture Hours: 16.000

Prerequisite: BSC2011, BSC2011L

Course User Fee: \$10.00

BSC4422 BIO MTHDS&APPL 3 3.000 credits

Duplicate. It will be removed.

Lecture Hours: 48.000

Course User Fee: \$25.00

BSC4422 Biotechnology Methods and Applications III 3.000 credits

This course will explore biotechnology as a science and its implications in modern society. Students will learn how to make well-designed and controlled experiments. Students will also demonstrate knowledge of data acquisition and interpretation.

Lecture Hours: 48.000

Prerequisite: BSC2427, BSC2427L, BCH3023, BCH3023L, PCB3060, PCB3060L, Corequisite: BSC4422L

Course User Fee: \$25.00

BSC4422L Biotechnology Methods and Applications III Lab 2.000 credits

This course provides students with hands-on laboratory experiences to supplement the BSC4422 lecture course. Students will learn how to perform advanced molecular biotechniques that build on previous knowledge. They will perform diagnostic assays, western blots, purifications, etc. and determine how to correlate findings with the basic research or clinical data.

Lab Hours: 64.000

Prerequisite: BSC2427, BSC2427L, BCH3023, BCH3023L, PCB3060, PCB3060L, Corequisite: BSC4422

Course User Fee: \$110.00

BSC4434 Bioinformatics for Biologists 4.000 credits

The student will be introduced to the basic concepts and tools that scientists use to analyze biological information. Students will learn, through the examination of literature, development of projects and use of available web-based tools, how to store, retrieve and analyze genetic information.

Lecture Hours: 64.000

Prerequisite: BSC2011, BSC2011L, PCB3060, PCB3060L

Course User Fee: \$55.00

BSC4940 Senior Specialty Internship 3.000 credits

This course will provide students with hands-on experience in the biological science workplace by conducting an internship. The experience prepares the individual for their first position in-field.

Prerequisite: Pre-completion of BSBS core curriculum and approval by BS-BS faculty. Special fee.

Lab Hours: 144.000

Course User Fee: \$181.00

BUL**BUL2131 Legal Environment 3.000 credits**

Law in relation to the proper conduct of business includes a consideration of the nature and sources of law, its legal environment and history. The topics of business torts, crimes, contracts, and forms of organizations are also covered.

Lecture Hours: 48.000

Course User Fee: \$10.00

BUL2241 Business Law 1 3.000 credits

Law in relation to the proper conduct of business, including a consideration of the nature and source of law, courts and courtroom procedure, contracts, sales of goods, negotiable instruments and secured transactions. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

BUL2242 Business Law 2 3.000 credits

Emphasis on the laws affecting agencies, the formation and operation of partnership and corporation, personal and real property, insurance, surety ship, estates and bankruptcy, and a general review of government regulations affecting usual business operations. Prerequisite: BUL2241. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

BUL4320 Business Law 3.000 credits

This course covers the basics of business law for accountants which include the topics of business organizations, securities law and corporate governance, agency and employment, sales and lease contracts, creditors' rights and bankruptcy, professional liability, government regulation, and property. The course also focuses on the nature of legal relationships in business with a particular emphasis on the government regulation of business, ethical, global, and corporate issues while addressing core curriculum requirements.

Prerequisites: ACG3113 and ACG3343. Students must pass this course with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: ACG3113 AND ACG3343

BUL4461 Law of International Trade 3.000 credits

Students will learn to interpret laws of international trade, licensing, and customer's regulation. This course also addresses U.S. customs regulations, classification of merchandise, application of tariff rules, duty free treatment, importing and exporting, liquidation inspection, search and seizure, and fines and penalties. In addition, the course will review the general rules and how to interpret the harmonized tariff schedule. Prerequisites: MAN2021 and GEB3358.

Lecture Hours: 48.000

Prerequisite: GEB3358

CAI**CAI1001C Artificial Intelligence (AI) Thinking 3.000 credits**

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAI1320C Introduction to Prompt Engineering 3.000 credits

This course introduces students to prompt engineering, focusing on foundational concepts and practical applications across various tools. Students will learn to craft prompts that enhance creativity, precision, and efficiency, leveraging AI to optimize workflows and productivity.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAI2100C Machine Learning Foundations 3.000 credits

Students are introduced to machine learning concepts and Python applications, including data acquisition, supervised, unsupervised, and reinforced learning. In addition, students will develop and deploy artificial intelligence (AI) models utilizing classification algorithms.

Prerequisites: CAI1001C, COP1047C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI1001CCOP1047C or equivalent knowledge of Python programming language.

Course User Fee: \$63.00

CAI2300C Introduction to Natural Language Processing 3.000 credits

Students will learn the fundamental concepts of Natural Language Processing (NLP) and text processing. In addition, focus will be on knowledge and skills necessary to create a language recognition application. Pre/Corequisite: CAI2100C. Lecture Hours: 32.000, Lab Hours: 32.000
Pre/Corequisite: CAI2100C
Course User Fee: \$63.00

CAI2820C Artificial Intelligence Applications Solutions 3.000 credits

A lower division course for students majoring in Applied Artificial Intelligence (AI). Students will demonstrate competence to scope, acquire/explore data, model, evaluate, and deploy an AI/machine learning solution in a team environment. Students will create and present a code or no-code AI solution. Must be taken during the last semester before graduation. Prerequisites: CAI2300C and CAI2840C. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: CAI2840C AND CAI2300C
Course User Fee: \$63.00

CAI2840C Introduction to Computer Vision 3.000 credits

Students will learn fundamental concepts in Computer Vision (CV) and image processing, including introduction to necessary proprietary and open-source Python libraries. Pre/Corequisite: CAI2100C. Lecture Hours: 32.000, Lab Hours: 32.000
Pre/Corequisite: CAI2100C
Course User Fee: \$63.00

CAI3303C Natural Language Processing 3.000 credits

Students will further explore Natural Language Processing domains. Focus on deep learning application and use of the latest pre-trained models on real world applications. Prerequisite: CAI2300C. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: CAI2300C
Course User Fee: \$63.00

CAI3821C Computational Methods and Applications for Artificial Intelligence I 3.000 credits

Computational data analysis is an essential part of artificial intelligence (AI). This course is designed to help students develop programming skills for AI applications. Students will learn core concepts of computational methods to solve data analysis problems, AI algorithmic methodologies, and how to test AI models. Prerequisites: CAI2100C, COP1047C, MAC1105, and STA2023. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: CAI2100C AND COP1047C AND STA2023 AND MAC1105
Course User Fee: \$63.00

CAI3822C Computational Methods and Applications for Artificial Intelligence II 3.000 credits

This course is designed for students to acquire a deeper understanding of computational methods used in the applications of artificial intelligence (AI) with programming. The topics of this course will be a continuation of those covered in Computational Methods and Applications for Artificial Intelligence I, with added emphasis on case studies using machine learning. Prerequisite: CAI3821C. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: CAI3821C
Course User Fee: \$63.00

CAI4420C Applied Decision and Optimization Theory 3.000 credits

Students will learn how to formulate and optimize queuing, linear and nonlinear programming models. Students will understand how to apply decision-making logic and optimization techniques to artificial intelligence models and conduct decision analysis of real-world problems. Prerequisite: CAI4505C. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: CAI4505C
Course User Fee: \$63.00

CAI4505C Artificial Intelligence 3.000 credits

The course provides an in-depth survey of important concepts, problems, and techniques in artificial intelligence, including heuristic and adversarial search, constraint satisfaction problems, logical reasoning, reasoning with uncertainty, and machine learning. A particular focus and a unifying theme of the course will be the notion of intelligent agents and their implementation. Prerequisites: COP3530 and CAI3822C. Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: COP3530 AND CAI3822C
Course User Fee: \$63.00

CAI4510C Machine Intelligence 3.000 credits

The course will cover advanced modeling techniques, including ensemble learning, extended linear models and kernel methods (PCA, support vector machines), probabilistic graphical models, Bayesian networks, mixture and latent variable models, biologically inspired computing (neural networks), feature selection and feature engineering techniques, Markov models, and temporal modeling to find patterns over time. Prerequisites: COP3530 and CAI3822C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: COP3530 AND CAI3822C

Course User Fee: \$63.00

CAI4525C Artificial Intelligence Systems Automation 3.000 credits

Students will learn how to use automated tools to develop enterprise automation projects. Students will understand how to create variables, arguments, workflows, build and publish automated processes and effective user interface automations.

Prerequisites: CAI4505C and CAI4510C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI4505C AND CAI4510C

Course User Fee: \$63.00

CAI4830C Simulation for Applied Artificial Intelligence 3.000 credits

Students will learn how to choose the right constructs of the modeling language to create a representation of a real-world system that is suitable for risk-free dynamic experiments. In addition, the students will learn how to build and deploy simulation models using the three major paradigms in simulation modeling for AI: agent-based, system dynamics, and discrete-event.

Prerequisite: CAI4505C.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: CAI4505C

Course User Fee: \$63.00

CAI4950C Artificial Intelligence Capstone 3.000 credits

Students will demonstrate competence to scope, acquire/explore data, model, evaluate, and deploy an AI/Machine Learning solution in a team environment. Students will create and present an AI solution. Course must be taken during the last semester before graduation. Prerequisites: CAI4510C, CAI4830C, and CAI4420C; Pre/Corequisite: CAI4525C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAI4510C AND CAI4830C AND

CAI4420C AND Pre/Corequisite: CAI4525C

Course User Fee: \$63.00

CAP1788 Introduction to Data Analytics 4.000 credits

This course offers a broad introduction to data analytics and the role it plays in modern organizations. Students will use Excel and Tableau to import, clean, transform, and visualize data. Students will also learn how to effectively communicate findings to decision makers.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CAP2047 User Interface Design 4.000 credits

This course is for students majoring in game development. It covers designing and developing interfaces for games. Students will learn how to use different input/output hardware devices, how to create and use existing interfaces for different types of hardware, and the development process for different types of gaming systems. Prerequisites: DIG1710 and DIG1729C; Pre/Corequisite: COP2335.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: DIG1710, DIG1729C AND Pre/Corequisite: COP2335

Course User Fee: \$129.55

CAP2048 Game Development Project I 3.000 credits

This capstone course is for students majoring in Game Development and Game Animation. Students will work in cross disciplinary teams to develop a working animated game or film. Students will learn how to apply the skills and knowledge they have acquired in a real-world working development environment. Prerequisites: CAP2047 or DIG1302, COP1334 or DIG1111, COP2335 or DIG1437, DIG1430, DIG1710 or DIG1132; Pre/Corequisite: DIG1712 or DIG2113.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAP2047, Corequisite: DIG2626, Pre/Corequisite: DIG1712

Course User Fee: \$129.55

CAP2743C Power BI - Data Visualization and Analysis 4.000 credits

This is the second of two introductory courses aimed at preparing students for the Microsoft Power BI certification exam. Students will learn how to create informative data visualizations and leverage Power BI's analytic capabilities to provide meaningful business insights. Prerequisite: CAP2791C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP2791C

Course User Fee: \$63.00

CAP2761C Intermediate Analytics 4.000 credits

This course offers an introduction to the SQL language and how it can be used by data analysts to query databases and gain valuable insights.

Students will also gain extensive hands-on experience with Microsoft SQL Server, Azure, and GitHub. Prerequisites: CAP1788, CGS1540C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1540C

Course User Fee: \$63.00

CAP2791C Power BI - Data Preparation and Modeling 4.000 credits

This is one of two introductory courses aimed at preparing students for the Microsoft Power BI certification exam. Students will learn how to get data from different sources, clean and transform the data, and design a data model that will be used in analysis and reports. Prerequisites: CAP1788 and CGS1540C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND CGS1540C

Course User Fee: \$63.00

CAP2920C Game Development Project II 3.000 credits

This capstone course is for students majoring in Game Development and Game Animation. Students will work in cross disciplinary teams to develop a working 5-10 min interactive game experience.

Students will learn how to apply the skills and knowledge they have acquired in a real world working development environment. Prerequisites: CAP2048, DIG1712 or DIG 2113.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CAP2048 AND DIG1430 AND

Corequisite: DIG1705

Course User Fee: \$129.55

CAP3321C Data Wrangling 4.000 credits

This course offers a broad introduction to data wrangling, data retrieval, and the use of Python in data analytics. Students will use Python and other business intelligence tools to retrieve data from various sources, clean the dataset, and prepare it for data analysis tasks. Prerequisites: CAP1788 and CAP2704.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND IDC2132

Course User Fee: \$63.00

CAP3330 Programming R for Statistics 4.000 credits

This upper division course is for students majoring in data analytics. Students will learn the R programming language and use it to perform intermediate-level statistical analysis. Techniques used in data analysis, such as analysis of variance and regression, will be emphasized. Prerequisite: STA2023.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: STA2023

Course User Fee: \$63.00

CAP3770 Predictive Analytics Algorithms 4.000 credits

This course is for students majoring in Data Analytics. Students will learn the fundamental algorithms used in data mining and analysis.

Students will learn various methods and techniques used in data mining, clustering and classification.

Prerequisite: STA2023.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: STA2023

Course User Fee: \$63.00

CAP3794C Data Analytics in the Cloud 4.000 credits

This course focuses on the role of a data analyst using cloud solutions. Students learn to utilize cloud-based tools and strategies for collecting, storing, preparing, analyzing, and visualizing data for analytics. Throughout the course, students will explore use cases from real-world applications, which will enable them to make informed decisions to effectively build data pipelines in a cloud environment. Prerequisites: CAP1788, CGS1540C, and CTS1145.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND CGS1540C AND CTS1145

Course User Fee: \$63.00

CAP4631C Machine Learning for Data Analytics I 4.000 credits

This upper division course is for students majoring in data analytics. Students will learn why machine learning is crucial for data analytics and why regression analysis is a foundation of supervised machine learning. Using Python programming, students will use a variety of packages to create regression models that make predictions.

Prerequisites: COP1047C, STA3164 or CAP3330.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND STA3164 OR CAP3330

Course User Fee: \$63.00

CAP4633C Machine Learning for Data Analytics II 4.000 credits

This upper division course is for students majoring in data analytics. In this second-level course, students will use the Python programming language to create additional machine learning models for classification. In addition, students will explore various applications of multi-layer neural networks. Prerequisites: CAP4631C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP4631C

Course User Fee: \$63.00

CAP4744 Data Visualization 4.000 credits

This course focuses on creating complex and informative visualizations using Tableau. Students will learn how to formulate business questions, organize complex and large datasets, analyze and interpret data, deliver actionable insights, and publish the results. Coverage includes preparation for intermediate-level Tableau certifications.

Prerequisites: CAP1788 and CAP2761C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND IDC2132

Course User Fee: \$63.00

CAP4767 Data Mining 4.000 credits

Students will learn how to extract useful data from an organization's data warehouse and analyze it to uncover hidden patterns and relationships. An emphasis will be placed on unsupervised data mining techniques, as well as the use of the RapidMiner software. Prerequisites: CAP1788 and CAP2761C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND CAP2761C

Course User Fee: \$63.00

CAP4784 Big Data 4.000 credits

Course Description: This course focuses on the processing of massive datasets, both structured and unstructured. Students will learn how to use Databricks and Spark to manage and analyze large datasets from a variety of sources. In addition, student will gain an understanding of how Databricks supports the end-to-end data science workflows that allow users to extract and share business insights. Prerequisites: CAP1788, CAP2761C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788 AND IDC2132

Course User Fee: \$63.00

CAP4910 Data Analytics Capstone 4.000 credits

This upper division course is for students majoring in data analytics. Students will initiate a business-driven data analytics solution to a real-world problem utilizing acquired skills in statistical analysis, machine learning, data mining, and data visualization. Must be taken during the last semester before graduation. Prerequisite: CAP4631C.

Lecture Hours: 48.000, Lab Hours: 32.000 - 28.000

Prerequisite: CAP4631C

Course User Fee: \$63.00

CAP4936 Special Topics in Data Analytics 4.000 credits

This upper division course is for students majoring in data analytics. The course centers around topics of current interest or of special interest to students or instructors. Topics or focus may vary from semester to semester. Prerequisites: CAP1788, CAP2761C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP1788, CAP2761C

Course User Fee: \$63.00

CCJ**CCJ1010 Introduction to Criminology 3.000 credits**

Theories and causes of criminal and delinquent behavior, including its variations, ramifications, explanations and measures of prevention, control and treatment.

Lecture Hours: 48.000

CCJ1020 Introduction to Criminal Justice 3.000 credits

This course will explore and define the various components of the criminal justice system. It is designed for the entry level criminal justice practitioner and beginning student interested in entering the field.

Lecture Hours: 48.000

CCJ1191 Human Behavior in Criminal Justice 3.000 credits

Human behavior and how it relates to the duties and responsibilities of the criminal justice practitioner.

Lecture Hours: 48.000

CCJ2053 Criminal Justice Ethics and Professionalism 3.000 credits

This course will provide students with an overview of moral, ethical, and professional issues and dilemmas facing individuals and organizations within the criminal justice system. Students will learn to define and implement ethical and professional standards by examining what they will be confronted with and how to respond appropriately.

Prerequisite: PHI 2600.

Lecture Hours: 48.000

Prerequisite: PHI2600

CCJ2358 Criminal Justice Reporting 3.000 credits

This course prepares students through instruction and practice to properly prepare written reports common to the criminal justice community.

Students will learn a variety of criminal justice scenarios presented and students will be instructed as to proper report format and presentation.

Prerequisite: ENC1101.

Lecture Hours: 48.000

Prerequisite: ENC1101

CCJ2650 Narcotics and Drug Education 3.000 credits

The general problems created by illegal use of narcotics and dangerous substances, with emphasis upon classification, description and history of drugs, etiology of addiction, extent of drug use and its relationship to criminal behavior and methods of control.

Lecture Hours: 48.000

CCJ2760 Cannabis Policy and Regulation 3.000 credits

In this course students will gain knowledge of the history, the control, and the regulations of cannabis as a recurring legal and social problem. Course completion will result in an understanding of the changing state-level law reforms, the ability to analyze the consequences of new legal, political, and practical issues, with a particular focus on the implications and impact of Florida's new cannabis laws. Students will examine the social and historical backdrop of cannabis usage and regulation, and will assess the reforms and debates impacting the control and regulations of cannabis distribution and use in Florida.

Lecture Hours: 48.000

CCJ2840 Introduction to Criminal Justice Research with AI 3.000 credits

This course provides an introduction to core research methods utilized in the field of Criminology and Criminal Justice. Students will learn foundational research concepts, including research design, data collection methods, and data analysis techniques. A key focus will be on the ethical and effective use of Artificial Intelligence (AI) tools to assist in various stages of the research process, such as data cleaning, analysis, and interpretation. Students will gain practical experience in applying these methods through hands-on projects and develop critical thinking skills necessary for evaluating research findings and understanding the social implications of criminal justice research.

Recommended: Students are encouraged to take ENC1101 or CCJ1020 prior to registration.

Lecture Hours: 48.000

Recommended Preparation: ENC1101 OR CCJ1020

CCJ3032 Crime and the Media 3.000 credits

An examination of the inter-relationship among the mass media, crime, and criminal justice. Includes media and the social construction of crime and criminal justice; media effects on attitudes toward crime and justice; and media as a cause of crime.

Lecture Hours: 48.000

CCJ3663 Female Crime and Delinquency 3.000 credits

A study of females in society and the criminal justice system. Includes the female delinquent, females as criminals, females as victims, and the impact of females as professionals in the Criminal Justice System. Prerequisites: CCJ1191, CJJ2002.

Lecture Hours: 48.000

Prerequisite: CCJ1191, CJJ2002

CCJ3666 Victimology 3.000 credits

A comprehensive study of victimization; analysis of contemporary victim-assistance and victim compensation programs and related research; review of the historical importance of victim restitution as a basis for punitive criminal law.

Lecture Hours: 48.000

CCJ3700 Methods of Research in Criminal Justice 3.000 credits

Evaluates the application of research methodologies as applied to the study of Public Safety Management. Prerequisite: STA2023

Lecture Hours: 48.000

Prerequisite: STA2023

CCJ4054 Ethics in The Criminal Justice System 3.000 credits

An in depth study of moral, ethical, legal, and professional issues and dilemmas facing individuals and organizations within the Criminal Justice systems. Prerequisites: CCJ2053 or PHI2600.
Lecture Hours: 48.000

CCJ4450 Criminal Justice Administration 3.000 credits

An analysis of leadership styles, management principles, supervisory techniques, policies and procedures within Law Enforcement agencies.
Prerequisite: CCJ1020.
Lecture Hours: 48.000
Prerequisite: CCJ1020

CCJ4641 Organized Crime 3.000 credits

An analysis of organized crime in today's society, as well as, past, present, and future perspectives of the topic.
Lecture Hours: 48.000

CCJ4651 Drugs and Crime 3.000 credits

An analysis of the interrelationship among drug usage, crime and the criminal justice system.
Lecture Hours: 48.000

CCJ4660 Crime, Violence, and Schools 3.000 credits

An examination of comprehensive and proven theoretical models of explaining, predicting, and preventing school-based violence.
Lecture Hours: 48.000

CCJ4678 Race, Gender, Ethnicity & Crime 3.000 credits

Focuses on the challenges and controversies of managing and treating special offender populations such as juvenile, elderly, disabled, mentally ill, pregnant inmates, etc.
Lecture Hours: 48.000

CCJ4941 INTERNSHIP FIELD 3.000 credits

The purpose of the internship program is to provide BAS Public Safety Management students with experience working in the field of criminology and criminal justice. This allows them to network with professionals who might later serve as references or points of contact and it also allows students to gain actual experience which can be included on a resume or job application. Each internship placement will differ with regard to what part of the criminal justice system the intern will be exposed to. (3 credits)
Lecture Hours: 240.000
Pre/Corequisite: CCJ1020BAS student, at least 30 BAS credits earned prior to start of internship, 3.0 GPA AND Prerequisite: CCJ1010 AND CCJ4450

CCJ4941 Internship Field 3.000 credits

The purpose of the internship program is to provide BAS Public Safety Management students with experience working in the field of criminology and criminal justice. This allows them to network with professionals who might later serve as references or points of contact and it also allows students to gain actual experience which can be included on a resume or job application. Each internship placement will differ with regard to what part of the criminal justice system the intern will be exposed to.
Internship Hours: 144.000
Pre/Corequisite: CCJ1020BAS student, at least 30 BAS credits earned prior to start of internship, 3.0 GPA AND Prerequisite: CCJ1010 AND CCJ4450

CCJ4943 INTERN CORRECTIONS 15.000 credits

Students will participate in the FDLE State-Mandated Certification Training Program in Corrections. (240 hr. Internship)
Lecture Hours: 560.000

CEN**CEN2211 C/C++ Programming for Embedded Devices 4.000 credits**

This course teaches the principles of programming in the C/C++ languages for embedded devices. The student will learn how to create programs to control open-source hardware for building digital devices that can sense and control the physical world around them.
Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$63.00

CEN2212C Introduction to Programming the Internet of Things (IoT) 4.000 credits

This course teaches the principles of programming Internet of Things devices using a computer language. The student will learn fundamental programming concepts and systematic design techniques. At the end of the course, the student will be able to write programs that control development boards, with sensors, connected to the Internet. Prerequisite: CEN2211.
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: CEN2211
Course User Fee: \$63.00

CEN4025C Software Engineering II 4.000 credits

This upper division course is for students majoring in the B.S. in Information Systems Technology or the B.S. in Electrical and Computer Engineering Technology programs. This course covers in-depth topics in software process structures, process models, requirements modeling with use-cases and class based methods. Students will also learn design concepts including abstraction, OOD concepts, component-level and architectural design, user interface analysis and design, and design patterns. Prerequisite: CET3383C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: Pre-requisite requirement CET3383C

Course User Fee: \$63.00

CEN4065C Software Architecture and Design 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology or B.S. in Electrical and Computer Engineering Technology. The course covers the software engineering process of building the software architecture and designing the software product according to design criteria. This course covers in-depth topics in software process models, software architecture, requirements modeling with use-cases and class-based methods. Students will also learn design concepts including abstraction, component-level and OO architectural design, user interface analysis and design, and design patterns, and software project management. Prerequisite: COP3530.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP3530

Course User Fee: \$63.00

CEN4090C Software Engineering Capstone 4.000 credits

This upper division course, for students majoring in the BS-IST Software Engineering track, requires students to demonstrate their competence to analyze, design, develop, and test a software system. Student(s) will create, implement, and present a software project plan that includes the following work products: software requirements specification, design specification document, code, unit tests and project system test plan to create an operational system. This course should be taken during the last semester before graduation.

Prerequisite: CEN4065C; Pre/Corequisite: COT4400.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CEN4065C, Pre/Corequisite: COT4400

Course User Fee: \$63.00

CET**CET1110C Digital Circuits 4.000 credits**

This is an introductory course in digital circuits for engineering and other technology majors. Students learn how to apply electronic principles to digital computer circuits and systems. Students will also learn how to simplify logic circuits, build digital circuits, and perform other laboratory activities. Pre/Co-requisite: MAC1105. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

CET1171 Introduction to Computer Service and Maintenance 3.000 credits

This course is designed as an introduction for students new to IT. Students will learn about the history, design, construction, and maintenance of microcomputers, including the proper handling and use of computer components and tools; how to assemble and disassemble computers; how to perform preventive maintenance; how to identify and upgrade components; how to interpret error messages, and how to perform basic troubleshooting.

Lecture Hours: 48.000

Course User Fee: \$63.00

CET1178C A+ Computer Hardware Service 3.000 credits

This is an intermediate level course that prepares students for A+ hardware certification. Students will learn how to: install, configure, and upgrade components; diagnose and troubleshoot computer systems, identify, test, and troubleshoot motherboards, processors, memory, and printers, and connect network equipment. Prerequisite: CET1171. Special fee. A.S. credit only.

Lecture Hours: 48.000

Course User Fee: \$63.00

CET1487C Network+ 3.000 credits

This is an intermediate level course designed for students preparing for the hardware component of the Network+ certification. Students will learn how to install, configure, manage, troubleshoot and upgrade network devices including network interface cards, switches, hubs, wireless access points, routers, and patch panels. They will also learn about the construction, installation, testing and repair of the physical layer of the network, including wired cables, fiber optic media, wireless transmitters and antennas. Demonstrated knowledge of microcomputer fundamentals and system components is required. Recommended Preparation: CET1178C or A+ certification.

Laboratory fee. A.S. degree credit only.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

CET2113C Advanced Digital Circuits 4.000 credits

This is a second level course in digital circuits for students majoring in electronics and related engineering technologies that extends the application of sequential and combinational logic circuits and other digital applications. Students will learn to program, operate, and interface with a micro-computer and its elements. Prerequisites: CET1110C, COP2270. Laboratory fee. Lecture Hours: 32.000, Lab Hours: 64.000

CET2123C Microprocessors 4.000 credits

This course is intended for students majoring in Electronics or Computer Engineering Technology. Students will learn to apply digital principles to the understanding of microprocessor parameters and characteristics (addressing range and models, instruction set, architecture, input/output, interrupts, and programming). Students will build practical microprocessor and/or microcontroller based systems to perform a variety of engineering applications. Prerequisites: CET1110C, COP 2270. Special Fee. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: CET1110C AND COP2270
Course User Fee: \$63.00

CET2588C Server + Service and Maintenance 3.000 credits

This course is designed for students preparing for the hardware component of the Server + certification. Students will learn how to install, configure, and upgrade workstations and servers, configure and test network and peripheral equipment, and diagnose and troubleshoot advanced computer systems. Laboratory fee. College readiness in reading and math required. Recommended Preparation: CET1178C or A+ certification. Special fee. Lecture Hours: 48.000
Course User Fee: \$63.00

CET2664C Electronic Security 4.000 credits

This is an introductory electronic security course for students who are studying cybersecurity, electronics or computer engineering technologies. The student will study information and communication security in computer systems and networks. Both information flow and information integrity policies will be considered. Topics will include authentication, protection, security models, cryptography, applications, and public policy, along with case studies. Prerequisites: COP2270, CET2369C. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: COP2270 OR CET2369C

CET2880C Digital Forensics 4.000 credits

This is an introductory digital forensics course for students who are studying cybersecurity, electronics or computer engineering technologies. In this course, students will learn the setup and use of an investigator's laboratory, how to perform data acquisition, web forensics, email forensics, mobile forensics, network analysis, and file recovery. Lecture Hours: 32.000, Lab Hours: 64.000

CET3126C Computer Architecture 4.000 credits

This course is intended for upper division students majoring in Electrical and Computer Engineering Technology as well as Information Systems Technology. This course introduces the study of advanced microprocessor design. Students will learn the basic organization of computer systems including instruction-set architecture, execution pipeline, memory hierarchy, virtual memory, and I/O subsystems. Students also learn advanced processor microarchitecture issues such as dynamic instruction scheduling, branch prediction, lock-up free caches, instruction-level parallelism, multiple instruction fetch/issuing, speculative execution, etc. to improve computer processor performance. Students will experimentally verify microarchitecture designs using industry standard microarchitecture simulators. Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$63.00

CET3383C Software Engineering I 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology or B.S. in Electrical and Computer Engineering Technology. The student will learn the basic principles and concepts of software engineering; system requirements; modeling and testing; object oriented analysis and design; testing and validation; configuration management; and the analysis, design and programming of extensible software systems. Prerequisite: CET2369C or COP2800. Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: CET2369C, COP2800
Course User Fee: \$63.00

CET4190C Applied Digital Signal Processing 4.000 credits

This is an upper division level course for students majoring in electronics engineering technology. Students will learn how to model digital signal processing (DSP) systems, apply the Z transform, and develop algorithms for convolution, correlation, the Discrete Fourier Transform (DFT), and the Fast Fourier Transform (FFT). Students will apply these concepts in the design and implementation of digital filters and DSP algorithms in an embedded system. Prerequisites: COP2270, EET4732C, EET2323C or EET2351C. Special Fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: COP2270 AND EET4732C AND EET2323C OR EET2351C

Course User Fee: \$63.00

CET4663C Electronic Security 3.000 credits

This is an upper division course for students who are majoring in electronics engineering technologies. The student will learn information and communication security principles for computer systems and networks, including authentication, protection, security models, cryptography, applications, and public policy, along with case studies. Prerequisites: CET2123C, COP2270.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: CET2123C AND COP2270

Course User Fee: \$63.00

CGS**CGS1005C Computing Fundamentals for Entrepreneurship 4.000 credits**

This interactive discovery course for non-computer majors teaches how to apply computational thinking to solve real world problems. Students will learn basic computer programming, web design, mobile application development, project management and desktop publishing through the use of case studies and scenarios that simulate real world business applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1021 scientific computing 4.000 - 32.000 credits

This course will be deleted, Effective Fall 2025 (2267).

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CGS1060C Introduction to Computer Technology and Applications 4.000 credits

This course provides the technology skills required for personal, academic and professional success. Students will learn essential computing concepts and skills including mobile productivity, cloud services, security, ethics, general programming concepts, email, web, operating systems, and the use of an office suite. The course satisfies the College's computer competency requirement.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1540C Database Concepts Design 4.000 credits

This course is designed for computer science majors and non-majors who require a fundamental knowledge of databases and database management systems. Students will learn how to design, implement and use databases to maintain and manipulate data. Students should have knowledge of basic computer concepts or seek faculty advisement.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Students should have knowledge of basic computer concepts or seek faculty advisement.

Course User Fee: \$63.00

CGS1560 A+ Computer Operating Systems 4.000 credits

This is a comprehensive course in the use of operating systems for microcomputers suitable for students seeking preparation for A+ operating system certification. Students will learn how to install, configure, use, manage, and troubleshoot the Disk Operating System (DOS), Microsoft Windows, and other microcomputer operating systems. Recommended CGS 1060C or computer experience. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS1700 Introduction to Operating Systems 4.000 credits

This course examines the role of operating systems as the interface between the hardware, the software and the users of a computer system. It explores the concepts such as processes and threads, file systems, virtual memory, interrupt handling, virtualization and security.

Lecture Hours: 48.000, Lab Hours: 32.000

CGS2091 Professional Ethics and Social Issues n Cs 4.000 credits

This course is designed to provide computer science majors and others with an introduction to professional ethics & social issues in Computer Science. Students will learn theories associated with the legal, ethical, and social issues relevant to information technology, and the roles and responsibilities of computer professionals in today's technological society. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS2108 Advanced Desktop Applications 4.000 credits

This is an advanced level course for major and non-major students who have completed CGS 1060C, Introduction to Computer Technology and Applications. Students will learn advanced computer skills using software applications, such as word processing, spreadsheets, database, presentation graphics, and communications and scheduling software. Students will also learn advanced file management techniques, deal with security issues, and troubleshoot hardware and software.

Prerequisite: CGS1060C. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CGS3763 Operating System Principles 4.000 credits

This upper division course, for students majoring in Information Systems Technology, introduces fundamental operating system topics and includes both computer system and operating system structure. Students will learn how processes, threads, concurrent programming, interrupt handling, CPU scheduling and process synchronization, and I/O system memory management affect the system structure. Additionally, students will learn how virtual memory, deadlocks, file system, and command interpreter relate to client/server systems. Prerequisite: COP1334. Special Fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1334

Course User Fee: \$63.00

CHI**CHI1120 Elementary Mandarin Chinese 1 4.000 credits**

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Mandarin Chinese. Note: students must pass this course with a "C" or better to continue to CHI1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

CHI1121 Elementary Mandarin Chinese 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Mandarin Chinese. Note: students must pass this course with a "C" or better to continue to CHI2220.

Lecture Hours: 64.000

Prerequisite: CHI1120

Course User Fee: \$15.00

CHI2220 Intermediate Mandarin Chinese 1 4.000 credits

A continuation of CHI1121. Students will learn Chinese language and culture through a systematic review of reading and writing skills with emphasis on oral as well as written presentations. Students will also learn the use of Chinese radicals and characters rather than pinyin. Prerequisite: CHI1121 or equivalent.

Lecture Hours: 64.000

Prerequisite: CHI1121

Course User Fee: \$15.00

CHI2221 Intermediate Mandarin Chinese 2 4.000 credits

A continuation of CHI2220. Students will learn advanced grammar, together with the introduction of more complex reading materials and an increase in the number of radicals and characters. Students will also be exposed to cross-cultural awareness.

Prerequisite: CHI2220 or equivalent.

Lecture Hours: 64.000

Prerequisite: CHI2220

Course User Fee: \$15.00

CHM

CHM1020 General Education Chemistry 3.000 credits

This course provides students with an introduction to chemical principles and applications for the non-science major. Students will engage in problem solving and critical thinking while applying chemical concepts. Topics will include the scientific method of problem solving, classification of matter, atomic theory, the periodic table, gases, chemical reactions, energy, and chemical bonds. Student learning outcomes: students will be able to distinguish between physical and chemical properties and changes; students will recognize components of gaseous chemistry; students will recognize components of aqueous chemistry including properties of water, solutions, and acids and bases; students will correlate the design of the periodic table to periodic trends and physical and chemical properties elements; and students will write and interpret chemical formula and write balanced chemical equations.

Lecture Hours: 48.000

Course User Fee: \$15.00

CHM1025 Introductory Chemistry 3.000 credits

This course will provide beginning students with certain basic knowledge and skills, which will enable them to be successful in the first semester of General Chemistry I, CHM1045. The students will learn elementary principles of modern chemistry, including basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature. Prerequisite: MAT1033.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$15.00

CHM1025L Introductory Chemistry Lab 1.000 credits

This course is an optional beginning chemistry laboratory course, which has been designed for those students who have little or no background in chemistry and are enrolled in CHM1025. The students will reinforce what they learn in CHM1025. Students will learn about basic measurements, chemical bonding, chemical reactions, stoichiometry, concentration of solutions, and chemical nomenclature.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1025

Course User Fee: \$30.00

CHM1033 Chemistry for Health Sciences 3.000 credits

This course emphasizes chemistry topics related to allied health. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry and their applications to physiological functions. Prerequisite: MAT1033; Corequisite: CHM1033L.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: CHM1033L

Course User Fee: \$15.00

CHM1033L Chemistry for Health Sciences Lab 1.000 credits

This course emphasizes chemistry topics related to the allied health sciences. Students will learn the essentials of inorganic chemistry, organic chemistry, biochemistry, and their application to physiological functions in a laboratory setting. Prerequisite: MAT1033; Corequisite: CHM1033.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: CHM1033

Course User Fee: \$30.00

CHM1045 General Chemistry and Qualitative Analysis 3.000 credits

This course is designed for students pursuing careers in the sciences or who need a more rigorous presentation of chemical concepts than is offered in an introductory course. Students will engage in problem-solving and critical thinking while applying chemical concepts. Topics will include the principles of chemistry including atomic theory, electronic and molecular structure, measurement, stoichiometry, bonding, periodicity, thermochemistry, nomenclature, solutions, and the properties of gases. Student learning outcomes: students will apply the law of conservation of matter and energy; students will implement rules of significant numbers to all measurements; students will explain the fundamental properties of matter, including but not limited to atomic and electronic structure, and periodicity; students will apply IUPAC rules of nomenclature; students will predict molecular geometry and properties from bonding theories; and students will predict and explain the products of chemical reactions (e.g., acid-base, oxidation-reduction, precipitation, dissociation). Prerequisite: CHM1025 or a passing score on the CART exam, MAC1105; Corequisite: CHM1045L.

Lecture Hours: 48.000

Prerequisite: CHM1025 or approved proficiency exam score., Corequisite: CHM1045L, Prerequisite: MAC1105

Course User Fee: \$15.00

CHM1045L General Chemistry Lab 2.000 credits

CHM 1045L is the first semester general chemistry laboratory course. Students will learn to introduce the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., stoichiometry, gas laws, atomic structure, and quantitative analysis). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science majors.

Lab Hours: 64.000

Prerequisite: CHM1025 or a passing score on the CART exam. OR MAC1105, Corequisite: CHM1045
Course User Fee: \$30.00

CHM1046 General Chemistry and Qualitative Analysis 3.000 credits

CHM1046 is the second course in the CHM1045-1046 sequence. Students will learn major topics in modern chemistry, including but not limited to thermodynamics, kinetics, solutions equilibria including acids, bases, and other ionic equilibria, and electrochemistry.

Lecture Hours: 48.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046L

Course User Fee: \$15.00

CHM1046L General Chemistry and Qualitative Analysis Lab 2.000 credits

CHM 1046L is the second semester general chemistry laboratory course. Students will learn the basic laboratory techniques involved in general chemistry and to re-enforce and illustrate several of the important topics in general chemistry (e.g., qualitative and quantitative analysis, equilibrium, thermodynamics, and kinetics). The enrollment generally consists of pre-medical, pharmacy, medical technology, physical therapy, engineering, and science students. Prerequisites: CHM1045, CHM1045L; Corequisite: CHM1046.

Lab Hours: 64.000

Prerequisite: CHM1045, CHM1045L, Corequisite: CHM1046

Course User Fee: \$30.00

CHM2200 Survey of Organic Chemistry 3.000 credits

This one-semester course briefly examines the structure, synthesis, nomenclature and reactivity of selected mono-and-poly-functional organic compounds. Theories that relate the structure of organic molecules to their chemical reactivity will be presented as a unifying principle. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2200L.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2200L

CHM2200L Survey of Organic Chemistry Laboratory 1.000 credits

Experiments and exercises will be conducted to introduce students to the basic laboratory techniques that are used in organic chemistry and that re-enforce and illustrate several important topics in organic chemistry. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2200. Special fee.

Lab Hours: 32.000

Course User Fee: \$46.00

CHM2210 Organic Chemistry 1 3.000 credits

In Organic Chemistry 1, students will learn about aliphatic hydrocarbons and their derivatives.

Lectures are supplemented by laboratory preparation of representative compounds.

Lecture Hours: 48.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210L

Course User Fee: \$15.00

CHM2210L Organic Chemistry 1 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2210. Topics such as nomenclature, preparations, reactions and electronic and structural features of alkanes, alkenes, alkynes, alkyl halides, aromatic hydrocarbons and other organic compounds will be performed in a laboratory setting. Prerequisites: CHM1046, CHM1046L; Corequisite: CHM2210.

Lab Hours: 64.000

Prerequisite: CHM1046, CHM1046L, Corequisite: CHM2210

Course User Fee: \$30.00

CHM2211 Organic Chemistry 2 3.000 credits

In organic chemistry 2, students will learn about nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds.

Lecture Hours: 48.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211L

Course User Fee: \$15.00

CHM2211L Organic Chemistry 2 Laboratory 2.000 credits

Students will learn to reinforce and illustrate topics learned in CHM2211. Topics include nomenclature, preparation reactions, and electronic and structural features of alcohols, ethers, phenols, aldehydes, ketones, carboxylic acids, acid anhydrides, amides, esters, and other organic compounds will be performed in a laboratory setting.

Lab Hours: 64.000

Prerequisite: CHM2210, CHM2210L, Corequisite: CHM2211

Course User Fee: \$45.00

CHS**CHS1522C Forensic Science 1 4.000 credits**

An introductory course in the principles and techniques of forensic science. Students will learn how forensic science pertains to crime scene investigation and crime laboratory analysis.

Lecture Hours: 48.000, Lab Hours: 32.000

CHS2523 Forensic Science 2 3.000 credits

This is a continuation of Forensic Science 1.

Students will learn topics which include but are not limited to: drug identification and toxicology; document analysis; death determination; soil examination methodology; forensic anthropology; tool marks and casts/impressions. Prerequisite: CHS1522C.

Lecture Hours: 48.000

CIS**CIS1321 Introduction to Systems Analysis and Design 4.000 credits**

This course introduces computer science and non-majors to fundamental skills of analysis and design of management information systems. Students learn the concept of charting, investigating, documenting and reporting using current information systems, system analysis tools and system design tools. The related concept of management, organization, computers, information processing and the system approach are combined and applied to case studies. Prerequisite: CGS1060C. Knowledge of business accounting is recommended. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS1531 Introduction to Secure Scripting 4.000 credits

This course provides students with the knowledge and skills to: create secure scripts and programs using system shells and programming languages; implement and debug algorithms to solve problems; automate and perform administrative tasks; manage data handling, and backup and storage.

Lecture Hours: 48.000, Lab Hours: 32.000

CIS1949 CO-Op Work Experience 1: CIS 1.000 - 4.000 credits

This course is designed as a work experience for students majoring in computer information systems programs. Students will learn to apply the skills and knowledge that they have acquired through their program of study in a real work environment.

Prerequisite: Successful completion of required program course work. Department approval required.

Lecture Hours: 16.000 - 64.000

Prerequisite: Successful completion of required program course work. Department approval required.

CIS2010C Introduction to Internet of Things Networking and Security 4.000 credits

This course explores Internet of Things (IoT) networks' design, implementation, and security, providing a well-rounded understanding of protocols and standards governing the Internet and IoT. Students gain experience performing security assessments and vulnerability testing of IoT systems using industry-standard tools, focusing on identifying and mitigating risks to build secure and resilient IoT solutions.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS2011C Introduction to Cloud Computing for Internet of Things 4.000 credits

This course explores the intersection of cloud computing and the Internet of Things (IoT). Students learn how cloud platforms support storing, analyzing, and managing vast amounts of data created by IoT devices. From understanding cloud infrastructure to implementing IoT solutions using cloud services, the course provides practical insights into optimizing IoT performance using the power of cloud computing.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CEN2211

Course User Fee: \$63.00

CIS2012C Introduction to Data Analytics and AI for the Internet of Things 4.000 credits

This course equips students with the knowledge and skills necessary to navigate the complex landscape of data analytics and artificial intelligence within the context of the Internet of Things (IoT), empowering them to contribute to the development of smart, data-driven IoT solutions.

Prerequisites: CIS 2010C, CIS 2011C.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS2010C AND CIS2011C

Course User Fee: \$63.00

CIS2013C Internet of Things Applications 4.000 credits

In this course, students learn about the practical applications of the Internet of Things (IoT) across various sectors and industries. They explore real-world use cases, ranging from smart homes and smart cities to the latest developments in the field. The course offers insights into how IoT solutions are designed, developed, and deployed, equipping students with the knowledge to create innovative IoT applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS2010C AND CIS2011C

Course User Fee: \$63.00

CIS2331 Systems Analysis, Design and Implementation 5.000 credits

This course is designed for computer and information technology majors. Students learn to investigate, analyze, plan, design, implement and document an information system using the traditional Systems Development Life Cycle (SDLC).

Lecture Hours: 64.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS2350 Cybersecurity Analysis (CySA+) 4.000 credits

This course provides students an intermediate skills-level approach to cybersecurity analysis. Students learn to identify the phases of an attack, the motivations of the adversary, the resources and techniques they use, the intended effect, or end-game, and how to mitigate threats. Topics include intrusion detection and response, analytics and advanced threat visibility. Prerequisites: CTS1120, CTS1134. Laboratory Fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120, CTS1134

Course User Fee: \$63.00

CIS2430C Remote IT and Security Management 4.000 credits

This course equips students with essential tools and strategies for Remote IT and Security Management (RITSM). It covers proactive security, business continuity, disaster recovery, and the use of Managed Service Provider (MSP) tools. Participants explore RITSM tools and technologies such as RMM, IT Glue, and Datto Networking. Prerequisites: CTS1134 and CTS1120. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120 AND CTS1134

CIS2619 Secure Software Development 4.000 credits

This course provides an introduction to Secure Software Development in modern languages such as Java, C and C++. Common weaknesses exploited by attackers are discussed, as well as mitigation strategies to prevent those weaknesses. Students practice programming and analysis of software systems through testing and static analysis.

Prerequisite: COP2800. Corequisite: COP2805C

Laboratory fee. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800, Corequisite: COP2805C

Course User Fee: \$63.00

CIS2900 Directed Study in IT 1.000 credits

This course, for majors in Information Technology, provides students the opportunity to pursue projects under faculty supervision through self-directed research, experiences, and presentations. Prerequisite: Successful completion of required program course work. Department approval required.

Lab Hours: 32.000

Recommended Preparation: Successful completion of required program course work. Department approval required

CIS2949 CO-OP WRK EXP 2: CIS 1.000 - 4.000 credits

This course is designed as a second-level capstone for students majoring in computer information systems programs. Students apply advanced skills and knowledge that they have acquired through their first capstone course in a real work environment. Prerequisite: Successful completion of required program course work. Department approval required.

Lecture Hours: 16.000 - 64.000

Prerequisite: CIS1949

CIS2991C Cloud Developer 4.000 credits

This course focuses on the role of a software developer in the cloud. A student will demonstrate an understanding of core cloud services, uses, and basic architecture and best practices. They will demonstrate proficiency in application lifecycle management, developing, deploying, and debugging cloud-based applications, serverless computing and containers by using the primary cloud platform tools for development including IDE's, distributed version control systems, CI/CD pipelines, including security best practices. Use of API's, command line interfaces and software development kits (SDKs) will be emphasized.

Prerequisites: COP 1047C and CTS 2375C

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2375C AND COP1047C

CIS2992C Cloud DevOps 4.000 credits

This course focuses on the DevOps role in cloud. A student will learn the concepts and skills needed to perform a DevOps engineer role. Students will gain expertise in provisioning, operating, and managing distributed application systems in the cloud. The student will evaluate and perform the following tasks in cloud-based systems: implement and manage continuous delivery systems and methodologies, implement and automate security controls, governance processes, and compliance validation, define and deploy monitoring, metrics, and logging systems, implement systems that are highly available, scalable, and self-healing, design, manage, and maintain tools to automate operational processes. Prerequisites: COP 1047C and CTS 2375C. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND CTS2375C

CIS3215 Ethics in CyberSecurity 4.000 credits

This course provides the study of the risk factors for digital and ethical misconduct, and it explores ethics, relevant laws, regulations, policies, standards, moral, and social issues and responsibilities faced by CyberSecurity professionals. Coverage includes examination of CyberSecurity policies; Federal Laws and Authorities and International Standards; ethical and legal compliance and enforcement; business issues; contractual management of assets and liabilities; and issues involving privacy, disclosure, free speech and individual rights.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS3360 Principles of Information Security 4.000 credits

This upper division course provides an overview of information systems security principles, practices, methods, and tools. Students will learn about the threats against an organization's digital assets, as well as the tools and methods to mitigate those threats. Topics include cryptography, authentication and access control systems, infrastructure security, and ethical issues. Prerequisite: CTS1134 or CTS1650

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CIS3361 Information Security Management 4.000 credits

This course covers how to manage, design, oversee and assess an organization's information security. Students will learn how to develop an information security strategy, how to write information security policies, and how to manage information risk. Other topics include security program development and management, business continuity planning and disaster recovery planning. Prerequisite(s): CIS 3360 or CTS 1120. Laboratory fee. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS3360 OR CTS1120

Course User Fee: \$63.00

CIS3368 Data Security & Governance 4.000 credits

This upper division course is for students majoring in Data Analytics. Students will gain an understanding of how analytics can be applied to a variety of security-related problems across organizations. In addition, students will explore various ethical, legal, and data governance issues that affect data analysts. (3 hr. lecture; 1 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

CIS3510 Information Technology Project Management 4.000 credits

This upper division course, for students majoring in Information Systems Technology, covers the general aspects of project management and emphasizes the pertinent considerations that apply to information technology projects. Students will learn the principles, processes and practices of information technology project management, including techniques for planning, organizing, scheduling, and controlling IT projects with a substantial focus on cost estimation and risk management. Students must enroll in this course one semester before taking CTS 4955.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CIS3641C Cloud Developer 4.000 credits

This course focuses on software development with cloud technologies. Students learn about core cloud services, basic architecture, and best practices used in the development and maintenance of applications in a cloud platform. Proficiency is gained in application lifecycle management, serverless computing, and containers using primary cloud platform tools, including IDE's, distributed version control systems, CI/CD pipelines, and security. API usage, command line interfaces, and software development kits (SDKs) will be emphasized. Prerequisites: COP 1047C and CTS 1145 (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND CTS1145

Course User Fee: \$63.00

CIS3644C Cloud Security 4.000 credits

This course equips students with essential principles, tools, and techniques to effectively secure, configure, and operate cloud-based information systems. Students will learn cloud services essentials, such as computing, storage, databases, and networking. Additionally, they'll delve into security tools, key management services, digital signatures, certificates, authentication protocols, SSO services, firewalls, and access control lists. Prerequisites: CTS 1120 and CTS 2375C (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120 AND CTS2375C

Course User Fee: \$63.00

CIS3651C Cloud System Administration and Operations 4.000 credits

This course offers a comprehensive exploration into the world of cloud system administration using cloud computing as the cornerstone of study. This intensive course seeks to equip students with hands-on knowledge and skill development in cloud infrastructure, tooling, computing, networking, storage, security, and deployment. From the fundamentals of Systems Operations on the cloud to intricate deployment strategies using CloudFormation, students will navigate various cloud services, ensuring a solid foundation in cloud administration. With a balance of theory, lab work, and activities, students will prepare to address real-world challenges helpful in the transition to cloud-based roles. Prerequisites: COP 1047C and CTS 1145. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C AND CTS1145

Course User Fee: \$63.00

CIS4204 Ethical Hacking I 4.000 credits

This upper division course introduces students to penetration testing techniques. Students will learn how to footprint, scan, sniff, enumerate networks, and hack systems. Other topics include denial of service attacks, social engineering, malware, vulnerability analysis and relevant laws.

Prerequisites: CIS3360

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CIS3360

Course User Fee: \$63.00

CIS4347 Information Storage Management 4.000 credits

This upper division course, for students majoring in Information Systems Technology, introduces challenges and solutions for data storage and data management. Students will learn how to manage advanced storage systems, protocols, and architectures, including storage area networks (SAN), network attached storage (NAS), fiber channel networks, internet protocol SANS (IPSAN), ISCSI, and content-addressable storage (CAS). Prerequisite: CTS1134 or CTS1650. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CIS4364 Intrusion Detection and Incident Response 4.000 credits

This upper division course addresses the underlying principles and techniques for detecting and responding to current and emerging cybersecurity threats. Students will learn how to handle various types of malware, email, web, network, cloud and internal network incidents, as well as risk assessment methodologies, and policies related to incident handling. Prerequisite: CIS 3360. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS3360
Course User Fee: \$63.00

CIS4366 Computer Forensics 4.000 credits

This upper division course provides the student with knowledge and skills to conduct formal incident investigations. Students will learn how to collect and analyze evidence from Windows and Linux computer systems. Other topics include legal issues, evidence analysis, and report writing. Prerequisite(s): CIS 3360 or CET 2880C. Laboratory fee. (3 hr. lecture; 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS3360 OR CET2880C
Course User Fee: \$63.00

CIS4378 Ethical Hacking II 4.000 credits

This upper division course is a continuation of Ethical Hacking I. Students will focus on how web applications, wireless networks, and mobile platforms can be hacked, and how intrusion detection systems (IDS), firewalls, and honeypots can be evaded. Other topics include cloud computing security, Internet of Things (IoT) security, and cryptography. Prerequisite: CIS4204. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS4204
Course User Fee: \$63.00

CIS4388 Advanced Computer Forensics 4.000 credits

This upper division course is a continuation of Computer Forensics. The course examines forensics techniques necessary to investigate and analyze network traffic. The course covers packet capture and analysis, log file analysis, and flow analysis. Other topics include mobile forensics, cloud forensics, malware forensics, database forensics, and investigating email crimes and web attacks. Prerequisite: CIS 4366 (3 hr. lecture, 2 hr. lab). Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS4366
Course User Fee: \$63.00

CIS4617 Knowledge Management 4.000 credits

This upper division course, for students majoring in Information Systems Technology, explores how an enterprise gathers, organizes, shares, and analyzes its knowledge in terms of resources, documents, and people skills. Students will learn how to gather, organize, refine and disseminate information needed in a small business or corporation using technical applications to house and mine the data. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: COP4723
Course User Fee: \$63.00

CIS4891 Cyber Security Capstone 4.000 credits

This upper division course, for students majoring in Information Systems Technology, requires students to demonstrate their competence to analyze, design, develop, and test an information system in a team environment. Students will learn how to create and present an information technology (IT) solution proposal that includes: design documentation, implementation plan, and project test plan to create an operational information system. Must be taken during the last semester before graduation. Prerequisite: CIS 3361. Pre/Corequisites: CIS4364, CIS4378, CNT3409C, CIS4388. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CIS3361, Pre/Corequisite: CIS4364, CIS4378, CIS4388, CNT3409C
Course User Fee: \$63.00

CJC**CJC0104 Correctional Officer Basic Defensive Driver Training 0.530 credits**

This course is a combination of classroom and practical exercises designed to evaluate the corrections recruit's ability to operate an emergency vehicle. Students will learn the psychological factors affecting vehicle operations, the elements of emergency driving skills, and skid-pan recoveries. For School of Justice Basic Correctional Officer students only. Special fee. Lecture Hours: 16.000

CJC1000 Introduction to Corrections 3.000 credits

A comprehensive view of the historical and philosophical treatment programs and developments in the field of adult institutional corrections. Emphasis is on understanding the offender in the correctional system; an examination of the correctional client and staff, agencies and recidivism; (3 hr. lecture). Lecture Hours: 48.000

CJC1005 Operations & Procedures In Correctional Institutions 3.000 credits

A basic survey of the operational routines that prevail in correctional facilities and the procedures used by officers in upholding these routines. The focus is on the preliminary knowledge needed by correctional officers before they can acquire the skills and techniques to perform job-related tasks. A.S. degree credit only. (3 hr. lecture)
Lecture Hours: 48.000

CJC1162 Parole and Probation 3.000 credits

The history, current practices and the consideration of philosophical concepts in the areas of probation and parole; (3 hr. lecture)
Lecture Hours: 48.000

CJC2301 Interpersonal Skills For Correctional Officers 3.000 credits

The interpersonal skills needed by officers to understand the incarcerated society is explored, with emphasis on supervision methods. Inmate adjustment and the various segments of inmate society are studied. This course is limited to School of Justice students only.
Lecture Hours: 48.000

CJC2350 Correctional Operations 3.000 credits

The operation of correctional facilities is studied, including the intake of new inmates, all aspects of their daily care, and institutional procedures. This course is limited to School of Justice students only.
Lecture Hours: 48.000

CJC4163 Advanced Probation & Parole 3.000 credits

A study of the process in which a convicted person can be released into society by means of probation or parole.
Lecture Hours: 48.000
Prerequisite: CJC1162

CJC4310 CORRECTIONAL THEORY 3.000 credits

An analysis of corrections relative to punishment and rehabilitation strategies utilized at penal institutions throughout the United States. Prerequisite: CJC 1000. (3 hr. Lecture)
Lecture Hours: 48.000
Prerequisite: CJC1000

CJC4311 CONTEMP ISSUES CO 3.000 credits

Focuses on and analyzes of major changes in incarceration philosophies and policies, prison populations, and operational costs. Prerequisite: CJC 1000. (3 Hr. Lecture)
Lecture Hours: 48.000

CJC4351 CORRECTIONAL OPS 3.000 credits

Focuses on challenges the correctional staff faces in their critical role in the day-to-day operations of a correctional facility. Prerequisite: CJC 1 000. (3 hr. lecture)
Lecture Hours: 48.000

CJE

CJE1003 Career Exploration In Criminal Justice 1.000 - 3.000 credits

To provide an overview of the various careers in criminal justice, and to help students define their career interests and physical abilities.
Lecture Hours: 16.000 - 48.000

CJE1567 FRST RESPOND/PUB SAF 3.000 credits

Provides training in emergency medical care for public safety officers who are apt to be the first persons responding to an accident or crime of violence. The focus is on the specific emergency situations a public safety officer is likely to confront, and the role of the public safety officer within the community's emergency medical service system. (3 hr. lecture)
Lecture Hours: 48.000

CJE1640 Crime Scene Technology 1 3.000 credits

This is an introductory course in Crime Scene Technology. Students will learn the techniques, materials and instrumentation used in securing, searching, recording, collecting, and examining physical evidence. There will be special emphasis on the tools, instruments, and techniques used in the studies of crime scene reconstruction, fingerprints, firearms, tool marks, and blood stain pattern analysis.
Lecture Hours: 48.000

CJE1642 Crime Scene Technology 2 3.000 credits

This course covers advanced principles, theories and applications in crime scene technology. Students will learn collection procedures of weapons, arson, gunshot residue, blood spatter, and recovery of human remains. Data analysis, reporting and plan of action development are emphasized.
Lecture Hours: 48.000
Prerequisite: CJE1640

CJE1673 Crime Scene Photography 1 3.000 credits

This is an introductory study of the history of photography including basic photography skills. Students will learn camera operations, exposure control, relational photographs and flash control for crime scene and evidentiary documentation. (3 hr. lecture)
Lecture Hours: 48.000
Course User Fee: \$9.00

CJE1680 Introduction to Computer Crimes 3.000 credits

This course provides the student with an overview of crimes involving the use of computer technology and the internet. Students will learn how computer related crimes are committed and how they are investigated. Topics covered will include computer crime scene management and the legal issues involved in the prosecution of computer crimes and legislation enacted to protect the public.. (3 hr. lecture)

Lecture Hours: 48.000

CJE1772 Crime Scene Photography 2 3.000 credits

This course expands upon concepts; knowledge and skills taught in Crime Scene Photography 1. Students will learn to include specialty light sources, darkroom techniques and procedures, filters and specialized equipment including black and white and color enlargers.

Lecture Hours: 48.000

Prerequisite: CJE1673

CJE1801 DEF TCT SKL-CJ PRACT 3.000 credits

Training of officers in the rationale and methodology of taking people into custody, searching subjects, using restraint devices, and utilizing the proper techniques and amount of force. For Institute of Criminal Justice students only. (1 hr. lecture ; 4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

CJE2302 Management of Police Functions 1.000 - 3.000 credits

The administration of line activities of law enforcement agencies, with emphasis on the patrol functions and the prevention of crime, including traffic, investigations, juvenile, vice, and other specialized units.

Lecture Hours: 16.000 - 48.000

CJE2304 POLICE SUPERVISION 3.000 credits

An introduction to basic theory pertaining to supervisory responsibilities and assignments. Practical application will be demonstrated through the case-study method. (3 hr. lecture)

Lecture Hours: 48.000

CJE2401 Race and the Criminal Justice System 3.000 credits

This course will provide a focused review of the history of policing racial and ethnic communities in America and provide context to understanding fears, attitudes, and perceptions of Police by racial and ethnic communities. It is designed for the entry-level criminal justice practitioner and beginning student interested in entering the field. (3 hr. lecture)

Lecture Hours: 48.000

CJE2452 CRIM JUSTICE COMM 3.000 credits

The report writing process from the interview, statement taking and note taking, through the final report product is covered, with practical exercises included. The differences between interviewing and interrogating are explored. Interpersonal communication skills are covered, along with radio and telephone procedures. Objectives are addressed as specified by the Criminal Justice Standards and Training Commission. Prerequisite: ENC 1101. (3 hr. lecture)

Lecture Hours: 48.000

CJE2552 law enf traffic 3.000 credits

Studies traffic enforcement and control with the inclusion of DUI offenses and enforcement. This course is limited to School of Justice students only. (3 hr. lecture)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

CJE2590 law enf patrol 3.000 credits

Theories, history, and development of police patrol are explored. Also addressed are the skills and techniques that are needed by officers on a daily basis to perform patrol tactics and respond to various types of calls. Methods of approach to various high-risk situations are explored, with practical exercises included. Unusual occurrence events, including firefighting and crowd control, are also addressed. This course is limited to School of Justice students only. (3 hr. lecture)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

CJE2600 Criminal Investigation 3.000 credits

Fundamentals of criminal investigation, theory and practice, including crime scene search; preservation, collection and transportation of physical evidence; interviewing, interrogating; statement taking; and case preparation, with investigation of specific offenses; relationship with the police science laboratory.

Lecture Hours: 48.000

CJE2601 Law Enforcement Investigations for Police Officers 3.000 credits

Fundamentals of criminal investigation, theory and practice, including crime scene search, preservation, collection and transportation of physical evidence are topics included in this course. Techniques are developed from the initial observation methods through the processing of the crime scene and case preparation. Florida's computer network is studied as an information source. This course is limited to School of Justice Basic Law Enforcement students only.

Lecture Hours: 48.000

CJE2644 Crime Scene Safety 3.000 credits

This course provides the fundamentals of protecting and preserving the crime scene and identifies the essential techniques of properly handling physical evidence. Students will learn the understanding of various hazards and safety issues and provides basic techniques for preserving evidence as it relates to various hazardous chemical and biological materials. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$14.00

CJE2671 Basic Fingerprinting 3.000 credits

This course provides a foundation in basic fingerprinting. Students will learn topics which include classification, identification, filing and rolling of fingerprints, problems and practices associated with post mortem fingerprinting and proper presentation of fingerprint evidence. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$11.00

CJE2672 Fingerprint Development 3.000 credits

This course provides a continuation of Basic Fingerprinting. Students will learn different methods involved in detection, enhancement, and recovery of latent fingerprints. Techniques will involve chemical and mechanical methods on substrates and evaluation for proper application in both theory and practices.

Lecture Hours: 48.000

Prerequisite: CJE2671

CJE3110 Law Enforcement Systems 3.000 credits

An analysis of the different law enforcement systems in Criminal Justice. Focuses on the different law agencies and their mission at the local, state, and federal levels.

Lecture Hours: 48.000

CJE3115 Police and Society 3.000 credits

Identifies police roles and philosophies, the nature of police work, community policing, and the debates pertaining to police discretion, community relations, and police misconduct.

Lecture Hours: 48.000

CJE3444 Crime Prevention 3.000 credits

Provides students with strategies of how to develop, implement and maintain a crime prevention program. Includes the history of crime prevention, homeland security programs, public speaking, media relations, crime against the elderly, sexual assault programs, youth crime prevention, and telemarketing fraud and scams.

Lecture Hours: 48.000

CJE3574 Interpersonal Communications For Law Enforcement 3.000 credits

An examination of the communication process and how it affects the relationship between the police and the people they serve.

Lecture Hours: 48.000

CJE4012 Criminal Profiling 3.000 credits

This course will examine criminal profiling undertaken by criminal justice professionals, which consists of gathering, reviewing, and analyzing evidence pertaining to violent crimes.

Lecture Hours: 48.000

CJE4310 POLICE ADMIN 3.000 credits

An analysis of corrections relative to punishment and rehabilitation strategies utilized at penal institutions throughout the United States. Prerequisite: CJC 1000. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CCJ1020

CJE4615 ADV CRIMINAL IVEST 3.000 credits

Lecture Hours: 48.000

CJE4647 Advanced Crime Scene Investigations 3.000 credits

A study of advanced search techniques, crime scenes reconstruction, computer sketching, laser mapping. DNA evidence, trajectory, and blood spatter evidence.

Lecture Hours: 48.000

Prerequisite: CJE2600

Course User Fee: \$59.00

CJE4648 Crime Scene Safety 3.000 credits

A study of how to properly handle crime scenes and hazardous crime scenes relative to various hazardous materials, to include chemical and biological.

Lecture Hours: 48.000

Course User Fee: \$71.00

CJE4650 Advanced Crime Scene Investigations 3.000 credits

A study of advanced search techniques, crime scenes reconstruction, computer sketching, laser mapping, DNA evidence, trajectory, and blood spatter evidence.

Lecture Hours: 48.000

Course User Fee: \$105.00

CJE4668 Computer Crime 3.000 credits

Synthesizes knowledge of crime elements, legal issues, investigative techniques, and computer skills used in the prevention and investigation of computer-generated crime.

Lecture Hours: 48.000

CJE4675 Modern Fingerprint Technology 3.000 credits

A study of the detection, preservation, and removal of fingerprint evidence pertaining to latent, patent, and plastic prints.

Lecture Hours: 48.000

Prerequisite: CJE2600

Course User Fee: \$32.00

CJJ**CJJ2002 JUVEN DELINQUENCY 3.000 credits**

Lecture Hours: 48.000

CJK**CJK0002 Introduction to Law Enforcement 0.400 credits**

In this course, students will learn to communicate effectively using empathy, courtesy, and professionalism while serving the community. The student will be able to understand the importance of self-talk and self-awareness in preparation for using the core communication competencies. For School of Justice students only.

Clock Hours: 12.000

Course User Fee: \$91.00

CJK0013 Interactions In A Diverse Community 1.330 credits

The student will learn the common communication traits of individuals based upon their cultures, experiences, physical and psychological conditions, and how specific situations can challenge an officer's effort to perform his or her duties. The student will learn different communication styles and to adapt to meet the needs of the people they are interacting with, eliminating potential miscommunications that may jeopardize an officer's effectiveness or safety. For School of Justice students only. (40 contact hrs.)

Lecture Hours: 40.000

CJK0016 Communications 0.800 credits

In this course, students will learn to communicate effectively using empathy, courtesy, and professionalism while serving the community. The student will be able to understand the importance of self-talk and self-awareness in preparation for using the core communication competencies. For School of Justice students only.

Clock Hours: 24.000

CJK0018 Legal 2.130 credits

In this course, students will learn legal rules and concepts, amendments, and law. The student will learn the basics of law, civil and criminal liability. For School of Justice students only.

Clock Hours: 64.000

CJK0019 Interviewing and Report Writing 1.800 - 1.870 credits

The student will learn note-taking, interviewing, elements and principles of effective report writing. The student will also learn to take statements from victims, witnesses, and suspects; write clear, concise and accurate incident and arrest reports. For School of Justice students only.

Clock Hours: 56.000

CJK0020 Law Enforcement Vehicle Operations 1.600 credits

Students will learn the physiological and psychological factors affecting vehicle operations. This course stresses the importance of vehicle maintenance, environmental conditions affecting driving, and elements of basic driving skills including skids and other causes of accidents. Students will demonstrate hands-on basic driving skills. For School of Justice students only.

Clock Hours: 48.000

Course User Fee: \$623.00

CJK0021 Serving Your Community 1.130 credits

The student will learn a community's relationship with law enforcement can greatly impact how law enforcement officers do their jobs. Students will also learn how to respond to people in crisis and high risk groups. For School of Justice students only.

Clock Hours: 34.000

CJK0031 CMS FRST AID C/J OFF 1.330 credits

This course prepares criminal justice recruits for a variety of medical emergencies with minimal medical supplies. Students will learn to initiate treatment for a variety of medical emergencies, understand and perform CPR, and know when to activate EMS and perform basic life support until help arrives. CPR and First Responder certification cards are issued upon successful completion. Basic training for School of Justice students only. (40 contact hrs.)
Lecture Hours: 40.000

Course User Fee: \$237.00

CJK0040 Criminal Justice Firearms 2.670 credits

Students will learn how to use both handguns and shotguns. Students must qualify with both weapons under both daylight and night conditions. Students must also demonstrate ability for both accuracy and decision-making. Students are also introduced to chemical weapons and their effects.

Course User Fee: \$1574.00

CJK0041 COM SERV OFC/POL SRV 3.660 credits

This course prepares students to become Community Service Officers/Police Service Aides by providing them with the basic knowledge needed to conduct preliminary property crimes investigations. For School of Justice students only. Special fee.

(110 contact hrs)

Lecture Hours: 110.000

CJK0051 Criminal Justice Defensive Tactics 2.670 credits

Students will learn how to physically defend themselves, physically control persons under arrest, and know what level of force is appropriate under differing circumstances. Students will learn escalate, de-escalate, or disengage techniques and when to use them. Additionally, a physical conditioning program is part of this course. For School of Justice students only.

Clock Hours: 80.000

Course User Fee: \$335.00

CJK0060 PATROL 1.900 credits

This course provides potential police officers with the knowledge, skills and abilities to function as a Patrol Officer in a law enforcement agency. Various methods of patrol activity, officer safety, and techniques will be examined. For School of Justice students only. (57 contact hrs.)

Lecture Hours: 57.000

CJK0063 Fundamentals of Patrol 1.330 credits

The student will learn law enforcement techniques and tactics that you will use while on patrol. The student will also learn how to respond to non-criminal calls and conduct structure and area searches, and provide resources that officers use while on patrol. For School of Justice students only.
Clock Hours: 40.000

CJK0065 Calls For Service 1.200 credits

The student will learn to respond to a variety of calls for service, which will vary between noncriminal and criminal incidents. This chapter provides an overview of the more common calls for service that an officer may encounter. For School of Justice students only (36 contact hrs.)

Lecture Hours: 36.000

CJK0070 INVESTIGATIONS 1.770 credits

This course provides training for new recruits in the search and location of physical evidence, along with the reproduction and identification collection, preservation and transporting of evidence to the crime laboratory. A basic understanding of the investigation of crimes needed by the street officer in their initial involvement of a crime scene will be provided. In addition, the fundamentals of interviewing, interrogation and statement taking will be addressed. For School of Justice students only. (53 contact hrs.)

Lecture Hours: 53.000

CJK0072 Crimes Against Persons 1.600 credits

The student will learn to conduct an investigation with attention to accuracy, detail, and professionalism. The student will learn to identify and respond to different types of crimes against persons. For School of Justice students only.

Clock Hours: 48.000

CJK0073 Crimes Involving Property and Society 0.400 credits

The student will learn the different types of crimes such as theft, burglary, white collar crimes and animal cruelty. The student will also learn about loitering, disorderly behavior, illicit drugs and vice crimes. For School of Justice students only.

Clock Hours: 12.000

CJK0075 INVEST OFFENSES 1.330 credits

This course includes the causes and effects of domestic violence; common facts and misconceptions about suicide and risks procedures for prevention and intervention and an officers responsibilities; identifying signs of adult, elder and child abuse and the proper procedure for reporting each. This course also includes methods and skills for conducting an initial investigation, a death: The definition, characteristics and situation in which an officer may encounter Sudden Infant Death Syndrome (SIDS); procedures for crime scene management; evidence collection and handling; developing information; and preparing and investigation report. (40 contact hrs.)

Lecture Hours: 40.000

CJK0077 Criminal Investigations 1.660 credits

The students will learn how to respond to an initial call for service, always preserve life first and then work to preserve the scene for investigation. This chapter will identify the key elements of crimes most frequently encountered during a shift to help you avoid some common mistakes made in the field. For School of Justice students only. (50 contact hrs.)

Lecture Hours: 50.000

CJK0079 Crime Scene Follow-up Investigations 1.130 credits

The student will learn evidence rules and concepts, securing the crime scene and documentation and handling procedures. The student will also learn to follow-up and review of initial information procedures and how to prepare for court. For School of Justice students only.

Clock Hours: 34.000

CJK0090 TACTICAL APPLICATION 1.800 credits

This course includes the Florida court system structures and how courts relate to law enforcement; instruction in the first response to emergency situations and rescue; general information involving law enforcement officers dealing with bomb explosives, bomb threats and weapons of mass destruction. Students will learn skills to perform different law enforcement functions while assigned to a special detail such as indoor or outdoor public events; the elements of unlawful assemblies and riots; and types of force that can be used in riotous situations and riot control procedures. For School of Justice students only. (54 contact hrs.)

Lecture Hours: 54.000

CJK0093 Critical Incidents 1.470 credits

The student will learn about events that can put lives at risk and cause major damage to property and the environment. The student will also learn about the causes of these events and how an effective response can affect a community's ability to recover from an event. For School of Justice students only.

Clock Hours: 44.000

CJK0095 C/J SPECIAL TOPICS 0.670 credits

Lecture Hours: 20.000

CJK0096 Criminal Justice Officer Physical Fitness Training 2.000 credits

The student will learn the benefits of maintaining physical fitness to include nutrition and diet. The student will also learn the effects of and how to deal with stress, how to build up muscular and cardiovascular endurance and perform the exercises as required. For School of Justice students only.

Clock Hours: 60.000

Course User Fee: \$65.00

CJK0109 ST EX...CORR OFC CRT 0.700 credits

This course is designed to provide substantive course review of the Criminal Justice Standards and Training basic correctional officer curriculum. Diligent use of review materials in this course will serve as excellent preparation for the FDLE correctional officer certification exam. This course is for SJSA correctional officers only. (21 contact hours)

Lecture Hours: 21.000

CJK0110 Parking Enforcement Specialist 0.530 credits

This course prepares agency employees for Parking Enforcement by teaching forms, parking law, court structure, human behavior and practical exercise. For Southeast Florida Institute of Criminal Justice students only.

Lecture Hours: 15.900 - 16.000

Course User Fee: \$48.64

CJK0111 Interviewing and Report Writing in Corrections 0.530 credits

Students will master the fundamentals of conducting interviews and preparing accurate, detailed reports in correctional settings. The course covers investigative interviewing techniques, documentation standards, evidence collection, and the preparation of incident reports, disciplinary reports, and other official documentation.

Clock Hours: 16.000

CJK0132 Basic Security Officer Training 1.330 credits

This course is designed to prepare students to apply for Class "D" Private Security Officer license.

Students will learn regulatory compliance, emergency procedures, ethics and entrepreneurship, courtroom procedures, traffic direction and crowd control, and more. Special Fee.

Course User Fee: \$25.00

CJK0134 Private Security Guard Training 2: Class "G" License 0.940 credits

Students will learn the basic knowledge and proficiency skills needed to safely handle and discharge a firearm. Students will also receive extensive educational material on firearms safety and the mechanical operation of firearms, various legal aspects of firearms ownership, possession, and use, with special emphasis on the legal issues involved in the carrying and use of a firearm as a licensed Florida security officer. The content and format of this course is established by section 493.6105(5), Florida Statutes, and Rule 5N-1.132, Florida Administrative Code. Additional firearms-related information is available in Rules 5N-1.129, 5N-1.130, and 5N-1.131, F. A.C.

Lecture Hours: 28.200

Prerequisite: To apply for a Class "G" license, you must be at least 18 years old. AND You must be either a U.S. citizen or a permanent legal resident alien to qualify for a Class "G" license, and you must provide documentation confirming your citizenship status. AND You must be either a U.S. citizen or a permanent legal resident alien to qualify for a Class "G" license, and you must provide documentation confirming your citizenship status. AND No felony convictions AND Recommended Preparation: Must Possess a valid Florida Class "D" security license or Florida class "CC" Private Investigator intern license or Florida class "C" Private Investigator license or a valid Florida Manager Class "M" or "MA" or "MB" license.

Course User Fee: \$142.00

CJK0212 CROSS-OVER CO TO LE 0.260 credits

Basic recruit students will learn the required Firearms High-Liability Proficiency Skills using a handgun (revolver or semi-automatic pistol for both daylight and night) and a long gun (shotgun or semi-automatic rifle/carbine), pursuant to Rule 11B-3.5.0024, F.A.C. (8.0 contact hrs.)

Lecture Hours: 8.000

CJK0221 CORRECTNL CROSS-OVER 1.560 credits

The students will learn and understand the importance of ethics, values, and professionalism, both in their personal lives and in their role as law enforcement officers. (47 contact hrs.)

Lecture Hours: 47.000

CJK0222 CORRECTNL CR-OV TO LE 1.860 credits

Students will learn to take statements from victims, witnesses, and suspects; write clear concise and accurate incident and arrest reports; and will engage in note taking skills such as grammar, spelling and proper sentence structure. For School of Justice students only. (56 contact hrs.)

Lecture Hours: 56.000

CJK0223 CORRTNL CR-OV LE HUM 1.060 credits

This course provides the student with a basic understanding of human relations with an emphasis on the student's ability as a Police Officer to influence others in a positive manner using interpretation skills. The student will learn the important role that interpersonal skills play in the relationship between the police and community. For School of Justice students only. (32 contact hrs.)

Lecture Hours: 32.000

CJK0301 Introduction to Corrections 1.066 credits

The student will learn the responsibilities of a correctional officer to provide safety for him/herself, public, staff, and inmates. The student will learn an overview on safety and security concerns, identification, manipulation and deception, contraband, and searches in a correctional setting. This course is for School of Justice students only.

CJK0306 Communications for Correctional Officers 1.070 credits

This course develops essential communication skills needed in correctional environments. Students learn effective verbal and non-verbal communication techniques, conflict resolution strategies, professional documentation methods, and appropriate interaction protocols with inmates, colleagues, and external agencies.

Clock Hours: 32.000

CJK0321 Intake and Release 0.530 credits

Students will study the processes and procedures involved in receiving new inmates and processing releases. The course covers booking procedures, classification systems, orientation processes, property management, record-keeping requirements, and coordination with courts and other agencies during intake and release operations.

Clock Hours: 16.000

CJK0324 Supervision in a Correctional Facility 1.070 credits

This course examines the principles and practices of supervising inmates within institutional settings. Students learn observation techniques, behavioral management strategies, rule enforcement, recreational supervision, work program oversight, and methods for maintaining order while promoting positive outcomes.

Clock Hours: 32.000

CJK0326 Supervising Correctional Populations 0.830 credits

Students will explore advanced supervision techniques for managing diverse correctional populations. The course addresses special population needs, risk assessment methods, case management principles, program coordination, and strategies for working with individuals who have varying security levels and behavioral challenges.

Clock Hours: 25.000

CJK0327 Shift Management and Safety 0.670 credits

This course focuses on the principles of managing correctional facility operations during assigned shifts. Students learn security procedures, safety protocols, routine operational checks, emergency preparedness, and effective coordination with other shifts and departments to maintain institutional security and order.

Clock Hours: 20.000

CJK0336 Incidents and Emergencies in Correctional Facilities 0.670 credits

This course prepares students to respond effectively to various incidents and emergency situations within correctional facilities. Students learn emergency response protocols, crisis management techniques, evacuation procedures, medical emergencies, disturbance control, and coordination with emergency services and administration.

Clock Hours: 20.000

CJK0340 Criminal Justice Officer Physical Fitness Training 1.000 credits

Students will learn the benefits of maintaining physical fitness to include nutrition and diet. Students will learn of the effects of stress and how to deal with it; how to build up muscular and cardiovascular endurance and perform the exercises as required. For School of Justice students only. Special fee.

Clock Hours: 30.000

Course User Fee: \$57.00

CJK0355 Legal for Correctional Officers 0.730 credits

Students will study the legal framework governing correctional operations, including constitutional rights of inmates, statutory requirements, liability issues, and court decisions that impact daily correctional practices. The course covers search and seizure procedures, disciplinary processes, and legal standards for institutional operations.

Clock Hours: 22.000

CJK0400 Traffic Incidents 0.400 credits

The student will learn the necessary information about traffic statutes and procedures and will lay the foundations for the practice of excellent traffic enforcement. The student will learn all aspects of law enforcement related to traffic enforcement, vehicles, roadways, and pedestrians. For School of Justice students only.

Clock Hours: 12.000

CJK0401 Traffic Stops 0.800 credits

The student will learn the goal of every traffic stop is to promote driver education and safer roadways. The student will also learn to exercise procedural justice and remain professional at all times when issuing a uniform traffic citation or warning, making an arrest, or finding that someone needs help. For School of Justice students only.

Clock Hours: 24.000 - 247.000

Course User Fee: \$44.00

CJK0402 Traffic Crash Investigations 1.000 credits

The student will learn about conducting a traffic crash investigations using a systematic approach. The student will learn about responding to, assessing, and protecting the scene; gathering information and evidence, documenting a crash. For School of Justice students only.

Clock Hours: 30.000

CJK0403 DUI Traffic Stops 0.800 credits

The student will learn how a driver that is under the influence of drugs or alcohol poses a hazard to themselves and others. The student will also learn how a law enforcement officer detects impaired driving, administers field sobriety tests, makes arrests when appropriate, and records the evidence of a DUI (driving under the influence) offense. For School of Justice students only.

Clock Hours: 24.000

Course User Fee: \$4.00

CJK0412 NPS ENFORCEMENT OP 1.030 credits

a

Lecture Hours: 31.000

Course User Fee: \$150.00

CJK0412 NPS ENFORCEMENT OP 1.030 credits

a

Lecture Hours: 31.000

CJK0415 NPS DRIVING 1.200 credits

a

Lecture Hours: 36.000

Course User Fee: \$488.00

CJK0415 NPS DRIVING 1.200 credits

a

Lecture Hours: 36.000

CJK0416 NPS PHYSICAL TECHN 1.730 credits

a

Lecture Hours: 52.000

Course User Fee: \$300.00

CJK0416 NPS PHYSICAL TECHN 1.730 credits

a

Lecture Hours: 52.000

CJK0421 Dart-Firing Stun Gun 0.130 credits

The student will learn the basics of the conducted electrical weapon (CEW), particularly the dart-firing stun gun. The student will learn information about its practical use based on statutory requirements.

For School of Justice students only.

Clock Hours: 4.000

CJK0441 PUB SER AIDE BAS TRN 13.000 credits

THE STUDENT IS PREPARED FOR INSTITUTE CERTIFICATION AS A PUBLIC SERVICE AIDE THROUGH COMPLETION OF CRIMINAL JUSTICE ASSISTING COMPETENCIES. TOPICS INCLUDE ACCIDENT INVESTIGATION, BASIC LAW, INVESTIGATIVE PROCEDURES, COMMUNICATION SKILLS, AND HUMAN BEHAVIOR. FOR INSTITUTE OF CRIMINAL JUSTICE STUDENTS ONLY. SPECIAL FEE. (390 CONTACT HRS.)

Lecture Hours: 390.000

CJK9995 CRIMINAL JUSTICE 15.000 credits

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Lecture Hours: 240.000

CJL**CJL1000 Street Law 3.000 credits**

This course will cover the evaluation, debate, and critical analysis of law and legal issues that affect individuals, their families, and communities. Students will learn about practical aspects of civil, criminal, constitutional, family, immigration, and consumer law in a diverse society with an orientation toward civic involvement in the local community.

Lecture Hours: 48.000

CJL1100 CRIMINAL LAW 3.000 credits

Historical background and foundations of American criminal law, including United States Constitutional requirements, Federal and State court organization and jurisdiction, criminal law basics, Florida statutes, rules or evidence and procedure. (3 hr. lecture)

Lecture Hours: 48.000

CJL2062 Constitutional Law and Legal Procedure or Evidence 3.000 credits

An examination of the United States and Florida Constitutions, with emphasis on leading cases dealing with arrest, search and seizure, confessions and the rules of evidence.

Lecture Hours: 48.000

CJL2080 COMPARATIVE LEGAL ST 3.000 credits

An introduction and comparative study of English and American systems of criminal justice, with particular reference to the protection of the liberty of the individual. Overview of legal systems of other nations, selected to afford a comparative perspective on the Anglo-American tradition. (3 hr. lecture). Offered through Overseas Study Program. (3 hr. lecture)

Lecture Hours: 48.000

CJL2102 Criminal Procedure & Evidence 1 3.000 credits

This course explores the history, principles and applications of criminal law procedures for criminal justice officers. This course is limited to the School of Justice students only. (3 hr. lecture)

Lecture Hours: 48.000

CJL2105 legal 3 for leo 2.000 credits

Lab Hours: 32.000, Clinical Hours: 1.000 - 30.000

CJL2130 CRIMINAL PROC&EVID 2 3.000 credits

Criminal Procedure and Evidence as they relate to the law enforcement profession will be examined. Constitutional provisions applicable to arrest, search and seizure, and interrogation will be covered. In addition, evidentiary principles will be taught emphasizing those provisions applicable to law enforcement. (3 hr. lecture)

Lecture Hours: 48.000

CJL2610 Courtroom Presentation 3.000 credits

This course introduces students to proper courtroom presentation and procedures. Students will learn the appropriate techniques for proper attire, grooming, speaking, listening and stress control during courtroom proceedings, visual aid preparation, and presentations of all evidence (commonly referred to as "scientific evidence") collected at the crime scene are also included.

Lecture Hours: 48.000

CJL2990 Cannabis Policy and Regulation 3.000 credits

This course will provide an overview of Florida's cannabis laws and regulations, federal regulations and scheduling of cannabis. Students will learn the social and historical usage and regulation of marijuana as well as the reforms and debates having an impact on the usage, control and regulation in Florida. The course will emphasize legal, political, ethical and practical issues. (3 hr. lecture)

Lecture Hours: 48.000

CJL3044 Civil Law 3.000 credits

A study of civil liability for damages caused by breach of an imposed duty, which includes intentional torts, negligence, strict liability, product liability, civil nuisance, defamation, civil wrongful invasion of privacy, and damages.

Lecture Hours: 48.000

CJL3564 Judicial Policy Making 3.000 credits

An analysis of the components, policies, and procedures of the court structure of the United States and various components. An analysis of local, state, and federal courts in the Criminal Justice System.

Lecture Hours: 48.000

CJL4064 CORR ADMIN & LAW 3.000 credits

An overall view of the nature, philosophy, operations and goals of secure and non-secure correctional institutions and programs. Prerequisite: CJC 100 0. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: CJC1000

CJL4133 Criminal Evidence 3.000 credits

A study of evidentiary principles and rules of evidence, and their application in a courtroom setting.-

Lecture Hours: 48.000

CJL4170 CORR LEGAL SYSTEM 3.000 credits

An analysis of contemporary legal decisions regarding the rights and responsibilities of prisoners, correctional administrators, and correctional officers. Prerequisite CJL1100. (3 hr. lecture)

Lecture Hours: 48.000

CJL4514 CRIMINAL SENTENCING 3.000 credits

An examination of the various pre-trial and post-trial community based treatment and supervision programs. (3 hr. lecture)

Lecture Hours: 48.000

CLP**CLP1006 PSY OF PERSONAL EFF 3.000 credits**

Lecture Hours: 48.000

CLP2000 Dynamics of Behavior 3.000 credits

Analysis of mechanisms of adjustment, motivation, frustration and conflict, learning personality and psychotherapy. Emphasis is on the psychological processes of the normal individual functioning in society rather than on behavior disorders.

Lecture Hours: 48.000

CLP2001 BASIC HUMAN DEVEL 2.000 - 3.000 credits

Lecture Hours: 32.000 - 48.000

CLP2140 Psychopathology 3.000 credits

This course examines the major categories of mental disorders. Diagnostic criteria, treatment methods, cultural factors, public attitudes, community resources, ethical issues and legislation applicable to individuals with mental disorders are studied. The impact of mental disorders on individuals, families and society are discussed.

Lecture Hours: 48.000

Prerequisite: PSY2012

CLP2431 Field Experience in Behavioral Analysis 3.000 credits

The student will learn the process and importance of observing, documenting, and interpreting the behavior of individuals with special needs. Students will learn and apply various methods to document behavior and create behavioral treatment plan.

Prerequisite: EAB 1001. (20 hours field experience).

(1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: EAB1001

CLP2470 Introduction to Childhood Psychopathology 3.000 credits

This course will satisfy a requirement for the Specialized Registered Behavior Technician Associate in Science program. This course is an introduction to the various forms of abnormal behavior in infancy, childhood, and adolescence within the context of traditional and contemporary psychological theory. Diagnostic criteria, treatment methods, cultural factors, public attitudes, community resources, ethical issues and legislation applicable to childhood mental disorders are studied. The impacts of mental disorders on individuals, families and society are discussed.

Prerequisite: PSY 2012. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: PSY2012

CNT

CNT1512 INTRO WIRELESS NTRWK 4.000 credits

This course provides the student with a complete foundation of knowledge for entering into or advancing in the wireless networking industry. Topics include: an introduction to wireless LANs; RF theory; spread spectrum technologies; wireless LAN infrastructure devices; antennas and accessories; wireless LAN standards; and wireless LAN organizations to link budget math, troubleshooting, and performing a site survey. This course delivers hands-on training that benefits the novice as well as the experienced network professional. Prerequisites: CGS 1060 and CTS 1134. Laboratory fee. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CNT3409C Network Security 4.000 credits

This upper division course introduces students to current and emerging threats to the security of computer networks, as well as tools and techniques for the prevention, detection and recovery from such attacks. Topics include firewalls, intrusion detection and intrusion prevention systems, virtual private networks, remote authentication and authorization systems, and security protocols. Prerequisite(s): CIS 3360. Laboratory fee. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 27.000 - 32.000

Prerequisite: CIS3360

Course User Fee: \$63.00

CNT3526C Wireless and Mobile Networking 4.000 credits

This is upper division course for students majoring in Information Systems Technology introduces students to wireless and mobile network architecture, protocols and technologies. In particular, students will learn about Wireless Local Area Networks (WLANs), Wireless Personal Area Networks (WPANs) and Wireless Metropolitan Area Networks (WMANs) technologies. Other topics include antenna concepts, cellular networks, the 802.11 network architecture, and wireless security. Prerequisites: CTS 1134 or CTS 1650.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CNT4603 System Administration and Maintenance 4.000 credits

This upper division course, for students majoring in Information Systems Technology, explores UNIX and Microsoft Windows systems and their administration and maintenance within the network setting. Students will learn how to install, maintain, and extend multi-user computer systems and how to develop administrative policies and procedures. Students will also learn how to apply troubleshooting and problem solving skills to resolve user and system issues.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1134 OR CTS1650

Course User Fee: \$63.00

CNT4702 Network Design and Planning 4.000 credits

This upper division course, for students majoring in Information Systems Technology, presents network design using layering. Students will learn how to apply cabling, topology, and architecture to design systems. Students will also learn how design impacts network performance and control issues such as congestion control, error control, and contention resolution. Prerequisite: CIS 3360.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP**COP1047C Introduction to Python Programming 4.000 credits**

This is a course in Python programming available for students at all levels. Students will learn the syntax and rules of the Python language, including how to code, compile, and execute programs. Students study program design, structured modular programming arrays, report generation, and file processing.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP1120 Introduction to COBOL Programming 4.000 credits

This is an introductory course in COBOL programming recommended for students majoring in Information Technology and Computer Information Systems. Students will learn how to design, code, compile, and execute structured programs for business applications. Recommended preparation: CGS1060 or experience working with computers and knowledge of elementary algebra. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Pre/Corequisite: CGS1060

Course User Fee: \$63.00

COP1332 Introduction to Visual Basic Programming 4.000 credits

This course introduces computer science and non-major students to fundamental programming skills using the Visual Basic Integrated Development environment. Students will learn program design, the fundamentals of event driven object-oriented programming, arrays, validation of user input, and how to create menu driven programs and multiple form applications. This course may be taken by those not majoring in Computer Information Systems.

Lecture Hours: 48.000, Lab Hours: 32.000

Pre/Corequisite: CGS1060C

Course User Fee: \$63.00

COP1334 Introduction to C++ Programming 4.000 credits

This course is designed for students in technology majors who require a foundation in computer programming. Students will learn the syntax and rules of the C++ language, including how to code, compile, debug and execute programs. Students will learn program design, structured and modular programming, arrays, and file processing. No previous computer courses are required although CGS 1060C is recommended. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2041C Python Programming 2 4.000 credits

This is an advanced level programming course using Python. Students will learn how to code, compile, and execute programs. Topics include problem solving, object-oriented programming, database programming, data visualization, and secure software development. Prerequisite: COP1047C; (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1047C

Course User Fee: \$63.00

COP2270 C for Engineers 4.000 credits

This course is intended for students majoring in Computer Engineering Technology, Electronics Engineering Technology, or any engineering discipline. Students will learn the "C" programming language, MATLAB, and the Engineering Problem Solving Method to analyze, design, code, compile and execute programs that solve engineering related problems. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105, Recommended

Preparation: CGS1060 OR knowledge of computer skills

Course User Fee: \$63.00

COP2333 ADV PROG CONC VIS BA 4.000 credits

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2335 C++ Programming 2 4.000 credits

This course delves further into the application of C++ programming techniques including classes, object orientation, function and operator overloading, inheritance and dynamic polymorphism, templates, exception handling, standard template library, data structures, and other advanced C++ techniques. Prerequisite: COP 1334 (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1334

Course User Fee: \$63.00

COP2658 iPhone Application Development 2 4.000 credits

This intermediate iOS course teaches the principles of iPhone application development for majors in Computer Science, Computer Information Systems, and related disciplines. Students will learn how to create mobile applications that can be deployed to iPhone smartphones, tablets or simulators utilizing Cocoa and XCode for development. Emphasis will be placed on learning the underlying iPhone framework and components in order to create quality mobile applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2664

Course User Fee: \$63.00

COP2660 Android Application Development 1 4.000 credits

This course teaches the principles of Android application development for majors in Computer Science, Computer Information Systems, and related disciplines. Students will learn how to create mobile applications for deployment to Android smartphones, tablets or simulators utilizing open source software (Java, Eclipse IDE, Android Plug-In and Android SDK) for development. Emphasis will be placed on the underlying Android framework to create quality applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

COP2662 Android Application Development 2 4.000 credits

This course for majors in Computer Science, Computer Information Systems, and related disciplines teaches how to develop advanced Android applications. Students will learn how to create applications utilizing the advanced capabilities of Android smartphones, including interfacing the application to the device's content providers' databases, GPS and location based services, notifications, background threads, audio, video, SMS, motion sensors and network connectivity. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: Prerequisite: COP 2660

Course User Fee: \$58.00

COP2664 iPhone Application Development 1 4.000 credits

This is an introduction to iOS programming course using the Objective C computer language, recommended for Computer Science and Computer Information Systems majors. Students will learn to code, compile and execute mobile iOS applications while learning advanced programming concepts and object oriented programming design concepts and principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP1332 OR COP1334

Course User Fee: \$63.00

COP2800 Java Programming 4.000 credits

This is an intermediate level programming course using the Java computer language. Students will learn to code, compile and execute programs, while learning advanced programming concepts and object-oriented programming and design principles.

Lecture Hours: 48.000, Lab Hours: 32.000

Corequisite: COP1047C OR COP1334 OR COP2270

Course User Fee: \$63.00

COP2805C Java Programming 2 4.000 credits

This is an advanced level programming course using Java. Students will learn how to code, compile and execute programs. Topics include object serialization, Java Collection, sorting/searching algorithms, multithreading, and Java databases.

Prerequisite: COP 2800. (3hr. lecture; 2hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

COP2823 ASP/SCRIPT LANG PROG 4.000 credits

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP2842 DEV WEBSTS PHP/MYSQL 4.000 credits

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

COP3530 Data Structures 4.000 credits

This upper division course is for students majoring in B.S. in Information Systems Technology. Students will learn the fundamentals of data structures using the Java programming language. The students will learn to design, implement and use data structures to organize and store data in a computer so that it can be accessed and modified efficiently.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800

Course User Fee: \$63.00

COP4656 Mobile Applications Development 4.000 credits

This course is for students majoring in Information Systems Technology, and covers project-oriented development of applications for mobile computing devices. Students will learn how to develop mobile applications utilizing memory management, user interface design, user interface building, input methods, data handling, network techniques, URL loading, and GPS and motion sensing. Students will develop a project that produces a professional-quality deployable mobile application.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2800 AND COP2654 OR COP2660

Course User Fee: \$63.00

COP4723 Database Administration 4.000 credits

This upper division course, for students majoring in Information Systems Technology, builds a deeper understanding of how databases work, including topics in database theory and architecture, data modeling, query languages, and security. Students will learn the fundamentals of SQL, including how to create and maintain database objects, and how to store, retrieve, and manipulate data, and the basics of managing the database environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1540

Course User Fee: \$63.00

COP4807 Web Programming with Java 4.000 credits

This is upper division course for students majoring in Information Systems Technology introduces students to the design, implementation and testing of web-based applications using the Java language. In particular, students will learn about the three-tier architecture, the Model View Controller architecture, servlets, and Java Server Pages, JDBC/JPA, and Web Services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP3530

Course User Fee: \$63.00

COP4834 Data-Driven Web Applications 4.000 credits

This upper division course, for students majoring in Information Systems Technology, utilizes three-tier application development to build web-based applications that use database systems. Students will learn how to integrate a web scripting language such as Node.js and a database server such as MongoDB and/or MySQL to build a data-driven web application system.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP3530

Course User Fee: \$63.00

CPO**CPO2100 Comparative European Government 3.000 credits**

This course discusses the structures and functioning of the systems of government in Western Europe. Students will analyze major policy issues and approaches to resolving them, and cultural differences and interdependence among the Western European countries in policy determination. Recommended prerequisite: a prior course in History or Social Science. Offered first semester. Given in English. Offered through overseas study program.

Lecture Hours: 48.000

CPO2408 Comparative Middle Eastern and North African Governments 3.000 credits

The Middle East & North Africa (MENA) has uninterruptedly been at the forefront of international geopolitics for almost a century. This enduring geopolitical visibility has important cultural, societal, economic, and security implications at national, regional, and international levels. This course is designed to tackle some of these issues by providing students with a historico-cultural, politico-economic, and geo-strategic survey of the Middle East.

Lecture Hours: 48.000

CPO2441 Islam in the Modern World 3.000 credits

The survey course is designed to grant students a broad-based exposure across a variety of conceptual debates, historical events, and policy issues at the intersection of Islam and the modern world and the paths forward. The issues especially religiously-inspired and informed violence/conflict and debates surrounding them have assumed prominence in the 21st century with the resurgence of religion in politics and the subsequent struggle for Islam. These issues will be contextualized in normative, empirical and historical frameworks to equip students with a sophisticated and multi-dimensional conceptual toolbox to analyze them.

Lecture Hours: 48.000

CRW**CRW2001 Creative Writing 1 3.000 credits**

Imaginative writing in selected genres.

Lecture Hours: 48.000

CRW2002 Creative Writing 2 3.000 credits

Imaginative writing in selected genres.

Lecture Hours: 48.000

CTE**CTE1050 Introduction to Fashion Design and Related Industries 3.000 credits**

In this course students will learn the basic design process, history, characteristics and global interrelationships of the fashion industry segments. The course explores how fashion, textile and apparel producers, retailers and related industries use merchandising and market their products within the industry and ultimately, to the consumer.

Lecture Hours: 48.000

CTE1401 Textiles 3.000 credits

This is a survey course designed for students majoring in fashion-related curriculum or with a general interest in textile materials. Students will learn basic elements of the transformation from fiber of textiles into finished goods. The course provides insights into textile manufactures with a primary focus on general textile applications relative to end-use consumer products. Students will learn the terminology needed for effective communication throughout the fashion supply chain, gain insight and appreciation for the relative value of textile products and the appropriateness of specific textile uses.

Lecture Hours: 48.000

CTE1401L Introductory Textile Science Lab 1.000 credits

The laboratory CTE 1401L course complements the Introductory Science CTE 1401 course. Students will learn the methods for basic identification of textile materials and rudimentary analysis techniques. The laboratory is also designed to support and parallel the concepts discussed in the lectures.

Lab Hours: 32.000

CTE1721C Fashion Design 1 3.000 credits

This introductory course on flat patterns explores the two-dimensional method of creating basic slopers in order to execute designs. Students will learn the two methods of making a basic set of slopers.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE1401 AND CTE1401L

Course User Fee: \$241.00

CTE1743C Patternmaking Level 1 3.000 credits

This course will focus on the development of basic blocks, dart manipulation, bodice, skirt and sleeve styling methods in accordance with approved garment trade practices. Students will learn to draft and manipulate the various garment components and develop patterns and samples.

Lecture Hours: 16.000, Lab Hours: 64.000

Course User Fee: \$241.00

CTE1801 Introduction to Fashion Merchandising 3.000 credits

This introductory class provides an exposure to merchandising and terminology. Students will learn the entrepreneurs who influence the industry, career possibilities and an overview of the components of a manufacturer's or retailer's promotional techniques.

Lecture Hours: 48.000

CTE1841C Apparel Evaluation & Production 3.000 credits

This is an introductory course in the apparel development process. Students will learn to facilitate the communication and coordination of pre-product development tasks achieved through linking design, costing, and manufacturing technology in the production setup for each design. Students will learn how outsourcing affects the product development process in editing garment designs and the line development calendar.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: CTE1401 AND CTE1401L

CTE1942 Fashion Industry Internship 3.000 credits

This internship course provides eligible students with placement in premier fashion settings and with the professional and practical experiences needed to further their education in a variety of fashion-based positions. Students control the internship selection process and work with an internship coordinator in the revisions of their resumes and schedule of interviews within the network of fashion companies. Fashion internships are available in the areas of design, merchandising, buying, show-room, and fashion public relations. (144 hr. Internship)

Internship Hours: 144.000

Recommended Preparation: Departmental Approval & Completion of the Program Core Requirements

CTE2111C Digital Fashion Portfolio 3.000 credits

In this course students will learn to integrate computer aided design tools in the creative process of developing a digital fashion collection portfolio. Visual presentation and specific applications will be used.

Lecture Hours: 16.000, Lab Hours: 32.000

Recommended Preparation: CAD/Photoshop/ Illustrator

CTE2120 Portfolio Collection Development 3.000 credits

In this course students will learn the basics for the development of a presentation portfolio collection. Students will research fabrics to use in the design of collection concepts for specific market areas. Various formats will be explored using digital and analog media. Retail market research and the internet may be applied as a basis for a design presentation.

Lecture Hours: 48.000

Prerequisite: CTE1401 AND CTE1401L

CTE2301 Product Development 3.000 credits

In this course students will learn the concepts and methods by which retailers create special, store-branded merchandise for targeted customer segments. The process of product development, from research to production to distribution, is studied.

Lecture Hours: 48.000

Prerequisite: CTE1401 AND CTE1401L

CTE2310C Basic Clothing Construction Methods 3.000 credits

Students will learn the basic elements of sewing utilized and incorporated into all designs in the garment industry. These garment structures form the fundamentals of sewing and are integrated into the construction methods used by each company in the applications to a specific design.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: CTE1721C

Course User Fee: \$390.00

CTE2330C Clothing Construction Methods Level 2 3.000 credits

This course focuses on the principles and techniques of applied clothing constructions. Students will learn to apply the principles and techniques of clothing construction using original student design and pattern work.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2310C, Corequisite: CTE2745C

Course User Fee: \$241.00

CTE2342C Clothing Construction Methods Level 3 3.000 credits

This course advances students' knowledge about materials, core properties and construction techniques that inform their choices made for silhouette creation and hands-on prototype development. Students will explore more advanced principles and techniques of unstructured draping using soft fabric for the creative interpretation and artistic development of contemporary designs. Emphasis on proportion, balance and shape as related to design aesthetics. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2330C, Corequisite: CTE2749C

CTE2388 Principles of Contemporary Retailing 3.000 credits

In this course students will learn the operational segments of the fashion industry and their functions. The course focus is on the contributions employees add to sales productivity and customer satisfaction in retail establishments and on the exploration of new technologies and their impact on consumers' shopping experiences.

Lecture Hours: 48.000

Prerequisite: CTE1050

CTE2610 Fashion Forecasting and Research 3.000 credits

In this course students will learn to explore and apply forecast research methods in preparation for developing, planning, purchasing, or merchandising apparel lines and collections. Using the case study method, trend research is evaluated through the use of scholarly texts, articles, databases, and relevant websites to identify opportunities for growth and profitability in a fashion business.

Lecture Hours: 48.000

Prerequisite: CTE1050 AND MAR1011

CTE2722C Fashion Design 2 3.000 credits

This course focuses on the use and development of basic slopers. Students will learn to proceed from basic applications and principles of patternmaking to the advanced process of design development. Combining the bodice and sleeve slopers to develop a kimono/dolman sloper is one example of the application of basic slopers to the comprehensive understanding of principles of pattern and design development.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: CTE1721C

Course User Fee: \$390.00

CTE2732 Fashion Illustration Technology 3.000 credits

This course introduces the basics of fashion illustration technology targeted specifically for the fashion industry. Using various computer software and tools, including but not limited to Photoshop, Fashion CAD, C-Design, Adobe Illustrator, and Digital Fashion Pro, students will learn the techniques to conceptualize, create, and manage fashion designs and collections, and create industry-standard presentations.

Lecture Hours: 48.000

CTE2745C Patternmaking Level 2 3.000 credits

This course focuses on the integration of flat patterns and draping. Students will learn the foundations of Draping as a design development and execution technique. How to move from the form to the flat and on to the body. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE1743C, Corequisite: CTE2330C

Course User Fee: \$241.00

CTE2749C Patternmaking Level 3 3.000 credits

This course reinforces the students' understanding of the art of draping and patternmaking and diverse methods the industry uses to create production patterns. Students will analyze draping and drafting techniques to create an awareness of which method is most applicable for a given situation. Analytical thinking and hands-on class experiences will strengthen students' skills, thus enabling them to expand on their creativity and provide a proper fit for their creations. Must be met with a grade of "C" or higher.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2745C, Corequisite: CTE2342C

CTE2760C Creative Design 3.000 credits

In this capstone course, the student will focus on the acquisition of a multidisciplinary methodology needed to produce a collection. The student will also survey the various steps of building a collection, from conception and range planning through to design, execution and presentation. A minimum of 6 looks will be produced by the end of the semester. Prerequisite: CTE2342C and CTE2722C and CTE2749C. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: CTE2722C AND CTE2342C AND CTE2749C

Course User Fee: \$241.00

CTE2800 Textile, Apparel & Retail Analysis 3.000 credits

In this course students will learn about textile marketing of sustainable apparel and the textile value chains from product concept to the consumer. A variety of topics on global value chains, market analysis, product development, manufacturing, market and sourcing are explored. The global impact of trade and sourcing constraints are examined. Through readings, case studies and in-class industry presentations, students will explore a comprehensive array of contemporary issues, both social and regulatory, that help in understanding the complex value and supply chain.

Lecture Hours: 48.000

Prerequisite: CTE2388

CTE2802 Fashion Merchandising Strategies 3.000 credits

In this course students will gain comprehensive knowledge of the merchandising environment, including the functions and objectives of the merchandising team, the principles and techniques of today's buyers, planners, product developers, and account executives.

Lecture Hours: 48.000

Prerequisite: CGS1060C

CTE2836 Global Merchandising 3.000 credits

In this course students will learn the merchandising practices used around the world in fashion apparel companies, both in retail and wholesale. American merchandising theory is used as a base of comparison in the consideration of various religions, cultures, legal systems, and other global systems.

Lecture Hours: 48.000

Corequisite: CTE2802

CTS**CTS0050 intro to microcomp 2.500 credits**

This course introduces the student to modern micro computer hardware and software. The topics covered include operation of microcomputer hardware and peripherals, operating system commands, word processing software and database management software. The 75 contact hours encompass both lecture and laboratory components. Special fee. (75 contact hrs)

Lab Hours: 75.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CTS0050 Introduction to Microcomputers 2.500 credits

This course introduces the student to modern micro computer hardware and software. The topics covered include operation of microcomputer hardware and peripherals, operating system commands, word processing software and database management software. The 75 contact hours encompass both lecture and laboratory components. Special fee. (75 contact hrs)

Lab Hours: 75.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CTS0094 des directory servs 2.500 credits

This course provides students with the knowledge and skills necessary to design a Windows directory services infrastructure in an enterprise network. Strategies are presented to assist the student in identifying the information technology needs of an organization, and then designing a directory services structure that meets those needs. Prerequisite: CTS 0093. Special fee. (75 contact hrs.)

Lab Hours: 75.000

Course User Fee: \$63.00

CTS1111 Linux + 4.000 - 32.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based work-stations and servers. Prerequisite: CGS 1060 or computer experience is required. (3 hr. lecture; 2 hr. lab)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CTS1111 Linux + 4.000 credits

This course is designed to help students prepare for the CompTIA Linux+ Certification Exam and to teach the skills needed to administer GNU/Linux-based workstations and servers. Students learn how to plan, install, maintain, document, and troubleshoot GNU/Linux operating system services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1060C

Course User Fee: \$63.00

CTS1120 Cybersecurity Fundamentals 4.000 credits

This course provides a foundation of knowledge in the information technology security field. Students will learn general network security concepts; compliance and operational security; threats and vulnerabilities; application, data, and host security; access control and identity management; cryptography. Hands-on training benefits the novice as well as the experienced network professional.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Note: Students are strongly encouraged to complete CTS 1134 prior to enrollment in CTS 1120. Students who enroll concurrently should ensure they have a foundational understanding of networking concepts, including IP addressing, protocols, and network services. In mini-term formats, students should complete CTS 1134 during the first half of the term and CTS 1120 during the second half.

Course User Fee: \$63.00

CTS1131 A+ Computer Essentials & Support 4.000 credits

This is an intermediate level course designed for students preparing for A+ certification as a support technician. Students will learn how to install, configure, upgrade and replace computer system components; how to troubleshoot processors, memory, storage devices, adapter cards, peripherals and other system components; how to install, configure and troubleshoot operating systems, laptops, portable devices, printers, scanners, network devices, security measures and virtualization and cloud computing; and how to provide professional IT support and customer service. Prerequisite: CGS1560. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1560

Course User Fee: \$63.00

CTS1134 Networking Technologies 4.000 credits

This course will provide an introduction to the technical areas of network connectivity, data communications, and communication protocols. Emphasis on understanding the foundation of networking technologies and data communication concepts. Topics covered will include an exploration of computer networking development, the OSI reference model, data signaling, data translation, standards for communications and data transmissions, network topologies and access methods. Laboratory fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1145 Cloud Essentials 4.000 credits

This course provides knowledge of Cloud computing concepts, services, architecture, system integration, connectivity, data center migration, administration, security, and technical support. Coverage includes preparation for the CompTIA Cloud Essentials certification examination. (3 hr. lecture; 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS1328 Supporting Microsoft Clients 4.000 credits

This course is intended for students preparing for IT careers as desktop and network support specialists and server administrators, as well as candidates for industry certification. Students will learn how to implement and maintain a Microsoft client operating system.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Recommended: CGS1060C or equivalent computer experience, knowledge or skills.

Course User Fee: \$63.00

CTS1437 MICROSOFT SQL ADMINISTRATION 4.000 - 32.000 credits

An introductory database administration course, in which students learn to install, administer, and optimize an enterprise-level database system. Emphasis on using SQL to define databases, tables, stored procedures, and constraints. Prerequisite: CGS 1541. Laboratory fee. A.S. degree credit only. (3 hr. lecture; 2 hr. lab)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CTS1650 CCNA 1: Cisco Fundamentals 4.000 credits

This is the first course of the four-course Cisco curriculum that will prepare students for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn networking concepts and practices, network terminology and protocols, the OSI reference model, cabling, cabling tools, routers, router and switch configurations, LAN/WAN topologies, IP addressing, and network standards.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Knowledge of computer operating systems and hardware is required.

Course User Fee: \$63.00

CTS1651 CCNA 2: Routing and Switching 4.000 credits

This is the second course of the four-course Cisco curriculum that will prepare the student for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn the architecture, components and operation of routers and switches, LAN (Local Area Networks) switch protocols and operations, VLANs (Virtual Local Area Networks), network routing protocols and concepts, static and dynamic routing, router and switch configuration and troubleshooting, and IP Address services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1650

Course User Fee: \$63.00

CTS1800 Introduction to Web Page Design 4.000 credits

This introductory course covers the basics of web design and development. Students will learn about the World Wide Web, Hypertext Markup Language (HTML), Extensible Hypertext Markup Language (XHTML), Cascading Style Sheets (CSS) and JavaScript using popular web authoring tools such as Dreamweaver. Students will also learn the basic functions of HTML, XHTML, CSS and JavaScript and how to develop and maintain a website.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CGS1060 or equivalent experience and skills

Course User Fee: \$63.00

CTS1801 Multimedia and Animation 4.000 credits

This course introduces computer science and non-majors to the tools and techniques to create multimedia and animated presentations. Students will learn how to make appropriate hardware and software decisions, how to select and use various authoring systems and tools, and how to publish their work to the Web. Prerequisites: CGS1060.

Laboratory fee A.S. degree credit only.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1060

Course User Fee: \$63.00

CTS2143C Server Administration 4.000 credits

This course is designed for students preparing for IT careers as server administrators, and focuses on server hardware and operating systems, system management and administration, security and disaster recovery, infrastructure and network services, data storage, and troubleshooting. This course is intended to prepare students for the CompTIA Server+ certification.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Server operating systems experience.

Course User Fee: \$63.00

CTS2148C IT Project Management 4.000 credits

This course prepares students to effectively plan, implement, coordinate and manage small to medium size information technology-related projects. The concepts and tools covered in this course offers practical knowledge on managing project scope, planning and scheduling, risk assessment, and identification of project resources. This course also prepares students for the CompTIA Project+ examination, an industry standard IT certification. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2153 Supporting Windows Users & Applications 4.000 credits

This is an advanced course designed to help students prepare for the Microsoft Certified IT Professional Support Technician Certification. Students will learn how to install, configure and manage Windows applications in a networked Windows environment and how to support enterprise users. Students will also deploy Windows and applications using various methods, resolve installation and compatibility issues, establish group policies and user profiles, perform support functions, troubleshoot user and application issues, secure the desktop and network from unauthorized use, install software upgrades and updates, perform systems monitoring and documentation, and develop customer service skills.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1328

Course User Fee: \$63.00

CTS2154 it help desk support 4.000 - 32.000 credits

This course is designed to prepare students as entry-level help desk computer support technicians. Students will learn skills needed to support computer users within the business organization and to provide exceptional customer service, including how to identify the appropriate tools, technologies, and processes to assess and meet computer user needs, essential communications skills, the IT function within the business organization, and career opportunities in computer user support. Prerequisites: CGS 1060, CGS 2108. Laboratory fee. A.S. degree credit only. (3 hr. lecture; 2 hr. lab)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

CTS2192C Microsoft Azure Administration 4.000 credits

This course is intended for students preparing for IT careers as cloud administrators, as well as candidates for industry certification. Students will learn the administration skills and knowledge required to implement, manage, and monitor identity, governance, storage, and compute virtual networks in the Microsoft Azure cloud environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS1145 or equivalent knowledge in Microsoft Azure Fundamentals

Course User Fee: \$63.00

CTS2302 DES NETWORK INFR DIR 4.000 credits

This course will provide the knowledge and skills necessary to plan and implement Windows 2000 Directory Services in an enterprise environment. The course focuses on planning the Active Directory replication, managing network traffic, optimizing Active Directory, deploying directory services, controlling schema modifications, synchronizing Exchange Server and Windows 2000 directories, and upgrading directory services from Windows NT 4.0. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2303 Windows Server Administration 4.000 credits

This course is intended for students preparing for IT careers as network support specialists as well as server and cloud administrators. Students will learn the fundamental administration skills and knowledge required to deploy and support Windows Server core services.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CGS1060 and CTS1134 or equivalent knowledge

Course User Fee: \$63.00

CTS2306 Windows Networking Services 4.000 credits

This course is intended for students preparing for IT careers as network support specialists and server administrators, as well as candidates for industry certification. Students will learn to administer the tasks required to maintain a Windows Server infrastructure.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS1134 and CTS2303 or equivalent knowledge

Course User Fee: \$63.00

CTS2314 Network Defense and Countermeasures 4.000 credits

In this course, students will take an in-depth look at network defense concepts and techniques.

Coverage includes network defensive concepts; policy development; problem-solving; and implementation of firewalls, DMZ, VPN, IDS, NAT and proxy servers.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS1120, CTS1134

CTS2334 Windows Identity Services 4.000 credits

This course is intended for students preparing for IT careers as network support specialists and server administrators, as well as candidates for industry certification. Students will learn how to perform the advanced configuration tasks required to deploy, manage, and maintain a Windows Server infrastructure.

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: CTS2303 and CTS2306 or equivalent knowledge

Course User Fee: \$63.00

CTS2375C Cloud Infrastructure and Services 4.000 credits

This course helps students develop technical expertise in Cloud computing and prepares them for industry certification. Students will learn the essentials of Cloud computing, business security and compliance considerations, migrating to the Cloud, architecting a Cloud server, and how to troubleshoot Cloud services.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2145

Course User Fee: \$63.00

CTS2404 DIST APPL VISUAL BAS 4.000 credits

This course will teach Microsoft Visual Basic programmers how to build N-tier client/server solutions for Microsoft Windows using Windows DNA and COM+ technologies. It includes developing distributed applications that conform to the Microsoft Solution Framework, and is designed to teach Visual Basic programmers, who currently develop desktop applications, how to build n-tier, client/server solutions. Also it will prepare students to take Microsoft's Certification Exam for Distributed Applications with Microsoft Visual Basic, it is a required course for MCSD and elective for MCDBA. Prerequisites: COP 2333. Laboratory fee. A.S. degree credit only. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2433 MICROSOFT SQL IMPLMN 4.000 credits

A comprehensive course in learning how to design and implement enterprise database solutions using SQL. Working through a system of modular lessons and hands-on labs to comprehend SQL Architecture. Prerequisite: CGS 1546. Laboratory fee. A.S. degree credit only. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

CTS2440 Introduction to Oracle: SQL And PL/SQL 4.000 credits

This is an introductory level course for students majoring in the Oracle Database Administrator and/or Solutions Developer programs. Students will learn the fundamentals of SQL and PL/SQL programming languages including the concepts of relational databases, how to create and maintain database objects, and how to store, retrieve, and manipulate data. Students will also learn to create PL/SQL blocks of application code that can be shared by multiple forms, reports, and data management applications.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CGS1060C

Course User Fee: \$63.00

CTS2441 Introduction to Oracle Database Administration 4.000 credits

This course is designed to give students who are preparing to become Oracle database administrators (DBA) a firm foundation in basic administrative tasks. Students will learn through instructor-led learning, structured hands-on practices, and challenge-level exercise labs, the necessary knowledge and skills to set up, maintain and troubleshoot an Oracle database.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2440

Course User Fee: \$63.00

CTS2442 Intermediate Oracle Database Administration 4.000 credits

This is the second course in Oracle database administration. Students will learn basic network administration, including techniques to backup and to recover an Oracle database. The skills developed in this class will help prepare students for the Oracle database administrator (DBA) certification exam.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2441

Course User Fee: \$63.00

CTS2444 Oracle Database Performance Tuning 4.000 credits

This course teaches students tuning steps which can be used to improve database performance. Students will learn through a combination of demonstrations, lectures, and lab exercises, gaining practical experience tuning an Oracle database. Students will also learn how to recognize, troubleshoot and resolve common performance-related problems in administering an Oracle database.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CTS2441

Course User Fee: \$63.00

CTS2450 bus intel: serv/data 4.000 - 32.000 credits

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000
Course User Fee: \$63.00

CTS2451 BUS INTELL:SERV/REPT 4.000 credits

Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

CTS2463 C# Web Application Development 4.000 - 32.000 credits

This course is designed to provide AS degree students majoring in computer information technology, database technology, or Internet services technology with skills necessary for web-based programming. Students will learn C# programming for ASP.NET, including database skills and problem-solving, using modular design techniques. The skills developed in this class will help prepare students for MCTS certification. Prerequisites: COP 1332 or COP 1334. Laboratory fee. (3 hr. lecture; 2 hr. lab)
Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000
Prerequisite: COP1332 OR COP1334
Course User Fee: \$63.00

CTS2466C Internet of Things (IoT) Development with C# 4.000 credits

This course teaches the principles of Internet of Things applications development using the C# language. The student will learn how to write programs in C# and deploy the applications to devices running Windows IoT Core. The student will also develop advanced working programs that connect the devices to cloud services. Prerequisite: CEN2211. (2 hr. lecture 4 hr. lab).
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: CEN2211

CTS2652 CCNA 3: Advanced Routing and Switching 4.000 credits

This is the third course of the four-course Cisco curriculum that will prepare the student for professional certification as a Cisco Certified Network Associate (CCNA). Students will learn how to create virtual local area networks (VLANs), configure inter VLAN routing, and implement wireless network access and VLAN security. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS1651
Course User Fee: \$63.00

CTS2653 CCNA 4: Connecting Networks 4.000 credits

This is the fourth and final course of the four-course Cisco curriculum that will prepare the student for certification as a Cisco Certified Network Associate (CCNA). Students will learn how to implement a hierarchical network design, configure wide area networks (WANs), including point-to-point and frame relay connections, implement IP addressing services such as Network Address Translation, VPN and broadband solutions, monitoring and troubleshooting enterprise networks. Lecture Hours: 48.000, Lab Hours: 32.000
Pre/Corequisite: CTS2652
Course User Fee: \$63.00

CTS2670 Check Point Security Administration 4.000 credits

This course, designed for students specializing in network security, prepares students for the Check Point Certified Security Administrator (CCSA) certification exam. Students will learn how to install security gateways; configure rules on servers; create a rule base; assign user permissions; schedule backups and upgrades; monitor and troubleshoot common network traffic. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: CTS1120, CTS1134

CTS2960 Cloud Computing Capstone 4.000 credits

This course requires students to demonstrate their competence to analyze, design, develop, and test a cloud based complex system. Each student will create and present a cloud based solution proposal that includes: design documentation, implementation plan, cloud resources required, projected cost analysis, basic security plan and project test plan to create an operational cloud based system solution. Must be taken during the last semester before graduation and with a departmental permission. Prerequisite: Departmental Approval. (3 hr. lecture, 2 hr. lab)
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: Departmental Approval
Course User Fee: \$63.00

CTS3452 Business Intelligence 4.000 credits

This course is for students majoring in Data Analytics. Students will learn how to organize, manage and analyze massive amounts of data on servers. Students will learn how to create reports and present information to optimize business decisions and performance. Lecture Hours: 48.000, Lab Hours: 32.000
Course User Fee: \$63.00

CTS4955 Networking Capstone 4.000 credits

This upper division course, for students majoring in the BS-IST Networking track, requires students to demonstrate their competence to analyze, design, develop, and test an information system in a team environment. Each team will create and present an information technology (IT) solution proposal that includes: design documentation, implementation plan, and project test plan to create an operational information system. Students will also implement a proof-of-concept in a real or simulated/virtualized environment.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: Student must be classified as a Senior and have completed at least 3 of the 5 concentration courses to obtain departmental approval.

Course User Fee: \$63.00

DAA**DAA1100 MODERN DANCE 1 2.000 - 3.000 credits**

Beginning exploration of techniques, creative aspects, and theoretical concepts of modern dance which includes but is not limited to proper body alignment and mechanics of breathing and phrasing, verbal movement vocabulary, including structural improvisation. No previous experience required. (1 hr. lecture; 2-4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA1101 INTER MODERN DANCE 2.000 - 3.000 credits

Further development of modern dance techniques, creative aspects, and theoretical concepts emphasizing components based on Graham Cunningham and Limon techniques. Prerequisite: Completion of DAA 1100 or permission of the department. (1 hr. lecture; 2-4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA1104 Modern 1 2.000 - 3.000 credits

Beginning exploration of techniques, creative aspects, and theoretical concepts of modern dance which includes but is not limited to proper alignment and mechanics of breathing and phrasing, verbal and movement vocabulary, including structural improvisation, and exercises utilizing Laban's movement analysis. No previous experience required. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA1105 Intermediate Modern 2.000 - 3.000 credits

Further development of modern dance techniques, creative aspects, theoretical concepts emphasizing components based on Graham, Cunningham and Limon techniques. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA1104

DAA1200 BALLET DANCE 1 2.000 - 3.000 credits

Lecture Hours: 80.000

DAA1201 INTER BALLET DANCE 2.000 - 3.000 credits

The continued development of various aspects of ballet technique terminology. Prerequisite: DAA 1200 or permission of the department. (1 hr. lecture; 2-4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA1204 Ballet 1 2.000 - 3.000 credits

Beginning exploration of techniques and theoretical concepts of ballet increasing awareness of proper alignment, balance, coordination and application of various musical meters. No previous experience required. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA1205 Intermediate Ballet 2.000 - 3.000 credits

Continuing exploration of techniques and theoretical concepts of ballet placing further emphasis on precision of lines and exactness of movement. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA1204

DAA1290 BAL FOR THEATER 1 1.000 - 3.000 credits

Lecture Hours: 96.000

DAA1291 BALLET FOR THEATER 1 1.000 - 3.000 credits

A continuation of the systematic training of the body through a progressive study of the traditional classic ballet vocabulary. More barre exercises and simple adagio jumps and turns will further the concentration on flexibility and coordination. Prerequisite: DAA 1290. (2-6 hr. lab.)

Lecture Hours: 32.000 - 96.000

DAA1330 AFRO CARIBBEAN DANCE 1.000 credits

Lecture Hours: 32.000

DAA1504 Jazz Dance 1 2.000 - 3.000 credits

This course is designed to introduce the student to the vocabulary and technique of jazz dance, incorporating a fusion of styles from popular, Afro-Caribbean, and contemporary modern jazz choreographers. For majors only. Audition required. May be repeated for credit.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA1680 repertory 1 0.000 - 64.000 credits

A special workshop course designed to provide the student with experience relative to the performance of dance concerts. Works choreographed by students as well as faculty will be featured. (1 hr. lecture; 2-4 hr. lab)

Lab Hours: 16.000, Clinical Hours: 1.000 - 30.000

DAA2106 Modern2 2.000 - 3.000 credits

Further development of modern dance techniques, creative aspects and theoretical concepts emphasizing components based on Graham, Cunningham and Limon techniques. The use of improvisation as an introduction to basic principles of form and their application to dance composition will be emphasized. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA1104or Departmental Permission.
Course User Fee: \$5.00

DAA2107 Advanced Modern 2 2.000 - 3.000 credits

Further development of modern dance techniques, creative aspects and theoretical concepts based on Graham, Cunningham, and Limon techniques. May be repeated for credit. Dance Majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA2106or Departmental Permission

DAA2202 BALLET DANCE 2 2.000 - 3.000 credits

The continued development of various aspects of ballet technique and terminology. Prerequisite: DAA 1201 or permission of the department. (1 hr. lecture; 2-4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA2206 BALLET 2 2.000 - 3.000 credits

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA2207 Advanced Ballet 2.000 - 3.000 credits

Continuing exploration of techniques and theoretical concepts of ballet placing further emphasis on precision of line and exactness of movement. May be repeated for credit. Dance majors only.

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

Prerequisite: DAA1205or Departmental Permission.

DAA2570 MODERN DANCE TECH/TH 1.000 - 3.000 credits

Music theatre students will be receiving training of the body through the study of modern dance vocabulary as developed by the originators of this dance from on the twentieth century. In the first semester concentration will be put on alignment, rhythm and phrasing, introducing the students to the fundamentals of jazz techniques. (2-6 hr. lab)

Lecture Hours: 64.000 - 96.000

DAA2571 MOD DAN FOR THEATR 2 1.000 - 3.000 credits

Music theatre students will continue receiving training of the body through the study of modern dance vocabulary. In the second semester emphasis will be on developing carriage, rhythm and more advanced phrasing through jazz techniques and styles. Prerequisite: DAA 2190. (2-6 hr. lab)

Lab Hours: 32.000 - 96.000

DAA2610 Dance Composition and Improvisation 1 2.000 - 3.000 credits

Individual experience in developing movement phrases and combinations based on solving problems within a form and a movement framework, as well as the movement imagery designed to develop the dancer's creative imagination. Individuals will experience composition using the basic elements of movement theory in an improvisational framework.

Lecture Hours: 16.000 - 32.000, Lab Hours: 32.000 - 64.000

DAA2611 DANCE COMP/IMPROV 2 2.000 - 3.000 credits

Further exploration of choreographic tools with emphasis on group forms, usage space, and orchestrations of movement. The formal study of compositional principles of choreographic invention with emphasis is on developing personal style.

Prerequisite: DAA 2700. (1 hr. lecture; 2-4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA2680 GEN PERFORMANCE REP 2.000 - 3.000 credits

Dance works in both ballet and many different styles of modern and ethnic dance vocabularies are studied. Works include both standard repertory and commissioned dances. Students work with choreographers, directors and reconstructors of classic works, giving the dancer the experience of being choreographed on and being directed in repertory works. The works learned are performed by the students in workshop and public performances throughout the year. (1 hr. lecture; 2-4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAA2681 GEN PERFORMANCE REP 2.000 - 3.000 credits

A continuation of DAA 2480. Prerequisite DAA 2480. (1 hr. lecture; 2-4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000 - 64.000

DAN**DAN2100 Dance Appreciation 3.000 credits**

This course provides an overview of dance as an art form, entertainment, and social activity and includes works from the Western canon to examine its significance from ancient times to the 21st century. Students will analyze dance genres such as ballet, modern dance, jazz dance, and world dance forms, exploring their connections to literature, music, and philosophy while considering the roles of dancers, choreographers, and audiences. Through this study, students will develop critical thinking skills and gain understanding of how dance reflects and influences culture across time periods and societies.

Lecture Hours: 48.000

DAN2130 Dance History 1 3.000 credits

In this course, students will engage in a comprehensive study of the origins and development of dance as an art form, tracing its evolution from its inception in primitive cultures to the present day. The course will explore the cultural, historical, and social contexts that shaped the various styles and techniques of dance throughout history. Students will analyze and critique significant dance works, including those from the western canon, to gain a deeper understanding of the artistic, aesthetic, and expressive elements of dance. Writing intensive course.

Lecture Hours: 48.000

DAN2131 Dance History 2 3.000 credits

In this course, students will examine the dance through the ages, from the stone age participatory dances to the spectator dances of the orient, the classical period in Greece and Rome, and the early Middle Ages, including works from the western canon. The course will conclude with the historical development of dance forms from the late Middle Ages through the renaissance into the 20th century, with an emphasis on the dance as a spectator event and a participatory art in relationship to other art forms, including works from the Western Canon.

Lecture Hours: 48.000

Prerequisite: DAN2130

DAN2430 LABAN MOV ANAL 1 3.000 credits

An introduction to Rudolf Laban's basic principles of effort, shape and space harmony. The class will explore ways varying movement dynamics, and will assist the student in discovering the many ways that the body can shape itself and project into space. Prerequisite: Permission of department chairperson. (3 hr. lecture)

Lecture Hours: 48.000

DAN2431 LABAN MOV ANAL 2 3.000 credits

XXX

Lecture Hours: 64.000

DAN2630 LIT/MAT OF MUS/DAN 1 2.000 - 3.000 credits

This course serves to develop the personal musical interest of choreographers and dance artists. The composition and performance of simple musical works will be taught. Actual hands on skills with dance accompaniment will be developed. (2-3 hr. lecture)

Lecture Hours: 32.000 - 48.000

DEA**DEA0020 Pre-Clinical Procedures Theory 3.000 credits**

This course is an introduction to dentistry and the dental assisting profession including duties and responsibilities, orientation to the dental office, basic concepts of microbiology, introduction to the basic skills in dental assisting including, but not limited to infection control procedures, instrument identification and four-handed dentistry.

Clock Hours: 90.000

DEA0020L Pre-Clinical Practice Lab 3.000 credits

This course introduces students to the basic clinical dental assisting skills as theorized in DEA 0020. Concentration of study will include areas such as reception and dismissal of patients, patient management, establishing and maintaining records, obtaining, and recording medical/dental histories and vital signs, charting, planning appointments, assisting with, or performing various dental procedures, and utilizing practice management system.

Clock Hours: 90.000

Course User Fee: \$76.53

DEA0031 Oral Pathology 1.000 credits

This course introduces principles of general pathology with an emphasis on conditions related to the oral cavity. There is a concentration on recognition of normal and abnormal conditions of the oral cavity and surrounding tissues.

Clock Hours: 30.000

DEA0132 Dental Nutrition 1.000 credits

This course is designed to integrate nutrition into the diagnosis, care, and treatment of dental patients demonstrating the relationship between diseases, diets, and oral health. An emphasis on dietary assessment methods in relation to dental health is explored.

Clock Hours: 30.000

DEA0800L Clinical Practice 1 5.000 credits

This course introduces students to the basic clinical dental assisting skills as theorized in DEA 0020. Students will be required to participate and observe clinical and administrative activities in the campus dental clinic and rotate through other community clinics/dental offices where they will get an opportunity to work with a varied patient population. Concentration of study will include reception and dismissal of patients, patient management, establishing and maintaining records, obtaining and recording medical/dental histories and vital signs, charting, planning appointments, assisting with or performing various dental procedures, and utilizing practice management system.

Clock Hours: 150.000

DEA0804L Clinical Practice 2 6.000 credits

This course is a continuation of DEA 0800L. Students will be required to participate and observe clinical and administrative activities in the campus dental clinic and rotate through other community clinics/dental offices where they will get an opportunity to work with a varied patient population as they work on their competence in performing chairside assisting, expanded functions, laboratory, and basic business office procedures.

Clock Hours: 180.000

DEA0930 Dental Assisting Seminar 1.000 credits

This course provides learners with a review of materials commonly covered on the Dental Assisting National Board (DANB). Topics include but are not limited to infection control, general chair side procedures, and radiographic techniques. The seminar will provide a mock board examination, and opportunities for review, problem solving and strengthening skills.

Clock Hours: 30.000

Course User Fee: \$16.05

DEH**DEH1002 Pre-Clinical Dental Hygiene 2.000 credits**

Introduction to procedures relevant to the practice of dental hygiene.

Lecture Hours: 32.000

Corequisite: DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L

DEH1002L Pre-Clinical Dental Hygiene Laboratory 2.000 credits

Laboratory for DEH 1002.

Clinical Hours: 96.000

Corequisite: DEH 1002, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L

Course User Fee: \$120.00

DEH1133 Dental Anatomy, Histology and Physiology 2.000 credits

This course covers specific tissues of the oral cavity, head, neck and their embryonic development. The students will learn structure, morphology and function of the primary and permanent dentitions.

Lecture Hours: 32.000

Corequisite: DEH 1002, DEH 1002L, DEH 1710, DEH 1720, DES 1200, DES 1200L

DEH1400 General and Oral Pathology 3.000 credits

Processes of inflammation, necrosis, retrograde changes, diseases caused by bacteria, viruses, and other organisms. Emphasis will be placed on differentiating between normal and abnormal conditions of the oral cavity.

Lecture Hours: 48.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600

DEH1710 Oral Health Literacy 1.000 credits

Students will learn the concepts of oral health literacy. Students will also identify how to improve patient's oral health literacy and the barriers that impede oral health care.

Lecture Hours: 16.000

Corequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1720, DES 1200, DES 1200L

DEH1720 Preventative Dentistry 2.000 credits

This is a foundation course in dental hygiene preventive care. Students will learn the concepts of oral health and how to prevent future disease. Students will become engaged in developing their own prevention strategies by selecting with a rationale, appropriate oral health devices used for self-care.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DES 1200, DES 1200L

DEH1800 Dental Hygiene 1 2.000 credits

Theory of the removal of hard and soft deposits from the teeth, and other related postoperative and preventive procedures.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800L, DEH 1811, DEH 2202, DES 1600

DEH1800L Dental Hygiene 1 Clinic 3.000 credits

Clinic for DEH 1800.

Clinical Hours: 144.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800, DEH 1811, DEH 2202, DES 1600

Course User Fee: \$135.00

DEH1802L Dental Hygiene 2 Clinic 1.000 credits

Continuation of clinical skills from DEH 1800L.

Clinical Hours: 48.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602

Course User Fee: \$75.00

DEH1804L Dental Hygiene 3 Clinic 1.000 credits

Designed to further student's knowledge and skills through clinical experiences more difficult than those experienced in DEH 1802L.

Clinical Hours: 48.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602

Course User Fee: \$75.00

DEH1811 Professional Issues 2.000 credits

This course is designed to provide the dental hygiene student with an understanding of the political, social, environmental and professional issues that affect the practice of dental hygiene. These issues include: a) cultural diversity, b) legal and ethical responsibilities, c) sexual harassment, d) child abuse, e) problem solving, f) communication style.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 2202, DES 1600

DEH1840L Advanced Radiographic & Clinical Assessment Techniq 1.000 credits

A laboratory course introducing advanced digital radiographic techniques, the intraoral camera, periodontal probing and dental charting software and other clinical assessment tools. These skills will enable the student to provide comprehensive patient treatment and enhance their ability to interpret intraoral conditions.

Clinical Hours: 48.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1804L, DEH 2300, DEH 2300L, DEH 2602

Course User Fee: \$60.00

DEH1940L DENTAL HYGIENE 1 OLS 1.000 credits

DEH 1940L runs concurrently with DEH 1800L and is designed to enhance students basic clinical skills and critical thinking abilities. Special emphasis is placed on collaborative learning techniques, effective decision-making, proper time management and self-assessment as students interact with their peers and apply their skills and knowledge in the treatment of clinical patients. (3 hr. clinic)

Lab Hours: 48.000

Course User Fee: \$181.00

DEH2202 Nutrition and Dental Health 2.000 credits

This course provides a study of nutrients, their nature, source, and utilization. Students will learn the relationship between diet and oral health care and oral manifestations of nutritional deficiencies.

Lecture Hours: 32.000

Prerequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200, DES 1200L, Corequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DES 1600

DEH2300 Pharmacology and Pain Control 1.000 credits

This course introduces the student to a broad range of pharmacological concepts including drug categories, drug action, and adverse drug effects. Dental prescriptions such as the antibiotics, antifungals and antivirals will be studied. Students will learn the common medical conditions affecting dental hygiene care, such as cardiovascular disease, endocrine and neurological disorders as well as their drug management will be examined.

Lecture Hours: 16.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300L, DEH 2602

DEH2300L Pharmacology and Pain Control Laboratory 1.000 credits

This course is designed to prepare the dental hygiene student for the safe and effective administration of local anesthesia. Students will learn about the psychology of pain management, pharmacology of anesthetic agents, emergency precautions, and a review of anatomy and physiology as they relate to the administration of anesthetic agents. This course will include online and clinical instruction.

Lab Hours: 32.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2602

Course User Fee: \$100.00

DEH2602 Periodontology 1 1.000 credits

This course will introduce the student to the concepts of non-surgical periodontal therapy, risk factors in periodontal diseases, classifications of periodontal diseases, the components of the comprehensive periodontal assessment and care plan. Ultrasonic periodontal debridement will be studied. Furthermore, the course will include the study of behavior motivation, the dental hygiene human needs conceptual model, the phases of self-care education and the importance of case presentation in modifying client self-care.

Lecture Hours: 16.000

Prerequisite: DEH 1400, DEH 1800, DEH 1800L, DEH 1811, DEH 2202, DES 1600, Corequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L

DEH2603 Periodontology 2 2.000 credits

Etiology, classification, diagnosis, treatment and maintenance of the periodontal patient.

Lecture Hours: 32.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH 2603L, DEH 2701, DEH 2806, DEH 2806L

DEH2603L Periodontology 2 Laboratory 1.000 credits

Laboratory for DEH 2603.

Lab Hours: 32.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH 2603, DEH 2701, DEH 2806, DEH 2806L

Course User Fee: \$100.00

DEH2701 Community Dental Health 1 3.000 credits

Public Health Dentistry and the role of the dental hygienist.

Lecture Hours: 48.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH 2603, DEH 2603L, DEH 2806, DEH 2806L

DEH2702L Community Dental Health 2 Clinic 2.000 credits

Provides the student an opportunity for application of the principles of public and community dentistry. Lab Hours: 64.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2808, DEH 2808L, DEH 2810L, DES 2100, DES 2100L

DEH2806 Dental Hygiene 4 2.000 credits

Continuation of dental hygiene theory and practice with special emphasis on gingival curettage and root planning.

Lecture Hours: 32.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH2806L, DEH 2603, DEH 2603L, DEH 2701, DEH 2806L

Course User Fee: \$75.00

DEH2806L Dental Hygiene 4 Clinic 4.000 credits

Clinic for DEH 2806.

Clinical Hours: 192.000

Prerequisite: DEH 1802L, DEH 1804L, DEH 1840L, DEH 2300, DEH 2300L, DEH 2602, Corequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806

Course User Fee: \$150.00

DEH2808 Dental Hygiene Practice Management and Leadership 2.000 credits

Students will learn the basic dental and behavioral sciences within the practice of dental hygiene. Special emphasis is given to Florida laws governing dental hygiene practices.

Lecture Hours: 32.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808L, DEH 2810L, DES 2100, DES 2100L

DEH2808L Dental Hygiene 5 Clinic 4.000 credits

Ongoing experience in total dental hygiene care of the periodontally involved patient.

Clinical Hours: 192.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808, DEH 2810L, DES 2100, DES 2100L

Course User Fee: \$150.00

DEH2810L Interprofessional Practice and Education Lab 1.000 credits

Students will learn to examine the Dental Hygiene scope of practice through professional identity by way of interprofessional collaboration. Students will also develop knowledge and skills to serve on health professional teams to improve health outcomes.

Lab Hours: 32.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808, DEH 2808L, DES 2100, DES 2100L

DEH3703 Oral Health Care: Determinants and Directions 3.000 credits

This course introduces theoretical frameworks for understanding how social, economic, environmental, and political factors influence oral health outcomes and dental hygiene practice. Students will examine the relationship between these determinants and oral health, as well as their impact on access to care and community health status. The course emphasizes the development of policy at the community, state, and national levels, and includes analysis of current research and case studies relevant to dental hygiene and public health. Students will gain skills in assessing social issues affecting oral health and in understanding how to advocate for effective dental public health policies.

Lecture Hours: 48.000

DEH3814 Dental Hygiene and Population Health Considerations 3.000 credits

This course provides an in-depth exploration of the health and healthcare needs specific to groups at elevated risk for oral health challenges. Students will examine the unique circumstances affecting these groups at the psychosocial and physiological levels and their impact on oral health. Through analysis of current trends, students will identify factors contributing to compromised oral health and explore strategies aimed at reducing risk and supporting health promotion, maintenance, and restoration within the context of dental hygiene practice.

Lecture Hours: 48.000

DEH4730 Dental Hygiene Educational Concepts 3.000 credits

This course is designed to introduce the principles of learning and teaching, especially as they apply to allied health professions such as staff training and patient education. Topics include understanding various learning theories, learning styles, motivational strategies, collaborative learning, the influence of diversity, and using technology to support learning. The course introduces students to the career option of education as a profession and provides a foundation for understanding the relationship between planning, implementing, and assessing student learning. Dental hygiene and adult education theory and concepts are the major components of this specialization.

Lecture Hours: 48.000

DEH4740 Introduction to Health Concepts in Dental Hygiene 3.000 credits

This foundation course introduces the study of dental global health and its theories, perspectives, methods, and evidence. The course offers a basic overview of the methods for measuring population health, analytic tools for decision-making, and the evidence base for the effectiveness, risks, and efficiency of interventions, programs, and policies. The course focuses on challenges, diseases, and conditions of major public health importance, their patterns of global mortality and morbidity, their social, economic, and political determinants and consequences, their inequalities and inequities, and the population-based interventions as well as social and institutional responses to these challenges structured through communities, national health systems as well as international agencies, consortia, and alliances.

Lecture Hours: 48.000

DEH4813 Dental Hygiene: Emerging Topics and Techniques 3.000 credits

This course examines current societal and professional issues and their impact on dental hygiene practice. Students will examine the roles of the dental hygienist and discuss the dental hygienists' role in increasing access to dental care. Students will research and compare traditional and alternative practice models and propose changes to improve dental care delivery. It also discusses changing technology in dentistry and dental hygiene; political advocacy; demographic shifts; ethics and professionalism; interprofessional education; and global perspectives of dental hygiene.

Lecture Hours: 48.000

DEH4851 Dental Hygiene Practice Management 3.000 credits

This course is designed to assist the dental hygienist in understanding and applying the principles of management related to a dental hygiene practice. The concepts of patient, business office, and dental team management will be explored. Strategies will be presented on the marketing and promoting of dental services and products to the general public.

Lecture Hours: 48.000

DEH4854 Leadership in Dental Hygiene 3.000 credits

This course focuses on the theories, concepts and principles of leadership. Emphasis will be on the development of leadership skills related to personal behavior, communication, organization and self-evaluation. This course explores opportunity to develop leadership roles appropriate to the dental hygiene profession.

Lecture Hours: 48.000

DEH4892 Principles of Professionalism in Dental Hygiene 3.000 credits

The student will be able to demonstrate knowledge of the dental hygiene code of ethics with emphasis on cultural competence as related to all professional encounters.

Lecture Hours: 48.000

DEH4930 Dental Hygiene Special Topics 3.000 credits

This online course is designed to help the dental hygiene student identify and explore current issues and topics related to dentistry and specifically to dental hygiene. The course will take the format of a study club and will involve student participation in the identification and discussion of the topic selected.

Lecture Hours: 48.000

DEH4947 Dental Hygiene Capstone 5.000 credits

This course is designed to provide the student with an opportunity to apply their knowledge of a chosen professional role through teaching, internship, observation or work experience. This course will use selected sites on and off campus to develop the expanding role of the dental hygienist in the health care system.

Lecture Hours: 48.000

DEP**DEP2000 Human Growth and Development 3.000 credits**

This course examines the physical, cognitive, social and emotional development of human beings from conception to death. Students will learn about theories of development and key issues in the field and apply research in developmental psychology throughout the prenatal, infancy, childhood, adolescence, and adulthood periods of the lifespan.

Lecture Hours: 48.000

DEP2100 Child Growth And Development 3.000 credits

This course in Child Growth and Development is designed especially for the student interested in the human life span from birth through the first eight years. The course is intended to acquaint the student with basic theoretical models of development and such specific topics as heredity, teratogenic agents, learning, intelligence, socialization, personality, sex role identification, language acquisition and moral development.

Lecture Hours: 48.000

DEP2402 Psychology of Adulthood and Aging 3.000 credits

This course will examine adulthood and aging from a multidisciplinary and diversity perspective. This course will explore the biological, physical, psychological, cognitive, and cultural changes during adulthood and aging including contemporary aspects of work-family life, and retirement. Aspects of death and dying will also be discussed from a multidisciplinary perspective. (3 hr. lecture)

Lecture Hours: 48.000

DES**DES0021 Dental Anatomy 3.000 credits**

This course provides a study of the anatomy of the human dentition, its development and function. A detailed study of bony, muscular, circulatory, and nervous systems of the head and neck is included. An introduction to oral histology will also be offered.

Clock Hours: 90.000

DES0103 Dental Materials 2.000 credits

This course provides a study of the properties, manipulation, and care of materials used in the prevention and treatment of oral disease. It will include the study of physical, mechanical, chemical, and biological characteristics of materials.

Clock Hours: 60.000

DES0103L Dental Materials Lab 3.000 credits

This course provides a study of the properties, manipulation, and care of materials used in the prevention and treatment of oral disease. It will include the study of physical, mechanical, chemical, and biological characteristics of materials.

Clock Hours: 90.000

Course User Fee: \$50.53

DES0205 Dental Radiology 2.000 credits

This course is designed to provide students with a study of the principles of radiology, theory, and operation of X-ray equipment, darkroom procedures, film storage, and the hazards and precautions involved in dental radiology.

Lab Hours: 3.000, Clock Hours: 6.000 - 60.000

DES0205L Dental Radiology Lab 2.000 credits

This course is designed to provide students with a study of the principles of radiology, theory, and operation of X-ray equipment, darkroom procedures, film storage, and the hazards and precautions involved in dental radiology.

Clock Hours: 60.000

Course User Fee: \$88.60

DES0206 Dental Radiology II 1.000 credits

This course is a continuation of DES 0205. This course focuses on identification and interpretation of normal and abnormal dental pathology, accessory techniques and tomographic imaging systems.

Clock Hours: 30.000

DES0206L Dental Radiology II Lab 1.000 credits

This course is a continuation of DES 0205L. This course focuses on identification and interpretation of normal and abnormal dental pathology, accessory techniques and tomographic imaging systems. Prerequisite: DES0205L.

Clock Hours: 30.000

Course User Fee: \$88.60

DES0501 Dental Practice Management 1.000 credits

This course introduces students to the principles of dental office management. Areas of study includes reception procedures, telephone techniques, appointment book control, inventory control, purchasing, filing systems, bookkeeping, and banking procedures, tax records, legal and ethical aspects of dentistry.

Clock Hours: 30.000

DES0602 Pharmacology / Dental Office Emergencies 1.000 credits

This course is designed to provide students with the basic concepts and considerations regarding pharmacology and pharmaceutical preparations used in dentistry and by patients; chemical and physical properties, actions, dose, methods of administration, therapeutic usage, side effects, indications and contraindications will be explored. In addition, this course also examines the anticipation and prevention of emergency situations in the dental office. The relationship of health and medical histories will be emphasized as well as recognizing and assisting with emergency care of dental patients.

Clock Hours: 30.000

DES0830 Expanded Functions Theory 1.000 credits

This course is designed to provide students with the foundational theory prior to gaining laboratory and clinical proficiency in the expanded functions legally allowable in the state of Florida.

Clock Hours: 30.000

DES0830L Expanded Functions Lab 1.000 credits

This course is designed to provide students with the foundational theory prior to gaining laboratory and clinical proficiency in the expanded functions legally allowable in the state of Florida

Clock Hours: 30.000

Course User Fee: \$50.53

DES1200 DENTAL RADIOLOGY 3.000 credits

Lecture Hours: 16.000, Lab Hours: 64.000

DES1200L Dental Radiology Laboratory 2.000 credits

Laboratory for DES 1200.

Lab Hours: 64.000

Corequisite: DEH 1002, DEH 1002L, DEH 1133, DEH 1710, DEH 1720, DES 1200

Course User Fee: \$90.00

DES1600 DENTAL OFFICE EMERGENCY 2.000 credits

Lecture Hours: 32.000

DES2100 DEN MATERIAL/SPECIAL 2.000 credits

Physical properties of dental materials and their use in the oral cavity. Prerequisite: DES 1030, 2806L; corequisite: DES 2130L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

DES2100L Dental Materials Laboratory 1.000 credits

Laboratory for DES 2100.

Lab Hours: 32.000

Prerequisite: DEH 2603, DEH 2603L, DEH 2701, DEH 2806, DEH 2806L, Corequisite: DEH 2702L, DEH 2808, DEH 2808L, DEH 2810L, DES 2100

Course User Fee: \$190.00

DIG**DIG1111 Digital Character Design 3.000 credits**

This course, for students majoring in Animation and Game Art, covers the observation and translation of three-dimensional form into two-dimensional drawings. Students will learn the interpretation of the human body, based on major masses organized by gestural lines. Students will create original characters and create design elements to support them. Students will transition to draw on digital tablets, knowledge or proficiency in Adobe Photoshop recommended. (3 hr. lecture)

Lecture Hours: 48.000, Other Hours: 3.000

Course User Fee: \$129.70

DIG1111 Digital Character Design 3.000 credits

This course, for students majoring in Animation and Game Art, covers the observation and translation of three-dimensional form into two-dimensional drawings. Students will learn the interpretation of the human body, based on major masses organized by gestural lines. Students will create original characters and create design elements to support them. Students will transition to draw on digital tablets, knowledge or proficiency in Adobe Photoshop recommended. (3 hr. lecture)

Lecture Hours: 48.000, Other Hours: 3.000

Course User Fee: \$129.55

DIG1132 Digital Art and Design 3.000 credits

This course is for students majoring in Animation and Game Art and introduces environmental design. Students will learn the concepts, hardware, and software related to digital image acquisition, image editing, manipulation, color management basics, masking, layering, retouching, scanning and output, and color theory as it relates to digital media.

Knowledge or proficiency in Adobe Photoshop and Illustrator recommended. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$129.70

DIG1132 Digital Art and Design 3.000 credits

This course is for students majoring in Animation and Game Art and introduces environmental design. Students will learn the concepts, hardware, and software related to digital image acquisition, image editing, manipulation, color management basics, masking, layering, retouching, scanning and output, and color theory as it relates to digital media.

Knowledge or proficiency in Adobe Photoshop and Illustrator recommended. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$129.55

DIG1302 3D Modeling 4.000 credits

This course, for students majoring in Animation & Game Art introduces students to the basic tools, techniques and applications for feature 3D animation and game development. Students will learn how to manipulate objects, build models, employ lighting, design movement, work with materials and textures and render a final image. (4 hr. lecture)

Lecture Hours: 64.000

Course User Fee: \$129.70

DIG1302 3D Modeling 4.000 credits

This course, for students majoring in Animation & Game Art introduces students to the basic tools, techniques and applications for feature 3D animation and game development. Students will learn how to manipulate objects, build models, employ lighting, design movement, work with materials and textures and render a final image. (4 hr. lecture)

Lecture Hours: 64.000

Course User Fee: \$129.55

DIG1430 Storyboarding 3.000 credits

This course is for students majoring in Animation and Game Art. It introduces the necessary tasks in the storytelling phase of an animation project. Students will learn how to develop and design visual storyboards and how to sell their storyboard ideas. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$129.70

DIG1430 Storyboarding 3.000 credits

This course is for students majoring in Animation and Game Art. It introduces the necessary tasks in the storytelling phase of an animation project. Students will learn how to develop and design visual storyboards and how to sell their storyboard ideas. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$129.55

DIG1437 Narrative Storytelling 3.000 credits

This course is for students majoring in Animation and Game Art. It introduces the conceptual structure and design of visual storytelling. Students will learn principles of animation, mechanics, cinematics, character development, personality of place, exaggeration and effects, adapting movement for the animation medium, and how to create the illusion of life. Pre/Corequisite of DIG 1430 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000, Lab Hours: -2.000, Clinical Hours: -1.000, Practicum Hours: -2.000

Pre/Corequisite: DIG1430

Course User Fee: \$129.70

DIG1437 Narrative Storytelling 3.000 credits

This course is for students majoring in Animation and Game Art. It introduces the conceptual structure and design of visual storytelling. Students will learn principles of animation, mechanics, cinematics, character development, personality of place, exaggeration and effects, adapting movement for the animation medium, and how to create the illusion of life. Pre/Corequisite of DIG 1430 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000, Lab Hours: -2.000, Clinical Hours: -1.000, Practicum Hours: -2.000

Pre/Corequisite: DIG1430

Course User Fee: \$129.55

DIG1705 3D Programming 1 4.000 credits

This course, provides students with a foundation in 3D programming which will allow them to develop programs using popular graphics libraries such as DirectX, OpenGL, and GLSL. Students will learn basic image processing, geometric transformations, geometric modeling of curves and surfaces, 3D viewing, shaders, and ray tracing.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: MAC1105, COP2335

Course User Fee: \$129.70

DIG1710 Introduction to Game Development 4.000 credits

This course is an introduction to the computer game design and development industry. Students will learn about game development careers, game development and design processes, marketing themes, copyright laws, game company structures, programming languages used by different types of games, the impact of video games on modern society, general programming concepts, how to create game design documentation, and how to use common game development environments.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$129.70

DIG1710 Introduction to Game Development 4.000 credits

This course is an introduction to the computer game design and development industry. Students will learn about game development careers, game development and design processes, marketing themes, copyright laws, game company structures, programming languages used by different types of games, the impact of video games on modern society, general programming concepts, how to create game design documentation, and how to use common game development environments.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$129.55

DIG1729C Game Engines 4.000 credits

This course is an introduction to game engines and their uses. Students will learn the basic techniques for creating interactive applications and how this techniques can be used for Virtual Reality (VR) and Augmented Reality (AR) projects. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$129.70

DIG1729C Game Engines 4.000 credits

This course is an introduction to game engines and their uses. Students will learn the basic techniques for creating interactive applications and how this techniques can be used for Virtual Reality (VR) and Augmented Reality (AR) projects. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$129.55

DIG1772C Introduction to Virtual & Augmented Reality Technologies 4.000 credits

This course introduces students to basic concepts, history and tools commonly used for Virtual Reality (AR) and Augmented Reality (AR) technologies. Students will learn the history of VR/AR, its current role in the industry, developer tools, and physical principles. Prerequisite: DIG 1729C. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: DIG1729C

Course User Fee: \$98.00

DIG1772C Introduction to Virtual & Augmented Reality Technologies 4.000 credits

This course introduces students to basic concepts, history and tools commonly used for Virtual Reality (AR) and Augmented Reality (AR) technologies. Students will learn the history of VR/AR, its current role in the industry, developer tools, and physical principles. Prerequisite: DIG 1729C. (3 hr. lecture; 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: DIG1729C

Course User Fee: \$129.55

DIG2113 Post Production & Editing 4.000 credits

This course, for students majoring in Animation & Game Art, equips students with skills required in post-production editing. Students will learn how to combine computer-generated imagery with matte painting and backgrounds and the core principles of proper compositing, color correction, and editing. Pre/Corequisite of DIG 1430 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG1430

Course User Fee: \$129.70

DIG2113 Post Production & Editing 4.000 credits

This course, for students majoring in Animation & Game Art, equips students with skills required in post-production editing. Students will learn how to combine computer-generated imagery with matte painting and backgrounds and the core principles of proper compositing, color correction, and editing. Pre/Corequisite of DIG 1430 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG1430

Course User Fee: \$129.55

DIG2304 Character Animation 3.000 credits

This course, for students majoring in Animation & Game Art, equips students with the skills needed to create animated characters. Students will learn in-depth character design, development, rigging, and animation techniques, how to create segmented and solid model mesh of bipeds and quadrupeds, and techniques used to create facial expressions and lip syncing. Prerequisite of DIG 2790 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG2790

Course User Fee: \$129.70

DIG2304 Character Animation 3.000 credits

This course, for students majoring in Animation & Game Art, equips students with the skills needed to create animated characters. Students will learn in-depth character design, development, rigging, and animation techniques, how to create segmented and solid model mesh of bipeds and quadrupeds, and techniques used to create facial expressions and lip syncing. Prerequisite of DIG 2790 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG2790

Course User Fee: \$129.55

DIG2318 Animation Studio 1 3.000 credits

This course is for students majoring in Animation and Game Art. Students will learn to design and implement a project involving computer animation, game production, VFX or scientific/architecture visualization. Students will work in collaboration with faculty and industry mentors. Prerequisite of DIG 1430; Prerequisite of DIG 1302; Prerequisite of DIG 1437 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG1430 AND DIG1302 AND DIG1437

Course User Fee: \$129.70

DIG2318 Animation Studio 1 3.000 credits

This course is for students majoring in Animation and Game Art. Students will learn to design and implement a project involving computer animation, game production, VFX or scientific/architecture visualization. Students will work in collaboration with faculty and industry mentors. Prerequisite of DIG 1430; Prerequisite of DIG 1302; Prerequisite of DIG 1437 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG1430 AND DIG1302 AND DIG1437

Course User Fee: \$129.55

DIG2319 Animation Studio 2 3.000 credits

This is a capstone course for students majoring in Animation and Game Art. Building on skills learned in Animation Studio 1, students will learn enhanced skills in the areas of 3D modeling, texturing, lighting, and animation. Working in groups, students develop a project plan and produce a short, 3D animated movie. Prerequisite of DIG 2318 with a Grade of C or better, or equivalent or Prerequisite of CAP 2048 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG2318 OR CAP2048

Course User Fee: \$97.00

DIG2319 Animation Studio 2 3.000 credits

This is a capstone course for students majoring in Animation and Game Art. Building on skills learned in Animation Studio 1, students will learn enhanced skills in the areas of 3D modeling, texturing, lighting, and animation. Working in groups, students develop a project plan and produce a short, 3D animated movie. Prerequisite of DIG 2318 with a Grade of C or better, or equivalent or Prerequisite of CAP 2048 with a Grade of C or better, or equivalent (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: DIG2318 OR CAP2048

Course User Fee: \$129.55

DIG2370 Character Modeling & Rigging 4.000 credits

This course focuses on the modeling and rigging of characters for performance. Students will solve complex issues of character articulation with an emphasis on skeleton, skin, and binding techniques. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: DIG1302

Course User Fee: \$97.00

DIG2370 Character Modeling & Rigging 4.000 credits

This course focuses on the modeling and rigging of characters for performance. Students will solve complex issues of character articulation with an emphasis on skeleton, skin, and binding techniques. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: DIG1302

Course User Fee: \$129.55

DIG2391C Animation Studio 3 4.000 credits

This is a capstone course for students majoring in Animation and Game Art. Students develop a project plan and produce a short, 3D animated movie. Students also create a website for the project, social media and market campaigns, and submit the short animated movie to festivals. Pre/Corequisite of DIG 2319 with a Grade of C or better, or equivalent or Pre/Corequisite of CAP 2920C with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG2319 OR CAP2920C

Course User Fee: \$97.00

DIG2391C Animation Studio 3 4.000 credits

This is a capstone course for students majoring in Animation and Game Art. Students develop a project plan and produce a short, 3D animated movie. Students also create a website for the project, social media and market campaigns, and submit the short animated movie to festivals. Pre/Corequisite of DIG 2319 with a Grade of C or better, or equivalent or Pre/Corequisite of CAP 2920C with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG2319 OR CAP2920C

Course User Fee: \$129.55

DIG2396C Motion Capture 4.000 credits

This course is for students majoring in Animation and Game Art. Students will learn to digitize motion and clean-up and editing techniques. They will also learn how to set up motion capture and shooting, data tracking, skeleton retargeting, as well as animation correction and enhancements. Pre/Corequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG1302

Course User Fee: \$95.00

DIG2396C Motion Capture 4.000 credits

This course is for students majoring in Animation and Game Art. Students will learn to digitize motion and clean-up and editing techniques. They will also learn how to set up motion capture and shooting, data tracking, skeleton retargeting, as well as animation correction and enhancements. Pre/Corequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Pre/Corequisite: DIG1302

Course User Fee: \$129.55

DIG2625 Network Programming for Game Development 4.000 credits

This course is for students majoring in game development. It introduces network programming and communication in a distributed computing environment for game development. Students will learn network technologies, architecture, protocols, programming across different environments.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: COP2335

Course User Fee: \$97.00

DIG2626 Artificial Intelligence for Game Development 4.000 credits

This course covers key aspects of Artificial Intelligence (AI) for students majoring in game development. Students will learn the origins and history of Artificial Intelligence, current and future uses of AI, AI methods algorithms such as: path planning, stimulus-response agents, agent architectures, decision-making systems, game trees, neural networks, and genetic algorithms. Students will create and modify existing games to include an AI system. Prerequisites: CAP 2047 and COP 2335. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP2047, COP2335

Course User Fee: \$97.00

DIG2626 Artificial Intelligence for Game Development 4.000 credits

This course covers key aspects of Artificial Intelligence (AI) for students majoring in game development. Students will learn the origins and history of Artificial Intelligence, current and future uses of AI, AI methods algorithms such as: path planning, stimulus-response agents, agent architectures, decision-making systems, game trees, neural networks, and genetic algorithms. Students will create and modify existing games to include an AI system. Prerequisites: CAP 2047 and COP 2335. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP2047, COP2335

Course User Fee: \$129.55

DIG2712 Level Building & Design 4.000 credits

This course is intended for students majoring in Game Development and Design. Students will learn how to develop game environments in industry standard engines, how to create documentation to plan out effective game play experiences, and the requirements to create virtual worlds. Prerequisites: CAP 2047; COP 2335; DIG 1430. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP2047 AND COP2335 AND DIG1430

Course User Fee: \$97.00

DIG2712 Level Building & Design 4.000 credits

This course is intended for students majoring in Game Development and Design. Students will learn how to develop game environments in industry standard engines, how to create documentation to plan out effective game play experiences, and the requirements to create virtual worlds. Prerequisites: CAP 2047; COP 2335; DIG 1430. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: CAP2047 AND COP2335 AND DIG1430

Course User Fee: \$129.55

DIG2714 SYS ANAL GAME DEVELP 4.000 credits

Lecture Hours: 48.000, Lab Hours: 32.000

DIG2717C Game Systems Design 4.000 credits

This is a core course for students majoring in game development and design. Students will learn how to develop game systems such as combat, economy, and social society. Students will also learn how to model and test systems before incorporating them into development, and how to use probability to create more interesting gameplay. Prerequisites: CAP 2047, DIG 1712, and MAC 1105. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000 - 58.000, Lab Hours: 32.000, Practicum Hours: 1.000

Prerequisite: CAP2047 AND DIG1712 AND MAC1105

Course User Fee: \$97.00

DIG2717C Game Systems Design 4.000 credits

This is a core course for students majoring in game development and design. Students will learn how to develop game systems such as combat, economy, and social society. Students will also learn how to model and test systems before incorporating them into development, and how to use probability to create more interesting gameplay. Prerequisites: CAP 2047, DIG 1712, and MAC 1105. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 48.000 - 58.000, Lab Hours: 32.000, Practicum Hours: 1.000

Prerequisite: CAP2047 AND DIG1712 AND MAC1105

Course User Fee: \$129.55

DIG2771 3D PROGRAMMING 2: VIRTUAL REALITY 4.000 credits

This course is for students majoring in game development and covers key aspects of advanced 3D programming. Students will learn how to program special effects and create realism for games by using: illumination, shading, reflections, collision detection/ reaction, light mapping, sound, music, alpha blending, fog, and applying basic Newtonian physics to objects.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: DIG1705 AND Pre/Corequisite: COP2335

Course User Fee: \$97.00

DIG2776C Virtual Reality Platform Development 4.000 credits

Students will learn the fundamentals of Virtual Reality (VR) by gaining practical experience using state-of-the-art technology. This course mixes together knowledge from a variety of correlated topics, including computer graphics, tracking systems, and perceptual psychology.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: DIG1729C AND DIG1772C

DIG2777C Augmented Reality Platform Development 4.000 credits

This course provides a comprehensive curriculum that targets the key areas of augmented reality (AR). Students will learn how to enhance real-life objects and environments with digitally generated image overlays.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: DIG1729C AND DIG1772C

DIG2790 Texturing & Environment Design 4.000 credits

This course is for students majoring in Animation & Game Art. Students will learn advanced 3D animation job skills used in creating 3D feature animation and game development, including advanced texturing, lighting and rendering a final image. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: DIG1302

Course User Fee: \$97.00

DIG2790 Texturing & Environment Design 4.000 credits

This course is for students majoring in Animation & Game Art. Students will learn advanced 3D animation job skills used in creating 3D feature animation and game development, including advanced texturing, lighting and rendering a final image. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent. Prerequisite of DIG 1302 with a Grade of C or better, or equivalent (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: DIG1302

Course User Fee: \$129.55

DIG3255C Advanced Sound Design 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn advanced audio production, emphasizing audio recording, mixing, editing, overdubbing, and aesthetics.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: RTV1240C

Course User Fee: \$100.00

DIG3347C Advanced Cinematography 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn the technical and aesthetic principles of advanced cinematography techniques.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: RTV2246C OR FIL2515C

Course User Fee: \$100.00

DIG3940 Upper Division Internship 3.000 credits

In this upper division internship for BAS students in Film, Television & Digital Production students will learn to apply their knowledge and skills at an established film or television entertainment company (144 hr. Internship).

Other Hours: 144.000

Recommended Preparation: BAS Students ONLY.

Must have at least a 3.0 GPA in order to register.

Must obtain professor permission in order to register.

DIG4345C Digital Effects and Compositing 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn the theory and practice of video compositing and motion graphics.

Lecture Hours: 32.000, Lab Hours: 32.000

Recommended Preparation: BAS Students ONLY

Course User Fee: \$100.00

DSC

DSC1002 Domestic and International Terrorism 3.000 credits

This course is a study of domestic and international terrorism as it relates to domestic security. Through focused topics, students will learn about terrorist organizations and motivations, investigating terrorism threats, conducting vulnerability assessments of potential terrorist targets, and the role of government agencies in response to a terrorist incident and recovery afterwards.

Lecture Hours: 48.000

DSC1006 Introduction to Homeland Security 3.000 credits

This course will introduce students to the vocabulary and important components of Homeland Security. Students will learn about the agencies associated with Homeland Security and their interrelated duties and relationships.

Lecture Hours: 48.000

DSC1590 Introduction to Intelligence Studies 3.000 credits

This course will provide a comprehensive overview of intelligence for the purpose of national security for entry-level intelligence practitioners and beginning students. The student will learn security issues, define critical terms and review the history of intelligence as practiced in the United States.

Lecture Hours: 48.000

DSC2242 Transportation and Border Security 3.000 credits

This course introduces students to global supply chains and intermodal transportation systems. Students will learn the threats to these systems, their vulnerabilities and potential for terrorist attacks, and the measures being undertaken to secure them.

Lecture Hours: 48.000

DSC2501 Effective Communication Skills for Security Professionals 3.000 credits

This course will provide a focused review and practice of the basic elements necessary for effective writing in any situation or any type of report specifically within the intelligence community. It is designed for the entry-level criminal justice practitioner and beginning student interested in entering the field. Pre-requisite:

ENC1101

Lecture Hours: 48.000

Prerequisite: ENC1101

DSC2590 Intelligence Analysis and Security Management 3.000 credits

This course examines intelligence analysis and its indispensable relationship to the security management of terrorist attacks, man-made disasters and natural disasters. Students will learn substantive issues regarding intelligence support of Homeland Security measures implemented by the United States and explore how the intelligence community operates.

Lecture Hours: 48.000

DSC4012 Terrorism 3.000 credits

A study of domestic and international terrorism, using current events and past incidents for analysis, to include the events, the responses, and the outcomes.

Lecture Hours: 48.000

DSC4014 domestic/int'l terr 3.000 credits

A study of the causes and effects of domestic and international terrorist events. Prerequisite: DSC 4012. (3 hr. lecture)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

DSC4214 CATASTROPHIC MGMT 3.000 credits

An analysis and evaluation of domestic and international terrorism, the events, the responses, and the outcomes. (3 hr. lecture)

Lecture Hours: 48.000

DSC4215 EME PLAN/SEC MEASURE 3.000 credits

A study of empirical vs theoretical approaches; human behavior in disasters; myths and realities; group disaster behavior; community social systems, and disaster; cultures, demographics and disaster behavior distinctions; and model-building in sociological disaster research. Corequisite: DSC 4214. (3 hr. lecture)

Lecture Hours: 48.000

Corequisite: DSC4214

EAB**EAB1001 Foundation in Registered Behavior Technician 3.000 credits**

The Foundation in Registered Behavior Technician (RBT) course is designed especially for students who are interested in completing the RBT examination through the BACB. The course will cover measurement, assessment, skill acquisition, behavior reduction, documentation and reporting, and professional conduct and scope of practice, as well as all subtasks listed in the RBT checklist and the Professional and Ethical Compliance Code for Behavior Analysts by the BACB. (3 hr. lecture) (special fee)

Lecture Hours: 48.000

Course User Fee: \$10.00

EAP

EAP0100 SPEECH/LISTENING 1 3.000 credits

Students develop the ability to understand frequently used words in oral contexts and understand and respond appropriately to simple phrases and questions. (3 hr. lecture)

Lecture Hours: 48.000

Corequisite: EAP0100L

EAP0100L SPEECH/LISTENING 1 LAB 1.000 credits

The student will practice oral production and aural comprehension of spoken American English. This practice will be related, but not limited to the material taught in EAP 0100.

Lab Hours: 32.000

Corequisite: EAP0100

Course User Fee: \$25.00

EAP0120 READING LEVEL 1 3.000 credits

Students develop the ability to comprehend limited written materials. (3 hr. lecture)

Lecture Hours: 48.000

EAP0140 WRITING LEVEL 1 3.000 credits

Students develop the ability to write appropriate phrases and short sentences on personal topics.

Lecture Hours: 48.000

Corequisite: EAP0140L

EAP0140L WRITING LEVEL 1 LAB 1.000 credits

This lab will provide support and additional practice as well as focus on a multi-skilled approach to writing as students develop their abilities in meeting the competencies of EAP 0140.

Lab Hours: 32.000

Corequisite: EAP0140

Course User Fee: \$25.00

EAP0160 GRAMMAR LEVEL 1 3.000 credits

Students develop the ability to understand and use basic, high frequency grammatical structures. (3 hr. lecture)

Lecture Hours: 48.000

EAP0200 SPEECH/LISTENING 2 3.000 credits

Students continue to develop the ability to understand frequently used words in oral contexts and understand and appropriately respond to simple phrases and questions. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: EAP0100, Corequisite: EAP0200L

EAP0200L SPEECH/LISTENING 2 LAB 1.000 credits

This course continues to give practice in oral production and aural comprehension of spoken American English. This practice will be related to, but not limited to, the material taught in EAP0200.

Lab Hours: 32.000

Prerequisite: EAP0100, Corequisite: EAP0200

Course User Fee: \$25.00

EAP0220 READING LEVEL 2 3.000 credits

Students continue to develop the ability to comprehend text appropriate to the level with emphasis on developing reading skills and vocabulary.

Lecture Hours: 48.000

Prerequisite: EAP0120

EAP0240 WRITING LEVEL 2 3.000 credits

Students continue to develop writing skills in the context of guided discourse on personal topics with an emphasis on logical thought and mechanics.

Lecture Hours: 48.000

Prerequisite: EAP0140, Corequisite: EAP0240L

EAP0240L WRITING LEVEL 2 LAB 1.000 credits

This lab will provide additional practices as well as focus on multi-skills as students develop their abilities in meeting the competencies of EAP 0240.

Lab Hours: 32.000

Corequisite: EAP0240, Prerequisite: EAP0140

Course User Fee: \$25.00

EAP0260 GRAMMAR LEVEL 2 3.000 credits

Students continue to develop control of basic grammatical structures and statement/question patterns.

Lecture Hours: 48.000

Prerequisite: EAP0160

EAP0300 SPEECH/LISTENING 3 3.000 credits

Students develop the speaking and listening skills necessary for participating in classroom discussions with an emphasis on clarification through rewording and asking questions.

Lecture Hours: 48.000

Prerequisite: EAP0200, Corequisite: EAP0300L

EAP0300L SPEECH/LISTENING 3 LAB 1.000 credits

Students practice speaking and listening skills necessary for participating in classroom discussions with an emphasis on clarification through rewording and asking questions.

Lab Hours: 32.000

Prerequisite: EAP0200, Corequisite: EAP0300

Course User Fee: \$25.00

EAP0320 READING LEVEL 3 3.000 credits

Students develop the ability to read texts on familiar and basic academic topics with an emphasis on vocabulary expansion and application of critical reading skills.

Lecture Hours: 48.000

Prerequisite: EAP0220

EAP0340 WRITING LEVEL 3 3.000 credits

Students develop the ability to write basic, structured academic paragraphs on familiar topics and execute other academic writing tasks.

Lecture Hours: 48.000

Prerequisite: EAP0240, Corequisite: EAP0340L

EAP0340L WRITING LEVEL 3 LAB 1.000 credits

Students develop the ability to write basic, structured academic paragraphs on familiar topics and execute other academic writing tasks.

Lab Hours: 32.000

Prerequisite: EAP0240, Corequisite: EAP0340

Course User Fee: \$25.00

EAP0360 GRAMMAR LEVEL 3 3.000 credits

Students develop the ability to use intermediate-level grammatical structures appropriate to classroom discussion and the writing of academic paragraphs with an emphasis on increased accuracy.

Lecture Hours: 48.000

Prerequisite: EAP0260

EAP0385 Intermediate 1 - Integrated Writing & Grammar 6.000 credits

Students will learn how to write paragraphs using intermediate grammar and rhetorical structures.

Lecture Hours: 96.000

Course User Fee: \$25.00

EAP0386 Intermediate 1 - Integrated Reading, Speech & Listening 6.000 credits

Students will learn to develop proficiency in reading level-appropriate text by listening to short oral discourse and discussing academic materials.

Emphasis is on vocabulary expansion and application of strategies that assist in comprehension and communication.

Lecture Hours: 96.000

Prerequisite: EAP0220 AND EAP0200 OR

Recommended Preparation: equivalent proficiency

Course User Fee: \$25.00

EAP0400 SPEECH/LISTENING 4 3.000 credits

Students continue to develop the speaking and listening skills necessary for participating in classroom discussions with an introduction to oral presentation and critical listening skills.

Lecture Hours: 48.000

Prerequisite: EAP0300, Corequisite: EAP0400L

EAP0400L SPEECH/LISTENING 4 L 1.000 credits

Students continue to practice speaking and listening skills necessary for participating in classroom discussions with an introduction to oral presentation and critical listening skills.

Lab Hours: 32.000

Prerequisite: EAP0300, Corequisite: EAP0400

Course User Fee: \$25.00

EAP0420 READING LEVEL 4 3.000 credits

Students continue to develop academic reading abilities including text on contemporary and social topics with an emphasis on extensive reading and the enhancement of critical reading skills.

Lecture Hours: 48.000

Prerequisite: EAP0320

EAP0440 WRITING LEVEL 4 3.000 credits

Students develop the ability to write more sophisticated, structured academic paragraphs in various rhetorical modes and execute other academic writing tasks.

Lecture Hours: 48.000

Prerequisite: EAP0340, Corequisite: EAP0440L

EAP0440L WRITING LEVEL 4 LAB 1.000 credits

Students continue to practice developing to write more sophisticated, structured academic paragraphs in various rhetorical modes and execute other academic writing tasks.

Lab Hours: 32.000

Prerequisite: EAP0340, Corequisite: EAP0440

Course User Fee: \$25.00

EAP0460 GRAMMAR LEVEL 4 3.000 credits

Students develop the ability to use intermediate-level grammatical structures appropriate to classroom discussion, oral presentation, and writing of more sophisticated academic paragraphs with an emphasis on increased accuracy.

Lecture Hours: 48.000

Prerequisite: EAP0360

EAP0485 Intermediate 2 - Integrated Writing & Grammar 6.000 credits

Students will learn to refine paragraphs using intermediate grammar and rhetorical structures. Co-requisite: (1) approved college-level course recommended.

Lecture Hours: 96.000

Prerequisite: EAP0340 AND EAP0360 AND EAP0385 OR Recommended Preparation: equivalent proficiency.

Course User Fee: \$25.00

EAP0486 Intermediate 2 - Integrated Reading, Speech & Listening 6.000 credits

Students will learn to develop speaking, listening, and academic reading skills through discussions, presentations, and analysis with an emphasis on oral fluency, critical reading, and vocabulary expansion. Co-requisite: (1) approved college level course.

Lecture Hours: 96.000

Prerequisite: EAP0386 OR EAP0300 AND EAP0320
OR Recommended Preparation: equivalent proficiency

Course User Fee: \$25.00

EAP0493 ACC INT SPC/GRA 6.000 credits

In this accelerated alternative course for EAP 0300, 0360, 0400, and 0460, students will learn intermediate-level grammar and vocabulary, and will enhance their oral communication proficiencies via brief lectures, oral presentations, and classroom discussions, improving spoken fluency, accuracy, and interpersonal skills in English. Prerequisites: EAP 0220 and 0240 or appropriate COMPASS score; Corequisite: EAP 0494. (6 hr. lecture)

Lecture Hours: 96.000

EAP0494 ACC INTERM READNG/WR 8.000 credits

In this alternative course for EAP 0300, 0360, 0400, and 0460, students will learn intermediate grammatical structures necessary for participating in classroom discussions with an introduction to oral presentations and critical listening skills, emphasizing clarification, rewording and asking questions. Prerequisites: EAP 0200 and 0260 or appropriate COMPASS score; corequisite: EAP 0431. (8 hr. lecture)

Lecture Hours: 128.000

EAP1500 SPEECH/LISTENNG LEV5 3.000 credits

Students develop the communication, organization, and pronunciation skills necessary for effective academic presentations and discussions with an introduction to lecture note taking. (3 hr. lecture)

Lecture Hours: 48.000

Corequisite: EAP1500L, Prerequisite: EAP0400

EAP1500L SPCH/LISTNG LEV 5 LA 1.000 credits

Students develop communication, organization, and pronunciation skills necessary for effective academic presentation and discussion with an introduction to lecture note-taking.

Lab Hours: 32.000

Prerequisite: EAP0400, Corequisite: EAP1500

Course User Fee: \$25.00

EAP1501 ACCENT REDUCTION 1 3.000 credits

Students develop the ability to write basic structured academic essays with an emphasis on accuracy and cohesiveness and execute other academic writing tasks. (1-3 hr. lecture)

Lecture Hours: 48.000

EAP1501L ACCENT REDUCTION 1 LA 1.000 credits

Insert

Lab Hours: 32.000

Course User Fee: \$25.00

EAP1502 ACCENT REDUCTION 2 3.000 credits

Students improve their pronunciation of American English including stress, rhythm, and intonation. The phonetic structure of vowel sounds is systematically analyzed, and students are given practice in correctly pronouncing these sounds and patterns in context. (3 hr. lecture)

Lecture Hours: 48.000

EAP1502L ACCENT REDUCTION 2 LA 1.000 credits

Insert

Lab Hours: 32.000

Course User Fee: \$25.00

EAP1520 READING LEVEL 5 3.000 credits

Students develop the ability to comprehend advanced academic readings on a variety of topics. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: EAP0420

EAP1540 WRITING LEVEL 5 3.000 credits

Students develop the ability to write more sophisticated, structured academic paragraphs and academic essays in various rhetorical modes and execute other academic writing tasks. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: EAP0440, Corequisite: EAP1540L

EAP1540L WRITING LEVEL 5 LAB 1.000 credits

Students develop the ability to write basic structured academic essays with an emphasis on accuracy and cohesiveness and execute other academic writing tasks.

Lab Hours: 32.000

Prerequisite: EAP0440, Corequisite: EAP1540

Course User Fee: \$25.00

EAP1560 GRAMMAR LEVEL 5 3.000 credits

Students develop the ability to use complex grammatical structures appropriate for academic presentations, discussions, and essays at this level. 3 credits

Lecture Hours: 48.000

Prerequisite: EAP0460

EAP1581 ADVANCED 1 ENGLISH 6.000 credits

Insert

Lecture Hours: 96.000

EAP1585 Advanced 1 - Integrated Writing & Grammar 6.000 credits

Students will learn to write essays by developing advanced grammar and rhetorical structures.
Prerequisites: EAP0485, or EAP0440 and EAP0460, or equivalent proficiency. Co-requisites: At least (1) approved college level course
Lecture Hours: 96.000
Prerequisite: EAP0485 OR EAP0440 AND EAP0460
Course User Fee: \$25.00

EAP1586 Advanced 1 - Integrated Reading, Speech & Listening 6.000 credits

Students will learn to comprehend academic and other authentic reading materials and effectively participate in college-level oral/aural tasks by applying appropriate learning strategies. Co-requisite: (1) approved college level course.
Lecture Hours: 96.000
Prerequisite: EAP0486 OR EAP0420 AND EAP0400 OR Recommended Preparation: equivalent proficiency
Course User Fee: \$25.00

EAP1600 SPEECH/LISTENING LEV6 3.000 credits

Students further develop communication skills necessary for full participation in mainstream college classrooms including comprehension of extensive discourse. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: EAP1500, Corequisite: EAP1600L

EAP1600L SPEECH/LISTENING LEV6 L 1.000 credits

Students further develop communication skills necessary for full participation in mainstream college classrooms including comprehension of extensive discourse.
Lab Hours: 32.000
Prerequisite: EAP1500, Corequisite: EAP1600
Course User Fee: \$25.00

EAP1620 READING LEVEL 6 3.000 credits

Students develop the ability to comprehend and interpret authentic college-level text in content areas by applying appropriate reading strategies. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: EAP1520

EAP1640 WRITING LEVEL 6 3.000 credits

Students develop the ability to write a variety of college-level essays with sophistication, fluency, and accuracy and execute other academic writing tasks. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: EAP1540, Corequisite: EAP1640L

EAP1640L WRITING LEVEL 6 LAB 1.000 credits

Students further develop the ability to write a variety of college-level essays with sophistication, fluency and accuracy, and execute other academic writing tasks.
Lab Hours: 32.000
Prerequisite: EAP1540, Corequisite: EAP1640
Course User Fee: \$25.00

EAP1660 GRAMMAR LEVEL 6 3.000 credits

Students develop the ability to use complex grammatical structures appropriate for effective academic presentations, discussions, and essays at this level. (3 credits)
Lecture Hours: 48.000
Prerequisite: EAP1560

EAP1683 COMB ACCEL ADV RE/WR 6.000 credits

This is an accelerated alternative course for EAP courses 1520, 1540, 1620, and 1640. Students will learn to complete college-level reading and writing assignments. Prerequisite: EAP 0420, 0440 or appropriate placement score on COMPASS exam (87-92 on reading subtest) and writing sample; corequisite EAP 1689 Combined Accelerated Advanced Speech, Listening, and Grammar. (6 hr. lecture)
Lecture Hours: 96.000
Prerequisite: EAP0420 AND EAP0440 OR appropriate placement score on COMPASS exam (87-92 on reading subtest) and writing sample., Corequisite: EAP1689 Combined Accelerated Advanced Speech, Listening, and Grammar

EAP1685 Advanced 2 - Integrated Writing & Grammar 6.000 credits

Students will learn to refine essays by developing advanced grammar & rhetorical structures.
Prerequisites: EAP1540 and EAP1560, or EAP1585, or equivalent proficiency. Co-requisites: At least (1) approved college level course.
Lecture Hours: 96.000
Prerequisite: EAP1585 OR EAP1540 AND EAP1560 OR Recommended Preparation: or Recommendation of Chair
Course User Fee: \$25.00

EAP1686 Advanced 2 - Integrated Reading, Speech & Listening 6.000 credits

Students will learn to comprehend academic and other authentic reading materials and effectively participate in college-level oral/aural tasks by applying appropriate learning strategies.
Lecture Hours: 96.000
Prerequisite: EAP1586 OR EAP1500 AND EAP1520 OR Recommended Preparation: or Permission of Chair
Course User Fee: \$25.00

EAP1689 CMB ACL ADV SP/LI/GR 6.000 credits

This is an accelerated alternative course for EAP courses 1500, 1560, 1600, and 1660. Students will learn oral communication and lexico-grammatical skills necessary for college-level courses. EAP 0420 and 0440 or appropriate placement score on COMPAS S exam (81-88 on grammar subtest and 83-91 on listening subtest) and writing sample; corequisite: EAP 1683 Combined Accelerated Advanced Reading and Writing. (6 hr. lecture) Lecture Hours: 96.000
Prerequisite: EAP0420 AND EAP0440 OR appropriate placement score on COMPASS exam (81-88 on grammar subtest and 83-91 on listening subtest) and writing sample., Corequisite: EAP1683 Combined Accelerated Advanced Reading and Writing

ECO**ECO2013 Principles of Economics (Macro) 3.000 credits**

In this course, students will learn the foundations of macroeconomics as the branch of economics concerned with how decision-making, in an environment of scarcity, maps onto the aggregate economy. Students will examine theories and evidence related the following core set of topics: national income determination, money, monetary and fiscal policy, macroeconomic conditions, international trade and the balance of payments, and economic growth and development. Student learning outcomes: students will recognize that all decisions happen in an environment of scarcity; students will examine theories and evidence regarding how changes in aggregate measurements are related to economic performance; students will recognize the relationships between the components of the national income accounts; students will analyze theory and evidence regarding fiscal and monetary policies and how they affect the economy; students will identify theories of long-term economic growth and examine evidence for those theories. This is a writing intensive course. Lecture Hours: 48.000

Course User Fee: \$15.00

ECO2013 PRIN/ECONOMICS-MACRO 3.000 credits

Lecture Hours: 48.000

Course User Fee: \$15.00

ECO2023 Principles of Economics (Micro) 3.000 credits

Theory of markets, price mechanism, production, distribution and resource allocation; application of marginal analysis and equilibrium theory to the price and output decisions of the individual firm in pure competition, monopolistic competition, oligopoly and monopoly; agriculture; labor, rent interest and profits theory 3 credits Pre-requisite: MAT1033 with a "C" or higher MAC1105 is not a pre requisite but is it highly recommended. (3 hr. lecture) Prerequisite: MAT1033 Recommended Preparation:

Lecture Hours: 48.000

Prerequisite: MAT1033Any Mathematics level higher than MAT 1033 will satisfy this requirement. MAC1105 is not a pre requisite but it is highly recommended , MAC1105

Course User Fee: \$15.00

ECO2071 ECO INSTI-ELEM ED 1 3.000 credits

This course is designed for Elementary Teachers. It provides coverage of major micro-economic concepts and their infusion into the K-12 curriculum through an activity oriented approach. This course will include those economic concepts required in the minimum Student Performance Standards for Social Studies. These concepts will be handled through various methodologies appropriate for the elementary curriculum. The latest economic education materials will be utilized. (3 hr. lecture)

Lecture Hours: 48.000

ECO2301 History of Economics Ideas and their Consequences 3.000 credits

An interdisciplinary study with major elements of economics, philosophy, history, sociology, anthropology, and political science that begins in the agricultural landscape of the 1700s and brings one forward into the age of the corporate giant and the nuclear warfare of modern industrial society.

Lecture Hours: 48.000

ECO2949 CO-OP WRK EXP 2: ECO 3.000 credits

This is a course designed to continue training in a student's field of study through work experience . Students are graded on the basis of documentation of learning acquired as reported by student and employer. Prerequisite: Co-op Department approval and completion of 1949 Co-op Work Experience. Students will be assigned specific course prefixes related to their academic major prior to registration . All students must contact the Co-operative Education Office to obtain registration approval. (3 hr . lecture)

Lecture Hours: 48.000

ECW

ECW2010 Teaching with AI: Designing Human-Centered Classrooms 3.000 credits

This course invites learners to explore the integration of artificial intelligence through a pedagogical lens, where technology serves as a tool for reimagining teaching and learning. Learners will critically engage with AI tools by grounding their use in learning theory and evidence-based design practices. Through inquiry, collaboration, and hands-on experimentation, learners will examine how AI can amplify creativity, personalize learning, and promote human-centeredness.

Lecture Hours: 48.000

ECW4012 Leading the Future: AI Advocacy and Innovation in Education 3.000 credits

This course empowers educators to lead ethical, inclusive AI integration in schools through systems thinking, human-centered advocacy, and collective leadership. Students will explore inclusive policy, responsible implementation, and community empowerment, while designing initiatives that promote lifelong leadership, voice, and agency for meaningful and sustainable local impact.

Lecture Hours: 48.000

ECW4013 Teaching with AI: Designing Human-Centered Classrooms 3.000 credits

This course invites learners to explore the integration of artificial intelligence through a pedagogical lens, where technology serves as a tool for reimagining teaching and learning. Learners will critically engage with AI tools by grounding their use in learning theory and evidence-based design practices. Through inquiry, collaboration, and hands-on experimentation, learners will examine how AI can amplify creativity, personalize learning, and promote human-centeredness.

Lecture Hours: 48.000

ECW4014 Reimagining Assessments to Maximize Student Potential 3.000 credits

The course will explore the interconnected relationship between lesson planning, assessment, feedback, and classroom management while maintaining pedagogical integrity. The student will develop reflective, inclusive, and ethical practices that support differentiated instruction, student engagement, and responsible AI use across diverse learning environments.

Lecture Hours: 48.000

EDF1005 Introduction to the Education Profession 3.000 credits

This course provides an overview of Florida's educator certification pathways and career options in education in accordance with Florida statutes. Topics include: key educational philosophies; professional ethics; legal responsibilities, including mandatory reporting for child abuse and neglect; the Florida educator accomplished practices (FEAP); key instructional practices associated with effective teaching of students with exceptionalities; Florida's state academic standards; and historical, legal and financial foundations of the Florida education systems. Students will explore effective teaching strategies in classroom management, educational technology and the foundations of all student learning needs. A minimum of 15 hours of required clinical experience in a variety of school settings is included.

Lecture Hours: 48.000

Course User Fee: \$18.00

EDF2085 Introduction to Diversity 3.000 credits

The student will explore the role of teachers as agents of social change and examine their own attitudes towards diversity and exceptionalities. The student will engage in learning opportunities that include cross-cultural dialogue and critical reflection on social justice and oppression based on race, ethnicity, gender, sexual orientation, religion, (dis)ability status, and social class. 15 hours of approved Service Learning will be required.

Lecture Hours: 48.000

Course User Fee: \$18.00

EDF2091 curr iss in edu 3.000 credits

This course is designed to present an overview of the changing role of the paraprofessional in education. It provides current information about employment requirements as well as state, district and school policies. It explores the legal and ethical issues related to the paraprofessional involvement with teachers, students, parents, and administration. Students taking this course will become familiar with a variety of documentation used in the classroom environment to record learner behavior in grades K-12. 10 hours of field experience required. (3 hr. lecture)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

EDF

EDF2130 Human Development and Learning for Educators 3.000 credits

The student will investigate child and adolescent development, including theories and principles of learning. The student will define typical and atypical human growth and development across the lifespan, with emphasis on major developmental issues, and how these interplay to holistically shape development. The student will examine literature on developmental processes, learning theories and concepts related to instructional practices and the crucial role that educators play in fostering the mindsets and skill sets that support optimal development throughout the lifespan of their diverse learners. This course will satisfy the requirement for a course in child and adolescent development for teacher certification.
Prerequisite: Departmental Permission

EDF2144 Maximizing Student Potential in the School Context 3.000 credits

The student will explore various aspects of learning and brain development in schools. Students will be introduced to the latest neuroscience education strategies used in schools to optimize their learning potential from infancy through adolescence.
Lecture Hours: 48.000

EDF3115 Child Development for ECE Teachers 3.000 credits

This course provides an overview of human life from fertilization through eight years of age. The student will examine growth and developmental characteristics during the prenatal, infancy, and early childhood periods. The student will learn to analyze typical and atypical development, developmental theories, learning theories, brain research, attachment, and relationships.
Lecture Hours: 48.000
Course User Fee: \$18.00

EDF4430 Measurement and Assessment in Education 3.000 credits

The student will learn research-based principles of assessment. The student will select specific standards and competencies and develop formative and summative traditional and alternative assessments. The student will interpret assessment data that will improve academic achievement and ensure equity in the application of quantitative and qualitative assessments.
Lecture Hours: 48.000
Prerequisite: EDG3321
Course User Fee: \$18.00

EDF4433 Data-Driven Instructional Change 3.000 credits

This course on data-driven decision-making is designed for current teachers. The student will learn to set measurable goals, collect and analyze data, implement instructional interventions, and align practice to school improvement.
Lecture Hours: 48.000
Prerequisite: Bachelor's Degree in Education.

EDF4951 Florida Teacher Certification Examination (FTCE) Seminar for Pre-Service Educators 2.000 credits

The Florida Teacher Certification Examination (FTCE) Seminar- is designed to address Florida teaching skills, knowledge base and competencies in direct alignment to the Florida Effective Accomplished Practices also known as (FEAPS). The instructional focus is embodied in three examination types known as the General Knowledge Test (GKT), the Professional Educator Certification Examination (PEC) and the Subject-Area Examination (SAE). The GKT is representative of the minimum basic skills (math, reading, language and essay) necessary for all teachers. The PEC is representative the foundational knowledge representative of effective teaching and learning constructs while the SAE is in direct alignment to the teacher or prospective teacher's area(s) of expertise (subject-area). The instructional focus during the Seminar will be individualized based on deficient skill areas and/or "unsuccessful" prior examination attempts.
Lecture Hours: 32.000

EDF4991 Brain-Based Teaching: Reading and the Brain 3.000 credits

The student will learn how the brain processes information and acquires the ability to read. The student will apply educational neuroscience and research-based pedagogy to the instruction of P-12 content areas.
Lecture Hours: 48.000
Prerequisite: Teacher Certification.

EDF4993 Brain-Based Teaching: The Bilingual Brain 3.000 credits

The student will learn how P-12 English Language Learners' (ELLs) brain processes information. The student will acquire research-based and best practices for teaching, differentiating instruction, and assessing ELLs.
Lecture Hours: 48.000
Prerequisite: Teacher Certification.

EDF4994 Brain-Based Teaching: Mathematics and the Brain 3.000 credits

The student will learn how the brain processes information and acquires the ability to perform mathematical processes. The student will integrate educational neuroscience, cognitive research-based, instructional practices, and mathematics assessment into the P-12 classroom.

Lecture Hours: 48.000

Prerequisite: Teacher certification

EDG**EDG2311 Substitute Training 1.000 credits**

Provides students with the necessary knowledge, skills, and dispositions to successfully serve as temporary instructors for the Miami-Dade County Public School (M-DCPS) Board. The course provides best practices in classroom management and effective teaching strategies; key items of M-DCPS Board policy and Florida statutes; and the Code of Ethics and Principles of Professional Conduct of the Education Profession in Florida.

Lecture Hours: 16.000

EDG2316 INTRO MATH/SCI PARA 3.000 credits

Course will be deleted.

Lecture Hours: 48.000

EDG2370 INTRO REA/LANG PARA 3.000 credits

It will be deleted.

Lecture Hours: 48.000

EDG2943 Educational Service Field Work 1.000 - 3.000 credits

The student will learn to compile the necessary documents and complete the process of obtaining a state and/or national early childhood credential. The student must complete the four courses in either the Infant/Toddler Specialization or the Preschool Specialization in order to take this course.

Lecture Hours: 16.000 - 48.000

Prerequisite: EEC1000, EEC1200, EEC1311, EEC2202, or EEC2002, EEC2520, EEC2523, EEC2527

EDG3321 General Teaching Skills 3.000 credits

The student will learn how human development and learning theories, developments in educational neuroscience, and current research-based pedagogy apply to the teaching and learning process. The student will incorporate principles and skills of effective instructional planning, including differentiated instruction, aligning instructional teaching skills to practice, selecting appropriate formative assessments, and developing learning experiences that require students to demonstrate a variety of skills.

Lecture Hours: 48.000

Course User Fee: \$18.00

EDG3443 Classroom and Behavior Management 3.000 credits

The student will examine strategies for fostering a productive learning environment, informed by research-based frameworks and learning theories. The student will learn strategies for observing, assessing, modifying behavior, and communicating with stakeholders. The student will learn to create a learning environment that encourages positive interactions, self-regulation, and resiliency. Fifteen hours of clinical experience is required.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

EDG4343 Instructional Strategies for P-12 Teachers 3.000 credits

In this teacher certification course, the student will utilize research-based instructional design models to create lesson plans and instruction that aligns with state standards. The student will learn to incorporate educational theories and educational neuroscience to develop strategies for inclusive P-12 classrooms serving diverse populations.

Lecture Hours: 48.000

EDG4376 Integrating Language Arts and Social Sciences 3.000 credits

The student will use knowledge, skills, and dispositions from the social sciences to organize and provide integrated instruction in the major themes, concepts, and modes of inquiry in grades K-12. The student will plan and integrate language arts and social science strategies and content to create accessibility of the curriculum to all learners.

Lecture Hours: 48.000

Prerequisite: EDG3321, Pre/Corequisite: EDF4430, RED3393

Course User Fee: \$18.00

EEC**EEC1000 INTRO/EARLY CHILD ED 3.000 credits**

This course iteration will be deleted.

Lecture Hours: 48.000

EEC1000 Introduction to Early Childhood Education 3.000 credits

Students will be introduced to the field of early childhood education with a focus on infants, toddlers, and preschool-aged children from birth through age five, highlighting play as a developmentally appropriate and research-supported approach to early learning. The course examines the purpose and scope of early childhood education and the roles of children, families, and educators. Students gain an overview of foundational theories and philosophies, family and societal influences, learning environments, classroom routines, and guidance practices that support development across domains. Common early childhood program types and curriculum models serving birth–five settings are also explored. Ten hours of service learning in an early childhood setting are required. Fingerprinting through MDC’s school of education must be completed before starting this course. See the MDC SOE Work-Force handbook.

Lecture Hours: 48.000

EEC1200 ERLY CHDHD CURR 1 3.000 credits

The student will learn developmentally appropriate curriculum planning and its impact on children’s total development. The student will examine the importance of dramatic play, proper room arrangement, outdoor environments, advances in technology, scheduling, classroom management, and activity planning for first and second language development, early literacy, and social studies.

Lecture Hours: 48.000

Course User Fee: \$10.00

EEC1200 Early Childhood Curriculum 1 (Birth - Age 5) 3.000 credits

Students will design and implement developmentally appropriate curriculum for children from birth to age five using the NAEYC Professional Standards as a guiding framework. Emphasis is placed on foundational curriculum planning grounded in children’s interests, with focus on development, social studies, language and literacy, and creative expression through the arts. Course content includes observation-based planning, learning centers, classroom and outdoor environments, daily routines, and play-based experiences. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. This course is paired with EEC1311, which extends curriculum planning through integrated and inquiry-based applications. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000

EEC1308 Classrooms for All Young Children 3.000 credits

The student will design and modify learning environments and learning approaches to meet the needs of all children. The student will use technology where applicable and select appropriate materials and activities to promote student independence, self-regulation, and developmental progress.

Lecture Hours: 48.000

EEC1311 Curriculum II: Integrated, Inquiry-Based Curriculum (Birth to Age 5) 3.000 credits

Students will design and implement integrated, developmentally appropriate curriculum for children from birth to age five, building on foundational curriculum planning introduced in EEC 1200. Emphasis is placed on arts-integrated STEM learning through inquiry-based curriculum planning, with focus on scientific inquiry, mathematical thinking, building and engineering, technology, and maker spaces for creative and artistic expression. Course content includes inquiry-based planning, learning centers, enriching classroom and outdoor environments, and play-based investigations that extend children's thinking across domains. Key early learning concepts, including play, cause and effect, and communication, are integrated to support developmentally appropriate practice, early learning, and school readiness. Ten (10) hours of service learning in an approved early childhood setting are required.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000 AND EEC1200

Course User Fee: \$10.00

EEC1522 Infant and Toddler Environments 3.000 credits

This course provides the student with information on planning the physical facilities, equipment, and materials for quality infant and toddler environments. Course content allows for observations and examination of how the physical environment affects development of children and supports individual differences utilizing appropriate and culturally responsive strategies. The student will learn about curriculum planning, promoting social and emotional development, language and literacy, child abuse and neglect, and comprehensive family support services in relation to establishing quality environments for children.

Lecture Hours: 48.000

EEC1713 Helping All Young Children 3.000 credits

The student will learn how to utilize a holistic approach in guiding all young children to become independent learners. The student will learn to utilize positive behavior support strategies to help improve executive functioning skills in all young children. The student will identify strategies to involve the family.

Lecture Hours: 48.000

EEC1752 Knowing and Understanding All Young Children credits

The student learn the process, principles, and patterns of child development. The student will identify the strengths and challenges of children with typical and atypical development. The student will learn to identify indicators of child abuse and neglect.

Lecture Hours: 48.000

EEC1753 Observing and Assessing All Young Children 3.000 credits

The student will learn the process and importance of observing, documenting, and interpreting the behavior of all young children. The student will use observation instruments to understand all young children's growth and development. The student will also develop suggested intervention measures that meet the needs of the children and the families they serve.

Lecture Hours: 48.000

EEC2002 Operation of an Early Childhood Facility 3.000 credits

This introductory course provides an overview for early childhood administrators to develop and enhance their leadership role in designing and implementing quality early care and education programs. The student will study the following topics: organizational leadership and management, programming, and financial and legal issues. This course meets the requirements for the Education Program Administrator Foundational Level Credential.

Lecture Hours: 48.000

EEC2201 Developing Curriculum for Infants and Toddlers 3.000 credits

This course provides the Early Childhood professional with information about developing appropriate curriculum for infants and toddlers based on health, safety, physical, social, emotional, cognitive, and language development. The student will focus on developmentally appropriate curriculum and learning opportunities based on daily classroom routines.

Lecture Hours: 48.000

EEC2202 Early Childhood Curricula 3.000 credits

The student will review historical perspectives and theories utilized in designing early childhood curriculums and examine best practices for early childhood education (birth- age 5). The student will examine philosophies, approaches to teaching and theories of learning which have influenced curriculum development in early childhood education. The student will analyze characteristics of foundational curriculum models and current trends and their influence on the development of early childhood curriculum implementation and program policies.

Lecture Hours: 48.000

Pre/Corequisite: EEC1000

EEC2224 Emergent Literacy 3.000 credits

The student will learn about language and literacy development for young children birth through five years of age. Students will examine the development of listening and understanding, speaking, vocabulary, and emergent reading and writing skills. The student will analyze the teacher's role and methods of creating a literacy-rich environment that engages children in creative, developmentally appropriate experiences and the importance of involving families in the development of these skills.

Lecture Hours: 48.000

EEC2225 Creative Arts for Preschool Education 3.000 credits

The student will research the principles, methods, and materials for fostering play and process-oriented arts exploration in early childhood settings birth-age five. The student will investigate the importance of how developmentally appropriate creative arts (visual arts, drama, music and movement) learning invitations foster total development. The student will analyze the role of the teacher in designing and planning culturally relevant integrated arts experiences and to foster divergent thinking for young children birth through age five.

Lecture Hours: 48.000

EEC2271 Working with Young Children with Special Needs and their Families 3.000 credits

This course provides the student with an overview of young children birth through five years of age with special needs and their families. The student will learn the possible causes and characteristics of exceptionalities, federal laws, and methods of observation, referral process, educational intervention, resources, and advocacy (20 hours in a B-4 SPED setting).

Lecture Hours: 48.000

EEC2302 Early Childhood Comprehension 3.000 credits

The student will learn about the early childhood teacher's role in increasing comprehension in early childhood education (ages 3-5). Topics will support a curriculum that builds an understanding of language acquisition, vocabulary development, and instructional strategies for increasing children's comprehension of literary text.

Lecture Hours: 48.000

EEC2401 Family and Community Interaction through Literacy 3.000 credits

The student will learn about how language and literacy practices help establish positive and productive relationships with the families and children in their classroom and program. The student will use literacy experiences to develop, trusting, supportive reciprocal relationships with young children, their families, and their communities. The student will understand how to promote community connections in the early childhood (birth - age 5) setting through language and literacy practices.

Lecture Hours: 48.000

EEC2407 FAC SOCIAL DEVELOPMT 3.000 credits

This course is designed to utilize educational neuroscience research on teaching and learning to assist students in understanding how young children learn. The student will examine how learning is affected by social, emotional, and intellectual engagement, and how development in these areas is inextricably linked.

Lecture Hours: 48.000

Course User Fee: \$10.00

EEC2407 Foundations of Learning and Executive Function in Early Childhood (Birth - 5) 3.000 credits

Students will build foundational knowledge of executive-function life skills, including focus and self-control, perspective taking, communication, making connections, critical thinking, taking on challenges, and self-directed learning, as the basis for learning in early childhood (Birth–5). The course examines how these skills interact with interpersonal development and cognitive growth, drawing on neuroscience and research on child development. Reciprocal family partnerships, observation-based reflection, and ethical considerations guide the analysis of young children’s interactions and abilities. Emphasis is placed on research-based inhibitory control strategies that support neurocognitive development. Students will connect these executive-function skills and regulation strategies to developmentally appropriate teaching methods across home and early education settings. Ten hours of service learning in an approved early childhood setting are required. See the School of Education workforce handbook.

Lecture Hours: 48.000

EEC2520 Early Childhood Organization Leadership and Management 3.000 credits

The student will develop skills needed by early childhood program administrators to manage a high-quality center. The student will learn about organizational structure of a center; ethics and professionalism; leadership strategies, skills, and competencies; self-reflection and work environment; quality improvement; staff recruitment, evaluation, and retention. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential. (Requires employment at an early childhood center.)

Lecture Hours: 48.000

Prerequisite: EEC2002

Course User Fee: \$10.00

EEC2523 Programming and Management for Early Childhood Administrators 3.000 credits

The student will learn about developmentally appropriate environments, curriculum, and professional standards for early childhood care program administrators. The student will develop an understanding of child observation, assessment, documentation, and referrals and their importance. The student will learn about program evaluation, health, safety, healthy nutrition practices, and the importance of partnerships with families. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential.

Lecture Hours: 48.000

Course User Fee: \$10.00

EEC2527 Legal and Financial Issues in Childcare 3.000 credits

The student will learn about the legal and financial issues related to operating a successful early childhood center. The student learns about financial planning, personnel cost and allocation, budgeting and accounting. The student will develop knowledge about compensation and benefit, risk management, marketing and public relations, regulatory requirements, and legal issues related to childcare management. This course meets the requirements for the Florida Advanced Level Child Care and Education Administrator Credential.

Lecture Hours: 48.000

Prerequisite: EEC2002

EEC2601 Early Childhood Observation and Assessment 3.000 credits

The student will learn how to implement a system of ongoing child observation and assessment for children from birth to age five. The student will understand the importance of and strategies for planning and preparing for an effective system of ongoing child assessment to inform their teaching, improve children’s learning, and provide information and opportunities for individualizing ongoing assessment strategies for children who are dual language learners and children with special needs. The student will learn how to use a variety of assessment tools related to language, literacy and school readiness.

Lecture Hours: 48.000

EEC3211 STEM for Early Childhood Education I 3.000 credits

The student will learn to use scientific and mathematical educational research-based methods and strategies to teach inquiry and problem-solving skills and plan activities for young children that foster exploration in the sciences. (Ten hours of clinical experience required in an approved prekindergarten: 1 observation required).

Lecture Hours: 48.000

Prerequisite: EEC3301 and 10 hrs. of Field Experience Required

Course User Fee: \$18.00

EEC3212 Integrated Social Sciences, Humanities, and Arts 3.000 credits

This course provides an overview of the role of social studies, the humanities, and the fine arts in early childhood classrooms. Students will learn to plan, implement, and assess developmentally appropriate activities that reflect the family and community influences on young children. Special fee. (10 hours of clinical experience and 1 observation required in grade 1–3 classroom with ESOL students).

Lecture Hours: 48.000

Prerequisite: EEC3301

Course User Fee: \$18.00

EEC3213 Language, Literature, and Emergent Literacy 3.000 credits

This course focuses on the development of language and literacy in children ages birth through age five. The student will learn the process of emergent literacy including vocabulary and language development, phonological awareness, alphabetic and print knowledge, and comprehension. The student will learn to design developmentally appropriate and effective emergent literacy activities in relation to the early learning and developmental standards. (Ten hours of clinical experience required in an approved pre-kindergarten inclusion setting.) This course focuses on the development of language and literacy in children ages birth through age five. The student will learn the process of emergent literacy including vocabulary and language development, phonological awareness, alphabetic and print knowledge, and comprehension. The student will learn to design developmentally appropriate and effective emergent literacy activities in relation to early learning and developmental standards. (Ten hours of clinical experience required in an approved pre-kindergarten least restrictive setting.)

Lecture Hours: 48.000

Pre/Corequisite: EEC1000

EEC3301 Teaching Methods in Early Childhood Education 3.000 credits

This course will introduce instructional models to design lesson plans and instruction based on state standards. The student will incorporate educational theories to develop strategies for early childhood serving all learners. (Ten hours of clinical experience required in an approved kindergarten-third grade setting.)

Lecture Hours: 48.000

Pre/Corequisite: EDF3115 and 10 hrs. of Field Experience Required.

Course User Fee: \$18.00

EEC3412 Family and Community Partnerships in Early Childhood Education 3.000 credits

This course focuses on family engagement and community involvement in early childhood education. The student will identify strategies for establishing and maintaining reciprocal relationships with all parents, families, and communities. The student will explore ways to connect students, parents, families, the school, and the community as a partnership to support student learning and well-being.

Lecture Hours: 48.000

Prerequisite: EEC1000

EEC3613 Observation and Assessment of Young Children 3.000 credits

This course will provide an overview of the process and importance of observing, documenting, and interpreting the behavior of young children. The student will learn and apply various methods to document the ongoing development of children and the value of using this information to support and plan for learning experiences. (Ten hours of clinical experience in an approved birth-age 1 least restrictive setting).

Lecture Hours: 48.000

Prerequisite: EEC1000

EEC4219C Science, Technology, and Mathematics (STEM) Methods for ECE II 3.000 credits

The student will learn to use scientific and mathematical research-based methods and strategies to teach inquiry and problem-solving skills and plan activities for young children that foster exploration in the nature of science, mathematics, and technology. (Ten hours of clinical experience required in a kindergarten-third grade setting with ESOL students during math and science instruction: 1 observation).

Lecture Hours: 48.000

Prerequisite: EEC3211

EEC4268 Practicum in Early Childhood Education 3.000 credits

The teacher candidate will use research methods to plan and implement an action research project to meet the needs of struggling readers. The student will attend professional development experiences designed to develop knowledge and pedagogy, prepare for teaching interviews, and entry into the profession.

Lecture Hours: 48.000

Recommended Preparation: Departmental Permission

Course User Fee: \$18.00

EEC4936 Student Teaching Seminar: ECE 2.000 credits

The student will discuss and reflect on their development and mastery of the Florida Educator Accomplished Practices (FEAPS) during the completion of their student teaching experience in a classroom setting.

Lecture Hours: 32.000

Prerequisite: EEC4268, LAE4211, Corequisite: EEC4940

EEC4940 Student Teaching in Early Childhood Education 10.000 credits

The student will complete a full-time differentiated student teaching experience consisting of a traditional internship or teaching residency in a classroom setting. The student will learn and experience all the educational and professional responsibilities common to teachers within their area of expertise.

Internship Hours: 480.000

Corequisite: EEC4936

EET**EET1015C Direct Current Circuits 4.000 credits**

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn electrical components, inductance, capacitance, and elementary network analysis. Students will additionally learn to verify and apply basic theories and principles through hands-on, laboratory experiments utilizing modern testing equipment. Pre/corequisite: MAT 1033. Laboratory fee. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 48.000

Pre/Corequisite: MAT1033

Course User Fee: \$63.00

EET1025C Alternating Current Circuits 4.000 credits

This course is intended for students majoring in electronics engineering technology and related disciplines. Students will learn inductance, capacitance, vector notation, AC circuits, impedance, phase shift, networks, transformers, and resonance. Students will apply and verify theories and principles through hands-on, laboratory experiments utilizing modern testing equipment.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1015C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

EET1033C Electrical Fundamentals 4.000 credits

This course is designed for students obtaining a CCC or AS degree in Engineering Technology, and related disciplines. The student will learn the basic concepts of electronics principles including Direct Current, Alternating Current, Series and Parallel circuits topologies, Basic electronics components, electronics measure tools and software simulation tools.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$103.00

EET1082 Introduction to Electronics 3.000 credits

Learn by building practical electronic circuits. Survey course suitable for both majors and non-majors. Instructors and tutors are available to assist in project completion. Topics include: schematics, pictorials, amplifiers, oscillators, burglar alarms, radios, digital circuits. Students will develop individual career plans and learn about employment opportunities within the field.

Lecture Hours: 48.000

Course User Fee: \$63.00

EET1141C Electronics 1 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology or related fields. Students will learn how to apply electronic principles to analog circuits and systems, including semiconductor diodes, applying the fundamental theory of transistors and other solid-state devices; analysis of amplifiers, oscillators, and other applications using a sinusoidal wave. Students also learn basic safety procedures to follow when working in an electronics laboratory and with electronic circuits and systems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1025C AND MAC1114 OR MAC1147

Course User Fee: \$63.00

EET2101C Electronics 2 4.000 credits

This course is intended for students majoring in electronics or computer engineering technologies. Students will learn how to apply electronic principles to analog circuits including transistor amplifiers, feedback and frequency response of linear circuits, operational amplifiers, MOSFET and oscillators.

Lecture Hours: 32.000, Lab Hours: 48.000

Prerequisite: EET1141C

Course User Fee: \$63.00

EET2323C Analog Communications 4.000 credits

This course is designed for students majoring in Electronics Engineering Technology, Telecommunications Engineering Technology, and related disciplines. Students will learn the principles of radio wave transmission and reception, including AM and FM transmitters, receivers, single sideband, television and digital data transmission lines, wave propagation antennas and microwaves. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1141C

Course User Fee: \$63.00

EET2351C Digital and Data Communications 4.000 credits

This course is intended for students majoring in Electronics or Computer Engineering Technology. It provides a theoretical and practical background in the basic concepts and applications of digital and data communications. Students will learn analog-to-digital (A/D) and digital-to-analog (D/A) conversions; data communications codes and standards; wired and wireless digital communications; modulation, transmission impairment, the telephone system, modems, multiplexers, and electrical interface standards.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2123C

Course User Fee: \$63.00

EET2527C mtr strts contrls br 3.000 - 32.000 credits

This course is designed for students specializing in industrial equipment maintenance covering AC and DC power distribution in the plant. Students learn operating principles, troubleshooting, repair, and maintenance of switch gear, motor control centers, breaker panel power, control, and instrument cable, raceways, protective devices and grounding as related to the generating station. Hands-on, laboratory exercises reinforce each major concept studied. Prerequisites: EET 1141C, EET 2515C. A.S. degree credit only. Special fee. (2 hr. lecture; 2 hr. lab) Lab Hours: 32.000, Clinical Hours: 1.000 - 30.000 Prerequisite: EET1141C, EET2515C Course User Fee: \$63.00

EET2547C trnsfrms & power dis 3.000 - 32.000 credits

This course is designed for students specializing in industrial equipment maintenance. Students acquire an understanding of the components and devices used to distribute power, and how to protect major elements involved in power distribution. Students learn about the uses and maintenance of fuses, circuit breakers, reclosures, and relay coordination; how to protect against lightning and other abnormal conditions; and the protection of transformers, motors, and generators. Prerequisite: EET 2515C ; corequisite: EET 2527C. A.S. degree credit only. Special fee. (2 hr. lecture; 2 hr. lab). Lab Hours: 32.000, Clinical Hours: 1.000 - 30.000 Prerequisite: EET2515C, Corequisite: EET2527C Course User Fee: \$63.00

EET3716C Advanced Systems Analysis 4.000 credits

This is an upper division level course for students majoring in electronics engineering technology designed to prepare students to perform electrical circuit systems analysis using Laplace transforms and partial fraction expansion. Students will learn theorems, Fourier series, frequency response and bode plots, and their application towards practical systems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1025C AND MAC2312

EET4158C Linear Integrated Circuits and Devices 4.000 credits

This is an upper division level course for students majoring in electronics engineering technology designed to provide students with practical skills and knowledge needed for application of operational amplifiers, comparators, phase-locked loops, timers, regulators, other integrated circuits in electronic systems. Students learn to apply these skills towards the design of amplifiers, active filters, oscillators, differentiators, integrators and other miscellaneous integrated circuit based systems .

Prerequisite: EET 3716C. Laboratory fee. (2 hr.

lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET3716C

Course User Fee: \$63.00

EET4165C Senior Design 1 3.000 credits

This project-based course is designed to synthesize students' knowledge of the analysis, design, manufacturing, and testing of electronic systems. Students will design experiments, explore professional ethics, practice professional oral and written communications, conduct project feasibility studies, and perform project scheduling. Students learn about human factors, intellectual property, and liability issues.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: Department approval required

Course User Fee: \$63.00

EET4166C Senior Design 2 2.000 credits

Senior Design 2 is a project-based experience course in which students apply all of the skills they have acquired to analyze, design, simulate, synthesize, and test a complete system.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: EET4165C

Course User Fee: \$63.00

EET4730C Feedback Control Systems 4.000 credits

This upper division course for students majoring in electronics and computer engineering technology, is designed to introduce students to the analysis of circuit networks and control systems. Students learn about stability and compensation considerations, using root locus, the Nichols chart, and Bode plots; simulation techniques; and how to apply these principles to build and test control systems.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET3716C

Course User Fee: \$63.00

EET4732C Signals & Systems 4.000 credits

This course is intended for upper division students majoring in Electronics Engineering Technology. Students will learn the theory and the mathematical techniques used in analyzing continuous-time linear systems. Students will learn continuous-time signal and systems analysis, the input-output relationships of linear time-invariant (LTI) systems, transient and steady state analysis, frequency domain analysis and Fourier analysis. Students will analyze and characterize LTI systems using Laplace transforms.

Prerequisite: EET 3716C.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET3716C

Course User Fee: \$63.00

EEV**EEV0162 Low Voltage Technician 1 5.000 credits**

This course is an overview of the trade with an introduction to construction safety and hand and power tools used in the construction industry.

Students will learn basic written and verbal communication skills, basic mathematics, construction methods and techniques, building codes, and how to read construction drawings.

Clock Hours: 150.000

Prerequisite: TABE REQUIREMENT A student enrolling in a Career Technical Education (CTE) program of 450 or more clock hours is required to complete a basic skills test within the first six weeks of admission into the program. The Tests of Adult Basic Education (TABE) test provided by MDC satisfies this requirement. The minimum passing scores vary among the CTE programs. Students should speak with their program advisor to validate score expectations and testing exemptions.

Students who complete the CTE program must meet basic skills competencies before a certificate of completion is awarded. TABE Scores:

Mathematics:10 Language:10 Reading: 10

EEV0163 Low Voltage Technician 2 5.000 credits

This course focuses on alternating current (AC) and direct current (DC) circuits and electronic devices, including an overview of applicable test equipment, cable and terminations used in the installation of low-voltage systems. Further instruction on specific trade-related drawings, codes and standards will be discussed. OSHA 10 or OSHA 30 Construction Industry training will be provided.

Clock Hours: 150.000

Prerequisite: EEV0162

EEV0164 Low Voltage Technician 3 5.000 credits

This course is an overview of hard-wired cabling, fiber, wireless networks, maintenance, and repair of low-voltage systems, plus project management.

Clock Hours: 150.000

Pre/Corequisite: EEV0163

EEV0165 Low Voltage Technician 4 5.000 credits

This course is an overview of audio, video, broadband, media management, telecommunication systems, and residential/commercial building networks.

Clock Hours: 150.000

Pre/Corequisite: EEV0164

EEV0166 Low Voltage Technician 5 5.000 credits

This course is an overview of intrusion detection, fire alarm systems, fiber optics, nurse call, CCTV, and access control systems.

Clock Hours: 150.000

Pre/Corequisite: EEV0165

EEV0562 microcomp main/rep 2 2.500 credits

course description

Lab Hours: 75.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$63.00

EEX**EEX1130 Reading Difficulties and Dyslexia 2.000 credits**

This course is designed in accordance with Rule 6A-6.053, F.A.C., which provides classroom teachers with the strategies needed to provide instruction for students exhibiting characteristics of dyslexia. The curriculum emphasizes the use of explicit, systematic, sequential, and multi-sensory approaches to reading instruction, ensuring a structured and supportive learning environment.

Lecture Hours: 32.000

EEX2000 Introduction to Exceptional Student Education 3.000 credits

The student will learn about the history, legal issues, and legislation related to exceptional student education in K-12. The student will learn about the differing types of exceptionalities, supports, and accommodations provided for students with differing learning profiles from birth to adulthood.

Lecture Hours: 48.000

Prerequisite: EDF1005

Course User Fee: \$18.00

EEX3071 Designing Classrooms for Students with Exceptionalities 3.000 credits

The student will learn current research-based instructional strategies, and legal and ethical issues necessary in addressing the needs of learners. The student will make informed decisions in adapting, accommodating, and modifying the curriculum.

Fifteen hours of clinical experience in a K-12 least restrictive setting are required.

Lecture Hours: 48.000

Prerequisite: EEX2000 AND EDG3321

Course User Fee: \$18.00

EEX3120 Language Development and Communication Disorders 3.000 credits

The student will learn about typical language and speech development, characteristics and manifestations of communication disorders, and educational neuroscience research concerning first and second language acquisition. The student will learn about effective strategies and accommodations that can be used in planning instruction for P - 12 students with speech and language disorders and differences.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

EEX3201 Young Children with Special Needs 3.000 credits

This course provides an overview of issues related to young children (birth through age eight) with special needs. The student will learn about possible causes and characteristics of exceptionalities, federal laws pertaining to students with disabilities, the referral process, educational interventions, available resources, and advocacy (Ten hours of clinical experience required in an approved birth - age four least restrictive setting).

Lecture Hours: 48.000

EEX3226 Assessment of All Young Children 3.000 credits

The student will utilize guidelines and techniques for observing, assessing, evaluating, and planning curriculum for young children. The student will learn to use formal and informal assessments to evaluate social-emotional, cognitive, language, and motor development and will use data to plan for instruction. (Ten hours of clinical experience required: 3 hours to include an observation of an evaluation of a young child at Early Steps, 3 hours to include a tour of the Mailman Center, Early Steps, and the FFAST Lab, and the remaining 4 hours are to be completed in an approved kindergarten-third grade inclusion setting with ESOL students).

Lecture Hours: 48.000

Prerequisite: EEC2271, EEC2601, Pre/Corequisite: EEC3301, EDF3115

Course User Fee: \$18.00

EEX3603 Positive Behavior Supports in All Settings 3.000 credits

This course provides a holistic approach to guiding young children's behavior and facilitating character education. The student will learn to utilize strategies that emphasize the importance of relationships to learning, self-awareness, and character based on best practices in character-based classroom management while focusing on individual needs of each child. (10 clinical experience hours in an approved pre-kindergarten least restrictive classroom).

Lecture Hours: 48.000

Pre/Corequisite: EEC3301, EEX3201 and 10 hrs. of Field Experience Required

Course User Fee: \$18.00

EEX4012 Introduction to Brain-Based Teaching Strategies 1.000 credits

The student will learn how the brain processes information and how to best engage the brain during learning. In this overview course, the student will acquire research-based, brain-friendly strategies that focus on students with disabilities, English language learners, reading, and mathematics.

Lecture Hours: 16.000

Recommended Preparation: Bachelor's degree.

EEX4024 Legal Issues of Working with Students with Exceptionalities 3.000 credits

The student will learn about the history, governing legislation, and current status of special education in the US. Through case analyses and simulations, the student will correlate and evaluate P-12 classroom and administrative practices and issues with current legislation and mandates.

Lecture Hours: 48.000

EEX4094 Nature and Needs of Students with Autism Spectrum 3.000 credits

The student will learn basic skills, knowledge, and strategies associated with teaching students with autism spectrum disorders (ASD). The student will acquire content and pedagogical knowledge to provide effective instructional practices to students with ASD. Six hours of clinical experience hours are required. Special fee.

Lecture Hours: 48.000

Prerequisite: BS in Exceptional Student Education (ESE) plus experience working with students with varying exceptionalities.

Course User Fee: \$25.00

EEX4221 Assessment in Special Education 3.000 credits

The student will study, analyze, and administer informal and formal assessments to K-12 learners with special needs. The student will learn to prepare and present assessment data for use in instructional planning and developing individualized educational plans for K-12 learners with disabilities.

Lecture Hours: 48.000

Prerequisite: EDG3321 AND EDF4430

Course User Fee: \$18.00

EEX4232 Assessment and Diagnosis of Autism Spectrum Disorders 3.000 credits

The student will learn assessment instruments and strategies used for the referral, diagnosis, and remediation of academic and behavioral difficulties of students with autism spectrum disorders. The student will learn to utilize assessment instruments for instructional planning and evaluating learning outcomes. Six hours of clinical experience are required. Special fee.

Lecture Hours: 48.000

Prerequisite: EEX4094 AND Recommended

Preparation: Professional Teaching certificate and experience working with students with exceptionalities OR Bachelor's degree and experience working with individuals with exceptionalities

Course User Fee: \$25.00

EEX4294 Differentiated Instruction in Mixed-Ability Classrooms 3.000 credits

The student will learn the educational neuroscientific basis for providing differentiated instruction in mixed-ability classrooms. The student will utilize research-based instructional and assessment strategies to create differentiated instruction to meet the needs of all learners in P-12 classrooms.

Lecture Hours: 48.000

EEX4613 Behavior Supports and Management for Students with Autism Spectrum Disorders 3.000 credits

The student will learn disability specific assessment tools used to evaluate social, emotional, and behavioral skills of students with autism spectrum disorders. The student will learn intervention strategies for teaching positive behavior support and appropriate adaptive behavior. Six hours of clinical experience are required. Special fee.

Lecture Hours: 48.000

Prerequisite: BS in Exceptional Student Education (ESE) plus experience working with students with varying exceptionalities.

Course User Fee: \$25.00

EEX4761 Assistive Technology and Communication Systems for Students with Autism Spectrum Disorders 3.000 credits

The student will learn about assistive technology (AT) strategies, including its use for improving the communication and functional capabilities of students with autism spectrum disorders. The student will learn about instructional and assistive technology devices used to support students with autism spectrum disorders. Six hours of clinical experience are required. Special fee.

Lecture Hours: 48.000

Corequisite: EEX4094 AND Recommended Preparation: Professional Teaching certificate and experience working with students with exceptionalities OR Bachelor's degree and experience working with individuals with exceptionalities

Course User Fee: \$25.00

EEX4833 Practicum in Exceptional Student Education 3.000 credits

The teacher candidate will plan, develop, and implement literacy pedagogic methods that meet the needs of students with differing profiles in a K-12 setting. The teacher candidate will utilize action research methodology, assessment principles, educational neuroscience research, and best practices to determine the effectiveness of a literacy strategy. Reading course competency 5. (Sixty hours of clinical experience are required).

Lecture Hours: 48.000

Prerequisite: EDF4430, EDG3321, EDG3443, EDG4376, EEX3071, EEX3120, RED3393, TSL3243, Pre/Corequisite: EEX4221 OR MAE4360 OR RED4519 OR SCE4362 OR TSL4311

EEX4930 Seminar in Exceptional Student Education 3.000 credits

The teacher candidate will engage in professional dialogue and explore professional learning opportunities related to teaching students with exceptionalities. This seminar course is taken in conjunction with a full time, supervised teaching experience and provides an opportunity to examine and reflect on the daily experiences of becoming a highly effective teacher.

Lecture Hours: 48.000

Prerequisite: All Coursework, State Examinations, and E-Portfolio completed and passed., Corequisite: EEX4940

Course User Fee: \$18.00

EEX4931 Introduction to Autism Spectrum Disorders 1.000 credits

The student will identify the legislation associated with Autism Spectrum Disorders (ASD), discuss the characteristics of ASD, understand the nature and needs of ASD, identify positive behavioral interventions, appropriate curricula and assessment modifications, as well as discuss the impact of home, school, and community relationships on students with ASD.

Lecture Hours: 16.000

Recommended Preparation: Bachelor's degree in Education

EEX4932 Advanced Topics in Exceptional Student Education 3.000 credits

The student will explore current issues and trends in Exceptional Student Education (ESE). The student will learn about the foundations of ESE, including instructional practices, differentiated instruction, positive behavioral supports, assessment, the transition process, and assistive technology. This course will help prepare students for the Florida Teacher Certification ESE exam.

Lecture Hours: 48.000

Prerequisite: EEX3071

EEX4940 Internship in Exceptional Student Education 9.000 credits

The teacher candidate will engage in the educational and professional responsibilities common to teachers in exceptional student education. This internship experience reinforces and augments teaching strategies that students have developed through their coursework and clinical experiences. Students participate in a full time, supervised teaching experience.

Internship Hours: 432.000

Prerequisite: All coursework and state exams completed and passed, Corequisite: EEX4930

EEX4992 Brain-Based Teaching: The Exceptional Brain 3.000 credits

The student will learn how the typical and atypical brain processes information. The student will acquire research-based and best practices for teaching, differentiating instruction, and assessing P-12 students with and without identified exceptionalities.

Lecture Hours: 48.000

Prerequisite: Teacher certification

EGN**EGN1008C Introduction to Engineering 3.000 credits**

An introduction to the opportunities, challenges, and required skills of the engineering profession. Students will explore the different disciplines of engineering, their function in industry, and required education. Professional issues such as registration, ethics, and safety are discussed. Projects and activities are used to develop problem solving, communication and computer skills.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$63.00

EGN1949 CO-OP WRK EXP 1 1.000 - 4.000 credits

This is a course designed to provide training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Prerequisite: Co-Op Department approval. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Cooperative Education Office to obtain registration approval. (3 hr. lecture)

Lecture Hours: 16.000 - 64.000

EGN2200 Computer Applications in Engineering 3.000 credits

An introduction to fundamental concepts and skills of mathematical programming and computer-aided design. This course explores the use of computer software to solve engineering problems and bring ideas from a concept to a model.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1114 OR MAC1147

EGN2322 Engineering Mechanics - Dynamics 4.000 credits

This course provides students with the skills they need to analyze and solve problems involving bodies in motion through the application of vector mechanics and Newton's laws. Students will learn kinematics, kinetics, energy of particles, rigid bodies in 2-D and 3-D motion, and vibrations.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: EGN2312 AND Pre/Corequisite: MAC2313

EGS**EGS1010 Applied Research Methods 1.000 - 3.000 credits**

This course is designed for STEM majors. Students will learn basic research practices: research methods, experimentation, validation, technical writing, and presentations. Using the Affinity Research Group model, students will work in groups to conduct theory-based STEM research, develop poster presentations, and write conference and journal publications.

Lecture Hours: 16.000 - 48.000

EME**EME2040 Creativity, Innovation, and Technology for the 21st Century Learner 3.000 credits**

The student will learn to manage a productive and safe technology environment by promoting creativity and innovation in the classroom. The student will gain 21st century knowledge, skills, and attitudes for applying technology across multiple disciplines and grade levels.

Lecture Hours: 48.000

Prerequisite: CGS1060C

Course User Fee: \$18.00

EME2071 Educating Young Children for Digital Literacy in the 21st Century 3.000 credits

This course provides an overview of appropriate use of technology in early childhood settings (ages 3-8). The student will learn how to make informed decisions regarding the intentional use of technology in ways that support young children's learning and development.

Lecture Hours: 48.000

Course User Fee: \$25.00

**EME4610 Introduction to Instructional Design
3.000 credits**

This is the first in a series of five courses that leads to a certificate in Instructional Design. The student will develop an overview of the field of instructional design as it relates to training, development, and education. The student will compare and contrast instructional design models, learning theories, and current technologies.

Lecture Hours: 48.000

Prerequisite: Bachelor degree

**EME4611 Instructional Design Development 1
3.000 credits**

In this course, the student will develop and launch a prototype of a training, development, or educational opportunity using different types of media tools.

Lecture Hours: 48.000

Recommended Preparation: Bachelor Degree., Pre/
Corequisite: EME4683

**EME4612 Instructional Design Development 2
3.000 credits**

In this course in Instructional Design, the student will develop assessments and an evaluation plan for a training, development, or educational opportunity. The student will conduct User Acceptance Testing using established criteria, and utilize feedback to modify his or her prototype.

Lecture Hours: 48.000

Recommended Preparation: Bachelor degree, Pre/
Corequisite: EME4611

**EME4671 Instructional Design Analysis 3.000
credits**

In this course, the student will evaluate and analyze training, development, and education opportunities and project development. The student will develop a needs analysis for an instructional design project based on research-based best practices in the field.

Lecture Hours: 48.000

Recommended Preparation: Bachelor degree, Pre/
Corequisite: EME4610

**EME4683 Instructional Design Application 3.000
credits**

In this course, the student will utilize best practices of learning design, learning theories, and instructional strategies for adult learners to write objectives, develop an outline, and create storyboards that encompass the scope of a training, development, or educational opportunity.

Lecture Hours: 48.000

Recommended Preparation: Bachelor degree, Pre/
Corequisite: EME4671

EMS**EMS0110 Emergency Medical Technician 10.000
credits**

The Emergency Medical Technician (EMT) course prepares students to provide pre-hospital assessment and care for patients of all ages with a variety of medical conditions and traumatic injuries. Areas of study include an introduction to emergency medical services systems, roles and responsibilities of EMTs, anatomy and physiology, medical emergencies, trauma, special considerations for working in the pre-hospital setting, and providing patient transportation.

Lecture Hours: 300.000

**EMS1059 Emergency Medical Responder 1.000
credits**

Provides training in emergency medical care for those who may be first to respond to an accident. The course meets the basic requirements of the U.S. Department of Transportation. Recommended for students who are not required to be certified EMTs.

Lecture Hours: 16.000

Corequisite: EMS1059L

EMS1059L RESPONDER EM CRE LA 1.000 credits

Lab Hours: 32.000

**EMS1119 Emergency Medical Technician 4.000
credits**

A review of basic life support theory. Areas of emphasis include the prehospital environment, preparatory information, patient assessment, medical emergencies, behavioral emergencies, OB/GYN emergencies, trauma emergencies, pediatric emergencies and EMS operations.

Lecture Hours: 64.000

Corequisite: EMS1119L, EMS1431

**EMS1119L Emergency Medical Technician Lab
and Clinic 3.000 credits**

Practical application of the content covered in EMS 1119 with an emphasis on cardiopulmonary resuscitation, splinting, bandaging, patient movement, and other skills as recommended by the U.S. Department of Transportation for the EMT-A level practitioner.

Lab Hours: 96.000

Corequisite: EMS1119, EMS1431

Course User Fee: \$55.00

**EMS1431 EMT Hospital/Field Experience 3.000
credits**

Practice in local emergency departments and rescue agencies under professional supervision. This course meets the skills recommended by the U.S.

Department of Transportation.

Clinical Hours: 144.000

Corequisite: EMS1119, EMS1119L

EMS2311 Emergency Medical Services Operations 2.000 credits

Advanced theory of management operations currently used nationally by comprehensive emergency medical service systems. Legal issues as related to various aspects of the system, personnel policies, provider versus client roles, disaster planning, communications, budgeting and evaluation of the system will be discussed.

Lecture Hours: 32.000

Prerequisite: MNA1345

EMS2395 EMER MED SER SEMINAR 1.000 credits

Current topics and trends for the Emergency Medical Services (EMS) provider. Support materials requested. A.S. degree credit only. (1 hr. lecture)

Lecture Hours: 16.000

EMS2601 Paramedic Lecture 1 8.000 credits

EMS2601 is the first course in the sequence necessary for completion of the Paramedic Certificate program. The course is designed to reinforce concepts and clinical skills learned at the EMT level and to integrate this knowledge beginning with advanced life support concepts and skills. Emphasis is placed on EMS systems, illness and injury prevention, medical-legal issues, patient assessment, airway management and ventilation, pathophysiology, pharmacology, shock, decision-making, and the management of trauma-related injuries. This course includes Modules 1–4 of the 1998 DOT National Standard Curriculum for Paramedic Programs.

Lecture Hours: 128.000

Corequisite: EMS2601L

Course User Fee: \$96.00

EMS2601L Paramedic Laboratory 1 4.000 credits

A review of basic life support practice and an introduction to advanced life support practice. Areas of emphasis include patient assessment, trauma emergencies, obstetric emergencies, gynecological emergencies, pediatric emergencies and psychiatric emergencies. Students will be expected to master the techniques of patient assessment, intravenous techniques and endotracheal intubation.

Lab Hours: 128.000

Corequisite: EMS2601

Course User Fee: \$75.00

EMS2602 Paramedic Lecture 2 8.000 credits

EMS 2602 is the second course in the sequence necessary for the completion of the Paramedic Certificate Program. This course is designed to reinforce and expand upon the material and skills learned in Paramedic 1 level and to integrate prior learning with enhanced life support concepts and skills. Emphasis is placed on patient assessment and recognition of significant findings, pre-hospital diagnosis and differential diagnosis, treatment strategies, anatomy and physiology, pathophysiology, and the management of various emergencies, patients with special challenges, assessment based management, and EMS operations. This course includes Modules 5–8 of the 1998 DOT National Standard Curriculum for Paramedic Programs.

Lecture Hours: 128.000

Prerequisite: EMS2601L, EMS2601, EMS2664,

Corequisite: EMS2602L, EMS2665

EMS2602L Paramedic Laboratory 2 4.000 credits

Continuation of advanced life support practice. Areas of emphasis include patient assessment, trauma emergencies, obstetric emergencies, gynecological emergencies, pediatric emergencies and psychiatric emergencies. Students will be expected to master the techniques of patient assessment, intravenous techniques, endotracheal intubation, and advanced life support.

Lab Hours: 128.000

Corequisite: EMS2602, EMS2665

Course User Fee: \$75.00

EMS2659 EMS-Field Internship and Conference 8.000 credits

A supervised clinical experience on an Advanced Life Support (ALS) vehicle. The student obtains increasing patient care responsibilities as a working member of the EMS team under the direct supervision of a designated preceptor.

Clinical Hours: 384.000

Prerequisite: EMS2602, EMS2602L, EMS2665

EMS2664 Paramedic Clinic 1 3.000 credits

EMS 2664 is designed to allow the students "hands-on" practice of the skills and theories learned in EMS 2601 and 2601L. Clinical experience will take place in many areas including the emergency department, operating room and medical examiner's office. All patient care experience will be practiced under the direct supervision of a medical professional (Paramedic, Nurse, Physician, etc.).

Clinical Hours: 144.000

Corequisite: EMS2601, EMS2601L

EMS2665 Paramedic Clinic 2 3.000 credits

EMS 2665 is designed to allow the students "hands-on" practice of the skills and theories learned in EMS 2602 and 2602L. Clinical experience will take place in many areas including the emergency department, operating room and critical care unit. All patient care experience will be practiced under the direct supervision of a medical professional (paramedic, nurse, physician, etc.).

Clinical Hours: 144.000

Prerequisite: EMS2664, Corequisite: EMS2602, EMS2602L

EMS9995 Emergency Medical Responder 2.000 credits

This course is designed for students who have earned the Emergency Medical Responder (EMR) credential and provides academic recognition of the knowledge, skills, and competencies demonstrated through certification attainment. The course validates proficiency in emergency patient assessment, basic life support, trauma care, medical emergencies, patient stabilization, and emergency response procedures. Credit awarded through this course supports academic progression and recognizes documented achievement of industry-recognized competencies in prehospital emergency care.

Clinical Hours: 96.000

EMS9996 Emergency Medical Technician (EMT) - Basic 10.000 credits

This course provides academic credit for students who have successfully completed a State of Florida-approved Emergency Medical Technician (EMT-Basic) Applied Technology Diploma (ATD) program and are enrolled in the related Associate in Science degree within the School of Health Sciences. In accordance with TMOP 1100-816125, the course recognizes competencies attained through prior educational and certification experiences, including emergency patient assessment, airway management, trauma care, medical emergency response, patient transportation, and prehospital care procedures. Credit awarded through this course supports academic progression and acknowledges demonstrated achievement of industry-recognized emergency medical services competencies.

Other Hours: 160.000

ENC**ENC0015 Developmental Writing I 4.000 credits**

Developmental Writing I is a college preparatory writing course. Students will learn to address effective sentence and paragraph development. Lab time required. Prerequisite: Non-demonstration of readiness through placement testing or alternate methods.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$5.00

ENC0025 Developmental Writing II 4.000 credits

Developmental Writing II is an intermediate college preparatory writing course. Students will learn to address effective sentence, paragraph, and essay development using standard edited American English. Lab time required. Prerequisite: Non-demonstration of readiness through placement testing or alternate methods.

Lecture Hours: 32.000, Lab Hours: 64.000 - 61.000

Course User Fee: \$5.00

ENC0027 Introduction to College Writing through Reading 3.000 credits

This course introduces students to college level composition and reading. Students will use the writing process to compose effective sentences, paragraphs, and essays using standard edited American English in response to various reading materials. Students will demonstrate proficiency in literal and critical comprehension by using a variety of reading strategies. Prerequisite: Non-demonstration of readiness through placement testing or alternate methods.

Lecture Hours: 48.000

Course User Fee: \$5.00

ENC1101 English Composition 1 3.000 credits

This course introduces students to rhetorical concepts and audience-centered approaches to writing including composing processes, language conventions and style, and critical analysis and engagement with written texts and other forms of communication. Learning outcomes: students will apply rhetorical knowledge to communicate for a range of audiences and purposes; students will employ critical thinking to analyze forms of communication; and students will engage in writing processes that involve drafting, revising, and reflecting. Students who complete this course will be awarded the Fundamentals of Written Communication Digital Badge.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education reading and writing requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$15.00

ENC1102 English Composition 2 3.000 credits

This is the second required general core course in college-level writing. Students will learn the conventions of standard edited American English. Students will compose informative and persuasive essays, write responses to a variety of literary genres and/or non-fiction, and produce a documented paper based on research. Writing Intensive Course and must be completed with a grade of "c" or better.

Lecture Hours: 48.000

Prerequisite: ENC1101

Course User Fee: \$15.00

ENC1112 Essential Elements of English Grammar 1.000 - 3.000 credits

This course is designed for students whose writing demonstrate a need for continued instructional support. Course content is individualized based on specific student needs. This course is repeatable. Prerequisites: Demonstration of readiness through placement testing or alternate methods or ENC0025 with a grade of "S".

Lecture Hours: 16.000 - 48.000

Prerequisite: Placement by Scholastic Assessment Test (SAT) verbal subtest score OR American College Testing (ACT) English subtest score OR Computerized Placement Test (CPT) English subtest score OR ENC0025

ENC1113 Writing Skills Review 1.000 - 3.000 credits

This course is designed for students whose writing and/or English language skills test scores demonstrate a need for continued instructional support and features self-assessment grammar diagnostics. Students will learn the principles of composition via the writing process: planning, drafting, revising, finishing and editing. Note: This variable credit course is repeatable. Prerequisites: Non-demonstration of readiness through placement testing or alternate methods or ENC0025 with a grade of "S".

Lecture Hours: 16.000 - 48.000

Prerequisite: ENC0025

Course User Fee: \$15.00

ENC1210 TECH REPORT WRITING 3.000 credits

Lecture Hours: 48.000

ENC2300 Advanced Composition and Communication 3.000 credits

This writing-based course addresses techniques of critical thinking, persuasion, and argumentation. Students will refine their composition skills and develop their oral communication skills by examining and discussing a range of issues.

Prerequisites: ENC1101, 1102 or equivalent with a grade of "C" or better. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101 AND ENC1102

ENG**ENG2012 Literary Theory 3.000 credits**

This course introduces students to the study of literary criticism. Students will discuss, analyze, and write about literature through the lens of contemporary critical theory.

Lecture Hours: 48.000

Prerequisite: ENC1101

ENL**ENL2012 English Literature 1 3.000 credits**

A survey of major British writers from Chaucer through the 18th century. Required of English majors.

Lecture Hours: 48.000

Prerequisite: ENC1102

ENL2022 English Literature 2 3.000 credits

A survey of major British writers from the 18th century through the contemporary period. Required of English majors.

Lecture Hours: 48.000

Prerequisite: ENC1102

ENT**ENT1501 Fundamentals of Changemaking and Social Innovation 3.000 credits**

This course introduces students to the work of changemaking and the field of social innovation. Students will explore principles of social innovation and social change, while developing the skills to analyze social issues, generate solutions to those issues, and become an effective social change agent.

Lecture Hours: 48.000

Course User Fee: \$43.00

ENT2201 Introduction to Lean Start-Up 3.000 credits

The student will learn how successful start-ups find a repeatable, scalable business model that creates value for themselves and customers. In this experientially driven course, student teams develop, validate and refine a business model by creating a hypothesis, designing products/services, testing the hypotheses, and reflecting on what has been learned. Prerequisite: GEB2112.

Lecture Hours: 48.000

Prerequisite: GEB2112

Course User Fee: \$69.00

ENT2212 Entrepreneurial Leadership 3.000 credits

The student will learn key skills and traits of successful entrepreneurs. Students will explore the notion of values-based business formation, personal strength and weakness identification, leadership for team building, project and personal time management, and story-telling.

Lecture Hours: 48.000

Course User Fee: \$100.00

ENT2270 Family Business Management 3.000 credits

This course covers special issues facing entrepreneurial and family businesses such as choice of organizational form, business planning, tax and compensation planning, business valuation, and succession strategies. Time is also devoted to the unique challenges often found in family business context, such as dealing with family conflicts, how to motivate and evaluate employees when a mix of family and non-members are involved, and planning for succession. (3 hr. lecture)

Lecture Hours: 48.000

ENT2421 Funding Your Venture 3.000 credits

This course focuses on critical skills necessary to develop appropriate funding strategies for new venture creation and growth. Students will explore a variety of ways to raise capital and gain an understanding of investors' expectations and how to evaluate the advantages and pitfalls of various sources of capital. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$43.00

ENT2502 Starting and Growing a Social Venture credits

The course explores the start-up, growth, and management of social entrepreneurship. Social ventures share attributes but also differ from for-profits in intent and practice. The student will learn the elements of integration, innovation; development and management of a business within and existing corporate culture. Ideation, venture creation, resource acquisition, and growth management are also addressed.

Lecture Hours: 48.000

Course User Fee: \$100.00

EPI**EPI0001 Classroom Management 3.000 credits**

The student will learn how to plan and develop a positive and productive learning environment in P-12 settings grounded in research-based character development models and learning theories. The student will learn to create a learning environment that encourages positive interactions and self-regulation. Ten hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Recommended Preparation: (1) Bachelor degree earned from an accredited institution of higher education in any area except Education; (2) Cumulative bachelor degree GPA of 2.5 or higher on a 4.0 scale; and (3) Admitted to the Educator Preparation Institute program.

EPI0002 Instructional Strategies 3.000 credits

The student will connect human developmental theories and current educational neuroscience research to the planning of instructional activities for students in P-12 settings. The student will learn to develop lesson plans that incorporate varied teaching strategies and questioning skills at all levels, differentiated instruction, and appropriate formative and summative assessments. The student will incorporate learning experiences that require P-12 students to demonstrate a variety of skills. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Prerequisite: EPI0001 or an equivalent course.

EPI0003 Technology 3.000 credits

The student will learn about the use of instructional technology to teach P-12 students. The student will learn how technology can support classroom management, administration, and teaching. The student will learn the legal and ethical use of technology. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Prerequisite: EPI0002

EPI0004 The Teaching and Learning Process 3.000 credits

The teacher candidate will learn the formal philosophies of education: perennialism, essentialism, progressivism, and reconstructionism, and their application to the P-12 teaching and learning process. The teacher candidate will develop a substantive understanding of the six components of reading as a process: oral language, phonological awareness, phonics, fluency, vocabulary, and comprehension, select and administer appropriate assessments, and analyze data to inform reading instruction. Reading competencies 1 and 3. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Prerequisite: EPI0002

EPI0010 Research-Based Practices in Reading 3.000 credits

The teacher candidate will differentiate instruction for all learners by applying the principles of research-based strategies and integrating the six components of reading: phonological awareness, phonics, oral language, fluency, vocabulary, and comprehension. Teacher candidates will have a broad knowledge of students from differing profiles to understand and apply evidence-based instructional practices by differentiating process, product, and context. Reading competencies 2 and 4. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 48.000

Prerequisite: EPI0004

EPI0020 Professional Foundations 1.000 credits

The student will learn to evaluate his/her role as a productive member of the teaching profession. The student will learn to reflect on his/her development and mastery of the pre-professional Florida educator accomplished practices during the completion of the clinical education experience in a P-12 classroom setting. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 16.000

Prerequisite: EPI0004

EPI0030 Empowering All Learners 2.000 credits

This course focuses on effective strategies for identifying and meeting the varied needs of students in the classroom. Students will examine teaching practices, legal and ethical responsibilities, and assessment methods to ensure all learners are supported. The course covers approaches to differentiated instruction, fostering positive learning environments, and utilizing assessment data to guide instructional decisions. Participants will also learn how to collaborate with families and communities to support student learning. Through reflective practice, students will develop actionable plans to enhance teaching performance and improve learner outcomes.

Lecture Hours: 32.000

Prerequisite: EPI0002

EPI0940 Field Experience II 2.000 credits

This is the second of two field experience courses that mark the culmination of the educator preparation institute program. In this course, the teacher will observe the classroom and develop and teach a unit lesson in a P-12 setting. Building on previous formal classroom observations and actionable feedback on their lesson plans and mini-teaches, the teacher candidate will refine their instructional practices. They will also demonstrate knowledge of the components of reading and the ability to apply effective instructional reading practices. The teacher candidate will be formally observed teaching in a P-12 setting. Reading course competency 5. Sixty hours of clinical experience are required for successful completion of this course.

Lecture Hours: 32.000

Prerequisite: EPI0004

EPI0945 Field Experience I 1.000 credits

The student will observe effective teaching/learning techniques in P-12 settings with to collect and analyze observational data, and also plan/ implement teaching strategies that meet the needs of all learners. A formal observation/assessment of the students teaching performance will be conducted. Fifteen hours of clinical experience are required for successful completion of this course.

Lecture Hours: 16.000

Prerequisite: EPI0004

ESC

ESC1000 General Education Earth Science 3.000 credits

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, lithosphere, biosphere, and exosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize earth's connections with humans. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize earth science data to draw scientifically valid conclusions; students will recognize the different time scales associated with different earth processes; students will effectively describe interactions between humans and the earth's spheres; and students will apply their understanding of earth science principles to complex global and local issues.
Lecture Hours: 48.000
Course User Fee: \$10.00

ETC2450 Concrete Construction 3.000 credits

The use of concrete in construction to include foundations, columns, beams, slabs, hydraulic conduits.
Lecture Hours: 48.000
Prerequisite: ETG2502

ETD**ETD1110 Technical Drawing 4.000 credits**

Introduces students to the principles of instrument drawing, orthographic projection, visualization, specialized computer processes and introductory computer aided drawing (CAD). Students develop drawing and sketching techniques common to industry.
Lecture Hours: 32.000, Lab Hours: 64.000

ETD1340 Computer Aided Drawing & Design 3.000 credits

This course is recommended for all engineering students as an introduction to the basic concepts of drafting and designing using a computer. Students will learn industry standard drafting and design practices using AutoCAD in a laboratory environment. Laboratory fee.
Lecture Hours: 32.000, Lab Hours: 32.000
Pre/Corequisite: MAC1105
Course User Fee: \$63.00

ETD1542 Structural Drafting 4.000 credits

Development of structural, fabrication and erecting drawings. Course involves study of structural shapes, their properties, and methods of developing connections, as well as study of common reinforced concrete practices. Laboratory fee.
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ETD1110
Course User Fee: \$63.00

ETI**ETI1040 Introduction to Bioscience Manufacturing 3.000 credits**

This course introduces students to the field of bioscience manufacturing. Topics will include basic principles of the industry, large-scale process development and the future of the bioscience industry. Current Good Manufacturing Practices (cGMPs), and the nature and delivery system of products will also be discussed.
Lecture Hours: 48.000

ETI1043C Introduction to Aerospace 3.000 credits

This course includes an overview of aerospace industry terminology and acronyms, along with practical activities involving tools, procedures, and standard practices. Additionally, it covers inspection procedures, workplace rules and regulations, safety measures, maintenance protocols, and career options in the aeronautical technician workforce. (2 hr. lecture, 2 hr. lab)
Lecture Hours: 16.000, Lab Hours: 64.000
Course User Fee: \$63.00

ETI1152C Mechanical Measurement and Instrumentation 3.000 credits

This course provides the basic foundation for mechanical measurement techniques used in manufacturing environments. The course will integrate the concepts, principles, and techniques of mechanical measurement with the use of various types of instruments including micrometers, verniers, calipers, gages, and other types of measuring equipment.
Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$63.00

ETI1172 Introduction to Quality Assurance 3.000 credits

This course describes the role and aspects of quality systems and Regulatory affairs in research laboratories, regulated companies, and firms that comply with voluntary standards. Topics include stages in development and submission of drugs and medical devices, patents legislation, and quality systems such as auditing, standard procedures, good manufacturing and laboratory practices. Lecture Hours: 48.000

ETI1622 Concepts of Lean and Six Sigma 3.000 credits

This course is designed for students who are preparing for careers in the manufacturing industry. Students will learn the basic concepts, frameworks, and techniques used in six sigma, including total quality philosophies, the calculation of six sigma and other vital statistics, tools of lean six sigma, and knowledge of various methodologies. Lecture Hours: 48.000

ETI1644 Advanced Manufacturing Supply Chain 3.000 credits

This course is designed to provide students who are preparing to be manufacturing support technologists with the fundamental concepts of advanced manufacturing supply chain management (SCM) principles. Students will learn how to use manufacturing planning and control systems to coordinate material, labor, capacity and other resources to optimize manufacturing operations. Students also learn the key features of automated systems that can be used to manage the supply chain process. Lecture Hours: 48.000

ETI1701 Industrial Safety 3.000 credits

This course provides the student with the knowledge and skills to recognize hazardous situations in industrial plants and the precautions to be observed and practiced to perform work activities safely. Among the topics covered are industrial safety hazards, electrical safety, working with chemicals, gases, and solvents, protective equipment, and safe working conditions. Lecture Hours: 48.000

ETI1830C Materials and Processes 1 3.000 credits

In this course, students explore the physical properties and characteristics of materials and commodities commonly utilized in the aerospace industry. It delves into materials compatibility, basics of metallurgy, and treatment processes, offering a foundation of how these elements interact and are applied within aerospace contexts. (1 hr. lecture, 4 hr. lab)
Lecture Hours: 16.000, Lab Hours: 64.000
Course User Fee: \$63.00

ETI1851C Applied Mechanics 4.000 credits

This course adopts a practical approach to teaching the identification, usage, and maintenance of tools and equipment utilized in aerospace systems. It covers blueprint reading, geometric dimensioning, tolerancing, and using both English and metric measuring systems. (2 hr. lecture, 4 hr. lab)
Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$63.00

ETI1852C Aerospace Test and Measurements 4.000 credits

This course covers the principles and applications of primarily nondestructive testing techniques used to examine materials and components of spacecraft systems. Students learn to interpret procedures, operate testing equipment, and apply safety measures while conducting visual, dye penetrant, magnetic particle, ultrasonic, and Eddy-current inspections. Emphasis is placed on accurate measurement, tool selection, and result evaluation. Prerequisite: ETI 1830C. (2 hr. lecture, 4 hr. lab)
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ETI1830C
Course User Fee: \$63.00

ETI1853C Aerospace Safety and Quality 3.000 credits

This course delves into the identification of hazards, the use of personal protective equipment, and the implementation of safe practices and measures for the protection of personnel, property, and equipment in the aerospace sector. It also addresses safety procedures, including compliance with OSHA regulations and the handling of hazardous materials. The course additionally explores fundamental principles of quality assurance engineering as they pertain to work processes. It includes the use of computer-based tools for analysis and reporting. (2 hr. lecture, 2 hr. lab)
Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$63.00

**ETI1855C Aerospace Structural Fabrication 1
3.000 credits**

This course covers essential skills in metalworking, machining, and aerospace fastener applications. Students will learn blueprint interpretation, sheet metal repair, and the use of tools like brakes, shears, and rivets, while gaining hands-on experience in project construction and layout design, following industry standards. Prerequisites: ETI 1830C and ETI 1851C. (1 hr. lecture, 4 hr. lab)
Lecture Hours: 16.000, Lab Hours: 64.000
Prerequisite: ETI1830C AND ETI1851C
Course User Fee: \$63.00

**ETI2321C Aerospace Systems and Propulsion
4.000 credits**

This course introduces aeronautical and spacecraft systems, including hydraulic, pneumatic, electrical, propulsion, mechanical, HVAC, and ECLSS (Environmental Control and Life Support Systems). It covers system interactions with computer and data acquisition systems, emphasizing safety, power sources, assembly, troubleshooting, and integration for aerospace operations. Prerequisites: AFR 1100C (2 hr. lecture, 4 hr. lab)
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ETI1043C
Course User Fee: \$63.00

**ETI2404 Advanced Manufacturing Technology
3.000 credits**

This is a course for students intending to work in manufacturing environments. Students will learn the basic concepts about advanced manufacturing operations and processes, including sourcing materials, production planning and process monitoring, and control to distribution activities. Students also review the facility and regulatory requirements needed to support manufacturing operations. Activities may include facility tours and site visits.
Lecture Hours: 48.000

**ETI2670 Engineering Economic Analysis 3.000
credits**

This course is designed for students who are majoring in any engineering discipline. Students will learn the basic methods of engineering cost analysis including equivalence, value measurement, interest relationships and decision support theory and techniques as applied to capital projects. Various problem solving methods will be used for decision making, multiple alternatives and uncertainty.
Lecture Hours: 48.000
Prerequisite: MAC1105

ETI2832C Materials and Processes 2 3.000 credits

This course provides information in aerospace applications of non-metallic materials. The use of adhesives, coatings, sealing, and issues with delamination, and faulty bonds are covered.
Prerequisite: ETI 1830C (1 hr. lecture, 4 hr. lab)
Lecture Hours: 16.000, Lab Hours: 64.000
Prerequisite: ETI1830C
Course User Fee: \$98.00

**ETI2856C Aerospace Structural Fabrication 2
4.000 credits**

This course introduces the student to the theory, materials, and basic manufacturing processes of composite materials. The course focuses on basic composite theory, including fiber reinforcements, matrix systems, fabrication techniques, and safety. It also introduces common core materials that are used in composites manufacturing, and outlines inspection and repair of composites structures including damage detection and repair instructions.
Prerequisites: ETI 1855C and ETI 2832C (2 hr. lecture, 4 hr. lab)
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: ETI1855C AND ETI2832C
Course User Fee: \$98.00

**ETI2951C Aerospace Technician Capstone 3.000
credits**

This course guides students through a capstone project, emphasizing project management, testing, troubleshooting, and documentation. Students evaluate requirements, conduct testing, resolve issues, and complete failure analysis. The course culminates with a comprehensive project presentation and a "lessons learned" review.
Prerequisites: ETI 1855C and ETI 2832C; Pre/
Corequisites: ETI 1852C and ETI 2856C (1 hr. lecture, 4 hr. lab)
Lecture Hours: 16.000, Lab Hours: 64.000
Prerequisite: ETI1855C AND ETI2832C AND Pre/
Corequisite: ETI1852C AND ETI2856C
Course User Fee: \$98.00

ETI4480C Applied Robotics 4.000 credits

This is an upper division level course designed as an introduction to robotics programming and includes robotic applications for multifunction part manipulation and motion with stepper and servo-motors. Students learn topics related to robotic design including robotic vision, motion planning, sensing and sensors, actuators, navigation systems, mobility, forward and inverse kinematics, and path planning. Laboratory activities provide hands-on application of concepts and theories. Prerequisite: CET2123C or CET3383C.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: CET2123C OR CET 3383C

Course User Fee: \$63.00

ETM**ETM1315C Applied Pneumatics and Hydraulics 3.000 credits**

This course prepares students to perform mechanical maintenance on industrial equipment and devices. Students learn the theory and application of fluid mechanics, how to calibrate metering devices, and conduct elementary hydraulic tests. Pre/Corequisite: MAC1105.

Lecture Hours: 32.000, Lab Hours: 32.000

ETP**ETP2040C Electric Power Distribution 4.000 credits**

This is a required course for all students pursuing a certificate in solar photovoltaic energy generation. The student will acquire an understanding of multi-phase power transmission, how to connect to the electric grid and the major components used in electric power distribution including: power transformers, circuit breakers, transmission lines, reclosures, relay coordination, fuses, motors and generators. The student will learn about the maintenance, troubleshooting and protection of these devices against lightning and other abnormal conditions through hands-on laboratory experiments utilizing modern testing and simulation equipment. Laboratory fee.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1033C

Course User Fee: \$103.00

ETP2410C Design, Installation and Operation of Solar PV Systems 4.000 credits

This is a required course for all students pursuing a certificate in solar photovoltaic energy generation. Students will learn about solar photovoltaic (PV) systems and the different components comprising the system including: PV panels, controllers and the batteries used with PV systems. Students will learn to size, install, maintain, troubleshoot and repair the PV system through hands-on laboratory experiments utilizing modern testing and simulation equipment. Prerequisite: EET1033C.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: EET1033C

Course User Fee: \$103.00

ETP2501C Introduction to Alternative and Renewable Energy 3.000 credits

This course is an introductory course to the concepts of renewable energy. It is designed to prepare students to enter the field of renewable energy and green technology. This course focuses on Solar photo-voltaics, solar power and tracking systems, charge controllers and inverters, Wind power systems, Biomass and geothermal power generation. Prerequisite: EET1033C.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: EET1033C

Course User Fee: \$103.00

ETP3240 Power Systems 3.000 credits

In this course, students will learn renewable energy theory and applications. This course focuses on solar photo-voltaics, solar power and tracking systems, charge controllers and inverters, wind power systems, biomass and geothermal power generation. In addition, this course covers the integration with electrical grid, production and end user systems. Prerequisite: EET1025C.

Lecture Hours: 48.000

ETP3320 Introduction to Renewable Energy Technology 3.000 credits

In this course, students will learn renewable energy theory and applications. This course focuses on solar photo-voltaics, solar power and tracking systems, charge controllers and inverters, wind power systems, biomass and geothermal power generation. In addition, this course covers the integration with electrical grid, production and end user systems. Prerequisite: EET2101C.

Lecture Hours: 48.000

Prerequisite: EET2101C

ETS

ETS1603C Introduction to Robotics 4.000 credits

This is an introductory level course designed as an introduction to robotics and robotic applications. Students learn topics related to robotic design including robotic terminology, robotic programming, sensing and sensors, actuators, modeling and sensing, robotic platforms, and the application of artificial intelligence in robotics. Laboratory activities provide hands-on application of concepts and theories.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$63.00

ETS2114C Lithography Methods in Semiconductor Manufacturing 4.000 credits

This course covers photolithography in semiconductor fabrication, focusing on pattern generation, statistical process control, and process fundamentals. Students gain hands-on experience with photomasks, spin-coating, exposure systems, and chemical development, while exploring various lithography methods and evaluating key process variables for successful pattern transfer.

Prerequisites: ETS 2160C, ETS 2163C. (2 hr. lecture 4 hr. lab).

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETS2160C AND ETS2163C

Course User Fee: \$98.00

ETS2163C Cleanroom Safety and Introduction to Cleanroom Processing 3.000 credits

Students learn cleanroom protocols focused on safety and contamination control. Semiconductor manufacturing processes are introduced, including photolithography, etching, deposition, and doping. Students explore characterization techniques such as microscopy and ellipsometry through hands-on cleanroom experience, emphasizing process optimization and compliance with industry standards. Recommended: Students are recommended to have completed CHM 1025. (1 hr. lecture 4 hr. lab).

Lecture Hours: 16.000, Lab Hours: 64.000

Recommended Preparation:

CHM1025 Recommended: Students are recommended to have completed CHM 1025.

Course User Fee: \$98.00

ETS2165C Semiconductor Characterization and Packaging 4.000 credits

This course introduces advanced semiconductor metrology and packaging processes. Students will gain hands-on experience with tools like scanning electron microscope, ellipsometers, and profilometers, focusing on thin-film measurement, wafer probing, and packaging techniques while addressing safety, quality standards, and emerging trends in nanomanufacturing. Prerequisites: ETS 2160C, ETS 2163C. (2 hr. lecture 4 hr. lab).

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETS2160C AND ETS2163C

Course User Fee: \$98.00

ETS2168C Semiconductor Plasma Techniques 4.000 credits

This course introduces students to the principles and applications of plasma in semiconductor manufacturing, with a focus on etching and deposition processes. Students will explore wet and dry etching methods, including reactive ion etching (RIE) and inductively coupled plasma (ICP), while gaining hands-on experience in plasma characterization, process optimization, and plasma-enhanced chemical vapor deposition (PECVD).

Prerequisites: ETS 2160C, ETS 2163C. (2 hr. lecture 4 hr. lab).

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ETS2160C AND ETS2163C

Course User Fee: \$98.00

ETS2632C Introduction to Computer Aided Manufacturing 3.000 credits

An introduction into the fundamentals of Computer Integrated Manufacturing as it relates to theory, operation, setup, safety, and practices. Students will learn the application of Computer Aided Drawing (CAD) and Computer Aided Manufacturing (CAM) software to develop prototypes. Prerequisite: MAC1105.

Lecture Hours: 32.000, Lab Hours: 32.000

Pre/Corequisite: MAC1105

Course User Fee: \$63.00

ETS2673C Programmable Logic Controls 4.000 credits

This course is intended for students majoring in Electronics Engineering Technology and Industrial Engineering Technology. Students will learn the principals of Programmable Logic Controls (PLC) including hardware, programming, and troubleshooting. Students will also develop advanced working programs, and troubleshoot hardware and software communication problems. Prerequisite: CET110C
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: CET1110C
Course User Fee: \$63.00

EUH**EUH2032 History of The Holocaust 3.000 credits**

This is a foundation course in Holocaust Studies. Students will learn the historical origins, execution and consequences of the Holocaust. This course also examines the Holocaust's place in context of genocides past and present.
Lecture Hours: 48.000

EVR**EVR1001 Introduction to Environmental Science 3.000 credits**

This course is a survey of basic chemical, biological, and physical principles of environmental science and their applications to environmental issues. This course is appropriate for students in a wide range of disciplines or programs. Student learning outcomes: students will apply critical thinking to analysis and interpretation of environmental information and model output; students will apply the scientific method to explain natural experiences and phenomena; students will explain the basic chemical, biological, and physical principles of environmental science; and students will use empirical evidence to describe the historical and modern context of environmental problems and their solutions. Special Fee.
Lecture Hours: 48.000
Course User Fee: \$20.00

EVR1930 ENVIRONMENTAL SEMINR 2.000 credits

Lab Hours: 64.000

FES**FES4003 PUB POLICY EME MGMT 3.000 credits**

An exploration of public policy used in emergency management, including how policy is made and conveyed. (3 hr. lecture)
Lecture Hours: 48.000

FES4823 INTER EME MGMT PLAN 3.000 credits

An analysis of technology applications and its role in emergency planning, responses, recovery, and mitigation. Prerequisite: CGS 1060. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: CGS1060

FFP**FFP0021 Fire Fighter Minimum Standards 16.400 credits**

This program offers a broad foundation of knowledge and skills to prepare the student for employment as a frontline, entry-level firefighter. The program content includes, but is not limited to, orientation to the fire service, fire alarms and communication, vehicles, apparatus and equipment, fire behavior, portable extinguishers, fire streams, fundamentals of extinguishment, ladders, hoses, tools and equipment, forcible entry, salvage, overhaul, ventilation, rescue, protective breathing equipment, first responder emergency medical techniques, water supplies, principles of in-service inspections, safety, controlled burning, and employability skills. Students who successfully complete the program will be required to pass a written and practical exam to obtain certification as a Firefighter 2 from the Florida Bureau of Fire Standards and Training. Students must be a minimum of 18 years old to participate. Fire Academy Students Only.
Lecture Hours: 492.000
Course User Fee: \$653.00

FFP0077 FIRST RESPONDER 1.500 credits

A training course for students who will provide basic life support to victims of emergencies, to minimize patient discomfort and prevent further injury. This course is a required part of fire fighter training. Special fee. (45 contact hours)
Lecture Hours: 45.000
Course User Fee: \$50.00

FFP1000 Introduction to Fire Science 3.000 credits

This course provides an overview of fire protection and emergency services; career opportunities in fire protection and related fields; culture and history of emergency services; fire loss analysis; organization and function of public and private fire protection services; fire departments as part of local government; laws and regulations affecting the fire service; fire service nomenclature; specific fire protection functions; basic fire chemistry and physics; introduction to fire protection systems; introduction to strategy and tactics; and life safety initiatives. (3 hr. lecture)
Lecture Hours: 48.000

FFP1301 Fire Service Hydraulics 3.000 credits

This course covers the relationship between flow, pressure, and mathematical hydraulic formulas. The course includes pump theory, pump rating, and pressure and vacuum gauges. (3 hr. lecture)
Lecture Hours: 48.000

FFP1302 Apparatus Operations 3.000 credits

This course covers the laws, rules and driving techniques for emergency vehicles as well as a review of hydraulics. Fireground evolutions and driving make up the practical part of the course. The evolution portion of the course includes the use of pre-connected lines, tandem pumping, drafting relays, and master streams. The student should have a basic understanding of fire stream hydraulics prior to entering this course. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: FFP1301

FFP1505 Fire Prevention Practices 3.000 credits

The purpose of this course is to place emphasis on fire prevention through inspection, code enforcement, the use of model building and fire prevention codes, including detailed information on the legal, economic, and political aspects of the fire inspection process. (3 hr. lecture)
Lecture Hours: 48.000

FFP1510 Codes and Standards 3.000 credits

The purpose of this course is to familiarize students or current inspectors with the Florida Fire Prevention Code which includes NFPA 101, NFPA 1, FSS 633 and FAC 69A. Students will learn how to research the codes and apply them to inspection activities. (3 hr. lecture)
Lecture Hours: 48.000

FFP1540 Private Fire Protection Systems 1 3.000 credits

This is a study of private fire protection and detection systems such as sprinkler and standpipe systems, chemical extinguishing systems, and detection systems and devices. Each system is discussed as to its need, construction, preventative maintenance and individual use. (3 hr. lecture)
Lecture Hours: 48.000, Practicum Hours: 3.000

FFP1710 Supervision-Leadership for Fire Officers 3.000 credits

Exploring effective leadership in high-stress fire service environments. (Pre-Officer Training). (3 hr. lecture)
Lecture Hours: 48.000

FFP1740 Fire Service Course Delivery 3.000 credits

The curriculum draws from many recognized authorities in exploring the methods and mechanics of imparting information and adult learning principles. The course emphasizes techniques that have wide application in teaching situations, as well as devices for specific training areas. The course also stresses measuring teaching effectiveness, and the use of media and visual aids. (3 hr. lecture)
Lecture Hours: 48.000

FFP1810 Firefighting Tactics & Strategy 1 3.000 credits

This course will provide students with a basic understanding of factors involved in coping with a fire emergency and determining the best use of available resources in protecting lives and property. The course emphasizes the changing nature of an emergency situation and the ways in which the fire officer can evaluate the effectiveness of their proposed incident action plan. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: , FFP1000

FFP2120 Building Construction for the Fire Service 3.000 credits

This course will provide the student with a basic understanding of building design and construction methods which will aid in decision making processes related to fire prevention and control. Students will gain an understanding of construction principles and practices related to fire loads and behavior to make more effective, safer, and timely decisions to protect people and property from potential and actual fires. (3 hr. lecture)
Lecture Hours: 48.000

FFP2521 Construction Documents and Plans Review 3.000 credits

This curriculum is geared towards teaching the student how to assimilate information contained in working, drawing, and specifications as they relate to the fire inspector. The curriculum includes how to interpret conventional graphic communications. Accepted standards and conventions are introduced. Symbols, abbreviations, principles of technical projection, as well as a review of geometry are included. Related worksheets are used to allow for applied experience of finding and interpreting information from authentic drawings. (3 hr. lecture)
Lecture Hours: 48.000

FFP2741 Fire Service Course Design 3.000 credits

The curriculum draws from many recognized authorities in exploring the methods and mechanics of imparting information and adult learning principles. The course emphasizes techniques that have wide application in teaching situations, as well as devices for specific training areas. The course also stresses measuring teaching effectiveness, and the use of media and visual aids. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: FFP1740

FFP9996 Fire Fighting and Protection - Firefighter II 3.000 credits

This course provides academic credit for students who have successfully completed Firefighter II training through an approved fire science training provider. In accordance with TMOP 1100:815720, the course serves as a mechanism for awarding college credit to Miami Dade College (MDC) students based on documented completion of approved fire science training. The course recognizes competencies attained in fire suppression, rescue operations, hazardous materials response, fire prevention, emergency services, and firefighter safety. Credit awarded through this course supports academic progression and acknowledges industry-recognized skills and professional preparation in the field of fire science and emergency services.

Lecture Hours: 48.000

FIL**FIL1030 History of Film 3.000 credits**

In this introductory course students will learn about the history of motion pictures, with an emphasis on American and European films.

Lecture Hours: 48.000

FIL1055 American Independent Film 3.000 credits

In this introductory course, students will learn about the American independent film movement with an emphasis on American directors and producers.

Lecture Hours: 48.000

FIL1060 Survey of Documentary Film 3.000 credits

In this introductory course students will learn the history of nonfiction films, with an emphasis on American and European filmmakers.

Lecture Hours: 48.000

FIL1100 Screenwriting 1: Introduction to Story Structure 3.000 credits

Lecture Hours: 48.000

FIL1420C Film Production 1: Introduction to the Filmmaking 4.000 credits

An introductory overview of the art and technology of narrative motion picture production. Students will learn basic production techniques, from pre-production through production to final screening.

Lecture Hours: 32.000, Lab Hours: 64.000

Corequisite: FIL2552C

Course User Fee: \$150.00

FIL1431C Film Production 2: Cinematography and Sound 4.000 credits

An introductory course in which students will learn cinematography and sync-sound motion picture production.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: FIL1420C AND Corequisite: FIL2553C

Course User Fee: \$150.00

FIL2131 Screenwriting 2: Character Development and Advanced Story Structure 3.000 credits

In this workshop-style intermediate level course students will learn about character development and various story structures for narrative motion picture screenplays.

Lecture Hours: 48.000

Prerequisite: FIL1100

FIL2413 Screenwriting 3 3.000 credits

An advanced course in which the fundamentals of story structure and character development introduced in Screenwriting 1 and Screenwriting 2 are refined. The student will learn how to write an outline for a feature-length motion picture.

Lecture Hours: 48.000

Prerequisite: FIL2131

FIL2480C Film Production 3: Directing 4.000 credits

An intermediate practicum in motion picture direction through the analysis of various directors and their cinematic styles. Students will learn the role of the director by interpreting dramatic material, effectively guiding acting performances, and communicating a story visually.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: FIL1431C AND FIL2553C, RTV1240C

Course User Fee: \$25.00

FIL2515C Film Production 4: Producing the Short Film 4.000 credits

An advanced course in film production. Students will learn to apply the fundamentals of film production as introduced in Film Production 1, 2, & 3 to the production of a portfolio-quality narrative short film.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: FIL2480C

Course User Fee: \$250.00

FIL2552C Editing Level 1: Introduction to Editing 3.000 credits

An introductory course in which students will learn the practice of editing digital media.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$35.00

FIL2553C Editing Level 2: Intermediate Editing and Visual Effects 3.000 credits

An intermediate course in which students will learn video editing with an emphasis on sound design and visual effects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2552C

Course User Fee: \$35.00

FIL2560C Editing Level 3: Advanced Editing: Color Correction and Finishing 3.000 credits

An advanced course in which students will learn the practice of color correcting and finishing fiction and non-fiction projects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2553C

Course User Fee: \$35.00

FIL2611 Film Business Marketing Distribution Exhibition 3.000 credits

Lecture Hours: 48.000

FIL2945 Film Internship 3.000 credits

Students will learn to apply the various skills gained throughout the program in a semester long immersion at an established film entertainment company.

Lecture Hours: 240.000

Prerequisite: FIL2480C AND Recommended

Preparation: Professor Permission

FIL2951 Film Festival Experience 3.000 credits

An immersive experience in a film festival. Students engage directly with films and filmmakers through their attendance to festival screenings and professional panels and discussions. Students become acquainted with a festival's organization and develop their professional skills by participating as volunteers in a multi-cultural community event.

Lecture Hours: 48.000

Recommended Preparation: Registration with Professor Approval Only.

FIL3602 Production Management 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn the theory and practice of managing film and television production with an emphasis on pre-production processes and software.

Lecture Hours: 48.000

Prerequisite: FIL2611 OR MMC2000

FIL3651 Business Proposals for Film & Television 3.000 credits

In this upper division course for BAS students in Film, Television & Digital Production students will learn the theory and practice of business plans/ grant proposals in media production funding.

Lecture Hours: 48.000

Prerequisite: FIL2611 OR MMC2000

FIL4164 Advanced Writing for Film and Television 3.000 credits

In this upper division course in Film, Television & Digital Production students will learn the process of completing a long form motion picture or television script.

Lecture Hours: 48.000

Prerequisite: FIL2131 OR RTV2300

FIL4585C Production Workshop 1 4.000 credits

In this upper division workshop for BAS students in Film, Television & Digital Production students will learn to apply industry-standard pre-production and production techniques to Production Workshop projects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: DIG3347C AND FIL3602

Course User Fee: \$200.00

FIL4586C Production Workshop 2 4.000 credits

In this upper division workshop for BAS Students in Film, Television & Digital Production students will learn to apply industry-standard post-production techniques to complete Production Workshop 1 projects.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL4585C

Course User Fee: \$200.00

FIN

FIN1930 Special Topic Seminar 1.000 credits

This course centers around topics of current interest or of special interest. Topics or focus may vary from semester to semester. (1 hr. lecture)

Lecture Hours: 16.000

Course User Fee: \$35.00

FIN2000 Principles of Finance 3.000 credits

The creation, allocation, and utilization of money, and the effect of monetary policy upon individuals, business, national and international economics. This course provides a basis for further study of monetary theory, banking, finance and securities.

Lecture Hours: 48.000

FIN2010 Investments in Stocks and Bonds 3.000 credits

The basic principles of the stock market as they affect the individual investor in stocks and bonds. Investment in these securities is studied from the standpoint of the short-term and long-term investors.

Lecture Hours: 48.000

Course User Fee: \$10.00

FIN2031 Risk Management and Compliance 3.000 credits

This course offers an analysis of the risks faced by investors and savers interacting through both financial institutions and financial markets. It will provide insight at the risks and opportunities of doing business in today's financial markets and the challenges presented by both regulators and market participants. The course will review the increasing integration of foreign and domestic financial markets. (3hr lecture)

Lecture Hours: 48.000

Course User Fee: \$70.00

FIN2032 Fundamentals of Wealth Management, Institutions, Markets and Products 3.000 credits

This course is a standard introduction to the financial services profession, financial markets, and financial institutions. It touches on nearly every aspect of financial services. It assists professionals in understanding concepts, markets, products, regulations and the application of financial planning and development of wealth management skills.

Lecture Hours: 48.000

Prerequisite: ACG2021, ACG2021L

Course User Fee: \$70.00

FIN2051 International Financial Management 3.000 credits

The student will learn basic concepts and principles of international finance, with consideration of the financial environment, transactions, and flows. Exchange rates, risks, and government policies affecting business are analyzed as well as management policies and decisions.

Lecture Hours: 48.000

Course User Fee: \$20.00

FIN2100 Personal Finance 1.000 - 3.000 credits

A study of economic and personal goals including personal budgeting, credit budgeting, borrowing money, banking facilities, the nature of investments, life insurance, casualty insurance, home ownership, stocks and bonds, and retirement plans.

Lecture Hours: 16.000 - 48.000

Course User Fee: \$20.00

FIN2642 Financial Analysis and Valuation 3.000 credits

This course will cover financial institutions, financial Investments, financial planning and analysis and international financial perspectives. It addresses core principles of value creation, merger, analysis of historical financial performance, cash flows and the identification of sources of value. (3 hr. lecture)

Lecture Hours: 48.000, Clinical Hours: 1.000

Course User Fee: \$70.00

FIN3400 Finance for Non-Financial Managers 3.000 credits

The students will learn to apply their financial skills and decision-making abilities to address financial issues in a business environment. They will learn how financial markets influence their decisions and the role of financial intermediaries in these markets. Emphasis will be placed on financial and capital budgeting, working capital management, short and long term financing, valuation of the firm, and overall capital structure. The course will require the student to apply the time value of money through the use of present and future value scenarios. This course must be met with a grade of "C" or higher. (3 hour lecture)

Lecture Hours: 48.000

Course User Fee: \$60.00

FIN3403 Financial Management 3.000 credits

Students will learn the sources and uses of funds as they relate to financial decision making in the corporate form of enterprise. Emphasis is placed on working capital management, capital budgeting techniques, short and long term financing, and capital structure and the value of the firm. This course emphasizes the use of formulas and spreadsheets through Excel.

Lecture Hours: 48.000

Prerequisite: ACG2071, ACG2071L, QMB2100

FIN4303 Financial Markets and Institutions 3.000 credits

Students will learn the importance of financial markets and the role financial intermediaries play. Emphasis will be upon the objectives and policies of financial intermediaries within the constraints of the law and regulatory authorities. Prerequisites: grade of "C" or higher in FIN 3400, ECO 2013, and ACG 2071 or equivalents. Special fee. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: FIN3400, ACG2071, ECO2013

Course User Fee: \$45.00

FOS

FOT

FOT2220 Localization and Project Management 3.000 credits

Software and website localization is a new and growing area of translation and is closely linked to the concepts of globalization and internationalization in electronic communications. In this introductory course students will learn how to adapt software for international markets as well as to how to translate websites. Students will also be introduced to basic concepts in localization project management.

Lecture Hours: 48.000

Prerequisite: FOT2802 AND FOT2825

Course User Fee: \$25.00

FOT2701 Simultaneous Conference Interpretation 3.000 credits

This course builds on the foundation established in the previous Simultaneous Interpretation Strategies course. Students will continue developing their simultaneous interpretation skills through exercises such as shadowing, decalage, paraphrasing, etc. Through a variety of authentic recording materials, students will practice the simultaneous interpretation mode in the context of international organizations and conferences so as to acquire smooth delivery techniques while forming professional habits both in conference booths. Extensive practice in simultaneous interpretation will be provided both in class and online.

Lecture Hours: 48.000

Prerequisite: FOT2824

Course User Fee: \$25.00

FOT2802 Introduction to Translation 3.000 credits

Develops the ability to do accurate written translations in general. Includes the application of contrastive structures and grammar rules of source and target languages; translation of idiomatic expressions and an introduction to legal and technical vocabulary; the use of bilingual dictionaries and glossaries. The demands of translation as a profession and its code of ethics are stressed.

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2821 INTRO TO INTERPRETON 3.000 credits

The acquisition and development of the abilities to convert an oral message from the source language into another consecutive oral message in the target language. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2821 Introduction to Interpretation 3.000 credits

The acquisition and development of skills to sight translate a text from the source to the target language. Training in the conceptualization and analysis of an oral message to transfer its content to another language consecutively. Preparatory exercises for the development of simultaneous interpretation skills.

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2822 COURT INTERPRETING SK 3.000 credits

Continuation of SPT 2810 including deepening and broadening the type of exercise of SPT 2810 and gradual introduction to simultaneous interpretation. Oral translation with notes and of conversations, radio or tape passages. Extensive practice in the process of hearing, understanding, remembering and speaking for simultaneous oral interpretation. Participation in an internship or practical training program. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2822 Court Interpreting Skills 3.000 credits

Continuation of FOT 2821 including deepening and broadening the type of exercise of FOT 2821 and gradual introduction to simultaneous interpretation. Oral translation with notes and conversations, radio or tape passages. Extensive practice in the process of hearing, understanding, remembering and speaking for simultaneous oral interpretation. Participation in an internship or practical training program.

Lecture Hours: 48.000

FOT2823 CONS INTERPRETATION 3.000 credits

This course builds on the foundation established in Introduction to Interpretation (FOT2810) and acquaints the students with the practice and application of consecutive interpretation (English/Spanish). Development of active listening, concentration and retention skills as well as the ability to perceive essential meaning for subsequent recall is emphasized. This course also explores basic note taking techniques and provides practice in monolateral and bilateral consecutive interpretation. Prerequisite: FOT 2810. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2823 Consecutive Interpretation 3.000 credits

This course builds on the foundation established in Introduction to Interpretation (FOT2810) and acquaints the students with the practice and application of consecutive interpretation (English/Spanish). Development of active listening, concentration and retention skills as well as the ability to perceive essential meaning for subsequent recall is emphasized. This course also explores basic note taking techniques and provides practice in monolateral and bilateral consecutive interpretation.

Lecture Hours: 48.000

Prerequisite: FOT2821

Course User Fee: \$15.00

FOT2824 Simultaneous Interpretation Strategies 3.000 credits

This course builds on the foundation established in previous interpretation courses while introducing the students to simultaneous interpretation (English/Spanish) by providing preparatory exercises such as shadowing, lagging, paraphrasing etc. Through a variety of recorded materials, students practice the simultaneous interpretation mode so as to acquire smooth delivery techniques while forming good professional habits.

Lecture Hours: 48.000

Prerequisite: FOT2821, FOT2823

Course User Fee: \$15.00

FOT2824 SIM INTERP STRATEGIE 3.000 credits

This course builds on the foundation established in previous interpretation courses while introducing the students to simultaneous interpretation (English/Spanish) by providing preparatory exercises such as shadowing, lagging, paraphrasing etc. Through a variety of recorded materials, students practice the simultaneous interpretation mode so as to acquire smooth delivery techniques while forming good professional habits. Prerequisites: FOT 2810, 2815. (3 hr. lecture)

Lecture Hours: 48.000

FOT2825 Computer Assisted Translation 1 3.000 credits

Examines the types of translation software currently used in the translation/interpretation profession as well as the commercial use and business application of these. Description and application of tools such as translation memory, electronic dictionaries, desktop-publishing systems, and website translation technologies are covered.

Lecture Hours: 48.000

Prerequisite: FOT2802

Course User Fee: \$25.00

FOT2825 Computer Assisted Translation 1 3.000 credits

Examines the types of translation software currently used in the translation/interpretation profession as well as the commercial use and business application of these. Description and application of tools such as translation memory, electronic dictionaries, desktop-publishing systems, and website translation technologies are covered. Prerequisite: CGS 1060. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$25.00

FOT2826 LEGAL TRANSLATION 3.000 credits

Continuation of FOT 2802. Written translations of multi-page documents and/or articles containing legal, technical and other specialized vocabulary from the source language into the target language. Firsthand translation experience by participating in a "translator's bureau," or an "internship" or practical training program. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2826 Legal Translation 3.000 credits

Continuation of FOT 2802. Written translations of multi-page documents and/or articles containing legal, technical and other specialized vocabulary from the source language into the target language. Firsthand translation experience by participating in a "translator's bureau," or an "internship" or practical training program.

Lecture Hours: 48.000

FOT2827 MEDICAL TRANSLATION 3.000 credits

This course further develops translation strategies while familiarizing the student with the characteristics of medical and health-related discourse in both English and Spanish. Included is the acquisition of medical and hospital/clinic terminology and the analysis of related linguistic structures so students can engage in translating texts from English into foreign language and vice versa. Prerequisites: FOT 2800, 2803. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2827 Medical Translation 3.000 credits

This course further develops translation strategies while familiarizing the student with the characteristics of medical and health-related discourse in both English and Spanish. Included is the acquisition of medical and hospital/clinic terminology and the analysis of related linguistic structures so students can engage in translating texts from English into foreign language and vice versa.

Lecture Hours: 48.000

Prerequisite: FOT2802

FOT2828 MEDICAL INTERPRETATION 3.000 credits

This course develops the techniques, practices and knowledge needed to function as interpreters in a medical environment. Interpreting models such as sight, consecutive and simultaneous - as they apply to the medical setting - are revisited. Medical vocabulary/terminology in English and foreign language as well as code of ethics will also be introduced. Prerequisites: FOT 2810, 2815, and FOT 2816 (recommended). (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

FOT2828 Medical Interpretation 3.000 credits

This course develops the techniques, practices and knowledge needed to function as interpreters in a medical environment. Interpreting models such as sight, consecutive and simultaneous — as they apply to the medical setting — are revisited. Medical vocabulary/terminology in English and foreign language as well as code of ethics will also be introduced.

Lecture Hours: 48.000

Prerequisite: FOT2821, FOT2823, FOT2824

FOT2829 FIN & BUS TRANSLATION 3.000 credits

Lecture Hours: 48.000

FOT2829 Financial and Business Translation 3.000 credits

This course further develops translation strategies while familiarizing the students with the characteristics of financial and business discourse in both English and Spanish. Included is the learning of special terminology and related linguistic structures so students can engage in the translation of texts containing financial/business or economic discourse from English into Spanish and vice versa. As in legal translation, students engage in terminology research and glossary development through the use of specialized bilingual financial and business dictionaries and other pertinent sources. Prerequisites: FOT 2802, 2803. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

FRE**FRE1120 Elementary French 1 4.000 credits**

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in French. Note: Students must pass this course with a C or better to continue to FRE1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

FRE1121 Elementary French 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in French. Note: Students must pass this course with a C or better to continue to FRE2220.

Lecture Hours: 64.000

Prerequisite: FRE1120

Course User Fee: \$15.00

FRE2220 Intermediate French 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of French through a systematic review (using an integrated, multimedia approach) of reading grammar, and writing skills with emphasis on oral and written communication.

Lecture Hours: 64.000

Prerequisite: FRE1121

Course User Fee: \$15.00

FRE2221 Intermediate French 2 4.000 credits

This is a continuation of Intermediate French 1. Students will learn to understand, speak, read, and write French. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: FRE1121

Course User Fee: \$15.00

FSE**FSE1000 INTRO MORT SCIENCE 3.000 credits**

Lecture Hours: 48.000

FSE1080 Funeral Law 3.000 credits

Federal, state and municipal statutes, rules, regulations and ordinances pertaining to funeral service; torts, contract and administrative laws, and financial disclosures pertinent to funeral operations and management.

Lecture Hours: 48.000

FSE1105 Funeral Service Chemistry 3.000 credits

A survey of the basic principles of chemistry as they relate to funeral service. Especially stressed are the chemical principles and precautions involved in sanitation, disinfection, public health and embalming practice.

Lecture Hours: 48.000

FSE2060 Funeral Directing 3.000 credits

Study of various religious, fraternal, military, traditional, nontraditional and humanistic variations of funeral ceremonies, including cultural, ethnic and geographic customs.

Lecture Hours: 48.000

FSE2061 Funeral Service Counseling and Ethics 3.000 credits

This course emphasizes the psychological and sociological dynamics of death, dying, and bereavement and the funeral director's role in counseling families through the process. A clear understanding of the grief process and its variations among individuals as influenced by psychological factors is addressed. Ethical and moral decisions involving end of life and the funeral arrangement process are discussed. The course also addresses the funeral director as a facilitating agent for effective mourning through personal interaction and the design and implementation of the funeral. Lecture Hours: 48.000

FSE2100 Embalming 1 3.000 credits

Designed as an introductory course outlining the basic skills, aptitudes, and personal qualifications necessary for successful embalmers. The subject in embalming is taught so that the student is trained in all phases of the embalming process. Prerequisite: BSC1084 or FSE1001
Lecture Hours: 48.000, Lab Hours: 48.000
Pre/Corequisite: BSC1084 OR FSE1001

FSE2100L Embalming 1 Lab 2.000 credits

Laboratory for FSE 2100. Laboratory fee. Corequisite: FSE 2100. A.S. degree credit only. (4 hr. lab)
Lab Hours: 64.000
Pre/Corequisite: FSE2100
Course User Fee: \$490.00

FSE2106 Funeral Service Microbiology 3.000 credits

This course is a survey of the basic principles of microbiology as it relates to Funeral Science. It emphasizes the importance of sanitation, disinfection, and public health in the embalming practice.(3 hr. lecture)
Lecture Hours: 48.000

FSE2120C Restorative Art 4.000 credits

Anatomical study of human features; familiarization with instruments, human proportions, special materials and techniques.
Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: FSE2100, FSE2100L

FSE2140 Embalming 2 3.000 credits

This course is a continuation of Embalming 1 lecture and places emphasis on embalming considerations and procedures for pathogenesis and advanced decomposition. Students will learn to use specialized chemicals, treatment of post-mortem cases and advanced embalming techniques. Corequisite: FSE2140L. (3 hr. lecture)
Lecture Hours: 48.000
Prerequisite: FSE2100, Recommended Preparation: Completed FSE2100L - Embalming 1 Lab - with a "C" or better.

FSE2140L Embalming 2 Laboratory 2.000 credits

This course is a continuation of FSE2100 and complements the lecture co-requisite FSE2140. Students will learn advanced procedures to enhance their foundational knowledge of embalming technique as learned in the prerequisite classes FSE2100/FSE2100L. Learning the fundamentals of general embalming technique as outlined by the American Board of Funeral Service Education Standards will enable students to work in a preparation room as embalming apprentices to licensed embalmers. Lab Hours: 64.000
Prerequisite: FSE2100, FSE2100L, Corequisite: FSE2140
Course User Fee: \$490.00

FSE2160 Funeral Service Pathology 3.000 credits

General, systemic and forensic pathology with emphasis on analysis of pre-and post-mortem histology, cytology and etiology. Students will learn the causative factors relating to death and determination of cause of death.
Lecture Hours: 48.000

FSE2160C PATH FUNERAL SER 3.000 credits

Lecture Hours: 48.000, Lab Hours: 32.000

FSE2200 Funeral Service Accounting 3.000 credits

An introduction to basic principles of accounting theory. This subject covers financial statements and their analysis, journalizing, receivables, payables, deferrals, and accruals. Inventory costing models, depreciation models and payroll accounting are included. Applications to funeral home operations are made throughout the subject material. Lecture Hours: 48.000

FSE2201 FUNERAL HOME OP II 3.000 credits

Lecture Hours: 48.000

FSE2202 Funeral Service Business Management 3.000 credits

The role and function of an effective manager is explored. Emphasis is placed on the management functions of planning, organizing, motivating, directing, and controlling. How to purchase a small business is also covered.

Lecture Hours: 48.000

FSE2203 FUN HOME APPLICATION 4.000 credits

Lecture Hours: 64.000

FSE2930L Funeral Service Professional Review 2.000 credits

This course is offered for the Funeral Service Education student who is in their final semester of study. The course is a comprehensive review of the assessed material on the National Board Exam to help prepare the student for the examination and professional licensure. A grade of "C" or better is required to successfully complete the class. Pre/Corequisite: FSE1000, FSE1080, FSE2060, FSE2061, FSE2100, FSE2106, FSE2120C, FSE2140, FSE2201, FSE2202, FSE2203, FSE2160.

Lab Hours: 64.000

Pre/Corequisite: FSE1000 AND FSE1080 AND FSE2060 AND FSE2061 AND FSE2100 AND FSE2106 AND FSE2120 AND FSE2140 AND FSE2160 AND FSE2201 AND FSE2203

FSE2932 Funeral Science Professional Review 2 1.000 credits

This course is for the funeral Science student who is graduating and taking the National Board Examination at the end of the semester that this course is being offered. The course is a review of the Arts section of the Funeral Science courses in order to help prepare the student for the National Board Examination.

Lecture Hours: 32.000

Prerequisite: Departmental Permission Required

FSS**FSS1100 Foodservice Purchasing 3.000 credits**

This is an introductory course where students will learn the principals of menu planning for various types of facilities and service as well as menu layout, selection, development and pricing structures. Students will learn the principals and practices concerned with the purchase and receipt of food, supplies and equipment for various food service operations. Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 48.000

Prerequisite: FSS1200 AND FSS1202C

FSS1200 Culinary Terminology and Procedures 3.000 credits

This is an introductory course in which students will learn to integrate industry vocabulary, terminology, knowledge, skills, and practices required for careers in the food service industry. The students will learn standard kitchen phrases, how to identify and describe equipment, recipe reading, costing, conversion formulas, product identification, measurements and basic cooking procedures.

Corequisite: FSS1202C.

Lecture Hours: 48.000

Corequisite: FSS1202C

FSS1202C Food Production 1 - Foundational Skills 3.000 credits

This course is an introductory kitchen lecture/lab experience in which students will be provided hands-on orientation to tools, equipment, recipe production, measurements, knife cut techniques and basic cooking procedures. Students will practice classic cooking methods, product identification and the functions of the production kitchen in a "green" team environment. Corequisite: FSS1200. A.S. degree credit only.

Lecture Hours: 16.000, Practicum Hours: 96.000

Corequisite: FSS1200

Course User Fee: \$1981.97

FSS1204C Food Production 2 - American Regional Cuisine 3.000 credits

The lecture/practicum course reinforces the knowledge and skills learned in the preceding classes and helps students build confidence in the techniques of basic cookery. The development of knife skills is accented. American Regional Cuisine explores the use of indigenous ingredients in the preparation of traditional and contemporary American specialties. The importance and basics of "sustainable agriculture" are introduced. The concepts of mise en place, timelines, plate presentation, and teamwork in a production setting are introduced and accentuated. Timing and organization skills are emphasized. Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200 AND FSS1202C

Course User Fee: \$1981.99

FSS1246C Basic Baking - Foundational Skills 3.000 credits

This is an introductory course in which students will learn the fundamentals of baking science, terminology, equipment, ingredients, weights, measures, formula conversion and storage. Students will learn the functions of various baking ingredients and execute baking recipes, including competencies relating to doughs, breads, cookies, pies, puff pastries, sweet & savory pastry fillings, quick breads, cakes and basic decorating techniques.

Prerequisites: FSS1200, FSS1202C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C

Course User Fee: \$1981.98

FSS1801 Culinary Sustainability and Practices 3.000 credits

This course will engage students in growing and producing food, processing, distribution, and end user/purchaser aspects of food systems. Students will explore how to reduce the carbon footprint of food service operations and understand the importance of bringing seasonal food to the table at its peak of freshness and height of nutritional value. With hands-on approach and a focus on biodiversity, sustainability, healthy food and animal welfare, students will develop critical thinking to understanding local and global food systems. The course includes on-site visits with farmers, food processors, and experts in our local food system to engage in health and sustainability practices related to food safety, water and waste systems, food marketing, and the heritage food movement.

Prerequisite: FSS2248C.

Lecture Hours: 48.000

Prerequisite: FSS2248C

Course User Fee: \$583.82

FSS2205C Food Production 3 - Contemporary Cuisine and Modernist Methods 3.000 credits

This lecture/lab course is an introduction to alternative methods of food production using science as the platform. Students will further develop their understanding of modernist and molecular cuisine through hands-on approach, utilizing state-of-art equipment and techniques.

Prerequisites: FSS1200, FSS1202C, FSS1204C, FSS1246C, FSS2242C, FSS2248C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200 AND FSS1202C AND FSS1204C AND FSS1246C AND FSS2242C AND FSS2248C

Course User Fee: \$1981.96

FSS2242C International Cuisine 3.000 credits

This is an advanced course in which students will reinforce their knowledge of equipment, vocabulary and theories learned in prerequisite classes.

Students will learn to use global ingredients, explore international cooking techniques and methods and be exposed to equipment specific to a myriad of world regions. Students will have the opportunity to develop international menus and prepare dishes from many countries. Prerequisites: FSS1200, FSS1202C, FSS1204C, FSS1246C, FSS2248C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C, FSS1204C, FSS1246C, FSS2248C

Course User Fee: \$1981.97

FSS2248C Garde Manger 3.000 credits

This is a foundational lecture/lab course in the preparation of cold food, hors d'oeuvres, display platters, charcuteries, smoking techniques, butchery and seafood. Students will prepare edible buffet centerpieces. Students will also have hands-on experiences in ice carving. Production methods and safe food handling techniques are re-emphasized.

Prerequisites: FSS1200, FSS1202C, FSS1246C.

Lecture Hours: 16.000, Practicum Hours: 96.000

Prerequisite: FSS1200, FSS1202C, FSS1204C

Course User Fee: \$1981.95

FSS2381 MCI Café Management Internship 4.000 credits

This required internship is designed to provide hands-on culinary training through industry work experience. Students will be located in an approved site to reinforce their skills while being exposed to various stations in a food services operation. Students will learn to actively participate in various aspects of the operation including cooking, cost controls, and sanitation programs. Prerequisite: FSS2205C. .

Practicum Hours: 192.000

Prerequisite: FSS2205C

Course User Fee: \$1981.97

FSS2940 Culinary Management Externship 4.000 credits

This course is designed to provide advanced hands-on culinary training for culinary students through work experience in an approved commercial foodservice or hospitality establishment. Students on externship will receive feedback from their supervisor and keep an externship manual to record and reflect on their work experience.

Internship Hours: 192.000

Course User Fee: \$249.05

FSS2950L Culinary Competition 3.000 credits

This advanced course will strengthen the students ability to utilize various culinary methods learned in prerequisite courses. The student will learn about the American Culinary Federation certification tests, guidelines and competitions. In addition, students will utilize their skills by participating in on site and recorded demonstrations. The class will reinforce and develop timing, teamwork and menu development. Prerequisites: FSS1200, FSS1202C, FSS1204C, FSS2242C, FSS2248C.

Lab Hours: 88.000

Prerequisite: FSS1200, FSS1202C, FSS1204C, FSS2242C, FSS2248C

Course User Fee: \$1981.95

GEB**GEB1000 Business Career Strategies 3.000 credits**

This course is designed to enable students to thrive in a competitive business environment. Students will learn business etiquette, alternative career pathways, personal financial management, and budgets. Additionally, students will learn interviewing, networking, and career development skills.

Lecture Hours: 48.000

Course User Fee: \$24.00

GEB1011 Principles of Business 3.000 credits

The student will learn the major disciplines of business including general business, business ethnics, forms of business ownership, economics, management and leadership, human relations marketing, information systems, accounting, financial management, money and banking, and business law.

Lecture Hours: 48.000

Course User Fee: \$20.00

GEB1432 Applied Artificial Intelligence (AI) in Business 3.000 credits

This course illustrates how Artificial Intelligence (AI) is currently applied in industries and across business functions to support business strategy formulation, implementation and evaluation. This course introduces the data-driven outcomes and ethical issues to be considered when implementing responsible AI in business and industry. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$35.20

GEB1949 General Business Internship 1 3.000 credits

This internship provides students with an opportunity to gain business experience while receiving academic credit. Students will learn to make connections between their internship experiences, academic coursework, and career goals. Students are required to complete 144 hours of internship in an organizational setting.

Other Hours: 144.000

GEB2100 Introduction to Business Analytics 3.000 credits

This course is for students majoring in business, marketing, business intelligence, computer science and other majors and introduces how information and technology are used in organizations to create market advantage. Students will learn about the operational units of an organization and the information requirements to support the organization.

Lecture Hours: 48.000

GEB2112 Introduction to Entrepreneurship 3.000 credits

Students will learn that start-ups are not smaller versions of big businesses. They are unique. This foundation course in business entrepreneurship covers the attributes of successful entrepreneurs, opportunity identification, opportunity analysis and development, as well as an overview of the key activities and functions which start-ups must address.

Lecture Hours: 48.000

Course User Fee: \$110.00

GEB2350 INTRO INTERNAT BUS 3.000 credits

Lecture Hours: 48.000

Course User Fee: \$20.00

GEB2350 Introduction to International Business 3.000 credits

Provides an overview of the cultural environment of international business and the institution which affects business today. International economic, political, and trade issues are analyzed in the context of socio-economic goals and policies of the nations involved.

Lecture Hours: 48.000

GEB2893 STR POLICY ISS NON-P 3.000 credits

This course provides a culminating experience for the student involved in the non-profit sector to integrate course work with current issues in the non-profit field. Three to four topics relevant to the management and boards are previewed with professionals from these areas as guest speakers. An understanding of the case study method will be required in order for students to prepare a case study for their non-profit organization and propose a solution. Issues to be included are: resource development, financial management, technology, and capacity building. Students will present their findings in a formal presentation to industry professionals. It is recommended that students take this course in their final semester. (Annually the topics selected for inclusion will be reviewed.) (3 hr. lecture)

Lecture Hours: 48.000

GEB2949 General Business Internship 2 3.000 credits

This internship is a continuation of GEB 1949 and provides students with an opportunity to gain business experience while receiving academic credit. Students will learn to make connections between their internship experiences, academic coursework, and career goals. Students are required to complete 144 hours of internship in an organizational setting. Other Hours: 144.000
Prerequisite: GEB1949

GEB3213 Advanced Communications in Business 3.000 credits

Student will develop effective and efficient oral and written communications skills that can be applied in professional business settings. Topics include formal and informal writing, preparation of reports, creation of business proposal, written correspondence, and presentations. 3 hours Lecture, 3 credits

Lecture Hours: 48.000

Course User Fee: \$35.00

GEB3358 International Negotiations and Transactions 3.000 credits

This course presents business negotiations in selected regions of the world. Students will learn the skills of negotiation by analyzing international business cases and developing solutions to situations. Topics include overseas market research using both domestic and international sources and the application of theoretical and practical business knowledge to foreign situations and trade negotiations. The importance of culture, language, and values in international negotiations is emphasized.

Lecture Hours: 48.000

Prerequisite: TRA1154

Course User Fee: \$69.00

GEB3522 Applied Business Analytics 3.000 credits

This upper division course is for students majoring in Data Analytics. Students will learn how to design and develop business analytic solutions to real-world problems using case studies. Students will gain experience working in small teams in deadline-driven environments and will present their results in class.

Lecture Hours: 48.000

Prerequisite: GEB2100

Course User Fee: \$50.00

GEB4363 Import Export Management 3.000 credits

Students will learn the functions and range of traffic management services performed by freight forwarders, including changes in governmental restrictions, rules, and regulations applicable to different countries, ports, and trade routes. Students will also learn the documentation needed to facilitate and coordinate the movement of goods in international trade and supply chain management systems. Prerequisites: MAN 2021 and TRA 2010.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA2010

Course User Fee: \$69.00

GER**GER1120 Elementary German 1 4.000 credits**

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in German. Note: Students must pass this course with a C or better to continue to GER1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

GER1121 Elementary German 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in German. Note: Students must pass this course with a C or better to continue to GER2220.

Lecture Hours: 64.000

Prerequisite: GER1120

Course User Fee: \$15.00

GER2220 Intermediate German 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of German through a systematic review (using an integrated, multimedia approach) of reading, grammar, and writing skills with emphasis on oral and written communication.

Lecture Hours: 64.000

Prerequisite: GER1121

Course User Fee: \$15.00

GER2221 Intermediate German 2 4.000 credits

This is a continuation of Intermediate German 1. Students will learn to understand, speak, read, and write German. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: GER2220

Course User Fee: \$15.00

GIS**GIS1040 Introduction to GIS Technology 4.000 credits**

This course teaches the principles of Geographic Information Systems (GIS). The student will learn fundamental GIS concepts, GIS models and formats, as well as map creation and analysis. At the end of the course, the student will be able to perform basic operations in GIS software. (3 hr. lecture, 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$63.00

GIS2045 Intermediate GIS Technology 4.000 credits

This course teaches intermediate-level concepts of Geographic Information Systems (GIS). The student will acquire an understanding of discrete geocoding and georeferencing, data input, working with spatial data-bases, and data creation. At the end of the course, the student will be able to perform intermediate-level operations in GIS software.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: GIS1040

Course User Fee: \$63.00

GIS2046 Advanced GIS Technology 4.000 credits

This course teaches advanced concepts of Geographic Information Systems (GIS). The student will learn how to use vector, raster and 3D data, geospatial structures, and write GIS functions using a programming language. At the end of the course, the student will be able to perform advanced operations in GIS software.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: GIS2045

Course User Fee: \$63.00

GIS2047 Applications of GIS Technology 4.000 credits

This course teaches the principles of urban analytics and disaster management through what-if scenario modeling in which risks are evaluated and managed in order to support better decision-making. By the end of the course, the student will apply a remote sensing technique to generate GIS data.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: GIS2045

Course User Fee: \$63.00

GLY**GLY1010 Physical Geology 3.000 credits**

Using the scientific method, critical thinking skills and data analysis, this course will examine the fundamental processes of the earth system, composed of an atmosphere, hydrosphere, cryosphere, lithosphere, biosphere, and exosphere through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize lithospheric connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize geoscience data to draw scientifically valid conclusions; students will recognize the different time scales associated with different geologic processes; students will describe interactions between humans and earth's spheres; and students will apply their understanding of geologic principles to complex issues.

Lecture Hours: 48.000

Course User Fee: \$5.00

GLY1010L Physical Geology Laboratory 1.000 credits

Laboratory for GLY 1010. Studies of common minerals and rocks and topographic and geologic maps along with aerial photography.

Lab Hours: 32.000

Corequisite: GLY1010

Course User Fee: \$10.00

GLY1100 Historical Geology 3.000 credits

This is a historical based course in geology. The student will learn about the history of the earth, the evolution of life, radiometric dating, and the history of modern geologic ideas on earth development.
Lecture Hours: 48.000

GLY1100L HISTORCL GEOLOGY LAB 1.000 credits

Lab Hours: 32.000

GLY4701C geomorphology 4.000 - 32.000 credits

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

GRA**GRA0430 DESKTOP PUBLISHING 4.000 credits**

Desktop publishing is the production of high quality printed publications using relatively inexpensive equipment: personal computers, desktop scanners, and laser printers. This class explores the qualities and abilities of Aldus PageMaker, and industry-standard page layout program. Class lectures are supported with audiovisual presentation and extensive handouts. Lab classes consist of a series of typical page layout jobs. Special fee. (120 contact hours)

Lecture Hours: 120.000

Course User Fee: \$25.00

GRA0446 PRIN OF TYPOGRAPHY 4.000 credits

Typography is the art of designing printed matter using type as a medium. The history and development of typography, the use of printer's measurements and the aesthetic uses of type will be covered in the lecture form. The production of learned through hands-on project assignments. Instruction also will include industry standard typesetting equipment and desktop publishing personal computers and software. Special fee. (120 contact hours)

Lecture Hours: 120.000

Course User Fee: \$25.00

GRA0452 H T PROCES GRAPH ART 4.000 credits

A halftone is a reproduction of a continuous tone photograph that has been converted into dots of various sizes so it can be reproduced by any of the major printing processes. The various size dots are so small and numerous that they fool the eye into seeing shades of gray similar to a continuous tone photo. Numerous hands-on projects will cover the use of halftone screens and the manipulation of tones by controlled exposures and development procedures. Prerequisite: GRA 0451. Special fee. (120 contact hours)

Lecture Hours: 120.000

Course User Fee: \$35.00

GRA0457 COLOR ELEC SCANNING 3.000 credits

Insert

Lecture Hours: 90.000

Course User Fee: \$100.00

GRA0465 DIGITL GRAPHIC PAINT 4.000 credits

STUDENTS, WORKING FROM PHOTOGRAPHS, REPRESENT THE NATURAL WORLD ON THE NEWEST ARTISTIC MEDIA: THE PERSONAL COMPUTER. FRACTAL DESIGN'S PAINTER SOFTWARE ENABLES STUDENT TO USE A WIDE VARIETY OF DIGITAL TOOLS AND SURFACES TO CREATE ELECTRONIC ILLUSTRATIONS. (120 CONTACT HRS.)

Lecture Hours: 120.000

Course User Fee: \$44.00

GRA0472 OFFSET STRIP 2 COLOR 4.000 credits

This is a vocational credit course that is an advanced course in film assembly for multi-color and 4 color process film assembly using the emulsion-up method. Hands-on projects will range from simple mechanically separate (fake color) projects to 4-color process separations for an 8 page brochure. This course is highly recommended because of the increased demand for color within the advertising field. Special fee. (120 contact hours)

Lecture Hours: 120.000

Course User Fee: \$100.00

GRA0481 PAPER IN GRAPHICS 1.500 credits

THIS COURSE IS A REVIEW OF THE VARIOUS TYPES AND SPECIFICATIONS OF PAPER THAT ARE USED FOR VARIOUS TYPES OF GRAPHIC PRODUCTION TASKS. THE COURSE IS APPROPRIATE ALSO FOR UPGRADING FOR PERSONS INVOLVED IN PURCHASING DEPARTMENTS.

Lecture Hours: 45.000

Course User Fee: \$30.00

GRA0840 WEB PAGE DESIGN ONE 4.000 credits

An introduction to the technologies and techniques of designing for the World Wide Web. This course covers all the key elements of Web design from concept to completion. The course also covers a basic introduction to WYSIWYG HTML editors. Special fee. (120 contact hrs.)

Lecture Hours: 120.000

Course User Fee: \$40.00

GRA1111C Graphic Design Fundamentals 4.000 credits

The basics of graphic design comprise this introductory course. The student will analyze the principles and practices of graphic design, and evaluate their creative solutions to challenging design problems. Topics include design vocabulary, elements and principles, the design process, proportion and composition, creativity and visualization, as well as color and typography basics. Assignments are designed to emulate real-world projects, including brainstorming, design, execution, critique, revision and implementation.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$20.00

GRA1113C Visual Identity & Branding Design 4.000 credits

Studio projects involving visual identity and branding design will comprise this intermediate course. The student will examine color theory, branding, and best practices to develop unique and impactful brands, including sustainability and brand value. The student will design logos and style manuals. Trademarks and copyrights will be examined.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1111C, GRA1206C, GRA2117C

Course User Fee: \$20.00

GRA1206C Typography Fundamentals 4.000 credits

The student will design with type in this introductory course. The student will examine readability vs. legibility, the history of type, type anatomy and how to classify type into groups. The student will examine typographic layout design and how the principals of design apply. Studio projects will introduce the student to expressiveness through typographic creativity, symbolism and abstraction to express a given topic or design solution.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$25.00

GRA1280C Digital Imaging Fundamentals 4.000 credits

The student will learn that leading industry-standard image editing software is used to prepare photos for print and digital media in this introductory course. Topics include: resolutions, compositing, masking, correcting and enhancing, as well as applying styles and effects. Assignments are designed to emulate real-world projects, including composite images for graphic design, photographic editing, and web pages.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$35.00

GRA1750 Web Design Fundamentals 3.000 credits

The basics of web design comprise this introductory course. An introduction to the history of the web, including current design trends, a survey of Internet architecture, as well as user experience fundamentals and usability will comprise this introductory-level course. The student will examine the structure and styling of web pages.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$118.00

GRA1751 Fixed-Layout Web Design 4.000 credits

The student will design fixed-width layout web pages in this intermediate-level course. The student will apply knowledge and skills to the design and construction of single web pages and multipage websites using a visual editor. Topics include styling, linking, incorporating media, production workflows, and the creation of web forms. Assignments emulate real-world projects, including ideation, design, creation, and revision, leading to portfolio-ready projects.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1750

Course User Fee: \$118.00

GRA1752 Motion Graphics for Web Design 4.000 credits

This intermediate-level course examines web vector graphics and their ability to display illustrations and animations on a web page. An in-depth examination of web vector graphics which when combined with transitions and transformations, enables web designers to create dynamic online images and animations that are compatible across desktop and mobile web browsers. The student will create portfolio-ready web projects and animations.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1750, GRA2117C

Course User Fee: \$118.00

GRA1753 MOTION DESIGN 1 4.000 credits

Best design practices and the twelve principles of animation are taught in this introductory course. Students will explore hand-drawn techniques and software tools used to create professional Motion Design projects in 2-D and 3-D environments. At the end of this course, a 15 second animated product advertisement will be produced. Students will demonstrate knowledge of outputting files formatted correctly for film, television and web application environments.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1280C, Pre/Corequisite: GRA2117C

Course User Fee: \$118.00

GRA1754 Responsive Web Design 4.000 credits

The student will design fluid-layout and responsive pages in this intermediate-level course. Responsive web pages adapt to various devices and user preferences, and examines the advantages of responsive design, as well as the latest industry research reporting on statistics for mobile device use globally. Topics include: fluid grids, responsive typography and images, CSS media queries, "mobile first" workflows, usability, tools and resources.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1751

Course User Fee: \$118.00

GRA1949 CO-OP WRK EXP 1: GRA 1.000 - 3.000 credits

Lecture Hours: 48.000

GRA2117C Digital Illustration Fundamentals 4.000 credits

The student will create illustrations using industry-standard vector-based illustration software. Topics include creating vector-based illustrations, leading to an illustration portfolio. Emphasis will be on both representational as well as commercial art styles. Assignments are designed to emulate real-world projects, including illustration, design, and typography.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

GRA2121C Publication Design 4.000 credits

Studio projects focusing on periodical and catalog designs comprise this intermediate-level course. The student will apply knowledge and skills toward the design and creation of newsletters, magazines, catalogs, booklets, press media kits and interactive e-books. Emphasis will be on graphic design principles, composition, layout, pagination, style balance format and project planning.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1111C, GRA1206C, GRA2117C,

Pre/Corequisite: GRA1280C

Course User Fee: \$25.00

GRA2151C Advanced Digital Illustration and Imaging 4.000 credits

This is an advanced course in digital illustration and imaging, building on existing skills to design and create portfolio-ready projects. Topics include: perspective illustration, advanced masking and compositing techniques, "comping" for layout presentations, outdoor advertisements or 3-D package designs. The student will explore optimizing graphics for web, animations and video.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1280C, GRA2117C

GRA2156C User Interface and Experience Design 4.000 credits

In this intermediate-level course, the student will examine the design and creation of mobile applications. The student will create visual designs for mobile apps, as well as analyze how to build and distribute them. The student will apply skills and knowledge toward creating portfolio-ready design projects. Prerequisites: GRA1751 and GRA1754. (2 hr. lecture; 4 hr. lab.)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1751, GRA1754

Course User Fee: \$50.00

GRA2162C MOTION DESIGN 2 4.000 credits

Students master the leading motion design software's interface and tools to create animated advertisements, broadcast graphics and title sequences in this intermediate-level course. This course presents the fundamentals of good design and creatively applying those basic principals to produce a 30 second animated piece. Projects include knowledge of outputting and formatting final files for Film, TV and Web application environments for client delivery. (2 hr. lecture; 4 hr. lab)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1753

Course User Fee: \$50.00

GRA2168C VISUAL EFFECTS & COMPOSITING 4.000 credits

Students will explore the world of compositing and visual effects in this intermediate-level course. This course will offer skills in creating transition effects, compositing and motion tracking. Projects include knowledge of outputting and formatting final files for Film, TV and Web application environments for client delivery.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA2162C

Course User Fee: \$50.00

GRA2190C COMMUNICATIONS DES 1 3.000 - 4.000 credits

Problems in advertising design involving: layout, lettering, current studio media and reproduction processes. Prerequisite: ART 101.

Lecture Hours: 16.000 - 32.000, Lab Hours: 64.000

GRA2191C COMMUNICATIONS DES 2 3.000 - 4.000 credits

Advanced problems in commercial art concentrating on layout, mechanical art for reproduction and illustration technique.

Prerequisite: ART 2230C. (1-2 hr. lecture; 4 hr. lab)

Lecture Hours: 16.000 - 32.000, Lab Hours: 64.000

GRA2203C Portfolio and Business Practices For Designers 3.000 credits

This advanced course enables students to create a portfolio for self-promotion to prospective employers and clients. The student will revise and assemble projects accomplished throughout their career tracks. The student will also analyze best practices involved in the profession: pricing freelance assignments, contracts, intellectual property rights, and other professional requirements. Course is taken in the semester prior to graduation.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: GRA2121C, GRA2151C

Course User Fee: \$3.00

GRA2207C Capstone Project 4.000 credits

The student will apply cumulative knowledge and integrative skills to the design and creation of a professional-level project. The graphic/web design assignment shall meet the requirements specified in a case study or real-world project. Course is taken in the semester prior to graduation.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA2151C

Course User Fee: \$50.00

GRA2305C SPECIAL TOPICS IN GRAPHIC DESIGN 3.000 credits

An advanced course featuring in-depth focus on special topics of the professor's choice within the field of graphic design. Topics include but are not limited to graphic novel design and illustration, informational graphics, as well as sustainability and social responsibility. Its end-goal is a detailed analysis of concepts presented and the creation of portfolio-ready projects. Prerequisite: GRA2151c. (2 hr. lecture; 4 hr. lab.)

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: GRA2151C

Course User Fee: \$100.00

GRA2545C Package Design 4.000 credits

The student will create package designs for use in retail sales, merchandising, and point-of-purchase in this introductory-level course. Assignments are designed to emulate real-world projects, including package construction, design, illustration and typography, leading to portfolio-ready projects.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA2151C

Course User Fee: \$50.00

GRA2546C Advertising Design 4.000 credits

The student will design and create effective advertising campaign concepts for print, outdoor and digital media in this intermediate-level course. Studio projects will emphasize originality, brainstorming, ad layouts, creative copywriting and persuasion, culminating in portfolio-ready projects.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1113C AND GRA2151C

Course User Fee: \$50.00

GRA2727 Dynamic Web Design 4.000 credits

The student will design and create dynamic web pages. This course presents PHP programming language to enable students to reduce website maintenance, change content depending on date and time, collect user input from an online form, and link to a MySQL database to display product information. Studio projects culminate in portfolio-ready artifacts.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA1754, GRA2156C

Course User Fee: \$118.00

GRA2755 EMERGING TECHNOLOGIES FOR MULTIMEDIA WEB DESIGN 4.000 credits

An examination of emerging web technologies encompasses this course. Topics may include (but are not limited to): content management systems, new visual editing applications, and the latest trends in web design. Studio projects culminate in portfolio-ready artifacts. Course is taken in the semester prior to graduation. Prerequisite: GRA2727. (2 hr. lecture; 4 hr. lab.)

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA2727

Course User Fee: \$118.00

GRA2765C DIGITAL MOTION GRAPHICS FOR BROADCAST DESIGN 4.000 credits

Students use industry-standard software to create visual effects, compositing and motion graphics for television production. Time-based design elements of space, pacing, motion and interaction comprise this intermediate-level course. Assignments enable students to create portfolio-ready projects that include: logos, lower-thirds, and credit rolls that incorporate 3-D elements.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: GRA2168C

GRA2949 CO-OP WRK EXP 2: GRF 3.000 credits

Lecture Hours: 48.000

GRA2990 WEB DESIGN 4 4.000 credits

Lecture Hours: 64.000

HFB

HFB2990 CBE Food and Beverage 1.000 - 12.000 credits

The HFB 2930 CBE Food and Beverage special topics course is designed to assess learner mastery of the competencies and skills necessary for a successful career in the food and beverage industry. The competency-based approach allows learner the flexibility to self-direct their learning experience. As a result, the learning is much more individualized and can be significantly accelerated. Learners enrolled HFB 2930 CBE Food and Beverage also have the opportunity to achieve industry-recognized academic credentials, and will be prepared for positions such as restaurant server, room service attendant or banquet set-up staff. Other Hours: 16.000 - 192.000

HFT**HFT1000 Introduction to Hospitality 3.000 credits**

In this introductory course, the student will learn a basic understanding of the lodging, travel, meeting planning, club management, food and beverage, gaming and cruise industry by tracing the industry's growth and development, reviewing the organization of hospitality operations, and by focusing on industry opportunities and future trends. Career opportunities, interview and resume writing skills are discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1210 Human Resources 3.000 credits

Provides information relating to the recruitment and selection of new staff, the handling of difficult employees, motivating employees and conducting performance evaluations. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1212 SAFETY & SANITATION 3.000 credits

HFT 1212 Safety and Sanitation Students will learn industry standards on sanitation as it relates to food preparation and kitchen operations. Students will recognize proper sanitation techniques and explain how to implement sanitation programs in food service operations (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$125.00

HFT1212 Safety and Sanitation 3.000 credits

Students will learn industry standards on sanitation as it relates to food preparation and kitchen operations. Students will recognize proper sanitation techniques and explain how to implement sanitation programs in food service operations.

Lecture Hours: 48.000

Course User Fee: \$125.00

HFT1220 Supervisory Development 3.000 credits

This introductory course will assist students in learning basic supervisory skills. The students will learn how to conduct proper employee evaluations, as well as how to apply discipline. Effective communication techniques, use of delegation and diversity in the workforce are discussed. Issues concerning employee relations and current topics in management are addressed.

Lecture Hours: 48.000

HFT1300 EXECUTIVE HOUSEKEEPG 3.000 credits

HFT 1300 Executive Housekeeping In this introductory course the student will learn the fundamentals of housekeeping management. This course describes the management functions, tools and practices required in today's lodging and institutional house keeping departments. Special attention to environmental, and safety implementations. Design and architectural elements and their relation to housekeeping will be discussed (3 hr. lecture)

Lecture Hours: 48.000, Clinical Hours: -4.000,

Internship Hours: 4.000

Course User Fee: \$15.00

HFT1300 Executive Housekeeping 3.000 credits

In this introductory course, the student will learn the fundamentals of housekeeping management. This course describes the management functions, tools and practices required in today's lodging and institutional housekeeping departments. Special attention to environmental and safety implementations. Design and architectural elements and their relation to housekeeping will be discussed.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1454 Food and Beverage Cost Controls 3.000 credits

Covers the principles and procedures involved in an effective food and beverage control system, including standards determination, the operation budget, income and cost control, menu pricing, and computer applications.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT1611 resp beverage vendor 1.000 credits

Lab Hours: 16.000, Clinical Hours: 1.000 - 30.000

HFT1631 Risk Management and Security 3.000 credits

Provides the opportunity to examine issues surrounding the need for individualized security and surveillance programs, risk management and review systems. The student will examine a wide variety of security and safety equipment procedures and discuss guest protection, internal security for asset protection, and OSHA regulations that apply to lodging properties.

Lecture Hours: 48.000

HFT1841 Dining Room Service 3.000 credits

Provides students with the opportunity to acquire knowledge of advanced service techniques, including guest satisfaction, food, wine and beverage serving, types of menus, table service techniques, tableside cooking, napkin folding, table setting, safety, sanitation, emergency procedures, restaurant technology, and service styles. Students will gain experience in cash and non-cash handling, forecasting sales, and merchandising techniques. Corequisite: HFT 1000. (3 hr. lecture)

Lecture Hours: 48.000

HFT1841L DINING ROOM SERV LAB 1.000 credits

Supplements the classroom theory portion of Dining Room Service HFT 1841 by having students create, plan, develop and participate in an actual dining room service experience. The student will work in tandem with industry professionals, faculty, and other students to outline, design, and bring to fruition an event. The student will become certified in ServSafe prior to the culmination of the event, so as to be in compliance with State mandated statute 509.039. Corequisite: HFT 1841. Special fee. (2 hr. lab)

Lab Hours: 32.000

HFT1852 Menu and Facilities Planning 3.000 credits

This course is a study of the factors involved in planning effective menus for a variety of food service operations. Students will learn to develop a firm working knowledge of menu-writing techniques and participate in actual menu design and facilities layout of a food service establishment.

Lecture Hours: 48.000

Prerequisite: HFT1000

Course User Fee: \$40.00

HFT1949 CO-OP WRK STDY ITRN1 3.000 credits

HFT 1949 Co-Op Work Study Internship 1

Prerequisite: Co-op department approval. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Cooperative Education Office to obtain registration approval. A.S. degree credit only. A.S degree only. (3 hr. lecture)

Lecture Hours: 48.000

Recommended Preparation: Prerequisite: Co-op department approval. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Cooperative Education Office to obtain registration approval. A.S. degree credit only. A.S degree only. (3 hr. lecture)

Course User Fee: \$25.00

HFT1949 Co-Op Work-Study Internships I 3.000 credits

Provides a supervised cooperative education experience in the hospitality industry that combines academic learning with practical workplace training. Students gain hands-on experience in hospitality operations, customer service, professional communication, and industry best practices while working in approved hospitality settings. Emphasis is placed on skill development, workplace professionalism, and career readiness through structured experiential learning.

Lecture Hours: 48.000

Course User Fee: \$25.00

HFT2223 TRAINING/SUPER DEVE 3 3.000 credits

Lecture Hours: 48.000

HFT2241 Leadership and Quality Assurance Management 3.000 credits

Provides an analysis of management issues related to the "personal touch" in customer service and quality assurance in the hospitality industry. Emphasis is placed on the importance of contemporary management and business practices to keep up with the demands of an ever-changing industry.

Lecture Hours: 48.000

HFT2261 Restaurant Management 3.000 credits

Familiarizes students with the general principles of food production management, work scheduling, and preparation supervision. Emphasis is placed on purchasing and financing, planning and equipping a kitchen, pricing and menu design, and marketing and promoting restaurants.

Lecture Hours: 48.000

Prerequisite: HFT1000

HFT2410 OFFICE PROC/LODG OP. 4.000 credits

HFT 2410 Front Office Procedures and Lodging Operations In this intermediary course the student will learn about various positions in the rooms division. Focus in hotel front office procedures involved in reservations, registering and checking out guests. Guest services, revenue management, accounting procedures and the operation of a PMS (Property Management System) are discussed. Special attention to hotel operations including security, house keeping and engineering (3 hr. lecture; 2 hr. lab)

Lecture Hours: 80.000

Course User Fee: \$15.00

HFT2410 Front Office Procedures and Lodging Operations 4.000 credits

In this intermediary course, the student will learn about various positions in the rooms division. Focus on hotel front office procedures involved in reservations, registering and checking out guests. Guest services, revenue management, accounting procedures and the operation of a PMS (Property Management System) are discussed. Special attention to hotel operations including security, housekeeping and engineering.

Lecture Hours: 48.000, Lab Hours: 32.000

Course User Fee: \$15.00

HFT2421 Managerial Accounting for Hospitality 3.000 credits

Presents managerial accounting concepts and explains how they apply to specific operations within the hospitality industry.

Lecture Hours: 48.000

HFT2449 E-Business For The Hospitality Industry 3.000 credits

Prepares student to manage information systems within their organizations. Emphasis is on selecting the right computer systems technology and maximizing available technology in order to promote and sell services. Introduces the use of automation in the broad hospitality sector and exams technological applications ranging from distribution systems (GDS, CRS and Web based), Property Management Systems, and EPOS systems to developments in telecommunications, and assesses their effect on the tourism sector. While a broad interpretation of both technology and tourism will be used, particular emphasis will be placed on the hospitality sector (i.e. hotels and catering) and on distribution technology. A combination of lecture, case studies, seminars, visiting lecture and lab sessions are used.

Lecture Hours: 48.000

Prerequisite: HFT1000

HFT2500 Marketing of Hospitality Services 3.000 credits

In this intermediary course, the students will learn basic knowledge and practical experience which will enable them to develop strategic marketing plans for any hospitality business. Special focus on the marketing mix as it applies to hospitality. Marketing trends and case studies are discussed.

Lecture Hours: 48.000

HFT2501 Hospitality Sales 3.000 credits

In this intermediary course, the student will learn sales presentation skills and the importance of sales in an organization. Topics discussed are the organization of the sales department, including the responsibilities of all involved in the sales process. Emphasis on sales presentations, looking for leads, negotiating skills, building rapport, and closing the sale. Role-playing and other activities will be used to enhance learning.

Lecture Hours: 48.000

HFT2750 Event and Meeting Management 3.000 credits

In this intermediary course, the student will learn the complete set of skills necessary to adequately perform as an event planner, hotel banquet manager and convention and meeting planner. Actual events will be used to reinforce the general rules of table service, booking functions, staffing banquets/conventions, and responsibilities of a host venue as they apply to buffets and banquets. Prepares students in trade show administration, meeting management, and legal issues associated with banquets and conventions.

Lecture Hours: 48.000

HFT2772 Introduction to Cruise Line Industry 3.000 credits

Provides students with an introduction to the cruise line industry, its evolution and relationship to other segments of tourism and hospitality, sales and marketing methods, management, and strategic planning are major topics. Corequisite: HFT 1000. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: HFT 2000

HFT2773 Cruise Line Sales and Marketing 3.000 credits

Provides an introduction to the sales and marketing functions of the cruise industry. Students will gain an understanding of how cruise lines position themselves in the competitive business environment and the sales and marketing techniques used to attract customers and group business. Students will gain an understanding of yield management and the issues surrounding travel agents during the sales process.

Lecture Hours: 48.000

Prerequisite: HFT2772, Corequisite: HFT2774, HFT2775

HFT2774 Shipboard Operations 3.000 credits

Provides an understanding of shipboard operations on cruise ships and their relationship to the shoreside headquarters office. Students will gain knowledge of the history of cruise ships and the activities and facilities that make cruise line operations complementary both to the industry and the guest. This course will focus on the ship as a hotel for passengers with the wining and dining aspects of service, as well as, casino practices on board.

Lecture Hours: 48.000

Prerequisite: HFT2772, Corequisite: HFT2773, HFT2775

HFT2775 Shoreside Operations 3.000 credits

Provides a basic understanding of the shoreside office operations and sales procedures of the cruise line and how they relate to the general operations of the cruise ship itself. Students will acquire knowledge of pier, airport, ground services and hotel operations and create elements for cruise line sales.

Lecture Hours: 48.000

Prerequisite: HFT2772, Corequisite: HFT2773, HFT2774

HFT2800 Food and Beverage Management 3.000 credits

Provides a basic understanding of the principles of food production and service management, menu planning, serving, purchasing, labor, food/bar service and costs, storage, beverage management, sales promotions, entertainment, and liability laws.

Lecture Hours: 48.000

Course User Fee: \$15.00

HFT2949 CO-OP WRK STDY ITRN2 3.000 credits

HFT 2949 Co-Op Work Study Internship 2 In this intermediary course the student will continue learning and training in students' field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Prerequisite: Co-op department approval and completion of 1949 Co-op work experience. In addition to the above the student keep a portfolio of required materials that they obtain during the course of their study. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: HFT1949

Course User Fee: \$25.00

HFT2949 Co-Op Work-Study Internships 2 3.000 credits

Provides an advanced supervised cooperative education experience in the hospitality industry that builds on foundational workplace skills and knowledge. Students apply hospitality management principles in professional settings while enhancing competencies in operations, customer service, leadership, communication, and problem-solving. Emphasis is placed on increased responsibility, professional growth, and career development through structured experiential learning in approved hospitality organizations.

Other Hours: 48.000

Course User Fee: \$25.00

HFT3263 Restaurant Management 3.000 credits

Students will examine restaurant management and operations. Topics include operational issues, customer satisfaction, restaurant trends and challenges, new technologies, financial accountability, cost controls, service issues, regulatory and safety compliance, leadership, training and staff development. Upon completion, students will be able to articulate knowledge of all aspects of a restaurant operation.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT3603 Hospitality Law 3.000 credits

Students will examine the legal rights and responsibilities related to the hospitality industry. Topics include federal and state regulations, historical and current practices, safety and security, risk management, loss prevention, liability, relevant torts and contracts. Upon completion, students will be able to articulate knowledge of the legal issues in the hospitality industry, and methods of preventing or minimizing organizational liability.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT4064 Bar and Beverage Management 3.000 credits

Students will examine the management of beverages served in hospitality operations. Topics include history and trends; service, procurement and storage; knowledge and control of wines and fermented/distilled beverages; and non-alcoholic beverages, coffees, and teas. Upon completion, students will be able to articulate knowledge of responsible alcohol service and knowledge of beverages consumed in a hospitality operation.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT4253 Hotel Management 3.000 credits

Students will examine the working components of a hotel and their interrelationships. Topics include hotel organization, management and leadership, human resources, food and beverage service, meeting industry, marketing and selling, communications, management companies, and ethics in hospitality management. Upon completion, students will be able to articulate knowledge of the responsibilities of each hotel department and how they interact to provide the overall guest experience.

Lecture Hours: 48.000

Course User Fee: \$35.00

HFT4468 Revenue Management in the Hospitality Industry 3.000 credits

Students will examine the basics of revenue management in the hotel industry. Topics include pricing and length-of-stay tools, revenue management performance measures, capacity management, demand and revenue forecasting, discounting, reservation systems, and pricing execution. Upon completion, students will be able to articulate knowledge of how to accurately forecast hotel guest arrivals, examine pricing models in accordance with revenue management principles, and how to manage overbooking.

Prerequisite: ACG2021 , ACG2021L, or HFT2421.

Lecture Hours: 48.000

Pre/Corequisite: HFT2421, Prerequisite: ACG2021, ACG2021L

Course User Fee: \$35.00

HFT4809 Food Service Management 3.000 credits

Students will examine food service management and operations. Topics include the management process, marketing, sales techniques, cost and pricing strategies, food purchase controls, production and service, sanitation and safety, management of bar and beverage design and financial management. Upon completion, students will be able to articulate knowledge of the financial relationship of the food and beverage aspect of the hotel industry.

Lecture Hours: 48.000

Course User Fee: \$35.00

HIM**HIM0009 Introduction to Health Information Technology 3.000 credits**

This course introduces students to health informatics and information management. Students will learn about the health care delivery system, communication skills, legal and ethical responsibilities, HIPAA and health records, and terminology related to health informatics. Other topics include developing leadership and teamwork skills and the application of critical thinking skills in a variety of presented scenarios. (90 contact hours)

HIM0061 MEDICAL REC TRANS 1 1.500 credits

This course covers the basic foundations of medical transcription to include role, ethics and legal responsibilities of the transcriptionist. Equipment, types of medical reports, quality control and reference materials are also discussed. Special fee . (45 contact hours)

Lecture Hours: 45.000

Course User Fee: \$10.00

HIM0274 HLTH INS CLAIM/DLQNT 1.500 credits

This course reveals how insurance claims are developed and processed from the health care provider's office to the insurance company. Delinquent claims and solving common billing problems are explored . Various health plans are discussed. Prerequisite s: HIM 0228, 0228L; corequisites: HIM 0230, 0230L. Special fee. (45 contact hrs.)

Lecture Hours: 45.000

Course User Fee: \$45.00

HIM1000 Introduction to Health Information Technology 2.000 credits

This course examines the role and functions of a health information technician. Students will learn about the structure, organization, and maintenance of the medical record. Students will also learn about the organization and function of various types of health care facilities, the responsibilities of national, state and local health agencies, and the organization and mission of the American Health Information Management Association.

Lecture Hours: 32.000

HIM1110 Health Information Technology and Data Collection 2.000 credits

This course is designed to provide the skills necessary to function in a health information management department. Students will learn health record data collection and informatics. Students will also learn about the various components and approaches of the electronic health record.

Lecture Hours: 32.000

Prerequisite: HIM1000, HIM2472, Corequisite: HIM1110L

HIM1110L Health Information Technology and Data Collection lab 3.000 credits

This course is designed to apply basic requirements imposed by regulatory agencies to health record data. Students will learn how clinical data repositories store health information. Students will also learn the concepts relating to confidentiality, ethics, and release of information. Prerequisite: HIM 1000, 2472; corequisite HIM 1110.

Lab Hours: 96.000

Prerequisite: HIM1000 AND HIM2472 AND Corequisite: HIM1110

HIM1300 Health Care Facility and Delivery Systems 2.000 credits

This course will examine healthcare complexities, the function of various types of health facilities, accreditation standards, Medicare law, and the American health delivery system. Students will learn the components of Medicare, Medicaid, Health Insurance Organizations (HMO's), and the federal laws that govern them.

Lecture Hours: 32.000

HIM1800 Professional Practice Experience 1 2.000 credits

This course will provide the student with a supervised professional practice experience in a health care setting, utilizing electronic health records and reports to manage health information data. Students will learn an in-depth knowledge of applying the minimum basic requirements for handling records imposed by regulatory agencies. Prerequisite: H IM 1000; corequisite: HIM 1110, 1110L.

Clinical Hours: 96.000

Course User Fee: \$10.00

HIM2012 Legal Aspects of Health Care 2.000 credits

This course provides basic knowledge of the United States of America court system and the interconnection between the health care system and the federal government. The student will learn concepts relating to the Health Insurance Portability Accountability Act (HIPAA), ethics, release of health information, record retention, and the legalities of electronic health records.

Lecture Hours: 32.000

Prerequisite: ENC1101

HIM2211C Health Information Technologies 2.000 credits

This course is designed to examine computer technology used to collect and store health information. The student will learn a variety of applications used to maintain and secure health care data.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: HIM1110, HIM1110L, HIM1800

Course User Fee: \$85.00

HIM2214C Health Statistics 2.000 credits

This course will focus on the definitions for analysis, interpretation, and display of healthcare data. The student will learn the acceptable terminology and basic definitions for reporting health statistics. Emphasis is placed on the use of the formulas necessary for computing standard rates, percentages, and averages from patient data.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: HIM1110, HIM1110L, Corequisite: HIM2512C

Course User Fee: \$20.00

HIM2222 ICD Coding Systems 2.000 credits

This course is designed to teach the current edition of International Classification of Diseases and Procedures codes. Students will learn code convention and format, usage of anatomy, physiology, and clinical disease processes. Pharmacology, present on admission indicators, correct code assignment and sequencing will be discussed.

Lecture Hours: 32.000

Prerequisite: BSC2085, BSC2085L, HIM2472,

Corequisite: HIM2222L

HIM2222L ICD Coding Systems Laboratory 3.000 credits

This course is designed to apply diagnoses and operations into numerical designations (codes) utilizing the International Classification of Diseases and Procedures. Students will learn correct sequencing, analysis of clinical disease processes, diagnosis and procedural terminology, and application of pharmacology in current coding systems. Encoding software is utilized.

Lab Hours: 96.000

Prerequisite: BSC2085, BSC2085L, HIM2472,

Corequisite: HIM2222

Course User Fee: \$25.00

HIM2234 ADV CODING&REIMB SYS 2.000 credits

Lecture Hours: 32.000

HIM2234L Advanced Coding & Reimbursement Systems Laboratory 1.000 credits

This course is designed to apply and compute Prospective Payment Systems categories. Students will learn to apply health record documentation to identify and validate correct code and payment assignments. The focus is on the computation of MS-DRGs, APCs and a Case-Mix Index using encoder, grouper, and electronic billing software for reimbursement.

Lab Hours: 32.000

Prerequisite: HIM2222, HIM2222L, Corequisite: HIM2234

Course User Fee: \$15.00

HIM2253C Current Procedural Terminology/ CPT-4 2.000 credits

This course provides an in-depth knowledge of coding and reporting using Current Procedural Terminology classification. Students will learn to read and interpret ambulatory health record documentation to classify and assign services and procedure codes, use encoder and grouper software, HCPCS, APCs, and RBRVS. Prerequisites: BSC 2085, 2085L and HIM 2472.

Lecture Hours: 32.000

Course User Fee: \$15.00

HIM2400C Diversified Non-Hospital Health Records 2.000 credits

This course emphasizes the importance of quality record-keeping practices, data flow, and management of health information systems in a non-acute care setting. The student will learn the documentation requirements based on Federal and State statutes, accreditation standards, Medicare Conditions of Participation, payment systems, funding, Health Insurance Portability Accountability Act, and the evolution of the electronic health record. Prerequisites: HIM 1110, 1110L, 1800.

Lecture Hours: 32.000

HIM2433 pathophysiology & ph 3.000 credits

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

HIM2472 Medical Terminology 3.000 credits

Analysis of medical terms through learning basic roots, prefixes and suffixes permitting the student to have a working knowledge of the language of medicine. Prerequisite: Permission of department chairperson.

Lecture Hours: 48.000

HIM2500 Data Management & Quality Assessment 2.000 credits

This course examines the basic principles of quality assessment, quality improvement and utilization review. Students will learn about the accreditation process, risk management, managed care models, and the methodologies and relationships of these key areas within a health care facility. Prerequisites: HIM 1110, 1110L, 2211C. Corequisite: HIM 2500L.

Lecture Hours: 32.000

Course User Fee: \$75.00

HIM2500L Data Management & Quality Assessment Laboratory 1.000 credits

The application of the basic principles of quality assessment: quality improvement and utilization review. The student will learn to generate models for the evaluation of different types of medical care. Activities will center on the accreditation process, managed care, and risk management. The methodologies and relationships of these key areas within a health care facility are emphasized.

Lab Hours: 32.000

Prerequisite: HIM 1110, HIM 1110L, Corequisite: HIM 2500

HIM2512C Supervision & Organization for Health Information Technology 2.000 credits

This course examines the basic principles of management and organizational life in a health information management department and the interrelationships within the healthcare organization. Students will learn about the supervisory role of the health information professional, including basic motivation and communication principles essential to the practice of health information management. Students will also learn how to identify and use specific motivational and communication techniques in health information supervision. Prerequisites: HIM 1800

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$15.00

HIM2652C elec health record 3.000 - 32.000 credits

Lab Hours: 16.000, Clinical Hours: 1.000 - 30.000

HIM2810 Professional Practice Experience 2 2.000 credits

This course is a supervised professional practice experience in a health care setting utilizing health records and reports to perform medical coding functions. Students will learn to assign Medicare Severity-Diagnosis Related Groups/Ambulatory Payment Classifications using the organization's information systems, encoder and grouper software. Students will also learn about the process for sequencing International Classification of Disease (ICD) and Current Procedural Terminology (CPT) coding systems. Prerequisite: HIM 2222, 2222L; corequisite: HIM 2234, 2234L.

Lecture Hours: 96.000

Course User Fee: \$15.00

HIM2813 Professional Practice Experience 2.000 credits

This course is an advanced coding/billing professional practice. Students will learn advanced coding and abstracting of actual inpatient and outpatient health records, with an emphasis on compliance and improving accuracy and productivity.

Clinical Hours: 96.000

Prerequisite: HIM2222 AND HIM2222L AND HIM2234 AND HIM2234L AND HIM2253C

HIM2820 Seminar and Professional Practice Experience 3 2.000 credits

This course is designed to provide students with structured learning experiences necessary for them to enter the health information management field. Students will learn preparation skills for the national examination by analyzing major examination topics offered in the curriculum. A professional practice experience will be a component of this course.

Prerequisite: HIM 2810; corequisites: HIM 2500, 2500L.

Lecture Hours: 96.000

Course User Fee: \$10.00

HIM9995 Health Information Technology 26.000 credits

This course is a placeholder awarding equivalent college credit for non-collegiate training based on Technical Manual of Procedure Number 1100: 816125. This course requires special permission and students must contact the department chairperson for registration/approval.

Lecture Hours: 416.000

HLP**HLP1080 Wellness 2.000 credits**

This course enables students to assess their present aerobic fitness level, lung capacity, percentage of body fat, flexibility and strength. From data collected, the student will be able to set personal wellness goals. Lectures, demonstrations, and multi-media materials will be used to provide the scientific basis for meeting one's personal wellness goals. (2 hr. lecture/lab)

Lecture Hours: 32.000

HLP1081 Fitness & Wellness for Life 3.000 credits

In this course students will learn the roles of exercise, physical activity, diet, and stress management in achieving optimal wellness. Students will explore current developments in health and complete lab assignments, which will assist in the determination of their current health status. Individualized exercise and dietary protocols based on these assessments will be developed. Special fee.

Lecture Hours: 48.000

Course User Fee: \$10.00

HLP1083 WEIGHT MANAGEMENT 3.000 credits

This course is designed for students to develop an understanding of the role of exercise and nutrition as it applies to the implementation of a weight management plan. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$10.00

HOS

HOS1010 Horticulture 1 3.000 credits

This is an introductory course on the principles of horticulture. Students will learn plant structure and function, plant propagation, plant nutrients and fertilizers, potting media, soils, pruning, and plant pests. A survey of various fields in ornamental horticulture will also be covered.

Lecture Hours: 48.000

HOS1011 Horticulture 2 3.000 credits

The student will learn the maintenance and management aspects of horticulture business (nursery facility or landscape maintenance and design) including irrigation systems, plant growing facilities, plant propagation equipment, and landscape maintenance equipment. Hands-on practice in programming of plant production crops and nursery design in our nursery.

Lecture Hours: 48.000

Prerequisite: HOS1010

HOS2005 Hydroponic Systems 3.000 credits

This course will provide an overview of the different types of hydroponic systems. Students will learn about set-up requirements, maintenance, nutrient formulations, and growing plants in a soilless culture. The course will emphasize knowledge and skills required to run small hydroponic systems. (3 Hr. Lecture)

Lecture Hours: 48.000

Course User Fee: \$35.60

HRM

HRM2990 CBE Rooms Division 1.000 - 13.000 credits

The HRM 2930 CBE Rooms Division special topics course is designed to assess learner mastery of the competencies and skills necessary for a successful career in the lodging industry. The competency-based approach allows learner the flexibility to self-direct their learning experience. As a result, the learning is much more individualized and can be significantly accelerated. Learners enrolled in the HRM 2930 CBE Rooms Division course also have the opportunity to achieve industry-recognized academic credentials, and will be prepared for positions such as front desk agent, guest relations agent or reservation clerk.

Other Hours: 16.000 - 208.000

HSA

HSA1102 Current Issues in Health 3.000 credits

This course covers current issues within the health care system. Emphasis focuses on varied areas of health care and is appropriate for persons directly or indirectly involved in the provision of health care or health education. (3 hr. lecture)

Lecture Hours: 48.000

HSA1380 Health Care Quality Management 3.000 credits

This course examines various efforts being introduced to improve the quality and efficiency of the health care system. Topics covered range from issues of medical error reduction, quality improvements in medical records, and utilization review, models for continuous quality improvement in physician-patient relations. (3 hr. lecture)

Lecture Hours: 48.000

HSA2001 Interprofessional Team Based Health Care 2.000 credits

This course is an introduction to interprofessional health care delivery. Reviewed are concepts of interprofessional collaboration practices in relation to team formation, ethical consideration with team management, effective communication approaches, and patient care processes. (3 hr. lecture)

Lecture Hours: 32.000

HSA2181 Health Services Management Concepts 3.000 credits

This course provides preparation and fundamental skills required to enter the field of medical office management. Emphasis will be placed on health care delivery in a medical office setting. Topics include an introduction to health care management information systems, an introduction to essential policies and regulations, and an introduction to evaluation of performance and process outcomes.

Lecture Hours: 48.000

HSA2532 Medical Documentation in Health Care 1.000 credits

Medical Documentation in Health Care will introduce the student to documentation in the written patient chart or electronic medical record. Through case discussions and in-class writing assignments, the student will acquire the necessary skills to document in the patient's medical record utilizing medico-legal principles and evaluation and management criteria. Patient confidentiality, billing, and coding will also be discussed.

Lecture Hours: 16.000

HSA3533 Medical Documentation in Health Care II 1.000 credits

Medical Documentation in Health Care II is a continuation of HSA 2532. This course will teach the student to documentation in the written patient chart or electronic medical record for special populations. Through case discussions and in-class writing assignments, the student will acquire the necessary skills to document in the patient's medical record utilizing medico-legal principles and evaluation and management criteria. Patient confidentiality, billing, and coding will also be discussed. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822C, 1823, 1831. (1 hour lecture)

Lecture Hours: 16.000

Prerequisite: HSA2532, PAS1801C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

HSC**HSC0003 Introduction to Healthcare 2.500 credits**

An introduction to the health care environment, this course focuses on the health care team and delivery systems. Students will learn about legal responsibilities, ethical issues, safety, infection control, communication, interpersonal behaviors, wellness, and disease. Corequisite: HSC0003L

Clock Hours: 75.000

Corequisite: HSC0003L

Course User Fee: \$20.00

HSC0003L Introduction to Healthcare Lab 0.500 credits

This course focuses on the performance of basic health care skills. Students will apply body mechanics and ergonomics, standard precautions used in infection control procedures and perform and record vital signs.

Corequisite: HSC0003

HSC3XXX Introduction to Public Health 1.000 credits

This course is designed to give students a foundation in the core functions of population-based health consisting of assessment, policy development and assurance. Students will gain an understanding of public health as a broad field which applies the benefits of current biomedical, environmental, and behavioral knowledge in ways that maximize the health status of all populations. This course is blended.

Lecture Hours: 16.000

Prerequisite: PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, HSC 2ZZZ Medical Documentation in Health Care

HSC3ZZZ Medical Documentation in Health Care II credits

Medical Documentation in Health Care II is a continuation of HSC 2ZZZ. This course will teach the student to documentation in the written patient chart or electronic medical record for special populations. Through case discussions and in-class writing assignments, the student will acquire the necessary skills to document in the patient's medical record utilizing medico-legal principles and evaluation and management criteria. Patient confidentiality, billing, and coding will also be discussed.

Prerequisite: PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, HSC 2ZZZ Medical Documentation in Health Care

HSC1149 General Pharmacology for Health Care Professions 3.000 credits

This course will provide learners with a basic introduction to pharmacology concepts and principles associated with management of common diseases and treatment options. It presents information on major drug classifications, indications for use of common medications, and potential contraindications adversely affecting medical care outcomes.

Lecture Hours: 48.000

HSC2060 Artificial Intelligence Applications in Healthcare 3.000 credits

An introduction to the health care environment and delivery systems. In this course, students will explore how Artificial Intelligence (AI) is impacting healthcare professions. Students will gain a fundamental understanding of ethical implications, legal principles, and applications of AI in various health settings.

Lecture Hours: 48.000

Course User Fee: \$28.02

HSC2400 Basic Emergency Care 2.000 credits

The prevention of common accidents and the care of emergencies in the home and school. Meets American Red Cross certification requirements in standard and advanced First Aid.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$17.00

HSC2401 INST TRN-1ST AID/CPR 3.000 credits

Lecture Hours: 64.000

HSC 2810 Professional Practice Experience 4.000 credits

This course provides an opportunity for the student to observe basic principles and techniques of managers in a health care setting. Students apply knowledge from previous health service management courses in completing course assignments. This course has a service learning component at an area health care agency. (3 hr. lecture)

Lecture Hours: 64.000

HSC3057 Introduction to Research Methods in Health Care 3.000 credits

This course will provide an overview of research methods used in healthcare. Students will learn the use of effective inquiry through research strategies that address healthcare issues with logical and observational rigor. Students will learn the rudiments relative to the evaluation of research literature, research design and the application of research methods to the clinical setting. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: STA2023

HSC3202 Introduction to Public Health 1.000 credits

This course is designed to give students a foundation in the core functions of population-based health consisting of assessment, policy development and assurance. Students will gain an understanding of public health as a broad field which applies the benefits of current biomedical, environmental, and behavioral knowledge in ways that maximize the health status of all populations. This course is blended. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822C, 1823, 1831. (1 hour lecture)

Lecture Hours: 16.000

Prerequisite: HSA2532, PAS1800C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

HSC3243 Teaching Skills for Health Care 3.000 credits

This course emphasizes significant problems instructors have related to content and design of curricula or courses. Students will learn the theoretical underpinnings of teaching and learning in health science disciplines from a variety of perspectives - educational, psychological, developmental, and social. (3 hr. lecture)

Lecture Hours: 48.000

HSC3655 Theoretical Foundations of Health Care Ethics 3.000 credits

This course will cover the Theoretical Foundations of Health Care Ethics. The student will learn how to apply the core principles of bioethics to the medical and health care decision-making process. (3 cr. Lecture)

Lecture Hours: 48.000

HSC3701 Leadership and Management in Healthcare 3.000 credits

This is a leadership and management course that will examine leadership as a process with a tri-fold focus: the leader, the followers, and the situation. The student will learn leadership theories and build leadership skills. (3 hr. lecture)

Lecture Hours: 48.000

HSC4942 Community Service-Learning Practicum 3.000 credits

This course is designed to engage students in meaningful healthcare related community service learning activities. Students will learn to explore new collaborations in service-learning, cultural diversity and community healthcare development. These experiences will help to and prepare them for lives as civically-engaged local, national, and global citizens. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: HSC3057, HSC 3231

HUM**HUM1020 Humanities 3.000 credits**

In this course, students will learn about the creative ideas and accomplishments of various cultures in various fields of humanities that may include art, architecture, drama, history, music, literature, philosophy, and religion. The course will include cultural expressions from the western canon and may also include expressions from around the globe. Student learning outcomes: students will demonstrate knowledge of arts and ideas and synthesize information from various sources; students will analyze and interpret selected expressions of arts and ideas; students will compare and contrast selected expressions of arts and ideas; and students will identify contextual influences on the development of interdisciplinary arts and ideas.

Lecture Hours: 48.000

Course User Fee: \$5.00

HUM2513 ARTS AND HUMANITIES 3.000 credits

Selected examples of Art including painting, sculpture, architecture, literature and the performing arts to illustrate the variety of art in relation to man's perception of self, nature and God. Intended primarily for use in overseas academic programs. May be repeated for credit. (6 hr. lab)
Lecture Hours: 96.000

HUN**HUN1201 ESSENTIALS OF HUMAN NUTRITION 3.000 credits**

The Essentials of Human Nutrition is a general education course designed to acquaint students with the specific role of carbohydrates, fats, proteins, vitamins, minerals, and water in daily life. Students will learn how the human body systems manage the breakdown, assimilation, and excretion of nutrients and their metabolic wastes. Students will also learn the relationships between food and optimal health including physical fitness and the relationships between nutritional imbalances and diseases.

Lecture Hours: 48.000

Course User Fee: \$10.00

HUS**HUS1001 INTRO/HUMAN SERVICES 3.000 credits**

An introduction to an overview of the field of Human Services, including the role of the human services worker as it relates to various agencies, counseling, interviewing and managing. (3 hr. lecture)

Lecture Hours: 48.000

HUS1302 BASIC COUNSEL SKILLS 3.000 credits

Development of the skills of observation, recording, reporting, interviewing and counseling. These skills are presented in the context of general counseling theory. (3 hr. lecture)

Lecture Hours: 48.000

HUS1318 Domestic Abuse and Family Violence 3.000 credits

This course is designed to educate human services workers for the evaluation, counseling and outreach skills necessary for working with victims of domestic violence. The dynamics of partner violence, child abuse, and elder abuse will all be explored.

Lecture Hours: 48.000

HUS1421 Assessment and Treatment Planning in Addictions 3.000 credits

This course is designed to familiarize students with the core functions of Assessment and Treatment Planning for the chemically dependent client. Emphasis on treatment planning will be accomplished drawing from the Florida Certification Board for addiction professionals and the Department of Children's and Family Services guidelines. Prerequisites: HUS2493, PSB2442.

Lecture Hours: 48.000

Prerequisite: HUS2493, PSB2442

HUS1423 Group Counseling in Substance Abuse 3.000 credits

This course stresses development of effective group counseling leadership skills including organizing, implementing, and evaluating group counseling programs. The course includes actual group experiences. Prerequisite: PSB2442.

Lecture Hours: 48.000

Prerequisite: PSB2442

HUS1428 Addiction Treatment Delivery Systems 3.000 credits

This course is designed to survey the modalities of addiction treatment. The course will study federal and state systems as well as private-not-for-profit and private-for-profit programs. All of these will be described using examples drawn from local agencies, the diverse populations they serve, and the politics and economics of the systems. This course will also present a critical exploration of the history and theory defining problems of addiction treatment and the characteristics and career issues of an addiction treatment services worker.

Lecture Hours: 48.000

HUS1440 Family Issues in Chemical Dependency 3.000 credits

This course is designed to analyze the effects of chemical abuse on the family system. Emphasis will be placed on family roles and dynamics; characteristics of children (including adult children) of chemical abusers; theories of codependence; and adaptations made individually and socially by family members. Critical issues and strategies in family treatment will be explored.

Lecture Hours: 48.000

**HUS1475 Addiction Counseling and the Law
3.000 credits**

This course is designed to introduce addiction counseling students to the vocabulary, agencies and processes required to work with clients involved in both the criminal and civil justice systems. This course focuses on the relationship between the law and Human Services institutions, patterns of lawmaking and law-breaking, the legal structures and processes, and law as an instrument of public policy, social control and social change. The roles and functions of police, courts and correctional services will be surveyed. Common civil issues that affect clients in recovery will be explored. In addition, this course will enable students to explain the legal basis for alcohol and other drug services in Florida. State statutes pertaining to alcohol and drugs and their administrative rules will be reviewed. Confidentiality requirements, compliance standards, and professional ethics will be presented. Prerequisite: PSB2442.

Lecture Hours: 48.000

Prerequisite: PSB2442

**HUS1480 HIV/AIDS and the Substance Abuser
3.000 credits**

This course is designed to educate prospective addiction counselors for the evaluation, counseling and outreach skills necessary for working with HIV disease and AIDS. The course will explore not only how this disease affects one personally, but also how this pandemic has affected many psychosocial aspects of society.

Lecture Hours: 48.000

**HUS2303 COUNSELING TECHNIQUE 3.000
credits**

Lecture Hours: 48.000

HUS2493 Addiction Counseling 3.000 credits

This course is designed to enable students to master the TAP 21 competencies: clinical evaluation, treatment planning, referral, service coordination, counseling, client, family, and community education, documentation and professional and ethical responsibilities. Additionally, the course will teach the student the process of identifying problems, establishing goals and deciding on a client treatment plan. Students will learn how to respond to an individual's needs during acute emotional and physical distress. Prerequisite: PSB2442

Lecture Hours: 48.000

Prerequisite: PSB2442

**HUS2500 Issues & Ethics in Human Services
3.000 credits**

This course is designed to familiarize students with the ethical problems that emerge from counseling the chemically dependent client. Emphasis will be placed on the following: the history and theory of ethics in health care; professionals' and patients' rights and responsibilities; the relationship between ethics and law; confidentiality and truth-telling in clinical relationships; technology; diagnostic testing and treatment; treatment of terminal illness; distribution of scarce medical resources and access to health care and systems payment. Prerequisite: PSB2442.

Lecture Hours: 48.000

**HUS2820 Field Experience in Human Service
3.000 credits**

Volunteer work as counseling paraprofessionals in a community agency under supervision. Students meet regularly with the Field Coordinator.

Prerequisites: HUS1001, HUS1302, HUS2303. (120 hrs. per term)

Other Hours: 144.000

Prerequisite: HUS1001, HUS1302, HUS2303

IDH**IDH1001 HONORS LEADRSHP SEM1 1.000 -
3.000 credits**

Rigorous, in-depth exploration of selected honors topics. The topic and content are arranged by the instructor, department chairperson and campus honors coordinators. These seminars will consist of small groups that meet on a regular basis and be offered in any subject area. (1-3 hr. lecture)

Lecture Hours: 16.000 - 48.000

**IDH1002 HONORS LEADRSHP SEM2 1.000 -
3.000 credits**

Rigorous, in-depth exploration of selected honors topics. The topic and content are arranged by the instructor, department chairperson and campus honors coordinators. These seminars will consist of small groups that meet on a regular basis and be offered in any subject area. (1-3 hr. lecture)

Lecture Hours: 16.000 - 48.000

**IDH2003 HONORS LEADRSHP SEM3 1.000 -
3.000 credits**

Hours taken by students to complete a capstone (thesis) project under the supervision of an advisor and a committee, which will produce a piece of work that students may take with them to upper division institution to demonstrate their ability to apply the principles learned and the quality of their work. (1-3 hr. lecture)

Lecture Hours: 16.000 - 48.000

IDH2004 HONORS LEADRSHP SEM4 1.000 - 3.000 credits

Rigorous, in-depth exploration of selected honors topics. The topic and content are arranged by the instructor, department chairperson and campus honors coordinators. These seminars will consist of small groups that meet on a regular basis and be offered in any subject area. (1-3 hr. lecture)

Lecture Hours: 16.000 - 48.000

IDS**IDS1044 LEADERSHIP SEMINAR 3.000 credits**

Lecture Hours: 48.000

IDS1107 Tools for Success 1.000 credits

This course is for students majoring in science, technology, engineering and mathematics fields (STEM). Students will learn writing, research, presentation, and technological skills necessary for success in STEM-related disciplines. Course topics include learning styles, collaborative skills, power study techniques and will use related technologies related to STEM.

Lecture Hours: 16.000

IDS1153 earth lit&sustainabil 3.000 credits

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

IDS1911 Interdisciplinary Undergraduate Research 1.000 credits

This undergraduate course provides students in all disciplines and majors with transformative learning experiences and immersion in hands-on research. By engaging in collaborative, real-world projects, students address challenges that require the acquisition, analysis, and synthesis of knowledge from multiple disciplines. The course equips participants with leadership skills, cultural competence, and an understanding of sustainability and ethical research practices. Students adopt research methodologies while applying both qualitative and quantitative techniques to practical problems. This approach not only enhances their analytical capabilities but also fosters critical thinking and decision-making skills, crucial for future leaders. Through this immersive experience, students learn to bridge the gap between theoretical concepts and real-world applications, preparing them for the multifaceted challenges of the modern professional landscape. Note: This course prepares students for continued studies, ensuring the successful transition to four-year and graduate institutions.

Lecture Hours: 16.000

IDS2123 Leadership in Science, Tech, Engineering and Math 1.000 credits

In this course students will research their career interests and interview professionals in Science, Technology, Engineering and Mathematics (STEM). Students will learn to identify, compare, and evaluate upper division degree programs and prepare applications for admission to these programs. Students will write successful application essays and develop interview skills for transfer. (1 hr. lecture)

Lecture Hours: 16.000

IDS2124 SKILLS FOR TRANSFER SUCCESS 1.000 credits

This course is for students in science, technology, engineering and mathematics (STEM) for matriculation to the upper division. Students will learn to research, write, coordinate and present grants and scholarships in conjunction with the college application process. Students will document all of their efforts in an electronic portfolio. (1 hr. lecture)

Lecture Hours: 16.000

IND**IND1020 Interior Design 1 4.000 credits**

Student's projects develop the ability to plan simple interior floor plans and elevations.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ARC1115

Course User Fee: \$25.00

IND1100 History of Interiors 1 3.000 credits

Lecture Hours: 48.000

Course User Fee: \$10.00

IND1130 History of Interiors 2 3.000 credits

Emphasis is on perfecting water color, casein and reproducible drawing techniques through the presentation of interior plans, elevations and perspectives. Projects also provide experience in assembling collages.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: IND1300, Corequisite: IND2210

Course User Fee: \$25.00

IND1200 Interior Design 2 4.000 credits

Problems in room planning, correlation of color schemes and furnishings.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: IND1020

Course User Fee: \$25.00

IND1300 Interior Design Presentations 1 2.000 credits

An introductory course in the use of various media for presentation of plans, schemes, and interior perspective renderings.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: IND1020, Corequisite: IND1200

Course User Fee: \$25.00

IND2201 Design Principles for Kitchen and Bath 3.000 credits

A specialized design course for kitchen and baths. Students will learn the basic elements and principles of creating spaces for clients, blending architectural styles, colors and themes with function and purpose.

Lecture Hours: 48.000

Prerequisite: ARC2461, IND1200, IND1300

Course User Fee: \$10.00

IND2210 Interior Design 3 4.000 credits

Projects provide practice in planning traditional and contemporary interiors including working drawings and specifications.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: IND1200, Corequisite: IND2330

Course User Fee: \$25.00

IND2220 Interior Design 4 4.000 credits

Emphasis is on perfecting watercolor, casein and reproducible drawing techniques through the presentation of interior plans, elevations and perspectives. Projects also provide experience in assembling collages.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: IND2210

Course User Fee: \$25.00

IND2330 Interior Design Presentations 2 3.000 credits

Emphasis is on perfecting water color, casein and reproducible drawing techniques through the presentation of interior plans, elevations and perspectives. Projects also provide experience in assembling collages.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: IND1300, Corequisite: IND2210

Course User Fee: \$25.00

IND2421 Introduction to Furniture Design 3.000 credits

Hands-on course dealing with the theoretical and practical considerations for designing and building furniture and cabinetry. Students will learn basic principles of designing furniture through researching history, theory, materials, methods and design considerations associated with the construction of furniture. Laboratory fee. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: IND1200 OR ARC1302

Course User Fee: \$10.00

IND2430 Lighting Design 3.000 credits

A survey of utilitarian interior lighting and exterior architectural lighting including fundamentals and basic physic laws, practical applications to interior and exterior spaces and lighting design considering different levels of space utilization and fixture efficiency.

Lecture Hours: 48.000

Prerequisite: IND1200

Course User Fee: \$10.00

IND2500 Professional Practices 3.000 credits

Duties and responsibilities relative to employment and business practices.

Lecture Hours: 48.000

Prerequisite: Sophomore standing level or equivalent.

Course User Fee: \$10.00

IND2608 Sustainable Design 3.000 credits

Introduction to basic theories and practices of ecological design in interiors. Students will learn to take responsibility for their work by understanding the impact their designs have on our environment, and ways to mitigate this impact by gaining a practical understanding of the field of sustainable design.

Lecture Hours: 48.000

Prerequisite: ARC1126C, ARC2461, IND1200, IND1300

Course User Fee: \$10.00

INP**INP2390 Psychology of Work 3.000 credits**

Applies the understanding of effective human relations to work situations. Personal dynamics for success are also considered. Students will be taught how to influence behavior on the job as they apply their knowledge and interpersonal skills to specific experiences in the workplace.

Lecture Hours: 48.000

INR

INR1949 CO-OP WRK EXP 1: INR 3.000 credits

This is a course designed to continue training in a student's field of study through work experience . Students are graded on the basis of documentatio n of learning acquired as reported by student and employer. Prerequisite: Co-op Department approval. Students will be assigned specific course prefixe s related to their academic major prior to registr ation. All students must contact the Co-operative Education Office to obtain registration approval. (3 hr. lecture)
Lecture Hours: 48.000

INR2002 International Relations 3.000 credits

The nature of international relations, the causes of leading international problems, foreign policies of world powers, international political organizations, and the origins of war and peace in the international arena.
Lecture Hours: 48.000

INR2440 International Law and Organization 3.000 credits

International law and problems in world politics; a review of man's attempt to control international politics through international law and organizations, including the League of Nations, the United Nations, NATO, and the European Union. A prior course in History or Social Science is desirable. Offered second semester. Given in English. Offered through Overseas Study Program.
Lecture Hours: 48.000

INR2949 CO-OP WRK EXP 2: INR 3.000 credits

This is a course designed to continue training in a student's field of study through work experience . Students are graded on the basis of documentatio n of learning acquired as reported by student and employer. Prerequisite: Co-op Department approval and completion of 1949 Co-op work experience. Stud ents will be assigned specific course prefixes rel ated to their academic major prior to registration . All students must contact the Co-operative Educa tion Office to obtain registration approval. (3 hr . lecture)
Lecture Hours: 48.000

INS**INS2200 Introduction to Conflict Resolution and Peace Studies 3.000 credits**

Introduction to Conflict Resolution and Peace Studies is an interdisciplinary course exploring foundational concepts, methodologies, and practical approaches to interstate and intra-state conflicts and how to tackle them. The course has no specific prerequisites and will appeal to a broad audience as it analyzes a broad spectrum of perspectives in the field of conflict and peace studies. By contextualizing the challenge and transmitting the knowledge of peace in normative, empirical and historical frameworks, the course will equip students with a multi-dimensional conceptual toolbox to analyze it while inspiring them to create a more equitable and harmonious world.

INS2300 Global Environmental Politics 3.000 credits

Global Environmental Politics is an interdisciplinary course exploring the evolving relationship between humanity and the environment from diverse conceptual, normative, and empirical perspectives. The course has no specific prerequisites and will appeal to a broad audience as it analyzes a broad spectrum of perspectives in the field. The course will explore the most pressing contemporary environmental challenges, including climate change, loss of biodiversity, desertification, trade in dangerous waste, and over-exploitation of natural resources as well as various political and policy responses to deal with them.

INS2310 Religion and Global Politics 3.000 credits

The interdisciplinary course introduces students to the increasingly visible role of religious identity in international and global politics. It offers a macro-level overview of the intersection of religion and politics in a historical context and sheds light on the intertwined histories and trajectories connecting them. It explores how this nexus has been shaped by belief and value systems as well as the historical socio-economic and political circumstances and conditions. Among the themes addressed in the course are religious diversity and human rights, interfaith dialogue and conflict resolution, and religion in the age of globalization.
Lecture Hours: 48.000

INT

INT1000 INTERPRETING 1 3.000 credits

Provide an overview of the occupational career of interpreting for the deaf. Included are the role of the interpreter, Code of Ethics issues and evaluation systems for determining competency. Also covered are the various settings for interpreting and interpreting for special populations such as deaf/blind, oral and MLS. Prerequisite: SPA 1381 (3 hr. lecture)

Lecture Hours: 48.000

INT1202 INTERPRETING 3 3.000 credits

In-depth discussion and application of techniques and principles for interpreting situations in legal, medical, oral and deaf/blind. Prerequisite: EHD 1401, SPA 2382C. A.S. degree credit only (3 hr. lecture)

Lecture Hours: 48.000

INT1240 voice to sign interp 3.000 credits

In-depth discussion and application of techniques and principles for interpreting situations in educational, social service, free-lance interpreting and the business aspects of interpreting. Prerequisites: ASL 2160C, INT 1000. (3 hr. lecture)

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

INT1400 ED INTERPRETING 3.000 credits

Course decals specifically with interpreting in educational settings. Content includes sign language systems, the Educational Code of Ethics and grammatical features of English and American Sign Language. Prerequisite: EHD 1400 Interpreting 1 and SPA 2333 Intro to Sign Language Systems.

Lecture Hours: 48.000

INT1480 INTERP SPEC SET&POPU 3.000 credits

The course examines various settings in which interpreters work. These include social service and rehabilitation, employment-related, mental health and substance abuse treatment, religious, performing arts legal and other settings. Also considered are specific deaf and hard of hearing consumers who present unique challenges for interpreters such as oral deaf persons, people who are both deaf and blind and those who would be classified as having minimal language skills (MLS). The course includes lecture and skill building opportunities. Prerequisites: EHD 1400, SPA 2382. (3 hr. lecture)

Lecture Hours: 48.000

INT1941 INTRPRTNG INTRNSHP P 3.000 credits

The course includes field observation and supervised practical interpreting experience in a one-to-one interpreting situation in the community. The student will be assigned to a practicing interpreter who will act as a mentor for the duration of the internship. A minimum of 130 hours will be spent at the host site and an additional five hours in consultation with college staff and the interpreter/mentor. Prerequisites: EHD 1402, 1409. (3 hr. lecture)

Lecture Hours: 48.000

IPM**IPM2112 Principles of Entomology 3.000 credits**

This is an introductory course on the principles of entomology. Students will learn to identify characteristics of arthropods, the insect orders, and the growth cycle of insects. Students will also address insect pests specific to South Florida and methods to responsibly manage plant pests.

Lecture Hours: 48.000

IPM2301 Pesticide Applications 3.000 credits

Students will learn government regulations with regards to pesticide mixing and application, and safety equipment. Preparation for the restricted use applicators license exam will be covered.

Lecture Hours: 48.000

IPM2635 Introduction to Plant Pathology 3.000 credits

The students will learn to identify diseases that affect plants and management practices for different types of plant diseases. Environmental factors contributing to a plant's susceptibility to a particular disease will also be discussed. Methods of prevention, eradication, and control will be given for each specific disease.

Lecture Hours: 48.000

ISM**ISM4011 Management Information Systems 3.000 credits**

The student will use information technology software to assist in making decisions of a business nature. The course will examine the use of computer systems and information technology and their applications to make more effective business decisions. The course will include the latest terminology, techniques and applications of information systems in a business organization. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$90.00

ISS

ISS1120 The Social Environment 3.000 credits

Lecture Hours: 48.000

ISS1161 The Individual in Society 3.000 credits

Lecture Hours: 48.000

ISS1301 Intro. to Social Research 3.000 credits

This course is a general introduction to research methodology in the Social Sciences, paying particular attention to research design, data collection and data analysis. This is a course in Applied Social Sciences that will provide students with a survey understanding of social scientific research. This course will include examination of empirical research, including literature reviews, theory, methodology, data collection, data analysis and presentation of results. (3 hr. lecture)
Lecture Hours: 48.000

ISS2270 Multicultural Communications and Relations 3.000 credits

This course uses an interdisciplinary approach to examine the complex interactions among ethnicity, race, gender, age, and class, as well as other ways in which we differ as they pertain to shaping personal awareness, understanding, and skills that will allow them to interact more effectively with diverse populations, age, groups, and lifestyles and to think through and value human diversity. This course has an overriding principle based on the concept of human rights. (3 hr. lecture)
Lecture Hours: 48.000

ITA

ITA1120 Elementary Italian 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Italian. Note: Students must pass this course with a C or better to continue to ITA1121.
Lecture Hours: 64.000
Course User Fee: \$15.00

ITA1121 Elementary Italian 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Italian. Note: Students must pass this course with a C or better to continue to ITA2220.
Lecture Hours: 64.000
Prerequisite: ITA1120
Course User Fee: \$15.00

ITA2220 Intermediate Italian 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of Italian through a systematic review (using an integrated, multimedia approach) of reading, grammar, and writing skills with emphasis on oral and written communication.
Lecture Hours: 64.000
Prerequisite: ITA1121
Course User Fee: \$15.00

ITA2221 Intermediate Italian 2 4.000 credits

This is a continuation of Intermediate Italian 1. Students will learn to understand, speak, read, and write Italian. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.
Lecture Hours: 64.000
Prerequisite: ITA2220
Course User Fee: \$15.00

JOU

JOU1100 BASIC REPORTING 3.000 credits

A course in journalistic writing in which students will master the elements of reporting with an emphasis on the modern news story. They will learn to recognize elements of news, style, structure of news stories, news sources, and the mechanics of newspaper production. Required of all journalism majors.
Lecture Hours: 48.000

JOU1946 JOURNALISM INTRNSHIP 1.000 credits

Lab Hours: 48.000 - 96.000

JOU1949 CO-OP WRK EXP 1: JOU 3.000 credits

This course is designed to provide training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Prerequisite: Co-Op Department approval. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Cooperative Education Office to obtain registration approval. (3 hr. lecture)
Lecture Hours: 48.000

JOU2200 EDITING & MAKEUP 3.000 credits

Lecture Hours: 48.000

JPN

JPN1120 Elementary Japanese 1 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Japanese. Note: Students must pass this course with a C or better to continue to JPN1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

JPN1121 ELEM JAPANESE 2 4.000 credits

In this course, the student will be able to develop socially and culturally appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Japanese. Note: students must pass this course with a C or better to continue to 2220.

Lecture Hours: 64.000

Prerequisite: JPN1120

Course User Fee: \$15.00

JPN2220 Intermediate Japanese 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of Japanese through a systematic review (using an integrated, multimedia approach) of reading, grammar, and writing skills with emphasis on oral and written communication.

Lecture Hours: 64.000

Prerequisite: JPN1121

Course User Fee: \$15.00

JPN2221 Intermediate Japanese 2 4.000 credits

This is a continuation of Intermediate Japanese 1. Students will learn to understand, speak, read, and write Japanese. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: JPN2220

Course User Fee: \$15.00

LAE**LAE4211 Methods and Resources for Literacy Development in Young Children 3.000 credits**

The student will utilize a variety of assessment tools to measure and evaluate literacy in a K-3 setting. The student will learn to create and administer informal assessments, evaluate results, and differentiate instruction encompassing the reading components.

Lecture Hours: 48.000

Prerequisite: RED3009, Corequisite: EEC4268

Course User Fee: \$18.00

LAH**LAH2021 Colonial Latin America 3.000 credits**

This course covers the history of Latin America, from the pre-Columbian civilizations of the region, to the Wars of Independence. Students will learn about the development of the political, social, economic and cultural structures of colonial Latin America to 1825.

Lecture Hours: 48.000

LAH2023 HISTORY OF CARIBBEAN 3.000 credits

This course introduces students to the economic social, political, and cultural history of the Caribbean and its peoples. Students will learn of the changes and continuities that have affected Caribbean development. (3 hr. lecture)

Lecture Hours: 48.000

LAH2025 History of Cuba 3.000 credits

Historical analysis of the development of Cuban society, its culture and institutions.

Lecture Hours: 48.000

LDE**LDE2000 PLANTING DESIGN 1 4.000 credits**

Basic principles of design, on-the-job sketching and plan presentation as done by nurseries.

Prerequisites: BIO 215 or permission of the instructor.

Lecture Hours: 32.000, Lab Hours: 64.000

LDE2310 Irrigation Design & Maintenance 3.000 credits

The students will learn the design, maintenance, and installation of nursery and landscape irrigation systems. All types of nursery systems will be covered including field, shade house, and greenhouse. Both sprinkle and low volume drip systems will be surveyed for appropriateness in nursery and landscape uses. Includes occasional weekend hands-on activities.

Lecture Hours: 48.000

LIN**LIN2011 Introduction to Linguistics 3.000 credits**

The course provides an exploration of basic linguistic concepts including phonology, morphology, syntax, pragmatics and semantics, how a language is organized and functions, and will establish a connection between Linguistics and Translation Theory. The course will also introduce students to the study of how cultural and social factors affect human communication.

Lecture Hours: 48.000

LIN2606 INTRO TO SOCIOLINGUI 3.000 credits

This course introduces students to the study of how social and cultural factors affect human communication. Topics such as language attitudes and policies, dialects vs. standards, class variation, and race and gender will be discussed. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

LIS**LIS2004 Digital Research Strategies 1.000 credits**

LIS 2004, Digital Research Strategies, equips students with information literacy skills to locate, evaluate, and ethically use information in an evolving digital landscape. Students will apply critical thinking and research techniques to explore traditional and emerging research tools. (1 hr. lecture)

Lecture Hours: 16.000

LIT**LIT2000 Introduction to Literature 3.000 credits**

In this course, students will be assigned readings representative of a broad range of literary genres and cultures. These readings will cover a variety of literary movements and historical eras. The readings will include selections from the western canon. Written analysis of literary works may be required. Students will be provided with opportunities to practice critical interpretation. Student learning outcomes: students will identify a variety of literary movements, historical eras, and/or cultural contexts; and students will demonstrate critical thinking and analytical skills. Writing Intensive Course.

Lecture Hours: 48.000

Prerequisite: ENC1101

LIT2090 Contemporary Literature 3.000 credits

This course is a survey of contemporary prose and poetry.

Lecture Hours: 48.000

LIT2120 A Survey of World Literature 3.000 credits

This course examines world literature from the mid-renaissance to the present and includes works from the Western canon. Students will analyze literary works that exemplify human experience, exploring their connections to historical contexts, ideas, and literary developments. Through this study, students will develop critical thinking skills and gain understanding of how literature reflects and influences societies across time periods and societies. This is a writing intensive course.

Lecture Hours: 48.000

Prerequisite: ENC1102or equivalent.

LIT2480 Issues in Literature & Culture 3.000 credits

Students will learn about literature as a socio-cultural response by writers to the world in which they live; they will connect literary texts to cultural issues through completion of oral and written assignments, critical analysis and practical investigation. Fulfills Gordon Rule writing requirement. Prerequisite: ENC 1102. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: ENC1102ENC1102

MAC**MAC1105 College Algebra 3.000 credits**

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications. Student learning outcomes: students will solve an equation or an inequality using an appropriate technique; students will define and describe functions, their properties, and graphs; students will manipulate functions to simplify expressions and find new functions; students will use transformations to write an equation for a function and to graph a function; and students will model and solve real world problems using functions. Prerequisite: MAT1033 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAT1033

Course User Fee: \$18.00

MAC1105L College Algebra Co-Requisite Lab 1.000 credits

Students will remediate high school mathematics knowledge to prepare themselves for college algebra concepts, while reinforcing concepts being learned in college algebra and applying their knowledge to real world applications. In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of equations, functions, and their graphs. Emphasis will be placed on quadratic, exponential, and logarithmic functions. Topics will include solving equations and inequalities, definition and properties of a function, domain and range, transformations of graphs, operations on functions, composite and inverse functions, basic polynomial and rational functions, exponential and logarithmic functions, and applications.

Lab Hours: 32.000

Corequisite: MAC1105

MAC1106 Integrated College and Precalculus Algebra 5.000 credits

The student will learn to analyze linear, quadratic, polynomial, rational, radical, absolute value, composite, inverse, piecewise, exponential, and logarithmic functions, conic sections, systems of equations/inequalities, matrices and determinants, sequences & series, the Binomial Theorem, and applications of mathematical modeling including exponential growth and decay. Prerequisite: MAT1033 with a grade of "B" or better.

Computational Course.

Lecture Hours: 80.000

Prerequisite: MAT1033 or appropriate placement scores.

Course User Fee: \$10.00

MAC1114 Trigonometry 3.000 credits

This course will cover the analysis and graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, work with vectors, graph both polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC 1105 or MAC 1106 with a grade of "C" or better. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

MAC1140 Pre-Calculus Algebra 3.000 credits

This course will cover properties and graphs of algebraic, exponential, and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices, and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, and corresponding applications and modeling. Prerequisite: MAC 1105 with a grade of "C" or better or equivalent. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105

Course User Fee: \$18.00

MAC1147 Pre-Calculus Algebra and Trigonometry 5.000 credits

The course will cover the properties and graphs of algebraic, exponential and logarithmic functions, piecewise-defined functions, the Fundamental Theorem of Algebra, solutions of polynomial equations, conic sections, systems of equations, matrices and determinants, arithmetic and geometric sequences and series, the Binomial Theorem, graph of trigonometric functions and inverse trigonometric functions, the fundamental trigonometric identities, solutions to conditional trigonometric equations, solutions for both right and oblique triangles, operations on complex numbers in trigonometric form, vectors, graphs of polar and parametric equations, and solutions of applications and modeling problems related to the above topics. Prerequisite: MAC1105 with a grade of "C" or better or departmental permission.

Computational Course.

Lecture Hours: 80.000

Prerequisite: MAC1105

Course User Fee: \$18.00

MAC2233 Business Calculus 3.000 credits

This course introduces the basic concepts of differential and integral calculus for students majoring in business administration and related fields. Topics include limits, continuity, differentiation and integration of polynomials, logarithmic and exponential functions with applications to business, economics, and the life sciences. Prerequisite: MAC 1105 or MAC1106 with a grade of C or better or departmental permission. Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC1105 OR MAC1106

Course User Fee: \$18.00

**MAC2311 Calculus and Analytical Geometry 1
5.000 credits**

In this course, students will develop problem solving skills, critical thinking, computational proficiency, and contextual fluency through the study of limits, derivatives, and definite and indefinite integrals of functions of one variable, including algebraic, exponential, logarithmic, and trigonometric functions, and applications. Topics will include limits, continuity, differentiation and rates of change, optimization, curve sketching, and introduction to integration and area. Student learning outcomes: students will calculate a limit, derivative, or integral using appropriate techniques; students will determine the continuity and differentiability of a function; students will use limits and derivatives to analyze relationships between the equation of a function and its graph; students will apply differentiation techniques to model and solve real world problems; and students will use integrals and the fundamental Theorem of Calculus to analyze the relationship between the integral of a function and the related area. Prerequisites: MAC1106 and MAC1114, or MAC1114 and MAC1140, or MAC1147 with a grade of "C" or better or departmental permission. Computational Course. Lecture Hours: 80.000
Prerequisite: MAC1147 or MAC1114 and MAC1140 both with a minimum grade of C, or MAC1106 and MAC114 both with a minimum grade of C
Course User Fee: \$18.00

**MAC2312 Calculus and Analytical Geometry 2
4.000 credits**

This second semester calculus course, the student will examine techniques of integration, applications of integration in STEM subjects, sequences and series, representation of functions by Taylor series, parametric equations, calculus in polar coordinates, and improper integrals. Prerequisite: MAC 2311 with a grade of "C" or better. Computational Course. Lecture Hours: 64.000
Prerequisite: MAC2311
Course User Fee: \$18.00

**MAC2313 Calculus and Analytic Geometry 3
4.000 credits**

The student will examine topics in analytic geometry in three dimensions, vectors and vector functions, curves and surfaces in three-space, partial differentiation and applications to optimization, multiple integrals and their applications, vector fields, line integrals and surface integrals, Green's Theorem, and the Divergence and Stokes' theorems. Prerequisite: MAC 2312 with a grade of "C" or better. Computational Course. Lecture Hours: 64.000
Prerequisite: MAC2312
Course User Fee: \$18.00

MAD**MAD1100 Discrete Mathematics for Computer Science 3.000 credits**

This course introduces students to the principles of discrete mathematics that apply to computer science. Topics include set theory, logic, Boolean algebra, number theory, vectors and matrices, combinatorics, probability, relations, functions, and basic graph theory. Computational Course. Lecture Hours: 48.000
Prerequisite: MAC1105
Course User Fee: \$18.00

MAD2104 Discrete Mathematics 3.000 credits

This course introduces the student to the principles of discrete mathematics that apply to computer science. The student will examine set theory, logic, Boolean algebra, number theory, vectors and matrices, combinatorics, probability, relations, functions, and basic graph theory. Computational Course. Lecture Hours: 48.000
Prerequisite: MAC1106 OR MAC1140
Course User Fee: \$18.00

MAD3107 Discrete Structures 3.000 credits

Topics include sets, logic, switching circuits, Boolean Algebra, combinatorics, probability, mathematical proofs, mathematical induction, functions, relations, and graph theory. Credit is not also given for MAD 2104. Lecture Hours: 48.000
Prerequisite: MAC2312

MAE

MAE3951 Problem-Based Learning in Mathematics Education 2.000 credits

The teacher candidate will learn and apply the principles of project-based learning by designing and implementing projects to explore real-world problems, questions, and challenges in the field of education. The student will develop their technology, critical, creative, and communication skills by producing products to share their findings and proposed solutions. (2-hour lecture; this course requires approved clinical hours).

Lecture Hours: 32.000

Prerequisite: Departmental Approval Required.

MAE4360 Methods of Teaching Mathematics 3.000 credits

The teacher candidate will utilize theory and educational neuroscience research in developing knowledge and pedagogy essential for K-12 mathematics instruction which accommodates the needs of all learners. The problem-solving approach will be used to design, implement, and assess mathematics instruction and curriculum. Fifteen hours of clinical experience in a mathematics classroom are required.

Lecture Hours: 48.000

Prerequisite: EDG3321, Pre/Corequisite: EDF4430

Course User Fee: \$25.00

MAE4940 Advanced Topics in Mathematics Education Practicum 3.000 credits

The teacher candidate will plan and implement mathematics instruction that meets the needs of all learners. The student will utilize action research methodology, assessment principles, educational neuroscience research, and best practices to identify and address issues related to mathematics learning in grades 6-12. Sixty (60) hours of clinical experience in a 6-12 mathematics classroom are required.

Lecture Hours: 48.000

Prerequisite: MAE4360, Select 6 credits (2 courses) from the following course option: MAD3103, MAS3105, MAS3301, MAS4203, MHF4404, MTG3214.

MAE4942 Seminar in Mathematics Education 3.000 credits

The teacher candidate will engage in professional dialogue and explore professional learning opportunities related to teaching in the secondary mathematics setting. This seminar course is taken in conjunction with a full time, supervised teaching experience and provides an opportunity to examine and reflect on the daily experiences of becoming a highly effective teacher.

Lecture Hours: 48.000

Prerequisite: All Coursework, State Examinations, and E-Portfolio completed and passed., Corequisite: MAE4945

Course User Fee: \$18.00

MAE4945 Internship in Secondary Mathematics Education 9.000 credits

The teacher candidate will complete a full time, supervised teaching experience. The student will learn and experience all of the educational and professional responsibilities common to mathematics teachers.

Internship Hours: 432.000

Prerequisite: All coursework and state exams completed and passed, Corequisite: MAE4942

MAN**MAN2021 Principles of Management 3.000 credits**

The student will learn to analyze the major functions of management. Emphasis is placed on learning how to manage organizations for excellence in both performance and employee satisfaction. Major topics include but are not limited to business ethics and social responsibility, strategic and operational planning, decision making, organization structure and behavior, managing groups and teams, communication and information technology.

Lecture Hours: 48.000

Course User Fee: \$15.00

MAN2300 Human Resources Management 3.000 credits

Reviews how the personnel/human resources department contributes to overall planning and profitability of an organization. Major topics include typical personnel functions: recruitment and selection, training, performance appraisal, job analysis, and compensation and benefits administration. Class discussions will focus on changing value systems in the work force and the resulting challenges for managers.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAN2604 Managing in a Multi-Cultural Environment 3.000 credits

This course will introduce opportunities and problems encountered by managers operating in a diverse environment either within or outside their home country's borders. Discussions will cover the environment of multinational management as well as planning, organizing, staffing, leading and controlling in both domestic and multinational companies. Current events and cultural issues that significantly affect international business will also be examined.

Lecture Hours: 48.000

MAN3025 Organization Management 3.000 credits

The student will learn the major functions of supervision and management including the functions of planning, staffing, directing and controlling. Emphasis is placed on learning how to communicate more effectively with employees and how to increase leadership effectiveness. Major topics include goal setting and attainment, organizational structure, decision-making, strategic planning, managing stress, and ethical behavior and ethical role modeling. Cases that present the student with opportunities to make supervisory and management decisions, along with timely feedback on their effectiveness, will be used. Must pass this course with a grade of C or higher.

Lecture Hours: 48.000

Prerequisite: MAN2021

Course User Fee: \$60.00

MAN3065 Business Ethics 3.000 credits

In this course the student will learn how personal values and ethics influence the decision-making capacity of the manager within an organization. The student will learn about ethical theories and the roles they play in the social and corporate behavior of an organization. Issues such as diversity in the workplace, intuition, technology, and the global environment of business, will be included in case examples. The student will acquire an understanding of how an organization can institutionalize its ethical system. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3240 Organizational Behavior 3.000 credits

The student will learn about social behavioral sciences that can be applied to supervision and management. Major topics include motivation, conflict, corrective actions and rewards, job related stress, organizational dynamics, the evolving global environment, and the responsibility to stakeholders and the planetary environment. The student organizational behavior and how to integrate behavioral concepts will be reviewed in an effective managerial decision-making process. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: MAN2021

Course User Fee: \$60.00

MAN3301 Human Resource Management 3.000 credits

The student will learn the functions of Human Resource Management including human resource planning, strategic development of human resources, recruitment techniques, selection and hiring processes, compensation systems, development of policy and procedures for effective and ethical human resource management, performance review and evaluation systems, working effectively with organized labor, retention of employees, and current issues in human resource management. The course will also include an exploration of human resources within the global business environment of a boundless organization. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAN3322 Human Resources Information Systems 3.000 credits

This course examines the role of human resources information system (HRIS) in today's organizations and human resources departments. The student will address topics such as human resource information systems design, acquisition, and implementation. The role of these systems in talent acquisition and management is also examined. Prerequisites:

MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN3504 Production Operations and Logistics Management 3.000 credits

This course examines the integration of production operations and logistics management and how they enable an organization to compete successfully in business. Students will learn the relationships that exist between operations and the supply chain, including operations and supply chain strategies, business processes, project management, product design and development, and inventory management. This course uses case study methodology. Prerequisite: TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021, TRA1154 with a minimum grade of C

MAN3506 Operations Management 3.000 credits

The course emphasizes the application of operational decision-making techniques to improve process, productivity, and the effective utilization of resources within organizations. Students will learn to recognize the trade-offs associated with operations management decisions and their effect on resource allocation. Topics include production processes, operations strategies, quantitative techniques, quality, performance, capacity planning, efficiency, forecasting, resource management, statistical process control, project management, and supply chains. Prerequisite: QMB2100 or STA2023.

Lecture Hours: 48.000

Prerequisite: QMB2100 OR STA2023

MAN3554 Safety and Risk Management 1.000 credits

This course will focus on safety and risk management with emphasis on how it applies to supply chain management. Students will learn the processes used to characterize and manage risk, as well as maintain a safe operating environment with the protection of personnel, assets, and services. Hazard characteristics of products, appropriate modes of transport, carrier selection and qualifications, packaging and container specifications, driver training requirements, financial issues, security precautions, warehousing risk, OSHA, hazardous materials handling, and customer assessment will be addressed in this course.

Lecture Hours: 16.000

MAN3562 Purchasing, Inventory and Warehouse Management 3.000 credits

This course explores the integration of production operations and logistics management. It will also examine purchasing, Inventory and Warehouse Management concepts and theory. Special emphasis is placed on the relationships that exist between operations and the supply chain to include operations and supply chain strategies, business processes, project management, product design and development, inventory management and effective warehouse and stores management.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3577 Procurement for Major Projects 3.000 credits

Students will learn the principles of procurement for major projects concepts and theory, including advanced purchasing and cost savings techniques, contracts administration from award to completion, strategic procurement management.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3578 Global Procurement Management 3.000 credits

Students will learn Global Procurement Management concepts and theory, including a good understanding in the differences in cultures, law, currency, communications and buying from foreign suppliers, international procurement is a challenge, buying internationally and to help support international procurement efforts.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3583 Project Management 3.000 credits

Students will learn project management concepts and theory, including attributes for a project life cycle, global project management, benefits of project management, project management environments, planning and managing risk, project planning controls, and terminating a project.

Lecture Hours: 48.000

MAN3731 Assessing and Managing Project Risk 3.000 credits

This course explores project uncertainty, and how to manage risk to keep the project on track and meet project goals. Project risk management is a vital part of all projects, which requires a purposeful strategy to avoid any setbacks. In this course there will be an examination of what the risk management process involves through the identification, assessment, and response to project risks.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN3786 Sustainable Enterprise Planning 3.000 credits

Students will learn the assessment tools, design and construction considerations, and operating planning requirements for sustainable enterprises. Students will also learn the ecological and economic benefits of sustainability practices, including construction, operations, supply chain decisions, recycling, reusing, and reconditioning to preserve the environment by increasing revenues. Best practices, case studies, evolving trends and experimental efforts are also covered. Prerequisite: MAN2021.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN3888 Project Leadership 3.000 credits

Students will learn the fundamentals of management vs. leadership, monitoring/controlling, delegation, personal traits, communication, and people management when in a leadership role. The specifics include techniques and strategies using the company's vision and project charter needed to effectively manage, lead and direct a team to ensure reaching project objectives. As part of this process students will determine their personal leadership traits and approach.

Lecture Hours: 48.000, Lab Hours: 3.000

Course User Fee: \$62.76

MAN4113 Managing in a Multifaceted Environment 3.000 credits

Managing in a Multifaceted Environment is a thorough course that offers a management perspective on making the most of the various qualities of today's workforce. This course explores the value that comes from human capital and other distinguishing factors in the modern workplace. Furthermore, it delves into new leadership styles as solutions for effectively managing and leading people. Students will gain insights into utilizing the strengths of the workforce and creating a dynamic work environment.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4120 Leadership Challenges and Supervision 3.000 credits

The student will learn to analyze leadership theories and will acquire an awareness of the dynamics of supervisory and managerial decision-making.

Emphasis will be placed on team building, crisis management, social and environmental responsibility, developing and communicating a vision, developing a full set of managerial and leadership skills. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair, MAN4894

Course User Fee: \$70.00

MAN4162 Customer Relations for Managers 3.000 credits

Students will learn Customer Relations for Managers skills by exploring the dynamics of building solid and lasting relationships with customers. Topics will include doing business in a global environment, cultural diversity, the diversity of customs and global etiquette, negotiation tactics, global promotional tactics, and acceptable professional and corporate behaviors in a global business environment. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair., MAN4894

Course User Fee: \$70.00

MAN4330 Compensation Management 3.000 credits

This course is a study of the strategic use of compensation system for the purposes of attracting, retaining, and motivating a competitive workforce.

The student will address topics such as designing compensation systems, bases for pay, employee benefit programs, laws affecting compensation practices, and compensation challenges for various employee groups. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4335 Employee Benefit Planning 3.000 credits

This course focuses on the both wage and nonwage related benefits made available to employees by the firm and various related social and governmental programs. The student will examine various state and federal laws regarding employee benefits planning. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4350 Professional Development 1.000 credits

Students will learn to implement basic business etiquette, work habits, and career planning strategies required for successful transition into the business profession. Focus is on setting professional goals, preparing for a job search, networking, finding job leads, applying for jobs, interviewing for jobs, following up, and evaluating job offers. This course must be met with a grade of "C" or higher.

Prerequisite: MAN2021.

Lecture Hours: 16.000

Prerequisite: MAN2021

MAN4352 Effective Employee Training 3.000 credits

This course focuses on professional development activities as performed by human resources specialists or organizational specialists. The student will analyze the benefits of employee training, establish employee development programs, and address the theories, issues, practices and problems regarding employee training and development.

Prerequisites: MAN3025, MAN3301

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4361 Organizational Staffing 3.000 credits

This course provides the student with an overview of the staffing function in organizations, including the topics of job analysis, forecasting, recruitment, selection, retention and turnover. It serves as an introductory course for the prospective human resources manager and as a survey of responsibility and activities of any manager with supervisory responsibilities. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4402 Employment Law and Regulation 3.000 credits

This course analyzes the federal and state regulation of the employment relationship, including wage and hour laws, EEO, and Affirmative Action programs. The student will address human resource issues such as employee benefits, insurance, workers' compensation, safety, health, employees' personal rights and collective bargaining legislation. Prerequisites: MAN3025, MAN3301.

Lecture Hours: 48.000

Prerequisite: MAN3025, MAN3301

Course User Fee: \$35.00

MAN4520 Quality Management 3.000 credits

This course provides an understanding of various theories of quality management (QM). Students will learn how organizations can develop excellence through the adoption of continuous improvement and process management. The course analyzes and uses various process management techniques, continuous improvement tools, and strategies to improve quality. The conceptual and analytical skills acquired in this course enable students to provide leadership in shaping a culture for quality within an organization and determining the effectiveness of quality initiatives such as Total QM, Six Sigma, Process Capability, Process Control, and Customer Relationships. Prerequisite: MAN2021.

Lecture Hours: 48.000

Prerequisite: MAN2021

MAN4523 Production Information Systems 3.000 credits

This course presents the fundamental aspects of computer technology required by the systems that provide data to, and derive information from, production in manufacturing. Students will learn the techniques to organize, store, manipulate data, report, derive and analyze production information, basics networking used in production, as well as various forms of information systems.

Lecture Hours: 48.000

MAN4552 Supply Chain Analytics & Decision Making 3.000 credits

Through this course students will focus on the science of learning from data to support decision making. The objective of this course is to explore and apply basic statistical concepts and procedures that are used to collect, analyze, summarize, and report data in a supply chain management role.

Course topics covered include methods for: collecting, analyzing, summarizing data; making statistical inferences about populations and exploring the relationship between variables. The course introduces analytical tools and techniques that provide a cause and effect understanding linking operational plans with corporate objectives.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4570 Purchasing for Industry 3.000 credits

Students will learn purchasing for industry concepts and theory, including negotiation skills: mastering the art of deal making strategies, contracts: bidding, evaluation, negotiation and award, purchasing management, tendering and supplier selection, vendor qualification: managing performance and contract compliance.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4593 Supply Chain Management Theory & Methodology 2.000 credits

This course presents a range of advanced topics in integrated logistics and supply chain management. Students will learn new theoretical and methodological developments in the field of supply chain. Specific topics vary depending upon current industry developments and will incorporate the use of speakers in the classroom. Prerequisites:

MAN2021, MAN3583, MAN4523.

Lecture Hours: 32.000

Prerequisite: MAN2021, MAN3583, MAN4523

MAN4597 Global Supply Chain Management 3.000 credits

This course presents an overview of the management of sourcing, operations, and distribution processes along a supply chain in domestic and international markets. Students will learn how firms gain a competitive advantage through supply chain activities. Topics include: supply chain network design, purchasing, forecasting, inventory management, globalization and outsourcing, logistics, and information technology. Prerequisite: MAN3506.

Lecture Hours: 48.000

Prerequisite: MAN2021, MAN3506

MAN4719 Challenges in the Digital Supply Chain 3.000 credits

Through this course, students will analyze the social, ethical, and legal issues in supply chain management. Students will examine the importance of fairness, accountability, and transparency of data policies in various supply chain operations. Various issues governing the collection, storage, access, use, and sharing of data will also be explored.

Prerequisites: MAN3506, QMB2100 or STA2023.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4720 Strategic Management Decision Making 3.000 credits

The student will learn the designing, planning, and implementation of strategic decision making in a business organization by identifying problems and designing possible solutions, formulating plans, goals, and feedback mechanisms. Through case studies, the student will conduct internal and external assessments to analyze effective strategic choices for companies. Must pass course with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: Senior status or permission of department chair., MAN4894

Course User Fee: \$70.00

MAN4732 Business Intelligence for Supply Chain 3.000 credits

This course will enable students to synthesize their learning from program coursework. The global context within which firms operate will provide the foundation for understanding global supply chain strategy, from market entry through to the integration of demand management and supply management processes to meet corporate objectives. The course will utilize exercises and a simulation to allow students to analyze and synthesize program coursework. The course will help students understand how to best use program knowledge to maximize value creation for employers and in their own career.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4741 Change & Innovation Management 3.000 credits

In this course, the student will study the concepts and strategies for change management and diffusion of innovation. Topics covered include, but are not limited to, implementing change, becoming a learning organization, innovation process management, and technology forecasting.

Lecture Hours: 48.000

Course User Fee: \$35.00

MAN4887 Project Planning and Control Systems for Supply Chain Management 3.000 credits

Students will learn about the complex supply-chain transformation that is required to manage resources from many different departments. Additionally, an exploration of how supply-chain project managers ensure internal and external stakeholder alignment, mitigate large amounts of risk, and implement communication, risk mitigation, and change management plans to ensure a successful project. Finally, the student will gain an integrated view of supply-chain transformation that incorporates elements of change management, test plan development, project management techniques, and establishing effective project management teams.

Lecture Hours: 48.000

Course User Fee: \$62.76

MAN4894 Applied Case Studies in Management 3.000 credits

This course emphasizes the identification of strategic management issues, evaluation of strategic goals, internal and external environment of the organization, as well as differentiating, categorizing and assessing strategic choices. Students will learn to apply strategic management process through strategy formulation, implementation and evaluation utilizing the case study methodology. This course must be met with a grade of "C" or higher.

Lecture Hours: 48.000

Prerequisite: MAN4720, MAN2021

Course User Fee: \$63.00

MAN4900 Capstone Project in Supervision and Management 3.000 credits

Students will apply and integrate the knowledge and skills learned throughout the program by completing a capstone project. Student will identify a business opportunity; conduct an industry feasibility assessment and operational mapping; evaluate market condition; construct a marketing plan, and formulate a financial plan for implementing and managing an existing or new business venture; and evaluate the role of ethic, social, and environmental responsibility within a business. Must pass course with a grade of "C" or higher. Prerequisite: Senior Status or Departmental Permission.

Lecture Hours: 48.000

Prerequisite: Senior Status or Departmental Permission.(3 hr. lecture)

Course User Fee: \$90.00

MAN4940 Field Study and Research 2.000 credits

Students will learn to apply information and skills studied in core and concentration program courses through various special projects, field research, or internships designed for cogitative learning in a student-centered manner which requires the command, analysis, and synthesis of knowledge and skills. Prerequisites: MAN2021, MAN3583, TRA1154.

Lecture Hours: 32.000

Prerequisite: MAN2021MAN3583, TRA1154

MAN4941 Management Internship 3.000 credits

The student will learn management techniques by becoming an employee or intern (on a paid or unpaid basis) at either a not-for-profit or for-profit organization. The student will be required to work at least 144 hours required by the state to earn the credit for the internship. The student will work with their MDC Faculty and Supervising Employer to establish a set of assignments / learning goals that will be achieved during the semester. This course must be met with a grade of "C" or higher.

Prerequisite: Senior Status or Departmental Permission.

Other Hours: 144.000

Prerequisite: Senior status and permission of department chair., Pre/Corequisite: MAN4900

Course User Fee: \$45.00

MAP**MAP2302 Introduction to Differential Equations 3.000 credits**

The student will analyze ordinary differential equations, methods of solution of first order linear and nonlinear equations and applications, homogeneous and non-homogeneous linear equations with constant coefficients, differential operator methods, higher order linear equations; the Laplace transform and its properties, elementary existence theorems, series solutions, numerical solutions of first order equations, initial and boundary value problems, vibrations and waves, and an introduction to autonomous systems. Computational Course. Prerequisite: MAC 2312 with a "C" or better or equivalent.

Lecture Hours: 48.000

Prerequisite: MAC2312

Course User Fee: \$18.00

MAR

MAR1011 Principles of Marketing 3.000 credits

This introductory course emphasizes key concepts and issues underlying the modern practice of marketing. The course provides the student with a clear understanding of marketing's role in the 21st Century and introduces the student to both traditional and contemporary ways of marketing. The student will learn how marketers deliver value by satisfying customer needs and wants, determine which target markets to serve, and decide which goods and services are needed to serve these markets.

Lecture Hours: 48.000

Course User Fee: \$10.00

MAR1440 Fundamentals of Negotiations 3.000 credits

This course introduces the negotiation process and identifying consumer motivations. Students will analyze the various positions of negotiations from gaining leverage to making considerations. Students will examine a long-term sales strategy and how a balanced negotiation plan impacts it. (3 hr. lecture)

Lecture Hours: 48.000, Lab Hours: 5.000

Course User Fee: \$47.44

MAR1502 Sales and Consumer Behavior 3.000 credits

This course introduces the process consumers use in making purchase decisions. Students will analyze how developing a sales personality factors into producing prospects, leads, and ultimately generating sales. Students will identify how both buyer and seller behaviors interact in the sales process. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$47.44

MAR1720 Marketing in a Digital World 3.000 credits

This course provides an introduction to new marketing concepts and the broad spectrum of digital marketing. Students will learn to utilize digital tools, social networks and media, strategies, and tactics to create awareness and to promote products and services to consumers and businesses.

Lecture Hours: 48.000

Course User Fee: \$60.00

MAR1930 Introduction to Salesforce Marketing Cloud 1.000 credits

This course introduces students on how to build and personalize a 1:1 marketing campaign that can help connect and deliver the right message, to the right person, at the right time. Students will learn how to manage the most utilized marketing channels in the industry and discover new marketing tools.

Lecture Hours: 16.000

Course User Fee: \$33.69

MAR1931 Fundamentals to Social Media and Search Engine Marketing 1.000 credits

This course provides an introduction on how to use social media and search engines for marketing and learning how to grow the brand value of companies. Students will be exposed to the latest social media trends and search engine marketing techniques.

Lecture Hours: 16.000

Course User Fee: \$33.69

MAR1932 Email Marketing Fundamentals 1.000 credits

This course introduces email marketing using Mailchimp with powerful marketing tips and techniques that will help students jump forward and build a list of targeted subscribers. Students will learn a variety of applications ranging from creating email lists, marketing campaigns, sending emails and how to read reports and analytics.

Lecture Hours: 16.000

Course User Fee: \$33.69

MAR1933 Fundamentals of Marketing Analytics 1.000 credits

This course provides an introduction on how to build and define a brand architecture and how to measure the impact of marketing efforts on brand value over time. As an introduction to Google Analytics, students will learn how to create an account, setup and navigate the Google Analytics interface.

Lecture Hours: 16.000

Course User Fee: \$33.69

MAR2101 Social Media Marketing 3.000 credits

This course addresses the changes in marketing as a result of social media. The student will explore in-depth the role of social media in marketing. The student will focus on developing an effective marketing plan utilizing social media, with an emphasis on the importance of building an online customer community and developing customer engagement and loyalty.

Lecture Hours: 48.000

MAR2150 International Marketing 3.000 credits

Students will learn the four P's of product, price, place (distribution), and promotion as they relate to a global marketing strategy. The concepts are introduced within the international trade framework, as well as the cultural, social, economic, regulatory, and political environments affecting global marketing efforts.

Lecture Hours: 48.000

MAR2419 Technology in Sales 3.000 credits

This course introduces the various technological tools used in the sales process. Students will explore sales management software, data management, artificial intelligence sales technology, and automated sales functions. Students will analyze the benefits of utilizing technology and the competitive advantage it provides to a sales force. Prerequisites: MAR1440, MKA1160, MAR1502, MKA1022, MKA2024. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MKA1160

Course User Fee: \$47.44

MAR2520 Hispanic Marketing Communications 3.000 credits

Students will learn how firms can formulate strategies to attract the U.S. Hispanic market and its various sub-segments. Emphasis is on strategy formulation and integrated marketing communications, including the use of case studies.

Lecture Hours: 48.000

MAR2703 Marketing Content, Branding and Strategy 3.000 credits

This course provides an introduction into branding and content strategy. Students will explore concepts, such as: integrated marketing communications, consumer psychology, buying motivations, appropriate content, branding, and marketing strategy as it relates to digital marketing. Students will have the opportunity to practice their writing and communications skills in the development of digital marketing content.

Lecture Hours: 48.000

Course User Fee: \$57.00

MAR2704 Marketing Web Analytics 3.000 credits

This course helps students to develop a deep understanding of the digital analytics landscape. The student will learn how to identify which metrics are best to measure web, mobile, social and marketing channels, and how organizations use web analytics to obtain higher profits, improved customer relationships, and measurable value. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$57.00

MAR2935 Fundamentals of Google Analytics 1.000 credits

This course illustrates the advanced analytical tools of Google Analytics to provide students with an understand of how to use data to drive digital business and marketing campaigns. The course will introduce topics ranging from performance measurements, basic metric definition, socio-demo data and website traffic evaluation.

Lecture Hours: 16.000

Course User Fee: \$33.69

MAR2952 Digital Marketing Capstone 3.000 credits

This course provides a real-world approach to designing, implementing, managing and analyzing a digital marketing campaign or strategy. The student will engage in a comprehensive field experience designed to gain insights into the creative processes and demonstrate competence by applying the knowledge and concepts of marketing.

Lecture Hours: 48.000

MAR3325 Digital Advertising 3.000 credits

In this course students learn about the principles and practices of digital advertising, as well as hands-on experience with the tools and platforms used in the industry. Students will learn about search engine advertising, social media advertising, e-commerce advertising, video advertising and effective implementation of advertising campaigns. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAR3803 Marketing for Managers 3.000 credits

Students will learn how managers implement the marketing elements within a strategic planning framework. Emphasis is on decision-making, and the development and execution of marketing strategies related to product and brand development, channels of distribution, pricing, and promotional efforts under varying marketplace conditions. This course incorporates the use of case studies. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$45.00

MAR4203 Supply Chain Marketing 3.000 credits

Students will learn the management of traditional and emerging marketing channels, with emphasis on legal, economic, and ethical considerations in wholesale and retail inventory control, raw goods, finished product transportation and relationship management. Prerequisite: MAN3506. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MAN3506

MAR4233 Social Media Marketing Application 3.000 credits

This course introduces students to social media and marketing functions and strategies that are essential to consumer involvement, community engagement, and customer relationship management. Students will learn about the dynamics of social marketing, trends in new technology, and integrated marketing through online platforms. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAR4327 Search Engine Optimization (SEO) and Search Engine Marketing (SEM) 3.000 credits

In this course students will learn how to optimize websites for search engines and search engine ranking. Students will learn how to use social media to complement and enhance traditional SEO strategies. Students will apply their knowledge of search and search optimization tools and technics using leading industry certification. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAR4674 Marketing Analytics 3.000 credits

A study of the metrics and systems needed to receive a return on every sale and marketing investment made. This course focuses on common calculations used in business to determine market share, penetration, brand/category development, awareness, margin, price per unit, break-even. The focus of this course is to enable students to take the vast amount of "data" available within companies and turn that data into "information" making them that much more valuable to potential employers.

Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAR4721 Digital Marketing Strategy 3.000 credits

Students will learn the process of creating a plan and developing a strategy for promoting a business or organization using digital channels. Students will learn how to apply their knowledge and skills in a marketing campaign project. Students will plan, implement and evaluate a digital marketing campaign using case studies and collaborative projects. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAR4860 Customer Relationship Management 3.000 credits

In this course students will learn how to effectively manage relationships with customers to increase customer satisfaction, loyalty, and retention. They will learn about customer behavior, communication strategies, data analysis techniques and CRM tools and technics to better understand and meet the needs of their customers. Prerequisite: MAR 3803.

Lecture Hours: 48.000

Prerequisite: MAR3803

Course User Fee: \$61.40

MAS**MAS2103 Elementary Linear Algebra 3.000 credits**

This course introduces the student to linear algebra and its applications. The student will examine linear systems of equations, matrices, determinants, vectors spaces, inner product spaces, linear transformations, linear independence and basis, eigenvalues and eigenvectors, decomposition theorems, and elements of proof writing.

Computational Course.

Lecture Hours: 48.000

Prerequisite: MAC2311

Course User Fee: \$18.00

MAS3105 Linear Algebra 3.000 credits

This course is designed for students who are majoring in secondary mathematics education. Major topics include systems of linear equations, matrices, determinants, vector spaces, linear transformations, eigenvectors and eigenvalues, inner-product spaces and orthogonality.

Lecture Hours: 48.000

Prerequisite: MAC2312

MAS3301 Algebraic Structures 3.000 credits

This course is designed for students who are majoring in secondary mathematics education, mathematics, science or engineering. Topics include set theory, basic properties of the integers, groups, rings, fields and the homomorphisms of these algebraic structures.

Lecture Hours: 48.000

Prerequisite: MAC2312

MAS4203 Number Theory 3.000 credits

Topics include mathematical induction, divisibility, the Euclidean algorithm, primes, the Fundamental Theorem of Arithmetic, number-theoretic functions, congruence, linear Diophantine equations, linear congruences, the Chinese Remainder Theorem, and the theorems of Euler, Fermat, and Wilson.

Lecture Hours: 48.000

Prerequisite: MAC2312

MAT

MAT0018 DEVELOPMENTAL MATH 1 4.000 credits

The student will learn operations with whole numbers, integers, fractions, decimals, percents and their applications; simplifying and evaluating algebraic expressions; ratios and proportions; solving linear equations in one variable and graphing solutions to linear inequalities. This course does not satisfy the college level mathematics requirements. Placement test scores or referral determine admission.

Lecture Hours: 64.000

Course User Fee: \$10.00

MAT0022C DEVELOPMENTAL MATH C 5.000 credits

This course combines Developmental Mathematics I and II. The student will learn operations on signed numbers, solving linear equations and inequalities in one variable, operations on polynomials, factoring, integer exponents, radicals, graphing, and applications. This course does not satisfy the college level mathematics requirements. Placement test scores or referral determine admission.

Lecture Hours: 80.000

Course User Fee: \$10.00

MAT0028 DEVELOPMENTAL MATH 2 4.000 credits

The student will learn topics which include operations with signed numbers; solving linear equations and inequalities in one variable; operations with polynomials, factoring, integer exponents, radicals, rational expressions, graphing and applications of these topics. This course does not satisfy the college level mathematics requirements.

Lecture Hours: 64.000

Prerequisite: MAT0018

Course User Fee: \$10.00

MAT0029 Developmental Mathematics for Statistics 3.000 credits

Students will investigate ratios, proportions, scaling, modeling with equations and inequalities, tables, graphs, linear functions, and non-linear functions, in preparation for Statistics. Students will learn the language of mathematics and mathematical symbols, procedural fluency, strategic competence, adaptive reasoning, quantitative investigative techniques, and questioning and solution-building skills.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAT0057 DEVELOPMENTAL MATHEMATICS (MODULES 3.0) 3.000 credits

Students will learn to strengthen arithmetic, geometry, and algebra skills. This course does not satisfy the college level mathematics requirements. Prerequisite: Placement test scores or departmental permission.

Lecture Hours: 48.000

Course User Fee: \$20.00

MAT1033 Intermediate Algebra 3.000 credits

The student will learn the concepts of linear equations, quadratic equations, rational equations, radical equations, rational expressions and equations, complex numbers, graphing linear equations and inequalities in one and two variables, and related applications.

Lecture Hours: 48.000

Prerequisite: MAT0022C OR MAT0028 OR MAT0057 or by placement score, or eligible exemption.

Course User Fee: \$18.00

MAT1033L Intermediate Algebra Recitation Hall 0.000 credits

The student will receive individualized, small group or whole group instruction to deepen their conceptual understanding of mathematics. The student will reinforce and apply content knowledge with effective problem-solving techniques and non-cognitive activities to make mathematics meaningful and relevant to their fields of study while strengthening the concepts needed to achieve the objectives of MAT1033. This lab session is associated and funded by the First in the World Grant, CCAT Math Project.

Lab Hours: 1.000

Corequisite: MAT1033

MCB

MCB2010 Microbiology 3.000 credits

This course introduces basic principles of morphology, physiology, biochemistry and genetics of microorganisms. The students will learn representative types of microorganisms including bacteria, algae, protozoa and viruses and the roles of various microorganisms in health and disease, modes of transmission and the effects of their activities in our biosphere. Students are strongly recommended to take the laboratory component MCB2010L. Prerequisites: BSC2010/2010L or BSC2085/2085L, CHM1033/1033L or CHM1045/1045L

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L

Course User Fee: \$9.00

MCB2010L Microbiology Laboratory 2.000 credits

This Laboratory course complements MCB2010 lecture topics. The student will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, counting, identification, and control of microbes.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L OR BSC2085 AND BSC2085L, CHM1033 AND CHM1033L OR CHM1045 AND CHM1045L, Corequisite: MCB2010

Course User Fee: \$55.00

MCB3023 Principles of Microbiology 3.000 credits

This course offers an introduction to the principles of microbiology. Students will learn the taxonomy, biochemistry, genetics, and ecology of microorganisms and will have an understanding of the impact of microorganisms on the advancement of the biological sciences. Special fee.

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L AND BSC2011 AND BSC2011L AND CHM2211 AND CHM2211L AND Corequisite: MCB3023L

Course User Fee: \$25.00

MCB3023L Principles of Microbiology Lab 2.000 credits

This Laboratory course accompanies MCB3023. Students will learn and have direct experience with fundamental techniques for observation, isolation, cultivation, enumeration, biochemistry, identification, genetics, and control of microbes. Special fee.

Lab Hours: 64.000

Corequisite: MCB3023, Prerequisite: BSC2010 AND BSC2010L AND BSC2011 AND BSC2011L AND CHM2211 AND CHM2211L

Course User Fee: \$110.00

MCB4503 Virology 3.000 credits

This course will cover general virology, including virus structure, replication cycles, infection and mode of transmission of human diseases. Student will learn the major families of the bacterial (bacteriophages), plant and animal viruses and how they influence infection. Special fee.

Lecture Hours: 48.000

Prerequisite: MCB3023, MCB3023L

Course User Fee: \$25.00

MEA**MEA0204 Theoretical Aspects of Clinical Skills 1.000 credits**

This course is designed to develop and further support students' knowledge and ability to organize and work efficiently and effectively in both performing and assisting with clinical procedures performed in medical offices. Emphasis will be on the role and responsibility of the Medical Assistant. Clock Hours: 30.000

MEA0204L Application of Clinical Skills 2.000 credits

This course is designed to develop and support students' ability to perform and assist in basic clinical skills. Emphasis will be on the role and responsibility of the medical assistant in performing sterile techniques and the use of organization and efficiency in performing and assisting with patient examination, sterile procedures, and diagnostic procedures and treatment performed in medical offices.

Clock Hours: 60.000

Course User Fee: \$3.00

MEA0231 Anatomy & Physiology and Medical Terminology 2.300 credits

This course is designed to introduce the student to basic anatomy and physiology and to develop the ability to communicate verbally and in writing within the medical field. Special fee.

Clock Hours: 69.000

Course User Fee: \$10.00

MEA0234 Pathophysiology & Disease for Medical Assistants 4.000 credits

This course is designed to introduce students to common diseases and medical conditions which affect patients who present themselves to medical offices for diagnosis and treatment. Emphasis will be on the role and responsibility of the Medical Assistant in prevention, diagnosis, and treatment. Clock Hours: 120.000

MEA0242 Pharmacology for the Medical Assistant 3.000 credits

This course is designed to introduce students to principles of pharmacology and provide a basis to comprehend the role and responsibility of Medical Assistants in administering medication. Emphasis will be placed on calculation of dosages, frequently used drugs, and classification of drugs as they relate to the body systems.

Course User Fee: \$10.00

MEA0254 Physician Office Laboratory Procedures 2.000 credits

Theoretical concepts of specimen collection and processing. This course focuses on the fundamentals of diagnostic tests, including urinalysis, basic office bacteriology, hematology, and blood chemistry. The principles of aseptic techniques, infection control, and safety procedures are discussed. Compliance with quality assurance practices are emphasized.

Clock Hours: 60.000

MEA0254L Physician Office Laboratory Procedure Applications 2.000 credits

A clinical laboratory course designed for the Medical Assistant student to practice specimen collection, microscopy and urinalysis. Includes basic office bacteriology, hematology, and blood chemistry. The student will apply principles of aseptic techniques and infection control.

Clock Hours: 60.000

Course User Fee: \$80.00

MEA0258 Radiology for the Medical Assistant 3.000 credits

This course focuses on the basic principles of x-ray, film handling and processing, radiographic technique, and radiation biology. The course prepares the student to take the examination given by the Florida Department of Professional Regulations (DPR) for the Basic Radiographer License.

Clock Hours: 90.000

Course User Fee: \$15.00

MEA0322 Office Mgmt. & Professional Issues for the MA 3.000 credits

Office management procedures, including planning and organization; financial and medical record keeping procedures; billing and collection; processing insurance claims using procedural and diagnostic coding. Legal and ethical responsibilities, credentialing and other professional issues of Medical Assisting.

Course User Fee: \$15.00

MEA0334C Medical Coding/Insurance Billing with Collections 4.000 credits

Processing health insurance claims using procedural and diagnostic coding. The student will learn and apply current government regulations affecting third-party reimbursement. Billing, electronic claims transmission, and collection systems are emphasized.

Clock Hours: 120.000

MEA0343 Computers in the Medical Office 3.000 credits

The application of computer concepts to medical office practices. The student will keyboard documents using word processing software. Emphasis will be on operating transcription equipment and transcribing medical records. The student will also be introduced to electronic spreadsheet and database applications.

Clock Hours: 90.000

Course User Fee: \$25.00

MEA0540 Electrocardiography/Emergency Procedures 2.000 credits

The nature and purpose of the electrocardiograph (EKG); maintenance of equipment and materials needed; preparation of the patient and the procedure for taking and mounting the EKG record and monitoring the record for abnormal or erratic tracings. The maintenance of emergency equipment and implementing emergency procedures in the medical office.

Clock Hours: 60.000

Course User Fee: \$10.00

MEA0802 Clinical Externship for the Medical Assistant 3.000 credits

This course is designed to provide students with experience in the practice of the clinical aspect of medical assisting. Students will be assigned to a physician's office or clinics where they will provide direct patient care under the guidance of an experienced Medical Assistant.

Clock Hours: 90.000

Course User Fee: \$10.00

MEA0810 Administrative Externship for the MA 3.000 credits

The student is assigned to a physician's office, clinic, laboratory, or other community health care facility. Emphasis is on integrating basic administrative skills demonstrated in previous courses.

MEA0832 Diagnostic Externship in Medical Assistant 3.000 credits

This course is designed to provide students with experience in the diagnostic aspect of Medical Assisting. Students will be assigned to a physician's office or clinics where they will perform diagnostic clinical laboratory procedures, electrocardiographic and basic x-ray procedures under the guidance of an experienced Medical Assistant.

Clock Hours: 90.000

MET**MET1010 INTRO TO WEATHER 3.000 credits**

Lecture Hours: 48.000

MGF**MGF1106 MATH FOR LIB ARTS 1 3.000 credits**

Topics include sets, logic, Euclidean geometry, probability, and statistics.

Lecture Hours: 48.000

Prerequisite: MAT1033 OR MAT0029 OR MGF1107

Course User Fee: \$18.00

MGF1107 MATH FOR LIB ARTS 2 3.000 credits

This course introduces the student to the concepts of financial mathematics, linear and exponential growth, numbers and number systems, history of mathematics, elementary number theory, voting techniques, and graph theory.

Lecture Hours: 48.000

Course User Fee: \$18.00

MGF1130 Mathematical Thinking 3.000 credits

In this course, students will utilize multiple means of problem solving through student-centered mathematical exploration. The course is designed to teach students to think more effectively and increase their problem-solving ability through practical application and divergent thinking. This course is appropriate for students in a wide range of disciplines/programs. Student learning outcomes: students will determine efficient means of solving a problem through investigation of multiple mathematical models; students will apply logic in contextual situations to formulate and determine the validity of logical statements using a variety of methods; students will apply mathematical concepts visually and contextually to represent, interpret and reason about geometric figures; students will recognize the characteristics of numbers and utilize numbers along with their operations appropriately in context; and students will analyze and interpret representations of data to draw reasonable conclusions.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MGF1131 Mathematics in Context 3.000 credits

Through this course, students will experience the practicality of mathematics in a global society. Students will engage in the applications of tools and techniques of mathematics in a variety of contextual situations from everyday life. This course is appropriate for students in a wide range of disciplines/programs. This course fulfills the Gordon Rule computational requirement and must be completed with a grade "C" or better.

Lecture Hours: 48.000

Prerequisite: Student must meet the Developmental Education mathematics requirements in State Rule 6A-10.0315 (by course, placement score, or eligible exemption).

Course User Fee: \$19.10

MHF**MHF2209 IB FURTHER MAT CR EX 3.000 credits
XXX**

Lecture Hours: 48.000

MHF4404 History of Mathematics 3.000 credits

A study of the development of mathematics from ancient civilizations to the present time.

Lecture Hours: 48.000

Prerequisite: MAC2312

MKA

MKA0242 intr expprt/impprt mkt 2.500 credits

This is a practical course designed to assist the participant enter the field of importing and exporting in a metropolitan that is of the major international marketing areas in the world. A step-by-step application of procedures is followed. Special fee. (75 contact hours)

Lab Hours: 75.000, Clinical Hours: 1.000 - 30.000

Course User Fee: \$20.00

MKA0949 COOP EDU WK EXP 3.000 - 6.000 credits

Lecture Hours: 90.000 - 180.000

MKA1021 FUND OF SELLING 3.000 credits

Lecture Hours: 48.000

Course User Fee: \$20.00

MKA1021 Fundamentals of Selling 3.000 credits

The nature and requirements of selling, including a consideration of buyer motivations and selling theory in relation to various buyer-seller situations. (3hr. Lecture)

Lecture Hours: 48.000

MKA1022 Relationship Selling 3.000 credits

This course introduces the role that relationship building plays in sales. Students will explore the communication process and the importance of interpersonal skills in developing a relationship for an effective sales strategy. Students will develop a sales presentation based around consumer needs.

Lecture Hours: 48.000

Course User Fee: \$47.44

MKA1160 Customer Relationship Management 3.000 credits

This course provides an introduction into Customer Relationship Management (CRM). Students will be exposed to concepts ranging from identifying customer needs, communication channels, customer acquisition cost, demographics and target marketing. Students will learn the role CRM plays in a sales strategy. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$47.44

MKA1161 Introduction to Customer Service 3.000 credits

A survey course which examines the attitudinal, behavioral and procedural basics which are common across all customer service sectors. An extensive vocabulary of customer service terms will be developed and students will understand their practical application in today's business environment. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$20.00

MKA1511 Principles of Advertising and Copywriting 3.000 credits

Techniques and behavioral factors used in advertising and copyrighting which best motivate the consumer. Principles are applied in clear, concise written expression of various appeals used in selling goods and services. (3 hr. lecture)

Lecture Hours: 48.000

MKA1531 AD LAYOUT/PRODUCTION 3.000 credits

A study of the principles of effective advertising layout and production techniques. Laboratory sessions emphasize use of color, art work, choice of type face and methods and techniques of producing ads for various media. Prerequisite: MKT 138 or permission of Department Chairman.

Lecture Hours: 48.000, Other Hours: -2.000

MKA2024 Organizational Sales Management 3.000 credits

This course introduces the role of a sales manager and sales team within an organization. Students will examine the management process of recruiting, leading, and setting sales targets and quotas for a sales team. Students will explore sales analytics that measure the effectiveness of an organizational sales strategy.

Lecture Hours: 48.000

Course User Fee: \$47.44

MLS

MLS3150 Special Topics in Medical Laboratory Sciences 3.000 credits

This course stresses the importance of evidence based practice in the medical laboratory sciences field. Students will be presenting case studies to the faculty and peers in the program. Instruction will emphasize professional, legal and ethics issues affecting the medical laboratory science field.

Students will review the material covered in the program to prepare for the comprehensive assessment. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$191.14

MLS3997 Validated Medical Laboratory Technology Course 27.000 credits

This course awards articulated credit for students who have successfully completed an Associate in Science (AS) degree in Histotechnology. It recognizes prior learning and competencies in histological techniques, including tissue processing, sectioning, staining, and laboratory safety, as applied in clinical practice. Emphasis is placed on validating skills aligned with program outcomes and professional standards in histotechnology.

MLS4181C Immunohistochemistry 3.000 credits

This course introduces the various techniques that are used in the preparation and evaluation of immunohistochemistry slides. Procedures and terminology related to immunohistochemistry are also discussed and strategies for troubleshooting problems are presented. (2 hr. lecture, 3 hr. clinical)
Lecture Hours: 32.000, Clinical Hours: 48.000

Course User Fee: \$499.15

MLS4193 Clinical Molecular Diagnostics 3.000 credits

Clinical molecular diagnostics course provides an introduction to molecular analysis of biological markers in clinical samples to aid in the diagnosis, monitoring and treatment of diseases (3 Hour Lecture).

Lecture Hours: 48.000

Course User Fee: \$41.20

MLS4195C Enzyme Histochemistry 3.000 credits

Introduction to advanced techniques and special procedures. Students will learn procedures for, muscle enzyme Histochemistry and molecular histology. The course will include tissue preparation, staining technology, quality control and troubleshooting. (2 hr. lecture, 3 hr. clinical)

Lecture Hours: 32.000, Clinical Hours: 48.000

Course User Fee: \$499.15

MLS4196C In Situ Hybridization (FISH) 3.000 credits

This course will explore the theoretical concepts used in fluorescence in-situ hybridization (FISH) testing. Commonly used FISH methodologies, necessary equipment, and the enumeration of FISH signals will also be discussed. (2 hr. lecture, 6 hr. clinical)

Lecture Hours: 32.000, Clinical Hours: 48.000

Course User Fee: \$499.15

MLS4198 Immunohistochemistry Clinical Concentration Capstone 5.000 credits

This clinical capstone course will introduce the students to the basic immunohistochemistry techniques as applied to the routine anatomical pathology laboratory.

Clinical Hours: 240.000

Course User Fee: \$499.15

MLS4221 Clinical Urinalysis 3.000 credits

The study of body fluids for physical health and identification of abnormalities in relation to disease states. (3 hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4306 Clinical Hematology 3.000 credits

This course is the study of the composition and function of blood; diseases related to blood disorders. Students will receive the necessary skills in the application of hematology diagnostic procedures, interpretation, problem solving and correlation of laboratory findings with disease states (3 Hour Lecture).

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4335 Clinical Hemostasis 3.000 credits

This course Provides an overview application of hemostasis (coagulation), as it relates to the medical laboratory. Presents coagulation laboratory principles with hemostasis diagnostic procedures, interpretation, problem solving and correlation of laboratory findings and results in accordance to the disease states. (3 Hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4461C Clinical Diagnostic Microbiology 3.000 credits

Clinical Diagnostic Microbiology provides concepts in bacteriology identification methods, rapid identification methods for parasites and fungi and an overview of virology methodology.

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4506 Clinical Immunology 3.000 credits

Clinical Immunology will provide an overview of immunology concepts and the theory of some immunologic procedures. The immunologic manifestation of infectious disease and immune disorders will also be covered. (3 Hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4552 Clinical Immunohematology 3.000 credits

The study of concepts related to the blood group systems, blood antigens and antibodies. The student will analyze the principles, procedures and clinical significance of testing in genetics, pre-transfusion testing, adverse effects of transfusions, donor selection and components, and hemolytic disease of the newborn. (3 hour lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4621 Clinical Biochemistry 4.000 credits

This course provides the student with the knowledge and understanding of clinical disorders and how biochemical factors and laboratory methods are used for the investigation, diagnosis and management of patients. (4 hour lecture)

Lecture Hours: 64.000

Course User Fee: \$41.00

MLS4630 Clinical Chemistry 3.000 credits

The study of the concepts and principles of Clinical Chemistry. Analytes and lab values are correlated to normal homeostasis and disease states. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4705 Laboratory Operations and Management 3.000 credits

This course provides students with Quality Management skills necessary in the medical laboratory. Students are exposed to organizational structure along with principles for leadership and managerial decision making and process improvement along with different principles used in Laboratory Instrumentation. (3 Hour Lecture)

Lecture Hours: 48.000

Course User Fee: \$74.90

MLS4910 Advances in Histotechnology Capstone 7.000 credits

This course will support the educational development of the histotechnology students by providing an opportunity for in-depth learning in one of the following domains Cytopreparatory techniques; Digital pathology; Electron microscopy; Micro-wave (MOHS) pathology; Ocular pathology histotechnology techniques; which will result in a scholarly project underlying its relevance in today's advanced pathology laboratories.

Lecture Hours: 112.000

Course User Fee: \$499.15

MLT**MLT0041 Phlebotomy Theory 0.500 credits**

This course covers the theory of phlebotomy techniques by venipuncture and skin puncture. This includes basic anatomy and physiology of the circulatory system, types of tubes to select for various blood tests, possible interfering substances, hospital hierarchy, professionalism, risk factors for Hepatitis, AIDS, and all sexually transmitted diseases, infection control guidelines, and employability skills. Special fee.

Course User Fee: \$10.00

MLT0048 Phlebotomy Practicum 1.500 credits

This course is designed to prepare students to draw blood by venipuncture and capillary puncture and to prepare them for employment in a hospital laboratory, blood center, or other health care facility. Students are taught safe and efficient work practices in obtaining adequate and correct blood specimens, labeling specimens, and transporting specimens correctly to the appropriate laboratory sections. The Center for Disease Control (CDC) guidelines for HIV/AIDS, Hepatitis B and other diseases are stressed.

Clock Hours: 45.000

MLT0061 Practical Aspects of Phlebotomy 0.500 credits

This course covers the collection of blood by venipuncture, skin puncture and donor room techniques. This includes the handling, labeling, transporting, and logging-in of specimens as well as the demonstration of correct infection control techniques. Special fee.

Lab Hours: 15.000

MLT1040L Fundamentals of Laboratory Operations 1.000 credits

Collection of blood by venipuncture, skin puncture and donor room techniques. This includes handling of specimens, professional ethics, basic anatomy and physiology of the circulatory system, medical terminology and safety practices, including those for AIDS patients.

Lecture Hours: 32.000

Course User Fee: \$184.16

MLT1191 Histotechnology 1 3.000 credits

This course will introduce students to the fundamental principles of histologic technology. These include the principles of fixation, processing for paraffin-embedding, microtome sectioning, staining and cover-slipping and laboratory safety.

Lecture Hours: 48.000

MLT1191L Histotechnology 1 Lab 3.000 credits

This course will introduce students to fundamental laboratory skills and safety concepts in histologic technology. It includes laboratory aspects of specimen preparation, fixation, sectioning and routine staining. The student will also be introduced to the basic principles of record keeping, use and maintenance of laboratory equipment and quality control.

Lab Hours: 96.000

Course User Fee: \$100.00

MLT1191L Histotechnology 1 Lab 4.000 credits

This course will introduce students to the fundamental skills and safety concepts in Histologic technology. It includes laboratory aspects of specimen preparation, fixation, sectioning and routine staining. The student will also be introduced to the basic principles of record keeping, use and maintenance of laboratory equipment and quality control.

Lab Hours: 64.000

MLT1195C Tissue Identification 1 3.000 credits

This course will introduce students to the study of human organs and tissues for the purpose of developing histotechnological skills. It will include recognition, composition, and function of organs and tissues. Macroscopic and microscopic laboratory examination and evaluation of specimens will be included.

Lecture Hours: 32.000, Lab Hours: 32.000

MLT1196 Laboratory Safety and Regulations 3.000 credits

This course will introduce students to the rules and regulations governing safety in the histotechnology laboratory. It will also introduce students to the federal regulations pertaining to the histotechnology laboratory and methods of compliance. Prominent safety issues to be covered include the biological and chemical hazards in histology laboratory, formaldehyde standard, hazardous waste disposal and minimization.

Lecture Hours: 48.000

MLT1210C Clinical Urinalysis with Lab 2.000 credits

Theoretical concepts and practice in the collection and analysis of urine and other body fluids by combination of didactic and laboratory instruction. Performance of routine urinalysis procedures, including microscopy with identification of related disease states.

Lecture Hours: 48.000

Course User Fee: \$65.00

MLT1300 Clinical Hematology 2.000 credits

Didactic study of blood cells to include the origin, morphology, function and dysfunction of cells and related disease states of the blood. Theoretical concepts and principles of routine hematology procedures, quality control and instrumentation.

Lecture Hours: 32.000

Corequisite: MLT1300L

MLT1300L CLIN HEMATOLOGY LAB 2.000 credits

Add course description.

Lab Hours: 64.000

Course User Fee: \$80.00

MLT1300L Clinical Hematology Laboratory 2.000 credits

Manual and automated procedures in hematology. This includes blood cell counts and other basic hematologic procedures in the simulated laboratory and in the clinical setting.

Lab Hours: 64.000

Corequisite: MLT1300

Course User Fee: \$90.00

MLT1330 Clinical Coagulation 1.000 credits

Didactic study of hemostasis, various clotting mechanisms, and related disease states.

Lecture Hours: 16.000

Prerequisite: MLT1300, Corequisite: MLT 1330L

MLT1330L Clinical Coagulation Laboratory 1.000 credits

Performance of selected coagulation assays by manual and automated methods. The significance of test results to assess hemostasis in health and disease is included.

Lab Hours: 32.000

Prerequisite: MLT1300L, Corequisite: MLT1330

Course User Fee: \$40.00

MLT1500 Clinical Immunology/Serology 2.000 credits

Theoretical concepts of the human immune system in health and disease. Relationships to immunohematology, infection, and serological procedures are analyzed.

Lecture Hours: 32.000

Prerequisite: BSC 2085, BSC 2086, Corequisite: MLT1500L

MLT1500L Clinical Immunology/Serology Laboratory 1.000 credits

Performance of serological procedures that are identified in MLT 1500. The clinical significance of test results to disease states is included.

Lab Hours: 32.000

Prerequisite: BSC 2085L, BSC 2086L, Corequisite: MLT1500

Course User Fee: \$49.00

MLT1610 Clinical Chemistry 1 2.000 credits

Theoretical concepts and principles of carbohydrate, nonprotein nitrogen, and electrolyte chemistry analyses with emphasis on their relationships to various disease states. Analytical procedures to assess liver function and acid-base balance are also included.

Lecture Hours: 32.000

Prerequisite: CHM1025, Corequisite: MLT1610L

MLT1610L CLN CHM 1 LAB/CL EXP 2.000 credits

Lab Hours: 64.000

Course User Fee: \$40.00

MLT1610L Clinical Chemistry 1 Laboratory 2.000 credits

Performance of chemistry procedures on body fluids with emphasis on manual and automated instrumentation.

Lab Hours: 64.000

Prerequisite: CHM1025L, Corequisite: MLT1610

Course User Fee: \$40.00

MLT1752 Quality Control Laboratory Mathematics 2.000 credits

Emphasis on mathematical computations related to procedures in the clinical laboratory including dilutions, solutions, colorimetry, hematology math, enzymatic calculations, calculations relating to renal function tests, and mathematical principles related to ionic solutions. The student will also be given specific statistical tools necessary for quality control procedures as well as interpretations of Levy-Jennings charts and troubleshooting tools.

Lecture Hours: 32.000

MLT1840L Histotechnology Practicum 1 5.000 credits

This is a clinical experience in which students will learn the techniques of processing human tissue for histological purposes.

Practicum Hours: 240.000

Prerequisite: MLT 2180C, MLT 2198, MLT 2198L

MLT2180C Infectious Diseases & Control Practices 3.000 credits

This course will focus on the principles of transmission and control of diseases with an emphasis on infectious tissue specimens.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: MCB2010, MCB2010L

MLT2192 Histotechnology 2 3.000 credits

This course is a continuation of Histotechnology 1. Students will be introduced to advanced processing techniques of human tissue for anatomical pathology and concepts of instrumentation.

Lecture Hours: 48.000

Prerequisite: MLT1191

Course User Fee: \$100.00

MLT2192L HISTOTECHNOLOGY 2_LAB 2.000 credits

This course is a continuation of Histotechnology Lab 1. Students will be introduced to more complex laboratory techniques in histotechnology. Prerequisite: MLT 1191L; corequisite: MLT 2192. (2 hr. lecture; 4 hr. lab)

Lab Hours: 64.000

MLT2192L Histotechnology 2 Laboratory 3.000 credits

This course is a continuation of Histotechnology Lab 1. Students will be introduced to more complex laboratory techniques in histotechnology.

Lab Hours: 96.000

Prerequisite: MLT1191, MLT1191L, Corequisite: MLT2197C

MLT2197C Tissue Identification 2 4.000 credits

This course will provide the students with the correlations between histotechnological procedures and diseases processes. Students will study the changes in tissue that are associated with various disease states, and will learn the usefulness of selected special stains and techniques in identifying disease processes.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: MLT1195C

MLT2198 Histochemistry 3.000 credits

This course will introduce students to organic chemistry of stains and special stains, dyes, hydrocarbons; aromatics, alcohols, ethers, aldehydes, ketones, carbonyl compounds, amines and amides.

Lecture Hours: 48.000

Prerequisite: MLT2192, MLT2192L, Corequisite: MLT2198L

MLT2198L HISTOCHEMISTRY LAB 2.000 credits

This course will introduce students to biochemicals used in histology with emphasis on laboratory preparation and use of histochemical and immunohistochemical stains. (4 hr. lab)

Lab Hours: 64.000

Course User Fee: \$100.00

MLT2198L Histochemistry Laboratory 3.000 credits

This course will introduce students to biochemicals used in histology with emphasis on laboratory preparation and use of histochemical and immunohistochemical stains. Prerequisite: CHM 1033L; corequisite: MLT 2198. Laboratory fee. (4 hr. lab)

Lab Hours: 96.000

Prerequisite: CHM 1033, CHM 1033L, MLT 2192, MLT 2192L, Corequisite: MLT2198

MLT2403 Clinical Microbiology 2 2.000 credits

This course will provide a working knowledge of clinical bacteriology and should complement the Microbiology 2 Lab. The student will be exposed to some of the indigenous flora and the pathogenicity of microorganisms as they affect various body sites. Specimen transport, collection, laboratory identification techniques, and antimicrobial therapy also provides the knowledge base necessary for working in a clinical setting.

Lecture Hours: 32.000

Prerequisite: MLT2440, Corequisite: MLT2403L

MLT2403L Clinical Microbiology Lab 2 2.000 credits

This course is designed to complement the Microbiology 2 lecture and provide students with the necessary knowledge base and laboratory skills to effectively identify microorganisms associated with infectious diseases.

Lab Hours: 64.000

Prerequisite: MLT2440L, Corequisite: MLT2403

Course User Fee: \$50.00

MLT2440 Clinical Microbiology 1 1.000 credits

This course will provide an overview of clinical mycology and parasitology. Topics will include both parasites and fungi and will cover life cycles, epidemiology, and etiology. Emphasis will be given to the most commonly encountered mycoses and parasitic infestations. This course should be taken concurrently with Clinical Microbiology 1 Lab.

Lecture Hours: 16.000

Corequisite: MLT 2400L

MLT2440L Clinical Microbiology Lab 1 1.000 credits

This course provides a practical overview of mycology and parasitology. Students will also obtain hands-on experience working with formalin preserve ova and parasites. They will also obtain the knowledge necessary to be able to identify at least the genus level of the most commonly encountered yeasts and fungi using microscopic and macroscopic techniques. This course should be taken concurrently with Clinical Microbiology.

Lab Hours: 32.000

Corequisite: MLT2440

Course User Fee: \$50.00

MLT2525 Immunohematology 2.000 credits

Theoretical concepts involving blood group systems, hemolytic diseases, and blood bank procedures relating to transfusion and component therapy.

Lecture Hours: 32.000

Prerequisite: MLT1500, Corequisite: MLT2525L

MLT2525L Immunohematology Laboratory 2.000 credits

Performance of basic blood typing, blood bank assays on prepared specimens, and appropriate quality control procedures. Interpretation of results is included.

Lab Hours: 64.000

Prerequisite: MLT1500L, Corequisite: MLT2525

Course User Fee: \$45.00

MLT2620 Clinical Chemistry 2 2.000 credits

Theoretical concepts and principles of proteins, enzymes, and lipids with emphasis on their relationship to various disease states.

Lecture Hours: 32.000

Prerequisite: MLT1610, Corequisite: MLT2620L

MLT2620L CLINICAL CHEM 2 LAB 1.000 credits

Lab Hours: 32.000

Course User Fee: \$83.00

MLT2620L Clinical Chemistry 2 Laboratory 1.000 credits

Performance on those analyses identified in MLT 2620 including electrophoresis and quality control.

Lab Hours: 32.000

Prerequisite: MLT1610L, Corequisite: MLT2620

Course User Fee: \$83.00

MLT2624L Special Techniques In Clinical Chemistry 2.000 credits

The principles and performance of radioimmunoassay, EMIT, ELISA, and toxicological techniques for thyroid function, hormones, and toxic substances.

Lab Hours: 64.000

Prerequisite: MLT 1610, MLT 1610L, Corequisite: MLT 2620, MLT 2620L

Course User Fee: \$62.00

MLT2807L Hospital Practicum: Immunohematology 3.000 credits

A supervised laboratory rotation in a clinical immunohematology facility. This provides the student an opportunity for the practice of skills previously learned and for the acquisition of new procedural skills. The development of interpersonal skills and the transition from student to professional are emphasized.

Clinical Hours: 144.000

Prerequisite: MLT 2525, MLT 2525L, Corequisite: MLT 2930

MLT2809L Hospital Practicum: Hematology 3.000 credits

A supervised laboratory rotation in a clinical hematology facility. This provides the student an opportunity for the practice of skills previously learned and for the acquisition of new procedural skills. The development of interpersonal skills and the transition from student to professional are emphasized.

Clinical Hours: 144.000

Prerequisite: MLT 1300, MLT 1300L, MLT 1330, MLT 1330L, Corequisite: MLT 2930

MLT2810L Hospital Practicum: Chemistry 3.000 credits

A supervised laboratory rotation in a clinical chemistry facility. The development of interpersonal skills the transition from student to professional are emphasized. This provides the student an opportunity for the practice of skills previously learned and for the acquisition of new procedural skills.

Clinical Hours: 144.000

Prerequisite: MLT 2620, MLT 2620L, MLT 2624L, Corequisite: MLT 2930

MLT2811L Hospital Practicum: Microbiology 3.000 credits

A supervised laboratory rotation in a clinical microbiology facility. This provides the student an opportunity for the practice of skills previously learned and for the acquisition of new procedural skills.

Clinical Hours: 144.000

Prerequisite: MLT 2403, MLT 2403L, Corequisite: MLT 2930

MLT2841L Histotechnology Practicum 2 5.000 credits

This clinical experience will introduce the students to the basic techniques of microtomy, staining and preparation of human tissue for anatomical pathology.

Practicum Hours: 240.000

Prerequisite: MLT1840L, Corequisite: MLT2931

MLT2930 MED LAB TECH SEMINAR 2.000 credits

Clinical correlations, professional issues, update s in Medical Laboratory Technology with student's reports on recent professional journal articles, and the use of microcomputers in the laboratory. Co requisite: MLT 2807L, 2809L, 2810L, 2811L. A.S. de gree credit only. (2 hr. seminar)

Lecture Hours: 32.000

MLT2930 Medical Laboratory Technology Seminar 2.000 credits

Clinical correlations, professional issues, updates in Medical Laboratory Technology with student's reports on recent professional journal articles, and the use of microcomputers in the laboratory.

Lecture Hours: 32.000

Prerequisite: MLT 2807L, MLT 2809L, MLT 2810L, MLT 2811L

Course User Fee: \$35.00

MLT2931 Histotechnology Seminar 3.000 credits

This course will prepare students for career entry. Emphasis will be placed on current topics in histotechnology, legal and ethical responsibilities of health care professionals, knowledge of the health care delivery system, including health policies and financing and employability skills.

Lecture Hours: 48.000

Prerequisite: MLT1840L, Corequisite: MLT2841L

MMC**MMC2000 Introduction to Mass Communications 3.000 credits**

Lecture Hours: 48.000

MNA**MNA1053 CONDO & ASSOC LAW 3.000 credits**

Lecture Hours: 48.000

MNA1130 Business for Financial Services 1.000 credits

This course teaches business professionals a structured approach to create clear, effective, professional business writing, including e-mail, memos, letters, and reports. (1 hr. lecture)

Lecture Hours: 16.000

Course User Fee: \$45.00

MNA1345 EFFECTIVE SUPERVISIO 3.000 credits

pending

Lecture Hours: 48.000

MNA2120 HUM RELATIONS IN BUS 3.000 credits

pending

Lecture Hours: 48.000

MSL**MSL1001 1 YR BASIC ARMY ROTC 2.000 credits**

U.S. Defense Establishment (Army); organization, branches and function of the Army; fundamentals of leadership.

Lecture Hours: 32.000

MSL1002 1 YR BASIC ARMY ROTC 1.000 - 2.000 credits

Introduction to basic leadership principles and traits, army command and staff officer duties, awards, decorations, individual military tactics, field discipline, patrolling techniques, radio telephone procedures, rappelling and river crossing. Requires outdoor leadership laboratory and at least one weekend field training exercise.

Lecture Hours: 16.000 - 32.000

MSL2101 2 YR BASIC ARMY ROTC 2.000 credits

Instruction in squad and platoon marching drills, development of physical fitness training programs, conduct on military training and inspections, leadership techniques, advanced map reading, rappelling and river crossing techniques. Requires outdoor leadership laboratory and at least one weekend field training exercise.

Lecture Hours: 32.000

MSL2102 2 YR BASIC ARMY ROTC 2.000 credits

Lecture Hours: 32.000

MSL3201 Leadership and Problem Solving 3.000 credits

This is a leadership & problem solving course for ROTC Cadets. Students will learn to examine skills that underlie effective problem solving, analyze military missions and plan military operations, and execute squad battle drills. Prerequisite: Cadets Eligible to Contract per ROTC Enrollment Officer and/or MAN2021.

Lecture Hours: 48.000

MSL3202 Leadership and Ethics 3.000 credits

This course explores leader responsibilities that foster an ethical command climate. Students will learn to develop Cadet leadership competencies, and apply principles and techniques of effective written and oral communication. Prerequisite: Cadets Eligible to Contract per ROTC Enrollment Officer and/or MAN2021.

Lecture Hours: 48.000

MSS**MSS0156 Anatomy & Physiology for Massage Therapy 2.500 credits**

This course will focus on the relationship between the anatomical and physiological effects of massage therapy on the body. Students will focus on the structure of organs, muscles, bones and tissues. Primary focus will center on the musculoskeletal system and innervations.

Course User Fee: \$62.50

MSS0156L Anatomy & Physiology for Massage Therapy Lab 2.500 credits

This course will examine the practical application and physiological effects of massage therapy on the body. Students will focus on the structure of organs, muscles, bones and tissues. Primary focus will center on the musculoskeletal systems and innervations as well as clinical pathologies related to those systems.

Course User Fee: \$62.50

MSS0215 History and Standards for Massage Therapy 1.000 credits

This course examines the history and development of massage therapy, basic legal concepts related to health care employment, and legal requirements for practice as a Massage Therapist in the State of Florida.

Course User Fee: \$25.00

MSS0250 Introduction to Massage Therapy 1.000 credits

This course focuses on the theories and principles of therapeutic massage. The Massage Therapist/Client Relationship, the effects of massage on the systems of the body, massage facilities, equipment/supplies, and furniture requirements will be discussed.

Course User Fee: \$25.00

MSS0250L Introduction to Massage Therapy Laboratory 6.000 credits

Laboratory for MSS 0250. This course provides opportunities for the practical application of the theories and principles of therapeutic massage.

Course User Fee: \$150.00

MSS0281 Allied Modalities 3.500 credits

A study of the advanced theories and techniques for massage therapy. Content includes: oriental bodywork, reflexology, trager approach, rolfing, craniosacral therapy, infant massage, pregnancy massage and aromatherapy.

Course User Fee: \$25.00

MSS0300 Hydrotherapy Modalities 1.000 credits

This course focuses on the history and development of hydrotherapy, application of equipment used, and the associated standards.

Course User Fee: \$25.00

MSS0300L Hydrotherapy Modalities Laboratory 1.500 credits

This course presents an opportunity for the students to safely and effectively apply various types of hydrotherapy and evaluate their effectiveness.

Course User Fee: \$62.50

MSS0803C Massage Therapy Clinical Practicum 3.000 credits

This course provides the student with the opportunity to practice and further develop an understanding of various massage techniques in a clinical placement setting under supervision of a licensed Massage Therapist.

Course User Fee: \$75.00

MSS0995 Massage Therapy - Accelerated 13.500 - 15.000 credits

This course is designed to provide PSAV credit for students with training and State of Florida licensure as a Physical Therapist or Physical Therapist Assistant. Students must provide documentation of a current state license and be a graduate of an accredited program. This course requires special permission and students must contact the program coordinator for registration approval.

MSS0996 Massage Therapy - Transitional 8.000 credits

This course is designed to provide PSAV credit for students with training and State of Florida licensure as an Allied Health Professional or Registered Nurse. Students must provide documentation of a current state license and be a graduate of an approved Associate Degree program. This course requires special permission and students must contact the program coordinator for registration approval.

MTB**MTB1103 Business Mathematics 3.000 credits**

Students will learn the basic arithmetic processes related to mathematics and computations used in businesses, to include cash and trade discounts, commissions, markups and markdowns, depreciation, simple and compound interest, bank discounts, payroll records, payroll taxes, charts and graphs, and consumer applications.

Lecture Hours: 48.000

Course User Fee: \$10.00

MTB1322 Technical Mathematics 2 3.000 credits

Applications of algebra, trigonometry, and analytic geometry are needed in technical programs.

Lecture Hours: 48.000

MTG**MTG2204 Geometry for Educators 3.000 credits**

This course will introduce basic elements of Euclidean geometry that includes measurement and properties of plane and solid figures, sets, logic, and proofs.

Lecture Hours: 48.000

Prerequisite: MAC1105

MTG3214 Euclidean Geometry 4.000 credits

This course encompasses a range of geometry topics and pedagogical ideas for the teaching of geometry including properties of shapes, defined and undefined terms, postulates and theorems, logical thinking and proofs, constructions, patterns, and sequences, the coordinate plane, axiomatic nature of Euclidean geometry, and basic topics of non-Euclidean geometries. Prerequisite: MAC 1147. (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: MAC1147

Course User Fee: \$25.00

MUC**MUC1201 COMPOSITION 1 2.000 credits**

Lecture Hours: 32.000

MUC1202 COMPOSITION 2 2.000 credits

Lecture Hours: 32.000

MUC2001 EXPER COMPOSITION 3.000 credits

An exploratory course providing experience with 20th century compositional techniques through listening, analysis, composition and performance. May be repeated for credit. By permission of instructor.

Lecture Hours: 48.000

MUC2101 COMPOSITION SKILLS 3 2.000 credits

This course is a continuance of the composition workshop at a more advanced level. Students receive private lessons in music composition. Students are encouraged to apply their theoretical skills to a diverse media, including writing for a variety of small ensembles. This will culminate into a mini recital at the end of the term which will also help prepare the student to effectively coordinate and organize performances of his or her own works in front of an academic and general audience. In the process the student learns to work with a variety of performers and appreciate exposure and feedback from a diverse group of people. (2 hr. lecture)

Lecture Hours: 32.000

MUC2102 COMPOSITION SKILLS 4 2.000 credits

This course is a continuance of Composition Skills 3 at a more advanced level. Students receive private lessons in music composition. Students are encouraged to apply their theoretical skills to a diverse media, including writing for a variety of small ensembles. This will culminate into a mini recital at the end of the term which will also help prepare the student to effectively coordinate and organize performances of his or her own works in front of an academic and general audience. In the process the student learns to work with a variety of performers and appreciate exposure and feedback from a diverse group of people. (2 hr. lecture)
Lecture Hours: 32.000

MUC2601 INTRO TO SONGWRITING 3.000 credits

This course explores the art and craft of popular songwriting. Students will learn the basics of lyric writing, chord progressions, melodic creation, and structure as they apply to popular song. (3 hr. lecture)
Lecture Hours: 48.000

MUE**MUE1430 VOICE TECHNIQUES 1.000 credits**

Lecture Hours: 32.000

MUH**MUH2111 Survey of Music History 1 3.000 credits**

This course introduces the history of musical styles from antiquity through the Baroque period and includes works from the Western canon. Students will examine the development of music in relation to cultural, historical, and philosophical contexts. Through analysis of musical works and their significance, students will develop critical thinking skills and gain understanding of how music reflects and influences people across time periods.
Lecture Hours: 48.000
Course User Fee: \$15.00

MUH2112 Survey of Music History 2 3.000 credits

In this course, students will be introduced to the history of musical styles from the Baroque period through the present. Through the examination of representative literature, students will explore the evolution of musical styles, genres, and forms, gaining a deeper understanding of the cultural and artistic influences that shaped them. The course will encompass a wide range of musical works, including compositions from the Western canon, allowing students to appreciate the richness of musical expression across different periods and regions. This is a Writing intensive course.
Lecture Hours: 48.000
Course User Fee: \$15.00

MUL**MUL1010 MUSIC APPRECIATION 3.000 credits**

The development of the various styles, forms, and idioms, in music. The emphasis is given to the student's ability to understand and enjoy music. (3 hr. lecture)
Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

MUL2380 Jazz and Popular Music in America 3.000 credits

A survey of the development of popular and jazz music with an emphasis on musical styles and outstanding artists. (3 hr. lecture)
Lecture Hours: 48.000

MUM**MUM1622L SOUND REINF/FUND LAB 1.000 credits**

This course is designed to provide students with "hands on" experience in conjunction with music school performance activities. Corequisite: MUM 1662. (2 hr. lab)
Lab Hours: 32.000

MUM1662 SOUND REINFORCE FUND 3.000 credits

Sound reinforcement fundamentals is a course designed to provide students with background in live sound reinforcement, concert sound practices, and general PA work associated with sound engineering. Corequisite: MUM 1622L. (3 hr. lecture)
Lecture Hours: 48.000

MUM1930 Introduction to DJ 3.000 credits

This course is designed to teach fundamental techniques of playing pre-recorded music using turntables, DJ software, and digital audio controllers. Through hands-on learning, students explore both the creative and technical facets of DJing for a variety of audiences, preparing them to pursue professional DJ careers.

Lecture Hours: 48.000

MUM1949 Music Industry Internship 3.000 credits

This course is designed to provide students with practical, hands-on experience in their chosen field of study—such as Music Business, Music Production, or a related field—through on-the-job training and faculty guidance. Students are evaluated on the basis of documented learning outcomes as reported by both the student and the employer. Prerequisite: MUM2700, MUM 2702, and MUM 2703; or department approval.

Lecture Hours: 48.000

Prerequisite: MUM2700 AND MUM2702 AND MUM2703

MUM2030 Commercial Music Performance 3.000 credits

A performance experience with concentration on repertoire, style and management of commercial engagements. Includes transposition, harmonization and show reading. Prerequisites: MUT 1111 and MUT 1241; or permission of instructor. May be repeated for credit.

Lecture Hours: 48.000

Prerequisite: MUT1111 AND MUT1241

MUM2600 Sound Recording 1 3.000 credits

An introduction to techniques, practices and procedures in making eight-track recordings. The student will gain experience with acoustical balancing, editing and over-dubbing in a wide variety of sound situations.

Lecture Hours: 48.000

Corequisite: MUM2600L

MUM2600L Sound Recording 1 Lab 1.000 credits

Participation in MUM 2600L offers students directed "hands on" experience coinciding with lectures in MUM 2600.

Lab Hours: 32.000

Corequisite: MUM2600

Course User Fee: \$15.00

MUM2601 Sound Recording 2 3.000 credits

This course explores advanced multi-track recording skills and audio production techniques. Emphasis is on mixing board skills, microphone techniques, use of outboard equipment and live 2 track recording.

Prerequisites: MUM 2600 and MUM 2600L;

Corequisite: MUM 2601L. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MUM2600 AND MUM2600L AND

Corequisite: MUM2601L

MUM2601L SOUND RECORDNG 2 LAB 1.000 credits

Corequisite for MUM 2601. Advanced Sound Recording. Participation in MUM 2601L offers students directed "hands on" experience paralleling lectures in MUM 2601. Special fee. (2 hr lab)

Lab Hours: 32.000

Course User Fee: \$15.00

MUM2603 Foundations of Audio Wiring 2.000 - 3.000 credits

Foundations of Audio Wiring is a hands-on course that provides students with practical knowledge and experience in audio wiring, soldering, and electronic troubleshooting. Designed for those pursuing careers in audio engineering, live sound, and studio maintenance, this course covers essential wiring techniques, tools, and safety procedures for working with audio equipment. Prior knowledge of basic electrical concepts is recommended but not required.

Lecture Hours: 32.000 - 48.000

Course User Fee: \$55.00

MUM2604 MULTITRACK MIX TECH 1.000 credits

This course deals with the application of signal processing gear to multi-track master recording mix down to 2 track stereo mastering machines; includes editing and packaging. Prerequisites: MUM 2600, 2600L. (2 hr. lab)

Lecture Hours: 32.000

Course User Fee: \$10.00

MUM2605 MULTITRK PROD TECH 1 1.000 credits

Multi-track production technique offers students with a background in multi-track recording an opportunity to sharpen their skills in recording, mixing, editing, and audio production. Prerequisites: MUM 2600, 2600L. Must precede MUM 2606 and 2607. (1 hr. lecture)

Lecture Hours: 16.000

MUM2606 MULTITRK PROD TECH 2 1.000 credits

Multi-track production technique offers students with a background in multi-track recording an opportunity to sharpen their skills in recording, mixdown editing, and audio production. Prerequisites: MUM 2600, 2600L, 2605. (1 hr. lecture)
Lecture Hours: 16.000

MUM2623C MIDI Electronic Music 1 3.000 credits

This course is designed to acquaint music students with basic applications of Musical Instrument Digital Interface (MIDI) for the purpose of composition and performance and learning pre-production concepts with multi-track recording studio. Emphasis will be placed on keyboards, outboard gear, drum machines, and computer-assisted operations. Special fee.

Lecture Hours: 32.000, Lab Hours: 16.000 - 32.000
Course User Fee: \$10.00

MUM2624C MIDI ELECTRONIC MUS2 2.000 - 3.000 credits

This course is designed to provide music students further study in the application of the Musical Instrument Digital Interface (MIDI). Emphasis will be placed on advanced techniques in sequencing, routing, synchronization, composition and arranging. Prerequisite: MUM 2623. Special fee. (1-2 hr. lecture; 2 hr. lab)

Lecture Hours: 16.000 - 32.000, Lab Hours: 32.000
Course User Fee: \$10.00

MUM2640L MULTITRACK MIX TECH 1.000 credits

This course deals with the application of signal processing gear to multi-track master recording mix down to 2 track stereo mastering machines includes editing and packaging. Prerequisites: MUM 2600, 2600L. Laboratory fee. (2 hr. lab)

Lab Hours: 32.000
Course User Fee: \$10.00

MUM2670 Avid Pro Tools 101/110 Fundamentals 3.000 credits

This course provides an in-depth introduction to Avid's Pro Tools, the industry-standard software for digital audio production. Aimed at beginners and intermediate users, the course covers the essential features and functionalities of Pro Tools, enabling students to create, edit, and produce professional-quality audio projects. Upon completion of the course, students are eligible to take the Pro Tools 101 and Pro Tools 110 exams to earn the Pro Tools Specialist certification. Prerequisite: MUM2600 and MUM2600L, or RTV1240C.

Lecture Hours: 48.000

Prerequisite: MUM2600 AND MUM2600L AND RTV1240C

Course User Fee: \$58.00

MUM2700 Music Business 1 3.000 credits

This course is an introduction to the foundational aspects of the music industry, focusing on the essential principles and applications of music copyright law, contracts, publishing, licensing, and management. Students will explore the roles of agencies and organizations in the industry, along with the processes of record production, marketing and distribution. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$20.00

MUM2702 Music Business 2 3.000 credits

This course provides an in-depth exploration of entrepreneurship, employability, and management principles within key areas of the music industry, including music equipment sales, concert promotion, independent record labels, record production, music licensing, and publishing. Students will gain an understanding of the business models and strategic approaches that drive success in each sector, with an emphasis on career skill development such as business planning, marketing, financial management, and contract negotiation. Prerequisite MUM 2700. (3 hr. lecture).

Lecture Hours: 48.000

Prerequisite: MUM2700

Course User Fee: \$20.00

MUM2703 Music Business 3 3.000 credits

This course provides hands-on training in essential computer applications and online tools relevant to the music industry. Students will learn to use digital tools to create professional promotional materials, presentations, and documents typically used in music industry transactions. (6 hr. lab)

Lab Hours: 96.000

Course User Fee: \$25.00

MUM2704 Music Business 4 3.000 credits

This course provides hands-on experience with cross-platform software applications and online resources used within the Music Business environment. Students will create their own website for marketing their own Music Business enterprise.

Prerequisite: MUM 2703. (6 hr. lab)

Lab Hours: 96.000

Prerequisite: MUM2703

Course User Fee: \$25.00

MUN**MUN1120 CONCERT BAND 1.000 credits**

Lab Hours: 48.000

MUN1210 SYMPHONY ORCHESTRA 1.000 - 3.000 credits

Lab Hours: 32.000 - 96.000

MUN1310 COLLEGE CHOIR 1.000 credits

Lecture Hours: 48.000

MUN1340 CHAMBER SINGERS 1.000 credits

Lab Hours: 48.000

MUN1391 GOSPEL ENSEMBLE 1.000 credits

Provides an opportunity to study and perform music of Black composers with emphasis placed on contemporary gospel idioms. This course is open to all students with the permission of the instructor. May be repeated for credit. (3 hrs. per week)

Lecture Hours: 48.000

MUN1420 CHAMBER MUSIC, WOOD 1.000 credits

Lab Hours: 48.000

MUN1430 CHAM MUS BRASS ENS 1.000 - 3.000 credits

Lecture Hours: 48.000 - 144.000

MUN1460 CHAMBER MUSIC, STRNG 1.000 credits

The performance of ensemble literature involving strings or other instruments in combination with strings. Particular attention given to literature of the five major periods. Open to all students with the permission of the instructor. It may be repeated for credit.

Lab Hours: 48.000

MUN1480 GUITAR ENSEMBLE 1.000 - 3.000 credits

Lecture Hours: 120.000

MUN1710 STAGE BAND 1.000 credits

Please insert rationale.

Lab Hours: 48.000

Course User Fee: \$10.00

MUN1720 VOCAL JAZZ/POP ENSEM 1.000 credits

Lecture Hours: 48.000

MUN2341 VOCAL ENSEMBLE 2.000 - 3.000 credits

An in-depth performance experience including classical and popular choral literature. Extensive public performance schedule provides professional training. May be repeated for credit. By permission of the instructor.

Lab Hours: 80.000 - 120.000

MUN2410 STRING ENSEMBLE 2.000 - 3.000 credits

Extended rehearsal schedule provides acquisition of specialized ensemble performance techniques. Literature includes classical and popular. May be repeated for credit. By permission of instructor.

Lab Hours: 80.000 - 120.000

MUN2711 JAZZ ENSEMBLE 2.000 credits

A performing group providing advanced skill in reading and interpreting jazz literature. May be repeated for credit. By permission of the instructor.

Lab Hours: 80.000

MUN2712 STUDIO JAZZ 1.000 credits

XXX

Lecture Hours: 48.000

MUO**MUO1501 OPERA WORKSHOP 1.000 - 3.000 credits**

Please insert rationale.

Lab Hours: 48.000 - 144.000

Course User Fee: \$10.00

MUS**MUS1211 DICTION/SINGING 1 2.000 - 3.000 credits**

Lecture Hours: 32.000 - 48.000

MUS1241 DICTION/SINGING 2 2.000 - 3.000 credits

Lecture Hours: 32.000 - 48.000

MUS2304 Introduction to DJ 3.000 credits

This course is designed to teach fundamental techniques of playing pre-recorded music using turntables, DJ software, and digital audio controllers. Through hands-on learning, students explore both the creative and technical facets of DJing for a variety of audiences, preparing them to pursue professional DJ careers.

Lecture Hours: 48.000

Course User Fee: \$60.00

MUT

MUT1001 FUND MUSIC THEORY 3.000 credits

Please insert rationale.

Lecture Hours: 48.000

Course User Fee: \$15.00

MUT1003 FUND MUSC THEORY LAB 1.000 - 3.000 credits

The development of basic aural skills through sight singing and ear training exercises. Corequisite:

MUT 1001. (2-6 hrs. per week)

Lecture Hours: 32.000 - 96.000

Course User Fee: \$10.00

MUT1111 Theory 1 3.000 credits

The techniques of writing four-part chord progressions using root position and inversions of the primary and secondary triads and the dominant and supertonic seventh; also, non-harmonic tones, melodic writing, and an introduction modulation.

Lecture Hours: 48.000

Prerequisite: MUT1001 or passing score on departmental placement exam., Corequisite: MUT1241

Course User Fee: \$15.00

MUT1112 THEORY 3.000 credits

Lecture Hours: 48.000

MUT1241 Sightsinging & Ear Training 1 Year 1.000 credits

The development of aural skill by means of rhythmic and melodic dictation and sightsinging.

Lab Hours: 32.000

Corequisite: MUT1111, MUT1112

Course User Fee: \$10.00

MUT1242 Sightsinging & Ear Training 2 Year 1.000 - 2.000 credits

The development of aural skills by means of rhythmic and melodic dictation and sightsinging.

Lab Hours: 16.000 - 32.000

Prerequisite: MUT1241, Corequisite: MUT1112

Course User Fee: \$10.00

MUT2116 Theory 3 3.000 credits

The continuation of modulation and the presentation of diatonic sevenths, secondary dominants, altered chords, augmented and Neapolitan sixths; melodic and harmonic analysis of selected works; ninth, eleventh, and thirteenth chords, and instrumental part writing.

Lecture Hours: 48.000

Prerequisite: MUT1112, Corequisite: MUT2246

Course User Fee: \$15.00

MUT2117 Theory 4 3.000 credits

The continuation of modulation and the presentation of diatonic sevenths, secondary dominants, altered chords, augmented and Neapolitan sixths; melodic and harmonic analysis of selected works; ninth, eleventh, and thirteenth chords, and instrumental part writing.

Lecture Hours: 48.000

Prerequisite: MUT2116, Corequisite: MUT2247

Course User Fee: \$15.00

MUT2238 INT JAZZ KEYBD HARM 1.000 credits

Please insert rationale.

Lecture Hours: 32.000

Course User Fee: \$10.00

MUT2239 JAZZ KEYBD HARMONY 2 1.000 credits

Please insert rationale.

Lecture Hours: 32.000

Course User Fee: \$10.00

MUT2246 Sightsinging and Ear Training 2 Yr. 1.000 - 2.000 credits

Develops aural and visual skills by means of rhythmic, melodic and harmonic dictation and sightsinging. Emphasis is on chromatic materials.

Lab Hours: 32.000 - 64.000

Prerequisite: MUT1242, Corequisite: MUT2117

Course User Fee: \$10.00

MUT2247 Sight Singing and Ear Training 2 1.000 - 2.000 credits

Develops aural and visual skills by means of rhythmic, melodic and harmonic dictation and sight singing. Emphasis is on chromatic materials.

Lab Hours: 32.000 - 64.000

Prerequisite: MUT2246, Corequisite: MUT2117

Course User Fee: \$10.00

MUT2351 INTRO POP MUS ARRANG 3.000 credits

A skills course providing basic experience with instrumental, ranges, transpositions two and three-part writing. Prerequisite: MUS 113 or permission of instructor; corequisite: MUS 232.

Lecture Hours: 48.000

MUT2352 POP MUS ARRANG 2 3.000 credits

A continuation of MUS 234 with the addition of four, five, and six-part writing. Concentration on scoring techniques. Prerequisite: MUS 234; corequisite: MUS 233.

Lecture Hours: 48.000

MUT2641 INTRO JAZZ IMPROV 1 3.000 credits

Please insert rationale.

Lecture Hours: 48.000

Course User Fee: \$15.00

MUT2642 JAZZ IMPROVIZATION 2 3.000 credits

Please insert rationale.

Lecture Hours: 48.000

Course User Fee: \$15.00

MVB**MVB1011 PRE-APPLIED TRUMPET 2.000 credits**

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVB1012 PRE-APPLIED HORN 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

MVB1013 PRE-APPLIED TROMBONE 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

MVB1014 PRE-AP BARITONE HORN 2.000 credits

Please insert rationale

Lecture Hours: 16.000

Course User Fee: \$110.00

MVB1015 PRE-APPLIED TUBA 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVB1211 TRUMPET-SEC 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVB1212 HORN-SECONDARY 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVB1213 TROMBONE-SEC 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVB1214 BARITON HORN-SEC 1YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVB1215 TUBA-SEC 1 YR 1.000 credits

Please insert rationale

Lecture Hours: 8.000

MVB1311 TRUMPET-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB1312 HORN-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB1313 TROMBONE-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB1314 BARITON HORN-PRN 1YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB1315 TUBA-PRINCIPAL 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB2221 TRUMPET-SEC 2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVB2224 BARITON HORN-SEC 2YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVB2225 TUBA-SEC SECOND YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVB2321 TRUMPET-PRIN 2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB2322 FRENCH HORN-PRIN 2YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB2323 TROMBONE-PRIN 2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB2324 BARITON HORN-PRN 2YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVB2325 TUBA-PRIN 2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ**MVJ1010 PRE-APPL JAZZ PIANO 2.000 credits**

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fee. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVJ1011 PRE-APPL JAZZ VOICE 2.000 credits

Private vocal for those music students who are not prepared to perform at the college music major level. Special fee. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVJ1013 PRE-APPL JAZZ GUITAR 2.000 credits

Please insert rationale.

Lecture Hours: 16.000

Course User Fee: \$110.00

MVJ1014 PRE-APPL ELEC BASS 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVJ1016 PRE-APPL JAZZ SAX 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVJ1017 PRE-APP JAZZ TRUMPET 2.000 credits

Please insert rationale.

Lecture Hours: 16.000

Course User Fee: \$110.00

MVJ1018 PRE-AP JAZZ TROMBONE 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVJ1019 PR-AP JAZZ PERCUSSION 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVJ1210 JAZZ PIANO SEC 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVJ1211 JAZZ VOICE SEC-1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVJ1212 JAZZ VIOLIN SEC-1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr per week)

Lecture Hours: 8.000

MVJ1213 JAZZ GUITAR SEC-1 YR 1.000 credits

Private instruction in first year secondary instrument. Prerequisite: Departmental approval. Special . May be repeated for credit. (1 hr. per week)

Lecture Hours: 8.000

MVJ1214 ELECT BASS SEC-1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ1215 JAZZ FLUTE SEC- 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

MVJ1216 JAZZ SAX SEC 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ1217 JAZZ TRUMPET SEC-1YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ1218 JAZZ TROMBONE SC-1YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ1219 JAZZ PRC-DRM SEC 1YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ1310 JAZZ PIANO PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ1311 JAZZ VOICE PRIN-1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ1313 JAZZ GUITAR PRIN-1YR 2.000 - 3.000 credits

Private instruction in first year principal instrument. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ1314 ELECT BASS PRIN-1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ1315 JAZZ FLUTE PRIN-1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ1316 JAZZ SAXO PRIN-1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ1317 JAZZ TRUMPET PRN-1YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ1318 JAZZ TROMBONE PR-1YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ1319 JAZZ PRC-DRM PRN 1YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2220 JAZZ PIANO SEC 2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ2221 JAZZ VOICE SEC-2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ2222 JAZZ VIOLIN SEC-2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ2223 JAZZ GUITAR SEC-2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ2224 ELECT BASS SEC-2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ2225 JAZZ FLUTE SEC-2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ2226 JAZZ SAX SEC- 2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ2229 JAZZ PRC-DRM-SEC 2YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVJ2320 JAZZ PIANO PRIN-2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2321 JAZZ VOICE PRIN-2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2322 JAZZ VIOLIN PRN-2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2323 JAZZ GUITAR PRIN-2YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2324 ELECT BASS PRIN-2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2326 JAZZ SAX PRIN-2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2327 JAZZ TRUMPET PRN 2YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2328 JAZZ TROMBONE PR-2YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVJ2329 JAZZ PRC-DRM PRN 2YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVK**MVK1011 PRE-APPLIED PIANO 2.000 credits**

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fee. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVK1012 PRE-APPL HARPSICHORD 2.000 credits

XXX

Lecture Hours: 16.000

Course User Fee: \$110.00

MVK1013 PRE-APPLIED ORGAN 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVK1111 PIANO-CLASS 1.000 credits

The secondary area of piano with emphasis on sight-reading, melody harmonization, and ensemble playing. Required of all music students except piano majors. May be repeated for credit. (2 hr. per week)

Lecture Hours: 32.000

MVK1112 Class Piano 2 1.000 credits

A continuation of MVK 1111.

Lab Hours: 32.000

Prerequisite: MVK1111 or placement by exam.

MVK1211 PIANO-SEC 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVK1212 HARPSICHORD-SEC 1YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVK1213 ORGAN-SEC 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVK1312 HARPSICHORD-PRIN 1YR 2.000 - 3.000 credits

Lecture Hours: 16.000 - 24.000

MVK2121 CLASS PIANO 3 1.000 credits

Further development of elementary keyboard techniques and musicianship, enhancing skills previously developed: Prerequisite MVK 1112 or placement by exam. (2 hr. lab)

Lecture Hours: 32.000

MVK2122 CLASS PIANO 4 1.000 credits

A continuation of MVK 2121. Prerequisite MVK 2121 or placement by exam. May be repeated for credit. (2 hr. lab)

Lecture Hours: 32.000

MVK2221 PIANO-SEC 2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVK2222 HARPSICHORD-SEC 2YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVK2223 ORGAN-SECONDARY 2 YR 1.000 credits

Private instruction in the secondary areas of piano or organ. Special fee.

Lecture Hours: 8.000

Course User Fee: \$60.00

MVK2321 PIANO-PRIN 2 YR 2.000 - 3.000 credits

Private instruction in second year principal instrument. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVK2322 HARPSICHORD-PRIN 2YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVK2323 ORGAN-PRIN 2YR 2.000 - 3.000 credits

Lecture Hours: 16.000 - 24.000

MVO**MVO1214 RECORDER/SEC 1 YR 1.000 credits**

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVP**MVP1011 PRE-APPL PERCUSSION 2.000 credits**

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVP1211 PERCUSSION-SEC 1 YR 1.000 credits

XXX

Lecture Hours: 8.000

Course User Fee: \$60.00

MVP1311 PERCUSSION-PRIN 1 YR 2.000 - 3.000 credits

Please insert rationale.

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVP2221 PERCUSSION-SEC 2 YR 1.000 credits

Private instruction in the secondary area of percussion. Special fee.

Lecture Hours: 8.000

Course User Fee: \$60.00

MVP2321 PERCUSSION-PRIN 2 YR 2.000 - 3.000 credits

Please insert rationale.

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS**MVS1011 PRE-APPLIED VIOLIN 2.000 credits**

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fee. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVS1012 PRE-APPLIED VIOLA 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVS1013 PRE-APPLIED CELLO 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVS1014 PRE-APPL STRING BASS 2.000 credits

Please insert rationale.

Lecture Hours: 16.000

Course User Fee: \$110.00

MVS1015 PRE-APPLIED HARP 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVS1016 PRE-APPLIED GUITAR 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

**MVS1017 PRE-APPL BASS GUITAR 2.000 credits
XXX**

Lecture Hours: 16.000

Course User Fee: \$110.00

MVS1110 CLASS STRINGS 1.000 credits

Lecture Hours: 32.000

MVS1211 VIOLIN-SEC FIRST YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS1212 VIOLA-SEC FIRST YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS1213 CELLO-SEC FIRST YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS1214 BASS-SEC FIRST YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS1215 STRING (SECONDARY IN 1.000 credits

Lab Hours: 8.000

MVS1216 GUITAR-SEC FIRST YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS1311 VIOLIN-PRIN FIRST YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS1312 VIOLA-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in the principal areas of string instruments. Other students may take the course only with the permission of the instructor. Special fee.

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS1313 CELLO-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS1314 BASS-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS1315 HARP-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS1316 GUITAR-PRIN FIRST YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS2223 CELLO-SEC 2 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS2224 BASS-SEC SECOND YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS2225 HARP-SEC SECOND YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS2226 GUITAR-SEC SECOND YR 1.000 credits

Private instruction in the secondary areas of Violin, Viola, Cello, String Bass or Guitar. Required for applied majors, optional for music education majors. Courses must be taken in sequence. Special fee.

Lecture Hours: 8.000

Course User Fee: \$60.00

MVS2321 VIOLIN-PRIN 2 YEAR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS2322 VIOLA-PRIN 2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS2323 CELLO-PRIN 2 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVS2324 BASS-PRIN 2 YR 2.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 24.000

Course User Fee: \$110.00

MVS2326 STRING (PRINCIPAL IN 2.000 credits

Private instruction on Violin, Viola, Cello, String Bass, or Guitar in the principal area. Required each term for music majors. Courses must be taken in sequence. Prerequisite: performing ability equivalent to MUS 903 and department approval. Special fee.

Lab Hours: 16.000

Course User Fee: \$110.00

MVV**MVV1011 PRE-APPLIED VOICE 2.000 credits**

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fee. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVV1111 VOICE (CLASS) 1.000 credits

Lab Hours: 32.000

MVV1211 VOICE-SEC 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVV1311 VOICE (PRIVATE) 2.000 credits

Private instruction in the principal area of percussion. Other students may take the course only with the permission of the instructor. Special fee.

Lab Hours: 16.000

Course User Fee: \$110.00

MVV2221 VOICE-SEC 2 YR 1.000 credits

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Lecture Hours: 8.000

Course User Fee: \$60.00

MVV2321 VOICE-PRIN SECOND YR 2.000 - 3.000 credits

Please insert rationale.

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW**MVW1011 PRE-APPLIED FLUTE 2.000 credits**

Please insert rationale.

Lecture Hours: 16.000

Course User Fee: \$110.00

MVW1012 PRE-APPLIED OBOE 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fee. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVW1013 PRE-APPLIED CLARINET 2.000 credits

Private instrumental for those music students who are not prepared to perform at the college music major level. Special fees. (1 hr. per week)

Lecture Hours: 16.000

Course User Fee: \$110.00

MVW1014 PRE-APPLIED BASSOON 2.000 credits

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Lecture Hours: 16.000

Course User Fee: \$110.00

MVW1015 PRE-APPL SAXOPHONE 2.000 credits

Please insert rationale.

Lecture Hours: 16.000

Course User Fee: \$110.00

MVW1211 FLUTE-SEC FIRST YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVW1212 OBOE-SEC FIRST YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVW1213 CLARINET-SEC 1 YR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVW1215 SAXOPHONE-SEC 1 YEAR 1.000 credits

Private instruction in the secondary area of woodwind instruments. Required for applied majors, optional for music education majors. Courses must be taken in sequence. Special fee.

Lecture Hours: 8.000

Course User Fee: \$60.00

MVW1311 FLUTE-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW1312 OBOE-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW1313 CLARINET-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW1314 BASSOON-PRIN 1 YEAR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW1315 SAXOPHONE-PRIN 1 YR 2.000 - 3.000 credits

Private instruction in the principal areas of woodwind instruments. Other students may take the course only with the permission of the instructor. Special fee.

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW2221 FLUTE-SEC 2 YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVW2223 CLARINET-SEC 2 YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVW2224 BASSOON-SEC 2 YEAR 1.000 credits

Private instruction in a secondary instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1/2 hr. per week)

Lecture Hours: 8.000

Course User Fee: \$60.00

MVW2225 SAXOPHONE-SEC 2 YEAR 1.000 credits

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Lecture Hours: 8.000

Course User Fee: \$60.00

MVW2321 FLUTE-PRIN SECOND YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW2322 OBOE-PRIN SECOND YR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW2323 CLARINET-PRIN 2 YEAR 2.000 - 3.000 credits

Private instruction in a principal instrument or voice. Required each term for music majors. Courses in each area must be taken in sequence. Prerequisite: Departmental approval. Special fee. May be repeated for credit. (1 hr. per week)

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

MVW2325 SAXOPHONE-PRIN 2 YR 2.000 - 3.000 credits

Private instruction in the principal areas of woodwind instruments. Required each term for music majors. Courses must be taken in sequence. Prerequisite: performing ability equivalent to MUS 903 and department approval. Special fee.

Lecture Hours: 16.000 - 24.000

Course User Fee: \$110.00

NMT**NMT1002 Introduction Nuclear Medicine 2.000 credits**

This course is designed to provide an introduction to the field of Nuclear Medicine. Students will learn about the history of the profession, terminology, hospital and patient safety, infection control, patient assessment, accessing and utilizing the patient's medical record, critical thinking, Nuclear Medicine protocols, and patient education. Prerequisites: CHM 1033, 1033L. Co-requisite: NMT 1002L.

Lecture Hours: 32.000

NMT1002L Introduction to Nuclear Medicine Laboratory 2.000 credits

Provides an overview of the field of nuclear medicine. Students will focus on medical terminology, the history of nuclear medicine, basic concepts of radiochemistry, radiation safety, medical law, patient care and positioning, and hospital administration. Field trips to nuclear medicine training facilities are included.

Prerequisites: CHM 1033, 1033L; Corequisites: NMT 1312C, NMT 1705C, NMT 2613. (4 hr. lab)

Lab Hours: 64.000

Prerequisite: CHM1033 AND CHM1033L AND

Corequisite: NMT1312 AND NMT2613

NMT1312C Radiation Protection and Radiobiology 2.000 credits

This course covers all local, state and federal regulations related to Nuclear Medicine. Students will learn the appropriate protection procedures to limit exposure, the performance of area surveys and wipe tests, the proper decontamination procedures, the disposal of radioactive waste procedures, and personnel monitoring of radiation exposure.

Corequisites: NMT 1002L, 2613. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000

Corequisite: NMT1002L AND NMT2613

NMT1705C Nuclear Medicine Pre-Clinical 2.000 credits

This nuclear medicine technology course prepares students to attend to patients, and evaluate data from patient records, make dose calculations, prepare radio-pharmaceuticals, perform in-vivo and in-vitro diagnostic studies, and perform quality control procedures. Prerequisites: BSC 2085/L, BSC 2086/L, CHM 1033/L; Corequisites: NMT 1002L, NMT 1312C, NMT 2613. (1 hr. lecture, 3 hr. clinical)

Lecture Hours: 16.000, Clinical Hours: 48.000

Course User Fee: \$293.31

NMT1713C Nuclear Medicine Methodology 1 2.000 credits

This course will include the imaging parameters necessary to obtain images for the basic procedures performed in a Nuclear Medicine department. Students will learn about imaging procedures related to the following systems: skeletal, central nervous, cardiovascular genitourinary, respiratory and gastrointestinal. Instrumentation necessary to produce the required images as well as patient management during the procedures will be addressed. Prerequisites: BSC 2085, 2085L, 2086, 2086L, CHM1033, 1033L; Co-requisites: NMT 2102, NMT 2534C, NMT 2804C. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 32.000

NMT2102 Nuclear Medicine Administration 1.000 credits

The student will learn the administrative duties required of a nuclear medicine technologist. Areas covered include patient scheduling, radioisotope ordering; scheduling and testing; communication; patient and clinician satisfaction. Prerequisites: NMT 1002L, NMT 1312C, NMT 1705C, NMT 2613; Corequisites: NMT 1713C, NMT 2534C, NMT 2804C. (1 hr. lecture)

Lecture Hours: 16.000

NMT2130C Nuclear Medicine Radiopharmacy 3.000 credits

Students will learn how to maintain radiopharmaceutical laboratory records and materials, obtain a generator eluate, prepare radiopharmaceuticals and perform quality control tests, as well as dispose of radioactive waste appropriately. The ordering of pharmaceuticals in appropriate dosage and effective time frames will also be included. Prerequisites: NMT 1002L, NMT 1312C, NMT 1705C, NMT 2613; Corequisites: NMT 2130C, NMT 2723C, NMT 2814C. (1 hr. lecture, 2 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

NMT2534C Instrumentation, Quality Assurance and Control 3.000 credits

This course will integrate and correlate the principles of electrical and nuclear physics associated with the operation, calibration, quality assurance and control testing of imaging systems and radiation detection devices employed in nuclear medicine. The student will learn the various types of devices used to provide information obtained from diagnostic images. Prerequisites: NMT 1002L, NMT 1312C, NMT 2613 and PHY1020; Corequisites: NMT 1713C, 2102, 2804C. (1 hr. lecture, 4 hr. lab)

Lecture Hours: 16.000, Lab Hours: 64.000

NMT2613 Nuclear Medicine Physics and Mathematical Applications 3.000 credits

Students will learn the basic concepts of atomic, nuclear and radiation physics with an emphasis on the interactions of radiation with matter. Alpha, beta, and gamma sources are explained as well mathematical principles associated with the types of radiation sources. Prerequisites: MAC 1105 and PHY 1020; Corequisites: NMT 1002L, NMT 1312C. (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: MAC1105 AND PHY1020 AND

Corequisite: NMT1002L AND NMT1312

**NMT2723C Nuclear Medicine Methodology 2
2.000 credits**

A continuation of Nuclear Medicine Procedures 1, students will learn the imaging parameters necessary to obtain images as well as the use of instrumentation necessary to produce the required images performed in a nuclear medicine department. Exposure to patient management during the procedures will also be addressed. Prerequisites: NMT 1713C, NMT 2102, NMT 2534C, NMT 2804C; Corequisite: NMT 2130C, NMT 2814C. (1 hr. lecture, 3 hr. clinical)
Lecture Hours: 16.000, Clinical Hours: 48.000
Prerequisite: NMT1713 AND NMT2804C,
Corequisite: NMT2573 AND NMT2814C

**NMT2733C Nuclear Medicine Methodology 3
2.000 credits**

A continuation of Nuclear Medicine Procedures 2, students will learn the imaging parameters necessary to obtain images as well as the use of instrumentation necessary to produce the required images performed in a nuclear medicine department. Exposure to patient management during the procedures will also be addressed. Prerequisites: BSC 2085/L, BSC 2086/L, NMT 1713C, NMT 2723C; Corequisites: NMT 2779C, NMT 2824C. (1 hr. lecture, 3 hr. clinical)
Lecture Hours: 16.000, Clinical Hours: 48.000
Course User Fee: \$239.31

NMT2779C Multi-Modalities and Cross-Sectional Anatomy 2.000 credits

This course educates the student upon proper recognition and interpretation of cross-sectional anatomy. The student will also compare and analyze images from complementary modalities. It is crucial for the nuclear medicine technologist to understand three-dimensional imaging in order to enhance patient care and be an asset to the facility. Prerequisites: NMT 2130C, NMT 2723C, NMT 2814C; Corequisites: NMT 2733C, NMT 2824C. (1 hr. lecture, 2 hr. lab)
Lecture Hours: 16.000, Lab Hours: 32.000
Prerequisite: NMT 2130C, NMT 2723C, NMT 2814C,
Corequisite: NMT 2733C, 2824C
Course User Fee: \$239.38

NMT2804C Nuclear Medicine Clinic Practice & Conference 1 3.000 credits

This course will introduce the student to the fundamentals of clinical nuclear medicine primarily through hospital involvement. The student will learn practical experience in a Nuclear Medicine department by performing the principles taught in class. Prerequisites: NMT 1002L, NMT 1312C, NMT 1705C, NMT 2613; Corequisites: NMT 1713C, NMT 2102, NMT 2534C. (144 hrs. clinic)
Clinical Hours: 144.000
Prerequisite: NMT 1002L, NMT 1312C, NMT 1705C, NMT 2613, Corequisite: NMT 1713C; NMT 2102, NMT 2534C
Course User Fee: \$25.00

NMT2814C Nuclear Medicine Clinic Practice & Conference 2 5.000 credits

This course is a continuation of NMT 2804C Clinic 1 and will provide the student the opportunity to participate in the fundamentals of clinical nuclear medicine in the hospital involvement. The student will learn practical experience in a Nuclear Medicine department by performing the principles taught in class.
Clinical Hours: 240.000
Prerequisite: NMT2804C NMT 2130, NMT 2534C, NMT 2613, NMT 2804C
Course User Fee: \$25.00

NMT2824C Nuclear Medicine Clinical Education 3 5.000 credits

This is the final course in the series of three clinical courses. Students will learn to apply all didactic competencies in the Nuclear Medicine department setting, as well as perform all procedures from the two Nuclear Medicine Procedures courses with minimal supervision. The ARRT Competency Requirements must be completed in this course. Prerequisites: NMT 2814C.
Clinical Hours: 240.000
Prerequisite: NMT2814C
Course User Fee: \$25.00

NMT2834C Nuclear Medicine Clinical Education 4 5.000 credits

This is the final course in the series of four clinical courses. Students will learn to apply all didactic competencies in the Nuclear Medicine department setting, as well as perform all procedures from the two Nuclear Medicine Procedures courses with minimal supervision. The ARRT Competency Requirements must be completed in this course. Clinical Hours: 240.000
Prerequisite: NMT2824C
Course User Fee: \$239.31

NMT2932C Nuclear Medicine Seminar 3.000 credits

The student will learn to incorporate all theory related to the production of a nuclear medicine image. The student will also learn about radiation protection, instrumentation, physics, pharmacology, and Quality Assurance/Quality Control.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: NMT1312NMT 2733C, NMT 2779C, NMT 2824C, Corequisite: NMT2824C

Course User Fee: \$25.00

NSP**NSP3685 End-of-Life Nursing Care 3.000 credits**

The purpose of this course is to provide healthcare professionals an overview of End-of-Life palliative and hospice care. Students will learn pain and symptom management, ethical and cultural considerations, assess psychosocial and emotional concerns of the patient and family, and describe the care of the patient during the last hours. Course is restricted to BSN students, requires departmental permission.

Lecture Hours: 48.000

NUR**NUR1002 Transition to Registered Nursing 2.000 credits**

This course introduces the student with selected prior health care experience and education to the profession of nursing, the roles basic to nursing practice, nursing process and the implementation of health-promoting activities to meet patient needs.

Prerequisite: Admission to the School of Nursing.

Co-requisites: NUR 1002L, 1141.

Lecture Hours: 32.000

Course User Fee: \$125.00

NUR1002L Transition to Medical Surgical Nursing Clinical Lab 2.000 credits

This lab introduces students with prior health care education to the nursing profession. Students will learn the nurse's role in meeting short and long term needs of the patient through preventive, therapeutic and palliative care. Students will also explore nursing care of the adults with moderate alterations in health within a body systems framework. Pre-Req: Program Admission; Co-Req: NUR 1002 and NUR 1141. (192 Hr. Clinical)

Clinical Hours: 96.000

Course User Fee: \$116.00

NUR1006 Health Care Professions to RN Transition 5.000 credits

This course is designed for Medical Assistants, Respiratory Therapist, and Paramedics entering into the Transition program and will introduce the framework of the nursing process to non-nurse health care providers desiring to transition into the role of the registered nurse. Fundamental nursing concepts related to role transition and the introduction of nursing practices will be emphasized along with essential nursing core competencies including the knowledge, skills, and abilities to provide safe nursing practice. This course also provides the opportunity for participants to observe, review and practice in nursing skills in a simulated laboratory setting. Critical Thinking and Clinical Analysis will be introduced along with the Nursing Process. (NUR 1XXX is required for Licensed Medical Assistant, Paramedic and/or Respiratory Therapist students only). 80.00 hr. lecture

Lecture Hours: 80.000

NUR1008 Transition to Medical Surgical Nursing 5.000 credits

This course introduces students with prior health care education to the nursing profession. Students will learn the nurse's role in meeting short- and long-term needs of the patient through preventive, therapeutic and palliative care. Students will also explore nursing care of the adults with moderate alterations in health within a body systems framework.

Lecture Hours: 80.000

NUR1025 Fundamentals of Nursing 3.000 credits

This course introduces students to the profession of Nursing. The student will learn the roles basic to the practice of nursing, the nursing process, and how nurses are involved in health promoting activities to meet client needs. Pre-Req: Program Admission; Co-Req: NUR 1025C, 1025L, 1060C, and 1141.

Lecture Hours: 48.000

Course User Fee: \$113.00

NUR1025C Fundamentals of Nursing Skills Lab 1.000 credits

Students will learn of the opportunities for the explanation, demonstration, and practice of care provider activities essential to the basic practice of nursing. Learning experiences are provided in the skills laboratory. Pre-Req: Program Admission; Co-Req: NUR 1025, 1025L, 1060C, 1141

Lab Hours: 32.000

Course User Fee: \$105.00

NUR1025L Fundamentals of Nursing Clinical Lab 2.000 credits

Students will learn of the opportunities for the explanation, demonstration, and practice of care provider activities essential to the basic practice of nursing. Learning experiences are provided in the skills laboratory. Pre-Req: Program Admission; Corequisite: NUR 1025, 1025C, 1141, 1060C.

Clinical Hours: 96.000

Course User Fee: \$8.00

NUR1060C Adult Health Assessment 3.000 credits

This course is designed to provide students with the necessary skills to perform an in-depth nursing history and a complete physical examination on an adult client. The focus will be on clients with minimal or no alterations in their health state.

Students will be introduced to and will demonstrate the techniques used in physical examination.

Prerequisite: Admission to the School of Nursing;

Corequisites: NUR 1025, 1025C, 1025L, 1141.

Lecture Hours: 32.000 - 33.000, Lab Hours: 32.000,

Practicum Hours: 2.000

Prerequisite: Admission to the program

Course User Fee: \$109.00

NUR1141 Nursing Math & Pharmacology 3.000 credits

Students will learn about medications and their effects on different body systems. The conceptual and mathematical operations necessary for safe and effective administration of intravenous medications, preparing medications that come in powdered form and adjusting medication administration based on medical protocols are discussed. Concepts of medications including history, nomenclature, sources of drug information, federal drug laws and standards, medication classifications, pharmacokinetics, pharmacodynamics, variables affecting medication actions and effects, and methods of delivery will be addressed. Co-Req: NUR 1025, 1025C, 1025L, 1060C.

Lecture Hours: 48.000, Practicum Hours: 1.000

Course User Fee: \$75.00

NUR1142 Introduction to Nursing Math & Pharmacology 1.000 credits

Students will learn concepts of medications including history, nomenclature, sources of drug information, federal drug laws and standards, medication classifications, pharmacokinetics, pharmacodynamics, variables affecting medication actions and effects, and methods of delivery. Pre-Req: Program Admission; Co-Req: NUR 1025, 1025C, 1060C or NUR 1002, 1002L.

Lecture Hours: 16.000

NUR1211 Medical-Surgical Nursing 3.000 credits

This course provides an introduction to the adult nursing care. Students will learn the nurse's role in meeting the short and long term needs of the client and community through preventive, therapeutic and palliative care are discussed. Pre-Req: NUR 1025, 1025C, 1060C, 1142; Co-Req: NUR 1211L, 1214C.

Lecture Hours: 48.000

Course User Fee: \$121.00

NUR1211L Medical Surgical Nursing Clinical 2.000 credits

Students will learn how to apply concepts of adult health nursing. Experiences in both in-patient and community settings will be provided. Students are encouraged to participate in projects emphasizing preventive aspects of nursing care. Prerequisites:

NUR 1025, 1025C, 1025L, 1060C, and 1141

Clinical Hours: 96.000

Course User Fee: \$87.00

NUR1214C Medical Surgical Nursing Skills Lab 1.000 credits

Students will have the opportunities for the explanation, demonstration, and practice of skills related to adult health nursing. Learning experiences are provided in the School of Nursing Skills Laboratory. Pre-Req: NUR 1025, 1025C, 1025L, 1060C, 1141; CO-REQ: NUR 1211, 1211L.

Lecture Hours: 8.000, Lab Hours: 16.000

Course User Fee: \$200.00

NUR2212 ADV MED-SURGICAL NUR 3.000 credits

This course explores the medical surgical nursing care of clients with complex alterations in health. Students will learn advanced concepts in medical-surgical nursing which will be discussed within a body systems framework focusing on the nurse's role in meeting the needs of the client, family, and community. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1002, 1002L; NUR 1142, 2310, 2310L, 2420 2420L, 2680L

Lecture Hours: 48.000

Course User Fee: \$377.00

NUR2212 Medical Surgical Nursing III 3.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1008, 1002L; NUR 1141, 2310, 2310L, 2420 2420L, 2680L
Lecture Hours: 48.000
Course User Fee: \$450.00

NUR2212L Medical Surgical Nursing III Clinical Lab 2.000 credits

This course provides students with the opportunity to apply advanced concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health. Prerequisites: NUR 1211, 1211L, 1214C or NUR 1002, 1008, 1002L; NUR 1141, 2310, 2310L, 2420, 2420L, 2680L
Clinical Hours: 96.000
Course User Fee: \$150.00

NUR2220 Medical Surgical Nursing II 3.000 credits

This course provides students with the opportunity to apply concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health.
Lecture Hours: 48.000
Prerequisite: NUR1025 AND NUR1025C AND NUR1025L AND NUR1060C AND NUR1141 AND NUR1211 AND NUR1211L AND NUR1214C AND Corequisite: NUR2220L

NUR2220L Medical Surgical Nursing II Clinical 1.000 credits

This course provides students with the opportunity to apply concepts of medical surgical nursing. Students will learn to provide health care delivery in both in-patient and community settings. Students will focus on the nurse's role in meeting the needs of the client, family, and community. Students are encouraged to actively participate in projects assisting clients in preventive care and maintenance of health.
Clinical Hours: 48.000
Prerequisite: NUR1025 AND NUR1025C AND NUR1025L AND NUR1060C AND NUR1141 AND NUR1211 AND NUR1211L AND NUR1214C AND Corequisite: NUR2220

NUR2310 Pediatric Nursing 2.000 credits

This course provides a family centered approach to the nursing care of pediatric clients and their families. Students will learn the nurse's role in meeting the short and long term needs of the pediatric client, family, and community through preventative, therapeutic and palliative care, with recognition for the multicultural aspects of client needs. Prerequisites: 1211, 1211L, 1214C or 1002, 1002L; 1142; corequisites: NUR 2310L, 2420, 2420L, 2680L. Special fee. (2 hr. lecture)
Lecture Hours: 32.000
Course User Fee: \$126.00

NUR2310L Pediatric Nursing Clinical Lab 1.000 credits

This course allows the student to apply the nursing process to the care of clients in selected pediatric clinical settings. Students will learn to observe cultural diversity and implement care to the pediatric client, family, and community through preventive, therapeutic and palliative measures. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2420, 2420L, 2520, 2520L, 2680L. Laboratory fee.
Clinical Hours: 48.000
Course User Fee: \$86.00

NUR2420 Obstetrical Nursing 2.000 credits

This course provides a family centered approach to the nursing care of obstetrical clients and their families. Students will learn to assess the pregnant client, to implement caring behaviors for the laboring client, educate the postpartum client, manage the care of the newborn and collaboration of care for the high risk client. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142. Corequisites: NUR 2310, 2310L, 2420L, 2680L. Special fee. (2 hr. lecture)
Lecture Hours: 32.000

NUR2420L Obstetrical Nursing Clinical Lab 1.000 credits

This course provides an introduction to obstetrical nursing clinical practice. Students will learn to apply the nursing process to the care of clients in selected obstetrical clinical settings. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2310, 2310L, 2420, 2680L. Laboratory fee.

Clinical Hours: 48.000

Course User Fee: \$96.00

NUR2520 Psychiatric Nursing 2.000 credits

This course introduces students to the basic concepts of psychiatric nursing. Students will learn to provide care in in-patient and community settings, focusing on the nurse's role in meeting the needs of the patient, family, and the community.

Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; 1142. Co-requisites: NUR 2520L. Special fee. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$62.00

NUR2520L Psychiatric Nursing Clinical Lab 1.000 credits

This course provides the student opportunities to apply concepts of psychiatric nursing. Students will learn psychiatric procedures for both in-patient and community settings, focusing on the nurse's role on meeting the needs of the client, family, and community. Students actively participate in projects assisting clients in preventative care and maintenance of mental health. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L, 1142

Clinical Hours: 48.000

Course User Fee: \$98.00

NUR2680L Community Health Nursing Lab 1.000 credits

This laboratory course assists the students in applying knowledge of community health resources. Students will learn to manage community health resources to support the delivery of care to the childbearing/child-rearing families. Special emphasis is placed on the understanding of cultural influences on the health practices and beliefs within the family. Prerequisites: NUR 1211, 1211L, 1214C or 1002, 1002L; co-requisites: NUR 2310, 2310L, 2420, 2420L.0L. Special fee. (48 hr. clinical)

Clinical Hours: 48.000

Course User Fee: \$81.00

NUR2811C Professional Nursing Leadership 4.000 credits

This course provides the student with the theoretical and clinical knowledge necessary for actualization of the role of the registered professional nurse, with emphasis on delegation and supervision. Prerequisites: NUR 2212, 2212L

Lecture Hours: 16.000, Clinical Hours: 144.000

Course User Fee: \$175.00

NUR2960 Senior Seminar 1.000 credits

This course provides a forum for the discussion, and preparation for the NCLEX-RN exam. Explain applicant requirements for the state board of nursing (and licensure eligibility requirements) and examination registration.

Lecture Hours: 16.000

Prerequisite: NUR2811C

NUR3069 Health Assessment Across the Lifespan Theory 2.000 credits

This theory-based course introduces students to the principles and practices of comprehensive health assessment across the lifespan. Emphasis is placed on health history-taking, clinical reasoning, cultural sensitivity, and ethical standards. Students will learn to interpret subjective data, prioritize patient needs, and plan individualized health promotion strategies. The course culminates in the submission of a comprehensive health history and a narrated PowerPoint presentation featuring an age-appropriate health promotion intervention. (2.0 hours)

Lecture Hours: 32.000

NUR3069L Health Assessment Across the Lifespan Lab 1.000 credits

This hands-on lab course builds on the theory-based course by guiding students through the performance of comprehensive head-to-toe assessments across the lifespan. Students apply therapeutic communication, cultural humility, and ethical standards while practicing physical examination techniques and documentation. The course culminates in a video submission of a complete subjective and objective assessment, demonstrating clinical judgment and evidence-based practice.

Clinical Hours: 48.000

Corequisite: NUR3069 Graduate from an accredited RN Program

NUR3125 Pathophysiology in Nursing Practice 3.000 credits

This course builds upon prior knowledge in assessment, management of patients with commonly occurring diseases, and evaluation patient outcomes. The course will integrate the principles and rationales of pathophysiology at the cellular and systemic level using the nursing process. It will provide an in-depth theoretical and clinical principles of diseases encountered in healthcare settings. Health disparities resulting from social determinants of health, genetics, and stress related causes will also be explored.

Lecture Hours: 48.000

NUR3165 Evidence-Based Nursing Practice 3.000 credits

This introductory course emphasizes developing skills to translate empirical evidence into professional nursing practice, strongly focusing on evidence-based practice (EBP) knowledge. Students will explore best practice guidelines and empirical dissemination to improve patient and organizational outcomes in healthcare delivery. Throughout the course, students will use technology to access, retrieve, organize, and evaluate literature according to levels of evidence to enhance nursing practice. A key component of this course is the responsible use of Artificial Intelligence (AI) in gathering reliable information. Students will learn about AI applications in nursing, ethical considerations, and the importance of transparency, accountability, and fairness in AI systems (3-hour lecture)

Lecture Hours: 48.000 - 52.000

NUR3178 Complementary and Alternative Health Care 3.000 credits

This is an upper-division course in Complementary and Alternative Health Care. This course evaluates complementary and alternative health care in diverse populations across the lifespan and around the globe. The course introduces students to complementary and alternative practice and formulates a paradigm shift from traditional patient care to holistic nursing care. The course addresses different complementary and alternative treatment practices through evidence-based research. The course involves the analysis of current knowledge and practice to illness prevention, health promotion, health restoration, patient education, and empowerment.

Lecture Hours: 48.000

NUR3289 Foundations of Gerontology 3.000 credits

This course focuses on the special healthcare needs of the geriatric population. Students will learn the physical, physiological, psychosocial and geropharmacologic implications related to aging. The trends in the changing demographics and the social consequences of aging will be identified. Established geriatric assessment and evaluative tools will also be discussed.

Lecture Hours: 48.000

Prerequisite: Admission to the BSN - RN program N9100

Course User Fee: \$125.00

NUR3674 Faith Based Community Nursing 3.000 credits

This course will provide education in faith-based community nursing to registered nurses. Students will learn the skills needed to integrate the care of mind, body, and spirit in faith-based communities. The intentional care of the spirit is part of the process in preventing and minimizing illness in a faith-based community.

Lecture Hours: 48.000

Prerequisite: RN with at least 2 years of experience and baccalaureate degree standing. Departmental permission required. (3 hr. lecture)

NUR3805 Foundations & Role Transitions of the Professional Nurse credits

This course will introduce students to the theory and principles for delivery of professional nursing practice and role transition. The course was designed to enhance the transition of registered nurses to the baccalaureate role and builds on existing knowledge and professional experience. The course explores concepts of professional nursing as a way to guide and promote professional practice. Core components of this course focuses on standards of nursing practice as it relates to historical and theoretical framework for practice, moral, legal issues, evidence based practice, quality and safety, leadership, economics, technology, information management and trends in the future of healthcare.

Lecture Hours: 48.000

Corequisite: NUR3069

Course User Fee: \$9.00

NUR3805 Foundations & Role Transitions of the Professional Nurse 3.000 credits

This course will introduce students to the theory and principles for delivering professional nursing practice. The course is designed to enhance the transition of registered nurses to the baccalaureate role and build on existing knowledge and professional experience. The course explores concepts of professional nursing as a way to guide and promote professional practice. The core components of this course focus on standards of nursing practice as it relates to the historical and theoretical framework for practice, moral, legal issues, evidence-based practice, quality and safety, leadership, economics, technology, information management, and trends in the future of healthcare. A minimum grade of "C" or better is required.

Corequisite: NUR 3041. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$9.00

NUR3846 FDNT OF PROF NURSING 3.000 credits

This course explores the evolution of professional nursing knowledge and theories. Concepts are analyzed in relation to conceptual theoretical frameworks within Nursing. Students will integrate philosophies and theories in the delivery of healthcare and theories are introduced as a foundation for the delivery of healthcare in a multicultural/global environment. Minimum grade of "C" or better required. Corequisite: NUR 3069. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$6.00

NUR3997 VALIDATED NURSING CR 30.000 credits

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Lecture Hours: 480.000

NUR4146 Pharmacology for Nursing 3.000 credits

This course is designed for the RN – BSN student and builds on the prior knowledge of entry level registered nurses. The focus is on the pharmacological treatment of various illnesses in diverse individuals across the lifespan. The emphasis is on the principles of pharmacokinetics, pharmacodynamics and pharmacogenetics in the safe treatment of selected illnesses.

Lecture Hours: 48.000

NUR4636 Community Health Nursing 3.000 credits

This course focuses on the holistic aspects of community nursing care applied to diverse global populations across the lifespan. The course introduces students to community nursing practice and formulates a paradigm shift from individual patients to the global community, addressing the history, evolution, theoretical framework, and purpose of community health nursing practice with an introduction to epidemiological principles, concepts of community assessment, health promotion, maintenance and education. The course involves the analysis of current knowledge and practice to illness prevention, health promotion, health restoration, community education and empowerment. Minimum grade of "C" or better required.

Lecture Hours: 48.000

Prerequisite: NUR3069, NUR3805, Corequisite: NUR4636L

Course User Fee: \$13.00

NUR4636L Community Health Nursing Practicum 3.000 credits

This course focuses on the clinical application of Community Health Nursing Theory. Students will utilize the nursing process in the delivery of healthcare within the community environment. Students will assess the individual, family, and/or community, develop a plan of care, and deliver care to an individual, family and/or community within a multicultural environment. Minimum grade of "C" or better required.

Clinical Hours: 144.000

Corequisite: NUR4636

NUR4667 Worldwide Trends in Nursing Care 3.000 credits

This course integrates the nursing process and theory to provide culturally sensitive, competent, and safe nursing care to diverse cultures. The goal of the course is to improve health outcomes by promoting quality of care for diverse populations through the development of self-evaluation and awareness of ethical, cultural practices, values, beliefs, including biases and prejudices. The course also provides the students with a worldview of the influence of culture on health care practices, barriers to transcultural communication, political, economic, social, and demographic issues in a diverse global society. A minimum grade of "C" or better.

Prerequisite: NUR 3069, NUR 3069L, NUR 3805 (3 hr. lecture)

Lecture Hours: 48.000

Prerequisite: NUR3069 AND NUR3069L AND NUR3805

NUR4945L Transforming Practice Through Innovation and Leadership Capstone 3.000 credits

This capstone practice experience empowers RN to BSN students to synthesize all previously acquired competencies gained throughout the program to a real-life clinical, patient, or population problem in a healthcare setting. Focus is on the application of nursing theory and principles, quality and safety standards, evidence-based practice guidelines, leadership skills, interprofessional collaboration, and innovative technologies to a quality improvement project. Students will present their findings through the development of an evidence-based poster.

Clinical Hours: 144.000

Prerequisite: NUR4636 AND NUR4636L

NUR9995 Nursing Statewide Industry Certification 10.000 credits

This course is designed for students who have earned an approved Statewide Nursing Industry Certification and provides academic recognition of the knowledge, skills, and competencies demonstrated through credential attainment. The course validates proficiency in nursing principles, patient care, clinical decision-making, safety practices, and professional standards within healthcare settings. Credit awarded through this course supports academic progression and recognizes documented achievement of industry-recognized nursing competencies. See Advisor to apply Technical Manual of Procedure (TMOP): 1100-815848.

OCB

OCB1010 Introduction to Marine Biology 3.000 credits

An introduction to the biology of the seas. Emphasis is placed on the variety of marine organisms and their structural, physiological, and behavioral adaptations within specific marine environments. Special attention is directed to marine communities, e.g., coral reefs and shallow grass flats, and the factors limiting the distribution of organisms within those communities. Discussions will also be directed towards geological, chemical and physical characteristics of the world's oceans. (3 hr. lecture)

Lecture Hours: 48.000

OCB1010L Introduction to Marine Biology Laboratory 1.000 credits

An optional laboratory class for OCB 1010. This laboratory course stresses understanding, familiarization, and identification of local marine organisms and study of local marine communities through field trips to selected local marine habitats and hands-on laboratory activities. An introduction to field collection methods and various sampling techniques is presented. (2 hr. lab.)

Lecture Hours: 32.000

Pre/Corequisite: OCB1010

Course User Fee: \$23.00

OCE

OCE1001 Introduction to Oceanography 3.000 credits

Using the scientific method, critical thinking skills, and data analysis, this course will examine the fundamental processes of the ocean system, composed of an atmosphere, hydrosphere, lithosphere, and biosphere, through time. The course will also explore interactions between these spheres, including critical analysis of scientific theories and emphasize oceanic connections with humanity. Student learning outcomes: students will use critical thinking to recognize the rigorous standards of scientific theories; students will analyze and synthesize oceanographic data to draw scientifically valid conclusions; students will recognize the different time scales associated with different ocean processes; students will describe interactions between humans and the ocean realm; and students will apply their understanding of oceanographic principles to various marine issues.

Lecture Hours: 48.000

Course User Fee: \$10.00

OPT

OPT1110 Physical & Geometrical Optics 4.000 credits

Behavior of light energy as it passes through air, plastic, glass and water with emphasis on how light is modified by prism and curved lens surfaces. These principles relate to the effect these ophthalmic devices have in correcting the errors of human vision.

Lecture Hours: 64.000

Corequisite: OPT 1205, OPT 1330

OPT1150 Ophthalmic Lenses 2.000 credits

Characteristic of unifocal and multifocal lens reference points for proper lens selection to meet visual needs of the patients. Emphasis is on accurate positioning of the optical centers and selected multi-focal addition design. ANSI and FDA standards; prescription ordering; verification procedures; absorptive lenses; and invisible and progressive multi-focals are presented.

Lecture Hours: 32.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1331, OPT 1331L, OPT 2505, OPT 2070L

OPT1205 Ocular Anatomy, Physiology & Pathophysiology 3.000 credits

The structure and function of the systems of the human body, emphasizing the anatomy, physiology and pathophysiology of the human eye. Visual recognition of common eye disorders is also discussed.

Lecture Hours: 48.000

Corequisite: OPT 1110, OPT 1330

OPT1330 Clinical Data Collection 1 2.000 credits

Techniques necessary in a clinical environment for the collection of patient case history, entrance visual acuity, basic visual skills of ocular mobility and accommodation, color discrimination, depth perception and binocular fusion. Emphasis is also placed on gaining familiarity with the medical terminology as it relates to the visual system.

Lecture Hours: 32.000

Corequisite: OPT 1110, OPT 1205

OPT1331 Clinical Data Collection 2 2.000 credits

Techniques necessary in a clinical environment for the collection of subjective and objective patient diagnostic information including visual field plotting, tonometry, lensometry, keratometry, and sphygmomanometry.

Lecture Hours: 32.000

Prerequisite: OPT1330 OPT 1110, OPT 1205, OPT 1330, Corequisite: OPT 1150, OPT 1331L, OPT 2505, OPT 2070L

OPT1331L Clinical Data Collection 2 Laboratory 1.000 credits

Laboratory for OPT1331 in which students are under the supervision of a licensed practitioner.

Lab Hours: 32.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1150, OPT 1331, OPT 2505, OPT 2070L

Course User Fee: \$10.00

OPT1331L CLIN DATA COLL 2 LAB 1.000 credits

Collecting data during an eye exam is essential for maintaining patient records and providing the proper eyecare. The course teaches the techniques necessary in a clinical environment for the collection of patient case history, entrance visual acuity, visual skills of ocular motility and accommodation, color discrimination, depth perception and binocular fusion. Emphasis is also placed on gaining familiarity with medical terminology as it relates to the visual system. Students will be taught legal, ethical and professional behaviors while serving as a health care provider.

Lab Hours: 32.000

Prerequisite: OPT1330

OPT1450 Ophthalmic Dispensing Procedures 1 1.000 credits

Basic procedures of ophthalmic dispensing such as frame selection, measurement and laboratory ordering. Emphasis will be placed on common ophthalmic frame materials; crown glass and CR-39 lenses; absorptive lenses; and frame alignment, adjustment and repair. The student will demonstrate skills necessary for entry level ophthalmic dispensing in Vision Care Clinic.

Lecture Hours: 16.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1450L

OPT1450L OPHTH DISP PROC 1 L 1.000 credits

Lab Hours: 32.000

Course User Fee: \$15.00

OPT1450L Ophthalmic Dispensing Procedures 1 Laboratory 1.000 credits

Laboratory for OPT 1450.

Lab Hours: 32.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1450

OPT2060 Ophthalmic Management Policy & Procedures 2.000 credits

Procedures and terminology used in the handling of patients, correspondence, legal and ethical principles, inter- and intra-professional relationships, and office management. Develop feasibility report of opening a retail ophthalmic dispensary. The history of opticianry, optometry and ophthalmology is traced. Special emphasis is on a comprehensive review of the curriculum.

Lecture Hours: 32.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2377L, OPT 2422C, OPT 2802L, OPT 2831L, OPT 2876L

OPT2070L Computers for Vision Care 1.000 credits

This course introduces students to the use of computers in ophthalmic practice. Students will learn computer basics and the use of application software for maintaining patient records and billing. Elements of ophthalmic coding are included.

Lab Hours: 32.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330,

Corequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 2505

Course User Fee: \$15.00

OPT2375 Refractometry 1.000 credits

Students will learn the basic principles of refractometry, theoretical aspects of retinoscopy, and the use of cross cylinders. Students will be able to describe various refractive problems and their solutions.

Lecture Hours: 16.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L,

Corequisite: OPT 2375L, OPT 2420, OPT 2420L, OPT 2506, OPT 2506L, OPT 2800L

OPT2375L Refractometry Laboratory 1 1.000 credits

Students will practice theoretical concepts of refractometry using a retinoscope, auto-refractor, and cross cylinders in a laboratory setting.

Lab Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L,

Corequisite: OPT 2375, OPT 2420, OPT 2420L, OPT 2506, OPT 2506L, OPT 2800L

Course User Fee: \$30.00

OPT2376L Refractometry Lab 2 1.000 credits

This course is designed to provide the student with the hands on experience of hand neutralizing a pair of glasses, retinoscopy, and the use of the phoropter and the Snellen chart.

Lab Hours: 32.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L

Course User Fee: \$26.00

OPT2377L Refractometry 3 Laboratory 1.000 credits

This laboratory course will continue to advance the skills already introduced in the previous laboratories 1 & 2. The improved skills will enhance the student's ability to determine the refractive status of the eye and be able to practice these skills on patients in the clinic.

Lab Hours: 32.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2422C, OPT 2802L, OPT 2831L, OPT 2876L

OPT2420 Eyewear Fabrication 1 2.000 credits

Theory of ophthalmic surfacing procedures.

Students acquire knowledge to arrange single vision lenses; use lensometers and lens clock; operate project-o-makers for single vision lens layout; select or fabricate frame patterns; and utilize several systems for edging lenses for ophthalmic frames.

Lecture Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L,

Corequisite: OPT 2375, OPT 2375L, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L

OPT2420L Eyewear Fabrication 1 Laboratory 1.000 credits

Laboratory for OPT 2420.

Lab Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L,

Corequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2506, OPT 2506L, OPT 2800L

Course User Fee: \$15.00

OPT2420L EYEWEAR FABRIC 1 LAB 1.000 credits

This laboratory course will introduce students to the process of making eyewear. It will also be the students' first exposure to the equipment they will use in their future fabrication laboratory experience.

The course allows students to gain practical experience in ophthalmic finishing procedures.

Students will fabricate lenses, use edgers, blockers, polishing wheels, and other machinery. They practice lens layout, pattern making, lensometry and utilize several systems for edging lenses for ophthalmic frames.

Lab Hours: 32.000

Corequisite: OPT2420

OPT2421C Eyewear Fabrication 2 3.000 credits

Advanced techniques in measurement, fabrication, and verification of unifocal and multifocal lenses. Students fabricate finished eyewear from written specifications ensuring that current ANSI and FDA standards are exceeded.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2376L, OPT 2801L, OPT 2830C, OPT 2875L

Course User Fee: \$25.00

OPT2422C Eyewear Fabrication 3 3.000 credits

A continuation of OPT 2421. Theory of evaluation and analysis of eyewear for accuracy and quality. Advanced techniques in operation of automated lens analyzer and lens edgers, and maintenance of equipment.

Lecture Hours: 16.000, Lab Hours: 64.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2377L, OPT 2802L, OPT 2831L, OPT 2876L

Course User Fee: \$25.00

OPT2451 Ophthalmic Dispensing Procedures 2 1.000 credits

Theory and terminology of advanced ophthalmic dispensing. Emphasis will be placed on new technology in ophthalmic frame materials; multifocal lenses including progressive power and blended bifocals; and high index lenses. The process of analyzing the patient's specific needs for the proper frame and lens selection is highlighted.

Lecture Hours: 16.000

Prerequisite: OPT1450, OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070, Corequisite: OPT 2451L

OPT2451L Ophthalmic Dispensing Procedures Laboratory 1.000 credits

Laboratory for OPT 2451.

Lab Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2451

Course User Fee: \$10.00

OPT2505 Contact Lenses 1 3.000 credits

Basic principles of contact lens fitting, emphasizing soft lenses. Topics include lens-relate terminology, anatomy and physiology, patient examination, soft lens materials, design, parameters, handling, fitting and care. Includes introduction to rigid lenses.

Lecture Hours: 48.000

Prerequisite: OPT 1110, OPT 1205, OPT 1330, Corequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 2070L

OPT2506 Contact Lenses 2 2.000 credits

Principles of contact lens fitting, emphasizing rigid lenses. Topics include materials, design parameters, verification, handling, fitting and care. Considers advanced and specialty design and ocular complications.

Lecture Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2506L, OPT 2800L

OPT2506L Contact Lenses 2 Lab 1.000 credits

Practical procedures designed to apply technical skills of contact lens application and removal, verification of the contact lens prescription, modification of hard and hard gas permeable contact lenses, and other skills discussed in previous lecture coursework.

Lab Hours: 32.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2506, OPT 2800L

Course User Fee: \$36.00

OPT2800L Vision Care Clinic 1 2.000 credits

Introductory clinic designed to apply technical skills acquired in previous course work. Recording of clinical data, administrative procedures and techniques in patient handling under close staff supervision.

Clinical Hours: 96.000

Prerequisite: OPT 1150, OPT 1331, OPT 1331L, OPT 1450, OPT 1450L, OPT 2505, OPT 2070L, Corequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2506, OPT 2506L

Course User Fee: \$25.00

OPT2801L Vision Care Clinic 2 4.000 credits

Development of skills in patient reception, clinical data collection, assisting clinician, and ophthalmic dispensing. This is an opportunity to follow the patient through the entire cycle of vision care under close supervision of the clinical staff.

Clinical Hours: 192.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2376L, OPT 2421C, OPT 2830C, OPT 2875L

Course User Fee: \$25.00

OPT2802L Vision Care Clinic 3 4.000 credits

Development of additional skills in visual fields, tonography, ocular photodocumentation, vision therapy/orthoptics, low vision, aseptic techniques, eye emergencies, and assisting in triage and laboratory diagnosis of external eye disease. On and off-campus clinics will be utilized under the close supervision of optometrists and ophthalmologists. Clinical Hours: 192.000 - 19.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2377L, OPT 2422C, OPT 2831L, OPT 2876L

OPT2830C Contact Lenses Clinic 1 2.000 credits

Observe and assist an optometrist in the initial fitting and follow-up care of rigid and soft contact lenses for patients referred from the Vision Care Clinic when conventional eyewear will not suffice. Familiarization with the instructions for lens handling, cleaning, care and storage of contact lenses.

Lab Hours: 64.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2875L

Course User Fee: \$10.00

OPT2831L Contact Lense Clinic 2 1.000 credits

Use of the soft contact lens instruments to confirm all the parameters for replacement lens. Particular attention is devoted to the patient that is having problems with contact lenses after long-term wear due to corneal changes and sensitivity to solutions under direct supervision of an optometrist.

Clinical Hours: 48.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2377L, OPT 2422C, OPT 2802L, OPT 2876L

Course User Fee: \$10.00

OPT2875L Ophthalmic Dispensing Practicum 1 2.000 credits

Externship in an approved finishing laboratory of a retail ophthalmic dispensing establishment. The student will gain a working knowledge of ophthalmic frame and lens stock, inventory system, layout and blocking, chemical and heat treating, edging, tinting assembly and alignment of eyewear according to the written prescription.

Practicum Hours: 96.000

Prerequisite: OPT 2375, OPT 2375L, OPT 2420, OPT 2420L, OPT 2451, OPT 2451L, OPT 2506, OPT 2506L, OPT 2800L, Corequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C

OPT2876L Ophthalmic Dispensing Practicum 2 2.000 credits

Externship in an approved retail ophthalmic dispensing establishment in the area of frame styling, ordering of appropriately designed lenses, adjustment, repair and dispensing of eyewear. The student will gain a working knowledge of administrative management procedures of the practice.

Practicum Hours: 96.000

Prerequisite: OPT 2376L, OPT 2421C, OPT 2801L, OPT 2830C, OPT 2875L, Corequisite: OPT 2060, OPT 2377L, OPT 2422C, OPT 2802L, OPT 2831L

ORH**ORH1251 Nursery Practices 1 3.000 credits**

The student will learn the techniques and practices in commercial production of ornamental plants.

Emphasis on types of nurseries.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: HOS1010

ORH1510 LANDSCAPE PLANT ID 1 3.000 credits

Lecture Hours: 48.000

ORH2932 SPEC TOPC IN LNDSCPE 1.000 credits

Special topics in landscaping offers horticulture students the opportunity of enriching their education with aspects of the field not covered in the A.S. program. Topics will be offered in the areas of irrigation, appropriate landscaping, recent innovations, pests and pesticides, etc. A.S. degree credit only. (1 hr. lecture)

Lecture Hours: 16.000

ORH2949 Landscape Technology Internship 3.000 credits

The internship will provide students with hands-on work experience in horticulture. Landscape, or related technology industries. Students will learn employability skills, and the specific skills and safety requirements necessary for effective work in this environment.

Other Hours: 144.000

OST**OST2335 Business Writing 3.000 credits**

Covers the procedures for writing effective business letters and memorandums, a review of grammar, and the proper format of today's business correspondence. Students learn how to prepare inquiry letters, direct and indirect response letters, application letters and resumes, and short reports.

Lecture Hours: 48.000

Course User Fee: \$15.00

PAS

PAS1800C Physical Diagnosis I 2.000 credits

A course which provides the students with the critical basis for and clinical exposure to techniques used in the proper performance and recording of the physical examination of patients.

Lecture Hours: 16.000, Clinical Hours: 48.000

Prerequisite: BSC 2086, BSC 2086L, CHM 1046, CHM 1046L, MCB 2010, MCB 2010L

Course User Fee: \$560.00

PAS1801C Physical Diagnosis 2 2.000 credits

This course will build upon the skills learned in Physical Diagnosis I and will prepare the physician assistant student with the essential skills for entry into clinical practice. Students will learn to perform medical interviews and physical examinations on live and professional patients, and high-fidelity simulators. Prerequisite: PAS1800C, PAS1803, PAS1831, PAS2936. (1 hr. lecture, 3 hr. clinical)

Lecture Hours: 16.000, Clinical Hours: 48.000

Prerequisite: PAS1800C, PAS1803, PAS1831, PAS2936

Course User Fee: \$560.00

PAS1803 Clinical Anatomy and Physiology 2.000 credits

This course is designed for students accepted into the Physician Assistant Program. The course will review basic Anatomy and Physiology principles, while integrating important clinical concepts. Students will learn to transition from Anatomy and Physiology to Pathophysiology.

Lecture Hours: 32.000

Prerequisite: BSC2085 AND BSC2085L AND BSC2086 AND BSC2086L AND CHM1045 AND CHM1045L AND CHM1046 AND CHM1046L AND MCB2010 AND MCB2010L

PAS1811C Introduction to Medicine I for PAS 5.000 credits

Students will learn the signs, symptoms, and pathophysiology of common diseases affecting pediatric, adult, and geriatric patients. In addition, diagnosis, therapeutic interventions, and follow-up will be learned. Patient education and preventative medicine will also be included. 3 hr. lecture, 2 hr. clinical)

Lecture Hours: 48.000, Clinical Hours: 96.000

Prerequisite: PAS 1800C, PAS 1803, PAS 1831, PAS 2936

Course User Fee: \$1400.00

PAS1812 Behavioral and Community Medicine 1.000 credits

This course utilizes a biopsychosocial system approach to identify social determinants within the health care delivery system. Students will participate in outreach projects to implement concepts learned in the classroom within a community environment.

Prerequisites:

Lecture Hours: 16.000

Prerequisite: PAS1800C AND PAS1831 AND PAS2936, PAS 2YYY

Course User Fee: \$280.00

PAS1813 Pathophysiological Basis of Disease I 2.000 credits

The clinical pathophysiology course serves as a transition from the basic medical sciences to clinical medicine. Students will learn the study of disease, both congenital and acquired with an emphasis on providing understanding of pathologic physiology in conjunction with information regarding medical history and laboratory data to solve case based clinical problems. Students are taught how pathophysiology translates into patient signs, symptoms. Prerequisites: PAS1800C, 2936. (2 hr. clinical)

Clinical Hours: 96.000

Prerequisite: PAS1800C AND PAS2936

PAS1820C Clinical Medicine 2 for Physician Assistants 5.000 credits

The second course in the sequence. Students will learn the signs, symptoms, and pathophysiology of common diseases affecting pediatric, adult, and geriatric patients. In addition, diagnosis, therapeutic interventions, and follow-up will be learned. Patient education and preventative medicine will also be included. Prerequisite: HSA2532, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823. (4 hr. lecture, 3 hr. clinical)

Lecture Hours: 64.000, Clinical Hours: 48.000

Prerequisite: HSA2532, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823

PAS1822C Electrocardiography 1.000 credits

Students will learn the basic knowledge to understand, interpret and utilize an electrocardiogram (EKG) as a diagnostic tool. This course provides the foundation for learning to interpret 12-lead EKG tracings and applying those principles to evaluate the EKG tracings of common cardiac disease.

Lab Hours: 32.000

Prerequisite: PAS 1800C, PAS 1803, PAS 1831, PAS 2936

Course User Fee: \$560.00

PAS1823 Pharmacology 1 4.000 credits

This course will focus on the preparation, uses and action of drugs. Students will learn about Pharmacokinetics and Pharmacodynamics. Students will be introduced to Clinical Pharmacology following an organ system approach, covering the major classes of therapeutically relevant drugs, and how they work at the molecular and cellular levels. Prerequisites: PAS1800C, PAS1803, PAS1831, and PAS2936 (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: HSA2532, PAS1801C, PAS1803, PAS1831, PAS2936

Course User Fee: \$560.00

PAS1824 Pathophysiological Basis of Disease II 2.000 credits

This course is a continuation of PAS 1813 Pathophysiological Basis of Disease I. This course establishes scientific core knowledge and bridges the basic medical sciences with clinical medicine. The course covers the pathophysiology of human diseases that appear as a result of structural and functional alterations of the human body systems. The course begins with the study of human anatomy and physiology pertinent to the pathological conditions presented and progresses to the pathophysiological topics needed by physician assistant students. The course will enhance decision-making ability when working as a PA in clinical practice.

Lab Hours: 32.000, Clinical Hours: 1.000 - 30.000

Prerequisite: PAS1813, PAS1823, PAS1812, PAS1822C, PAS1811, PAS1801C, HSA2532

Course User Fee: \$560.00

PAS1831 Clinical Diagnostic Imaging 1.000 credits

The course introduces the PA student to principles of diagnostic imaging and a familiarity with the many diagnostic techniques available, their values and limitations, and how they may be best used in the management of the patient in clinical practice.

Lecture Hours: 16.000

Prerequisite: BSC2085 AND BSC2085L AND BSC2086 AND BSC2086L AND CHM1045 AND CHM1045L AND CHM1046 AND CHM1046L AND MCB2010, MCB2010L

Course User Fee: \$280.00

PAS2936 Contemporary Issues for the PA 1.000 credits

In this course the student will examine current issues, challenges, and practices influencing the physician assistant profession. Historical perspectives will be examined and the contributions of physician assistants in the healthcare system will be researched. The student will learn to use evidenced based medicine to research topics including the changing nature of health care delivery system in the United States; demographic, economic, ethical, and political factors influencing the practice of the physician assistant.

Lecture Hours: 16.000

PAS3010C Surgical Problems & Procedures 5.000 credits

During this course the student will be exposed to the various aspects of general, orthopedic, cardio vascular, thoracic, ENT, neurologic, urologic, and pediatric surgical problems, their diagnosis and treatment. Laboratory components of this course will include learning fundamental techniques necessary in preoperative and postoperative care, including nasogastric intubation, central venous line placement, arterial and venous punctures and sterile techniques. Prerequisites: PAS 1821, 1824, 3000C. (4 2/3 hr. lecture; 1 hr. lab)

Lab Hours: 75.200

Prerequisite: PAS1821 AND PAS1824 AND PAS3000C AND ***xxxxPAS 3027

Course User Fee: \$1400.00

PAS3019 Pathophysiological Basis of Disease III 2.000 credits

This course establishes scientific core knowledge and bridges the basic medical sciences with clinical medicine. The course covers the pathophysiology of human diseases that appear as a result of structural and functional alterations of the human body systems. The course begins with the study of human anatomy and physiology pertinent to the pathological conditions presented and progresses to the pathophysiological topics needed by physician assistant students. The course will enhance decision-making ability when working as a Physician Assistant (PA) in clinical practice.

Prerequisite: PAS 1821, 1824, 3038C, 3075. (2 hour lecture)

Lecture Hours: 32.000

Prerequisite: DEH2810L, HSA3533, HSC3202, PAS1820C, PAS1824, PAS3038C, PAS3075

PAS3027 PHARMACOTHERAPEUTICS 4.000 credits

The second course in the sequence PAS 1823, 3LLC. The study of the use of drugs to treat disease, including contraindication and incompatibilities; drug interactions; side effects and their treatment, and dosages and calculations. Prerequisites: PAS HSA 1803, PAS 1800C, 1801, 1811, 1812, 1813, 1822C, 1823, 1831 (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: HSA2532 AND PAS1800C AND PAS1801C AND PAS1811C AND PAS1812 AND PAS1813 AND PAS1822C AND PAS1823 AND PAS1831

PAS3038C Physical Diagnosis III 2.000 credits

This course will build upon skills learned in Physical Diagnosis I and II and will prepare the physician assistant student with the essential skills for entry into clinical practice. Students will learn to perform medical interviews and physical examinations on professional patients and high-fidelity simulators. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822C, 1823, 1831. (1 hour lecture; 3 hour clinical)

Lecture Hours: 16.000, Clinical Hours: 48.000

Prerequisite: HSA2532, PAS1800C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

PAS3042C Clinical Medicine III for Physician Assistants 5.000 credits

Students will learn the signs and symptoms, diagnosis, therapeutic and non-therapeutic interventions of common diseases affecting pediatric, adult, and geriatric patients. Patient education and preventive medicine will also be included. Prerequisites: DEH 2801L, HSA 3533, HSC 3202, PAS 1824, 1820C, 3038C, 3075. (4 hour lecture; 3 hour clinical)

Lecture Hours: 64.000, Clinical Hours: 48.000

Prerequisite: DEH2810L, HSA3533, HSC3202, PAS1820C, PAS1824, PAS3038C, PAS3075

PAS3070 Clinical Pharmacotherapeutics 3.000 credits

This course will provide a team-based, active, and applied learning environment to solve patient cases related to the management of pharmacotherapy. Students will work within specified learning groups to solve patient cases. Students will learn from their future colleagues by working together to develop treatment plans and answer questions which are designed to promote active learning and critical thinking skills. Following each case, facilitated discussion will occur to promote student learning and retention of material. Real world examples will be incorporated into discussions and explanations of patient cases. Prerequisites: DEH 2801L, HSA 3533, HSC 3202, PAS 1824, 1820C, 3038C, 3075. (3 hour lecture)

Lecture Hours: 48.000

Prerequisite: PAS1824, DEH2810L, HSA3533, PAS3038C, PAS3075, HSC3202, PAS1820C

PAS3075 Pharmacotherapeutics 4.000 credits

The second course in the sequence pas 1823, 3070. The study of the use of drugs to treat disease, including contraindication and incompatibilities; drug interactions; side effects and their treatment, and dosages and calculations. Prerequisite: HSA 2532, PAS 1800C, 1801C, 1811C, 1812, 1813, 1822C, 1823, 1831. (4 hr. lecture)

Lecture Hours: 64.000

Prerequisite: HSA2532, PAS1800C, PAS1801C, PAS1811C, PAS1812, PAS1813, PAS1822C, PAS1823, PAS1831

Course User Fee: \$1120.00

PAS3140 Genetics 2.000 credits

This course will prepare the PA student for human genetics and medical genomics, including Mendelian and complex diseases, chromosomal abnormalities, heritability, genetic variations and methods for investigating associations with diseases, gene-environment interactions, population genetics, cancer genetics, and epigenetics. Prerequisites: DEH 2801L, HSA 3533, HSC 3202, PAS 1824, 1820C, 3038C, 3075. (2 hour lecture)

Lecture Hours: 32.000

Prerequisite: DEH2810L, HSA3533, HSC3202, PAS1820C, PAS1824, PAS3038C, PAS3075

PAS4191L Internal Medicine Clerkship 4.000 credits

The clinical course focuses on basic medical practice. The student is exposed to common medical problems encountered on in-patient and out-patient medical services. Emphasis is placed on the history and physical examination and the process required in the proper work-up and management of the patient. Patient care experience in the various subdivisions of internal medicine including oncology, hematology, neurology, nephrology, gastroenterology, rheumatology, pulmonology, cardiology, and infectious diseases may be required. Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C. (192 hr. Clinical)
Clinical Hours: 192.000
Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C

PAS4290 Surgery Clerkship 2.000 credits

During the clinical course the student will be exposed to a variety of clinical problems routinely seen on the surgical service. Emphases will be placed on preoperative, intraoperative and postoperative management of the patient. In the operating room the student will practice aseptic technique, operating room principles, and assist in surgery. Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C. (96 hr. Clinical)
Clinical Hours: 96.000
Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C
Course User Fee: \$560.00

PAS4300 PEDIATRICS CLERKSHIP 4.000 credits

this clinical course in pediatric care settings will introduce students to childhood illnesses and normal variations of growth and development. students will perform histories and physical examinations and manage patients in the newborn nursery, pediatric out-patient clinic and emergency room.(4 credit hours) Pre-Req PAS 3010C, 3029, 3CCC, 3HHHC, 3LLL.
Clinical Hours: 288.000
Prerequisite: PAS3010C AND PAS3029 AND PAS3CCC AND PAS3HHHC AND PAS3LLL

PAS4493 Family Medicine Clerkship 4.000 credits

This clinical course introduces the student to the family practice setting where emphasis is placed on the common diseases treated by primary care practitioners in conjunction with other members of the health care team. The student is exposed to rural epidemiology, cultural diversity, and problems that affect delivery of health care in rural and under-served areas. Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C. (192 hr. clinical)
Clinical Hours: 192.000
Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C
Course User Fee: \$1120.00

PAS4500 OBSTETRICS/GYNECOLOGY CLERKSHIP 2.000 credits

During this clinical course the student will participate on the obstetrical service managing pregnancy, labor and delivery and be introduced to pre-and postnatal complications. The student will also participate in the management of common gynecological problems. (2 credit hours) Pre-Req PAS 3010C, 3029, 3CCC, 3HHHC, 3LLL.
Clinical Hours: 144.000
Prerequisite: PAS3CCC AND PAS3HHHC AND PAS3LLL AND PAS3010C AND PAS3029

PAS4690 Emergency Medicine Clerkship 2.000 credits

This clinical course in an emergency care setting will provide opportunities for the student to manage the acutely ill and traumatized patient. The student will learn to perform history and physical examination on the acutely ill patient with emphasis being placed on the management and support measures necessary in situations which are life threatening. Prerequisites: PAS 3019, 3042C, 3070, 3140, 3203C.
Clinical Hours: 96.000
Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C
Course User Fee: \$560.00

PAS4940 PSYCHIATRY CLERKSHIP 2.000 credits

This clinical course in a psychiatric care setting will allow students to participate in daily rounds and become knowledgeable about the use of psychotropic medications for psychiatric disorders. Group therapy sessions will be a major part of the learning experience. (2 credit hours) Pre-Req PAS 3010C, 3029, 3CCC, 3HHHC, 3LLL.
Clinical Hours: 144.000
Prerequisite: PAS3CCC AND PAS3HHHC AND PAS3LLL AND PAS3010C AND PAS3029

PAS4943 Selective Clerkship 2.000 credits

This clinical course provides the opportunity for students to become familiar with a medical specialty of their choice. The student will have the opportunity to familiarize with the role of the physician assistant in that specialty and reinforce the most common examination techniques, diagnoses and treatments within the selected specialty. Prerequisites: PAS 3019, 3042C, 3070, 3140, and 3203C
Clinical Hours: 96.000
Prerequisite: PAS3019, PAS3042C, PAS3070, PAS3140, PAS3203C
Course User Fee: \$560.00

PCB**PCB2033 Introduction To Ecology 3.000 credits**

Lecture Hours: 48.000

PCB3043 Fundamentals of Ecology 3.000 credits

This is a foundations course in ecology. In this course, students will learn the basic principles of ecology at organismal, population, community, and ecosystem levels, including consideration of Florida's ecosystems and human impact on those systems.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L

PCB3060 Principles of Genetics 3.000 credits

This course is an introduction to the mechanisms of transmission of hereditary information. Students will learn the classical Mendelian principles of heredity, deviation of Mendelian principles, genetic analysis, linkage and mapping, genetics of populations, gene regulation, mutation, the genetic bases of cancer and other genetic disorders will also be studied.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L, Corequisite: PCB3060L

PCB3060L Principles of Genetics Laboratory 2.000 credits

This laboratory course is designed to complement PCB 3060 Principles of Genetics. Students will learn hands-on skills with emphasis on laboratory principles, techniques, and instrumentation within the field of genetics. Special fee.

Lab Hours: 64.000

Prerequisite: BSC2010 AND BSC2010L AND BSC2011 AND BSC2011L, Corequisite: PCB3060

Course User Fee: \$110.00

PCB4023 Molecular and Cell Biology 3.000 credits

Students will learn the structure and function of cells and biological membranes, signal transduction pathways, cell cycle and cell division, the flow of genetic information and the regulation of gene expression. Exploration of laboratory techniques and discussion of the cellular basis of human diseases will also occur. Special fee.

Lecture Hours: 48.000

Prerequisite: BSC2010 AND BSC2010L AND BSC2011 AND BSC2011L

Course User Fee: \$25.00

PCB4097 Human Physiology 3.000 credits

The student studies the physiology of organism's major organ systems with emphasis on humans. Student will learn the principles of physics, cell biology, and anatomy in order to explain how the different organs systems work individually and in the context of the whole organism. Special fee.

Lecture Hours: 48.000

Prerequisite: PHY2054 AND PHY2054L AND BCH3023 AND PCB4024

Course User Fee: \$25.00

PCB4233C Fundamentals of Immunology 4.000 credits

Students will learn the immunological processes and concepts as they pertain to human health, disease prevention, development, and treatment. Its primary emphasis is on the cellular and non-cellular components of the immune system, and the ways in which these components interact to provide immunity. This is a combination lecture and lab course. Special fee.

Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: MCB3023 AND MCB3023L

Course User Fee: \$55.00

PCB4674 Evolution 3.000 credits

Students will learn the theory of evolution as it pertains to different fields of modern biology including the theory of natural selection, the evidence for evolution, microevolution, speciation, macroevolution, the origin of life on Earth, major evolutionary trends, and the evolution of humans. Special fee.

Lecture Hours: 48.000

Prerequisite: BSC2010, BSC2010L, BSC2011, BSC2011L, PCB3060, PCB3060L

Course User Fee: \$25.00

PET

PET1170 Fundamentals of Athletic Coaching and Management 3.000 credits

This course will introduce students to the basic principles of athletic coaching and team management. Students will discuss the fundamentals of developing an individual athlete and team, effective athlete training programming, and factors which impact the operations of athletic organizations. Students will develop an understanding of how policy and coaching decisions influence an organization at all levels (Credit hours 3.00).

Lecture Hours: 48.000

Course User Fee: \$55.83

PET1173 Strategies of Coaching and Competition 3.000 credits

This course will provide students with the tools and resources necessary to effectively evaluate team and athlete performance. In addition, the utilization of strategic thinking, game strategy, and available resources will be applied to coaching scenarios. Students will leverage their aggregate knowledge of coaching and competition to develop and adapt coaching and management plans to fulfill the needs of participants (Credit hours: 3.00).

Course User Fee: \$55.83

PGY

PGY2110C Color Photography 1 4.000 credits

An introductory course in the making of Type C photographic prints, including the darkroom techniques of developing color film, color filtering, color balance and density control. There will be an exploration of significant contributions to the aesthetics of color photography. Students must provide their own cameras, film and photographic paper.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2401C

Course User Fee: \$50.00

PGY2111C Color Photography 2 4.000 credits

Deals primarily with printing methods used in printing color negatives. Concentrated practice is given in light, color balancing, exposure and processing of color printing materials; the techniques of producing matched multi-size prints are demonstrated.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2110C

Course User Fee: \$50.00

PGY2112C COLOR PHOTOGRAPHY 3 3.000 - 4.000 credits

Lecture Hours: 96.000

Course User Fee: \$50.00

PGY2211 PORTRAIT/STILL PHOTO 3.000 - 4.000 credits

Lecture Hours: 80.000 - 96.000

PGY2222 Fashion Photography 4.000 credits

The production of commercially viable photographs illustrating clothes as desirable objects as well as recent trends in fashion industry are studied. An awareness of mood, make-up, and dramatic impact is stressed.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2239

Course User Fee: \$25.00

PGY2238 Illustrative Photography 1 4.000 credits

The use of the camera to illustrate either an original concept or a concept provided by an art director for clients such as magazines, manufacturing concerns, advertising agents, newspapers, technical publications and schools. The creative approach is stressed in planning and production-effective color and black/white illustrations.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2401C

PGY2239 Illustrative Photography 2 4.000 credits

A sophisticated level of photographic illustration is reached and emphasis is given to conceptual and visual continuity. Concepts, methods and techniques necessary to produce slide presentations for variety of clients are stressed. Seminars and conferences prepare students for the business aspects of the illustration and advertising markets.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY2238

Course User Fee: \$25.00

PGY2401C Introduction to Photography 4.000 credits

Fundamentals of black and white photography as an art medium with emphasis on composition, design and processing. Students will supply their own camera, film and paper.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: ART 1203C, ART 1300C, or Equivalent

Course User Fee: \$50.00

PGY2404C Intermediate Photography 4.000 credits

Emphasis on achieving more technical control of black and white photography with introduction to larger format photography utilization of studio aspects such as strobe, quartz lighting and view camera controls continued development of aesthetics.

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

PGY2470C PORTFOLIO PREP 3.000 - 4.000 credits

Provides graduating students individual guidance and direction in the preparation of their portfolio s. Emphasis is given to the realization of new photographic images. Prerequisite: PGY 2111C, 2210, 2221, 2222. Laboratory fee. (1-2 hr. lecture; 4 hr. lab) Lecture Hours: 80.000 - 96.000

Course User Fee: \$25.00

PGY2470C Photography Portfolio 4.000 credits

This course is the culmination of artistic and technical experiences acquired in previous photography courses. Students explore portfolio creation through a semester-long analysis, crafting an artist statement, and studio/critique experiences. The course addresses professional practices such as the creation of a resume/CV, a website, and a social media profile. Students gain insights into strategic approaches to pursue opportunities such as transfer applications, seeking exhibitions/grants, entrepreneurship, and job interview readiness.

Lecture Hours: 32.000, Lab Hours: 64.000

Prerequisite: PGY 2111C, PGY 2211, PGY 2222, PGY 2238, PGY 2239

Course User Fee: \$25.00

PGY2475 ADV PHOTOGRAPHY 3.000 - 4.000 credits

Lecture Hours: 16.000 - 32.000, Lab Hours: 64.000

Course User Fee: \$50.00

PGY2800C DIGITAL PHOTOGRAPHY 4.000 credits

Lecture Hours: 32.000, Lab Hours: 64.000

PHI

PHI1100 Introduction to Logic 3.000 credits

This course serves as a solid foundation for the study of philosophy, focusing on both the fundamental principles of valid reasoning and the exploration of the Western Canon. Students will delve into the rich tradition of philosophical thought, honing their analytical skills through the application of various techniques.

Lecture Hours: 48.000

PHI2010 Introduction to Philosophy 3.000 credits

In this course, students will be introduced to the nature of philosophy, philosophical thinking, major intellectual movements in the history of philosophy, including topics from the western philosophical tradition, and various problems in philosophy. Students will strengthen their intellectual skills, become more effective learners, and develop broad foundational knowledge. Student learning outcomes: students will develop critical thinking skills; students will demonstrate an understanding of classical western philosophical views; and students will analyze, explain, and evaluate foundational concepts of epistemology, metaphysics, and ethics. Writing Intensive Course. Lecture Hours: 48.000

PHI2070 Introduction to Eastern Philosophy 3.000 credits

This is a foundation course in philosophy. Students will learn various philosophies of from Eastern and Western Canons. Philosophers from various traditions such as Buddhism, Confucianism, Hinduism and Taoism will be discussed and analyzed.

Lecture Hours: 48.000

PHI2600 Introduction to Ethics 3.000 credits

This course provides students with an understanding of ethics and morality and includes works from the Western canon. Students will evaluate ethical theories and apply them to classical and contemporary issues, exploring the intersection of ethics with aspects of human development, including literature, history, and art. Through this process, students will develop critical thinking skills and gain understanding of how ethical reasoning shaped and influences human thought and behavior. This is a writing intensive course.

Lecture Hours: 48.000

PHI2680 Artificial Intelligence and Ethics 3.000 credits

This course acquaints students with ethics as relevant to the design, implementation, and administration of artificial intelligence and emerging technologies. It will explore issues such as the biases of algorithms, autonomous warfare, the risks of surveillance technologies, and the effects of technologies on employment. In addition, this course will also delve into the critical examination of the Western Canon's influence on the development and ethical implications of artificial intelligence and emerging technologies.

Lecture Hours: 48.000

PHI3633 Biomedical Ethics 3.000 credits

This is a foundation course in biomedical ethics and ethical theory. Students will learn to use methods of effective reasoning to apply to topics in biomedical ethics. These topics may include, but are not limited to, genetic engineering, stem cell research, human cloning, euthanasia, and clinical research ethics.

Lecture Hours: 48.000

PHT**PHT1102C Anatomy for the Physical Therapist Assistants 4.000 credits**

Regional description of the musculoskeletal landmarks utilized in implementing and documenting assessment and treatment procedures in physical therapy. Prerequisite: Departmental Approval, Corequisites: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L. (3 hr. lecture and 2 hr. lab)

Lecture Hours: 48.000, Lab Hours: 32.000

Recommended Preparation: Departmental Approval, Corequisite: PHT1201 AND PHT1201L AND PHT1211 AND PHT1211L

Course User Fee: \$438.81

PHT1201 Introduction to Physical Therapy 2.000 credits

This course introduces students to the survey and history of the physical therapy profession. The role and responsibilities of the physical therapist assistant as they interact with patients and other health care workers are discussed. It incorporates discussion on legal and ethical issues, levels of supervision, and the medical record. An overview of common pathologies is presented. Prerequisites: BSC 2085, BSC 2085L, PHY 1004, PHY 1004L; Corequisites: PHT 1102C, PHT 1201L, PHT 1211, PHT 1211L. (2 hr. lecture)

Lecture Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, Corequisite: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L.

PHT1201L Introduction to Physical Therapy Laboratory 1.000 credits

Laboratory course for PHT1201 Introduction to Physical Therapy. This course includes laboratory practice for basic patient care skills and treatment procedures including body mechanics, vital signs, transfers, gait training, ROM, flexibility, manual resistance. Case scenarios to identify proper transfer techniques and gait training patterns are utilized. Principles of effective documentation and data collection are introduced.

Lab Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, Corequisite: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L.

Course User Fee: \$20.00

PHT1211 Disabilities and Therapeutic Procedures 1 2.000 credits

This course introduces theory and practical applications of physical therapy biophysical agents. The physiologic effects, indications, contraindications, and intervention parameters of patient care interventions including electrotherapeutic agents, compression therapies, cryotherapy, hydrotherapy, superficial and deep thermal agents, traction, and therapeutic massage are presented.

Lecture Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, Corequisite: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L.

PHT1211L Disabilities and Therapeutic Procedures 1 Lab 2.000 credits

Laboratory course for PHT1211 Disabilities and Therapeutic Procedures 1. This course includes laboratory practice of technical skills relating to practical applications of physical therapy biophysical agents including electrotherapeutic agents, compression therapies, cryotherapy, hydrotherapy, superficial and deep thermal agents, and traction. Students develop basic skills in therapeutic massage. Practical application of each intervention is emphasized.

Lab Hours: 64.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, Corequisite: PHT 1201, PHT 1201L, PHT 1211, PHT 1211L.

Course User Fee: \$25.00

PHT2120 Applied Kinesiology 2.000 credits

Students learn concepts and principles of applied kinesiology and biomechanics with emphasis on the function of the musculoskeletal system as it pertains to human motion. Anatomical structures and movement concepts covered in PHT 1102C

Anatomy for PTA are expanded upon. Goniometric measurement, manual muscle testing skills, plumb-line analysis, data collection skills, palpation skills, understanding of special tests, and concepts of reliability and validity as it relates to goniometry are emphasized.

Lecture Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L ;

Corequisite: PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C

PHT2120L Applied Kinesiology Laboratory 1.000 credits

Laboratory course for PHT 2120 Applied Kinesiology. This course provides laboratory practice of technical skills relating to practical applications of foundational concepts and principles pertaining to Applied Kinesiology. Students learn goniometric measurement, manual muscle testing skills, plumb-line analysis, palpation skills, data collection, and develop an understanding of special tests.

Lab Hours: 32.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L ,

Corequisite: PHT 2120, PHT 2224, PHT 2224L, PHT 2801C

Course User Fee: \$25.00

PHT2162 Survey of Neurological Deficits 3.000 credits

This course introduces the etiology, pathophysiology, and clinical manifestations of common neurological diseases/conditions treated in physical therapy. Neuroanatomy of the central and peripheral nervous system is reviewed. Reflex integration as well as normal growth and development are discussed. Students have an end of course project-based topic assignment that pertains to commonly seen neurological conditions in physical therapy.

Lecture Hours: 48.000

Prerequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C , Corequisite: PHT 2701, PHT 2701L, PHT 2810

PHT2224 Disabilities and Therapeutic Procedures II 3.000 credits

This course introduces foundational concepts and principles pertaining to therapeutic exercise prescription and interventions for the upper and lower extremity, spine, and Women's Health across the lifespan and continuum of care. Applied science of exercise and techniques, exercise interventions by body region, peripheral joint mobilization, and medical and surgical procedures and protocols are emphasized. Students engage in active and collaborative learning and utilize case scenarios for practical application.

Lecture Hours: 48.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L ,

Corequisite: PHT 2120, PHT 2120L, PHT 2224L, PHT 2801C

PHT2224L Disabilities and Therapeutic Procedures II Lab 2.000 credits

Laboratory course for PHT 2224 Disabilities and Therapeutic Procedures 2. This course is designed to provide students with technical skill practice and application of foundational concepts and principles pertaining to therapeutic exercise prescription and interventions for the upper and lower extremity, spine, and Women's Health across the lifespan and continuum of care. Students apply exercise and therapeutic interventions by body region, practice documentation, prescribe home exercise programs, and are introduced to peripheral joint mobilization for the upper and lower extremities.

Lab Hours: 64.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L, Corequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2801C

Course User Fee: \$25.00

PHT2701 Rehabilitation Procedures 3.000 credits

This course presents treatment techniques related to physical therapy interventions and rehabilitation concepts and principles for neurological and other medical conditions across the lifespan and continuum of care.

Lecture Hours: 48.000

Prerequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C, Corequisite: PHT 2162, PHT 2701L, PHT 2810

**PHT2701L Rehabilitation Procedures Laboratory
2.000 credits**

Laboratory course for PHT 2701 Rehabilitation Procedures. This course provides laboratory practice of technical skills relating to physical therapy interventions and rehabilitation concepts and principles for neurological and other medical conditions across the lifespan and continuum of care.

Lab Hours: 64.000

Prerequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C, Corequisite: PHT 2162, PHT 2701L, PHT 2810

Course User Fee: \$25.00

**PHT2801C Clinical Practice and Conference I
2.000 credits**

The students will be exposed to clinical experiences in supervised patient care activities in a variety of clinical facilities including general hospitals and physical therapy clinics.

Clinical Hours: 96.000

Prerequisite: BSC 2086, BSC 2086L, PHT 1102C, PHT 1201, PHT 1201L, PHT 1211, PHT 1211L, Corequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L

Course User Fee: \$389.50

**PHT2810 Clinical Practice and Conference II
7.000 credits**

Intermediate clinical experiences in selected patient care activities under the supervision of a licensed physical therapist.

Clinical Hours: 336.000

Prerequisite: PHT 2120, PHT 2120L, PHT 2224, PHT 2224L, PHT 2801C, Corequisite: PHT 2162, PHT 2701, PHT 2701L

**PHT2820 Clinical Practice and Conference III
7.000 credits**

Advanced clinical experiences in patient care activities under the direct supervision of a licensed physical therapist.

Clinical Hours: 336.000

Prerequisite: PHT 2162, PHT 2701, PHT 2701L, PHT 2810, Corequisite: PHT 2931

Course User Fee: \$117.00

**PHT2931 Seminar for Physical Therapist
Assistants 3.000 credits**

Recognition of the expected current competency levels, and ethical and legal responsibilities of the physical therapist assistant in the health care system. A comprehensive course designed to enhance students' understanding of the curricular content. It provides a thorough review of the curriculum, ensuring that students consolidate their knowledge and skills. Special topics are presented. The role of the PTA including ethical and legal responsibilities are emphasized. The course prepares students for licensure requirements and preparation for the National Physical Therapy Examination (NPTE-PTA) and the Florida Laws and Rules examination.

Lecture Hours: 48.000

Prerequisite: PHT 2162, PHT 2701, PHT 2701L, PHT 2810, Corequisite: PHT 2820

Course User Fee: \$30.00

PHY**PHY1004 Physics With Applications 1 3.000
credits**

Emphasizes the basic concepts and principles and their practical applications. Designed specifically for students in technical studies and for others wishing to strengthen their physics background before taking advanced courses.

Lecture Hours: 48.000

Prerequisite: MAT1033, Corequisite: PHY1004L

Course User Fee: \$10.00

**PHY1004L Physics with Applications 1 Lab 1.000
credits**

Laboratory for PHY 1004.

Lab Hours: 32.000

Prerequisite: MAT1033, Corequisite: PHY1004

Course User Fee: \$20.00

PHY1005 PHYSICS W/APPLIC 3.000 credits

Lecture Hours: 48.000

Course User Fee: \$10.00

PHY1020 General Education Physics 3.000 credits

This course offers a comprehensive survey of physics, covering a wide range of topics including motion, Newton's laws, energy, sound, heat, electricity, magnetism, and optics. Emphasizing a conceptual understanding of physics, the course integrates critical thinking skills and real-world applications. Student learning outcomes: students will critically evaluate everyday phenomena using the scientific method; students will explain the basis of physical principles (such as conservation laws) and how they apply to everyday phenomena; students will interpret information conveyed in diagrams and graphs; and students will perform simple calculations relevant to real world problems. Lecture Hours: 48.000

PHY1025 Basic Physics 3.000 credits

This course will help students to facilitate the transition from high school to college/university physics. The course will emphasize problem-solving techniques. Topics may include units of measure, particle mechanics, conservation laws, and basic field concepts.

Lecture Hours: 48.000

Prerequisite: MAC1105

PHY2048 Physics with Calculus 1 4.000 credits

This calculus-based course serves as the first in a two-part series, covering topics like kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Designed for science and engineering majors, the course integrates critical thinking, analytical skills, and real-world applications. Student learning outcomes: students will solve analytical problems describing different types of motion, including translational, rotational, and simple harmonic motion; students will apply Newton's Laws, and conservation laws to solve analytical problems of mechanics; students will identify and analyze relevant information presented in various formats such as graphs, tables, diagrams, and/or mathematical formulations; and students will solve real-world problems using critical thinking skills and knowledge developed from this course.

Lecture Hours: 64.000

Prerequisite: PHY 1025 or PHY 2053 and PHY 2053L, and (MAC 2311 or MAC 2312 or MAC 2313 or MAD 2104 or MAP 2302 or MAS 2103), Corequisite: PHY2048L

Course User Fee: \$10.00

PHY2048L Physics with Calculus 1 Lab 1.000 credits

This laboratory course complements PHY2048 by providing hands-on experience in the principles of kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Students will engage in practical experiments that reinforce theoretical concepts covered in the parent course, allowing them to apply analytical problem-solving techniques to real-world scenarios. Through a series of structured laboratory activities, students will develop critical thinking and analytical skills while collecting and analyzing data related to motion and forces. This course is designed for science and engineering majors, preparing them for advanced studies and professional applications in physics and related fields.

Lab Hours: 32.000

Prerequisite: MAC2311, PHY1025 OR PHY2053 AND PHY2053L, Corequisite: PHY2048

Course User Fee: \$20.00

PHY2049 Physics with Calculus 2 4.000 credits

Foundation course for physical science and engineering majors. PHY 2048 covers classical mechanics and thermodynamics. PHY 2049 includes electricity, magnetism, waves and optics.

Lecture Hours: 64.000

Prerequisite: PHY2048, PHY2048L, Corequisite: MAC2312, PHY2049L

Course User Fee: \$10.00

PHY2049L Physics with Calculus 2 Lab 1.000 credits

Laboratory for PHY 2049.

Lab Hours: 32.000

Prerequisite: PHY2048, PHY2048L, Corequisite: MAC2312, PHY2049

Course User Fee: \$20.00

PHY2053 Physics without Calculus 1 3.000 credits

The student will learn the basic principles of physics, including mechanics, sound and thermodynamics.

Lecture Hours: 48.000

Prerequisite: MAC1114 OR MAC1147 OR MAC2241 OR PHY1025, Corequisite: PHY2053L

PHY2053L Physics (Without Calculus) 1 Lab 1.000 credits

This laboratory course is designed to complement PHY2053. Students will engage in hands-on experiments that reinforce theoretical concepts related to kinematics, dynamics, energy, momentum, rotational motion, fluid dynamics, oscillatory motion, and waves. Through practical applications of algebra and trigonometry, learners will develop analytical skills while conducting experiments that illustrate key physical principles. The course emphasizes critical thinking and problem-solving as students collect and analyze data, interpret results, and communicate their findings effectively. This lab experience aims to deepen students' understanding of the scientific concepts covered in the parent course and their relevance to real-world situations.

Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147, Corequisite: PHY2053

Course User Fee: \$20.00

PHY2054 Physics (Without Calculus) 2 3.000 credits

An introduction to the basic principles of physics. PHY 2053 covers mechanics, sound and thermodynamics. PHY 2054 includes electricity, magnetism and optics.

Lecture Hours: 48.000

Prerequisite: PHY2053, PHY2053L, Corequisite: PHY2054L

Course User Fee: \$10.00

PHY2054L Physics (Without Calculus) 2 Lab 1.000 credits

The physics lab courses are one-credit courses designed to be taken in conjunction with a physics lecture. A different experiment is performed each week with topics chosen to correspond with the material being studied in the lecture. Each experiment is designed to be completed in about 2 contact hours.

Lab Hours: 32.000

Prerequisite: PHY2053, PHY2053L, Corequisite: PHY2054

Course User Fee: \$20.00

PLA**PLA1949 Paralegal Studies Practicum 3.000 credits**

This is a course designed to continue training in a student's field of study through work experience. Students are graded on the basis of documentation of learning acquired as reported by student and employer. Students will be assigned specific course prefixes related to their academic major prior to registration. All students must contact the Co-operative Education Office to obtain registration approval.

Other Hours: 144.000

Prerequisite: Departmental Permission Required

PLA2003 Fundamentals of Law 3.000 credits

This course provides students with an overview of the American legal system. It explores the basic concepts of law in society including the different sources of law. The federal, state and county court systems are examined along with judicial interpretation of the law. The course also covers the distinctions between procedural and substantive law, civil versus criminal and a court of equity and a court of law. The roles of paralegals are discussed with an emphasis given in their professional relationships, functions, career opportunities and ethical obligations.

Lecture Hours: 48.000

Pre/Corequisite: ENC1101, Corequisite: LCO0999, PLA2931

Course User Fee: \$25.00

PLA2114 Legal Writing 3.000 credits

This course provides knowledge and understanding of how to present legal research and analysis in proper written format. As legal research is an integral part of legal writing, the course will reinforce the skills used in legal research. It will also cover basic writing skills, the process of legal analysis, methodology involved in drafting a Memorandum of law, practice in drafting pleadings, and various types of specific law office correspondence.

Lecture Hours: 48.000

Prerequisite: ENC 1101, PLA 2003, PLA 2120C, PLA 2931

Course User Fee: \$50.00

PLA2120C Legal Research and Analysis 4.000 credits

Legal Research and Analysis is an introduction to the legal system, the law library system, the ethical responsibilities and skills of a paralegal when conducting research for primary and secondary sources of law in the United States. Students must earn a minimum of a C to pass this course. Pre-requisite: ENC1101; Pre/Co-requisites: PLA2003 and PLA2931 with a grade of C or better, and Departmental approval for non-Paralegal Studies majors. (3 hr. lecture, 2 hr. lab).

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: ENC1101 AND Pre/Corequisite: PLA2003, PLA2931

Course User Fee: \$45.84

PLA2227 Civil Law & Litigation 3.000 credits

This course will introduce students to a comprehensive exploration of civil procedure, pleadings, evidence, motions, damages, appeals, and alternative dispute resolution. Students will gain skills in the application of law when preparing documents before, during and post trial. A minimum of a C is required to pass this course.

Prerequisites: ENC 1101, PLA 2003, PLA 2120C, and PLA 2931. (3 hr. lecture).

Lecture Hours: 48.000

Prerequisite: ENC1101 AND PLA2003 AND PLA2931, PLA2120C

Course User Fee: \$45.84

PLA2273 TORTS 3.000 credits

2019 Course Fee Audit

Lecture Hours: 48.000

Course User Fee: \$50.00

PLA2273 Torts 3.000 credits

This course provides an examination of the theories governing tort law and the use of various pre-litigation tools. Topics covered include intentional torts, negligence and strict liability. The course also requires students to utilize the knowledge obtained to draft documents employed in practice.

Lecture Hours: 48.000

Prerequisite: ENC 1101, PLA 2203, PLA 2120C, PLA 2114, PLA 2227, PLA 2931

Course User Fee: \$50.00

PLA2303 Criminal Law & Litigation 3.000 credits

This course focuses on the substantive areas of criminal law including the offenses, elements, defenses and parties to a criminal law proceeding. It also emphasizes the role of the criminal justice system in adjudicating, enforcing and sentencing criminal defendants. It examines the Florida Rules of Criminal Procedure and provides practice in drafting documents required in the conduct of a criminal trial.

Lecture Hours: 48.000

Prerequisite: ENC 1101, PLA 2203, PLA 2120C, PLA 2114, PLA 2227, PLA 2931

Course User Fee: \$50.00

PLA2600 Wills, Trust, Estate 3.000 credits

Wills, Trusts, and Estates is a study of the laws governing wills and interstate succession. The course provides practice in drafting a simple will and trust. It also examines the procedures and rules involved in probate administrations and explains the ethical obligations of attorneys and paralegals who are involved in this area of practice.

Lecture Hours: 48.000

Prerequisite: PLA 2114, REE 2040

Course User Fee: \$50.00

PLA2610 Introduction to Real Property 4.000 credits

This is an introductory course that provides an overview of the general concepts of real property law applied in most states, including substantive legal issues, ownership, terms and various legal forms, the mechanics of various residential, some commercial transactions, and mortgage foreclosures. Prerequisite: REE2040; Corequisite: PLA2631C (3 hr. lecture)

Lecture Hours: 64.000

Prerequisite: REE2040, Corequisite: PLA2631C

Course User Fee: \$45.84

PLA2631C Real Estate Transactions and Settlement 4.000 credits

This course is a combination of both class lecture and lab. The lecture segment includes an overview of intermediate real estate law topics with a specific emphasis on Florida real estate transactions.

Students who complete this course will understand the fundamental concepts and sequences followed in a real estate transaction from initial contract review, through settlement (closing). The lab segment is transaction and problem-oriented. This segment trains students in the use of the industry settlement software to complete title examinations, updates, issuance of title policies & endorsements, and various federal and state regulations that affect real estate settlements. Students may also perform various duties typically assigned to a paralegal in a real estate transaction. Prerequisite: REE2040; Corequisite: PLA2610 (3 hr. lecture, 2 hr. lab). Lecture Hours: 48.000, Lab Hours: 32.000 Prerequisite: REE2040, Corequisite: PLA2610
Course User Fee: \$45.84

PLA2704 Professionalism & Civility in the Law 3.000 credits

This class will provide students guidance as they interact in a real-world setting as legal professionals while strengthening their communication skills. This course will cultivate the students' leadership capacities at the intra-personal, interpersonal, and industry levels. The coursework will proceed through experiential learning that is grounded in theory and includes reflection. The study material will develop students' knowledge in the context of the aspiration, relational, and practical aspects of professionalism and civility in the legal field (3 credits)

Lecture Hours: 48.000

Course User Fee: \$45.84

PLA2736 Electronic Discovery, Investigations, and Evidence 3.000 credits

This course will examine the electronic discovery (e-discovery) process primarily following the 9 stages of the Electronic Discovery Reference Model (EDRM). It will include analysis of e-discovery case law including past and current cases; the Federal Rules of Civil Procedure related to e-discovery; different types of e-discovery including litigation, government investigations, and internal audits; and ethical issues related to e-discovery. Additionally, it will have a practical aspect that will teach e-discovery skills related to e-discovery technology, including review platforms. Students must earn a grade of C or better to pass this course.

Prerequisites: ENC1101, PLA2003, PLA2120C, PLA2227, and PLA2931.

Lecture Hours: 48.000

Prerequisite: PLA2003, ENC 1101, PLA 2203, PLA 2120C, PLA 2114, PLA 2227, PLA 2931

Course User Fee: \$45.84

PLA2763 Law Office Management 3.000 credits

A survey of economical and efficient law office practices and procedures including the proper use of law office equipment; business data processing law office management, personnel selection, training and management; employer/employee relationships; correct utilization of time and space; correct time keeping and billing procedures.

Lecture Hours: 48.000

Prerequisite: PLA2003

Course User Fee: \$50.00

PLA2800 Family Law 3.000 credits

An examination of the legal aspects of domestic relations. This course focuses upon dissolution of marriage law with emphasis on pleadings, discovery, and property settlements. Other areas of family law such as adoption and annulment will be reviewed.

Lecture Hours: 48.000

Prerequisite: ENC 1101, PLA 2203, PLA 2120C, PLA 2114, PLA 2227, PLA 2931

Course User Fee: \$50.00

PLA2872 Artificial Intelligence (AI) and the Law 3.000 credits

Introduction to the intersection of artificial intelligence and the law, focusing on its challenges, opportunities, ethics, and impact on the law, the legal industry, and society. (3 credits)

Lecture Hours: 48.000

Recommended Preparation: Students are encouraged to enroll in ENC1101 prior to registering for PLA2872.

PLA2890 Human Rights Law & Procedure 3.000 credits

This course will address theoretical and practical questions the evolution and development of the laws addressing international human rights issues. Emphasis will be placed on the core issues, debates, struggles, failures, and successes of legal claims. (3 hr. lecture)

Lecture Hours: 48.000

PLA2930 Special Topics in Law 1.000 - 3.000 credits

This course will explore a selection of emerging trends in the law. May be repeated for credit.

Lecture Hours: 16.000

PLA2931 Legal Specialty Seminar 1.000 credits

Intensive practical and theoretical training is provided in a seminar format. The seminar topics cover current and timely legal issues and are addressed by practicing attorneys. The topics are announced at the beginning of the Fall and Winter semesters. Corequisite: PLA 2003. (1 hr. lecture)

Lecture Hours: 16.000

Corequisite: PLA2003

PLA2933 Legal Seminar: Real Estate Closing 1.000 credits

Intensive practical and theoretical training is provided in a seminar format. The seminar topics cover current and timely legal issues and are addressed by practicing attorneys. The topics are announced at the beginning of the Fall and Winter semesters. Prerequisite: REE 2040. (16 hr. Seminar)

Lecture Hours: 16.000

Prerequisite: REE2040

PLA2934 Legal Seminar: Immigration Law 1.000 credits

Intensive practical and theoretical training is provided in a seminar format. The seminar topics cover current and timely legal issues and are addressed by practicing attorneys. The topics are announced at the beginning of the Fall and Winter semesters.

Lecture Hours: 16.000

PLA2935 Legal Seminar: Intellectual Property Law 1.000 credits

Intensive practical and theoretical training is provided in a seminar format. The seminar topics cover current and timely legal issues and are addressed by practicing attorneys. The topics are announced at the beginning of the Fall and Winter semesters.

Lecture Hours: 16.000

PLA2940 LEGAL ASTNG INTERNSP 1.000 - 3.000 credits

Lecture Hours: 16.000 - 48.000

POR**POR1120 Elementary Portuguese 1 4.000 credits**

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Portuguese. Note: Students must pass this course with a C or better to continue to POR1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

POR1121 Elementary Portuguese 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Portuguese. Note: students must pass this course with a "C" or better to continue to POR2220.

Lecture Hours: 64.000

Prerequisite: POR1120

Course User Fee: \$15.00

POR2220 INTER PORTUGUESE 1 4.000 credits

In this course, the student will be able to develop socially and culturally appropriate proficiency in interpersonal, interpretive, and presentational modes at the intermediate mid-level in Portuguese. Note: Student must pass this course with a C or better to continue to POR 2221. (Four-hour lecture)

Prerequisite: POR 1121 or equivalent

Lecture Hours: 64.000

Prerequisite: POR1121

POS**POS2041 AMERICAN FEDERAL GOV 3.000 credits**

Lecture Hours: 48.000

POS2112 STATE & LOCAL GOV AM 3.000 credits

A survey of typical state and local government organization and political practices in America with special emphasis upon the governmental organization and the major contemporary political problems of the State of Florida and of Florida communities. Prerequisite or corequisite: SSE 102.

Lecture Hours: 48.000

POS2293 Islam and America 3.000 credits

Islam and the United States offers a survey treatment of the historically charged relationship between America and the Muslim World and the path forward. The course analyzes the intersection of religion, identity politics, geopolitics and economics of Islam-US relations in a historical context. These issue areas will be contextualized in normative, empirical and historical frameworks to equip the student with a sophisticated and multi-dimensional conceptual toolbox to analyze them.

Lecture Hours: 48.000

POT

POT2014 European Political Theory 1 3.000 credits

This course covers the more important trends in European political thought from Plato to the present. It examines those ideas which have contributed to the shaping of the political cultures of Western and Eastern Europe. It discusses the historical evolution of key concepts of politics such as freedom, order, political obligations, justice, consent, rights and duties, power and authority. A prior course in Government, History or Philosophy is desirable. Given in English. Level 1. Offered through Overseas Study Program.
Lecture Hours: 48.000

PRN

PRN0005C Fundamentals of Nursing 5.000 credits

This course provides an introduction to nursing and roles of the practical nurse in health care settings; as well as profession related and client-care concepts. Emphasis is placed on the knowledge and skills needed to provide safe, quality care. The theoretical foundation for basic assessment and nursing skills is presented, and the student is given an opportunity to demonstrate these skills in a laboratory setting. An introduction to the nursing process provides the student with a beginning framework for clinical decision making.
Lecture Hours: 60.000, Clinical Hours: 90.000
Course User Fee: \$451.06

PRN0006C Professional Role Transition 5.000 credits

This course facilitates the transition of the student to the role of PN. Emphasis is placed on issues related to nursing and healthcare, as well as skills necessary to provide care to multiple patients and assign tasks to other PN and unlicensed personnel. Concepts related to leadership and management are presented, as well as career development options that enhance career mobility. Standards of practice and the importance of practicing according to state regulations and statutes are examined. Clinical experiences provide the student the opportunity to apply theoretical concepts.
Lecture Hours: 25.000, Clinical Hours: 125.000
Prerequisite: PRN0210C, PRN0082C
Course User Fee: \$451.06

PRN0021C Anatomy and Physiology and Medical Terminology 1 3.500 credits

This course is designed to provide the student with basic knowledge of normal human body structure and function. The student will learn major systems, organs and terminology necessary for the provision of safe and effective nursing care.
Lecture Hours: 80.000, Lab Hours: 25.000
Course User Fee: \$451.06

PRN0035C Pharmacology 3.000 credits

The purpose of this course is to examine pharmacotherapeutic agents used in the treatment of illness and the promotion, maintenance, and restoration of wellness in diverse individuals. It focuses on drug classification, concepts, and principles of pharmacology, with special consideration for the nursing role in developing a comprehensive approach to the clinical application of drug therapy through the use of the nursing process. Nursing implications relative to the utilization of drug therapy are examined. Safety and legal implications of drug administration are discussed.
Lecture Hours: 90.000
Course User Fee: \$451.06

PRN0082C Anatomy and Physiology and Medical Terminology 3.500 credits

This course is designed to provide the student with an increasingly complex knowledge of normal human body structure and function. The student will learn major systems, organs and terminology necessary for the provision of safe and effective nursing care.
Lecture Hours: 80.000, Lab Hours: 25.000
Prerequisite: PRN0005C, PRN0021C, PRN0035C
Course User Fee: \$451.06

PRN0130C Special Populations 5.000 credits

This course examines the application of nursing to the care of special populations throughout the life cycle. Emphasis is placed on nursing care of clients with psychosocial disorders, childbearing women, newborns, and children to promote independence, wellness, and maximizing quality of life.
Lecture Hours: 75.000, Clinical Hours: 75.000
Prerequisite: PRN0082C, PRN0210C
Course User Fee: \$451.06

PRN0210C Adult Health I 10.000 credits

This course focuses on the care of adult/older adult patients with health alterations that require medical and/or surgical intervention. Emphasis is placed on the care of patients with alterations in selected body functions. Concepts of client-centered care, informatics, safe practice, and professionalism are emphasized throughout the course. Clinical experiences provide the student an opportunity to apply theoretical concepts and implement safe patient care to adults in a variety of settings. Lecture Hours: 150.000, Clinical Hours: 150.000
Prerequisite: PRN0005C AND PRN0021C, PRN0035C
Course User Fee: \$451.06

PRN0211C Adult Health II 10.000 credits

This course focuses on the care of adult/older adult patients with health alterations that require medical and/or surgical intervention. Emphasis is placed on the care of patients with alterations in selected body functions. Concepts of client-centered care, informatics, safe practice, and professionalism are emphasized throughout the course. Clinical experiences provide the student an opportunity to apply theoretical concepts and implement safe patient care to adults in a variety of settings. Lecture Hours: 150.000, Clinical Hours: 150.000
Prerequisite: PRN0210C, PRN0082C
Course User Fee: \$451.06

PSB**PSB2041 Behavioral Neuroscience 3.000 credits**

This course presents an approach to the study of the biology of human behavior and aims to introduce the field of Biopsychology. It is designed to provide each student with comprehensive exposure to the nervous system and how it governs various behaviors. This is an introduction course that explores how our brains develop, how they work, and how they are changed by life experiences. Topics include neural communication, localization of brain function, neural systems, and control of behavior. Upon completion of the course, the student will have a solid foundation regarding the biological basis of behavior upon which to build on in more advanced courses of study. Lecture Hours: 48.000

PSB2442 Psychology of Addiction 3.000 credits

This course will examine psychological, medical, pharmacological, legal, economic and sociological aspects of addiction to and use of various chemicals. The course will take an in-depth look at narcotic sedatives, and stimulants including alcohol, cocaine, heroin, cannabis, caffeine and tobacco. Lecture Hours: 48.000

PSC**PSC1121 GEN EDUC PHYS SCIEN 3.000 credits**

Lecture Hours: 48.000

PSC1515 Energy in the Natural Environment 3.000 credits

Investigation of the physical environment using energy as a theme to demonstrate the impact of science and technology on the environment and on the lives of people. Special fee. Lecture Hours: 48.000
Course User Fee: \$10.00

PSY**PSY2012 Introduction to Psychology 3.000 credits**

In this course, students will gain an introduction to the scientific study of human behavior and mental processes. Topics may be drawn from historical and current perspectives in psychology. Student learning outcomes: Students will be able to identify basic psychological theories, terms, and principles from historical and current perspectives; Students will be able to recognize real-world applications of psychological theories, terms, and principles; students will be able to recognize basic strategies used in psychological research; Students will be able to draw logical conclusions about behavior and mental processes based on empirical evidence. Lecture Hours: 48.000

PSY2050 Introduction to Forensic Psychology 3.000 credits

This course explores the interaction between psychology and the legal system. Students will learn the foundations, history, and terminology of forensic psychology and the influence of media and cultural issues. Students will also learn about the forensic psychologists role in criminal and civil proceedings, public policy, law enforcement, and victimization. Lecture Hours: 48.000
Prerequisite: PSY2012

PSY2800 Psychology of Genocide 3.000 credits

Students will learn the psychological, social, and cultural roots of genocide, human cruelty, and mass violence. Students will examine the various factors influencing such acts, and the emotional and psychological impact upon victims, perpetrators, rescuers, and society. Lecture Hours: 48.000
Prerequisite: ENC1101

**PSY2940C Mental Health - Aging Practicum
3.000 credits**

This Capstone course will provide students with field experience on neuroscience and aging through a practicum (non-paid), in a for profit or not for profit setting that serves older adult populations. The students will have the opportunity to apply the knowledge, skills and abilities gained through the Neuroscience and Aging CCC. Departmental Permission Required. (144 hours practicum).
Practicum Hours: 144.000

PTN

PTN0003 Intro to Pharmacy Practice & Medical Terminology 3.000 credits

This course is an orientation to the overall functions and services of a hospital pharmacy. Students will learn medical abbreviations, terminology, chemical symbols, formulas, and incompatibilities.

Prerequisite: HSC0003, HSC0003L, Corequisite: PTN0006

Course User Fee: \$10.00

**PTN0004 Pharmacy Practitioner Applications
3.000 credits**

This course focuses on pharmacy practitioner applications. Students will learn to develop skills relating to the specific, technical, manipulative and clerical tasks involved with the preparation and distribution of medications under the supervision of Licensed Pharmacists.

Clock Hours: 90.000

Prerequisite: HSC0003, HSC0003L, Corequisite: PTN0021

PTN0006 Pharmacy Calculations 3.000 credits

This is a course in Pharmacy Calculations. Students will learn to define systems of measurement, convert from one system to another, and calculate pharmacology problems.

Clock Hours: 90.000

Prerequisite: HSC0003, HSC0003L, Corequisite: PTN0003

PTN0021 Drug Classifications 3.000 credits

This course covers the major classifications of pharmaceuticals, standards for quality and purity of drugs, and authoritative information on dosage and administration. Students will learn about poisons, placebos, and the sources from which medications are produced.

Prerequisite: HSC0003, HSC0003L, Corequisite: PTN0004

Course User Fee: \$15.00

PTN0041 Pharmacy Technician Hospital Field Experience 10.000 credits

This course covers clinical hospital training to develop the student's knowledge and skills on the job. Students will learn how to properly prepare doses of medications and intravenous admixtures.

Prerequisite: HSC0003, HSC0003L, PTN0003, PTN0004, PTN0006, PTN0021, Corequisite: PTN0049

PTN0049 PH STR RET ST FLD EX 10.000 credits

Other Hours: 300.000

PUR

PUR2003 Public Relations 3.000 credits

This course provides students with a broad spectrum of topics related to the Public Relations profession. Current practices or organized programs used in business to earn public acceptance and good will for products, services, personnel, and policies are explored, studied and experienced. The course employs a hands-on approach to applying public relations techniques in hypothetical business situations. Students prepare press releases, brochures, and other collateral materials.

Lecture Hours: 48.000

QMB

QMB2100 Basic Business Statistics 3.000 credits

The application of basic statistical methods to business problems. Emphasis is on learning to select the appropriate statistical method of solving a given business problem, applying the chosen method, and interpreting the solution.

Lecture Hours: 48.000

Prerequisite: Acceptable score on the Algebra Placement test or equivalent.

RAT

RAT1001 INTRO RAD ONCOLOGY 2.000 credits

Lecture Hours: 32.000

RAT1021 P&P OF RAD THERAPY 1 2.000 credits

Lecture Hours: 32.000

RAT1211 HUMAN DISEASE 1.000 credits

Lecture Hours: 16.000

RAT1242 Clinical Oncology & Pathology 2.000 credits

Malignant conditions, etiology, and methods of treatment. Patient management, treatment planning, patient prognosis, treatment results, and the use and effect of combined therapies will be discussed. Contributing factors, growth and biologic behavior of neoplastic diseases as well as specific types of tumors and tumor sites will also be discussed.

Lecture Hours: 32.000

RAT1614C Radiation Therapy Physics 1 2.000 credits

A basic radiation physics course containing fundamental principles and concepts. The course includes radiation production, properties, and characteristics as well as structure of the atom and matter, electrostatics, magnetism, electrodynamics, and the electromagnetic spectrum.

Lecture Hours: 16.000, Clinical Hours: 48.000, Other Hours: 48.000

RAT1615 Radiation Therapy Medical Imaging 3.000 credits

This course is an introductory study to radiographic processes which includes the processes behind computed tomography, magnetic resonance imaging, nuclear medicine, positron emitting tomography, and ultrasound as it pertains to simulation, detection, and diagnosis of cancer.

Lecture Hours: 48.000

RAT1619 ELM OF TREATMENT PLN 2.000 credits

Lecture Hours: 32.000

RAT1657 Radiation Protection/Quality Assurance 1.000 credits

The student will learn to present basic principles of radiation protection and safety in radiation therapy. Radiation health and safety requirements of federal and state regulatory agencies, accreditation agencies and health care organizations are included. Specific responsibilities of the radiation therapist are discussed, examined and evaluated.

Lecture Hours: 16.000

RAT1801L INTRO TO CLINIC 2.000 credits

Clinical Hours: 96.000

RAT1804L RAT CLINIC 1 5.000 credits

Clinical Hours: 240.000

RAT1814L CLINIC 2 8.000 credits

Clinical Hours: 384.000

RAT1824L Clinical Education III 8.000 credits

Continuation of advanced patient treatment competencies under the supervision of an ARRT Certified Radiation Therapy Technologist.

Clinical Hours: 384.000

Prerequisite: RAT1814

RAT1840 CLIN APPL OF ANATOMY 1.000 credits

Lecture Hours: 16.000

RAT2022 P&P RAD THERAPY 2 2.000 credits

Lecture Hours: 32.000

RAT2061 Radiation Therapy Seminar 3.000 credits

This course will provide the opportunity for the radiation therapy student to evaluate their cumulative knowledge through comprehensive testing, refinement of accumulated knowledge, and retention of all aspects of radiation therapy. The course challenges the student to be prepared for the American Registry of Radiologic Technologist (ARRT) comprehensive national examination upon completion of graduation.

Lecture Hours: 48.000

RAT2123 Patient Care in Radiation Therapy 3.000 credits

This course presents patient care procedures relevant to the radiologic health science student. Content includes specific knowledge and skills required for patient care such as pharmacology, vital sign monitoring, recognizing side effects of treatment, nutritional needs of patients and psychological needs of patients.

Lecture Hours: 48.000

RAT2240C Radiation Oncology Sectional Anatomy 1.000 credits

An introduction to the concept of disease and general pathology. The types of growth, causative factors and biological behavior of neoplastic diseases are stressed. Pharmacology with emphasis on the radiation therapy patient is included in this course.

Clinical Hours: 48.000

RAT2241 RADIOBIOLOGY 2.000 credits

Lecture Hours: 32.000

RAT2243 CLIN ONCOLOGY & NEOP 2.000 credits

Lecture Hours: 32.000

RAT2618 RAD THERAPY PHYSICS2 2.000 credits

Lecture Hours: 32.000

RAT2690 INTEGRATION RAD THER 2.000 credits

This course integrates anatomy, clinical oncology and neoplasms, radiation physics, radiation biology, and radiation protection as they relate to the treatment planning process. Basic concepts used to develop the treatment plan for patients with particular needs will be discussed. Prerequisite: RAT 2022. (2 hr. lecture)

Lecture Hours: 32.000

RAT2834L Clinical Education IV 6.000 credits

This course includes clinical rotations through the radiation therapy department. Students will be provided the opportunity to apply theory learned from the previous semester in the various areas of the treatment process.

Clinical Hours: 288.000

Prerequisite: RAT1824L

REA**REA0007 READING 1 4.000 credits**

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$5.00

REA0017 Developmental Reading 2 4.000 credits

The Reading Lab provides intensive tutorial assistance for basic to advanced level students. This encompasses word recognition, pronunciation, reading rate, and technical reading. A tutorial study lab is available to assist with any college level course work area. Special Sections for Learning Disabled Students available. College preparatory, may not be used to satisfy graduation requirements. (1. 5-9 clock hrs.)

Lecture Hours: 32.000, Lab Hours: 64.000

Course User Fee: \$5.00

RED**RED3009 Emergent and Early Literacy 3.000 credits**

The teacher candidate will acquire the knowledge to apply the principles of explicit, systematic, and sequential evidence-based reading instruction and scaffold student learning while integrating the six components of reading (oral language, phonological awareness, phonics, fluency, vocabulary, and comprehension) in a K-2 setting. The teacher candidate will understand that teaching reading is an ongoing strategic process resulting in students comprehending various text. Reading competencies 1 and 2.

Lecture Hours: 48.000

Prerequisite: EEC3213, EEC3412

Course User Fee: \$18.00

RED3013 Foundations of Reading Instruction 3.000 credits

The teacher candidate will develop an understanding of reading components as a systematic process including oral language, phonological awareness, phonics, fluency, vocabulary, and comprehension. The teacher candidate will apply the principles of explicit, systematic and sequential evidence-based reading instruction and scaffold student learning while integrating the six components of reading in a K-12 setting. (Fifteen hour of clinical experience in a K-3 classroom is required.)

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

Course User Fee: \$18.00

RED3393 Differentiated Instruction in Content Reading 3.000 credits

The teacher candidate will differentiate instruction for individual students based on their strengths and needs by applying the principles of research-based strategies and integrating the six components of reading: phonological awareness, phonics, oral language, fluency, vocabulary, and comprehension in the content area classroom. The teacher candidate will have a broad knowledge of students with differing profiles to understand and apply evidence-based instructional practices by differentiating process, product, and context. Reading competencies 2 and 4 (fifteen hours of clinical experience are required in a math or science 3-12 classroom.)

Lecture Hours: 48.000

Prerequisite: EDG3321, : Optional Prerequisite: RED3013.

Course User Fee: \$18.00

RED4033 Teaching Foundations of Reading Instruction 3.000 credits

The student will develop a foundational understanding of the six components of reading as a systematic process: oral language, phonological awareness, phonics, fluency, vocabulary, and comprehension. The student will recognize the principles, techniques, and procedures required to develop the foundational reading skills that are essential to increasing reading proficiency in students from grades K-12. This is the first of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 1.

Lecture Hours: 48.000

RED4342 Application of Research Based Practices in Reading Education 3.000 credits

Building on the foundation of RED4033, the student will apply the principles of evidence-based research in comprehensive reading instruction. The student will use the reading skills, techniques, and strategies that facilitate reading comprehension. The student will apply the principles of research-based reading instruction, and integrate the six components of reading to facilitate the comprehension of different texts. This is the second of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 2.

Lecture Hours: 48.000

Prerequisite: RED4033 Bachelor's Degree

RED4519 Diagnosis and Instructional Intervention in Reading 3.000 credits

The teacher candidate will apply the principles of explicit, systematic and sequential evidence-based reading instruction and scaffold student learning while integrating the six components of reading. Teacher candidates will engage in a systematic problem-solving process to identify and remediate reading difficulties, using the results of informal reading assessments, to plan interventions. Reading competencies 3 and 4 (fifteen hours of clinical experience required in an approved first-third grade setting with ESOL students.)

Lecture Hours: 48.000

Prerequisite: RED3393

Course User Fee: \$18.00

RED4541 Foundations of Assessment 3.000 credits

The student will select and administer appropriate assessments and analyze data to inform reading instruction to meet the needs of all students. The student will engage in a systematic problem-solving process to remediate reading difficulties in emergent, beginning, and fluent readers and will plan effective instructional interventions. This is the third of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 3.

Lecture Hours: 48.000

Prerequisite: RED4342 Bachelor's Degree

RED4654 Foundations and Applications of Differentiated Instruction 3.000 credits

The student will apply research-based best practices in educational neuroscience research related to the instruction of reading. Using knowledge of diverse learners, the student will apply intensive, explicit, multisensory sequential approaches and instructional practices by differentiating the process, product, or context of their reading instruction, and providing scaffolds to enhance comprehension in all content areas. This is the fourth of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 4.

Lecture Hours: 48.000

Prerequisite: RED4541 Bachelor's Degree

RED4854 Reading Practicum 3.000 credits

The student will synthesize and apply knowledge of the six components of reading, formal and informal reading assessments, skills related to data analysis, and differentiation of instruction in order to plan and implement a comprehensive, intensive, effective, systematic, multisensory, research-based reading plan of instruction for all students. This is the fifth of five courses that lead to the Florida Department of Education Reading Endorsement. This course corresponds to Competency 5.

Lecture Hours: 48.000

Prerequisite: RED4654 Bachelor's Degree

REE**REE0032 PRINC & PRACTICES 2 2.500 credits**

Lecture Hours: 75.000

REE0044 real estate sales 2.100 credits

Principles & Practices 1 is the beginning course for a student wishing to enter the Real Estate business and receive a Real Estate License. This is a survey course that looks into the legal relationship between salesman and client, salesman and customer, and salesman and broker. It is a study of many of the mechanical principles of the Real Estate business such as deeds, surveys, financing and appraising. Special fee. (63 contact hours)

Lab Hours: 63.000, Clinical Hours: 1.000 - 30.000

REE0801 REAL ES PST LIC BR 1 1.000 credits

THIS IS THE MANAGEMENT PORTION (PART 1) OF THE STATE REQUIRED POST-LICENSING COURSE FOR REAL ESTATE BROKERS. THE OBJECTIVE IS TO PROVIDE THE LICENSED BROKERS WITH ADVANCED KNOWLEDGE OF THE MANAGEMENT AND OPERATION OF A BROKERAGE OFFICE.

Lecture Hours: 30.000

Course User Fee: \$20.00

REE2040 RL EST PRIN & PRAC 1 4.000 credits
XXX

Lecture Hours: 64.000

Course User Fee: \$15.00

REE2040 Real Estate Principles and Practices (P&P 1) 4.000 credits

Topics include real property, liens, titles, contracts, tax factors, mortgages, property evaluation, real estate market, licensing requirements, legal aspects of the real estate business, and property management. Completion of this course is required by the Florida Real Estate Commission for approval to take the State Examination.

Lecture Hours: 64.000

REE2085 POST-LIC ED/SALESPER 3.000 credits
XXX

Lecture Hours: 48.000

REE2200 REAL ESTATE FINANCE 3.000 credits

A course designed to develop an understanding of the finance industry as it relates to real estate.

Detailed information covering legal aspects of mortgages, brokerage regulation, ethics and all major sources of funds for real estate financing will be covered. Prerequisite: RES 260 or possession of valid real estate salesman's license.

Lecture Hours: 48.000

REE2270 MORTGAGE BNK & BROK 3.000 credits

Lecture Hours: 48.000

REE2304 Commercial Real Estate 3.000 credits

In this course, the student will learn specific financial tools and techniques required to assess the feasibility of commercial real estate investment, funding and financing options, risks associated with real estate transactions and factors impacting real estate investment decisions. The student will learn how the commercial real estate process works, the different interested parties involved in the transaction, the essentials of risk and return, financing options and different phases in a commercial real estate transaction.

Lecture Hours: 48.000

Course User Fee: \$70.00

REL

REL1210 LIT OF THE OLD TESTM 3.000 credits

An examination of the historical sources and material in the Old Testament, with emphasis on its literary and cultural importance.

Lecture Hours: 48.000

REL1240 Religion of the New Testament 3.000 credits

In this course, students will not only explore the historical sources and material in the New Testament, but also examine its significance within the Western Canon, shedding light on its literary and cultural impact across different societies and time periods.

Lecture Hours: 48.000

REL2300 Survey of World Religions 3.000 credits

A survey of the origins, beliefs and contemporary practices of the world's religions: Hinduism, Islam, Taoism, Zen Buddhism, Judaism, Christianity and Confucianism. Attention is given to the interactions between specific religions and the cultures in which they are practiced.

Lecture Hours: 48.000

RET

RET1024 Introduction to Respiratory Care 3.000 credits

This is an introductory course to the Respiratory Care discipline. Students will learn the history of the profession, terminology, hospital and patient safety, infection control, patient assessment, accessing and utilizing the patient's medical record, critical thinking, Respiratory Care protocols, and patient education. Prerequisite: ENC 1101; Corequisite: RET 1484. (2 hr. lecture, 1 hr. clinical)

Lecture Hours: 32.000, Clinical Hours: 48.000

Prerequisite: ENC1101 AND Corequisite: RET1484

RET1024C Fundamentals of Respiratory Care 3.000 credits

This is an introductory course to the Respiratory Care discipline. Students will learn the history of the profession, essential medical terminology, hospital and patient safety, infection control, patient assessment, accessing and utilizing the patient's medical record, critical thinking, Respiratory Care protocols, and patient education.

Lecture Hours: 32.000, Clinical Hours: 48.000

Prerequisite: ENC1101, Corequisite: RET2274C

RET1024L INTRO TO RESP CRE LA 1.000 credits

Laboratory for RET 1024. Corequisite: RET 1024.

Laboratory fee. (2 hr. lab)

Lab Hours: 32.000, Practicum Hours: 2.000

Course User Fee: \$61.00

RET1484 Respiratory Care Pathophysiology 1 2.000 credits

This is an introductory course in the study of pulmonary and cardiovascular anatomy, physiology and pathology. Students will learn terminology, disease classification, diagnostic techniques and related physiological concepts.

Lecture Hours: 32.000

Prerequisite: ENC1101, Corequisite: RET1024C

RET2264 Advanced Modalities and Monitoring 2.000 credits

This is an advanced course relating to critical care. Students will learn advanced techniques in invasive and non-invasive monitoring, electrocardiographic monitoring and interpretation, alternatives to conventional ventilation and advanced cardiovascular support systems.

Lecture Hours: 32.000

Prerequisite: RET2284, Corequisite: RET2714, RET2714L

RET2274 RESP CARE THEORY 1 2.000 credits

Theory of supplemental oxygen and humidity in respiratory pathology. Special emphasis is given to the medical, surgical, and pediatric patients and their cardiopulmonary physiology as it relates to therapeutic oxygen techniques. Corequisites: RET 1024, 1484, 2274L. (2 hr. lecture)

Lecture Hours: 32.000

RET2274C Respiratory Care Equipment & Procedures 3.000 credits

An overview of oxygen and aerosol therapy equipment and procedures normally used for respiratory therapy. Especially emphasized are methods used in medical, surgical, and pediatric patients and their cardiopulmonary physiology as it relates to therapeutic oxygen techniques.

Lecture Hours: 32.000, Clinical Hours: 48.000

Corequisite: RET1024C

RET2274L RESP CARE THEO LAB 1 1.000 credits

Laboratory for RET 2274. Corequisite: RET 2274.

Laboratory fee. (2 hr. lab)

Lab Hours: 32.000

Course User Fee: \$61.00

RET2275 RESP CARE THEORY 2 2.000 credits

Emphasis on pressure breathing modalities, chest physiotherapy, and incentive devices

Lecture Hours: 32.000

RET2275C Advanced Respiratory Care Equipment & Procedures 3.000 credits

This course reviews airway equipment and procedures normally used for respiratory therapy. Especially emphasized are methods used in airway management, bronchial hygiene therapies, hyperventilation therapies, and airway clearance techniques.

Lecture Hours: 32.000, Clinical Hours: 48.000

Prerequisite: RET2274C

RET2275L RESP CARE THEO LAB 2 1.000 credits

Laboratory for RET 2275.

Lab Hours: 32.000

Course User Fee: \$61.00

RET2284 Principles of Mechanical Ventilation 2.000 credits

A continuation of RET 2275. A concentrated course of study which focuses on the theoretical operation, application and procedures related to critical care and mechanical ventilation.

Lecture Hours: 32.000

Prerequisite: RET2275C, RET2284L

RET2284L Principles of Mechanical Ventilation Laboratory 2.000 credits

Laboratory for RET 2284. This course will provide an in depth study of the operation of mechanical ventilation devices and associated monitors. Patient safety, troubleshooting and application are stressed.

Lab Hours: 64.000

Corequisite: RET2284

Course User Fee: \$35.00

RET2285L PRESS BREATH APPLC 2.000 credits

Lab Hours: 64.000

RET2350 Respiratory Care Pharmacology 2.000 credits

This course is designed to provide training in the basic principles of the administration of medications including dosage and solutions. The drugs administered by respiratory therapists are covered in-depth, along with an introduction to the general pharmacological classifications of other drugs that may be administered to pulmonary patients.

Lecture Hours: 32.000

RET2414 Pulmonary Studies 2.000 credits

In-depth study of diagnostic techniques in the field of pulmonary medicine which includes lung volumes, static and dynamic mechanics of breathing, ventilation, distribution of gases, diffusion and arterial blood gas sampling and handling. Corequisite: RET 2414L. (2 hr. lecture)

Lecture Hours: 32.000

Corequisite: RET2414L

RET2414L Pulmonary Studies Laboratory 1.000 credits

Laboratory for RET 2414. Simulated clinical settings of diagnostic techniques used to evaluate pulmonary functions.

Lab Hours: 32.000

Prerequisite: RET2414

Course User Fee: \$20.00

RET2503 Respiratory Care Pathophysiology 2 3.000 credits

This is a foundation course on cardiopulmonary disease. The student will learn the pathogenesis, diagnosis, treatment and rehabilitation of the diseases included in the course.

Lecture Hours: 48.000

RET2601 Respiratory Care Seminar 2.000 credits

This is an advanced course focuses on clinical and nonclinical issues. Students will learn concepts including, but not limited to, clinical research, legal and ethical concerns, home care, extended care, rehabilitation and management. ACLS and PALS certification obtained.

Lecture Hours: 32.000

Course User Fee: \$90.00

RET2714 Perinatal and Pediatric Respiratory Care 2.000 credits

This course is designed to provide training in perinatal and pediatric respiratory care. Students will learn assessment and therapeutic techniques related to critical care.

Lecture Hours: 32.000

Corequisite: RET2264, RET2714L

RET2714L Perinatal & Pediatric Respiratory Care Laboratory 1.000 credits

This is an introductory laboratory course that will explore assessment and Respiratory Care therapeutics of the perinatal and pediatric patient populations. The student will learn to apply physical assessment techniques, oxygen aerosol and humidity therapies, therapeutic procedures, airway management, resuscitation and management of mechanical ventilation.

Lab Hours: 32.000

Corequisite: RET2264, RET2714L

RET2832 Respiratory Care Clinic 1 2.000 credits

This is an introductory clinical practice course. The student will learn psychomotor skills related to basic respiratory care and patient care procedures including patient charting, vital signs, infection control and non-pressurized oxygen adjuncts.

Clinical Hours: 96.000

RET2833 Respiratory Care Clinic 2 5.000 credits

In conjunction with RET 2274, 2274L and RET 1024, 1024L, RET 2832 is designed to allow the student to develop psychomotor skills related to basic respiratory care and patient care procedures (patient charting, vital signs, infection control and non-pressurized oxygen adjuncts). During the rotation, the student is provided with the opportunity to apply and discuss the theory and techniques as presented in corequisite courses.

Clinical Hours: 240.000

RET2834 Respiratory Care Clinic 3 8.000 credits

This course is a continuation of RET 2833. Training will be provided on the clinical application of procedures and techniques relating to respiratory critical care.

Clinical Hours: 384.000

Prerequisite: RET2284, RET2284L, Corequisite: RET2714, RET2714L

RET2835 Respiratory Care Clinic 4 8.000 credits

This course is designed to provide the student with the clinical application of adult, pediatric, and neonatal intensive respiratory care. Procedures and techniques presented in RET 2280, 2714, 2264 as it relates to their clinical application will be emphasized.

Clinical Hours: 384.000

Prerequisite: RET2834, Corequisite: RET2601

RET9995 Respiratory Care 23.000 - 232.000 credits

This course is designed for students who currently hold the Certified Respiratory Therapist (CRT) Credential (TMOP 1100:815732) and provides academic recognition of the knowledge, skills, and competencies acquired through professional credential attainment. The course validates expertise in respiratory care, including patient assessment, airway management, oxygen therapy, mechanical ventilation, diagnostic testing, and therapeutic interventions. Credit awarded through this course supports academic progression and recognizes demonstrated competency in the respiratory care profession.

Lecture Hours: 640.000

RTE

**RTE1000 Orientation to The Imaging Sciences
2.000 credits**

This course is an introduction to the overall field of Imaging Sciences, radiography in particular. Students will learn the basic principles of radiation protection as it applies to the various modalities in imaging and treatment, a history of the imaging sciences, ethical/legal issues, professional behavior, medical terminology, and math/physics.
Lecture Hours: 32.000

RTE1418 RADIOLOGIC TECHN I 3.000 credits

An introductory course providing a firm base upon which the student can build an understanding of the technical aspects of the profession. Includes medical and technical terminology, film history and construction, darkroom chemistry and procedures, use of radiographic accessories, manipulation of technical factors and other related material. The content is correlated with clinical practice beginning the second term. Prerequisite: permission of the program coordinator; corequisites: RTE 170, 172.
Lecture Hours: 32.000, Lab Hours: 32.000

RTE1503 Radiographic Positioning 1 3.000 credits

Basic routine positioning of the chest, abdomen, upper and lower extremities, digestive and urinary systems.
Lecture Hours: 48.000
Corequisite: RTE1503L

**RTE1503L Radiographic Positioning Laboratory 1
1.000 credits**

Laboratory for RTE 1503
Lab Hours: 32.000
Corequisite: RTE1503

RTE1513 Radiographic Positioning 2 3.000 credits

Positioning of the bony pelvis, shoulder girdle, bony thorax, spinal column, skull and facial bones.
Lecture Hours: 48.000
Prerequisite: RTE1503, RTE1503L, Corequisite: RTE1513L

**RTE1513L Radiographic Positioning Laboratory 2
1.000 credits**

Laboratory for RTE 1513
Lab Hours: 32.000
Prerequisite: RTE1503, RTE1503L, Corequisite: RTE1513
Course User Fee: \$25.00

RTE1613 RADIATION & HLTH PHYSC 4.000 credits

Theoretical foundation of physical phenomena which characterize the interactions of radiation with matter. Includes mathematics, basic atomic structure, electron configurations, properties of radiation and radioactive decay particles, and the application of these phenomena in particular radiographic procedures. Particular emphasis is on the use of the above material to provide maximum radiation safety for the patient. Prerequisite: permission of the program coordinator; corequisites: RTE 171, 172.
Lecture Hours: 48.000, Lab Hours: 32.000

RTE1804 Radiographic Clinic 1 5.000 credits

The first in a series of six clinical courses. Under direct supervision of faculty and clinical staff, performance of basic diagnostic radiographic procedures is carried out.
Clinical Hours: 240.000
Corequisite: RTE1503, RTE1503L
Course User Fee: \$40.00

RTE1814 Radiographic Clinic 2 5.000 credits

The student will be evaluated on competency performances in routine fluoroscopic, and in urographic procedures. This is the second of six clinical education courses.
Clinical Hours: 240.000
Prerequisite: RTE1503, RTE1503L, RTE1804
Course User Fee: \$25.00

RTE1824 Radiographic Clinic 3 5.000 credits

The student continues to rotate, under supervision, through different units of a Radiology Department. Development of a capability to assist in diagnostic procedures at a more complex level.
Clinical Hours: 24.000 - 240.000
Prerequisite: RTE1814

**RTE2010 New Imaging Modalities In Radiology
1.000 credits**

This course will enable the students to compare and contrast the current imaging modalities with the emerging technologies available in Radiology departments. Included in this course will be pictorial archiving and communications systems (PACs), digital imaging, and fusion imaging.
Lecture Hours: 16.000
Course User Fee: \$83.00

RTE2385 RADIATION BIOLOGY 2.000 credits

Lecture Hours: 32.000

RTE2457 Radiologic Technology 2 3.000 credits

A more in-depth study of radiographic exposure factors as they relate to specialized procedures and equipment.
Lecture Hours: 48.000

RTE2563 Radiographic Positioning 3 2.000 credits

Radiographic procedures which utilize contrast media, sterile techniques, and/or specialized equipment and accessories.

Lecture Hours: 32.000

Prerequisite: RTE1513, RTE1513L

RTE2571 Principles of Computed Tomography 3.000 credits

This course provides the radiologic technologist advanced imaging techniques of computed tomography. This introduction to the CT scanning technology will include history and development, equipment, terminology, patient preparation and care, and the principles of image formation, acquisition, and production. Corequisite: RTE2762.

Lecture Hours: 48.000

Corequisite: RTE2762

RTE2575 MRI Principles and MR Safety 3.000 credits

This course is an introduction to the basic principles of MRI, including signal production, basic concepts in image formation and image contrast and resolution. It is intended to provide a solid foundation in MRI technology as well as an understanding of the MRI equipment and its appropriate and safe use. A practical approach to using the correct imaging sequences and parameters will be discussed. Emphasis will be placed in MRI Screening and MRI Safety.

Corequisites: RTE2762, RTE2576.

Lecture Hours: 48.000

Corequisite: RTE2762, RTE2576

RTE2576 MRI Procedures & Patient Care 3.000 credits

This course is designed to introduce the student to the different image acquisition techniques and common scan sequences used in MRI imaging to describe normal and abnormal anatomy. Emphasis will be placed on basic patient care skills and safe use of equipment used in MRI. Corequisites: RTE2762, RTE2575.

Lecture Hours: 48.000

Corequisite: RTE2762, RTE2575

RTE2577C Magnetic Resonance Imaging (MRI) Practicum 6.000 credits

This course is designed to provide the students with hands-on experience for body imaging. Students will learn to position the patients correctly inside the Magnetic Resonance Imaging (MRI) machine and program the machine to create axial, coronal, and sagittal images of the brain and spinal column.

Prerequisites: RTE2762, RTE2575, RTE 2576.

Practicum Hours: 288.000

Prerequisite: RTE2762 AND RTE2575 AND RTE2576

RTE2762 Cross Sectional Anatomy 3.000 credits

This course is designed to provide the students a thorough knowledge of Axial, coronal. and sagittal imaging planes of our body. Students will also learn to identify various anatomical structures in axial, coronal, and sagittal images of our body.

Lecture Hours: 48.000

RTE2834 Radiographic Clinic 4 5.000 credits

Performance of procedures of increasing levels of complexity and responsibility including specialized diagnostic procedures. At this level the program faculty and clinical supervisor will determine if the student can perform procedures with less supervision.

Clinical Hours: 240.000

Prerequisite: RTE1824

Course User Fee: \$66.00

RTE2844 Radiographic Clinic 5 8.000 credits

The fifth in a series of six clinical education courses. During this clinical course the student will perform standard quality assurance tests on radiographic equipment and accessories. In addition, the student will have competency evaluations to include a gastrointestinal series and either paranasal sinuses or facial bone studies.

Clinical Hours: 384.000

Prerequisite: RTE2834

Course User Fee: \$25.00

RTE2854 Radiographic Clinic 6 3.000 credits

The student will complete the competencies required by the American Registry of Radiologic Technologists to become eligible to apply to sit for the certification exam. The student will socialize into radiography practice by beginning to work more independently of a radiographer. The student will use organizational skills to provide care to patient clients assigned to them during radiographic exams. During this course the student will be assigned to one rotation during hours other than the normal working hours of the radiology department to gain competency in procedures not usually available during the day.

Clinical Hours: 144.000

Prerequisite: RTE2844

RTE2940C Computed Tomography Clinical Education 3.000 credits

The course provides the radiologic technologist practical, first-hand experience of scanning procedures and techniques at a supervised clinical site; theories learned in RTE 2571 will be applied. Students will observe, assist, and perform Computed Tomography under the supervision and guidance of a qualified CT Technologist.
Prerequisites: RTE2571, RTE2762.
Clinical Hours: 144.000
Prerequisite: RTE2571 AND RTE2762
Course User Fee: \$29.36

RTV**RTV1000 Fundamentals of Broadcasting 3.000 credits**

In this introductory course for television and radio broadcasting, students will learn about the foundations of the the American broadcast system.
Lecture Hours: 48.000

RTV1240C Sound Design 3.000 credits

In this introductory Sound Design course students will learn an overview of sound recording and audio post-production.
Lecture Hours: 32.000, Lab Hours: 32.000
Course User Fee: \$15.00

RTV1241C Television Production 1 4.000 credits

An introductory overview of the practices and procedures used in a television studio. Students will learn basic operation of studio and control room equipment and work towards the completion of broadcast standard projects.
Lecture Hours: 32.000, Lab Hours: 64.000
Course User Fee: \$25.00

RTV1242C Television Production 2 4.000 credits

An intermediate course in television studio production. Students will learn to implement the fundamentals as introduced in Television Production 1 to the production of scripted programs.
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: RTV1241C
Course User Fee: \$25.00

RTV2205C TELEVISION WORKSHOP 3.000 credits

Lecture Hours: 80.000

RTV2243C Directing 3.000 credits

An introductory overview of television directing. Students will learn the responsibilities of the television director in coordinating production elements and television control room techniques.
Lecture Hours: 32.000, Lab Hours: 32.000
Prerequisite: RTV1242C

RTV2245C Electronic Field Production 1 4.000 credits

An introductory course in which students will learn single-camera field production and electronic news-gathering.
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: RTV1242C
Course User Fee: \$40.00

RTV2246C Electronic Field Production 2 4.000 credits

In this intermediate workshop style class students will learn about documentary production.
Lecture Hours: 32.000, Lab Hours: 64.000
Prerequisite: RTV2245C
Course User Fee: \$40.00

RTV2300 Broadcast Writing 3.000 credits

In this intermediate workshop style class students will learn broadcast writing emphasizing news, documentary, commercials, and long-form programming.
Lecture Hours: 16.000, Lab Hours: 32.000

RTV2940 Television Internship 3.000 credits

In this lower division internship designed for AS students in Television Production, students will learn to apply skills and knowledge learn in the program at an established television entertainment company (144 hr. Internship).
Other Hours: 144.000
Prerequisite: RTV1242C

RTV2942 Spring Television Practicum 3.000 credits

In this semester long on-site training with MDC Television students will learn from Television Professionals by working on current MDC TV programming (144 hr. Practicum).
Other Hours: 144.000
Prerequisite: RTV1242C

RTV3203 The Telemundo Academy 3.000 credits

The Telemundo Academy is a project-based learning course that provides students with a fundamental understanding of the concepts and processes of short-form content production for social media and mobile video platforms. The course will be divided into four phases: development, pre-production, production, and post-production. The first portion of the course will focus on pre-production given that the quality of the final projects will be correlated to students' understanding of all elements and practices in this phase. Students will learn how research, script development, and storyboarding are critical to transforming their original concepts into successful productions. The course will cover additional fundamental skills related to: communications, leadership, marketing, and digital and social media best practices.

Lecture Hours: 48.000

Course User Fee: \$80.00

RTV3256C Advanced Post Production 3.000 credits

An advanced course in which students will learn complex digital media post production techniques.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2553C

RTV3277C Television Studio Production Workshop 3.000 credits

In this upper division workshop class for BAS students in Film, Television & Digital Production students will learn advanced television studio production techniques for pre-scripted programs.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2515C OR RTV2246C

Course User Fee: \$75.00

RTV3408 Ethics & Research for Non-Fiction Scripts 3.000 credits

In this upper division course students will learn research methods applicable to documentary production.

Lecture Hours: 48.000

Recommended Preparation: Student must be in BAS Program AND Prerequisite: FIL2131 OR RTV2300

RTV3810C Broadcast Design & on Air Promotions 3.000 credits

In this upper division course in BAS in Film, Television & Digital Production students will learn about on-air promotions with specific emphasis on layout, color, and composition.

Lecture Hours: 32.000, Lab Hours: 32.000

Recommended Preparation: BAS Students ONLY

Course User Fee: \$100.00

RUS**RUS1120 Elementary Russian 1 4.000 credits**

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Russian. Note: Students must pass this course with a C or better to continue to RUS1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

RUS1121 Elementary Russian 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Russian.

Lecture Hours: 64.000

Prerequisite: RUS1120

Course User Fee: \$15.00

SBM**SBM1000 Small Business Management 3.000 credits**

Students will learn that growing a business involves shifting from the search for and validation of a business model to executing and scaling the business model. The skill sets are different and more formal management is often required. The course reviews the context and complexity of scaling a business and executing a business model.

Lecture Hours: 48.000

Course User Fee: \$90.00

SCE**SCE4362 Methods of Teaching Science 3.000 credits**

The teacher candidate will learn to design and implement science instruction utilizing the national framework for K-12 science education and educational neuroscience to provide all students with high-quality science education. The student will learn about the theoretical knowledge and skills essential for facilitating science instruction in a variety of classroom settings. Fifteen hours of clinical experience are required.

Lecture Hours: 48.000

Corequisite: EDF4430

Course User Fee: \$35.00

SCE4363 Addressing Scientific Misconceptions in the Secondary Science Classroom (Practicum) 3.000 credits

The teacher candidate will design, implement, and examine the alignment of their personal instructional practices to the national framework for K-12 science education utilizing the action research process. The student will focus on identifying, analyzing, and addressing misconceptions in science in grades 6-12. Sixty hours of clinical. Lecture Hours: 48.000
Prerequisite: EEX3071, SCE4362, Corequisite: RED3393

Course User Fee: \$18.00

SCE4943 Seminar in Science Education 3.000 credits

The teacher candidate will engage in professional dialogue and explore professional learning opportunities related to teaching in the secondary science setting. This seminar course is taken in conjunction with a full time, supervised teaching experience and provides an opportunity to examine and reflect on the daily experiences of becoming a highly effective teacher.

Lecture Hours: 48.000

Prerequisite: All coursework (EDG 3321, EDF 4430, BSC 1084 or BSC 2020, RED 3393, SCE 4362, EEX 3071, MCB 2010/MCB 2010L or MCB 3023/MCB 3023L, PCB 3043, PCB 3060, TSL 4324C, EDG 3443, SCE 4363, state exams, and E-Portfolio completed and passed., Corequisite: SCE4945

Course User Fee: \$18.00

SCE4945 Internship in Science Education 9.000 credits

The teacher candidate will complete a full time, supervised teaching experience. The student will learn and experience all of the educational and professional responsibilities common to science teachers.

Internship Hours: 432.000

Prerequisite: All coursework (EDG 3321, EDF 4430, BSC 1084 or BSC 2020, RED 3393, SCE 4362, EEX 3071, MCB 2010/MCB 2010L or MCB 3023/MCB 3023L, PCB 3043, PCB 3060, TSL 4324C, EDG 3443, SCE 4363) and state exams completed and passed, Corequisite: SCE4943

SCM

SCM1114 Applied Advanced Technologies & AI in Supply Chain Management 3.000 credits

This course provides an in-depth exploration of Industry 4.0 and Artificial Intelligence (AI) applications within the supply chain industry, focusing on strategic support, optimization, data-driven decision-making, and organizational readiness. Students will discover how AI technologies enhance operational efficiency, reduce costs, and improve overall supply chain performance while ensuring responsible and ethical implementation. Through case studies and practical exercises, learners will gain the skills needed to leverage AI for innovative solutions and competitive advantage in supply chain management. Lecture Hours: 48.000

SCY

SCY0010 SURETY AGENT 4.000 credits

insert

Lecture Hours: 120.000

SCY0051 PRIVATE INVESTIGATOR INTERN CR A 0.800 credits

This course requires twenty four hours of training as required by Section 493.6203(b) F.S. for Private Investigator Interns. Students will learn topics which include Florida Statutes and Florida Administrative Code, the Intern/Sponsor Relationship, Ethics, Liability, Surveillance, Report Writing, Equipment, Interviewing, Sources of Information, the Computer and Investigations, and Restrictions on Records. (24 contact hrs.)

Lecture Hours: 24.000

Course User Fee: \$48.00

SCY0052 PRIVATE INVESTIGATOR INTERN CR B 0.530 credits

This course requires sixteen hours of training as required by Section 493.6203(b) F.S. for Private Investigator Interns. Students will learn topics which include locating people and performing background investigations, evidence, executive protection, anti-terrorism, courtroom and formal hearing demeanor, pretrial responsibilities, and the investigator as a witness. Prerequisite: SCY 0051 (16 contact hrs.)

Lecture Hours: 16.000

Course User Fee: \$35.00

SLS

SLS0341 EMPLOYABILITY SKILLS 1.000 credits

insert

Lecture Hours: 30.000

SLS1106 First Year Experience Seminar 1.000 credits

This course is designed to provide students a forum for transitioning into college. Students will learn to develop the skills required for success in college and beyond. This course is intended for first time in college students, who are seeking an Associate in Arts degree program.

Lecture Hours: 16.000

Course User Fee: \$10.00

SLS1125 STUDENT SUPPORT SEM 3.000 credits

This three credit course is intended for students that have been placed on Academic Warning. It is designed to help students be more successful academically by focusing on performance in a learning environment. This will include social, cultural, psychological and academic aspects of the individual and the role they play in the learning environment. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$15.00

SLS1401 Psychology of Career Adjustment 1.000 - 6.000 credits

For students who have not decided, are having difficulty deciding, or need clarification in making a career choice. A format for a systematic investigation for career and life planning is included. It is concerned with "who you are," "where are you going," "how to get there," and "what's out there that fits you."

Lecture Hours: 16.000 - 96.000

Course User Fee: \$15.00

SLS1502 COLLEGE STUDY SKILLS 1.000 - 3.000 credits

insert

Lecture Hours: 16.000 - 48.000

SLS1505 College Survival Skills 1.000 credits

This is an introductory self-discovery course designed to help students make the transition to college. Students will learn the knowledge and skills necessary for success, including knowledge of academic policies and procedures, effective study strategies, and making sound academic and career choices.

Lecture Hours: 16.000

Course User Fee: \$5.00

SLS1510 Preparing for Student Success 3.000 credits

This course provides an orientation to college life and helps develop academic, career, and personal goals. Students will learn college success strategies, goal-setting, learning style assessments, as well as general and discipline-specific study skills in the context of various theoretical, practical, and experiential perspectives.

Lecture Hours: 48.000

Course User Fee: \$15.00

SON**SON1000L Introduction to Sonography 1 1.000 credits**

An introduction to the physical principles of diagnostic ultrasound. Bases of imaging with ultrasound are discussed as well as clinical units in the various areas of specialization. In conjunction with the lectures, supervised laboratory classes are conducted to familiarize students with operations of the equipment in each of the clinical areas.

Lab Hours: 32.000

Corequisite: SON 1111C, SON 1121C

SON1001L Introduction to Sonography 2 1.000 credits

This second introductory course will cover the past present and future of sonography. After the historical landmarks are identified, the focus will be on the current diversity of applications of diagnostic medical sonography. Students will also discover future trends and developments on the technology horizon of the profession.

Lab Hours: 32.000

Prerequisite: SON1000L

SON1005L Basic Sonography 2.000 credits

This course is designed to cover the essential of the profession of Diagnostic Medical Sonography. Topics include: professionalism, medical ethics, hospital administration, sonographic terminology, quality assurance, photographic principles, related radiological specialties and scanning techniques. Laboratory experience will include equipment use and quality assurance techniques.

Lab Hours: 64.000

Prerequisite: SON1000L

Course User Fee: \$15.00

**SON1006L Professional Aspects of Sonography
1.000 credits**

An introduction to the professional aspects of sonography. Topics include: medical ethics and law, hospital administration, quality assurance/quality control and management. Laboratory experience includes actual phantom scanning conducting equipment QA protocols, and participation on a mock ethics board.

Lab Hours: 32.000

**SON1100L Principles of Protocols Of Imaging
2.000 credits**

An introduction to radiographic film, its handling & processing and the various radiographic specialties. Laboratory experience includes: film composition and identification, rapid processing, photographic techniques, reading H&D curves, performing sensitometry and identifying film artifacts. During radiographic specialties, there will be an introduction to CT, MRI, and the areas of radiologic technology in order to discover how these modalities compliment sonography.

Lab Hours: 64.000

SON1111C Abdominal Sonography 1 2.000 credits

An in-depth course designed to cover all aspects of clinical abdominal ultrasound studies. Subject matter includes: review of normal anatomy (ultrasonic appearance), indications for ultrasound studies, clinical presentation and data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls.

Lecture Hours: 16.000, Lab Hours: 32.000

Corequisite: SON1000L

SON1112C Abdominal Sonography 2 2.000 credits

An in-depth course designed to cover all aspects of clinical abdominal ultrasound studies. Subject matter includes: review of normal anatomy (ultrasonic appearance), indications for ultrasound studies, clinical presentation and data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON1111C

Course User Fee: \$15.00

**SON1113L Sonography Cross Sectional Anatomy
2.000 credits**

A thorough course aimed at teaching the student to understand anatomical relationships and recognize structures on cross-sectional and sagittal diagrams, photographs of gross anatomy and sonography. The laboratory conducted in conjunction with the classroom lectures is designed to identify all normal anatomical landmarks in multiple planes in actual scanning situations.

Lab Hours: 64.000

SON1115L Duplex Abdominal Sonography 1.000 credits

This course is designed to cover aspects of duplex abdominal sonography applications. Topics include: the aorta and its branches, the IVC and its tributaries, and the portal system. Subject matter includes: etiology, pathophysiology, clinical presentations, sonographic appearance and differential diagnosis of diseases.

Lab Hours: 32.000

Prerequisite: SON1112C

**SON1121C Obstetrics/Gynecology Sonography 1
2.000 credits**

An in-depth course designed to present all aspects of clinical OB/GYN ultrasound studies. Subject matter includes: review of normal anatomy (ultrasound appearance), indications for ultrasonic studies, clinical presentation, clinical data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls.

Lecture Hours: 16.000, Lab Hours: 32.000

Corequisite: SON1000L

**SON1122C Obstetrics/Gynecology Sonography 2
2.000 credits**

An in-depth course designed to cover all aspects of clinical OB/GYN ultrasound studies. Subject matter includes: review of normal anatomy (ultrasound appearance), indications for ultrasonic studies, clinical presentation, clinical data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON1121C

SON1141C Small Parts Sonography 2.000 credits

An in-depth course designed to cover all aspects of clinical abdominal ultrasound studies. Subject matter includes: review of normal anatomy (ultrasonic appearance), indications for ultrasound studies, clinical presentation and data, pathophysiological basis of disease, ultrasonic manifestations of diseases, recognition of adequate images and scanning pitfalls. Prerequisite: SON 1112C. (2 hr. lab.)
Lab Hours: 48.000

SON1145L Pediatric Sonography 1.000 credits

This course is designed to cover aspects of pediatric ultrasound examinations. Topics include: Liver, biliary, spleen, renal, adrenal, gastrointestinal, scrotum, and musculoskeletal structures. Subject matter includes: etiology, pathophysiology, clinical presentations, sonographic appearance and differential diagnosis.
Lab Hours: 32.000
Prerequisite: SON1141C

SON1804 Clinic 1 2.000 credits

This is the first in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded a hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.
Clinical Hours: 96.000
Corequisite: SON1000L

SON1814 Clinic 2 2.000 credits

This is the second in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.
Clinical Hours: 96.000
Prerequisite: SON1804

SON1824 Clinic 3 3.000 credits

This is the third in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.
Clinical Hours: 144.000
Prerequisite: SON1814

SON2139L Cardiovascular Principles 1.000 credits

An introductory course to techniques other than echocardiography utilized in the diagnosis of cardiovascular disease. Topics discussed include physical examination, electrocardiogram, Phonocardiogram, cardiac catheterization, and nuclear medicine cardiology.
Lab Hours: 32.000
Prerequisite: SON2400C, Corequisite: SON2401C

SON2151C Neurosonography 2.000 credits

A comprehensive course designed to examine sonographic imaging of the neonatal and infant brain, with an introduction to ultraoperative brain and spinal cord imaging. Emphasis is placed on normal brain anatomy, congenital and malformations and acquired pathologic conditions.
Lecture Hours: 16.000, Lab Hours: 32.000
Prerequisite: SON1113L, SON1141C
Course User Fee: \$15.00

SON2171C VASCULAR SONOGRAPHY 2.000 credits

This course is designed to cover aspects of Clinical Vascular Technology. Topics include the pathophysiological levels of disease, clinical presentation and data, hemodynamic of blood flow, anatomy and physiology of the vascular system and anatomical appearance. Prerequisite: SON 2161C.
Lecture Hours: 48.000

SON2400C Echocardiography 1 2.000 credits

An in-depth course designed to present all aspects of clinical cardiovascular ultrasound studies. Topics discussed are: pathophysiological basis of diseases, clinical presentation and clinical data, Doppler and echocardiographic findings in disease, hemodynamic relationships, scanning pitfalls and differential diagnosis.
Lecture Hours: 16.000, Lab Hours: 32.000
Prerequisite: SON1000L, SON1113L

SON2401C Echocardiography 2 2.000 credits

An in-depth course designed to cover all aspects of clinical cardiovascular ultrasound studies. Topics discussed are pathophysiological basis of diseases, clinical presentation and clinical data, Doppler and echocardiographic findings in disease, hemodynamic relationships, scanning pitfalls and differential diagnosis.
Lecture Hours: 16.000, Lab Hours: 32.000
Prerequisite: SON2401C

SON2614C Acoustical Physics and Instrumentation 1 2.000 credits

The course will present a review of fundamental physics and in-depth study of the physical principles of diagnostic ultrasound. Topics discussed include: properties of sound waves, interaction of sound waves with matter, generation of ultrasound and principles of Doppler ultrasound.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON1005L

SON2618C Acoustical Physics and Instrumentation 2 2.000 credits

Physical principles of Ultrasound Instrumentation -A course designed to familiarize the student with the physical principles and modes of operation of diagnostic ultrasound equipment. Subject matter includes: transducers, display systems, component parts of a scanning system, real-time scanners, Doppler equipment, quality control, routine maintenance and recent developments.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON2614C

SON2619C Doppler Principles and Instrumentation 2.000 credits

This course presents a review of fundamental physics and an in-depth study of Doppler Physical Principles of Diagnostic Ultrasound. Topics also include Doppler Instrumentation, equipment, display systems, quality control, and hemodynamics of blood flow.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: SON2614C

Course User Fee: \$15.00

SON2834 Clinic 4 2.000 credits

This is the fourth in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 96.000

Prerequisite: SON1824

SON2844 Clinic 5 3.000 credits

This is the fifth in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 144.000

Prerequisite: SON2834

SON2854 Clinic 6 3.000 credits

This is the last in a series of six (6) clinics in which the student is assigned to a medical facility. The student is afforded hands-on experience in sonography under the supervision of a clinical instructor, sonographer or physician.

Clinical Hours: 144.000

Prerequisite: SON2844

SON2930L Seminar in Sonography 1.000 credits

Students will participate in the various types of continuing education. This may include: society meetings, seminars, conferences and in-services.

Lab Hours: 32.000

SON2934L FILM CRITIQUE 4 1.000 credits

insert

Lab Hours: 32.000

Course User Fee: \$30.00

SON2935L Diagnostic Ultrasound Imaging: Advanced Techniques and Case Analysis 1.000 credits

An extensive laboratory aimed at teaching the student to recognize quality images, anatomy, patient positioning, pathology, and scanning technique errors as well as artifacts. For each class, the student will present a case from their current rotation of the teaching file. The presentation will include all technical and clinical information as well as the final interpretation by the supervising physician.

Lab Hours: 32.000

SOP**SOP2002 Social Psychology 3.000 credits**

Combines a knowledge of psychology and sociology, in an interdisciplinary approach to the study of human interaction. Main themes deal with the nature of attitudes, how attitudes may be changed, the processes of interaction and the nature of group structures.

Lecture Hours: 48.000

SOP2772 Human Sexuality 3.000 credits

This course examines the biological, theoretical, social, psychological, and cultural aspects of human sexuality. Students will learn about sexual anatomy, sexual response cycle, sex and gender development, sexual attraction and relationships, reproductive health issues, and social issues in sexuality.

Lecture Hours: 48.000

SPC

SPC1017 Introduction to Communication 3.000 credits

Students will develop conceptual knowledge and practical skills needed for effective verbal and nonverbal communication in interpersonal, intercultural, small-group, and public communication contexts enabling students to communicate successfully in personal, professional, educational, social, and civic settings. This course fulfills the Gordon Rule requirement. (3 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$7.00

SPC2601 Advanced Public Speaking 3.000 credits

For students who have had a basic course in speech or previous experience in public speaking. The course provides participation in such areas as contest, community and on-campus speaking, and speech criticism. Students receive instruction in audience analysis and rhetorical principles and strategies.

Lecture Hours: 48.000

Prerequisite: SPC2608

SPC2608 Introduction to Public Speaking 3.000 credits

SPC2608 is a course in which students will practice speaking to audiences as well as listening to and critically analyzing oral communication. Through oral and written communication, students will learn communication theory as applied to a variety of communication situations and social interactions. Gordon Rule assigned.

Lecture Hours: 48.000

Prerequisite: SPC2608 Placement by Scholastic Assessment Test (SAT) verbal subtest score; American College Testing (ACT) English subtest score; Computerized Placement Test (CPT) English subtest score; or ENC0025 with a grade of "S."

Course User Fee: \$15.00

SPC2940 PEER TEACH/SPE COMM 3.000 credits

Provides the opportunity for outstanding speech students to advance their skills by functioning as student teachers in speech courses which they have completed successfully. Prerequisite: Permission of the department. (3 hr. lecture)

Lecture Hours: 48.000

SPM**SPM1000 Introduction to Sports Management 3.000 credits**

This course will provide students with the fundamental knowledge of the field of sports management. The course will provide an overview of sports across different levels, key events within the sports industry, the stakeholders involved, and jobs within sports. Students will learn skills and competencies that will lend towards a career in sports management.

Lecture Hours: 48.000

Course User Fee: \$64.65

SPM1160 Introduction to Esports 3.000 credits

This course is designed to provide students with a foundational understanding of the esports industry. Students will begin by learning the history of gaming and esports, transition into an overview of the modern esports industry, and learn how the key stakeholders influence the esports ecosystem. The interdisciplinary nature of esports and the potential career pathways will be outlined. No course prerequisites.

Lecture Hours: 48.000

Course User Fee: \$55.83

SPM2105 Sports Events and Facility Management 3.000 credits

This course examines the different kinds of sporting events and the facilities in which they are hosted. Through the examination of events and facilities, students will learn the basics of event and facility management and apply them in a sports-specific environment. The course will prepare students for event planning, facility and event operations, event management, and other essential competencies for career readiness.

Lecture Hours: 48.000

Course User Fee: \$64.67

SPM2151 The Business of Sports 3.000 credits

This course is designed to provide an overview of the financial, management, and business components of sports. Students will learn how concepts such as ticket pricing, sales, marketing, media, and sponsorship affect amateur and professional sporting events. By examining past events in sports, best business practices will be identified and discussed.

Lecture Hours: 48.000

Course User Fee: \$64.67

SPM2164 The Business of Esports 3.000 credits

This course is designed to introduce students to the fundamentals of business and the sources of revenue within the esports industry. Utilizing past business failures and successes within the industry as case studies, the students will learn the best business practices and how to leverage esports revenue sources for success. Additionally, current, and future business trends within esports will be discussed (Credit hours: 3.00).

Lecture Hours: 48.000

Course User Fee: \$55.83

SPM2167 Esports Event Management 3.000 credits

This course will provide students with the basic concepts of event management. General event management concepts will be applied to esports-specific events to better prepare students to run both online and in-person events. Pertinent event management concepts such as determining the purpose of an event, different event formats, sponsorship, marketing, staffing, and risk management will be covered (Credit hours: 3.00).

Lecture Hours: 48.000

Course User Fee: \$55.83

SPM2201 Sports Ethics and Leadership 3.000 credits

This course will provide students with an understanding of effective leadership and help students begin to develop their own ethical framework. Students will examine ethical issues in sport such as the use of performance-enhancing drugs, cheating, match-fixing, etc. Additionally, students will learn how to apply different leadership styles and theories to develop a cohesive team culture and effectively navigate contemporary ethical issues within sport (Credit hours: 3.00).

Lecture Hours: 48.000

Course User Fee: \$55.83

SPM2950 Esports Management Capstone 3.000 credits

Course Description: This course employs project-based learning to create an opportunity for students to bolster their portfolio. Students will select an area of interest in esports and construct a business plan or other approved project. Key concepts learned in other esports, and business courses will be applied to exhibit the culmination of esports and management knowledge (Credit hours: 3.00).

Lecture Hours: 48.000

Course User Fee: \$55.83

SPN**SPN1120 Elementary Spanish 1 4.000 credits**

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice mid-level in Spanish. Note: Students must pass this course with a C or better to continue to SPN1121.

Lecture Hours: 64.000

Course User Fee: \$15.00

SPN1121 Elementary Spanish 2 4.000 credits

In this course, students will develop appropriate proficiency in interpersonal, interpretive, and presentational modes at the novice high level in Spanish. Note: Students must pass this course with a C or better to continue to SPN2220.

Lecture Hours: 64.000

Prerequisite: SPN1120

Course User Fee: \$15.00

SPN2220 Intermediate Spanish 1 4.000 credits

Students will learn to understand, speak, read, write, and gain cultural awareness of Spanish through a systematic review (using an integrated, multimedia approach) of reading, grammar, and writing skills with emphasis on oral and written communication.

Lecture Hours: 64.000

Prerequisite: SPN1121

Course User Fee: \$15.00

SPN2221 Intermediate Spanish 2 4.000 credits

This is a continuation of Intermediate Spanish 1. Students will learn to understand, speak, read, and write Spanish. Students will also learn to develop cross-cultural awareness through a systematic review of reading and writing skills with emphasis on oral as well as written expression.

Lecture Hours: 64.000

Prerequisite: SPN2220

Course User Fee: \$15.00

SPN2340 SPN/NATIVE SPEAKER 1 3.000 credits

Writing, spelling and punctuation, sentence structure and reading selections for vocabulary expansion as they are relevant to the training of individual students. Prerequisite: oral ability to communicate in Spanish or permission of department chairperson. (3 hr. lecture)

Lecture Hours: 48.000

SPN2341 SPN/NATIVE SPEAKER 2 3.000 credits

A continuation of SPN 2340. Prerequisite: SPN 2340 or equivalent. (3 hr. lecture)

Lecture Hours: 48.000

SPT

**SPT2842 Contrastive Analysis Spanish/English
3.000 credits**

Comparison/Contrastive study of the phonology, morphology and syntax of Spanish and English. Recommended for students of translation and interpretation.

Lecture Hours: 48.000

Course User Fee: \$15.00

**SPT2842 Contrastive Analysis Spanish/English
3.000 credits**

Comparison/Contrastive study of the phonology, morphology and syntax of Spanish and English. Recommended for students of translation and interpretation.

Lecture Hours: 48.000

SPW

SPW2010 SEL READ/SPN LIT 3.000 credits

SPW 2010 explores Peninsular Spanish literature, focusing on Spanish culture, society, and humanistic expressions through various genres such as poetry, essay, novel, drama, and short story. Selected readings from different literary periods from the Middle Ages to the present will be studied. The course will be conducted in Spanish. Oral and written fluency in Spanish and/or permission from faculty or Department Chairperson.

Lecture Hours: 48.000

Recommended Preparation: Oral and written fluency in Spanish and/or permission from faculty or Department Chairperson.

**SPW2010 Introduction to Masterpieces of Spain
3.000 credits**

This course presents major works of Spanish literature from medieval times through the present and includes works from the Western canon. Students will study representative texts that demonstrate the evolution of Spanish literary forms, themes, and artistic achievement. Through critical reading and analysis, students develop skills in literary interpretation while examining works that have significantly influenced Spanish intellectual tradition.

Lecture Hours: 48.000

Recommended Preparation: Oral and written fluency in Spanish and/or permission from faculty or Department Chairperson.

SPW2020 SEL READ/LATN AM LIT 3.000 credits

SPW 2020 explores Latin-American Spanish literature through reading, discussion and written analysis of poetry, drama, and the novel, focusing on Spanish culture, society, and humanistic expressions through the various genres. The course will be conducted in Spanish. Oral and written fluency in Spanish and/or permission from faculty or Department Chairperson.

Lecture Hours: 48.000

Recommended Preparation: Oral and written native fluency in Spanish and/or permission from faculty or Department Chairperson.

**SPW2020 Introduction to Masterpieces of Latin
America 3.000 credits**

This course introduces students to significant works of Latin American literature alongside influential texts that shaped its development and includes works from the Western canon. Students will analyze diverse literary voices and movements that characterize Latin American artistic expression. The course emphasizes critical thinking through examination of texts that reflect the philosophical, historical, and intellectual foundations of Latin American literary tradition.

Lecture Hours: 48.000

Recommended Preparation: Oral and written native fluency in Spanish and/or permission from faculty or Department Chairperson.

STA

STA2023 Statistical Methods 3.000 credits

In this course, students will utilize descriptive and inferential statistical methods in contextual situations, using technology as appropriate. The course is designed to increase problem-solving abilities and data interpretation through practical applications of statistical concepts. This course is appropriate for students in a wide range of disciplines and programs. Student learning outcomes: students will visualize and summarize data using descriptive statistics; students will apply basic probability concepts to draw reasonable conclusions; students will employ concepts of random variables, sampling distributions, and central limit theorem to analyze and interpret representations of data; students will choose an appropriate method of inferential statistics, including confidence intervals and hypothesis testing, to make broader decisions based on sample data; and students will model linear relationships between quantitative variables using correlation and linear regression. Prerequisite: Satisfactory score/grade on an approved placement method. Lecture Hours: 48.000
Recommended Preparation: Completion of MGF1131 prior to registration.
Course User Fee: \$18.00

STA3164 Statistical Methods II 4.000 credits

This course is for students majoring in Data Analytics, Systems Engineering, and related disciplines which require advanced skills in statistical analysis. Students will learn how to perform tests of variance, analysis of variance, analysis of covariance, regression, correlation, and non-parametric statistics. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: STA2023
Course User Fee: \$50.00

STA4210 Regression Analysis 4.000 credits

This course is for students majoring in data analytics, systems engineering, and related disciplines who require advanced in statistical analysis. Students will learn the principles and procedures of correlations and regression analysis and how to allocate information in data sets using statistical software. Lecture Hours: 48.000, Lab Hours: 32.000
Prerequisite: STA3164
Course User Fee: \$50.00

STS0013 Central Sterile Processing Technician 13.660 credits

This course focuses on the application of central sterile processing including proper care, disinfection, assembly, sterilization processes, and sterile storage for instrumentation following surgical procedures. The course will also include the review of basic anatomy, physiology, microbiology, relevant equipment, supplies, and techniques regarding different surgical procedures. Hands-on-experience in the Sterile Processing Department at local hospitals will be included in the course. Clock Hours: 410.000
Course User Fee: \$49.50

STS0019 Central Sterile Service Material Management 5.000 credits

This course is intended to teach the role of the central sterile processing technician in the sterile processing department and related areas will be covered. The course introduces anatomy and physiology to enable the SPD student to better understand each job function that they will perform. As well as microbiology to include the characteristics and activities of microorganisms and various significant aspects of infectious disease that occur in humans and infection control. This course will also cover professional management, communication skills and teamwork, ethical and moral issues in the healthcare setting, and legal issues and risk management. Clock Hours: 150.000

STS1302 Introduction to Surgical Technology 2.000 credits

This course is intended to teach the role of the surgical technologist and central sterile supply technician in the operating room, delivery room and related areas will be covered. The student will learn the basic knowledge of equipment, supplies and instrumentation including the physical environment of the surgical suite. (2 hr. lecture)
Lecture Hours: 32.000
Course User Fee: \$28.65

STS

**STS1303 Fundamentals of Surgical Technology
3.000 credits**

This course introduces the discipline of surgical technology and the role of the surgical technologist in preventing perioperative disease transmission and microbiology to include the characteristics and activities of microorganisms. It surveys the various microbial groups, especially the bacteria, viruses and fungi with emphasis on pathogenic forms. Various significant aspects of infectious disease that occur in humans are also covered. The course will also include an introduction to principles of perioperative care, asepsis/infection control, proper disinfection, assembly, and sterilization processes for instrumentation following surgical procedures. (2 hr. lecture)

Lecture Hours: 48.000

Course User Fee: \$28.65

**STS1304L Operating Room Techniques
Laboratory 3.000 credits**

This course will provide the student with information on operating room furniture, equipment, and supplies used during surgery. The preparation of the operating room, surgical scrub, gowning and gloving, development of the sterile field, patient positioning, surgical skin prep, and draping will be included and will be part of the lab practical exam. Students will be required to simulate a surgical procedure from start to finish. Students must complete this course with a grade of "C" or higher in order to start the clinical rotations.

Lab Hours: 96.000

Course User Fee: \$46.83

**STS1307 Surgical Equipment and
Instrumentation 3.000 credits**

This course prepares the student for the scrub role during surgical procedures, identification of basic surgical instrumentation, passing of the instruments, proper care, reprocessing of instruments following a surgical procedure, and utilizing equipment in the operating room. Students will also learn about surgical wounds, proper tissue handling techniques, wound closure, wound healing, suture material, and stapling devices.

Lecture Hours: 48.000

Course User Fee: \$46.83

**STS1308 Perioperative Patient Care Concepts
2.000 credits**

This course will introduce the student to the needs of the surgical patient, with a focus on the special needs patient care concept, which includes adult, pediatric, geriatric, and bariatric patients. Discussion will also include types of consents, transportation and transfer of the surgical patient, preoperative patient routines, positioning, safety issues, and death and dying will be included. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$28.65

STS1323 Surgical Procedures I 3.000 credits

This course is designed to prepare students for surgical procedures by providing instruction on diagnostic procedures and relevant equipment, supplies, and techniques. The course will also include the review of surgical anatomy, physiology, and pathophysiology in relation to general surgery, endoscopic surgery, gynecological and obstetrical surgery, genitourinary surgery, orthopedic surgery, and ophthalmic surgery.

Lecture Hours: 48.000

Course User Fee: \$46.83

**STS1327L Principles and Practices of Surgical
Technology Laboratory 2.000 credits**

This course is an introduction to surgical instrumentation, to include identification, classifications, selection, passing, proper care, and handling equipment and supplies. The course will include proper care, disinfection, assembly, and sterilization processes for instrumentation following surgical procedures. The decontamination of the operating room will be covered. Hands-on-experience in the Sterile Processing Department at local hospitals will be included in the course.

Lab Hours: 64.000, Clinical Hours: 4.000

Course User Fee: \$46.83

**STS1925C Endoscopy Technician Theory 1 and
Lab 3.000 credits**

Upon completion of the course, the student will have a basic understanding of the care and handling of endoscopic equipment, including storage, testing, decontamination, disinfection and preparation for use. Modules include anatomy and physiology review relevant to endoscopy procedures, essential communication, roles and responsibilities of the endoscopy team, equipment, risk management, safety, reprocessing/infection control, and emergency preparedness.

Lecture Hours: 16.000, Lab Hours: 32.000

Course User Fee: \$39.45

STS1926C Endoscopy Technician Theory 2 and Lab 3.000 credits

This course is designed to prepare students for the role of Gastrointestinal (GI) technician regarding flexible endoscopic procedures, including the specialties of gastroenterology and pulmonology.

Lecture Hours: 16.000, Lab Hours: 32.000

Prerequisite: STS1925C

Course User Fee: \$40.45

STS1931 Surgical Technology Special Topics Seminar 3.000 credits

This course provides a review of all materials covered throughout the five-semester program. Certification examination process are covered.

Employability skills are also covered.

Lecture Hours: 48.000

Course User Fee: \$332.05

STS2179 Surgical Biomedical Fundamentals 3.000 credits

This course teaches the skills necessary to function as a surgical technologist in the operating room, including principles of aseptic technique, and a basic understanding of robotics and their use in the operating room setting. In addition, an understanding of the principles of physics and electricity as it relates to the operating room environment will be covered.

Lecture Hours: 48.000

Course User Fee: \$28.65

STS2324 Surgical Procedures II 3.000 credits

This course is designed to prepare students for specialty surgical procedures including the review of surgical anatomy, physiology, pathophysiology, relevant equipment, supplies, and techniques regarding otorhinolaryngological surgery (ear, nose, and throat/ENT), oral and maxillofacial surgery, plastic and reconstructive surgery, cardiothoracic surgery, peripheral vascular surgery, neurosurgery, transplant and trauma surgery. The depth of coverage is determined by the current edition of the core curriculum for surgical technologists published by the Association of Surgical Technologists.

Lecture Hours: 48.000

Course User Fee: \$46.83

STS2340 Surgical Pharmacology 3.000 credits

This course introduces general pharmacological concepts and principles in the management of patient care. Effective administration of therapeutic drugs, indications, and contraindications are discussed, including effects of medication on body systems, drug classifications and their principle action. Correct drug and dose identification is emphasized, including medications handled by the surgical technologist on the sterile field.

Lecture Hours: 48.000

Course User Fee: \$28.65

STS2360 Professional Skills for the Surgical Technologist 2.000 credits

This course will cover professional management, communication skills and teamwork, ethical and moral issues in the healthcare setting, and legal issues and risk management. (2 hr. lecture)

Lecture Hours: 32.000

Course User Fee: \$46.83

STS2944 Surgical Clinical I 3.000 credits

This course is the first in a series of three clinical courses. The course will introduce the student to the surgical clinical environment, provide the ability to begin utilizing skills acquired during previous lab courses, and implement the principles of aseptic technique while participating in the role of a surgical technologist.

Clinical Hours: 144.000

Course User Fee: \$46.83

STS2945 Surgical Clinical II 3.000 credits

This course is the second in a series of three clinical courses. In this course students will concentrate their clinical experience on circulating and scrubbing in on all phases of specialty surgical cases, assisting in the use of special instruments, procedures, and surgical equipment. The student will also focus on obtaining the required surgical cases in the appropriate role as a surgical technologist. (96 hr. clinical)

Clinical Hours: 144.000

Course User Fee: \$46.83

STS2946 Surgical Clinical III 3.000 credits

This course is the third in a series of three clinical courses. In this course students will concentrate their clinical experience on scrubbing in on general and specialty surgical cases, assisting in the use of special instruments, procedures, and surgical equipment. The student will also focus on obtaining the required surgical cases in the appropriate role as a surgical technologist.

Clinical Hours: 144.000

Course User Fee: \$46.83

SUR

SUR0001 CONSTRUCTION SURVEY 4.000 credits

insert

Lecture Hours: 120.000

SUR1001C Construction Survey 3.000 credits

Practice of surveying as related to the building and construction industry. Includes a combination of classroom instruction and practical field problems with the tape, level and transit.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147

Course User Fee: \$55.00

SUR1101C Surveying 1 4.000 credits

The theories and practices in surveying and the use of the principal types of surveying instruments in horizontal and vertical planes. Problems include the measurement of distance; the use of compass, sextant, transit traverse, stadia, and basic mapping. Field and laboratory practice are required.

Laboratory fee.
Lecture Hours: 48.000, Lab Hours: 32.000

Prerequisite: MAC1114 OR MAC1147

Course User Fee: \$55.00

SUR1202 Surveying 2 4.000 credits

Advanced study in route, land, and mapping surveying to include triangulation, astronomic observations, topographic and photogrammetric mapping. Field demonstrations and surveys performed with many modern types of survey instruments. Laboratory fee.

Lecture Hours: 32.000 - 30.000, Lab Hours: 64.000 - 6.000

Prerequisite: SUR1001C

Course User Fee: \$55.00

SUR2301C Topography and Mapping 4.000 credits

Field and engineering graphic methods of map construction. Mapping principles and their application in producing topographic, land survey, hydrographic, and photographic maps. Field work includes control and detail surveys using transit-tape and transit-stadia, plane table and alidade applications. Measuring areas of straight or irregular boundaries with the planimeter.

Lecture Hours: 96.000

Prerequisite: Permission of Department Chairperson.

SUR2400C Land Surveying 1 3.000 credits

The theory and practice of land surveying, subdivision, filing and recording deeds; United States Government survey of public land; Florida laws governing land surveys, descriptions, coordinate systems and professional licensing. Field surveys are performed.

Lecture Hours: 64.000

Prerequisite: SUR1101C

SUR2404C Land Surveying 2 3.000 credits

Emphasizes the practice of surveying and the writing of legal descriptions. Areas studied include legal principles of retracement under Federal rules, combination of sequence and simultaneous conveyances, locating reversion rights, riparian and littoral owners, state statutes and regulations, standards of practice, field and office guidelines, performing the survey, legal authority and liability of the surveyor.

Lecture Hours: 64.000

Prerequisite: SUR2400C

SUR2462C Subdivisions 3.000 credits

Actual plant construction from tract description to final record plat. Research current plat laws and regulations, meaning of legal parts, necessary computations and graphical representation.

Lecture Hours: 64.000

SYG

SYG2000 Introduction to Sociology 3.000 credits

This course engages in a scientific study of society providing an overview of sociology as a social science. It includes its development as a discipline and methodology. It examines culture as a basis for human behavior, how it is acquired and its norms obeyed. It explores issues of social inequality, including social inequality, "race" and ethnicity, and gender. The class also examines issues of social change and social institutions along issues associated with demographic changes and urbanization.

Lecture Hours: 48.000

SYG2010 Social Problems 3.000 credits

An analysis of the major contemporary and recurring social problems, emphasizing scientific search for variables involved and exploring alternative solutions.

Lecture Hours: 48.000

**SYG2110 INTRODUCTION TO SOCIAL RESEARCH
3.000 credits**

A general introduction to research methodology in the Social Sciences, paying particular attention to research design, data collection and data analysis (3 hr. lecture).

Lecture Hours: 48.000

SYG2230 Multi-Ethnic America 3.000 credits

An introduction to the theory and problems of minority groups in American society. The focus is on structural inequality, institutional discrimination, and the changing patterns of prejudice and discrimination. (3 hr. lecture)

Lecture Hours: 48.000

SYG2430 Marriage and The Family 3.000 credits

The family as a social institution — its origin and development, its forms and functions, its interrelation with other social institutions, and its role in contemporary civilization. Areas of study include factors contributing to or acting against successful, stable marriage.

Lecture Hours: 48.000

TAX**TAX2000 Income Tax 3.000 credits**

Federal income tax fundamentals with emphasis on individual returns. Topics considered include gross income, capital gains and losses, deductions and exemptions, and tax credits.

Lecture Hours: 48.000

Course User Fee: \$15.00

**TAX2002 Taxation Practices and Procedures
3.000 credits**

This course will cover Internal Revenue Service taxation practices and procedures. Topics covered will convey knowledge of IRS rules and penalties, rules for representing taxpayers before the IRS and in the courts, rules and requirements associated with the tax return filing process and records maintenance rules and basic tax research skills.

Lecture Hours: 48.000

Course User Fee: \$22.00

TAX2010 Business Taxes & Returns 3.000 credits

A practical course on the various tax reports and forms required in an accounting office. Topics include payroll deposits, payroll returns, corporate tax return, annual report, tangible and intangible tax returns, sales taxes, employment forms and licenses.

Lecture Hours: 48.000

Course User Fee: \$20.00

**TAX2021 Taxation of Business Organizations
3.000 credits**

This course will cover federal income taxation of corporations, S Corporations, limited liability companies and partnerships. Topics covered include determining the tax consequences of income, expenses, distributions, redemptions and liquidations for business entities. Practical application of the tax law will be emphasized along with analysis of tax procedures.

Lecture Hours: 48.000

Course User Fee: \$22.00

TAX2401 Tax of Estates, Gifts and Trusts 3.000 credits

This course covers definitions and operations of various fiduciary forms of wealth transfer including but not limited to fiduciary accounting principles and concepts; record keeping requirements; and various tax reporting requirements, forms and calculations.

Lecture Hours: 48.000

Course User Fee: \$22.00

TAX4001 Federal Income Tax I 3.000 credits

A survey of federal income tax with emphasis of taxation of individuals and the ethics of income tax accounting. Students will gain the basic theory, concepts, practice, and methods of determining taxable income and tax liabilities. Prerequisite: ACG3103.

Lecture Hours: 48.000

Course User Fee: \$98.00

TAX4011 Federal Income Tax II 3.000 credits

A survey of federal income tax with emphasis of taxation of business entities and the ethics of income tax accounting, student of the basic theory, concepts, practice and methods of determining the taxable income and tax liabilities. Prerequisite: TAX4001.

Lecture Hours: 48.000

Prerequisite: TAX4001

Course User Fee: \$98.00

**TCN0300 Telecommunications Fundamentals
3.830 credits**

This course introduces students to basic concepts in telecommunications, including safety practices, electrical fundamentals, and industry standards. Students will learn about telecommunications systems architecture, cable types, and basic troubleshooting techniques. The course also covers customer service skills and professional conduct in the telecommunications industry.

Course User Fee: \$96.92

TCN0310 Network Cabling and Installation 3.830 credits

This course provides a comprehensive overview of network cabling and installation techniques essential for building robust telecommunications infrastructures. Students will explore various types of cabling, including twisted pair, coaxial, and fiber optic, while gaining hands-on experience in installation practices, cable termination, and testing procedures.

Course User Fee: \$96.92

TCN0312 Fiber Optics and Splicing Techniques 3.830 credits

This hands-on course provides students with a comprehensive understanding of fiber optic technology and essential splicing techniques crucial for modern telecommunications networks. Students will explore the fundamental principles of fiber optic transmission, including light propagation, cable types, and optical components.

Course User Fee: \$96.92

TCN0320 Advanced Telecommunications Systems 3.840 credits

This comprehensive course delves into cutting-edge telecommunications technologies and advanced network architectures. Students will explore the intricacies of modern communication systems, including 5G networks, Internet of Things (IoT) infrastructures, and software-defined networking (SDN).

Clock Hours: 115.000

Course User Fee: \$96.92

THE**THE2000 Theatre Appreciation 3.000 credits**

In this course, students will explore dramatic structure, techniques, and various organizational elements. The course provides an introduction to theatre as a collaborative art form through the critical analysis of its historical context, production, theory, and connections to theatrical literature, including the Western Canon. Student learning outcomes: students will identify the basic principles of theatrical performance, design, technology, organization, and management; students will assess the significance of the human condition as expressed through the performing arts; students will explore and interpret works of art utilizing creative and critical thinking skills; students will demonstrate college-level writing; and students will define, compare and contrast theater as both an expressive art form and a commercial industry. Writing Intensive Course.

Lecture Hours: 48.000

THE2083 THEATRE PROBLEMS 1.000 - 3.000 credits

This is an advanced course for theatre majors who have already earned credit in a required subject or who have demonstrated that they are capable of an advanced, highly specialized work in a particular area of requirements and objectives. Possible areas of study include advanced scene work; intensive training in particular acting methods; playwriting; and directing. Students are assigned to a teacher, who will design, supervise, and evaluate their projects. May be repeated for credit. (2-6 hr. lab)
Lecture Hours: 16.000 - 48.000

TPA**TPA1200 STAGECRAFT 3.000 credits**

A basic study of technical theater practices with emphasis on scenery construction, rigging, and prop construction. This course may be taken concurrently with DRA 122.

Lecture Hours: 32.000, Lab Hours: 32.000

TPA1215 AUDIO-VIS MULTI-MEDIA 4.000 credits

This course presents the principles and practice of unpacking, unloading, setting up and operating visual aids for conference and convention, and A/V for industrial shows, conventions, concerts and special events. Also covered is media using recorded sound (A/V) and media accompanying live presentations (V/A) including 8, 16, 35 and 70 mm. movie, single and multi-media. Students will practice this technology in labs and in performance environment, under performance conditions. Prerequisite: Permission of department. (2 hr. lecture; 4 hr. Lab)

Lecture Hours: 96.000

TPA1225 AUTOMATON & COMPUTER 3.000 credits

This course presents the principles and practices of automated robotics lighting (intelligent lighting), automated machinery, rigging, wagons, turntables, lifts, event sequencing between pyro, multi-media, sound and stage lights, automated show control of up to ninety-nine elements of production and computerize control of light and sound. Prerequisite: Permission of department. (2 hr. lecture; 2 hr. lab)

Lecture Hours: 64.000

TPA1248 Makeup for the Stage 3.000 credits

An introduction to the art and techniques of makeup as used by the actor, theatrical designer, and technician. Special emphasis is given to straight makeup, age makeup, hair, character extension, and stylization.

Lecture Hours: 48.000

TPA1253 ENTRNMNT TECH: TEC 1 3.000 credits

This course presents the principles and practice of stage rigging, stage carpentry, road crew and gripping. Students will practice the use of hardware, knots, hemp, counter-weight and motorized flying system for scenery, curtains and ground rigging, temporary and permanent stages, sound stages or on location, expositions and/or special outdoor events. Also covered are the principles and practices of the installation and operation of wagons, winches, chain hoists and trusses, lighting equipment, sound for on-stage or studio performance, gripping for motion pictures or video production. Occupational health and safety issues are discussed and practiced. (2 hr. lecture; 2 hr. lab)
Lecture Hours: 64.000

TPA1254 ENTRNMNT TECH:TECH2 3.000 credits

This course is an advanced course in entertainment technology and continuation of the principles and practices covered in Entertainment Technology Technician 1. (2 hr. lecture; 2 hr. lab)
Lecture Hours: 64.000

TPA1255 CNCRT STUDIO # STG L 4.000 credits

Lecture Hours: 96.000

TPA1260 CONCRT & STAGE SOUND 4.000 credits

This course presents the principles and practices of the installation and strike of sound technology for a variety of entertainment venues: theater, dance and opera, rock and roll concert tours, orchestras, choirs and music festivals, theme parks, themed entertainment and industrial shows, special events, casino and cruise line shows. Special attention is paid to venues for performances outdoors, indoors and on sound stages. Also covered are the principles and practices associated with the installation and operation of film studio, location sound and sound studio set-up technology.
Lecture Hours: 32.000, Lab Hours: 64.000

TPA1274 PROP PRAC NON-ELECTR 3.000 credits

This course provides the student: the principles and practice of unloading, installing, presetting operating, striking, storing, loading and packing properties, practicals, physical effects and set dressing; the preparation, care and handling and cleanup of food used during a live performance and filming; the preparation, preset/strike, organization and storage of properties before, during and after performance and film shots. Students learn how to take instructions from management, designers and decorators for the installation and running for furniture, dressing and effects and executing cues for their movement and operation. Prerequisite: Permission of department. (2 hr. lecture; 2 hr. lab)
Lecture Hours: 64.000

TPA1275 SPC EFF-ELEC LASER&P 3.000 credits

This course presents the principles and practice of operating scenic, mechanical, sound, and lighting special effects including laser light and pyrotechnics. Also covered are the standard practices, rules, regulations, procedures, guidelines and precautions for the safe operation of currently available devices used in industry today and those invented or special events. Prerequisite: Permission of department. (2 hr. lecture; 2 hr. lab)
Lecture Hours: 64.000

TPA2233 MAINSTAGE COST/MAKE 1.000 credits

Practical experience in theatrical costuming and makeup through participation in a major theatrical production. May be repeated for credit. Prerequisite: Permission of department chairperson. (2 hr. lab)
Lecture Hours: 32.000

TPA2276 ENTERTN TECH:CRAFT 1 3.000 credits

This course presents the principles and practices of woodworking, welding, smithing, casting, weaving, paperhanging, painting, ceramics, plaster, sewing and plastics technology for the entertainment industry. State of the art tool technology, shop and field practice, health and safety standards will be emphasized. These crafts are entertainment industry oriented with a perspective that states that objects created are to be used for production. Prerequisite: TPA 1213 or department permission. (2 hr. lecture; 2 hr. lab)
Lecture Hours: 64.000

TPA2277 CRAFTS 2 3.000 credits

xxx

Lecture Hours: 64.000

TPA2292 PRODUCTION LAB 1.000 - 3.000 credits

Lab Hours: 96.000

TPA2600 INTRO STAGE MGNT 3.000 credits

Lecture Hours: 48.000

TPP**TPP1100 ACTING 1 3.000 credits**

Lecture Hours: 48.000

TPP1110 BASIC PRIN ACTING 3.000 credits

XXX

Lecture Hours: 48.000

TPP1120 Improvisation Ensemble 3.000 credits

The student will develop the skills of improvisation for use in role development and for performance.
Lecture Hours: 48.000

TPP1150 Scene Study 1 1.000 - 3.000 credits

This course teaches the aspiring young theatre professional how to analyze a play in terms of the author's personal statement, the historical and social context within which it was written, the particular style used by the author, and the many options open to director and actor for bringing the work to stage life. A substantial portion of class time will be devoted to oral reading and interpretation of text.

Lecture Hours: 16.000 - 48.000

TPP1160 VOICE & MOVEMENT 1 3.000 credits

XXX

Lecture Hours: 48.000

TPP1161 VOICE & MOVEMENT 1 3.000 credits

XXX

Lecture Hours: 48.000

TPP1170 BEG CHARACTERIZATION 3.000 credits

A course which builds upon the centered foundation of creating a role developed in TPP 1100 and TPP 1110. The student uses a subjective approach to creating a character which differs from him/her physically, culturally and psychologically. He/she att empts ever greater degrees of transformation. Prer equisite: TPP 1110. (3 hr. lecture)

Lecture Hours: 48.000

TPP1190 STUDIO THEATRE-CAST 1.000 credits

XXX

Lab Hours: 32.000

TPP1250 MUSICAL THEATRE 2.000 - 3.000 credits

XXX

Lecture Hours: 80.000

TPP1606 PLAYWRITING 1/2 3.000 credits

The process of exploring playwriting styles and techniques is continued. A one-act play of significant length and complexity will be the semester project. (3 hr. lecture)

Lecture Hours: 48.000

TPP1700 Voice for the Stage 3.000 credits

The study and application of voice production, breathing, articulation, accents and movement in the actor's delivery. Emphasis is on clarity, precision, properly phrased and meaningful communication from the performer to the audience.

Lecture Hours: 48.000

TPP2111 ACTING 2 3.000 credits

XXX

Lecture Hours: 48.000

TPP2112 Acting 2 3.000 credits

Emphasis on building a characterization. The art of improvisation, with reference to its function in the preparation of a role, is included. Prerequisite: TPP1110.

Lecture Hours: 48.000

TPP2151 Advanced Scene Study 3.000 credits

In this course the theatre student learns to analyze plays with a heavy focus on particular characters and major scenes. Emphasis will be placed on works of prime importance in the history of the theatre, both past and present, so that the aspiring actor can begin to experience some of the problems involved in approaching a significant role. Each student is required to research the performance history of the roles and scenes studied as well as to uncover the subtexts and the inner line of character development. Attention will be given to both Stanislavsky and improvisation techniques as methods by which the actor comes closer to the full reality of a part. May be repeated for credit.

Lecture Hours: 48.000

TPP2152 Scene Study 3 3.000 credits

This course is the culmination of a sequence. In it the advanced acting student learns how to analyze the longer one-act or shorter full-length play and to develop the through-line of one character as a preparation for an in-class performance. The student also learns how to work with the director and to relate acquired acting techniques to the stylistic requirements of a given script.

Lecture Hours: 48.000

TPP2162 Voice & Movement 2 3.000 credits

An intense two-semester course in precision techniques of voice production and bodily flexibility integrating them with specific acting exercises with an emphasis on demonstrating the automatic, non-conscious application of acquired voice and movement skills. Prerequisite: TPP1161.

Lecture Hours: 48.000

TPP2163 Voice & Movement 2 Continuation 3.000 credits

Continuation of TPP2162. Prerequisite: TPP2162

Lecture Hours: 48.000

TPP2191 Mainstage Production - Cast 1.000 - 3.000 credits

Participation in a major theatrical production as a member of the cast. Main stage productions will be presented publicly to the student body and community. May be repeated for credit.

Prerequisite: Permission of department chairperson.

Lab Hours: 32.000 - 96.000

TPP2256 Musical Theatre 2 3.000 credits

A continuation of TPP1250 in which the student is expected to develop further the performing skills of singing, dancing, and acting.

Lecture Hours: 48.000

TPP2260 Acting for the Camera 1 3.000 credits

Acting students will attend lecture/lab to acquire the technical knowledge and training necessary for acting in the film and television industry. They will acquire a knowledge of the working procedure and terminology used in these media.

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: TPP1100or Departmental Permission.

TPP2300 Introduction to Play Directing 3.000 credits

Introduction to the basics of play directing, composition, picturization, business and movement. The course will offer the student a method of analysis and rehearsal scheduling. Prerequisites: TPP1110, TPA1200.

Lecture Hours: 48.000

TRA**TRA0080 Tractor Trailer Truck Driver 10.130 credits**

The Tractor Trailer Truck Driver course prepares students for entry into the trucking and logistics industry. Students explore career opportunities and requirements of a professional tractor trailer driver. Students study vehicle safety, accident prevention, operating regulations, cargo handling, documentation procedures, pre-trip preparation, vehicle inspection, maintenance, service, control procedures, backing, coupling, uncoupling, maneuvering, road and hazardous driving skills, and licensing requirements. Pre/Corequisite: Must have a valid Florida Driver's License and good driving record; Must have Commercial Learner's Permit; Must pass DOT Health Examination; Must pass DOT Drug Test.

Clock Hours: 320.000

Pre/Corequisite: Must have a valid Florida Driver's License and good driving record; Must have Commercial Learner's Permit; Must pass DOT Health Examination; Must pass DOT Drug Test.

TRA0080L On-the-Road Practice and Driving Skills - Class A 0.530 credits

This course focuses on on-the-road practice, honing driving skills crucial for Tractor Trailer Truck Drivers. Students will engage in practical exercises involving backing, coupling, uncoupling, maneuvering, and road driving. Special attention will be given to hazardous driving skills, ensuring students are adept at handling various challenging situations on the road. This is a hands-on training course conducted on the road, providing practical experience for students pursuing a Class 'A' Commercial Driver's License.

Clock Hours: 16.000

TRA0084 Truck Driver Heavy Florida Class "B" 4.600 credits

The Truck Driver Heavy Florida Class "B" course prepares students for entry into the trucking and logistics industry. Students explore career opportunities and requirements of a professional class "B" truck driver. Students study vehicle safety, accident prevention, operating regulations, cargo handling, documentation procedures, pre-trip preparation, vehicle inspection, maintenance, service, control procedures, backing, maneuvering, road and hazardous driving skills, and licensing requirements. Prerequisites/Corequisite Information: Must have a valid Florida Driver's License and good driving record, must have Commercial Learner's Permit, must pass DOT Health Examination, must pass DOT Drug Test.

Clock Hours: 138.000

Pre/Corequisite: Prerequisites/Corequisite Information: • Must have a valid Florida Driver's License and good driving record • Must have a Commercial Learner's Permit • Must pass DOT Health Examination • Must pass DOT Drug Test

TRA0084L On-the-Road Practice and Driving Skills - Class B 0.400 credits

This course focuses on on-the-road practice for the Truck Driver Heavy Florida Class "B" License. The course prepares students to study vehicle safety, accident prevention, operating regulations, cargo handling, documentation procedures, pre-trip preparation, vehicle inspection, maintenance, service, control procedures, backing, maneuvering, road and hazardous driving skills, and licensing requirements. Prerequisites/Co-requisite Information: Must have a valid Florida Driver's License and good driving record, must have Commercial Learner's Permit, must pass DOT Health Examination, must pass DOT Drug Test.

Clock Hours: 12.000

TRA0701 TRNSPRT/GEOGRPH CONS 1.000 credits

This course will be removed.

Lecture Hours: 30.000

TRA1154 Introduction to Supply Chain Management 3.000 credits

This course is an introduction to the concepts, principles and techniques in the field of supply chain management (SCM) with particular emphasis on the economic significance of distribution to business and the U.S. economy. Students will learn the interrelationship between logistics and other areas of business, noting how the SCM pipeline can significantly impact customer loyalty by adding value.

Lecture Hours: 48.000

TRA1410 Introduction to Rail Freight Operations 3.000 credits

This is an introductory course in Rail Freight Operations. Students will learn the advantages and disadvantages of freight movement by rail, how rail lines are organized and operate, including the use of intermodal transfer facilities and on-dock rail. Topics include shipping documents, shipment tracking, management of human resources and equipment, and an overview of hazardous materials shipments and security issues. Corequisites: TRA1420, TRA1430.

Lecture Hours: 48.000

Corequisite: TRA1430 AND TRA1420

TRA1420 Introduction to Trucking Operations 3.000 credits

This is an introductory course in Trucking Operations and the movement of goods via highways and roadways. Students will learn U.S. Department of Transportation requirements, documents for shipping, vehicle and shipment tracking, scheduling, management of human resources and equipment, just-in-time implications and integration with other transportation modes. Hazardous materials shipments and security issues will be discussed. Corequisites: TRA1410, TRA1430.

Lecture Hours: 48.000

Corequisite: TRA1430 AND TRA1410

TRA1430 Introduction to Port Freight Operations 3.000 credits

This is an introductory course in Port Freight Operations. Students will learn how Seaports and Inland Ports are organized and operate, how freight is moved domestically and internationally, including the integration of port operations with other modes of transportation. Topics include break-bulk handling during loading, discharging, in-transit carriage, on-dock rail, harbor drayage, equipment and cargo management, and an overview of hazardous materials shipments and security. Corequisites: TRA1410, TRA1420.

Lecture Hours: 48.000

Corequisite: TRA1410 AND TRA1420

TRA2010 Introduction to Transportation and Logistics 3.000 credits

This course surveys the organization and operations of the commercial transportation industry and its impact on the bottom line of today's modern businesses. Students will learn to review regulations and processes affecting transportation and logistics functions as well as explore the industry job market and look at technologies and current issues shaping transportation and logistics. A.S. degree only.

Lecture Hours: 48.000

TRA2156 Operations Management for Transportation 3.000 credits

This course includes the skills necessary for a supervisory role in logistics. Students will learn the roles and responsibilities in managing different types of transportation operations. Topics include human resources, design and management of production operations, productivity, capacity planning, resource management, just-in-time systems, hazardous materials management, planning and project management. Prerequisite: MAN2021.

Lecture Hours: 48.000

Prerequisite: MAN2021

TRA2321 Transportation Public Policy, Law, and Regulations 3.000 credits

Students will learn the transportation regulatory environment, including the various levels of government regulations. Review of security, environmental requirements, regulatory research and labor laws are also covered. Prerequisite: TRA 2010.

Lecture Hours: 48.000

Prerequisite: TRA2010

TRA2402 Intermodal Transportation Operations and Project Management 3.000 credits

Students will learn the fundamental elements necessary to plan, implement and control efficient and market-responsive integrated transportation systems. Topics include strategic, operational, and project management roles of transportation in supply chains. Emphasis is placed on services pricing, carrier selection, equipment and shipment planning, intermodal operations, financial/budgetary constraints, security and distribution services. Prerequisite: AVM2120, TRA2010; Corequisites: TRA1410, TRA1420, TRA1430.

Lecture Hours: 48.000

Prerequisite: AVM2120 AND TRA2010, Corequisite: TRA1430 AND TRA1410 AND TRA1420

TRA2702 International Logistics and Transportation 3.000 credits

International logistics concerns the flow of materials into, through and out of an international corporation as it relates to materials management, storage, inventory locations, physical distribution and documentation. This course will emphasize international transportation infrastructure and modes such as ocean, airfreight, intermodal movement, truck and rail. Choices among these modes will be explored considering such factors as transit time, packaging, risks, predictability and cost. The roles of freight forwarders and custom brokers in moving international cargo and operation of foreign trade zones will be discussed. Prerequisites: TRA2010, TRA2321.

Lecture Hours: 48.000

Prerequisite: TRA2010, TRA2321

TRA2945 Transportation & Logistics Capstone I 1.000 credits

In this Transportation & Logistics Capstone course, students will learn to incorporate the major concepts presented in the transportation courses through the application of special projects, internship and/or examination. The course will assess the student's understanding of major concepts in transportation and logistics.

Prerequisite: Departmental Approval and Completion of the Program Core Requirements.

Lecture Hours: 16.000

Prerequisite: Departmental Approval & Completion of the Program Core Requirements

TRA3034 Transportation and Traffic Management 3.000 credits

This course covers developments leading to national and federal regulations, division of territories, official descriptions, etc. Students will learn the scope of authority of territorial associations, factors controlling traffic flows, basic governing classification rules, principals of freight rates and tariffs, and elements of rate making.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA3132 Purchasing and Inventory Management 3.000 credits

This course provides a comprehensive introduction to the purchasing/procurement and supply chain management field. Students will learn purchasing and supply chain issues in a variety of settings, from process industries to high-tech manufacturing and services as well as public institutions. Emphasis is on the purchasing process as it relates to such topics as inventory control procedures, price/cost analysis, laws and ethics, negotiations, vendor selection and the development of vendor relationships.

Prerequisites: MAN2021, TRA1154.

Lecture Hours: 48.000

Prerequisite: MAN2021TRA1154

TRA4234 Warehouse Management 3.000 credits

Students will learn warehousing functions, facility operations, and operational productivity improvements and measurements with the inclusion of concepts from marketing, finance, statistics, operations management, and human resources. The course presents an integrated business approach to the detailed operational aspects of logistics facilities such as warehouse and distribution centers. The course will use real-world warehouse data to design the layout and operations requirements for a local warehousing facility and to identify the operational equipment used in warehousing.

Prerequisites: MAN2021, MAN3065, QMB2100.

Lecture Hours: 48.000

Prerequisite: MAN2021 AND MAN3065 AND QMB2100

TSL**TSL1084 Introduction to Esol Principles and Practices 3.000 credits**

Course will be removed.

Lab Hours: 48.000, Clinical Hours: 1.000 - 30.000

TSL2082 Introduction to Teaching English to Speakers of Other Languages (TESOL) 3.000 credits

In this introductory course, the student will understand the process of acquiring a second language, and the social-emotional impact it has on the learner. The student will examine the laws and policies in place to support English language learners (ELLs), as well as research-based strategies to facilitate language development, literacy skills and content knowledge.

Lecture Hours: 48.000

TSL3080 ESOL in Early Childhood Education I 3.000 credits

The student will explore theories, research, and practices of English learners, including legal issues that have influenced the field. The student will analyze theories of first and second language acquisition, and concepts of global citizenship. (Ten hours of clinical experience requires in an approved kindergarten-third grade least restrictive environment with ESOL students.)

Lecture Hours: 48.000

Pre/Corequisite: EEC3301

Course User Fee: \$18.00

TSL3240 Applied Linguistics 3.000 credits

Duplicate course. It will be removed from the catalog.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval

TSL3240 Applied Linguistics 3.000 credits

This course examines applied linguistics and its practical application to instruction for English Language Learners, aligned with the 2025 Florida Teacher Standards for English for Speakers of Other Languages (ESOL) Endorsement. Students will explore the structure of language: phonology, morphology, syntax, semantics, and pragmatics; and analyze major theories of second language acquisition. Instruction places emphasis on language development across listening, speaking, reading, and writing, with attention to academic language and literacy growth. Students will apply linguistic knowledge to design linguistically responsive lessons, differentiate instruction, and support access to grade-level content. By the end of the course, students will be prepared to implement research-based ESOL practices that promote language proficiency and academic success for English Language Learners (ELLs). A minimum of 15 hours of structured field experience is required to provide practical application of course concepts in classroom settings. This three-credit lecture course is required for the Florida add-on ESOL Endorsement.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval

TSL3243 ESOL I: Second Language Acquisition and Communication 3.000 credits

The teacher candidate will critically examine and apply theoretical frameworks related to first and second language acquisition, literacy development, and the structural features of language; evaluate the impact of varied communication styles on student learning; and analyze legal and policy considerations pertinent to the education of English language learners (ELLs). Fifteen hours of clinical experience in a classroom with ELLs are required. Fifteen hours of clinical experience are required.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321, EEX3120

TSL3520C Cultural Dimensions of ESOL 3.000 credits

Duplicate course. It will be removed.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval.

TSL3520C Communication and Understanding for English Language Learners (ELLs) 3.000 credits

This course offers a comprehensive exploration of communication and understanding of appropriate language, roles, rights, and responsibilities of U.S. citizens within the United States, particularly those found in Florida. Through critical analysis and reflection, students will deepen their understanding of the intricate relationship between language, learning and responsibility of civic engagement. The course is designed to equip future educators with the knowledge and skills necessary to effectively support English Language Learners. Aligned with the 2025 Florida Teacher Standards for English for Speakers of Other Languages (ESOL) Endorsement, this course emphasizes practices that promote meaningful engagement in the U.S. educational settings. A minimum of 15 hours of structured field experience is required to provide practical application of course concepts in classroom settings. This three-credit lecture course is required for the Florida Add-on ESOL Endorsement.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval.

TSL4140C TESOL Curriculum & Materials 3.000 credits

Duplicate. It will be removed from catalog.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval.

TSL4140C ESOL Curriculum and Materials Development 3.000 credits

This course provides comprehensive knowledge and application of TESOL theories, principles, and current research in the analysis, planning, design, and evaluation of curriculum and instructional materials for English Language Learners (ELLs). Aligned with the new 2025 Florida Teacher Standards for English for Speakers of Other Languages (ESOL) Endorsement, students will learn to design standards-aligned, linguistically appropriate lessons that integrate English language acquisition with grade-level academic content. Emphasis is placed on differentiating instruction to meet the needs of ELLs at varying English proficiency levels, including students with interrupted or limited formal education. Students will examine and utilize standards-aligned resources, student language proficiency data, and evidence-based practices to select and adapt materials that promote access to rigorous content. The course also focuses on the effective integration of instructional technologies and digital tools to enhance academic language development and content mastery. A minimum of 15 structured hours of field experience is required to provide practical application of curriculum planning and materials adaptation in classroom settings. This three-credit lecture course is required for the Florida add-on ESOL Endorsement.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval.

TSL4310 ESOL in Early Childhood Education II 3.000 credits

In this second TESOL course, the student will apply theories, principles, and current research on curriculum, instructional methods, and assessment for English learners in early childhood settings. The student will select modifications appropriate for language and content area teaching and learning. (Ten (10) hours of clinical experience in an approved first through third grade setting with ESOL students.)

Lecture Hours: 48.000

Prerequisite: TSL3080 and 10 hrs. of Field Experience Required

Course User Fee: \$18.00

TSL4311 ESOL II: Methods of Teaching and Assessing ESOL Students 3.000 credits

The teacher candidate will apply TESOL instructional methods and strategies, as well as analyze, plan, design, and evaluate curriculum and materials. The teacher candidate will select, develop, and adapt assessment instruments and examine standardized ESOL measurement tools. Fifteen hours of clinical experience with English language learners are required.

Lecture Hours: 48.000

Prerequisite: TSL3243

TSL4324C ESOL Strategies for Content Area Teachers 3.000 credits

The student will learn topics related to teaching content area subjects to English Language Learners. The student will plan and implement curriculum, instruction, and assessment activities to meet the needs of culturally and linguistically diverse learners. Fifteen hours of clinical experience are required.

Pre/Corequisite: EDG3321.

Lecture Hours: 48.000

Pre/Corequisite: EDG3321

TSL4340C TESOL Methods 3.000 credits

This iteration will be removed from catalog.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval

TSL4340C Methods of Teaching ESOL 3.000 credits

This course provides in-depth knowledge and application of TESOL theories, principles, and research to prepare students to implement effective, standards-based instruction for English Language Learners (ELLs). Aligned to the 2025 Florida Teacher Standards for English for Speakers of Other Languages (ESOL) Endorsement, students will examine legal requirements for identification, instruction, and progress monitoring of ELLs, and apply evidence-based practices in English language acquisition. Content in this course will emphasize planning and delivering instruction that supports listening, speaking, reading, and writing development across varying English proficiency levels while ensuring access to grade-level academic content. Students will design and implement standard-aligned lessons that integrate Florida's English Language Development (ELD) Standards with academic content objectives. This course focuses on differentiated instructional strategies, structured academic talk, literacy development, scaffolding techniques, and data-informed decision-making to promote both language proficiency and academic achievement. A minimum of 15 hours of structured field experience is required to provide practical application of course concepts in classroom settings. This three-credit lecture course is required for the Florida add-on ESOL Endorsement.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval

TSL4441C ESOL Testing & Evaluation 3.000 credits

This iteration will be removed from catalog.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval

TSL4441C Testing and Evaluation of ESOL 3.000 credits

This course provides knowledge and practical application of principles, and current research related to the testing and evaluation of English Language Learners (ELLs) in K-12 setting. The course is aligned to the 2025 Florida Teacher Standards for English for Speakers of Other Languages (ESOL) Endorsement with an emphasis placed on the selection, development, adaptation, and interpretation of linguistically appropriate assessment instruments, including state and classroom-based assessments. Students will examine formative, summative, diagnostic, and performance-based assessments and learn to distinguish between language proficiency and academic achievement to support appropriate evaluation and instructional decision-making. This course will also address the legal, ethical and professional responsibilities related to testing and evaluation of ELLs, including appropriate accommodation, achievement, and learning gains ELL students, and effective communication of assessment results. A minimum of 15 structured hours of supervised field experience is required, during which students will apply assessment practices with ELL students and analyze authentic data to inform instruction. This three-credit lecture course is required for the Florida add-on ESOL Endorsement.

Lecture Hours: 48.000

Recommended Preparation: School of Education Approval

VIC**VIC1000 Visual Communications 3.000 credits**

An introductory course in which students will learn the visual aspects of film, video, photography and graphic arts, specifically dealing with design elements and principles.

Lecture Hours: 32.000, Lab Hours: 32.000

Course User Fee: \$10.00

VIC1205C Video Compositing and Motion Graphics 1 3.000 credits

This course is an introduction to visual effects for film and television. The student will learn basic level techniques of still and motion graphic design in visual effect compositing for film and video using Photoshop and After Effects. Prerequisite: FIL 2552C. Laboratory fee. (2 hr. lecture; 2 hr. lab)

Lecture Hours: 32.000, Lab Hours: 32.000

Prerequisite: FIL2552C

WOH**WOH2003 History of Genocide 3.000 credits**

This course includes works from the Western canon and is designed to explore the history, causes, and consequences of genocide by identifying patterns of intentional mass killings. Students will examine case studies to understand the characteristics of both historical and contemporary incidents classified as genocide. Through this analysis, learners will gain insight into the social and behavioral science concepts that inform understanding of tragic events and their implications for society today.

Lecture Hours: 48.000

WOH2012 History of World Civilization to 1789 3.000 credits

This course covers the history of World Civilizations from the prehistoric period to the 18th century. Students will learn of the major political, social, economic, and cultural structures of civilizations and their development through 1789.

Lecture Hours: 48.000

WOH2022 History of World Civilization From 1789 3.000 credits

The student will examine the historical development of world civilizations since 1789. Students will learn of historical processes and developments in social, cultural, political, and economic contexts since the 18th century.

Lecture Hours: 48.000

ZOO**ZOO3021 Survey of Animal Diversity 3.000 credits**

This course presents zoology as a scientific discipline. Students will learn the basic principles of zoological nomenclature, taxonomy, systematic, and the basic understanding of the relationships of animals to one another, to humans, their environment and to society.

Lecture Hours: 48.000

Prerequisite: BSC2011, BSC2011L

Course User Fee: \$10.00

ZOO3021L Survey of Animal Diversity**Laboratory 1.000 credits**

This laboratory course provides hands-on experience with the concepts covered in the lecture ZOO3021. Students will learn the basic principles of zoological nomenclature, taxonomy, and systematic; and the basic understanding of the relationships of animals to other organisms and to one another.

Prerequisites: BSC2010, 2010L, 2011, 2011L;

Corequisite: ZOO3021

Lab Hours: 32.000

Prerequisite: BSC2010, BSC2010L, BSC2011, BSC2011L, Corequisite: ZOO3021

Course User Fee: \$40.00

ACADEMIC PROGRAMS – PROGRAM SHEETS AND CURRICULUM GUIDES

Baccalaureate Degrees (BAS, BS)

Applied Artificial Intelligence	Leadership and Management Innovation - Digital Marketing
Biological Sciences - Biopharmaceutical Sciences	Leadership and Management Innovation - Hospitality Management
Biological Sciences - Biotechnology	Leadership and Management Innovation - Human Resource Management
Clinical Laboratory Science	Leadership and Management Innovation
Cybersecurity	Leadership and Management Innovation - Accounting
Data Analytics	Nursing (RN to BSN)
Dental Hygiene	Public Safety Management - Crime Scene Investigation
Early Childhood Education	Public Safety Management - Criminal Justice
Electrical and Computer Engineering Technology	Sciences in Supply Chain Management
Exceptional Student Education (K-12)	Secondary Mathematics Education
Film, Television & Digital Production	Secondary Science Education - Biology
Health Science - Histotechnology	Supply Chain Management – Procurement Management
Health Science - Physician Assistant Studies	Supply Chain Management – Project Management
Information Systems Technology - Networking	Supply Chain Management – Supply Chain Analytics
Information Systems Technology - Software Engineering	

Certificate of Professional Preparation (CPP)

Cyber Defense	Instructional Design and Technology
Educator Preparation Institute	Networking
Future-Ready Teaching: AI Applications in the Classroom	Reading Endorsement
	Software Engineering
	Vulnerability Assessment

Associate in Arts Degree Pathways

Accounting	Engineering - Civil
Agriculture	Engineering - Computer
Anthropology	Engineering - Electrical
Architecture	Engineering - Geomatics (Surveying and Mapping)
Art or Art Education	Engineering - Industrial
Atmospheric Science & Meteorology	Engineering - Mechanical
Biology	Engineering - Ocean
Building Construction	English/Literature & English Education
Business Administration	Environmental Sciences
Chemistry	Exercise Science
Computer Arts Animation	Foreign Language
Computer Information Systems	Forestry
Computer Science	Geology
Criminal Justice Administration	Graphic or Commercial Arts
Dance	Health Services Administration
Dietetics	History
Drama or Drama Education	Hospitality Administration/Travel & Tourism
Economics	Interior Design
Engineering - Architectural	International Relations
Engineering - Biomedical	Landscape Architecture
Engineering - Chemical	Mass Communications/Journalism
	Mathematics
	Music or Music Education

Philosophy
Physics
Political Science
Pre-Bachelor of Arts
Pre-Law
Pre-Medical Science/Dentistry
Pre-Medical Technology
Pre-Nursing
Pre-Occupational Therapy
Pre-Optometry
Pre-Pharmacy
Pre-Physical Therapy

Pre-Veterinary Medicine
Psychology
Public Administration
Recreation
Social Work
Sociology
Speech Pathology & Audiology
Teaching (Early Childhood)
Teaching (Exceptional Student Education)
Teaching Secondary (Biology)
Teaching Secondary (Mathematics Education)

Associate in Science

Accounting Technology
Aerospace Technician
Animation & Game Art
Applied Artificial Intelligence
Architectural Design and Construction Technology
Aviation Administration
Aviation Maintenance Management
Biomedical Engineering Technology
Biotechnology
Building Construction Technology
Business Administration
Business Intelligence Specialist
Civil Engineering Technology
Clinical Laboratory Science
Computer Crime Investigation
Computer Engineering Technology
Computer Information Technology
Computer Programming and Analysis - Business Application Programming
Computer Programming and Analysis - Internet of Things (IoT) Applications
Computer Programming and Analysis - Mobile Applications Development
Crime Scene Technology - Crime Scene Investigation
Crime Scene Technology - Forensic Science
Criminal Justice Technology
Culinary Arts Management
Cybersecurity
Database Technology - Oracle Database Administration
Dental Hygiene
Diagnostic Medical Sonography
Electronics Engineering Technology
Emergency Medical Services
Entrepreneurship
Fashion Design
Fashion Merchandising
Film Production Technology
Financial Services - Banking
Financial Services - Wealth Management
Financial Technology
Fire Science Technology
Funeral Service Education

Game Development & Design
 Graphic Design Technology
 Graphic Internet Technology
 Health Information Technology
 Health Science - Health Services Management
 Health Sciences
 Histologic Technology
 Hospitality & Tourism Management
 Interior Design Technology
 Landscape & Horticulture Technology
 Marketing
 Music Business - Management and Marketing
 Music Business - Performance and Production
 Networking Services Technology - Enterprise Cloud Computing
 Networking Services Technology - Network Infrastructure
 Nuclear Medicine Technology
 Nursing - R.N. (Accelerated)
 Nursing - R.N. (Generic Full-Time)
 Nursing - R.N. (Generic Part-Time)
 Nursing - R.N. (Transitional Full-Time)
 Nursing - R.N. (Transitional Part-Time)
 Opticianry
 Paralegal Studies - ABA Approved
 Photographic Technology
 Physical Therapist Assistant
 Professional Pilot Technology
 Radiation Therapy
 Radio & Television Broadcast Programming
 Radiography
 Respiratory Care
 Respiratory Care (Accelerated) - CRT to RRT
 Sign Language Interpretation
 Social and Human Services - Addictions Studies
 Social and Human Services - Generalist
 Sport Management
 Surgical Technology
 Teaching and Learning: Early Childhood Education
 Translation/Interpretation Studies
 Transportation and Logistics
 Veterinary Technology

Advanced Technical Certificate (ATC)

Biotechnology
 Biotechnology - Bioinformatics
 Biotechnology - Chemical Technology
 Certified Flight Instructor
 Compensation and Benefits
 Computed Tomography

Hospitality Management
 Human Resources Management
 Magnetic Resonance Imaging
 Museum Studies
 Procurement Management
 Project Management
 Supply Chain Analytics

College Credit Certificate (CCC)

Accounting and Budgeting
 Accounting Technology Management

Addiction Studies
 Air Cargo Management

Airline Maintenance Procedures and Records Management
 Airport Management
 American Sign Language
 Artificial Intelligence Awareness
 Artificial Intelligence Practitioner
 Audio Technology
 Automation
 Biotechnology
 Building Construction Specialist
 Business Industry Operations
 Business Intelligence Professional
 Business Operations - Business Management
 Business Operations - Human Resources
 Business Operations - Marketing
 Business Operations - Small Business
 Business Specialist - Business Management
 Business Specialist - General Business
 Business Specialist - Human Resources
 Business Specialist - International Business
 Business Specialist - Marketing
 Business Specialist - Small Business
 Chef Apprentice
 Cisco Certified Network Associate (CCNA)
 Commercial Transport Pilot
 Computer Aided Design Assistant
 Computer Aided Design Operator
 Computer Programmer - Business Applications
 Computer Programmer - Mobile Applications Development
 Crime Scene Technician
 Cruise Line Operations
 Culinary Arts Management Operations
 Cybersecurity Analyst
 Digital Marketing Specialist
 Digital Marketing Strategy
 Early Childhood Education - Administrator
 Emergency Medical Technician
 Endoscopic Technician
 Engineering Technology Support Specialist
 Entrepreneurship Operations and Management
 Enterprise Cloud Computing
 Film Production Fundamentals
 Florida Funeral Director

Food and Beverage Management
 Food and Beverage Operations
 Food and Beverage Specialist
 Geographic Information Systems
 Graphic Design Support
 Health Care Services Specialist
 Healthcare Informatics Specialist
 Help Desk Support Technician
 Homeland Security
 Horticulture Professional
 Horticulture Specialist
 Information Technology Support
 Intelligence Studies
 Intermodal Freight Transportation
 International Freight Transportation
 Internet of Things (IoT) Applications
 Lean Manufacturing
 Lending
 Logistics and Transportation Specialist
 Mechatronics
 Medical Coder/Biller
 Mental Health - Neuroscience and Aging
 Microcomputer Repairer/Installer
 Network Server Administration
 Network Systems Developer
 Oracle Certified Database Administrator
 Paramedic
 Passenger Service Agent
 Patternmaking and Construction
 Rapid Prototyping Specialist
 Real Estate Legal Specialization Certificate
 Rooms Division Management
 Rooms Division Operations
 Rooms Division Specialist
 Sales and Customer Management
 Solar Energy Systems Specialist
 Sport Management Operations
 Tax Specialist
 Television Studio Production
 Translation and Interpretation
 UAV/Aviation Operations Management
 Virtual and Augmented Reality Technologies

Career Technical Certificate (CTE/VCC)

Advanced Automotive Service Technology - Tesla Technician
 Aircraft Structural Assembly and Fabrication
 Apprenticeship
 Central Sterile Processing Technician
 Commercial Class B Driving
 Commercial Vehicle Driving

Correctional Officer - State
 Dental Assisting Technology and Management
 Fire Fighter Minimum Standards
 Fire Fighter/Emergency Medical Technician - Combined
 Florida Law Enforcement Academy
 Licensed Practical Nursing (LPN)
 Low Voltage Technician
 Massage Therapy - Accelerated Option
 Massage Therapy - Generic Option

Massage Therapy - Transitional Option
Medical Assisting
Pharmacy Technician
Phlebotomy Technician

Private Investigator Intern
Private Security Officer
Telecommunications and Fiber Optics Technician

COLLEGEWIDE ACADEMIC SCHOOLS

The College has adopted a management approach to the delivery of occupational and technical education, including respective transfer options through a system of collegewide schools. The primary objective is to serve students more effectively and efficiently, provide more accessible programs countywide, and be more responsive to the needs of business and industry.

Benjamín León School of Nursing

The Benjamín León School of Nursing (BLSON) is accredited by the Accreditation Commission for Education in Nursing (ACEN) (3390 Peachtree Road NE, Suite 1400, Atlanta, Georgia 30326, 404-975-5000, info@acenursing.org) and Commission on Collegiate Nursing Education (CCNE) (655 K Street, NW, Suite 750 Washington, DC 20001, 202-887-6791). The BLSON offers an Associate in Science in Nursing (ASN) and a Bachelor of Science in Nursing (BSN) degree to provide students and practicing nurses with a high-quality, accessible, cost-effective and seamless academic program designed to meet the critical workforce need for baccalaureate-prepared nurses in the state of Florida. Students entering the BSN program must have an earned Associate in Science in Nursing (ASN) from a regionally accredited institution/regionally accredited ASN program and an active license as a registered professional nurse (RN). The Benjamín León School of Nursing also offers the Associate in Science degree in Nursing, leading to eligibility to apply for the licensing examination for registered nurse practice (NCLEX-RN). The Associate in Science program offers four options designed to meet the needs of individual learners (generic, transitional bridge and accelerated and part-time tracks) and all combine class work with clinical nursing experiences in local hospitals and agencies. Students entering the associate degree program should possess college-level cognitive, communication and computational skills. Specific general education and science courses are included in the curriculum; selected courses are required before admission to these health care programs.

Miguel B. Fernandez Family School of Global Business, Trade and Transportation

The Miguel B. Fernandez Family School of Global Business, Trade and Transportation offers a full range of academic programs to prepare students for careers in business or to start a successful business of their own. Course offerings are available in a wide number of disciplines including accounting, business administration, economics, entrepreneurship, management, marketing, international business, supply chain and logistics, and financial services. The School of Business has a long tradition of preparing students to meet the needs of the local workforce and partnering with industry to offer students cutting-edge instruction in various fields. The School has become known for excellence in providing customized training to meet industry needs. School of Business courses are offered at the North, Kendall, Wolfson, Homestead, Padrón, Hialeah and West Campus, as well as through MDC Online.

Academic options include:

- Bachelor of Applied Science (BAS) in Leadership and Management Innovation with emphasis on critical thinking and problem solving skills in high demand among employers worldwide. The BAS in Leadership and Management Innovation provides knowledge in a range of organizational settings such as personnel management, organizational behavior, international and small business, finance, business ethics, and leadership. Graduates in this field will develop effective interpersonal skills, foster decision-making and entrepreneurial thinking, and become familiar with diverse business environments.
- Bachelor of Applied Science (BAS) in Supply Chain Management provides knowledge to the supply chain and explores the risks, logistics, economics, regulatory issues, resource allocation, production planning, inventory management and other functions basic to business. Due to the globalization of business, graduates in this field can find employment within the supply chain, logistics and transportation fields.
- Associate in Arts (AA) with a pathway to a degree in accounting, business administration or economics. The Associate in Arts degree is designed for students wishing to transfer to colleges or universities for upper-division coursework. The areas of concentration parallel university coursework and prepare students to enter their junior year at four-year institutions upon completion of the AA degree.
- Associate in Science (AS) in a broad range of business functions including accounting, entrepreneurship, management and marketing. The Associate in Science is designed to prepare students for immediate employment. Credits earned for many courses in these programs are acceptable to upper-division colleges or universities should the student decide to pursue a four-year degree.
- College Credit Certificates including accounting and financial services, entrepreneurship, international business, management and marketing. Focusing on a specific job or set of skills, these programs require fewer

credits than an associate degree and are Florida Department of Education Certified College Credit programs. The credits granted will apply to the related AS degree.

- Eig-Watson aviation programs provide students with the education and skills required for a successful aviation career. Associate in Science degrees include Aviation Administration, Aviation Maintenance Management, Professional Pilot Technology, Transportation and Logistics. Related College Credit Certificates include Air Cargo Management, Airline/ Aviation Management, Airport Management, Intermodal Freight Transportation and International Freight Transportation; an Advance Technical Certificate in Certified Flight Instructor is also offered. Additionally, the School offers an aircraft dispatcher course and flight simulation training.
- Training techniques and simulation equipment to provide students with a hands-on approach to their education at Miami International Airport, Miami Executive Airport, and the Homestead Campus. The following flight training courses are offered through contracted flight providers:
 - ATF 2210 Commercial Pilot Flight (3 credits)
 - ATF 2305 Instrument Pilot Flight (3 credits)
 - ATF 2400 Multi-Engine Pilot Flight (1 credit)
 - ATF 2501 Flight Instructor Flight Training (3 credits)
- Miami Culinary Institute. At the Miami Culinary Institute, you will gain the real-world and hands-on knowledge to create a perfect blend of classic skills and innovative techniques used by some of the world's best chefs. Our two-year Associate in Science degree in Culinary Arts Management, has been formulated to prepare students of all ages for dynamic careers in culinary arts. Our core philosophy of Food Culture Innovation requires an examination and understanding of the culture built upon our interaction with food. As culinary professionals, we understand the role we play and the responsibility we must own in elevating our community's expectations about the food they eat. We help our students understand the value of tracing our foods to their source to evaluate how a particular farm, ranch or fishery impacts our environment, community and economy. We examine how food distribution not only contributes to a significant portion of the world's greenhouse gasses, but also how we can make choices that will improve that process. We analyze how food is prepared to ensure that the best nutrition possible is delivered on each plate. Miami Culinary Institute is training the next great culinary professionals and arming them with the tools to innovate the way we interact with food.
 - Associate in Science in:
 - Culinary Arts Management
 - College Credit Certificate in:
 - Culinary Arts Management Operations
 - Chef's Apprentice
- Miami Hospitality Center. MDC's four-year and two-year academic programs offer a range of degree pathways that prepare you to enter the modern workforce or pursue higher educational goals.
 - Bachelor of Applied Science in:
 - Supervision and Management-Hospitality Concentration
 - Associate in Arts pathway in:
 - Hospitality & Tourism Management
 - Associate in Science in:
 - Hospitality Administration/Travel & Tourism
 - College Credit Certificates in:
 - Cruise Line Operations
 - Food & Beverage Management
 - Food & Beverage Operations
 - Food & Beverage Specialist
 - Rooms Division Management
 - Rooms Division Operations
 - Rooms Division Specialist

School of Continuing Education and Professional Development

The School of Continuing Education and Professional Development's (SCEPD) mission is to create highquality workforce training, adult education programs, and personal enrichment courses that are accessible to the

community. The School provides innovative educational experiences that are responsive to the needs and interests of industry and life-long learners of all ages. Through the Continuing Education departments located on each campus, the school offers noncredit courses in recreational, continuing workforce education and adult education categories. Recreational courses cover a huge range of topics from aerobics to Zen, and they serve individuals wanting to enrich their cultural experiences, pursue interests or learn alongside others with similar interests. Continuing workforce education courses are just-in-time courses intended to help students improve their professional or occupational skills. The topics covered include computer workshops, certification courses, preparing oral presentations, building contractor license exam preparation, as well as several hundred workrelated topics. Adult education courses prepare the student to pass the GED test or master the basic skills needed for success in one of the College's accredited programs. The School of Continuing Education & Professional Development endeavors to provide classes both on and off campus. The majority of classes are conducted in the evenings and on weekends at times that are most convenient to the students enrolling. In its effort to meet the diverse needs of a large, multifaceted community, the school also welcomes suggestions and requests for courses that are not being offered.

School of Education

Teaching is a vital and dynamic profession. A career in teaching offers the opportunity to influence children and shape the future. Trends in population growth, an aging teacher workforce and the demand for class size reduction will result in ample professional opportunities for prospective teachers. The School of Education provides training and professional development opportunities for pre-service teachers as well as practicing professionals. The School offers a wide variety of programs in Early Childhood Education, K-12 Teacher Education preparation, alternative pathways to certification and teacher recertification.

- Early Childhood Education: Students may earn College Credit Certificates (CCC) with an Infant/Toddler Specialization or Preschool Specialization that may lead to the National Child Development Associate credential and/or the Florida Child Care Professional Certificate (FCCPC). Students may earn a College Credit Certificate with an Administrator Specialization that may lead to the Director's Credential Level 1 or 2. Students may also earn a CCC in Inclusion or a CCC in Intervention Studies. All CCC lead directly to the AS degree program. Students may earn an Associate in Science (AS) degree in early childhood education which will prepare students for immediate employment as early childhood practitioners or professionals in both the public school system or private school sector. All AS degrees are accredited by the National Association for the Education of Young Children. The AS degree leads seamlessly to the Bachelor of Science in Early Childhood Education from birth to grade three with endorsements in English for Speakers of Other Languages (ESOL), Prekindergarten Disabilities, and Reading.
- K-12 Teacher Education Programs: Students may earn an Associate in Arts with the requisite courses needed for a baccalaureate degree in teaching. Students may complete a Bachelor of Science degree in ESE, secondary math or secondary biology, chemistry, earth/space science, or physics. The School's courses meet state certification requirements as a state-approved teacher preparation program. Students who complete the A.A. with requisite courses needed for a baccalaureate degree in teaching may transfer to state university colleges of education or private institution with junior-level standing. K-12 Teacher Education programs are in the areas of: Exceptional Student Education (K-12), Mathematics Education (Grades 6-12), Biology Education (Grades 6-12), Chemistry Education (Grades 6-12), Earth/Space Science Education (Grades 6-12) and Physics Education (Grades 6-12). The baccalaureate programs in education are designed to prepare future teachers to enter the teaching profession immediately after graduation. Upon program completion students meet all Florida Department of Education requirements including the successful completion of the certification exams, an e-Portfolio of artifacts that demonstrate Florida Educator Accomplished Practices, clinical experience in a variety of settings and grade levels, and a semesterlong internship. Professional development workshops also are provided.
- Center for Professional Development: The CPD is the post-baccalaureate branch of the School of Education. Students may complete the substitute teacher training or a variety of teacher certification and recertification courses as well as endorsements in Autism Spectrum Disorders, ESOL, Gifted, Prekindergarten Disabilities, and Reading. Career changers who hold a bachelor's degree can become a state-certified teacher holding a Professional Certificate from Florida Department of Education in one year through the Educator Preparation Institute.

School of Engineering, Technology, and Design

The School of Engineering, Technology, and Design provides the dynamic knowledge, skill, hands-on training and

industry connection to turn your dreams and imagination into achievable steps to realize the success you desire. Our vision is to provide students with unmatched opportunities and access to an outstanding curriculum taught by world-class faculty within the information technology and engineering fields. Our programs are strategically formulated to exceed the critical demands of the high-technology marketplace, providing students with a wide variety of options for success in the 21st century. The School of Engineering, Technology, and Design is a Cisco Regional Networking Academy offering CCNA classes that lead to valuable industry certifications. These courses are available at most campuses. The School also provides instruction using official curriculum from companies like Microsoft, AWS and Oracle. With input from industry partners, including NextEra Energy, Baptist Hospital, Amazon, Visa, IBM, SAS, Dell, Oracle and Microsoft, our courses deliver the knowledge and skills that the nation's top employers are looking for. The Engineering Department offers pathways to Associate in Arts programs in ten different fields. It also offers three unique Associate in Science programs in Computer Engineering Technology, Electronics Engineering Technology and Industrial Engineering Technology that provide the skills for high paying jobs. Students looking for short term programs with recognized credentials have access to College Credit Certificates in Mechatronics, Microcomputer Repairer/Installer, Rapid Prototyping Specialist, Engineering Technology Support Specialist and Computer Specialist. The Technology department offers Associate in Arts degrees in Computer Science and Computer Information Systems, and Associate in Science degrees in emerging technologies. These include: Animation and Game Art, Business Intelligence, Computer Information Technology, Computer Programming and Analysis for Business, Mobile Apps and IoT, Cybersecurity, Database Technology, Game Development and Design, and Network Services Technology with tracks in Cloud Computing and Network Security. At MDC's School of Engineering, Technology, and Design, you can earn a bachelor's degree in fields that are in high demand. Upon completion of an Associate in Arts degree or an Associate in Science, students can transition to the following baccalaureates:

- Bachelor of Science in Data Analytics: This was the first undergraduate program in the state of Florida in data analytics. It prepares students for the booming field of big data across all industries. Students learn how to find patterns, apply statistics, and create data visualizations all necessary skills to acquire positions as Data Analysts, Database Architects and Business Intelligence Analysts.
- Bachelor of Science in Electrical and Computer Engineering Technology: This program opens plenty of doors to a variety of technology-related fields. Graduates are trained as engineering practitioners, ready to take on roles as Electronics Engineers, Manufacturing Engineers and Project Engineers.
- Bachelor of Science in Information Systems Technology: The field of information systems continues to grow, offering graduates a wide range of lucrative and rewarding career opportunities. The program offers students the skills and knowledge to direct and control computerized information resources within an organization. Two concentrations are available for students to choose from: Networking and Software Engineering.

Additional disciplines under the School of Engineering, Technology, and Design include:

- Architecture and Interior Design. Offering programs in:
 - Associate in Arts degree with pathways in:
 - Architecture
 - Building Construction
 - Interior Design
 - Landscape Architecture
 - Associate in Science degrees in:
 - Architectural Design and Construction Technology
 - Building Construction Technology
 - Interior Design Technology
 - College Credit Certificate in:
 - Computer Aided Design Assistant
 - Computer Aided Design Operator
- Entertainment & Design Technology. Offering design and media production programs in:
 - Bachelor of Applied Science in Film, Television & Digital Production
 - Associate in Arts degree with pathways in:
 - Computer Art Animation
 - Associate in Science degrees in:

- Film Production Technology
- Radio & Television Broadcasting
- Music Business
- Graphic Design Technology
- Graphic Internet Technology
- Photographic Technology
- College Credit Certificate in:
 - Film Production Fundamentals
 - Television Studio Production
 - Graphic Design Support
 - Audio Technology

School of Health Sciences

The Medical Campus is committed to assisting qualified students interested in pursuing careers in the health science professions. Health Science professionals provide more than 60 percent of all health care administered in the United States. The School of Health Sciences offers more than 30 challenging vocational, certificate and degree programs, such as respiratory care, opticianry, clinical laboratory sciences and health information management. Programs in the School of Health Sciences prepare students for employment in a wide variety of settings including hospitals, clinics, research centers, long term care facilities, physician's offices, and wellness centers. In collaboration with more than 300 health care facilities throughout Miami-Dade County, students receive the necessary theory, laboratory experience and clinical practice. Students use state-of-the-art equipment and are supervised by licensed professional faculty. Health Science programs are fully accredited through their respective state and national associations. Most programs have limited access. Program completion affords the graduate the opportunity to seek employment in high-demand professions while receiving a competitive salary. Interested students are encouraged to contact the Medical Campus at 305-237-4141 to receive current information regarding program requirements, application procedures and selection process for the specific Health Science program of interest. Program offerings within the School of Health Science include:

Bachelor of Applied Science in Health Sciences (BAS)

- Physician Assistant Studies Concentration
- Clinical Laboratory Science
- Histotechnology Concentration

Bachelor of Science (BS)

- Dental Hygiene

Associate in Science (AS) degrees in:

- Clinical Laboratory Science
- Dental Hygiene
- Diagnostic Medical Sonography Specialist
- Emergency Medical Services
- Health Service Management
- Health Information Technology
- Histologic Technology
- Nuclear Medicine Technology
- Opticianry
- Physical Therapist Assistant
- Physician Assistant
- Radiation Therapy

- Radiography
- Respiratory Care
- Surgical Technology
- Veterinary Technology

College Credit Certificates (CCC) in:

- Emergency Medical Technician
- Endoscopic Technician
- Medical Coder/Biller
- Paramedic
- Healthcare Informatics Specialist
- Health Care Services Specialists

Career Technical Certificates (CTC) in:

- Central Sterile Processing Technician
- Dental Assisting
- Firefighter/EMT
- Massage Therapy
- Medical Assisting
- Pharmacy Technician
- Phlebotomy Technician

Advanced Technical Certificate (ATC) in:

- Computed Tomography
- Magnetic Resonance Imaging

School of Justice, Public Safety, and Law Studies

The School of Justice, Public Safety, and Law Studies, located on the North Campus, offers degree, certificate, and workforce training programs for students pursuing careers in criminal justice, public safety, and related fields, while also supporting the professional development of current practitioners. Committed to expanding access to education, the School of Justice recognizes prior training and professional experience when applicable and maintains partnerships with four-year institutions to support students in continuing their education beyond the associate and baccalaureate levels. The School of Justice operates in cooperation with federal, state, county, and local agencies and includes an Assessment Center that provides testing and evaluation services for public safety agencies and applicants. In addition, through the School of Justice, MDC delivers specialized training and professional development programs, including equivalency training and Fire Safety Inspector preparation for the State Certification Examination. Program offerings within the School of Justice include:

Bachelor of Applied Science (BAS) in Public Safety Management

- Criminal Justice
- Crime Scene Investigation

Associate in Art (AA) pathway in Criminal Justice Administration

Associate in Science (AS) degrees in:

- Criminal Justice Technology
- Crime Scene Technology
- Forensic Science
- Paralegal Studies
- Fire Science Technology
- Computer Crime Investigation

College Credit Certificates (CCC) in:

- Crime Scene Technician
- Homeland Security
- Intelligence Studies
- Real Estate Legal Specialization

Career Technical Certificates (CTC) in:

- Firefighter Certification
- Firefighter/Emergency Medical Technician (EMT)
- Private Investigator Intern
- Private Security Officer

Basic Recruit Training (BRT) programs in:

- Law Enforcement
- Corrections

School of Science

Offering a variety of degrees and certificate programs, Miami Dade College's School of Science builds a strong foundation in the study of natural sciences. Whether starting a journey to an advanced degree or building skills for today's workforce, the School of Science offers a modern approach to learning that opens doors to discovery and success. A Top Choice - Students choose the school's programs because of the quality curriculum, undergraduate research experiences, individual attention, small class sizes, internship opportunities and highly qualified and motivated instructors. Engaging Industry - The School of Science collaborates with industry principally through student internships, a required component of the Bachelor of Science in Biological Sciences. An example is internship in a local biotechnology company. Students typically spend a semester working with an industry partner. Research Focused - Students at the School of Science are afforded the rich experience of authentic undergraduate

scientific research, culminating in presentations at a number of regional, national and international conferences. The year-round STEM Research Institute (SRI) offers students a semester long Spring Program and a 10-week Summer Program. Students apply and if accepted receive compensation for their work. Most research opportunities are within MDC, but there are a handful of options conducted at collaborating institutions in South Florida and beyond, including the University of Florida, Florida Atlantic University, Nova Southeastern University, St. Thomas University and the University of Miami. Talented Teachers - Full-time faculty members at the School of Science are talented facilitators of knowledge and are highly qualified in various science fields. Faculty selection is a very competitive process as many exceptional instructors want to work here. The school's faculty is inventive in reaching students and leveraging technology and new teaching methods, resulting in higher success rates in rigorous STEM courses. Furthermore, the school's professors are frequent recipients of the Endowed Teaching Chair award for excellence at MDC. Many get invited to present the results of their research endeavors at scientific conferences.

- School of Science Programs:
 - Bachelor of Science in Biological Sciences Degrees (BS)
 - BSBS – Biopharmaceutical Science Concentration
 - BSBS – Biotechnology Concentration
 - BSBS – Science Education Concentration
 - Associate in Arts Pathways (AA)
 - Atmospheric Science & Meteorology
 - Biology
 - Chemistry
 - Environmental Science
 - Geology
 - Physics
 - Associate in Science Degrees (AS)
 - Biotechnology
 - Biotechnology – Bioinformatics
 - Biotechnology – Chemical Technology
 - Funeral Services
 - Landscape & Horticulture Technology
 - Advanced Technical Certificates (ATC)
 - Biotechnology
 - Biotechnology – Bioinformatics
 - Biotechnology – Chemical Technology
 - College Credit Certificates (CCC)
 - Biotechnology
 - Florida Funeral Director
 - Horticulture Professional
 - Horticulture Specialist

SPECIAL ACADEMIC AND OTHER PROGRAMS

In meeting its commitment to serve the community, Miami Dade College offers a variety of programs, both on and off campus, to meet the specific educational needs of the groups involved. These may take the form of specially structured programs on campus, courses, seminars or workshops offered at times and locations that best serve public interests and needs. For example, MDC offers:

- Assistance to companies and governmental agencies in conjunction with their own training programs;
- Workshops, seminars and institutes in cooperation with business, professional or other groups;
- Recreation, personal improvement and cultural activities;
- Postsecondary occupational career offerings to serve business, industry, the professions and governmental agencies.

MDC Apprenticeship Program

MDC is the first academic institution in Florida to receive the designation of Program Sponsor by the Florida Department of Education (FLDOE). MDC was also deemed Apprenticeship Ambassador by the U.S. Department of Labor. As a Program Sponsor, MDC is authorized to register occupations, employers, and apprentices to participate in a Registered Apprenticeship Program. Apprenticeships are an Earn-as-You-Learn model as students are employed, earning a wage, while attending school. Registered Apprenticeship Programs (RAP) assist employers to develop and apply industry standards to training programs for registered apprentices. These programs can increase productivity and improve the quality of the workforce. Apprentices who complete a RAP are accepted by the industry as journeyworkers. By providing on-the-job training (OJT), related technical instruction (RTI), and guaranteed wage increases, employers who train apprentices retain more highly qualified employees and improve productivity. Certifications earned through registered apprenticeship programs are recognized nationwide.

Benefits of a Registered Apprenticeship Program:

- **Free Tuition:** Begin your career training with no tuition fees, making education accessible while you focus on gaining real-world experience.
- **Earn While You Learn:** Apprentices receive paid wages and benefits starting on day one of the program, with scheduled pay increases as new skills and milestones are achieved.
- **Classroom Training:** Participants receive free related technical instruction designed to complement hands-on experience and strengthen industry knowledge.
- **Hands-On Job Training:** Work directly in the field with real-world projects and responsibilities while learning from experienced professionals for at least one year.
- **Professional Mentorship:** Each apprentice is paired with an experienced mentor who provides guidance, supervision, and ongoing support throughout the program to ensure steady growth and career readiness.
- **Industry-Recognized Certification:** Upon successful completion, apprentices earn a National Credential from the Florida Department of Education, recognized by the U.S. Department of Labor, helping open doors to long-term career opportunities.

Miami Dade College currently offers 23 Registered Apprenticeship and Pre-Apprenticeship programs designed to help students gain real-world experience while developing in-demand skills. These programs provide career pathways in a variety of high-growth industries, including Automotive, Aviation, Business/Finance, Construction, Education, Health Sciences, Hospitality, Information Technology, and Logistics and Transportation. MDC continues to grow its workforce training opportunities and plans to expand apprenticeship programs in emerging and high-demand fields such as Advanced Manufacturing, Creative Design and Energy.

Cooperative Education

Cooperative Education provides an opportunity for students to obtain career-related work experience and academic credit for such work. It enables students to apply classroom theory to actual work situations. In many instances, it helps students earn needed cash to meet education costs. It gives students work experience that employers look for and it may turn into permanent employment. Job opportunities are available in many career fields. Transfer students may continue their Cooperative Education program at many four-year colleges and universities. While enrolled at MDC, this work experience may be part time or full time, paid or voluntary, and may continue for one or two terms. The program is flexible and tailored to meet student and employer needs.

The volunteer plan provides for one term of six hours or more per week for 12 weeks minimum, and for 10 hours or more per week for 12 weeks during a second term. Through Cooperative Education, students may earn three elective credits per term for two terms. Application for the program should be made to the Cooperative Education liaison at each campus discipline. A minimum GPA of 2.0 is required.

Credit for Specialized Training

College credit for specialized non-collegiate occupational training may be granted to students enrolled in occupational programs. This credit is granted upon validation of the non-collegiate instruction by the appropriate academic department. A processing fee of \$15 per course, up to a maximum of \$50 for any single application, will be charged for the evaluation of non-collegiate instruction. Agreements to recognize specialized non-collegiate occupational training must have been previously approved in accordance with College curriculum procedures.

Certified Professional Secretary (CPS)

Students passing the complete national examination of the Certified Professional Secretary Examination (CPS) and the CPS Exam Prep courses may be granted credit toward an Office Administration Associate in Science degree at Miami Dade after official score reports are received from the International Association of Administrative Professionals (IAAP). The credit will appear on the student's permanent record as earned credit only, without any indication of grades.

Industry Certifications

Miami Dade College may award college credit to eligible students who have earned a recognized appropriate industry certification which will be applied toward the specific academic degree. Evaluations of industry certification(s) are made after the student has been admitted to the College and provides a valid industry certification based on the State Board of Education, which has an inventory of approved a statewide industry certifications. For a complete list of the approved statewide Career and Technical Education articulation agreements visit

<https://www.mdc.edu/industrycertifications/>.

MDC WORKS: Career Studio

MDC WORKS Career Studio is a professional development resource that prepares MDC students for the workforce by giving them the skills, information and contacts to successfully launch their careers and create meaningful futures. Launched in 2018 in collaboration with the U.S. Department of Labor and the Florida Department of Education, MDC WORKS Career Studio was the first program of its kind to integrate interactive, virtual and augmented reality into its career exploration process at an academic institution in Florida. MDC WORKS is free and automatically available to all MDC students. Students activate their personal Career Portal by signing in with their MDC email address and password. Once the portal is activated, students must confirm their registration via e-mail and then they will have access to apply to the thousands of job and internship postings and networking resources. The MDC WORKS Career Studio provides the following services to students and alumni up to one year after their last class at MDC:

- Career Coaching
- Career Readiness Workshops & Resources
- Resume Assistance
- Interview Preparation & Practice
- Networking Events
- Job & Internship Postings
- On-Campus & Virtual Employer Information Sessions
- Interviews with Employers
- Career Fairs and Hiring Events
- Online Career Portal to stay organized

MDC WORKS Career Studio serves all 8 Miami Dade College Campuses and assists all students who are seeking career readiness, internship and job placement assistance who are registered in associate and bachelor degree programs, as well as, GED, ESL and ESOL class's in-person, virtually, and over the phone at 305-237-9675 (WORK), mdcworks@mdc.edu, www.mdcworks.edu. In addition, MDC WORKS Career Studio offers a plethora of career readiness resources, support and downloadable resume templates on their website at www.mdcworks.edu.

The KICK-START Program

Learn | Upskill | Succeed

MDC's Kick-Start program provides robust financial assistance for short-term, industry-recognized certificate programs that prepare students for high-demand and high-wage careers across a wide range of growing sectors. By covering tuition and related educational costs, the program removes financial barriers and opens doors for individuals seeking to begin a new career, strengthen their current expertise, or adapt their skills to meet the needs of an evolving workforce. Through The Kick-Start Program, students gain access to flexible, career-focused training designed to build job-ready skills quickly and efficiently. Whether learners aim to enter the workforce for the first time, transition into a more rewarding field, or advance within their current profession, the program offers a clear pathway to opportunity and upward mobility. By supporting personal and professional growth, The Kick-Start Program helps students enhance their long-term career prospects, achieve greater economic stability, and contribute meaningfully to the region's talent pipeline and economic development. For more information, visit: www.mdc.edu/kickstart or email us at: kickstart@mdc.edu

Center for Economic Education (*Padrón Campus*)

The mission of the Center for Economic Education is to work closely with the educational communities in Miami-Dade and Monroe counties to develop greater awareness for economic literacy. Among the most popular of the Center's programs are the four recertification credit courses offered to area teachers in grades K-12. Of these, the national Stock Market Game is played in grades 5-12 in each of the major semesters. The Free Enterprise Bank Program, available to grades K-12, provides real money for class business activities. The Center works with area educational administrators to create and assist in the development of curriculum materials. These materials have included a tourism and development program, a Civics Teachers Resource Guide, Elementary Program of the Economics of the Stanford Achievement Test and many more program examples at each of the major grade levels. It is the Center's goal to provide the latest and best materials and programs in economic education to our schools. Through these opportunities the Center seeks to promote greater understanding on the part of our young people about the economy in which they live and the economic climate in which they will work.

Center for Financial Training (*Wolfson Campus*)

The Center for Financial Training Southeastern (CFTSE) is a local training provider of the American Bankers Association (ABA). As the largest industry-sponsored adult education program in the world for financial services professionals, CFTSE benefits more than 3,500 financial services professionals locally and is one of 20 centers located throughout the United States. CFTSE is a unique source for commercial banking and financial industry training and education. CFTSE is a nonprofit educational organization that conducts college credit courses (live classes, guided self-study and online), seminars, webinars, computer workshops and customized and contract training. Students can earn CFTSE and/or American Bankers Association (ABA) diplomas and certificates that are recognized throughout the industry and accepted as college credit. Students can also earn Banking College Credit Certificates. CFTSE has established an academic partnership with Miami Dade College, enabling CFTSE students to achieve degree status while completing their financial services studies. CFTSE courses are offered at all MDC campuses, and at certain financial institutions. All courses are open to the public however, special fees are charged by CFTSE for certification and materials. The fee structure varies depending on whether the student is a member or nonmember of CFTSE. The fee is charged in addition to MDC tuition and is paid to CFTSE. CFTSE also offers special programs in partnership with MDC, to include financial literacy workshops and the Future Bankers' Camp. The Future Bankers' Camp is a partnership between CFTSE, MDC - School of Business and Miami Dade County Public Schools - Academy of Finance.

Cloud Computing Center (*Padrón Campus*)

The Cloud Computing Center serves as a Cloud Computing hub where technology leaders, industry experts and students can engage and collaborate. The Center houses the College Credit Certificate and Associate of Science degrees in Enterprise Cloud Computing, and exposes students and existing IT professionals to industry leading cloud platforms and industry certifications needed to fill increasing demand of IT cloud jobs in the workplace. The Center also hosts Cloud Computing accelerated training programs, summer camps, forums, speakers and events as Miami Dade College continues to serve the community.

Cybersecurity Center of the Americas (*Wolfson Campus*)

The Cybersecurity Center of the Americas is an all-in-one resource for those interested in building a career in cybersecurity, offering the most advanced education and hands-on experience in cybersecurity defense systems. The unique, customizable technology in place at the Center caters to all skill levels, allowing for a flexible learning environment for trainees of all backgrounds. The holistic training center provides programs for trainees of all skill levels, from beginners looking to enter the world of cybersecurity, to security experts looking to improve

their skills and advance their careers. Dedicated to education and serving the community, the Cybersecurity Center of the Americas also hosts forums, speakers, and events.

Continuing Education

The School of Continuing Education and Professional Development is committed to the philosophy that learning is a life-long process and that the many years spent in formal education do not complete our learning experience. This philosophy serves as the foundation of our learning experience and values the knowledge we acquire daily and use for the rest of our lives. Campuses offer recreation and leisure courses and activities for those who wish to enrich their cultural lives or improve their personal efficiency and professional skills. No record of previous education is necessary and little or no homework is required. No grades are given through Continuing Education, no academic credit gained and attendance standards are voluntary. Continuing Workforce Education training courses are offered to improve employment-related skills for postlicensing and for professional licensing. Training is listed on a student's transcript. The transcript can be used in lieu of continuing education units (CEU) to show evidence of participation in professional development to employers, and licensing or certification agencies (see below). For additional information, contact the campus Continuing Education department. The Adult Education program offers students the opportunity to learn basic skills to earn a GED or to pursue further training through the College's vocational programs. The College offers courses both on and off campus to meet the needs of the community, and makes every effort to begin a course when an adequate number of people request it.

Continuing Education Units (CEU)

Miami Dade provides students with the opportunity to obtain continuing education units (CEUs) for certain noncredit courses. The CEU program encourages long-range education goals and lifelong learning, and permits adult students to aggregate a number of continuing education courses to meet their personal needs. The CEU is used as the basic means for recognizing an individual's participation in, and for recording an institution's offering of continuing workforce education courses. A CEU is defined as 10 contact hours of participation in an organized, continuing education experience under responsible sponsorship, capable direction and qualified instruction. Transcripts indicating completion of continuing workforce education courses designated for CEUs will be provided.

Contract Training for Business and Industry

Through the School of Continuing Education and Professional Development, business, industry and government can benefit from workshops and courses offered at the job site or at any of our campuses. These contract training programs are designed to meet the educational and training needs of community businesses and organizations by reaching beyond traditional academic curriculum and offering courses and workshops that focus on practical application. Offered in credit and noncredit formats, these programs are available at times and locations convenient to the participants. Program topics include computers, management, customer service, communications, foreign languages and English as a Second Language, business English, writing and math and many others. All programs may be customized to the specific needs of the client, with job-related materials included in the curriculum.

Weekend College

Weekend College is designed for students unable to attend weekday or evening classes, but it is not restricted to these individuals; students wishing to complement their schedules with additional courses are encouraged to enroll. Weekend College offers a selection of core, distribution and elective credit courses to satisfy degree and certification program requirements.

Earth Ethics Institute (*Collegewide; Located on Wolfson Campus*)

Earth Ethics Institute (EEI) is an Earth Literacy and sustainability academic initiative at Miami Dade College (MDC) offering workshops, conferences, courses, and support for MDC administrators, faculty, staff and students as well as the greater South Florida community. The mission of the Earth Ethics Institute is to foster Earth Literacy in the course objectives of each discipline and all campus operations at Miami Dade College, as well as in the South Florida Community and the extended Earth community beyond. Earth Literacy includes an understanding of cosmology and ecological principles as the basis for sustainable living. The cosmological context is the story of the universe, as contemporary science describes the developmental process out of which Earth and all life emerge. The Global Sustainability and Earth Literacy Studies (GSELS) Learning Network is EEI's most recent initiative. GSELS provides inclusive educational opportunities for the Miami Dade College community to explore global citizenship, ecological sustainability, and civic engagement, through understanding planetary challenges and limits and by developing values, skills, and behaviors that promote prosperity and communities of well-being. In addition, the GSELS project is replicable, requires very little funding, and hopes to serve as a

national model of best practices. GSELS acknowledges the interconnections and interdependence of the personal, social, economic, cultural, environmental, and political aspects of our world. Collaboratively, students, faculty, administrators, and staff explore the significance of human activity within an evolving Universe and Earth. Through shared leadership, this nurturing learning community facilitates the emergence of awareness, knowledge, skills and solutions necessary to create sustainable systems that support a healthy and just economy, society, culture and environment, while fostering values of Earth ethics, social justice, cultural diversity, and civic engagement. GSELS draws on several international documents, including the four pillars of life-long learning detailed by UNESCO, The Earth Charter, and Thomas Berry's "12 Principles of Understanding the Universe and the Role of the Human in the Universe Process." Additionally, GSELS is grounded in the principles of ecology, and environmental, sustainability and global education concepts. GSELS course criteria provide the basics to acquire the knowledge and skills needed to cope and constructively engage with the 21st century, including these eight guiding principles of global citizenship (from a consensus of experts in nine countries, East and West, as cited in Sustainable Education by Stephen Sterling):

1. looking at problems in a global context
2. working cooperatively and responsibly
3. accepting cultural differences
4. thinking in a critical and systemic way
5. solving conflicts non-violently
6. changing lifestyles to protect the environment
7. defending human rights
8. participating in the political process

EEL Programs for Faculty and Staff - Global Sustainability and Earth Literacy Studies

Earth Ethics Institute grew out of two earlier Miami Dade College programs, Life Lab and the Environmental Demonstration Center. It now offers a series of professional development workshops and programs for Miami Dade College administrators, faculty and staff interested in infusing ecological concepts and a cosmological context into their professions. Through Earth Literacy, one deepens his or her understanding of the interdependent human-Earth relationship and thus broadens the sense of responsibility inherent in the practice of every profession and vocation. Hundreds of MDC faculty and staff have participated in EEI workshops, featuring topics such as incorporating sustainability in existing and new curriculum, biophilia, culture and cosmology, ethics, technology and sustainability, and regenerative, interactive and sustainable design. MDC administrators, faculty and staff are also invited to participate in immersion field trips to explore the unique ecology and hydrology of South Florida. The Institute also collaborates with Genesis Farm in New Jersey, Narrow Ridge Earth Literacy Center in Tennessee, St. Thomas University and Florida International University in Miami in offering courses in Earth Literacy. Earth Ethics Institute certifies faculty who wish to participate in the GSELS Learning Network. Miami Dade College faculty, who currently hold a Master's Degree and have taken an EEI professional development workshop or course are encouraged to participate and begin exploring GSELS in their courses. Faculty who currently hold a Master's Degree can become GSELS-certified faculty in one of four ways:

1. 36 EEI CTD Professional Development hours
2. 3 Graduate Credits (one course) in Earth Literacy or Sustainability Leadership/Sustainable Education Studies

Environmental Center (*Kendall Campus*)

The Environmental Center provides noncredit courses to children and adult community members and to our work force. Enrollment is open to everyone, and there are no prior education levels, transcripts or tests required. Most classes meet weekends or evenings and are scheduled on and off campus for convenient access. The Center has many programs:

1. Landscape/gardening/home improvement courses encourage the public to utilize environmentally appropriate landscape materials and to maintain their home and landscape in ways that minimize environmental impact. Short-term training certification preparation and opportunities to participate in segments of credit courses improve the skill of landscape professionals.
2. Hands-on, interactive environmental education field trip programs are available for school groups, Kindergarten - grade 9.

3. Nature-based teacher-planning day/holiday camps serve the needs of working parents while sensitizing children in pre-kindergarten through seventh grade to the natural world. Children participate in nature games, crafts, outdoor activities and cooperative games.
4. Scout Days provide Boy and Girl Scout groups opportunities to participate in nature-based activities designed to meet badge requirements as well as to implement Eagle Scout and Gold Award projects.

Field trips, day camps and scout days are held at our Environmental Center, which includes a pine rockland, a lake, a floating dock, chickee huts, butterfly gardens, a butterfly house, organic vegetable sand gardens, a composting demonstration exhibit and an Everglades waterflow demonstration exhibit. The Center also offers courses on the use of natural/alternative healing methods, skills for life change and courses in nontraditional spirituality. Initiatives include Native American cultural programs, expanded pine rockland research, development of community service project opportunities for high school students, weekend recreational and educational programs for adults and families.

Funeral Service Education (North Campus)

The Funeral Service Education program was the first public community college program in the southeastern United States to offer a degree in funeral service education. The school has embalming and restorative arts laboratories enabling students to do all training on campus. An on-campus chapel gives students a unique opportunity to work in all aspects of funeral preparation, including embalming, dressing, cosmeticizing and casketing decedents for viewing and final services. The Funeral Services Education degree program at Miami Dade College is accredited by the American Board of Funeral Service Education Inc. (ABFSE), 992 Mantua Pike, Suite 108, Woodbury Heights, NJ 08097 (816) 233-3747. Web: <http://www.abfse.org>. The Funeral Service Education Program requires a separate application to be admitted to the program. Miami Dade College requires that students register for the National Board Exam as administered by the International Conference of Funeral Service Examining Boards in order to graduate with the Associate in Sciences degree in their final semester. The program also offers a college credit certificate to obtain licensure as a funeral director only in the State of Florida. This academic program is designed to meet specific state or professional needs. It is not accredited by the American Board of Funeral Service Education. Students graduating from this program are not eligible to take the National Board Examination or any state board examination for which graduation from an ABFSE accredited program is required. National Board Examination pass rates, graduation rates, and employment rates for this and other ABFSE accredited programs are available at www.abfse.org. To request a printed copy of this program's rates, go to the Funeral Service Education Office (Building 3142) by e-mail at funeralservices@mdc.edu, or by telephone 305- 237-1244. The annual passage rate of first time takers on the National Board Exam (NBE) for the most recent three year period for this institution and all ABFSE accredited funeral service education programs is posted on the ABFSE website (www.abfse.org). Funeral service graduates from MDC are qualified to practice in most states provided they have met the requirements for licensure in the given state of choice. The school provides continuing education required for license renewal of Florida funeral directors, embalmers and direct disposers, and conducts special seminars for the enrichment of funeral services personnel.

The Center @ MDC (Wolfson Campus)

The Center @ MDC is a cultural and academic initiative that promotes reading, writing and theater throughout the year by consistently presenting high-quality literary activities open to all in South Florida. Housed at the Wolfson Campus, The Center serves MDC and K-12 students, as well as the larger South Florida community. Center programs include many reading and writing initiatives, in addition to the prominent Teatro Prometeo, a community theater, and Miami Book Fair the largest literary gathering in the U.S. Center initiatives:

- Visiting Writers participate in readings, teach workshops and conduct residencies that help students and others to deepen their understanding of literature and sharpen their creative writing abilities.
- The Center's reading campaigns for children, students and adults include The Big Read (a national program funded by the National Endowment for the Arts), One Picture Book, One Community and Story Time! All encourage an appreciation for books, while enhancing the reading and comprehension skills of people of all ages. Within the College, Current Voices in Literature provides thousands of free books every year, along with supporting materials, to students in various disciplines.
- Noncredit Creative Writing Courses are offered throughout the year, giving aspiring writers the opportunity to receive critique and encouragement from published authors with extensive teaching experience. The twice-yearly Miami Writers Institute is a conference that features workshops with bestselling and award-winning authors and publishing professionals.

- The Miami Writers Institute is a twice-yearly conference for writers featuring three and four days of intensive workshops with bestselling and awardwinning authors and publishing professionals.
- The Center's Miami: City of Refuge program provides a safe haven for writers persecuted or threatened with imprisonment or death in their home countries.
- The Miami Book Fair is the largest and finest literary gathering in the U.S. Over eight days in November, more than 400 authors from all over the world present books, and dozens of events and activities celebrate literature and encourage literacy. The Book Fair features a country each year, organizing symposia and festivities that promote cultural understanding.
- Teatro Prometeo is a community theater with the mission of preserving the Spanish language and Hispanic culture. It offers a conservatory-style program that features a rigorous, well-rounded curriculum of study. Prometeo presents full productions and dramatic readings throughout the year. Courses and workshops are offered year-round.

Creative Writing Workshops

Creative writing workshops offer writers in our community a chance to share their work with a supportive, yet critical community of writers whose goal is continual development. All workshops are noncredit and open to everyone in the community. Twice a year, the Center's Writers Institute offers four days of intensive workshops on poetry, fiction, nonfiction, publishing and more. These are complemented by readings and festive gatherings.

Literacy Initiatives

The Center's literacy-based initiatives include One Book, One Community; One Picture Book, One Community; First Readers; El Club de Lectores; and The Big Read, a nationwide reading initiative funded by the National Endowment for the Arts in partnership with Arts Midwest and the Institute of Museum and Library Services. They encourage an appreciation for books with the goal of fostering dialogue in the community and enhancing the reading skills of children and adults.

Miami Book Fair

The acclaimed Miami Book Fair, the nation's finest and largest literary gathering for more than three decades, held in November to MDC's Wolfson Campus in downtown Miami. Attended by more than 200,000 people over eight days, Miami Book Fair features more than 600 authors and poets from around the world with readings from new works in English, Spanish and Creole and panel discussions on many topics. Other highlights include author readings, storytelling, and interactive experiences for children and teens, as well as panels on comics and graphic novels. Founded in 1984 by Miami Dade College and partners, Miami Book Fair engages the community through inclusive, accessible programs that promote reading and support writers year-round. The annual eight-day festival has grown into the largest and most comprehensive community-rooted literary gathering in the United States generating discourse on contemporary literature and current issues of international importance.

The Honors College

The Honors College is a collegewide community of student and faculty scholars who collaborate in an intellectually stimulating, enriching, challenging and supportive environment. Housed at Wolfson, North, Kendall, Homestead, and Padrón campuses, The Honors College provides an academically rich curriculum with special scholarship, and enrichment and service opportunities. The Honors College encourages critical and creative thinking and intellectual curiosity in an array of programs and disciplines. Students study in small class settings and work closely with Honors faculty. The Honors College expects its students to take advantage of the many enrichment opportunities provided. These include cultural and community activities, leadership development programs, internships, national tours, study abroad programs and colloquia. The Padrón Campus offers the Honors Dual Language Program, which mirrors the rigorous curriculum of the other campuses. This program offers courses in English or Spanish for students who demonstrate mastery of both languages. Students receive personalized guidance in preparing applications for competitive scholarship awards and transfer admission to prestigious private and public universities. In addition, The Honors College offers exemplary models of learning, a notable speakers series, discipline-specific honors seminars and student forums. Components of the program include:

The Honors College Fellows award, a stackable scholarship covering in-state tuition, fees and expenses;

1. Opportunities to attend an array of cultural events;
2. Attendance and participation of students and faculty at the annual meetings of the National Collegiate Honors Council, as well as the Regional and Florida Collegiate Honors Council meetings;

3. Transfer admission and scholarship opportunities by upper-division colleges and universities upon graduation from The Honors College;
4. Membership in campus chapters of Phi Theta Kappa International Honor Society for students with a GPA of 3.5 or higher;
5. Opportunities to participate in international study experiences and internships abroad;
6. Recognition as a graduate of The Honors College at commencement and designation on transcript and diploma with 36 credits in honors courses and a 3.5 GPA or higher;
7. Internships and Service Learning opportunities provided in related fields of study.

Additionally, the Honors Dual Language program offers:

1. A global perspective in all classes
2. Proficiency in two languages
3. Opportunity for a global experience as an exchange student or intern.

All of the activities associated with The Honors College are designed to inspire and challenge students in their studies and to provide support and encouragement in their quest for knowledge, academic and personal fulfillment. Students should contact the Dean of The Honors College or the Honors Director on the corresponding campus for specific information. Students may also visit the website for additional information at www.mdc.edu/honorscollege.

MEED Program

The MEED Program (Model for Enhanced Employment Development) has served students with disabilities in Miami-Dade County with distinction for more than 25 years and has received a congratulatory Proclamation from the Office of the Mayor saluting its success. The Program has been redesigned as a national model in employability training, enhanced with the development of digital technology skills and achievement of excellence in professional skills. The MEED Program's goal is to open doors to competitive employment opportunities. There are two distinct elements of the Program which include (1) employment assistance; e.g., effective résumé development, strategic job searching, defining accommodative needs in the workplace, etc., and (2) employment development; e.g., working with business and industry and agencies throughout the County to expand inclusive employment opportunities. To learn more about the MEED Program, students are invited to visit the MEED webpage or call 305-237-3072.

Outreach Program

The College endeavors to provide college credit and noncredit classes to residents of Miami-Dade County who find it more convenient to attend a neighborhood center than to travel to a campus. These courses are fully accredited and follow the same curriculum as on-campus courses. Classes are held in community schools, businesses, municipal agencies and other close-to-home locations. The smaller classes provide opportunities for increased interaction with instructors. Students who attend outreach classes also find a strong network of support from fellow classmates.

New World School of the Arts (Wolfson Campus)

New World School of the Arts is a comprehensive college program and full-time high school preparing students for professional careers in dance, music, theater and the visual arts. The program, created by the Florida Legislature in 1984 as a Center of Excellence in the Arts, is an educational partnership of the University of Florida, Miami Dade College and Miami-Dade County Public Schools. Through its sponsoring institutions, New World School of the Arts awards the Bachelor of Music, Bachelor of Fine Arts degrees and Associate in Arts, as well as high school diplomas. Students are admitted on the basis of talent and commitment as demonstrated through audition or portfolio presentation. The school is located at Wolfson Campus in downtown Miami.

MDC Online

MDC Online has provided award winning online credit courses to students since 1999. The online courses and programs offered by MDC Online are an excellent alternative to classroom-based instruction for the students of Miami Dade College. We take pride in a diverse community of engaged, mature students. Our online courses use a learning management system (LMS) called Canvas and the internet to deliver instruction and to support class work. Our courses use the latest interactive technology, and provide forums for group study and support. While online courses do not require time on campus, the courses are equivalent to face-to-face courses, meeting the same competencies and learning outcomes. Students can take one course or pursue an entire two-year or four-year degree through MDC Online. MDC Online's course offerings and degree programs allow you to maintain your personal and career schedule while earning the same regionally accredited degree that a student at our MDC campuses are awarded. At MDC Online, a staff of technical and functional professionals supports the development of new online courses, the teaching of existing online courses, the students in online courses, and the technology

used in these courses.

Earn a Degree on Your Time, Online!

MDC Online courses and degree programs were created by making the MDC student's online learning experience our top priority – offering convenience, flexibility and support. These high-quality online courses and degree programs are designed to match workforce needs to ensure that our graduates acquire the knowledge, skills and credentials for a best-fit job.

Online Program Standards

MDC Online subscribes to the following best practice distance learning tenants as prescribed in the following publications:

- Distance and Correspondence Education Policy Statement (09/2020), Southern Association of Colleges and Schools Commission on Colleges (SACSCOC)
- Principles of Good Practice-The Foundation for Quality of Southern Regional Education Board's Electronic Campus
- Five Pillars of Quality Online Education, Online Learning Consortium
- The Quality Matters Higher Education Rubric, Sixth Edition
- MDC Online is an active member of the Florida Virtual Campus

For more information and assistance, please contact: MDC Online - Student Support Center @ 305-237-3800 or Email Us @ online@mdc.edu

Fitness, Aquatic and Wellness Center (North, Kendall and Wolfson Campuses)

The College has several Wellness Centers, located on the North, Kendall and Wolfson campuses. These programs are designed to meet the wellness needs of faculty/staff, students and the community. The centers have the capability to perform various health/fitness assessments, including sub-maximal cardiovascular testing, blood pressure measurement, body composition, muscular strength and flexibility. Each center has a variety of cardiovascular and strength training equipment as well as an array of free-weights. The MDC Fitness, Aquatic and Wellness Center offers:

- Convenience - Work out before or after work or class, or even on lunch breaks
- Flexibility - The Wellness Center is open Monday - Friday and has hours to suit all types of schedules
- Affordability - We offer unbeatable monthly, term, or yearly membership rates

Our Fitness & Wellness Centers also offer a wide array of group exercise classes. Whether you want to improve your cardiorespiratory endurance, increase your muscle mass, or lose those extra pounds, our fitness classes will help you accomplish your goals. In addition, the Kendall and North campus also have a fully equipped, state of the art Aquatic Facility that is available to all students and employees with an active Aquatic Membership. For more information about each campus facility visit the appropriate link below:

- Kendall Campus Fitness, Aquatic and Wellness Center - <https://www.mdc.edu/kendallfitness/>
- North Campus Aquatic and Fitness Center - <https://www.mdc.edu/northfitness/>
- Wolfson Campus Fitness and Wellness Center - <https://www.mdc.edu/wolfsonfitness/>

Study Abroad Programs

Miami Dade College is a member institution of the College Consortium for International Studies (CCIS). A cooperative consortium arrangement affords reciprocal access for MDC students to participate in study abroad programs offered by other member institutions and earn college credit. The CCIS is a nationwide partnership of more than 100 membership colleges and universities worldwide, including two- and four-year, public and private. This partnership offers American undergraduates a choice of more than 100 study-abroad programs in more than 20 countries. CCIS offers semester and summer programs. MDC students benefit from other institutional partnerships the college has with international organizations, such as CIEE, that sponsor events and scholarships for our students. Miami Dade College also offers faculty-led short-term study abroad programs in numerous disciplines to various countries around the globe. Program offerings vary yearly. Traditionally, MDC faculty have led programs in the following disciplines:

- Anatomy & Physiology
- Anthropology
- Art & Humanities

- Culinary Sciences
- Environmental Sciences
- Graphic Design
- Literature
- Multicultural Communications & Relations
- Spanish Literature
- World Languages

Participation is not automatic. Students must apply through the MDC Office of International Education located at the Wolfson Campus. Virtual advising appointments are available. Most programs require a minimum 2.5 GPA. No previous study or knowledge of another language is required for most programs. If a student is eligible for financial aid, this aid may be used for study abroad. After acceptance to a program, the restricted registration for courses abroad is completed with the assistance and authorization of the Office of International Education. Some programs offer a "homestay" option (living with a local family or individual), which accelerates language acquisition and provides in-depth knowledge of the host culture. Course content is usually country-based and many courses are fully compatible with the MDC curriculum. Course descriptions and information on the classes offered in each pro-gram are detailed during the application process. For more information about the study abroad programs and scholarship opportunities, please visit www.mdc.edu/studyabroad.

Time-Saving Degree Opportunities

Miami Dade College encourages students to accelerate their education by providing time-saving programs to shorten the time necessary to complete an Associate degree. The articulated acceleration mechanism includes dual enrollment, early admission, advanced placement, credit-by-examination and the International Baccalaureate Program among others. These accelerated options can save a student valuable time and money because they provide an alternative way of earning credit at MDC and the opportunity to earn a degree more quickly.

Prior Learning Assessment (PLA)

Save money by earning credits for the things you already know. Through military or corporate training, previous learning, or even volunteer experiences, you may already have the knowledge you need to get ahead. Through PLA opportunities, you can prove that knowledge and fulfill the requirements of your degree program. In short, PLA is the evaluation and assessment of an individual's life learning for college credit, certification, or advanced standing toward further education or training. For additional information relating to PLA, contact the Office of Prior Learning at pla@mdc.edu or visit <https://www.mdc.edu/pla/>.

Competency Based Education Programs

MDC Accelerate competency-based education (CBE) programs are designed in an accelerated and flexible online format for working professionals. Each program was designed with input from industry advisors. While enrolled in the program, students will have opportunities to network with business leaders. How Does It Work? The first step is to submit an application for admission. After taking the CBE readiness assessment and meeting with a department chair or advisor, the next step is to register for a CBE program. Once in the program, students complete online performance assessments to demonstrate competency. By demonstrating competency, a student sets his or her own pace through the course. MDC Accelerate allows students to spend more time with the material they need and less time with the material they already know. To get started, apply to MDC and register for your CBE program of interest.

Foreign-Trained Professionals Program

Let your foreign professional experience work for you. With a longstanding commitment to serving multiple and diverse communities, Miami Dade College has initiated the Foreign-Trained Professionals Program (FTP). Designed to assist foreign nationals, as well as U.S. nationals with foreign credentials, FTP helps Foreign-Trained professionals obtain the necessary U.S. credentials to continue their careers. Courses and programs are offered on campuses and online. Benefits of FTP include:

- Evaluation of foreign credentials
- College credit awarded for applicable foreign work experience
- Opportunity to earn an MDC certificate or degree
- Quick return to your career of choice

Dual Enrollment and Early Admission (See Special Admissions Categories)

The Dual Enrollment program allows high school students (or home education students) to earn college credit and credit toward a high school diploma simultaneously. The college credit may be applied toward a postsecondary diploma, or a certificate or degree at a Florida public institution. The Dual Enrollment program is an opportunity to take challenging courses and accelerate education opportunities. Students who successfully complete dual enrollment courses will save time in obtaining their college degree, and save money as well, because these students are exempt from the payment of registration, tuition and laboratory fees. To enroll in courses through the dual enrollment program, students must demonstrate readiness for college-level coursework. Eligibility criteria take both GPA and assessment of communication and computation skills into consideration. The high school must grant permission for the student to enroll in these courses, thereby agreeing to accept these college courses to meet high school graduation requirements. Early admission is a form of dual enrollment through which eligible high school students enroll at the college on a full-time basis. The courses these students take are creditable toward a high school diploma and the certificate or associate degree. Students selected for early admission or dual enrollment may begin their studies in any term, provided that they complete the dual enrollment admission, advisement and registration procedures and receive permission from their high school.

Alternative Ways of Earning Credit Through Standardized Examination

- Advanced Placement (AP)
- American Council on the Teaching of Foreign Languages (ACTFL)
- Cambridge Advanced International Certificate of Education Examination (AICE)
- Caribbean Advanced Proficiency Examination (CAPE)
- Certified Professional Secretary Examination (CPS)
- College-Level Examination Program (CLEP)
- DANTES Subject Standardized Tests (DSSTs)
- Defense Language Proficiency Test (DLPT)
- International Baccalaureate (IB)
- UExcel

Miami Dade College awards college credit for standardized examinations that document the required knowledge and competencies for one or more subject areas. Evaluations of examinations are made after the student has been admitted to the College. Official score reports must be sent directly from the testing agencies to the College's Transcript Processing Services Office. Awarded credit based on the College's approved course equivalents will appear on the student's permanent record and on the student's official College transcript as earned credit only. There will be no indication of grades or quality points and duplicate credit is not awarded (State Rule 6A-10.024[8]). Miami Dade College uses the minimum scores, credits, and guidelines for awarding credit for exams established by the State of Florida's Articulation Coordinating Committee (ACC). For additional information please visit the Testing Criteria Credit-by-Exam website (<https://www.mdc.edu/main/testing/criteria/credit-by-exam.aspx>).

Institutional (Departmental) Credit-by-Examination

Students who have been admitted to the College may receive credit for courses through departmental examinations. Applications for this type of credit are available from the Office of Prior Learning Assessment and must be approved first by the appropriate academic department. Subsequently, the registration must be completed at the Registrar's Office and fees need to be paid by each term's published deadline. Credits for departmental examination are not included in any computation of credit load for full-time or part-time student status. Institutional credit-by-examination will become a part of the student's permanent record at the conclusion of the term in which it is awarded. Grades of A, B, C or D will be assigned for college credits earned by examination and will be computed in the student's GPA. A nonrefundable fee of \$30 per credit will be charged for each examination administered.

INSTITUTIONAL ENGAGEMENT AND TECHNOLOGY

Computer Services

Miami Dade College provides students and faculty with a state-of-the-art computing and telecommunication infrastructure. The College's campuses and centers are interconnected by a highspeed gigabit fiber network backbone supporting voice, video and data. The network currently provides 10Gbps bandwidth connection to the Internet from diverse sites using multiple service providers. Wireless connectivity for mobile computing is available

in classrooms, libraries, conference centers, and outdoor locations. All classrooms are augmented with a variety of technological tools including computers and digital projectors that can enhance the learning experience. In addition to extensive computing facilities at each College location, the College also offers a wide array of online services for students. The MyMDC student portal and mobile app allow students to manage their college experience via web or mobile phone. The portal and mobile app provide self-services in admissions, orientation, registration, advising, financial aid, transcript requests, term grades, credit card payments, and many more. The Blackboard Learning Management System facilitates the creation and delivery of online instructions.

Institutional Advancement (*District Office*)

The College secures essential support for student scholarships, STEM education, arts and culture, entrepreneurship, workforce training, and other programs and projects through the Miami Dade College Resource Development Department, the Office of Alumni Relations, and the Miami Dade College Foundation, Inc.

Resource Development Department

The Resource Development Department identifies external sources of grant funding to support the programs and priorities of the College. The department works with College faculty, staff, and leadership to develop, prepare, and submit innovative grant proposals to public and private funding sources designed to promote excellence in teaching, learning, and institutional effectiveness. Resources obtained through grant awards help fund new and existing programs, special projects, student services, curriculum development, professional staff development, the construction of new facilities, exchange programs, research, new equipment, and student scholarships. The Resource Development Department also encourages public-private partnerships, and collaboration with industry, community-based organizations, and other educational institutions. In addition to handling all preaward processes and post-award compliance assistance for the College, the department serves as the sponsored programs entity, with all grant applications going through it for approval of the College's authorized organizational representative and College Board of Trustees.

Miami Dade College Office of Alumni Relations

The Office of Alumni Relations advances the goals, objectives and priorities of Miami Dade College by generating private financial support, building and maintaining relationships with alumni and donors, and engaging alumni and students to foster a lifelong intellectual and emotional connection between MDC and its graduates. Alumni are vital to the long-term success of Miami Dade College and supporting its strategic goals. With the Miami Dade College Foundation, the Office of Alumni Relations fosters and cultivates a common bond of pride, affinity and connectivity among alumni, students, prospective students and friends of the College through quality programs, services, events and programmatic initiatives.

Miami Dade College Foundation Inc.

Miami Dade College Foundation was chartered by the state of Florida in 1965 as a nonprofit 501(c)3 direct support organization of Miami Dade College. Governed and guided by an independent Board of Directors of more than a dozen community leaders, the Foundation raises awareness and financial resources for Miami Dade College to maintain open-door access to anyone who seeks an education, and to provide innovative and multicultural academic and cultural programs, all of which contribute to the vitality of our community. The Foundation ensures the mission of Miami Dade College is accomplished by pursuing funding opportunities that support MDC's 2015-2020 Strategic Plan priorities:

- Students Access and Success
- Educational Quality
- Institutional Agility

As state support for higher education continues to decline, the Foundation's efforts to identify alternative funding sources are vital to the future of MDC. The Foundation manages relationships with and seeks funding opportunities from individuals, private and family foundations, civic organizations and corporations. Gifts from these sources have established and continue to support scholarships, new programs, direct faculty support and critical capital improvement funds. Contributions to the Foundation are tax deductible under Section 170 of the Internal Revenue Code and are administered according to gift agreements and donor intentions. Numerous donations from many generous sources have contributed to the growth of the Foundation's endowment, which is approximately \$132 million. The endowment is comprised of more than 1,000 scholarship and program support donor accounts that

directly benefit the College.

Endowed Teaching Chairs

The Miami Dade College Endowed Teaching Chair program is dedicated solely to recognizing excellence in teaching and the first of its kind at a community college. The Endowed Teaching Chairs represent our institution's highest recognition of our faculty. Recipients of this award, past and present, have demonstrated to their peers the absolute definition of excellence in every aspect of teaching. Further, they have made student learning their top priority and, in doing so, help fulfill the mission of Miami Dade College. The Endowed Teaching Chairs have been made possible through the generous support of individuals, corporations and organizations committed to the art of teaching and are managed by the Miami Dade College Foundation. The Endowed Teaching Chair program awards each recipient \$22,500 over three years, allowing faculty to explore new teaching methods, develop new projects, purchase specialized or innovative teaching materials, enhance their technological expertise and further their own knowledge for the benefit of students. Since the program's inception in 1992, the Foundation has awarded more than 300 Endowed Teaching Chairs. A gift of an Endowed Teaching Chair is among the most important contributions that can be made to the College and the thousands who are educated at MDC.

BOARD OF TRUSTEES

Michael Bileca

Chair, MDC District Board of Trustees



Michael Bileca is a healthcare entrepreneur who in 2005 founded a Miami-based Dental Service Organization (DSO), Towncare Dental Partnership. Mr. Bileca grew the organization from start-up to 55 locations with 100 affiliated doctors supported by 1,000 employees. The organization merged with Dental Care Alliance in 2013. Mr. Bileca is currently an executive at Dental Care Alliance which supports over 260 affiliated practices throughout the United States.

In addition to his private sector experience, Mr. Bileca served in various capacities in the Florida House of Representatives from 2010 to 2018, where he maintained a focus on education. In his last term, he served as Chairman of the Education Committee. During his tenure, he sponsored and passed major education reform legislation in Florida.

Mr. Bileca is now the Executive Director of the Dennis Bileca Institute for Character and Excellence, a foundation that primarily funds schools focusing on virtue-based education with an emphasis on a classical liberal arts curriculum.

Mr. Bileca served on the Public Health Trust (PHT) Financial Recovery Board when it was instituted in 2011 to help turn around the struggling Jackson Health System. During his first year on the board of the PHT, and under the leadership of CEO Carlos Migoya, the System went from an \$82 million dollar loss to a surplus. In all the subsequent years that Mr. Bileca served as a Board member, the PHT/Jackson earned a financial surplus while maintaining its distinction as a world-class provider of quality medical care.

Mr. Bileca received an undergraduate degree in Business from Tulane University with a minor in philosophy and an MBA from Northwestern University and its acclaimed Kellogg School of Management.

Roberto Alonso*Vice Chair, MDC District Board of Trustees*

Roberto Alonso has always known the importance of a good education, strong family values and hard work. He has been a leader in education technology, helping to develop solutions that have impacted millions of students nationwide in K-12 education as a co-founder of Hero K12, a business founded to have a profound, positive impact on student outcomes, making school a better place to learn and grow. In 2017, Mr. Alonso and his team had a vision for an EdTech portfolio of integrated solutions to improve schools. They raised the capital and developed a school choice platform named SchoolMint, a student enrollment software that allows parents to easily merge their child's educational needs with the proper school environment. After nearly 20 years of traveling the country, helping thousands of schools and numerous school districts across the nation, Trustee Alonso decided to dedicate more time to his family and community. He returned to his family's roots in Miami real estate where he now works at Costa Realtors as the Vice President of Business Development.

Mr. Alonso is also heavily vested in his community, providing leadership as a board member of the Miami-Dade County Planning Advisory Board, President of Loch Lomond Homeowners Association, Past Vice-Chair of the Town of Miami Lakes' Charter Review Committee, Past Chair of the Town of Miami Lakes' Public Safety Committee, and Past President of the Royal Oaks Homeowners Association. He added to his community involvement and philanthropic endeavors through the founding of Miami Lakes Cars for a Cure, a non-for-profit organization that produces community events to raise funds for local cancer research, where he also serves as President and Event Operations Manager. Trustee Alonso was born in Hialeah, Florida, and raised in the surrounding area by his parents, both Cuban immigrants who were provided the chance for freedom in the United States thanks to Operation Pedro Pan which allowed unaccompanied minors to flee the newly seated communist regime in 1962. Mr. Alonso's father, Roberto Alonso, Sr., attended MDC for one year, having to end his enrollment in order to work multiple jobs to help his parents and family "make ends meet." This was the only higher education experience that he had, but he continued to better himself and instill in Roberto and his sisters the importance of an education beyond just high school.

Mr. Alonso graduated from Monsignor Pace High School in northwest Miami-Dade County and began attending MDC classes while working a full-time job, attaining an Associate in Arts in 2002. He later went on to FIU to earn a Bachelor in Business Administration, with a focus on Management Information Systems.

Jose Felix Diaz



Trustee Diaz is an Executive Vice President and Managing Partner at Ballard Partners, a Partner at Panza Maurer Law, and a Co-Founder of the Miami Tech Summit. He was previously elected as a member of the Florida House of Representatives, representing both Districts 115 and 116. Diaz earned his bachelor's degree in political science and English from the University of Miami and his juris doctor from Columbia University.

Marcell Felipe



Marcell Felipe the founder of Marcell Felipe Attorneys. He is an international tax and corporate lawyer who has advised governments, heads of state, industry leaders and many others. In 2006, Mr. Felipe co-founded a company which today owns Miami TV stations Channel 41 and 48. He has since sold his stake in the company, but continues to provide advice to its management and shareholders. Mr. Felipe is also the founder of the Inspire America Foundation, an organization dedicated to promoting democracy in Cuba and the Americas. In 2018, he was elected to serve as Chairman of the new American Museum of the Cuban Diaspora in Miami. Previously, he has served on the Board of the Miami Dade County Housing Finance Authority and in the Board of several non-profit and for-profit organizations.

Mr. Felipe graduated from the University of Pittsburgh School of Law, where he served as the Business Manager for the University of Pittsburgh Law Review and Journal of Commerce for the Summer 1996 Editions. Mr. Felipe completed Harvard Law School's Program of Instruction for Lawyers in 1995 and served as an intern to the Honorable W. Standish of the US District Court for Western Pennsylvania. He is a member of the Florida Bar and is a native Spanish speaker.

Ismare Monreal



Ismare Monreal is acting vice president and dean of students at Johnson & Wales University. A proud MDC alumna, she is an experienced leader having spent more than 21 years as a higher education administrator implementing inclusive program strategies, championing collaboration, refining operations and driving organizational growth.

She has worked extensively on issues of representation, access to education, community service, and servicing at-risk populations. Her areas of focus include nonprofit organizations, student affairs, leadership development, and admissions. Previously, she was a senior legislative assistant at the Florida House of Representatives.

Ms. Monreal has volunteered with the United Way of Miami-Dade, Archdiocese of Miami Marriage and Family Ministry and A Safe Haven for Newborns. She earned an associate degree from MDC, a bachelor's from Florida International University and master's in business administration from the University of Phoenix.

Juan Segovia



Trustee Segovia is a Homicide Detective in the Cold Case Unit and a Crisis and Hostage Negotiator for the Miami-Dade Police Department. He is the recipient of the 2014 Employee Excellence Award and the Distinguished Officer of the Month Award. Segovia earned his bachelor's degree in criminal justice and his master's degree in international disaster management from Florida International University.

David J. Smith



David J. Smith is a Special Assistant to the Combatant Commander and Speechwriter for United States Southern Command. He also serves in the United States Marine Corps Reserve as a Judge Advocate and Civil Affairs Officer. Previously, Smith worked in private practice and as an Assistant State Attorney in Miami-Dade County. He earned his bachelor's degree in political science and LL.M. from the University of Miami and Juris Doctor from Suffolk University Law School.

ADMINISTRATION AND FACULTY

District Administration - Executive

PUMARIEGA, MADELINE, College President, College President, Wolfson Campus, BA, St. Thomas University, MED, Florida Atlantic University.
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Hialeah Campus

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 Homestead Campus, BA,
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 DALBERRY, BARBARA, Grant
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 FRAGA, YILIAN, Dir TRIO Stu
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Kendall Campus

Administration - Executive

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Kendall Campus

Administration

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North Campus Administration

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